

EEBus SPINE Technical Specification

Resource Specification

Version 1.3.0

Cologne, 2023-08-31

EEBus Initiative e.V.

Deutz-Mülheimer-Straße 183
51063 Cologne
GERMANY

Rue d'Arlon 25
1050 Brussels
BELGIUM

Phone: +49 221 / 715 938 - 0
Fax: +49 221 / 715 938 - 99

info@eebus.org
www.eebus.org

District court: Cologne
VR 17275

Terms of use for publications of EEBus Initiative e.V.**General information**

The specifications, particulars, documents, publications and other information provided by the EEBus Initiative e.V. are solely for general informational purposes. Particularly specifications that have not been submitted to national or international standardisation organisations by EEBus Initiative e.V. (such as DKE/DIN-VDE or IEC/CENELEC/ETSI) are versions that have not yet undergone complete testing and can therefore only be considered as preliminary information. Even versions that have already been published via standardisation organisations can contain errors and will undergo further improvements and updates in future.

Liability

EEBus Initiative e.V. does not assume liability or provide a guarantee for the accuracy, completeness or up-to-date status of any specifications, data, documents, publications or other information provided and particularly for the functionality of any developments based on the above.

Copyright, rights of use and exploitation

The specifications provided are protected by copyright. Parts of the specifications have been submitted to national or international standardisation organisations by EEBus Initiative e.V., such as DKE/DIN-VDE or IEC/CENELEC/ETSI, etc. Furthermore, all rights to use and/or exploit the specifications belong to the EEBus Initiative e.V., Deutz-Mülheimer-Straße 183, 51063 Cologne, Germany and can be used in accordance with the following regulations:

The use of the specifications for informational purposes is allowed. It is therefore permitted to use information evident from the contents of the specifications. In particular, the user is permitted to offer products, developments and/or services based on the specifications.

Any respective use relating to standardisation measures by the user or third parties is prohibited. In fact, the specifications may only be used by EEBus Initiative e.V. for standardisation purposes. The same applies to their use within the framework of alliances and/or cooperations that pursue the aim of determining uniform standards.

Any use not in accordance with the purpose intended by EEBus Initiative e.V. is also prohibited.

Furthermore, it is prohibited to edit, change or falsify the content of the specifications. The dissemination of the specifications in a changed, revised or falsified form is also prohibited. The same applies to the publication of extracts if they distort the literal meaning of the specifications as a whole.

It is prohibited to pass on the specifications to third parties without reference to these rights of use and exploitation.

It is also prohibited to pass on the specifications to third parties without informing them of the authorship or source.

Without the prior consent of EEBus Initiative e.V., all forms of use and exploitation not explicitly stated above are prohibited.

Table of contents

Table of contents.....	3
List of figures	8
List of tables	13
1 Introduction.....	22
1.1 References.....	22
1.1.1 EEBUS SPINE documents	22
1.1.2 Other EEBUS documents	22
1.1.3 Other documents	22
1.1.4 Websites	22
1.1.5 Normative References.....	22
1.2 Terms and definitions.....	23
1.3 Document structure	25
1.4 How to read this document.....	26
1.4.1 Graphical representation	26
1.4.2 Namespaces	29
1.4.3 "datagram" vs. specific function – XMLs with different roots and scopes	29
1.4.4 Simplification of XML examples	30
1.4.5 Used requirement keywords.....	31
1.4.6 Presence indications in Feature Types	31
1.4.7 Presence indications in SmartEnergyManagementPs.....	31
2 Overall model hierarchy concept overview	32
2.1 Resource Type Definition Concept	33
2.1.1 Introduction.....	33
2.1.2 Device Model Facets	34
2.1.3 Vendor specific extensions.....	34
2.1.4 Feature Group	35
2.1.5 Specific usage	35
2.2 Class Concept	35
2.2.1 Standard Classes.....	35
2.2.2 Complex Classes	36
2.2.3 SPINE class hierarchy (Class / Sub-Class / Function-Group / Function / Element).....	36
3 Common technical details.....	37
3.1 Introduction.....	37
3.2 Time information (absolute / relative / recurring).....	37
3.3 List Data concept	37
3.4 Identifier concept	38
3.4.1 Introduction.....	38
3.4.2 Identifier rules	38
3.5 Multidimensional data representation with "flat" list-based data model.....	41
3.6 Restricted function exchange for list based functions	44
3.7 Relations between thematically connected SPINE classes	44
3.7.1 Relations between Alarm / ElectricalConnection / HVAC / LoadControl / Measurement / Setpoint / SupplyCondition / TariffInformation / Threshold / TimeTable.....	45
3.7.2 Relations between DirectControl / HVAC / LoadControl / OperatingConstraints / PowerSequences (SmartEnergyManagementPs) / TaskManagement	46

47	3.8	On the use of "label" and "description"	47
48	3.9	Empty elements as "tags"	48
49	3.10	Common data types	49
50	3.10.1	General	49
51	3.10.2	Time-related	62
52	3.10.3	Address-related	68
53	3.11	Result.....	70
54	4	Device model definitions.....	72
55	4.1	Device Types.....	72
56	4.1.1	ChargingStation	72
57	4.1.2	Dishwasher	72
58	4.1.3	Dryer	72
59	4.1.4	ElectricitySupplySystem	72
60	4.1.5	EnergyManagementSystem	72
61	4.1.6	EnvironmentSensor	72
62	4.1.7	Generic	73
63	4.1.8	HeatGenerationSystem	73
64	4.1.9	HeatSinkSystem	73
65	4.1.10	HeatStorageSystem	73
66	4.1.11	HVACController	73
67	4.1.12	Inverter.....	74
68	4.1.13	SubMeter.....	74
69	4.1.14	Washer	74
70	4.2	Entity Types	74
71	4.2.1	Battery.....	74
72	4.2.2	BatterySystem	74
73	4.2.3	CEM	74
74	4.2.4	ChargingOutlet	74
75	4.2.5	Compressor	75
76	4.2.6	ControllableSystem	75
77	4.2.7	DeviceInformation.....	75
78	4.2.8	DHWCircuit.....	75
79	4.2.9	DHWStorage	75
80	4.2.10	Dishwasher	75
81	4.2.11	Dryer	75
82	4.2.12	ElectricalImmersionHeater.....	76
83	4.2.13	ElectricalStorage	76
84	4.2.14	ElectricityGenerationSystem	76
85	4.2.15	ElectricityStorageSystem	76
86	4.2.16	EV.....	76
87	4.2.17	EVSE	76
88	4.2.18	Fan	76
89	4.2.19	GasHeatingAppliance	77
90	4.2.20	Generic	77
91	4.2.21	GridConnectionPointOfPremises	77
92	4.2.22	GridGuard	77

93	4.2.23	HeatingBufferStorage.....	77
94	4.2.24	HeatingCircuit.....	77
95	4.2.25	HeatingObject.....	78
96	4.2.26	HeatingZone	78
97	4.2.27	HeatPumpAppliance.....	78
98	4.2.28	HeatSinkCircuit	78
99	4.2.29	HeatSourceCircuit.....	78
100	4.2.30	HeatSourceUnit	78
101	4.2.31	Household	79
102	4.2.32	HVACController	79
103	4.2.33	HVACRoom	79
104	4.2.34	InstantDHWHeater	79
105	4.2.35	Inverter	79
106	4.2.36	OilHeatingAppliance.....	79
107	4.2.37	Pump	80
108	4.2.38	PV.....	80
109	4.2.39	PVESHybrid	80
110	4.2.40	PVString	80
111	4.2.41	PVSystem.....	80
112	4.2.42	RefrigerantCircuit	80
113	4.2.43	SmartEnergyAppliance	80
114	4.2.44	SolarDHWStorage.....	81
115	4.2.45	SolarThermalCircuit.....	81
116	4.2.46	SubMeterElectricity.....	81
117	4.2.47	TemperatureSensor.....	81
118	4.2.48	Washer	81
119	4.3	Feature Types	81
120	4.3.1	ActuatorLevel	81
121	4.3.2	ActuatorSwitch	83
122	4.3.3	Alarm	85
123	4.3.4	Bill	86
124	4.3.5	DataTunneling	92
125	4.3.6	DeviceClassification	94
126	4.3.7	DeviceConfiguration.....	97
127	4.3.8	DeviceDiagnosis.....	106
128	4.3.9	DirectControl	109
129	4.3.10	ElectricalConnection.....	112
130	4.3.11	Generic	120
131	4.3.12	HVAC.....	121
132	4.3.13	Identification	129
133	4.3.14	IncentiveTable	131
134	4.3.15	LoadControl	144
135	4.3.16	Measurement	149
136	4.3.17	Messaging.....	157
137	4.3.18	NetworkManagement	159
138	4.3.19	NodeManagement	159

139	4.3.20	OperatingConstraints	159
140	4.3.21	PowerSequences	165
141	4.3.22	Sensing.....	165
142	4.3.23	Setpoint	170
143	4.3.24	SmartEnergyManagementPs.....	173
144	4.3.25	StateInformation	220
145	4.3.26	SupplyCondition	222
146	4.3.27	TariffInformation	226
147	4.3.28	TaskManagement.....	226
148	4.3.29	Threshold.....	230
149	4.3.30	TimeInformation.....	233
150	4.3.31	TimeSeries	236
151	4.3.32	TimeTable	242
152	5	Class descriptions	246
153	5.1	Introduction.....	246
154	5.2	Complex Classes	246
155	5.2.1	IncentiveTable	246
156	5.2.2	NodeManagement	255
157	5.2.3	SmartEnergyManagementPs.....	255
158	5.3	Standard Classes.....	276
159	5.3.1	ActuatorLevel	276
160	5.3.2	ActuatorSwitch	278
161	5.3.3	Alarm	281
162	5.3.4	Bill	283
163	5.3.5	BindingManagement	292
164	5.3.6	DataTunneling	297
165	5.3.7	DeviceClassification	300
166	5.3.8	DeviceConfiguration	304
167	5.3.9	DeviceDiagnosis.....	313
168	5.3.10	DirectControl	318
169	5.3.11	ElectricalConnection.....	322
170	5.3.12	HVAC.....	334
171	5.3.13	Identification	347
172	5.3.14	LoadControl	351
173	5.3.15	Measurement	363
174	5.3.16	Messaging.....	376
175	5.3.17	NetworkManagement	380
176	5.3.18	OperatingConstraints	405
177	5.3.19	PowerSequences	414
178	5.3.20	Sensing.....	441
179	5.3.21	Setpoint	447
180	5.3.22	SubscriptionManagement	454
181	5.3.23	StateInformation	459
182	5.3.24	SupplyCondition	461
183	5.3.25	TariffInformation	467
184	5.3.26	TaskManagement.....	492

185	5.3.27	Threshold.....	498
186	5.3.28	TimeInformation.....	503
187	5.3.29	TimeSeries	508
188	5.3.30	TimeTable	516
189	5.3.31	UseCaseInformation	524
190	5.3.32	Version.....	527
191		Annex A - SPINE XSDs	529
192	A.1	Complex Classes	529
193	A.1.1	IncentiveTable	529
194	A.1.2	NodeManagement	537
195	A.1.3	SmartEnergyManagementPs.....	549
196	A.2	Standard Classes.....	559
197	A.2.1	ActuatorLevel	559
198	A.2.2	ActuatorSwitch	560
199	A.2.3	Alarm	561
200	A.2.4	Bill	562
201	A.2.5	BindingManagement	568
202	A.2.6	DataTunneling	569
203	A.2.7	DeviceClassification	570
204	A.2.8	DeviceConfiguration	571
205	A.2.9	DeviceDiagnosis.....	574
206	A.2.10	DirectControl	576
207	A.2.11	ElectricalConnection.....	578
208	A.2.12	HVAC.....	583
209	A.2.13	Identification	588
210	A.2.14	LoadControl	590
211	A.2.15	Measurement	593
212	A.2.16	Messaging.....	598
213	A.2.17	NetworkManagement	599
214	A.2.18	OperatingConstraints	604
215	A.2.19	PowerSequences	607
216	A.2.20	Sensing.....	614
217	A.2.21	Setpoint	616
218	A.2.22	StateInformation	618
219	A.2.23	SubscriptionManagement	620
220	A.2.24	SupplyCondition	621
221	A.2.25	TariffInformation	624
222	A.2.26	TaskManagement	631
223	A.2.27	Threshold.....	634
224	A.2.28	TimeInformation.....	636
225	A.2.29	TimeSeries	637
226	A.2.30	TimeTable	640
227	A.2.31	UseCaseInformation	642
228	A.2.32	Version.....	643
229	A.3	Further XSDs	644
230	A.3.1	CommandCommonDefinitions.....	644

231	A.3.2	CommandFrame	650
232	A.3.3	CommonDataTypes	651
233	A.3.4	Datagram	665
234	A.3.5	Result	666
235	A.3.6	Overview example	666
236	Annex B - Resource Definitions Overview		668
237	B.1	Introduction	668
238	B.2	Device Types	668
239	B.3	Entity Types	668
240	B.4	Feature Types	671
241	B.5	Complex Classes	672
242	B.6	Standard Classes	673
243	B.7	Identifier list	674
244			

245 List of figures

246	Figure 1: XSD "simple type"	26
247	Figure 2: XSD "complex type" with no content	26
248	Figure 3: XSD "complex type" with content	27
249	Figure 4: XSD element with type	27
250	Figure 5: XSD sequence	27
251	Figure 6: XSD choice	27
252	Figure 7: XSD all	27
253	Figure 8: Optional vs. mandatory	28
254	Figure 9: Lists / multiplicity	28
255	Figure 10: XSD expanded view	28
256	Figure 11: Datagram structure (red line: "way" to the functions)	29
257	Figure 12: Payload structure (red line: "way" to the functions)	30
258	Figure 13: Dependencies of the standardised types and defined data structures	32
259	Figure 14: SPINE device model example	33
260	Figure 15: SPINE class hierarchy	36
261	Figure 16: Absolute or relative time type	37
262	Figure 17: Non-list data function example	37
263	Figure 18: List data function example	38
264	Figure 19: Excerpt of the list element "measurementData" of the SPINE function	
265	measurementListData	42
266	Figure 20: SPINE identifiers can create a multidimensional space of data.	42
267	Figure 21: The flat data structure of Figure 19 can also be rearranged (virtually) to emphasize the	
268	priorities of the identifiers.	44
269	Figure 22: Relations between Alarm / ElectricalConnection / HVAC / LoadControl / Measurement /	
270	Setpoint / SupplyCondition / TariffInformation / Threshold / TimeTable	46
271	Figure 23: Relations between DirectControl / HVAC / LoadControl / OperatingConstraints /	
272	PowerSequences (SmartEnergyManagementPs) / TaskManagement	47
273	Figure 24: Empty elements example	48
274	Figure 25: ScaledNumberType	51

275	Figure 26: ScaledNumberSetType	51
276	Figure 27: ScaledNumberRangeType	52
277	Figure 28: PossibleOperationsType	62
278	Figure 29: FunctionPropertyType	62
279	Figure 30: TimePeriodType	63
280	Figure 31: TimestampIntervalType	63
281	Figure 32: DaysOfWeekType	65
282	Figure 33: AbsoluteOrRecurringTimeType	67
283	Figure 34: RecurrenceInformationType	68
284	Figure 35: DeviceAddressType	69
285	Figure 36: EntityAddressType	69
286	Figure 37: FeatureAddressType	70
287	Figure 38: resultData function overview	70
288	Figure 39: IncentiveTable example (single incentiveSlot) for different stacked tiers	133
289	Figure 40: IncentiveTable example (single incentiveSlot) for different unique tiers	134
290	Figure 41: IncentiveTable function-group overview	246
291	Figure 42: incentiveTableData function overview	247
292	Figure 43: IncentiveTableIncentiveSlotType overview	248
293	Figure 44: incentiveTableDescriptionData function overview	249
294	Figure 45: IncentiveTableDescriptionTierType overview	251
295	Figure 46: incentiveTableConstraintsData overview	254
296	Figure 47: SmartEnergyManagementPs function-group overview	256
297	Figure 48: Concept of a power sequence	257
298	Figure 49: smartEnergyManagementPsData function overview, part 1	261
299	Figure 50: smartEnergyManagementPsData function overview, part 2	261
300	Figure 51: smartEnergyManagementPsData function overview, part 3	262
301	Figure 52: smartEnergyManagementPsData function overview, part 4	263
302	Figure 53: smartEnergyManagementPsConfigurationRequestCall function overview	272
303	Figure 54: smartEnergyManagementPsPriceData function overview	273
304	Figure 55: smartEnergyManagementPsPriceCalculationRequestCall function overview	275
305	Figure 56: ActuatorLevel function-group overview	276
306	Figure 57: actuatorLevelData function overview	276
307	Figure 58: actuatorLevelDescriptionData function overview	277
308	Figure 59: ActuatorSwitch function-group overview	279
309	Figure 60: actuatorSwitchData function overview	279
310	Figure 61: actuatorSwitchDescription function overview	280
311	Figure 62: Alarm function-group overview	281
312	Figure 63: alarmListData function overview	281
313	Figure 64: Bill function-group overview	283
314	Figure 65: billListData function overview	284
315	Figure 66: billListData function detail 1, "total"	284
316	Figure 67: billListData function detail 2, "position"	285
317	Figure 68: billConstraintsListData function overview	290
318	Figure 69: billDescriptionListData function overview	291
319	Figure 70: BindingManagement function-group overview (data)	293
320	Figure 71: BindingManagement function-group overview (call)	293

321	Figure 72: bindingManagementEntryListData function overview	293
322	Figure 73: bindingManagementRequestCall function overview	295
323	Figure 74: bindingManagementDeleteCall function overview	296
324	Figure 75: DataTunneling function-group overview (call).....	297
325	Figure 76: dataTunnelingCall function overview.....	299
326	Figure 77: DeviceClassification function-group overview	300
327	Figure 78: deviceClassificationManufacturerData function overview	301
328	Figure 79: deviceClassificationUserData function overview	303
329	Figure 80: DeviceConfiguration function-group overview	304
330	Figure 81: deviceConfigurationKeyValueListData function overview	305
331	Figure 82: deviceConfigurationKeyValueDescriptionListData function overview	307
332	Figure 83: deviceConfigurationKeyValueConstraintsListData function overview.....	310
333	Figure 84: deviceConfigurationKeyValueConstraintsListData function detail	311
334	Figure 85: DeviceDiagnosis function-group overview.....	313
335	Figure 86: deviceDiagnosisStateData function overview	314
336	Figure 87: deviceDiagnosisHeartbeatData function overview	316
337	Figure 88: deviceDiagnosisServiceData function overview.....	317
338	Figure 89: DirectControl function-group overview	318
339	Figure 90: directControlActivityListData function overview	319
340	Figure 91: directControlDescriptionData function overview	321
341	Figure 92: ElectricalConnection function-group overview.....	322
342	Figure 93: electricalConnectionParameterDescriptionListData function overview.....	323
343	Figure 94: electricalConnectionPermittedValueSetListData function overview.....	326
344	Figure 95: electricalConnectionCharacteristicListData function overview	328
345	Figure 96: electricalConnectionStateListData function overview	331
346	Figure 97: electricalConnectionStateDescriptionData function overview	333
347	Figure 98: HVAC function-group overview.....	335
348	Figure 99: hvacSystemFunctionListData function overview	335
349	Figure 100: hvacSystemFunctionOperationModeRelationListData function overview.....	337
350	Figure 101: hvacSystemFunctionSetpointRelationListData function overview	338
351	Figure 102: hvacSystemFunctionPowerSequenceRelationListData function overview.....	339
352	Figure 103: hvacSystemFunctionDescriptionListData function overview.....	340
353	Figure 104: hvacOperationModeDescriptionListData function overview.....	342
354	Figure 105: hvacOverrunListData function overview.....	344
355	Figure 106: hvacOverrunDescriptionListData function overview	345
356	Figure 107: Identification function-group overview	347
357	Figure 108: identificationListData function overview	347
358	Figure 109: sessionIdentificationListData function overview	349
359	Figure 110: sessionMeasurementRelationListData function overview	350
360	Figure 111: LoadControl function-group overview	352
361	Figure 112: loadControlNodeData function overview	352
362	Figure 113: loadControlEventListData function overview	353
363	Figure 114: loadControlStateListData function overview	355
364	Figure 115: loadControlLimitListData function overview	357
365	Figure 116: loadControlLimitConstraintsListData function overview	359
366	Figure 117: loadControlLimitDescriptionListData function overview	361

367	Figure 118: Measurement function-group overview	363
368	Figure 119: measurementListData function overview	364
369	Figure 120: measurementSeriesListData function overview	368
370	Figure 121: measurementConstraintsListData function overview	371
371	Figure 122: measurementDescriptionListData function overview	372
372	Figure 123: measurementThresholdRelationListData function overview	375
373	Figure 124: Messaging function-group overview	376
374	Figure 125: messagingListData function overview	376
375	Figure 126: NetworkManagement function-group overview (call)	380
376	Figure 127: NetworkManagement function-group overview (data)	380
377	Figure 128: networkManagementAddNodeCall function overview	382
378	Figure 129: networkManagementRemoveNodeCall function overview	384
379	Figure 130: networkManagementModifyNodeCall function overview	385
380	Figure 131: networkManagementScanNetworkCall function overview	387
381	Figure 132: networkManagementDiscoverCall function overview	388
382	Figure 133: networkManagementAbortCall function overview	389
383	Figure 134: networkManagementProcessStateData function overview	390
384	Figure 135: networkManagementJoiningModeData function overview	392
385	Figure 136: networkManagementReportCandidateData function overview	393
386	Figure 137: networkManagementDeviceDescriptionListData function overview, part 1	394
387	Figure 138: networkManagementDeviceDescriptionListData function overview, part 2	395
388	Figure 139: networkManagementEntityDescriptionListData function overview, part 1	399
389	Figure 140: networkManagementEntityDescriptionListData function overview, part 2	399
390	Figure 141: networkManagementFeatureDescriptionListData function overview, part 1	401
391	Figure 142: networkManagementFeatureDescriptionListData function overview, part 2	402
392	Figure 143: networkManagementFeatureDescriptionListData function overview, part 3	403
393	Figure 144: OperatingConstraints function-group overview	406
394	Figure 145: operatingConstraintsInterruptListData function overview	407
395	Figure 146: operatingConstraintsDurationListData function overview	408
396	Figure 147: operatingConstraintsPowerDescriptionListData function overview	409
397	Figure 148: operatingConstraintsPowerRangeListData function overview	411
398	Figure 149: operatingConstraintsPowerLevelListData function overview	412
399	Figure 150: operatingConstraintsResumImplicationListData function overview	413
400	Figure 151: Concept overview of power sequences: Power P over time t; sequences are constituted	
401	of slots	415
402	Figure 152: Example of a device that first consumes and later on generates power. In this case	
403	"consumption" is assigned a positive value.	416
404	Figure 153: Example of a sequence with only one slot with some possible parameter boundaries..	417
405	Figure 154: PowerSequences function-group overview (data)	418
406	Figure 155: PowerSequences function-group overview (call)	418
407	Figure 156: powerTimeSlotScheduleListData function overview	420
408	Figure 157: powerTimeSlotValueListData function overview	422
409	Figure 158: powerTimeSlotScheduleConstraintsListData function overview	424
410	Figure 159: powerSequenceAlternativesRelationListData function overview	426
411	Figure 160: powerSequenceDescriptionListData function overview	427
412	Figure 161: powerSequenceStateListData function overview	430

413	Figure 162: powerSequenceScheduleListData function overview	432
414	Figure 163: powerSequenceScheduleConstraintsListData function overview	433
415	Figure 164: powerSequencePriceListData function overview	435
416	Figure 165: powerSequenceSchedulePreferenceListData function overview	436
417	Figure 166: powerSequenceNodeScheduleInformationData function overview	437
418	Figure 167: powerSequenceScheduleConfigurationRequestCall function overview.....	439
419	Figure 168: powerSequencePriceCalculationRequestCall function overview	440
420	Figure 169: Sensing function-group overview.....	441
421	Figure 170: sensingListData function overview	442
422	Figure 171: sensingDescriptionData function overview	445
423	Figure 172: Setpoint function-group overview	447
424	Figure 173: setpointListData function overview	448
425	Figure 174: setpointConstraintsListData function overview.....	450
426	Figure 175: setpointDescriptionListData function overview.....	452
427	Figure 176: SubscriptionManagement function-group overview (data)	454
428	Figure 177: SubscriptionManagement function-group overview (call)	454
429	Figure 178: subscriptionManagementEntryListData function overview	455
430	Figure 179: subscriptionManagementRequestCall function overview	456
431	Figure 180: subscriptionManagementDeleteCall function overview.....	458
432	Figure 181: StateInformation function-group overview	459
433	Figure 182: stateInformationListData function overview	459
434	Figure 183: SupplyCondition function-group overview	462
435	Figure 184: supplyConditionListData function overview	462
436	Figure 185: supplyConditionDescriptionListData function overview.....	465
437	Figure 186: supplyConditionThresholdRelationListData function overview.....	466
438	Figure 187: TariffInformation relations, sub-class level.....	468
439	Figure 188: TariffInformation relations, function level.....	468
440	Figure 189: TariffInformation Sub-class overview	469
441	Figure 190: TariffInformation, sub-class Tariff, function-group overview	470
442	Figure 191: tariffOverallConstraintsData function overview	471
443	Figure 192: tariffListData function overview.....	472
444	Figure 193: tariffBoundaryRelationListData function overview	473
445	Figure 194: tariffTierRelationListData function overview	474
446	Figure 195: tariffDescriptionListData function overview	476
447	Figure 196: TariffInformation, sub-class TierBoundary, function-group overview.....	477
448	Figure 197: tierBoundaryListData function overview	478
449	Figure 198: tierBoundaryDescriptionListData function overview.....	480
450	Figure 199: TariffInformation, sub-class Commodity, function-group overview.....	482
451	Figure 200: commodityListData function overview	482
452	Figure 201: TariffInformation, sub-class Tier, function-group overview	484
453	Figure 202: tierListData function overview.....	484
454	Figure 203: tierIncentiveRelationListData function overview.....	486
455	Figure 204: tierDescriptionListData function overview	487
456	Figure 205: TariffInformation, sub-class Incentive, function-group overview.....	488
457	Figure 206: incentiveListData function overview.....	489
458	Figure 207: incentiveDescriptionListData function overview	491

459	Figure 208: TaskManagement function-group overview	492
460	Figure 209: taskManagementJobListData function overview.....	493
461	Figure 210: taskManagementJobRelationListData function overview	495
462	Figure 211: taskManagementJobDescriptionListData function overview	496
463	Figure 212: taskManagementOverviewData function overview	498
464	Figure 213: Threshold function-group overview.....	499
465	Figure 214: thresholdListData function overview	499
466	Figure 215: thresholdConstraintsListData function overview.....	500
467	Figure 216: thresholdDescriptionListData function overview	502
468	Figure 217: TimeInformation function-group overview (data).....	504
469	Figure 218: TimeInformation function-group overview (call).....	504
470	Figure 219: timeInformationData function overview	504
471	Figure 220: timeDistributorData function overview	506
472	Figure 221: timePrecisionData function overview	507
473	Figure 222: timeDistributorEnquiryCall function overview	508
474	Figure 223: TimeSeries function-group overview	509
475	Figure 224: timeSeriesListData function overview	509
476	Figure 225: timeSeriesDescriptionListData function overview	512
477	Figure 226: timeSeriesConstraintsListData function overview	514
478	Figure 227: TimeTable function-group overview	517
479	Figure 228: timeTableListData function overview	517
480	Figure 229: timeTableConstraintsListData function overview	521
481	Figure 230: timeTableDescriptionListData function overview.....	523
482	Figure 231: UseCaseInformation function-group overview	524
483	Figure 232: useCaseInformationListData function overview	525
484	Figure 233: Version function-group overview.....	527
485	Figure 234: specificationVersionListData function overview	528
486		

487 List of tables

488	Table 1: ScaledNumberSetType	51
489	Table 2: ScaledNumberRangeType	52
490	Table 3: Enumeration CommodityTypeEnumType	52
491	Table 4: Enumeration EnergyDirectionEnumType value rules.....	53
492	Table 5: Enumeration EnergyModeEnumType	53
493	Table 6: Enumeration EnergyModeEnumType value rules.....	54
494	Table 7: Enumeration UnitOfMeasurementEnumType	56
495	Table 8: Enumeration CurrencyEnumType	57
496	Table 9: Enumeration ScopeTypeEnumType	59
497	Table 10: TimePeriodType.....	63
498	Table 11: TimestampIntervalType.....	63
499	Table 12: RecurringIntervalEnumType	64
500	Table 13: MonthType	64
501	Table 14: DayOfWeekType	65
502	Table 15: OccurrenceEnumType	66

503	Table 16: AbsoluteOrRecurringTimeType	67
504	Table 17: RecurrenceInformationType	68
505	Table 18: resultData function detailed description of elements	71
506	Table 19: Values specified for element errorNumber.....	71
507	Table 20: actuatorLevelData Element rules	82
508	Table 21: Element "function" value rules.....	82
509	Table 22: actuatorLevelDescriptionData Element rules.....	83
510	Table 23: actuatorSwitchData Element rules.....	84
511	Table 24: Element "function" value rules.....	84
512	Table 25: actuatorSwitchDescriptionData Element rules	84
513	Table 26: alarmListData Element rules	86
514	Table 27: Element "alarmType" value rules	86
515	Table 28: billListData Element rules	89
516	Table 29: Element "billType" value rules	89
517	Table 30: Element "costType" value rules.....	89
518	Table 31: Element "positionType" value rules	90
519	Table 32: billConstraintsListData Element rules.....	90
520	Table 33: billDescriptionListData Element rules	91
521	Table 34: dataTunnelingCall Element rules.....	94
522	Table 35: deviceClassificationManufacturerData Element rules	96
523	Table 36: Element "powerSource" value rules.....	97
524	Table 37: deviceClassificationUserData Element rules	97
525	Table 38: deviceConfigurationKeyValueListData Element rules	98
526	Table 39: Element "value" sub-element rules.....	99
527	Table 40: deviceConfigurationKeyValueDescriptionListData Element rules.....	100
528	Table 41: Element "keyName" value rules.....	104
529	Table 42: Element "valueType" value rules.....	105
530	Table 43: deviceConfigurationKeyValueListData Element rules	105
531	Table 44: deviceDiagnosisStateData Element rules	107
532	Table 45: Element "operatingState" value rules	108
533	Table 46: Element "powerSupplyCondition" value rules	108
534	Table 47: deviceDiagnosisHeartbeatData Element rules	109
535	Table 48: deviceDiagnosisServiceData Element rules.....	109
536	Table 49: directControlActivityListData Element rules	111
537	Table 50: Element "activityState" value rules	112
538	Table 51: directControlDescriptionData Element rules	112
539	Table 52: electricalConnectionParameterDescriptionListData Element rules	114
540	Table 53: Element "voltageType" value rules	114
541	Table 54: Element "acMeasuredPhases" value rules.....	115
542	Table 55: Element "acMeasuredInReferenceTo" value rules	115
543	Table 56: Element "acMeasurementType" value rules.....	115
544	Table 57: Element "acMeasurementVariant" value rules.....	115
545	Table 58: electricalConnectionPermittedValueSetListData Element rules	116
546	Table 59: electricalConnectionCharacteristicListData Element rules	117
547	Table 60: Element "characteristicContext" value rules.....	117
548	Table 61: Element "characteristicType" value rules	118

549	Table 62: electricalConnectionStateListData Element rules	119
550	Table 63: electricalConnectionDescriptionListData Element rules	120
551	Table 64: hvacSystemFunctionListData Element rules	122
552	Table 65: hvacSystemFunctionOperationModeRelationListData Element rules	123
553	Table 66: hvacSystemFunctionSetpointRelationListData Element rules	124
554	Table 67: hvacSystemFunctionPowerSequenceRelationListData Element rules	124
555	Table 68: hvacSystemFunctionDescriptionListData Element rules	125
556	Table 69: Element "systemFunctionType" value rules.....	125
557	Table 70: hvacOperationModeDescriptionListData Element rules.....	125
558	Table 71: Element "operationModeType" value rules.....	126
559	Table 72: hvacOverrunListData Element rules	127
560	Table 73: Element "overrunStatus" value rules	127
561	Table 74: hvacOverrunDescriptionListData Element rules	128
562	Table 75: Element "overrunType" value rules	128
563	Table 76: Relation between "overrunStatus" and "isOverrunActive"	128
564	Table 77: identificationListData Element rules	130
565	Table 78: Element "identificationType" value rules.....	130
566	Table 79: sessionIdentificationListData Element rules	131
567	Table 80: sessionMeasurementRelationListData Element rules.....	131
568	Table 81: incentiveTableData Element rules.....	138
569	Table 82: incentiveTableDescriptionData Element rules	141
570	Table 83: Element "tier. tierDescription. tierType" value rules	141
571	Table 84: Element "tier. boundaryDescription. boundaryType" value rules	142
572	Table 85: Element "tier. incentiveDescription. incentiveType" value rules.....	142
573	Table 86: incentiveTableConstraintsData Element rules	144
574	Table 87: loadControlNodeData Element rules	145
575	Table 88: loadControlLimitListData Element rules.....	146
576	Table 89: loadControlLimitConstraintsListData Element rules	147
577	Table 90: loadControlLimitDescriptionListData Element rules	148
578	Table 91: Element "limitType" value rules.....	148
579	Table 92: Element "limitCategory" value rules	149
580	Table 93: measurementListData Element rules	151
581	Table 94: Element "valueType" value rules.....	151
582	Table 95: Element "valueSource" value rules	152
583	Table 96: Element "valueTendency" value rules.....	152
584	Table 97: Element "valueState" value rules	152
585	Table 98: measurementSeriesListData Element rules	154
586	Table 99: measurementConstraintsListData Element rules.....	154
587	Table 100: measurementDescriptionListData Element rules.....	155
588	Table 101: Element "measurementType" value rules.....	156
589	Table 102: measurementThresholdRelationListData Element rules	157
590	Table 103: messagingListData Element rules.....	158
591	Table 104: Element "type" value rules.....	159
592	Table 105: operatingConstraintsInterruptListData Element rules	160
593	Table 106: operatingConstraintsDurationListData Element rules	161
594	Table 107: operatingConstraintsPowerDescriptionListData Element rules.....	162

595	Table 108: operatingConstraintsPowerRangeListData Element rules	163
596	Table 109: operatingConstraintsPowerLevelListData Element rules	163
597	Table 110: operatingConstraintsResumeImplicationListData Element rules.....	164
598	Table 111: sensingListData Element rules.....	166
599	Table 112: Element "state" value rules	168
600	Table 113: sensingDescriptionData Element rules.....	169
601	Table 114: Element "sensingType" value rules	169
602	Table 115: setpointListData Element rules	171
603	Table 116: setpointConstraintsListData Element rules	172
604	Table 117: setpointDescriptionListData Element rules.....	173
605	Table 118: Element "setpointType" value rules.....	173
606	Table 119: Server messages overview. The "write (restricted)" denotes if the message is intended for	
607	partial modification.	174
608	Table 120: Description of complex class function "smartEnergyManagementPsData" for	
609	powerSequences information	205
610	Table 121: Dedicated "selectors" of "smartEnergyManagementPsData" to request a specific	
611	powerSequences information.	206
612	Table 122: Description of complex class function	
613	"smartEnergyManagementPsConfigurationRequestCall" for powerSequences configuration request	
614	call	208
615	Table 123: Description of "restricted write" operation for powerSequences configuration.....	212
616	Table 124: Description of "restricted write" operation for powerSequences state change.....	215
617	Table 125: Description of smartEnergyManagementPsPriceData for powerSequences price	
618	information.....	216
619	Table 126: Dedicated "selectors" of "smartEnergyManagementPsPriceData" to request a specific	
620	powerSequences price information.	217
621	Table 127: Description of "restricted write" operation for a power sequence price modification. ...	218
622	Table 128: Description of complex class function	
623	"smartEnergyManagementPsPriceCalculationRequestCall" for powerSequences price calculation	
624	request call	220
625	Table 129: stateInformationListData Element rules	221
626	Table 130: Element "stateInformation" value rules.....	222
627	Table 131: Element "category" value rules	222
628	Table 132: supplyConditionListData Element rules	224
629	Table 133: Element "eventType" value rules	224
630	Table 134: Element "originator" value rules	224
631	Table 135: Element "gridCondition" value rules	224
632	Table 136: supplyConditionDescriptionListData Element rules	225
633	Table 137: supplyConditionThresholdRelationListData Element rules	226
634	Table 138: taskManagementJobListData Element rules	227
635	Table 139: taskManagementJobRelationListData Element rules	228
636	Table 140: taskManagementJobDescriptionListData Element rules	229
637	Table 141: Element "jobSource" value rules.....	229
638	Table 142: taskManagementOverviewData Element rules	230
639	Table 143: thresholdListData Element rules	231
640	Table 144: thresholdConstraintsListData Element rules.....	231

641	Table 145: thresholdDescriptionListData Element rules.....	232
642	Table 146: Element "thresholdType" value rules.....	232
643	Table 147: timeInformationData Element rules.....	234
644	Table 148: timeDistributorData Element rules	234
645	Table 149: timePrecisionData Element rules	235
646	Table 150: timeSeriesListData Element rules.....	238
647	Table 151: timeSeriesDescriptionListData Element rules	239
648	Table 152: Element "timeSeriesType" value rules	240
649	Table 153: timeSeriesConstraintsListData Element rules	242
650	Table 154: timeTableListData Element rules.....	243
651	Table 155: timeTableConstraintsListData Element rules	244
652	Table 156: timeTableDescriptionListData Element rules	244
653	Table 157: Element "timeSlotTimeMode" value rules.....	245
654	Table 158: incentiveTable detailed description of elements	247
655	Table 159: IncentiveTableIncentiveSlotType detailed description of elements	249
656	Table 160: incentiveTableDescriptionData function detailed description of elements	250
657	Table 161: IncentiveTableDescriptionTierType detailed description of elements	252
658	Table 162: Enumeration TierTypeEnumType	252
659	Table 163: Enumeration TierBoundaryTypeEnumType	253
660	Table 164: Enumeration IncentiveTypeEnumType	253
661	Table 165: incentiveTableConstraints detailed description of elements.....	255
662	Table 166: Description of complex class function "smartEnergyManagementPsData"	271
663	Table 167: Dedicated "selectors" of "smartEnergyManagementPsData".	272
664	Table 168: Description of complex class function	
665	"smartEnergyManagementPsConfigurationRequestCall".....	273
666	Table 169: Description of smartEnergyManagementPsPriceData.	274
667	Table 170: Dedicated "selectors" of "smartEnergyManagementPsPriceData".	274
668	Table 171: Description of complex class function	
669	"smartEnergyManagementPsPriceCalculationRequestCall"	275
670	Table 172: actuatorLevelData function detailed description of elements.....	276
671	Table 173: Enumeration ActuatorLevelFctEnumType.....	277
672	Table 174: actuatorLevelDescriptionData function detailed description of elements	278
673	Table 175: actuatorSwitchData function detailed description of elements	279
674	Table 176: Enumeration ActuatorSwitchFctEnumType	279
675	Table 177: actuatorSwitchDescription function detailed description of elements	280
676	Table 178: alarmListData function detailed description of elements.....	282
677	Table 179: Enumeration AlarmTypeEnumType	282
678	Table 180: billListData function detailed description of elements	287
679	Table 181: Enumeration BillTypeEnumType	287
680	Table 182: Enumeration BillCostTypeEnumType	287
681	Table 183: Enumeration BillPositionTypeEnumType	288
682	Table 184: billConstraintsListData function detailed description of elements.....	290
683	Table 185: billDescriptionListData function detailed description of elements.....	292
684	Table 186: bindingManagementEntryListData function detailed description of elements.....	294
685	Table 187: bindingManagementRequestCall function detailed description of elements	295
686	Table 188: bindingManagementDeleteCall function detailed description of elements.....	296

687	Table 189: dataTunnelingCall function detailed description of elements	300
688	Table 190: deviceClassificationManufacturerData function detailed description of elements.....	302
689	Table 191: Enumeration PowerSourceEnumType.....	302
690	Table 192: deviceClassificationUserData function detailed description of elements.....	304
691	Table 193: deviceConfigurationKeyValueListData function detailed description of elements.....	305
692	Table 194: Complex type DeviceConfigurationKeyValueValueType	306
693	Table 195: deviceConfigurationKeyValueDescriptionListData function detailed description of	
694	elements.....	308
695	Table 196: Enumeration DeviceConfigurationKeyNameEnumType	309
696	Table 197: Enumeration DeviceConfigurationKeyValueTypeEnumType	309
697	Table 198: deviceConfigurationKeyValueConstraintsListData function detailed description of	
698	elements.....	312
699	Table 199: deviceDiagnosisStateData function detailed description of elements	315
700	Table 200: Enumeration DeviceDiagnosisOperatingStateEnumType	315
701	Table 201: Enumeration PowerSupplyConditionEnumType	315
702	Table 202: deviceDiagnosisHeartbeatData function detailed description of elements	316
703	Table 203: deviceDiagnosisServiceData function detailed description of elements	318
704	Table 204: directControlActivityListData function detailed description of elements.....	320
705	Table 205: Enumeration DirectControlActivityStateEnumType.....	320
706	Table 206: directControlDescriptionData function detailed description of elements.....	321
707	Table 207: electricalConnectionParameterDescriptionListData function detailed description of	
708	elements.....	324
709	Table 208: Enumeration ElectricalConnectionVoltageTypeEnumType.....	324
710	Table 209: Enumeration ElectricalConnectionPhaseNameEnumType.....	325
711	Table 210: Enumeration ElectricalConnectionAcMeasurementTypeEnumType	325
712	Table 211: Enumeration ElectricalConnectionMeasurandVariantEnumType.....	325
713	Table 212: electricalConnectionPermittedValueSetListData function detailed description of elements	
714		326
715	Table 213: electricalConnectionStateListData function detailed description of elements.....	328
716	Table 214: Enumeration ElectricalConnectionCharacteristicContextEnumType.....	329
717	Table 215: Enumeration ElectricalConnectionCharacteristicTypeEnumType.....	329
718	Table 216: electricalConnectionStateListData function detailed description of elements.....	331
719	Table 217: electricalConnectionDescriptionListData function detailed description of elements	334
720	Table 218: hvacSystemFunctionListData function detailed description of elements	336
721	Table 219: hvacSystemFunctionOperationModeRelationListData function detailed description of	
722	elements.....	337
723	Table 220: hvacSystemFunctionSetpointRelationListData function detailed description of elements	
724		338
725	Table 221: hvacSystemFunctionPowerSequenceRelationListData function detailed description of	
726	elements.....	339
727	Table 222: hvacSystemFunctionDescriptionListData function detailed description of elements	341
728	Table 223: Enumeration HvacSystemFunctionTypeEnumType	341
729	Table 224: hvacOperationModeDescriptionListData function detailed description of elements	342
730	Table 225: Enumeration HvacOperationModeTypeEnumType	343
731	Table 226: hvacOverrunListData function detailed description of elements	344
732	Table 227: Enumeration HvacOverrunStatusEnumType.....	344

733	Table 228: hvacOverrunDescriptionListData function detailed description of elements	346
734	Table 229: Enumeration HvacOverrunTypeEnumType	346
735	Table 230: identificationListData function detailed description of elements	348
736	Table 231: Enumeration IdentificationTypeEnumType	348
737	Table 232: sessionIdentificationListData function detailed description of elements	349
738	Table 233: sessionMeasurementRelationListData function detailed description of elements	350
739	Table 234: loadControlNodeData function detailed description of elements	352
740	Table 235: loadControlEventListData function detailed description of elements	354
741	Table 236: Enumeration LoadControlEventActionEnumType	354
742	Table 237: loadControlStateListData function detailed description of elements	356
743	Table 238: Enumeration LoadControlEventStateEnumType	356
744	Table 239: loadControlLimitListData function detailed description of elements	358
745	Table 240: loadControlLimitConstraintsListData function detailed description of elements	360
746	Table 241: loadControlLimitDescriptionListData function detailed description of elements	362
747	Table 242: Enumeration LoadControlLimitTypeEnumType	362
748	Table 243: Enumeration LoadControlLimitCategoryEnumType	362
749	Table 244: measurementListData function detailed description of elements	365
750	Table 245: Enumeration MeasurementValueTypeEnumType	365
751	Table 246: Enumeration MeasurementValueSourceEnumType	366
752	Table 247: Enumeration MeasurementValueTendencyEnumType	366
753	Table 248: Enumeration MeasurementValueStateEnumType	366
754	Table 249: measurementSeriesListData function detailed description of elements	369
755	Table 250: measurementConstraintsListData function detailed description of elements	371
756	Table 251: measurementDescriptionListData function detailed description of elements	373
757	Table 252: Enumeration MeasurementTypeEnumType	374
758	Table 253: measurementThresholdRelationListData function detailed description of elements	375
759	Table 254: messagingListData function detailed description of elements	377
760	Table 255: Enumeration MessagingTypeEnumType	377
761	Table 256: networkManagementAddNodeCall function element description	382
762	Table 257: networkManagementRemoveNodeCall function element description	384
763	Table 258: networkManagementModifyNodeCall function element description	386
764	Table 259: networkManagementScanNetworkCall function element description	387
765	Table 260: networkManagementDiscoverCall function element description	388
766	Table 261: networkManagementProcessStateData function element description	391
767	Table 262: Enumeration NetworkManagementProcessStateStateType	391
768	Table 263: networkManagementJoiningModeData function element description	392
769	Table 264: networkManagementReportCandidateData function element description	393
770	Table 265: networkManagementDeviceDescriptionListData function element description	396
771	Table 266: Enumeration NetworkManagementFeatureSetType	396
772	Table 267: Enumeration NetworkManagementStateChangeType	397
773	Table 268: networkManagementEntityDescriptionListData function element description	400
774	Table 269: networkManagementFeatureDescriptionListData function element description	404
775	Table 270: operatingConstraintsInterruptListData function element description	407
776	Table 271: operatingConstraintsDurationListData function element description	408
777	Table 272: operatingConstraintsPowerDescriptionListData function element description	410
778	Table 273: operatingConstraintsPowerRangeListData function element description	411

779	Table 274: operatingConstraintsPowerLevelListData element description.....	412
780	Table 275: operatingConstraintsResumeImplicationListData function element description.....	414
781	Table 276: powerTimeSlotScheduleListData function element description.....	421
782	Table 277: powerTimeSlotValueListData function element description	422
783	Table 278: Enumeration PowerTimeSlotValueTypeEnumType	422
784	Table 279: powerTimeSlotScheduleConstraintsListData function element description	425
785	Table 280: powerSequenceAlternativesRelationListData function element description	426
786	Table 281: powerSequenceDescriptionListData function element description	428
787	Table 282: Enumeration PowerSequenceScopeEnumType	429
788	Table 283: powerSequenceStateListData function element description.....	431
789	Table 284: Enumeration PowerSequenceStateEnumType	431
790	Table 285: powerSequenceScheduleListData function element description	432
791	Table 286: powerSequenceScheduleConstraintsListData function element description	433
792	Table 287: powerSequencePriceListData function element description	435
793	Table 288: powerSequenceSchedulePreferenceListData function element description	436
794	Table 289: powerSequenceNodeScheduleInformationData function element description.....	438
795	Table 290: powerSequenceScheduleConfigurationRequestCall function element description	439
796	Table 291: powerSequencePriceCalculationRequestCall function element description	441
797	Table 292: sensingListData function detailed description of elements	442
798	Table 293: Enumeration SensingStateEnumType	444
799	Table 294: sensingDescriptionData function detailed description of elements	446
800	Table 295: Enumeration SensingTypeEnumType.....	446
801	Table 296: setpointListData function detailed description of elements.....	449
802	Table 297: setpointConstraintsListData function detailed description of elements	450
803	Table 298: setpointDescriptionListData function detailed description of elements	452
804	Table 299: Enumeration SetpointTypeEnumType	453
805	Table 300: subscriptionManagementEntryListData function detailed description of elements	455
806	Table 301: subscriptionManagementRequestCall function detailed description of elements.....	457
807	Table 302: subscriptionManagementDeleteCall function detailed description of elements	458
808	Table 303: stateInformationListData function detailed description of elements.....	460
809	Table 304: Enumeration StateInformationFunctionalityEnumType	460
810	Table 305: Enumeration StateInformationFailureEnumType	461
811	Table 306: Enumeration StateInformationCategoryEnumType.....	461
812	Table 307: supplyConditionListData function detailed description of elements.....	463
813	Table 308: Enumeration SupplyConditionEventTypeEnumType	463
814	Table 309: Enumeration SupplyConditionOriginatorEnumType.....	464
815	Table 310: Enumeration GridConditionEnumType	464
816	Table 311: supplyConditionDescriptionListData function detailed description of elements	465
817	Table 312: supplyConditionThresholdRelationListData function detailed description of elements ..	466
818	Table 313: tariffOverallConstraintsData function detailed description of elements.....	472
819	Table 314: tariffListData function detailed description of elements	473
820	Table 315: tariffBoundaryRelationListData function detailed description of elements	474
821	Table 316: tariffTierRelationListData function detailed description of elements.....	475
822	Table 317: tariffDescriptionListData function detailed description of elements.....	476
823	Table 318: tierBoundaryListData function detailed description of elements.....	478
824	Table 319: tierBoundaryDescriptionListData function detailed description of elements	481

825	Table 320: Enumeration TierBoundaryTypeEnumType	481
826	Table 321: commodityListData function detailed description of elements.....	483
827	Table 322: tierListData function detailed description of elements.....	485
828	Table 323: tierIncentiveRelationListData function detailed description of elements	486
829	Table 324: tierDescriptionListData function detailed description of elements	487
830	Table 325: Enumeration TierTypeEnumType	487
831	Table 326: incentiveListData function detailed description of elements	489
832	Table 327: Enumeration IncentiveValueTypeEnumType	490
833	Table 328: incentiveDescriptionListData function detailed description of elements.....	491
834	Table 329: Enumeration IncentiveTypeEnumType	492
835	Table 330: taskManagementJobListData function detailed description of elements	493
836	Table 331: taskManagementJobRelationListData function detailed description of elements.....	495
837	Table 332: taskManagementJobDescriptionListData function detailed description of elements.....	497
838	Table 333: Enumeration TaskManagementJobSourceEnumType	497
839	Table 334: taskManagementOverviewData function detailed description of elements.....	498
840	Table 335: thresholdListData function detailed description of elements.....	499
841	Table 336: thresholdConstraintsListData function detailed description of elements	501
842	Table 337: thresholdDescriptionListData function detailed description of elements	502
843	Table 338: Enumeration ThresholdTypeEnumType	503
844	Table 339: timeInformationData function detailed description of elements.....	505
845	Table 340: timeDistributorData function detailed description of elements.....	506
846	Table 341: timePrecisionData function detailed description of elements.....	507
847	Table 342: timeSeriesListData function detailed description of elements	509
848	Table 343: Complex type TimeSeriesSlotType	510
849	Table 344: timeSeriesDescriptionListData function detailed description of elements.....	513
850	Table 345: Enumeration TimeSeriesTypeEnumType	513
851	Table 346: timeSeriesConstraintsListData function detailed description of elements.....	515
852	Table 347: timeTableListData function detailed description of elements	518
853	Table 348: timeTableConstraintsListData function detailed description of elements	522
854	Table 349: timeTableDescriptionListData function detailed description of elements	523
855	Table 350: Enumeration TimeSlotTimeModeEnumType	524
856	Table 351: useCaseInformationListData function detailed description of elements.....	526
857	Table 352: specificationVersionListData function detailed description of elements	528
858	Table 353: Well known device types.....	668
859	Table 354: Well known entity types.....	671
860	Table 355: Well known feature types	672
861	Table 356: Well known complex classes	673
862	Table 357: Well known standard classes	674
863	Table 358: Identifier overview	675
864		
865		

1 Introduction

The SPINE data model is described within this technical specification (please keep in mind that some information may only be specified in the SPINE XSDs, see below). It contains the SPINE resource model, consisting of the device types, entity types and feature types as well as descriptions of the (standard and complex) classes as they are defined through the SPINE XSDs that are part of each release of the SPINE specifications. Additionally, some general rules, relations and data type descriptions are included.

The underlying SPINE data model of this document is identified by the XML namespace "http://docs.eebus.org/spine/xsd/v1".

1.1 References

1.1.1 EEBUS SPINE documents

[Introduction]	EEBus_SPINE_TR_Introduction.pdf
[ProtocolSpecification]	EEBus_SPINE_TS_ProtocolSpecification.pdf
[DataModelXSDs]	EEBus_SPINE_TS_ActuatorLevel.xsd, ..., EEBus_SPINE_TS_Version.xsd

1.1.2 Other EEBUS documents

[SHIPSpecification]	EEBus_SHIP_TS_Specification_V1.0.1.pdf
[SPINEUseCaseXSDs]	XSDs delivered within the EEBUS Use Case Package

1.1.3 Other documents

[SGReady]	2012-11-01_SG_Ready_Regularien_Version1.1.pdf
------------------	---

1.1.4 Websites

[EEBus]	http://www.eebus.org	Official EEBus Initiative e.V. website.
[W3C]	http://www.w3.org/	Official World Wide Web Consortium website.
[W3Schools]	http://www.w3schools.com/	Tutorials and examples for XML and others.

1.1.5 Normative References

[RFC2119]	IETF RFC 2119: 1997, Key words for use in RFCs to indicate requirement levels Please see section 1.4.5 for details.
------------------	--

1.2 Terms and definitions

Binding

Concept for connecting functionally matching features. For details, please refer to [ProtocolSpecification].

(Standard or Complex) Class

Set of SPINE functions used to model data for a specific functionality. A class can be considered as a topic where functions are defined for. For example the SPINE class "Measurement" is a collection of SPINE functions that are used to describe measurement values.

Classifier

Specifies whether a message serves to read, reply, write, etc.

Client

Role that specifies that a node uses data from a "server" or can change it.

Command

The functional part of a Message.

Complex Class

SPINE class that is build up by parts of SPINE standard classes and combines them in a new, ordered way.

(SPINE) Data model

Definition of the possible data that can be used for SPINE communications. Defined as XSD.

Device (specific node)

SPINE node that can include a set of entities. It has a "Device Type". With regards to the hierarchy of SPINE nodes a device is a root node for all functionalities offered by a device.

"device" (address information)

SPINE address part for the (physical) device.

Device Type

Specific type of physical device (e.g. "WashingMachine", "HeatPump", "FridgeFreezer", etc.).

Discovery

Process of finding appropriate partners for communication. Dependent on the context this can be either finding other devices or examination of a device's potential functionalities. For details, please refer to [ProtocolSpecification].

DSO

Distribution System Operator of an electricity grid.

Element

Item (or "attribute") of a SPINE function. Holds one information (e.g. "timestamp", "value", etc.) or contains further sub-elements.

- 934 **Entity** (specific node)
935 SPINE node that can include a set of (child) entities or features. It has an "Entity Type". With regards
936 to the hierarchy of SPINE nodes an entity is a child element of a device or another entity.
- 937 **"entity"** (address information)
938 SPINE address part for the (logical) entity.
- 939 **Entity Type**
940 Specific type of logical device (e.g. "Freezer" is one logical part of a physical device "FridgeFreezer").
- 941 **EV**
942 Electric Vehicle
- 943 **EVSE**
944 Electric Vehicle Supply Equipment
- 945 **Feature** (specific node)
946 SPINE node that can include a set of functions (of a class). It has a "Feature Type". With regards to
947 the hierarchy of SPINE nodes a feature is a child element of an entity.
- 948 **"feature"** (address information)
949 SPINE address part for one feature.
- 950 **Feature Type**
951 Defines optional or mandatory rules and a general behaviour of the underlying Class (standard or
952 complex).
- 953 **(SPINE) Function**
954 A (SPINE) function is the smallest structure to model "actual data" ("functional data"). I.e. functions
955 usually consist of child elements that each hold an information (e.g. "timestamp", "value", etc.).
956 Information between communication partners is exchanged via the exchange of a function (as part of
957 a so-called "payload").
- 958 **Header**
959 SPINE Header, including elements for addressing, unique identification of messages, timestamp, etc.
- 960 **Message**
961 One SPINE transfer from a sender to a receiver.
- 962 **(XML) Namespace**
963 XML namespaces provide a simple method for qualifying element and attribute names used in XML
964 documents by associating them with namespaces identified by URI references (source: www.w3.org).
- 965 **Node**
966 Common term for a SPINE instance that has a SPINE address. Dependent on the situation a node can
967 be either a device or an entity (of a specific device) or a feature (of a specific device-entity).
- 968 **Payload**
969 SPINE Payload, containing the functional SPINE data.

Role

Each Feature has a functional role, usually either "server" (data owner) or "client". For some special features (NodeManagement, e.g.) the role "special" is defined.

Scope (Type)

Some feature types define scope types for identifying specific functionalities unambiguously (e.g. *outsideAirTemperature*).

Server

Role that specifies that a node offers own data to be read or written by a node with role client. A server can notify its data to other nodes (with role client).

SPINE

Smart **P**remises **I**nteroperable **N**eutral-message **E**xchange

Standard Class

All basic/standard functions are defined in standard classes. Functions of standard classes follow very simple patterns and do not have deeply nested data structures.

Subscription

Enables the receiving of messages of interest from another device without polling it. For details, please refer to [ProtocolSpecification].

Use Case

Textual description of a re-usable functionality consisting of one or more messages of one or more participating actors. May be visualized with a sequence diagram. E.g. "A CEM shifts the energy usage of a washing machine."

User Story

Complete (but specific) business case described from the perspective of a user. Can be separated into several use cases. E.g. "The user wants to get the laundry done by 8:00pm."

XML (Extensible Markup Language)

Human- and machine-readable markup language containing data. Used to model SPINE messages.

XSD (XML Schema Definition)

Definition format for XMLs, written in XML. Specifies how a well-formed XML (in regards to this XSD) can be built. The SPINE data model is defined in XSD and supplementary documents (as not every rule can be specified with XSD only). Other formats than XML can be derived from an XSD, too (e.g. JSON).

1.3 Document structure

The main focus of this document is the description of the SPINE resources and the data model. Before going into technical details, chapter 2 presents a coarse overview of the overall concept. Chapter 3 collects a number of technical low-level details and can be left out at first reading. Chapter 4 defines types of different facets of a device (device types, entity types, feature types). These types contain also rules which are relevant to establish an interoperable use of exchanged data. Finally, chapter 5 describes SPINE classes with so-called "functions". These "functions" are the basic data

structures that are used within "messages" to represent or modify the content of a functionality or data point. This chapter describes the potential purpose and use of functions, whereas chapter 4 (and esp. the feature types defined in section 4.3) impose rules on their use.

1.4 How to read this document

1.4.1 Graphical representation

1.4.1.1 Introduction

Documentation provided by the EEBus Initiative e.V. sometimes makes use of graphical representations of parts of the SPINE data models in order to ease the readability. All graphical representations are created using a specific commercial tool (oXygen). Please note that these graphical representations are not standardized. Furthermore, the explanations given below are tailored to the needs of the SPINE data models. In general, the explanations are not exhaustive.

1.4.1.2 Explanation of graphical XSD representation

The graphical XSD explanation uses a set of images, which are described to accurately understand the diagrams. This section gives an overview of these images.

1.4.1.2.1 XSD "simple type"

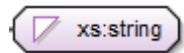


Figure 1: XSD "simple type"

This image represents an XSD "simple type" (in this case a string). Simple types are signed with a left upper corner triangle. A simple type can either be given by the W3C (starting with "xs:" in this document; see also 1.4.2) or can be self-defined. Self-defined simple types can only be restrictions of given W3C simple types, such as making a string an enumeration (only allowing certain words/literals) or restrict a number to a certain range. The kind of restriction is not graphically represented, but can only be read in the XSD file itself.

1.4.1.2.2 XSD "complex type" with no content

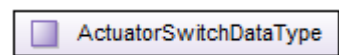
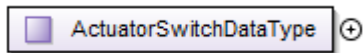


Figure 2: XSD "complex type" with no content

This image represents an XSD "complex type". Complex types are signed with a square. A complex type can consist of different other complex and/or simple types, which are embedded. A complex type might, however, also be completely empty, which is the case shown in this image. An empty "complex type" has no effect if used, other than to be a placeholder for easier extensibility.

1044 1.4.1.2.3 XSD "complex type" with content

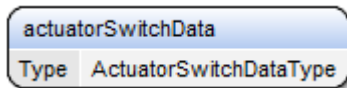


1046 Figure 3: XSD "complex type" with content

1047 This image shows an XSD "complex type" with a content (shown by the "+").

1048

1049 1.4.1.2.4 XSD element with type



1050

1051 Figure 4: XSD element with type

1052 This element represents an XML element "actuatorSwitchData". The contents of this element are
1053 defined by the type "ActuatorSwitchDataType". An element can be defined using a type, or can be
1054 directly defined. Using a type has the advantage of being easily extensible.

1055 Please note that figure above may vary. This depends whether the type is empty and whether it is
1056 referenced from a separate XSD.

1057

1058 1.4.1.2.5 XSD sequence



1059

1060 Figure 5: XSD sequence

1061 This image shows an XSD "sequence". The element connected on the left (not shown in the image) is
1062 the one, which contains this sequence (the "parent"). The elements connected on the right are
1063 contained in that sequence (the "children"). A sequence is an ordered list of elements on the right,
1064 contained in an element connected on the left. (An unordered list is an "all").

1065

1066 1.4.1.2.6 XSD choice



1067

1068 Figure 6: XSD choice

1069 This image shows an XSD "choice". The element connected on the left shall contain exactly one of the
1070 elements connected on the right (none of these elements are shown here).

1071

1072 1.4.1.2.7 XSD all



1073

1074 Figure 7: XSD all

This image shows an XSD "all". The element connected on the left can contain any of the elements connected on the right ("children"; none of these elements are shown here). This is similar to an XSD sequence. The difference is that each "child" is implicitly optional (i.e. can be absent in an XML) and that the order of the children in an XML is arbitrary.

1.4.1.2.8 Optional vs. mandatory

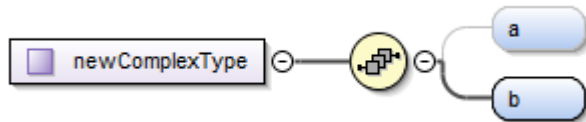


Figure 8: Optional vs. mandatory

There are different styles for the display of connections between building blocks. These styles are used to display whether a building block is optional or mandatory, among others. The image shows a complex type "newComplexType", consisting of a sequence of an optional element "a" (light grey line) and a mandatory element "b" (dark line).

1.4.1.2.9 Lists / multiplicity

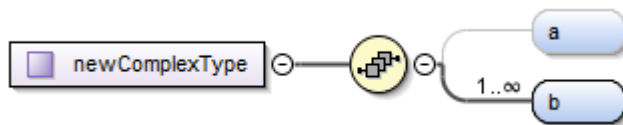


Figure 9: Lists / multiplicity

Elements might also be present multiple times. The shown image shows a complex type "newComplexType", containing a sequence of an optional element "a", as well as 1 up to infinite elements "b".

1.4.1.2.10 Example: "Expanded view"

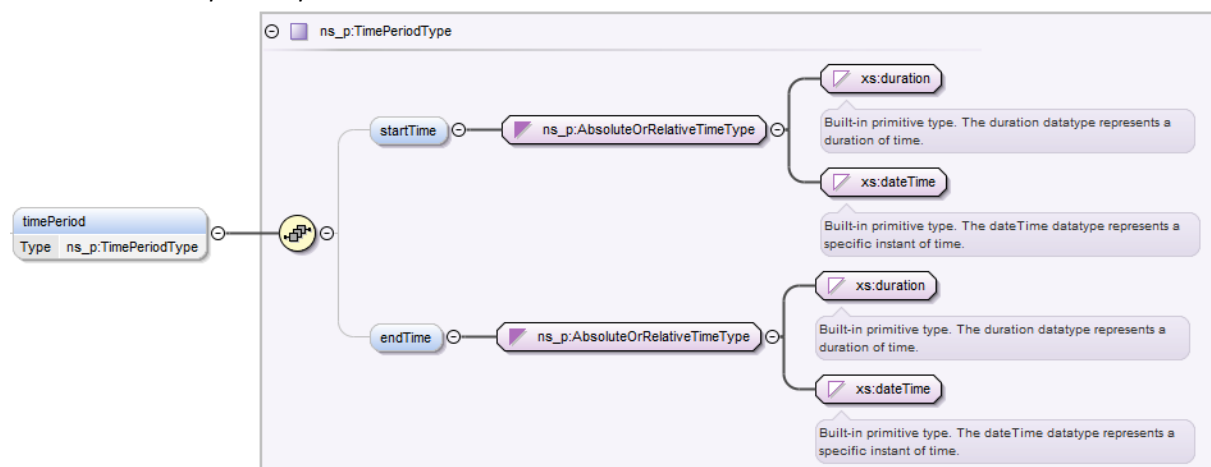


Figure 10: XSD expanded view

This image shows an element called "timePeriod", which is based upon the Type "TimePeriodType". Attached to it (on the right, embedded in a purple box) is the definition of the type "TimePeriodType". The type is a sequence of two optional elements called "startTime" and "endTime", which are both using the "AbsoluteOrRelativeTimeType" format. There are also informative comments shown in this image, which are embedded in little speech bubbles / balloons.

An XML part conforming to this XSD at least follows the structure of:

```
<timePeriod>
  <startTime>2006-05-04T18:13:51.0Z</startTime>
  <endTime>2006-05-04T18:13:51.0Z</endTime>
</timePeriod>
```

1.4.2 Namespaces

The SPINE data models make use of many types defined by W3C. These types are referenced within the SPINE data models under a so-called namespace of the W3C XML Schema. The SPINE data models currently denote this namespace using the prefix "xs:". Thus, a W3C XML Schema type like "dateTime" finally leads to the reference "xs:dateTime".

It shall just be noted that the SPINE data models put the definition of their own types and elements under namespaces of the EEBus Initiative e.V.

1.4.3 "datagram" vs. specific function – XMLs with different roots and scopes

The SPINE data models provide definitions for different levels/layers. The subsequent figure shows a (partly collapsed) graphical representation of the SPINE XML schema definition of "datagram", with a specific branch highlighted in red for a subsequent example.

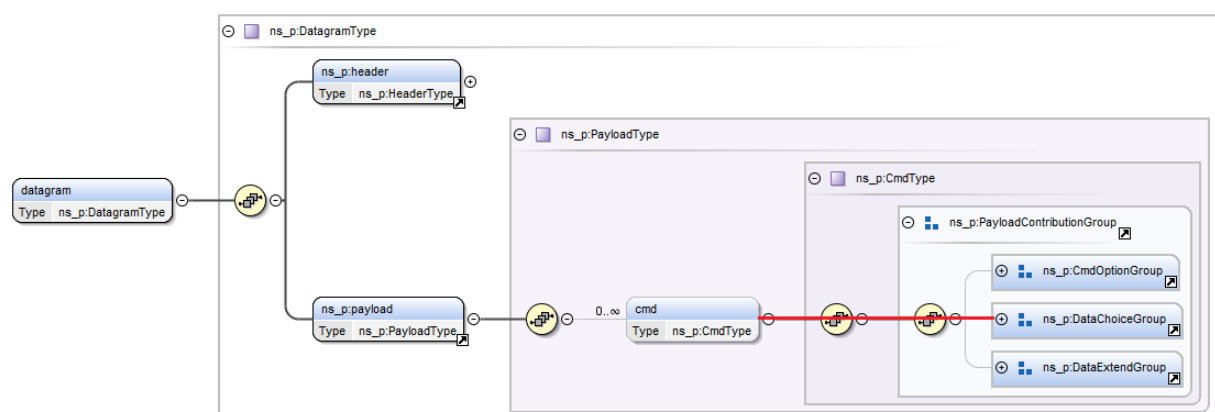


Figure 11: Datagram structure (red line: "way" to the functions)

The next figure shows a graphical excerpt of possible children of the highlighted branch:

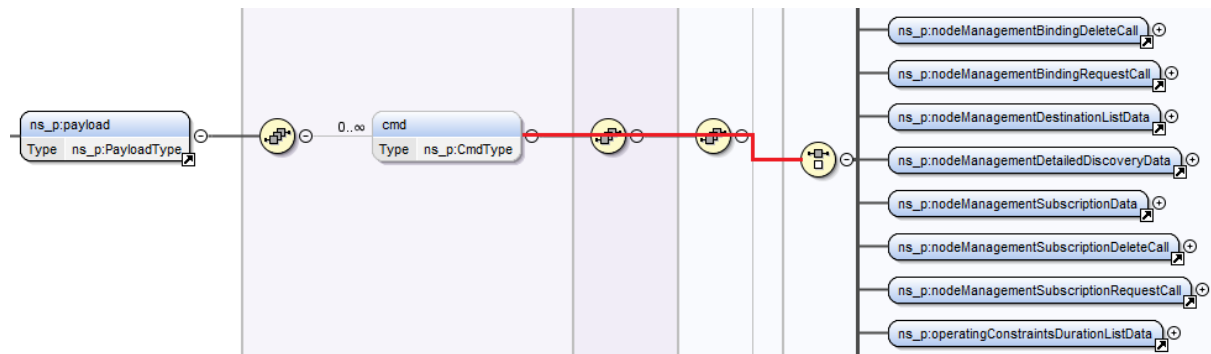


Figure 12: Payload structure (red line: "way" to the functions)

These children are also called "functions" within the SPINE specification.

It is possible to create valid XMLs with either "datagram" or any of payload's child elements (like a concrete function as from the second picture) as root element. It depends on scope and "layer" (in terms of the ISO-OSI layer model, e.g.) which kind of XML is required.

Throughout this document most explanations of classes will make use of XMLs with specific functions as root tag. However, it shall be noted that also XMLs with "datagram" as root tag will be discussed and that different kinds of "payload" contents will be explained.

1.4.4 Simplification of XML examples

Supposed the SPINE data model of "datagram" is available for download from "http://docs.eebus.org/spine/xsd/v1/EEBus_SPINE_TS_Datagram.xsd". The beginning of a valid SPINE XML with a root tag ("datagram" in the subsequent example) could look like this:

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
  xsi:schemaLocation="http://docs.eebus.org/spine/xsd/v1
EEBus_SPINE_TS_Datagram.xsd">
...
```

The schemaLocation information of the SPINE data models is not finally decided yet, is not always mandatory within an XML (see W3C definitions for details) and is not required for the explanation of the SPINE data models. Hence, it is usually left out throughout this document:

```
<?xml version="1.0" encoding="UTF-8"?>
<datagram xmlns="http://docs.eebus.org/spine/xsd/v1">
...
```

The W3C recommendation does not oblige the XML declaration which contains the XML version information, but it clearly recommends its use. However, this part is not of interest for the presentation of examples.

The XML namespace is omitted within this document, too, as there is no need to mention it in all examples.

Thus, the example is further simplified:

```
<datagram>
```

1158 . . .

1159 As most examples in this document are specific for a single function, the information of the header is
1160 not of interest and therefore not included in the examples. The <datagram> declaration as well as
1161 the subjacent element <payload> is not needed either. Most examples begin with the function itself,
1162 or with the <function> element, if "restricted function exchange" (with "<filter>") is used before the
1163 actual function.

1164 Please see [ProtocolSpecification] for more information about the structure of SPINE XMLs.

1165

1166 **1.4.5 Used requirement keywords**

1167 The following keywords are used:

- 1168 - SHALL
- 1169 - SHALL NOT
- 1170 - SHOULD
- 1171 - SHOULD NOT
- 1172 - MAY

1173 They apply only if written in capital letters!

1174 For the meaning of the keywords, please refer to [RFC2119].

1175

1176 **1.4.6 Presence indications in Feature Types**

1177 If no presence indication keyword is stated for an element, the keyword "MAY" SHALL be applied (in
1178 other words: if not stated otherwise, an element is optional).

1179

1180 **1.4.7 Presence indications in SmartEnergyManagementPs**

1181 The following abbreviations on the presence of elements or the support of entities or features are
1182 used:

- 1183 - M = mandatory
- 1184 - R = recommended
- 1185 - O = optional
- 1186 - C = "choice", i.e. a presence depends also on the selection from multiple possibilities

1187 meaning that a functionality SHALL be supported (mandatory), SHOULD be supported
1188 (recommended) or MAY be supported (optional). If "choice" is used, the element description needs
1189 to be considered to find out which alternative applies.

1190 These abbreviations are used exactly the same way as stated in the section "Element presence
1191 indications" of the document [ProtocolSpecification]. The following information is a brief excerpt
1192 from this document:

In case of elements (of data model definitions), the presence indications "M" and "C" are always meant relative to the respective parent element. I.e. if a parent element is optional ("O") and a child is mandatory ("M") the child element can only be present if the parent element is present as well. The presence indications from the data model definitions describe general requirements on the presence of elements. These requirements can be strengthened (but not weakened) by process rules.

For more information and examples please consider the above mentioned document. In general, the design concept of section "Data model specialization: ..." of the document [ProtocolSpecification] applies here as well.

2 Overall model hierarchy concept overview

The following figure gives an overview on the miscellaneous levels that contribute to the definition and use of interoperable functionality.

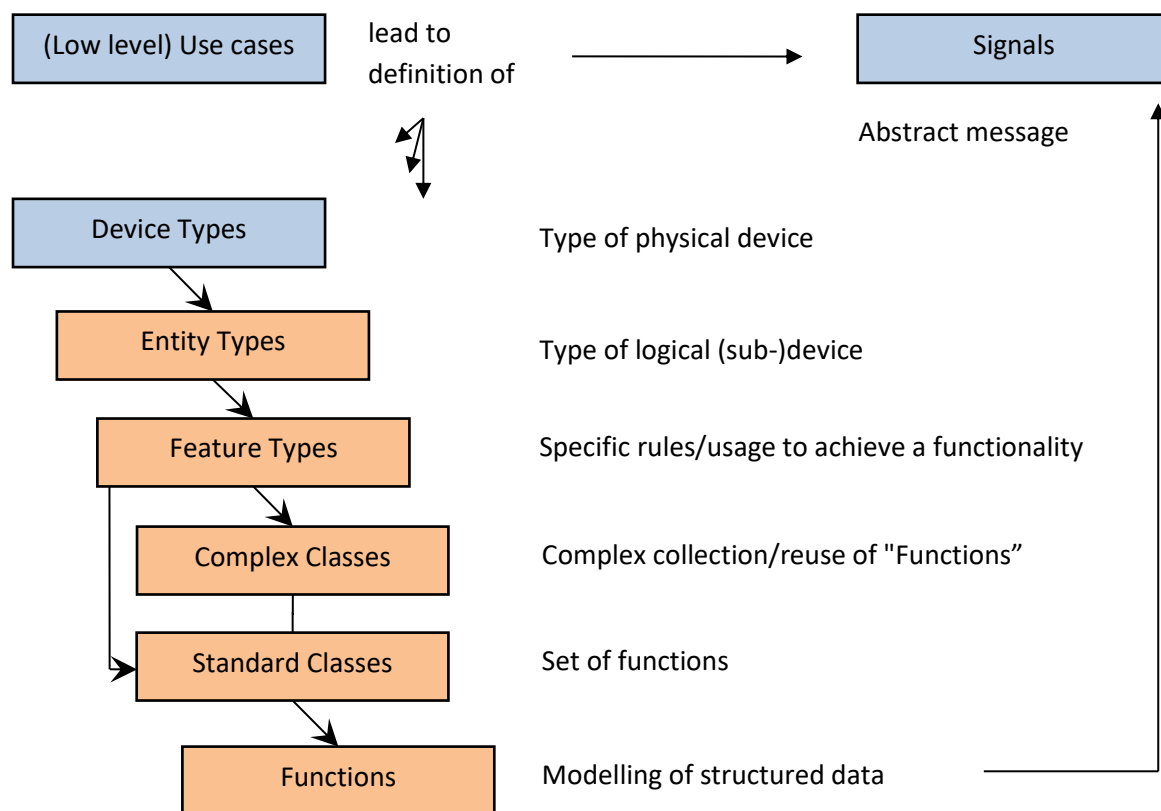


Figure 13: Dependencies of the standardised types and defined data structures

Apart from user stories, the most abstract level deals with use cases and abstract signals exchanged between actors. Starting from the top of Figure 13, each level defines its own set of definitions for its own hierarchy of the problem. The device facets "device type", "entity type", and "feature type" define logical structures of a device and rules to achieve a required functionality. These facets are introduced in more detail in section 2.1.2. The required functionality makes use of well-defined data structures (so-called "functions") which are organised in so-called classes. The class concept is explained in section 2.2. The data within the functions is expressed with so-called "elements". The

1215 rules and the functions together permit the modelling of signals. This is used in document
 1216 [ProtocolSpecification] to model a neutral message framework.

1217

1218 2.1 Resource Type Definition Concept

1219 2.1.1 Introduction

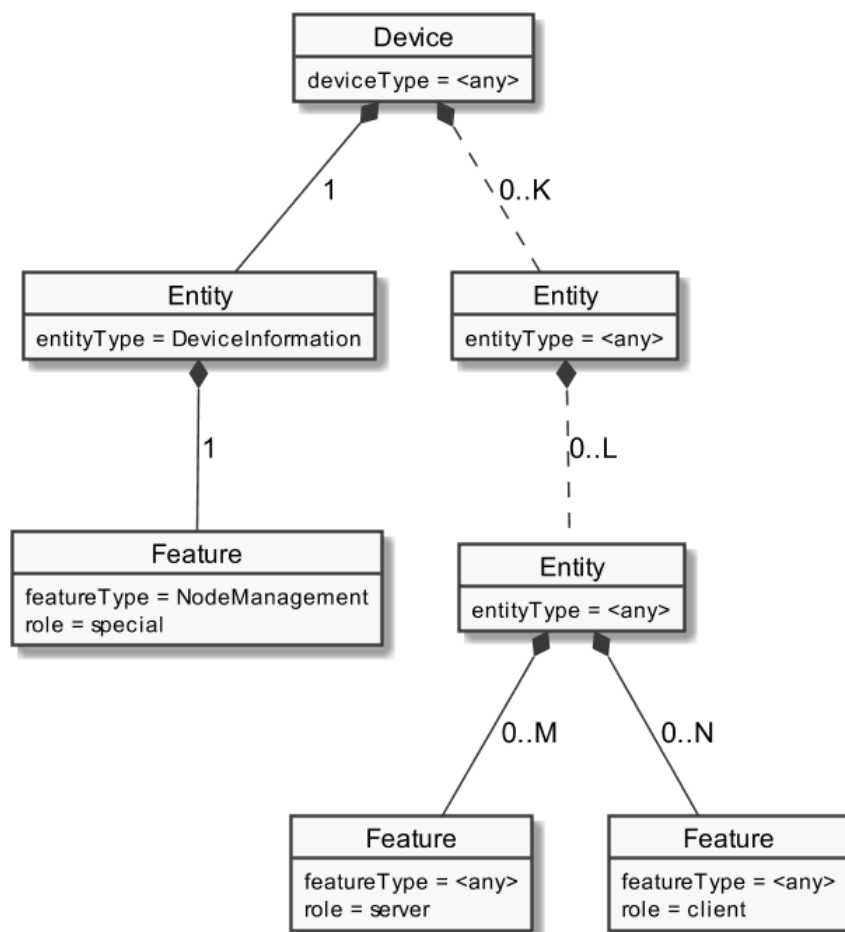
1220 The SPINE device model consists of three layers

- 1221 - Device
- 1222 - Entity (with sub-Entities)
- 1223 - Feature

1224 where the following applies:

- 1225 - 1 Device has 1..K Entities
- 1226 - 1 Entity has 0..L sub-Entities
- 1227 - 1 Entity has 1 parent (Device or Entity)
- 1228 - 1 Entity has 0..M Features
- 1229 - 1 Feature has 1 parent (Entity).

1230 These relations can be combined. The following figure gives an example for this:



1231

1232 Figure 14: SPINE device model example

1233 The dotted line in the figure indicates that further Entities MAY be present between the depicted
1234 instances.

1235 In the following sub-sections these components are described more in detail.

1236

1237 **2.1.2 Device Model Facets**

1238 **2.1.2.1 Device Type**

1239 The description of the (physical) device (that can be seen as kind of a root instance within the SPINE
1240 device model) includes a short description of the device. As the Device Type is only of informative
1241 character, no further information is needed.

1242 Vendors may use official Device Types as defined in section 4.1 or create own Device Types that then
1243 have to be marked as vendor specific ones (see section 2.1.3). Any kind of Device Type may contain
1244 any combination of Entity Types (official or vendor specific ones).

1245

1246 **2.1.2.2 Entity Type**

1247 In contrast to the only informative Device Type, the Entity Type is more important for the
1248 interoperable working of SPINE. Within the Use Case definitions, Entity Types are mapped to the Use
1249 Case Actor's functionality. Hence, a discovery of the correct Entity Type (and its path, see below) may
1250 be of relevance.

1251 As already denoted in sub-section 2.1.1, Entities may be nested, meaning an Entity may have other
1252 Entities as child. This can lead to a tree structure of Entities. This hierarchy can be retrieved from the
1253 detailed discovery (see [ProtocolSpecification] for details).

1254

1255 **2.1.2.3 Feature Type**

1256 Feature Types define optional or mandatory rules for the underlying Class (mostly with the same
1257 name, e.g. the Feature Type Measurement is based on the underlying Class Measurement), their
1258 functions and the elements and values within the functions. As most rules must be deriveable from
1259 the elements and values presented on the SPINE Features, the corresponding rules are provided for
1260 each element and value within the tables of the "Feature Types" section.

1261

1262 **2.1.3 Vendor specific extensions**

1263 The Device-/Entity-/Feature-Type structure mentioned in the sections above relies on definitions
1264 provided by the EEBus Initiative e.V. for ensuring interoperability. If a vendor needs type definitions
1265 that are not already provided, a vendor specific extension allows the definition of new types that add
1266 further functionality. These extensions are completely vendor specific and will not be supported by
1267 the EEBus Initiative e.V. at all. A vendor may bring these definitions into the EEBUS working-groups
1268 and, if of interest, they potentially could become part of a future SPINE release.

1269 These types defined by the vendors themselves SHALL be marked as vendor specific. For details
1270 please see [ProtocolSpecification], section "Rules for vendor specific extensions".

1271

1272 **2.1.4 Feature Group**

1273 Some features may need another feature to fulfil a functionality in a senseful way. E.g. the Setpoint
1274 Class should be linked to some Measurement instance, so the desired value (from Setpoint) can be
1275 compared to the actual one (Measurement). Therefore, feature groups can be used. They are always
1276 entity-wide defined. Two entities may define the same feature group for a different meaning. If two
1277 or more features shall be linked, the same feature group is assigned to them (e.g. "#1"). More than
1278 one feature group may be defined for one feature (e.g. if a Measurement instance is linked to a
1279 Setpoint instance as well as to a SmartEnergyManagementPs feature). If the features that reference
1280 each other are only defined once on an entity, no feature group is needed (picking up the previous
1281 example: if there is only one Measurement and one Setpoint instance on an entity, no ambiguity is
1282 given). If, without the feature groups, there would be an ambiguity, feature groups SHALL be used to
1283 prevent the ambiguity and identify the associated features.

1284 **Examples:**

- 1285 - #1
- 1286 - #1#2
- 1287 - #2#5

1288

1289 **2.1.5 Specific usage**

1290 This functionality is deprecated.

1291

1292 **2.2 Class Concept**

1293 **2.2.1 Standard Classes**

1294 The SPINE data model relies on the standard classes (like ActuatorSwitch, Measurement, etc.). They
1295 define the functions with all their elements, some examples and a general description what can be
1296 communicated with this functionality. No (or just very few) rules are given in the class sections as the
1297 definition of rules is the scope of the device model (see section 2.1.2).

1298 Please note that some classes (or at least function type definitions) are also used in so-called
1299 complex classes (see section 2.2.2). In fact, for some classes there may even be no standard Feature
1300 Type defined if they are only used within a complex class. In this version of the specification this is
1301 the case for the classes

- 1302 - BindingManagement
- 1303 - SubscriptionManagement
- 1304 - Version

1305 as they are primarily re-used within the complex class NodeManagement.

1306

1307 **2.2.2 Complex Classes**

1308 Complex classes combine one or more functionalities from standard classes in new functions that are
 1309 treated the same way as functions from standard classes. They do not define new elements as leaves
 1310 (in a tree structure), but only order the standard class functions in another way. Lists may be added
 1311 and (leave) elements may be deleted.

1312

1313 **2.2.3 SPINE class hierarchy (Class / Sub-Class / Function-Group / Function / Element)**

1314 To identify different levels of SPINE data-structures, the following naming conventions are defined:

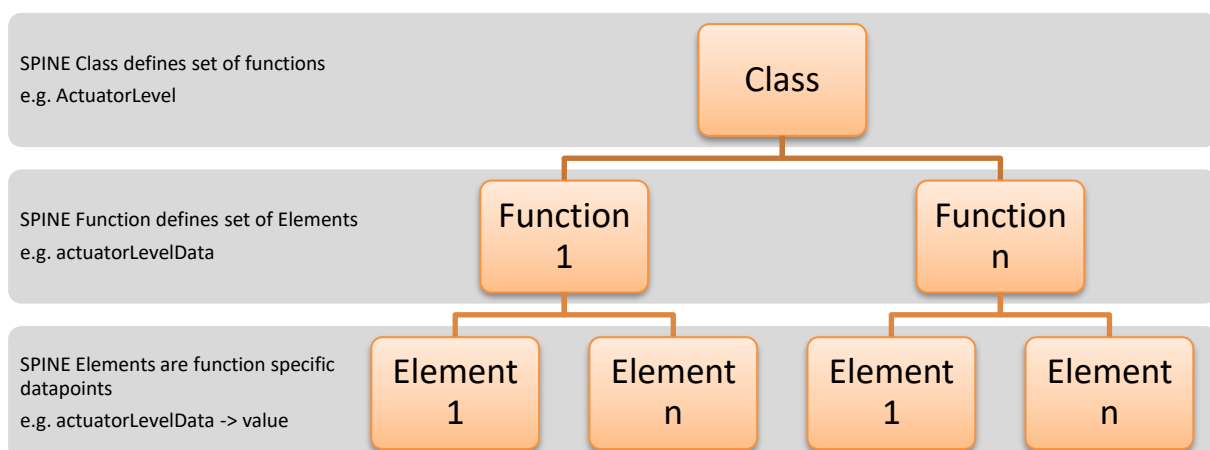
1315 **Class:** A SPINE class is the highest level of data-grouping. E.g. all functions relating to
 1316 *PowerSequences* are described within the *PowerSequences* class. The first letter of the SPINE class
 1317 name is always capitalized.

1318 **Sub-Class:** Classes may have sub-classes. They are introduced with a short sub-section in the
 1319 corresponding class description. E.g. the *PowerSequences* class contains the sub-classes
 1320 *PowerTimeSlot*, *PowerSequence* and *PowerSequenceNode*. These are not marked as own groups in
 1321 the XSD. The sub-classes are not used in any identifier and are not relevant for interoperability
 1322 considerations. The first letter of the SPINE sub-class name is always capitalized.

1323 **Function-Group:** A function-group consists of the function itself (xListData or xData) and (in case of a
 1324 ListData) its corresponding xListDataSelectors. The first letter of the SPINE functions-group name is
 1325 always lowercase.

1326 **Function:** The resources that are used for communication are called functions (e.g.
 1327 *powerTimeSlotValueListData*). The first letter of the SPINE function name is always lowercase.

1328 **Element:** Elements contain the actual data information (e.g. value, timestamp, etc.). The first letter
 1329 of the element name is always lowercase.



1330

1331 *Figure 15: SPINE class hierarchy*

1332

3 Common technical details

3.1 Introduction

The subsequent sections define and discuss miscellaneous technical details that are required by classes.

3.2 Time information (absolute / relative / recurring)

Nearly all time information is of type: *AbsoluteOrRelativeTimeType* (see below).

Exceptions:

- *timeInformationData* -> utc

- *timePrecisionData* -> lastSyncAt

The simple type *AbsoluteOrRelativeTimeType* is a union of *xs:dateTime* and *xs:duration*. So it is possible to either model absolute times (*xs:dateTime* format, e.g. 2014-04-28T17:43:14.0Z) or relative time information (*xs:duration* format, e.g. P2Y8M15DT7H32M17S).

Unless specified otherwise, an *xs:duration* based time of *AbsoluteOrRelativeTimeType* always relates to "now".

Absolute times SHALL always be stated as UTC-Time ("Z" at the end of the time information).

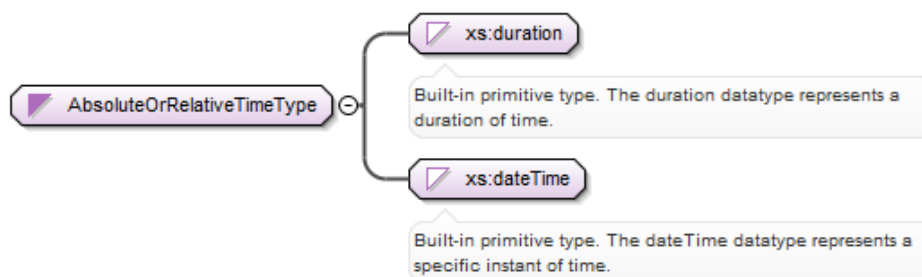


Figure 16: Absolute or relative time type

For recurring times, another type (*AbsoluteOrRecurringTimeType*, see section 3.10.2.13) is defined.

3.3 List Data concept

Some SPINE classes provide lists of data instances. These lists always consist of 0 (hence, may be empty) to unlimited count of single data instances (and nothing else). A function can either contain a (non-list) data as resource or the list data. If a list data is defined in a SPINE class, only this list data may be used as resource, not its non-list data (which is only used as reference in the list data).

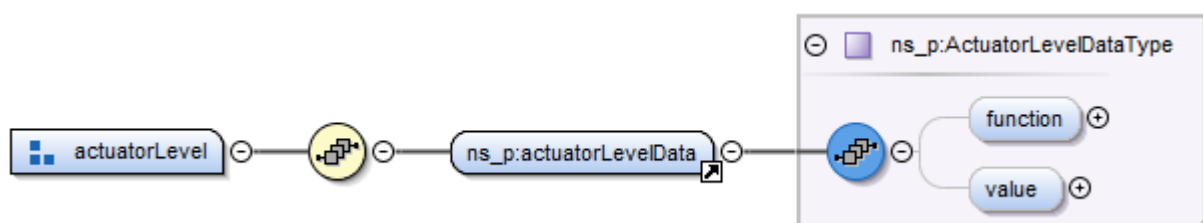


Figure 17: Non-list data function example



Figure 18: List data function example

3.4 Identifier concept

3.4.1 Introduction

Identifiers are used esp. in ListData-functions to identify single list entries unambiguously. They are also used for references within the class itself (to relate different kinds of information) as well as for references from other classes (e.g. the Setpoint class (see section 5.3.21) references the Measurement class (see section 5.3.15), to announce for which actual measurand a specific setpoint is used for). References are only possible within the same entity, not between different entities or devices.

A special kind of identifier is a SPINE address, composed of the elements "device", "entity", and "feature" (the types are defined in section 3.10.3). Among others, the element "entity" can occur multiple times (i.e. as list) within some address definitions. As already explained in [ProtocolSpecification], section "Address level details", a SPINE address must always be seen as indivisible information if it is used as information unit (only the "device" part may be omitted in particular cases). Unlike other identifiers, a SPINE address identifies nodes across all entities and devices. See also section 5.3.17.1.1. For its use in a selectors, please see [ProtocolSpecification], section "Selectors with address elements".

Note: For some functions or use cases a specific data instance is only uniquely identified with the use of multiple identifiers (e.g. within the Power Sequences class a "slot" with number 2 of "sequence" 5 is different to "slot" 2 of "sequence" 7; this typically requires the use of sequenceId AND slotNumber in order to identify a specific slot). In some cases a data instance may carry all these identifiers "at the same place" (in the data model next or close to each other, e.g.). In other cases the identifiers may be organized rather hierarchically in a data structure (this finally depends on the chosen structure of a function; it is esp. true for so-called "complex classes", if a "parent" identifier is placed closer to the root of a function's data structure than a "child" identifier).

An overview of all identifiers can be found in section B.7.

Feature Types define which kind of Identifier is associated to a particular element (see section 3.4.2.1).

3.4.2 Identifier rules

3.4.2.1 Identifier keywords

In the Feature Types the following keywords are used for identifiers:

- 1394 - PRIMARY IDENTIFIER: A primary identifier is unique for the whole Feature. Primary identifiers are
1395 used only as "primary" or "top-level" identifier of a Function. MAY be combined with further SUB
1396 IDENTIFIERS if needed to identify a list entry uniquely (for lists with more than one dimension).
1397 Note: A Feature Type may define two or more PRIMARY IDENTIFIERS. In this case the Feature Type
1398 describes a relation between the PRIMARY IDENTIFIERS.
- 1399 - SUB IDENTIFIER: One or more SUB IDENTIFIERS MAY be used within a function to identify the
1400 further dimensions of list entries (if the list has more than one dimension (e.g. *parameterId* within
1401 *electricalConnectionParameterDescriptionListData*)).
- 1402 - FOREIGN IDENTIFIER: Used to refer to other functionality on the same entity. It is not used to
1403 create further dimensions of list entries.
1404 Usually, a FOREIGN IDENTIFIER contains a value that is also set as PRIMARY IDENTIFIER in another
1405 Function. This is the most common relation between list entries of different Features. However,
1406 there can also be Use Cases that make use of FOREIGN IDENTIFIERS only, i.e. without an explicit
1407 instance of a list entry where a given FOREIGN IDENTIFIER value is set as PRIMARY IDENTIFIER.
1408 Such list entries with the same FOREIGN IDENTIFIER values express a relationship between these
1409 list entries as well.

1410 The according rules only apply if the identifier keyword is written in capital letters.

1411

1412 **3.4.2.2 Order of identifiers**

1413 The denoted order of the identifiers in the function description within the Feature Types determines
1414 the order of the identifiers.

1415 There might be a variance of that in some classes where a SUB IDENTIFIER appears before a PRIMARY
1416 IDENTIFIER in the data model. In this case, the order is

- 1417 - PRIMARY IDENTIFIER
- 1418 - SUB IDENTIFIER
- 1419 - [eventually further SUB IDENTIFIERS]

1420 It MAY be allowed to skip identifiers (e.g. if a Feature Type defines three identifiers, the one in the
1421 middle MAY be omitted in some cases). If so, it is defined within the Feature Type.

1422

1423 **3.4.2.3 timestamp identifier rules**

1424 If timestamp is used as identifier, some special rules need to be considered:

- 1425 - If used as part of a notification or reply, it may be of type *xs:duration* or *xs:dateTime*.
- 1426 - If used as part of a write operation:
 - 1427 ○ If the time in timestamp shall express "now", this can be modelled in two ways:
 - 1428 ▪ Using *xs:duration*: duration of 0 seconds (e.g. PT0S)
 - 1429 ▪ Using *xs:dateTime*: the current time and date

1430 As the time base of the communication partners client and server may differ, only
1431 relative time representation SHOULD be used for write operations related to "now"
1432 to avoid misinterpretations. Please keep in mind that the server may provide the
1433 times as absolute times in notify or reply messages.

- 1434 ○ For times others than "now", the client SHALL only use time representations
1435 (absolute or relative) supported by the server. The support currently may only be
1436 discovered by analyzing the messages sent by the server.
- 1437 ○ For historical values that are writeable, only absolute time representation SHOULD
1438 be used.
- 1439 - Normally a "binary comparison" of the identifier with the ones stored on the own device
1440 finds the appropriate data. But as timestamp can be modelled using absolute or relative time
1441 representation, a more complex comparison has to be chosen!
- 1442 - The feature type may define further rules (i.e. restrictions) on the use of timestamp. This
1443 may affect its use as identifier as well as a restriction of its value range (e.g. if only
1444 *xs:duration* shall be used but not *xs:dateTime*).

1445

1446 **3.4.2.4 Further rules**

1447 All primary identifiers SHALL be set if not stated otherwise in the feature type. Non-primary
1448 identifiers SHOULD be set, if available and if not stated otherwise in the feature type.

1449 Primary identifiers are those that are used to reference the classes functionality within the Class
1450 itself as well as from other features (e.g. measurementId within the Measurement class, referenced
1451 from ElectricalConnection) and are unique within the feature.

1452 Each xListData entry xData SHALL be uniquely identifiable within the list by the respective PRIMARY
1453 IDENTIFIERS and SUB IDENTIFIERS.

1454

1455 **3.4.2.5 Optional identifiers with default values**

1456 Some identifiers in some Feature Types (e.g. "valueId" in the Feature Type "Bill") are defined with a
1457 default value and can be omitted in a message if the feature contains data of at maximum one
1458 identifier value of the respective list. In this case, default values for omitted identifiers apply.
1459 However, there are typical pitfalls implementers need to consider:

- 1460 1. In general, identifiers SHALL be evaluated if they are present! Esp. extensions with new use
1461 cases/applications can easily lead to a situation where a formerly absent identifier element
1462 suddenly is set because there are now two or more identifier values used!
- 1463 2. As soon as an Identifier is set within one list entry, all list entries SHALL set the according
1464 Identifier.
- 1465 3. As mentioned above, default values are defined for some identifiers. This must be considered
1466 properly:
 - 1467 a. We assume there is a Feature "A" with a Function that omits a particular PRIMARY or SUB
1468 IDENTIFIER in its messages as the feature server applies internally the default value for the
1469 list entries. Now we consider an other Feature "B" with a Function that contains a FOREIGN
1470 IDENTIFIER to refer to respective list entries of Feature "A"'s Function with the omitted
1471 identifier: In this case the FOREIGN IDENTIFIER in "B" must be set to this default value.
 - 1472 b. The rule in item 2 above applies regardless of the value of the identifier. However,
1473 implementations should in particular consider cases briefly mentioned above, where
1474 suddenly two identifier values are used: In addition to the list elements with the default

1475 identifier value (usually "1") there are now also list elements with a new identifier value ("8",
1476 e.g.). In this case also for the formerly omitted Identifiers the corresponding list entries
1477 SHALL then contain the identifier explicitly.

1478 A typical implementation error is to ignore the possibility of multiple identifier values and the
1479 implementation of wrong default values. In this case, client features may not be able to analyze
1480 Function (data) changes properly. This can especially occur for the case described above in item 1.

1481 Generally, it is recommended to never omit identifiers if they are set to the default value, if possible,
1482 as it is more explicit and less prone to errors.

1483

1484 **3.4.2.6 Optional identifiers without default values**

1485 A Feature Type may also define an optional Identifier that has no default value. The following rules
1486 apply:

- 1487 1. If the Identifier is omitted:
1488 In this case the Identifier SHALL be omitted in all list entries. This means it is not possible then to
1489 submit or refer to only particular entries of this list.
- 1490 2. If the Identifier is present:
1491 In this case the Identifier SHALL be present in all list entries. As usual, the Identifier is then used
1492 to identify particular list entries.

1493 A typical implementation error is to ignore that a given list with such an Identifier may sometimes
1494 omit the Identifier and sometimes may have it set. Switching from one list type to the other always
1495 denotes a complete replacement of the former list by the new one.

1496

1497 **3.5 Multidimensional data representation with "flat" list-based data model**

1498 Many applications need to work with multidimensional data. The functions of SPINE standard classes,
1499 on the other hand, usually are modelled as lists (i.e. one-dimensional objects) of rather flat data
1500 structures. However, this list concept is comparable to common table-based databases (SQL, e.g.)
1501 that are also frequently used for the management of multidimensional data. As there are similar
1502 concepts, most of the SPINE standard classes can as well be used for the modelling of
1503 multidimensional data. This section gives a brief example.

1504 For this example we will use an excerpt of the function "measurementListData" of the SPINE
1505 Measurement class. It can take a list of "measurementData" elements, each comprising of the
1506 elements

- 1507 • measurementId
- 1508 • valueType
- 1509 • timestamp
- 1510 • value
- 1511 • evaluationPeriod
- 1512 • and some more elements

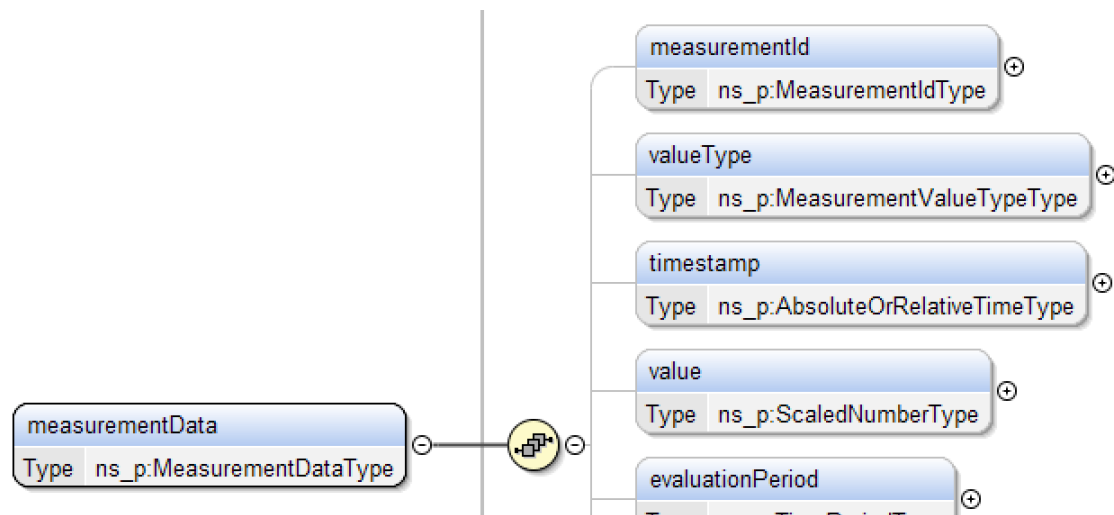
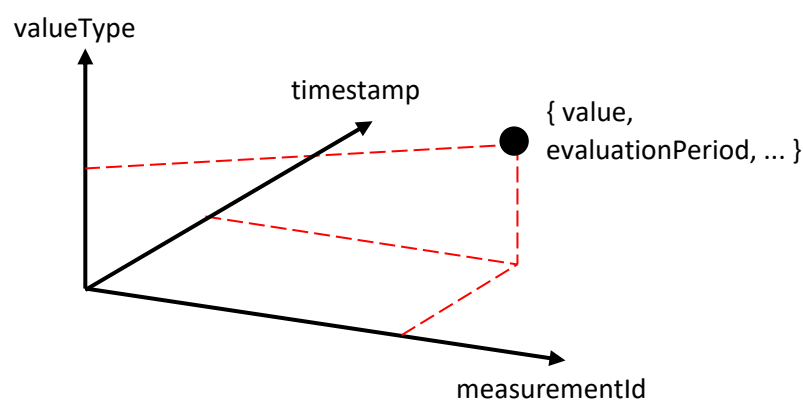


Figure 19: Excerpt of the list element "measurementData" of the SPINE function measurementListData

In the Measurement class definition, the first three elements "measurementId", "valueType", and "timestamp" are marked as so-called "identifiers". This means a particular measurementData instance is identified by these elements (just to give a comparison: In table-based databases such kinds of elements are frequently called "keys"). Mathematically, an identifier can be seen as the axe of a single dimension. The combination of distinct identifiers then creates a multidimensional space. At each position of this space the remaining elements ("value", "evaluationPeriod", etc.) can be placed with proper data:



SPINE identifiers can create a multidimensional space of data.

As can be seen in Figure 19, the "position" in the multidimensional space is part of the measurementData structure. Therefore, a list of measurementData instances can build a complete multidimensional space with data. Among others, a particular value of measurementId can then contain data (value, evaluationPeriod, etc.) for multiple valueType values and multiple timestamp values. The following XML shows an example of a list of measurementData instances (for brevity the example just contains the element "value" and skips "evaluationPeriod" and further elements):

```
<measurementListData>
  <measurementData>
    <measurementId>1</measurementId>
    <valueType>value</valueType>
    <timestamp>2018-07-03T17:16:28.0Z</timestamp>
    <value>
```

```

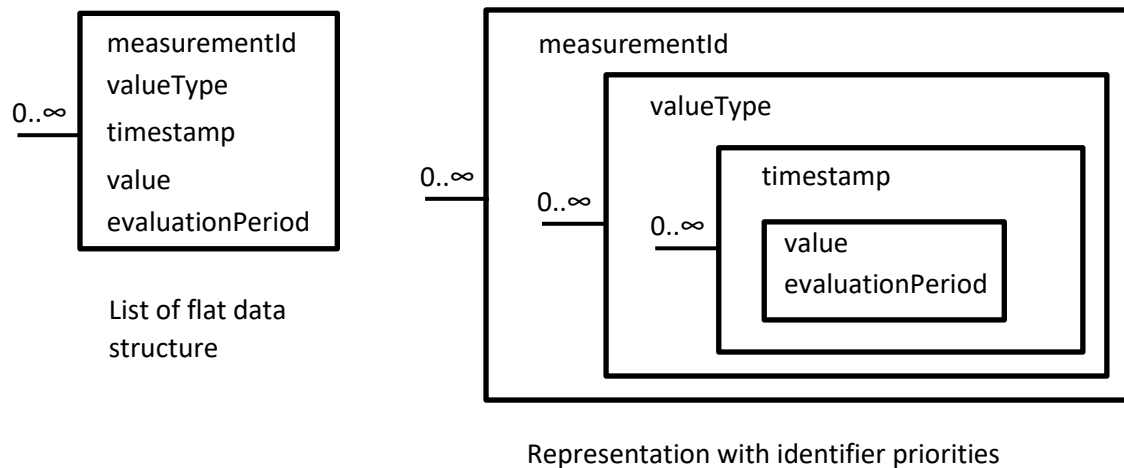
1536         <number>15</number>
1537         <scale>-1</scale>
1538     </value>
1539 </measurementData>
1540 <measurementData>
1541     <measurementId>1</measurementId>
1542     <valueType>value</valueType>
1543     <timestamp>2018-07-03T17:16:33.0Z</timestamp>
1544     <value>
1545         <number>21</number>
1546         <scale>-1</scale>
1547     </value>
1548 </measurementData>
1549 <measurementData>
1550     <measurementId>1</measurementId>
1551     <valueType>minValue</valueType>
1552     <timestamp>2018-07-03T17:16:33.0Z</timestamp>
1553     <value>
1554         <number>1</number>
1555     </value>
1556 </measurementData>
1557 <measurementData>
1558     <measurementId>2</measurementId>
1559     <valueType>value</valueType>
1560     <timestamp>2018-07-03T17:16:33.0Z</timestamp>
1561     <value>
1562         <number>55</number>
1563     </value>
1564 </measurementData>
1565 </measurementListData>

```

Semantically, this example XML models data for two different measurementId values, where

1. measurementId 1 has data for two different valueTypes:
 - a. For valueType "value" there is data available for two different timestamps.
 - b. For valueType "minValue" there is data available for only one timestamp.
2. measurementId 2 just has data for a single valueType = "value" at a particular timestamp.

Please note that SPINE identifiers have a "priority" within a SPINE class (or Feature Type, resp.), as explained in section 3.4. This means that a list-based SPINE function cannot be created in a way that identifiers of lower priority are used without the mandatory higher priority identifiers. With regards to the example above this means it would be invalid to create a measurementData instance where "valueType" is set but "measurementId" is absent. The following figure compares the flat list-based structure from Figure 19 with a virtual arrangement that considers the priorities of the identifiers.



Representation with identifier priorities

Figure 21: The flat data structure of Figure 19 can also be rearranged (virtually) to emphasize the priorities of the identifiers.

Note: FOREIGN IDENTIFIERS SHALL NOT be used for creating the above described multidimensional space.

3.6 Restricted function exchange for list based functions

As defined in [ProtocolSpecification], section "Restricted function exchange with cmdOptions", in the communication between server and client not the complete server data has always to be considered but can be restricted considering some given rules as long as it is supported by the server (see [ProtocolSpecification], section "Minimum restricted function exchange support").

Further rules concerning the SPINE resources are defined in this section as default rules for all feature types that make use of SPINE standard class functions:

- If a server feature supports "write" operations with cmdControl "partial": The server feature SHALL accept received "xListData" containing at least one "xData". The server feature MAY accept xListData with more than one xData.
- If a server feature supports "write" operations with cmdControl "delete": The server feature SHALL accept the deletion request (using "<SELECTORS>") for at least a single "xData" of its "xListData". The server feature MAY support the deletion of more than one xData.
- If a server feature supports "read" operations with cmdControl "partial": The server feature SHALL accept the read request (using "<SELECTORS>") for at least a single "xData" of its "xListData". The server feature MAY return more than one xData.
- Complete function exchange SHALL be supported in any case (operations on the complete xListData).

Note: Feature types may still impose more restrictions than the above ones!

3.7 Relations between thematically connected SPINE classes

Some SPINE classes are thematically connected. The relations are indicated by the use of identifiers. To get a better understanding of the relations, the following diagrams visualize them. For details about the classes and the identifiers please consider the corresponding sections in this

1606 documentation. Relations are only possible within the same entity, not between different entities or
1607 devices.

1608

1609 **3.7.1 Relations between Alarm / ElectricalConnection / HVAC / LoadControl / Measurement /** 1610 **Setpoint / SupplyCondition / TariffInformation / Threshold / TimeTable**

1611 The **Measurement** class is the central functionality of these related classes. It can be used to
1612 measure a lot of different measurands, describe the measurand and model constraints, etc.

1613 If a setpoint shall be set, only the setpoint itself has to be stated in the **Setpoint** class. Via the
1614 *measurementId*, the type of measurand, that a setpoint is set for, is denoted. The period in time, this
1615 setpoint shall be applied can be specified in a time table from the **TimeTable** class. With the
1616 *timeTableId*, the reference is given.

1617 The **HVAC** class references the **Setpoint** class as well as the **TimeTable** class.

1618 If thresholds shall be set - either to activate an alarm if the threshold is exceeded or to do previously
1619 configured actions (implementation specific) - the **Threshold** class is used. The value that shall be
1620 measured to check if it is over- or underrun is the one from the Measurement class. The Threshold
1621 class does not reference the Measurement class, instead, in the Measurement class a relation
1622 function exists that lists all relevant thresholds.

1623 If a threshold is exceeded, the **Alarm** class notifies all subscribed clients. To identify which threshold
1624 was exceeded, the Alarm class references the *thresholdId*.

1625 The **SupplyCondition** class can reference to thresholds, too, to eventually generate events in case of
1626 exceeding some supply concerning limits.

1627 Specifically for electrical measurands, the **ElectricalConnection** class describes all parameters
1628 needed. The values themselves are modelled with the Measurement class.

1629 The **LoadControl** class enables a client to set limits on the server instance. To model further
1630 information about the measurand that is limited, the Measurement class is used.

1631 The **TariffInformation** class references the **Measurement** class as well as the **TimeTable** class to
1632 model simple or complex tariff related functionality.

1633

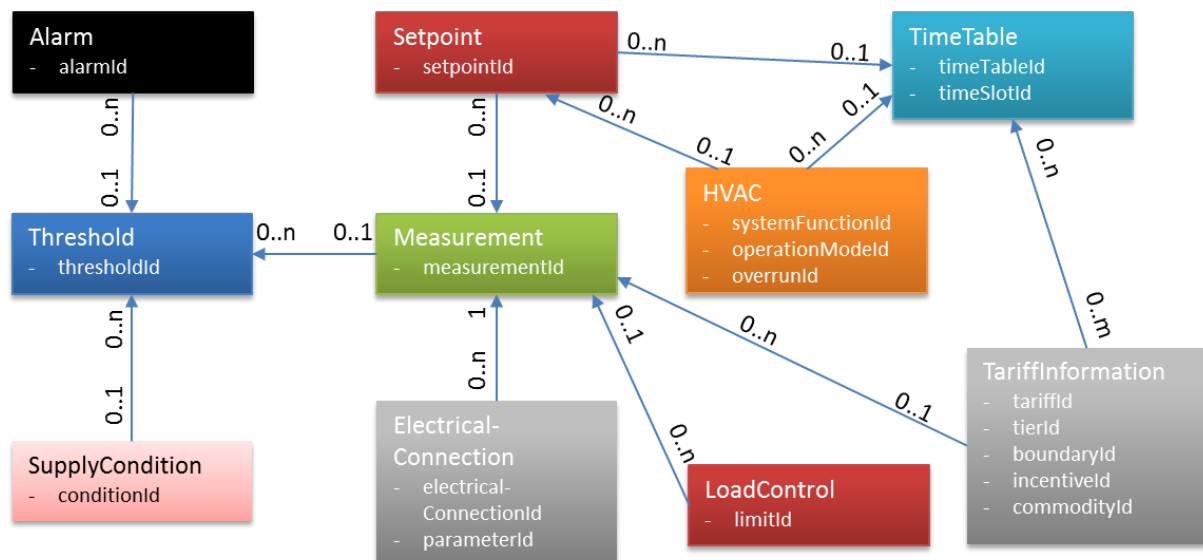


Figure 22: Relations between Alarm / ElectricalConnection / HVAC / LoadControl / Measurement / Setpoint / SupplyCondition / TariffInformation / Threshold / TimeTable

3.7.2 Relations between DirectControl / HVAC / LoadControl / OperatingConstraints / PowerSequences (SmartEnergyManagementPs) / TaskManagement

The **PowerSequences** (or the **SmartEnergyManagementPs** complex) class can be used to model sequences and slots, characterising a specific demand of energy that may be shifted in some range.

The **OperatingConstraints** class adds constraints to a power sequence to avoid misconfigurations of a device and models resume implications like additional energy or costs generated by pausing and then resuming some functionality. When using DirectControl (see below), the OperatingConstraints SHALL be considered, too, if linked with the same feature group (see section 2.1.4).

Pausing and reactivating devices can be done with the **DirectControl** class, which can relate to some power sequence that shall be controlled in a direct manner.

The **HVAC** class may reference power sequences that characterize specific functionality in more detail.

Another way the demand of a functionality may be affected is by some more or less external instruction (e.g. by the DSO) via the **LoadControl** class.

The **TaskManagement** class lists all the jobs from these classes (except for OperatingConstraints) and gives the needed relation in an extra function.

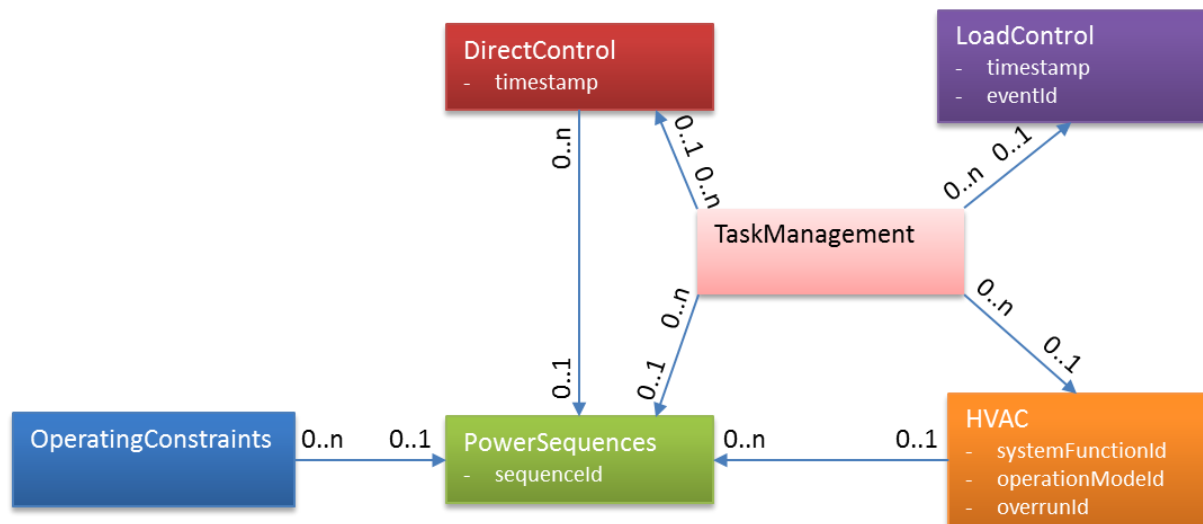


Figure 23: Relations between DirectControl / HVAC / LoadControl / OperatingConstraints / PowerSequences (SmartEnergyManagementPs) / TaskManagement

3.8 On the use of "label" and "description"

In most functions with rather static information, elements called "label" and "description" are defined. These two elements MAY contain human readable strings (they are always optional) that SHALL NOT be used for automatisms (there are other elements for this purpose).

In general, the label element should contain a very short string which could be displayed below an icon, e.g. The description may contain a longer string that is only shown if the user clicks on a corresponding button or similar.

The following examples show how the strings can be used:

label: ActuatorLevel TestDevice
description: Device for testing SPINE compliance.

label: SmartPlug_1
description: SmartPlug Test Installation 01.04.2012

label: mains connection
description: 1-phase mains connection of the device

label: battery connection
description: Backup battery connection of the device

label: TempSensor1
description: Temperature Sensor 1 (Bedroom)

3.9 Empty elements as "tags"

In some cases elements shall only be used as "tags" without any content as some kind of boolean information (element is present = true, element is not present = false). This can be achieved by assigning the element a certain complex type that has no content (see section 3.10.1.1).

An example is the *DaysOfWeekType* (see section 3.10.2.10) used (e.g.) for the element *daysOfWeek* within the *start-* and *endTime* of the *timeTableListData* function (see section 5.3.30.2):

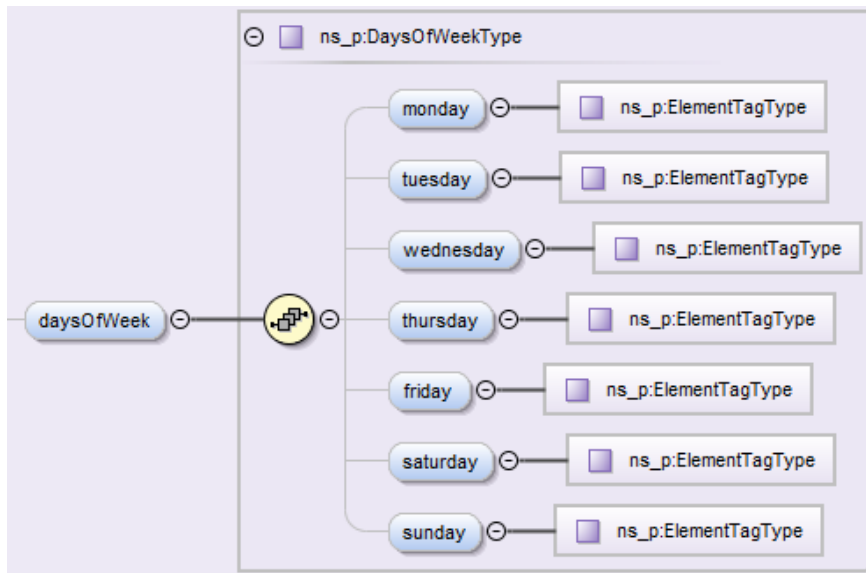


Figure 24: Empty elements example

An XML where (e.g.) the elements *monday* and *wednesday* are present would look like this:

```

...
<daysOfWeek>
  <monday/>
  <wednesday/>
</daysOfWeek>
...

```

In future versions of the SPINE specification, the type of an empty element may change and hence may include child elements, e.g. Please note that compatibility requirements and concepts always consider the content of a message (a concrete XML document, e.g.) and not the type names: The types are defined or used within XSD files, but type names are not explicitly mentioned in XML documents in this specification. Consequently, a subsequent version of the specification may well define the same (or properly extended) messages as a former version, but with partially renamed types as long as document/message compatibility is maintained. This concept permits also to extend formerly "empty" elements with child elements: If an element "foo" is defined as complex type with no further child element in version 1.0.0, then it is well permissive to assign "foo" a different complex type ("NewFooType", e.g.) with child elements in version 1.1.0.

3.10 Common data types

Some types are used by several classes. They are described in this section. For the exact naming and the complete list of all common data types please consider the underlying revision of the SPINE data model.

3.10.1 General

3.10.1.1 *ElementTagType*

Used for elements that have no content (empty tag).

A *complexType*, containing no elements.

3.10.1.1.1 *Default rules for ElementTagType*

If used as type for an element, the element SHALL have no further content in this SPINE version.

Future SPINE versions may add content that SHALL be ignored by devices implementing the SPINE version at hand.

3.10.1.2 *LabelType*

A default type used for all "label" elements.

A *simpleType* (restricted string).

3.10.1.2.1 *Default rules for LabelType*

The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.

3.10.1.3 *DescriptionType*

A default type used for all "description" elements.

A *simpleType* (restricted string).

3.10.1.3.1 *Default rules for DescriptionType*

The string-length SHOULD NOT be longer than 4096 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 4096 characters.

3.10.1.4 *SpecificationVersionType*

The SPINE specification version.

A *simpleType* (restricted string), with the pattern

[1-9] [0-9]*\.[0-9]+\.[0-9]+

1738 allowing version strings like "1.0.0", "2.5.3", etc. The maximum string length is limited to 128
1739 characters.

1740

1741 **3.10.1.5 EnumExtendType**

1742 Used for extending enumeration with vendor specific ones.

1743 Some enumerations may be extended by the vendors with own values.

1744 A *simpleType* (restricted string), with the pattern

1745 $_ (i: [1-9] [0-9]^* | n: [a-zA-Z0-9-]^+) _ [^\p{Cc} \p{Cf} \p{Z}]^+$

1746 For details please see section 2.1.3. The maximum string length is limited to 256 characters.

1747

1748 **3.10.1.5.1 Default rules for EnumExtendType**

1749 The values SHALL be marked as vendor specific by adding the IANA PEN code of the vendor ("i:") or
1750 some vendor name ("n:") surrounded by "_".

1751 The IANA PEN SHOULD be used!

1752

1753 **3.10.1.6 NumberType**

1754 Used to represent a numeric (integer) value. See also section 3.10.1.8!

1755 A *simpleType* (xs:long).

1756

1757 **3.10.1.7 ScaleType**

1758 Used to scale a numeric value. See also section 3.10.1.8!

1759 A *simpleType* (xs:short).

1760

1761 **3.10.1.8 ScaledNumberType**

1762 Numeric values (e.g. numerical values, setpoints, thresholds, etc.) use this number representation.

1763 A *complexType*, containing the elements *number* (type: *NumberType*, see section 3.10.1.6) and *scale*
1764 (type: *ScaleType*, see section 3.10.1.7). See Figure 25.

1765

1766 **3.10.1.8.1 Default rules for ScaledNumberType**

1767 Throughout this specification, this type and its elements are used as follows unless stated otherwise:

- 1768 1. This type represents a real number (subsequently called "scaledNumber") that is calculated from
1769 the type's elements as follows:

$$1770 \text{ scaledNumber} = \text{number} * 10^{\text{scale}}$$

1771

- 1772 2. If the element "scale" is omitted a default value of "0" SHALL be applied for "scale". Then, item 1
1773 applies, or (equivalently)

$$1774 \text{ scaledNumber} = \text{number}$$

3. The element "number" SHALL be set.

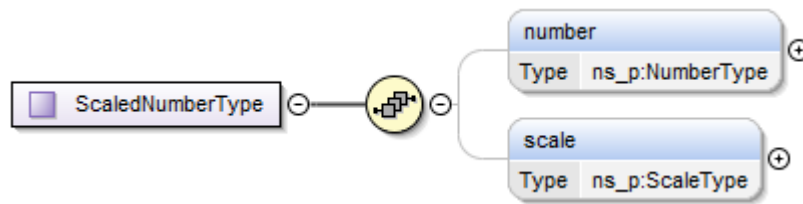


Figure 25: ScaledNumberType

3.10.1.9 ScaledNumberSetType

This type can be used to describe any sets of scaledNumbers. The set can have gaps compared to a simple value range and include many different values and value ranges.

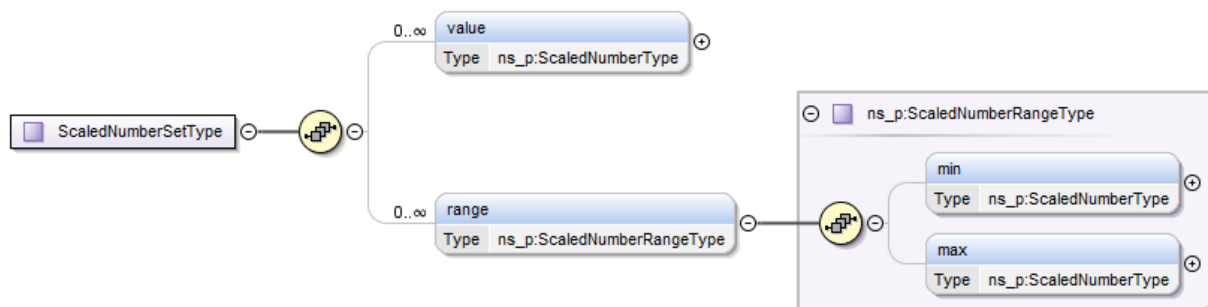


Figure 26: ScaledNumberSetType

Name	Type	Description
value (list)	Common data type "ScaledNumberType". See section 3.10.1.8.	Allows to describe numbers that are part of the set: Each "value" instance contributes its value as single number to the set.
range (list)	Common data type "ScaledNumberRangeType". See section 3.10.1.10.	Allows to describe number ranges that are part of the set. Each "range" instance contributes its value range to the set.

Table 1: ScaledNumberSetType

3.10.1.9.1 Default rules for ScaledNumberSetType

The set of lists MAY have gaps (e.g., the value list contains [1, 3, 7] and the range list contains [3..5, 8..13, 18..20]).

3.10.1.10 ScaledNumberRangeType

This type can be used to describe a value range. All ScaledNumbers between min and max are supported in this case, at least if no further constraints apply, e.g. a value step size.

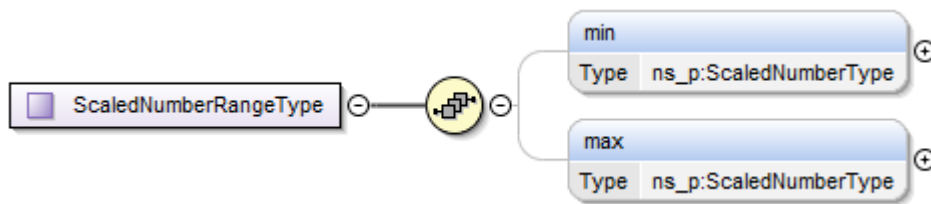


Figure 27: ScaledNumberRangeType

Name	Type	Description
min	Common data type "ScaledNumberType". See section 3.10.1.8.	The min element defines the lower bound of the ScaledNumberRange.
max	Common data type "ScaledNumberType". See section 3.10.1.8.	The max element defines the upper bound of the ScaledNumberRange.

Table 2: ScaledNumberRangeType

3.10.1.11 CommodityTypeEnumType

Union of:

- CommodityTypeEnumType (see section 3.10.1.12)
- EnumExtendType (see section 3.10.1.5)

3.10.1.12 CommodityTypeEnumType

The commodity, used for measurements, pricing, etc.

A *simpleType* (restricted string), containing enumerations for the commodity.

Value	Description
electricity	Electricity (AC or DC).
gas	Gas, as used by a gas heater.
oil	Oil, as used by an oil heater.
water	(Drinking) water.
wasteWater	(Domestic) waste water.
domesticHotWater	Domestic hot (drinking, e.g.) water.
heatingWater	Water circulating through the heating system of a building.
steam	Steam, as used in (e.g.) a steam boiler or within a coal-fired power station.
heat	Heat (e.g. in the form of hot water, hot air, etc.).
coolingLoad	Cooling load used to cool down goods or a room / building.
air	Air, as commonly available outside (no special gas or chemical compound).

Table 3: Enumeration CommodityTypeEnumType

3.10.1.13 EnergyDirectionType

Union of:

- EnergyDirectionEnumType (see section 3.10.1.14)

- EnumExtendType (see section 3.10.1.5)

3.10.1.14 EnergyDirectionEnumType

The energy direction. Can be applied for other measurands than energy, too (such as power or current).

A *simpleType* (restricted string), containing the enumerations *consume* and *produce*.

3.10.1.14.1 Default rules for EnergyDirectionEnumType

If this type is used for an Element, the related data SHALL be interpreted in a way as defined by the rules below.

Value	Rule
consume	Positive values SHALL be interpreted as consumption; negative values SHALL be interpreted as production.
produce	Positive values SHALL be interpreted as production; negative values SHALL be interpreted as consumption.

Table 4: Enumeration EnergyDirectionEnumType value rules

3.10.1.15 EnergyModeType

Union of:

- EnergyModeEnumType (see section 3.10.1.16)

- EnumExtendType (see section 3.10.1.5)

3.10.1.16 EnergyModeEnumType

Different modes for energy demand.

A *simpleType* (restricted string), containing the enumerations for the energy mode.

Value	Description
consume	Used in cases where the functionality consumes energy.
produce	Used in cases where the functionality produces energy.
idle	Used in cases where the functionality neither consumes nor produces energy.
auto	Used in cases where the functionality may consume or produce energy (or none of both), decided on its internal business logic.

Table 5: Enumeration EnergyModeEnumType

3.10.1.16.1 Default rules for EnergyModeEnumType

Value	Rule
consume	The related functionality SHALL only consume energy.

produce	The related functionality SHALL only produce energy.
idle	The related functionality SHALL neither consume nor produce energy.
auto	The related functionality MAY consume or produce energy or be in idle-mode based on its own decision.

Table 6: Enumeration EnergyModeEnumType value rules

3.10.1.17 UnitOfMeasurementType

Union of:

- UnitOfMeasurementEnumType (see section 3.10.1.18)

- EnumExtendType (see section 3.10.1.5)

3.10.1.18 UnitOfMeasurementEnumType

Unit, used for measurements, etc.

A *simpleType* (restricted *string*), containing several units.

In general the following format is used to model mathematical operations as strings:

 $x/y \equiv x$ divided by y (i.e. $x \div y$) $x^y \equiv x$ raised to the power of y (i.e. x^y) $x_y \equiv x$ multiplied by y (i.e. $x * y$)**Examples:** $m^3 \equiv m^3$ $m^3/h \equiv (m^3) \div h$ $m/s^2 \equiv m \div (s^2)$ $kg_m^2 \equiv kg * (m^2)$

Unit Name	Unit Description
unknown	Unknown unit
1	1 (unitless)
m	metre (SI unit, used for length)
kg	kilogram (SI unit, used for mass)
s	second (SI unit, used for time)
A	ampere (SI unit, used for electric current)
K	kelvin (SI unit, used for temperature)
mol	mole (SI unit, used for the amount of a substance)
cd	candela (SI unit, used for luminous intensity)
V	volt (used for voltage)
W	watt (used for power)
Wh	watt hour (used for energy)
VA	volt-ampere (used for apparent power)
VAh	volt-ampere hour (used for apparent energy)
var	volt-ampere reactive (used for reactive power)
varh	volt-ampere-reactive hours (used for reactive energy)
degC	degree Celsius (°C, used for temperature)
degF	degree Fahrenheit (°F, used for temperature)

Lm	lumen (used for total quantity of visible light emitted by a source)
lx	lux (used for illuminance and luminous emittance, measuring luminous flux per unit area)
Ohm	ohm (used for electrical resistance)
Hz	hertz (used for frequency)
dB	decibel (used to express the ratio of one value of a physical property to another on a logarithmic scale)
dBm	decibel with a reference value of one milliwatt
pct	percentage
ppm	parts-per-million
l	litre (used for volume)
l/s	litre per second (used for flow rate)
l/h	litre per hour (used for flow rate)
deg	degree of arc (used for planar angle; a full rotation is 360 degrees)
rad	radian (used for planar angle; a full rotation is 2π)
rad/s	radian per second (used for rotational speed (angular velocity))
sr	steradian or square radian (used for solid angles in three-dimensional geometry)
Gy	gray (used for ionizing radiation dose)
Bq	becquerel (used for radioactivity)
Bq/m ³	becquerel per cubic metre (used for specific radioactive activity in air and gases)
Sv	sievert (used for ionizing radiation dose)
Rd	rutherford (used for radioactive decay)
C	coulomb (used for electric charge)
F	farad (used for electrical capacitance)
H	henry (used for electrical inductance)
J	joule (used for energy)
N	newton (used for force)
N_m	newton metre (used for torque (also called <i>moment</i>))
N_s	newton second (used for impulse)
Wb	weber (used for magnetic flux)
T	tesla (used for magnetic flux density)
Pa	pascal (used for pressure)
bar	bar (used for pressure)
atm	standard atmosphere (used for pressure)
psi	pound-force per square inch (used for pressure)
mmHg	millimetre of mercury (used for pressure)
m ²	square metre (used for area)
m ³	cubic metre (used for volume)
m ³ /h	cubic metre per hour (used for (volumetric) flow rate)
m/s	metre per second (used for speed (scalar) and velocity (vector quantity which specifies both magnitude and a specific direction))
m/s ²	metre per second squared (used for acceleration)
m ³ /s	cubic metre per second (used for (volumetric) flow rate)
m/m ³	metre per cubic metre (used for fuel efficiency)
kg/m ³	kilogram per cubic metre (used for density)

kg_m	kilogram metre (used for payload-distance)
m^2/s	square metre per second (used for kinematic viscosity)
W/m_K	watts per meter-kelvin, W/(m·K) (used for thermal conductivity)
J/K	joule per kelvin (used for heat capacity (or thermal capacity) and (thermodynamic) entropy)
1/s	1 per second (used for frequency)
W/m^2	watt per square metre (used for heat flux density, irradiance)
J/m^2	joule per square metre (used for surface tension, stiffness)
S	siemens (used for electric conductance, electric susceptance and electric admittance)
S/m	siemens per metre (used for electrical conductivity)
K/s	kelvin per second (used for heating rate)
Pa/s	pascal per second (used for power density)
J/kg_K	joule per kilogram kelvin, J/(kg·K) (used for specific heat capacity)
Vs	volt-second (used for magnetic flux)
V/m	volt per metre (used for electric field strength)
V/Hz	volt per hertz (used for magnetic flux)
As	ampere second (used for electric charge)
A/m	ampere per metre (used for magnetization, magnetic field strength)
Hz/s	hertz per second (used for frequency drift)
kg/s	kilogram per second (used for mass flow rate)
kg_m^2	kilogram square metre (used for moment of inertia)
J/Wh	joule per watt-hour (used for heat rate)
W/s	watt per second (used for ramp rate)
ft^3	cubic foot (used for volume)
ft^3/h	cubic foot per hour (used for (volumetric) flow rate)
ccf	centum cubic feet = 100 ft ³ (used for volume)
ccf/h	centum cubic feet per hour = 100 ft ³ /h (used for (volumetric) flow rate)
US.liq.gal	US (liquid) gallon (used for (fluid) capacity, volume)
US.liq.gal/h	US (liquid) gallon per hour (used for flow rate)
Imp.gal	imperial (UK) gallon
Imp.gal/h	imperial (UK) gallon per hour (used for flow rate)
Btu	british thermal unit (used for heat)
Btu/h	british thermal unit per hour (used for power)
Ah	ampere hour (used for electric charge)
kg/Wh	kilogram per watt-hour (used for (e.g.) fuel economy or CO ₂ emission)

Table 7: Enumeration UnitOfMeasurementEnumType

3.10.1.19CurrencyType

Union of:

- CurrencyEnumType (see section 3.10.1.20)

- EnumExtendType (see section 3.10.1.5)

1863 **3.10.1.20 CurrencyEnumType**

1864 Currency as used by price information, e.g.

1865 A *simpleType* (restricted *string*), containing several currencies (as defined in ISO 4217:2015).

AED	BRL	CZK	HNL	KYD	MXV	RUB	TND	XBA
AFN	BSD	DJF	HRK	KZT	MYR	RWF	TOP	XBB
ALL	BTN	DKK	HTG	LAK	MZN	SAR	TRY	XBC
AMD	BWP	DOP	HUF	LBP	NAD	SBD	TTD	XBD
ANG	BYR	DZD	IDR	LKR	NGN	SCR	TWD	XCD
AOA	BZD	EGP	ILS	LRD	NIO	SDG	TZS	XDR
ARS	CAD	ERN	INR	LSL	NOK	SEK	UAH	XOF
AUD	CDF	ETB	IQD	LYD	NPR	SGD	UGX	XPD
AWG	CHE	EUR	IRR	MAD	NZD	SHP	USD	XPF
AZN	CHF	FJD	ISK	MDL	OMR	SLL	USN	XPT
BAM	CHW	FKP	JMD	MGA	PAB	SOS	UYI	XSU
BBD	CLF	GBP	JOD	MKD	PEN	SRD	UYU	XTS
BDT	CLP	GEL	JPY	MMK	PGK	SSP	UZS	XUA
BGN	CNY	GHS	KES	MNT	PHP	STD	VEF	XXX
BHD	COP	GIP	KGS	MOP	PKR	SVC	VND	YER
BIF	COU	GMD	KHR	MRO	PLN	SYP	VUV	ZAR
BMD	CRC	GNF	KMF	MUR	PYG	SZL	WST	ZMW
BND	CUC	GTQ	KPW	MVR	QAR	THB	XAF	ZWL
BOB	CUP	GYD	KRW	MWK	RON	TJS	XAG	
BOV	CVE	HKD	KWD	MXN	RSD	TMT	XAU	

1866 Table 8: Enumeration CurrencyEnumType

1867

1868 **3.10.1.21 ScopeTypeType**

1869 Union of:

1870 - ScopeTypeEnumType (see section 3.10.1.22)

1871 - EnumExtendType (see section 3.10.1.5)

1872

1873 **3.10.1.22 ScopeTypeEnumType**1874 Some feature types use scope types for specific functionalities. These are listed in the
1875 ScopeTypeEnumType.1876 A *simpleType* (restricted *string*), containing enumerations for the scope type.

Scope Type	Description
ac	Alternating current.
acCosPhiGrid	Alternating current cosine phi of the electricity grid.
acCurrentA	Alternating current current on phase A.
acCurrentB	Alternating current current on phase B.
acCurrentC	Alternating current current on phase C.
acFrequencyGrid	Alternating current frequency of the electricity grid.
acPowerA	Alternating current power on phase A.
acPowerB	Alternating current power on phase B.
acPowerC	Alternating current power on phase C.
acPowerLimitPct	Alternating current power limit in percentage.

acPowerTotal	Alternating current power total (all phases).
acVoltageA	Alternating current voltage on phase A.
acVoltageB	Alternating current voltage on phase B.
acVoltageC	Alternating current voltage on phase C.
acYieldDay	Alternating current yield one (this) day.
acYieldTotal	Alternating current yield total (since last reset).
dcCurrent	Direct current.
dcPower	Direct current power.
dcString1	Direct current of string 1 (no info here whether voltage/current/etc.!).
dcString2	Direct current of string 2 (no info here whether voltage/current/etc.!).
dcString3	Direct current of string 3 (no info here whether voltage/current/etc.!).
dcString4	Direct current of string 4 (no info here whether voltage/current/etc.!).
dcString5	Direct current of string 5 (no info here whether voltage/current/etc.!).
dcString6	Direct current of string 6 (no info here whether voltage/current/etc.!).
dcTotal	Direct current of all strings (no info here whether voltage/current/etc.!).
dcVoltage	Direct current voltage.
dhwTemperature	Domestic hot water temperature (for example in a domestic hot water storage).
flowTemperature	Flow temperature (for example of a heating circuit).
outsideAirTemperature	Temperature of the outside air.
returnTemperature	Return (backflow) temperature (for example of a heating circuit).
roomAirTemperature	Air temperature inside a room.
charge	(Battery-)Charge-related values.
stateOfCharge	State-of-charge-related values of a (battery) appliance.
discharge	(Battery-)Discharge-related values.
gridConsumption	Values related to the consumption from the electricity grid.
gridFeedIn	Values related to the electricity grid feed in.
selfConsumption	Self consumption (optimization).
overloadProtection	Overload protection of the local grid, e.g.
acPower	AC power.
acEnergy	AC energy.
acCurrent	AC current.
acVoltage	AC voltage.
batteryControl	Battery control (e.g. charge, discharge, idle).
simpleIncentiveTable	Simple incentive table (limiting the possible variations of the Feature Type to an easy to implement solution).
stateOfHealth	State of health.
travelRange	Travel range.
nominalEnergyCapacity	Nominal energy capacity.
acPowerReal	AC power real.
acPowerApparent	AC power apparent.
acPowerReactive	AC power reactive
acYieldMonth	AC yield month.

acYieldYear	AC yield year.
acFrequency	AC frequency.
acCosPhi	AC cos ϕ .
dcEnergy	DC energy.
insulationResistance	Insulation resistance.
stateOfEnergy	State of energy.
useableCapacity	Useable capacity.
dcChargeEnergy	DC charge energy.
dcDischargeEnergy	DC discharge energy.
loadCycleCount	Load cycle count.
componentTemperature	Component temperature.
gridLimit	Grid limit.
gridLimitFallback	Grid limit fallback.
acPowerApparentTotal	AC power apparent total
acPowerReactiveTotal	AC power reactive total.
acCurrentTotal	AC current total.
acEnergyConsumed	AC energy consumed.
acEnergyProduced	AC energy produced.
batteryAcPower	Battery AC power (total).
batteryAcPowerPhaseSpecific	Battery AC power (phase-specific).
batteryDcPower	Battery DC power.
pccPower	PCC (Point of Common Coupling) power.
activePowerLimit	Active power limit.
activePowerLimitPercentage	Active power limit percentage.
simpleCommittedIncentiveTable	Simple committed incentive table. It contains committed data.
simplePreliminaryIncentiveTable	Simple preliminary incentive table. This incentive table is not committed. It might contain data for simulation or test purposes, e.g.
committedPowerPlan	A power consumption/production plan that belongs to a committed scenario (e.g., based upon a simple committed incentive table).
preliminaryPowerPlan	A power consumption/production plan that belongs to a preliminary scenario (e.g., based upon a simple preliminary incentive table).
incentiveTableEnConsWithPoETF	Incentive values per time-period for the price of energy and transmission fee for the consumption of energy.
incentiveTableEnProdWithPoETF	Incentive values per time-period for the price of energy and transmission fee for the production of energy.
incentiveTableEnConsWithPoE	Incentive values per time-period for the price of energy for the consumption of energy.
incentiveTableEnProdWithPoE	Incentive values per time-period for the price of energy for the production of energy.
incentiveTableEnConsWithTF	Incentive values per time-period for the transmission fee for the consumption of energy.
incentiveTableEnProdWithTF	Incentive values per time-period for the transmission fee for the production of energy.
activePowerForecast	A power over time curve of the probable power consumption and production of appliances .

Table 9: Enumeration ScopeTypeEnumType

1879 3.10.1.23 RoleType

1880 If a role has to be stated, this type is used.

1881 A *simpleType* (restricted *string*), containing the enumeration values *client*, *server* or *special*.

1882

1883 3.10.1.24 FeatureGroupType

1884 Use for connecting related features (see section 2.1.4).

1885 A *simpleType* (restricted *string*), with the pattern

1886 (# [1-9] [0-9] *) *

1887 allowing feature group strings like "#1", "#1#5", etc. The maximum string length is limited to 128
1888 characters.

1889

1890 3.10.1.25 DeviceTypeType

1891 Union of:

1892 - DeviceTypeEnumType (see section 3.10.1.26)

1893 - EnumExtendType (see section 3.10.1.5)

1894

1895 3.10.1.26 DeviceTypeEnumType

1896 The type of the device.

1897 A *simpleType* (restricted *string*), containing the device types defined by the EEBus Initiative e.V. See
1898 Table 353 for all enumerations.

1899

1900 3.10.1.27 EntityTypeType

1901 Union of:

1902 - EntityTypeEnumType (see section 3.10.1.28)

1903 - EnumExtendType (see section 3.10.1.5)

1904

1905 3.10.1.28 EntityTypeEnumType

1906 The type of the entity.

1907 A *simpleType* (restricted *string*), containing the entity types defined by the EEBus Initiative e.V. See
1908 Table 354 for all enumerations.

1909

1910 3.10.1.29 FeatureTypeType

1911 Union of:

1912 - FeatureTypeEnumType (see section 3.10.1.30)

1913 - EnumExtendType (see section 3.10.1.5)

1914

1915 **3.10.1.30 FeatureTypeEnumType**

1916 The type of the feature.

1917 A *simpleType* (restricted *string*), containing the feature types defined by the EEBus Initiative e.V. See
1918 Table 355 for all enumerations.

1919

1920 **3.10.1.31 FeatureSpecificUsageType**

1921 Deprecated.

1922

1923 **3.10.1.32 FeatureHvacSpecificUsageEnumType**

1924 Deprecated.

1925

1926 **3.10.1.33 FeatureMeasurementSpecificUsageEnumType**

1927 Deprecated.

1928

1929 **3.10.1.34 FeatureSetpointSpecificUsageEnumType**

1930 Deprecated.

1931

1932 **3.10.1.35 FunctionType**

1933 Union of:

1934 - FunctionEnumType (see section 3.10.1.36)

1935 - EnumExtendType (see section 3.10.1.5)

1936

1937 **3.10.1.36 FunctionEnumType**

1938 A list of all SPINE functions defined by the EEBus Initiative e.V.

1939 A *simpleType* (restricted *string*), containing enumerations of all SPINE functions. Please consider the
1940 class sections where all SPINE functions are specified.

1941

1942 **3.10.1.37 PossibleOperationsClassifierType**

1943 If the element *partial* is used, filtering the further content is permitted. Otherwise only full
1944 operations (no filtering) are allowed.

1945 A *complexType*, containing the element *partial* (type: *ElementTagType*, see section 3.10.1.1).

1946

1947 **3.10.1.38 PossibleOperationsReadType**

1948 If read operations are permitted, this element is set. Elements from the base type can be used (see
1949 the corresponding section).

1950 A *complexType* with the base type *PossibleOperationsClassifierType* (see section 3.10.1.37).

3.10.1.39 PossibleOperationsWriteType

If write operations are permitted, this element is set. Elements from the base type can be used (see the corresponding section).

A *complexType* with the base type *PossibleOperationsClassifierType* (see section 3.10.1.37).

3.10.1.40 PossibleOperationsType

Defines which operations on a function are allowed.

A *complexType*, containing the elements *read* (type: *PossibleOperationsReadType*, see section 3.10.1.38) and *write* (type: *PossibleOperationsWriteType*, see section 3.10.1.39).

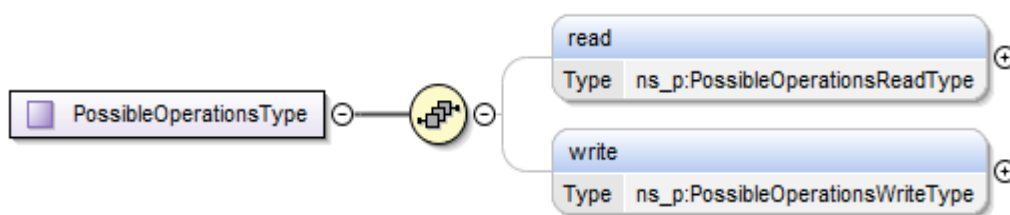


Figure 28: PossibleOperationsType

3.10.1.41 FunctionPropertyType

If a function together with its possible operations will be denoted, this complex type can be used.

A *complexType*, containing the two elements *function* (type: *FunctionType*, see section 3.10.1.35) and *possibleOperations* (type: *PossibleOperationsType*, see section 3.10.1.40).

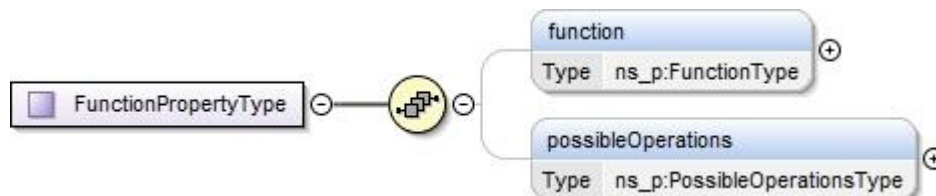


Figure 29: FunctionPropertyType

3.10.1.42 MaxResponseDelayType

Used to represent the maxResponseDelay.

A *simpleType* (xs:duration).

3.10.2 Time-related**3.10.2.1 TimePeriodType**

The typically used type for all time periods in the SPINE data model.

A *complexType* containing the two elements *startTime* and *endTime* (both of type *AbsoluteOrRelativeTimeType*, see section 3.10.2.3).

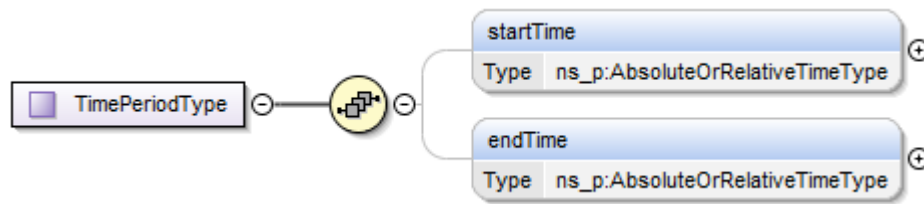


Figure 30: TimePeriodType

Name	Type	Description
startTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The startTime element defines the start time.
endTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The endTime element defines the end time.

Table 10: TimePeriodType

3.10.2.2 TimestampIntervalType

The typically used type for all timestamp intervals in selectors in the SPINE data model.

A *complexType*, containing the two elements *startTime* and *endTime* (both of type *AbsoluteOrRelativeTimeType*, see section 3.10.2.3).

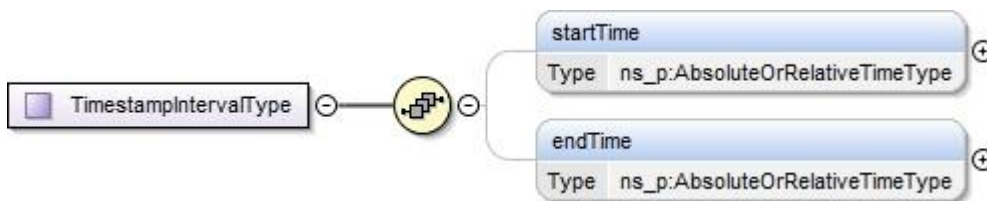


Figure 31: TimestampIntervalType

Name	Type	Description
startTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The startTime element defines the start time.
endTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The endTime element defines the end time.

Table 11: TimestampIntervalType

3.10.2.3 AbsoluteOrRelativeTimeType

The commonly used type for time information in the SPINE data model. Allows the modelling of absolute (*xs:dateTime*) or relative (*xs:duration*) times (see section 3.2).

A *union* with the two members (allowed types) *xs:duration* and *xs:dateTime*.

xs:dateTime is used to model absolute times.

1998 xs:duration is used to model relative times. If not stated otherwise a duration of 0 seconds (e.g.
 1999 "PT0S", or "PT0H", etc.) equals "now", while negative durations denote times in the past and positive
 2000 values denote times in the future relative to "now".

2001

2002 **3.10.2.4 RecurringIntervalType**

2003 Union of:

2004 - RecurringIntervalEnumType (see section 3.10.2.5)

2005 - EnumExtendType (see section 3.10.1.5)

2006

2007 **3.10.2.5 RecurringIntervalEnumType**

2008 Recurring interval length make use of this type.

2009 A *simpleType* (restricted *string*), containing enumerations for the recurring interval length.

yearly
monthly
weekly
daily
hourly
everyMinute
everySecond

2010 Table 12: RecurringIntervalEnumType

2011

2012 **3.10.2.6 MonthType**

2013 All month names are included in this type.

2014 A *simpleType* (restricted *string*), containing enumerations for the months.

january	july
february	august
march	september
april	october
may	november
june	december

2015 Table 13: MonthType

2016

2017 **3.10.2.7 DayOfMonthType**

2018 This type is used for stating the day of the month.

2019 A *simpleType* (restricted *xs:unsignedByte*), with the allowed value range from "1" to "31".

2020

2021 **3.10.2.8 CalendarWeekType**

2022 This type is used for stating the calendar week.

2023 A *simpleType* (restricted *xs:unsignedByte*), with the allowed value range from "1" to "53".

2024

2025 **3.10.2.9 DayOfWeekType**

2026 This type contains all weekdays as enumeration (only one selectable).

2027 A *simpleType* (restricted *xs:string*), containing enumerations for the weekdays.

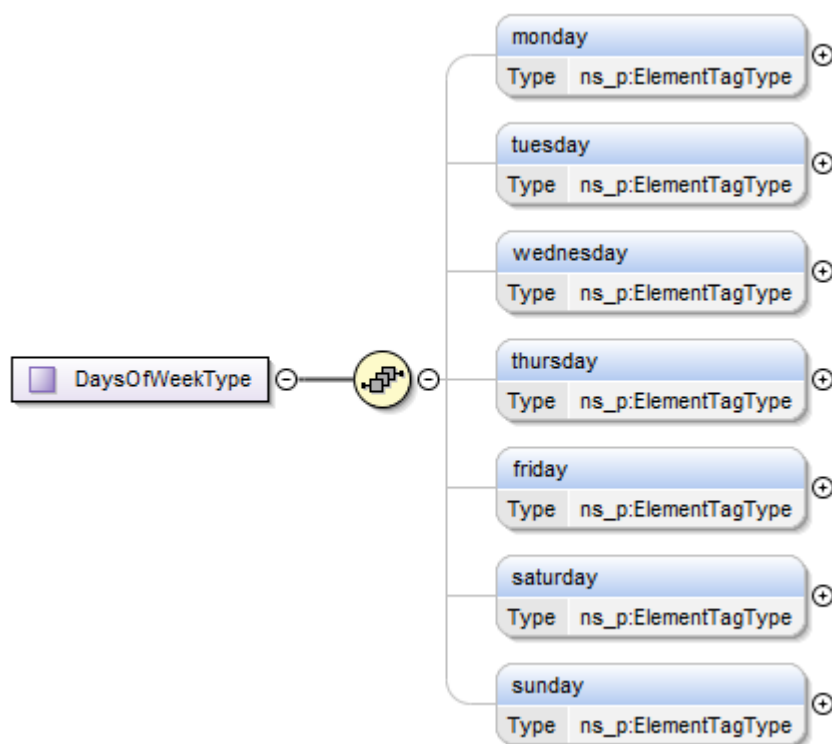
monday
tuesday
wednesday
thursday
friday
saturday
sunday

2028 Table 14: DayOfWeekType

2029

2030 **3.10.2.10 DaysOfWeekType**

2031 This type contains all weekdays as flags (more than one selectable).

2032 A *complexType*, containing the elements monday, tuesday, wednesday, thursday, friday, saturday,
2033 sunday.

2034

2035 Figure 32: DaysOfWeekType

2036

2037 **3.10.2.11 OccurrenceType**

2038 Union of:

2039 - OccurrenceEnumType (see section 3.10.2.12)

2040 - EnumExtendType (see section 3.10.1.5)

2041

2042 **3.10.2.12 OccurrenceEnumType**

2043 If the occurrence of something will be modelled, this type can be used.

2044 A *simpleType* (restricted *string*), containing enumerations for some occurrences.

Value	Description
first	First occurrence of the related event where this type is used.
second	Second occurrence of the related event where this type is used.
third	Third occurrence of the related event where this type is used.
fourth	Fourth occurrence of the related event where this type is used.
last	Last occurrence of the related event where this type is used.

2045 Table 15: OccurrenceEnumType

2046

2047 **3.10.2.13 AbsoluteOrRecurringTimeType**2048 If absolute (specific point in time) or recurring (relative point in time that is repeated) times will be
2049 modelled, this type is used.2050 A *complexType*, containing the elements described in Table 16.2051 The typical usage is that either *dateTime* is denoted or one of the following combinations:

- 2052 - month + dayOfMonth + time (specific day once per year)
- 2053 - dayOfMonth + time (specific day once per month)
- 2054 - calendarWeek + daysOfWeek + time (specific day of week once per year)
- 2055 - month + dayOfWeekOccurrence + daysOfWeek + time (first, second, ... occurrence of specific
- 2056 day of week in a specific month once per year, e.g. every second Thursday of July)
- 2057 - daysOfWeek + time (specific day of week once per week)
- 2058 - time (every day at a specific time)

2059 The *dayOfWeekOccurrence* element only makes sense in combination with *daysOfWeek* where
2060 *calendarWeek* is not set. It defines which week in a month was meant. For example, the *first* week, or
2061 the *second*, etc.

2062 Instead of only one day, combinations of several days may be stated in the element *daysOfWeek*
2063 (e.g. "monday" and "tuesday" and "friday" or "saturday" and "sunday").

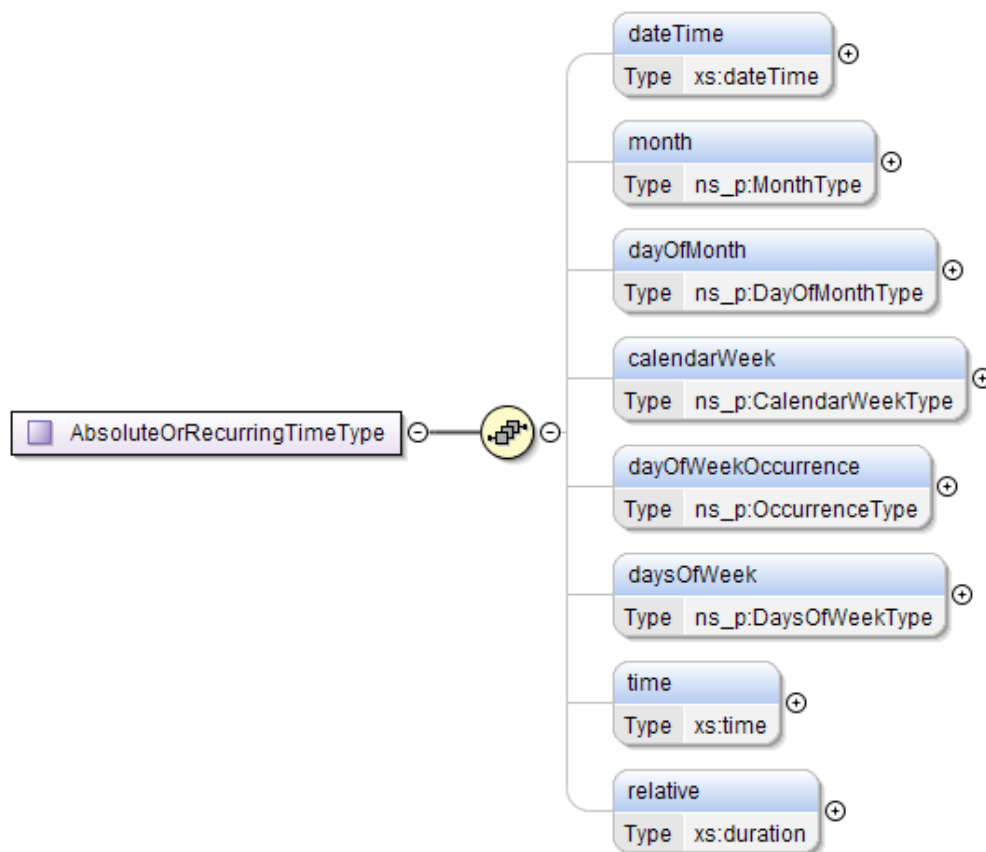


Figure 33: AbsoluteOrRecurringTimeType

Element	Type	Description
dateTime	Common data type "RecurringIntervalType". See section 3.10.2.4.	An absolute time.
month	Common data type "MonthType". See section 3.10.2.6.	The month information for the recurring time (e.g. "January").
dayOfMonth	Common data type "DayOfMonthType". See section 3.10.2.7.	The day of month information for the recurring time (e.g. "25").
calendarWeek	Common data type "CalendarWeekType". See section 3.10.2.8.	The calendar week information for the recurring time (e.g. "50").
dayOfWeekOccurrence	Common data type "OccurrenceType". See section 3.10.2.11.	The day of week occurrence information for the recurring time (e.g. "second", if the second occurrence of this weekday within a month is chosen).
daysOfWeek	Common data type "DaysOfWeekType". See section 3.10.2.10.	The days of week information for the recurring time (e.g. the elements "monday" and "wednesday" could be set).
time	xs:time (W3C standard type)	Time information either for a recurring time or a single event (e.g. "18:15:31Z").
relative	xs:duration (W3C standard type)	Relative time information either for a recurring time or a single event like a timer (e.g. "PT23H15M20S").

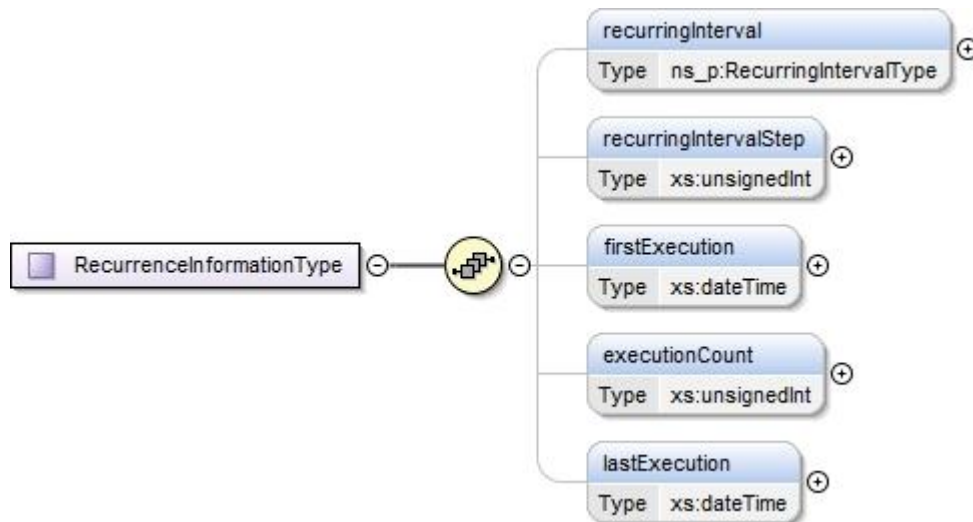
Table 16: AbsoluteOrRecurringTimeType

2067

2068 **3.10.2.14 RecurrenceInformationType**

2069 If the recurrence of some time information will be modelled, this type is used.

2070 A complexType, containing the elements described in Table 17.



2071

2072 *Figure 34: RecurrenceInformationType*

Element	Type	Description
<code>recurringInterval</code>	<i>Common data type "RecurringIntervalType". See section 3.10.2.4.</i>	With the <i>recurringInterval</i> the frequency of the recurrence is defined (e.g. "weekly").
<code>recurringIntervalStep</code>	<code>xs:unsignedInt</code> (W3C standard type)	If not every occurrence will be active, the <i>recurringIntervalStep</i> is used (e.g. if only every second occurrence will be active, <i>recurringIntervalStep</i> would have a value of "2").
<code>firstExecution</code>	<code>xs:dateTime</code> (W3C standard type)	If the recurring interval will not start with the next occurrence, the <i>firstExecution</i> element is used to specify this first time.
<code>executionCount</code>	<code>xs:unsignedInt</code> (W3C standard type)	To end the recurring execution, the <i>executionCount</i> can be used (e.g. after "10" occurrences, the interval ends).
<code>lastExecution</code>	<code>xs:dateTime</code> (W3C standard type)	To end the recurring execution, the <i>lastExecution</i> can be used, stating a specific point in time after that the interval will be not active any more (the <i>lastExecution</i> point in time does not have to be an interval occurrence).

2073 *Table 17: RecurrenceInformationType*

2074

2075 **3.10.3 Address-related**2076 **3.10.3.1 AddressDeviceType**

2077 Used to address a (physical) device.

2078 A *simpleType* (restricted *string*), containing the device address part, with the pattern

d:_(i:[1-9][0-9]*|n:[a-zA-Z0-9-]+)_[^\\p{Cc}\\p{Cf}\\p{Z}]+

For details please see [ProtocolSpecification], section "Rules for devices". The maximum string length is limited to 256 characters.

3.10.3.2 AddressEntityType

Used to address a logical device.

A *simpleType* (*unsignedInt*), containing an entity address part.

3.10.3.3 AddressFeatureType

Used to address a specific feature.

A *simpleType* (*unsignedInt*), containing the feature address part.

3.10.3.4 DeviceAddressType

Address type including only the device part.

A *complexType* with the element *device* (type: *AddressDeviceType*, see section 3.10.3.1).

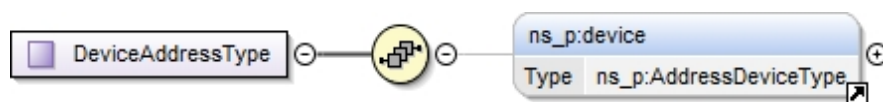


Figure 35: DeviceAddressType

3.10.3.5 EntityAddressType

Address type including the device part and a list of entities.

A *complexType* with the base type *DeviceAddressType* (see section 3.10.3.4) and a further list of element *entity* (type: *AddressEntityType*, see section 3.10.3.2).

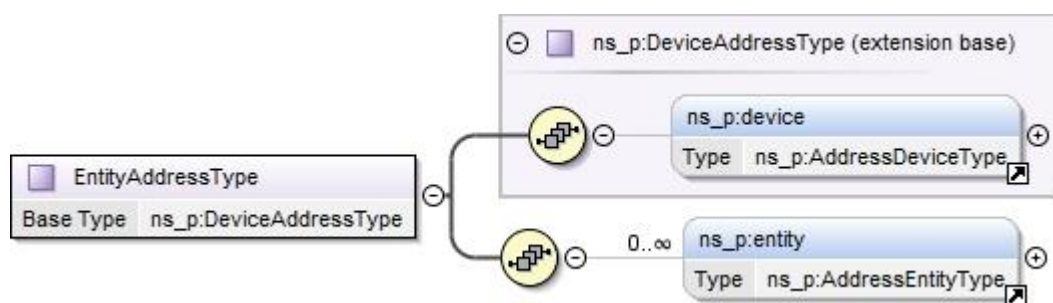


Figure 36: EntityAddressType

3.10.3.6 FeatureAddressType

Address type including the device part, a list of entities and the feature part.

A *complexType* with the base type *EntityAddressType* (see section 3.10.3.5) and a further element *feature* (type: *AddressFeatureType*, see section 3.10.3.3).

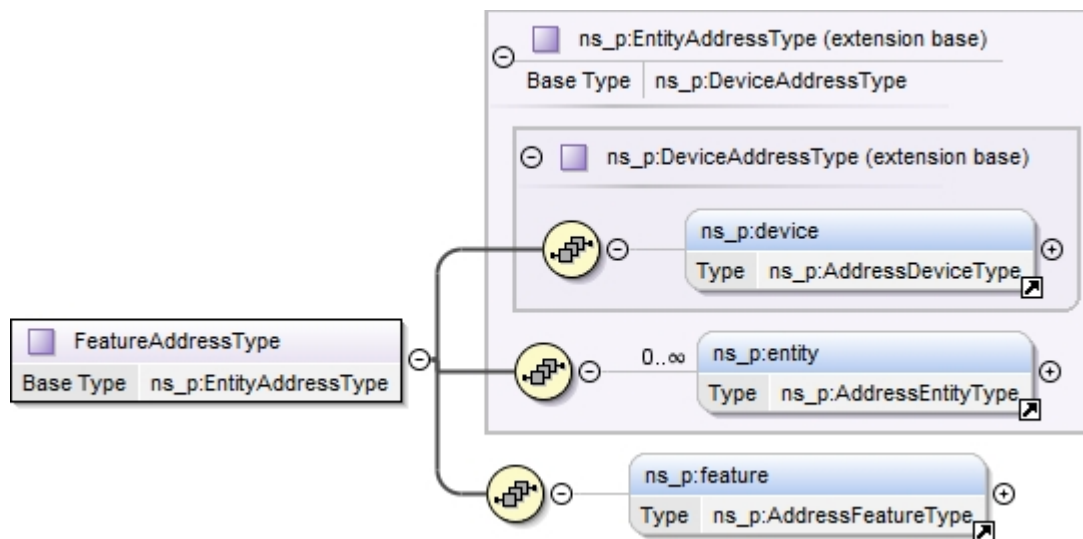


Figure 37: FeatureAddressType

3.11 Result

Note: This section discusses the SPINE function "resultData" and NOT the classifier "result"!

The result function concept is used for modelling acknowledges or errors for messages exchanged between communications partners. It is intended for these use cases:

- Notification of an error related to another function (e.g. as a reaction to a failed write operation).
- Substitution of a "regular reply" with an "error response". This applies if it is not possible to create a valid and proper reply that belongs to the function group of the corresponding read function.

Further information about the usage of result functions can be found in [ProtocolSpecification], section "Acknowledgement message".

The result function concept is modelled similar to (standard) classes. Its only function is called "resultData".

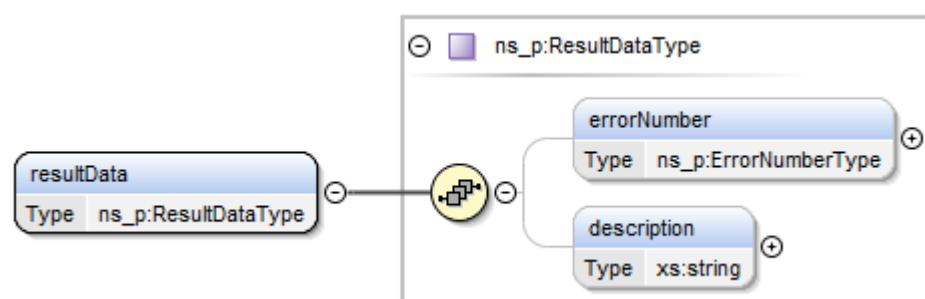


Figure 38: resultData function overview

The elements `errorNumber` and `description` are defined as follows:

Name	Type	Description
errorNumber	xs:unsignedInt (W3C standard type) See Table 19 for currently specified error numbers.	The error number specifies whether the result is "OK" (value "0") or an error (value ">0").
description	xs:string (W3C standard type)	Descriptive information on the result.

Table 18: resultData function detailed description of elements

Values specified for element **errorNumber**:

Value	Description
0	No Error.
1	General Error.
2	Timeout.
3	Overload.
4	Destination unknown.
5	Destination unreachable.
6	Command not supported.
7	Command rejected.
8	Restricted function exchange combination not supported.
9	Binding is necessary for this command.

Table 19: Values specified for element errorNumber

Further error numbers will be added in future releases.

Example of an error indication:

```

<resultData>
  <errorNumber>2</errorNumber>
  <description>Timeout</description>
</resultData>

```

4 Device model definitions

The subsequent sections define details on the use of "Device Type", "Entity Type" and "Feature Type". An overview of all types can be found in Annex B.

4.1 Device Types

4.1.1 ChargingStation

4.1.1.1 Descriptive information

Represents a charging station, typically used in the context of electrical vehicle charging.

4.1.2 Dishwasher

4.1.2.1 Descriptive information

Represents a dishwasher

4.1.3 Dryer

4.1.3.1 Descriptive information

Represents a dryer.

4.1.4 ElectricitySupplySystem

4.1.4.1 Descriptive information

Represents a set of sources or storages of electric power of a premises (for example a local PV system, the electricity grid connection point or a battery storage).

4.1.5 EnergyManagementSystem

4.1.5.1 Descriptive information

Represents an energy management system (EMS).

4.1.6 EnvironmentSensor

4.1.6.1 Descriptive information

Represents a sensor typically used for outdoor measurements.

2170 **4.1.7 Generic**

2171 **4.1.7.1 Descriptive information**

2172 This is a generic device, which can either be used for client-only functionality or for development
2173 purposes where no specific type of device can be stated yet.

2174

2175 **4.1.8 HeatGenerationSystem**

2176 **4.1.8.1 Descriptive information**

2177 This Device Type is used to represent the subsystem of an HVAC system that generates heat. It can
2178 be used to represent single heat generators like gas heating boilers or heat pumps, but also be used
2179 to represent a heat generator cascade (like a cascade of several gas heating boilers or a combination
2180 of heat pump appliances and gas heating boilers). It can also be used to represent a solar thermal
2181 system.

2182 The Device Type "HeatGenerationSystem" serves as a container for specific Entity Types that
2183 represent the single heat generators.

2184

2185 **4.1.9 HeatSinkSystem**

2186 **4.1.9.1 Descriptive information**

2187 Represents the end user relevant heat sinks of the HVAC system (heating circuits, domestic hot water
2188 (DHW) circuits, air conditioning system, ...) and the end user relevant system functions (space
2189 heating, space cooling, DHW, air conditioning, ...)

2190

2191 **4.1.10 HeatStorageSystem**

2192 **4.1.10.1 Descriptive information**

2193 Represents a system of heat storages consisting of one or more heat storage appliances or
2194 appliances that are attached to a heat storage (for example the Entity Type InstantDHWHeater (see
2195 section 4.2.34) (fresh water station attached to a heating buffer storage)). All heat storages of the
2196 HVAC system should be represented by this Device Type.

2197

2198 **4.1.11 HVACController**

2199 **4.1.11.1 Descriptive information**

2200 Represents the central controller appliance of an HVAC System.

2201 A device of Device Type HVACController is used to deliver overall information about the HVAC
2202 system.

2203

2204 4.1.12 Inverter**2205 4.1.12.1 Descriptive information**

2206 Represents a power Inverter.

2207

2208 4.1.13 SubMeter**2209 4.1.13.1 Descriptive information**

2210 A sub meter measures the demand or generation of a commodity (e.g. electricity).

2211

2212 4.1.14 Washer**2213 4.1.14.1 Descriptive information**

2214 Represents a washing machine.

2215

2216 4.2 Entity Types**2217 4.2.1 Battery****2218 4.2.1.1 Descriptive information and further definitions**

2219 Represents a single battery unit.

2220

2221 4.2.2 BatterySystem**2222 4.2.2.1 Descriptive information**

2223 A system consisting of one or more battery storages and one or more battery inverters.

2224

2225 4.2.3 CEM**2226 4.2.3.1 Descriptive information and further definitions**

2227 Abbreviation for customer energy manager and typically used to represent an energy manager within
2228 SPINE.

2229

2230 4.2.4 ChargingOutlet**2231 4.2.4.1 Descriptive information and further definitions**

2232 Represents a charging outlet of a charging station.

2233

2234 **4.2.5 Compressor**

2235 **4.2.5.1 Descriptive information and further definitions**

2236 Represents the compressor of (e.g.) an (electrical) heat pump appliance.

2237

2238 **4.2.6 ControllableSystem**

2239 **4.2.6.1 Descriptive information and further definitions**

2240 An appliance or a system of appliances controllable by an energy manager.

2241

2242 **4.2.7 DeviceInformation**

2243 **4.2.7.1 Descriptive information and further definitions**

2244 This entity has a special role and contains general information about the complete device.

2245

2246 **4.2.8 DHWCircuit**

2247 **4.2.8.1 Descriptive information and further definitions**

2248 Represents all elements for domestic hot water supply like cold water line, domestic hot water line,
2249 circulation line, tapping points.

2250 Should not include DHW storages, these should be represented as Entity Type "DHWStorage"

2251

2252 **4.2.9 DHWStorage**

2253 **4.2.9.1 Descriptive information and further definitions**

2254 Represents a heat storage appliance for storing domestic hot water.

2255

2256 **4.2.10 Dishwasher**

2257 **4.2.10.1 Descriptive information and further definitions**

2258 Represents a dishwasher

2259

2260 **4.2.11 Dryer**

2261 **4.2.11.1 Descriptive information and further definitions**

2262 Represents a dryer.

2263

2264 **4.2.12 ElectricalImmersionHeater**

2265 **4.2.12.1 Descriptive information and further definitions**

2266 Represents an electrical immersion heater (electrical resistance heater) for example as part of a heat
2267 pump appliance, a heat source unit, a DHW storage.

2268

2269 **4.2.13 ElectricalStorage**

2270 **4.2.13.1 Descriptive information and further definitions**

2271 Entity used to categorize an inverter as (pure) battery inverter (no (e.g.) PV string(s) connected to the
2272 inverter).

2273

2274 **4.2.14 ElectricityGenerationSystem**

2275 **4.2.14.1 Descriptive information and further definitions**

2276 A system that is capable of producing electric power through local generation (for example PV-
2277 systems, wind turbines or block heating and power plants).

2278

2279 **4.2.15 ElectricityStorageSystem**

2280 **4.2.15.1 Descriptive information and further definitions**

2281 A system that is capable of storing electric power and releasing the majority of it as electric power
2282 (for example a battery system or a capacitor).

2283

2284 **4.2.16 EV**

2285 **4.2.16.1 Descriptive information and further definitions**

2286 Represents an electrical vehicle.

2287

2288 **4.2.17 EVSE**

2289 **4.2.17.1 Descriptive information and further definitions**

2290 Abbreviation for electrical vehicle supply equipment, typically used to charge an electrical vehicle.

2291

2292 **4.2.18 Fan**

2293 **4.2.18.1 Descriptive information and further definitions**

2294 Represents an air fan, for example of an air water heat pump unit or a ventilation unit.

2295

2296 **4.2.19 GasHeatingAppliance**

2297 **4.2.19.1 Descriptive information and further definitions**

2298 Represents a heating appliance that generates heat (for example for heating or domestic hot water)
2299 by burning natural gas.

2300

2301 **4.2.20 Generic**

2302 **4.2.20.1 Descriptive information and further definitions**

2303 This is a generic entity, which can either be used for client-only functionality, or for development
2304 purposes, where no specific entity can be stated yet.

2305

2306 **4.2.21 GridConnectionPointOfPremises**

2307 **4.2.21.1 Descriptive information and further definitions**

2308 The electrical connection point between the utility electricity grid and the electric circuit of a
2309 premises.

2310

2311 **4.2.22 GridGuard**

2312 **4.2.22.1 Descriptive information and further definitions**

2313 An energy manager located inside or outside the house or premises that limits or controls appliances
2314 in a way that the public electricity grid remains stable.

2315

2316 **4.2.23 HeatingBufferStorage**

2317 **4.2.23.1 Descriptive information and further definitions**

2318 Represents a heat storage appliance for storing heating water.

2319

2320 **4.2.24 HeatingCircuit**

2321 **4.2.24.1 Descriptive information and further definitions**

2322 Represents a single heating circuit for space heating (or space cooling like active or passive cooling),
2323 including flow and return lines, optional mixing units (for mixed heating circuits) and the heat
2324 distribution:

- 2325 - directly to one or more heated rooms (see Entity Type "HVACRoom", section 4.2.33)
- 2326 - to one or more heating zones (see Entity Type "HeatingZone", section 4.2.26)

2327

2328 **4.2.25 HeatingObject**

2329 **4.2.25.1 Descriptive information and further definitions**

2330 Represents a heating object, e.g. a radiator.

2331

2332 **4.2.26 HeatingZone**

2333 **4.2.26.1 Descriptive information and further definitions**

2334 Represents a heating zone that provides heating energy (or cooling energy) to one or more heated
2335 rooms (see Entity Type HVACRoom).

2336

2337 **4.2.27 HeatPumpAppliance**

2338 **4.2.27.1 Descriptive information and further definitions**

2339 Represents either a complete heat pump appliance (for example a brine water heat pump, indoor; an
2340 air water heat pump, completely indoor or completely outdoor) or the indoor part of a refrigerant
2341 split heat pump combination.

2342

2343 **4.2.28 HeatSinkCircuit**

2344 **4.2.28.1 Descriptive information and further definitions**

2345 Represents a (hydraulic) circuit of a heat generator that delivers heat energy from the heat generator
2346 to a heat sink. For example Entity Type "HeatSinkCircuit" could represent a (hydraulic) circuit of a gas
2347 heating appliance that delivers heat energy to a heating circuit.

2348

2349 **4.2.29 HeatSourceCircuit**

2350 **4.2.29.1 Descriptive information and further definitions**

2351 Represents a (hydraulic) circuit of a heat generator (especially of heat pumps) that delivers heat
2352 energy from an ambient energy source to the heat generator. For example Entity Type
2353 "HeatSourceCircuit" could represent a brine circuit of a heat pump appliance that delivers heat
2354 energy from an ambient heat source to the heat pump appliance.

2355

2356 **4.2.30 HeatSourceUnit**

2357 **4.2.30.1 Descriptive information and further definitions**

2358 Represents an additional unit that provides the heat from the environmental heat source to the heat
2359 pump appliance. This could for example be:

- 2360 - an outdoor ventilation unit that transfers the heat energy from the outside air to the heat source
- 2361 circuit of the (indoor) heat pump appliance (i.e. air water split heat pump)

2362 - an outdoor ventilation unit that transfers the heat energy from the outside air to the refrigerant
2363 circuit of the (indoor) heat pump appliance (refrigerant split heat pump)

2364

2365 **4.2.31 Household**

2366 **4.2.31.1 Descriptive information and further definitions**

2367 This is a rather virtual entity representing a complete household that aggregates all relevant
2368 information of the household (e.g. the complete energy demand, etc.).

2369

2370 **4.2.32 HVACController**

2371 **4.2.32.1 Descriptive information and further definitions**

2372 Represents the central controller appliance of the HVAC System (as a SPINE Entity).

2373

2374 **4.2.33 HVACRoom**

2375 **4.2.33.1 Descriptive information and further definitions**

2376 Represents one physical room in a building/house that can be heated / cooled, ventilated or air
2377 conditioned.

2378

2379 **4.2.34 InstantDHWHeater**

2380 **4.2.34.1 Descriptive information and further definitions**

2381 Represents an appliance for instant generation of domestic hot water (for example by using heat
2382 from a heating buffer storage).

2383

2384 **4.2.35 Inverter**

2385 **4.2.35.1 Descriptive information and further definitions**

2386 Represents a power inverter.

2387

2388 **4.2.36 OilHeatingAppliance**

2389 **4.2.36.1 Descriptive information and further definitions**

2390 Represents a heating appliance that generates heat (for example for heating or domestic hot water)
2391 by burning oil.

2392

2393 **4.2.37 Pump**

2394 **4.2.37.1 Descriptive information and further definitions**

2395 Represents a pump that moves a medium (for example in a heating circuit).

2396

2397 **4.2.38 PV**

2398 **4.2.38.1 Descriptive information and further definitions**

2399 Entity used to categorize an inverter as (pure) PV inverter (no (e.g.) battery connected to the
2400 inverter).

2401

2402 **4.2.39 PVESHybrid**

2403 **4.2.39.1 Descriptive information and further definitions**

2404 Entity used to categorize an inverter as PV/battery hybrid inverter (battery or batteries and PV
2405 panel(s) connected to the inverter).

2406

2407 **4.2.40 PVString**

2408 **4.2.40.1 Descriptive information and further definitions**

2409 This Entity represents a (DC connected) PV panel connected to an inverter.

2410

2411 **4.2.41 PVSystem**

2412 **4.2.41.1 Descriptive information and further definitions**

2413 A system consisting of one or more PV inverters with one or more PV modules. The Entity PVSystem
2414 provides aggregated PV related system parts of the house or premises.

2415

2416 **4.2.42 RefrigerantCircuit**

2417 **4.2.42.1 Descriptive information and further definitions**

2418 Represents the refrigerant circuit of a heat pump appliance.

2419

2420 **4.2.43 SmartEnergyAppliance**

2421 **4.2.43.1 Descriptive information and further definitions**

2422 An appliance that allows a client to shift the load announced by the server can be modelled with this
2423 Entity Type. Non-shiftable loads (so-called "fixed forecasts") can be provided by this Entity, too.

2424

2425 **4.2.44 SolarDHWStorage**

2426 **4.2.44.1 Descriptive information and further definitions**

2427 Represents a heat storage appliance for storing domestic hot water. The heat energy can be provided
2428 especially by solar thermal energy (compare to Entity Type "DHWStorage").

2429

2430 **4.2.45 SolarThermalCircuit**

2431 **4.2.45.1 Descriptive information and further definitions**

2432 Represents a single circuit of a solar thermal system including a solar thermal collector field and flow
2433 and return pipes to/from the collector field and a solar circuit pump.

2434

2435 **4.2.46 SubMeterElectricity**

2436 **4.2.46.1 Descriptive information and further definitions**

2437 Represents an electrical sub meter.

2438

2439 **4.2.47 TemperatureSensor**

2440 **4.2.47.1 Descriptive information and further definitions**

2441 Represents a sensor that can provide a temperature measurement value.

2442

2443 **4.2.48 Washer**

2444 **4.2.48.1 Descriptive information and further definitions**

2445 Represents a laundry washer that washes clothes (laundry).

2446

2447 **4.3 Feature Types**

2448 **4.3.1 ActuatorLevel**

2449 **4.3.1.1 Dependencies to other Feature Types**

2450 None.

2451

2452 **4.3.1.2 Introduction**

2453 Independent from the *ActuatorSwitch* class, the *ActuatorLevel* class enables a user or application to
2454 model *LEVEL* commands (*start*, *up*, *percentageAbsolute*, *relative*, ...). This can be used to dim a light,
2455 set the speed of an electric motor, etc.

2456

2457 **4.3.1.3 Rules**

2458 None.

2459

2460 **4.3.1.4 actuatorLevelData**2461 **4.3.1.4.1 Description**

2462 This function is used for setting the desired operation mode of the node or getting the actual mode
 2463 respectively.

2464

2465 **4.3.1.4.2 Rules**

2466 None.

2467

2468 **4.3.1.4.3 Element rules**

Element	Rule	Description
ActuatorLevel. actuatorLevelData.		
function	SHALL always be set. Value rules of Table 21 SHALL be applied.	The element "function" contains the actuator level mode.
value	None.	The value is only needed for some functions. E.g. setting <i>function</i> to <i>percentageAbsolute</i> just makes sense together with a proper value.

2469 Table 20: actuatorLevelData Element rules

2470 Element "**function**" value rules:

Value	Rule	Description
start	None.	Actuator starts.
up	None.	Actuator moves up.
down	None.	Actuator moves down.
stop	None.	Actuator stops.
percentageAbsolute	None.	The percentage position or value of the actuator (e.g. 25 %).
percentageRelative	None.	Changes the current value by a percentage value relative to the current value.
absolute	None.	The absolute position or value of the actuator (e.g. 0.91 m).
relative	None.	Changes the current value by some value (e.g. plus 0.23 m).

2471 Table 21: Element "function" value rules

2472

2473 **4.3.1.5 actuatorLevelDescriptionData**2474 **4.3.1.5.1 Description**

2475 The rather static descriptive data of the *actuatorLevel* class can contain a default unit, among others.
 2476 This is useful for some *absolute* commands.

2477

2478 **4.3.1.5.2 Rules**

2479 None.

2480

2481 **4.3.1.5.3 Element rules**

Element	Rule	Description
ActuatorLevel. actuatorLevelDescriptionData.		
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the actuator.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the actuator.
levelDefaultUnit	None.	If an actuator can specify a unit for its level-controlling, it can be denoted in this element.

2482 Table 22: actuatorLevelDescriptionData Element rules

2483

2484 **4.3.1.6 Descriptive information and further definitions**

2485 None.

2486

2487 **4.3.2 ActuatorSwitch**2488 **4.3.2.1 Dependencies to other Feature Types**

2489 None.

2490

2491 **4.3.2.2 Introduction**2492 Basic on/off operations on a simple actuator can be modelled with the *ActuatorSwitch* class.2493 Whether the function turns a device itself *on* or *off*, or whether it switches a specific functionality,

2494 depends on the implementation. E.g. one could model the super freeze program of a freezer using

2495 *ActuatorSwitch* class. An *on* command would then activate the super freeze program and an *off*2496 command would deactivate it. This example shall just give an idea how *ActuatorSwitch* can be used

2497 for more purposes than only turning devices on and off.

2498 Please note a device's different features may be modelled with separate implementations of this

2499 class, each associated to a different node address. Continuing with the example above, a "defrost"

2500 function may be modelled on another node address with *ActuatorSwitch*.

2501

2502 **4.3.2.3 Rules**

2503 None.

2504

4.3.2.4 actuatorSwitchData**4.3.2.4.1 Description**

The *actuatorSwitchData* function models the functionality described in section 5.3.2.1.

4.3.2.4.2 Rules

None.

4.3.2.4.3 Element rules

Element	Rule	Description
ActuatorSwitch. actuatorSwitchData.		
function	None. Value rules of Table 24 SHALL be applied.	The only element in this function to model an <i>on</i> , <i>off</i> or <i>toggle</i> command.

Table 23: actuatorSwitchData Element rules

Element "**function**" value rules:

Value	Rule
on	If used in a write, the actuator SHALL be activated, if the write is accepted. If used in a notify or reply, the actuator SHALL be active.
off	If used in a write, the actuator SHALL be deactivated, if the write is accepted. If used in a notify or reply, the actuator SHALL NOT be active.
toggle	SHALL only be used in a write. If used in a write, an inactive actuator SHALL be activated and an active actuator SHALL be deactivated, if the write is accepted.

Table 24: Element "function" value rules

4.3.2.5 actuatorSwitchDescriptionData**4.3.2.5.1 Description**

The (typically) non-changing information on the actuator is modelled with this function.

4.3.2.5.2 Rules

None.

4.3.2.5.3 Element rules

Element	Rule	Description
ActuatorSwitch. actuatorSwitchDescriptionData.		
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the actuator.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the actuator.

Table 25: actuatorSwitchDescriptionData Element rules

2526

2527 **4.3.2.6 Descriptive information and further definitions**

2528 None.

2529

2530 **4.3.3 Alarm**2531 **4.3.3.1 Dependencies to other Feature Types**

2532 - Threshold

2533

2534 **4.3.3.2 Introduction**

2535 There are several reasons for generating an alarm. This might be due to exceeding (in a positive or
 2536 negative way) a threshold (modelled with the Threshold class, see section 5.3.27), or because of
 2537 some other occurrence that provokes the generation of an alarm. The Alarm class provides the
 2538 needed data model.

2539

2540 **4.3.3.3 Rules**

2541 None.

2542

2543 **4.3.3.4 alarmListData**2544 **4.3.3.4.1 Description**

2545 This function models data for an alarm. Depending on the type of alarm, different elements are used.
 2546 It can be used to notify an alarm or as a reply to a read command.

2547

2548 **4.3.3.4.2 Rules**

2549 None.

2550

2551 **4.3.3.4.3 Element rules**

Element	Rule	Description
Alarm. alarmListData. alarmData. (list)		
alarmId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the alarm.
thresholdId	If the alarm is related to a threshold, the value SHALL be set as FOREIGN IDENTIFIER. Otherwise it SHALL be omitted.	If the alarm relates to a threshold, the corresponding ID can be stated here.
timestamp	SHOULD be set.	The time of creation of the alarm.
alarmType	SHOULD be set. Value rules of Table 27 SHALL be applied.	The type of alarm is given here.

measuredValue	None.	The value that was measured which exceeded the threshold is denoted in this element.
evaluationPeriod	None.	If a time period can be stated in which the alarm is valid it is denoted within this Element and its sub-Elements.
scopeType	None.	Specifies a more detailed meaning of the alarm (e.g. exceeding of a threshold).
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the alarm.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the alarm.

Table 26: alarmListData Element rules

Element "**alarmType**" value rules:

Value	Rule
alarmCancelled	SHALL be used, if a previously generated alarm is cancelled.
underThreshold	SHALL be used, if the value is under a related threshold (<i>thresholdId</i> SHALL be set).
overThreshold	SHALL be used, if the value is over a related threshold (<i>thresholdId</i> SHALL be set).

Table 27: Element "alarmType" value rules

4.3.3.5 Descriptive information and further definitions

None.

4.3.4 Bill

4.3.4.1 Dependencies to other Feature Types

None.

4.3.4.2 Introduction

The Bill class can be used to issue bills, containing positions and their costs.

4.3.4.3 Rules

None.

4.3.4.4 billListData

4.3.4.4.1 Description

billData holds the bill data, including descriptive elements like currency, unit and billType.

2572 A billData has a "total" element, including total values and corresponding costs in its sub-values.
 2573 Additionally, positions of the total data can be provided, to describe the composition of the total
 2574 data. To link total costs and values in a position the valuelId and costId are used within a billData.

2575

2576 4.3.4.4.2 Rules

2577 The identifier positionId MAY be omitted, if only one position is included in the bill. Otherwise it
 2578 SHALL be set.

2579 The identifier valuelId MAY be omitted, if only one value is included in the bill. Otherwise it SHALL be
 2580 set.

2581 The identifier costId MAY be omitted, if only one cost is included in the bill. Otherwise it SHALL be
 2582 set.

2583

2584 4.3.4.4.3 Element rules

Element	Rule	Description
Bill. billListData. billData. (list)		
billId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a bill.
billType	Value rules of Table 29 SHALL be applied.	The type of the bill.
scopeType	None.	Specifies the scope of the bill.
total	None.	Total information of the bill (in contrast to the positions, see below).
total. timePeriod	None.	Allows to define a time, or time period for a bill.
total. value (list)	None.	Different values can be listed for a bill. E.g. energy values or just a number of items.
total. value. valuelId	This Element SHALL be interpreted as SUB IDENTIFIER. If there is only one "value" entry within "total", the "valuelId" MAY be omitted. Otherwise it SHALL be stated. If omitted, "valuelId" SHALL equal "1".	Allows the identification of a value within the bill.
total. value. unit	If the value has a unit, it SHOULD be stated here.	Allows definition of a unit for the value.
total. value. value	Sales from the perspective of the Feature Server SHALL be expressed with negative values and purchases with positive values. In "total" all position sales and purchases SHALL be accumulated.	Allows to express the total amount that was bought or sold of a certain commodity during the time or time period given in total. timePeriod.
total. cost (list)	None.	Different types of costs can be listed for a value. E.g. monetary costs like a price as well as environmental costs as CO2 emission.

total. cost. costId	This Element SHALL be interpreted as SUB IDENTIFIER. If there is only one "cost" entry within "total", the "costId" MAY be omitted. Otherwise it SHALL be stated. If omitted, "costId" SHALL equal "1".	Allows identification of costs within bill.
total. cost. costType	Value rules of Table 30 SHALL be applied.	Allows to describe the type of costs.
total. cost. valueId	If the cost is linked to a value, the corresponding valueId SHALL be set.	Allows to link cost to a value.
total. cost. unit	None.	Costs that are not monetary may need to describe a unit (e.g. m ³).
total. cost. currency	None.	Monetary costs may need to describe a currency.
total. cost. cost	None.	Total costs generated during the time or time period given in total. timePeriod.
total. label	Default rules of section 3.10.1.2.1 SHALL be applied.	Allows to express a user friendly label for the bill.
total. description	Default rules of section 3.10.1.3.1 SHALL be applied.	Allows to express a user friendly description for the bill.
position (list)	None.	For each total data different positions can be assigned that describe the composition of the total data.
position. positionId	This Element SHALL be interpreted as SUB IDENTIFIER. If there is only one "position" entry the "positionId" MAY be omitted. Otherwise it SHALL be stated. If omitted, "positionId" SHALL equal "1".	Identifier of a position within a specific bill.
position. positionType	Value rules of Table 31 SHALL be applied.	Allows to describe the type of the a position E.g. different types of energy consumption like self produced or electricity grid energy.
position. timePeriod	None.	Allow to define a time, or time period for a position if the time or time period is different from the overall time period described in total.
position. value (list)	None.	Different values can be listed for each position.
position. value. valueId	This Element SHALL be interpreted as SUB IDENTIFIER. If there is only one "value" entry for the "position", the "valueId" MAY be omitted. Otherwise it SHALL be stated. If omitted, "valueId" SHALL equal "1".	Identifier of a value within a specific bill (see also "total. value. valueId").
position. value. value	Allows to describe a value for each position. This can be helpful if some positions have negative while other positions have positive values, as this	Allows to describe a value for each position. This can be helpful if some positions have negative while other positions have positive values, as

	would can not be express with valuePercentage.	this would can not be express with valuePercentage.
position. value. valuePercentage	Percentage of the total value (see "total").	Percentage of the total value (see "total").
position. cost (list)	Different costs can be listed for each position	Different costs can be listed for each position
position. cost. costId	This Element SHALL be interpreted as SUB IDENTIFIER. If there is only one "cost" entry for the "position", the "costId" MAY be omitted. Otherwise it SHALL be stated. If omitted, "costId" SHALL equal "1".	Identifier of a cost within a specific bill (see also "total. cost. costId").
position. cost. cost	None.	The actual costs for this position. This can be helpful if some positions have negative while other positions have positive costs, as this can not be expressed with costPercentage.
position. cost. costPercentage	None.	Percentage of the total costs (see "total").
position. label	Default rules of section 3.10.1.2.1 SHALL be applied.	Allows to provide a user friendly label for each bill position.
position. description	Default rules of section 3.10.1.3.1 SHALL be applied.	Allows to provide user friendly information for each bill position.

2585 Table 28: billListData Element rules

2586 Element "**billType**" value rules:

Value	Rule	Description
chargingSummary	This summary SHOULD NOT be used for billing the customer as it may also contain approximated values.	This bill type allows to summarize a recharging cycle, e.g. containing total energy charged, total costs as well as billing positions for different types of energy.

2587 Table 29: Element "billType" value rules

2588 Element "**costType**" value rules:

Value	Rule	Description
absolutePrice	None.	Used for absolute prices.
relativePrice	None.	Used for relative prices, e.g. a price that shows how much cheaper the costs are related to the average tariff in place.
co2Emission	None.	In this case the cost value is used to express how much CO2 emission was emitted.
renewableEnergy	None.	In this case the cost value is used to express how much renewable energy was used.
radioactiveWaste	None.	In this case the cost value is used to express how much radioactive waste was generated.

2589 Table 30: Element "costType" value rules

2590 Element "**positionType**" value rules:

Value	Rule	Description
gridElectricEnergy	None.	This position type specifically describes the electricity grid energy portion of the total data.

selfProducedElectricEnergy	None.	This position type specifically describes the self produced energy portion of the total data.
----------------------------	-------	---

Table 31: Element "positionType" value rules

4.3.4.5 billConstraintsListData

4.3.4.5.1 Description

The billConstraintsListData function allows a server to define certain constraints for each linked billListData list entry. This is especially important if a particular billListData list entry is writeable and allows the server to communicate the corresponding constraints before a client performs a write command. The client can match the server constraints with own constraints and perform a corresponding write command.

4.3.4.5.2 Rules

None.

4.3.4.5.3 Element rules

Element	Rule	Description
Bill. billConstraintsListData. billConstraintsData. (list)		
billId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a bill. Allows linking of the different functions to the same bill. Allows linking of the bill within other features that are placed in the same entity.
positionCountMin	If a related billData exists, it SHALL NOT contain less positions.	Minimum amount of possible positions within the specified bill.
positionCountMax	If a related billData exists, it SHALL NOT contain more positions.	Maximum amount of possible positions within the specified bill.

Table 32: billConstraintsListData Element rules

4.3.4.6 billDescriptionListData

4.3.4.6.1 Description

The billDescriptionListData function is especially important if a linked billListData list entry is writeable and allows a server to define if a bill is writeable and which billTypes are supported for the corresponding billId, so that a client can perform corresponding write operations. Additionally, the server can request updates from a client with the updateRequired flag.

4.3.4.6.2 Rules

None.

2617 4.3.4.6.3 *Element rules*

Element	Rule	Description
Bill. billDescriptionListData. billDescriptionData. (list)		
billId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a bill. Allows linking of the different functions to the same bill. Allows linking of the bill within other features that are placed in the same entity.
billWriteable	If set to "true", a client MAY write the related billData entry. If there is no billData entry with a corresponding billId, the client MAY still write the complete billData entry and the server SHALL accept it, if the write command matches the rules given in the related billDescriptionData and billConstraintsData entries for the corresponding billId.	Indicates whether the related billData entry is writeable.
updateRequired	With updateRequired the server can request an update of writeable or changeable data related to the same PRIMARY IDENTIFIER from a client. The server SHALL ensure that only one responsible client is permitted to update the related data. To request an update the server SHALL set updateRequired to "true". Note: In this case, the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server SHALL set the updateRequired back to "false", as soon as "billDescriptionData" was updated successfully (if writeable or changeable) OR the update of the other writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false". Note: The client does not need to stop an ongoing update process (e.g. if multiple functions are written), when updateRequired is set back to "false".	Indicates whether the server requests an update for the bill.
supportedBillType (list)	None.	List of all supported types for this billId.
sessionId	MAY be set as FOREIGN IDENTIFIER, if a session is linked to this bill.	Link to the session this bill was created for.

2618 Table 33: billDescriptionListData Element rules

2619

2620 4.3.4.7 *Descriptive information and further definitions*

2621 None.

2622

2623 **4.3.5 DataTunneling**2624 **4.3.5.1 Dependencies to other Feature Types**

2625 None.

2626

2627 **4.3.5.2 Introduction**

2628 DataTunneling provides a mechanism to submit data of proprietary protocols using the SPINE data
2629 models (esp. addressing other nodes). However, the nature of proprietary definitions imposes
2630 restrictions:

2631 Unless stated otherwise, the term "payload" refers to the sub-element of "dataTunnelingCall" within
2632 this section (in contrast to the definition of "payload" as sub-element of "datagram" as defined in
2633 document [ProtocolSpecification]).

2634 Please note that the use of the above mentioned elements will always be proprietary. The SPINE
2635 specification will never define any specific protocol for dataTunneling.

2636 To put it in other words, the SPINE data tunnelling model is not intended for any kind of flexible
2637 extension of the SPINE classes. This is mainly because a so-called technology specific mapping could
2638 hardly be defined in a standardized way. Another reason is that it shall always be transparent that
2639 data tunnelling contains proprietary data. This should not be considered a limitation of SPINE as the
2640 EEBus Initiative e.V. committees may define specific classes with functions (like messagingListData,
2641 timeInformationData, e.g.) including mappings for specific protocols, if need be.

2642

2643 **4.3.5.3 Rules**

2644 None.

2645

2646 **4.3.5.4 dataTunnelingCall**2647 **4.3.5.4.1 Description**

2648 In general, several conditions must be fulfilled in order to make use of SPINE data tunnelling. This is
2649 explained with an example:

2650 A manufacturer "X" of a device "A" created a proprietary extension of a communications protocol
2651 (e.g. KNX, ZigBee, LON, ... – we call it "CP" here) in order to configure device "A" with special
2652 parameters. We assume there is no specific SPINE function defined so far for these parameters.
2653 There is also no mapping defined for it.

2654 Another manufacturer "Y" creates a device "B" that runs "SPINE applications" (i.e. applications
2655 communicating with an internal interface using XMLs of the SPINE data models). The device "B" can
2656 as well communicate via "CP".

2657 Now, manufacturer "Y" wants to make use of the proprietary configuration of device "A". Therefore,
2658 manufacturer "Y" needs to implement the proprietary extension of manufacturer "X"'s
2659 communications protocol as well. I.e. the "CP" implementation of device "B" must be extended
2660 properly. Furthermore, manufacturer "Y" needs to define how the proprietary protocol shall be
2661 represented as SPINE data tunnelling towards the "SPINE applications" running on device "B" (i.e.
2662 define the proprietary mapping) and of course needs to implement this mapping. Finally, the
2663 applications need to implement the recognition and use of the corresponding SPINE data tunnelling
2664 XMLs.

2665 The scenario above may look "unattractive" or even raise concerns. However, it shall be mentioned
2666 that proprietary extensions of standardized protocols are not unusual. This practice shall not be
2667 judged here.

2668 In the aforementioned scenario it is manufacturer "B" who implements a number of extensions in
2669 order to comply with the extensions of device "A". But it could also be possible that manufacturer
2670 "A" already provided some definitions and implementation for other manufacturers. Then,
2671 manufacturer "B" could integrate this solution and potentially implements less by its own.

2672 Some may find a comparison with Virtual Private Networks helpful (though VPNs have a different
2673 scope): Two computers "A", "B" shall each run a compatible (proprietary) program (i.e.
2674 mapping/implementation) that know how to communicate with each other (i.e. set up the
2675 communication and provide a specific authentication and encryption). There are manufacturers of
2676 VPN gateways that provide client software for different operating systems. This way a proprietary
2677 implementation is made available for others.

2678 The combination { `purposeld`, `channelld`, `sequenceld` } should be considered unique for a
2679 communication from "A" to "B". Within a system (e.g. house) several communications may be used
2680 in parallel. The combinations used in these communications are in general independent of each
2681 other.

2682 Please note that an instance of a `dataTunnelingCall` implementation permits bidirectional
2683 communication in general. I.e. if a `dataTunnelingCall` functionality of "B" submits a message to a
2684 `dataTunnelingCall` functionality of "A", it is also permissive that the `dataTunnelingCall` functionality of
2685 "A" submits `dataTunnelingCall` messages to the `dataTunnelingCall` functionality of "B". However,
2686 similar to a VPN, it may well be that the establishment for such message exchanges is
2687 "unidirectional" to some extent. I.e. it may be that the `dataTunnelingCall` functionality of "A"
2688 determines which kind of messages (or permissive values for "`purposeld`") are accepted on this
2689 functionality and that the functionality first of all acts as some kind of "listening port". This means the
2690 functionality of "A" may wait for an initial incoming message before it also sends `dataTunnelingCall`
2691 messages back.

2692 Please note that the example above does not prevent device "B" from implementing another
2693 `dataTunnelingCall` functionality as "listening port" with own permitted values for "`purposeld`" as well.
2694 In that case device "A" could send a proper initial message to this "listening functionality" of "B".

2695 SPINE data tunnelling is intended to be used for call-operations only. No read-reply or write
2696 procedures are supported.

2697

2698 4.3.5.4.2 *Rules*

2699 None.

2700

2701 4.3.5.4.3 *Element rules*

Element	Rule	Description
DataTunneling. dataTunnelingCall.		
header	SHALL be set.	A data tunneling specific header.
header. purposeld	SHALL be set as PRIMARY IDENTIFIER. The string-length SHOULD NOT be longer than 128 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 128 characters.	A string that identifies the purpose / kind of tunnelled data / protocol.
header. channelId	SHALL be set as SUB IDENTIFIER.	Between two nodes there might be several channels used in parallel using the same purposeld. channelId helps to separate these channels. It can also be used to differentiate subsequent (rather than parallel) sessions.
header. sequenceld	SHALL be set as SUB IDENTIFIER.	This helps to give the payload for each combination of { purposeld, channelId } a chronological order. Note: This "sequenceld" is data tunneling specific and should not be confused with "sequenceld" elements of other classes!
payload	SHALL be set.	The actual proprietary data.

2702 Table 34: dataTunnelingCall Element rules

2703

2704 **4.3.5.5 Descriptive information and further definitions**

2705 The DataTunneling Feature Type permits exchange of arbitrary data.

2706 Only "call" operations are permitted with this Feature Type.

2707 Unlike "conventional" features (i.e. features with functions where read/reply or write operations are
 2708 supported) the DataTunneling feature does not apply the "data owner" concept in a usual way at
 2709 first glance. However, the role concept still applies as follows:

2710 An instance of Feature Type DataTunneling and role "server" defines which kind of data is
 2711 exchanged. Specifically, the permitted values for the element "header. purposeld" (if used) are
 2712 determined by the server feature only.

2713

2714 **4.3.6 DeviceClassification**

2715 **4.3.6.1 Dependencies to other Feature Types**

2716 None.

2717

2718 **4.3.6.2 Introduction**

2719 Each device can have associated information useful for its identification and which kind of power it
 2720 requires. This information (mostly plaintext) is bundled in the *DeviceClassification* class. It is intended
 2721 to describe the physical device, but may be used for logical devices (i.e. *entities*), too.

2722

2723 **4.3.6.3 Rules**

2724 None.

2725

2726 **4.3.6.4 deviceClassificationManufacturerData**2727 **4.3.6.4.1 Description**

2728 Information which is usually NOT writable (hard coded in the device or preconfigured information,
 2729 e.g.), is modelled with *deviceClassificationManufacturerData*.

2730

2731 **4.3.6.4.2 Rules**

2732 No specific element is mandatory, but at least one SHALL be set.

2733 For visualisation purposes, the content of the element *vendorName* SHOULD be used as name of the
 2734 vendor only if the element *brandName* is not set. Otherwise *brandName* SHOULD be used.

2735

2736 **4.3.6.4.3 Element rules**

Element	Rule	Description
DeviceClassification. deviceClassificationManufacturerData.		
deviceName	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	The name of the (physical or logical) device as defined by the manufacturer.
deviceCode	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	A device code for the (physical or logical) device as defined by the manufacturer.
serialNumber	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	The serial number of the (physical or logical) device as defined by the manufacturer. Usually the same as printed on the case.
softwareRevision	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider	The software revision of the (physical or logical) device as defined by the manufacturer.

	the possibility that the receiver will shorten the string to 256 characters.	
hardwareRevision	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	The hardware revision of the (physical or logical) device as defined by the manufacturer.
vendorName	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	The name of the vendor of the (physical or logical) device as defined by the manufacturer.
vendorCode	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	A code for the vendor of the (physical or logical) device as defined by the manufacturer.
brandName	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	The name of the brand. Useful where the name of the brand and the vendor differs.
powerSource	Value rules of Table 36 SHALL be applied.	This element describes for which kind of primary power source the device is designed for. It is only used for informative purposes. It is NOT used to describe how or to which kind of power source the device is finally connected (this means it is not considered here whether the device's single mains is fed by a house's mains supply or a backup system, e.g.).
manufacturerNode Identification	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	A node identification for the (physical or logical) device as defined by the manufacturer or vendor. This could be used for the identification of a (physical or logical) device, even if it was removed from the network and rejoined later with changed node address.
manufacturerLabel	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the (physical or logical) device as defined by the manufacturer.
manufacturerDescription	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for the (physical or logical) device as defined by the manufacturer.

2737 Table 35: deviceClassificationManufacturerData Element rules

2738 Element "powerSource" value rules:

Value	Rule
unknown	None.

mainsSinglePhase	None.
mains3Phase	None.
battery	None.
dc	None.

Table 36: Element "powerSource" value rules

4.3.6.5 deviceClassificationUserData

4.3.6.5.1 Description

Information which is usually writable (changeable by a user / application) is modelled with *deviceClassificationUserData*.

4.3.6.5.2 Rules

No specific element is mandatory, but at least one SHALL be set.

4.3.6.5.3 Element rules

Element	Rule	Description
DeviceClassification. deviceClassificationUserData.		
userNodeIdentification	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	Similar to <i>manufacturerNodeIdentification</i> but defined by a user or application.
userLabel	Default rules of section 3.10.1.2.1 SHALL be applied.	Similar to <i>manufacturerLabel</i> but defined by a user or application.
userDescription	Default rules of section 3.10.1.3.1 SHALL be applied.	Similar to <i>manufacturerDescription</i> but defined by a user or application.

Table 37: deviceClassificationUserData Element rules

4.3.6.6 Descriptive information and further definitions

None.

4.3.7 DeviceConfiguration

4.3.7.1 Dependencies to other Feature Types

None.

4.3.7.2 Introduction

Some information cannot be represented by the "normal" SPINE classes, but are important for the interoperable functionality of a device, hence should not be modelled with the DataTunneling class

(see section 5.3.6) or other proprietary ways. For this purpose, the DeviceConfiguration class should be used. It provides a key-value based modelling of information. Some data, important for interoperability, is already defined by the EEBus Initiative e.V. and listed in Table 196.

The information is split into a rather static part (see section 4.3.7.5) and the actual value (see section 4.3.7.4).

4.3.7.3 Rules

None.

4.3.7.4 deviceConfigurationKeyValueListData

4.3.7.4.1 Description

The actual values belonging to a specific key are modelled within this function.

4.3.7.4.2 Rules

None.

4.3.7.4.3 Element rules

Element	Rule	Description
DeviceConfiguration. deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData. (list)		
keyId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different keys on one SPINE feature.
value	Exactly one of the child elements SHALL be set. This SHALL match with the content of element <i>valueType</i> within the key value description part (see section 5.3.8.3.2). Value rules of Table 39 SHALL be applied.	The actual value belonging to the <i>keyId</i> .
isValueChangeable	If set to "true" the server SHALL accept changes of the element "value" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	States whether the value belonging to the <i>keyId</i> is changeable by a client or not.

Table 38: deviceConfigurationKeyValueListData Element rules

Element "**value**" sub-element rules:

Element	Rule	Description
boolean	If the Element "boolean" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "boolean".	Used, if the <i>value</i> is of type <i>boolean</i> .

date	If the Element "date" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "date".	Used, if the <i>value</i> is of type <i>date</i> .
dateTime	If the Element "dateTime" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "dateTime".	Used, if the <i>value</i> is of type <i>dateTime</i> .
duration	If the Element "duration" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "duration".	Used, if the <i>value</i> is of type <i>duration</i> .
string	If the Element "string" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "string". The string-length of the Element SHOULD NOT be longer than 512 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 512 characters.	Used, if the <i>value</i> is of type <i>string</i> .
time	If the Element "time" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "time".	Used, if the <i>value</i> is of type <i>time</i> .
scaledNumber	If the Element "scaledNumber" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "scaledNumber".	Used, if the <i>value</i> is of type <i>scaledNumber</i> .
integer	If the Element "integer" is set, the "valueType" Element within the related deviceConfigurationDescriptionData entry SHALL be set to "integer".	Used, if the <i>value</i> is of type <i>long</i> .

Table 39: Element "value" sub-element rules

4.3.7.5 deviceConfigurationKeyValueDescriptionListData

4.3.7.5.1 Description

The rather static part of the device configuration keys is modelled with this function, including an identifier for the key, a name that defines the meaning of the key and further additional elements.

4.3.7.5.2 Rules

None.

4.3.7.5.3 Element rules

Element	Rule	Description
DeviceConfiguration. deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData. (list)		
keyId	SHALL be set as PRIMARY IDENTIFIER.	The key identifier.
keyName	Value rules of Table 41 SHALL be applied.	A certain key name. If one of the keys defined by the EEBus Initiative e.V. is used, the meaning is defined in Table 196. Vendors may define own key names (but then have to

		mark them as vendor specific, see [ProtocolSpecification], section "Rules for vendor specific extensions").
valueType	The <i>keyName</i> definitions specify which <i>valueType</i> SHALL be used. Value rules of Table 42 SHALL be applied.	The type of the "value" Element within the corresponding "deviceConfigurationKeyValueData" entry is denoted here.
unit	If set, the unit SHALL be applied to the value of the key.	The unit in which the value of the key is given. Not all keys need the element "unit".
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A label for this key.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for this key.

2792 Table 40: deviceConfigurationKeyValueDescriptionListData Element rules

2793 Element "**keyName**" value rules:

Value	Rule	Description
peakPowerOfPvSystem	This value SHALL be positive or 0. If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	The nominal peak power of the production of the installed PV System.
pvCurtailmentLimitFactor	This value SHALL be in range of 0 to 100. If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "pct".	The percentage of the peak power of the PV System that is permitted to be fed into the electricity grid through the electricity grid connection point.
asymmetricChargingSupported	If used, the content of the element "valueType" SHALL be "boolean" and the element "unit" SHALL be omitted.	This configuration parameter is used to indicate if it is possible to charge with different current per phase.
communicationsStandard	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. As content of the element "value. string" within the linked deviceConfigurationKeyValueData entry, only one of the following strings SHALL be used: - iso15118-2ed1 - iso15118-2ed2 - iec61851	Defines the used communications standard for the entity where the DeviceConfiguration feature is located.
inverterGridCode	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Information elements: - country (optional): contains the country (e.g. AT, DE, SE, UK, etc.)	Grid code of the Inverter.

	- technical connection rule (mandatory): contains the name of the connection rule itself (e.g. TORD4, VDE-AR-N4105, EN50438, G98/1-3, etc.) - year (optional): the year in which the connection rule was published - ancillary (optional): further information about the (sub part of the) rule (e.g. LV, etc.).	
pvStringAvailabilityStatus	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Possible values: - Unknown - o The status of the PV String(s) is unknown. - Disconnected - o The PV String(s) is/are disconnected. - Connected - o The PV String(s) is/are connected, but not ready yet. - Available - o The PV String(s) is/are connected and ready for operation (but currently not producing energy). - In operation - o The PV String(s) is/are producing energy.	Indicates the availability of the PV String(s).
batteryAvailabilityStatus	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Possible values: - Unknown - o The status of the Battery or Batteries is unknown. - Disconnected - o The Battery is (or the Batteries are) disconnected. - Connected - o The Battery is (or the Batteries are) connected, but not ready yet. - Available - o The Battery is (or the Batteries are) connected and ready for operation (but currently not consuming or producing energy). - In operation - o The Battery is (or the Batteries are) consuming or producing energy.	Indicates the availability of the Battery (or Batteries).

gridConnectionStatus	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Possible values: - Unknown - o The status of the Inverter's public grid connection is unknown. - Disconnected - o The Inverter is disconnected from the public grid. - Connected - o The Inverter is connected to the public grid.	Indicates whether the Inverter is connected to the public grid or not.										
timeToAcChargePowerMax	This value SHALL be positive or 0. If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "s".	The time (in seconds) currently required by the appliance to adjust the active power until the currently maximum possible charge power is reached.										
timeToAcDischargePowerMax	This value SHALL be positive or 0. If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "s".	The time (in seconds) currently required by the appliance to adjust the active power until the currently maximum possible discharge power is reached.										
tilt	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "deg".	Tilt of the appliance.										
azimuth	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "deg".	Azimuth of the appliance.										
batteryType	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Possible types are: <table><tr><th>Type</th><th>Value for Element "value. string"</th></tr><tr><td>Lithium-ion</td><td>lilon</td></tr><tr><td>Lead-acid</td><td>leadAcid</td></tr><tr><td>Redox flow</td><td>redoxFlow</td></tr><tr><td>Sodium-ion</td><td>sodiumIon</td></tr></table>	Type	Value for Element "value. string"	Lithium-ion	lilon	Lead-acid	leadAcid	Redox flow	redoxFlow	Sodium-ion	sodiumIon	Type of the Battery.
Type	Value for Element "value. string"											
Lithium-ion	lilon											
Lead-acid	leadAcid											
Redox flow	redoxFlow											
Sodium-ion	sodiumIon											
maxCycleCountPerDay	This value SHALL be positive or 0. If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "1".	Maximum number of cycles the battery permits per day.										
failsafeConsumptionActivePowerLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Failsafe value for the appliances active (real) power consumption.										

failsafeProductionActivePowerLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Failsafe value for the appliances active (real) power production.
failsafePositiveReactivePowerLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "var".	Failsafe value for the appliances positive reactive power.
failsafeNegativeReactivePowerLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "var".	Failsafe value for the appliances negative reactive power.
failsafePositiveCosPhiLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "1".	Failsafe value for the appliances positive cos phi value.
failsafeNegativeCosPhiLimit	The content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "1".	Failsafe value for the appliances negative cos phi value.
maxAcChargePower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Maximum AC charge power (aggregated) for all connected appliances.
maxAcDischargePower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Maximum AC discharge power (aggregated) for all connected appliances.
maxDcChargePower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Maximum DC charge power (aggregated) for all connected appliances.
maxDcDischargePower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Maximum DC discharge power (aggregated) for all connected appliances.
batteryActiveControlMode	If used, the content of the element "valueType" SHALL be "string" and the element "unit" SHALL be omitted. Possible values: - Power - PCC - Auto	Current active control mode of the Inverter
defaultAcPower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Default AC power value.
defaultDcPower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Default DC power value.
defaultPccPower	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Default AC power value at the PCC (Point of Common Coupling).
failsafeAcPowerSetpoint	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Failsafe value for the Charge/Discharge AC power setpoint.
failsafeDcPowerSetpoint	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Failsafe value for the Charge/Discharge DC power setpoint.
failsafePccPowerSetpoint	If used, the content of the element "valueType" SHALL be "scaledNumber" and the element "unit" SHALL be set to "W".	Failsafe value for the PCC power setpoint.

failsafeDurationMinimum	If used, the content of the element "valueType" SHALL be "duration" and the element "unit" SHALL be omitted. The appliance SHALL remain in failsafe state after loss of Heartbeats at least the time stated in this Element.	Minimum time the appliance remains in failsafe state after loss of Heartbeats.
dischargingBelowTargetEnergyRequestPermitted	If used, the content of the element "valueType" SHALL be "boolean" and the element "unit" SHALL be omitted.	States whether the appliance (e.g. EV, or any other appliance capable of storing energy) allows discharging its battery below the target SoC or not.
incentivesSimulationCyclesMax	If used, the content of the element "valueType" SHALL be "integer" and the element "unit" SHALL be omitted. The value SHALL be a non-negative integer.	The maximum number of successive simulation cycles permitted by an appliance.
incentivesSimulationConcurrent	If used, the content of the element "valueType" SHALL be "boolean" and the element "unit" SHALL be omitted.	Value "true" means an appliance is capable of executing a scenario with simulation data in parallel to a scenario with committed data. Otherwise, the value is "false".
incentivesTimeoutIncentiveRequest	If used, the content of the element "valueType" SHALL be "duration" and the element "unit" SHALL be omitted.	Maximum waiting period of an appliance to receive a valid update of a simple committed incentive table.
incentivesWaitIncentiveWriteable	If used, the content of the element "valueType" SHALL be "integer" and the element "duration" SHALL be omitted.	Waiting period of an appliance before it permits getting an updated simple committed incentive table.

2794 Table 41: Element "keyName" value rules

2795 Element "**valueType**" value rules:

Value	Rule	Description
boolean	SHALL be used, if the corresponding key value is of type xs:boolean.	The corresponding key value is of type xs:boolean.
date	SHALL be used, if the corresponding key value is of type xs:date.	The corresponding key value is of type xs:date.
dateTime	SHALL be used, if the corresponding key value is of type xs:dateTime.	The corresponding key value is of type xs:dateTime.
duration	SHALL be used, if the corresponding key value is of type xs:duration.	The corresponding key value is of type xs:duration.
string	SHALL be used, if the corresponding key value is of type xs:string.	The corresponding key value is of type xs:string.
time	SHALL be used, if the corresponding key value is of type xs:time.	The corresponding key value is of type xs:time.

scaledNumber	SHALL be used, if the corresponding key value is of type " <i>ScaledNumberType</i> " (see section 3.10.1.8).	The corresponding key value is of type " <i>ScaledNumberType</i> " (see section 3.10.1.8).
integer	SHALL be used, if the corresponding key value is of type xs:long.	The corresponding key value is of type xs:long.

Table 42: Element "valueType" value rules

4.3.7.6 deviceConfigurationKeyValueConstraintsListData

4.3.7.6.1 Description

In cases where a configuration value has some range, it could make sense to specify the upper and lower bounds that are possible for this value as well as a step size. This can be achieved with the deviceConfigurationKeyValueConstraintsListData function.

4.3.7.6.2 Rules

None.

4.3.7.6.3 Element rules

Element	Rule	Description
DeviceConfiguration. deviceConfigurationKeyValueConstraintsListData. deviceConfigurationKeyValueConstraintsData. (list)		
keyId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different keys on one SPINE feature.
valueRangeMin	If set, the child element SHALL match with the content of the Element <i>valueType</i> within the related deviceConfigurationDescriptionData entry. For rules on the sub-elements see Table 39.	The lower bound that is possible for the related value.
valueRangeMax	If set, the child element SHALL match with the content of the Element <i>valueType</i> within the related deviceConfigurationDescriptionData entry. For rules on the sub-elements see Table 39.	The upper bound that is possible for the related value.
valueStepSize	If set, the child element SHALL match with the content of the Element <i>valueType</i> within the related deviceConfigurationDescriptionData entry. For rules on the sub-elements see Table 39.	The step size that is possible for the related value.

Table 43: deviceConfigurationKeyValueListData Element rules

2810 **4.3.8 DeviceDiagnosis**2811 **4.3.8.1 Dependencies to other Feature Types**

2812 None.

2813

2814 **4.3.8.2 Introduction**

2815 This class contains information about the state of a device, heartbeat counting and service data.

2816

2817 **4.3.8.3 Rules**

2818 None.

2819

2820 **4.3.8.4 deviceDiagnosisStateData**2821 **4.3.8.4.1 Description**2822 The *deviceDiagnosisStateData* function provides information about dynamically changing state
2823 information.

2824

2825 **4.3.8.4.2 Rules**

2826 No specific element is mandatory, but at least one SHALL be set.

2827

2828 **4.3.8.4.3 Element rules**

Element	Rule	Description
DeviceDiagnosis. deviceDiagnosisStateData.		
timestamp	None.	The time of creation of the state.
operatingState	Value rules of Table 45 SHALL be applied.	The current operating state of the device.
vendorStateCode	The string-length SHOULD NOT be longer than 128 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 128 characters.	A vendor specific state. May contain any state the vendor uses for its devices.
lastErrorCode	The string-length SHOULD NOT be longer than 128 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 128 characters. Even if the device's "operationState" has a value of <i>normalOperation</i> again, the error	The last error code that occurred.

	code SHOULD remain in the element <code>lastErrorCode</code> .	
<code>upTime</code>	If used, the following rules apply: - If located on entity "DeviceInformation" (see section 4.2.7), the up time of the complete device SHALL be stated here. - If located on some other entity, only the up time of the specific functionality SHALL be stated. In case of nested entities, the up time of all "child"-entities SHALL be considered, too.	The duration the functionality is running since the last start.
<code>totalUpTime</code>	If used the following rules apply: - If located on entity "DeviceInformation" (see section 4.2.7), the total up time of the complete device SHALL be stated here. - If located on some other entity, only the total up time of the specific functionality SHALL be stated. In case of nested entities, the total up time of all "child"-entities SHALL be considered, too.	The total duration the functionality is running since installation (see <i>installationTime</i> element in <i>deviceDiagnosisServiceData</i> function).
<code>powerSupplyCondition</code>	Value rules of Table 46 SHALL be applied.	The condition of the power supply can be stated here.

2829 Table 44: *deviceDiagnosisStateData* Element rules

2830 Element "**operatingState**" value rules:

Value	Rule	Description
<code>normalOperation</code>	None.	The device does its normal operation without any errors or limitations.
<code>standby</code>	None.	The device is in standby mode and does nothing but waiting for user interaction and listens on its communications protocol for external commands.
<code>failure</code>	None.	A failure occurred and the device cannot run as desired. Its normal functionality is likely not given.
<code>serviceNeeded</code>	None.	The device needs some kind of service. Possibly the normal functionality is not given.
<code>overrideDetected</code>	None.	The device detected an override. Normal operation may be given or the device is in some safe mode.
<code>inAlarm</code>	None.	An alarm (or emergency) occurred and user interaction is needed. Normal operation may be given or the device is in some safe mode.
<code>notReachable</code>	None.	The device is not reachable via its non-SPINE communications protocol. This operating state can only be set by some technology gateway device that handles the normal communication to the device.

finished	None.	The device has finished its temporary operation. As long as it remains in state <i>finished</i> , the normal operation is not given.
temporarilyNotReady	None.	The device is temporarily not ready for normal operation (e.g. due to maintenance).
off	None.	The device is off. Hence, it cannot operate.

Table 45: Element "operatingState" value rules

Element "**powerSupplyCondition**" value rules:

Value	Rule	Description
good	None.	The power supply is in a good condition.
low	None.	The power supply has low capacity. It would make sense to check it (e.g. change battery).
critical	None.	The power supply is in critical state. If it is supplied by a battery, a change of the battery is advised.
unknown	None.	The power supply condition is unknown. A notification to a user could make sense.
error	None.	The power supply is in error condition.

Table 46: Element "powerSupplyCondition" value rules

4.3.8.5 deviceDiagnosisHeartbeatData

4.3.8.5.1 Description

The *deviceDiagnosisHeartbeatData* function enables sending a specific signal, which indicates a device is still running.

4.3.8.5.2 Rules

No specific element is mandatory, but at least one SHALL be set.

4.3.8.5.3 Element rules

Name	Rule	Description
DeviceDiagnosis. deviceDiagnosisHeartbeatData.		
timestamp	None.	The time of creation of the data.
heartbeatCounter	The value of the heartbeatCounter element SHALL be increased after every <i>heartbeatTimeout</i> (NOT with every sending of this function). The <i>deviceDiagnosisHeartbeatData</i> function can not only be sent initially by the device itself, but can be requested by another device, too. In this case, the element <i>heartbeatCounter</i> SHALL NOT be incremented and the <i>heartbeatTimeout</i> has (as always) its fixed value (i.e. not the remaining time to the next (automatic) notification by the device).	An incrementing counter of the heartbeat.

heartbeatTimeout	A notification of the deviceDiagnosisHeartbeatData function SHALL be sent to all subscribed clients after each heartbeatTimeout period.	The heartbeatTimeout element contains the period, in which the deviceDiagnosisHeartbeatData function is sent by the device (NOT the remaining time till the next sending).
------------------	---	--

Table 47: deviceDiagnosisHeartbeatData Element rules

4.3.8.6 deviceDiagnosisServiceData

4.3.8.6.1 Description

The deviceDiagnosisServiceData function informs about service information related to a device's installation or maintenance.

4.3.8.6.2 Rules

No specific element is mandatory, but at least one SHALL be set.

4.3.8.6.3 Element rules

Name	Rule	Description
DeviceDiagnosis. deviceDiagnosisServiceData.		
timestamp	None.	The time of creation of the data.
installationTime	None.	The point in time the device was installed (commissioned).
bootCounter	None.	The count of boot processes the device completed since <i>installationTime</i> .
nextService	None.	The point in time, the next service should be done, is stated by this element. Alternatively, the decrementing duration till the next service can be indicated.

Table 48: deviceDiagnosisServiceData Element rules

4.3.8.7 Descriptive information and further definitions

None.

4.3.9 DirectControl

4.3.9.1 Dependencies to other Feature Types

- OperatingConstraints
- PowerSequences
- SmartEnergyManagementPs

4.3.9.2 Introduction

Some kind of electricity consuming or producing devices require (in contrast to the PowerSequences Class, see section 5.3.19) a rather instantaneous energy consumption or generation control. This can be modelled with two primary information sets:

1. Notification of instantaneous consumption/generation as well as instantaneous control. This is covered by DirectControl and discussed in this section.
2. Information on the power/energy constraints and implications. This is covered by OperatingConstraints and discussed in section 4.3.20.

Information on and control of energy/power can be modelled using the directControlActivityListData Function. The directControlDescriptionData Function is used to model some general rather constant data.

4.3.9.3 Rules

None.

4.3.9.4 directControlActivityListData**4.3.9.4.1 Description**

The function *directControlActivityListData* combines some information on the activity of a given time. Apart from information on state, power and energy it can contain information whether power, energy, etc. can be modified by a client with a proper (partial) write command.

4.3.9.4.2 Rules

The current values SHALL be represented by a list entry with timestamp = "zero seconds" (e.g. "PT0S" or "PT0H"). The server SHALL only accept write commands on data instances with timestamp = "zero seconds" (e.g. "PT0S" or "PT0H") to adjust current values of the Function.

4.3.9.4.3 Element rules

Name	Rules	Description
DirectControl. directControlActivityListData. directControlActivityData. (list)		
timestamp	SHALL be set as PRIMARY IDENTIFIER. The most actual directControlActivityData SHALL have a duration (i.e. relative time to "now") of zero seconds.	The timestamp of this directControlActivityData.
activityState	Value rules of Table 50 SHALL be applied.	The state of the activity.
isActivityStateChangeable	If this element is set to <i>true</i> the server SHALL accept changes of the device's <i>activityState</i> by appropriate clients (e.g. the	Indicates whether the "activityState" can be modified by a client or not.

	bound client). Otherwise, the server SHALL decline change requests.	
energyMode	For further value rules see section 3.10.1.16.1.	Energy mode of the device.
isEnergyModeChangeable	If this element is set to <i>true</i> the server SHALL accept changes of the device's "energyMode" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Indicates whether the "energyMode" can be modified by a client or not.
power	Target value that SHOULD be followed as good as possible.	The overall electrical power value belonging to this activity information. Please note: Phase-specific information is not given here.
isPowerChangeable	If this element is set to <i>true</i> the server SHALL accept changes of the device's "power" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Indicates whether the "power" can be modified by a client or not.
energy	Target value that SHOULD be followed as good as possible.	The electrical energy value belonging to this activity information.
isEnergyChangeable	If this element is set to <i>true</i> the server SHALL accept changes of the device's "energy" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Indicates whether the "energy" can be modified by a client or not.
sequenceId	If the activity is related to a power sequence, the according sequenceId SHOULD be stated here as FOREIGN IDENTIFIER.	The related power sequence identifier (if available).

2893 Table 49: directControlActivityListData Element rules

2894 Element "activityState" value rules:

Value	Rule	Description
running	If <i>isActivityStateChangeable</i> is set to <i>true</i> , the activity MAY be paused or deactivated by a client (if <i>sequenceId</i> is set, further information from the related sequence SHALL be considered).	The activity is running.
paused	The activity is paused. If <i>isActivityStateChangeable</i> is set to <i>true</i> , the activity MAY be activated or deactivated by a client (if <i>sequenceId</i> is set, further information from the related sequence SHALL be considered).	The activity is paused.
inactive	The activity is inactive. If <i>isActivityStateChangeable</i> is set to <i>true</i> , the activity MAY be activated by a client (if <i>sequenceId</i> is set,	The activity is inactive.

	further information from the related sequence SHALL be considered).	
--	---	--

Table 50: Element "activityState" value rules

4.3.9.5 directControlDescriptionData**4.3.9.5.1 Description**

This function contains the rather static information available in this Feature Type.

4.3.9.5.2 Rules

None.

4.3.9.5.3 Element rules

Element	Rule	Description
DirectControl. directControlDescriptionData.		
positiveEnergyDirection	SHOULD be set. For further value rules see section 3.10.1.14.1.	The <i>positiveEnergyDirection</i> states whether energy consumption or production will be counted as positive value.
powerUnit	SHOULD be set, if a power is stated in the function <i>directControlActivityListData</i> .	This is the unit for all power values related to consumption or production.
energyUnit	SHOULD be set, if an energy is stated in the function <i>directControlActivityListData</i> .	This is the unit for all energy values related to consumption or production.

Table 51: directControlDescriptionData Element rules

4.3.9.6 Descriptive information and further definitions

None.

4.3.10 ElectricalConnection**4.3.10.1 Dependencies to other Feature Types**

- Measurement

4.3.10.2 Introduction

Electrical connections have some more complex parameters (e.g. which AC phase is measured, real/reactive/apparent measurement, amplitude/RMS, number of measured harmonic, etc.), which are modelled with this class. Some electrical connection specific state information (e.g. AC or DC, number of connected phases (only AC), positive energy direction, label, etc.) is modelled, too.

2919 Measuring electrical values is still done with the Measurement class (see section 5.3.15). But the
 2920 above-mentioned additional parameters are only described within the ElectricalConnection class.
 2921 The *measurementId* from the Measurement class is stated in the
 2922 *electricalConnectionParameterDescriptionListData* function, to have a proper reference.

2923 Characteristical information (like nominal values) are modelled within the
 2924 *electricalConnectionCharacteristicListData* Function.

2925

2926 **4.3.10.3 Rules**

2927 None.

2928

2929 **4.3.10.4 *electricalConnectionParameterDescriptionListData***

2930 **4.3.10.4.1 Description**

2931 The aforementioned parameters for the measurement of an electrical connection are described in
 2932 this function. The measurement values themselves are modelled in the Measurement class (section
 2933 5.3.15). Only a reference is given in this function.

2934

2935 **4.3.10.4.2 Rules**

2936 None.

2937

2938 **4.3.10.4.3 Element rules**

Element	Rule	Description
ElectricalConnection. <i>electricalConnectionParameterDescriptionListData</i> . <i>electricalConnectionParameterDescriptionData</i> . (list)		
<i>electricalConnectionId</i>	SHALL be set as PRIMARY IDENTIFIER.	Reference to the electrical connection, described in the <i>electricalConnectionDescriptionListData</i> function (see section 5.3.11.6).
<i>parameterId</i>	SHALL be set as SUB IDENTIFIER.	One electrical connection can have multiple parameters. The <i>parameterId</i> helps to distinguish them.
<i>measurementId</i>	The FOREIGN IDENTIFIER MAY be set. If set, the related electrical connection data SHALL be linked to a measurand or data of another Feature that uses the same <i>measurementId</i> .	This is a foreign identifier.
<i>voltageType</i>	If <i>powerSupplyType</i> in <i>electricalConnectionDescriptionData</i> has the value "dc", this element MAY be omitted, or, if stated, it SHALL be set to "dc". If	Specifies which kind of electricity is measured ("ac" or "dc").

	<i>powerSupplyType</i> is of value "ac", the <i>voltageType</i> can either have the value "ac" (<i>voltageType</i> then MAY be omitted), or "dc". In the latter case, the DC-offset of the AC connection is measured and the <i>voltageType</i> element SHALL be set accordingly. Value rules of Table 53 SHALL be applied	
acMeasuredPhases	In case of <i>powerSupplyType</i> = "dc", this element SHALL be omitted. Value rules of Table 54 SHALL be applied.	In case of <i>powerSupplyType</i> = "ac", this element states the phase, which is measured. Combinations of phases (e.g. "ab") are possible.
acMeasuredInReference To	In case of <i>powerSupplyType</i> = "dc", this element SHALL be omitted. Value rules of Table 55 SHALL be applied.	In case of <i>powerSupplyType</i> = "ac", this element states the phase, which <i>acMeasuredPhases</i> is measured against (e.g. "neutral").
acMeasurementType	In case of <i>powerSupplyType</i> = "dc", this element SHALL be omitted. Value rules of Table 56 SHALL be applied.	In case of <i>powerSupplyType</i> = "ac", this element states the kind of AC measurement is done (e.g. "real").
acMeasurementVariant	In case of <i>powerSupplyType</i> = "dc", this element SHALL be omitted. Value rules of Table 57 SHALL be applied.	In case of <i>powerSupplyType</i> = "ac", this element states the variation of the AC measurement (e.g. "amplitude").
acMeasuredHarmonic	In case of <i>powerSupplyType</i> = "dc", this element SHALL be omitted.	In case of <i>powerSupplyType</i> = "ac", this element states the harmonic, which is measured. A value of "1" indicates that the normal value is measured.
scopeType	None.	A certain meaning of the parameter combination.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A label for this electrical connection parameter.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for this electrical connection parameter.

Table 52: *electricalConnectionParameterDescriptionListData* Element rulesElement "**voltageType**" value rules:

Value	Rule	Description
ac	SHALL NOT be used if <i>powerSupplyType</i> in <i>electricalConnectionDescriptionData</i> has the value "dc".	Alternating current.
dc	None.	Direct current.

Table 53: Element "*voltageType*" value rules

2942 Element **"acMeasuredPhases"** value rules:

Value	Rule	Description
a	None.	Phase a (or L1).
b	None.	Phase b (or L2).
c	None.	Phase c (or L3).
ab	None.	Phase a and b (or L1 and L2).
bc	None.	Phase b and c (or L2 and L3).
ac	None.	Phase a and c (or L1 and L3).
abc	None.	Phase a, b and c (or L1, L2 and L3).
neutral	None.	The neutral conductor of a connection.
ground	None.	The ground conductor of a connection.
none	None.	No phase specified or available.

2943 Table 54: Element **"acMeasuredPhases"** value rules

2944 Element **"acMeasuredInReferenceTo"** value rules:

Value	Rule	Description
a	None.	Phase a (or L1).
b	None.	Phase b (or L2).
c	None.	Phase c (or L3).
ab	None.	Phase a and b (or L1 and L2).
bc	None.	Phase b and c (or L2 and L3).
ac	None.	Phase a and c (or L1 and L3).
abc	None.	Phase a, b and c (or L1, L2 and L3).
neutral	None.	The neutral conductor of a connection.
ground	None.	The ground conductor of a connection.
none	None.	No phase specified or available.

2945 Table 55: Element **"acMeasuredInReferenceTo"** value rules

2946 Element **"acMeasurementType"** value rules:

Value	Rule	Description
real	None.	AC measurement of the real part.
reactive	None.	AC measurement of the reactive part.
apparent	None.	AC measurement of the apparent part.
phase	None.	AC measurement of the phase.

2947 Table 56: Element **"acMeasurementType"** value rules

2948 Element **"acMeasurementVariant"** value rules:

Value	Rule	Description
amplitude	None.	Measurement of the amplitude.
rms	None.	Measurement of the RMS (Root Mean Square).
instantaneous	None.	Measurement of the instantaneous value.
angle	None.	Measurement of the angle.
cosPhi	None.	Measurement of the cosine phi.

2949 Table 57: Element **"acMeasurementVariant"** value rules

2950

4.3.10.5 electricalConnectionPermittedValueSetListData**4.3.10.5.1 Description**

If an electrical connection parameter has some defined value(s) or value range(s) that are allowed or possible, they can be modelled with this function. It enables the server to state single values, value ranges or combinations of both.

4.3.10.5.2 Rules

None.

4.3.10.5.3 Element rules

Element	Rule	Description
ElectricalConnection. electricalConnectionPermittedValueSetListData. electricalConnectionPermittedValueSetData. (list)		
electricalConnectionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the electrical connection, this rather static information belongs to.
parameterId	SHALL be set as SUB IDENTIFIER.	Relation to the according parameter.
permittedValuesSet (list)	At least one set of permitted values SHALL be stated.	The permittedValueSet allows to define an arbitrary set of permitted data. Please refer to the ScaledNumberSetType for more details.

Table 58: electricalConnectionPermittedValueSetListData Element rules

4.3.10.6 electricalConnectionCharacteristicListData**4.3.10.6.1 Description**

Characteristical information related to a specific electrical connection parameter (and hence to a measurand) are modelled with this function.

4.3.10.6.2 Rules

None.

4.3.10.6.3 Element rules

Element	Rule	Description
ElectricalConnection. electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData. (list)		
electricalConnectionId	SHALL be set as PRIMARY IDENTIFIER.	Electrical connection, this characteristic information belongs to.
parameterId	SHALL be set as SUB IDENTIFIER.	Relation to the according parameter this characteristic belongs to.
characteristicId	SHALL be set as SUB IDENTIFIER of parameterId.	Identifier of this characteristic for the given parameterId.

characteristicContext	SHALL be set. Value rules of Table 60 SHALL be applied.	Context, the characteristic is valid for.
characteristicType	SHALL be set. Value rules of Table 61 SHALL be applied.	The type of the characteristic.
value	SHALL be set.	The value for this characteristic.
unit	SHALL be set, if there could be ambiguity how to interpret the value without the unit.	The unit of this characteristic.

2972 Table 59: *electricalConnectionCharacteristicListData* Element rules

2973 Element "**characteristicContext**" value rules:

Value	Rule	Description
device	None.	The characteristic is valid for the Device where this information is located on, incl. all Entities.
entity	None.	The characteristic is valid for the Entity where this information is located on.
inverter	None.	The characteristic is valid for the inverter sub-Entities of the Entity where this information is located on.
pVString	None.	The characteristic is valid for the PV string sub-Entities of the Entity where this information is located on.
battery	None.	The characteristic is valid for the battery sub-Entities of the Entity where this information is located on.

2974 Table 60: Element "*characteristicContext*" value rules

2975 Element "**characteristicType**" value rules:

Value	Rule	Description
powerConsumptionMin	None.	The minimum (active) power the related functionality can consume.
powerConsumptionMax	None.	The maximum (active) power the related functionality can consume.
powerConsumptionNominalMin	None.	The nominal minimum (active) power the related functionality can consume, as specified in the manufacturer's data sheet.
powerConsumptionNominalMax	None.	The nominal maximum (active) power the related functionality can consume, as specified in the manufacturer's data sheet.
powerProductionMin	None.	The minimum (active) power the related functionality can produce.
powerProductionMax	None.	The maximum (active) power the related functionality can produce.
powerProductionNominalMin	None.	The nominal minimum (active) power the related functionality

		can produce, as specified in the manufacturer's data sheet.
powerProductionNominalMax	None.	The nominal maximum (active) power the related functionality can produce, as specified in the manufacturer's data sheet.
energyCapacityNominalMax	None.	The nominal capacity of the related functionality as specified in the manufacturer's data sheet.
contractualConsumptionNominalMax	None.	The nominal maximum (active) power the related functionality is allowed to consume due to contractual limitations.
contractualProductionNominalMax	None.	The nominal maximum (active) power the related functionality is allowed to produce due to contractual limitations.
apparentPowerProductionNominalMax	None.	The nominal maximum apparent power the related functionality can produce, as specified in the manufacturer's data sheet.
apparentPowerConsumptionNominalMax	None.	The nominal maximum apparent power the related functionality can consume, as specified in the manufacturer's data sheet.

Table 61: Element "characteristicType" value rules

4.3.10.7 electricalConnectionStateListData

4.3.10.7.1 Description

Some electrical connection specific state information is modelled in this function.

4.3.10.7.2 Rules

None.

4.3.10.7.3 Element rules

Element	Rule	Description
ElectricalConnection. electricalConnectionStateListData. electricalConnectionStateData. (list)		
electricalConnectionId	SHALL be set as PRIMARY IDENTIFIER.	Identifies the electrical connection, this state information belongs to.
timestamp	None.	Contains the point in time, the state information was generated.
currentEnergyMode	For further value rules see section 3.10.1.16.1.	Whether a device is consuming, producing or idle is stated by this element.

consumptionTime	SHOULD be set to the duration the device is in energy mode "consume" since the last reboot.	The duration the device is in energy mode "consume" since the last reboot (see section 5.3.9.2).
productionTime	SHOULD be set to the duration the device is in energy mode "produce" since the last reboot.	The duration the device is in energy mode "produce" since the last reboot (see section 5.3.9.2).
totalConsumptionTime	SHOULD be set to the duration, the device is in energy mode "consume" since its <i>installationTime</i> .	The duration the device is in energy mode "consume" since its <i>installationTime</i> (see section 5.3.9.4).
totalProductionTime	SHOULD be set to the duration, the device is in energy mode "producing" since its <i>installationTime</i> .	The duration the device is in energy mode "producing" since its <i>installationTime</i> (see section 5.3.9.4).

Table 62: electricalConnectionStateListData Element rules

4.3.10.8 electricalConnectionDescriptionListData

4.3.10.8.1 Description

Some electrical connection specific information, that are rather static, are modelled within this function.

4.3.10.8.2 Rules

None.

4.3.10.8.3 Element rules

Element	Rule	Description
ElectricalConnection. electricalConnectionDescriptionListData. electricalConnectionDescriptionData. (list)		
electricalConnectionId	SHALL be set as PRIMARY IDENTIFIER.	Electrical connection, this information belongs to.
powerSupplyType	SHOULD be set.	States whether the electrical connection is of type "ac" or "dc".
acConnectedPhases	If <i>powerSupplyType</i> = "ac", this element SHOULD be set.	If <i>powerSupplyType</i> ="ac", this element contains the number of phases, this electrical connection has (typically 1, 2 or 3).
acRmsPeriodDuration	None.	If the "rms" value of a parameter combination (see element <i>acMeasurementVariant</i> in section 5.3.11.2.2) is measured, this element states the duration, the RMS value is determined in.

positiveEnergyDirection	SHOULD be set. If it is omitted, it SHALL be interpreted as <i>positiveEnergyDirection</i> = "consume".	This element states whether energy consumption or production will be counted as positive value.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A label for this electrical connection.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for this electrical connection.

Table 63: *electricalConnectionDescriptionListData* Element rules

2997

2998

2999 **4.3.10.9 Descriptive information and further definitions**

3000 None.

3001

3002 **4.3.11 Generic**3003 **4.3.11.1 Dependencies to other Feature Types**

3004 None.

3005

3006 **4.3.11.2 Introduction**

3007 None.

3008

3009 **4.3.11.3 Rules**

3010 None.

3011

3012 **4.3.11.4 Function and Identifier rules**

3013 None.

3014

3015 **4.3.11.5 Element rules**

3016 None.

3017

3018 **4.3.11.6 Descriptive information and further definitions**

3019 The Generic Feature Type does not contain any information. It SHALL only be used as a client feature!

3020 Such a client MAY send any read, write or call messages to servers from this feature. This can be very

3021 useful if a client wants to communicate with different Feature Types from the same address.

3022 Additionally, unnecessary duplications of server Feature Types on client side are avoided (e.g. it is

3023 not needed to have a Measurement and a Setpoint client Feature; instead, only this Generic Feature
3024 is used by the client). Therefore, it is preferable that a client uses the Generic Feature Type.

3025

3026 **4.3.12 HVAC**

3027 **4.3.12.1 Dependencies to other Feature Types**

- 3028 - PowerSequences
- 3029 - Setpoint
- 3030 - SmartEnergyManagementPs
- 3031 - TimeTable

3032

3033 **4.3.12.2 Introduction**

3034 Heating, Ventilation and Air Conditioning (HVAC) devices have some special functionality which is
3035 covered by this class.

3036 All system functions (like heating, cooling, domestic hot water preparation (dhw), ventilation, etc.)
3037 have their specific HVAC operation modes (like on, off, auto, etc.). Relations between these are
3038 modelled as well as relations to (optional) power sequences, which represent a concrete run of a
3039 device's program.

3040 HVAC overruns (like an one time domestic-hot-water (DHW) preparation), which do have an impact
3041 on the normal operation mode of a HVAC system function, are modelled in further SPINE functions of
3042 the HVAC Class.

3043

3044 **4.3.12.3 Rules**

3045 None.

3046

3047 **4.3.12.4 hvacSystemFunctionListData**

3048 **4.3.12.4.1 Description**

3049 The hvacSystemFunctionListData function announces the current status of the HVAC system
3050 functions, including the current HVAC operation mode, a flag that indicates, if the HVAC operation
3051 mode can be changed by some external communications partner (client) and a flag that indicates
3052 whether an HVAC overrun is currently active or not.

3053

3054 **4.3.12.4.2 Rules**

3055 None.

3056

3057 4.3.12.4.3 Element rules

Element	Rule	Description
HVAC. hvacSystemFunctionListData. hvacSystemFunctionData. (list)		
systemFunctionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for a specific HVAC system function.
currentOperationModelId	SHALL be set as FOREIGN IDENTIFIER to the according operationModelId of the currently selected operation mode. If the function hvacSystemFunctionOperationModeRelationListData is used, only a operationModelId value related there to the according systemFunctionId SHALL be used here.	States which HVAC operation mode is currently active for this specific HVAC system function
isOperationModelIdChangeable	If set to "true" the server SHALL accept changes of the element "currentOperationModelId" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Denotes whether the current HVAC operation mode identifier is changeable (by some external communication partner) or not.
currentSetpointId	The setpointId SHALL be set here as FOREIGN IDENTIFIER, if a setpoint is linked to the system function. If the function hvacSystemFunctionSetpointRelationListData is used, only a setpointId value related there to the according systemFunctionId SHALL be used here.	States which setpoint is active for this specific HVAC system function
isSetpointIdChangeable	If set to "true" the server SHALL accept changes of the element "currentSetpointId" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Denotes whether the current setpoint identifier is changeable (by some external communication partner) or not.
isOverrunActive	If an overrun is active that has an effect on this system function, this element SHOULD be set to "true".	If an HVAC overrun is currently active, this element is set to <i>true</i> . The HVAC overrun itself is modelled with other functions (see section 5.3.12.8 and 5.3.12.9) and may be located on a different feature or even a different entity or device.

3058 Table 64: hvacSystemFunctionListData Element rules

3059

4.3.12.5 hvacSystemFunctionOperationModeRelationListData**4.3.12.5.1 Description**

Each HVAC system function has its own relation to all for this HVAC system function available HVAC operation modes. E.g. there may be three HVAC operation modes available on the feature: *on*, *off*, *auto* (see section 4.3.12.9). HVAC System function 1 only supports HVAC operation mode 1 (*on*) and 2 (*off*). HVAC System function 2 supports all three HVAC operation modes. The enumeration of the available HVAC system functions is described in section 4.3.12.8.3.

4.3.12.5.2 Rules

None.

4.3.12.5.3 Element rules

Element	Rule	Description
HVAC. hvacSystemFunctionOperationModeRelationListData. hvacSystemFunctionOperationModeRelationData. (list)		
systemFunctionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for a specific HVAC system function.
operationModelId (list)	SHALL be set as FOREIGN IDENTIFIER (at least one) to the according operationModelId of the referenced operation mode.	Lists the HVAC operation modes that are available (selectable) for a specific HVAC system function.

Table 65: hvacSystemFunctionOperationModeRelationListData Element rules

4.3.12.6 hvacSystemFunctionSetpointRelationListData**4.3.12.6.1 Description**

The setpoint class (see section 5.3.21) may be used to model some HVAC relevant setpoints like desired room temperature. An HVAC system function may relate to some of these setpoints. The relation is modelled with this SPINE function.

4.3.12.6.2 Rules

None.

4.3.12.6.3 Element rules

Element	Rule	Description
HVAC. hvacSystemFunctionSetpointRelationListData. hvacSystemFunctionSetpointRelationData. (list)		
systemFunctionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for a specific HVAC system function.
operationModelId	SHALL be set as SUB IDENTIFIER.	Identifier for a specific HVAC operation mode.

setpointId (list)	MAY be set as FOREIGN IDENTIFIER to the according setpointId of the referenced setpoint.	List of setpoints that are related with a specific HVAC system function.
-------------------	--	--

Table 66: hvacSystemFunctionSetpointRelationListData Element rules

4.3.12.7 hvacSystemFunctionPowerSequenceRelationListData**4.3.12.7.1 Description**

Power sequences (see section 5.3.19) describe tasks of a functionality in detail. An HVAC system function may relate to some of these power sequences. The relation is modelled with this function.

4.3.12.7.2 Rules

None.

4.3.12.7.3 Element rules

Element	Rule	Description
HVAC. hvacSystemFunctionPowerSequenceRelationListData. hvacSystemFunctionPowerSequenceRelationData. (list)		
systemFunctionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for a specific HVAC system function.
sequenceId (list)	MAY be set as FOREIGN IDENTIFIER to the according sequenceId of the referenced power sequence.	List of power sequences that are related with a specific HVAC system function.

Table 67: hvacSystemFunctionPowerSequenceRelationListData Element rules

4.3.12.8 hvacSystemFunctionDescriptionListData**4.3.12.8.1 Description**

The rather static information about the HVAC system functions is modelled within this SPINE function.

4.3.12.8.2 Rules

None.

4.3.12.8.3 Element rules

Element	Rule	Description
HVAC. hvacSystemFunctionDescriptionListData. hvacSystemFunctionDescriptionData. (list)		
systemFunctionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for a specific HVAC system function.
systemFunctionType	SHALL be set.	Denotes the type of the HVAC system function.

	Value rules of Table 69 SHALL be applied.	
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the HVAC system function.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the HVAC system function.

Table 68: hvacSystemFunctionDescriptionListData Element rules

Element "**systemFunctionType**" value rules:

Value	Rule	Description
heating	None.	HVAC system function for space heating.
cooling	None.	HVAC system function for space cooling.
ventilation	None.	HVAC system function for ventilation.
dhw	None.	HVAC system function for domestic hot water preparation.

Table 69: Element "systemFunctionType" value rules

4.3.12.9 hvacOperationModeDescriptionListData

4.3.12.9.1 Description

HVAC operation modes are used to describe which behaviour the HVAC system functions have. Each HVAC operation mode may be used by several HVAC system functions.

4.3.12.9.2 Rules

None.

4.3.12.9.3 Element rules

Element	Rule	Description
HVAC. hvacOperationModeDescriptionListData. hvacOperationModeDescriptionData. (list)		
operationModeId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the HVAC operation mode.
operationModeType	SHALL be set. Value rules of Table 71 SHALL be applied.	Denotes the type of the HVAC operation mode.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the HVAC operation mode.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the HVAC operation mode.

Table 70: hvacOperationModeDescriptionListData Element rules

Element "**operationModeType**" value rules:

Value	Rule	Description
auto	None.	HVAC System Function
		heating / cooling / dhw / ventilation
		Meaning
		HVAC System function is controlled depending on a time table (i.e. the current

			setpoint for the HVAC system function is changed depending on a time table).
on	None.	HVAC System Function	Meaning
		heating / cooling / dhw / ventilation	The HVAC System function is controlled (e.g. Setpoint) independent from time tables (i.e. the current setpoint for the HVAC system function is not depending on any time table).
off	None.	HVAC System Function	Meaning
		heating / dhw	Protection mode against freezing (no physically turning off). The (temperature) setpoint for freezing protection may be stated with a setpointId.
		cooling	Space cooling function not active.
		ventilation	Ventilation function not active (protection functions like humidity protection may be active).
eco	None.	HVAC System Function	Meaning
		Heating / cooling / dhw / ventilation	HVAC System function is controlled (e.g. current Setpoint) independent from time tables (i.e. the current setpoint for the HVAC system function is not depending on any time table). HVAC System function is controlled by a different setpoint as for "on".

Table 71: Element "operationModeType" value rules

4.3.12.10 hvacOverrunListData

4.3.12.10.1 Description

The normal HVAC operation may be overridden by some HVAC specific functionality. There are some HVAC overruns defined by the EEBus Initiative e.V., but vendors may add further vendor-specific HVAC overruns, too (which are then out of interoperability scope).

When the HVAC overrun is active, some (not all! See section 5.3.12.9) HVAC system functions have a specific behaviour, other than the normal operation would be. After the HVAC overrun is inactive again, the HVAC system returns to its normal operation.

4.3.12.10.2 Rules

None.

4.3.12.10.3 Element rules

Element	Rule	Description
HVAC. hvacOverrunListData. hvacOverrunData. (list)		
overrunId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the HVAC overrun.

overrunStatus	SHALL be set. Value rules of Table 73 SHALL be applied.	The current status of the HVAC overrun.
timeTableId	None.	Reference to a time table. Needed, if an overrun has some start and end point(s). The overrun is then activated and deactivated at the times stated within the time table.
isOverrunStatusChangeable	If set to "true" the server SHALL accept changes of the Element "overrunStatus" by appropriate clients (e.g. the bound client). Otherwise, the server SHALL decline change requests.	Whether the overrun status is changeable (element set to "true") by a client or not (element set to "false") is denoted in this element.

3136 Table 72: hvacOverrunListData Element rules

3137 Element "**overrunStatus**" value rules:

Value	Rule	Description
active	None.	HVAC overrun triggered, but not started yet.
running	None.	HVAC overrun running.
finished	MAY only be used as notification directly after HVAC overrun was finished. "overrunStatus" SHALL be changed to "inactive" afterwards. SHOULD NOT be used as part of a reply message!	HVAC overrun finished.
inactive	MAY be used as reply or as notification. The latter only, if HVAC overrun was not finished in a usual way (deactivated by user, e.g.).	HVAC overrun inactive.

3138 Table 73: Element "overrunStatus" value rules

3139

3140 **4.3.12.11 hvacOverrunDescriptionListData**3141 **4.3.12.11.1Description**

3142 Each HVAC overrun is defined by its hvacOverrunDescriptionListData list data entry. Here, the HVAC
 3143 overrun type is specified as well as the list of HVAC system functions that are affected by the HVAC
 3144 overrun (if this information is available).

3145

3146 **4.3.12.11.2Rules**

3147 None.

3148

3149 **4.3.12.11.3Element rules**

Element	Rule	Description
HVAC. hvacOverrunDescriptionListData. hvacOverrunDescriptionData. (list)		

overrunId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the HVAC overrun.
overrunType	SHALL be set. Value rules of Table 75 SHALL be applied.	The type of the overrun.
affectedSystemFunctionId (list)	If system functions are affected by this overrun, their according systemFunctionId SHOULD be stated here.	Identifiers of all HVAC system functions that are affected by this HVAC overrun.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the HVAC overrun.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the HVAC overrun.

Table 74: hvacOverrunDescriptionListData Element rules

Element "**overrunType**" value rules:

Value	Rule	Description
oneTimeDhw	None.	A one-time domestic hot water loading (independend from current HVAC operation mode or time table).
party	None.	During the "party" overrun, the HVAC system will stay in the normal day-operation mode and not turn off at the usual time. Example: HVAC system function "heating" is controlled by the day setpoint, independend from operation mode and time table, until HVAC overrun "party" is finished.
sgReadyCondition1	None.	The SGReady condition 1 will be applied. Please consider [SGReady].
sgReadyCondition3	None.	The SGReady condition 3 will be applied. Please consider [SGReady].
sgReadyCondition4	None.	The SGReady condition 4 will be applied. Please consider [SGReady].
oneDayAway	None.	Overrides the normal daily routine to a out-of-home configuration (e.g. eco-mode the whole day).
oneDayAtHome	None.	Overrides the normal daily routine to an at-home configuration (e.g. on-mode the whole day).
oneTimeVentilation	None.	Activates the ventilation once, independent from the configured routine.
hvacSystemOff	None.	Deactivates the HVAC system (frost protection is still active and the system is in a network-standby mode, in which it is reachable for re-configuration).

Table 75: Element "overrunType" value rules

4.3.12.12 Descriptive information and further definitions

Relation between "overrunStatus" element in function "hvacOverrunListData" and "isOverrunActive" element in function "hvacSystemFunctionListData":

overrunStatus	isOverrunActive
active	false
running	true
finished	false
inactive	false

Table 76: Relation between "overrunStatus" and "isOverrunActive"

The content of the according elements SHALL match the relation stated above.

3159

3160 **4.3.13 Identification**3161 **4.3.13.1 Dependencies to other Feature Types**

3162 None.

3163

3164 **4.3.13.2 Introduction**

3165 The Identification Class can be used to express certain identification values of the parent Entity.

3166 Additionally, information about sessions can be provided.

3167

3168 **4.3.13.3 Rules**

3169 None.

3170

3171 **4.3.13.4 identificationListData**3172 **4.3.13.4.1 Description**

3173 The identificationListData function allows to express certain identification values of the

3174 corresponding Entity (the parent Entity of the Identification Feature).

3175

3176 **4.3.13.4.2 Rules**

3177 None.

3178

3179 **4.3.13.4.3 Element rules**

Element	Rule	Description
Identification. identificationListData. identificationData. (list)		
identificationId	SHALL be set as PRIMARY IDENTIFIER.	The identificationId is used to identify the different list entries within IdentificationListData.
identificationType	If set, the value rules of Table 78 SHALL be applied.	The identificationType allows to describe a certain type of identification, e.g. a MAC address or an RFID tag.
identificationValue	The string-length SHOULD NOT be longer than 256 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 256 characters.	This holds the value of the identification and therefore is the essential element of IdentificationData.
authorized	None.	The "authorized" element can be used to directly authorize an identification value with a write command.

3180 *Table 77: identificationListData Element rules*

3181 Element "**identificationType**" value rules:

Value	Rule	Description
eui48	The identificationValue SHALL be interpreted as eui48 MAC address. The following pattern SHALL be used: (([A-F] [0-9]) {2} \-) {5} ([A-F] [0-9]) {2}	eui48 MAC address
eui64	The identificationValue SHALL be interpreted as eui64 MAC address. The following pattern SHALL be used: (([A-F] [0-9]) {2} \-) {7} ([A-F] [0-9]) {2}	eui64 MAC address
userRfidTag	None.	RFID tag used for identification.

3182 *Table 78: Element "identificationType" value rules*

3183

3184 **4.3.13.5 sessionIdentificationListData**

3185 **4.3.13.5.1 Description**

3186 The sessionIdentificationListData function is used to model information about so-called sessions
3187 (e.g., an EV's charging session or a washing machine's washing cycle). It can be related to measurands
3188 that are measured during a session, see section 4.3.13.6.

3189 A session may be a current one (it is still ongoing and has only a start time, but not an end time) or a
3190 past one (the session is completed and it has a start time as well as an end time).

3191 An appliance may provide only the latest session (which can be a current one or the latest past
3192 session) or store previous sessions as well.

3193

3194 **4.3.13.5.2 Rules**

3195 None.

3196

3197 **4.3.13.5.3 Element rules**

Element	Rule	Description
Identification. sessionIdentificationListData. sessionIdentificationData. (list)		
sessionId	SHALL be set as PRIMARY IDENTIFIER.	Unique identifier for each session.
identificationId	MAY be set as FOREIGN IDENTIFIER.	Reference to an appliance's identification information.
isLatestSession	If this session is the latest one, this Element SHALL be set to "true". Otherwise, it SHALL be set to "false".	States if this is the latest session. The latest session is either the current session, or (if no current session is available) the past session with the newest endTime.

timePeriod	For current sessions, the startTime SHALL be set. For past sessions, the startTime as well as the endTime SHALL be set.	Includes the startTime of the session and, if it is a past session, the endTime of the session.
------------	---	---

Table 79: sessionIdentificationListData Element rules

4.3.13.6 sessionMeasurementRelationListData

4.3.13.6.1 Description

The sessionMeasurementRelationListData function provides the relation between the session(s) and the measurands that are related to the respective session (e.g., periodically stored power values or the consumed energy during the session).

4.3.13.6.2 Rules

None.

4.3.13.6.3 Element rules

Element	Rule	Description
Identification. sessionMeasurementRelationListData. sessionMeasurementRelationData. (list)		
sessionId	SHALL be set as PRIMARY IDENTIFIER.	Unique identifier for each session.
measurementId (list)	SHALL be set as FOREIGN IDENTIFIER.	List of linked measurementIds for that session.

Table 80: sessionMeasurementRelationListData Element rules

4.3.14 IncentiveTable

4.3.14.1 Dependencies to other Feature Types

None.

4.3.14.2 Introduction

The IncentiveTable data model allows to define tiers with different incentives and boundaries. In case the incentive is price-based, it also could be called a pricing table, where a tier represents a kind of price level. These incentives may be used to motivate a building (via an energy management

3220 system) or an appliance to schedule its energy consumption (or production) towards a cost reduction
3221 or a reduced carbon footprint, e.g.

3222 Depending on the scope of a given IncentiveTable, different rules may apply for the relation or
3223 evaluation of the tiers. The following sub-sections give two brief examples.

3224 The IncentiveTable has three different functions:

- 3225 1. incentiveTableData: holds the actual values of the boundaries and incentives
- 3226 2. incentiveTableDescriptionData: holds data that describes the values of incentiveTable
- 3227 3. incentiveTableConstraintsData: allows a server to define certain constraints regarding the
3228 incentiveTableData, this is especially important if the IncentiveTable is writeable

3229

3230 4.3.14.2.1 Example with stacked tiers

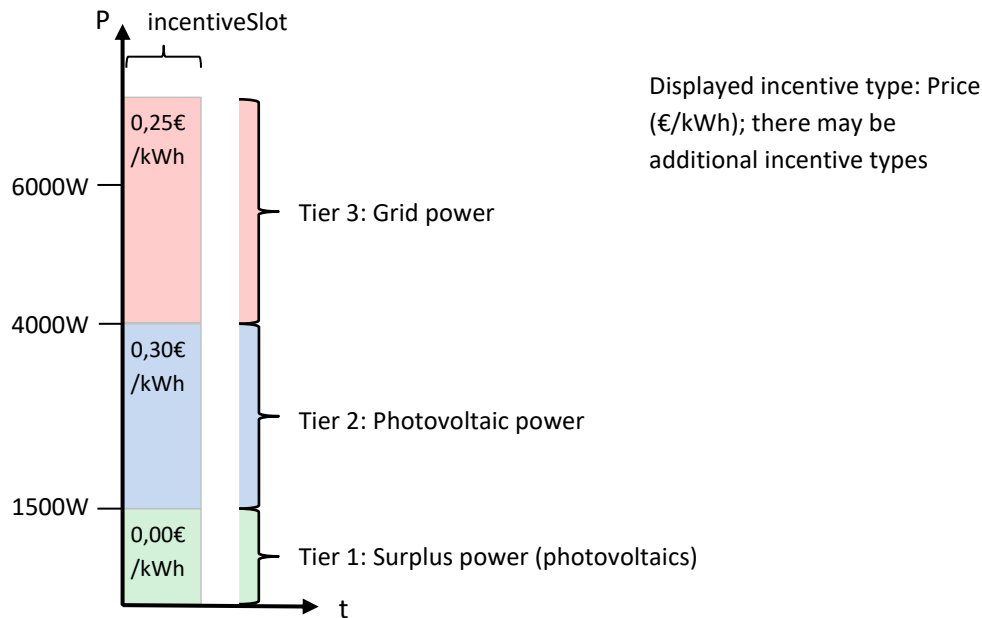
3231 This example discusses a typical application of an IncentiveTable towards appliances. Here, so-called
3232 "stacked tiers" (see section 4.3.14.4.2.1) are used to model the impact of a photovoltaic system on
3233 the total price properly.

3234 A photovoltaic system may sometimes produce more power than is consumed and fed into the
3235 electricity grid. This portion is called "surplus power" in Figure 39. This means this power is for free
3236 and can be offered by the energy management system to energy consuming appliances that have the
3237 flexibility to increase their consumption. If more than the surplus power is consumed the next tier in
3238 Figure 39 would be the one for the electricity grid feed-in power (i.e. not getting a revenue from
3239 feeding power into the grid is considered as cost in this example). If more power is consumed than
3240 the photovoltaic system produces, the tier for the electricity grid power consumption applies.

3241 Depending on the power consumed, up to three tiers contribute to the total price in this "stacked
3242 tiers" approach: Each tier's incentive(s) contribute according to their power share that is determined
3243 by the tier's (power) boundary, as shown in examples below Figure 39.

3244 Electricity grid power may also be cheaper than the grid feed-in power, e.g. in case the electricity grid
3245 has too much power available. The values in Figure 39 show such a situation for a given time period.

3246 As conditions change over time, incentives as well as boundaries may vary over time and be
 3247 expressed in respective time periods (each represented by an "incentiveSlot" in this data model).
 3248 Although in this example power was used, the IncentiveTable may also be used for other things.



3249
 3250 *Figure 39: IncentiveTable example (single incentiveSlot) for different stacked tiers*

3251 For the scenario described above some examples show how "stacked tiers" contribute to the price:

3252 Example 1: If the actual power value is 5000W:

- 3253 - 1500W are for free
- 3254 - 2500W cost 0,30€/kWh
- 3255 - and the last 1000W cost 0,25€/kWh.

3256 Example 2: If the actual power value is 1000W:

- 3257 - Only the lowest tier is used
- 3258 - the 1000W are for free.

3259 Example 3: If the actual power value is 2000W:

- 3260 - 1500W are for free
- 3261 - and the last 500W cost 0,30€/kWh.

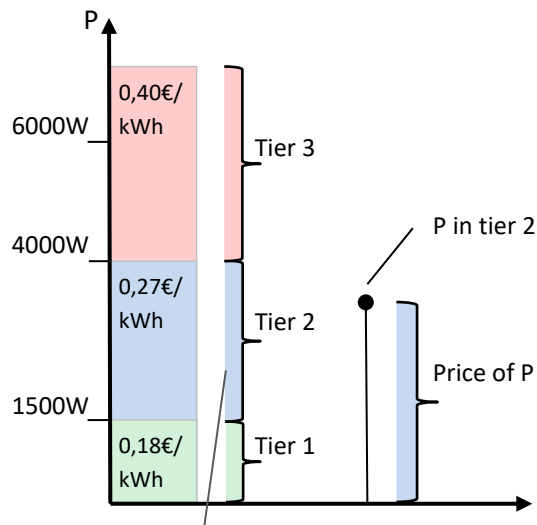
3262

3263 4.3.14.2.2 Example with unique tiers

3264 This example discusses a typical application of an IncentiveTable from the grid towards a building's
 3265 energy management system. Here, so-called "unique tiers" (see section 4.3.14.4.2.2) are used.

3266 An energy supplier may charge a building's power consumption dependent on the current power
 3267 consumption. Figure 40 shows an example where the energy supplier provides three tiers with

3268 different prices. In this example, the building's current power consumption P matches the power
 3269 boundary of tier 2. Hence, the power is charged with $(0,12\text{€}/\text{kWh}) \cdot P$.



Price: Selected by tier, but tier selected by P

3270

3271 *Figure 40: IncentiveTable example (single incentiveSlot) for different unique tiers*

3272

3273 4.3.14.3 Rules

3274 Each tier is uniquely defined by the value pair of "tariffId" and "tierId".

3275

3276 4.3.14.4 incentiveTableData

3277 4.3.14.4.1 Description

3278 The incentiveTable within the incentiveTableData function holds the values of the IncentiveTable.

3279 This includes the boundary values as well as the incentive values for each tier.

3280 The Function "incentiveTableData" models rather time dependent information of each tier within the
 3281 respective element "tier". However, the time dependency is not modelled within the "tier" elements
 3282 directly. Instead, each "tariff" is split into time slots which are expressed by the element
 3283 "incentiveSlot". This means each "incentiveSlot" entry finally contains time dependent data of one or
 3284 more "tier" entries.

3285

3286 4.3.14.4.2 Rules

3287 If the same tier is used within different "incentiveSlot" entries the same "tierId" value SHALL be used
 3288 in these "incentiveSlot" entries.

3289

3290 4.3.14.4.2.1 Stacked tiers

3291 If not stated otherwise by an application, tiers SHALL be interpreted as so-called "stacked tiers". The
 3292 following rules apply to stacked tiers:

Each tier defines its supply (difference between upperBoundaryValue and lowerBoundaryValue) that can be consumed or produced for the incentive(s) (e.g. price) given by the tier. If the boundaries of a tier are negative the supply can be produced. If the boundaries of a tier are positive the supply can be consumed. The supply of each tier can only be consumed or produced, if the supply of the tiers between this tier and zero are already consumed.

If (e.g.) the lowerBoundaryValue of tier x is set to 2 units and the upperBoundaryValue is 4 units, tier x can provide a supply of $4-2 = 2$ units. These 2 units can be consumed for the incentive(s) given by tier x. If the actual value of a device surpasses the upperBoundaryValue, the device consumes the whole 2 units of tier x for the corresponding incentive(s). If the actual value of a device is below the upperBoundaryValue but greater than or equal to lowerBoundaryValue (e.g. 3 units), the device consumes the difference between the actual value and the lowerBoundaryValue of tier x (e.g. $3-2 = 1$) for the incentive(s) given by the tier. As negative values are used for production, the meaning of the boundaries switch. For production the lowerBoundaryValue of a tier will have the higher distance to zero then the upperBoundaryValue. Therefore, the upperBoundaryValue for production has the same meaning as the lowerBoundaryValue for consumption and the lowerBoundaryValue for production has the same meaning as the upperBoundaryValue for consumption.

For more examples, please consider section 4.3.14.2.1.

3310

3311 4.3.14.4.2.2 Unique tiers

3312 If an application makes use of unique tiers, the following rules apply:

3313 The supply of only one unique tier SHALL be active each moment. The boundary
3314 (upperBoundaryValue and lowerBoundaryValue) of a tier denotes when the tier becomes active. The
3315 supply of the tier then holds for all consumed/produced units (i.e. not just those that belong to the
3316 boundary). If the boundaries of a tier are negative the supply can be produced. If the boundaries of a
3317 tier are positive the supply can be consumed.

3318 If (e.g.) the lowerBoundaryValue of tier x is set to 2 units and the upperBoundaryValue is 4 units, tier
3319 x can provide a supply of at least 2 and at maximum 4 units. These 2 to 4 units can be consumed for
3320 the incentive(s) given by tier x. Tier x applies if and only if the consumption of a device is below the
3321 upperBoundaryValue (4 in this example) but greater than or equal to lowerBoundaryValue (2 in this
3322 example). Then, the incentives of tier x contribute to the whole consumption of the device. If the
3323 actual consumption of a device is smaller than lowerBoundaryValue, tier x does not apply and
3324 instead a tier w with matching boundary would apply. If the actual value of a device equals or
3325 exceeds upperBoundaryValue, tier x does not apply and instead a tier y with matching boundary
3326 would apply.

3327 For another example, please consider section 4.3.14.2.2.

3328

3329 4.3.14.4.3 Element rules

Element	Rule	Description
IncentiveTable. incentiveTableData.		
incentiveTable (list)		

incentiveTable. tariff		
incentiveTable. tariff. tariffId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a incentiveTable. Allows linking of the different functions within the IncentiveTable Feature. Allows linking of the incentiveTable within other Features that are placed in the same entity.
incentiveTable. incentiveSlot (list)		
incentiveTable. incentiveSlot. timeInterval.	SHALL be set if an incentive has a time restriction or the incentiveTable contains more than one incentiveSlot entry. The timeInterval of different incentiveSlots within an incentiveTable SHALL NOT overlap in time.	Time period of the incentiveSlot.
incentiveTable. incentiveSlot. timeInterval. timeSlotId	MAY be used as SUB IDENTIFIER of tariffId to allow partial modification of the incentiveSlot list if slotIdSupport is set to "true" for this incentiveTable. SHALL NOT be used if the Element "slotIdSupport" within the Function "incentiveTableDescriptionData" is not set to "true". If used in a incentiveSlot it SHALL be used in all incentiveSlots of an incentiveTable. Please refer to section 3.4.2.6 for details.	Allows identification of an incentiveSlot within an incentiveTable.
incentiveTable. incentiveSlot. timeInterval. startTime.	SHALL be set.	
incentiveTable. incentiveSlot. timeInterval. startTime. dateTime		Absolute start time.
incentiveTable. incentiveSlot. timeInterval. startTime. relative		Relative start time containing a value relative to now. Positive values are in the future, negative ones are in the past and zero seconds represents "now".
incentiveTable. incentiveSlot. timeInterval. endTime.	SHALL be set for the last slot. SHALL be set if a time gap follows after this slot. MAY be set for all other slots.	
incentiveTable. incentiveSlot. timeInterval. endTime. dateTime		Absolute end time.

incentiveTable. incentiveSlot. timeInterval. endTime. relative		Relative end time containing a value relative to now. Positive values are in the future, negative ones are in the past and zero seconds represents "now".
incentiveTable. incentiveSlot. tier (list)		
incentiveTable. incentiveSlot. tier. tier		
incentiveTable. incentiveSlot. tier. tier. tierId	SUB IDENTIFIER of tariffId to identify a tier of a tariff. SHALL be set if the incentiveTable contains more than one tier. If the tierId is omitted, a default value of "1" SHALL be applied.	Allows identification of a tier within a incentiveTable.
incentiveTable. incentiveSlot. tier. boundary (list)	The boundary list defines the boundaries of a tier. The boundary range of different tiers within an incentiveSlot defined by lowerBoundaryValue and upperBoundaryValue SHALL NOT overlap for boundaries with the same boundaryType. The boundary range of a tier includes all values that are equal or greater than lowerBoundaryValue and lower than upperBoundaryValue.	
incentiveTable. incentiveSlot. tier. boundary. boundaryId	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.	Allows identification of a boundary within a tier.
incentiveTable. incentiveSlot. tier. boundary. lowerBoundaryValue	Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. Lower and upper boundaries of adjacent tiers: If lowerBoundaryValue is omitted and upperBoundaryValue is set, the lowerBoundaryValue SHALL be interpreted as if it is equal (without a gap) to the upperBoundaryValue of the next lower tier within the incentiveSlot. Gaps between boundaries of adjacent "tier" instances can only be modelled by setting the upperBoundaryValue of the lower "tier" AND the lowerBoundaryValue of the higher "tier" instance explicitly. Minus infinity lowerBoundary rule: If lowerBoundaryValue is omitted and there is no lower tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot,	This represents the lower end value range boundary.

	the omitted lowerBoundaryValue SHALL be interpreted as minus infinity.	
incentiveTable. incentiveSlot. tier. boundary. upperBoundaryValue	Within a boundary lowerBoundaryValue SHALL be smaller than upperBoundaryValue. Lower and upper boundaries of adjacent tiers: If upperBoundaryValue is omitted and lowerBoundaryValue is set, the upperBoundaryValue SHALL be interpreted as equal (without any gap) to the lowerBoundaryValue of the next higher tier within the incentiveSlot. Gaps between boundaries of adjacent "tier" instances can only be modelled by setting the upperBoundaryValue of the lower "tier" AND the lowerBoundaryValue of the higher "tier" instance explicitly. Infinity upperboundary rule: If upperBoundaryValue is omitted and there is no higher tier with a lowerBoundaryValue or upperBoundaryValue within the incentiveSlot, the omitted upperBoundaryValue SHALL be interpreted as infinity.	This represents the upper end value range boundary.
incentiveTable. incentiveSlot. tier. incentive (list)		
incentiveTable. incentiveSlot. tier. incentive. incentiveId	SHALL be used as SUB IDENTIFIER of tierId. SHALL be set if the tier contains more than one incentive entry. If the incentiveId is omitted, a default value of "1" SHALL be applied.	Allows identification of an incentive within a tier.
incentiveTable. incentiveSlot. tier. incentive. value		The value of the incentive.

Table 81: incentiveTableData Element rules

4.3.14.5 incentiveTableDescriptionData

4.3.14.5.1 Description

The incentiveTableDescription within the incentiveTableDescriptionData function holds the rather static description of the IncentiveTable. This includes the tier, boundary and incentive type as well as further elements.

4.3.14.5.2 Rules

None.

3341 4.3.14.5.3 Element rules

Element	Rule	Description
IncentiveTable. incentiveTableDescriptionData.		
incentiveTableDescription (list)		
incentiveTableDescription. tariffDescription.		
incentiveTableDescription. tariffDescription. tariffId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a incentive table. Allows linking of the data in the different functions of IncentiveTable. Allows linking of the incentive tables within other features that are placed in the same entity.
incentiveTableDescription. tariffDescription. measurementId	SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId	If a measurand is linked to the incentive table, its identifier is denoted here.
incentiveTableDescription. tariffDescription. tariffWriteable	SHALL be set to "true" if the incentive table is writeable by a client. Otherwise it may be omitted.	Whether the incentive table is writeable by a client or not can be denoted in this element.
incentiveTableDescription. tariffDescription. updateRequired	If updateRequired is "true", the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. When the client updates the server, the client SHALL ignore updateRequired until the writeable and changeable data related to the same PRIMARY IDENTIFIER was updated successfully. The server SHALL set updateRequired back to "false", as soon as the update of the writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. The server SHALL ensure that only one responsible client is able to update the related data. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".	Allows to request an update from the bound client.

incentiveTableDescription. tariffDescription. scopeType		A scopeType for the incentive table can be stated here.
incentiveTableDescription. tariffDescription. label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label on the incentive table.
incentiveTableDescription. tariffDescription. description	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for the incentive table.
incentiveTableDescription. tariffDescription. slotIdSupport	If omitted, a default of "false" SHALL be applied. If set to "false" no incentiveSlot SHALL contain a timeSlotId. If set to "true" timeSlotId MAY be used within the incentiveSlot as described for the element timeSlotId within the function incentiveTableData.	Indicates support of timeSlotId within the corresponding tariff in the Function incentiveTableData.
incentiveTableDescription. tier (list)		Allows to describe tiers and the incentives and boundaries that are used by the tier. Please refer to the detailed description of <i>IncentiveTableDescriptionTierType elements</i> . See Table 161.
incentiveTableDescription. tier. tierDescription.		
incentiveTableDescription. tier. tierDescription. tierId	SHALL be set as SUB IDENTIFIER of tariffId, if the incentiveTable contains more than one tier. If the tierId is omitted, a default value of "1" SHALL be applied.	Allows identification of a tier within a incentiveTable.
incentiveTableDescription. tier. tierDescription. tierType	Value rules of Table 83 SHALL be applied.	Allows to define the type of the tier.
incentiveTableDescription. tier. tierDescription. label	Default rules of section 3.10.1.2.1 SHALL be applied.	Allows to define user friendly name for the tier.
incentiveTableDescription. tier. tierDescription. description	Default rules of section 3.10.1.3.1 SHALL be applied.	Allows to define a user friendly description for the tier.
incentiveTableDescription. tier. boundaryDescription (list)		
incentiveTableDescription. tier. boundaryDescription. boundaryId	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one boundary entry. If the boundaryId is omitted, a default value of "1" SHALL be applied.	Allows identification of a boundary within a incentiveTable.
incentiveTableDescription. tier. boundaryDescription. boundaryType	Value rules of Table 84 SHALL be applied. There SHALL NOT be multiple boundaries with the same boundaryType for the same tier.	Allows to define the type of the boundary.

incentiveTableDescription. tier. boundaryDescription. switchToTierIdWhenLower	FOREIGN IDENTIFIER used to refer to the tier that is going to be active as soon as the lowerBoundaryValue is underrun.	Link to the tier that will become active if the lowerBoundaryValue is underrun.
incentiveTableDescription. tier. boundaryDescription. switchToTierIdWhenHigher	FOREIGN IDENTIFIER used to refer to the tier that is going to be active as soon as the upperBoundaryValue is exceeded.	Link to the tier that will become active if the upperBoundaryValue is exceeded.
incentiveTableDescription. tier. boundaryDescription. boundaryUnit	If set, the unit SHALL be applied to the lowerBoundaryValue and upperBoundaryValue.	Allows to define the unit of the boundary values.
incentiveTableDescription. tier. incentiveDescription (list)		
incentiveTableDescription. tier. incentiveDescription. incentiveId	SHALL be set as SUB IDENTIFIER of tierId, if the tier contains more than one incentive entry. If the incentiveId is omitted, a default value of "1" SHALL be applied.	Allows identification of an incentive within a incentiveTable.
incentiveTableDescription. tier. incentiveDescription. incentiveType	Value rules of Table 85 SHALL be applied.	Allows to define the type of the incentive.
incentiveTableDescription. tier. incentiveDescription. incentivePriority	Values SHALL be greater than zero. A value of "1" SHALL be interpreted as highest priority. The priority decreases with increasing value of incentivePriority.	Allows to define the priority of the incentive. E.g. the priority that the customer has assigned for this incentive.
incentiveTableDescription. tier. incentiveDescription. currency	If set, the currency SHALL be applied to the value of the incentive	Monetary incentives may need to describe a currency.
incentiveTableDescription. tier. incentiveDescription. unit	If set, the unit SHALL be applied to the value of the incentive	Allows definition of a unit, if the incentive has a unit.
incentiveTableDescription. tier. incentiveDescription. label	Default rules of section 3.10.1.2.1 SHALL be applied.	Allows to express a user friendly label for the incentive.
incentiveTableDescription. tier. incentiveDescription. description	Default rules of section 3.10.1.3.1 SHALL be applied.	Allows to express a user friendly description for the incentive.

3342 *Table 82: incentiveTableDescriptionData Element rules*

3343 Element "**incentiveTableDescription. tier. tierDescription. tierType**" value rules:

Value	Rule
fixedCost	The tier has a cost incentive that SHALL NOT vary over time
dynamicCost	The tier has a cost incentive that MAY vary over time

3344 *Table 83: Element "tier. tierDescription. tierType" value rules*

3345 Element "**incentiveTableDescription. tier. boundaryDescription. boundaryType**" value rules:

Value	Rule
powerBoundary	The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used. Positive powerBoundary values describe which power can be consumed with certain incentives, while negative powerBoundary values describe which power can be produced with certain incentives.
energyBoundary	The boundary SHALL be an electrical energy. E.g. if the stored self produced energy in a battery is consumed, more expensive energy has to be used. Positive energyBoundary values describe which energy can be consumed with certain incentives, while negative energyBoundary values describe which energy can be produced with certain incentives.
countBoundary	The boundary SHALL be a count. E.g. after a certain count the price of a good may be reduced.

3346 Table 84: Element "**tier. boundaryDescription. boundaryType**" value rules

3347 Element "**incentiveTableDescription. tier. incentiveDescription. incentiveType**" value rules:

Value	Rule	Description
absoluteCost	Each related value SHALL be an absolute cost with a specific currency. Positive values SHALL be interpreted as costs. Negative values SHALL be interpreted as profits. If also a unit is set, the cost SHALL contain the cost per unit. E.g. EUR/Wh.	Absolute costs with a currency and a unit.
relativeCost	The values SHALL have no currency and unit SHALL be set to percentage (pct). The value SHALL be in the range from 0% to 100%.	Relative costs that cannot be interpreted as some monetary costs.
renewableEnergyPercentage	The unit of the incentive SHALL be set to percentage (pct) in this case. The value SHALL be in the range from 0% to 100%.	Percentage of renewable energy of the total obtained energy. This is an ecological incentive that allows optimizing consumption of renewable energy instead of costs.
co2Emission	The unit of the incentive SHALL be set to "kg/Wh". Only positive values SHALL be used.	CO2 emission related to the obtained energy. This is an ecological incentive that allows optimizing the CO2 emission instead of costs.

3348 Table 85: Element "**tier. incentiveDescription. incentiveType**" value rules

3349

3350 **4.3.14.6 incentiveTableConstraintsData**3351 **4.3.14.6.1 Description**

3352 The incentiveTableConstraints within the incentiveTableConstraintsData function holds additional
 3353 constraints of the IncentiveTable, regarding how much tiers, boundaries, incentives and slots are
 3354 allowed.

3355

3356 **4.3.14.6.2 Rules**

3357 None.

3358

3359 **4.3.14.6.3 Element rules**

Element	Rule	Description
IncentiveTable. incentiveTableConstraintsData.		
incentiveTableConstraints (list)		
incentiveTableConstraints. tariff		
incentiveTableConstraints. tariff. tariffId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a incentiveTable. Allows linking of the different functions within the IncentiveTable Feature. Allows linking of the incentiveTable within other Features that are placed in the same entity.
incentiveTableConstraints. tariffConstraints		
incentiveTableConstraints. tariffConstraints. maxTiersPerTariff	If set the incentiveTable SHALL NOT include more tiers than given in maxTiersPerTariff.	Allows the server to define a maximum tier count for this IncentiveTable. This is especially important if a client wants to write an IncentiveTable.
incentiveTableConstraints. tariffConstraints. maxBoundariesPerTier	If set, the related "incentiveTableData. incentiveTable. incentiveSlot. tier" instance SHALL NOT include more "boundary" instances than given in maxBoundariesPerTier.	Allows the server to define a maximum count of "boundary" instances for tiers. This is especially important if a client wants to write an IncentiveTable.
incentiveTableConstraints. tariffConstraints. maxIncentivesPerTier	If set the tier within the incentiveTable SHALL NOT include more incentives than given in maxIncentivesPerTier.	Allows the server to define a maximum incentive count for tiers. This is especially important if a client wants to write an IncentiveTable.

incentiveTableConstraints. incentiveSlotConstraints		
incentiveTableConstraints. incentiveSlotConstraints. slotCountMax	If set the incentiveTable SHALL NOT include more slots than given in slotCountMax.	Allows the server to define a maximum slot count for this IncentiveTable. This is especially important if a client wants to write an IncentiveTable.

Table 86: incentiveTableConstraintsData Element rules

4.3.14.7 Descriptive information and further definitions

None.

4.3.15 LoadControl**4.3.15.1 Dependencies to other Feature Types****4.3.15.1.1 measurementId**

LoadControl can use the FOREIGN IDENTIFIER measurementId from the Measurement Feature Type to define a relation to functions of an ElectricalConnection or Measurement Feature.

If a measurementId is given within a LoadControlLimitDescriptionData entry the following holds valid:

- If there is an electricalConnectionPermittedValueSetData entry that relates to the same measurementId, then the Element "value" of the according loadControlLimitData entry SHOULD only contain values that are defined within the permittedValueSet of the according electricalConnectionPermittedValueSetData entry.
- If there is an electricalConnectionParameterDescriptionData entry that relates to the same measurementId, then the elements of the related electricalConnectionParameterDescriptionData entry SHOULD be considered for the according loadControlLimitData entry.
- If there is a measurementDescriptionData entry that relates to the same measurementId, then the Element "unit" of this measurementDescriptionData entry SHALL be ignored for the according loadControlLimitData entry.

4.3.15.2 Introduction

The LoadControl class has two different approaches to control the load of an appliance:

- Event-based load control with "loadControlEvent" (see section 5.3.14.3) and "loadControlState" (see section 5.3.14.4)
- Limit-based load control with "loadControlLimit" (see section 5.3.14.5) and "loadControlLimitDescription" (see section 5.3.14.7)

3390 The event-based load control is reserved for future use.

3391 The limit-based load control is used where limits (changeable by a client) shall affect the behaviour of
3392 a node automatically. E.g. the power demand of an appliance could be limited to a certain value to
3393 protect the household of an overload.

3394 Additionally, with the information given in loadControlNode (see section 5.3.14.2), a node can define
3395 whether it is (at the moment) remote controllable or not.

3396

3397 **4.3.15.3 Rules**

3398 None.

3399

3400 **4.3.15.4 loadControlNodeData**

3401 *4.3.15.4.1 Description*

3402 The loadControlNodeData includes node-wide information related to load control (e.g. whether the
3403 node is remote controllable).

3404

3405 *4.3.15.4.2 Rules*

3406 If the function is absent, a default of "true" SHALL be applied for the element
3407 *isNodeRemoteControllable*.

3408

3409 *4.3.15.4.3 Element rules*

Element	Rule	Description
LoadControl. loadControlNodeData.		
isNodeRemoteControllable	SHOULD be set. If set to "false", write operations on this feature SHALL be declined. If absent, a default value of "true" SHALL be applied.	Defines whether the node is (at the moment) remote controllable by a client, or not.

3410 *Table 87: loadControlNodeData Element rules*

3411

3412 **4.3.15.5 loadControlEventsListData**

3413 Reserved for future use.

3414

3415 **4.3.15.6 loadControlStateListData**

3416 Reserved for future use.

3417

4.3.15.7 loadControlLimitListData**4.3.15.7.1 Description**

The loadControlLimitListData function provides the value of the limit (and an optional time period for that limit) and further information whether the limit is active or not and if it is changeable by a client or not.

4.3.15.7.2 Rules

None.

4.3.15.7.3 Element rules

Element	Rule	Description
LoadControl. loadControlLimitListData. loadControlLimitData. (list)		
limitId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the limit.
isLimitChangeable	SHOULD be set. If set to "false", the timePeriod, value and isLimitActive element SHALL NOT be writeable by a client. If omitted or set to "true", the timePeriod, value and isLimitActive element SHALL be writeable by appropriate clients (e.g. the bound client).	States whether the limit may be changed by a client or not.
isLimitActive	SHOULD be set. If set to "false", the limit and its timePeriod and value element SHALL be ignored. If set to "true" or omitted, the timePeriod and value element SHALL be applied, at least if timePeriod or value are set.	Indicates whether the limit is currently active or not.
timePeriod	MAY be used to define a timePeriod for the value. In this case the value SHALL only be applied within the given timePeriod.	The period where the limit shall be active.
value	If <i>isLimitActive</i> is set to "true", the <i>value</i> SHALL be set. Otherwise the element MAY be omitted. If <i>isLimitActive</i> is set to "false", but <i>value</i> is set, the content of <i>value</i> SHALL be ignored. If in the related loadControlLimitDescriptionData entry (with the same PRIMARY IDENTIFIER limitId) measurementId is set, the rules of section 4.3.15.1 SHALL be applied.	The actual limit.

Table 88: loadControlLimitListData Element rules

4.3.15.8 loadControlLimitConstraintsListData**4.3.15.8.1 Description**

Constraints that must be kept when trying to change a limit, are modelled with the loadControlLimitConstraintsListData function, including a minimum value, a maximum value and the stepsize of the values between.

3436 4.3.15.8.2 Rules

3437 None.

3438

3439 4.3.15.8.3 Element rules

Element	Rule	Description
LoadControl. loadControlLimitConstraintsListData. loadControlLimitConstraintsData. (list)		
limitId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the limit.
valueRangeMin	SHOULD be set, if not already defined by other functionality (e.g. <i>electricalConnectionPermittedValueSetListData</i>). If set, the content of element <i>value</i> within <i>loadControlLimitListData</i> SHALL NOT contain a value below <i>valueRangeMin</i> .	This element allows to define a minimum limit for the element value within <i>loadControlLimitListData</i> .
valueRangeMax	SHOULD be set, if not already defined by other functionality (e.g. <i>electricalConnectionPermittedValueSetListData</i>). If set, the content of element <i>value</i> within <i>loadControlLimitListData</i> SHALL NOT contain a value above <i>valueRangeMax</i> .	This element allows to define a maximum limit for the element value within <i>loadControlLimitListData</i> .
valueStepSize	SHOULD be set, if there is a step size limitation.	The minimum step size between two different limit values.

3440 Table 89: loadControlLimitConstraintsListData Element rules

3441

3442 4.3.15.9 loadControlLimitDescriptionListData

3443 4.3.15.9.1 Description

3444 The rather static information about the available limit(s) are modelled within the
3445 loadControlLimitDescriptionListData function.

3446

3447 4.3.15.9.2 Rules

3448 None.

3449

3450 4.3.15.9.3 Element rules

Name	Rule	Description
loadControlLimitDescriptionListData. loadControlLimitDescriptionData. (list)		
limitId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the limit.
limitType	SHALL be set. Value rules of Table 91 SHALL be applied.	Whether the limit shall be used as a minimum or maximum is denoted within this element.
limitCategory	SHOULD be set. Value rules of Table 92 SHALL be applied.	There are different levels of how important the limit is (see enumeration description).
limitDirection	SHOULD be set.	If a device supports both, consumption and production, the

	If omitted, the limitDirection SHALL be interpreted as "consume". Default rules of section 3.10.1.14.1 SHALL be applied.	direction which will be limited is stated in this element (e.g. if only the consumption of a battery (i.e. it is recharging) will be limited, the enumeration "consume" will be set; if both, consumption and production will be limited, two limits have to be set).
measurementId	SHALL be set as FOREIGN IDENTIFIER, if a measurand or other feature is linked with the measurementId. If set, the rules of section 4.3.15.1 SHALL be applied.	This is a foreign identifier.
unit	SHOULD be set, if there could be ambiguity how to interpret the value without the unit.	The unit, which is used for the limit value, is denoted with this element.
scopeType	None.	A certain meaning of the limit.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A label for the limit.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	A description for the limit.

3451 Table 90: loadControlLimitDescriptionListData Element rules

3452 Element "limitType" value rules:

Value	Rule	Description
minValueLimit	The limit value SHALL be interpreted as a minimum limit. The corresponding value SHALL stay over or equal to the given value of a minValueLimit, if isLimitActive = true and limitType = obligation.	The limit set in the "value" Element of the according "loadControlLimitData" entry shouldn't be underrun (value $\geq$ minValueLimit).
maxValueLimit	The limit value SHALL be interpreted as a maximum limit. The corresponding value SHALL stay under or equal to the given value of a maxValueLimit, if isLimitActive = true and limitType = obligation.	The limit set in the "value" Element of the according "loadControlLimitData" entry shouldn't be overrun (value $\leq$ maxValueLimit).
signDependentAbsValueLimit	The limitType SHALL be interpreted in the following way (following the passive sign convention; if the active sign convention is applied in the Use Case, the logic SHALL be vice versa): - A positive value only limits the consumption. - A negative value only limits the absolute value of the production.	Depending on the applied sign convention and the sign of the limit set in the "value" Element of the according "loadControlLimitData" entry the consumption or production is limited.

3453 Table 91: Element "limitType" value rules

3454 Element "**limitCategory**" value rules:

Value	Rule	Description
obligation	If this category is used, the limit SHALL be kept under any circumstances. A loss of comfort MAY be risked for as long as needed.	Used for limits that are absolutely important to be hold. A loss of comfort could happen and will be accepted. The normal operation will be interrupted for keeping the limit (if needed).
recommendation	If this category is used, the limit SHOULD be kept wherever possible. A loss of comfort MAY be taken into account for a short period of time.	Used for limits that are quite important for the energy management. A loss of comfort could happen and will be accepted for a short period of time. The normal operation should not be interrupted.
optimization	If this category is used, the limit MAY be considered as good as possible. A loss of comfort SHOULD NOT be accepted.	Used for limits that just optimize the energy management. No loss of comfort will be accepted. The normal operation shall not be interrupted.

3455 *Table 92: Element "limitCategory" value rules*

3456

3457 **4.3.15.10 Descriptive information and further definitions**

3458 None.

3459

3460 **4.3.16 Measurement**

3461 **4.3.16.1 Dependencies to other Feature Types**

3462 - ElectricalConnection

3463 - Threshold

3464

3465 **4.3.16.2 Introduction**

3466 The SPINE Measurement class provides functionality for transmitting measured, calculated or
 3467 empirically obtained values from sensors. Mainly these are sensors that provide some kind of
 3468 information about the environment or physical parameters. Clients can use the information for
 3469 further actions or just visualize them for the information of a user.

3470 Due to the nature of measurement, the measured values are only readable. Some control systems
 3471 permit a user to select a preferred value (e.g. a desired room temperature). Such "desired" values
 3472 obviously require writable fields. This can be achieved with the "Setpoint" class (see section 5.3.21).

3473

3474 **4.3.16.3 Rules**

3475 None.

3476

3477 **4.3.16.4 measurementListData**3478 **4.3.16.4.1 Description**

3479 The *measurementListData* command is used to communicate measurement values. Typically, a
 3480 sensor sends this command as notification as soon as it changes more than a specific threshold. Of
 3481 course, it may be readable, too.

3482

3483 **4.3.16.4.2 Rules**

3484 This Function SHALL only be used for the current (last) value, not for historical ones (see section
 3485 4.3.16.5 for the Function *measurementSeriesListData* that may be used for that).

3486

3487 **4.3.16.4.3 Element rules**

Element	Rule	Description
Measurement. measurementListData. measurementData. (list)		
measurementId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different sensors on one SPINE feature.
valueType	SHOULD be set as SUB IDENTIFIER. Value rules of Table 94 SHALL be applied.	It is possible to model different types of measurement values. For example "value", "averageValue", "minValue", etc. (see the SPINE data model for all possible value types). There can be multiple measurementData entries with different valueTypes for the same measurementId. This allows to express different facets of the measurand.
timestamp	MAY be set as SUB IDENTIFIER. If set, the timestamp of the creation of the message SHOULD be used.	The time of creation of a measurement value element (i.e. the time it was measured, calculated, ...). Some sensors and communications technologies may already provide this information natively. In other cases the technology interface may set timestamp to the time of creation of a measurementData instance.
value	SHALL be set, if a value is available. Otherwise the whole list entry SHALL be omitted or the element <i>valueState</i> SHALL be set to "error". If <i>valueState</i> is set to "error", but <i>value</i> is set, the content of <i>value</i> SHALL be ignored. If <i>valueState</i> is set to "outOfRange" and <i>measurementConstraintsListData.valueRangeMax</i> is set and <i>value</i> is equal to or greater than <i>valueRangeMax</i> , <i>value</i> SHALL be interpreted as greater than <i>valueRangeMax</i> . If <i>valueState</i> is set to	The measurement value itself, i.e. the magnitude according to "valueType".

	"outOfRange" and <i>measurementConstraintsListData.valueRangeMin</i> is set and <i>value</i> is equal to or smaller than <i>valueRangeMin</i> , <i>value</i> SHALL be interpreted as lower than <i>valueRangeMin</i> . If "value" is set, <i>measurementDescriptionListData.measurementType</i> SHALL be set, too.	
evaluationPeriod	None.	Some <i>valueType</i> values can be detailed or even require information about the time period. For example, an <i>averageValue</i> is usually computed from measured values of a finite time period. For smart devices, the start and end time of the <i>evaluationPeriod</i> can be filled in directly. For bare ("simple") sensors, such "derived"/"accompanying" values (average, min, max, ...) could be calculated (if desired) by the technology interface component together with the corresponding <i>evaluationPeriod</i> .
valueSource	Value rules of Table 95 SHALL be applied.	Whether element "value" is measured, calculated or empirically obtained can be denoted with this element.
valueTendency	Value rules of Table 96 SHALL be applied.	This elements states whether the tendency is <i>rising</i> , <i>stable</i> or <i>falling</i> .
valueState	Value rules of Table 97 SHALL be applied.	A measurand may have a state.

3488 Table 93: *measurementListData* Element rules

3489 Element "**valueType**" value rules:

Value	Rule	Description
value	None.	The normal measurement value.
averageValue	None.	The average measurement value over the last time, stated in the element <i>evaluationPeriod</i> .
minValue	None.	The minimum measurement value of the last time, stated in the element <i>evaluationPeriod</i> .
maxValue	None.	The maximum measurement value of the last time, stated in the element <i>evaluationPeriod</i> .
standardDeviation	None.	Used to quantify the amount of variation or dispersion of a set of values.

3490 Table 94: Element "*valueType*" value rules

3491 Element "**valueSource**" value rules:

Value	Rule	Description
measuredValue	None.	The measurement value was measured with a dedicated sensor. The value is rather exact.

calculatedValue	None.	The measurement value was calculated with the values from some other parameters (e.g. water temperature, outside temperature, etc.) that have a calculable impact on the measurand. This value is not as exact as a measured value, but it differs not too much from the real value.
empiricalValue	None.	The measurement value is only an empirical one that can differ a lot from the real value. It can rely on some other parameters that have a known value, but there is no exactly calculable relation between them.

3492 Table 95: Element "valueSource" value rules

3493 Element "**valueTendency**" value rules:

Value	Rule	Description
rising	None.	The average measurement value is currently rising.
stable	None.	The average measurement value is currently stable.
falling	None.	The average measurement value is currently falling.

3494 Table 96: Element "valueTendency" value rules

3495 Element "**valueState**" value rules:

Value	Rule	Description
normal	None.	The measurement value could be measured / calculated / empirically determined as expected.
outOfRange	SHALL be used if the value is out of range	The measurement value is currently out of range of the sensor (as defined by "valueRangeMin" and "valueRangeMax", see section 5.3.15.4.2).
error	SHALL be used if the value has some error.	An error occurred while determining the measurement value.

3496 Table 97: Element "valueState" value rules

3497

3498 **4.3.16.5 measurementSeriesListData**3499 **4.3.16.5.1 Description**

3500 The *measurementSeriesListData* Function is similar to the *measurementListData* Function (see
 3501 section 4.3.16.4). It is used for historical data of measurands.

3502

3503 **4.3.16.5.2 Rules**

3504 If both, *measurementListData* and *measurementSeriesListData* are supported on one Measurement
 3505 Feature instance, and both are delivering data for the same measurand (*measurementListData* for
 3506 the current value and *measurementSeriesListData* for the historical data), they SHALL use the same
 3507 measurementId for all data that belong to this measurand.

3508

3509 **4.3.16.5.3 Element rules**

Element	Rule	Description
Measurement. measurementSeriesListData. measurementSeriesData. (list)		

measurementId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different measurands on one SPINE feature.
valueType	SHOULD be set as SUB IDENTIFIER. Value rules of Table 94 SHALL be applied.	It is possible to model different types of measurement values. For example "value", "averageValue", "minValue", etc. (see the SPINE data model for all possible value types). There can be multiple measurementData entries with different valueTypes for the same measurementId. This allows to express different facets of the measurand.
timestamp	SHALL be set as SUB IDENTIFIER. If set, the timestamp of the creation of the message SHOULD be used.	The time of creation of a measurement value element (i.e. the time it was measured, calculated, ...). Some sensors and communications technologies may already provide this information natively. In other cases the technology interface may set timestamp to the time of creation of a measurementData instance. There can be multiple measurementSeriesData entries for the same measurementId, but with different timestamps. This allows to express historical values or a measurement timeline.
value	SHALL be set, if a value is available. Otherwise the whole list entry SHALL be omitted or the element <i>valueState</i> SHALL be set to "error". If <i>valueState</i> is set to "error", but <i>value</i> is set, the content of <i>value</i> SHALL be ignored. If <i>valueState</i> is set to "outOfRange" and <i>measurementConstraintsListData.valueRangeMax</i> is set and <i>value</i> is equal to or greater than <i>valueRangeMax</i> , <i>value</i> SHALL be interpreted as greater than <i>valueRangeMax</i> . If <i>valueState</i> is set to "outOfRange" and <i>measurementConstraintsListData.valueRangeMin</i> is set and <i>value</i> is equal to or smaller than <i>valueRangeMin</i> , <i>value</i> SHALL be interpreted as lower than <i>valueRangeMin</i> . If "value" is set, <i>measurementDescriptionListData.measurementType</i> SHALL be set, too.	The measurement value itself, i.e. the magnitude according to "valueType".
evaluationPeriod	None.	Some <i>valueType</i> values can be detailed or even require information about the time period. For example,

		an <i>averageValue</i> is usually computed from measured values of a finite time period. For smart devices, the start and end time of the <i>evaluationPeriod</i> can be filled in directly. For bare ("simple") sensors, such "derived"/"accompanying" values (average, min, max, ...) could be calculated (if desired) by the technology interface component together with the corresponding <i>evaluationPeriod</i> .
valueSource	Value rules of Table 95 SHALL be applied.	Whether element "value" is measured, calculated or empirically obtained can be denoted with this element.
valueTendency	Value rules of Table 96 SHALL be applied.	This elements states whether the tendency is <i>rising</i> , <i>stable</i> or <i>falling</i> .
valueState	Value rules of Table 97 SHALL be applied.	A measurand may have a state.

Table 98: measurementSeriesListData Element rules

4.3.16.6 measurementConstraintsListData

4.3.16.6.1 Description

Some constraints concerning the possible values of a measurand are modelled within this function.

The constraints apply to values of "valueType" = "value". This implies that they hold valid for "minValue" and "maxValue", too. For "averageValue" only the Elements "valueRangeMin" and "valueRangeMax" apply. For "valueType" = "standardDeviation", the constraints are not valid.

4.3.16.6.2 Rules

None.

4.3.16.6.3 Element rules

Element	Rule	Description
Measurement. measurementConstraintsListData. measurementConstraintsData. (list)		
measurementId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different sensors on one SPINE feature.
valueRangeMin	MAY be used to express the minimal value that can be measured.	The minimum possible measurement value measurable by the feature.
valueRangeMax	MAY be used to express the maximum value that can be measured.	The maximum possible measurement value measurable by the feature.
valueStepSize	MAY be used if values can only be delivered in a certain stepsize.	The minimum step size between two different measurement values.

Table 99: measurementConstraintsListData Element rules

3524

3525 **4.3.16.7 measurementDescriptionListData**3526 **4.3.16.7.1 Description**

3527 Descriptive information on a specific measurement type is modelled with a
 3528 *measurementDescriptionListData* function. This information is rather constant in comparison to
 3529 measurement values. Typically, it is requested once at the beginning, when a node (an external
 3530 display, e.g.) is configured to present or use a specific measurement value. As usual, for such
 3531 scenarios it is recommended the the server notifies changes to all relevant communication partners,
 3532 because a server cannot expect clients to periodically request this information.

3533

3534 **4.3.16.7.2 Rules**

3535 None.

3536

3537 **4.3.16.7.3 Element rules**

Element	Rule	Description
Measurement. measurementDescriptionListData. measurementDescriptionData. (list)		
measurementId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different measurands on one SPINE feature.
measurementType	SHALL be set, if a measurement value is available in <i>measurementListData</i> .	To identify which type of measurand is measured (for example "temperature").
commodityType	SHOULD be set if a commodity is measured.	If a measurand of a commodity is measured, the type of commodity is stated here.
unit	SHALL be set, if there could be ambiguity how to interpret the value without the unit.	The unit, which is used for the value. It is always related to the normal value. Other <i>valueType</i> values (e.g. <i>averageDerivativeValue</i>) can have different units (derived from <i>unit</i>).
calibrationValue	None.	Some measurement sensors need a calibration value (e.g. set during production phase). It is automatically added to the internal measured value. I.e. the value stated in the <i>value</i> element in the <i>measurementListData</i> function is the already corrected value.
scopeType	None.	Specifies a more detailed meaning of the measurand (e.g. outsideTemperature).
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A (short) label can be useful to identify or group specific sensors.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	A (longer) description of a measurement device can be deposited here.

3538 Table 100: measurementDescriptionListData Element rules

3539 Element "**measurementType**" value rules:

Value	Rule	Description
acceleration	None.	The acceleration is measured.
angle	None.	An angle is measured.
angularVelocity	None.	The angular velocity is measured.
area	None.	An area is measured.
atmosphericPressure	None.	The atmospheric pressure is measured.
capacity	None.	The capacity is measured.
concentration	None.	A concentration of a substance is measured.
count	None.	A count is measured.
current	None.	The current is measured.
density	None.	The density is measured.
distance	None.	A distance is measured.
electricField	None.	The electric field is measured.
energy	None.	The energy is measured.
force	None.	The force is measured.
frequency	None.	The frequency is measured.
harmonicDistortion	None.	The harmonic distortion is measured.
heat	None.	The heat is measured.
heatFlux	None.	The heat flux is measured.
illuminance	None.	The illuminance is measured.
impulse	None.	An impulse is measured.
level	None.	A level is measured.
magneticField	None.	The magnetic field is measured.
mass	None.	A mass is measured.
massFlow	None.	The mass flow is measured.
particles	None.	The quantity of particles is measured.
percentage	None.	The percentage is measured.
power	None.	The power is measured.
powerFactor	None.	The power factor is measured.
pressure	None.	The pressure is measured.
radonActivity	None.	The radon activity is measured.
relativeHumidity	None.	The relative humidity is measured.
resistance	None.	The resistance is measured.
solarRadiation	None.	The solar radiation is measured.
speed	None.	The speed is measured.
temperature	None.	The temperature is measured.
time	None.	The time is measured.
torque	None.	The torque is measured.
unknown	None.	An unknown measurand is measured.
velocity	None.	The velocity is measured.
voltage	None.	The voltage is measured.
volume	None.	The volume is measured.
volumetricFlow	None.	The volumetric flow is measured.

Table 101: Element "measurementType" value rules

4.3.16.8 measurementThresholdRelationListData**4.3.16.8.1 Description**

The relation between a measurand and dedicated thresholds (see section 5.3.27) is modelled with this function.

4.3.16.8.2 Rules

None.

4.3.16.8.3 Element rules

Element	Rule	Description
Measurement. measurementThresholdRelationListData. measurementThresholdRelationData. (list)		
measurementId	SHALL be set as PRIMARY IDENTIFIER.	Enables the identification of different sensors on one SPINE address.
thresholdId (list)	SHALL be set as FOREIGN IDENTIFIER, if a threshold or other feature is linked with the thresholdId.	Related threshold identifier(s).

Table 102: measurementThresholdRelationListData Element rules

4.3.16.9 Descriptive information and further definitions

None.

4.3.17 Messaging**4.3.17.1 Dependencies to other Feature Types**

None.

4.3.17.2 Introduction

Messaging is basically intended for simple textual information transfer for displays and users. The textual information is not intended to be used for any automatisms. It should be used for textual information that are for example event based or otherwise will change over time.

Some devices may keep a list of messages rather than (or in addition to) a single message.

4.3.17.3 Rules

None.

3569 **4.3.17.4 messagingListData**3570 **4.3.17.4.1 Description**

3571 The messagingListData function contains the necessary elements to model a simple text message.

3572

3573 **4.3.17.4.2 Rules**

3574 None.

3575

3576 **4.3.17.4.3 Element rules**

Element	Rule	Description
Messaging. messagingListData. messagingData. (list)		
timestamp	SHOULD be set as SUB IDENTIFIER to the current timestamp at the creation of the message.	The time when the message was generated.
messagingNumber	SHALL be set as PRIMARY IDENTIFIER.	An identifier for one specific message. If a message is marked as obsolete (see below) this number can be used to identify the original message.
type	SHALL be set. Value rules of Table 104 SHALL be applied.	The type of the message.
text	If "type" has the value "obsolete", this element SHOULD be omitted. Otherwise it SHALL be set. The string-length SHOULD NOT be longer than 4096 characters. If it is longer, the sender SHALL consider the possibility that the receiver will shorten the string to 4096 characters.	The message itself.

3577 Table 103: messagingListData Element rules

3578 Element "**type**" value rules:

Value	Rule	Description
logging	Used for messages that should be stored in a log file.	Used for messages that are stored in a log file.
information	Messages that should be presented to the customer on a display; lower priority.	Messages that are presented to the customer on a display; lower priority.
warning	Messages that should be presented to the customer on a display; medium priority.	Messages that are presented to the customer on a display; medium priority.
alarm	Messages that should be presented to the customer on a display; high priority.	Messages that are presented to the customer on a display; high priority.
emergency	Messages that should be presented to the customer on a display; very high priority.	Messages that are presented to the customer on a display; very high priority.

obsolete	Previously sent messages that shall be marked as obsolete and eventually not be displayed anymore.	Used to mark previously sent messages as obsolete (so that they are not displayed anymore).
----------	--	---

Table 104: Element "type" value rules

3579

3580

3581 **4.3.17.5 Descriptive information and further definitions**

3582 None.

3583

3584 **4.3.18 NetworkManagement**

3585 Reserved for future use.

3586

3587 **4.3.19 NodeManagement**

3588 All rules and procedures are described in the [ProtocolSpecification].

3589

3590 **4.3.20 OperatingConstraints**3591 **4.3.20.1 Dependencies to other Feature Types**

3592 None.

3593

3594 **4.3.20.2 Introduction**

3595 This class is designed to model some constraints of an appliance (or functionality) with regards to its
 3596 energy consumption or production. I.e. this class is NOT intended to control a device. Rather, it
 3597 enables an appliance to express what needs to be considered especially for the case it permits being
 3598 controlled by another device. An example is a central energy management system that uses the class
 3599 DirectControl to control the appliance; in this case the energy management system needs to be
 3600 aware of the appliance's operating constraints.

3601

3602 **4.3.20.3 Rules**

3603 None.

3604

3605 **4.3.20.4 operatingConstraintsInterruptListData**3606 **4.3.20.4.1 Description**

3607 This function models basic constraints for an interrupt of a device's current task.

3608

3609 **4.3.20.4.2 Rules**

3610 If used, at least one constraint SHALL be set.

3611

3612 4.3.20.4.3 *Element rules*

Element	Rule	Description
OperatingConstraints. operatingConstraintsInterruptListData. operatingConstraintsInterruptData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If interrupt constraints are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
isPausable	If set to "true", a client MAY pause the related sequence, functionality or task of a server. Otherwise the element SHALL be set to "false" or omitted.	States whether a functionality or task is pausable (true) or not (false).
isStoppable	If set to "true", a client MAY stop the related sequence, functionality or task of a server. Otherwise the element SHALL be set to "false" or omitted.	States whether a functionality or task is stoppable (true) or not (false).
notInterruptibleAtHighPower	If set to "true", a client SHALL NOT pause or stop the related sequence, functionality or task of a server, if the power consumption or production is high. Otherwise the element SHALL be set to "false" or omitted.	States whether a functionality or task is NOT interruptible while it is consuming (or producing) energy (true) or whether it can be interrupted nevertheless (false).
maxCyclesPerDay	If set, the related sequence, functionality or task SHALL NOT restart more often within a day than stated in this element.	If a functionality or task may be started only a maximum number of times per day, this value is denoted here.

3613 Table 105: operatingConstraintsInterruptListData Element rules

3614

3615 4.3.20.5 *operatingConstraintsDurationListData*3616 4.3.20.5.1 *Description*

3617 This function models constraints on the minimum/maximum duration of active ("on") and pause
 3618 ("off") phases. Example: A compressor can be damaged if it is started and then switched off too early.

3619

3620 4.3.20.5.2 *Rules*

3621 If used, at least one constraint SHALL be set.

3622

3623 4.3.20.5.3 *Element rules*

Element	Rule	Description
---------	------	-------------

OperatingConstraints. operatingConstraintsDurationListData. operatingConstraintsDurationData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If duration constraints are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
activeDurationMin	If set, the related sequence, functionality or task SHALL be active without interruption for at least the time stated in this element.	This is the minimum duration a device has to run without interruption.
activeDurationMax	If set, the related sequence, functionality or task SHALL at maximum be active without interruption for as long as the time stated in this element.	This is the maximum duration a device can run without interruption.
pauseDurationMin	If set, a pause of the related sequence, functionality or task SHALL be at least as long as the time stated in this element.	This is the minimum duration a device has to pause after the end of an activity.
pauseDurationMax	If set, a pause of the related sequence, functionality or task SHALL be at maximum as long as the time stated in this element.	This is the maximum duration a device can pause after the end of an activity.
activeDurationSumMin	If set, the related sequence, functionality or task SHALL be in sum active for at least the time stated in this element.	This is the minimum duration a device has to run in total for a given task (summation of all active times).
activeDurationSumMax	If set, the related sequence, functionality or task SHALL be in sum active for maximum the time stated in this element.	This is the maximum duration a device can run in total for a given task (summation of all active times).

Table 106: operatingConstraintsDurationListData Element rules

4.3.20.6 operatingConstraintsPowerDescriptionListData

4.3.20.6.1 Description

This function models basic information relevant for *operatingConstraintsPowerRangeListData* and *operatingConstraintsPowerLevelListData*.

4.3.20.6.2 Rules

None.

3634 4.3.20.6.3 *Element rules*

Element	Rule	Description
OperatingConstraints. operatingConstraintsPowerDescriptionListData. operatingConstraintsPowerDescriptionData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If function elements are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
positiveEnergyDirection	SHALL be set, if there could be ambiguity how to interpret the value of the related sequence, functionality or task. For further value rules see section 3.10.1.14.1.	The <i>positiveEnergyDirection</i> states whether energy consumption or production is counted as positive value.
powerUnit	SHALL be set, if there could be ambiguity how to interpret the power value of the related sequence, functionality or task without the unit.	The unit for all power values in operatingConstraintsPowerRangeListData and operatingConstraintsPowerLevelListData.
energyUnit	SHALL be set, if there could be ambiguity how to interpret the energy value of the related sequence, functionality or task without the unit.	The unit for all energy values in operatingConstraintsPowerRangeListData and operatingConstraintsPowerLevelListData.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information.

3635 Table 107: operatingConstraintsPowerDescriptionListData Element rules

3636

3637 4.3.20.7 *operatingConstraintsPowerRangeListData*3638 4.3.20.7.1 *Description*

3639 This function can be used to model a power/energy range(!) a device can operate. An alternative
 3640 model for power/energy constraints is operatingConstraintsPowerLevelListData.

3641

3642 4.3.20.7.2 *Rules*

3643 If used, at least one range SHALL be set.

3644

3645 4.3.20.7.3 *Element rules*

Element	Rule	Description
---------	------	-------------

OperatingConstraints. operatingConstraintsPowerRangeListData. operatingConstraintsPowerRangeData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If the power range is related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
powerMin	If set, the related sequence, functionality or task SHALL at least consume or produce the given value of powerMin.	The minimum power a device can consume/produce.
powerMax	If set, the related sequence, functionality or task SHALL at maximum consume or produce the given value of powerMax.	The maximum power a device can consume/produce.
energyMin	If set, the related sequence, functionality or task SHALL at least consume or produce the given value of energyMin.	The minimum energy a device can consume/produce.
energyMax	If set, the related sequence, functionality or task SHALL at maximum consume or produce the given value of energyMax.	The maximum energy a device can consume/produce.

Table 108: operatingConstraintsPowerRangeListData Element rules

4.3.20.8 operatingConstraintsPowerLevelListData

4.3.20.8.1 Description

This function can be used to model a list of possible power values a device can operate. An alternative model for power constraints is operatingConstraintsPowerRangeListData, where a full range (not only discrete levels) can be modelled.

4.3.20.8.2 Rules

If used, at least one power value SHALL be set.

4.3.20.8.3 Element rules

Element	Rule	Description
OperatingConstraints. operatingConstraintsPowerLevelListData. operatingConstraintsPowerLevelData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If the power levels are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
power (list)	The values stated within the list SHALL define the allowed value set that MAY be used within the value of a related sequence, functionality or task.	Each "power" item contains a possible power level.

Table 109: operatingConstraintsPowerLevelListData Element rules

4.3.20.9 operatingConstraintsResumeImplicationListData**4.3.20.9.1 Description**

Pausing a device might be permitted according to above mentioned (duration) constraints. Nevertheless, it can be expected that it has implications if a device's task is paused temporarily and then resumed to continue the task. One possible implication is an additional mechanical stress which reduces the device's lifetime and could be interpreted as additional cost. Such additional costs or additional energy consumption can be modelled with this function in order to be considered for energy management. However, these are typically empirical (thus rather constant) data, therefore should not be considered to be precise.

4.3.20.9.2 Rules

None.

4.3.20.9.3 Element rules

Element	Rule	Description
OperatingConstraints.resumeImplicationListData. operatingConstraintsResumeImplicationData. (list)		
sequenceId	SHALL be set as FOREIGN IDENTIFIER, if the interrupt constraints are related to another feature defining sequenceId (see section 4.3.20.10 for details). Otherwise it SHALL be omitted.	If the resume implications are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
resumeEnergyEstimated	None.	Additional energy a device consumes before resuming to its normal operation (after a pause).
energyUnit	SHOULD be set, if there could be ambiguity how to interpret the value without the unit.	Unit for the energy stated in the element resumeEnergyEstimated.
resumeCostEstimated	None.	Additional costs for the resumption of a device to its normal operation.
currency	SHOULD be set, if "resumeCostEstimate" is set, too.	Currency for the price stated in the element resumeCostEstimated.

Table 110: operatingConstraintsResumeImplicationListData Element rules

4.3.20.10 Descriptive information and further definitions

If operatingConstraintsPowerRangeListData or operatingConstraintsPowerLevelListData is supported, operatingConstraintsPowerDescriptionListData SHALL be supported, too.

Section 5.3.18.1 mentions some recommendations to avoid ambiguities. Therefore, some rules are defined:

1. In every moment for each supported function "xListData" (with "x" as "prefix" placeholder for functions mentioned in section 4.3.20) exactly one of the following conditions SHALL be valid:
 - 1.1. "no sequenceld": xListData contains exactly one xData AND in xData the element "sequenceld" is NOT present.
 - 1.2. "all sequenceld": xListData contains one or more xData AND in every xData the element "sequenceld" is present. No two xData items have the same value in "sequenceld".
 - 1.3. "currently no sequence": xListData has no xData. This shall be treated as special case of item 1.2: In this case there is currently no related sequenceld.

Please note that these rules DO NOT permit to have an xListData where at least one xData has "sequenceld" set AND at least one xData has no "sequenceld" set.

2. Further rules on the use of elements apply:
 - 2.1. If an xData item has "sequenceld" set the other elements of xData are related to the corresponding power sequence except for the following case:
 - 2.2. The use of element "maxCyclesPerDay" of function "operatingConstraintsInterruptData" is optional. If the element is used it is NOT related to any power sequence even if the element "sequenceld" is present.

Remark: This means this specific constraint always applies to the feature and its related functionality in general.
3. Use of parallel instances of this Feature Type in the same entity for the same functionality: It is valid to have in the same entity one Feature Type instance that follows the "no sequenceld" rule (item 1.1 above) AND one Feature Type instance that follows the "all sequenceld" / "currently no sequence" rules (items 1.2 and 1.3 above) for the same functionality. However, in this case the xData child elements of these two instances SHALL be disjoint (i.e. they SHALL NOT contradict each other).
4. Remark: Esp. in case of the "no sequenceld" rule it is recommended to use the feature group concept (see section 2.1.4) in order to clearly relate the operating constraints to the actual/essential functionality.

4.3.21 PowerSequences

Reserved for future use.

4.3.22 Sensing

4.3.22.1 Dependencies to other Feature Types

None.

4.3.22.2 Introduction

The *Sensing* class is used for modelling data obtained by sensors in the meaning of more state-based sensors like contact sensors, switches and buttons, etc., in contrast to value-based sensors like temperature or humidity sensors that are treated with the *Measurement* class.

3720

3721 **4.3.22.3 Rules**

3722 None.

3723

3724 **4.3.22.4 sensingListData**3725 **4.3.22.4.1 Description**

3726 The list function of sensing can be used to model sensor information over time. This may be of
 3727 interest for a motion detector (when someone was at the backdoor), a contact sensor (when was the
 3728 window open in the last 24 hours), etc.

3729 Some sensors may only store the latest value and therefore only support one entry in the list. To
 3730 enable users and applications to query past changes, a communications technology interface may
 3731 cache states/values of simple sensors that only support the single data function. That is no
 3732 mandatory feature, however.

3733

3734 **4.3.22.4.2 Rules**

3735 If used, at least one constraint SHALL be set.

3736

3737 **4.3.22.4.3 Element rules**

Element	Rule	Description
Sensing. sensingListData. sensingData. (list)		
timestamp	SHOULD be set as PRIMARY IDENTIFIER to the timestamp at the creation of the message.	The creation time of the <i>sensingData</i> instance.
state	SHALL be set. Value rules of Table 112 SHALL be applied.	The state of the sensor is denoted in this element.
value	Whether this element SHALL be set or SHOULD NOT be set depends on the state (see Table 112 for details). If no information is given, as a default, "value" SHOULD NOT be set.	Some sensors deliver a value (e.g. a <i>levelSwitch</i>).

3738 *Table 111: sensingListData Element rules*3739 Element "**state**" value rules:

Value	Rule	Description
on	"value" SHOULD NOT be set.	"on" was detected, meaning that the according functionality was switched on.
off	"value" SHOULD NOT be set.	"off" was detected, meaning that the according functionality was switched off.
toggle	"value" SHOULD NOT be set.	"toggle" was detected, meaning that the according functionality was toggled (e.g. from "on" to "off" or vice versa).
level	"value" SHALL be set.	"level" was detected, meaning that the according functionality changed its level (e.g. from 2.4 to 8.2) that can be found in the Element "value".

levelUp	"value" SHALL be set.	"levelUp" was detected, meaning that the according functionality changed its level upwards.
levelDown	"value" SHALL be set.	"levelDown" was detected, meaning that the according functionality changed its level downwards.
levelStart	"value" SHOULD NOT be set.	"levelStart" was detected, meaning that the according functionality started to change its level.
levelStop	"value" SHOULD NOT be set.	"levelStop" was detected, meaning that the according functionality stopped to change its level.
levelAbsolute	"value" SHALL be set.	"levelAbsolute" was detected, meaning that the according functionality changed its level to an absolute value stated in the Element "value" (e.g. 1190).
levelRelative	"value" SHALL be set.	"levelRelative" was detected, meaning that the according functionality moved its level in comparison to the old one. The relative change can be found in the Element "value" (e.g. a change from 22 to 33 would result in a content of the Element "value" of 11).
levelPercentageAbsolute	"value" SHALL be set.	"levelPercentageAbsolute" was detected, meaning that the according functionality changed its level to an absolute percentage value stated in the Element "value" (e.g. 90%).
levelPercentageRelative	"value" SHALL be set.	"levelPercentageRelative" was detected, meaning that the according functionality changed its percentage level relative to the old one. The relative change can be found in the Element "value" (e.g. a change from 40% to 50% would result in a content of the Element "value" of 10).
pressed	"value" SHOULD NOT be set.	"pressed" was detected, meaning that the according functionality was pressed (e.g. a button, that was pressed).
longPressed	"value" SHOULD NOT be set.	"longPressed" was detected, meaning that the according functionality was pressed longer (e.g. a button, that was pressed longer). The timing difference between "pressed" and "longPressed" is up to the vendor.
released	"value" SHOULD NOT be set.	"released" was detected, meaning that the according functionality was released (e.g. a button, that was released).
changed	"value" SHOULD NOT be set.	"changed" was detected, meaning that the according functionality changed.
started	"value" SHOULD NOT be set.	"started" was detected, meaning that the according functionality started.
stopped	"value" SHOULD NOT be set.	"stopped" was detected, meaning that the according functionality stopped.
paused	"value" SHOULD NOT be set.	"paused" was detected, meaning that the according functionality paused.
middle	"value" SHOULD NOT be set.	"middle" was detected, meaning that the according functionality (e.g. a scroll bar or a throttle) is in position "middle".

up	"value" SHOULD NOT be set.	"up" was detected, meaning that the according functionality moves up.
down	"value" SHOULD NOT be set.	"down" was detected, meaning that the according functionality moves down.
forward	"value" SHOULD NOT be set.	"forward" was detected, meaning that the according functionality moves forward.
backwards	"value" SHOULD NOT be set.	"backwards" was detected, meaning that the according functionality moves backwards.
open	"value" SHOULD NOT be set.	"open" was detected, meaning that the according functionality is open (e.g. a door).
closed	"value" SHOULD NOT be set.	"closed" was detected, meaning that the according functionality is closed (e.g. a door).
opening	"value" SHOULD NOT be set.	"opening" was detected, meaning that the according functionality is opening (e.g. a door, that is changing from "closed" to "open").
closing	"value" SHOULD NOT be set.	"closing" was detected, meaning that the according functionality is closing (e.g. a door, that is changing from "open" to "closed").
high	"value" SHOULD NOT be set.	"high" was detected, meaning that the according functionality is at a high position or value.
low	"value" SHOULD NOT be set.	"low" was detected, meaning that the according functionality is at a low position or value.
day	"value" SHOULD NOT be set.	"day" was detected, meaning that the according functionality is in day-mode.
night	"value" SHOULD NOT be set.	"night" was detected, meaning that the according functionality is in night-mode.
detected	"value" SHOULD NOT be set.	"detected" was detected, meaning that the according functionality was detected (e.g. a motion sensor detected a motion).
notDetected	"value" SHOULD NOT be set.	"notDetected" was detected, meaning that the according functionality was detected (e.g. a motion sensor reports that no motion is detected).
alarmed	"value" SHOULD NOT be set.	"alarmed" was detected, meaning that the according functionality is in alarm mode (e.g. a smoke detector).
notAlarmed	"value" SHOULD NOT be set.	"notAlarmed" was detected, meaning that the according functionality is not in alarm mode (e.g. a smoke detector).

Table 112: Element "state" value rules

4.3.22.5 *sensingDescriptionData*

4.3.22.5.1 *Description*

The rather static (non-changing) elements of the *Sensing* class are combined in the *sensingDescriptionData* function. Typically, it is only requested once by a user or an application.

4.3.22.5.2 *Rules*

None.

3749

3750 4.3.22.5.3 *Element rules*

Element	Rule	Description
Sensing. sensingDescriptionData.		
sensingType	SHALL be set. Value rules of Table 114 SHALL be applied.	Which type of sensor is modelled is denoted with this element.
unit	SHALL be set, if there could be ambiguity how to interpret the sensing value without the unit.	The unit in which the values of the sensor are given.
scopeType	None.	Specifies a more detailed meaning of the sensor (e.g. shutter).
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the sensor.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the sensor.

3751 Table 113: sensingDescriptionData Element rules

3752 Element "**sensingType**" value rules:

Value	Rule	Description
switch	None.	A switch that has usually two states "on" and "off".
button	None.	A button that can be "pressed" (and eventually supports "longPressed" and "released")
level	None.	Some level functionality supporting one or more of the level-related sensing states.
levelSwitch	None.	Some level functionality supporting one or more of the level-related sensing states and a switch functionality ("on" / "off").
windowHandle	None.	A window handle supporting different positions.
contactSensor	None.	Sensor that can detect whether something is "open" or "closed".
occupancySensor	None.	Sensor that can detect whether someone is present in a room ("detected") or not ("notDetected").
motionDetector	None.	Sensor that can detect motion ("detected"/"notDetected").
fireDetector	None.	A sensor that detects fire ("detected"/"notDetected").
smokeDetector	None.	A sensor that detects smoke ("detected"/"notDetected").
heatDetector	None.	A sensor that detects heat ("detected"/"notDetected").
waterDetector	None.	A sensor that detects water ("detected"/"notDetected").
gasDetector	None.	A sensor that detects gas ("detected"/"notDetected").
alarmSensor	None.	A sensor providing alarm information ("alarmed" / "notAlarmed").
powerAlarmSensor	None.	A sensor reporting power alarms ("alarmed" / "notAlarmed").
dayNightIndicator	None.	A sensor detecting whether it is "day" or "night".

3753 Table 114: Element "sensingType" value rules

3754

3755 4.3.22.6 *Descriptive information and further definitions*

3756 None.

3757

3758 **4.3.23 Setpoint**3759 **4.3.23.1 Dependencies to other Feature Types**

- 3760 - Measurement
 3761 - TimeTable

3762

3763 **4.3.23.2 Introduction**

3764 Setting setpoint values is an often chosen way to control some functionality of a device. Especially in
 3765 the home automation and HVAC domain, several use cases depend on setpoints. This class handles
 3766 the setpoint itself. Relations to Measurement and TimeTable class enables the control of a
 3767 measurand for specific time frames.

3768

3769 **4.3.23.3 Rules**

3770 None.

3771

3772 **4.3.23.4 setpointListData**3773 **4.3.23.4.1 Description**

3774 The setpoint values and (if needed) the identifier of related measurands or time tables are modelled
 3775 within this function. It is possible to either define an exact value, that should be reached or a range
 3776 which shall not be fallen below or exceeded. Absolute or relative tolerances can be defined, too.

3777

3778 **4.3.23.4.2 Rules**

3779 At least one of the following Elements SHALL be set (combinations MAY be used):

- 3780 - value
 3781 - valueMin
 3782 - valueMax

3783

3784 **4.3.23.4.3 Element rules**

Element	Rule	Description
Setpoint. setpointListData. setpointData. (list)		
setpointId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the setpoint.
value	None.	The setpoint value itself.
valueMin	If set, the valueMin SHALL be interpreted as acceptable minimum setpoint value.	Instead of a fixed value (see above) a value range can be set. This would be the lower limit.

valueMax	If set, the valueMax SHALL be interpreted as acceptable maximum setpoint value.	Instead of a fixed value (see above) a value range can be set. This would be the upper limit.
valueToleranceAbsolute	If set, the valueToleranceAbsolute SHALL be added to the value to calculate the tolerated maximum and subtracted from the value to calculate the tolerated minimum. The valueToleranceAbsolute SHALL be positive.	If the value may vary some absolute value around the desired setpoint, the maximum variation allowed (in both directions) can be stated here.
valueTolerancePercentage	If set, the valueTolerancePercentage SHALL be multiplied with the value and then added to the value to calculate the tolerated maximum and the valueTolerancePercentage SHALL be multiplied with the value and subtracted from the value to calculate the tolerated minimum. The valueTolerancePercentage SHALL be positive or zero.	If the value may vary some percentage around the desired setpoint, the maximum variation allowed (in both directions) can be stated here.
isSetpointChangeable	If set to "true" the server SHALL accept changes of the setpoint by appropriate clients (e.g. the bound client). If set to "false" the server SHALL decline change requests. If absent, the server SHOULD accept changes.	States whether the setpoint is changeable by a client or not.
isSetpointActive	If set to "false", the setpoint is inactive. If set to "true" or omitted, the setpoint is active.	Indicates whether the setpoint is currently active or not.
timePeriod	MAY be used to define a timePeriod for the setpoint. If set, the setpoint SHALL only be applied within the given timePeriod.	The period where the setpoint shall be active.

Table 115: setpointListData Element rules

4.3.23.5 setpointConstraintsListData**4.3.23.5.1 Description**

The constraints regarding the possible setpoints allowed are modelled within this function.

4.3.23.5.2 Rules

If used, at least one constraint SHALL be set.

4.3.23.5.3 Element rules

Element	Rule	Description
Setpoint. setpointConstraintsListData. setpointConstraintsData. (list)		

setpointId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the setpoint, these constraints belong to.
setpointRangeMin	The setpoint value SHALL NOT be smaller than setpointRangeMin.	The minimum value, the setpoint can be set to.
setpointRangeMax	The setpoint value SHALL NOT be larger than setpointRangeMax.	The maximum value, the setpoint can be set to.
setpointStepSize	SHOULD be used if values are only supported in a certain stepsize. Values that do not match the step size SHALL be rounded by the server.	The minimum step size between two different values.

Table 116: setpointConstraintsListData Element rules

4.3.23.6 setpointDescriptionListData

4.3.23.6.1 Description

The rather static (non-changing) elements of the *Setpoint* class are combined in the *setpointDescriptionListData* function. Typically, it is only requested once by a user or an application.

4.3.23.6.2 Rules

None.

4.3.23.6.3 Element rules

Element	Rule	Description
Setpoint. setpointDescriptionListData. setpointDescriptionData. (list)		
setpointId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the setpoint.
measurementId	SHALL be set as FOREIGN IDENTIFIER, if the setpoint is linked to a measurement or another Feature that uses the same measurementId as FOREIGN IDENTIFIER. Otherwise it SHALL be omitted.	If the setpoint relates to a measurand, the corresponding ID can be stated here.
timeTableId	SHALL be set as FOREIGN IDENTIFIER to the according timeTableId, if the setpoint is linked to a timeTable or another Feature that uses the same timeTableId as FOREIGN IDENTIFIER. Otherwise it SHALL be omitted.	If the activation of the setpoint is time dependent, the ID of a corresponding time table can denoted in this element.
setpointType	SHOULD be set. If omitted, the setpoint SHALL be interpreted as <i>setpointType</i> "valueAbsolute". Value rules of Table 118 SHALL be applied.	This is the type of setpoint used for this specific setpoint.
unit	SHALL be set, if there could be ambiguity how to interpret the value without the unit.	The unit used for this setpoint (see SPINE data model for all possible values).
scopeType	None.	Specifies a more detailed meaning of the setpoint.

label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the setpoint.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the setpoint.

Table 117: setpointDescriptionListData Element rules

Element "**setpointType**" value rules:

Value	Rule	Description
valueAbsolute	None.	An absolute value, e.g. 28°C. A unit is needed in most cases.
valueRelative	None.	A relative value in percentage. The unit is "pct" in most cases.

Table 118: Element "setpointType" value rules

4.3.23.7 Descriptive information and further definitions

None.

4.3.24 SmartEnergyManagementPs

4.3.24.1 Functions and message purposes

The following table shows which messages purposes are required or optional (see column "M/O") for a server. It also shows which function is used for the given purpose and which direction and classifier applies:

Message purpose	Used complex function	M / O	Direction	Used classifiers	Explanation
powerSequences information	smartEnergyManagementPs Data	M	Out	reply, notify	Informs or updates a client about a forecast (fixed or with flexibilities).
powerSequences information request	smartEnergyManagementPs Data	M	In	read	Requests current information from a server.
powerSequences configuration request call	smartEnergyManagementPs ConfigurationRequestCall	O	Out	call	Informs the client that the server's power sequences need to be (re-)scheduled.
powerSequences configuration	smartEnergyManagementPs Data	M	In	write (restricted)	Is used to submit a new configuration to a server.
powerSequences state change	smartEnergyManagementPs Data	O	In	write (restricted)	Enables a power sequences client to change the current state

					of a power sequences server's sequence (if permitted by the server).
powerSequences price calculation request call	smartEnergyManagementPs PriceCalculationRequestCall	O	Out	call	Requests a price calculation from a client.
powerSequences price information	smartEnergyManagementPs PriceData	O	Out	reply, notify	Informs or updates a client about a price.
powerSequences price information request	smartEnergyManagementPs PriceData	O	In	read	Requests price information stored on the server.
powerSequences price change	smartEnergyManagementPs PriceData	O	In	write (restricted)	Enables a power sequences client to change the price information of a power sequences server's sequence (if permitted by the server).

Table 119: Server messages overview. The "write (restricted)" denotes if the message is intended for partial modification.

Details on the messages are defined in section 4.3.24.4.

4.3.24.2 Introduction

4.3.24.2.1 Scope of this Feature Type

The underlying SmartEnergyManagementPs complex class enables a client to shift the demand of a server within some given constraints. The corresponding rules for interoperable implementations are denoted within this feature type.

4.3.24.2.2 Theory of operation (informative)

Each power sequences server requires the establishment of a binding with a power sequences client (see 4.3.24.3.8) as only a bound power sequences client is permitted to configure a power sequences server. One power sequences client may be bound to many power sequences servers (see 4.3.24.3.8), while one power sequences server is only bound to exactly one power sequences client (see 4.3.24.3.9). This completes the basic setup for a power sequences server and client.

Please note: Binding does not imply subscription mechanisms! If a client is interested in data changes on the server (which is recommended), a subscription has to be established, too.

If the power sequences contain flexibilities and/or alternatives, the power sequences server issues a power sequences configuration request call. Once the power sequences client receives this message, it evaluates whether it will configure the according power sequences server or ignore the call

(temporarily or permanent). The power sequences client configures the server by writing a power sequences configuration to the power sequence server. If the power sequences client does not configure the power sequences server, the power sequences server follows the configuration it has initially exposed, i.e. its own preference. If the power sequences server receives a configuration it should take into account, it first checks whether the configuration is valid. If the received configuration is valid, i.e. meets the constraints the power sequences server has laid out, it takes over the configuration.

During runtime, the power sequences client and server may re-negotiate the configuration multiple times, even if the according power sequence is already running (if the power sequences server supports it). However, every time the data on either the power sequence information, configuration or state changes, the power sequences server updates the subscribed nodes.

Please note: Devices that can't freely choose between different modes of operations (e.g. because they cannot interrupt an already running process) would not publish the power sequences which they cannot switch to "currently" until the situation has been resolved.

Dependent on the use case, a power sequence needs to contain different kind of data. This means there are different types of sequences possible. The description of the supported sequence types and how they are recognised (evaluated) is given in detail in section 4.3.24.3.5.5.

In the subsequent sections terms and rules are defined for the use of the complex class within this Feature Type. These rules impose restrictions on the use of the class as follows:

- A sequence contains a minimum of one slot, but may contain as many slots as needed.
- Slots which appear in a certain order within a power sequence cannot be interchanged, i.e. a slot can never be shifted so that it is executed before a formerly subsequent slot. The slot order within a power sequence cannot be changed. Thus, each time shift to a slot moves the subsequent slots accordingly.
- Within a sequence, each pause or gap is treated as regular slot of a power sequence and thus holds the same data a regular slot holds (but with an energy value of "0").
- For fixed forecasts, no constraints are specified, because no shifting is permitted (see 4.3.24.3.5.5.2).
- etc.

4.3.24.3 Common specification details

4.3.24.3.1 Guidance to the concept and description

The concept permits numerous possibilities for the representation of a device's energy management related data. In order to achieve interoperability, a formal definition of terms and rules is given subsequently. First of all, definitions and rules related to "slots" are given, followed by definitions and rules on "sequences".

However, one key to achieve interoperability is to focus on those "variants" of device's energy management information that are supported by this specification. These are called "permitted sequence types" and explained in section 4.3.24.3.5.5. It is recommended to start with this section

3877 for a first reading at least. These sequence types definitions make use of the terms that are explained
3878 in the previous sections.

3879 Finally, rules on the evaluation of power sequences information and some necessary conventions are
3880 given.

3881

3882 *4.3.24.3.2 Slots*

3883 *4.3.24.3.2.1 Overview*

3884 Section 4.3.24.3.2 contains a number of basic terms and rules on the modelling of power sequence
3885 slots. In order to get a better overview and understanding of this section and the concepts it may be
3886 preferable to continue first with the following sections and then return to the other sections on
3887 demand:

- 3888 1. Section 4.3.24.3.2.9 discusses some general duration facets of slots.
- 3889 2. Section 4.3.24.3.2.7 discusses slots with configurable schedule.
- 3890 3. Section 4.3.24.3.2.5 describes some requirements on power/energy values.
- 3891 4. Section 4.3.24.3.2.8 describes an impact of power/energy consumption/generation through
3892 the modification of a slot's duration.

3893

3894 *4.3.24.3.2.2 Determined vs. undetermined slot*

3895 *4.3.24.3.2.2.1 Introduction (informative)*

3896 A "determined slot" can be used to build detailed power/energy curves. This provides a basis for
3897 time-dependent forecasts, process configuration (in advance) and autonomous process execution.
3898 An "undetermined slot" on the other hand can be used for processes where only few information is
3899 available in advance.

3900

3901 *4.3.24.3.2.2.2 Definitions*

3902 A slot is called "determined" if all of the following information is present:

- 3903 1. "alternatives. powerSequence. powerTimeSlot. schedule. slotNumber"
- 3904 2. "alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration"
- 3905 3. The slot is (explicitly or implicitly) activated or deactivated (see 4.3.24.3.2.4).
- 3906 4. The slot value forecast is determined (see 4.3.24.3.2.5).

3907 The value of "alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration" SHALL be
3908 greater than 0.

3909 Remark: The element "alternatives. powerSequence. powerTimeSlot. schedule. durationUncertainty"
3910 may as well be present in a determined slot, like other elements as well. However, this is relevant for
3911 the duration estimation only and not for the classification as determined slot.

3912 A slot is called "undetermined" if it is NOT "determined" AND all of the following conditions are
3913 fulfilled:

- 3914 1. "alternatives. powerSequence. powerTimeSlot. schedule. slotNumber" is present.
3915 2. "alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration" is NOT present.
3916 3. The slot value forecast is determined (see 4.3.24.3.2.5).

3917 Undetermined slots have no activation or deactivation information themselves. Rather, this kind of
3918 slot is designed for a specific purpose where the final execution is determined by its power
3919 sequence's state.

3920 A slot that is neither determined nor undetermined is malformed and not allowed in this
3921 specification.

3922 The element "alternatives. powerSequence. powerTimeSlot. schedule. timePeriod" SHALL NOT be
3923 used to make a slot schedule "determined". Rather, "timePeriod" can be used for supplementary
3924 purposes (e.g. display) only. If used, it SHALL NOT contradict regular schedule information (i.e.
3925 sequence start time, sum of preceding (and explicitly/implicitly activated) slot durations).

3926

3927 4.3.24.3.2.2.3 Process rules

- 3928 1. On numbering slot numbers in a power sequence:
3929 Each slot of a power sequence SHALL be assigned a slot number within the element
3930 "slotNumber". The slot numbers within the sequence SHALL be ascending according to the
3931 chronological order of the slots. The combination of sequence ID and slot number and
3932 (optionally) the current repetition number (see sections 4.3.24.3.2.10 and 4.3.24.3.5.3.2)
3933 SHALL remain unique at any given point in time.
3934 2. Some more process rules on the use of defaultDuration and its dependencies to some other
3935 values are described in section 4.3.24.3.2.7.3.

3936

3937 4.3.24.3.2.3 Duration uncertainty

3938 This section (4.3.24.3.2.3 and subsections) applies for determined slots only!

3939

3940 4.3.24.3.2.3.1 Introduction (informative)

3941 Slots exposed by a power sequences server typically have fixed durations. In some applications, the
3942 exact duration of a slot cannot be determined. Thus, a slot duration might be rather an average
3943 value, supplemented with an uncertainty of the duration.

3944 Note: A duration uncertainty does NOT model a configuration range of a duration. A duration
3945 uncertainty also does NOT express any potential interrupt of the slot. A duration uncertainty only
3946 models that a power sequences server cannot estimate a duration more precisely.

3947

3948 4.3.24.3.2.3.2 Definitions

3949 The duration of a determined slot SHALL be rather "exact" if NO duration uncertainty is given.

3950 A duration uncertainty is given if the following information is present:

- 3951 1. The slot schedule is determined.
3952 2. "alternatives. powerSequence. powerTimeSlot. schedule. durationUncertainty" is present.
- 3953 If "alternatives. powerSequence. powerTimeSlot. schedule. durationUncertainty" is present the value
3954 SHALL be positive.
- 3955 If a duration uncertainty is given, the slot's final duration (i.e. the measured duration when the slot
3956 was executed completely) SHOULD be in the range from "lower duration bound" to (defaultDuration
3957 + durationUncertainty). The value of "lower duration bound" is determined by (defaultDuration -
3958 durationUncertainty) provided that (defaultDuration - durationUncertainty) is not negative.
3959 Otherwise, "lower duration bound" is 0.
- 3960 If a slot provides information about a duration uncertainty and has a configurable duration (see
3961 4.3.24.3.2.7), the duration uncertainty is considered to be constant for all configurable durations of
3962 the slot.
- 3963
- 3964 *4.3.24.3.2.4 Activated vs. deactivated slot*
3965 This section (4.3.24.3.2.4) applies for determined slots only.
- 3966 Usually, a slot is explicitly or implicitly activated, hence contributes with its data to the schedule and
3967 execution of its power sequence. However, in case a slot is "optional" it can instead be marked as
3968 "deactivated". A deactivated slot does not contribute to the schedule and execution of a power
3969 sequence. I.e. a power sequence is scheduled and executed as if all deactivated slots have been
3970 removed from the sequence.
- 3971
- 3972 *4.3.24.3.2.5 Determined vs. undetermined slot value forecast*
3973 A slot can be assigned power or energy values to model a potential or expected forecast of a power
3974 sequences server's power or energy consumption or production.
- 3975 A slot value forecast is called "determined" if all of the following conditions are fulfilled:
- 3976 1. At least one instance of "alternatives. powerSequence. powerTimeSlot. valueList. value" is
3977 present for the given slot AND the child element "valueType" is present AND the child
3978 element "value" is present.
3979 2. The list of the slot's "alternatives. powerSequence. powerTimeSlot. valueList. value"
3980 instances fulfils the value types requirement (see subsequent definition).
- 3981 Otherwise the slot value forecast is called "undetermined".
- 3982 The value types requirement is defined as follows: Within each slot of a power sequence, different
3983 pairs of the elements "value" and "valueType" may be provided. The availability of these values and
3984 more important the value types depends on the process(es) the power sequences server exposes.
3985 However, a power sequences server SHALL at least provide one of the values and value types or
3986 tuples of values and value types contained in the following list:
- 3987
- power

- 3988 • powerMin
- 3989 • powerMax
- 3990 • energy
- 3991 • energyMin
- 3992 • energyMax
- 3993 • powerExpectedValue
- 3994 • powerExpectedValue + powerStandardDeviation
- 3995 • powerExpectedValue + powerStandardDeviation + powerSkewness
- 3996 • energyExpectedValue
- 3997 • energyExpectedValue + energyStandardDeviation
- 3998 • energyExpectedValue + energyStandardDeviation + energySkewness

3999 As an example, a power sequences server might expose all slots with a valueType of "power". A
 4000 power sequences server might as well expose all slots with the valueTypes "powerExpectedValue",
 4001 "powerStandardDeviation" and "powerSkewness". However, it is invalid to only provide the
 4002 valueType combination of "powerExpectedValue" and "powerSkewness" without
 4003 "powerStandardDeviation" for one or more slots.

4004 Please note that "expectedValue" refers to the term "expected value" (also "expectation") of
 4005 probability theory. It may be used to express the estimation of an average value.

4006 Note: See section 4.3.24.3.2.8 on a dependency between power/energy forecast and configurable
 4007 slot durations!

4008

4009 4.3.24.3.2.6 *Constrained vs. implicit slot schedule*

4010 This section (4.3.24.3.2.6 and subsections) applies for determined slots only!

4011

4012 4.3.24.3.2.6.1 *Introduction (informative)*

4013 This section is closely related to the question whether a slot schedule can be configured or not (see
 4014 4.3.24.3.2.7). In case of configurable slot schedules there is a need to model the constraints of such a
 4015 configuration. Non-configurable slots, on the other hand, do not require such constraints, hence
 4016 their schedule is given implicitly.

4017

4018 4.3.24.3.2.6.2 *Definitions*

4019 A slot schedule is called "constrained" if all of the following conditions are fulfilled:

- 4020 1. The slot is determined.
- 4021 2. "alternatives. powerSequence. powerTimeSlot. scheduleConstraints" is present with at least
 4022 one child element.

4023 A slot schedule is called "implicit" if all of the following conditions are fulfilled:

- 4024 1. The slot is determined.

4025 2. "alternatives. powerSequence. powerTimeSlot. scheduleConstraints" is absent or has no
4026 child element.

4027 An implicit slot schedule does NOT permit any direct variation of the slot's schedule itself. I.e. it is not
4028 possible to modify the slot's duration or start time or end time directly. This means the duration of
4029 the slot is not modifiable by a value assignment, and the slot's start time is determined implicitly (by
4030 the end of the preceding slot or by the sequence's start time if there is no preceding slot). It also
4031 means the slot is not optional, i.e. it cannot be deactivated individually (see 4.3.24.3.2.4).

4032 A constrained slot schedule denotes further rules that need to be considered for the configuration of
4033 the slot's schedule. Among others, this may include the possibility for the complete deactivation of
4034 the slot (see 4.3.24.3.2.4).

4035

4036 *4.3.24.3.2.7 Configurable vs. non-configurable schedule of a determined slot*

4037 This section (4.3.24.3.2.7 and subsections) applies for determined slots only!

4038

4039 *4.3.24.3.2.7.1 Introduction (informative)*

4040 Some devices can permit the configuration of some slot durations or even the disabling of slots of a
4041 given power sequence. Such slot schedules are called "configurable", whereas other slot schedules
4042 are called "non-configurable". The following definition explains the proper conditions. Furthermore,
4043 process rules that affect both kinds of slot schedules are discussed.

4044

4045 *4.3.24.3.2.7.2 Definitions*

4046 The schedule information of a determined slot is configurable if all of the following conditions are
4047 fulfilled:

- 4048 1. The slot schedule is constrained (see 4.3.24.3.2.6).
4049 2. The slot's sequence permits configuration.

4050 In this case the slot's "alternatives. powerSequence. powerTimeSlot. schedule" can be modified with
4051 the proper modification according to section 4.3.24.3.2.7.3. However, this requires the consideration
4052 of the whole sequence's schedule constraints (including all slot schedule constraints).

4053 The duration of a slot is configurable if all of the following conditions are fulfilled:

- 4054 1. The slot schedule is configurable.
4055 2. In "alternatives. powerSequence. powerTimeSlot. scheduleConstraints" the child elements
4056 minDuration AND maxDuration are present.

4057 If a slot's duration is configurable the following rules apply:

- 4058 1. minDuration and maxDuration SHALL be non-negative.
4059 2. minDuration SHALL be smaller than maxDuration.
4060 3. defaultDuration SHALL be in the range [minDuration, maxDuration].

Note: In case a duration uncertainty is given (see 4.3.24.3.2.3), the final duration of a slot (i.e. the measured duration once the slot was completed) may violate a boundary given by minDuration or maxDuration. This is because minDuration and maxDuration denote only configuration boundaries. This is independent from the capability of forecasting (i.e. estimating) a final duration from a given configuration.

The element "alternatives. powerSequence. powerTimeSlot. scheduleConstraints" is malformed if only one of the child elements minDuration and maxDuration is present.

A slot can be activated or deactivated if all of the following conditions are fulfilled:

1. The slot schedule is configurable.
2. In "alternatives. powerSequence. powerTimeSlot. scheduleConstraints" the child element optionalSlot is present and set to "true".

See also section 4.3.24.3.2.7.3 on process related restrictions to schedule configurability!

The schedule information of a determined slot with implicit schedule (see 4.3.24.3.2.6) is not configurable.

4.3.24.3.2.7.3 Process rules

This section discusses rules that affect determined slots with configurable as well as non-configurable schedule.

The schedule configuration of a power sequence's slots is modelled within the element "alternatives. powerSequence" of a "powerSequences configuration" message (see section 4.3.24.4.4), where each instance of the child element "powerTimeSlot. schedule" contains the schedule configuration of a single slot.

1. Uniqueness and order:

The slot schedule configuration list SHALL be unique and ordered: Within "alternatives. powerSequence" each slotNumber SHALL occur not more than once. The slots in the list SHALL be ordered by the slotNumber according to section 4.3.24.3.2.2.3.

2. Slots with non-configurable schedule:

Slots with non-configurable schedule do not need to be listed in "alternatives. powerSequence" unless stated otherwise (some of the following rules REQUIRE the presence for specific cases; however, the presence is not required in general). If listed anyhow, the slot's schedule configuration MAY be empty (except for slotNumber) and SHOULD NOT contradict the slot's schedule information. If evaluated, the schedule configuration of slots with non-configurable schedule SHALL be ignored.

3. Completeness of a slot's schedule configuration:

Within a message to configure one or more slots, each instance of "alternatives. powerSequence. powerTimeSlot. schedule" of a configurable slot SHALL be complete with regards to the configurable options. I.e. if a schedule configuration is given, it SHALL configure ALL schedule information the slot permits for configuration.

Example: If a slot can be deactivated and the duration can be modified it is not sufficient to configure only the activation or only the duration.

- 4101 4. Contributions to the schedule configuration list:
4102 Within "alternatives. powerSequence" there can be "alternatives. powerSequence.
4103 powerTimeSlot. schedule" instances for "past slots", the "current slot", and "future slots". A
4104 "past slot" is a slot that was already executed. The "current slot" is the slot that is currently
4105 executed. A "future slot" is a slot that is not executed yet.
- 4106 5. On "past slots":
4107 A "past slot" SHALL be treated as non-configurable slot, even if it was configurable before its
4108 execution. Its "alternatives. powerSequence. powerTimeSlot. schedule" instance MAY be
4109 part of "alternatives. powerSequence". However, such a slot's instance of "alternatives.
4110 powerSequence. powerTimeSlot. schedule" has no further functional relevance within this
4111 specification.
- 4112 6. On the "current slot":
4113 a. It SHALL NOT be possible to deactivate a "current slot". This means even if the slot's
4114 child element "optionalSlot" (see 4.3.24.3.2.7.2) is available and set to "true" it
4115 SHALL be interpreted as if the element is set to "false".
4116 Remark: This rule is only about the slot's specific flag "optionalSlot". It is
4117 independent from any possibility to modify a whole power sequence's state.
- 4118 b. If a configurable slot becomes the current slot it MAY remain configurable or it MAY
4119 become non-configurable (this is finally implementation specific to the power
4120 sequences server).
- 4121 c. The current slot's instance of "alternatives. powerSequence. powerTimeSlot.
4122 schedule" SHALL be listed in "alternatives. powerSequence" (regardless whether it is
4123 configurable or not; see rules above).
- 4124 d. If the power sequences server provides a "alternatives. powerSequence. state"
4125 instance of the power sequence with the current slot, the following rules apply:
4126 i. The child element "remainingSlotTime" SHALL be set to the duration the
4127 current slot still needs to be active. I.e. in case the power sequence itself is
4128 paused for a while (hence the current slot can also not be active) the value of
4129 remainingSlotTime SHALL NOT decrease.
- 4130 ii. The child element "elapsedSlotTime" MAY be present. If it is present it SHALL
4131 be set to the duration the current slot has been active. I.e. in case the power
4132 sequence itself is paused for a while (hence the current slot can also not be
4133 active) the value of elapsedSlotTime SHALL NOT increase.
- 4134 iii. The sum of remainingSlotTime's value and the duration the current slot has
4135 been active SHALL be at least approximately equal to the slot's
4136 defaultDuration. At least a deviation according to durationUncertainty is
4137 permitted.
- 4138 Note: This also means the current slot's defaultDuration SHALL always cover
4139 the time it began (i.e. the past) up to the time it will end (i.e. the future). I.e.
4140 this value SHALL NOT be reduced just because a part of the slot's overall
4141 activity was already completed. It also means that phases where the slot
4142 (sequence) was paused do not require an update of defaultDuration.
- 4143 7. On "future slots":
4144 All "future slots" instances of "alternatives. powerSequence. powerTimeSlot. schedule"
4145 SHALL be listed in "alternatives. powerSequence" (regardless whether they are configurable
4146 or not; see rules above).

Please note: Some of the rules describe situations where the configurability of a slot changes. Of course, this requires a proper information update from the power sequences server to the power sequences client in general. However, this update process is not subject of this section.

4.3.24.3.2.8 Dependency between power/energy and slot duration configuration

This section (4.3.24.3.2.8) applies for slots with configurable duration according to section 4.3.24.3.2.7 only!

According to section 4.3.24.3.2.5 miscellaneous power OR energy values can be used as forecast of a slot's consumption or production. It shall be noted that this requires special consideration in conjunction with slots of configurable duration. In general a slot with a configurable duration can be modelled with an energy forecast OR a power forecast, resulting in two fundamentally different consequences:

1. A slot with a configurable duration and an exposed "valueType" element stating an energy-related information, is expected to (nearly) have the same energy for all possible configurations of durations it offers. I.e. the power is reduced when extending the duration and increased when shortening the duration.
2. A slot with a configurable duration and an exposed "valueType" element stating a power-related information is expected to (nearly) have the same power for all possible configurations of durations it offers. I.e. the energy is reduced when shortening the duration and increased when extending the duration.

Note: A slot that provides a configurable duration SHALL NOT provide "valueType"s related to energy and power for the same slot!

4.3.24.3.2.9 Duration facets of determined slots, overview

This section (4.3.24.3.2.9) applies for determined slots only! It provides a brief overview on several aspects of a slot duration.

The following aspects can be modelled:

1. A slot's duration is "exact" or just given with an uncertainty: See 4.3.24.3.2.3.
2. A slot's duration is configurable (i.e. modifiable) or not: See 4.3.24.3.2.7.
3. A slot is optional (i.e. can be deactivated) or not: See 4.3.24.3.2.7.

Thus, for optional as well as non-optional slots the following combinations are possible:

1. A slot with an "exact" duration (i.e. without any duration uncertainty) and no configurable duration.
2. A slot with a (average) duration and a duration uncertainty and no configurable duration.
3. A slot with an "exact" duration (i.e. without any duration uncertainty) and a configurable duration.
4. A slot with a (average) duration and a duration uncertainty and a configurable duration.

4.3.24.3.2.10 Repeated slots vs. one-time slots

Within a given sequence, each phase is represented by a slot. Each slot can be assigned a unique number. In this situation, each slot is processed only one time.

However, some implementations favour a "repetition concept" where a sequence of phases repeats a number of times. This permits to model only the first occurrence of a phase. E.g. if a sequence has three phases "A", "B", "C" and needs to process the phase sequence { A, B, C } two times, i.e. { A, B, C, A, B, C }, it is possible to model this with three slots only (e.g. slotNumber values 1, 2, 3) and an additional repetition counter. This leads to the case where the slots are processed in the order { 1, 2, 3, 1, 2, 3 }, i.e. each slot is repeated. However, this concept does not contradict any rule of section 4.3.24.3.2. As this concept requires the consideration of a whole sequence, more details are discussed in section 4.3.24.3.5.3.

4.3.24.3.3 Sequences

4.3.24.3.3.1 Chronological classification of single sequence

This section defines some terms used to classify power sequences chronologically. It also imposes restrictions on the use of the power sequence's "state" information element ("alternatives. powerSequence. state").

1. "Past sequence":

A "past sequence" is a sequence that was relevant in the past. This means the sequence was processed or was at least offered. The sequence is neither "now" nor in the future offered or considered by the power sequences server. The power sequence's "state" information element SHALL be either "completed" or "invalid".

2. "Pre-announced sequence":

A "pre-announced sequence" is a sequence that can be relevant for a time not earlier than "now". However, it is not a "current sequence". The power sequence's "state" information element SHALL be either "scheduled" or "scheduledPaused" or "inactive".

3. "Current sequence":

A "current sequence" is a sequence that is currently processed by a power sequences server. The power sequence's "state" information element SHALL be either "running" or "paused".

Note: If a "pre-announced sequence" becomes a "current sequence" the natural "state" transition is as follows: "scheduled" changes to "running" and "scheduledPaused" changes to "paused".

Note: The above mentioned "relevance" does not only apply to sequences that were or will be executed. It also applies to sequences just offered as choice (to the bound power sequences client).

Note: The term "past sequence" does not say anything about the time frame the sequence was relevant for. A "past sequence" may have been a "pre-announced sequence" recently and notified with other "pre-announced sequences" within an alternatives group (see 4.3.24.3.3.2) by the power sequences server. Suddenly a sequence became "invalid" for any reason, hence is a "past sequence", now. This example shows a "past sequence" can belong to the same time frame as a "pre-announced sequence" (e.g.). Nevertheless, it is not offered any more by the power sequences server for a choice by the bound power sequences client.

4226

4227 *4.3.24.3.3.2 Chronological classification of alternatives group*

4228 Sequences belonging to the same "alternatives" element constitute an alternatives group (see
4229 section 4.3.24.2.2). This section defines some terms used to classify such a group chronologically. It
4230 also imposes rules on the creation of such groups.

4231 1. "Past group":

4232 A "past group" is a group that contains only "past sequences".

4233 2. "Pre-announced group":

4234 A "pre-announced group" is a group that contains only "pre-announced sequences".

4235 3. "Current group":

4236 A "current group" is a group that SHALL contain exactly one "current sequence".

4237 Furthermore, it MAY contain any number of "past sequences". It SHOULD NOT contain any
4238 "pre-announced sequence".

4239 Note: A power sequences server may face a condition where a previously notified "pre-announced
4240 sequence" is not valid anymore and becomes a "past sequence". According to the rules above this
4241 sequence SHALL NOT be part of any update of the alternatives group it formerly did belong to.
4242 However, it is still possible to notify the state update of the sequence only.

4243

4244 *4.3.24.3.3.3 Sequence schedule information*

4245 A sequence has "sequence schedule information" if all the following conditions are fulfilled:

- 4246 1. The information element "alternatives. powerSequence. schedule" is present and at least its
4247 child element "startTime" is present as well.

4248 Furthermore, the following rules apply:

- 4249 1. A "pre-announced sequence" SHALL NOT have a startTime referring to the past.
4250 2. If the child element "endTime" is available its value SHALL point to a time later than
4251 "startTime".

4252 Remark: For "sequence schedule modification" see 4.3.24.3.3.7.

4253

4254 *4.3.24.3.3.4 Sequence selection types*

4255 This section applies to "pre-announced groups" and "current groups" only.

4256 Esp. for a sensible control of a power sequences server it is important to identify if or which
4257 sequence of an alternatives group is "currently preferred or chosen" and if or which sequences are
4258 available as alternative (i.e. offered as choice to the bound power sequences client). The following
4259 selection types and rules are defined:

- 4260 1. "Chosen sequence":

- 4261 a. A sequence is called a "chosen sequence" if it has sequence schedule information. Its
 4262 state SHALL be "scheduled" or "scheduledPaused" if it is a "pre-announced
 4263 sequence". It SHALL be "running" or "paused" if it is a "current sequence".
 4264 b. If it is a "pre-announced sequence" the power sequences server SHALL execute the
 4265 sequence autonomously at the sequence's startTime (i.e. it becomes a "current
 4266 sequence") provided that the sequence is still valid and no other sequence of the
 4267 alternatives group has been selected (by the bound power sequences client (through
 4268 configuration) or by the power sequences server (for internal reasons or by user's
 4269 choice, e.g.)) as "chosen sequence".
 4270 Note: The autonomous start may well result in an immediate "paused" state if the
 4271 sequence's state was "scheduledPaused" before (see section 4.3.24.3.3.1).
 4272 c. Each "current sequence" SHALL be a "chosen sequence".
 4273 d. Within an alternatives group at maximum one sequence can be a "chosen
 4274 sequence".
- 4275 2. "Optional sequence":
 4276 a. A sequence that has no sequence schedule information and its state information
 4277 element set to "inactive" is called an "optional sequence".
 4278 b. An "optional sequence" SHALL NOT be executed by the power sequences server
 4279 autonomously. If, for some reason, there is a request or need to execute this
 4280 sequence it has to be modified to become a "chosen sequence".
 4281 c. Remark: Within a "pre-announced group" it is permissive to have only "optional
 4282 sequences" and no "chosen sequence".
- 4283 3. "Unavailable sequence":
 4284 a. A sequence that is neither a "chosen sequence" nor an "optional sequence" as called
 4285 an "unavailable sequence". It SHALL NOT have a sequence schedule information and
 4286 it SHALL be a "past sequence".
 4287 Please note: This section is NOT about "past groups".
 4288 b. An "unavailable sequence" SHALL NOT be executed by the power sequences server.

4289

4290 *4.3.24.3.3.5 Sequence modifiability*4291 *4.3.24.3.3.5.1 Introduction (informative)*

4292 Modifiability is required if a bound power sequences client wants to change any of the following data
 4293 of the power sequences server's sequence:

- 4294 1. The sequence overall schedule (information elements "startTime", "endTime").
 4295 2. The sequence state (interruptibility, i.e. pausing/resuming or aborting a current sequence).
 4296 3. The sequence's slot data (slot durations, activation/deactivation, energy/power).

4297 Dependent on the data further specific conditions may be required.

4298 Furthermore, modifiability is also required if a power sequences client wants to make any of the
 4299 offered "optional sequences" a "chosen sequence".

4300

4301 *4.3.24.3.3.5.2 Definition*

4302 A sequence is called "modifiable" if the sequence's information element
4303 "sequenceRemoteControllable" is available and set to "true". Otherwise it is not modifiable.

4304

4305 *4.3.24.3.3.6 Sequence interruptibility*4306 *4.3.24.3.3.6.1 Definition*

4307 A sequence is called "interruptible" if all of the following conditions are fulfilled:

- 4308 1. The sequence is modifiable.
- 4309 2. The information element "alternatives. powerSequence. operatingConstraintsInterrupt" is
4310 available AND at least one of the child elements "isPausable" and "isStoppable" are set to
4311 "true".

4312 Otherwise a sequence is not interruptible. This also means the absence of the information element
4313 "alternatives. powerSequence. operatingConstraintsInterrupt" assigns "isPausable" and
4314 "isStoppable" the value "false" implicitly.

4315 The process of interrupting/resuming a sequence may be further constrained by additional data as
4316 explained in 4.3.24.3.3.6.2.

4317

4318 *4.3.24.3.3.6.2 Process rules*

4319 Only "current sequences" and partly "pre-announced sequences" can be interrupted and resumed by
4320 the bound power sequences client. However, further conditions apply:

- 4321 1. A "current sequence" can be paused, resumed or aborted dependent on the interruptibility
4322 data (see 4.3.24.3.3.6.1) and the state of the sequence (see 4.3.24.4.5.1.2).
- 4323 2. A "chosen sequence" of a "pre-announced group" (i.e. a "chosen pre-announced sequence")
4324 can be changed as follows:
- 4325 a. Its state can be changed from "scheduled" to "scheduledPaused" if the sequence is
4326 interruptible and "isPausable" is set to true.
- 4327 b. Its state can be changed from "scheduledPaused" to "scheduled".
- 4328 c. It can be changed to become an "unavailable sequence", provided that the sequence
4329 is interruptible in general and "isStoppable" is set to "true". This requires the
4330 modification of the sequence's information element "state" to "invalid" (see
4331 4.3.24.4.5.1.2).

4332 Remark: This item just discusses "direct" state changes. It does not discuss "indirect" state
4333 changes (e.g. from the configuration of another sequence).

- 4334 3. There can be further constraints which need to be considered as explained below.

4335 The process of interrupting/resuming a power sequences server REQUIRES that the bound power
4336 sequences client advises the power sequences server to change its state. Consequently, a bound
4337 power sequences server accepts such commands only if it permits the proper interruptibility as
4338 stated above. Please note that this process is in general independent from a sequence configuration
4339 through the bound power sequences client.

4340 If a sequence is interruptible and further constraints are specified in "alternatives. powerSequence.
4341 operatingConstraintsInterrupt" or "alternatives. powerSequence. operatingConstraintsDuration",
4342 these constraints SHALL be considered by the bound power sequences client (esp. in order to
4343 prevent damage of the power sequences server).

4344 The information element "alternatives. powerSequence. operatingConstraintsResumeImplication"
4345 SHALL only be available if "alternatives. powerSequence. operatingConstraintsInterrupt" is available
4346 AND the child element "isPausable" is available. It SHALL NOT be considered if "isPausable" is set to
4347 "false".

4348

4349 *4.3.24.3.3.7 Sequence schedule modification*

4350 *4.3.24.3.3.7.1 Introduction (informative)*

4351 Section 4.3.24.3.3.3 describes "sequence schedule information" itself. This section (4.3.24.3.3.7)
4352 describes some rules to set or modify "sequence schedule information".

4353

4354 *4.3.24.3.3.7.2 Definition*

4355 A sequence is called to have a "modifiable sequence schedule" if all of the following conditions are
4356 fulfilled:

- 4357 1. The sequence is either a "chosen sequence" or an "optional sequence".
- 4358 2. The sequence is modifiable (see 4.3.24.3.3.5).
- 4359 3. The information element "alternatives. powerSequence. scheduleConstraints" is available.

4360 Otherwise a sequence's schedule information is not modifiable.

4361

4362 *4.3.24.3.3.7.3 Process rules*

4363 If a "chosen sequence" has a "modifiable sequence schedule" a bound power sequences client MAY
4364 modify the sequence's "sequence schedule information".

4365 If an "optional sequence" has a "modifiable sequence schedule" a bound power sequences client
4366 MAY assign the sequence a "sequence schedule information".

4367

4368 *4.3.24.3.3.8 Sequences configuration*

4369 *4.3.24.3.3.8.1 Introduction (informative)*

4370 This section discusses the relation and dependency of modifiability aspects as described in section
4371 4.3.24.3.3.5.1. Furthermore it describes how the configuration of one sequence may affect another
4372 sequence of the same alternatives group.

4373

4374 *4.3.24.3.3.8.2 Definition*

4375 A sequence is called "configurable" if all of the following conditions are fulfilled:

- 4376 1. The sequence has a "modifiable sequence schedule" according to section 4.3.24.3.3.7.
4377 2. The sequence belongs to a "current group" or "pre-announced group".

4378 Otherwise a sequence is called "not configurable".

4379

4380 4.3.24.3.3.8.3 Process rules

4381 Throughout this section only "current groups" and "pre-announced groups" are considered.

4382 Furthermore, only "chosen sequences" and "optional sequences" of these groups are considered and
4383 are called "applicable sequences" within this section. The following rules apply:

- 4384 1. Applies to "pre-announced groups" only: Each "optional sequence" SHALL be configurable.
4385 2. Applies to "pre-announced groups" only: If a bound power sequences client assigns a
4386 configurable "optional sequence" a "sequence schedule information" (see 4.3.24.3.3.7.3) the
4387 sequence becomes then a "chosen sequence". If there was another sequence of the
4388 alternatives group a "chosen sequence", this other sequence becomes an "optional
4389 sequence" or (implementation specific) a "past sequence" then.
4390 3. A "current sequence" SHALL NOT become an "optional sequence" through the configuration
4391 of another sequence that is an "optional sequence" of the alternatives group. I.e. once a
4392 sequence becomes a "current sequence" it remains the "current sequence" until it completes
4393 successfully or is aborted finally (if applicable).
4394 4. Each "applicable sequence" that is configurable MAY be interruptible according to section
4395 4.3.24.3.3.6.
4396 5. Each "applicable sequence" that is configurable MAY have configurable slots according to
4397 section 4.3.24.3.2.7. If a sequence needs to have configurable slots the sequence itself SHALL
4398 be configurable.
4399 6. A configurable sequence MAY become a non-configurable sequence or vice versa at any
4400 time. Whether this happens is up to the power sequences server and implementation
4401 specific. However, specific process rules may impose restrictions on such changes.

4402 Note: If a "chosen sequence" becomes an "optional sequence" (or even "unavailable sequence") its
4403 information changes. Among others, this can "remove" a sequence's schedule information (see
4404 section 4.3.24.3.3.4).

4405 Please note that the configuration of a sequence through the bound power sequences client is in
4406 general independent from a sequence's interruptibility. However, dependent on the implementation
4407 or further process rules it may well happen that the process of sequence configuration leads to a
4408 change of the sequence's interruptibility.

4409

4410 4.3.24.3.4 Overall energy management information

4411 This section describes briefly some "global information" and rules on its use.

4412 The information element "nodeScheduleInformation" contains information on the device's overall
4413 energy management state, i.e. regardless of a specific sequence or slot. Some of its child elements
4414 are rather static (constant) while others typically vary over time. However, the following rule applies
4415 in every case: The content of "nodeScheduleInformation" SHALL NOT contradict the sequence- or

4416 slot-specific information. This also means that the content does not "supersede" sequence- or slot-
4417 specific information. This is relevant esp. for the following information:

4418 1. Remote controllability

4419 A power sequences server is "remote controllable" exactly if it has at least one sequence that
4420 is configurable (see 4.3.24.3.3.8.2).

4421 2. Scheduling of single slots vs. multiple slots

4422 Even if a power sequence permits the configuration of slots it might be that the power
4423 sequences server requires individual configuration messages for each slot. I.e. it may not
4424 support the configuration of several slots within a single message. This is denoted by the
4425 element "supportsSingleSlotSchedulingOnly".

4426 3. One-time sequence selection

4427 A power sequences server may offer to select a sequence (i.e. an "optional sequence"
4428 becomes a "chosen sequence"). As long as the sequence did not start it might be possible to
4429 alter the selection, i.e. make another sequence a "chosen sequence". However, some device
4430 may not permit such multiple re-selections and permit just a one-time selection. The device's
4431 possibility for re-selection is denoted by the element "supportsReselection".

4432 The information element "nodeScheduleInformation" can contain more than the items mentioned
4433 above. However, these are explained in the definition of the element and are not relevant for
4434 process considerations.

4435

4436 *4.3.24.3.5 Rules for creation of information*

4437 This section primarily focuses on rules to be considered by the power sequences server. However,
4438 the rules are relevant for a power sequences client as well.

4439

4440 *4.3.24.3.5.1 On the use of "alternativesId"*

4441 As described in section 4.3.24.3.3.2 each "alternatives" element of the server's power sequences
4442 information contains the power sequences of the same alternatives group. Each of these groups can
4443 be identified with the element "alternatives. relation. alternativesId". Therefore, each alternatives
4444 group exposed by a power sequences server SHALL always have a unique value in "alternativesId",
4445 i.e. there SHALL never be concurrently valid, duplicate values of "alternativesId". Furthermore, a
4446 power sequences server SHALL NOT reuse values of "alternativesId" immediately after an
4447 alternatives group is not present anymore on the power sequences server.

4448 As power sequence IDs are unique across all alternatives groups (see section 4.3.24.3.5.2) the use
4449 and evaluation of alternativesId is required just in case of "restricted function exchange" (see
4450 [ProtocolSpecification]). This is in particular the case if a new power sequence is notified for an
4451 already existing alternatives group.

4452

4453 *4.3.24.3.5.2 On the use of (power) "sequenceId"*

4454 Each power sequence exposed by a power sequences server SHALL always have a unique value in its
4455 power sequence ID (element "sequenceId"), i.e. there SHALL never be concurrently valid, duplicate

power sequences IDs. Furthermore, a power sequences server SHALL NOT reuse power sequences IDs immediately after sequences with the according IDs have been completed. As an example: A washing machine should not expose its cycles with a fixed sequence ID of "1" all the time. Instead, it should increase the sequence ID with every new sequence. In general, a power sequence ID is just a simple identifier (to select a specific sequence from a list) and SHALL NOT be interpreted to carry any additional information, such as information about which sequence runs after another judging from ascending or descending IDs.

4463

4.3.24.3.5.3 On the embedding of slots in sequences

Sequences are used together with slots within this specification.

4466

4.3.24.3.5.3.1 Common rules

The following common rules apply:

1. Each sequence SHALL have at least one slot.
2. Each sequence SHALL EITHER have only determined slots OR exactly one undetermined slot (see 4.3.24.3.2.2).
3. Applies only to sequences with at least one determined slot: For at least one slot it SHALL NOT be possible to deactivate it and it SHALL always have a duration greater than 0 seconds. Remark: This requirement has no impact on the potential possibility to "deactivate" the sequence itself.

4476

4.3.24.3.5.3.2 Embedding slots in case of "repetition concept"

Some additional considerations are required in case of the "repetition concept" as briefly explained in section 4.3.24.3.2.10. Again we consider the example where a sequence consists of three different phases "A", "B", "C" that are processed two times, resulting in an overall phase cycle { A, B, C, A, B, C }. We assume this is represented with three slots 1, 2, 3 and accordingly the number of executions is set in the element "alternatives. powerSequence. description. repetitionsTotal" to two (please note that "repetition" here means "number of execution"). Then, the number of the current repetition has to be set in the element "alternatives. powerSequence. state. activeRepetitionNumber".

As just three explicit slot information entities are used in this example (instead of the effectively six phases/slots) this means that rules have to be defined and considered whether a slot's information belongs to the first repetition (i.e. first execution) only or to every repetition or is updated with each repetition. These rules are defined in the element "alternatives. powerSequence. slotInformation" of message "smartEnergyManagementPsData". As a rule of thumb, slot information of this message usually belongs to the current repetition unless stated otherwise. This means information for slot number 1 still belongs to the current repetition (in contrast to the next repetition) even if it was already processed and slot number 2 is currently processed. Constraints, on the other hand, usually (but not always) apply to each repetition. In general a configuration of slots shall be considered to apply to every repetition.

4496

4497 *4.3.24.3.5.4 Prevent group overlap in time*

4498 Pre-announced alternatives groups SHALL NOT overlap in time. No pre-announced alternatives group
4499 SHALL overlap in time with a current alternatives group.

4500 This means no schedule or schedule constraints of any sequence of alternatives group "A" shall
4501 overlap with any schedule or schedule constraints of any sequence of alternatives group "B".

4502 Remark: Of course, sequences of the same alternatives group may well overlap in time.

4503 As a consequence, some restrictions SHALL be applied if

- 4504 1. EITHER one current alternatives group and at least one pre-announced group,
4505 2. OR at least two pre-announced groups

4506 are created:

- 4507 1. All pre-announced sequences of the alternatives group that is later in time SHALL have a
4508 schedule with startTime AND endTime or they SHALL have sequence schedule constraints
4509 with earliestStartTime AND latestEndTime.

4510

4511 *4.3.24.3.5.5 Permitted sequence types*4512 *4.3.24.3.5.5.1 Introduction*

4513 The data models, rules (and messages described subsequently) are designed to permit a limited
4514 number of different sequence types or processes. These types are briefly described in the following
4515 sections. The description primarily makes use of terms defined in previous sections and is extended
4516 with specific restrictions or rules where required. This approach shall also help to compare the
4517 sequence types.

4518

4519 *4.3.24.3.5.5.2 Sequence type "fixed forecast"*

4520 This type is designed to notify a sequence where a power sequences client will not be given any kind
4521 of control. I.e. this sequence denotes a completely autonomous schedule of the power sequences
4522 server.

4523 A power sequence with this sequence type SHALL be initially announced as follows:

- 4524 1. It SHALL be notified by the power sequences server before its execution as "pre-announced
4525 sequence" unless the sequence was created suddenly and begins immediately. If it was
4526 created suddenly and begins immediately it SHALL be notified as "current sequence". In
4527 either case the sequence SHALL be the only sequence in the alternatives group (which is
4528 accordingly a "pre-announced group" or "current group", resp.).
4529 2. It SHALL be notified as "chosen sequence". Additionally, its "alternatives. powerSequence.
4530 schedule" child element "endTime" SHALL be set.

4531 Remark: Of course, slot duration uncertainties can have an impact on the precision of the
4532 whole sequence's end time as well.

- 4533 3. The sequence SHALL NOT be interruptible.
4534 4. The sequence SHALL NOT be configurable.
4535 5. The sequence SHALL have at least one slot.
4536 6. All slots of the sequence SHALL be determined.
4537 7. All slots of the sequence SHALL have an implicit slot schedule (hence are also not
4538 configurable).

4539 If the sequence becomes a "current sequence" it SHALL remain non-interruptible and non-
4540 configurable.

4541

4542 4.3.24.3.5.5.3 Sequence type "flexible chosen forecast"

4543 This type is designed to notify a sequence where a power sequences client usually will be given some
4544 control. I.e. this sequence denotes the preference or current configuration a power sequences server
4545 will execute autonomously unless the power sequences client takes its possibilities to modify the
4546 schedule or state.

4547 A power sequence with this sequence type SHALL be initially announced as follows:

- 4548 1. It SHALL be notified by the power sequences server before its execution as "pre-announced
4549 sequence". It SHALL only be notified within a "pre-announced group".
4550 2. It SHALL be notified as "chosen sequence". Additionally, its "alternatives. powerSequence.
4551 schedule" child element "endTime" MAY be set.
4552 3. The sequence MAY be interruptible.
4553 4. The sequence SHALL be configurable.
4554 5. The sequence SHALL have at least one slot.
4555 6. All slots of the sequence SHALL be determined.
4556 7. Each slot SHALL have EITHER a constrained schedule OR an implicit schedule. The general
4557 rules on configurable and non-configurable slots of section 4.3.24.3.2.7 apply.

4558 If the sequence becomes a "current sequence" its interruptibility and configurability MAY be changed
4559 by the power sequences server.

4560

4561 4.3.24.3.5.5.4 Sequence type "flexible optional forecast"

4562 This type is similar to the type of section 4.3.24.3.5.5.3 but denotes just an option. I.e. a power
4563 sequences server will NOT execute this sequence autonomously. However, a power sequences client
4564 can take its possibilities to modify the schedule or state in order to make this sequence a "chosen
4565 sequence" (see below also).

4566 A power sequence with this sequence type SHALL be initially announced as follows:

- 4567 1. It SHALL be notified by the power sequences server before its execution as "pre-announced
4568 sequence". It SHALL only be notified within a "pre-announced group".
4569 2. It SHALL be notified as "optional sequence".
4570 3. The sequence MAY be interruptible.
4571 4. The sequence SHALL be configurable.

- 4572 5. The sequence SHALL have at least one slot.
- 4573 6. All slots of the sequence SHALL be determined.
- 4574 7. Each slot SHALL have EITHER a constrained schedule OR an implicit schedule. The general
- 4575 rules on configurable and non-configurable slots of section 4.3.24.3.2.7 apply.
- 4576 See section 4.3.24.3.3.8 in order to modify the kind of the sequence or its state.
- 4577 In case this sequence becomes a "chosen sequence" please see section 4.3.24.3.5.5.3, as this change
- 4578 corresponds to the change of the sequence type.
- 4579
- 4580 *4.3.24.3.5.5.5 Sequence type "optional sequence-based immediate control"*
- 4581 This type is designed for devices that cannot forecast their consumption or production with
- 4582 "sufficient precision" but that can at least announce an optional task with upper or lower bounds
- 4583 (e.g. schedule constraints or min/max values). For this sequence type the client is given at least some
- 4584 kind of control in order to pause and resume this task (considering constraints imposed by the power
- 4585 sequences server). This sequence type is initially optional and requires a proper configuration from
- 4586 the power sequences client in order to start at the configured time. Afterwards the power sequences
- 4587 client can use its options on the interrupt/resumption of the sequence.
- 4588 A power sequence with this sequence type SHALL be initially announced as follows:
- 4589 1. It SHALL be notified by the power sequences server before its execution as "pre-announced
- 4590 sequence". It SHALL only be notified within a "pre-announced group".
- 4591 2. It SHALL be notified as "optional sequence".
- 4592 3. The sequence SHALL be interruptible.
- 4593 4. The sequence SHALL be configurable.
- 4594 5. The sequence SHALL have exactly one slot.
- 4595 6. The slot of the sequence SHALL be undetermined.
- 4596 Remark: This item and item 5 are in accordance with the rules in section 4.3.24.3.5.3.
- 4597 7. Remark: Due to item 6 the slot itself is not configurable.
- 4598 8. The child element "endTime" of "alternatives. powerSequence. schedule" SHALL NOT be set
- 4599 unless the sequence becomes a "past sequence". I.e. neither can the power sequences
- 4600 server announce "endTime" in advance nor after the configuration of the schedule nor shall
- 4601 the power sequences client try to configure "endTime".
- 4602 9. Remark: The absence of any duration prevents a price forecast (see 4.3.24.4.9.1).
- 4603 See section 4.3.24.3.3.8 in order to modify the kind of the sequence or its state.
- 4604 If the sequence becomes a "current sequence" it SHALL NOT be configurable any more. Its
- 4605 interruptibility MAY be changed by the power sequences server (temporarily).
- 4606 Please note: This sequence type typically requires at the power sequences client a strong
- 4607 consideration of constraints imposed by the power sequences server. Furthermore, this sequence
- 4608 type is typically more likely to be aborted at some point of time by the power sequences server
- 4609 autonomously.
- 4610

4611 *4.3.24.3.6 Rules for evaluation*

4612 This section primarily focuses on rules to be considered by the power sequences client. It is based on
4613 the sequence grouping concept and section 4.3.24.3.5.

4614 A power sequences client should analyse received information on sequences as follows:

- 4615 1. Order alternatives groups chronologically:
4616 Received alternatives groups SHALL be classified chronologically according to section
4617 4.3.24.3.3.2. Afterwards the chronological order of current or pre-announced groups shall be
4618 derived according to the rules defined in section 4.3.24.3.5.4 to prevent overlapping groups.
- 4619 2. Within each alternatives group consider rules of section 4.3.24.3.5.

4620

4621 *4.3.24.3.7 Conventions*

4622 This section describes some further conventions that SHALL apply in every case.

4623

4624 *4.3.24.3.7.1 Production vs. generation, sign convention:*

4625 The basic PowerSequences class defines the element "positiveEnergyDirection" in order to model
4626 whether positive or negative values (of power/energy) denote the production of energy. This
4627 element is not used explicitly within the complex class of this specification and is replaced by the
4628 following convention instead:

- 4629 1. Power and energy values less than zero (negative value sign) denote the production of
4630 energy.
- 4631 2. Power and energy values greater than zero (positive value sign) denote the consumption of
4632 energy.

4633 This also means an implicit assignment of "positiveEnergyDirection = consume".

4634

4635 *4.3.24.3.7.2 Update of time information elements*

4636 All time information that identifies a point in time SHALL be updated continuously with every
4637 message communicated (e.g. the startTime of a sequence). Time information that specifies a
4638 duration of some slot or sequence that do not change over time SHALL keep the same value (e.g. the
4639 defaultDuration of a slot).

4640

4641 *4.3.24.3.7.3 Contiguous sequence data*

4642 Within a message instance, information on a given power sequence SHALL be contiguous. I.e.
4643 information on the power sequence with ID "1" is not interrupted with information on the power
4644 sequence with ID "2". The contiguity also means each power sequence SHALL be split into as few
4645 data groups as possible (unless stated otherwise by a specific message definition).

4646

4647 *4.3.24.3.8 Role concept*

4648 If a device is a power sequences server and a power sequences client at the same time, the instances
4649 of the power sequences server and the power sequences client SHALL NOT be present on the same
4650 entity. There SHALL NOT be more than one instance of a power sequences server on the same entity
4651 as well. However, there MAY be multiple instances of power sequences clients on one entity.

4652

4653 *4.3.24.3.9 Binding considerations*

4654 Power sequences servers SHALL only accept one binding to a power sequences client for each
4655 instance of a power sequences server entity. Once the power sequence instance is bound to a client,
4656 the power sequences server SHALL reject additional binding requests for this power sequence
4657 instance.

4658 If a power sequences server detects that a bound power sequence client is not accessible any more
4659 (e.g. since it has been sold to another household), it is a vendor specific implementation to either
4660 delete the binding or continue waiting for the bound device. Thus, it might only be possible for
4661 device owners to re-bind the power sequences server with a factory reset of the device.

4662 Devices which are interested in data of a power sequences server, but don't configure the power
4663 sequences server flexibilities, SHALL only use data subscriptions or explicit read commands without
4664 bindings to receive updated information and SHALL never try to create bindings to a power
4665 sequences server.

4666 For more details about binding, please consider [ProtocolSpecification].

4667

4668 *4.3.24.3.10 General error handling*

4669 *4.3.24.3.10.1 Acknowledgement concept*

4670 The acknowledgement concept as specified in [ProtocolSpecification] applies. A bound power
4671 sequences server and bound power sequences client SHOULD explicitly request acknowledgement
4672 messages for message exchanged among each others (provided the message type permits
4673 acknowledgement request).

4674

4675 *4.3.24.3.10.2 Valid vs. invalid classifiers*

4676 Subsequent message specific sections contain sub-sections with the headline "When to send". These
4677 sub-sections determine explicitly under which condition a given message shall be sent and also which
4678 classifier(s) shall be used. These classifiers are subsequently considered valid for the respective
4679 messages and conditions. Other classifiers are considered invalid for the respective messages.

4680 A brief overview of valid classifiers is given in Table 119. Furthermore, sub-sections with the headline
4681 "When received" usually mention valid classifiers only.

4682

4.3.24.4 Message purpose specification details

This section defines details of the message purposes that were briefly announced in Table 119 of section 4.3.24.1.

4.3.24.4.1 powerSequences information

4.3.24.4.1.1 General data model description and rules

4.3.24.4.1.1.1 General aspects

With a power sequences information message a power sequences server exposes all power sequences (potentially) relevant for the power sequences client, which may be a combination of:

- Alternative processes to choose from:

This has already been introduced in section 4.3.24.2.2 as choice of power sequences: A device may have different operation modes or (more generally speaking) mutually exclusive processes, which it offers to a power sequences client. The client can only choose one of these mutually exclusive processes. These processes SHALL be exposed within one "alternatives" element.

- Single or multiple, sequenced processes:

A device may have different processes which are executed in a serialized manner, i.e. cannot be parallelized or run concurrently. These processes SHALL be exposed within different "alternatives" elements.

A power sequences server can also expose a combination of these options. As an example, we consider the case of an HVAC system, which knows that it has to run in the morning and in the evening. These processes cannot be run concurrently. However, it has 2 modes of heating: One with high power consumption, but shorter heating cycle and one with a lower power consumption and longer heating cycle. In this case, the HVAC system would expose 2 "alternatives" elements, each containing 2 power sequences. One of these pairs of power sequences would reflect the different power consumption modes during the morning whereas the second pair does the same for the evening.

4.3.24.4.1.1.2 Cancelling (deleting/aborting) offered power sequences

There might be situations in which a power sequences server offers one or more power sequences, but encounters a situation which renders one or more of the offered power sequence(s) invalid. In such situations the affected power sequences SHALL be cancelled/deleted. As an example, a washing machine might offer a power sequence representing a washing cycle, but a few seconds later the user of the washing machine cancels the washing cycle, because he has chosen the wrong washing program. If a power sequences server has offered one or more power sequences which are now invalid, it SHALL send a "powerSequences information" with a notify classifier and set the according power sequence to an "invalid" state. Afterwards it SHALL remove the according power sequence ID and respect the rules on reusing sequence IDs (see section 4.3.24.3.5.2). The deletion of the sequence MAY result in a delete-notification (see section "Restricted function exchange with cmdOptions" in [ProtocolSpec]), but it is not recommended to do so.

4723 In general, a power sequences server SHALL cancel all power sequences that have been announced
4724 and possibly also configured, but are no longer valid.

4725 Errors might occur if a power sequences client does not receive the notification(s) about cancelled
4726 power sequence(s), e.g. because the power sequences client is currently offline. However, in these
4727 situations the power sequences client SHOULD be aware of potentially lost packets and SHOULD
4728 apply appropriate mechanisms, such as explicitly requesting the power sequences available.

4729

4730 *4.3.24.4.1.1.3 Processes that might be run in parallel*

4731 Please note a single instance of a power sequences server SHALL NOT expose processes that
4732 (potentially or partly) run at the same time! To give an example: Consider a fridge/freezer
4733 combination with independent temperature control. I.e. for both the fridge temperature control as
4734 well as for the freezer temperature control a process can be exposed. However, as these processes
4735 are independent from each other (hence can also run in parallel) each process has to be exposed by
4736 an individual power sequences server instance. This means there has to be a power sequences server
4737 for cooling processes and another power sequences server for freezing processes. As stated in
4738 section 4.3.24.3.8, these power sequences server instances SHALL be located on different entities.

4739

4740 *4.3.24.4.1.1.4 Power sequences with and without flexibilities*

4741 Please note that power sequences might or might not contain flexibilities, i.e. offer time-shifting of a
4742 power sequence or slot or offer slots which have durations which can be reduced or extended, i.e.
4743 have a configurable duration. A power sequence with schedule flexibilities SHALL contain schedule
4744 constraints. A power sequence not containing flexibilities, i.e. a fixed forecast, SHALL NOT contain
4745 schedule constraints.

4746

4747 *4.3.24.4.1.1.5 Consistency of a power sequence*

4748 In case of a sequence with flexible schedule (and determined slots), a sequence's "earliest start time"
4749 and "latest end time" SHALL always frame the slots that are part of the sequence. I.e. the sum of
4750 durations of the slots in a sequence SHALL always fit into the frame given by these elements. Care
4751 has to be taken if slots can be extended regarding their duration, i.e. which have a configurable
4752 duration: Extending all durations to their maximum may exceed the time frame permitted by a
4753 sequence's "earliest start time" and "latest end time". I.e. such configurations are invalid and shall be
4754 rejected by the power sequences server. However, there SHALL at least be one possible
4755 configuration of the duration of all slots in a sequence that fits into the frame given by "earliest start
4756 time" and "latest end time" of the sequence.

4757 As an example, consider the slots 1, 2 and 3 of a sequence. All of these slots may have a typical
4758 duration of 5 minutes, but all of them can be extended so they have a duration of 15 minutes each.
4759 This results in a minimum length of 15 minutes and a maximum length of 45 minutes for the whole
4760 sequence. However, the power sequences server might be limited to an "earliest start time" and
4761 "latest end time", which only provide a duration of 35 minutes for the whole sequence. In this
4762 example, the power sequences client would be free to either extend the duration of all slots equally
4763 so they fit into the frame, but it might also only extend slot 1 and 2 to their maximum duration. The

4764 power sequences client might, however, also choose to not extend the duration of any of the slots at
4765 all.

4766

4767 4.3.24.4.1.1.6 Possible state values

4768 The states delivered in a power sequences state can have several, pre-defined values. This Feature
4769 Type uses as well the definition of PowerSequenceStateType (see Table 284, section 5.3.19.9.2), but
4770 restricts the permitted values. The following values are allowed within the scope of this specification:

- 4771 • "running": The according sequence is currently executed.
- 4772 • "paused": The according sequence is paused.
- 4773 • "scheduled": The according sequence will be executed by the power sequences server at the
4774 scheduled time. It will then usually be in the state "running".
- 4775 • "scheduledPaused": The according sequence will be executed by the power sequences server
4776 at the scheduled time – but it will then usually be in the state "paused".
- 4777 • "inactive": The according sequence has been offered as optional choice, but has not been
4778 configured.
- 4779 • "completed": The according sequence has been completed.
- 4780 • "invalid": The according sequence has been cancelled. Please consider section 4.3.24.4.1.1.2.

4781

4782 4.3.24.4.1.1.7 Data model description and rules

4783 The complex function "smartEnergyManagementPsData" of the complex class
4784 "SmartEnergyManagementPs" is used with the following restrictions and rules:

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsData.		
nodeScheduleInformation	M	
nodeScheduleInformation. nodeRemoteControllable	M	If nodeRemoteControllable is equal to "false" no power sequence is remote controllable, independent from the value of the element "sequenceRemoteControllable". If nodeRemoteControllable is set to "true", it depends on the value of the element "sequenceRemoteControllable" if a specific sequence is remote controllable, or not.
nodeScheduleInformation. supportsSingleSlotSchedulingOnly	M	If set to "true" the power sequences server applies a restriction on received "powerSequences configuration" messages: The element "alternatives. powerSequence" SHALL NOT contain more than one child element "powerTimeSlot". This restriction does not apply if supportsSingleSlotSchedulingOnly is set to "false".
nodeScheduleInformation. alternativesCount	M	

nodeScheduleInformation. totalSequencesCountMax	M	
nodeScheduleInformation. supportsReselection	M	
alternatives	O	SHALL only be left out if no power sequences are present within its group. Otherwise, it SHALL be present.
alternatives. relation	M	
alternatives. relation. alternativesId	M	PRIMARY IDENTIFIER of "alternatives" groups. See section 4.3.24.3.5.1.
alternatives. powerSequence	M	
alternatives. powerSequence. description	M	
alternatives. powerSequence. description. sequenceId	M	PRIMARY IDENTIFIER of powerSequence.
alternatives. powerSequence. description. description	O	Default rules of section 3.10.1.3.1 SHALL be applied.
alternatives. powerSequence. description. powerUnit	O	Element SHOULD be set. If omitted, "W" SHALL be assumed as unit for the value implicitly.
alternatives. powerSequence. description. energyUnit	O	Element SHOULD be set. If omitted, "Wh" SHALL be assumed as unit for the value implicitly.
alternatives. powerSequence. description. valueSource	O	If absent, the source of the forecasted values is undefined.
alternatives. powerSequence. description. taskIdentifier	O	
alternatives. powerSequence. description. repetitionsTotal	O	MAY be present. Absence of the element is equal to a presence with a value of 1 (one). SHALL be absent if the value is 1. If a power sequence executes its sequence of slots more than one time, the element SHALL be present and contain the total number of executions.
alternatives. powerSequence. state	M	
alternatives. powerSequence. state. state	M	Consider esp. section 4.3.24.3.3.1 on restrictions of permitted values from the enumeration!

alternatives. powerSequence. state. activeSlotNumber	O	If state is set to "running" or "paused" this element SHALL contain the currently active slot number. Otherwise it SHALL be omitted.
alternatives. powerSequence. state. elapsedSlotTime	O	If state is set to "running" or "paused" AND the slot is determined (see 4.3.24.3.2.2) this element MAY contain the time the slot has already been in "running" state (this also means the value remains constant during a "paused" state). Otherwise it SHALL be omitted.
alternatives. powerSequence. state. remainingSlotTime	O	If state is set to "running" or "paused" AND the slot is determined (see 4.3.24.3.2.2) this element SHALL contain the time the slot still needs to be in "running" state (this also means the value remains constant during a "paused" state). Otherwise it SHALL be omitted.
alternatives. powerSequence. state. sequenceRemoteControllable	M	Denotes whether the sequence is modifiable (value "true") or not (value "false"), see 4.3.24.3.3.5. Modifiability is required to configure sequences and slots (see section 4.3.24.4.4). It is also required to change a power sequence state (if supported; see 4.3.24.4.5).
alternatives. powerSequence. state. activeRepetitionNumber	O	SHALL be present if alternatives. powerSequence. description. repetitionsTotal is present and has a value greater than 1. Otherwise, it SHALL be absent.
alternatives. powerSequence. state. remainingPauseTime	O	This element SHALL ONLY be present if the power sequence is interruptible. Otherwise, it SHALL be omitted. In case of the sequence's interruptibility the following rules apply: If the element is absent this means there is no explicit pause duration restriction for the current slot; a value of 0s denoted the slot does not permit being paused.
alternatives. powerSequence. schedule	O	If the sequence has a schedule, this element with its sub-elements SHALL be present. Otherwise it SHALL be omitted.
alternatives. powerSequence. schedule. startTime	M	
alternatives. powerSequence. schedule. endTime	O	If the power sequence has an endTime, it SHALL be put here. Otherwise the element SHALL be omitted.
alternatives. powerSequence. scheduleConstraints	O	If there are constraints available, this element and its sub-elements SHALL be present. Otherwise it SHALL be omitted.

alternatives. powerSequence. scheduleConstraints. earliestStartTime	M	Only "xs:duration" value types SHALL be used to denote a relative time which relates to "now" as time 0.
alternatives. powerSequence. scheduleConstraints. latestEndTime	M	Only "xs:duration" value types SHALL be used to denote a relative time which relates to "now" as time 0.
alternatives. powerSequence. schedulePreference	O	MAY be present. A power sequences client should analyse this element when it configures the power sequences server. Note: There may be several options available which cannot be fulfilled altogether. This means the options can in general only be considered "as good as it can get".
alternatives. powerSequence. schedulePreference. greenest	O	MAY be present. Absence of this element is equal to the presence with value "false".
alternatives. powerSequence. schedulePreference. cheapest	O	MAY be present. Absence of this element is equal to the presence with value "false".
alternatives. powerSequence. operatingConstraintsInterrupt	O	If the sequence may be paused or stopped by the power sequences client, this element and the required sub-elements SHALL be present. Otherwise it SHALL be omitted.
alternatives. powerSequence. operatingConstraintsInterrupt. isPausable	O	If the sequence is pausable by the bound power sequences client, this element SHALL be present and set to true. Otherwise it SHALL be omitted or set to false (default value).
alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable	O	If the sequence is stoppable by the bound power sequences client, this element SHALL be present and set to true. Otherwise it SHALL be omitted or set to false (default value).
alternatives. powerSequence. operatingConstraintsInterrupt. maxCyclesPerDay	O	
alternatives. powerSequence. operatingConstraintsDuration	O	The element SHALL ONLY be present if at least one of its child elements is present. Otherwise it SHALL be omitted.
alternatives. powerSequence. operatingConstraintsDuration. activeDurationMin	O	
alternatives. powerSequence. operatingConstraintsDuration. activeDurationMax	O	

alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMin	O	
alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMax	O	
alternatives. powerSequence. operatingConstraintsDuration. activeDurationSumMin	O	
alternatives. powerSequence. operatingConstraintsDuration. activeDurationSumMax	O	
alternatives. powerSequence. operatingConstraintsResumeI mplication	O	The element SHALL ONLY be present if at least one of its child elements is present. Otherwise it SHALL be omitted.
alternatives. powerSequence. operatingConstraintsResumeI mplication. resumeEnergyEstimated	O	
alternatives. powerSequence. operatingConstraintsResumeI mplication. energyUnit	O	For electric energy, this element SHALL be omitted and Wh SHALL be assumed as unit for the value implicitly. For any other energy type, the unit SHALL be stated explicitly.
alternatives. powerSequence. operatingConstraintsResumeI mplication. resumeCostEstimated	O	
alternatives. powerSequence. operatingConstraintsResumeI mplication. currency	O	
alternatives. powerSequence. powerTimeSlot	M	
alternatives. powerSequence. powerTimeSlot. schedule	M	
alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	M	SUB IDENTIFIER within powerSequence.

alternatives. powerSequence. powerTimeSlot. schedule. timePeriod	O	SHALL ONLY be present if at least one of its child elements is present.
alternatives. powerSequence. powerTimeSlot. schedule. timePeriod. startTime	O	
alternatives. powerSequence. powerTimeSlot. schedule. timePeriod. endTime	O	The following equation SHALL apply: $endTime - startTime = defaultDuration$.
alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration	O	SHALL be present in case of "determined slot" and state the duration of the slot. If the slot has a configurable length, this element SHALL reflect the currently configured length.
alternatives. powerSequence. powerTimeSlot. schedule. durationUncertainty	O	May only be used if alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration is set. Otherwise it SHALL be omitted.
alternatives. powerSequence. powerTimeSlot. schedule. slotActivated	O	If the slot is optional this element SHALL be present and reflect the current status of the slot. If the slot is not optional, this element SHALL be absent.
alternatives. powerSequence. powerTimeSlot. schedule. description	O	The string-length SHALL NOT be longer than 60 characters.
alternatives. powerSequence. powerTimeSlot. valueList	M	
alternatives. powerSequence. powerTimeSlot. valueList. value	M (1..6)	The following restriction to the list size applies: 1..6 occurrences of "value" are permitted. If more than just one "value" in "valueList" is present, each "value" in "valueList" SHALL state a different valueType than each other "value" in "valueList" of this slot. See section 4.3.24.3.2.5 for details about allowed values.
alternatives. powerSequence. powerTimeSlot. valueList. value. valueType	M	
alternatives. powerSequence. powerTimeSlot. valueList. value. value	M	

alternatives. powerSequence. powerTimeSlot. scheduleConstraints	O	If a slot has information about constraints for a time slot, this element and the required sub-elements SHALL be present. Otherwise AND for fixed forecasts (i.e. sequences without any flexibility), this element SHALL be omitted. Note: In case of a power sequence with repetition of its sequence of slots, this element often applies to the first execution only! Details are explained in the description of each child element.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. earliestStartTime	O	Only "xs:duration" value types SHALL be used to denote a relative time which relates to "now" as time 0. Note: This element applies to the first execution of the slot number only!
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. latestEndTime	O	Only "xs:duration" value types SHALL be used to denote a relative time which relates to "now" as time 0. Note: This element applies to the first execution of the slot number only!
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. minDuration	O	If the slot has a configurable duration, this element SHALL be present and denote the minimum supported configuration. Please also see section 4.3.24.3.2.7. Note: This element applies to every execution (repetition) of the slot number.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. maxDuration	O	If the slot has a configurable duration, this element SHALL be present and denote the maximum supported configuration. Please also see section 4.3.24.3.2.7. Note: This element applies to every execution (repetition) of the slot number.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. optionalSlot	O	SHALL be present and set to "true" if the slot can be omitted. Otherwise the element SHALL be omitted or set to "false". I.e. the value "false" SHALL be assumed implicitly if the element is omitted. Note: This element applies to every execution (repetition) of the slot number.

Table 120: Description of complex class function "smartEnergyManagementPsData" for powerSequences information

4.3.24.4.1.2 When received

If a power sequences server receives this message, it SHALL consider this as erroneous receipt.

4790 *4.3.24.4.1.3 When to send*

4791 A power sequences server SHALL produce this message with a "notify" classifier and send it to the
 4792 subscribed nodes if the data for one of its power sequences on this server instance changes.

4793 A power sequences server SHALL produce a proper message with a "reply" classifier upon a received
 4794 "read" request according to the rules described in section 4.3.24.4.2. This reply may then be a
 4795 complete or restricted function exchange.

4796

4797 *4.3.24.4.1.4 Error considerations*

4798 See section 4.3.24.4.1.2.

4799

4800 *4.3.24.4.2 powerSequences information request*4801 *4.3.24.4.2.1 General data model description and rules*

4802 A power sequences information request is a request for the power sequences information of a power
 4803 sequences server.

4804 A power sequences server SHALL answer to power sequences information requests, even if most of
 4805 the power sequences communication relies on event based (unsolicited) transmissions.

4806 The request to get power sequences information is sent with "read" classifier and an "empty"
 4807 complex class function "smartEnergyManagementPsData".

4808 Optionally, the request MAY be modelled for "restricted function exchange" using a dedicated
 4809 selectors. This version of the specification RECOMMENDS to support the request for information of a
 4810 specific power sequence as shown in Table 121.

Element name	M/O/C (1.4.7)	Explanation
smartEnergyManagementPsDataSelectors	O	SHALL ONLY be present if a child element is required. Otherwise it SHALL be absent.
smartEnergyManagementPsDataSelectors . alternativesRelation	-	Not considered in this recommendation.
smartEnergyManagementPsDataSelectors . alternativesRelation. alternativesId	-	Not considered in this recommendation.
smartEnergyManagementPsDataSelectors . powerSequenceDescription	M	SHALL ONLY be present if the following child element is required. Otherwise it SHALL be absent.
smartEnergyManagementPsDataSelectors . powerSequenceDescription. sequenceId	M	SHALL be present and set to the sequenceId of the power sequence whose information is requested.
smartEnergyManagementPsDataSelectors . powerTimeSlotSchedule	-	Not considered in this recommendation.
smartEnergyManagementPsDataSelectors . powerTimeSlotSchedule. slotNumber	-	Not considered in this recommendation.

4811 Table 121: Dedicated "selectors" of "smartEnergyManagementPsData" to request a specific powerSequences information.

4812

4813 *4.3.24.4.2.2 When received*

4814 If a power sequences server receives a power sequences information request with a "read" classifier,
4815 it SHALL answer with a proper "power sequence information".

4816 If the request is modelled for "restricted function exchange" as shown in Table 121, a reply with
4817 "smartEnergyManagementPsData" (as described in section 4.3.24.4.1) is modified as follows: The
4818 response SHALL only contain information of the power sequence with the same sequenceId value of
4819 the "selectors" content.

4820

4821 *4.3.24.4.2.3 When to send*

4822 A power sequences server SHALL never produce this message.

4823

4824 *4.3.24.4.2.4 Error considerations*

4825 If the power sequences server gets a power sequence request containing an invalid sequenceId (a
4826 sequenceId which does not exist at the time of the request on this instance of the power sequences
4827 server), the power sequences server SHALL consider this as erroneous receipt.

4828

4829 *4.3.24.4.3 powerSequences configuration request call*4830 *4.3.24.4.3.1 General data model description and rules*

4831 This function can be sent from a power sequences server to its bound power sequences client if it
4832 wants one or "all" of its power sequences to be configured. It is intended as "trigger" for the bound
4833 power sequences client that may then initiate a power sequences configuration to configure the
4834 power sequences server's power sequences (see 4.3.24.4.4).

4835 The server's message uses the complex function

4836 "smartEnergyManagementPsConfigurationRequestCall" of the complex class

4837 "SmartEnergyManagementPs" with the following restrictions and rules:

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsConfigurationRequestCall.		
scheduleConfigurationRequest	M	
scheduleConfigurationRequest . sequenceId	O	If present, denotes the sequence ID of the power sequences server which SHOULD be configured. More precisely, it denotes the current power sequence preference of a single alternatives group. If "sequenceId" is not present, "all" power sequences of the power sequences server SHOULD be configured (more precisely: from each alternatives group a

		configuration of a proper power sequence should be done, if feasible).
--	--	--

Table 122: Description of complex class function "smartEnergyManagementPsConfigurationRequestCall" for powerSequences configuration request call

4840

4841 4.3.24.4.3.2 When received

4842 If a power sequences server receives this message, it SHALL consider this as erroneous receipt.

4843

4844 4.3.24.4.3.3 When to send

4845 A power sequences server SHALL only send this message with a "call" classifier to its bound power
4846 sequences client. It SHALL never send this message to clients that are only subscribed.

4847 Once a power sequences server has a bound power sequences client and at least one of the power
4848 sequences contains flexibilities, it MAY send a power sequences configuration request call. If none of
4849 the power sequences contains flexibilities, it SHALL NOT send a power sequences configuration
4850 request call.

4851 A power sequences server MAY send this message containing a "sequenceId" to explicitly trigger the
4852 configuration of a specific power sequence or any of the (permissive) sequences of the same
4853 alternatives group. However, this SHALL NOT be done for fixed forecasts.

4854 A power sequences server MAY also omit the "sequenceId" in the message. This denotes the request
4855 to get a configuration for each alternatives group except for those with fixed forecast.

4856

4857 4.3.24.4.3.4 Error considerations

4858 If a power sequences server encounters a failure with its transmission of the power sequences
4859 configuration request call (e.g. explicitly from a received error message, or from any internal
4860 transmission API), the power sequences server SHOULD consider that the bound power sequences
4861 client may currently be not reachable or able to process the information, hence the client will not
4862 send a configuration. The absence of a power sequences configuration from the client SHALL NOT be
4863 considered as a problem, because the power sequences client may need more time for the
4864 calculation or needs to collect some further information beforehand.

4865 If the bound power sequences client has not sent a power sequences configuration within 15
4866 minutes, the power sequences server MAY consider to re-announce its power sequences and do a
4867 power sequences configuration request call or run as it would do in stand-alone mode, if it lacks time
4868 to wait for a configuration.

4869 However, in which way a power sequences server handles errors is up to the manufacturer. For a
4870 good user experience, a power sequences server SHOULD, however, not behave in a different way
4871 than it would do in stand-alone mode. A washing machine SHOULD consider starting a cycle with a
4872 configuration defined on its own, while a battery storage MAY consider to idle or execute a
4873 maintenance cycle, e.g.

4874

4875 4.3.24.4.4 *powerSequences configuration*4876 4.3.24.4.4.1 *General data model description and rules*

4877 The power sequences configuration is for use with a "write" classifier together with proper
4878 restriction information from the power sequences client to a bound power sequences server only.

4879 A power sequences server may hold more than one power sequence with configurability at a time.
4880 However, a power sequences client SHALL only configure one power sequence within one command,
4881 i.e. configuring one power sequence after the other.

4882 If the power sequences server offered multiple power sequences which were mutual alternatives, a
4883 configuration of one of these power sequences SHALL be interpreted as a choice for this specific
4884 power sequence. A consecutive configuration of another of the alternative power sequences SHALL
4885 be interpreted as a superseding choice between the mutual alternative power sequences.

4886 There are 3 possibilities for configuring a power sequence:

- 4887 - Configuration option A: By configuring the start time of a whole sequence
4888 o If a power sequences server uses this method, it SHALL start the sequence at the
4889 given time and follow its own preference of the slot durations.
4890 o MAY only be used by the power sequences client to configure the bound power
4891 sequences server if the durations of all slots are already set (by the server as its own
4892 preference). Otherwise the client SHALL NOT use this configuration method.
- 4893 - Configuration option B: By configuring the end time of a whole sequence
4894 o If a power sequences server uses this method, it SHALL calculate the start time of the
4895 sequence by subtracting the expected durations of all slots (including all repetitions,
4896 if any) from the configured end time. Then, the server SHALL start the sequence at
4897 the calculated time and follow its own preference of the slot durations.
4898 Please note that the sum of expected durations SHOULD consider slot duration
4899 uncertainties in order to finish not later than the configured endTime.
- 4900 o MAY only be used by the power sequences client to configure the bound power
4901 sequences server if the durations of all slots are already set (by the server as its own
4902 preference). Otherwise the client SHALL NOT use this configuration method.
- 4903 - Configuration option C: By configuring all configurable slots of a sequence
4904 o If a power sequences server uses this method, it SHALL set the durations of all slots
4905 according to the given configuration, if all constraints are meeting the server's
4906 requirements.
4907 o If a power sequences client wants to use this method in its configuration message, it
4908 SHALL retain the order of the slot numbers it received from the power sequences
4909 server message with power sequences information. Additionally, at least the start
4910 time or the end time of the sequence SHALL be configured (since only durations are
4911 included within the slots).
4912 o Please also note that this is the only method to re-configure a power-sequence that
4913 has already begun (i.e. at least the first slot has been started), provided that this
4914 power sequence still offers flexibilities.
4915 o Configurable slots are:

- 4916 ▪ Slots whose duration can be changed
- 4917 ▪ Slots that are marked as optional (and hence can be set active or inactive)
- 4918 ▪ Slots with a combination of the above stated configurabilities

4919 Note: This section is just about the configuration of a sequence, not about a "state change": Apart
 4920 from a configuration a power sequences server's active power sequence may also be modified by a
 4921 state change (provided that this functionality is enabled). This is described in 4.3.24.4.5.

4922 The configuration message is modelled with a specific "write" operation to the power sequences
 4923 server's instance of "smartEnergyManagementPsData" (i.e. the server's powerSequences
 4924 information as described in section 4.3.24.4.1). The message SHALL be composed as follows:

- 4925 1. The classifier SHALL be set to "write".
- 4926 2. The destination of the message SHALL be the power sequences server's instance of the
 4927 complex function "smartEnergyManagementPsData".
- 4928 3. The message SHALL be modelled for "restricted function exchange" (this also means the
 4929 "payload" element "cmd. function" SHALL be set properly; in this case it SHALL be
 4930 "smartEnergyManagementPsData").
- 4931 4. The "cmdControl" SHALL contain "partial".
- 4932 5. A (curtailed) function smartEnergyManagementPsData" SHALL be given, containing only the
 4933 data to modify. The content is defined by Table 123.

4934

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsData.		
alternatives	M (1..1)	The following restriction to the list size applies: Exactly one occurrence of "alternatives" is permitted.
alternatives. powerSequence	M (1..1)	The following restriction to the list size applies: Exactly one occurrence of "powerSequence" is permitted within its parent element.
alternatives. powerSequence. description	M	
alternatives. powerSequence. description. sequenceId	M	States the sequenceId for this configuration.
alternatives. powerSequence. schedule	O	Its presence depends on its child elements.
alternatives. powerSequence. schedule. startTime	See right, (*1).	If used, SHALL state the relative start time of the whole power sequence. If used with configuration option A, elements alternatives. powerSequence. schedule. endTime and element alternatives.

		powerSequence. powerTimeSlot shall be omitted, since the whole power sequence is configured. (*1) Element presence: For configuration option A: M B: NV C: If endTime is used: O; Otherwise: M
alternatives. powerSequence. schedule. endTime	See right, (*1).	If used, SHALL state the relative end time of the whole power sequence. If used with configuration option B, elements alternatives. powerSequence. schedule. startTime and alternatives. powerSequence. powerTimeSlot SHALL be omitted, since the whole power sequence is configured. (*1) Element presence: For configuration option A: NV B: M C: If startTime is used: O; Otherwise: M
alternatives. powerSequence. powerTimeSlot	See right, (*1).	List of power time slot schedule configuration data. (*1) Element presence: For configuration option A: NV B: NV C: M
alternatives. powerSequence. powerTimeSlot. schedule	See right, (*1).	The power time slot schedule configuration of a slot. (*1) Element presence:

		For configuration option A: NV B: NV C: M
alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	M	States the slot number for which this configuration applies.
alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration	O	SHALL be present if the slot has a configurable length AND the slot duration shall be changed to this new value.
alternatives. powerSequence. powerTimeSlot. schedule. slotActivated	O	If the optionality of the slot is configurable and its activation status shall be changed to the new value, this element SHALL be present and denote whether the slot shall be used (slotActivated = "true") or not (slotActivated = "false"). Otherwise is SHALL be absent (i.e. the slot is mandatory).

Table 123: Description of "restricted write" operation for powerSequences configuration.

4.3.24.4.2 When received

A power sequences server SHALL only process power sequences configuration data received with a "write" classifier and proper restriction elements. Otherwise, a power sequences server SHALL consider this as erroneous receipt.

A power sequences server SHALL only accept power sequences configuration data that fit (with an "acceptable tolerance" as explained below) to the constraints of the according sequenceId. A power sequences server SHALL only accept power sequences configuration that configure all flexibilities that are available (durations, etc.) offered in one "write" operation for one power sequence.

If a power sequences server receives a power sequences configuration data with a configuration this power sequence cannot fulfil, it SHALL consider this as erroneous receipt and – if an acknowledgement message is to be sent – respond with a negative acknowledgement message with error number "7" (command rejected).

On "acceptable tolerance": The transmission of a configuration may be delayed "a bit too much" due to network problems, e.g. In this case the time configuration may violate the constraints given by the server "a bit". In such a case, the server SHOULD tolerate and adjust the received configuration silently as long as the violation is negligible. In general a tolerance of 30 seconds SHOULD be applied on time constraints.

Note: It may happen that a received power sequences configuration is valid according to the rules stated above but leads to no change at all of the sequence schedule (e.g. because received start/end

4956 times are identical to the schedule the server already announced). This SHALL NOT be considered an
4957 error.

4958

4959 *4.3.24.4.4.3 When to send*

4960 A power sequences server SHALL never produce this message.

4961

4962 *4.3.24.4.4.4 Error considerations*

4963 If the power sequences server gets a power sequence configuration containing an invalid sequenceId
4964 (a sequenceId which does not exist on this instance of the power sequences server), the power
4965 sequences server SHALL consider this as erroneous receipt.

4966 See section 4.3.24.4.3.4 for more error considerations.

4967

4968 *4.3.24.4.5 powerSequences state change*

4969 *4.3.24.4.5.1 General data model description and rules*

4970 *4.3.24.4.5.1.1 General aspects*

4971 The normal procedure for a power sequences client to control a bound power sequences server's
4972 power sequences is to initiate a powerSequences configuration as described in section 4.3.24.4.4. In
4973 some cases, a more direct way to control power sequences is needed. For this the "powerSequences
4974 state change" can be used. It enables a bound power sequences client to request a change of the
4975 state of a power sequences server's power sequence to "run", "pause" or "invalid" (i.e. terminate),
4976 dependent on the current state of the power sequence and some flags (see section 4.3.24.4.5.1.2 for
4977 details about the rules). This is based upon the definitions of section 4.3.24.3.3.6, among others.

4978 Please note that state changes may occur from other situations as well. Among others, the state of a
4979 currently "chosen sequence" changes if another sequence is configured to become a "chosen
4980 sequence". In this case, the previously "chosen sequence" may get the state "inactive" (e.g.).
4981 However, this is an "indirect" state change which is caused by (server) internal mechanisms
4982 according to this specification's rules. Such "indirect" state changes are NOT subject of this section.

4983

4984 *4.3.24.4.5.1.2 Rules for changing the state*

4985 The state of a power sequences server's power sequence MAY be changed by the bound power
4986 sequence client if the following aspects are considered:

- 4987 1. If the power sequence is in state scheduled AND the flag isPausable is set to true AND the
4988 flag sequenceRemoteControllable is set to true: The client then MAY request for the state
4989 change:
4990 ○ scheduledPaused

- 4991 2. If the power sequence is in state scheduledPaused AND the flag isPausable is set to true AND
 4992 the flag sequenceRemoteControllable is set to true: The client then MAY request for the
 4993 state change:
 4994 ○ scheduled
 4995 3. If the power sequence is in state running AND the flag isPausable is set to true AND the flag
 4996 sequenceRemoteControllable is set to true: The client then MAY request for the state
 4997 change:
 4998 ○ paused
 4999 4. If the conditions in item 3 are met and additionally the flag isStoppable is set to true, the
 5000 client MAY request for the state changes:
 5001 ○ paused
 5002 ○ invalid
 5003 5. If the power sequence is in state paused AND the flag isPausable is set to true AND the flag
 5004 sequenceRemoteControllable is set to true, the client MAY request for the state change:
 5005 ○ running
 5006 6. If the conditions in item 5 are met AND additionally the flag isStoppable is set to true, the
 5007 client MAY request for the state changes:
 5008 ○ running
 5009 ○ invalid
 5010 7. If the power sequence is in state running, paused, scheduled, scheduledPaused or inactive
 5011 AND the flag isStoppable is set to true, the client MAY request for the state change:
 5012 ○ invalid

5013 In all other cases, a (direct) state change SHALL NOT be requested by the bound power sequences
 5014 client.

5015

5016 4.3.24.4.5.1.3 Data model description and rules

5017 The "powerSequences state change" message is modelled with a specific "write" operation to the
 5018 power sequences server's instance of "smartEnergyManagementPsData" (i.e. the server's
 5019 powerSequences information as described in section 4.3.24.4.1). The message SHALL be composed
 5020 as follows:

- 5021 1. The classifier SHALL be set to "write".
 5022 2. The destination of the message SHALL be the power sequences server's instance of the
 5023 complex function "smartEnergyManagementPsData".
 5024 3. The message SHALL be modelled for "restricted function exchange" (this also means the
 5025 "payload" element "cmd. function" SHALL be set properly; in this case it SHALL be
 5026 "smartEnergyManagementPsData").
 5027 4. The "cmdControl" SHALL contain "partial".
 5028 5. A (curtailed) function "smartEnergyManagementPsData" SHALL be given, containing the data
 5029 to modify. The content is defined by Table 124.

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsData.		

alternatives	M (1..1)	The following restriction to the list size applies: Exactly one occurrence of "alternatives" is permitted.
alternatives. powerSequence	M (1..1)	The following restriction to the list size applies: Exactly one occurrence of "powerSequence" is permitted within its parent element.
alternatives. powerSequence. description	M	
alternatives. powerSequence. description. sequenceId	M	States the sequenceId for this state change.
alternatives. powerSequence. state	M	
alternatives. powerSequence. state. state	M	This element SHALL always be present and denotes the state change. See section 4.3.24.4.5.1.2 for values permitted in this state change.

Table 124: Description of "restricted write" operation for powerSequences state change.

4.3.24.4.5.2 When received

If a power sequences server receives a powerSequences state change from its bound power sequences client, it SHALL verify whether this state change is allowed within the rules denoted in 4.3.24.4.5.1.2. If it is not allowed, the power sequences server SHALL consider this as erroneous receipt.

4.3.24.4.5.3 When to send

A power sequences server SHALL never generate this message.

4.3.24.4.5.4 Error considerations

See section 4.3.24.4.5.2.

4.3.24.4.6 powerSequences price information

A power sequences server MAY provide information on the cost of one or more of its power sequences. The support of this information is optional.

The complex function "smartEnergyManagementPsPriceData" of the complex class "SmartEnergyManagementPs" is used with the following restrictions and rules:

Element name	M/O/C (1.4.7)	Explanation
--------------	------------------	-------------

SmartEnergyManagementPs. smartEnergyManagementPsPriceData.		
price	O	SHALL ONLY be present if proper child elements are present.
price. sequenceId	M	
price. price	M	
price. currency	M	

Table 125: Description of smartEnergyManagementPsPriceData for powerSequences price information.

4.3.24.4.6.1 When received

If a power sequences server receives this message with a "notify" or "reply" classifier, it SHALL consider this as erroneous receipt. For "write" classifier see section 4.3.24.4.8.

4.3.24.4.6.2 When to send

A power sequences server SHALL produce this message with a "notify" classifier and send it to subscribed nodes if the price information for one of its power sequences on this server instance changes.

A power sequences server SHALL produce a proper message with a "reply" classifier upon a received "read" request according to the rules described in section 4.3.24.4.7.2. This reply may then be a complete or restricted function exchange.

4.3.24.4.6.3 Error considerations

See section 4.3.24.4.7.2.

4.3.24.4.7 powerSequences price information request

4.3.24.4.7.1 General data model description and rules

A power sequences price information request is a request for the power sequences price information of a power sequences server.

A power sequences server SHALL answer to power sequences price information requests if the power sequences server support power sequences price information.

The request to get power sequences price information is sent with "read" classifier and an "empty" complex class function "smartEnergyManagementPsPriceData".

Optionally, the request MAY be modelled for "restricted function exchange" using a dedicated selectors. This version of the specification RECOMMENDS to support the request for price information of a specific power sequence as shown in Table 126.

Element name	M/O/C (1.4.7)	Explanation
smartEnergyManagement PsPriceDataSelectors	O	SHALL ONLY be present if a child element is required. Otherwise it SHALL be absent.
smartEnergyManagement PsPriceDataSelectors. price	M	SHALL ONLY be present if the following child element is required. Otherwise it SHALL be absent.
smartEnergyManagement PsPriceDataSelectors. price. sequenceld	M	SHALL be present and set to the sequenceld of the power sequence whose price information is requested.

Table 126: Dedicated "selectors" of "smartEnergyManagementPsPriceData" to request a specific powerSequences price information.

4.3.24.4.7.2 When received

If a power sequences server receives a power sequences price information request with a "read" classifier, it SHALL answer with a proper "power sequence price information".

If the request is modelled for "restricted function exchange" as shown in Table 126, a reply with "smartEnergyManagementPsPriceData" (as described in section 4.3.24.4.6) is modified as follows:
The response SHALL only contain price information of the power sequence with the same sequenceld value of the "selectors" content.

4.3.24.4.7.3 When to send

A power sequences server SHALL never produce this message.

4.3.24.4.7.4 Error considerations

If the power sequences server gets a power sequence request with a sequenceld where no price information is available, the power sequences server SHALL consider this as erroneous receipt.

4.3.24.4.8 powerSequences price information change

4.3.24.4.8.1 General data model description and rules

4.3.24.4.8.1.1 General aspects

A power sequences price information change is usually only initiated upon a power sequences price calculation request call. I.e. once a powerSequences server asked a bound powerSequences client for a price calculation (see section 4.3.24.4.9), the server awaits receiving a powerSequences price information change from the client. This information should contain the price for the requested sequence. Of course, the powerSequences client is then obliged to consider the server's information on the current sequence schedule (smartEnergyManagementPsData).

Further rules apply:

A power sequences price information change can only be considered to reflect a whole sequence's price if the proper power sequence information is complete. Once a sequence started, a power sequences server may remove past slots from the power sequences information, resulting in incomplete sequence information. As a consequence, the request for the price calculation and the execution and delivery of the price calculation SHOULD occur before the power sequence starts as otherwise a bound power sequences client may not have sufficient information to calculate a price for the whole sequence.

Once the price for a whole sequence was calculated and transmitted successfully, it SHOULD also be possible to request and execute a price calculation after the sequence started. As mentioned above, this is only feasible if the power sequences server keeps past slots in its sequence information.

4.3.24.4.8.1.2 Data model description

The "powerSequences price information change" message is modelled with a specific "write" operation to the power sequences server's instance of "smartEnergyManagementPsPriceData" (i.e. the server's powerSequences price information as described in section 4.3.24.4.6). The message SHALL be composed as follows:

1. The classifier SHALL be set to "write".
2. The destination of the message SHALL be the power sequences server's instance of the complex function "smartEnergyManagementPsPriceData".
3. The message SHALL be modelled for "restricted function exchange" (this also means the "payload" element "cmd. function" SHALL be set properly; in this case it SHALL be "smartEnergyManagementPsPriceData").
4. The "cmdControl" SHALL contain "partial".
5. A (curtailed) function "smartEnergyManagementPsPriceData" SHALL be given, containing only the data to modify. The content is defined by Table 127.

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsPriceData.		
price	M (1..1)	The following restriction to the list size applies: Exactly one occurrence of "price" is permitted.
price. sequenceId	M	The sequenceId whose price shall be modified.
price. price	M	The new price.
price. currency	M	The currency of the price.

Table 127: Description of "restricted write" operation for a power sequence price modification.

4.3.24.4.8.2 When received

If a power sequences server receives a power sequences price information change from a device that is NOT its bound power sequences client, it MAY consider this as erroneous receipt.

5135 A power sequences server SHOULD ONLY accept a power sequences price information change if it
 5136 was received from its bound power sequences client. Afterwards, further rules apply:

5137 If a power sequences server receives a power sequences price information change as valid message
 5138 upon its previous power sequence price calculation request, it SHALL evaluate the message
 5139 accordingly. Afterwards, it SHOULD NOT accept subsequently received power sequences price
 5140 information changes. Such "additional responses" may denote a price update information. However,
 5141 instead of evaluating these "additional responses" the power sequences server SHOULD consider this
 5142 as erroneous receipt, and it MAY initiate a new power sequences price calculation request.

5143 If a power sequences server receives a power sequences price information change without having
 5144 issued a power sequences price calculation request call before, it SHOULD consider this as erroneous
 5145 receipt.

5146 If a power sequences server receives a power sequences price information change containing a
 5147 power sequenceId it has not asked for, it SHALL consider this as erroneous receipt.

5148 Remark (informative): If a power sequences server receives a valid power sequences price
 5149 information change, the use of this information is manufacturer specific. Examples of use are
 5150 displaying it to a customer on a device display.

5151

5152 *4.3.24.4.8.3 When to send*

5153 A power sequences server SHALL never generate this message.

5154

5155 *4.3.24.4.8.4 Error considerations*

5156 See section 4.3.24.4.8.2.

5157

5158 *4.3.24.4.9 powerSequences price calculation request call*

5159 *4.3.24.4.9.1 General data model description and rules*

5160 A power sequences price calculation request call is a request from a power sequences server to a
 5161 bound power sequences client, in order to ask the power sequences client to calculate a price for a
 5162 certain sequence ID of the power sequences server. The answer is delivered by the power sequences
 5163 client using the "power sequences price calculation response call" message. The sequence needs to
 5164 have schedule information (see 4.3.24.3.3.3) and all slots need to be determined to enable the client
 5165 to calculate a price for it. Furthermore, the request SHOULD be sent "in time" in order to receive the
 5166 price calculation before the sequence starts. See section 4.3.24.4.8.1 for details!

5167 The server's message uses the complex function
 5168 "SmartEnergyManagementPsPriceCalculationRequestCall" of the complex class
 5169 "SmartEnergyManagementPs" with the following restrictions and rules:

Element name	M/O/C (1.4.7)	Explanation
SmartEnergyManagementPs. smartEnergyManagementPsPriceCalculationRequestCall.		

priceCalculationRequest	M	
priceCalculationRequest. sequenceId	M	

Table 128: Description of complex class function "smartEnergyManagementPsPriceCalculationRequestCall" for powerSequences price calculation request call

5172

5173 4.3.24.4.9.2 When received

5174 If a power sequences server receives this message, it SHALL consider this as erroneous receipt.

5175

5176 4.3.24.4.9.3 When to send

5177 A power sequences server SHALL produce this message with a "call" classifier if it is interested in the
 5178 price for a configured sequence it offers, i.e. a sequence which has a valid schedule (including
 5179 information about all slot's durations as stated in 4.3.24.4.9.1). It will get a calculation response
 5180 based on its offered sequence and the according schedule, provided that the bound client provides
 5181 this service and has sufficient information. Please consider also rules described in section
 5182 4.3.24.4.8.1!

5183 Note: A power sequences server SHALL never change the configuration for a power sequence that
 5184 has already been configured by its bound power sequences client, just to calculate different prices.

5185

5186 4.3.24.4.9.4 Error considerations

5187 See section 4.3.24.4.9.2.

5188 Please note that calculating prices is an optional feature.

5189 Additionally, if a device sends a power sequences price calculation request call to its bound power
 5190 sequences client and the binding partner returns a negative acknowledgement message, this might
 5191 happen due to the fact that the power sequences client temporarily cannot calculate prices, e.g.
 5192 because the client has not yet been updated with valid prices for the whole period of the sequence.
 5193 Thus, a power sequences server SHOULD be aware that it might encounter power sequences clients
 5194 which currently cannot calculate prices. For this case, the server SHALL be able to cope with errors
 5195 returned to price calculation requests it issues. How exactly it handles these errors is up to the
 5196 manufacturer.

5197

5198 4.3.25 StateInformation

5199 4.3.25.1 Dependencies to other Feature Types

5200 None.

5201

4.3.25.2 Introduction

Information about the state of an Entity can help understand which operational mode or functionality can be expected from the Entity or whether a failure occurred (which may lead to a non-normal operation).

Each entry within the state information list represents one possible functionality or failure. The functionalities or failures then are marked as active or inactive.

4.3.25.3 Rules

None.

4.3.25.4 stateInformationListData**4.3.25.4.1 Description**

This function delivers all state information available at the Entity. A state information is not only listed here when it is active, but always, as long as the state information is supported. Whether the state information is active or not is marked with the "isActive" flag (see section 4.3.25.4.3).

4.3.25.4.2 Rules

None.

4.3.25.4.3 Element rules

Element	Rule	Description
StateInformation. stateInformationListData. stateInformationData. (list)		
stateInformationId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the state information.
stateInformation	Value rules of Table 130 SHALL be applied.	The actual state information.
isActive	None.	If the isActive flag is set to true, the according stateInformation is active. Otherwise it is inactive.
category	Value rules of Table 131 SHALL be applied.	Each state information has a category that classifies the kind of state.
timeOfLastChange	None.	The timestamp of the last change of the stateInformation.

Table 129: stateInformationListData Element rules

Element "**stateInformation**" value rules:

Value	Rule	Description
externalOverrideFromGrid	None.	The appliance's functionality is overwritten by a command from grid-side.
autonomousGridSupport	None.	The appliance supports grid stability.

islandingMode	None.	The appliance works independent from the (electricity) grid.
balancing	None.	The battery is in balancing mode.
trickleCharging	None.	The battery is in trickle charging mode.
calibration	None.	The appliance is calibrating itself.
commissioningMissing	None.	The appliance waits for commissioning.
sleeping	None.	The appliance is in sleeping mode.
starting	None.	The appliance is starting.
mppt	None.	The inverter is in MPPT (Maximum Power Point Tracker) mode.
throttled	None.	The appliance throttled its consumption or production.
shuttingDown	None.	The appliance is shutting down.
manualShutdown	None.	The appliance is shutting down, initiated manually.
inverterDefective	None.	A defect of the inverter was detected.
batteryOvercurrentProtection	None.	The battery is in overcurrent protection mode.
pvStringOvercurrentProtection	None.	The PV string is in overcurrent protection mode.
gridFault	None.	A fault of the grid connection was detected.
groundFault	None.	A fault of the ground connection was detected.
acDisconnected	None.	The AC connection was disconnected.
dcDisconnected	None.	The DC connection was disconnected.
cabinetOpen	None.	The cabinet is open.
overTemperature	None.	The temperature is too high.
underTemperature	None.	The temperature is too low.
frequencyAboveLimit	None.	The frequency is above the limit.
frequencyBelowLimit	None.	The frequency is below the limit.
acVoltageAboveLimit	None.	The AC voltage is above the limit.
acVoltageBelowLimit	None.	The AC voltage is below the limit.
dcVoltageAboveLimit	None.	The DC voltage is above the limit.
dcVoltageBelowLimit	None.	The DC voltage is below the limit.
hardwareTestFailure	None.	A failure occurred during hardware test.
genericInternalError	None.	Some internal error was detected.

5224 Table 130: Element "stateInformation" value rules

5225 Element "category" value rules:

Value	Rule	Description
functionality	None.	The state information category for functionalities (normal operation).
failure	None.	The state information category for failures (most likely not normal operation).

5226 Table 131: Element "category" value rules

5227

5228 4.3.25.5 Descriptive information and further definitions

5229 None.

5230

5231 4.3.26 SupplyCondition

5232 4.3.26.1 Dependencies to other Feature Types

5233 None.

5234

5235 **4.3.26.2 Introduction**

5236 Practice shows that the supply of electricity, water, etc. can be temporarily impaired. Some possible
 5237 causes are announced maintenance work at the supply line, supply network overload, or depletion of
 5238 a local energy resource (battery, e.g.). The Supply Condition concept permits indication of the supply
 5239 condition by a household's SPINE data models.

5240 The concept is less intended for any automatisms. It is rather intended for indications to users in
 5241 order to prepare or apply measures in time.

5242

5243 **4.3.26.3 Rules**

5244 None.

5245

5246 **4.3.26.4 supplyConditionListData**5247 **4.3.26.4.1 Description**

5248 This function can be used to model the "events", i.e. the information on a specific supply condition.
 5249 These events can, e.g., be sent from a DSO to a household. In this case, the household would be the
 5250 data owner and the DSO would send (as a client) write commands to this resource.

5251

5252 **4.3.26.4.2 Rules**

5253 None.

5254

5255 **4.3.26.4.3 Element rules**

Element	Rule	Description
SupplyCondition. supplyConditionListData. supplyConditionData. (list)		
conditionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the condition.
timestamp	SHALL be set as SUB IDENTIFIER to the timestamp of the creation of the message.	The timestamp of the event.
eventType	SHALL be set. Value rules of Table 133 SHALL be applied.	The kind of the event.
originator	SHOULD be set. Value rules of Table 134 SHALL be applied.	The originator (in terms of instance, not in terms of address) of the information.
thresholdId	MAY be set as FOREIGN IDENTIFIER to the according thresholdId, if related to a threshold.	The identifier of this event's threshold (if available).
thresholdPercentage	Reached percentage of the threshold value. Only if	Relation of current situation to (configured) threshold (if available).

	<i>thresholdId</i> is set, <i>thresholdPercentage</i> MAY be set, too.	
relevantPeriod	None.	Esp. the announcement of maintenance can make use of this element in order to tell when a maintenance is planned.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the event.
gridCondition	None.	Indicates the condition of the electricity grid.

5256 Table 132: *supplyConditionListData* Element rules5257 Element "**eventType**" value rules:

Value	Rule	Description
thresholdExceeded	None.	The related threshold was exceeded (may be good or bad, see definition of the corresponding <i>thresholdType</i> element in section 5.3.27.4.2 for details). This means that the actual value is greater than the threshold.
fallenBelowThreshold	None.	The related threshold was fallen below (may be good or bad, see definition of the corresponding <i>thresholdType</i> element in section 5.3.27.4.2 for details). This means that the actual value is lower than the threshold.
supplyInterrupt	None.	The supply of the corresponding commodity was interrupted.
releaseOfLimitations	None.	"Back to normal", no limitations or threshold violations.
otherProblem	None.	Some problem with the supply without further details about the problem.

5258 Table 133: Element "*eventType*" value rules5259 Element "**originator**" value rules:

Value	Rule	Description
externDSO	None.	The originator of the current condition "event" is the DSO (distribution system operator).
externSupplier	None.	The originator of the current condition "event" is the energy supplier.
internalLimit	None.	Some internal limit at the premises generated the current condition "event".
internalService	None.	Due to an internal service at the premises, the current condition "event" was generated.
internalUser	None.	A user interaction is the source of the current condition "event".

5260 Table 134: Element "*originator*" value rules5261 Element "**gridCondition**" value rules:

Value	Rule	Description
consumptionRed	None.	The electricity grid condition for consumption from the grid is "red".
consumptionYellow	None.	The electricity grid condition for consumption from the grid is "yellow".
good	None.	The electricity grid condition is "good".
productionYellow	None.	The electricity grid condition for feeding into the grid is "yellow".
productionRed	None.	The electricity grid condition for feeding into the grid is "red".

5262 Table 135: Element "*gridCondition*" value rules

5263

5264 **4.3.26.5 supplyConditionDescriptionListData**5265 **4.3.26.5.1 Description**

5266 The more static information about a supply condition is modelled within this function.

5267

5268 **4.3.26.5.2 Rules**

5269 None.

5270

5271 **4.3.26.5.3 Element rules**

Element	Rule	Description
SupplyCondition. supplyConditionDescriptionListData. supplyConditionDescriptionData. (list)		
conditionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the condition.
commodityType	SHOULD be set.	The type of the commodity handled by this condition is stated here.
positiveEnergyDirection	SHOULD be set. The value "consume" expresses that positive values of energy or power denote energy consumption by the premises. The value "produce" expresses that positive values of energy or power denote energy production by the premises.	The <i>positiveEnergyDirection</i> states whether energy consumption or production will be counted as positive value.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the condition.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the condition.

5272 *Table 136: supplyConditionDescriptionListData Element rules*

5273

5274 **4.3.26.6 supplyConditionThresholdRelationListData**5275 **4.3.26.6.1 Description**

5276 Multiple thresholds can be associated to a condition. This leads to the relation

- 5277
- 1 conditionId -> 0..m thresholdId

5278 The supplyConditionThresholdRelationListData function can be used to model concrete relations.

5279

5280 **4.3.26.6.2 Rules**

5281 None.

5282

5283 4.3.26.6.3 *Element rules*

Element	Rule	Description
SupplyCondition. supplyConditionThresholdRelationListData. supplyConditionThresholdRelationData. (list)		
conditionId	SHALL be set as PRIMARY IDENTIFIER.	Identifier of the condition.
thresholdId (list)	Multiple thresholdIds of thresholds linked to this condition MAY be provided.	Related threshold identifier(s).

5284 Table 137: supplyConditionThresholdRelationListData Element rules

5285

5286 4.3.26.7 *Descriptive information and further definitions*

5287 None.

5288

5289 4.3.27 **TariffInformation**

5290 Reserved for future use.

5291

5292 4.3.28 **TaskManagement**5293 4.3.28.1 *Dependencies to other Feature Types*

- 5294 - DirectControl
- 5295 - HVAC
- 5296 - LoadControl
- 5297 - PowerSequences
- 5298 - SmartEnergyManagementPs

5299

5300 4.3.28.2 *Introduction*

5301 The TaskManagement Class lists all "tasks" that are currently relevant on an entity. This enables one
 5302 to have a quick look on the tasks of related classes, but not for their modification. Further
 5303 information has to be gathered with the corresponding classes supporting the specific functionality.

5304 In this version of the specification, tasks can be "jobs" like in DirectControl, PowerSequences /
 5305 SmartEnergyManagementPs, etc. (see the related function groups below).

5306 Remark: Future versions of this specification may add other kinds of tasks (i.e. tasks that are not
 5307 covered by "jobs").

5308

5309 4.3.28.3 *Rules*

5310 None.

5311

5312 **4.3.28.4 taskManagementJobListData**5313 **4.3.28.4.1 Description**

5314 All jobs that are available on the entity are listed here, together with their state and (if available) the
 5315 elapsed and remaining time.

5316

5317 **4.3.28.4.2 Rules**

5318 None.

5319

5320 **4.3.28.4.3 Element rules**

Element	Rule	Description
TaskManagement. taskManagementJobListData. taskManagementJobData. (list)		
jobId	SHALL be set as PRIMARY IDENTIFIER.	The <i>jobId</i> is used for indentifying different job information collected by this class.
timestamp	SHOULD be set to the timestamp of the creation of the message.	Point in time, this information was generated.
jobState	SHALL be set. Value rules of Table 50 SHALL be applied. Value rules of Table 73 SHALL be applied. Value rules of section 3.10.1.5.1 SHALL be applied.	Current state of the job.
elapsedTime	If set, this element SHALL contain the duration the job was already active.	The duration, this job was already active (normally, pause times are not counted).
remainingTime	If set, this element SHALL contain the duration the job still needs to be active.	The duration, this job will most likely be active before it ends.

5321 *Table 138: taskManagementJobListData Element rules*

5322

5323 **4.3.28.5 taskManagementJobRelationListData**5324 **4.3.28.5.1 Description**

5325 To identify where a job originates from, the *taskManagementJobRelationListData* function is used. If
 5326 the related class allows more than one job the corresponding identifier(s) are denoted.

5327

5328 **4.3.28.5.2 Rules**

5329 If this function is used, each list entry SHALL contain exactly one relation to a job. This means that it is
 5330 not allowed to set more than one of the following Elements within one list entry:

- 5331 - directControlRelated
- 5332 - hvacRelated
- 5333 - loadControlRelated

- 5334 - powerSequencesRelated
- 5335 - smartEnergyManagementPsRelated

5336

5337 4.3.28.5.3 *Element rules*

Element	Rule	Description
TaskManagement. taskManagementJobRelationListData. taskManagementJobRelationData. (list)		
jobId	SHALL be set as PRIMARY IDENTIFIER.	The <i>jobId</i> is used for indentifying different job information collected by this class.
directControlRelated	None.	Flag that indicates that the job is related to DirectControl.
hvacRelated	None.	Flag that indicates that the job is related to HVAC.
hvacRelated. overrunId	If the job is related to an overrun, this element SHALL be set to the FOREIGN IDENTIFIER of the corresponding overrun.	Identifier for the related overrun.
loadControlRelated	None.	Flag that indicates that the job is related to LoadControl.
loadControlRelated. eventId	If the job is related to an event, this element SHALL be set to the FOREIGN IDENTIFIER of the corresponding event.	Identifier for the related load control event.
powerSequencesRelated	None.	Flag that indicates that the job is related to PowerSequences.
powerSequencesRelated. sequenceId	If the job is related to a power sequence, this element SHALL be set to the FOREIGN IDENTIFIER of the corresponding power sequence.	Identifier for the related power sequence.
smartEnergyManagementPsRelated	None.	Flag that indicates that the job is related to SmartEnergyManagementPs.
smartEnergyManagementPsRelated. sequenceId	If the job is related to a SmartEnergyManagementPs sequence, this element SHALL be set to the FOREIGN IDENTIFIER of the corresponding sequence.	Identifier for the related power sequence.

5338 Table 139: taskManagementJobRelationListData Element rules

5339

5340 4.3.28.6 *taskManagementJobDescriptionListData*5341 4.3.28.6.1 *Description*

5342 The more static information about the jobs are communicated with this function.

5343

5344 4.3.28.6.2 *Rules*

5345 None.

5346

5347 4.3.28.6.3 *Element rules*

Element	Rule	Description
TaskManagement. taskManagementJobDescriptionListData. taskManagementJobDescriptionData. (list)		
jobId	SHALL be set as PRIMARY IDENTIFIER.	The <i>jobId</i> is used for indentifying different job information collected by this class.
jobSource	SHOULD be set. Value rules of Table 141 SHALL be applied.	Specifies what kind of reason caused this job.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the job.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the job.

5348 Table 140: taskManagementJobDescriptionListData Element rules

5349 Element "**jobSource**" value rules:

Value	Rule	Description
internalMechanism	None.	The job was started due to some device internal mechanism (e.g. a regular cleaning cycle).
userInteraction	None.	The user activated some functionality directly via the UI of the device.
externalConfiguration	None.	The job started, because it was triggered from some external device (e.g. an energy manager).

5350 Table 141: Element "jobSource" value rules

5351

5352 4.3.28.7 **taskManagementOverviewData**5353 4.3.28.7.1 *Description*

5354 A general overview is given with this function. It specifies whether a remote controlling is permitted
 5355 at that moment and if currently jobs are active.

5356

5357 4.3.28.7.2 *Rules*

5358 None.

5359

5360 4.3.28.7.3 *Element rules*

Element	Rule	Description
TaskManagement. taskManagementOverviewData.		
remoteControllable	SHOULD be set. If set to "false", write operations on features linked via <i>taskManagementJobRelationListData</i> to this feature SHALL be declined. If absent, a default value of "true" SHALL be applied.	States if remote controlling of the device or entity is permitted.

jobsActive	None.	Denotes whether jobs are active on the device or entity or not.
------------	-------	---

Table 142: taskManagementOverviewData Element rules

4.3.28.8 Descriptive information and further definitions

The TaskManagement Feature Type enables the brief listing of all tasks that are currently relevant for the feature's entity. A special case is the implementation of the Feature Type *TaskManagement* on an entity with the entity type *DeviceInformation*: In this case the feature covers tasks of all entities that are available on this device.

4.3.29 Threshold**4.3.29.1 Dependencies to other Feature Types**

None.

4.3.29.2 Introduction

Thresholds are helpful in several kind of use cases. E.g. a device or application can be configured to do some specific behaviour if a measured value is above or below a configured threshold. Some measurands need to stay in a certain value range, to ensure proper and safe working of devices. Automatic alarms after reaching a threshold help reacting on the occurrence immediately.

4.3.29.3 Rules

None.

4.3.29.4 thresholdListData**4.3.29.4.1 Description**

Each threshold (identified by its *thresholdId*) has a configured value. This function is used to read or write the actual value.

4.3.29.4.2 Rules

None.

4.3.29.4.3 Element rules

Element	Rule	Description
Threshold. thresholdListData. thresholdData. (list)		
thresholdId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the threshold.

thresholdValue	SHALL be set.	Value that should not be fallen below or exceeded (depends on the <i>thresholdType</i> , see section 5.3.27.4.2).
----------------	---------------	---

Table 143: thresholdListData Element rules

4.3.29.5 thresholdConstraintsListData**4.3.29.5.1 Description**

The value range from which a threshold value configuration may be chosen from may be restricted by a minimum and a maximum value. Between these, not every value, but only ones with a specific gap, may be used, specified with the step size.

4.3.29.5.2 Rules

If thresholds are writeable, this function SHOULD be supported, otherwise it MAY be supported.

4.3.29.5.3 Element rules

Element	Rule	Description
Threshold. thresholdConstraintsListData. thresholdConstraintsData. (list)		
thresholdId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the threshold.
thresholdRangeMin	If set, the threshold value SHALL NOT be set to a value below this minimum value.	Minimum value that may be used to configure the element <i>thresholdValue</i> (see section 5.3.27.2.2).
thresholdRangeMax	If set, the threshold value SHALL NOT be set to a value above this maximum value.	Maximum value that may be used to configure the element <i>thresholdValue</i> (see section 5.3.27.2.2).
thresholdStepSize	The threshold value SHALL ONLY contain multiples of thresholdStepSize.	Step size for the configuration of the element <i>thresholdValue</i> (see section 5.3.27.2.2).

Table 144: thresholdConstraintsListData Element rules

4.3.29.6 thresholdDescriptionListData**4.3.29.6.1 Description**

The rather static information about thresholds is defined within this function.

4.3.29.6.2 Rules

None.

4.3.29.6.3 Element rules

Element	Rule	Description
---------	------	-------------

Threshold. thresholdDescriptionListData. thresholdDescriptionData. (list)		
thresholdId	SHALL be set as PRIMARY IDENTIFIER.	Identifier for the threshold.
thresholdType	SHALL be set.	The type of the threshold.
unit	SHALL be set, if there could be ambiguity how to interpret the threshold without the unit.	The unit in which the value of the threshold is denoted.
scopeType	None.	Specifies a more detailed meaning of the threshold.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the threshold.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the threshold.

5413 Table 145: thresholdDescriptionListData Element rules

5414 Element "thresholdType" value rules:

Value	Rule	Description
goodAbove	None.	A positive reaction will follow after the value is greater than the threshold.
badAbove	None.	A negative reaction will follow after the value is greater than the threshold.
goodBelow	None.	A positive reaction will follow after the value is lower than or equal to the threshold.
badBelow	None.	A negative reaction will follow after the value is lower than or equal to the threshold.
minValueThreshold	None.	If the value is lower than this minimum threshold, some warning will be activated.
maxValueThreshold	None.	If the value is greater than this maximum threshold, some warning will be activated.
minValueThresholdExtreme	None.	If the value is lower than this "extreme minimum" threshold, some important warning will be activated.
maxValueThresholdExtreme	None.	If the value is greater than this "extreme maximum" threshold, some important warning will be activated.
sagThreshold	None.	If the value is lower than this sag threshold, some warning will be activated. Commonly used for electricity voltage sags.
swellThreshold	None.	If the value is greater than this swell threshold, some warning will be activated. Commonly used for electricity voltage swells.

5415 Table 146: Element "thresholdType" value rules

5416 Note: All threshold rules SHALL be understood in a mathematical way.

5417

5418 4.3.29.7 Descriptive information and further definitions

5419 None.

5420

5421 **4.3.30 TimeInformation**5422 **4.3.30.1 Dependencies to other Feature Types**

5423 None.

5424

5425 **4.3.30.2 Introduction**

5426 The exchange of time information is important for most smart grid and home automation systems.

5427 Please note the underlying revision of the SPINE data model does not specify a sophisticated

5428 synchronisation mechanism that includes a given precision. This is due to the complex of problems

5429 appearing with communication delays in the different technologies and hardware platforms.

5430 However, an application could measure all relevant delays and take care of a real synchronisation in

5431 the network using the *TimeInformation* class.

5432 Independent of any synchronisation aspect the "owner concept" applies here as well: The "owner" of

5433 any given data (a measured value of a sensor, e.g.) applies its own SPINE time to its own data. This is

5434 relevant for the case of a substantial difference between the SPINE time of two nodes.

5435 In order to avoid problems from such differences a typical sequence could be that a new device

5436 sends a *timeDistributorEnquiryCall* (with the *isTimeDistributor* element set to true) to all devices in5437 the SPINE network. All time distributors would send a *timeDistributorData* instance as notification to

5438 the device. The new device would pick the time distributor with the highest priority and send a read

5439 command to the *timeInformationData* function of this distributor. The time distributor would reply5440 with a *timeInformationData* containing its current time.

5441

5442 **4.3.30.3 Rules**

5443 None.

5444

5445 **4.3.30.4 timeInformationData**5446 **4.3.30.4.1 Description**5447 The *timeInformationData* command is used for exchanging the SPINE time.

5448

5449 **4.3.30.4.2 Rules**

5450 None.

5451

5452 **4.3.30.4.3 Element rules**

Element	Rule	Description
TimeInformation. timeInformationData.		
utc	SHALL be set if <i>isTimeDistributor</i> is set	The current time in UTC.

	"true". Otherwise it SHOULD be set.	
utcOffset	None.	The configured offset to UTC (e.g. PT2H [= +2 hours] for Central European Summer Time (CEST)).
dayOfWeek	None.	To model the actual day of week (Monday to Sunday) this element is used. It refers to the local time. If there is no utcOffset given, it is assumed, that the local time is equivalent to UTC-time.
calendarWeek	None.	A value between 1 and 53 stating the current calendar week number. This element refers to the local time. If there is no utcOffset given, it is assumed, that the local time is equivalent to UTC-time.

Table 147: *timeInformationData* Element rules

4.3.30.5 *timeDistributorData*

4.3.30.5.1 *Description*

One or some devices in a SPINE network should act as time distributor. The information thereof and the dedicated priority can be modelled with the *timeDistributorData* command.

4.3.30.5.2 *Rules*

None.

4.3.30.5.3 *Element rules*

Element	Rule	Description
TimeInformation. timeDistributorData.		
isTimeDistributor	If set to "true" the device SHALL be able to act as time distributor. Otherwise it SHALL be omitted or set to "false".	Denotes whether a device can act as a time distributor in the SPINE network or not.
distributorPriority	If <i>isTimeDistributor</i> is set to "false", this element SHOULD be absent. The time distributor with the highest priority SHOULD be used for time synchronization. There MAY be further time distributors with lower priority to have redundant time distributors that can be used as soon as the time distributor with the higher priority can currently not be used. Only values greater than zero SHALL be used. A value of "1" SHALL be interpreted as highest priority. The priority decreases with increasing value of distributorPriority.	A (configured) priority of the time distributor.

Table 148: *timeDistributorData* Element rules

5465

5466 **4.3.30.6 *timePrecisionData***5467 **4.3.30.6.1 *Description***5468 Information regarding the precision of the time of a device can be modelled using *timePrecisionData*.

5469

5470 **4.3.30.6.2 *Rules***

5471 The element "isSynchronised" SHALL be used with regards to the time distributor with the highest
 5472 priority. Time should not differ more than 10 seconds from the time retrieved from a commonly used
 5473 NTP server in the internet (e.g. ptbtime1.ptb.de). If a device has a *clockDrift* element set to 1, it
 5474 should at least get a sync every 10 days. After that time, it can not be sure to have a correct time
 5475 information and therefore sets "isSynchronised" to false.

5476

5477 **4.3.30.6.3 *Element rules***

Element	Rule	Description
TimeInformation. <i>timePrecisionData</i> .		
isSynchronised	SHOULD be set. If omitted, isSynchronised SHALL be interpreted as "false".	This is a Boolean value, stating whether the device's SPINE time information is synchronized with the SPINE network or not, meaning that it could receive the SPINE time from a time distributor and has still a "sufficiently precise time".
lastSyncAt	MAY be used to communicate when the last synchronization occurred.	This is the timestamp of the last synchronisation with a time distributor, or, in case of a time distributor itself, the timestamp of the last official date and time information (e.g. from an NTP server).
clockDrift	MAY be used to communicate the clock drift.	The drift of the internal clock (in seconds per day) is indicated by this element.

5478 *Table 149: timePrecisionData Element rules*

5479

5480 **4.3.30.7 *timeDistributorEnquiryCall***5481 **4.3.30.7.1 *Description***

5482 This function can be used for finding all time distributors in the network. Therefore, it can be sent to
 5483 all devices in the network that have their *isTimeDistributor* element in the function *timeDistributor*
 5484 (see section 5.3.28.3) set to true.

5485

5486 **4.3.30.7.2 *Rules***

5487 Devices that do not support the function *timeDistributor* SHOULD NOT be used as destination of this
 5488 call.

5489

5490 *4.3.30.7.3 Element rules*

5491 None.

5492

5493 **4.3.30.8 Descriptive information and further definitions**

5494 None.

5495

5496 **4.3.31 TimeSeries**5497 **4.3.31.1 Dependencies to other Feature Types**

5498 None.

5499

5500 **4.3.31.2 Introduction**

5501 TimeSeries is a versatile class that allows to express any kind of time dependent values as a time
 5502 series of values. It can be used to express electricity related values like power constraints or power
 5503 consumption over time, but also other values that are not electricity related like temperature or
 5504 humidity over time.

5505

5506 **4.3.31.3 Rules**

5507 None.

5508

5509 **4.3.31.4 timeSeriesListData**5510 *4.3.31.4.1 Description*

5511 The timeSeriesListData function is used to describe the value of a timeSeries over time.

5512

5513 *4.3.31.4.2 Rules*

5514 None.

5515

5516 *4.3.31.4.3 Element rules*

Element	Rule	Description
TimeSeries. timeSeriesListData. timeSeriesData. (list)		
timeSeriesId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a timeSeries.
timePeriod	If set, the time period information within all timeSeriesSlots SHALL NOT exceed the time period listed here.	Time period of the complete timeSeries (also called overall time period).

timeSeriesSlot (list)	A timeSeries SHOULD have at least one timeSeriesSlot.	Allows to define slots of a timeSeries.
timeSeriesSlot. timeSeriesSlotId	SHALL be set as SUB IDENTIFIER, if timeSeriesData contains more than one timeSeriesSlot entry. In case of more than one timeSeriesSlot entry, timeSeriesSlotId SHALL increase matching with the chronological order of the timeSeriesSlots. If timeSeriesSlotId is omitted it SHALL be interpreted as zero.	Allows identification of a timeSeriesSlot within a timeSeries.
timeSeriesSlot. timePeriod	timeSeriesSlot. timePeriod. startTime SHALL be used in timeSeriesSlots that have a time gap to the previous timeSeriesSlot. timeSeriesSlot. timePeriod. startTime SHALL be used in the first timeSeriesSlot if no startTime is provided in the timePeriod element with the overall time period of the timeSeries.	Allows to define a timePeriod of a timeSeriesSlot and to model time gaps between timeSeriesSlots.
timeSeriesSlot. duration	MAY be set if "timeSeriesType" in the respective "timeSeriesDescriptionData" (with the same value of "timeSeriesId") is set to "singleDemand". SHALL be set otherwise. SHALL only contain values greater than zero seconds. If (within the same timeSeriesSlot) timeSeriesSlot. timePeriod. startTime is set, the duration starts from there. If timeSeriesSlot. timePeriod. startTime is not set in the first timeSeriesSlot (slot with the lowest timeSeriesSlotId), the timePeriod. startTime of the overall timeSeries SHALL be set and the duration of the first timeSeriesSlot starts from there. If timeSeriesSlot. timePeriod. startTime is not set in a timeSeriesSlot that is not the first timeSeriesSlot, the duration starts after the previous timeSeriesSlot is finished.	Allows to define a duration of a timeSeriesSlot.
timeSeriesSlot. recurrenceInformation		If the timeSeriesSlot is a recurring one, the corresponding information can be found within this element and its sub-elements.

timeSeriesSlot. value	A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If minValue is set the content of value SHALL be greater. If maxValue is set the content of value SHALL be smaller.	Defines the expected value during a timeSeriesSlot.
timeSeriesSlot. minValue	A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value or maxValue is set the content of minValue SHALL be smaller.	Defines a lower value limit.
timeSeriesSlot. maxValue	A positive value SHALL relate to consumption and a negative value SHALL relate to production of the server. If value or minValue is set the content of maxValue SHALL be greater.	Defines an upper value limit.

Table 150: timeSeriesListData Element rules

4.3.31.5 timeSeriesDescriptionListData

4.3.31.5.1 Description

The timeSeriesDescriptionListData function is used to describe a timeSeries.

4.3.31.5.2 Rules

This is the first data that SHOULD initially be read of a timeSeries. If this data matches with the data that the client is expecting, the other functions of timeSeries become relevant.

4.3.31.5.3 Element rules

Element	Rule	Description
TimeSeries. timeSeriesDescriptionListData. timeSeriesDescriptionData. (list)		
timeSeriesId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a timeSeries.
timeSeriesType	Value rules of Table 152 SHALL be applied.	Allows to define which timeSeries type is supported.
timeSeriesWriteable	If set to "true", this timeSeriesData SHALL be writeable. If set to false or omitted, this timeSeriesData SHALL NOT be writeable.	Allows to define if the timeSeries is writeable.
updateRequired	If updateRequired is "true", the server expects the responsible client to update the writeable or changeable data related to the same PRIMARY IDENTIFIER. When the client updates the server, the client SHALL ignore updateRequired until the writeable and changeable data related to the same PRIMARY IDENTIFIER was updated successfully.	Request a write to timeSeriesData from the bound client.

	The server SHALL set updateRequired back to "false", as soon as the update of the writeable or changeable data related to the same PRIMARY IDENTIFIER was successful. The server SHALL ensure that only one responsible client is able to update the related data. However, also if updateRequired is set to "false" a server SHOULD in general allow updates of the data from the responsible client. The server MAY choose to withdraw the update request at any time by setting updateRequired back to "false".	
measurementId	MAY be set as FOREIGN IDENTIFIER to link data of other Features to the timeSeries.	Allows to link data of other Features to a timeSeries, e.g. Measurement or ElectricalConnection data
currency	If set, the currency SHALL be applied to the value in timeSeriesData	Allows to describe the currency of the value, if the value is a price.
unit	SHALL be set if the corresponding value has a unit. If set, the unit SHALL be applied to the value in timeSeriesData	Allows to describe the unit of the value, if the value has a unit.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	Allows to define user friendly name for the timeSeries.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Allows to define a user friendly description for the timeSeries.
scopeType	None.	Specifies a more detailed meaning of the time series.

5528 Table 151: timeSeriesDescriptionListData Element rules

5529 Element "**timeSeriesType**" value rules:

Value	Rule	Description
plan	"plan" SHALL be used to model how a certain value will change over time. value as well as minValue and maxValue MAY be used.	"plan" is used to model how a certain value changes over time.
singleDemand	"singleDemand" SHALL be used to model a single demand with a simple timeSeries that SHALL include only one slot.	"singleDemand" is used to model a single demand with a simple timeSeries (e.g. this can be used to model a simple energy demand).
constraints	"constraints" SHALL be used to model certain value constraints or limits over time. For this the minValue or maxValue or both SHALL be used.	"constraints" is used to model certain value constraints over time with a curve of minValue, maxValue or both.

energyRequest	"energyRequest" SHALL be used for energy consumption requests of an appliance.	"energyRequest" is used for energy consumption requests of an appliance.
dischargingEnergyRequest	"dischargingEnergyRequest" SHALL be used for energy production requests of an appliance.	"dischargingEnergyRequest" is used for energy production requests of an appliance (e.g., discharging a battery).
consumptionLimitCurve	"consumptionLimitCurve" SHALL be used for curves limiting the consumption.	"consumptionLimitCurve" is used for curves limiting the consumption.
productionLimitCurve	"productionLimitCurve" SHALL be used for curves limiting the production.	"productionLimitCurve" is used for curves limiting the production.

Table 152: Element "timeSeriesType" value rules

4.3.31.6 timeSeriesConstraintsListData

4.3.31.6.1 Description

Allows to define constraints for the timeSeries, e.g. what is the maximum amount of slots allowed or similar. This function can be especially valuable if timeSeriesData is writeable but the server has certain constraints. The client can then match its own constraints with the server constraints to create a proper write command.

4.3.31.6.2 Rules

None.

4.3.31.6.3 Element rules

Element	Rule	Description
TimeSeries. timeSeriesConstraintsListData. timeSeriesConstraintsData. (list)		
timeSeriesId	SHALL be set as PRIMARY IDENTIFIER.	Allows the identification of a timeSeries.
slotCountMin	If set and a corresponding timeSeriesData exists, the timeSeriesData SHALL NOT have less timeSeriesSlots.	The minimum amount of timeSeriesSlots for a specific time series is stated in this element.
slotCountMax	If set and a corresponding timeSeriesData exists, the timeSeriesData SHALL NOT have more timeSeriesSlots.	The maximum amount of timeSeriesSlots for a specific time series is stated in this element. Recurring timeSeriesSlots only count as one.
slotDurationMin	If set and a corresponding timeSeriesData exists, the	If the timeSeriesSlots of this time series have a minimum duration, it is denoted here.

	timeSeriesData SHALL NOT have timeSeriesSlots with shorter duration. If slotDurationStepSize is set, slotDurationMin SHALL be a multiple of slotDurationStepSize.	
slotDurationMax	If set and a corresponding timeSeriesData exists, the timeSeriesData SHALL NOT have timeSeriesSlots with longer duration. If slotDurationStepSize is set, slotDurationMax SHALL be a multiple of slotDurationStepSize.	If the timeSeriesSlots of this time series have a maximum duration, it is denoted here.
slotDurationStepSize	If set, the duration of a timeSeriesSlot SHALL be a multiple of slotDurationStepSize and also slotDurationMin and slotDurationMax SHALL be multiples of slotDurationStepSize.	timeSeriesSlots can have a specific duration step size. This element holds the corresponding value.
earliestTimeSeriesStartTime	If set, the start time of a timeSeries SHALL be later than or equal to earliestTimeSeriesStartTime.	This is the earliest start time the corresponding time series is allowed to start.
latestTimeSeriesEndTime	If set, the end time of a timeSeries SHALL be earlier than or equal to latestTimeSeriesEndTime.	This is the latest end time the corresponding times series is allowed to end.
slotValueMin	The elements value, maxValue or minValue of a timeSeriesSlot SHALL be equal to or higher than slotValueMin.	The values within the timeSeriesSlots of the corresponding time series are not allowed to be lower than this value.
slotValueMax	The value, maxValue or minValue of a timeSeriesSlot SHALL be equal to or lower than slotValueMax.	The values within the timeSeriesSlots of the corresponding time series are not allowed to be higher than this value.
slotValueStepSize	If slotValueStepSize is set and neither slotValueMin nor slotValueMax is set, the Elements value, maxValue and minValue of a timeSeriesSlot within the Function timeSeriesListData SHALL be a multiple of slotValueStepSize. If slotValueStepSize is set and slotValueMin is also set, the Elements value, maxValue and minValue (if set) of a timeSeriesSlot within the Function timeSeriesListData SHALL be slotValueMin plus a multiple of slotValueStepSize. If slotValueStepSize is set and slotValueMax is also set, the Elements	The minimum step size between two different values within the timeSeriesSlots of the corresponding time series.

	value, max Value and min Value of a timeSeriesSlot within the Function timeSeriesListData SHALL be slotValueMax minus a multiple of slotValueStepSize.	
--	--	--

Table 153: timeSeriesConstraintsListData Element rules

4.3.32 TimeTable

4.3.32.1 Dependencies to other Feature Types

None.

4.3.32.2 Introduction

All time dependent functionality like setpoint-configuration, etc. makes use of this elementary class. It provides a list of time tables, each containing some number of time slots. Each slot has a start and end time. The times can either be absolute (*dateTime*) or recurring (*daysOfWeek* plus *time* and, if required, the recurring interval step elements). The count of slots and the start and end times can be configurable or static (device specific). Other classes that make use of time tables only reference to the *timeTableId*, not to specific time slots.

4.3.32.3 Rules

None.

4.3.32.4 timeTableListData

4.3.32.4.1 Description

The list of time tables and time slots together with the start and end times are modelled in this function.

4.3.32.4.2 Rules

None.

4.3.32.4.3 Element rules

Element	Rule	Description
TimeTable. timeTableListData. timeTableData. (list)		
timeTableId	SHALL be set as PRIMARY IDENTIFIER.	The <i>timeTableId</i> is used for indentifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
timeSlotId	SHALL be set as SUB IDENTIFIER.	Each slot of a time table has its own slot ID.

recurrenceInformation	None.	Information about the recurrence of a slot.
startTime	None.	Start time of one slot.
endTime	None.	End time of one slot.

Table 154: *timeTableListData* Element rules**4.3.32.5 timeTableConstraintsListData****4.3.32.5.1 Description**

If a time table has some constraints, they are denoted with the help of this function (e.g. the minimum and maximum count of slots (may differ between the time tables)).

4.3.32.5.2 Rules

None.

4.3.32.5.3 Element rules

Element	Rule	Description
TimeTable. timeTableConstraintsListData. timeTableConstraintsData. (list)		
timeTableId	SHALL be set as PRIMARY IDENTIFIER.	The <i>timeTableId</i> is used for indentifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
slotCountMin	If set, the time table SHALL NOT have less slots.	The minimum amount of slots for a specific time table is stated in this element. Recurring slots only count as one.
slotCountMax	If set, the time table SHALL NOT have more slots.	The maximum amount of slots for a specific time table is stated in this element. Recurring slots only count as one.
slotDurationMin	If set, the corresponding time table SHALL NOT have slots with lower duration.	If the slots of this time table have a minimum duration, it is denoted here.
slotDurationMax	If set, the corresponding time table SHALL NOT have slots with higher duration.	If the slots of this time table have a maximum duration, it is denoted here.
slotDurationStepSize	If set, the duration of a slot SHALL be a multiple of slotDurationStepSize.	Slots can have a specific duration step size. This element holds the corresponding value.
slotShiftStepSize	If set, the start time of a slot SHALL be firstSlotBeginsAt + a multiple of slotShiftStepSize.	Slots can have a specific shift step size. This element holds the corresponding value.
firstSlotBeginsAt	If set, the start time of the first slot SHALL be firstSlotBeginsAt.	Especially time tables with fixed slot durations probably need this information to specify, where in time the slots are located (i.e. fixed 15 minute slots can start either at 00:00:00 (default) or e.g. at 00:10:00).

		This value only defines, when the time pattern (defined by the element <i>slotDurationStepSize</i>) begins on a day.
--	--	---

Table 155: *timeTableConstraintsListData* Element rules**4.3.32.6 timeTableDescriptionListData****4.3.32.6.1 Description**

The more static information about the time tables are described in the *timeTableDescriptionData* function.

4.3.32.6.2 Rules

None.

4.3.32.6.3 Element rules

Element	Rule	Description
TimeTable. timeTableDescriptionListData. timeTableDescriptionData. (list)		
timeTableId	SHALL be set as PRIMARY IDENTIFIER.	The <i>timeTableId</i> is used for indentifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
timeSlotCountChangeable	If set to "true", appropriate clients (e.g. the bound client) SHALL be allowed to add new slots to or remove slots from the timeTable. Otherwise the element SHALL be omitted or set to "false". The timings of added slots SHALL NOT overlap with existing slots.	States whether the amount of slots is changeable by a client for a specific time table or not.
timeSlotTimesChangeable	If set to "true", appropriate clients (e.g. the bound client) SHALL be able to change timings of existing slots. Otherwise it SHALL be omitted or set to "false".	States whether the times of the slots of a specific time table are changeable by a client or not.
timeSlotTimeMode	Value rules of Table 157 SHALL be applied.	If either <i>absolute</i> or <i>recurring</i> time slots or <i>both</i> modes are supported is stated in this element.
label	Default rules of section 3.10.1.2.1 SHALL be applied.	A short label of the time table.
description	Default rules of section 3.10.1.3.1 SHALL be applied.	Descriptive information on the time table.

Table 156: *timeTableDescriptionListData* Element rules

5592 Element "**timeSlotTimeMode**" value rules:

Value	Rule	Description
absolute	None.	Only absolute times are allowed for the time table.
recurring	None.	Only recurring times are allowed for the time table.
both	None.	Absolute and recurring times are allowed for the time table.

5593 *Table 157: Element "timeSlotTimeMode" value rules*

5594

5595 **4.3.32.7 Descriptive information and further definitions**

5596 None.

5597

5598

5 Class descriptions

5.1 Introduction

In this chapter, all Classes defined by the EEBUS Initiative e.V. are described. It should be seen as equivalent to the XSD definitions. Everything that can be found there is described within the sections of this chapter. Descriptions and rules for the Functions and Elements are only described within the Feature Types as they define a more specific meaning for a Class (see section 2.1.2.3).

Please refer to section 2.2 for details on the concepts of Standard Classes and Complex Classes.

5.2 Complex Classes

5.2.1 IncentiveTable

5.2.1.1 Introduction

The IncentiveTable data model allows to define tiers with different incentives and boundaries. In the case the incentive is price based, it also could be called a price table. However, there can also be other incentives, CO₂ Emission for example. Within the boundaries of a tier the incentive of the tier apply. This allows to distribute e.g. different types of power to different power consumers. E.g. a photovoltaic system produces more power than can be consumed and fed into the electricity grid (surplus power), in this case the power is for free and can be distributed from the CEM to consumers that have the flexibility to increase their consumption. When all surplus power is consumed the next tier would be the electricity grid feed-in power, which in this example would be cheaper than the electricity grid power. In most cases the electricity grid power would be more expensive and therefore would be the last tier. However, the electricity grid power can also get cheaper, e.g. in case the electricity grid has an excess of power available an incentive to flexible consumers could be given.

The incentiveTable has three different function groups:

1. incentiveTable: holds the actual values of the boundaries and incentives
2. incentiveTableDescription: holds data that describes the values of incentiveTable
3. incentiveTableConstraints: allows a server to define certain constraints regarding the incentiveTable, this is especially important if the IncentiveTable is writeable

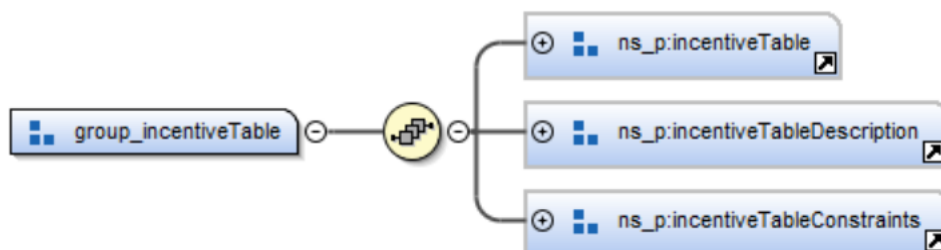


Figure 41: IncentiveTable function-group overview

The tariffId is used to identify the data of different IncentiveTables within each function.

5.2.1.2 *incentiveTableData*

5.2.1.2.1 *General*

The incentiveTable within the incentiveTableData function holds the values of the IncentiveTable.

This includes the boundary values as well as the incentive values for each tier.

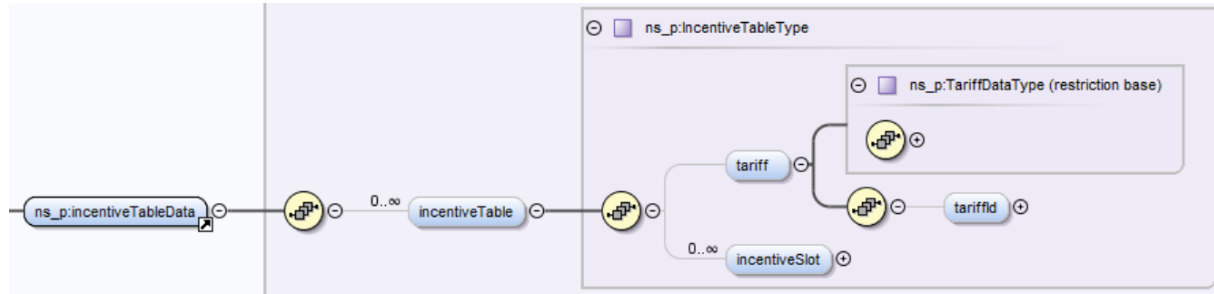


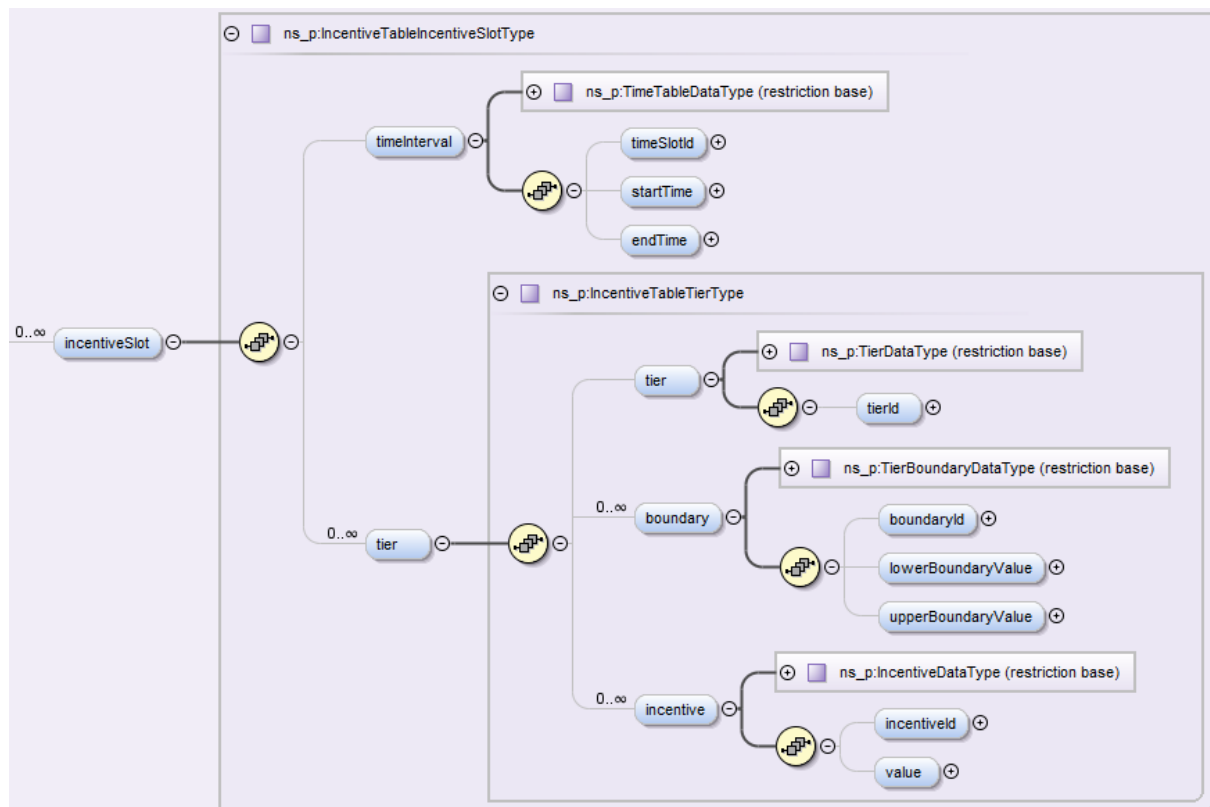
Figure 42: incentiveTableData function overview

5.2.1.2.2 *Detailed description of elements*

Element	Type	Description
incentiveTable.		
incentiveTable (list)		
incentiveTable. tariff		
incentiveTable. tariff. tariffid	Identifier "TariffidType" (see Table 358).	Allows the identification of a incentiveTable. Allows linking of the data in the different functions of IncentiveTable. Allows linking of the incentiveTables within other features that are placed in the same entity.
incentiveTable. incentiveSlot (list)	Complex type "IncentiveTableIncentiveSlotType". See Table 159.	An incentiveTable can include different incentiveSlots to describe changes of boundaries or incentives over time. Please refer to the detailed description of IncentiveTableIncentiveSlotType elements in Table 159.

Table 158: incentiveTable detailed description of elements

IncentiveTableIncentiveSlotType:



5641

5642 Figure 43: IncentiveTableIncentiveSlotType overview

Element	Type	Description
timeInterval.		Time period of the incentiveSlot.
timeInterval. timeSlotId	Identifier "TimeSlotIdType" (see Table 358).	Allows identification of an incentiveSlot within an incentiveTable.
timeInterval. startTime.		
timeInterval. startTime. dateTime	xs:dateTime (W3C standard type)	Absolute start time.
timeInterval. startTime. relative	xs:duration (W3C standard type)	Relative start time containing a value relative to now. Positive values are in the future, negative ones are in the past and zero seconds represents "now".
timeInterval. endTime.		
timeInterval. endTime. dateTime	xs:dateTime (W3C standard type)	Absolute end time.
timeInterval. endTime. relative	xs:duration (W3C standard type)	Relative end time containing a value relative to now. Positive values are in the future, negative ones are in the past and zero seconds represents "now".
tier (list)		
tier. tier		
tier. tier. tierId	Identifier "TierIdType" (see Table 358).	Allows identification of a tier within a incentiveTable.
tier. boundary (list)		
tier. boundary. boundaryId	Identifier "TierBoundaryIdType" (see Table 358).	Allows identification of a boundary within a tier.

tier. boundary. lowerBoundaryValue	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	This represents the lower end value range boundary.
tier. boundary. upperBoundaryValue	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	This represents the upper end value range boundary.
tier. incentive (list)		
tier. incentive. incentiveId	<i>Identifier "IncentiveIdType" (see Table 358).</i>	Allows identification of an incentive within a tier.
tier. incentive. value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The value of the incentive.

Table 159: IncentiveTableIncentiveSlotType detailed description of elements

5.2.1.2.3 Available selectors

- tariff. tariffId

5.2.1.3 incentiveTableDescriptionData

5.2.1.3.1 General

The incentiveTableDescription within the incentiveTableDescriptionData function holds the description of the IncentiveTable. This includes the tier, boundary and incentive type as well as further elements.

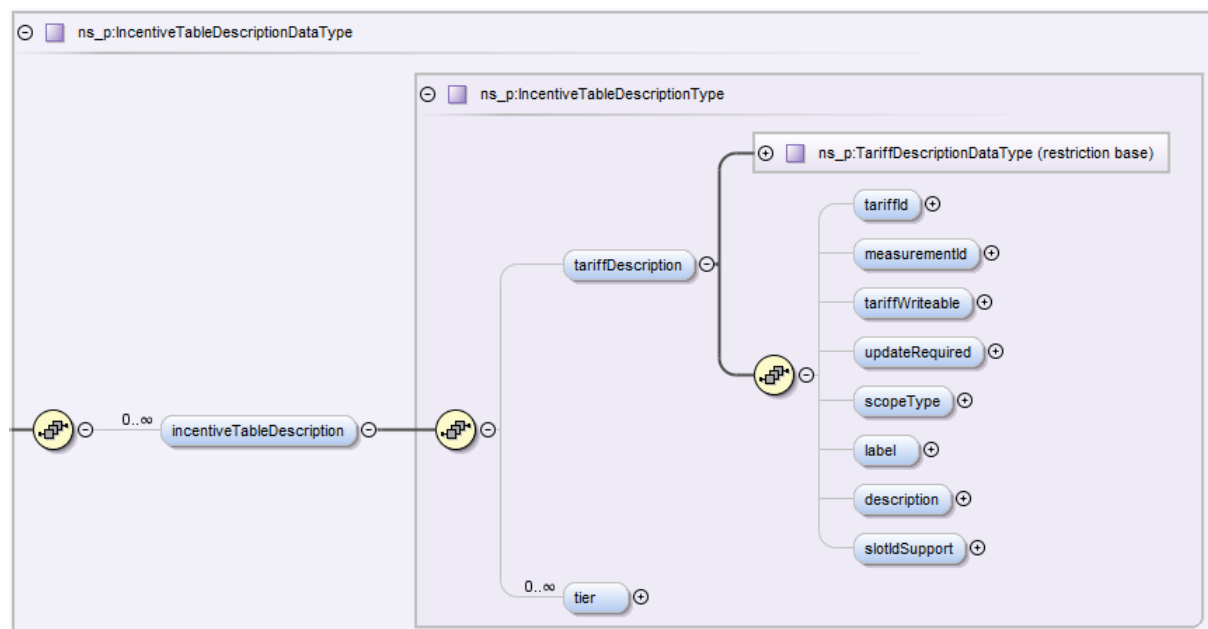


Figure 44: incentiveTableDescriptionData function overview

5.2.1.3.2 Detailed description of elements

Element	Type	Description
incentiveTableDescriptionData.		

incentiveTableDescription (list)		
incentiveTableDescription. tariffDescription.		
incentiveTableDescription. tariffDescription. tariffId	<i>Identifier "TariffIdType" (see Table 358).</i>	Allows the identification of a incentiveTable. Allows linking of the data in the different functions of IncentiveTable. Allows linking of the incentiveTables within other features that are placed in the same entity.
incentiveTableDescription. tariffDescription. measurementId	<i>Identifier "MeasurementIdType" (see Table 358).</i>	If a measurand is linked to the incentive table, its identifier is denoted here.
incentiveTableDescription. tariffDescription. tariffWriteable	xs:boolean (W3C standard type)	Whether the incentive table is writeable by a client or not can be denoted in this element.
incentiveTableDescription. tariffDescription. updateRequired	xs:boolean (W3C standard type)	Allows to request an update from the bound client.
incentiveTableDescription. tariffDescription. scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	A scopeType for the incentive table can be stated here.
incentiveTableDescription. tariffDescription. label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A short label on the incentive table.
incentiveTableDescription. tariffDescription. description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A description for the incentive table.
incentiveTableDescription. tariffDescription. slotIdSupport	xs:boolean (W3C standard type)	Indicates support of timeSlotId within the corresponding tariff in the Function incentiveTableData.
tier (list)	<i>Complex type "IncentiveTableDescriptionTierType". See Table 161.</i>	Allows to describe tiers and the incentives and boundaries that are used by the tier. Please refer to the detailed description of <i>IncentiveTableDescriptionTierType elements</i> .

5657 Table 160: incentiveTableDescriptionData function detailed description of elements

5658 IncentiveTableDescriptionTierType:

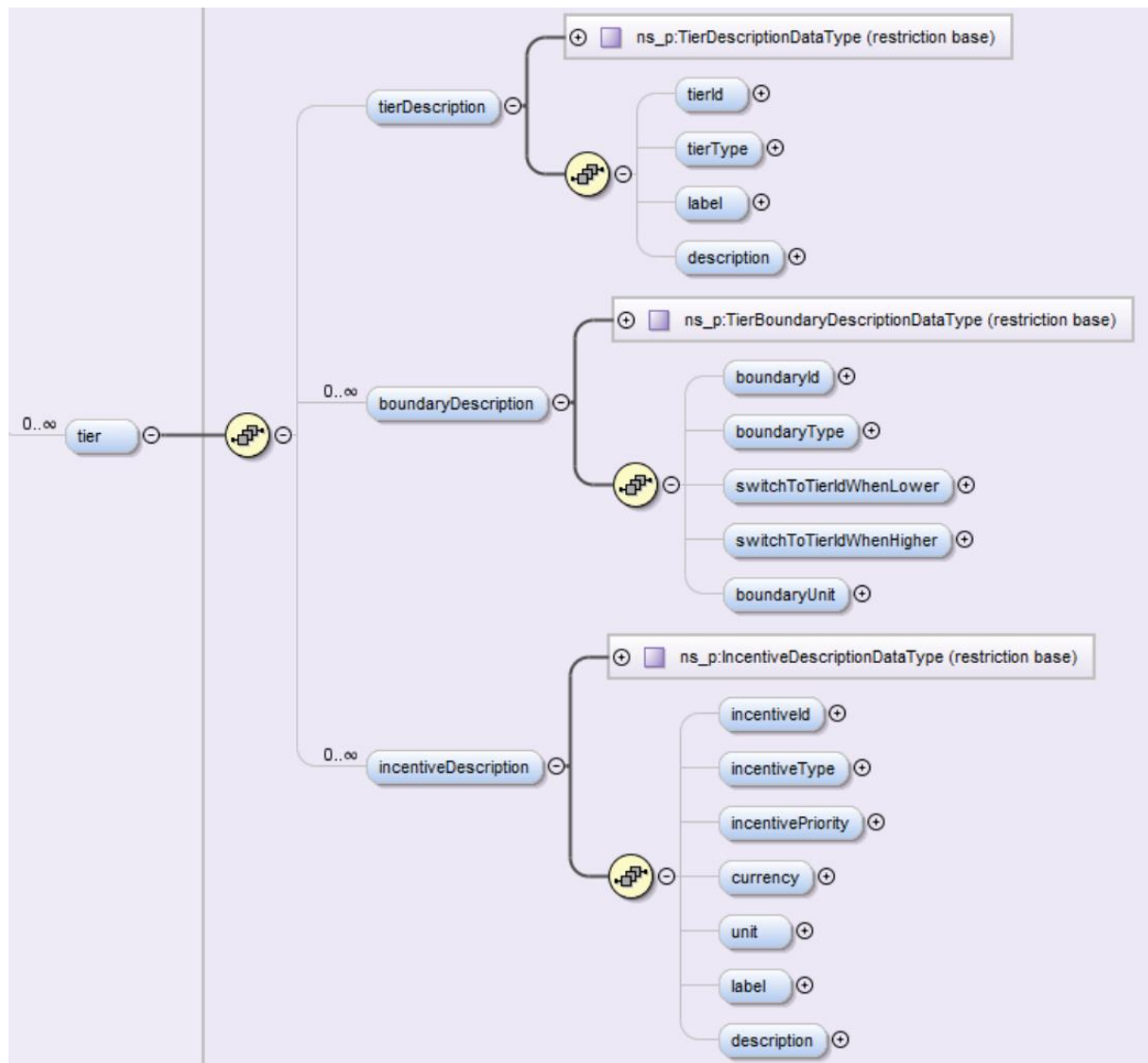


Figure 45: IncentiveTableDescriptionTierType overview

Element	Type	Description
tierDescription.		
tierDescription. tierId	Identifier "TierIdType" (see Table 358).	Allows identification of a tier within a incentiveTable.
tierDescription. tierType	Union "TierTypeType": - Enum (see Table 162): TierTypeEnumType - EnumExtendType (see section 3.10.1.5)	Allows to define the type of the tier
tierDescription. label	Common data type "LabelType". See section 3.10.1.2.	Allows to define user friendly name for the tier.
tierDescription. description	Common data type "DescriptionType". See section 3.10.1.3.	Allows to define a user friendly description for the tier.
boundaryDescription (list)		
boundaryDescription. boundaryId	Identifier "TierBoundaryIdType" (see Table 358).	Allows identification of a boundary within a tier.
boundaryDescription. boundaryType	Union "TierBoundaryTypeType": - Enum (see Table 163):	Allows to define the type of the boundary.

	TierBoundaryTypeEnumType - EnumExtendType (see section 3.10.1.5)	
boundaryDescription. switchToTierIdWhenLower	Identifier "TierIdType" (see Table 358).	Link to the tier that will become active if the lowerBoundaryValue is underrun.
boundaryDescription. switchToTierIdWhenHigher	Identifier "TierIdType" (see Table 358).	Link to the tier that will become active if the upperBoundaryValue is exceeded.
boundaryDescription. boundaryUnit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	Allows to define the unit of the boundary values.
incentiveDescription (list)		
incentiveDescription. incentiveId	Identifier "IncentiveIdType" (see Table 358).	Allows identification of an incentive within a tier.
incentiveDescription. incentiveType	Union "IncentiveTypeEnumType": - Enum (see Table 164): IncentiveTypeEnumType - EnumExtendType (see section 3.10.1.5)	Allows to define the type of the incentive.
incentiveDescription. incentivePriority	Simple type "IncentivePriorityType" (restriction of xs:unsignedInt)	Allows to define the priority of the incentive. E.g. the priority that the customer has assigned for this incentive.
incentiveDescription. currency	Common data type "CurrencyType". See section 3.10.1.19.	Monetary incentives may need to describe a currency.
incentiveDescription. unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	Allows definition of a unit, if the incentive has a unit.
incentiveDescription. label	Common data type "LabelType". See section 3.10.1.2.	Allows to express a user friendly label for the incentive.
incentiveDescription. description	Common data type "DescriptionType". See section 3.10.1.3.	Allows to express a user friendly description for the incentive.

5661 Table 161: IncentiveTableDescriptionTierType detailed description of elements

5662 Enumeration **TierTypeEnumType**:

Value	Rule	Description
fixedCost	The tier has a cost incentive that SHALL NOT vary over time	The fixed cost have to be payed, independent from the consumed energy.
dynamicCost	The tier has a cost incentive that MAY vary over time	The dynamic cost has a (linear) dependency to the consumed energy.

5663 Table 162: Enumeration TierTypeEnumType

5664 Enumeration **TierBoundaryTypeEnumType**:

Value	Rule	Description
-------	------	-------------

powerBoundary	The boundary SHALL be an electrical power. E.g. if the surplus power is consumed, more expensive power has to be used.	Provides boundaries for power.
energyBoundary	The boundary SHALL be an electrical energy. E.g. if the stored self produced energy in a battery is consumed, more expensive energy has to be used.	Provides boundaries for energy.
countBoundary	The boundary SHALL be a count. E.g. after a certain count the price of a good may be reduced.	Provides boundaries for counts.

Table 163: Enumeration TierBoundaryTypeEnumType

Enumeration IncentiveTypeEnumType:

Value	Rule	Description
absoluteCost	None.	Absolute costs with a currency and a unit.
relativeCost	None.	Relative costs that cannot be interpreted as some monetary costs.
renewableEnergyPercentage	None.	Percentage of renewable energy of the total obtained energy. This is an ecological incentive that allows optimizing consumption of renewable energy instead of costs.
co2Emission	None.	CO2 emission related to the obtained energy. This is an ecological incentive that allows optimizing the CO2 emission instead of costs.

Table 164: Enumeration IncentiveTypeEnumType

5.2.1.3.3 Available selectors

- tariffDescription. tariffId
- tariffDescription. scopeType

5.2.1.4 incentiveTableConstraintsData

5.2.1.4.1 General

The incentiveTableConstraints within the incentiveTableConstraintsData function holds additional constraints of the IncentiveTable, regarding how much tiers, boundaries, incentives and slots are allowed.

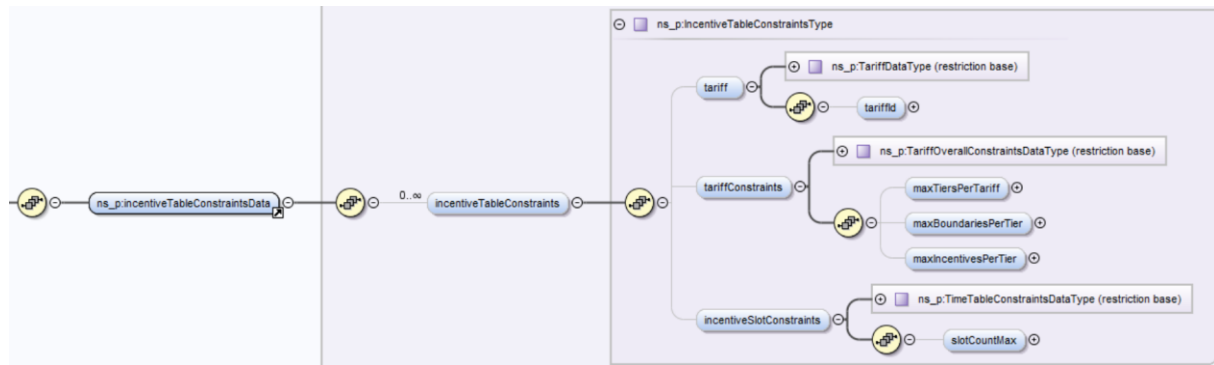


Figure 46: incentiveTableConstraintsData overview

5.2.1.4.2 Detailed description of elements

Element	Type	Description
incentiveTableConstraintsData.		
incentiveTableConstraints (list)		
incentiveTableConstraints. tariff		
incentiveTableConstraints. tariff. tariffId	Identifier "TariffIdType" (see Table 358).	Allows the identification of a incentiveTable. Allows linking of the different functions within the IncentiveTable Feature. Allows linking of the incentiveTable within other Features that are placed in the same entity.
incentiveTableConstraints. tariffConstraints		
incentiveTableConstraints. tariffConstraints. maxTiersPerTariff	Simple type "TierCountType" (restriction of "TierIdType" (see Table 358)).	Allows the server to define a maximum tier count for this IncentiveTable. This is especially important if a client wants to write an IncentiveTable.
incentiveTableConstraints. tariffConstraints. maxBoundariesPerTier	Simple type "TierBoundaryCountType" (restriction of "TierBoundaryIdType" (see Table 358)).	Allows the server to define a maximum count of "boundary" instances for tiers. This is especially important if a client wants to write an IncentiveTable.
incentiveTableConstraints. tariffConstraints. maxIncentivesPerTier	Simple type "IncentiveCountType" (restriction of "IncentiveIdType" (see Table 358)).	Allows the server to define a maximum incentive count for tiers. This is especially important if a client wants to write an IncentiveTable.
incentiveTableConstraints. incentiveSlotConstraints		
incentiveTableConstraints. incentiveSlotConstraints. slotCountMax	Simple type "TimeSlotCountType" (restriction of "TimeSlotIdType", see Table 347).	Allows the server to define a maximum slot count for this IncentiveTable. This is especially important if a client wants to write an IncentiveTable.

5682 *Table 165: incentiveTableConstraints detailed description of elements*

5683

5684 *5.2.1.4.3 Available selectors*

5685 - tariff. tariffId

5686

5687 **5.2.2 NodeManagement**

5688 See document [ProtocolSpecification], chapter "Functional commissioning" for details.

5689

5690 **5.2.3 SmartEnergyManagementPs**

5691 **5.2.3.1 Introduction**

5692 *5.2.3.1.1 Scope of this complex class*

5693 This complex class provides solutions for "smart energy management" use cases that are typical for
 5694 domestic homes: On the one hand, energy providers often offer their energy with a price that varies
 5695 over time (e.g. consumption at night is cheaper than at noon). The availability of renewable energies
 5696 from an own photovoltaics also varies over time. On the other hand, the residents certainly need to
 5697 run their white goods (do one's laundry etc.) and other machines, but may have the opportunity to
 5698 shift these tasks in order to reduce the overall cost. Such kind of optimisation is hard to achieve
 5699 manually in order to gain a real benefit. As a consequence, some kind of automation is required.

5700 This complex class provides a solution that can best be expressed with the definition of two roles (i.e.
 5701 device types) for these roles, and the information exchanged between them. The following roles are
 5702 assumed:

5703 1. A "smart appliance":

5704 This is kind of device is usually an energy consuming or producing device, i.e. a device whose
 5705 activity can be worth to be scheduled for a specific time in order to reduce the overall cost.
 5706 Typical devices are whitegoods, photovoltaics, or battery storages. Such devices need to be
 5707 "smart" as they have to be able to provide some information on their activity (esp. ahead of
 5708 time) and (if possible) the capability to shift their activity upon appropriate requests from a
 5709 role that is described next.

5710 2. An energy management system:

5711 A smart appliance usually cannot have the information required to perform an overall
 5712 optimization of the cost and possibility for energy consumption and production. Instead, this
 5713 is the responsibility of an energy management system. Such a system can gather activity
 5714 information from several appliances of the domestic home. It may even be able to fetch
 5715 information from an energy provider to get details of the tariff in order to execute up-to-date
 5716 price calculations. This collection of information can then be used to adjust the activities of
 5717 the connected appliances.

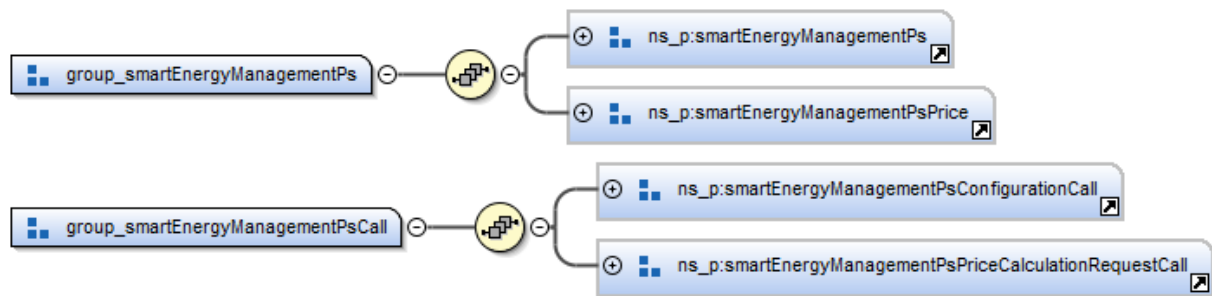


Figure 47: SmartEnergyManagementPs function-group overview

5.2.3.1.2 Re-use of standard classes: PowerSequences etc.

This complex class is mainly based upon definitions and concepts of the class PowerSequences (see section 5.3.19) which are used to express power or energy over time. In fact, this complex class primarily groups definitions from this and some other classes together into "useful collections". Among others, this helps to avoid many single messages and gives the corresponding data parts a more "direct" relation to each other. This way, the overall structure of the complex class basically still represents power sequences. Therefore, throughout this complex class the term "power sequences" is still used. Section 5.2.3.1.3 briefly introduces the power sequences concept of this complex class.

5.2.3.1.3 Theory of operation (informative)

The power sequences structures of this complex class model the energy consumption / generation / storage as a mathematical graph and additional information, such as constraints for time based shifts.

Throughout this complex class, two terms - from the inherited PowerSequences class - are of major interest: "sequence" and "slot". A sequence can be used to describe a "task". It is the most "coarse" view in this class, i.e. a sequence represents all single steps of a whole task. The single steps are represented by slots. Slots are the most "fine" view in this class with regards to the time resolution of a task.

Each sequence and slot can be associated with an identifier or number, respectively. The slot numbers of two sequences should be considered independent from each other. I.e. slot number 7 of sequence 1 describes a different slot than slot number 7 of sequence 2. This basically means a slot is only "uniquely" identified in combination with a sequence ID. As formal notation this leads to the following relation:

- 1 sequence has 1..m slots
- 1 slot belongs to exactly 1 sequence
- 1 pair of a sequenceId and slotNumber uniquely identifies a slot

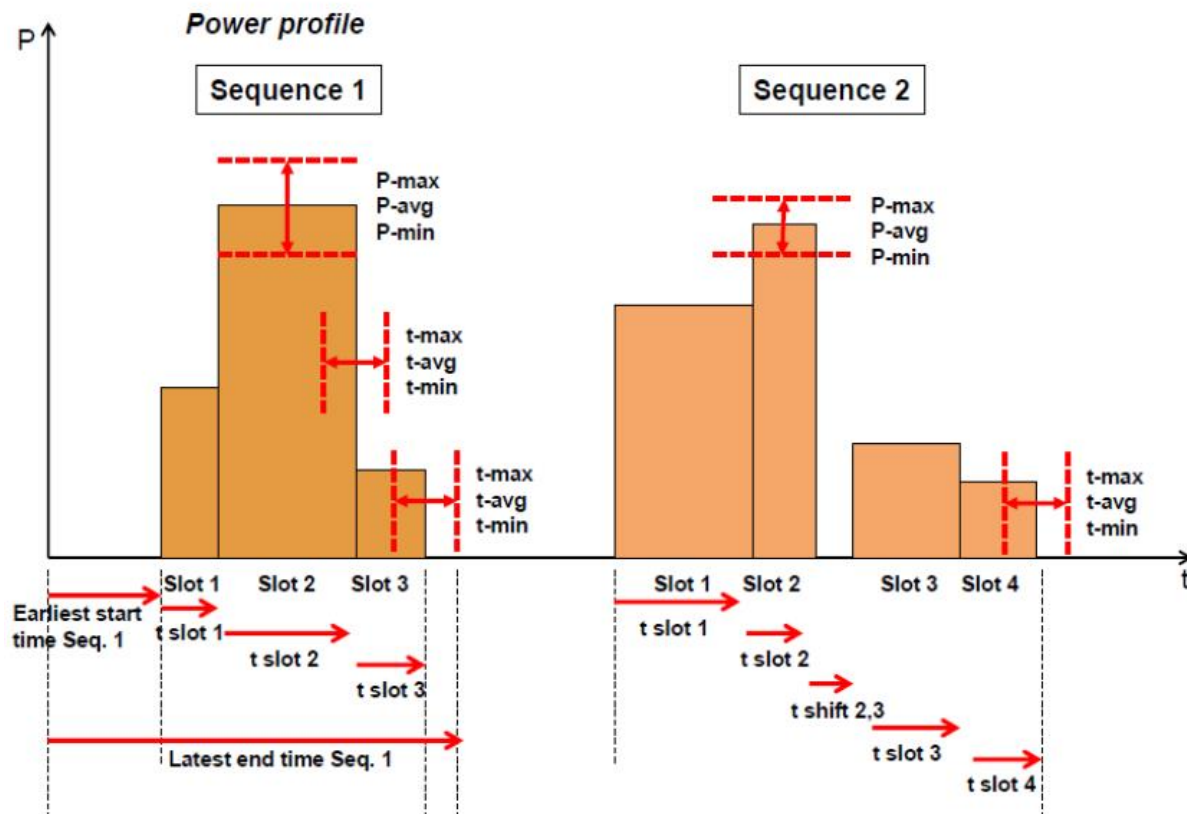


Figure 48: Concept of a power sequence

Each slot is, among other things, characterized by a time it applies for and a list of valid values, which apply for this time, such as average energy consumption, maximum energy consumption and others.

In addition to value properties, a slot and a sequence can be described by their constraints.

Constraints state whether a sequence and/or its single slots can be shifted in time. Additionally, if this is the case, the constraints state the boundaries for time shifts or duration configurations. This contributes to the definition of a power sequence's flexibility.

Some time-related information on a slot shall be considered:

- startTime: The currently scheduled start time of the slot.
- endTime: The currently scheduled end time of the slot.
- defaultDuration: The currently scheduled duration of this slot.
- durationUncertainty: The duration may have some uncertainty. The actual duration then may be in the range from defaultDuration - durationUncertainty to defaultDuration + durationUncertainty, e.g.
- earliestStartTime: The earliest time a slot could start. This determines the "lowest" boundary to shift a slot.
- latestEndTime: The latest time a slot has to be finished. This determines the "highest" boundary to shift a slot.

5767 Within a power sequence, there may be mandatory and optional slots. An example for an optional
5768 slot is a pause/intentional gap between 2 process steps of a device, where the pause might either
5769 take place or can be skipped.

5770 Within a sequence, the model permits in theory to have a "time gap" between two consecutive slots.
5771 However, for some reason it is usually better to express such a gap (or "pause") as regular slot with
5772 an energy value of "0".

5773 Each sequence and slot can be described using textual information, e.g. for displaying it on a
5774 smartphone or an in home display.

5775 A power sequences server can also request a price calculation for a sequence, e.g. to display a price
5776 on a device display to the user.

5777 In general, a so-called power sequences server (see section 5.2.3.1.4) can offer one or more power
5778 sequences. A power sequence can contain flexibility or can instead be used to forecast fixed
5779 consumption or generation behaviour without any flexibility. The granularity, i.e. the number of slots
5780 and value types provided for the slots a device models per power sequence, is a choice which is up to
5781 the manufacturer.

5782 A power sequences server can offer multiple power sequences for different purposes:

- 5783 1. The power sequences can model choices a so-called power sequences client (see 5.2.3.1.4)
5784 can take.
5785 As an example, consider a freezer which can run in normal mode or in deep freezing mode
5786 (which means increased power consumption, but the time frame for the power consumption
5787 is shorter). This choice can be offered to the power sequences client. In this complex class
5788 the power sequences of a choice are combined in a group called "alternatives" (see below).
- 5789 2. The power sequences can model behaviour over time.
5790 As an example, consider an HVAC system which knows that it has to run in the morning and
5791 in the evening. A power sequences server can publish 2 different power sequences, one for
5792 each time frame.
- 5793 3. The power sequences can model mixtures of choices and behaviour over time (1 and 2).

5794 The general concept of power sequences is applicable to many kinds of devices. The above
5795 mentioned "alternatives" groups are based on the same concept as the "alternatives" concept of the
5796 PowerSequences class (see section 5.3.19.1.3). Within this complex class an "alternatives" group is
5797 modelled in the root element "alternatives" of function smartEnergyManagementPsData.

5798 Summarizing, there are several kinds of data contributing to the extent of a power sequences
5799 server's flexibility:

- 5800 1. Several flags denoting whether the power sequences server permits configuration from a
5801 bound power sequences client.
- 5802 2. The number of power sequences of an "alternatives" group (i.e. whether a choice from
5803 multiple concurrent power sequences can be taken).
- 5804 3. The possibility for shifting a sequence or its slots (see above for explanation on constraints).

5805

5806 *5.2.3.1.3.1 Power sequences application behaviour guidance (informative)*

5807 In general, the SmartEnergyManagementPs complex class permits a lot of different use cases or
5808 applications to be realised. Thus, it is rather difficult to give a complete overview for all applications.
5809 Instead, this section gives a guideline, which covers the main aspects that are typical for most
5810 applications.

5811 A power sequence in general is one of the following:

- 5812 • An energy- or power-forecast for a process without any flexibilities (fixed forecast)
- 5813 • An energy- or power-forecast that can be shifted, chosen or configured by a power
5814 sequences client

5815 Anyhow, the basic application guideline is common for both types of power sequences.

5816 We assume a power sequences server and a power sequences client established a communication
5817 channel. At some point in time, a process is going to be started on the device which contains the
5818 power sequences server. The power sequences server thus exposes at least one power sequence for
5819 this process. Depending on the kind of action the device wants to start, the power sequences server
5820 exposes fixed forecast(s), i.e. power sequence(s) without any flexibilities, power sequences with
5821 flexibilities, which might also contain alternative choices, or a mixture of both.

5822 Once a power sequences client receives one or more power sequences from a server, it should start
5823 the evaluation of the received data. The first thing to determine is, whether the power sequences
5824 server has exposed power sequence(s) which allow for configuration, or whether they are fixed
5825 forecasts.

5826 Independent of the type of the power sequence(s), the power sequences server should notify
5827 changes of the power sequence(s) (configuration and state changes, e.g.) according to the current
5828 situations.

5829 If the power sequences contain flexibilities and/or alternatives, the power sequences server can send
5830 a power sequences configuration request call. Once the power sequence client receives this
5831 message, it can evaluate whether it will configure the according power sequences server or ignore
5832 the call (temporarily or permanently). If the power sequences client does not configure the power
5833 sequences server, the power sequences server follows the configuration it has initially exposed, i.e.
5834 its own preference. If the power sequences client wants to configure the server it can achieve this by
5835 writing a power sequences configuration to the power sequence server. If the power sequences
5836 server receives a configuration, it first checks whether the configuration is valid. If the received
5837 configuration is valid, i.e. meets the constraints the power sequences server has laid out, it takes
5838 over the configuration.

5839 During runtime, the power sequences client and server may re-negotiate the configuration multiple
5840 times, even if the according power sequence is already running (if the power sequences server
5841 supports it). However, every time the data on either the power sequence information, configuration
5842 or state changes, the power sequences server should update the subscribed nodes. Of course, this
5843 should also be done at the completion of a power sequence.

5844 If new processes are going to be considered, these mechanisms are then repeated.

5845

5846 *5.2.3.1.3.2 Pausing and resuming power sequences (informative)*

5847 Some devices may permit to pause and resume specific power sequences. Such sequences can be
5848 marked as pausable (via function smartEnergyManagementPsData, "alternatives. powerSequence.
5849 operatingConstraintsInterrupt. isPausable" = "true", eg.). The sequence needs to be remotely
5850 controllable, too (which may be expressed by "alternatives. powerSequence. state.
5851 sequenceRemoteControllable" = "true" in function smartEnergyManagementPsData, e.g.), when a
5852 pause command shall be executed. Of course, the device itself needs to permit power sequences
5853 configuration in general (which may be expressed by "nodeScheduleInformation.
5854 nodeRemoteControllable" = "true" in function smartEnergyManagementPsData, e.g.). The power
5855 sequences client then may pause a running sequence and resume it at a later time (provided that the
5856 constraints are met).

5857 Please note that a device may resume its actions on its own, to meet the configured constraints. It
5858 then SHALL inform all subscribed clients about the changed operation.

5859

5860 *5.2.3.1.4 Role concept*

5861 The SPINE Protocol Specification defines the terms "server" and "client" which apply here as well.
5862 With regards to this Smart Energy Management specification the terms also get a specific semantic
5863 which is briefly described in this section.

5864 A device offering time based shifts for its energy consumption/generation/storage or fixed forecasts
5865 or mode switches (between consumption/generation/storage) SHALL be called a power sequences
5866 server.

5867 A device providing configuration for the flexibilities another device offers and/or understanding fixed
5868 forecasts SHALL be called a power sequences client.

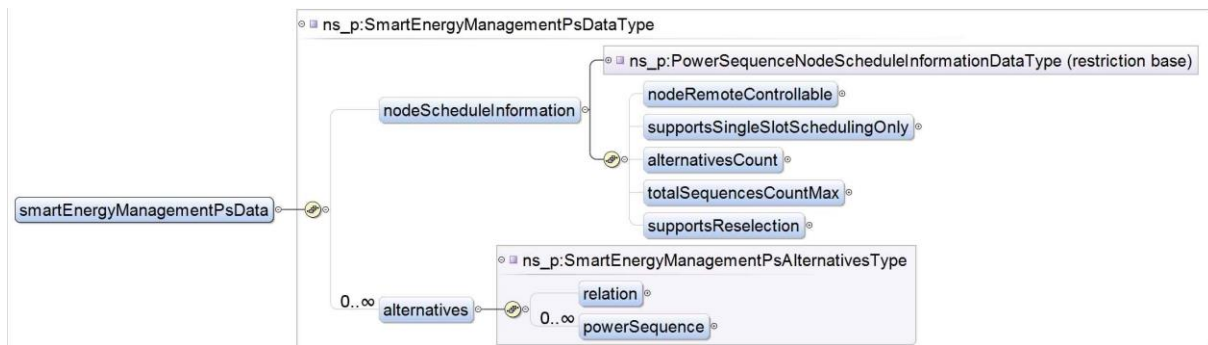
5869 Applications using this complex class may well benefit from binding or subscription mechanisms.
5870 Therefore, the following terms are defined:

- 5871 • A power sequences client which has 1 or more bindings to power sequences servers is called
5872 a "bound power sequences client".
- 5873 • A power sequences client which does not have a binding to a power sequences server but a
5874 subscription instead is called a "subscribed power sequences client".

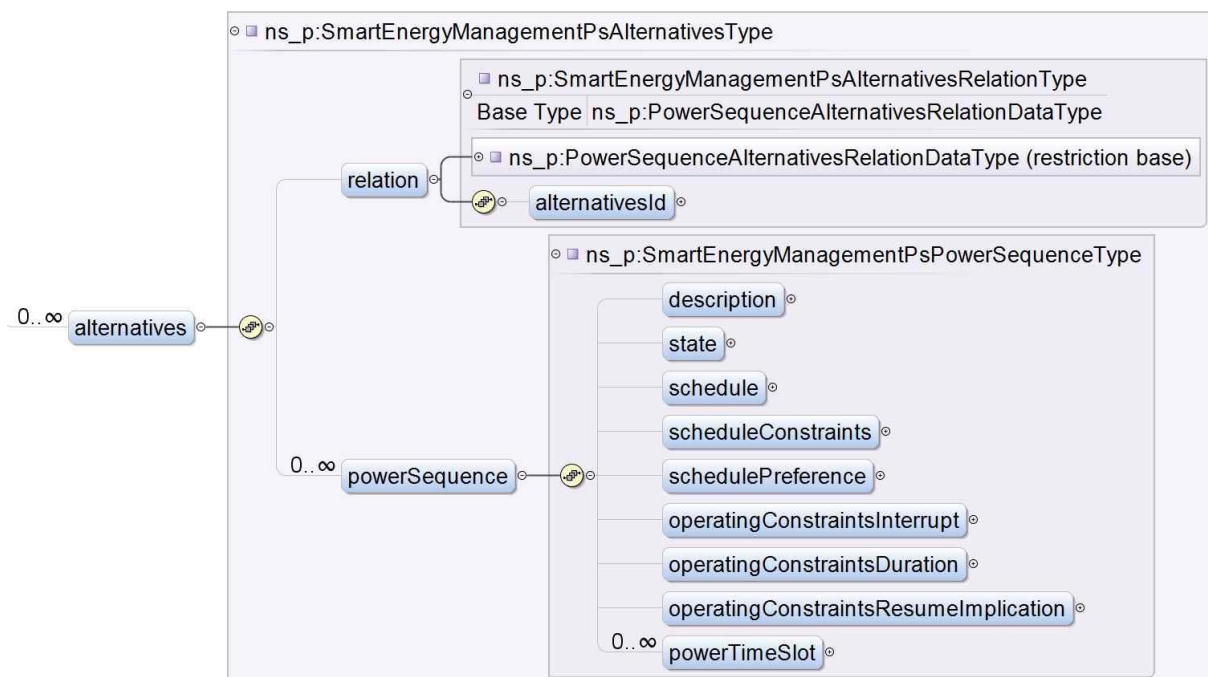
5875

5876 **5.2.3.2 Function specifications**5877 *5.2.3.2.1 smartEnergyManagementPsData*5878 *5.2.3.2.1.1 General*

5879 This function is modelled to keep all "operational" power sequences information.



5880

5881 *Figure 49: smartEnergyManagementPsData function overview, part 1*

5882

5883 *Figure 50: smartEnergyManagementPsData function overview, part 2*



Figure 51: smartEnergyManagementPsData function overview, part 3

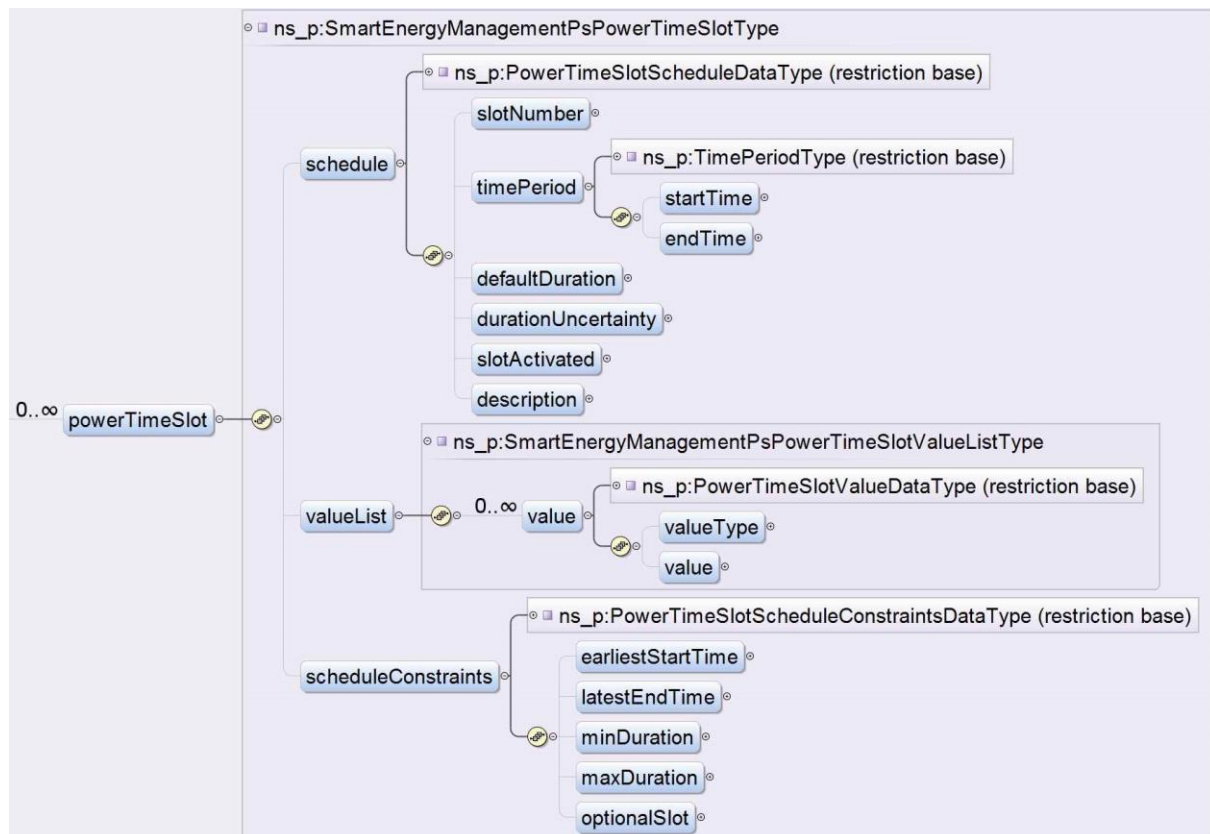


Figure 52: smartEnergyManagementPsData function overview, part 4

5.2.3.2.1.2 Detailed description of elements

The complex function "smartEnergyManagementPsData" of the complex class "SmartEnergyManagementPs" is described in the following table:

Element name	Data type	Explanation
smartEnergyManagementPsData.		
nodeScheduleInformation		Contains "global"/summary information (independent from a specific sequence). Note: This element is derived from the PowerSequences class function "powerSequenceNodeScheduleInformationData".
nodeScheduleInformation. nodeRemoteControllable	xs:boolean	Indicates whether the power sequences server (e.g., appliance) is configured for remote control (e.g., by an energy management system). This refers to the selection chosen by the user on the remote control feature of the device. If the value is "false", the remote energy management is disabled, otherwise it is enabled.
nodeScheduleInformation. supportsSingleSlotSchedulingOnly	xs:boolean	If set to "true" the power sequences server does NOT permit the modification of more than one slot per configuration

		command received from a client. I.e. in this case a client needs to submit individual configuration commands for each slot.
nodeScheduleInformation.alternativesCount	xs:unsignedInt	The number of "alternatives" groups. I.e. the number of occurrences of the element "alternatives".
nodeScheduleInformation.totalSequencesCountMax	xs:unsignedInt	The total number of sequences supported by the power sequences server. I.e. the sum of all power sequences across all "alternatives" group in this message (i.e. also regardless of the sequences' states).
nodeScheduleInformation.supportsReselection	xs:boolean	If set to "true" the power sequences server does not restrict the number of sequence re-selections by a power sequences client. I.e. within a given "alternatives" group the power sequences client may first choose one sequence, alter the selection by configuring another sequence later on, then alter the selection again, etc. (provided the process rules and data still permit configuration). If supportsReselection is set to "false", on the other hand, the power sequences server permits the power sequences client to select a sequence of an "alternatives" group only one time.
alternatives (list)		Each occurrence of "alternatives" announces a new group of mutually alternative sequences.
alternatives. relation		Note: This element is derived from the PowerSequences class function "powerSequenceAlternativesRelationData".
alternatives. relation. alternativesId	<i>Identifier</i> "AlternativesIdType" (see Table 358).	The endpoint-wide unique identifier of the "alternatives" group.
alternatives. powerSequence	List of data group (1..unbounded)	Each occurrence of "powerSequence" announces a new group of power sequences data.
alternatives. powerSequence. description		General descriptions on the power sequence. Note: This element is derived from the PowerSequences class function "powerSequenceDescriptionData".
alternatives. powerSequence. description. sequenceId	<i>Identifier</i> "PowerSequenceId"	The endpoint-wide unique sequence identifier.

	<i>Type" (see Table 358).</i>	
alternatives. powerSequence. description. description	xs:string (1..60 characters)	If used, should state a textual description for this power sequence.
alternatives. powerSequence. description. powerUnit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	The unit that applies to all power related values of this function.
alternatives. powerSequence. description. energyUnit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	The unit that applies to all energy related values of this function.
alternatives. powerSequence. description. valueSource	<i>Union "MeasurementValueSourceType": - Enum (see Table 246): MeasurementValueSourceEnumType - EnumExtendType (see section 3.10.1.5)</i>	The source (origin / foundation) of the forecasted values for this power sequence, such as "calculatedValue". Remark: This element shall express the reliability of the forecast.
alternatives. powerSequence. description. taskIdentifier	xs:unsignedInt	This element is used if a server wants to uniquely identify reoccurring types of sequences. As an example: Specific types of washing cycles with specific parameters should have the same taskIdentifier every time they are offered using power sequences. Note: This element should not be confused with "taskId" of the class TaskManagement.
alternatives. powerSequence. description. repetitionsTotal	xs:unsignedInt	If a power sequence executes its sequence of slots more than one time, the element contains the total number of executions.
alternatives. powerSequence. state		Information about the state of the power sequence. Note: This element is derived from the PowerSequences class function "powerSequenceStateData".

alternatives. powerSequence. state. state	<i>Union</i> "PowerSequenceStateType": - Enum (see Table 284): PowerSequenceStateEnumType - EnumExtendType (see section 3.10.1.5)	The current state of the sequence is stated in this element.
alternatives. powerSequence. state. activeSlotNumber	<i>Identifier</i> "PowerTimeSlotNumberType" (see Table 358).	Contains the currently active slot number.
alternatives. powerSequence. state. elapsedSlotTime	xs:duration	This element contains the time the slot has already been in "running" state (this also means the value remains constant during a "paused" state).
alternatives. powerSequence. state. remainingSlotTime	xs:duration	This element contains the time the slot still needs to be in "running" state (this also means the value remains constant during a "paused" state).
alternatives. powerSequence. state. sequenceRemoteControllable	xs:boolean	Denotes whether the sequence can be configured by a client (value "true") or not (value "false").
alternatives. powerSequence. state. activeRepetitionNumber	xs:unsignedInt	activeRepetitionNumber denotes the number of the current execution of the sequence of slots.
alternatives. powerSequence. state. remainingPauseTime	xs:duration	Denotes the duration the current slot (element activeSlotNumber) permits being paused.
alternatives. powerSequence. schedule		Information about the schedule of the power sequence. Note: This element is derived from the PowerSequences class function "powerSequenceScheduleData".
alternatives. powerSequence. schedule. startTime	xs:duration	The startTime of the power sequence.
alternatives. powerSequence. schedule. endTime	xs:duration	The endTime of the power sequence.
alternatives. powerSequence. scheduleConstraints		Information about the schedule constraints of the power sequence.

		Note: This element is derived from the PowerSequences class function "powerSequenceScheduleConstraintsData".
alternatives. powerSequence. scheduleConstraints. earliestStartTime	xs:duration	The earliest possible start time for this whole power sequence.
alternatives. powerSequence. scheduleConstraints. latestEndTime	xs:duration	The latest possible end time for this whole power sequence.
alternatives. powerSequence. schedulePreference		Contains information on the power sequences server's preference for its scheduling. Note: This element is derived from the PowerSequences class function "powerSequenceSchedulePreferenceData".
alternatives. powerSequence. schedulePreference. greenest	xs:boolean	If present and set to "true": The power sequences client should try to optimise the configuration towards the minimum usage of NON-renewableenergy.
alternatives. powerSequence. schedulePreference. cheapest	xs:boolean	If present and set to "true": The power sequences client should try to apply a configuration that minimises the user's energy bill for this power sequence.
alternatives. powerSequence. operatingConstraintsInterrupt		Information about the interruptibility of the power sequence. Note: This element is derived from the OperatingConstraints class function "operatingConstraintsInterruptData".
alternatives. powerSequence. operatingConstraintsInterrupt. isPausable	xs:boolean	If the sequence is pausable by the bound power sequences client, this element is set to true.
alternatives. powerSequence. operatingConstraintsInterrupt. isStoppable	xs:boolean	If the sequence is stoppable by the bound power sequences client, this element is set to true.
alternatives. powerSequence. operatingConstraintsInterrupt. maxCyclesPerDay	xs:unsignedInt	States the maximum amount of starts the device allows per day.

alternatives. powerSequence. operatingConstraintsDuration		Constraints about minimum or maximum durations of active- or pause-phases. Note: This element is derived from the OperatingConstraints class function "operatingConstraintsDurationData".
alternatives. powerSequence. operatingConstraintsDuration. activeDurationMin	xs:duration	This is the minimum duration the sequence has to run without interruption.
alternatives. powerSequence. operatingConstraintsDuration. activeDurationMax	xs:duration	This is the maximum duration the sequence can run without interruption.
alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMin	xs:duration	This is the minimum duration the sequence has to pause after the end of an activity.
alternatives. powerSequence. operatingConstraintsDuration. pauseDurationMax	xs:duration	This is the maximum duration the sequence can pause after the end of an activity.
alternatives. powerSequence. operatingConstraintsDuration. activeDurationSumMin	xs:duration	This is the minimum duration the sequence has to run in total (summation of all active times).
alternatives. powerSequence. operatingConstraintsDuration. activeDurationSumMax	xs:duration	This is the maximum duration the sequence can run in total (summation of all active times).
alternatives. powerSequence. operatingConstraintsResumeImplication		If resuming an active phase after a pause has implications on energy or price costs, this element and the required sub-elements contain the proper information. Note: This element is derived from the OperatingConstraints class function "operatingConstraintsResumeImplicationData".
alternatives. powerSequence. operatingConstraintsResumeImplication. resumeEnergyEstimated	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Additional energy the device will consume before resuming its normal operation (after a pause). This is only an estimated value which will not be added to the value stated in any slot value information.

alternatives. powerSequence. operatingConstraintsResumeIm plication. energyUnit	<i>Common data type "UnitOfMeasurem entType". See section 3.10.1.17.</i>	Unit for the energy stated in the element resumeEnergyEstimated.
alternatives. powerSequence. operatingConstraintsResumeIm plication. resumeCostEstimated	<i>Common data type "ScaledNumberTyp e". See section 3.10.1.8.</i>	Additional costs for the resumption of a device to its normal operation (after a pause).
alternatives. powerSequence. operatingConstraintsResumeIm plication. currency	<i>Common data type "CurrencyType". See section 3.10.1.19.</i>	Currency for the price stated in the element resumeCostEstimated.
alternatives. powerSequence. powerTimeSlot	List of data group (1..unbounded)	Each occurrence of "powerTimeSlot" announces a new "slot information group".
alternatives. powerSequence. powerTimeSlot. schedule		Information about the schedule of a slot. Note: This element is derived from the PowerSequences class function "powerTimeSlotScheduleData".
alternatives. powerSequence. powerTimeSlot. schedule. slotNumber	<i>Identifier "PowerTimeSlotNu mberType" (see Table 358).</i>	The powerSequenceId-wide unique slot identifier.
alternatives. powerSequence. powerTimeSlot. schedule. timePeriod		
alternatives. powerSequence. powerTimeSlot. schedule. timePeriod. startTime	xs:duration	The start time of the slot.
alternatives. powerSequence. powerTimeSlot. schedule. timePeriod. endTime	xs:duration	The end time of the slot.
alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration	xs:duration	The duration of the slot.
alternatives. powerSequence. powerTimeSlot. schedule. durationUncertainty	xs:duration	The uncertainty of the duration given in the element alternatives. powerSequence. powerTimeSlot. schedule. defaultDuration.

alternatives. powerSequence. powerTimeSlot. schedule. slotActivated	xs:boolean	If the slot is optional, i.e. "alternatives. powerSequence. powerTimeSlot. scheduleConstraints. optionalSlot" is set to "true", this element reflects the current status of the slot (true = the slot will be executed, false = the slot will not be executed).
alternatives. powerSequence. powerTimeSlot. schedule. description	The string-length SHALL NOT be longer than 60 characters.	A textual description for this slot.
alternatives. powerSequence. powerTimeSlot. valueList		Parent element of subsequent list.
alternatives. powerSequence. powerTimeSlot. valueList. value	List of "alternatives. powerSequence. powerTimeSlot. valueList. value" (0..unbounded)	Note: This element is derived from the PowerSequences class function "powerTimeSlotValueData".
alternatives. powerSequence. powerTimeSlot. valueList. value. valueType	<i>Union</i> "PowerTimeSlotVa lueTypeType": - Enum (see Table 278): PowerTimeSlotVal ueTypeEnumType - EnumExtendType (see section 3.10.1.5)	The value type of this value.
alternatives. powerSequence. powerTimeSlot. valueList. value. value	<i>Common data type</i> "ScaledNumberTyp e". See section 3.10.1.8.	A value according to the type expressed in alternatives. powerSequence. powerTimeSlot. valueList. value. valueType. The value's unit is stated in alternatives. powerSequence. description. powerUnit or alternatives. powerSequence. description. energyUnit.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints		Information about constraints for a time slot. Note: This element is derived from the PowerSequences class function "powerTimeSlotScheduleConstraintsData ".
alternatives. powerSequence. powerTimeSlot.	xs:duration	The earliest start time for this slot.

scheduleConstraints. earliestStartTime		
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. latestEndTime	xs:duration	The latest possible end time for this slot.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. minDuration	xs:duration	If the slot has a configurable duration, this element denotes the minimum supported configuration.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. maxDuration	xs:duration	If the slot has a configurable duration, this element denotes the maximum supported configuration.
alternatives. powerSequence. powerTimeSlot. scheduleConstraints. optionalSlot	xs:boolean	If present and set to a value of "true" it denotes the slot can be omitted via "alternatives. powerSequence. powerTimeSlot. schedule. slotActivated".

Table 166: Description of complex class function "smartEnergyManagementPsData"

5.2.3.2.1.3 Identifiers

- alternatives. powerSequence. description. sequenceId
- alternatives. powerSequence. powerTimeSlot. schedule. slotNumber

Please note: "alternatives. relation. alternativesId" is NOT an identifier as it does not open a further dimension in the data room. Instead it is kind of a group for one or more sequences.

5.2.3.2.1.4 Available selectors

The selectors definition for function "smartEnergyManagementPsData" is given in the following table:

Element name	Data range	Explanation
smartEnergyManagementPsDataSelectors		
smartEnergyManagementPsDataSelectors. alternativesRelation		Note: This element is derived from the PowerSequences class selectors definition "powerSequenceAlternativesRelationListDataSelectors".

smartEnergyManagementPsDataSelectors. alternativesRelation. alternativesId	Identifier "AlternativesIdType" (see Table 358).	Selector item: The proper alternatives group is selected.
smartEnergyManagementPsDataSelectors. powerSequenceDescription		Note: This element is derived from the PowerSequences class selectors definition "powerSequenceDescriptionListDataSelectors".
smartEnergyManagementPsDataSelectors. powerSequenceDescription. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	Selector item: The proper power sequence is selected.
smartEnergyManagementPsDataSelectors. powerTimeSlotSchedule		Note: This element is derived from the PowerSequences class selectors definition "powerTimeSlotScheduleListDataSelectors".
smartEnergyManagementPsDataSelectors. powerTimeSlotSchedule. slotNumber	Identifier "PowerTimeSlotNumberType" (see Table 358).	Selector item: The proper slot is selected.
smartEnergyManagementPsDataSelectors. powerTimeSlotValue		Note: This element is derived from the PowerSequences class selectors definition "powerTimeSlotValueListDataSelectors".
smartEnergyManagementPsDataSelectors. powerTimeSlotValue. valueType	Union "PowerTimeSlotValueTypeType": - Enum (see Table 278): PowerTimeSlotValueTypeEnumType - EnumExtendType (see section 3.10.1.5)	Selector item: The kind of the value ("power", "powerMin", "energy", ...).

Table 167: Dedicated "selectors" of "smartEnergyManagementPsData".

5.2.3.2.2 smartEnergyManagementPsConfigurationRequestCall

5.2.3.2.2.1 General

A power sequences server may want to explicitly inform (or remind) a power sequences client if it wants one or "all" of its power sequences to be configured. The server can express this by sending a power sequences configuration request call to the client.



Figure 53: smartEnergyManagementPsConfigurationRequestCall function overview

5.2.3.2.2.2 Detailed description of elements

The complex function "smartEnergyManagementPsConfigurationRequestCall" of the complex class "SmartEnergyManagementPs" is described in the following table:

Element name	Data range	Explanation
smartEnergyManagementPsConfigurationRequestCall.		
scheduleConfigurationRequest		Note: This element is derived from the PowerSequences class function "powerSequenceScheduleConfigurationRequestCall".
scheduleConfigurationRequest.sequenceld	Identifier "PowerSequenceIdType" (see Table 358).	States the sequence ID of the power sequences server which it prefers for configuration.

Table 168: Description of complex class function "smartEnergyManagementPsConfigurationRequestCall"

5.2.3.2.2.3 Identifiers

None.

5.2.3.2.2.4 Available selectors

None.

5.2.3.2.3 smartEnergyManagementPsPriceData

5.2.3.2.3.1 General

A power sequence may have a price, i.e. information on its cost. The complex function "smartEnergyManagementPsPriceData" of the complex class "SmartEnergyManagementPs" can be used to describe this information.

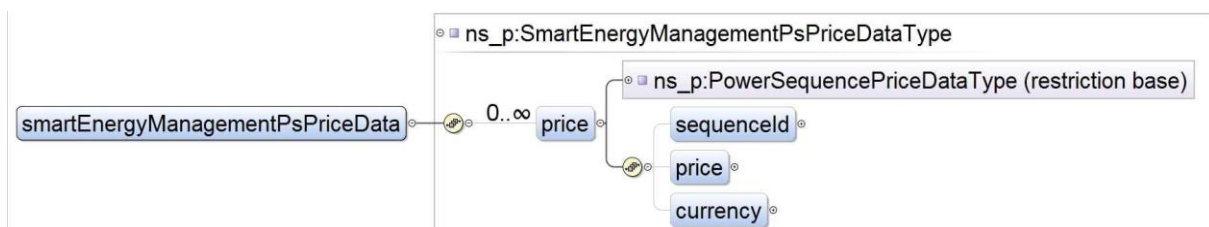


Figure 54: smartEnergyManagementPsPriceData function overview

5.2.3.2.3.2 Detailed description of elements

The complex function "smartEnergyManagementPsPriceData" of the complex class "SmartEnergyManagementPs" is described in the following table:

Element name	Data range	Explanation
smartEnergyManagementPsPriceData.		
price. (list)	List of price information (0..unbounded)	Each occurrence of this element contains information on one power sequence's cost. Note: This element is derived from the PowerSequences class function "powerSequencePriceData".
price. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequenceId the calculated price applies for.
price. price	Common data type "ScaledNumberType". See section 3.10.1.8.	The price (value) which has been calculated.
price. currency	Common data type "CurrencyType". See section 3.10.1.19.	The currency of the price.

Table 169: Description of smartEnergyManagementPsPriceData.

5.2.3.2.3.3 Identifiers

- sequenceId

5.2.3.2.3.4 Available selectors

The selectors definition for function "smartEnergyManagementPsPriceData" is given in the following table:

Element name	Data range	Explanation
smartEnergyManagementPsPriceDataSelectors		
smartEnergyManagementPsPriceDataSelectors. price		Note: This element is derived from the PowerSequences class selectors definition "powerSequencePriceListDataSelectors".
smartEnergyManagementPsPriceDataSelectors. price. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	Selector item: The price information of the proper power sequence is selected.

Table 170: Dedicated "selectors" of "smartEnergyManagementPsPriceData".

5.2.3.2.4 *smartEnergyManagementPsPriceCalculationRequestCall*

5.2.3.2.4.1 *General*

Similar to the function "powerSequencePriceCalculationRequestCall" of class PowerSequences, the complex function "smartEnergyManagementPsPriceCalculationRequestCall" permits to model a trigger for an external price calculation. I.e. the function is designed for the case that a power sequences server cannot calculate a price for one of its sequences on its own, but knows a power sequences client that is capable to do this. Then, the power sequences server may send a powerSequencePriceCalculationRequestCall with proper data to the client in order to trigger the price calculation at the client. As a result, the client may then write the price to the server. The calculated price can for example be shown on the power sequences server's device display or stored inside the device to build device internal historical information.



Figure 55: smartEnergyManagementPsPriceCalculationRequestCall function overview

5.2.3.2.4.2 *Detailed description of elements*

The complex function "smartEnergyManagementPsPriceCalculationRequestCall" of the complex class "SmartEnergyManagementPs" is described in the following table:

Element name	Data range	Explanation
smartEnergyManagementPsPriceCalculationRequestCall.		
priceCalculationRequest		Note: This element is derived from the PowerSequences class function "powerSequencePriceCalculationRequestCall".
priceCalculationRequest. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequenceId a power sequences server wants to have a price calculated for.

Table 171: Description of complex class function "smartEnergyManagementPsPriceCalculationRequestCall"

5.2.3.2.4.3 *Identifiers*

None.

5.2.3.2.4.4 *Available selectors*

None.

5.3 Standard Classes

5.3.1 ActuatorLevel

5.3.1.1 Introduction

Independent from the *ActuatorSwitch* class, the *ActuatorLevel* class enables a user or application to model *LEVEL* commands (*start*, *up*, *percentageAbsolute*, *relative*, ...). This can be used to dim a light, set the speed of an electric motor, etc. Its simple functions can easily be mapped to other technologies.

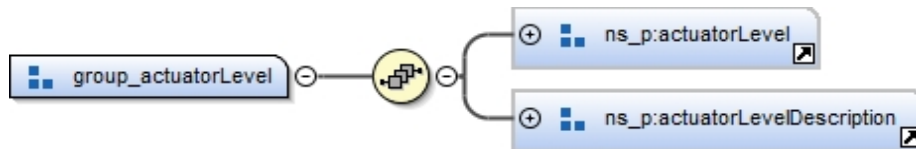


Figure 56: ActuatorLevel function-group overview

5.3.1.2 actuatorLevelData

5.3.1.2.1 General

This function is used for setting the desired operation mode of the node or getting the actual mode respectively.

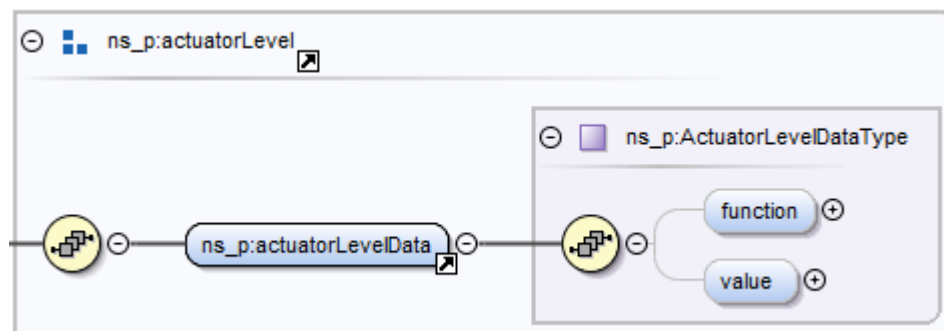


Figure 57: actuatorLevelData function overview

5.3.1.2.2 Detailed description of elements

Element	Type	Description
actuatorLevelData.		
function	Union "ActuatorLevelFctType": - Enum (see Table 173): ActuatorLevelFctEnumType - EnumExtendType (see section 3.10.1.5)	The function contains the mode.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	The value is only needed for some functions. E.g. setting <i>function</i> to <i>percentageAbsolute</i> just makes sense together with a proper value.

Table 172: actuatorLevelData function detailed description of elements

Enumeration **ActuatorLevelFctEnumType**:

Value	Description
start	Actuator starts.
up	Actuator moves up.
down	Actuator moves down.
stop	Actuator stops.
percentageAbsolute	The percentage position / value of the actuator (e.g. 25 %).
percentageRelative	Changes the current value by some percentage value (e.g. move plus 10 percent).
absolute	The absolute position / value of the actuator (e.g. 0.91 m).
relative	Changes the current value by some value (e.g. move plus 0.23 m).

Table 173: Enumeration ActuatorLevelFctEnumType

5.3.1.2.3 Available selectors

None.

5.3.1.2.4 Examples

5.3.1.2.4.1 Write

To dim a light (50 percent brightness), one could send the following XML together with a write classifier:

```

<actuatorLevelData>
  <function>percentageAbsolute</function>
  <value>
    <number>5</number>
    <scale>1</scale>
  </value>
</actuatorLevelData>

```

5.3.1.3 actuatorLevelDescriptionData

5.3.1.3.1 General

The non-changing descriptive data of the *actuatorLevel* class can contain a default unit, among others. This is useful for some *absolute* commands.

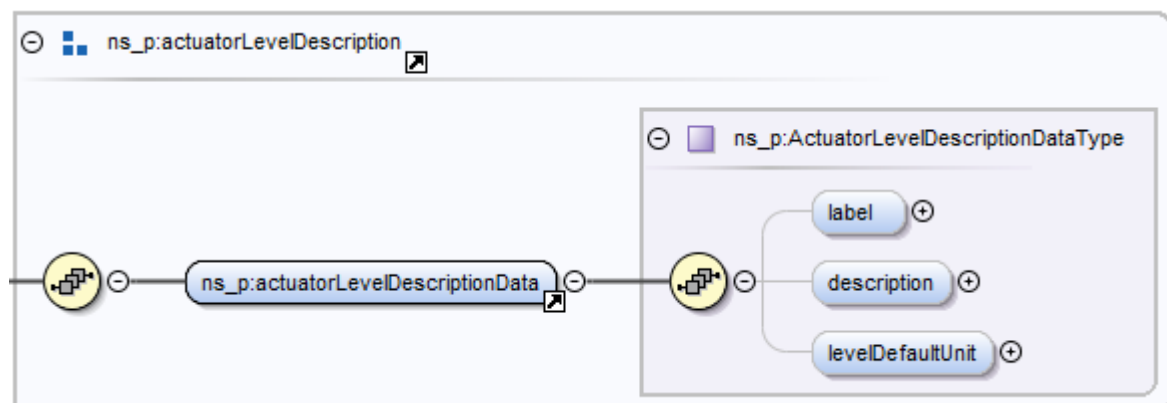


Figure 58: actuatorLevelDescriptionData function overview

6013

6014 5.3.1.3.2 Detailed description of elements

Element	Type	Description
actuatorLevelDescriptionData.		
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the actuator.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the actuator.
levelDefaultUnit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	If an actuator can specify a unit for its level-controlling, it can be denoted in this element.

6015 Table 174: actuatorLevelDescriptionData function detailed description of elements

6016

6017 5.3.1.3.3 Available selectors

6018 None.

6019

6020 5.3.1.3.4 Examples

6021 5.3.1.3.4.1 Read-reply

6022 To query the description of an actuator level, one could send a read command to the server. The
 6023 reply could be e.g.:

```

6024 <actuatorLevelDescriptionData>
6025   <label>ActuatorLevel TestDevice</label>
6026   <description>Device for testing SPINE compliance.</description>
6027   <levelDefaultUnit>W</levelDefaultUnit>
6028 </actuatorLevelDescriptionData>

```

6029

6030 5.3.2 ActuatorSwitch

6031 5.3.2.1 Introduction

6032 Basic on/off operations on a simple actuator can be modelled with the *ActuatorSwitch* class.

6033 Whether the function turns a device itself *on* or *off*, or whether it switches a specific feature,
 6034 depends on the implementation. E.g. one could model the super freeze program of a freezer using
 6035 *ActuatorSwitch* class. An *on* command would then activate the super freeze program and an *off*
 6036 command would deactivate it. This example shall just give an idea how *ActuatorSwitch* can be used
 6037 for more purposes than only turning devices on and off.

6038 Please note a device's different features may be modelled with separate implementations of this
 6039 class, each associated to a different node address. Continuing with the example above, a "defrost"
 6040 function may be modelled on another node address with *ActuatorSwitch*.

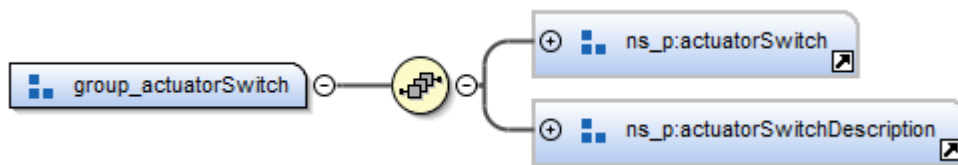


Figure 59: ActuatorSwitch function-group overview

5.3.2.2 actuatorSwitchData

5.3.2.2.1 General

The *actuatorSwitchData* function models the functionality described in section 5.3.2.1.

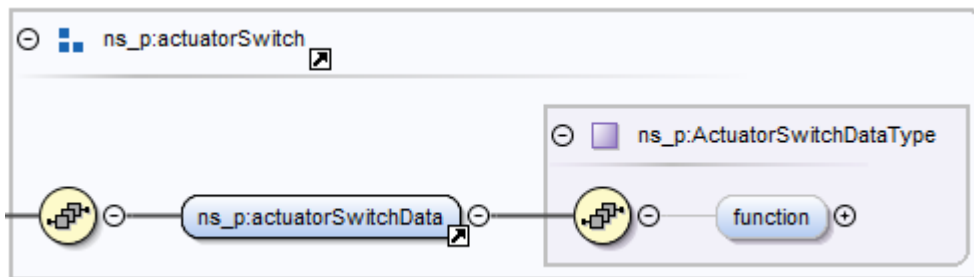


Figure 60: actuatorSwitchData function overview

5.3.2.2.2 Detailed description of elements

Element	Type	Description
actuatorSwitchData.		
function	Union "ActuatorSwitchFctType": - Enum (see Table 176): ActuatorSwitchFctEnumType - EnumExtendType (see section 3.10.1.5)	The only element in this function to model an <i>on</i> , <i>off</i> or <i>toggle</i> command.

Table 175: actuatorSwitchData function detailed description of elements

Enumeration **ActuatorSwitchFctEnumType**:

Value	Description
on	Turn on.
off	Turn off.
toggle	Toggle between on and off.

Table 176: Enumeration ActuatorSwitchFctEnumType

5.3.2.2.3 Available selectors

None.

5.3.2.2.4 Examples

5.3.2.2.4.1 Write

This example could turn on a device upon receipt with a write classifier:

```
<actuatorSwitchData>
  <function>on</function>
</actuatorSwitchData>
```

5.3.2.3 actuatorSwitchDescriptionData

5.3.2.3.1 General

The (typically) non-changing information on the actuator is modelled with this function.

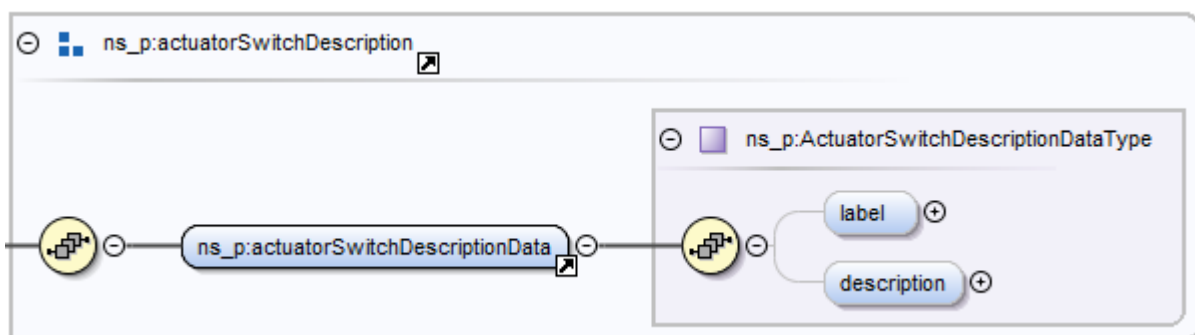


Figure 61: actuatorSwitchDescription function overview

5.3.2.3.2 Detailed description of elements

Element	Type	Description
actuatorSwitchDescriptionData.		
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the actuator.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the actuator.

Table 177: actuatorSwitchDescription function detailed description of elements

5.3.2.3.3 Available selectors

None.

5.3.2.3.4 Examples

5.3.2.3.4.1 Read-reply

One would like to know the contents of the label and description of a device. Therefore, a standard read-command is sent to the server. Then a proper reply could be:

```
<actuatorSwitchDescriptionData>
  <label>SmartPlug_1</label>
  <description>SmartPlug Test Installation 01.04.2012</description>
```

6084 </actuatorSwitchDescriptionData>

6085

6086 5.3.3 Alarm

6087 5.3.3.1 Introduction

6088 There are several reasons for generating an alarm. This might be due to exceeding (in a positive or
 6089 negative way) a threshold (modelled with the Threshold class, see section 5.3.27), or because of
 6090 some other occurrence that provokes the generating of an alarm (modelled in a later version of this
 6091 specification). The Alarm class provides the needed data model.



6092

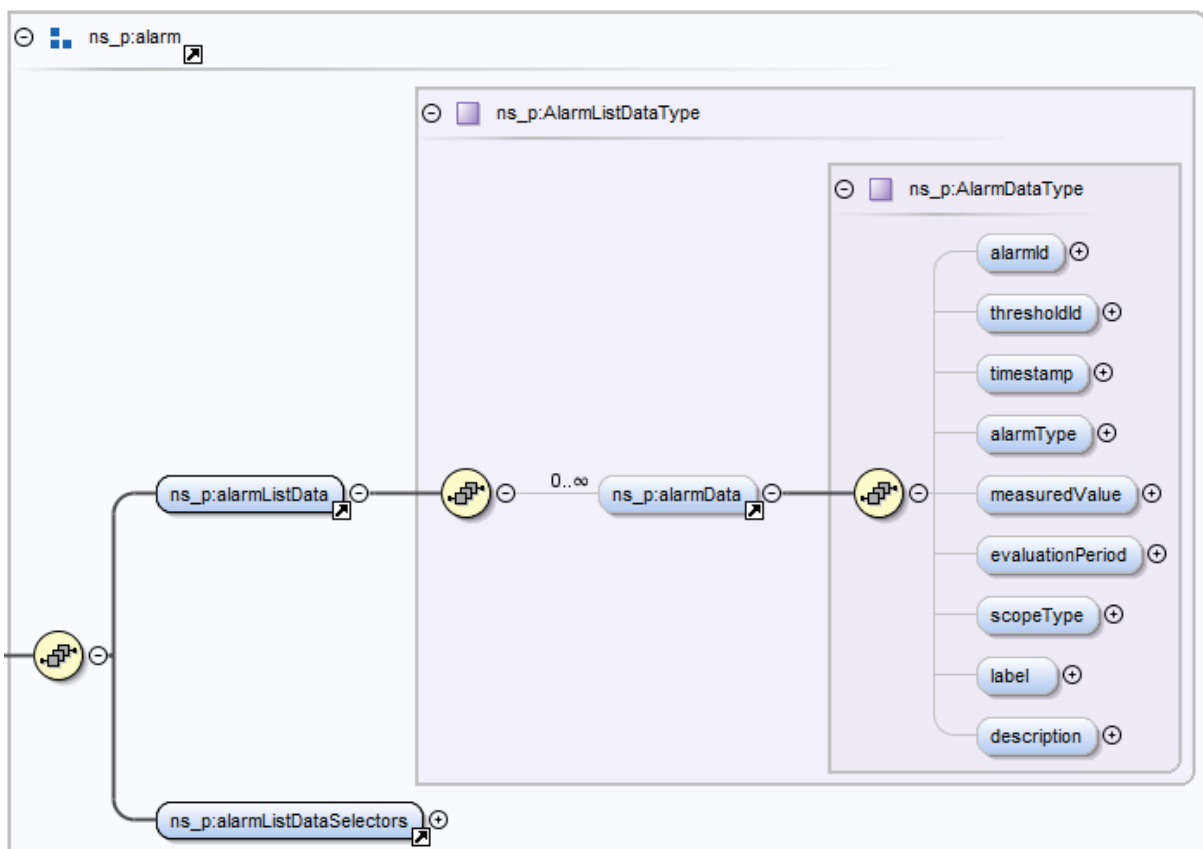
6093 Figure 62: Alarm function-group overview

6094

6095 5.3.3.2 alarmListData

6096 5.3.3.2.1 General

6097 This function models all the needed data for an alarm. Depending on the type of alarm, different
 6098 elements are used (see Feature Type "Alarm" for more details; section 4.3.3). It can be used to notify
 6099 an alarm or as a reply to a read command.



6100

6101 Figure 63: alarmListData function overview

6102

6103 5.3.3.2.2 Detailed description of elements

Element	Type	Description
alarmListData. alarmData. (list)		
alarmId	Identifier "AlarmIdType" (see Table 358).	Identifier of the alarm.
thresholdId	Identifier "ThresholdIdType" (see Table 358).	If the alarm relates to a threshold, the corresponding ID can be stated here.
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time of creation of the alarm.
alarmType	Union "AlarmTypeType": - Enum (see Table 179): AlarmTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of alarm is given here.
measuredValue	Common data type "ScaledNumberType". See section 3.10.1.8.	The value that was measured which exceeded the threshold is denoted in this element.
evaluationPeriod	Common data type "TimePeriodType". See section 3.10.2.1.	If a time period can be stated in which the alarm is valid it is denoted within this Element and its sub-Elements.
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	Specifies a more detailed meaning of the alarm (e.g. exceeding of a threshold).
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the alarm.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the alarm.

6104 Table 178: alarmListData function detailed description of elements

6105 Enumeration **AlarmTypeEnumType**:

Value	Description
alarmCancelled	A previously generated alarm is cancelled (e.g. because the value is in a "normal" range again).
underThreshold	The value is under a related threshold.
overThreshold	The value is over a related threshold.

6106 Table 179: Enumeration AlarmTypeEnumType

6107

6108 5.3.3.2.3 Available selectors

6109 - alarmId

6110 - scopeType

6111

6112 5.3.3.2.4 Examples

6113 5.3.3.2.4.1 Notify

6114 When a threshold is reached, the server would send a notification to all subscribed clients, with a
 6115 content like the following (the "filter/partial" part to announce the "restricted function exchange" is
 6116 simplified by "..." just to improve the readability):

```

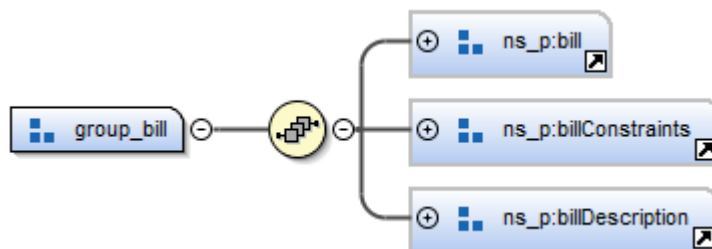
6117 ...
6118 <alarmListData>
6119   <alarmData>
6120     <alarmId>4</alarmId>
6121     <thresholdId>2</thresholdId>
6122     <timestamp>2015-12-01T18:27:32.4Z</timestamp>
6123     <alarmType>overValue</alarmType>
6124     <measuredValue>
6125       <number>711</number>
6126       <scale>-1</scale>
6127     </measuredValue>
6128     <label>Temperature reached</label>
6129     <description>Desired tea temperature reached by water
6130 boiler.</description>
6131   </alarmData>
6132 </alarmListData>
  
```

6133

6134 5.3.4 Bill

6135 5.3.4.1 Introduction

6136 The Bill Class can be used to issue bills, containing positions and their costs.



6137

6138 Figure 64: Bill function-group overview

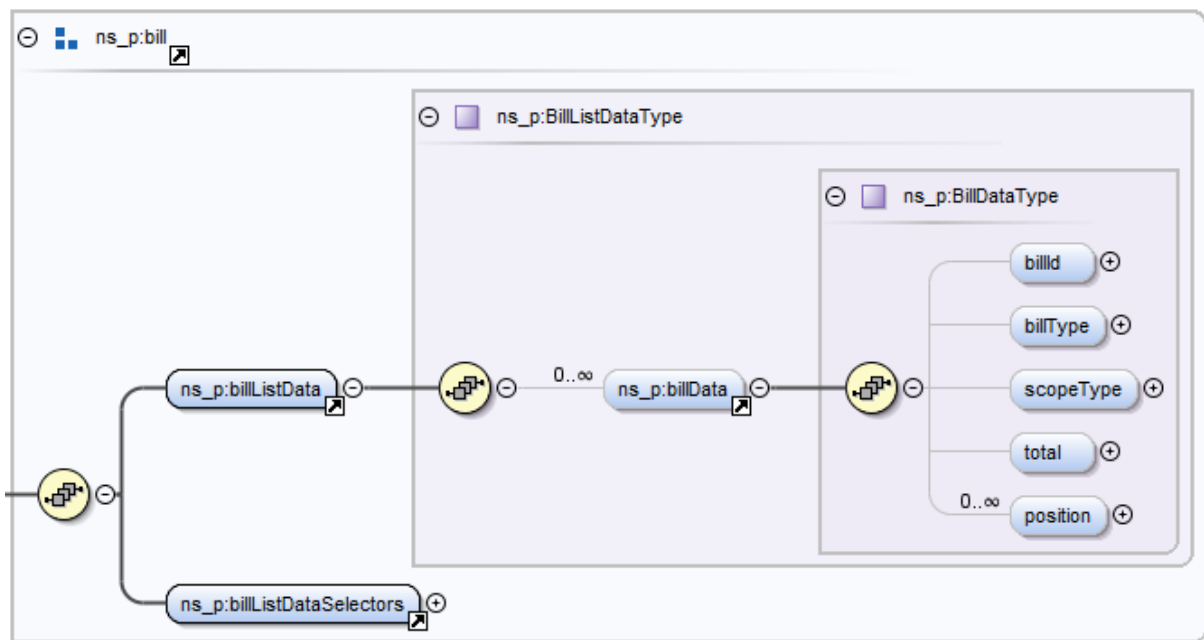
6139

6140 5.3.4.2 billListData

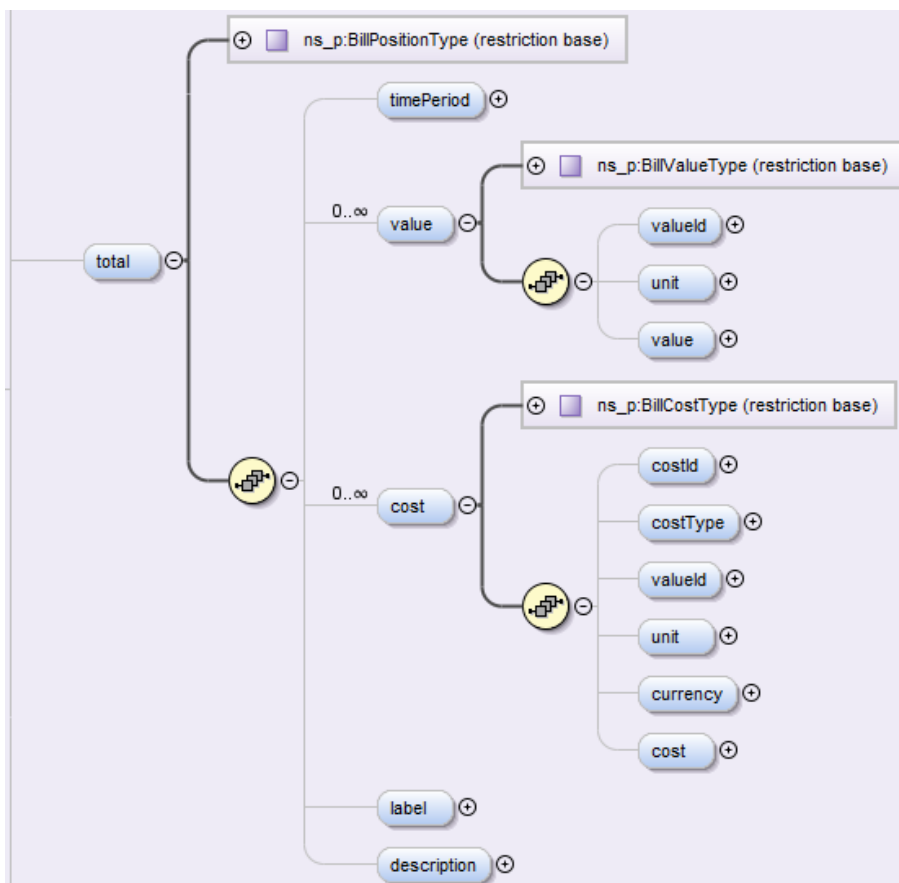
6141 5.3.4.2.1 General

6142 billData holds the bill data, including descriptive elements like currency, unit and billType. This is
 6143 necessary so the client can write a bill with a single write.

6144 A billData has a "total" element, including total values and corresponding costs in its sub-values.
 6145 Additionally positions of the total data can be provided, to describe the composition of the total
 6146 data. To link total costs and values in a position the valuelId and costId are used within a billData. The
 6147 billId allows to identify the overall bill in other functions of bill as well as outside of the bill resource.



6148

6149 *Figure 65: billListData function overview*

6150

6151 *Figure 66: billListData function detail 1, "total"*

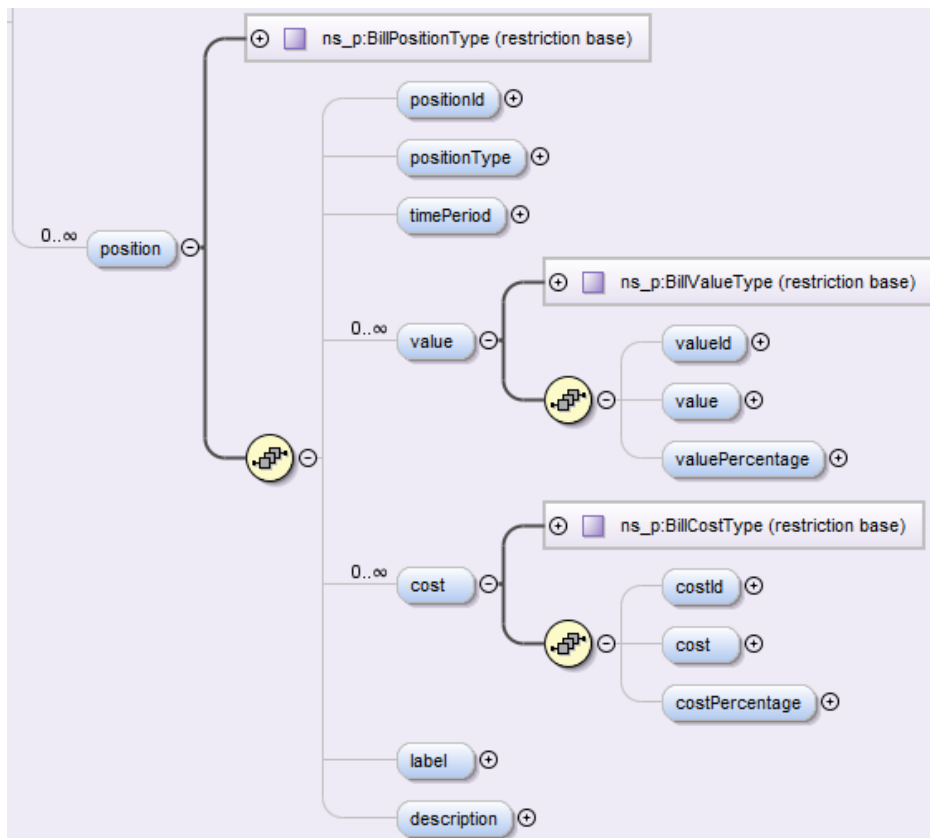


Figure 67: billListData function detail 2, "position"

5.3.4.2.2 Detailed description of elements

Element	Type	Description
billListData. billData. (list)		
billId	Identifier "BillIdType" (see Table 358).	Allows the identification of a bill. Allows linking of the different functions to the same bill. Allows linking of the bill within other features that are placed in the same entity.
billType	Union "BillTypeType": - Enum (see Table 181): BillTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the bill.
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	Specifies a more detailed meaning of the bill.
total	Restriction of "BillPositionType"	Total information of the bill (in contrast to the positions, see below).
total. timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	Allows to define a time, or time period for a bill.
total. value (list)	Restriction of "BillValueType"	Different values can be listed for a bill. E.g. energy values or just a number of items.
total. value. valueId	Identifier "BillValueIdType" (see Table 358).	Allows the identification of a value within the bill.

total. value. unit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	Allows definition of a unit for the value.
total. value. value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Allows to express the total amount that was bought or sold of a certain commodity during the time or time period given in total. timePeriod.
total. cost (list)	<i>Restriction of "BillCostType"</i>	Different types of costs can be listed for a value. E.g. monetary costs like a price as well as environmental costs as CO2 emission.
total. cost. costId	<i>Identifier "BillCostIdType" (see Table 358).</i>	Allows identification of costs within bill.
total. cost. costType	<i>Union "BillCostTypeType": - Enum (see Table 182): BillCostTypeEnumType - EnumExtendType (see section 3.10.1.5)</i>	Allows to describe the type of costs.
total. cost. valueId	<i>Identifier "BillValueIdType" (see Table 358).</i>	Allows to link cost to a value.
total. cost. unit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	Costs that are not monetary may need to describe a unit (.g. m³).
total. cost. currency	<i>Common data type "CurrencyType". See section 3.10.1.19.</i>	Monetary costs may need to describe a currency.
total. cost. cost	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Total costs generated during the time or time period given in total. timePeriod.
total. label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	Allows to express a user friendly label for the bill.
total. description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Allows to express a user friendly description for the bill.
position (list)	<i>Restriction of "BillPositionType"</i>	For each total data different positions can be assigned that describe the composition of the total data.
position. positionId	<i>Identifier "BillPositionIdType" (see Table 358).</i>	Identifier of a position within a specific bill.
position. positionType	<i>Union "BillPositionTypeType": - Enum (see Table 183): BillPositionTypeEnumType - EnumExtendType (see section 3.10.1.5)</i>	Allows to describe the type of the a position E.g. different types of energy consumption like self produced or electricity grid energy.
position. timePeriod	<i>Common data type "TimePeriodType". See section 3.10.2.1.</i>	Allow to define a time, or time period for a position if the time or time period is different from the overall time period described in total.
position. value (list)	<i>Restriction of "BillValueType"</i>	Different values can be listed for each position.
position. value. valueId	<i>Identifier "BillValueIdType" (see Table 358).</i>	Identifier of a value within a specific bill (see also "total. value. valueId").

position. value. value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Allows to describe a value for each position. This can be helpful if some positions have negative while other positions have positive values, as this would can not be express with valuePercentage.
position. value. valuePercentage	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Percentage of the total value (see "total").
position. cost (list)	<i>Restriction of "BillCostType"</i>	Different costs can be listed for each position
position. cost. costId	<i>Identifier "BillCostIdType" (see Table 358).</i>	Identifier of a cost within a specific bill (see also "total. cost. costId").
position. cost. cost	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The actual costs for this position. This can be helpful if some positions have negative while other positions have positive costs, as this can not be expressed with costPercentage.
position. cost. costPercentage	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Percentage of the total costs (see "total").
position. label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	Allows to provide a user friendly label for each bill position.
position. description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Allows to provide user friendly information for each bill position.

Table 180: billListData function detailed description of elements

Enumeration **BillTypeEnumType**:

Value	Description
chargingSummary	This bill type allows to summarize a charging cycle. In most cases it is used to communicate the total energy that was charged and the total costs, as well as different positions for different types of energy. The summary may also be sent during charging, also multiple times. This allows to keep the customer updated during the charging process.

Table 181: Enumeration BillTypeEnumType

Enumeration **BillCostTypeEnumType**:

Value	Description
absolutePrice	Used for absolute prices.
relativePrice	Used for relative prices, e.g. a price that shows how much cheaper the costs are related to the average tariff in place.
co2Emission	In this case the cost value is used to express how much CO2 emission was emitted.
renewableEnergy	In this case the cost value is used to express how much renewable energy was used.
radioactiveWaste	In this case the cost value is used to express how much radioactive waste was generated.

Table 182: Enumeration BillCostTypeEnumType

Enumeration **BillPositionTypeEnumType**:

Value	Description
-------	-------------

gridElectricEnergy	This position type specifically describes the electricity grid energy portion of the total energy.
selfProducedElectricEnergy	This position type specifically describes the self produced energy portion of the total energy.

Table 183: Enumeration BillPositionTypeEnumType

5.3.4.2.3 Available selectors

- billId
- scopeType

5.3.4.2.4 Examples

5.3.4.2.4.1 Notify

When there is an update for a specific bill, a server would send a notification to all subscribed clients that could look like this:

```

...
<billListData>
  <billData>
    <billId>1</billId>
    <billType>chargingSummary</billType>
    <total>
      <timePeriod>
        <startTime>2017-10-24T13:14:31.0Z</startTime>
        <endTime>2017-10-24T14:11:43.0Z</endTime>
      </timePeriod>
      <value>
        <valueId>1</valueId>
        <unit>Wh</unit>
        <value>
          <number>35</number>
          <scale>3</scale>
        </value>
      </value>
      <cost>
        <costId>1</costId>
        <costType>absolutePrice</costType>
        <valueId>1</valueId>
        <currency>EUR</currency>
        <cost>
          <number>96</number>
          <scale>-1</scale>
        </cost>
      </cost>
    </total>
    <position>
      <positionId>1</positionId>
      <positionType>gridElectricEnergy</positionType>
      <timePeriod>
        <startTime>2017-10-24T13:14:31.0Z</startTime>
        <endTime>2017-10-24T14:11:43.0Z</endTime>
      </timePeriod>
      <value>
        <valueId>1</valueId>
        <valuePercentage>

```

```

6211         <number>73</number>
6212         <scale>0</scale>
6213     </valuePercentage>
6214 </value>
6215 <cost>
6216     <costId>1</costId>
6217     <costPercentage>
6218         <number>95</number>
6219         <scale>0</scale>
6220     </costPercentage>
6221 </cost>
6222 </position>
6223 <position>
6224     <positionId>2</positionId>
6225     <positionType>selfProducedElectricEnergy</positionType>
6226     <timePeriod>
6227         <startTime>2017-10-24T13:14:31.0Z</startTime>
6228         <endTime>2017-10-24T14:11:43.0Z</endTime>
6229     </timePeriod>
6230 <value>
6231     <valueId>1</valueId>
6232     <valuePercentage>
6233         <number>27</number>
6234         <scale>0</scale>
6235     </valuePercentage>
6236 </value>
6237 <cost>
6238     <costId>1</costId>
6239     <costPercentage>
6240         <number>5</number>
6241         <scale>0</scale>
6242     </costPercentage>
6243 </cost>
6244 </position>
6245 </billData>
6246 </billListData>

```

6247

6248 **5.3.4.3 billConstraintsListData**

6249 **5.3.4.3.1 General**

6250 The billConstraintsListData function allows a server to define certain constraints for each linked
6251 billListData list entry. This is especially important if a particular billListData list entry is writeable and
6252 allows the server to communicate the corresponding constraints before a client performs a write.
6253 The client can match the server constraints with own constraints and perform a corresponding write
6254 command.

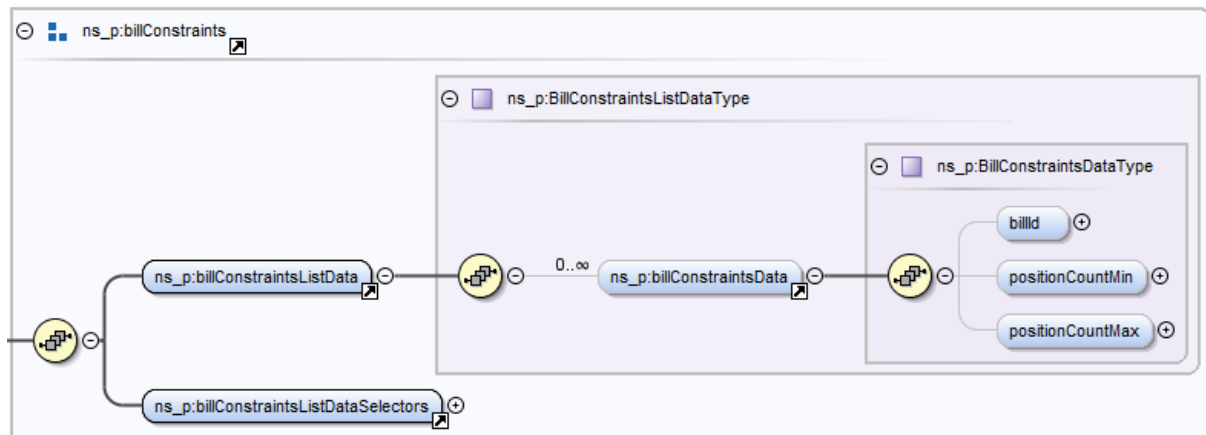


Figure 68: billConstraintsListData function overview

5.3.4.3.2 Detailed description of elements

Element	Type	Description
billConstraintsListData. billConstraintsData. (list)		
billId	Identifier "BillIdType" (see Table 358).	Allows the identification of a bill. Allows linking of the different functions to the same bill. Allows linking of the bill within other features that are placed in the same entity.
positionCountMin	Simple type "BillPositionCountType" (restriction of "BillPositionIdType" (see Table 358)).	Minimum amount of possible positions within the specified bill.
positionCountMax	Simple type "BillPositionCountType" (restriction of "BillPositionIdType" (see Table 358)).	Maximum amount of possible positions within the specified bill.

Table 184: billConstraintsListData function detailed description of elements

5.3.4.3.3 Available selectors

- billId

5.3.4.3.4 Examples

5.3.4.3.4.1 Read-reply

If one wants to know the constraints of a specific bill, he could send the following read command:

```

<function>billConstraintsListData</function>
<filter>
  <cmdControl>
    <partial/>
  </cmdControl>
  <billConstraintsListDataSelectors>
    <billId>1</billId>
  </billConstraintsListDataSelectors>
</filter>

```

6276 <billConstraintsListData/>

6277 The reply could look like this:

```

6278 <function>billConstraintsListData</function>
6279 <filter>
6280   <cmdControl>
6281     <partial/>
6282   </cmdControl>
6283 </filter>
6284 <billConstraintsListData>
6285   <billConstraintsData>
6286     <billId>1</billId>
6287     <positionCountMin>2</positionCountMin>
6288     <positionCountMax>4</positionCountMax>
6289   </billConstraintsData>
6290 </billConstraintsListData>

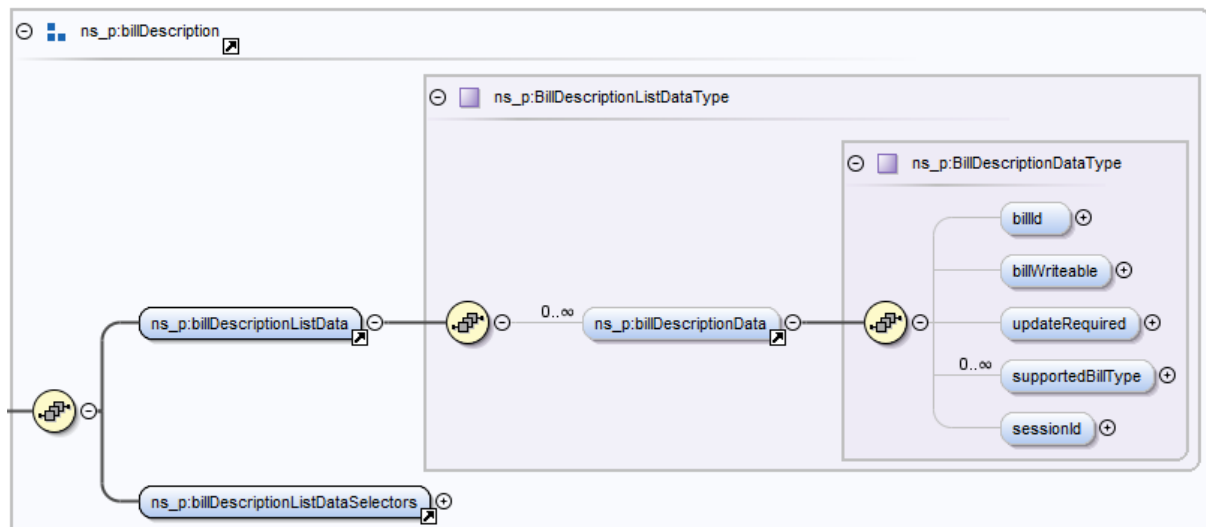
```

6291

6292 5.3.4.4 *billDescriptionListData*

6293 5.3.4.4.1 *General*

6294 The billDescriptionListData function is especially important if a linked billListData list entry is
 6295 writeable and allows a server to define if a bill is writeable and which billTypes are supported for the
 6296 corresponding billId, so that a client can perform corresponding writes. Additionally, the server can
 6297 request updates from a client with the updateRequired flag.



6298

6299 Figure 69: billDescriptionListData function overview

6300

6301 5.3.4.4.2 *Detailed description of elements*

Element	Type	Description
billDescriptionListData. billDescriptionData. (list)		
billId	Identifier "BillIdType" (see Table 358).	Allows the identification of a bill. Allows linking of the different functions to the same bill. Allows linking of the bill within other

		features that are placed in the same entity.
billWriteable	xs:boolean (W3C standard type)	Indicates whether the linked billData entry is writeable.
updateRequired	xs:boolean (W3C standard type)	Indicates whether the server requests an update for the bill.
supportedBillType (list)	<i>Union "BillTypeType":</i> - Enum (see Table 181): BillTypeEnumType - EnumExtendType (see section 3.10.1.5)	List of all supported types for this billId.
sessionId	<i>Identifier "SessionIdType" (see Table 358).</i>	Link to the session this bill was created for.

Table 185: billDescriptionListData function detailed description of elements

5.3.4.4.3 Available selectors

- billId

5.3.4.4.4 Examples

5.3.4.4.4.1 Read-reply

If one is interested in the descriptions of all bills, he could send a standard read command to the server. The reply could look like this:

```
<billDescriptionListData>
  <billDescriptionData>
    <billId>1</billId>
    <billWriteable>true</billWriteable>
    <supportedBillType>chargingSummary</supportedBillType>
  </billDescriptionData>
</billDescriptionListData>
```

5.3.5 BindingManagement

5.3.5.1 Introduction

Bindings are used to simplify communication between devices. Based on matching features, finding suitable communication partners is a lot easier than via manual configuration only.

Each binding exists between a binding client and a binding server. Only clients may initiate new bindings. After a successful establishment of a binding, the binding client may do actions on the binding server, depending on the bound functionality. E.g. a binding server accepts load management configurations by the client and updates the client when changes in its parameters occur. The exact definition of rules and proceedings is out of scope of this documentation.

The *BindingManagement* class provides functions for establishing, reading and deleting bindings.

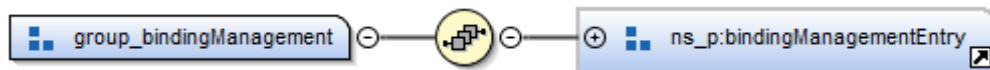


Figure 70: BindingManagement function-group overview (data)

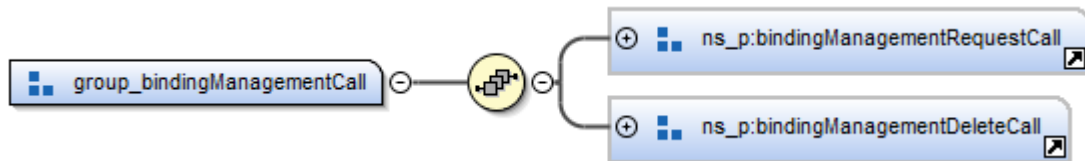


Figure 71: BindingManagement function-group overview (call)

5.3.5.2 bindingManagementEntryListData

5.3.5.2.1 General

All existing bindings can be listed in the *bindingManagementEntryListData* function. It may be used to store the entries in the device itself. If another device asks for the list of bindings, only the bindings with the requesting device should be replied.

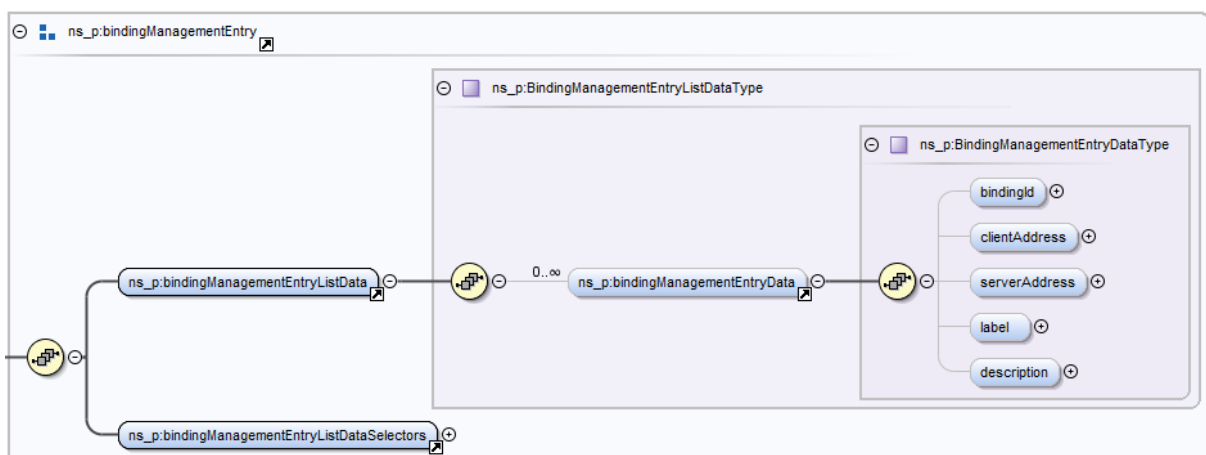


Figure 72: bindingManagementEntryListData function overview

5.3.5.2.2 Detailed description of elements

Element	Type	Description
bindingManagementEntryListData. bindingManagementEntryData. (list)		
bindingId	Identifier "BindingIdType" (see Table 358).	Identifier of the binding.
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in this binding.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in this binding.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the binding.

description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the binding.
-------------	---	---

Table 186: bindingManagementEntryListData function detailed description of elements

5.3.5.2.3 Available selectors

- bindingId
- clientAddress
- serverAddress

5.3.5.2.4 Examples

5.3.5.2.4.1 Read-reply

If one is interested in all bindings that are active between the own device and another one, he could send a standard read function to the device, which could reply something like this:

```
<bindingManagementEntryListData>
  <bindingManagementEntryData>
    <bindingId>0</bindingId>
    <clientAddress>
      <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
      <entity>2</entity>
      <feature>1</feature>
    </clientAddress>
    <serverAddress>
      <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
      <entity>1</entity>
      <feature>1</feature>
    </serverAddress>
    <label>SwitchLight_01</label>
  </bindingManagementEntryData>
  <bindingManagementEntryData>
    <bindingId>4</bindingId>
    <clientAddress>
      <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
      <entity>2</entity>
      <feature>4</feature>
    </clientAddress>
    <serverAddress>
      <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
      <entity>1</entity>
      <feature>2</feature>
    </serverAddress>
    <label>SwitchLight_02</label>
  </bindingManagementEntryData>
</bindingManagementEntryListData>
```

5.3.5.3 bindingManagementRequestCall

5.3.5.3.1 General

To initiate a new binding, an entity capable of node management, sends a *bindingManagementRequestCall* to an entity (on another device that should become the binding

partner) capable of node management, too, with the Function element *clientAddress* set to one of its features with the role *client* and the Function element *serverAddress* set to one of the other device's feature address with the role *server*. The receiving device checks whether it will accept or decline the request and responds with an acknowledgement message (see [ProtocolSpecification], section "Acknowledgement message") with the *resultData* function and an according *errorNumber* set to 0 (accept binding request) or some other value (definition of the error code is out of scope of this documentation; decline binding request).

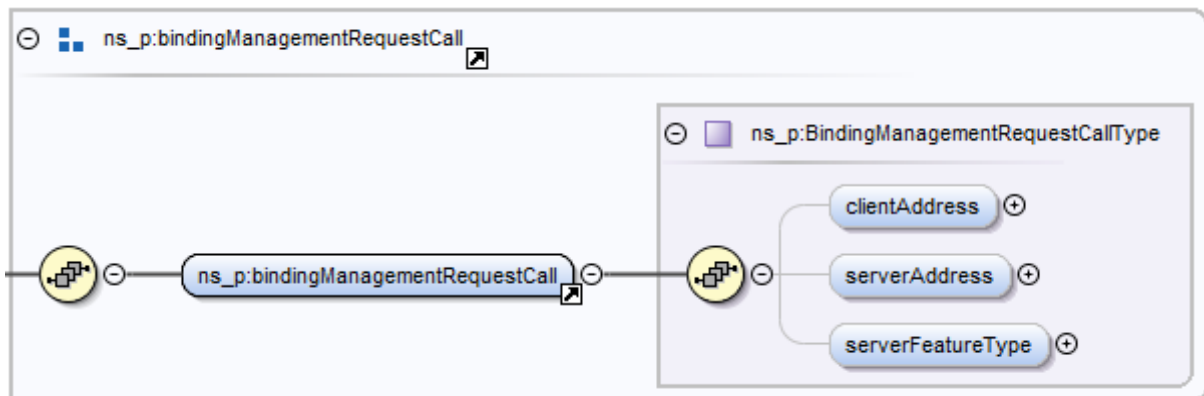


Figure 73: bindingManagementRequestCall function overview

5.3.5.3.2 Detailed description of elements

Element	Type	Description
bindingManagementRequestCall.		
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in the binding request.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in the binding request.
serverFeatureType	Common data type "FeatureTypeType". See section 3.10.1.29.	The <i>Feature Type</i> of the server feature. Stated for verification.

Table 187: bindingManagementRequestCall function detailed description of elements

5.3.5.3.3 Available selectors

None.

5.3.5.3.4 Examples

5.3.5.3.4.1 Call

If a new binding shall be initiated, the client device could send a command like this to the server device:

```
<bindingManagementRequestCall>
  <clientAddress>
```

```

6411         <device>d:_i:46925_CP\_pqt1-27638162683172637812356813</device>
6412         <entity>2</entity>
6413         <feature>1</feature>
6414     </clientAddress>
6415     <serverAddress>
6416         <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
6417         <entity>1</entity>
6418         <feature>1</feature>
6419     </serverAddress>
6420     <serverFeatureType>ActuatorSwitch</serverFeatureType>
6421 </bindingManagementRequestCall>

```

5.3.5.4 *bindingManagementDeleteCall*

5.3.5.4.1 *General*

A device may delete an existing binding with itself and another device. Therefore, this call is used. It may be used by clients and servers.

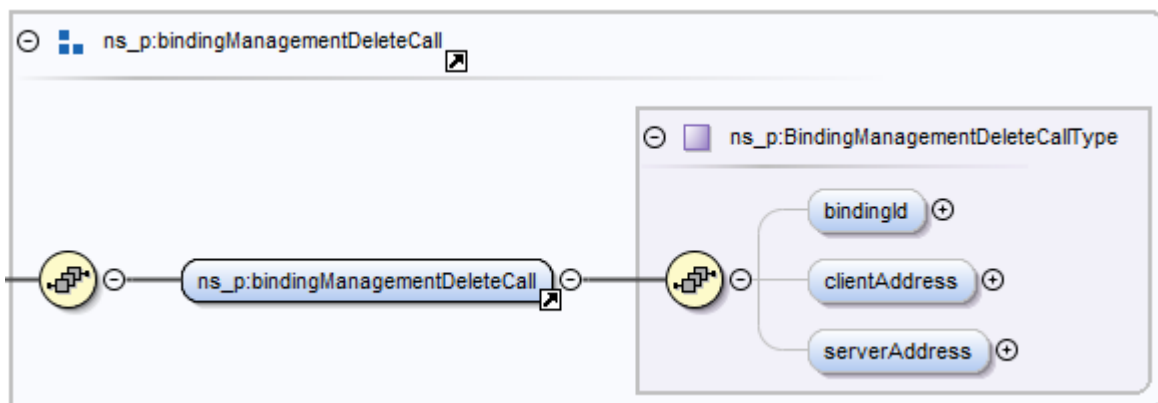


Figure 74: *bindingManagementDeleteCall* function overview

5.3.5.4.2 *Detailed description of elements*

Element	Type	Description
bindingManagementDeleteCall.		
bindingId	Identifier "BindingIdType" (see Table 358).	Identifier of the binding.
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in this binding.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in this binding.

Table 188: *bindingManagementDeleteCall* function detailed description of elements

5.3.5.4.3 *Available selectors*

None.

6436 5.3.5.4.4 Examples

6437 5.3.5.4.4.1 Call

6438 If an existing binding (with the bindingId "2") shall be deleted, one device could send a command like
6439 this to the other device:

```
6440 <bindingManagementDeleteCall>
6441   <bindingId>2</bindingId>
6442   <clientAddress>
6443     <device>d:_i:46925_CP\_pqt1-27638162683172637812356813</device>
6444     <entity>2</entity>
6445     <feature>1</feature>
6446   </clientAddress>
6447   <serverAddress>
6448     <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
6449     <entity>1</entity>
6450     <feature>1</feature>
6451   </serverAddress>
6452 </bindingManagementDeleteCall>
```

6453

6454 5.3.6 DataTunneling

6455 5.3.6.1 Introduction

6456 DataTunneling provides a mechanism to submit data of proprietary protocols using the SPINE data
6457 models (esp. addressing other nodes). However, the nature of proprietary definitions imposes
6458 restrictions:

6459 In the example below an XML makes use of SPINE dataTunneling sub-elements "purposeId",
6460 "channelId", "sequenceId", and "payload". Unless stated otherwise, the term "payload" refers to
6461 the sub-element of "dataTunnelingCall" within this section (in contrast to the definition of "payload"
6462 as sub-element of "datagram" as defined in document [ProtocolSpecification]).

6463 Please note that the use of the above mentioned elements will always be proprietary. The SPINE
6464 specification will never define any specific protocol for dataTunneling.

6465 To put it in other words, the SPINE data tunnelling model is not intended for any kind of flexible
6466 extension of the SPINE classes. This is mainly because a so-called technology specific mapping could
6467 hardly be defined in a standardized way. Another reason is that it shall always be transparent that
6468 data tunnelling contains proprietary data. This should not be considered a limitation of SPINE as the
6469 EEBus Initiative e.V. committees may define specific classes with functions (like messagingListData,
6470 timeInformationData, e.g.) including mappings for specific protocols, if need be.



6471

6472 Figure 75: DataTunneling function-group overview (call)

6473

6474 **5.3.6.2 dataTunnelingCall**

6475 5.3.6.2.1 General

6476 In general, several conditions have to be fulfilled in order to make use of SPINE data tunnelling. This
6477 shall be explained with an example:

6478 A manufacturer "X" of a device "A" created a proprietary extension of a communications protocol
6479 (e.g. KNX, ZigBee, LON, ... – we call it "CP" here) in order to configure device "A" with special
6480 parameters. We assume there is no specific SPINE function defined so far for these parameters.
6481 There is also no mapping defined for it.

6482 Another manufacturer "Y" creates a device "B" that runs "SPINE applications" (i.e. applications
6483 communicating with an internal interface using XMLs of the SPINE data models). The device "B" can
6484 as well communicate via "CP".

6485 Now, manufacturer "Y" wants to make use of the proprietary configuration of device "A". Therefore,
6486 manufacturer "Y" needs to implement the proprietary extension of manufacturer "X"'s
6487 communications protocol as well. I.e. the "CP" implementation of device "B" has to be extended
6488 properly. Furthermore, manufacturer "Y" needs to define how the proprietary protocol shall be
6489 represented as SPINE data tunnelling towards the "SPINE applications" running on device "B" (i.e.
6490 define the proprietary mapping) and of course needs to implement this mapping. Finally, the
6491 applications need to implement the recognition and use of the corresponding SPINE data tunnelling
6492 XMLs.

6493 The scenario above may look "unattractive" or even raise concerns. However, it shall be mentioned
6494 that proprietary extensions of standardized protocols are not unusual. This practice shall not be
6495 judged here.

6496 In the aforementioned scenario it is manufacturer "B" who implements a number of extensions in
6497 order to comply with the extensions of device "A". But it could also be possible that manufacturer
6498 "A" already provided some definitions and implementation for other manufacturers. Then,
6499 manufacturer "B" could integrate this solution and potentially implements less by its own.

6500 Some may find a comparison with Virtual Private Networks helpful (though VPNs have a different
6501 scope): Two computers "A", "B" each have to run a compatible (proprietary) program (i.e.
6502 mapping/implementation) that know how to communicate with each other (i.e. set up the
6503 communication and provide a specific authentication and encryption). There are manufacturers of
6504 VPN gateways that provide client software for different operating systems. This way a proprietary
6505 implementation is made available for others.

6506 The combination { purposeld, channelld, sequenceld } should be considered unique for a
6507 communication from "A" to "B". Within a system (e.g. house) several communications may be used
6508 in parallel. The combinations used in these communications are in general independent of each
6509 other.

6510 Please note that an instance of a dataTunnelingCall implementation permits bidirectional
6511 communication in general. I.e. if a dataTunnelingCall functionality of "B" submits a message to a
6512 dataTunnelingCall functionality of "A", it is also permissive that the dataTunnelingCall functionality of
6513 "A" submits dataTunnelingCall messages to the dataTunnelingCall functionality of "B". However,

similar to a VPN, it may well be that the establishment for such message exchanges is "unidirectional" to some extent. I.e. it may be that the dataTunnelingCall functionality of "A" determines which kind of messages (or permissive values for "purposeld") are accepted on this functionality and that the functionality first of all acts as some kind of "listening port". This means the functionality of "A" may wait for an initial incoming message before it also sends dataTunnelingCall messages back.

Please note that the example above does not prevent device "B" from implementing another dataTunnelingCall functionality as "listening port" with own permitted values for "purposeld" as well. In that case device "A" could send a proper initial message to this "listening functionality" of "B".

SPINE data tunnelling is intended to be used for call-operations only. No read-reply or write procedures are supported.

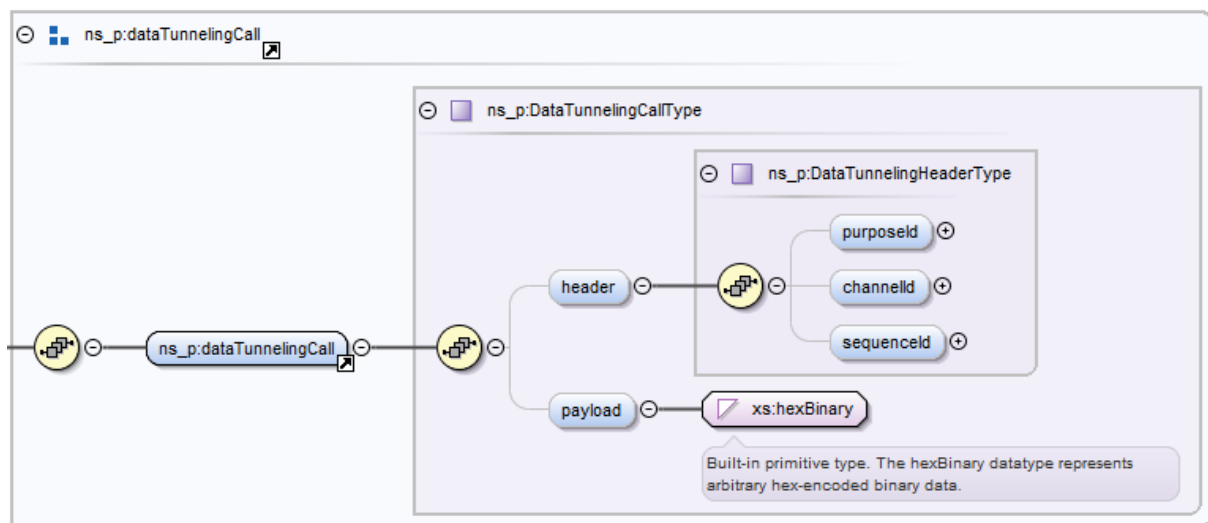


Figure 76: dataTunnelingCall function overview

5.3.6.2.2 Detailed description of elements

Element	Type	Description
dataTunnelingCall.		
header	Complex type "DataTunnelingHeaderType"	A data tunneling specific header.
header.purposeld	Simple type "PurposeldType" (restriction of xs:string)	A string that identifies the purpose / kind of tunnelled data / protocol.
header.channelId	Simple type "ChannelIdType" (restriction of xs:unsignedInt)	Between two nodes there might be several channels used in parallel using the same purposeld. channelId helps to separate these channels. It can also be used to differentiate subsequent (rather than parallel) sessions.
header.sequenceId	xs:unsignedInt (W3C standard type)	This helps to give the payload for each combination of { purposeld, channelId } a chronological order. Note: This "sequenceId" is data tunneling specific and should not be confused with "sequenceId" elements of other classes!

payload	xs:hexBinary (W3C standard type)	The actual proprietary data.
---------	----------------------------------	------------------------------

Table 189: dataTunnelingCall function detailed description of elements

5.3.6.2.3 Available selectors

None.

5.3.6.2.4 Examples

5.3.6.2.4.1 Call

Device "B" is configured to submit certain proprietary data to device "A". Due to a specific occasion it sends a call classifier together with the following XML:

```
<dataTunnelingCall>
  <header>
    <purposeId>FooManufacturer:BarProtocolWithVersion</purposeId>
    <channelId>7</channelId>
    <sequenceId>6425</sequenceId>
  </header>
  <payload>3c3f786d6c2076657273696f6e3d22312e302220656e636f64696e673d22555446
2d38223f3e0d0a3c6d6573736167654461746120786d6c6e733d22687474703a2f2f646f637
32e65656275732e6f72672f62657461312f312e302f787364223e0d0a202020203c74696d65
7374616d703e323030362d30352d30345431383a31333a35312e305a3c2f74696d657374616
d703e0d0a202020203c6d6573736167654e756d6265723e37323c2f6d6573736167654e756d
6265723e0d0a202020203c747970653e696e666f726d6174696f6e3c2f747970653e0d0a202
020203c746578743e55706461746520636f6d706c657465642e3c2f746578743e0d0a3c2f6d
657373616765446174613e0d0a</payload>
</dataTunnelingCall>
```

5.3.7 DeviceClassification

5.3.7.1 Introduction

Each device can have associated information useful for its identification and which kind of power it requires. This information (mostly plaintext) is bundled in the *DeviceClassification* class. It is intended to describe the physical device, but may be used for logical devices (= *entities*), too.

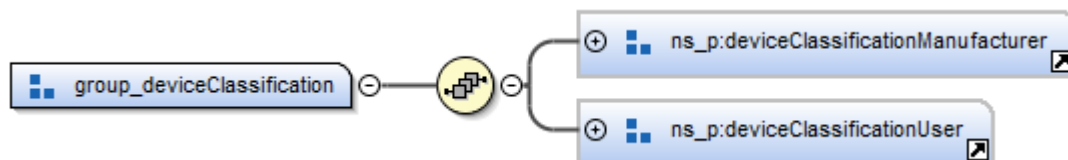


Figure 77: DeviceClassification function-group overview

5.3.7.2 *deviceClassificationManufacturerData*

5.3.7.2.1 *General*

Information which is usually NOT writable (hard coded in the device or preconfigured information, e.g.), is modelled with *deviceClassificationManufacturerData*.

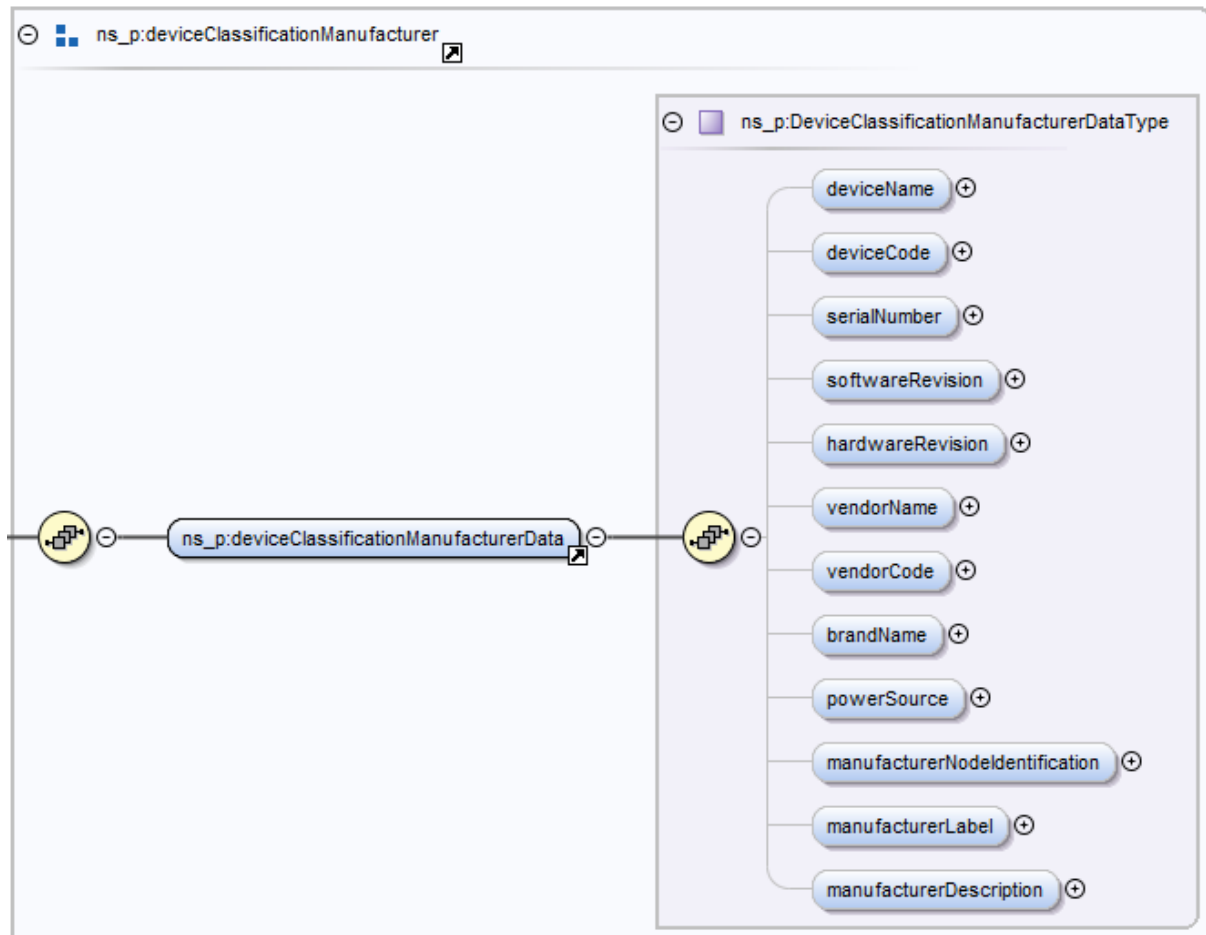


Figure 78: *deviceClassificationManufacturerData* function overview

5.3.7.2.2 *Detailed description of elements*

Please note that most of the subsequent elements permit any content. I.e. the underlying SPINE data model usually does not impose any format for the content of these elements. An exception applies if an enumeration simple-type is used (see *powerSource* element).

Element	Type	Description
deviceClassificationManufacturerData.		
deviceName	Simple type "DeviceClassificationStringType" (restriction of xs:string)	The name of the (physical or logical) device as defined by the manufacturer.
deviceCode	Simple type "DeviceClassificationStringType" (restriction of xs:string)	A device code for the (physical or logical) device as defined by the manufacturer.

serialNumber	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	The serial number of the (physical or logical) device as defined by the manufacturer. Usually the same as printed on the case.
softwareRevision	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	The software revision of the (physical or logical) device as defined by the manufacturer.
hardwareRevision	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	The hardware revision of the (physical or logical) device as defined by the manufacturer.
vendorName	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	The name of the vendor of the (physical or logical) device as defined by the manufacturer.
vendorCode	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	A code for the vendor of the (physical or logical) device as defined by the manufacturer.
brandName	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	The name of the brand. Useful where the name of the brand and the vendor differs.
powerSource	<i>Union "PowerSourceType":</i> <i>- Enum (see Table 191):</i> <i>PowerSourceEnumType</i> <i>- EnumExtendType (see section 3.10.1.5)</i>	This element describes for which kind of primary power source the device is designed for. It is only used for informative purposes. It is NOT used to describe how or to which kind of power source the device is finally connected (this means it is not considered here whether the device's single mains is fed by a house's mains supply or a backup system, e.g.).
manufacturerNodeIdentification	<i>Simple type</i> <i>"DeviceClassificationStringType"</i> <i>(restriction of xs:string)</i>	A node identification for the (physical or logical) device as defined by the manufacturer. This could be used for the identification of a (physical or logical) device, even if it was removed from the network and rejoined later with changed node address.
manufacturerLabel	<i>Common data type "LabelType".</i> <i>See section 3.10.1.2.</i>	A short label of the (physical or logical) device as defined by the manufacturer.
manufacturerDescription	<i>Common data type</i> <i>"DescriptionType". See section 3.10.1.3.</i>	A description for the (physical or logical) device as defined by the manufacturer.

6573 Table 190: deviceClassificationManufacturerData function detailed description of elements

6574 Enumeration **PowerSourceEnumType**:

Value	Description
unknown	Power source not known.
mainsSinglePhase	AC power connection, 1 phase.
mains3Phase	AC power connection, 3 phases.
battery	DC powered with a battery.
dc	DC powered with a constant energy source.

6575 Table 191: Enumeration PowerSourceEnumType

6576

6577 **5.3.7.2.3 Available selectors**

6578 None.

6579

6580 **5.3.7.2.4 Examples**6581 **5.3.7.2.4.1 Read-reply**6582 An example of a *deviceClassificationManufacturerData* instance, send as reply:

```

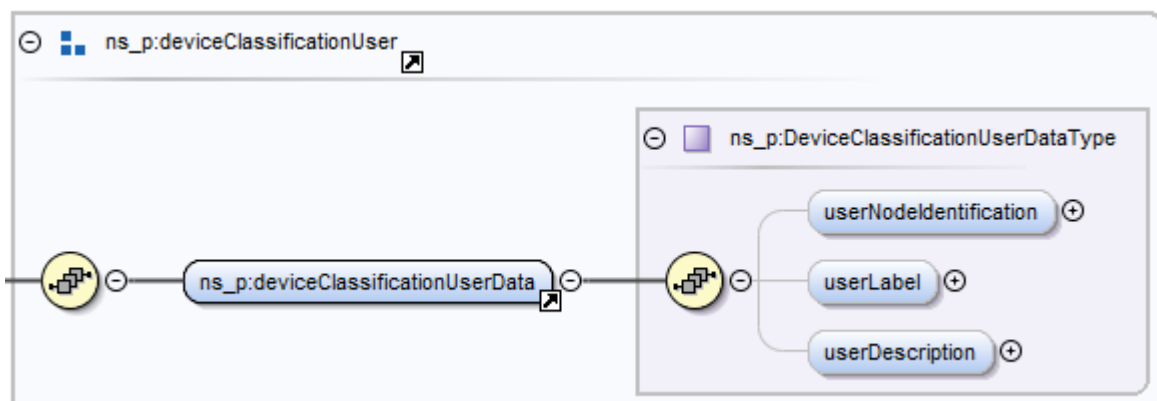
6583 <deviceClassificationManufacturerData>
6584   <deviceName>SPINETestDevice_A</deviceName>
6585   <deviceCode>SPINE_TD_A</deviceCode>
6586   <serialNumber>159753852456</serialNumber>
6587   <softwareRevision>1.2.0</softwareRevision>
6588   <hardwareRevision>1.0.1</hardwareRevision>
6589   <vendorName>EEBus Initiative e.V.</vendorName>
6590   <vendorCode>4545427573</vendorCode>
6591   <brandName>TestBrand</brandName>
6592   <powerSource>mainsSinglePhase</powerSource>
6593 <manufacturerNodeIdentification>SPINE_TD_A_1.2.0_159753852456_A</manufactur
6594 erNodeIdentification>
6595   <manufacturerLabel>SPINE_TD_A@EEBus_e.V.</manufacturerLabel>
6596   <manufacturerDescription>SPINETestDevice_A Installation @ EEBus
6597 Initiative e.V.</manufacturerDescription>
6598 </deviceClassificationManufacturerData>

```

6599

6600 **5.3.7.3 deviceClassificationUserData**6601 **5.3.7.3.1 General**

6602 Information which is usually writable (changeable by a user / application) is modelled with
 6603 *deviceClassificationUserData*.



6604

6605 *Figure 79: deviceClassificationUserData function overview*

6606

6607 **5.3.7.3.2 Detailed description of elements**

Element	Type	Description
deviceClassificationUserData.		

userNodeIdentification	Simple type "DeviceClassificationStringType" (restriction of xs:string)	Similar to <i>manufacturerNodeIdentification</i> but defined by a user or application.
userLabel	Common data type "LabelType". See section 3.10.1.2.	Similar to <i>manufacturerLabel</i> but defined by a user or application.
userDescription	Common data type "DescriptionType". See section 3.10.1.3.	Similar to <i>manufacturerDescription</i> but defined by a user or application.

Table 192: deviceClassificationUserData function detailed description of elements

5.3.7.3.3 Available selectors

None.

5.3.7.3.4 Examples

5.3.7.3.4.1 Read-reply

An example of a *deviceClassificationUserData* instance, send as reply:

```
<deviceClassificationUserData>
  <userNodeIdentification>TD_A_159753852456</userNodeIdentification>
  <userLabel>TD_A of EEBus e.V.</userLabel>
  <userDescription>Test Device A; Install Date:
15.07.13</userDescription>
</deviceClassificationUserData>
```

5.3.8 DeviceConfiguration

5.3.8.1 Introduction

Some information cannot be represented by the "normal" SPINE classes, but are important for the interoperable functionality of a device, hence should not be modelled with the DataTunneling class (see section 5.3.6) or other proprietary ways. For this purpose, the DeviceConfiguration class should be used. It provides a key-value based modelling of information. Some data, important for interoperability, is already defined by the EEBus Initiative e.V. and listed in Table 196.

The information is split into a rather static part (see section 5.3.8.3) and the actual value (see section 5.3.8.2).

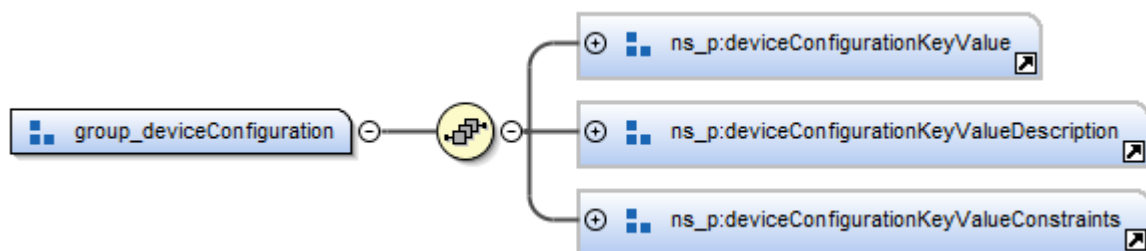
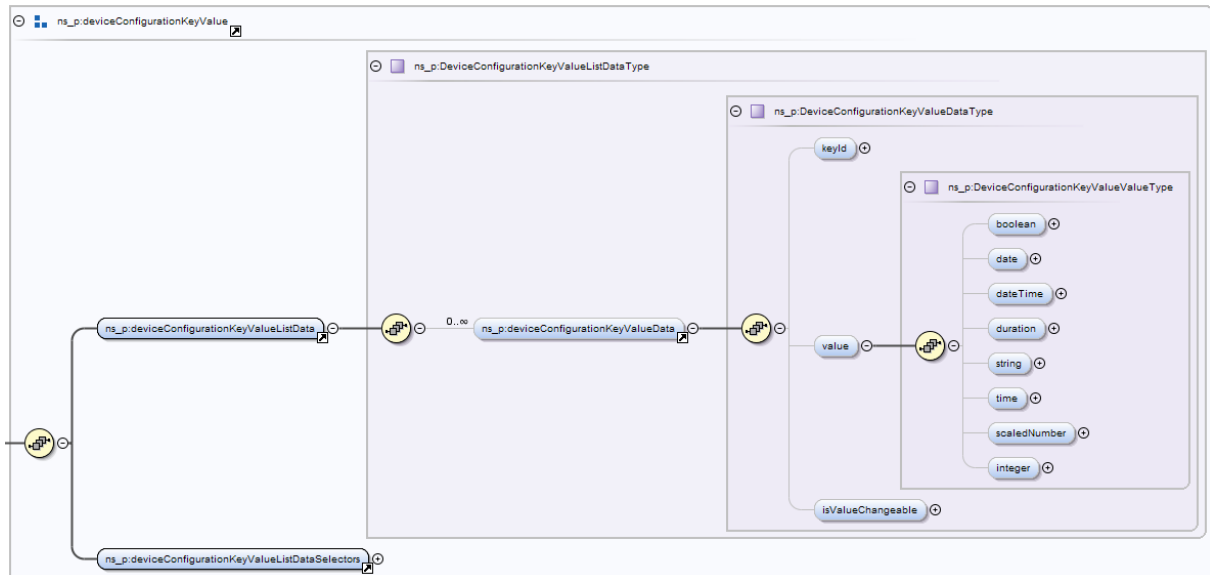


Figure 80: DeviceConfiguration function-group overview

6634

6635 **5.3.8.2 deviceConfigurationKeyValueListData**6636 **5.3.8.2.1 General**

6637 The actual values belonging to a specific key are modelled within this function.



6638

6639 *Figure 81: deviceConfigurationKeyValueListData function overview*

6640

6641 **5.3.8.2.2 Detailed description of elements**

Element	Type	Description
deviceConfigurationKeyValueListData. deviceConfigurationKeyValueData. (list)		
keyId	Identifier "DeviceConfigurationKeyIdType" (see Table 358).	Enables the identification of different keys on one SPINE feature.
value	Complex type "DeviceConfigurationKeyValueValueType". See Table 194.	The actual value belonging to the keyId.
isValueChangeable	xs:boolean (W3C standard type)	States whether the value belonging to the keyId is changeable by a client or not.

6642 *Table 193: deviceConfigurationKeyValueListData function detailed description of elements*6643 Complex type **DeviceConfigurationKeyValueValueType**:

Element	Type	Description
boolean	xs:boolean (W3C standard type)	Used, if the value is of type boolean.
date	xs:date (W3C standard type)	Used, if the value is of type date.
dateTime	xs:dateTime (W3C standard type)	Used, if the value is of type dateTime.
duration	xs:duration (W3C standard type)	Used, if the value is of type duration.

string	<i>Simple type</i> <i>"DeviceConfigurationKeyValueStringType"</i> <i>(restriction of xs:string)</i>	Used, if the <i>value</i> is of type <i>string</i> .
time	xs:time (W3C standard type)	Used, if the <i>value</i> is of type <i>time</i> .
scaledNumber	<i>Common data type "ScaledNumberType".</i> <i>See section 3.10.1.8.</i>	Used, if the <i>value</i> is of type <i>scaledNumber</i> .
integer	xs:long (W3C standard type)	Used, if the <i>value</i> is of type <i>long</i> .

Table 194: Complex type DeviceConfigurationKeyValueValueType

5.3.8.2.3 Available selectors

- keyId

5.3.8.2.4 Examples

5.3.8.2.4.1 Read-reply

If one is interested in the value of a specific key, he could send the following read command:

```
<function>deviceConfigurationKeyValueListData</function>
<filter>
  <cmdControl>
    <partial/>
  </cmdControl>
  <deviceConfigurationKeyValueListDataSelectors>
    <keyId>2</keyId>
  </deviceConfigurationKeyValueListDataSelectors>
</filter>
<deviceConfigurationKeyValueListData/>
```

If the receiving device has a proper value for this *keyId*, it would respond with the following function:

```
<deviceConfigurationKeyValueListData>
  <deviceConfigurationKeyValueData>
    <keyId>2</keyId>
    <value>
      <scaledNumber>
        <number>18</number>
        <scale>0</scale>
      </scaledNumber>
    </value>
    <isValueChangeable>false</isValueChangeable>
  </deviceConfigurationKeyValueData>
</deviceConfigurationKeyValueListData>
```

5.3.8.2.4.2 Notify

If a value changes, it will be notified to all subscribed clients:

```
<deviceConfigurationKeyValueListData>
  <deviceConfigurationKeyValueData>
    <keyId>2</keyId>
    <value>
      <scaledNumber>
        <number>32</number>
```

```

6684         <scale>0</scale>
6685     </scaledNumber>
6686 </value>
6687     <isValueChangeable>false</isValueChangeable>
6688 </deviceConfigurationKeyValueData>
6689 </deviceConfigurationKeyValueListData>

```

5.3.8.3 deviceConfigurationKeyValueDescriptionListData

5.3.8.3.1 General

The rather static part of the device configuration keys is modelled with this function, including an identifier for the key, a name that defines the meaning of the key and further additional elements.

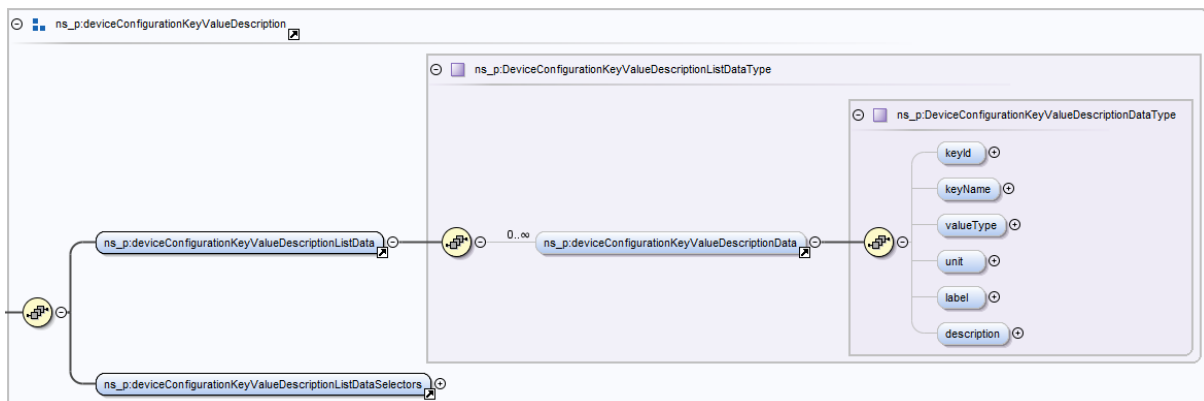


Figure 82: deviceConfigurationKeyValueDescriptionListData function overview

5.3.8.3.2 Detailed description of elements

Element	Type	Description
deviceConfigurationKeyValueDescriptionListData. deviceConfigurationKeyValueDescriptionData. (list)		
keyId	Identifier "DeviceConfigurationKeyIdType" (see Table 358).	The key identifier.
keyName	Union "DeviceConfigurationKeyNameType": - Enum (see Table 196): DeviceConfigurationKeyNameEnumType - EnumExtendType (see section 3.10.1.5)	A certain key name. If one of the keys defined by the EEBus Initiative e.V. is used, the meaning is defined in Table 196. Vendors may define own key names (but then have to mark them as vendor specific, see [ProtocolSpecification], section "Rules for vendor specific extensions").
valueType	Enumeration DeviceConfigurationKeyValueDataTypeType (see Table 197).	Different value types are possible, but only one may be chosen. It is denoted in this element.
unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	The unit in which the value of the key is given. Not all keys need the unit element.
label	Common data type "LabelType". See section 3.10.1.2.	A label for this key.

description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A description for this key.
-------------	--	-----------------------------

6699 Table 195: deviceConfigurationKeyValueDescriptionListData function detailed description of elements

6700 Enumeration **DeviceConfigurationKeyNameEnumType**:

Value	Description
peakPowerOfPvSystem	The nominal peak power of the production of the installed PV System.
pvCurtailementLimitFactor	The percentage of the peak power of the PV System that is permitted to be fed into the electricity grid through the electricity grid connection point.
asymmetricChargingSupported	This configuration parameter is used by the Entity EV to indicate the Entity SECC if it is possible to charge with different current per phase.
communicationsStandard	Defines the used communications standard for the entity where the DeviceConfiguration feature is located.
inverterGridCode	Grid code of the Inverter.
pvStringAvailabilityStatus	Indicates the availability of the PV String(s).
batteryAvailabilityStatus	Indicates the availability of the Battery (or Batteries).
gridConnectionStatus	Indicates whether the Inverter is connected to the public grid or not.
timeToAcChargePowerMax	The time (in seconds) currently required by the appliance to adjust the active power until the currently maximum possible charge power is reached.
timeToAcDischargePowerMax	The time (in seconds) currently required by the appliance to adjust the active power until the currently maximum possible discharge power is reached.
tilt	Tilt of the appliance.
azimuth	Azimuth of the appliance.
batteryType	Type of the Battery.
maxCycleCountPerDay	Maximum number of cycles the battery permits per day.
failsafeConsumptionActivePowerLimit	Failsafe value for the appliances active (real) power consumption.
failsafeProductionActivePowerLimit	Failsafe value for the appliances active (real) power production.
failsafePositiveReactivePowerLimit	Failsafe value for the appliances positive reactive power.
failsafeNegativeReactivePowerLimit	Failsafe value for the appliances negative reactive power.
failsafePositiveCosPhiLimit	Failsafe value for the appliances positive cos phi value.
failsafeNegativeCosPhiLimit	Failsafe value for the appliances negative cos phi value.

maxAcChargePower	Maximum AC charge power (aggregated) for all connected appliances.
maxAcDischargePower	Maximum AC discharge power (aggregated) for all connected appliances.
maxDcChargePower	Maximum DC charge power (aggregated) for all connected appliances.
maxDcDischargePower	Maximum DC discharge power (aggregated) for all connected appliances.
batteryActiveControlMode	Current active control mode of the Inverter
defaultAcPower	Default AC power value.
defaultDcPower	Default DC power value.
defaultPccPower	Default AC power value at the PCC (Point of Common Coupling).
failsafeAcPowerSetpoint	Failsafe value for the Charge/Discharge AC power setpoint.
failsafeDcPowerSetpoint	Failsafe value for the Charge/Discharge DC power setpoint.
failsafePccPowerSetpoint	Failsafe value for the PCC power setpoint.
failsafeDurationMinimum	Minimum time the appliance remains in failsafe state after loss of Heartbeats.
dischargingBelowTargetEnergyRequestPermitted	States whether the appliance (e.g. EV, or any other appliance capable of storing energy) allows discharging its battery below the target SoC or not.
incentivesSimulationCyclesMax	The maximum number of successive simulation cycles permitted by an appliance.
incentivesSimulationConcurrent	Value "true" means an appliance is capable of executing a scenario with simulation data in parallel to a scenario with committed data. Otherwise, the value is "false".
incentivesTimeoutIncentiveRequest	Maximum waiting period of an appliance to receive a valid update of a simple committed incentive table.
incentivesWaitIncentiveWriteable	Waiting period of an appliance before it permits getting an updated simple committed incentive table.

6701 Table 196: Enumeration DeviceConfigurationKeyNameEnumType

6702 Enumeration **DeviceConfigurationKeyValueTypeEnum**:

Value	Description
boolean	The corresponding key value is of type xs:boolean.
date	The corresponding key value is of type xs:date.
dateTime	The corresponding key value is of type xs:dateTime.
duration	The corresponding key value is of type xs:duration.
string	The corresponding key value is of type xs:string.
time	The corresponding key value is of type xs:time.
scaledNumber	The corresponding key value is of type "ScaledNumberType" (see section 3.10.1.8).
integer	The corresponding key value is of type xs:long.

6703 Table 197: Enumeration DeviceConfigurationKeyValueTypeEnum

6704

5.3.8.3.3 Available selectors

- keyId
- keyName

5.3.8.3.4 Examples

5.3.8.3.4.1 Read-reply

If one is interested in the rather static information of the device configuration keys, he could request the information with a read command. The device replies with information like this:

```
<deviceConfigurationKeyValueDescriptionListData>
  <deviceConfigurationKeyValueDescriptionData>
    <keyId>1</keyId>
    <keyName>peakPowerOfPvSystem</keyName>
    <valueType>scaledNumber</valueType>
    <unit>W</unit>
    <label>Peak Power of PV System</label>
    <description>The nominal peak power of the production of
the installed PV System</description>
  </deviceConfigurationKeyValueDescriptionData>
  <deviceConfigurationKeyValueDescriptionData>
    <keyId>2</keyId>
    <keyName>pvCurtailementLimitFactor</keyName>
    <valueType>scaledNumber</valueType>
    <unit>pct</unit>
    <label>PV Curtailement Limit Factor</label>
    <description>The percentage of the peak power of the PV System that
is permitted to be fed into the electricity grid through the electricity
grid connection point.</description>
  </deviceConfigurationKeyValueDescriptionData>
</deviceConfigurationKeyValueDescriptionListData>
```

5.3.8.4 deviceConfigurationKeyValueConstraintsListData

5.3.8.4.1 General

In cases where a configuration value has some range, it could make sense to specify the upper and lower bounds that are possible for this value as well as a step size. Therefore, the deviceConfigurationKeyValueConstraintsListData function can be used.

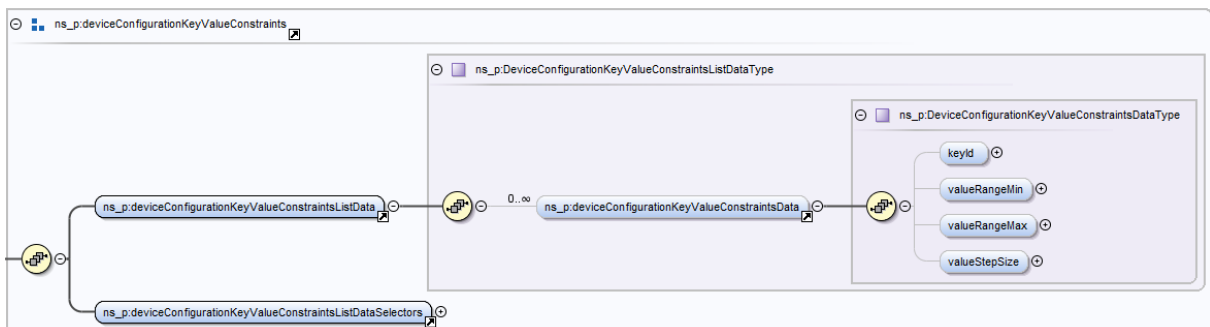


Figure 83: deviceConfigurationKeyValueConstraintsListData function overview

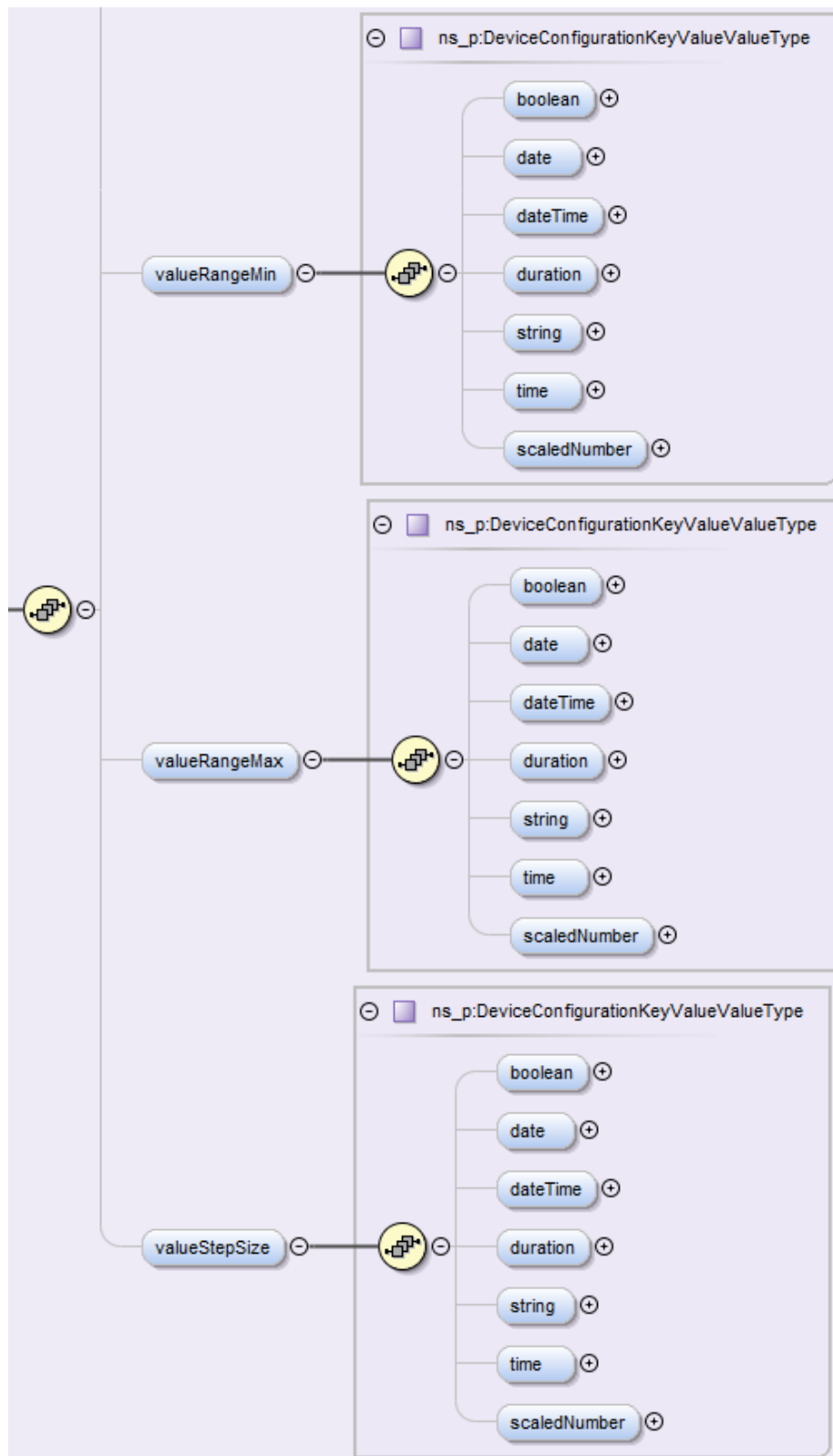


Figure 84: `deviceConfigurationKeyValueConstraintsListData` function detail

5.3.8.4.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

deviceConfigurationKeyValueConstraintsListData. deviceConfigurationKeyValueConstraintsData. (list)		
keyId	Identifier "DeviceConfigurationKeyIdType" (see Table 358).	Enables the identification of different keys on one SPINE feature.
valueRangeMin	Complex type "DeviceConfigurationKeyValueValueType". See Table 194.	The lower bound that is possible for the related value.
valueRangeMax	Complex type "DeviceConfigurationKeyValueValueType". See Table 194.	The upper bound that is possible for the related value.
valueStepSize	Complex type "DeviceConfigurationKeyValueValueType". See Table 194.	The step size that is possible for the related value.

Table 198: deviceConfigurationKeyValueConstraintsListData function detailed description of elements

5.3.8.4.3 Available selectors

- keyId

5.3.8.4.4 Examples

5.3.8.4.4.1 Read-reply

If one is interested in the constraints of all device configuration keys, he could request the information with a read command. The device replies with information like this:

```

<deviceConfigurationKeyValueConstraintsListData>
  <deviceConfigurationKeyValueConstraintsData>
    <keyId>1</keyId>
    <valueRangeMin>
      <scaledNumber>
        <number>5</number>
        <scale>3</scale>
      </scaledNumber>
    </valueRangeMin>
    <valueRangeMax>
      <scaledNumber>
        <number>7</number>
        <scale>3</scale>
      </scaledNumber>
    </valueRangeMax>
    <valueStepSize>
      <scaledNumber>
        <number>1</number>
      </scaledNumber>
    </valueStepSize>
  </deviceConfigurationKeyValueConstraintsData>
  <deviceConfigurationKeyValueConstraintsData>
    <keyId>2</keyId>
    <valueRangeMin>
      <scaledNumber>
        <number>0</number>
      </scaledNumber>
    </valueRangeMin>
    <valueRangeMax>

```

```

6784         <scaledNumber>
6785             <number>1</number>
6786             <scale>2</scale>
6787         </scaledNumber>
6788     </valueRangeMax>
6789     <valueStepSize>
6790         <scaledNumber>
6791             <number>5</number>
6792             <scale>-1</scale>
6793         </scaledNumber>
6794     </valueStepSize>
6795 </deviceConfigurationKeyValueConstraintsData>
6796 </deviceConfigurationKeyValueConstraintsListData>

```

5.3.9 DeviceDiagnosis

5.3.9.1 Introduction

This class contains information about the state of a device, heartbeat counting and service data.

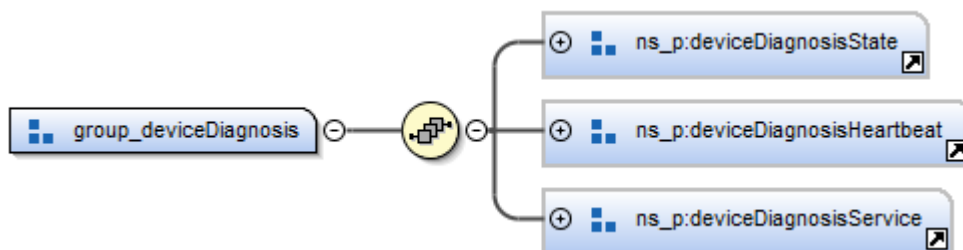


Figure 85: DeviceDiagnosis function-group overview

5.3.9.2 deviceDiagnosisStateData

5.3.9.2.1 General

The *deviceDiagnosisStateData* function provides information about dynamically changing state information.

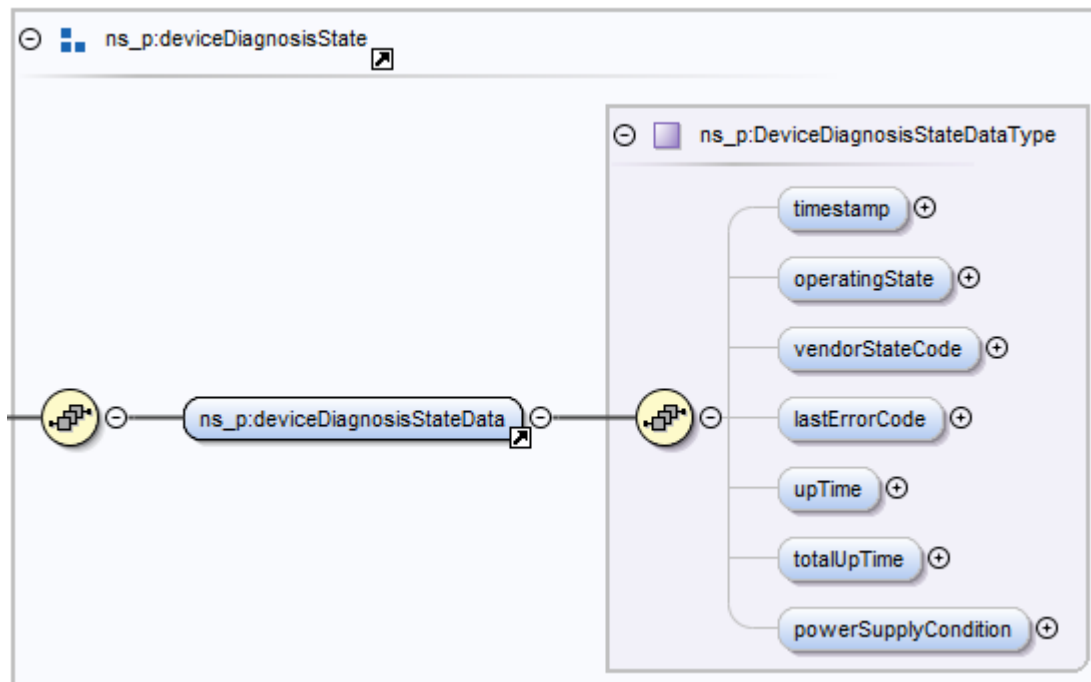


Figure 86: deviceDiagnosisStateData function overview

5.3.9.2.2 Detailed description of elements

Element	Type	Description
deviceDiagnosisStateData.		
timestamp	Common data type "AbsoluteOrRelativeTimeType" . See section 3.10.2.3.	The time of creation of the state.
operatingState	Union "DeviceDiagnosisOperatingStateType": - Enum (see Table 200): DeviceDiagnosisOperatingStateEnumType - EnumExtendType (see section 3.10.1.5)	The current operating state of the device.
vendorStateCode	Simple type "VendorStateCodeType" (restriction of xs:string)	A vendor specific state. May contain any state, the vendor uses for its devices.
lastErrorCode	Simple type "LastErrorCodeType" (restriction of xs:string)	The last error code that occurred is stored in this element. Even if the device is in <i>normalOperation</i> again, the error code remains here.
upTime	xs:duration (W3C standard type)	The duration the functionality is running since the last start.
totalUpTime	xs:duration (W3C standard type)	The total duration the functionality is running since installation (see <i>installationTime</i> element in <i>deviceDiagnosisServiceData</i> function).
powerSupplyCondition	Union "PowerSupplyConditionType":	The condition of the power supply can be stated here.

	- Enum (see Table 201): PowerSupplyConditionEnumType - EnumExtendType (see section 3.10.1.5)	
--	---	--

6812 Table 199: deviceDiagnosisStateData function detailed description of elements

6813 Enumeration **DeviceDiagnosisOperatingStateEnumType**:

Value	Description
normalOperation	The device does its normal operation without any errors or limitations.
standby	The device is in standby mode and does nothing but waits for user interaction and listens on its communications protocol for external commands.
failure	A failure occurred and the device cannot run as desired. Its normal functionality is likely not given.
serviceNeeded	The device needs some kind of service. Possibly the normal functionality is not given.
overrideDetected	The device detected an override. Normal operation may be given or the device is in some safe mode.
inAlarm	An alarm (or emergency) occurred and user interaction is needed. Normal operation may be given or the device is in some safe mode.
notReachable	The device is not reachable via its non-SPINE communications protocol. This operating state can only be set by some technology gateway device that handles the normal communication to the device.
finished	The device has finished its temporary operation. As long as it remains in state <i>finished</i> , the normal operation is not given.
temporarilyNotReady	The device is temporarily not ready for normal operation (e.g. due to maintenance).
off	The device is off. Hence, it cannot operate.

6814 Table 200: Enumeration DeviceDiagnosisOperatingStateEnumType

6815 Enumeration **PowerSupplyConditionEnumType**:

Value	Description
good	The power supply is in a good condition.
low	The power supply has low capacity. It would make sense to check it (e.g. change battery).
critical	The power supply is in critical state. If it is supplied by a battery, a change of the battery is advised.
unknown	The power supply condition is unknown. A notification to a user could make sense.
error	The power supply is in error condition.

6816 Table 201: Enumeration PowerSupplyConditionEnumType

6817

6818 5.3.9.2.3 Available selectors

6819 None.

6820

5.3.9.2.4 Examples

5.3.9.2.4.1 Notify

The current state of a device is notified to all subscribed clients:

```
<deviceDiagnosisStateData>
  <timestamp>2014-04-28T17:43:14.0Z</timestamp>
  <operatingState>normalOperation</operatingState>
  <vendorStateCode>heating up</vendorStateCode>
  <lastErrorCode>13</lastErrorCode>
  <upTime>P3DT4H7M14S</upTime>
  <totalUpTime>P2Y8M15DT7H32M17S</totalUpTime>
  <powerSupplyCondition>good</powerSupplyCondition>
</deviceDiagnosisStateData>
```

5.3.9.3 deviceDiagnosisHeartbeatData

5.3.9.3.1 General

The *deviceDiagnosisHeartbeatData* function enables sending a specific signal, which indicates a device is still running.

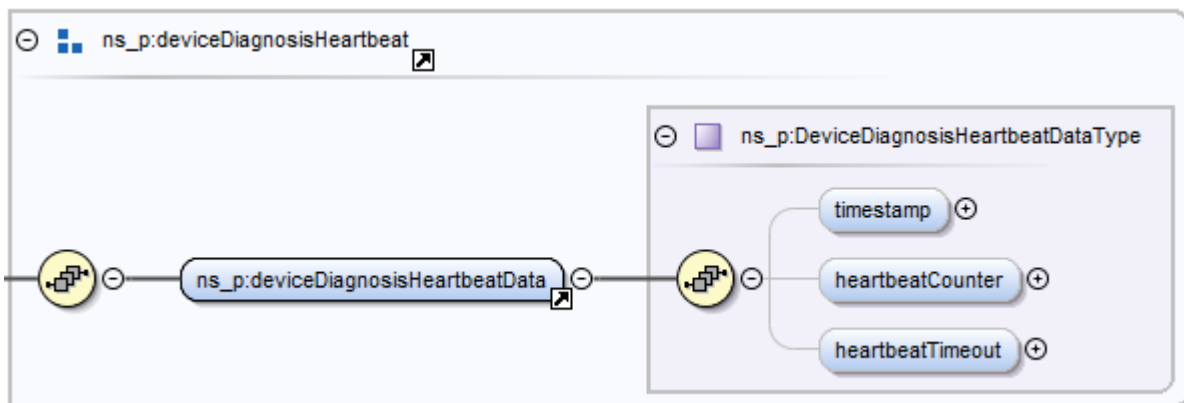


Figure 87: deviceDiagnosisHeartbeatData function overview

5.3.9.3.2 Detailed description of elements

Element	Type	Description
deviceDiagnosisHeartbeatData.		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time of creation of the data.
heartbeatCounter	xs:unsignedLong (W3C standard type)	An incrementing counter of the heartbeat.
heartbeatTimeout	xs:duration (W3C standard type)	The heartbeatTimeout element contains the period, in which the deviceDiagnosisHeartbeatData function is sent by the device (NOT the remaining time till the next sending).

Table 202: deviceDiagnosisHeartbeatData function detailed description of elements

5.3.9.3.3 Available selectors

None.

5.3.9.3.4 Examples

5.3.9.3.4.1 Read-reply

One is interested in the heartbeat information of a device and sends a standard read command to the device which responds with a *deviceDiagnosisHeartbeatData* function:

```
<deviceDiagnosisHeartbeatData>
  <timestamp>2014-04-28T17:54:38.0Z</timestamp>
  <heartbeatCounter>238</heartbeatCounter>
  <heartbeatTimeout>PT15M</heartbeatTimeout>
</deviceDiagnosisHeartbeatData>
```

5.3.9.4 deviceDiagnosisServiceData

5.3.9.4.1 General

The *deviceDiagnosisServiceData* function informs about service information related to a device's installation or maintenance.

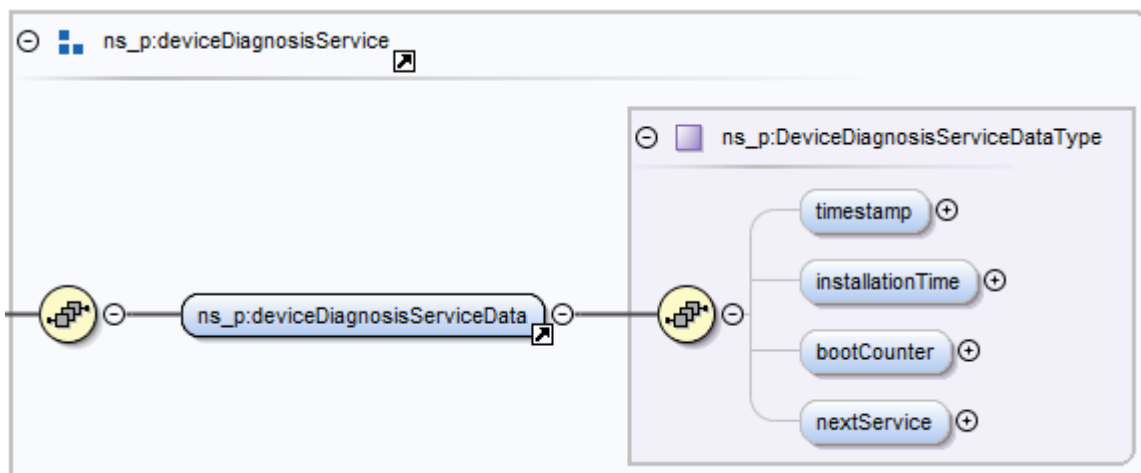


Figure 88: deviceDiagnosisServiceData function overview

5.3.9.4.2 Detailed description of elements

Element	Type	Description
deviceDiagnosisServiceData.		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time of creation of the data.
installationTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The point in time the device was installed (commissioned).
bootCounter	xs:unsignedLong (W3C standard type)	The count of boot processes the device completed since <i>installationTime</i> .

nextService	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The point in time, the next service should be done, is stated by this element. Alternatively, the decrementing duration till the next service can be indicated.
-------------	--	---

Table 203: deviceDiagnosisServiceData function detailed description of elements

5.3.9.4.3 Available selectors

None.

5.3.9.4.4 Examples

5.3.9.4.4.1 Read-reply

One is interested in the service information of a device and sends a standard read command to the device which responds with a *deviceDiagnosisServiceData* function:

```
<deviceDiagnosisServiceData>
  <timestamp>2014-04-28T17:59:05.0Z</timestamp>
  <installationTime>2011-08-15T11:05:08.0Z</installationTime>
  <bootCounter>83</bootCounter>
  <nextService>2015-08-15T11:05:08.0Z</nextService>
</deviceDiagnosisServiceData>
```

5.3.10 DirectControl

5.3.10.1 Introduction

Some kind of electricity consuming or producing devices require (in contrast to the PowerSequences Class, see section 5.3.19) a rather instantaneous energy consumption or generation control. This can be modelled with two primary information sets:

3. Notification of instantaneous consumption/generation as well as instantaneous control. This is covered by DirectControl and discussed in this section.
4. Information on the power/energy constraints and implications. This is covered by OperatingConstraints and discussed in section 5.3.18.

Information on and control of energy/power can be modelled using directControlActivity. directControlDescription is used to model some general constant data.

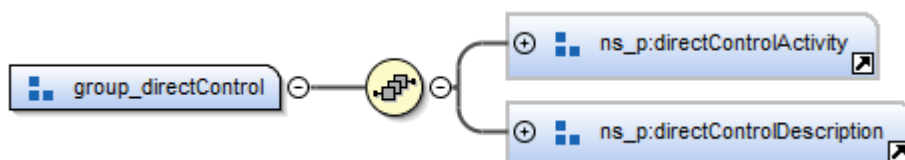


Figure 89: DirectControl function-group overview

5.3.10.2 *directControlActivityListData*

5.3.10.2.1 *General*

The function *directControlActivityListData* combines some information on the activity of a given time. Apart from information on state, power and energy it can contain information whether power, energy, etc. can be modified with a proper (partial) write command.

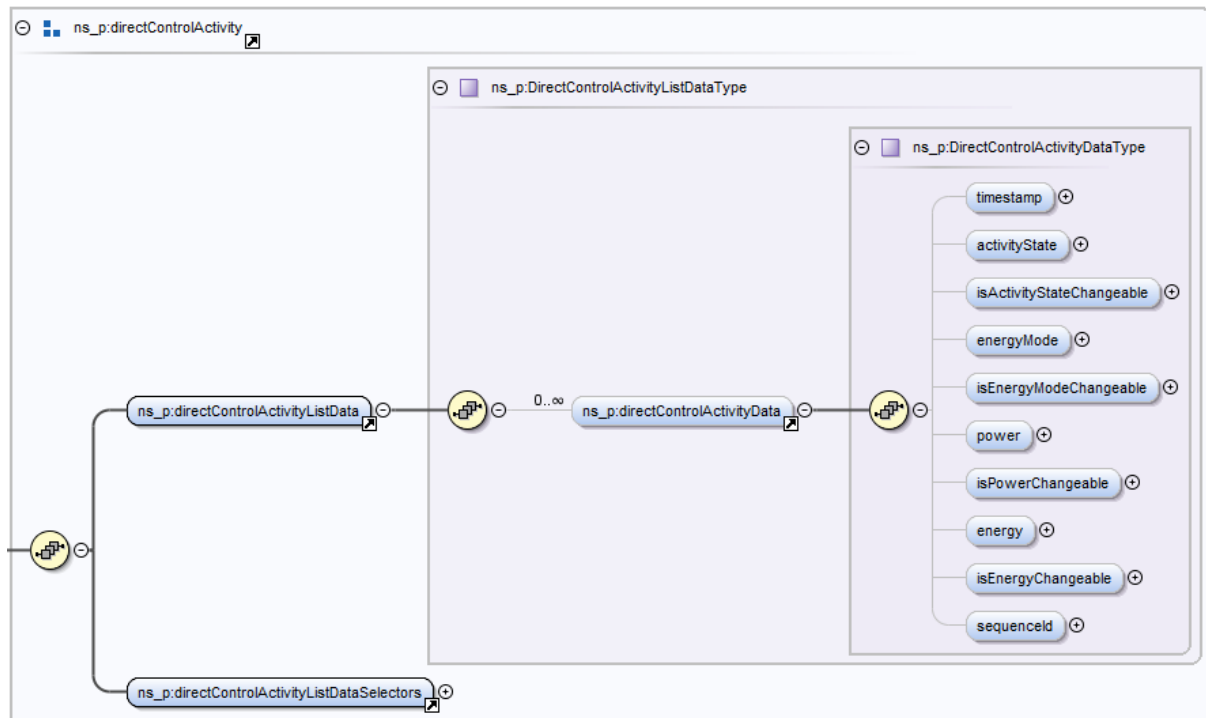


Figure 90: *directControlActivityListData* function overview

5.3.10.2.2 *Detailed description of elements*

Element	Type	Description
directControlActivityListData. directControlActivityData. (list)		
timestamp	<i>Common data type</i> <i>"AbsoluteOrRelativeTimeType". See section 3.10.2.3.</i>	The timestamp of this function instance.
activityState	<i>Union</i> <i>"DirectControlActivityStateType":</i> - Enum (see Table 205): DirectControlActivityStateEnumType - EnumExtendType (see section 3.10.1.5)	The state of the activity.
isActivityStateChangeable	xs:boolean (W3C standard type)	Indicates whether the "activityState" can be modified by a client or not.
energyMode	<i>Common data type</i> <i>"EnergyModeType". See section 3.10.1.15.</i>	Energy mode of the device.
isEnergyModeChangeable	xs:boolean (W3C standard type)	Indicates whether the "energyMode" can be modified by a client or not.

power	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The overall electrical power value belonging to this activity information. Please note: Phase-specific information is not given here.
isPowerChangeable	xs:boolean (W3C standard type)	Indicates whether the "power" can be modified by a client or not.
energy	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The electrical energy value belonging to this activity information.
isEnergyChangeable	xs:boolean (W3C standard type)	Indicates whether the "energy" can be modified by a client or not.
sequenceId	xs:unsignedInt (W3C standard type)	The related power sequence identifier (if available).

6904 Table 204: directControlActivityListData function detailed description of elements

6905 Enumeration **DirectControlActivityStateEnumType**:

Value	Description
running	The activity is running.
paused	The activity is paused.
inactive	The activity is inactive.

6906 Table 205: Enumeration DirectControlActivityStateEnumType

6907

6908 5.3.10.2.3 Available selectors

6909 - timestampInterval

6910

6911 5.3.10.2.4 Examples

6912 5.3.10.2.4.1 Notify

6913 When the activity state changes, a server will notify the new information to all subscribed devices
6914 with a message like this (the "filter/partial" part to announce the "restricted function exchange" is
6915 simplified by "..." just to improve the readability):

```

6916 ...
6917 <directControlActivityListData>
6918   <directControlActivityData>
6919     <timestamp>2015-11-23T12:05:04.3Z</timestamp>
6920     <activityState>running</activityState>
6921     <isActivityStateChangeable>true</isActivityStateChangeable>
6922     <energyMode>consume</energyMode>
6923     <isEnergyModeChangeable>false</isEnergyModeChangeable>
6924     <power>
6925       <number>2</number>
6926       <scale>3</scale>
6927     </power>
6928     <isPowerChangeable>false</isPowerChangeable>
6929     <sequenceId>1</sequenceId>
6930   </directControlActivityData>
6931 </directControlActivityListData>

```

5.3.10.3 directControlDescriptionData

5.3.10.3.1 General

This function contains the rather static information available in this class.

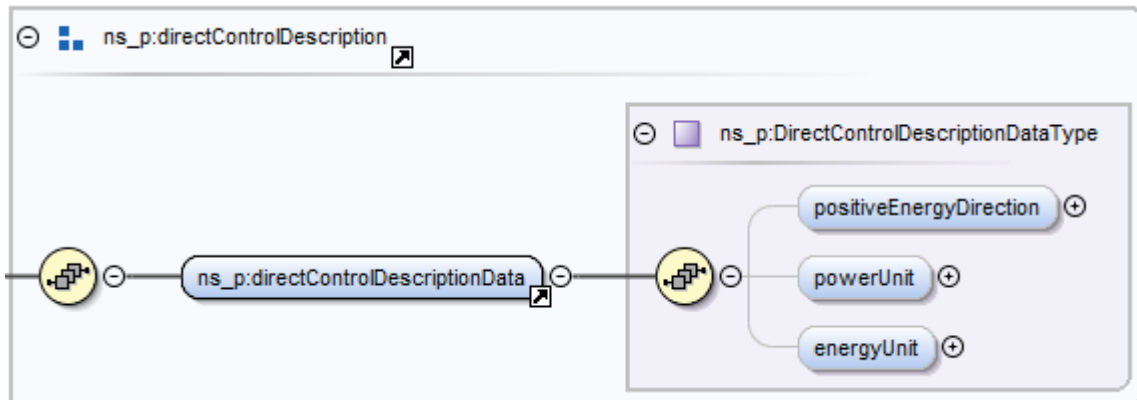


Figure 91: directControlDescriptionData function overview

5.3.10.3.2 Detailed description of elements

Element	Type	Description
directControlDescriptionData.		
positiveEnergyDirection	Common data type "EnergyDirectionType". See section 3.10.1.13.	The <i>positiveEnergyDirection</i> states whether energy consumption or production will be counted as positive value.
powerUnit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	This is the unit for all power values related to consumption or production.
energyUnit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	This is the unit for all energy values related to consumption or production.

Table 206: directControlDescriptionData function detailed description of elements

5.3.10.3.3 Available selectors

None.

5.3.10.3.4 Examples

5.3.10.3.4.1 Read-reply

If one is interested in the rather static information of the direct control, he could request the information with a read command. The device replies with information like this:

```

<directControlDescriptionData>
  <positiveEnergyDirection>consume</positiveEnergyDirection>
  <powerUnit>W</powerUnit>

```

6952 <energyUnit>Wh</energyUnit>
 6953 </directControlDescriptionData>

6954

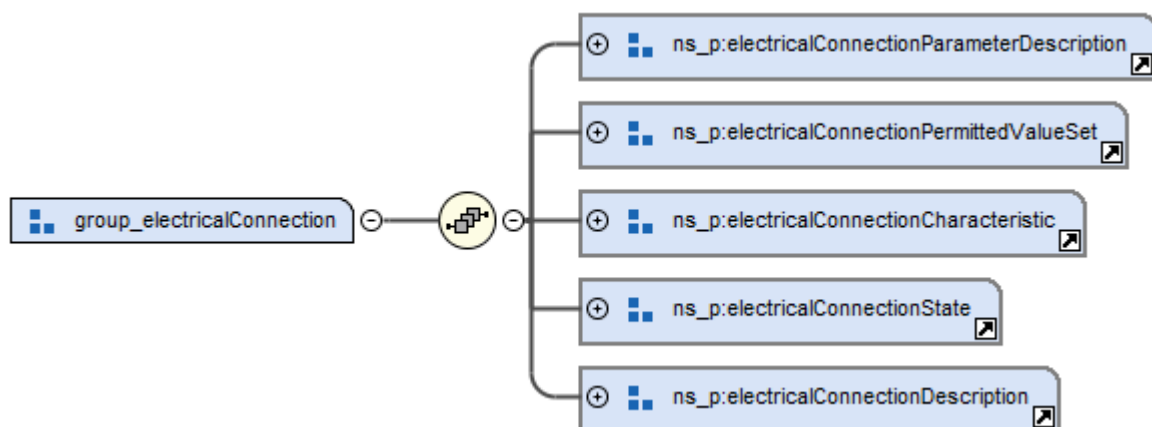
6955 **5.3.11 ElectricalConnection**

6956 **5.3.11.1 Introduction**

6957 Electrical connections have some more complex parameters (e.g. which AC phase is measured,
 6958 real/reactive/apparent measurement, amplitude/RMS, number of measured harmonic, etc.), which
 6959 are modelled with this class. Some electrical connection specific state information (e.g. AC or DC,
 6960 number of connected phases (only AC), positive energy direction, label, etc.) is modelled, too.

6961 Measuring electrical values is still done with the Measurement class (see section 5.3.15). But the
 6962 additional parameters are only described within the ElectricalConnection class. The *measurementId*
 6963 from the Measurement class is stated in the *electricalConnectionParameterDescriptionListData*
 6964 function, to have a proper reference.

6965 Characteristical information (like nominal values) are modelled within the
 6966 electricalConnectionCharacteristicListData Function.



6967

6968 Figure 92: *ElectricalConnection* function-group overview

6969

6970 **5.3.11.2 electricalConnectionParameterDescriptionListData**

6971 5.3.11.2.1 General

6972 The aforementioned parameters for electrical connection measurement are described in this
 6973 function. The value itself is modelled in the Measurement class (section 5.3.15). Only a reference is
 6974 given in this function.

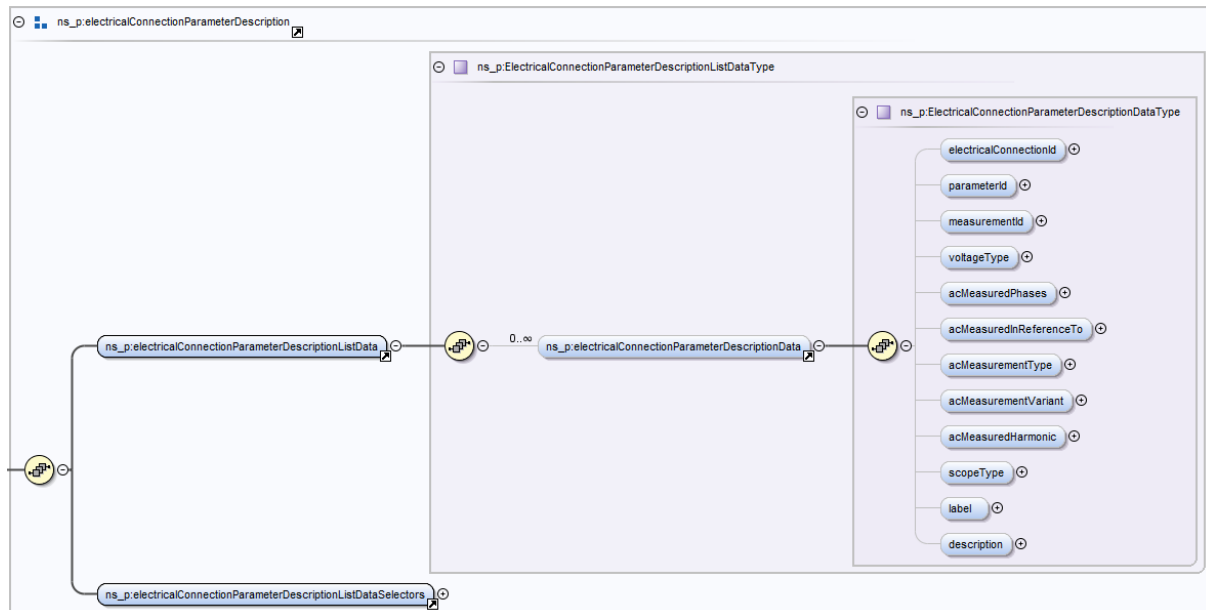


Figure 93: electricalConnectionParameterDescriptionListData function overview

5.3.11.2.2 Detailed description of elements

Element	Type	Description
electricalConnectionParameterDescriptionListData. electricalConnectionParameterDescriptionData. (list)		
electricalConnectionId	Identifier "ElectricalConnectionIdType" (see Table 358).	Reference to the electrical connection, described in the <i>electricalConnectionDescriptionListData</i> function (see section 5.3.11.6).
parameterId	Identifier "ElectricalConnectionParameterIdType" (see Table 358).	One electrical connection can have multiple parameter combinations, which can be measured. The <i>parameterId</i> helps to distinguish them.
measurementId	Identifier "MeasurementIdType" (see Table 358).	Reference to the Measurement class (see section 5.3.15), which contains the actual measured values for this parameter combination.
voltageType	Union "ElectricalConnectionVoltageTypeType": - Enum (see Table 208): ElectricalConnectionVoltageTypeEnumType - EnumExtendType (see section 3.10.1.5)	Specifies which kind of electricity is measured ("ac" or "dc").
acMeasuredPhases	Union "ElectricalConnectionPhaseNameType": - Enum (see Table 209): ElectricalConnectionPhaseNameEnumType - EnumExtendType (see section 3.10.1.5)	In case of <i>powerSupplyType</i> = "ac", this element states the phase, which is measured. Combinations of phases (e.g. "ab") are possible.

acMeasuredInReferenceTo	<i>Union</i> "ElectricalConnectionPhaseNameType": - <i>Enum</i> (see Table 209): ElectricalConnectionPhaseNameEnumType - EnumExtendType (see section 3.10.1.5)	In case of <i>powerSupplyType</i> = "ac", this element states the phase, which <i>acMeasuredPhases</i> is measured against (e.g. "neutral").
acMeasurementType	<i>Union</i> "ElectricalConnectionAcMeasurementType": - <i>Enum</i> (see Table 210): ElectricalConnectionAcMeasurementTypeEnumType - EnumExtendType (see section 3.10.1.5)	In case of <i>powerSupplyType</i> = "ac", this element states the kind of AC measurement is done (e.g. "real").
acMeasurementVariant	<i>Union</i> "ElectricalConnectionMeasurandVariantType": - <i>Enum</i> (see Table 211): ElectricalConnectionMeasurandVariantEnumType - EnumExtendType (see section 3.10.1.5)	In case of <i>powerSupplyType</i> = "ac", this element states the variation of the AC measurement (e.g. "amplitude").
acMeasuredHarmonic	xs:unsignedByte (W3C standard type)	In case of <i>powerSupplyType</i> = "ac", this element states the harmonic, which is measured. A value of "1" indicates that the normal value is measured.
scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	A certain meaning of the parameter combination.
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A label for this electrical connection parameter.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A description for this electrical connection parameter.

6979 Table 207: electricalConnectionParameterDescriptionListData function detailed description of elements

6980 Enumeration **ElectricalConnectionVoltageTypeEnumType**:

Value	Description
ac	Alternating current.
dc	Direct current.

6981 Table 208: Enumeration ElectricalConnectionVoltageTypeEnumType

6982 Enumeration **ElectricalConnectionPhaseNameEnumType**:

Value	Description
a	Phase a (or L1).
b	Phase b (or L2).
c	Phase c (or L3).
ab	Phase a and b (or L1 and L2).
bc	Phase b and c (or L2 and L3).
ac	Phase a and c (or L1 and L3).
abc	Phase a, b and c (or L1, L2 and L3).
neutral	The neutral conductor of a connection.
ground	The ground conductor of a connection.
none	No phase specified or available.

6983 *Table 209: Enumeration ElectricalConnectionPhaseNameEnumType*

6984 Enumeration **ElectricalConnectionAcMeasurementTypeEnumType**:

Value	Description
real	AC measurement of the real part.
reactive	AC measurement of the reactive part.
apparent	AC measurement of the apparent part.
phase	AC measurement of the phase.

6985 *Table 210: Enumeration ElectricalConnectionAcMeasurementTypeEnumType*

6986 Enumeration **ElectricalConnectionMeasurandVariantEnumType**:

Value	Description
amplitude	Measurement of the amplitude.
rms	Measurement of the RMS (Root Mean Square).
instantaneous	Measurement of the instantaneous value.
angle	Measurement of the angle.
cosPhi	Measurement of the cosine phi.

6987 *Table 211: Enumeration ElectricalConnectionMeasurandVariantEnumType*

6988

6989 *5.3.11.2.3 Available selectors*

- 6990 - electricalConnectionId
- 6991 - parameterId
- 6992 - measurementId
- 6993 - scopeType

6994

6995 *5.3.11.2.4 Examples*

6996 *5.3.11.2.4.1 Read-reply*

6997 If one only knows the *measurementId* of a specific measurand and is interested in the electrical
6998 connection parameters, he could send the following read command:

```

6999 <function>electricalConnectionParameterDescriptionListData</function>
7000 <filter>
7001   <cmdControl>
7002     <partial/>
7003   </cmdControl>
7004   <electricalConnectionParameterDescriptionListDataSelectors>
7005     <measurementId>6</measurementId>
7006   </electricalConnectionParameterDescriptionListDataSelectors>
7007 </filter>
7008 <electricalConnectionParameterDescriptionListData/>

```

7009 If the receiving device has a proper parameter description for this *measurementId*, it would respond
7010 with the following function:

```

7011 <function>electricalConnectionParameterDescriptionListData</function>
7012 <filter>
7013   <cmdControl>
7014     <partial/>
7015   </cmdControl>

```

```

7016 </filter>
7017 <electricalConnectionParameterDescriptionListData>
7018   <electricalConnectionParameterDescriptionData>
7019     <electricalConnectionId>1</electricalConnectionId>
7020     <parameterId>3</parameterId>
7021     <measurementId>6</measurementId>
7022     <voltageType>ac</voltageType>
7023     <acMeasuredPhases>a</acMeasuredPhases>
7024     <acMeasuredInReferenceTo>neutral</acMeasuredInReferenceTo>
7025     <acMeasurementType>real</acMeasurementType>
7026     <acMeasurementVariant>rms</acMeasurementVariant>
7027     <acMeasuredHarmonic>1</acMeasuredHarmonic>
7028     <scopeType>acYieldTotal</scopeType>
7029   </electricalConnectionParameterDescriptionData>
7030 </electricalConnectionParameterDescriptionListData>

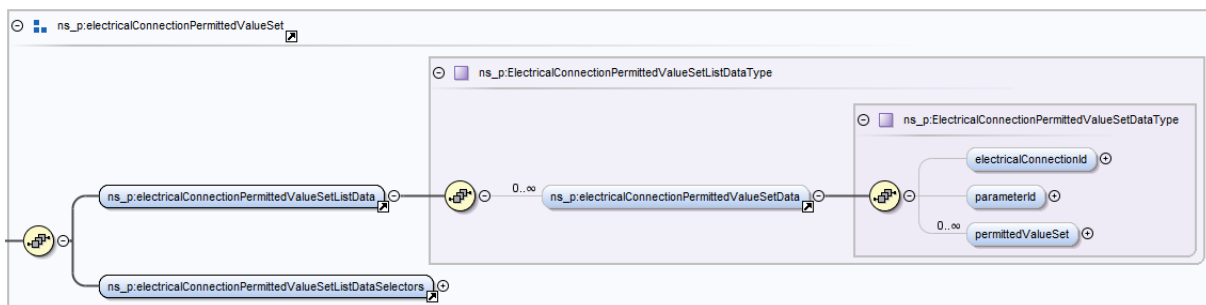
```

7031

7032 5.3.11.3 *electricalConnectionPermittedValueSetListData*

7033 5.3.11.3.1 *General*

7034 If an electrical connection parameter has some defined value(s) or value range(s) that are allowed or
 7035 possible, they can be modelled with this function. It enables the server to state single values, value
 7036 ranges or combinations of both.



7037

7038 Figure 94: *electricalConnectionPermittedValueSetListData* function overview

7039

7040 5.3.11.3.2 *Detailed description of elements*

Element	Type	Description
electricalConnectionPermittedValueSetListData. electricalConnectionPermittedValueSetData. (list)		
electricalConnectionId	Identifier "ElectricalConnectionId Type" (see Table 358).	Electrical connection, this static information belongs to.
parameterId	Identifier "ElectricalConnectionPa rameterIdType" (see Table 358).	Relation to the according parameter.
permittedValuesSet (list)	Common data type "ScaledNumberSetType ". See section 3.10.1.9.	The permittedValueSet allows to define an arbitrary set of permitted data. Compared to a simple value range the permittedValueSet also may have gaps. Please refer to the ScaledNumberSetType for more details.

7041 Table 212: *electricalConnectionPermittedValueSetListData* function detailed description of elements

7042

7043 **5.3.11.3.3 Available selectors**

- 7044 - electricalConnectionId
- 7045 - parameterId

7046

7047 **5.3.11.3.4 Examples**7048 **5.3.11.3.4.1.1 Read-reply**

7049 If one is interested in all electrical connection permitted value sets of a device, he could send the
 7050 read command to the device, which will respond with something like the following command:

```

7051 <electricalConnectionPermittedValueSetListData>
7052   <electricalConnectionPermittedValueSetData>
7053     <electricalConnectionId>1</electricalConnectionId>
7054     <parameterId>2</parameterId>
7055     <permittedValueSet>
7056       <range>
7057         <min>
7058           <number>0</number>
7059         </min>
7060         <max>
7061           <number>35</number>
7062           <scale>2</scale>
7063         </max>
7064       </range>
7065     </permittedValueSet>
7066   </electricalConnectionPermittedValueSetData>
7067   <electricalConnectionPermittedValueSetData>
7068     <electricalConnectionId>1</electricalConnectionId>
7069     <parameterId>3</parameterId>
7070     <permittedValueSet>
7071       <value>
7072         <number>0</number>
7073       </value>
7074       <range>
7075         <min>
7076           <number>5</number>
7077           <scale>1</scale>
7078         </min>
7079         <max>
7080           <number>2</number>
7081           <scale>3</scale>
7082         </max>
7083       </range>
7084     </permittedValueSet>
7085   </electricalConnectionPermittedValueSetData>
7086 </electricalConnectionPermittedValueSetListData>

```

7087

7088 **5.3.11.4 electricalConnectionCharacteristicListData**7089 **5.3.11.4.1 General**

7090 Characteristical information related to a specific electrical connection parameter (and hence to a
 7091 measurand) are modelled with this function.

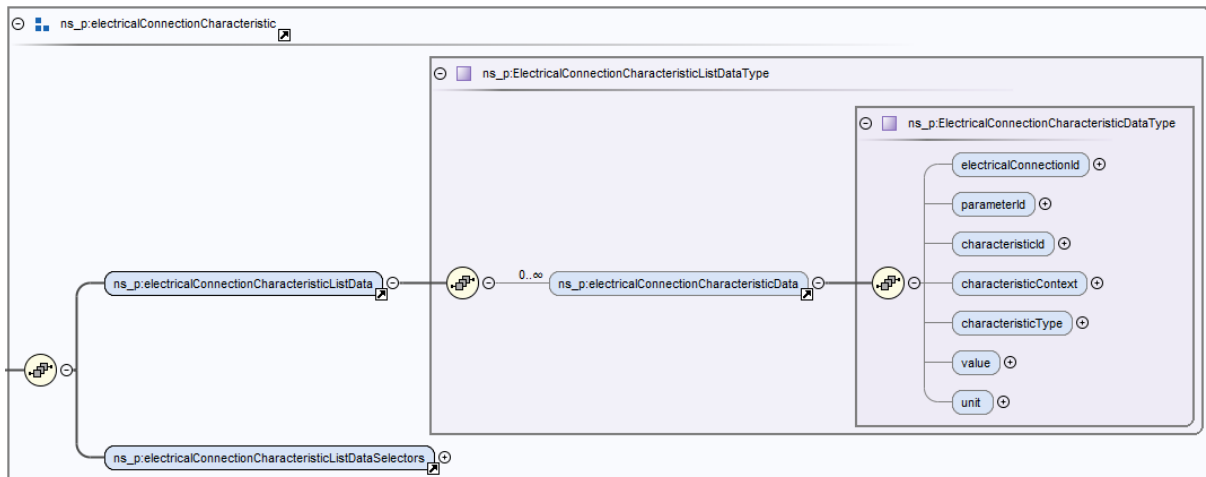


Figure 95: electricalConnectionCharacteristicListData function overview

5.3.11.4.2 Detailed description of elements

Element	Type	Description
electricalConnectionCharacteristicListData. electricalConnectionCharacteristicData. (list)		
electricalConnectionId	Identifier "ElectricalConnectionIdType" (see Table 358).	Electrical connection, this characteristic information belongs to.
parameterId	Identifier "ElectricalConnectionParameterIdType" (see Table 358).	Relation to the according parameter this characteristic belongs to.
characteristicId	Identifier "ElectricalConnectionCharacteristicIdType" (see Table 358).	Identifier of this characteristic for the given parameterId.
characteristicContext	Union "ElectricalConnectionCharacteristicContextType": - Enum (see Table 214): ElectricalConnectionCharacteristicContextEnumType - EnumExtendType (see section 3.10.1.5)	Context, the characteristic is valid for.
characteristicType	Union "ElectricalConnectionCharacteristicTypeType": - Enum (see Table 215): ElectricalConnectionCharacteristicTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the characteristic.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	The value for this characteristic.
unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	The unit of this characteristic.

Table 213: electricalConnectionStateListData function detailed description of elements

Enumeration **ElectricalConnectionCharacteristicContextEnumType**:

Value	Description
device	The characteristic is valid for the Device where this information is located on, incl. all Entities.
entity	The characteristic is valid for the Entity where this information is located on.
inverter	The characteristic is valid for the inverter sub-Entities of the Entity where this information is located on.
pVString	The characteristic is valid for the PV string sub-Entities of the Entity where this information is located on.
battery	The characteristic is valid for the battery sub-Entities of the Entity where this information is located on.

Table 214: Enumeration *ElectricalConnectionCharacteristicContextEnumType*Enumeration **ElectricalConnectionCharacteristicTypeEnumType**:

Value	Description
powerConsumptionMin	The minimum (active) power the related functionality can consume.
powerConsumptionMax	The maximum (active) power the related functionality can consume.
powerConsumptionNominalMin	The nominal minimum (active) power the related functionality can consume, as specified in the manufacturer's data sheet.
powerConsumptionNominalMax	The nominal maximum (active) power the related functionality can consume, as specified in the manufacturer's data sheet.
powerProductionMin	The minimum (active) power the related functionality can produce.
powerProductionMax	The maximum (active) power the related functionality can produce.
powerProductionNominalMin	The nominal minimum (active) power the related functionality can produce, as specified in the manufacturer's data sheet.
powerProductionNominalMax	The nominal maximum (active) power the related functionality can produce, as specified in the manufacturer's data sheet.
energyCapacityNominalMax	The nominal capacity of the related functionality as specified in the manufacturer's data sheet.
contractualConsumptionNominalMax	The nominal maximum (active) power the related functionality is allowed to consume due to contractual limitations.
contractualProductionNominalMax	The nominal maximum (active) power the related functionality is allowed to produce due to contractual limitations.
apparentPowerProductionNominalMax	The nominal maximum apparent power the related functionality can produce, as specified in the manufacturer's data sheet.
apparentPowerConsumptionNominalMax	The nominal maximum apparent power the related functionality can consume, as specified in the manufacturer's data sheet.

Table 215: Enumeration *ElectricalConnectionCharacteristicTypeEnumType*

7102 **5.3.11.4.3 Available selectors**

- 7103 - electricalConnectionId
- 7104 - parameterId
- 7105 - characteristicId
- 7106 - characteristicContext
- 7107 - characteristicType

7108

7109 **5.3.11.4.4 Examples**7110 **5.3.11.4.4.1 Read-reply**

7111 The data of a specific electrical connection characteristic can be requested with this read command:

```

7112 <function>electricalConnectionCharacteristicListData</function>
7113 <filter>
7114     <cmdControl>
7115         <partial/>
7116     </cmdControl>
7117     <electricalConnectionCharacteristicListDataSelectors>
7118         <electricalConnectionId>1</electricalConnectionId>
7119         <parameterId>2</parameterId>
7120         <characteristicId>3</characteristicId>
7121     </electricalConnectionCharacteristicListDataSelectors>
7122 </filter>
7123 <electricalConnectionCharacteristicListData/>

```

7124 If the requested device has proper information, it will respond them with the following command:

```

7125 <function>electricalConnectionCharacteristicListData</function>
7126 <filter>
7127     <cmdControl>
7128         <partial/>
7129     </cmdControl>
7130 </filter>
7131 <electricalConnectionCharacteristicListData>
7132     <electricalConnectionCharacteristicData>
7133         <electricalConnectionId>1</electricalConnectionId>
7134         <parameterId>2</parameterId>
7135         <characteristicId>3</characteristicId>
7136         <characteristicContext>entity</characteristicContext>
7137         <characteristicType>powerConsumptionNominalMax</characteristicType>
7138         <value>
7139             <number>36</number>
7140             <scale>2</scale>
7141         </value>
7142         <unit>W</unit>
7143     </electricalConnectionCharacteristicData>
7144 </electricalConnectionCharacteristicListData>

```

7145

7146 **5.3.11.5 electricalConnectionStateListData**7147 **5.3.11.5.1 General**

7148 Some electrical connection specific state information is modelled in this function.

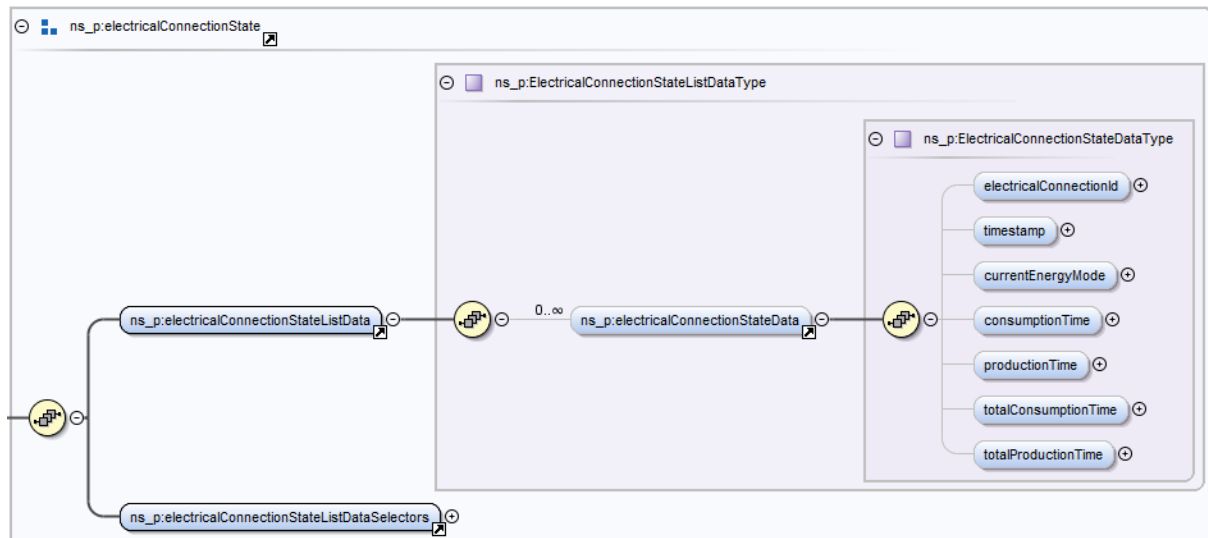


Figure 96: electricalConnectionStateListData function overview

5.3.11.5.2 Detailed description of elements

Element	Type	Description
electricalConnectionStateListData. electricalConnectionStateData. (list)		
electricalConnectionId	Identifier "ElectricalConnectionIdType" (see Table 358).	Electrical connection, this state information belongs to.
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	Point in time, the state information was generated.
currentEnergyMode	Common data type "EnergyModeType". See section 3.10.1.15.	Whether a device is consuming, producing or idle is stated by this element.
consumptionTime	xs:duration (W3C standard type)	The duration, the device is in energy mode "consume" since the last reboot (see section 5.3.9.2).
productionTime	xs:duration (W3C standard type)	The duration, the device is in energy mode "produce" since the last reboot (see section 5.3.9.2).
totalConsumptionTime	xs:duration (W3C standard type)	The duration, the device is in energy mode "consume" since its <i>installationTime</i> (see section 5.3.9.4).
totalProductionTime	xs:duration (W3C standard type)	The duration, the device is in energy mode "producing" since its <i>installationTime</i> (see section 5.3.9.4).

Table 216: electricalConnectionStateListData function detailed description of elements

5.3.11.5.3 Available selectors

- electricalConnectionId

7158 5.3.11.5.4 Examples

7159 5.3.11.5.4.1 Read-reply

7160 The state of a specific electrical connection can be requested with this read command:

```

7161 <function>electricalConnectionStateListData</function>
7162 <filter>
7163     <cmdControl>
7164         <partial/>
7165     </cmdControl>
7166     <electricalConnectionStateListDataSelectors>
7167         <electricalConnectionId>1</electricalConnectionId>
7168     </electricalConnectionStateListDataSelectors>
7169 </filter>
7170 <electricalConnectionStateListData/>

```

7171 If the requested device has proper information, it will respond them with the following command:

```

7172 <function>electricalConnectionStateListData</function>
7173 <filter>
7174     <cmdControl>
7175         <partial/>
7176     </cmdControl>
7177 </filter>
7178 <electricalConnectionStateListData>
7179     <electricalConnectionStateData>
7180         <electricalConnectionId>1</electricalConnectionId>
7181         <timestamp>2014-05-12T17:55:30.0Z</timestamp>
7182         <currentEnergyMode>consume</currentEnergyMode>
7183         <consumptionTime>PT2H10M15S</consumptionTime>
7184         <productionTime>PT0S</productionTime>
7185         <totalConsumptionTime>P1Y3M2DT21H14M57S</totalConsumptionTime>
7186         <totalProductionTime>PT0S</totalProductionTime>
7187     </electricalConnectionStateData>
7188 </electricalConnectionStateListData>

```

7189

7190 5.3.11.6 electricalConnectionDescriptionListData

7191 5.3.11.6.1 General

7192 Some electrical connection specific information, that are rather static, are modelled within this
7193 function.

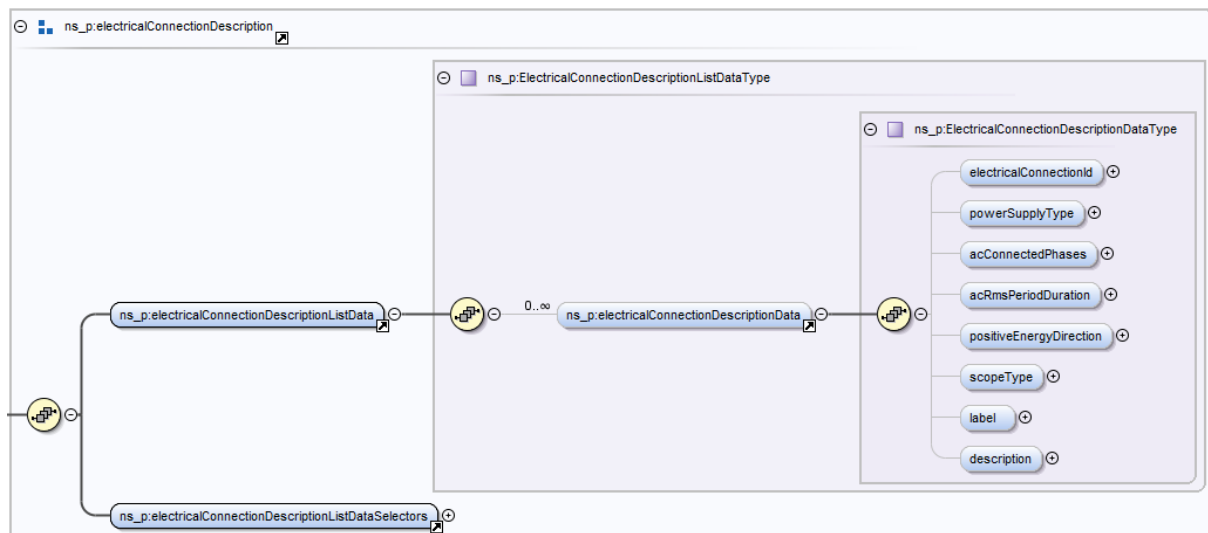


Figure 97: electricalConnectionStateDescriptionData function overview

5.3.11.6.2 Detailed description of elements

Element	Type	Description
electricalConnectionDescriptionListData. electricalConnectionDescriptionData. (list)		
electricalConnectionId	Identifier "ElectricalConnectionIdType" (see Table 358).	Electrical connection, this static information belongs to.
powerSupplyType	Union "ElectricalConnectionVoltageTypeType": - Enum (see Table 208): ElectricalConnectionVoltageTypeEnumType - EnumExtendType (see section 3.10.1.5)	States whether the electrical connection is of type "ac" or "dc".
acConnectedPhases	xs:unsignedInt (W3C standard type)	If powerSupplyType="ac", this element contains the number of phases, this electrical connection has (typically 1, 2 or 3).
acRmsPeriodDuration	xs:duration (W3C standard type)	If the "RMS" value of a parameter combination (see element <i>acMeasurementVariant</i> in section 5.3.11.2.2) is measured, this element states the duration, the RMS value is determined in.
positiveEnergyDirection	Common data type "EnergyDirectionType". See section 3.10.1.13.	This element states whether energy consumption or production will be counted as positive value.
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	A certain meaning of the electrical connection.
label	Common data type "LabelType". See section 3.10.1.2.	A label for this electrical connection.

description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A description for this electrical connection.
-------------	--	---

Table 217: *electricalConnectionDescriptionListData* function detailed description of elements

5.3.11.6.3 Available selectors

- `electricalConnectionId`
- `scopeType`

5.3.11.6.4 Examples

5.3.11.6.4.1 Read-reply

If one is interested in all electrical connection descriptions of a device, he could send the `read` command to the device, which will respond with something like the following command:

```
<electricalConnectionDescriptionListData>
  <electricalConnectionDescriptionData>
    <electricalConnectionId>1</electricalConnectionId>
    <powerSupplyType>ac</powerSupplyType>
    <acConnectedPhases>1</acConnectedPhases>
    <acRmsPeriodDuration>PT0.1S</acRmsPeriodDuration>
    <positiveEnergyDirection>consume</positiveEnergyDirection>
    <label>mains connection</label>
    <description>1-phase mains connection of the device</description>
  </electricalConnectionDescriptionData>
  <electricalConnectionDescriptionData>
    <electricalConnectionId>2</electricalConnectionId>
    <powerSupplyType>dc</powerSupplyType>
    <positiveEnergyDirection>consume</positiveEnergyDirection>
    <label>battery connection</label>
    <description>Backup battery connection of the device</description>
  </electricalConnectionDescriptionData>
</electricalConnectionDescriptionListData>
```

5.3.12 HVAC

5.3.12.1 Introduction

Heating, Ventilation and Air Conditioning (HVAC) devices have some special functionality which is covered by this class.

All system functions (like heating, cooling, domestic hot water preparation (dhw), ventilation, etc.) have their specific HVAC operation modes (like on, off, auto, etc.). Relations between these are modelled as well as relations to (optional) power sequences, which represent a concrete run of a device's program.

HVAC overruns (like an one time domestic-hot-water (DHW) preparation), which do have an impact on the normal operation mode of a HVAC system function, are modelled in further SPINE functions of the HVAC Class.

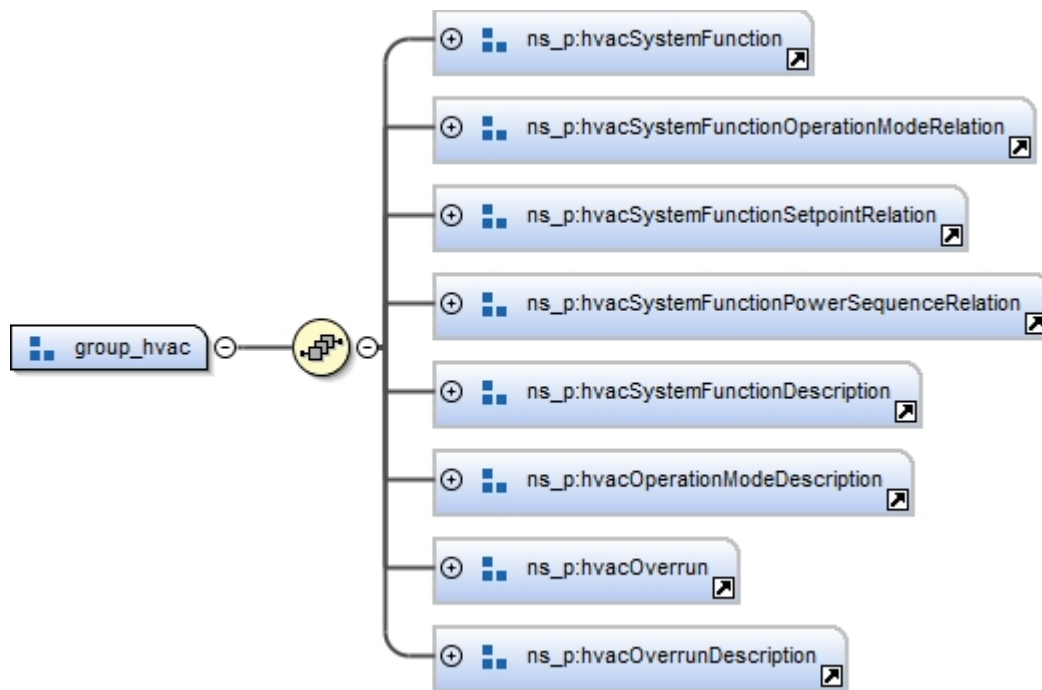


Figure 98: HVAC function-group overview

5.3.12.2 hvacSystemFunctionListData

5.3.12.2.1 General

The hvacSystemFunctionListData function announces the current status of the HVAC system functions, including the current HVAC operation mode, a flag that indicates, if the HVAC operation mode can be changed by some external communications partner (client) and a flag that indicates whether an HVAC overrun is currently active or not.

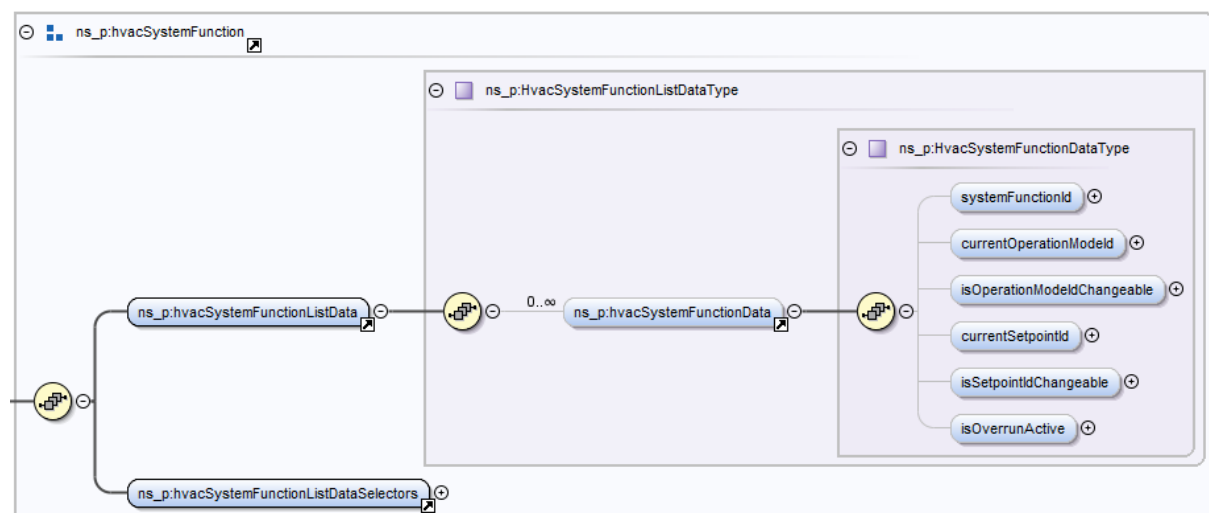


Figure 99: hvacSystemFunctionListData function overview

5.3.12.2.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

hvacSystemFunctionListData. hvacSystemFunctionData. (list)		
systemFunctionId	<i>Identifier</i> "HvacSystemFunctionIdType" (see Table 358).	Identifier for a specific HVAC system function.
currentOperationModeId	<i>Identifier</i> "HvacOperationModeIdType" (see Table 358).	States which HVAC operation mode is currently active for this specific HVAC system function
isOperationModeIdChangeable	xs:boolean (W3C standard type)	Denotes whether the current HVAC operation mode identifier is changeable (by some external communications partner) or not.
currentSetpointId	<i>Identifier</i> "SetpointIdType" (see Table 358).	States which setpoint is active for this specific HVAC system function
isSetpointIdChangeable	xs:boolean (W3C standard type)	Denotes whether the current setpoint identifier is changeable (by some external communications partner) or not.
isOverrunActive	xs:boolean (W3C standard type)	If an HVAC overrun is currently active, this element is set to <i>true</i> . The HVAC overrun itself is modelled with other functions (see section 5.3.12.8 and 5.3.12.9) and may be located on a different feature or even a different entity or device.

Table 218: hvacSystemFunctionListData function detailed description of elements

5.3.12.2.3 Available selectors

- systemFunctionId

5.3.12.2.4 Examples

5.3.12.2.4.1 Notify

When the *currentOperationMode* of a HVAC system function changes, the server sends a notify to all its subscribed clients (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..." just to improve the readability):

```

...
<hvacSystemFunctionListData>
  <hvacSystemFunctionData>
    <systemFunctionId>2</systemFunctionId>
    <currentOperationModeId>3</currentOperationModeId>
  </hvacSystemFunctionData>
</hvacSystemFunctionListData>

```

5.3.12.3 hvacSystemFunctionOperationModeRelationListData

5.3.12.3.1 General

Each HVAC system function has its own relation to all for this HVAC system function available HVAC operation modes. E.g. there may be three HVAC operation modes available on the feature: *on*, *off*, *auto* (see section 5.3.12.7). HVAC System function 1 only supports HVAC operation mode 1 (*on*) and 2 (*off*). HVAC System function 2 supports all three HVAC operation modes. The enumeration of the available HVAC system functions is described in section 5.3.12.6.2.

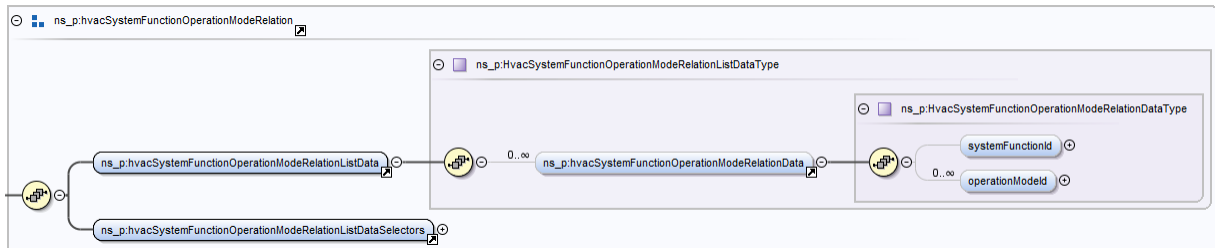


Figure 100: hvacSystemFunctionOperationModeRelationListData function overview

5.3.12.3.2 Detailed description of elements

Element	Type	Description
hvacSystemFunctionOperationModeRelationListData. hvacSystemFunctionOperationModeRelationData. (list)		
systemFunctionId	Identifier "HvacSystemFunctionIdType" (see Table 358).	Identifier for a specific HVAC system function.
operationModeId (list)	Identifier "HvacOperationModeIdType" (see Table 358).	Lists the HVAC operation modes that are available (selectable) for a specific HVAC system function.

Table 219: hvacSystemFunctionOperationModeRelationListData function detailed description of elements

5.3.12.3.3 Available selectors

- systemFunctionId

5.3.12.3.4 Examples

5.3.12.3.4.1 Read-reply

If one is interested in the supported HVAC operation modes of all HVAC system functions, he could send a standard read command to the server. The reply could look like this:

```
<hvacSystemFunctionOperationModeRelationListData>
  <hvacSystemFunctionOperationModeRelationData>
    <systemFunctionId>1</systemFunctionId>
    <operationModeId>1</operationModeId>
    <operationModeId>2</operationModeId>
    <operationModeId>3</operationModeId>
  </hvacSystemFunctionOperationModeRelationData>
</hvacSystemFunctionOperationModeRelationListData>
```

```

7297         <systemFunctionId>2</systemFunctionId>
7298         <operationModeId>1</operationModeId>
7299         <operationModeId>2</operationModeId>
7300     </hvacSystemFunctionOperationModeRelationData>
7301 </hvacSystemFunctionOperationModeRelationListData>

```

5.3.12.4 hvacSystemFunctionSetpointRelationListData

5.3.12.4.1 General

The setpoint class (see section 5.3.21) may be used to model some HVAC relevant setpoints like desired room temperature. An HVAC system function may relate to some of these setpoints. The relation is modelled with this SPINE function.

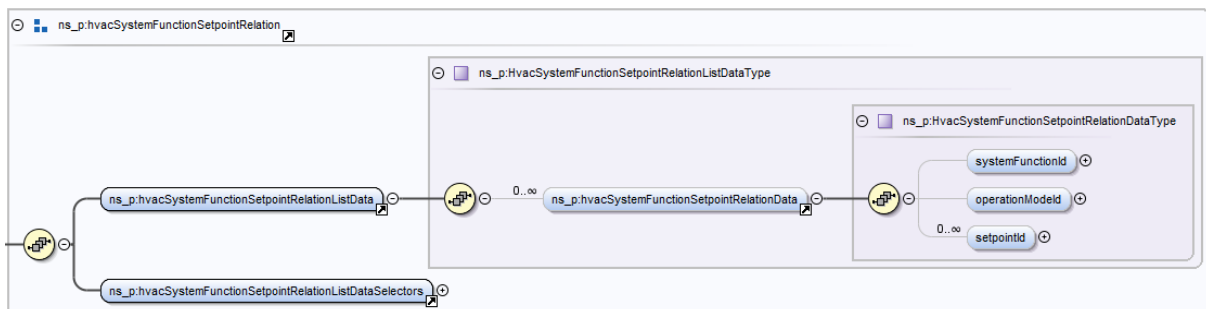


Figure 101: hvacSystemFunctionSetpointRelationListData function overview

5.3.12.4.2 Detailed description of elements

Element	Type	Description
hvacSystemFunctionSetpointRelationListData. hvacSystemFunctionSetpointRelationData. (list)		
systemFunctionId	Identifier "HvacSystemFunctionIdType" (see Table 358).	Identifier for a specific HVAC system function.
operationModeId	Identifier "HvacOperationModeIdType" (see Table 358).	Identifier for a specific HVAC operation mode.
setpointId (list)	Identifier "SetpointIdType" (see Table 358).	List of setpoints that are related with a specific HVAC system function.

Table 220: hvacSystemFunctionSetpointRelationListData function detailed description of elements

5.3.12.4.3 Available selectors

- systemFunctionId
- operationModeId

7318 5.3.12.4.4 Examples

7319 5.3.12.4.4.1 Read-reply

7320 If one is interested in the related setpoints of all HVAC system functions, he could send a standard
7321 read command to the server. The reply could look like this:

```

7322 <hvacSystemFunctionSetpointRelationListData>
7323   <hvacSystemFunctionSetpointRelationData>
7324     <systemFunctionId>1</systemFunctionId>
7325     <operationModeId>1</operationModeId>
7326     <setpointId>1</setpointId>
7327   </hvacSystemFunctionSetpointRelationData>
7328   <hvacSystemFunctionSetpointRelationData>
7329     <systemFunctionId>1</systemFunctionId>
7330     <operationModeId>2</operationModeId>
7331     <setpointId>2</setpointId>
7332     <setpointId>3</setpointId>
7333   </hvacSystemFunctionSetpointRelationData>
7334   <hvacSystemFunctionSetpointRelationData>
7335     <systemFunctionId>1</systemFunctionId>
7336     <operationModeId>3</operationModeId>
7337     <setpointId>4</setpointId>
7338   </hvacSystemFunctionSetpointRelationData>
7339 </hvacSystemFunctionSetpointRelationListData>

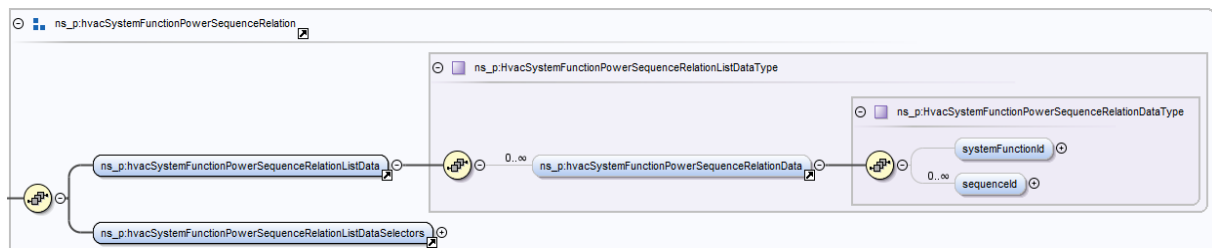
```

7340

7341 5.3.12.5 hvacSystemFunctionPowerSequenceRelationListData

7342 5.3.12.5.1 General

7343 Power sequences (see section 5.3.19) describe tasks of a functionality in detail. An HVAC system
7344 function may relate to some of these power sequences. The relation is modelled with this function.



7345

7346 Figure 102: hvacSystemFunctionPowerSequenceRelationListData function overview

7347

7348 5.3.12.5.2 Detailed description of elements

Element	Type	Description
hvacSystemFunctionPowerSequenceRelationListData. hvacSystemFunctionPowerSequenceRelationData. (list)		
systemFunctionId	Identifier "HvacSystemFunctionIdType" (see Table 358).	Identifier for a specific HVAC system function.
sequenceId (list)	Identifier "PowerSequenceIdType" (see Table 358).	List of power sequences that are related with a specific HVAC system function.

7349 Table 221: hvacSystemFunctionPowerSequenceRelationListData function detailed description of elements

7350

7351 **5.3.12.5.3 Available selectors**

7352 - systemFunctionId

7353

7354 **5.3.12.5.4 Examples**7355 **5.3.12.5.4.1 Read-reply**

7356 If one is interested in the related power sequences of all HVAC system functions, he could send a
 7357 standard read command to the server. The reply could look like this:

```

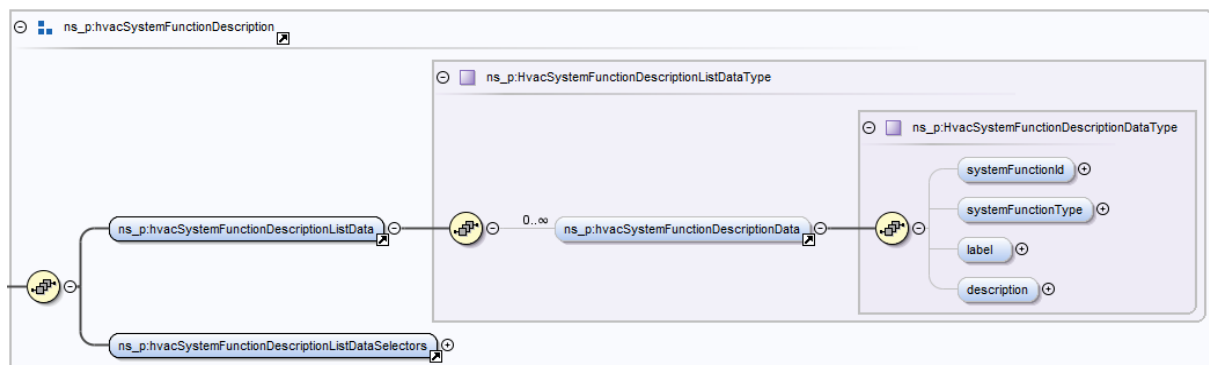
7358 <hvacSystemFunctionPowerSequenceRelationListData>
7359   <hvacSystemFunctionPowerSequenceRelationData>
7360     <systemFunctionId>1</systemFunctionId>
7361     <sequenceId>1</sequenceId>
7362     <sequenceId>2</sequenceId>
7363     <sequenceId>11</sequenceId>
7364   </hvacSystemFunctionPowerSequenceRelationData>
7365   <hvacSystemFunctionPowerSequenceRelationData>
7366     <systemFunctionId>2</systemFunctionId>
7367     <sequenceId>5</sequenceId>
7368   </hvacSystemFunctionPowerSequenceRelationData>
7369 </hvacSystemFunctionPowerSequenceRelationListData>

```

7370

7371 **5.3.12.6 hvacSystemFunctionDescriptionListData**7372 **5.3.12.6.1 General**

7373 The rather static information about the HVAC system functions are modelled within this SPINE
 7374 function.



7375

7376 *Figure 103: hvacSystemFunctionDescriptionListData function overview*

7377

7378 **5.3.12.6.2 Detailed description of elements**

Element	Type	Description
hvacSystemFunctionDescriptionListData. hvacSystemFunctionDescriptionData. (list)		
systemFunctionId	Identifier "HvacSystemFunctionIdType" (see Table 358).	Identifier for a specific HVAC system function.

systemFunctionType	<i>Union</i> <i>"HvacSystemFunctionTypeType":</i> <i>- Enum (see Table 223):</i> <i>HvacSystemFunctionTypeEnumType</i> <i>- EnumExtendType (see section 3.10.1.5)</i>	Denotes the type of the HVAC system function.
label	<i>Common data type "LabelType".</i> <i>See section 3.10.1.2.</i>	A short label of the HVAC system function.
description	<i>Common data type</i> <i>"DescriptionType". See section 3.10.1.3.</i>	Descriptive information on the HVAC system function.

Table 222: hvacSystemFunctionDescriptionListData function detailed description of elements

Enumeration **HvacSystemFunctionTypeEnumType**:

Value	Description
heating	HVAC system function for space heating.
cooling	HVAC system function for space cooling.
ventilation	HVAC system function for ventilation.
dhw	HVAC system function for domestic hot water preparation.

Table 223: Enumeration HvacSystemFunctionTypeEnumType

5.3.12.6.3 Available selectors

- systemFunctionId

5.3.12.6.4 Examples

5.3.12.6.4.1 Read-reply

If one is interested in the description of all HVAC system functions, he could send a standard read command to the server. The reply could look like this:

```

<hvacSystemFunctionDescriptionListData>
  <hvacSystemFunctionDescriptionData>
    <systemFunctionId>1</systemFunctionId>
    <systemFunctionType>heating</systemFunctionType>
    <label>Heating function</label>
  </hvacSystemFunctionDescriptionData>
  <hvacSystemFunctionDescriptionData>
    <systemFunctionId>2</systemFunctionId>
    <systemFunctionType>dhw</systemFunctionType>
    <label>Domestic hot water</label>
  </hvacSystemFunctionDescriptionData>
</hvacSystemFunctionDescriptionListData>

```

5.3.12.7 hvacOperationModeDescriptionListData

5.3.12.7.1 General

HVAC operation modes are used to describe which behaviour the HVAC system functions have. Each HVAC operation mode may be used by several HVAC system functions.

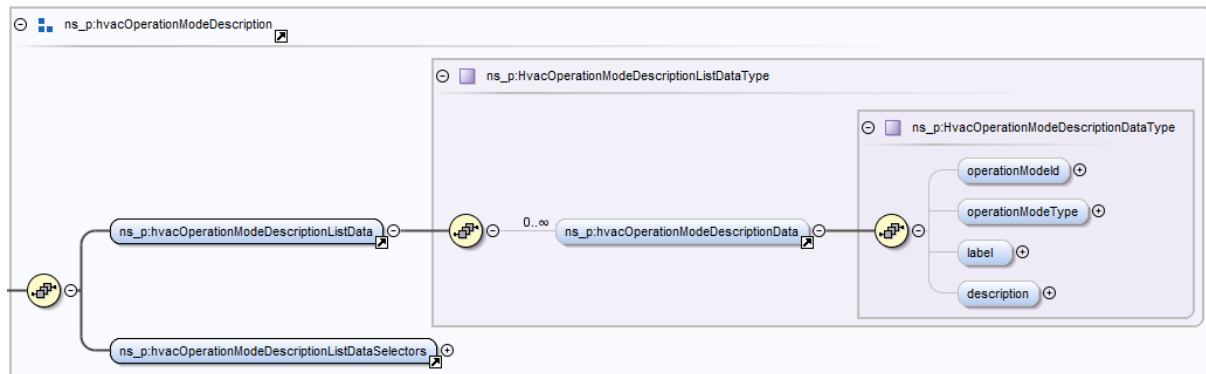


Figure 104: hvacOperationModeDescriptionListData function overview

5.3.12.7.2 Detailed description of elements

Element	Type	Description
hvacOperationModeDescriptionListData. hvacOperationModeDescriptionData. (list)		
operationModelId	Identifier "HvacOperationModelIdType" (see Table 358).	Identifier for the HVAC operation mode.
operationModeType	Union "HvacOperationModeTypeType": - Enum (see Table 225): HvacOperationModeTypeEnumType - EnumExtendType (see section 3.10.1.5)	Denotes the type of the HVAC operation mode.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the HVAC operation mode.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the HVAC operation mode.

Table 224: hvacOperationModeDescriptionListData function detailed description of elements

Enumeration HvacOperationModeTypeEnumType:

Value	Description	
auto	HVAC System Function	Meaning
	heating / cooling / dhw / ventilation	HVAC System function is controlled depending on a time table (i.e. the current setpoint for the HVAC system function is changed depending on a time table)
on	HVAC System Function	Meaning
	heating / cooling / dhw / ventilation	HVAC System function is controlled (e.g. Setpoint) independent from time tables (i.e. the current setpoint for the HVAC system function is not depending on any time table).
off	HVAC System Function	Meaning
	heating / dhw	Protection mode against freezing (no physically turning off). The (temperature) setpoint for freezing protection may be stated with a setpointId.
	cooling	Space cooling function not active.
	ventilation	Ventilation function not active (protection functions like humidity protection may be active).

eco	HVAC System Function	Meaning
	Heating / cooling / dhw / ventilation	HVAC System function is controlled (e.g. current Setpoint) independent from time tables (i.e. the current setpoint for the HVAC system function is not depending on any time table). HVAC System function is controlled by a different setpoint as for "on"

7413 *Table 225: Enumeration HvacOperationModeTypeEnumType*

7414

7415 5.3.12.7.3 Available selectors

7416 - operationModeId

7417

7418 5.3.12.7.4 Examples

7419 5.3.12.7.4.1 Read-reply

7420 If one is interested in the description of all HVAC operation modes, he could send a standard read
7421 command to the server. The reply could look like this:

```

7422 <hvacOperationModeDescriptionListData>
7423   <hvacOperationModeDescriptionData>
7424     <operationModeId>1</operationModeId>
7425     <operationModeType>on</operationModeType>
7426   </hvacOperationModeDescriptionData>
7427   <hvacOperationModeDescriptionData>
7428     <operationModeId>2</operationModeId>
7429     <operationModeType>off</operationModeType>
7430   </hvacOperationModeDescriptionData>
7431   <hvacOperationModeDescriptionData>
7432     <operationModeId>3</operationModeId>
7433     <operationModeType>auto</operationModeType>
7434   </hvacOperationModeDescriptionData>
7435 </hvacOperationModeDescriptionListData>

```

7436

7437 5.3.12.8 hvacOverrunListData

7438 5.3.12.8.1 General

7439 The normal HVAC operation may be overridden by some HVAC specific functionality. There are some
7440 HVAC overruns defined by the EEBus Initiative e.V., but vendors may add further vendor-specific
7441 HVAC overruns, too (which are then out of interoperability scope).

7442 When the HVAC overrun is active, some (not all! See section 5.3.12.9) HVAC system functions have a
7443 specific behaviour, other than the normal operation would be. After the HVAC overrun is inactive
7444 again, the HVAC system returns to its normal operation.

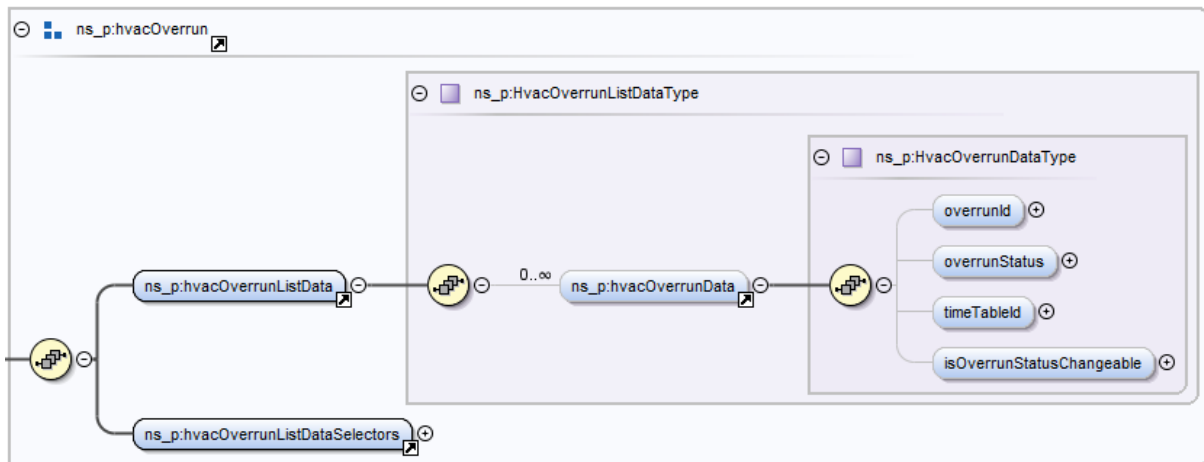


Figure 105: hvacOverrunListData function overview

5.3.12.8.2 Detailed description of elements

Element	Type	Description
hvacOverrunListData. hvacOverrunData. (list)		
overrunId	Identifier "HvacOverrunIdType" (see Table 358).	Identifier for the HVAC overrun.
overrunStatus	Union "HvacOverrunStatusType": - Enum (see Table 227): HvacOverrunStatusEnumType - EnumExtendType (see section 3.10.1.5)	The current status of the HVAC overrun.
timeTableId	Identifier "TimeTableIdType" (see Table 358).	Reference to a time table. Needed, if an overrun has some start and end point(s).
isOverrunStatusChangeable	xs:boolean (W3C standard type)	Whether the overrun status is changeable by a client or not is denoted in this element.

Table 226: hvacOverrunListData function detailed description of elements

Enumeration HvacOverrunStatusEnumType:

Value	Description
active	HVAC overrun triggered, but not started yet.
running	HVAC overrun running.
finished	HVAC overrun finished.
inactive	HVAC overrun inactive.

Table 227: Enumeration HvacOverrunStatusEnumType

5.3.12.8.3 Available selectors

- overrunId

5.3.12.8.4 Examples

5.3.12.8.4.1 Read-reply

If one is interested in the status of all HVAC overruns, he could send a standard read command to the server. The reply could look like this:

```
<hvacOverrunListData>
  <hvacOverrunData>
    <overrunId>1</overrunId>
    <overrunStatus>inactive</overrunStatus>
    <timeTableId>5</timeTableId>
  </hvacOverrunData>
  <hvacOverrunData>
    <overrunId>2</overrunId>
    <overrunStatus>active</overrunStatus>
  </hvacOverrunData>
</hvacOverrunListData>
```

5.3.12.9 hvacOverrunDescriptionListData

5.3.12.9.1 General

Each HVAC overrun is defined by its description part. Here, the HVAC overrun type is specified as well as the list of HVAC system functions that are affected by the HVAC overrun (if this information is available).

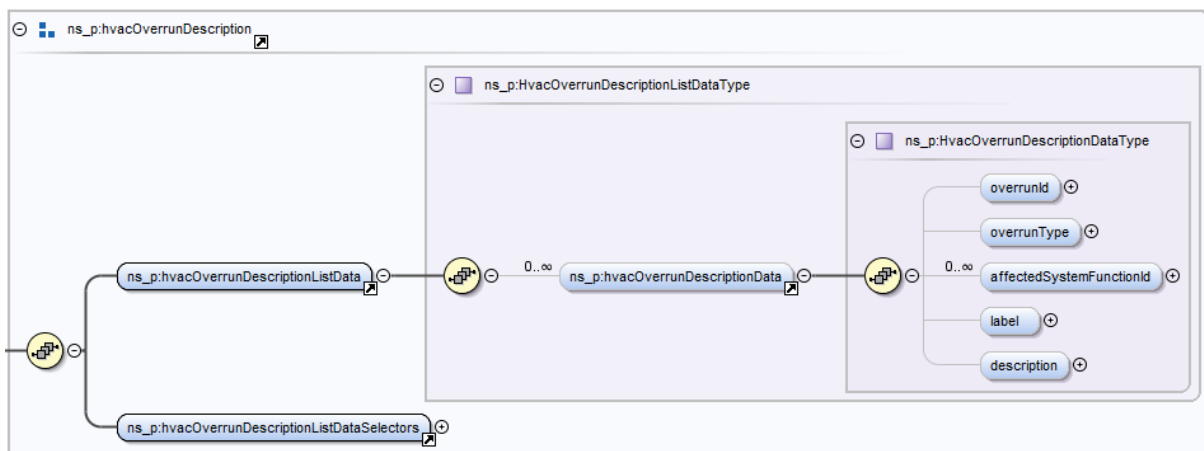


Figure 106: hvacOverrunDescriptionListData function overview

5.3.12.9.2 Detailed description of elements

Element	Type	Description
hvacOverrunDescriptionListData. hvacOverrunDescriptionData. (list)		
overrunId	Identifier "HvacOverrunIdType" (see Table 358).	Identifier for the HVAC overrun.
overrunType	Union "HvacOverrunTypeType": - Enum (see Table 229): HvacOverrunTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the overrun.

affectedSystemFunctionId (list)	Identifier "HvacSystemFunctionIdType" (see Table 358).	Identifiers of all HVAC system functions that are affected by this HVAC overrun.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the HVAC overrun.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the HVAC overrun.

7481 Table 228: hvacOverrunDescriptionListData function detailed description of elements

7482 Enumeration **HvacOverrunTypeEnumType**:

Value	Description
oneTimeDhw	A one-time domestic hot water loading (independend from current HVAC operation mode or time table).
party	During a party, the HVAC system will stay in the normal day-operation mode and not turn off at the usual time. Example: HVAC system function "heating" is controlled by the day setpoint, independend from operation mode and time table, until HVAC overrun "party" is finished.
sgReadyCondition1	The SGReady condition 1 will be applied. Please consider [SGReady].
sgReadyCondition3	The SGReady condition 3 will be applied. Please consider [SGReady].
sgReadyCondition4	The SGReady condition 4 will be applied. Please consider [SGReady].
oneDayAway	Overrides the normal daily routine to a out-of-home configuration (e.g. eco-mode the whole day).
oneDayAtHome	Overrides the normal daily routine to an at-home configuration (e.g. on-mode the whole day).
oneTimeVentilation	Activates the ventilation once, independent from the configured routine.
hvacSystemOff	Deactivates the HVAC system (frost protection is still active and the system is in a network-standby mode, in which it is reachable for re-configuration).

7483 Table 229: Enumeration HvacOverrunTypeEnumType

7484

7485 5.3.12.9.3 Available selectors

7486 - overrunId

7487

7488 5.3.12.9.4 Examples

7489 5.3.12.9.4.1 Read-reply

7490 If one is interested in the description of all HVAC overruns, he could send a standard read command
7491 to the server. The reply could look like this:

```

7492 <hvacOverrunDescriptionListData>
7493   <hvacOverrunDescriptionData>
7494     <overrunId>1</overrunId>
7495     <overrunType>party</overrunType>
7496     <affectedSystemFunctionId>1</affectedSystemFunctionId>
7497     <affectedSystemFunctionId>2</affectedSystemFunctionId>
7498     <label>label0</label>
7499     <description>description0</description>
7500   </hvacOverrunDescriptionData>
7501   <hvacOverrunDescriptionData>
```

```

7502         <overrunId>2</overrunId>
7503         <overrunType>oneTimeDhw</overrunType>
7504         <affectedSystemFunctionId>2</affectedSystemFunctionId>
7505         <label>label1</label>
7506         <description>description1</description>
7507     </hvacOverrunDescriptionData>
7508 </hvacOverrunDescriptionListData>

```

5.3.13 Identification

5.3.13.1 Introduction

The Identification Class can be used to express certain identification values of the parent Entity. Additionally, information about sessions can be provided.

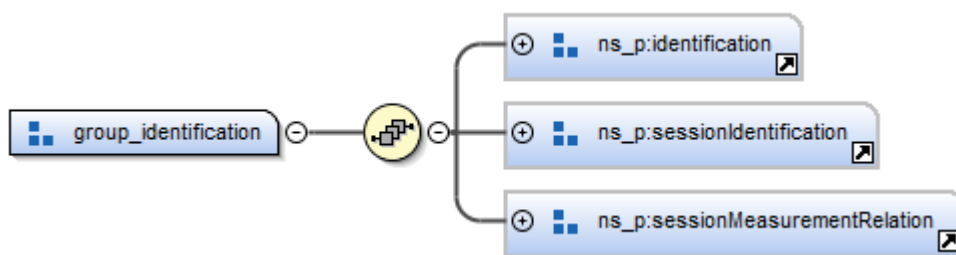


Figure 107: Identification function-group overview

5.3.13.2 identificationListData

5.3.13.2.1 General

The identificationListData function allows to express certain identification values of the corresponding Entity (the parent Entity of the Identification Feature).

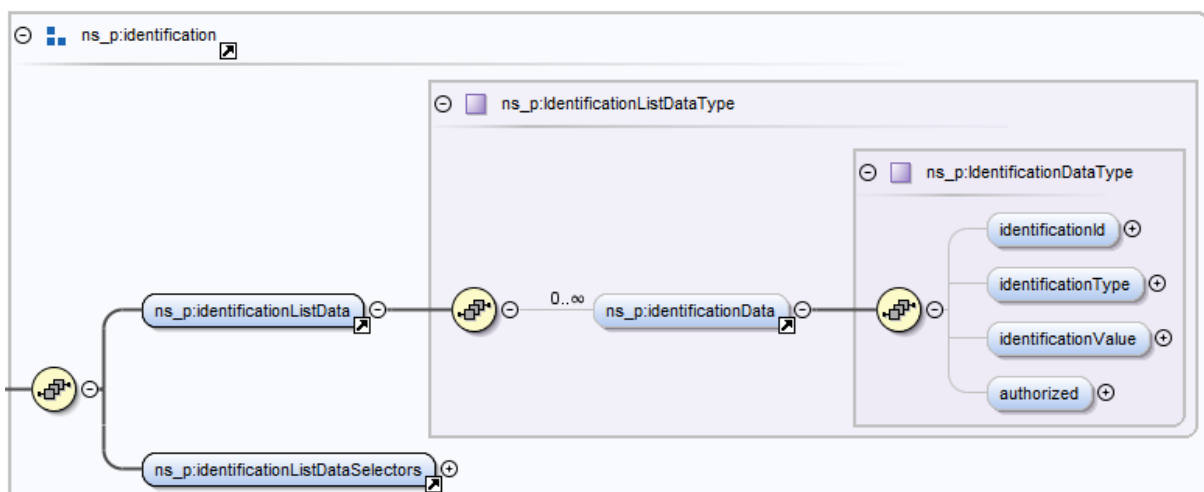


Figure 108: identificationListData function overview

5.3.13.2.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

identificationListData. identificationData. (list)		
identificationId	Identifier "IdentificationIdType" (see Table 358).	The identificationId is used to identify the different list entries within IdentificationListData.
identificationType	Union "IdentificationTypeType": - Enum (see Table 231): IdentificationTypeEnumType - EnumExtendType (see section 3.10.1.5)	The identificationType allows to describe a certain type of identification, e.g. a MAC address or an RFID tag.
identificationValue	Simple type "IdentificationValueType" (restriction of "xs:string")	This holds the value of the identification and therefore is the essential element of IdentificationData.
authorized	xs:boolean	The "authorized" element can be used to directly authorize an identification value with a write command.

7525 Table 230: identificationListData function detailed description of elements

7526 Enumeration **IdentificationTypeEnumType**:

Value	Description
eui48	eui48 MAC address
eui64	eui64 MAC address
userRfidTag	RFID tag used for identification.

7527 Table 231: Enumeration IdentificationTypeEnumType

7528

7529 5.3.13.2.3 Available selectors

7530 - identificationId

7531

7532 5.3.13.2.4 Examples

7533 5.3.13.2.4.1 Notify

7534 If the content of a list entry's elements changes, the information is sent to all subscribed resources:

```

7535 <identificationListData>
7536   <identificationData>
7537     <identificationId>1</identificationId>
7538     <identificationType>eui64</identificationType>
7539     <identificationValue>12-AB-34-CD-56-EF-78-90</identificationValue>
7540     <authorized>true</authorized>
7541   </identificationData>
7542 </identificationListData>

```

7543

5.3.13.3 sessionIdentificationListData

5.3.13.3.1 General

The sessionIdentificationListData function is used to model information about so-called sessions (e.g., an EV's charging session or a washing machine's washing cycle). It can be related to measurands that are measured during a session (see section 5.3.13.4).

A session may be a current one (it is still ongoing and has only a start time, but not an end time) or a past one (the session is completed and it has a start time as well as an end time).

An appliance may provide only the latest session (which can be a current one or the latest past session) or store previous sessions as well.

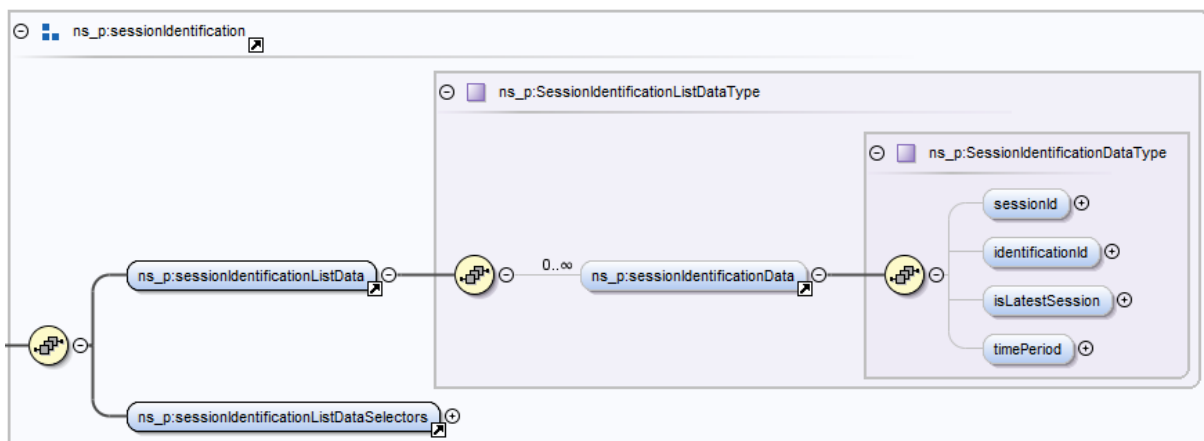


Figure 109: sessionIdentificationListData function overview

5.3.13.3.2 Detailed description of elements

Element	Type	Description
sessionIdentificationListData. sessionIdentificationData. (list)		
sessionId	Identifier "SessionIdType" (see Table 358).	Unique identifier for each session.
identificationId	Identifier "IdentificationIdType" (see Table 358).	Reference to an appliance's identification information.
isLatestSession	xs:boolean	States if this is the latest session. The latest session is either the current session, or (if no current session is available) the past session with the newest endTime.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	Includes the startTime of the session and, if it is a past session, the endTime of the session.

Table 232: sessionIdentificationListData function detailed description of elements

5.3.13.3.3 Available selectors

- sessionId
- identificationId

7562 - isLatestSession
7563 - timePeriod

7564

7565 5.3.13.3.4 Examples

7566 5.3.13.3.4.1 Notify

7567 If the content of a list entry's elements changes, the information is sent to all subscribed resources:

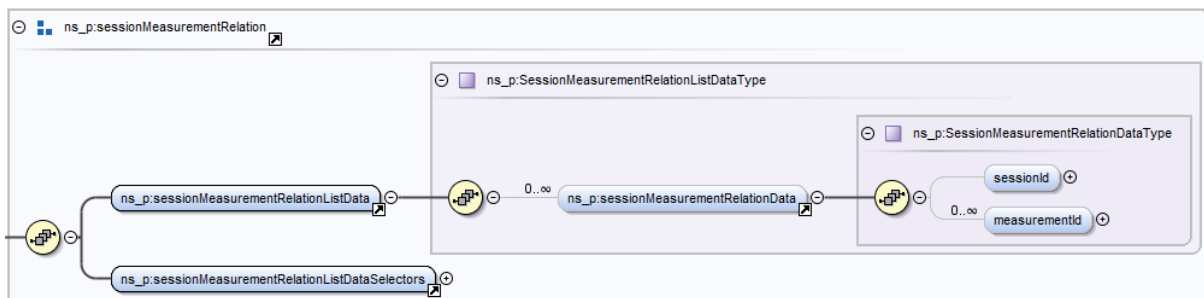
```
7568 <sessionIdentificationListData>
7569   <sessionIdentificationData>
7570     <sessionId>1</sessionId>
7571     <identificationId>2</identificationId>
7572     <isLatestSession>true</isLatestSession>
7573     <timePeriod>
7574       <startTime>2021-08-23T18:12:43.5Z</startTime>
7575       <endTime>2021-08-23T21:25:13.6Z</endTime>
7576     </timePeriod>
7577   </sessionIdentificationData>
7578 </sessionIdentificationListData>
```

7579

7580 5.3.13.4 sessionMeasurementRelationListData

7581 5.3.13.4.1 General

7582 The sessionMeasurementRelationListData function provides the relation between the session(s) and
7583 the measurands that are related to the respective session (e.g., periodically stored power values or
7584 the consumed energy during the session).



7585

7586 Figure 110: sessionMeasurementRelationListData function overview

7587

7588 5.3.13.4.2 Detailed description of elements

Element	Type	Description
sessionMeasurementRelationListData. sessionMeasurementRelationData. (list)		
sessionId	Identifier "SessionIdType" (see Table 358).	Unique identifier for each session.
measurementId (list)	Identifier "MeasurementIdType" (see Table 358).	List of linked measurementIds for that session.

7589 Table 233: sessionMeasurementRelationListData function detailed description of elements

7590

7591 5.3.13.4.3 Available selectors

- 7592 - sessionId
- 7593 - measurementId

7594

7595 5.3.13.4.4 Examples

7596 5.3.13.4.4.1 Notify

7597 If the content of a list entry's elements changes, the information is sent to all subscribed resources:

```

7598 <sessionMeasurementRelationListData>
7599   <sessionMeasurementRelationData>
7600     <sessionId>1</sessionId>
7601     <measurementId>3</measurementId>
7602     <measurementId>5</measurementId>
7603   </sessionMeasurementRelationData>
7604 </sessionMeasurementRelationListData>

```

7605

7606 5.3.14 LoadControl

7607 5.3.14.1 Introduction

7608 The LoadControl class has two different approaches to control the load of an appliance:

- 7609 - Event-based load control with "loadControlEvent" (see section 5.3.14.3) and
- 7610 "loadControlState" (see section 5.3.14.4)
- 7611 - Limit-based load control with "loadControlLimit" (see section 5.3.14.5) and
- 7612 "loadControlLimitDescription" (see section 5.3.14.7)

7613 The event-based load control is used to change the energy demand (load) or production of a
 7614 household by some external contract partner or energy manager. The household or device (owner)
 7615 may reject the change commands, but then may have to pay more to the external partner or receive
 7616 less reward. The concept defines two function-groups. One (*loadControlEvent*) that keeps the
 7617 received change commands (a household or device may store more than one command, which may
 7618 be active in different time periods; these commands are requestable by some other client side) and
 7619 another one (*loadControlState*) for the status of these commands (where more than one could be
 7620 requestable, too). Possible commands are: pausing / resuming the demand or production, decreasing
 7621 / increasing it, an emergency mode (where the electricity grid or local house power supply is in a
 7622 critical condition and only an immediate substantial reduction of the energy consumption or feeding
 7623 of energy into the electricity grid can prevent a blackout situation) and the "back-to-normal"
 7624 command.

7625 The limit-based load control is used where limits (changeable by a client) shall affect the behaviour of
 7626 a node automatically. E.g. the power demand of an appliance could be limited to a certain value to
 7627 protect the household of an overload.

7628 Additionally with the information given in loadControlNode (see section 5.3.14.2), a node can define
 7629 whether it is (at the moment) remote controllable or not.

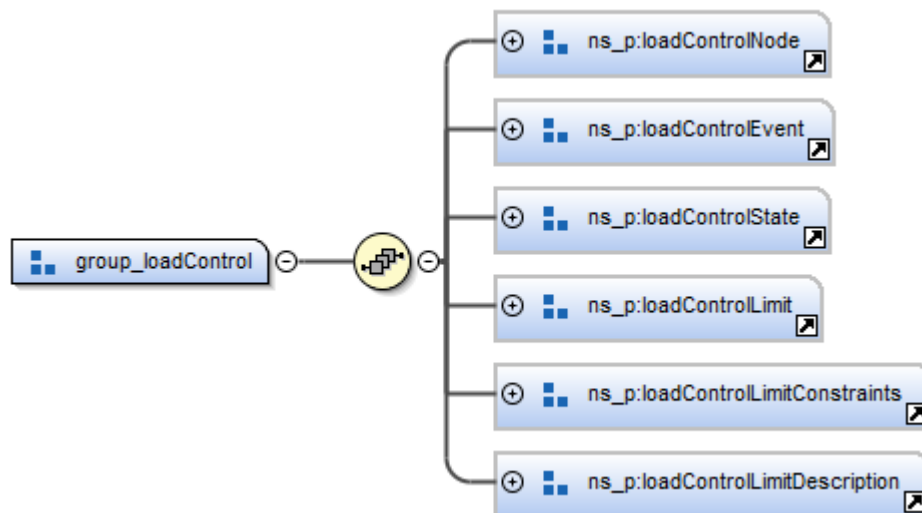


Figure 111: LoadControl function-group overview

5.3.14.2 loadControlNodeData

5.3.14.2.1 General

The loadControlNodeData includes node-wide information related to load control (e.g. whether the node is remote controllable).

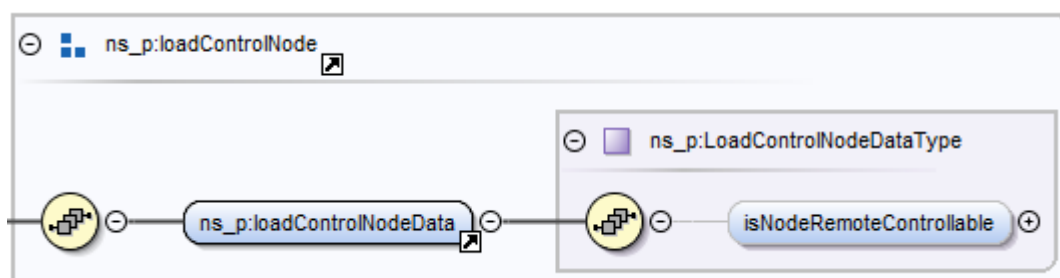


Figure 112: loadControlNodeData function overview

5.3.14.2.2 Detailed description of elements

Element	Type	Description
loadControlNodeData.		
isNodeRemoteControllable	xs:boolean (W3C standard type)	Defines whether the node is (at the moment) remote controllable by a client, or not.

Table 234: loadControlNodeData function detailed description of elements

5.3.14.2.3 Available selectors

None.

7646 5.3.14.2.4 Examples

7647 5.3.14.2.4.1 Notify

7648 If the content of the element `isNodeRemoteControllable` changes (e.g. due to a configuration change
7649 by the user), the information is sent to all subscribed resources:

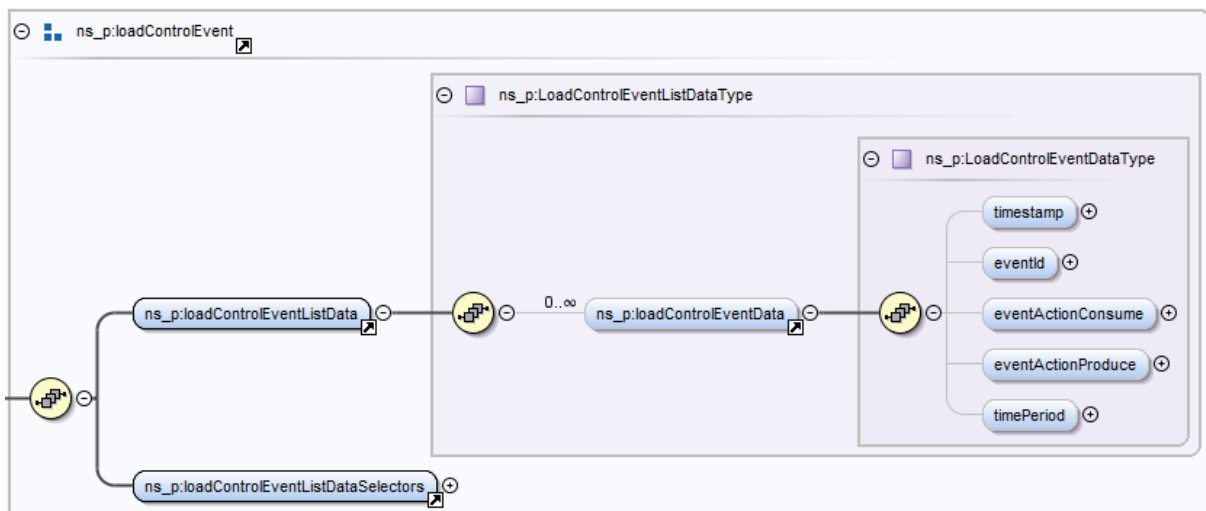
```
7650 <loadControlNodeData>
7651   <isNodeRemoteControllable>true</isNodeRemoteControllable>
7652 </loadControlNodeData>
```

7653

7654 5.3.14.3 loadControlEventListData

7655 5.3.14.3.1 General

7656 With the `loadControlEventListData` function, a household or device can show the commands (events)
7657 it received to change its load demand or energy production (or both). These commands contain
7658 rather simple instructions like "reduce", "increase", "pause", "resume", etc. This function can also be
7659 used to model a single command from another point (external contract partner or energy manager,
7660 e.g.) to the household or device (this could be a "write" operation to add a new
7661 "loadControlEventData" to the `loadControlEventListData` instance, e.g.).



7662

7663 Figure 113: loadControlEventListData function overview

7664

7665 5.3.14.3.2 Detailed description of elements

Element	Type	Description
loadControlEventListData. loadControlEventData. (list)		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time when the message was generated.
eventId	Identifier "LoadControlEventIdType" (see Table 358).	An incrementing number, enabling the linking of a <i>loadControlState</i> command to a <i>loadControlEvent</i> command.

eventActionConsume	<i>Union</i> "LoadControlEventActionType": - Enum (see Table 236): LoadControlEventActionEnumType - EnumExtendType (see section 3.10.1.5)	The action for consuming that shall be applied by the receiving device (feature server).
eventActionProduce	<i>Union</i> "LoadControlEventActionType": - Enum (see Table 236): LoadControlEventActionEnumType - EnumExtendType (see section 3.10.1.5)	The action for producing that shall be applied by the receiving device (feature server).
timePeriod	<i>Common data type</i> "TimePeriodType". See section 3.10.2.1.	The period of time, the event shall be active (e.g. from 12:00 to 14:00 or with relative times, from 1 hour in the future to 3 hours in the future).

7666 Table 235: loadControlEventListData function detailed description of elements

7667 Enumeration **LoadControlEventActionEnumType**:

Value	Description
pause	Demanding the pausing of consumption / production.
resume	Demanding the resumption of consumption / production.
reduce	Demanding the reduction of consumption / production.
increase	Demanding the increasing of consumption / production.
emergency	Demanding the reduction of consumption to the absolutely minimum level that is possible without risking anything. Not to be used for production.
normal	Demanding the return to the normal operation.

7668 Table 236: Enumeration LoadControlEventActionEnumType

7669

7670 5.3.14.3.3 Available selectors

7671 - timestampInterval

7672 - eventId

7673

7674 5.3.14.3.4 Examples

7675 5.3.14.3.4.1 Write (partial)

7676 If an energy provider wants a household to reduce its consumption and increase its production for a
 7677 certain period of time, it could send the following command to the CEM:

```

7678 <function>loadControlEventListData</function>
7679 <filter>
7680   <cmdControl>
7681     <partial/>
7682   </cmdControl>
7683 </filter>
7684 <loadControlEventListData>
7685   <loadControlEventData>
7686     <timestamp>2015-10-15T18:40:23.2Z</timestamp>
7687     <eventId>1</eventId>
7688     <eventActionConsume>reduce</eventActionConsume>

```

```

7689         <eventActionProduce>increase</eventActionProduce>
7690         <timePeriod>
7691             <startTime>PT1H10M</startTime>
7692             <endTime>PT3H15M</endTime>
7693         </timePeriod>
7694     </loadControlEventData>
7695 </loadControlEventListData>

```

5.3.14.4 loadControlStateListData

5.3.14.4.1 General

A device that supports being controlled by LoadControl mechanisms (see function "loadControlEventListData") should express the states it currently assigns to each event.

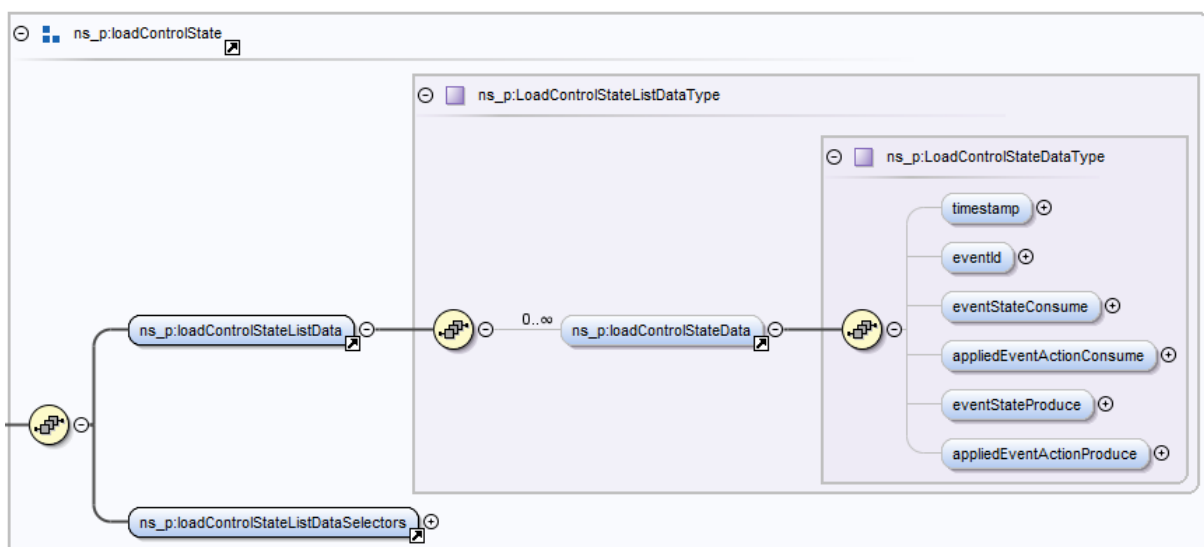


Figure 114: loadControlStateListData function overview

5.3.14.4.2 Detailed description of elements

Element	Type	Description
loadControlStateListData. loadControlStateData. (list)		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time when the message was generated.
eventId	Identifier "LoadControlEventIdType" (see Table 358).	An incrementing number, enabling the linking of a loadControlState command to a loadControlEvent command.
eventStateConsume	Union "LoadControlEventStateType": - Enum (see Table 238): LoadControlEventStateEnumType - EnumExtendType (see section 3.10.1.5)	Current state for consumption of the event.

appliedEventActionConsume	<i>Union</i> "LoadControlEventActionType": - Enum (see Table 236): LoadControlEventActionEnumType - EnumExtendType (see section 3.10.1.5)	The event action for consumption, actually applied by the server.
eventStateProduce	<i>Union</i> "LoadControlEventStateType": - Enum (see Table 238): LoadControlEventStateEnumType - EnumExtendType (see section 3.10.1.5)	Current state for production of the event.
appliedEventActionProduce	<i>Union</i> "LoadControlEventActionType": - Enum (see Table 236): LoadControlEventActionEnumType - EnumExtendType (see section 3.10.1.5)	The event action for production, actually applied by the server.

7705 Table 237: loadControlStateListData function detailed description of elements

7706 Enumeration **LoadControlEventStateEnumType**:

Value	Description
eventAccepted	The server accepted the event command. It has not started yet!
eventStarted	The server started the event, meaning that, e.g., the server has started to reduce its consumption or is increasing its production.
eventStopped	The server stopped the event due to reaching the end time, of the event (stated in the <i>timePeriod</i> element of the <i>loadControlEvent</i> command).
eventRejected	The server rejected the command due to some reason (that is not stated here). It depends on the contract of the household owner how often he may reject a <i>loadControlEvent</i> command.
eventCancelled	The server cancelled the event due to some reason (that is not stated here).
eventError	An error occurred during executing the event.

7707 Table 238: Enumeration LoadControlEventStateEnumType

7708

7709 5.3.14.4.3 Available selectors

7710 - timestampInterval

7711 - eventId

7712

7713 5.3.14.4.4 Examples

7714 5.3.14.4.4.1 Notify

7715 As a result of the load-reducing-command example from section 5.3.14.3.4.1, the CEM could inform
7716 the DSO with the following (complete) notification:

```

7717 <loadControlStateListData>
7718   <loadControlStateData>
7719     <timestamp>2015-10-15T18:40:23.9Z</timestamp>
7720     <eventId>1</eventId>
7721     <eventStateConsume>eventAccepted</eventStateConsume>

```



```

7722         <appliedEventActionConsume>reduce</appliedEventActionConsume>
7723         <eventStateProduce>eventAccepted</eventStateProduce>
7724         <appliedEventActionProduce>increase</appliedEventActionProduce>
7725     </loadControlStateData>
7726 </loadControlStateListData>

```

7727 Following this example, exact one hour and ten minutes later, the CEM would send the following
7728 (complete) notification to the DSO:

```

7729 <loadControlStateListData>
7730     <loadControlStateData>
7731         <timestamp>2015-10-15T19:50:23.9Z</timestamp>
7732         <eventId>1</eventId>
7733         <eventStateConsume>eventStarted</eventStateConsume>
7734         <appliedEventActionConsume>reduce</appliedEventActionConsume>
7735         <eventStateProduce>eventStarted</eventStateProduce>
7736         <appliedEventActionProduce>increase</appliedEventActionProduce>
7737     </loadControlStateData>
7738 </loadControlStateListData>

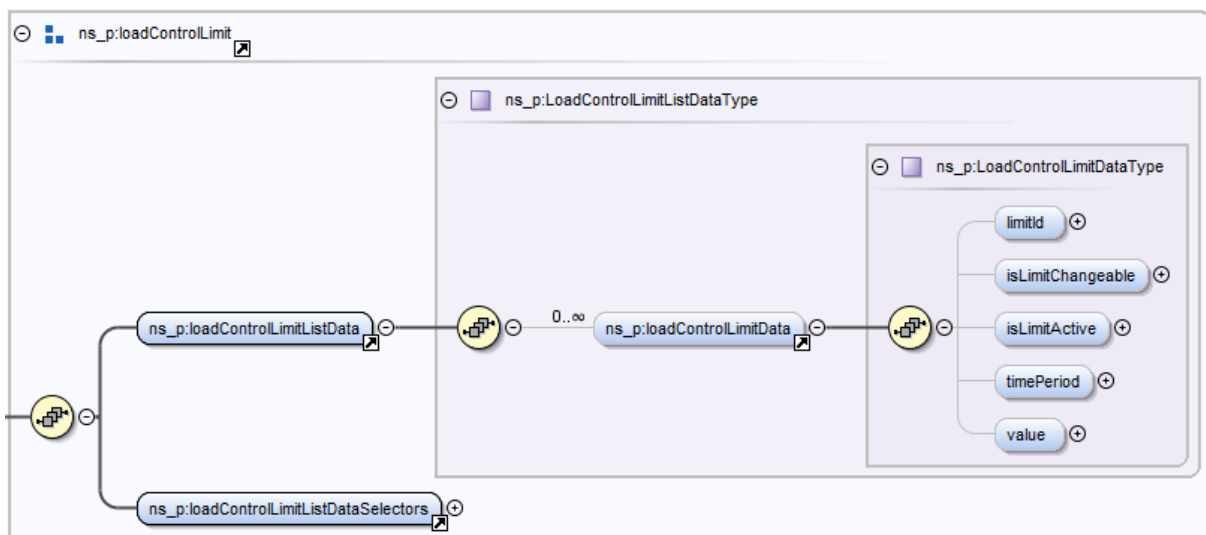
```

7739

7740 5.3.14.5 loadControlLimitListData

7741 5.3.14.5.1 General

7742 The loadControlLimitListData function provides the value of the limit (and an optional time period for
7743 that limit) and further information whether the limit is active or not and if it is changeable by a client
7744 or not.



7745

7746 Figure 115: loadControlLimitListData function overview

7747

7748 5.3.14.5.2 Detailed description of elements

Element	Type	Description
loadControlLimitListData. loadControlLimitData. (list)		
limitId	Identifier "LoadControlLimitIdType" (see Table 358).	Identifier for the limit.
isLimitChangeable	xs:boolean (W3C standard type)	States whether the limit may be changed by a client or not.

isLimitActive	xs:boolean (W3C standard type)	Indicates whether the limit is currently active or not.
timePeriod	<i>Common data type "TimePeriodType". See section 3.10.2.1.</i>	The period where the limit shall be active.
value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The actual limit.

7749 Table 239: loadControlLimitListData function detailed description of elements

7750

7751 5.3.14.5.3 Available selectors

7752 - limitId

7753

7754 5.3.14.5.4 Examples

7755 5.3.14.5.4.1 Read-reply

7756 If one is interested in the current value of a specific limit, he could send the following read command:

```

7757 <function>loadControlLimitListData</function>
7758 <filter>
7759   <cmdControl>
7760     <partial/>
7761   </cmdControl>
7762   <loadControlLimitListDataSelectors>
7763     <limitId>2</limitId>
7764   </loadControlLimitListDataSelectors>
7765 </filter>
7766 <loadControlLimitListData/>

```

7767 If the receiving device has a proper value for this *keyId*, it would respond with the following function:

```

7768 <loadControlLimitListData>
7769   <loadControlLimitData>
7770     <limitId>2</limitId>
7771     <isLimitActive>true</isLimitActive>
7772     <isLimitChangeable>true</isLimitChangeable>
7773     <timePeriod>
7774       <startTime>-PT13M30S</startTime>
7775       <endTime>PT16M30S</endTime>
7776     </timePeriod>
7777     <value>
7778       <number>15</number>
7779       <scale>1</scale>
7780     </value>
7781   </loadControlLimitData>
7782 </loadControlLimitListData>

```

7783

7784 5.3.14.5.4.2 Write (partial)

7785 If an client wants to limit the consumption of an appliance to a certain value that shall begin in 15
7786 minutes and end in 2 hours, it could send the following command to the load control resource:

```

7787 <function>loadControlLimitListData</function>
7788 <filter>

```

```

7789     <cmdControl>
7790         <partial/>
7791     </cmdControl>
7792 </filter>
7793 <loadControlLimitListData>
7794     <loadControlLimitData>
7795         <limitId>2</limitId>
7796         <timePeriod>
7797             <startTime>PT15M</startTime>
7798             <endTime>PT2H</endTime>
7799         </timePeriod>
7800         <value>
7801             <number>1</number>
7802             <scale>2</scale>
7803         </value>
7804     </loadControlLimitData>
7805 </loadControlLimitListData>

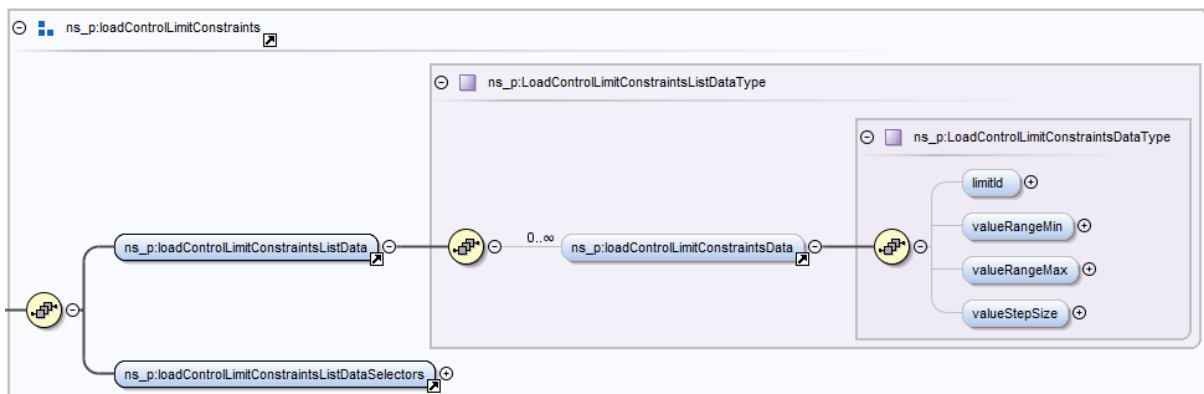
```

7806

7807 5.3.14.6 loadControlLimitConstraintsListData

7808 5.3.14.6.1 General

7809 Constraints that shall be held when trying to change a limit, are modelled with the
 7810 loadControlLimitConstraintsListData function, including a minimum value, a maximum value and the
 7811 stepsize of the values between.



7812

7813 Figure 116: loadControlLimitConstraintsListData function overview

7814

7815 5.3.14.6.2 Detailed description of elements

Element	Type	Description
loadControlLimitConstraintsListData. loadControlLimitConstraintsData. (list)		
limitId	Identifier "LoadControlLimitIdType" (see Table 358).	Identifier for the limit.
valueRangeMin	Common data type "ScaledNumberType". See section 3.10.1.8.	This element allows to define a minimum limit for the element value within loadControlLimitListData.
valueRangeMax	Common data type "ScaledNumberType". See section 3.10.1.8.	This element allows to define a maximum limit for the element value within loadControlLimitListData.

valueStepSize	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum step size between two different limit values.
---------------	---	---

Table 240: loadControlLimitConstraintsListData function detailed description of elements

5.3.14.6.3 Available selectors

- limitId

5.3.14.6.4 Examples

5.3.14.6.4.1 Read-reply

If one is interested in the value constraints of a specific limit, he could send the following read command:

```
<function>loadControlLimitConstraintsListData</function>
<filter>
  <cmdControl>
    <partial/>
  </cmdControl>
  <loadControlLimitConstraintsListDataSelectors>
    <limitId>2</limitId>
  </loadControlLimitConstraintsListDataSelectors>
</filter>
<loadControlLimitConstraintsListData/>
```

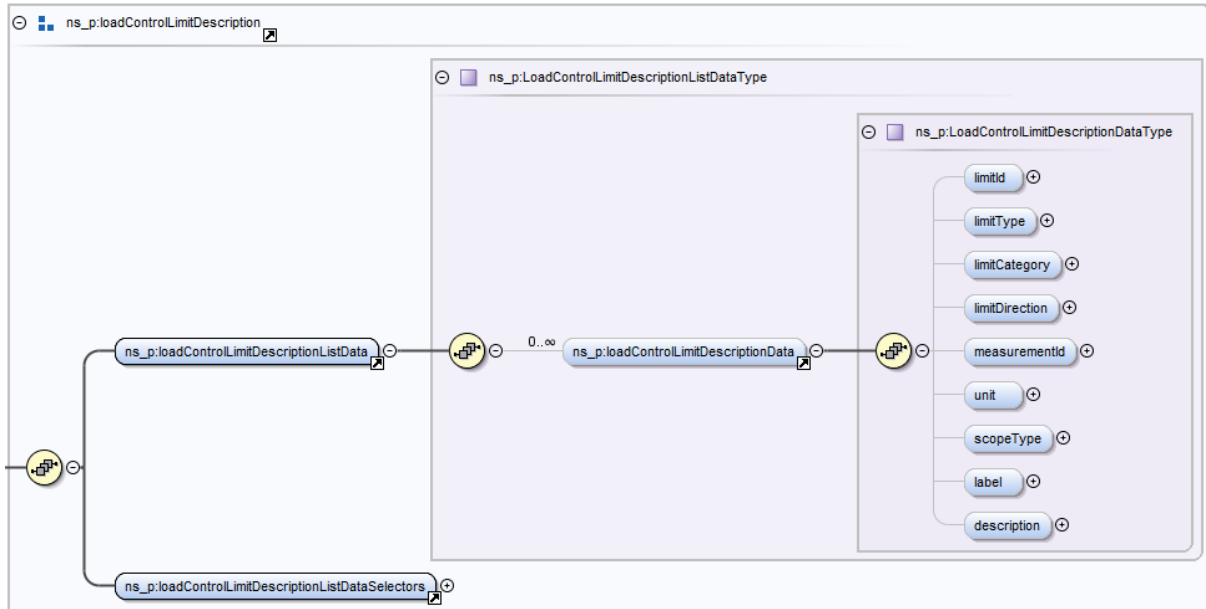
If the receiving device has a proper value for this *limitId*, it would respond with the following function:

```
<loadControlLimitConstraintsListData>
  <loadControlLimitConstraintsData>
    <limitId>2</limitId>
    <valueRangeMin>
      <scaledNumber>
        <number>0</number>
      </scaledNumber>
    </valueRangeMin>
    <valueRangeMax>
      <scaledNumber>
        <number>36</number>
        <scale>2</scale>
      </scaledNumber>
    </valueRangeMax>
    <valueStepSize>
      <scaledNumber>
        <number>5</number>
        <scale>-1</scale>
      </scaledNumber>
    </valueStepSize>
  </loadControlLimitConstraintsData>
</loadControlLimitConstraintsListData>
```

7860 5.3.14.7 loadControlLimitDescriptionListData

7861 5.3.14.7.1 General

7862 The rather static information about the available limit(s) are modelled within the
7863 loadControlLimitDescriptionListData function.



7864
7865 Figure 117: loadControlLimitDescriptionListData function overview

7866

7867 5.3.14.7.2 Detailed description of elements

Element	Type	Description
loadControlLimitDescriptionListData. loadControlLimitDescriptionData. (list)		
limitId	Identifier "LoadControlLimitIdType" (see Table 358).	Identifier for the limit.
limitType	Union "LoadControlLimitTypeType": - Enum (see Table 242): LoadControlLimitTypeEnumType - EnumExtendType (see section 3.10.1.5)	Whether the limit shall be used as a minimum or maximum is denoted within this element.
limitCategory	Union "LoadControlLimitCategoryType": - Enum (see Table 243): LoadControlLimitCategoryEnumType - EnumExtendType (see section 3.10.1.5)	There are different levels of how important the limit is (see enumeration description).
limitDirection	Common data type "EnergyDirectionType". See section 3.10.1.13.	If a device supports both, energy consumption and energy production, the direction which shall be limited is stated in this element (e.g. if only the consumption of a battery shall be limited, the enumeration "consume" will be set; if both, consumption and production shall

		be limited, two limits have to be set).
measurementId	<i>Identifier "MeasurementIdType" (see Table 358).</i>	If a measurand is linked to the limit, its identifier is denoted here.
unit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	The unit, which is used for the limit value, is denoted with this element.
scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	A certain meaning of the limit.
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A label for the limit.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A description for the limit.

Table 241: loadControlLimitDescriptionListData function detailed description of elements

Enumeration **LoadControlLimitTypeEnumType**:

Value	Description
minValueLimit	The limit set in the "value" shouldn't be underrun (value > minValueLimit).
maxValueLimit	The limit set in the "value" shouldn't be overrun (value < maxValueLimit).
signDependentAbsValueLimit	Depending on the applied sign convention and the sign of the limit set in the "value" Element of the according "loadControlLimitData" entry the consumption or production is limited.

Table 242: Enumeration LoadControlLimitTypeEnumType

Enumeration **LoadControlLimitCategoryEnumType**:

Value	Description
obligation	Used for limits that are absolutely important to be hold. A loss of comfort could happen and will be accepted. The normal operation will be interrupted for keeping the limit (if needed).
recommendation	Used for limits that are quite important for the energy management. A loss of comfort could happen and will be accepted for a short period of time. The normal operation should not be interrupted.
optimization	Used for limits that just optimize the energy management. No loss of comfort will be accepted. The normal operation shall not be interrupted.

Table 243: Enumeration LoadControlLimitCategoryEnumType

5.3.14.7.3 Available selectors

- limitId
- limitType
- limitDirection
- measurementId
- scopeType

7881 **5.3.14.7.4 Examples**7882 **5.3.14.7.4.1 Read-reply**7883 A *loadControlLimitDescriptionListData* instance sent as reply to a standard read command:

```

7884 <loadControlLimitDescriptionListData>
7885   <loadControlLimitDescriptionData>
7886     <limitId>1</limitId>
7887     <limitType>maxValueLimit</limitType>
7888     <limitCategory>obligation</limitCategory>
7889     <limitDirection>consume</limitDirection>
7890     <measurementId>3</measurementId>
7891     <unit>W</unit>
7892   </loadControlLimitDescriptionData>
7893   <loadControlLimitDescriptionData>
7894     <limitId>2</limitId>
7895     <limitType>maxValueLimit</limitType>
7896     <limitCategory>obligation</limitCategory>
7897     <limitDirection>produce</limitDirection>
7898     <measurementId>5</measurementId>
7899     <unit>W</unit>
7900   </loadControlLimitDescriptionData>
7901 </loadControlLimitDescriptionListData>

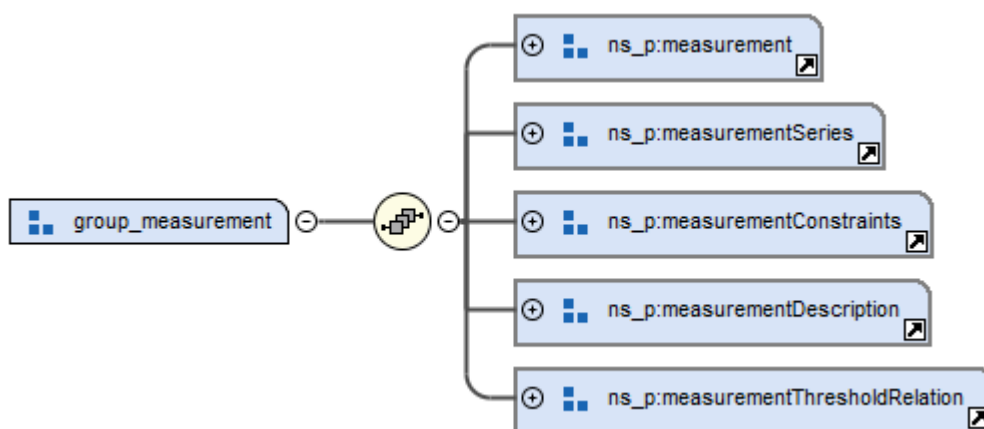
```

7902

7903 **5.3.15 Measurement**7904 **5.3.15.1 Introduction**

7905 The SPINE Measurement class provides functionality for transmitting measured, calculated or
 7906 empirically obtained values from sensors. Mainly these are sensors that provide some kind of
 7907 information about the environment or physical parameters. Clients can use the information for
 7908 further actions or just visualize them for the information of a user.

7909 Due to the nature of measurement, the measured values are only readable. Some control systems
 7910 permit a user to select a preferred value (e.g. a desired room temperature). Such "desired" values
 7911 obviously require writable fields. This can be achieved with the "Setpoint" class (see section 5.3.21).



7912

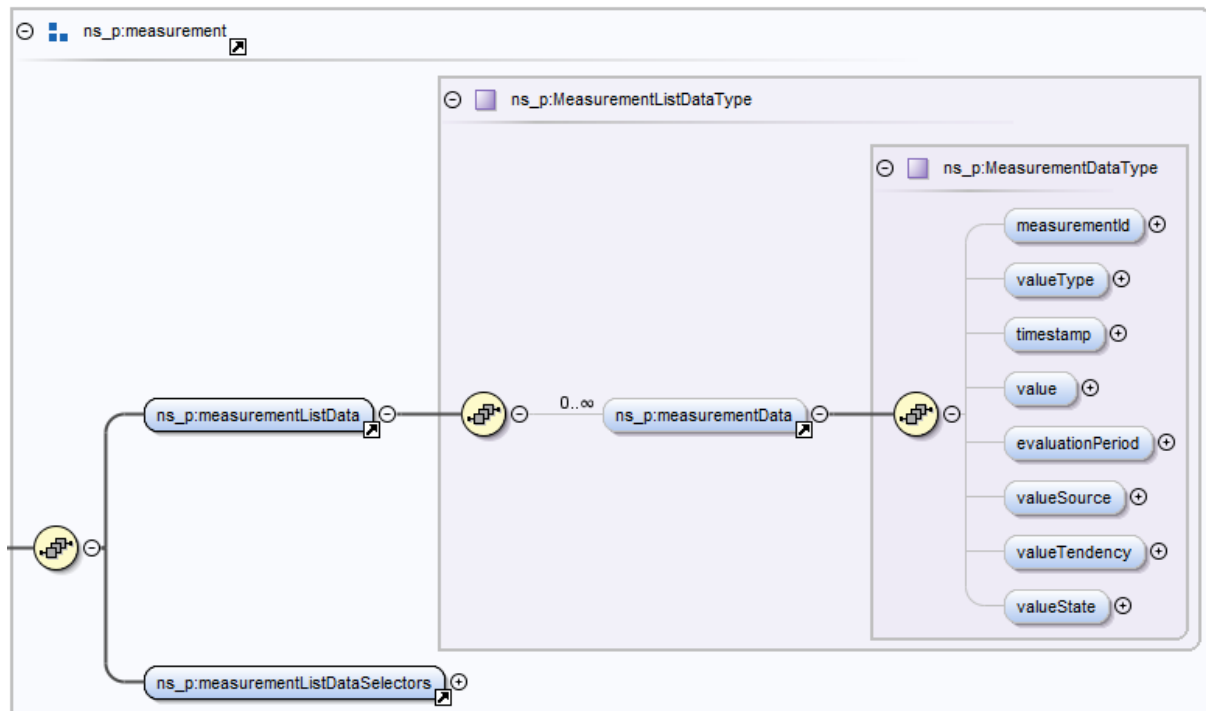
7913 *Figure 118: Measurement function-group overview*

7914

7915 5.3.15.2 *measurementListData*

7916 5.3.15.2.1 *General*

7917 The *measurementListData* command is used to communicate measurement values. Typically, a
 7918 sensor sends this command as notification as soon as it changes more than a specific threshold. Of
 7919 course, it may be readable, too.



7920
 7921 Figure 119: *measurementListData* function overview

7922

7923 5.3.15.2.2 *Detailed description of elements*

Element	Type	Description
measurementListData. measurementData. (list)		
measurementId	Identifier "MeasurementIdType" (see Table 358).	Enables the identification of different sensors on one SPINE feature.
valueType	Union "MeasurementValueTypeType": - Enum (see Table 245): MeasurementValueTypeEnumType - EnumExtendType (see section 3.10.1.5)	It is possible to model different types of measurement values. For example "value", "averageValue", "minValue", etc. (see the SPINE data model for all possible value types).
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time of creation of a measurement value element (i.e. the time it was measured, calculated, ...). Some sensors and communications technologies may already provide this information natively. In other cases the technology interface may set timestamp to the time of creation of a measurementData instance.

value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The measurement value itself, i.e. the magnitude according to "valueType".
evaluationPeriod	<i>Common data type "TimePeriodType". See section 3.10.2.1.</i>	Some <i>valueType</i> values can be detailed or even require information about the time period. For example, an <i>averageValue</i> is usually computed from measured values of a finite time period. For smart devices, the start and end time of the <i>evaluationPeriod</i> can be filled in directly. For bare ("simple") sensors, such "derived"/"accompanying" values (average, min, max, ...) could be calculated (if desired) by the technology interface component together with the corresponding <i>evaluationPeriod</i> .
valueSource	<i>Union "MeasurementValueSourceType":</i> - Enum (see Table 246): MeasurementValueSourceEnumType - EnumExtendType (see section 3.10.1.5)	Whether element "value" is measured, calculated or empirically obtained can be denoted with this element.
valueTendency	<i>Union "MeasurementValueTendencyType":</i> - Enum (see Table 247): MeasurementValueTendencyEnumType - EnumExtendType (see section 3.10.1.5)	This elements states whether the tendency is <i>rising</i> , <i>stable</i> or <i>falling</i> .
valueState	<i>Union "MeasurementValueStateType":</i> - Enum (see Table 248): MeasurementValueStateEnumType - EnumExtendType (see section 3.10.1.5)	A measurand may have a state.

7924 Table 244: measurementListData function detailed description of elements

7925 Enumeration **MeasurementValueTypeEnumType**:

Value	Description
value	The normal measurement value.
averageValue	The average measurement value over the last time, stated in the element evaluationPeriod.
minValue	The minimum measurement value of the last time, stated in the element evaluationPeriod.
maxValue	The maximum measurement value of the last time, stated in the element evaluationPeriod.
standardDeviation	Used to quantify the amount of variation or dispersion of a set of values.

7926 Table 245: Enumeration MeasurementValueTypeEnumType

7927 Enumeration **MeasurementValueSourceEnumType**:

Value	Description
measuredValue	The measurement value was measured with a dedicated sensor. The value is rather exact.
calculatedValue	The measurement value was calculated with the values from some other parameters (e.g. water temperature, outside temperature, etc.) that have a calculable impact on the measurand. This value is not as exact as a measured value, but it differs not too much from the real value.
empiricalValue	The measurement value is only an empirical one that can differ a lot from the real value. It can rely on some other parameters that have a known value, but there is no exactly calculable relation between them.

7928 Table 246: Enumeration MeasurementValueSourceEnumType

7929 Enumeration **MeasurementValueTendencyEnumType**:

Value	Description
rising	The average measurement value is currently rising.
stable	The average measurement value is currently stable.
falling	The average measurement value is currently falling.

7930 Table 247: Enumeration MeasurementValueTendencyEnumType

7931 Enumeration **MeasurementValueStateEnumType**:

Value	Description
normal	The measurement value could be measured / calculated / empirically determined as expected.
outOfRange	The measurement value is currently out of range of the sensor (as defined by "valueRangeMin" and "valueRangeMax", see section 5.3.15.4.2).
error	An error occurred while determining the measurement value.

7932 Table 248: Enumeration MeasurementValueStateEnumType

7933

7934 5.3.15.2.3 Available selectors

- 7935 - measurementId
- 7936 - valueType
- 7937 - timestampInterval

7938

7939 5.3.15.2.4 Examples

7940 5.3.15.2.4.1 Read-reply

7941 A *measurementListData* command with a measured value, sent as reply with only one entry (device
7942 is not storing historical values):

```

7943 <measurementListData>
7944   <measurementData>
7945     <measurementId>1</measurementId>
7946     <valueType>value</valueType>
7947     <timestamp>2021-07-15T12:00:00.0Z</timestamp>
7948     <value>
7949       <number>255</number>
7950       <scale>-1</scale>
7951     </value>
7952     <valueSource>measuredValue</valueSource>

```

```

7953         <valueTendency>rising</valueTendency>
7954         <valueState>normal</valueState>
7955     </measurementData>
7956 </measurementListData>

```

7957

7958 5.3.15.2.4.2 Notify

7959 A *measurementListData* command with a calculated average value (no other value type supported by
7960 the device), sent as (complete) notification:

```

7961 <measurementListData>
7962     <measurementData>
7963         <measurementId>1</measurementId>
7964         <valueType>averageValue</valueType>
7965         <timestamp>2021-07-15T12:13:14.5Z</timestamp>
7966         <value>
7967             <number>23</number>
7968         </value>
7969         <evaluationPeriod>
7970             <startTime>2021-07-15T11:00:00.0Z</startTime>
7971             <endTime>2021-07-15T12:00:00.0Z</endTime>
7972         </evaluationPeriod>
7973         <valueSource>calculatedValue</valueSource>
7974         <valueTendency>stable</valueTendency>
7975         <valueState>normal</valueState>
7976     </measurementData>
7977 </measurementListData>

```

7978

7979 5.3.15.2.4.3 Read-reply

7980 In this example the requested node supports only values of type "value" and offers this reply:

```

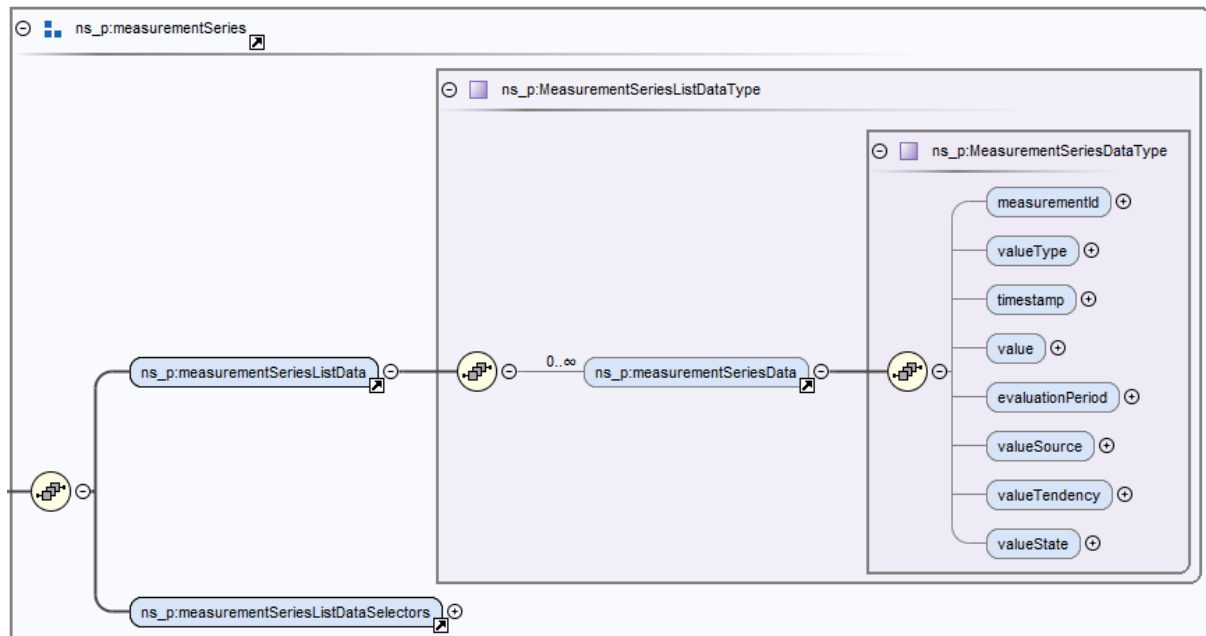
7981 <measurementListData>
7982     <measurementData>
7983         <measurementId>1</measurementId>
7984         <valueType>value</valueType>
7985         <timestamp>2021-07-15T12:10:00.0Z</timestamp>
7986         <value>
7987             <number>257</number>
7988             <scale>-1</scale>
7989         </value>
7990         <valueSource>measuredValue</valueSource>
7991         <valueTendency>falling</valueTendency>
7992         <valueState>normal</valueState>
7993     </measurementData>
7994     <measurementData>
7995         <measurementId>2</measurementId>
7996         <valueType>value</valueType>
7997         <timestamp>2021-07-15T12:10:00.0Z</timestamp>
7998         <value>
7999             <number>261</number>
8000             <scale>-1</scale>
8001         </value>
8002         <valueSource>measuredValue</valueSource>
8003         <valueTendency>stable</valueTendency>
8004         <valueState>normal</valueState>
8005     </measurementData>
8006 </measurementListData>

```

8007

8008 **5.3.15.3 measurementSeriesListData**8009 **5.3.15.3.1 General**

8010 The *measurementSeriesListData* Function is similar to the *measurementListData* Function (see
 8011 section 5.3.15.2). It is used for historical data of measurands.



8012

8013 *Figure 120: measurementSeriesListData function overview*

8014

8015 **5.3.15.3.2 Detailed description of elements**

Element	Type	Description
measurementSeriesListData. measurementSeriesData. (list)		
measurementId	Identifier "MeasurementIdType" (see Table 358).	Enables the identification of different sensors on one SPINE feature.
valueType	Union "MeasurementValueTypeType": - Enum (see Table 245): MeasurementValueTypeEnumType - EnumExtendType (see section 3.10.1.5)	It is possible to model different types of measurement values. For example "value", "averageValue", "minValue", etc. (see the SPINE data model for all possible value types).
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time of creation of a measurement value element (i.e. the time it was measured, calculated, ...). Some sensors and communications technologies may already provide this information natively. In other cases the technology interface may set timestamp to the time of

		creation of a measurementData instance. There can be multiple measurementSeriesData entries for the same measurementId, but with different timestamps. This allows to express historical values or a measurement timeline.
value	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	The measurement value itself, i.e. the magnitude according to "valueType".
evaluationPeriod	<i>Common data type "TimePeriodType". See section 3.10.2.1.</i>	Some <i>valueType</i> values can be detailed or even require information about the time period. For example, an <i>averageValue</i> is usually computed from measured values of a finite time period. For smart devices, the start and end time of the <i>evaluationPeriod</i> can be filled in directly. For bare ("simple") sensors, such "derived"/"accompanying" values (average, min, max, ...) could be calculated (if desired) by the technology interface component together with the corresponding <i>evaluationPeriod</i> .
valueSource	<i>Union "MeasurementValueSourceType":</i> - Enum (see Table 246): MeasurementValueSourceEnumType - EnumExtendType (see section 3.10.1.5)	Whether element "value" is measured, calculated or empirically obtained can be denoted with this element.
valueTendency	<i>Union "MeasurementValueTendencyType":</i> - Enum (see Table 247): MeasurementValueTendencyEnumType - EnumExtendType (see section 3.10.1.5)	This elements states whether the tendency is <i>rising</i> , <i>stable</i> or <i>falling</i> .
valueState	<i>Union "MeasurementValueStateType":</i> - Enum (see Table 248): MeasurementValueStateEnumType - EnumExtendType (see section 3.10.1.5)	A measurand may have a state.

Table 249: measurementSeriesListData function detailed description of elements

5.3.15.3.3 Available selectors

- measurementId
- valueType
- timestampInterval

8023 **5.3.15.3.4 Examples**8024 **5.3.15.3.4.1 Read-reply**8025 A *measurementSeriesListData* command with a measured value, sent as reply with historical values:

```

8026 <measurementSeriesListData>
8027   <measurementSeriesData>
8028     <measurementId>1</measurementId>
8029     <valueType>value</valueType>
8030     <timestamp>2020-08-17T12:00:00.0Z</timestamp>
8031     <value>
8032       <number>255</number>
8033       <scale>-1</scale>
8034     </value>
8035     <valueSource>measuredValue</valueSource>
8036     <valueTendency>rising</valueTendency>
8037     <valueState>normal</valueState>
8038   </measurementSeriesData>
8039   <measurementSeriesData>
8040     <measurementId>1</measurementId>
8041     <valueType>value</valueType>
8042     <timestamp>2020-08-17T12:15:00.0Z</timestamp>
8043     <value>
8044       <number>278</number>
8045       <scale>-1</scale>
8046     </value>
8047     <valueSource>measuredValue</valueSource>
8048     <valueTendency>rising</valueTendency>
8049     <valueState>normal</valueState>
8050   </measurementSeriesData>
8051   <measurementSeriesData>
8052     <measurementId>1</measurementId>
8053     <valueType>value</valueType>
8054     <timestamp>2020-08-17T12:30:00.0Z</timestamp>
8055     <value>
8056       <number>294</number>
8057       <scale>-1</scale>
8058     </value>
8059     <valueSource>measuredValue</valueSource>
8060     <valueTendency>rising</valueTendency>
8061     <valueState>normal</valueState>
8062   </measurementSeriesData>
8063 </measurementSeriesListData>

```

8064

8065 **5.3.15.4 measurementConstraintsListData**8066 **5.3.15.4.1 General**

8067 Some constraints concerning the possible values of a measurand are modelled within this function.

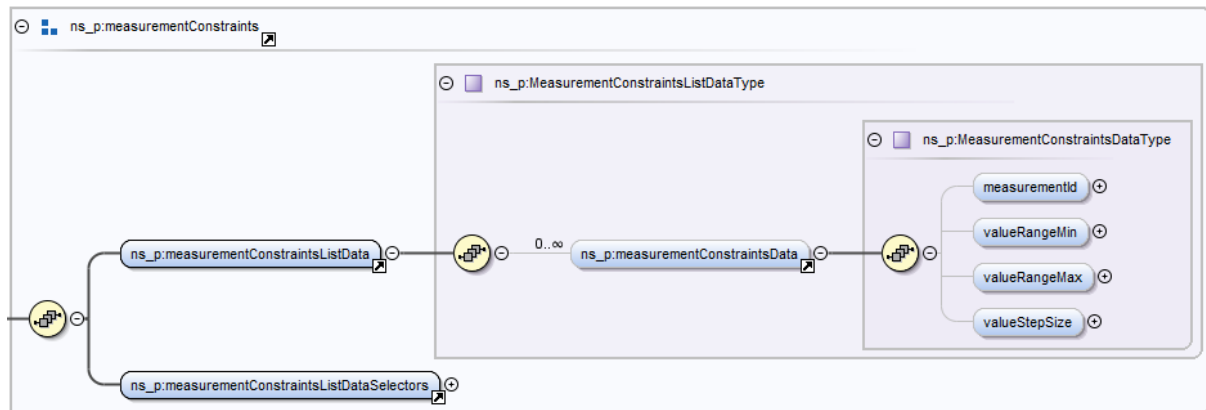


Figure 121: measurementConstraintsListData function overview

5.3.15.4.2 Detailed description of elements

Element	Type	Description
measurementConstraintsListData. measurementConstraintsData. (list)		
measurementId	Identifier "MeasurementIdType" (see Table 358).	Enables the identification of different sensors on one SPINE feature.
valueRangeMin	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum possible measurement value measurable by the feature.
valueRangeMax	Common data type "ScaledNumberType". See section 3.10.1.8.	The maximum possible measurement value measurable by the feature.
valueStepSize	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum step size between two different measurement values.

Table 250: measurementConstraintsListData function detailed description of elements

5.3.15.4.3 Available selectors

- measurementId

5.3.15.4.4 Examples

5.3.15.4.4.1 Read-reply

A measurementConstraintsListData instance from a temperature sensor, sent as reply to a standard read command:

```

<measurementConstraintsListData>
  <measurementConstraintsData>
    <measurementId>1</measurementId>
    <valueRangeMin>
      <number>-1</number>
      <scale>1</scale>
    </valueRangeMin>
    <valueRangeMax>

```

```

8089         <number>4</number>
8090         <scale>1</scale>
8091     </valueRangeMax>
8092     <valueStepSize>
8093         <number>1</number>
8094         <scale>-1</scale>
8095     </valueStepSize>
8096 </measurementConstraintsData>
8097 </measurementConstraintsListData>

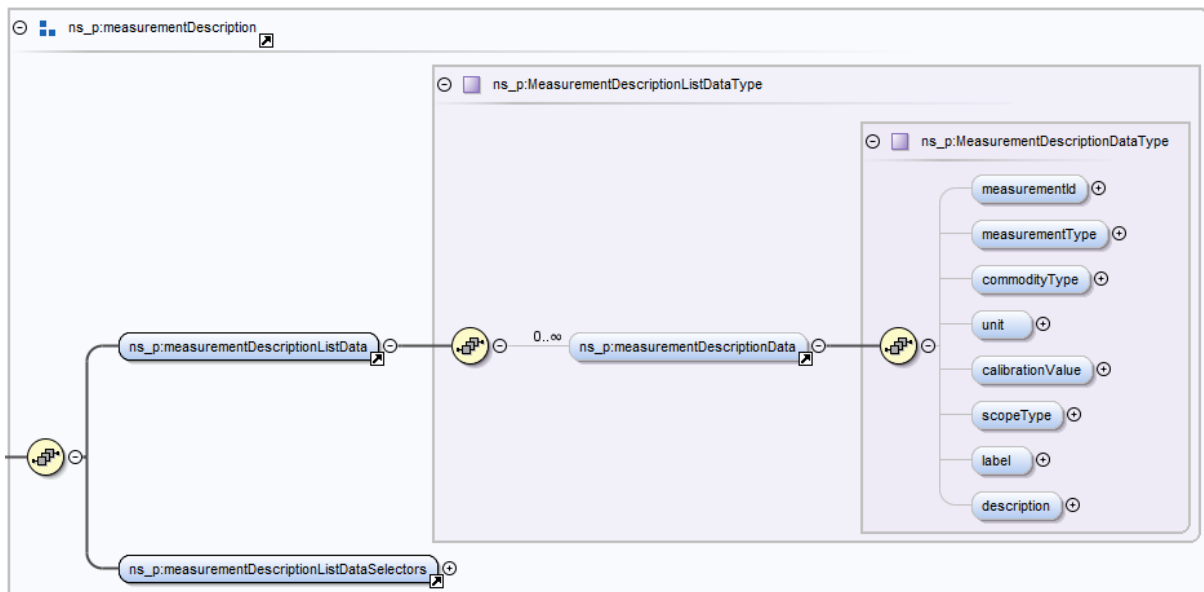
```

8098

8099 5.3.15.5 *measurementDescriptionListData*

8100 5.3.15.5.1 *General*

8101 Descriptive information on a specific measurement type is modelled with a
8102 *measurementDescriptionListData* function. This information is rather constant in comparison to
8103 measurement values. Typically, it is requested once at the beginning, when a node (an external
8104 display, e.g.) is configured to present or use a specific measurement value. As usual, for such
8105 scenarios it is recommended the the server notifies changes to all relevant communication partners,
8106 because a server cannot expect clients to periodically request this information.



8107

8108 Figure 122: *measurementDescriptionListData* function overview

8109

8110 5.3.15.5.2 *Detailed description of elements*

Element	Type	Description
measurementDescriptionListData. measurementDescriptionData. (list)		
measurementId	Identifier "MeasurementIdType" (see Table 358).	Enables the identification of different measurands on one SPINE feature.
measurementType	Union "MeasurementTypeType": - Enum (see Table 252): MeasurementTypeEnumType	To identify which type of measurand is measured (for example "temperature").

	- EnumExtendType (see section 3.10.1.5)	
commodityType	<i>Common data type "CommodityTypeType". See section 3.10.1.9.</i>	If a measurand of a commodity is measured, the type of commodity is stated here.
unit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	The unit, which is used for the value. It is always related to the normal value. Other <i>valueType</i> values (e.g. <i>averageDerivativeValue</i>) can have different units (derived from <i>unit</i>).
calibrationValue	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Some measurement sensors need a calibration value (e.g. set during production phase). It is automatically added to the internal measured value. I.e. the value stated in the <i>value</i> element in the <i>measurementListData</i> function is the already corrected value.
scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	Specifies a more detailed meaning of the measurand (e.g. <i>outsideTemperature</i>).
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A (short) label can be useful to identify or group specific sensors.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	A (longer) description of a measurement device can be deposited here.

8111 Table 251: *measurementDescriptionListData* function detailed description of elements

8112 Enumeration **MeasurementTypeEnumType**:

Value	Description
acceleration	The acceleration is measured.
angle	An angle is measured.
angularVelocity	The angular velocity is measured.
area	An area is measured.
atmosphericPressure	The atmospheric pressure is measured.
capacity	The capacity is measured.
concentration	A concentration of a substance is measured.
count	A count is measured.
current	The current is measured.
density	The density is measured.
distance	A distance is measured.
electricField	The electric field is measured.
energy	The energy is measured.
force	The force is measured.
frequency	The frequency is measured.
harmonicDistortion	The harmonic distortion is measured.
heat	The heat is measured.
heatFlux	The heat flux is measured.
illuminance	The illuminance is measured.
impulse	An impulse is measured.
level	A level is measured.

magneticField	The magnetic field is measured.
mass	A mass is measured.
massFlow	The mass flow is measured.
particles	The quantity of particles is measured.
percentage	The percentage is measured.
power	The power is measured.
powerFactor	The power factor is measured.
pressure	The pressure is measured.
radonActivity	The radon activity is measured.
relativeHumidity	The relative humidity is measured.
resistance	The resistance is measured.
solarRadiation	The solar radiation is measured.
speed	The speed is measured.
temperature	The temperature is measured.
time	The time is measured.
torque	The torque is measured.
unknown	An unknown measurand is measured.
velocity	The velocity is measured.
voltage	The voltage is measured.
volume	The volume is measured.
volumetricFlow	The volumetric flow is measured.

8113 *Table 252: Enumeration MeasurementTypeEnumType*

8114

8115 5.3.15.5.3 Available selectors

- 8116 - measurementId
- 8117 - measurementType
- 8118 - commodityType
- 8119 - scopeType

8120

8121 5.3.15.5.4 Examples

8122 5.3.15.5.4.1 Read-reply

8123 A *measurementDescriptionListData* instance from a temperature sensor, sent as reply to a standard
8124 read command:

```

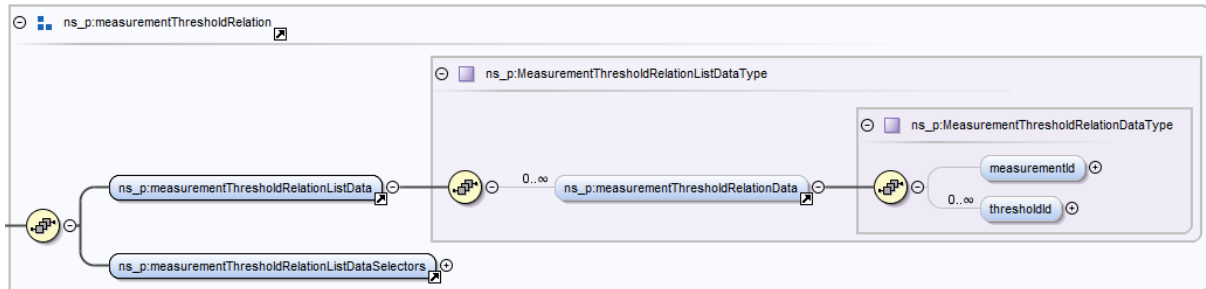
8125 <measurementDescriptionListData>
8126   <measurementDescriptionData>
8127     <measurementId>1</measurementId>
8128     <measurementType>temperature</measurementType>
8129     <commodityType>air</commodityType>
8130     <unit>degC</unit>
8131     <calibrationValue>
8132       <number>-48</number>
8133       <scale>-1</scale>
8134     </calibrationValue>
8135     <scopeType>outsideTemperature</scopeType>
8136     <label>Outside_Temp.</label>
8137     <description>Outside Air Temperature</description>
8138   </measurementDescriptionData>
8139 </measurementDescriptionListData>

```

8140

8141 **5.3.15.6 measurementThresholdRelationListData**8142 **5.3.15.6.1 General**

8143 The relation between a measurand and dedicated thresholds (see section 5.3.27) is modelled with
 8144 this function.



8145

8146 *Figure 123: measurementThresholdRelationListData function overview*

8147

8148 **5.3.15.6.2 Detailed description of elements**

Element	Type	Description
measurementThresholdRelationListData. measurementThresholdRelationData. (list)		
measurementId	Identifier "MeasurementIdType" (see Table 358).	Enables the identification of different sensors on one SPINE address.
thresholdId (list)	Identifier "ThresholdIdType" (see Table 358).	Related threshold identifier(s).

8149 *Table 253: measurementThresholdRelationListData function detailed description of elements*

8150

8151 **5.3.15.6.3 Available selectors**

- 8152 - measurementId
- 8153 - thresholdId

8154

8155 **5.3.15.6.4 Examples**8156 **5.3.15.6.4.1 Read-reply**

8157 A measurementThresholdRelationListData sent as a reply to a standard read command:

```

8158 <measurementThresholdRelationListData>
8159   <measurementThresholdRelationData>
8160     <measurementId>1</measurementId>
8161     <thresholdId>2</thresholdId>
8162     <thresholdId>3</thresholdId>
8163     <thresholdId>6</thresholdId>
8164   </measurementThresholdRelationData>
8165 </measurementThresholdRelationListData>

```

8166

5.3.16 Messaging

5.3.16.1 Introduction

Messaging is basically intended for simple textual information transfer for displays and users. The textual information is not intended to be used for any automatisms. It should be used for textual information that are for example event based or otherwise will change over time.

Some devices may keep a list of messages rather than (or in addition to) a single message.



Figure 124: Messaging function-group overview

5.3.16.2 messagingListData

5.3.16.2.1 General

The messagingListData function contains the necessary elements to model a simple text message.

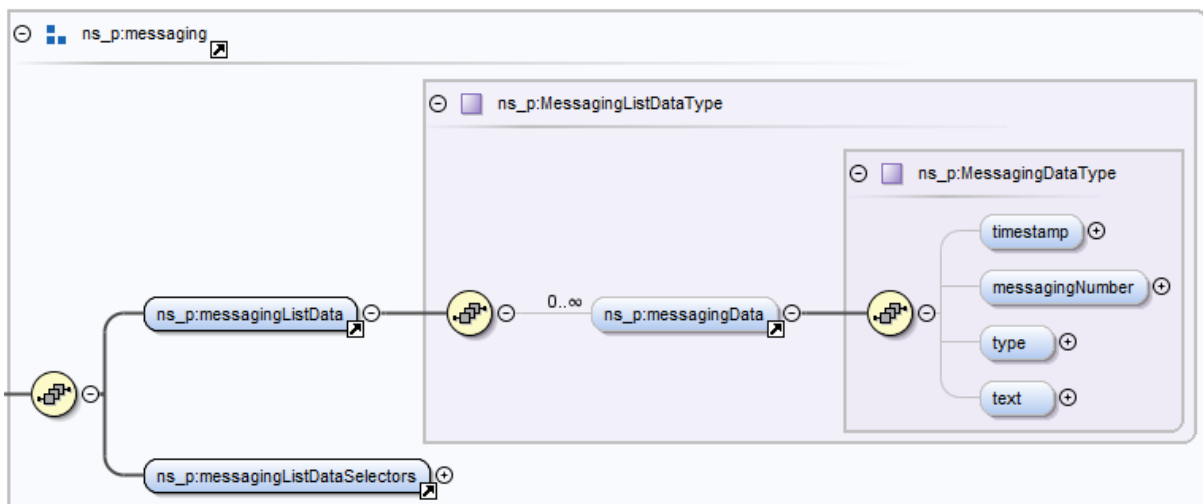


Figure 125: messagingListData function overview

5.3.16.2.2 Detailed description of elements

Element	Type	Description
messagingListData. messagingData. (list)		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The time when the message was generated.
messagingNumber	Identifier "MessagingNumberType" (see Table 358).	An identifier for one specific message. If a message is marked as obsolete (see below) this number can be used to identify the original message.
type	Union "MessagingTypeType": - Enum (see Table 255): MessagingTypeEnumType	The type of the message.

	- EnumExtendType (see section 3.10.1.5)	
text	<i>Simple type</i> <i>"MessagingDataTextType"</i> <i>(restriction of xs:string)</i>	The message itself.

Table 254: messagingListData function detailed description of elements

Enumeration **MessagingTypeEnumType**:

Value	Description
logging	Used for messages that are stored in a log file.
information	Messages that are presented to the customer on a display; lower priority.
warning	Messages that are presented to the customer on a display; medium priority.
alarm	Messages that are presented to the customer on a display; high priority.
emergency	Messages that are presented to the customer on a display; very high priority.
obsolete	Used to mark previously sent messages as obsolete (so that they are not displayed anymore).

Table 255: Enumeration MessagingTypeEnumType

5.3.16.2.3 Available selectors

- timestampInterval
- messagingNumber

5.3.16.2.4 Examples

5.3.16.2.4.1 Notify

A washing machine is configured to send messages to an external display. After a firmware update it sends this (complete) notification:

```
<messagingListData>
  <messagingData>
    <timestamp>2006-05-04T18:13:51.0Z</timestamp>
    <messagingNumber>72</messagingNumber>
    <type>information</type>
    <text>Update completed.</text>
  </messagingData>
</messagingListData>
```

Other possible reasons for messages to a display: Notification of end of washing cycle; notification of problem with water supply; etc.

5.3.16.2.4.2 Notify

An alarm device keeps a list of its events of the last month. It is configured to send new messages to an external display every 2 minutes (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..." just to improve the readability):

```
...
<messagingListData>
  <messagingData>
```

```

8213         <timestamp>2006-05-04T18:13:51.0Z</timestamp>
8214         <messagingNumber>2014</messagingNumber>
8215         <type>information</type>
8216         <text>Window 1st floor closed.</text>
8217     </messagingData>
8218     <messagingData>
8219         <timestamp>2006-05-04T18:14:23.0Z</timestamp>
8220         <messagingNumber>2015</messagingNumber>
8221         <type>information</type>
8222         <text>Alarm engaged.</text>
8223     </messagingData>
8224 </messagingListData>

```

8225

8226 5.3.16.2.4.3 Read-reply

8227 The alarm device is able to provide its complete message list upon request.

8228 A user selects the alarm device on an external display and queries its messages. The alarm device
8229 replies with a proper messagingListData. Extract with just the first and the last message:

```

8230 <messagingListData>
8231     <messagingData>
8232         <timestamp>2006-04-05T03:23:17.0Z</timestamp>
8233         <messagingNumber>1716</messagingNumber>
8234         <type>alarm</type>
8235         <text>Window ground floor opened.</text>
8236     </messagingData>
8237     ...
8238     <messagingData>
8239         <timestamp>2006-05-04T18:14:23.0Z</timestamp>
8240         <messagingNumber>2015</messagingNumber>
8241         <type>information</type>
8242         <text>Alarm engaged.</text>
8243     </messagingData>
8244 </messagingListData>

```

8245

8246 5.3.16.2.4.4 Read-reply

8247 The alarm device supports the selector *timestampInterval*. A user selects the alarm device on an
8248 external display and queries its messages of the last hour. The display sends a read classifier together
8249 with the corresponding filter:

```

8250 <function>messagingListData</function>
8251 <filter>
8252     <cmdControl>
8253         <partial/>
8254     </cmdControl>
8255     <messagingListDataSelectors>
8256         <timestampInterval>
8257             <startTime>-PT1H</startTime>
8258             <endTime>PT0S</endTime>
8259         </timestampInterval>
8260     </messagingListDataSelectors>
8261 </filter>
8262 <messagingListData/>

```

8263 The alarm device replies with a proper `messagingListData`. Extract with just the first and the last
8264 message:

```
8265 <function>messagingListData</function>
8266 <filter>
8267   <cmdControl>
8268     <partial/>
8269   </cmdControl>
8270 </filter>
8271 <messagingListData>
8272   <messagingData>
8273     <timestamp>2006-05-04T17:55:19.0Z</timestamp>
8274     <messagingNumber>2012</messagingNumber>
8275     <type>information</type>
8276     <text>Front door closed.</text>
8277   </messagingData>
8278   ...
8279   <messagingData>
8280     <timestamp>2006-05-04T18:14:23.0Z</timestamp>
8281     <messagingNumber>2015</messagingNumber>
8282     <type>information</type>
8283     <text>Alarm engaged.</text>
8284   </messagingData>
8285 </messagingListData>
```

8286

8287 5.3.16.2.4.5 Read-reply

8288 The alarm device supports the select option *messagingNumber*. A user selects the alarm device on an
8289 external display and queries the messages with `messagingNumber` "2011". The display sends a read
8290 classifier together with the Selector:

```
8291 <function>messagingListData</function>
8292 <filter>
8293   <cmdControl>
8294     <partial/>
8295   </cmdControl>
8296   <messagingListDataSelectors>
8297     <messagingNumber>2011</messagingNumber>
8298   </messagingListDataSelectors>
8299 </filter>
8300 <messagingListData/>
```

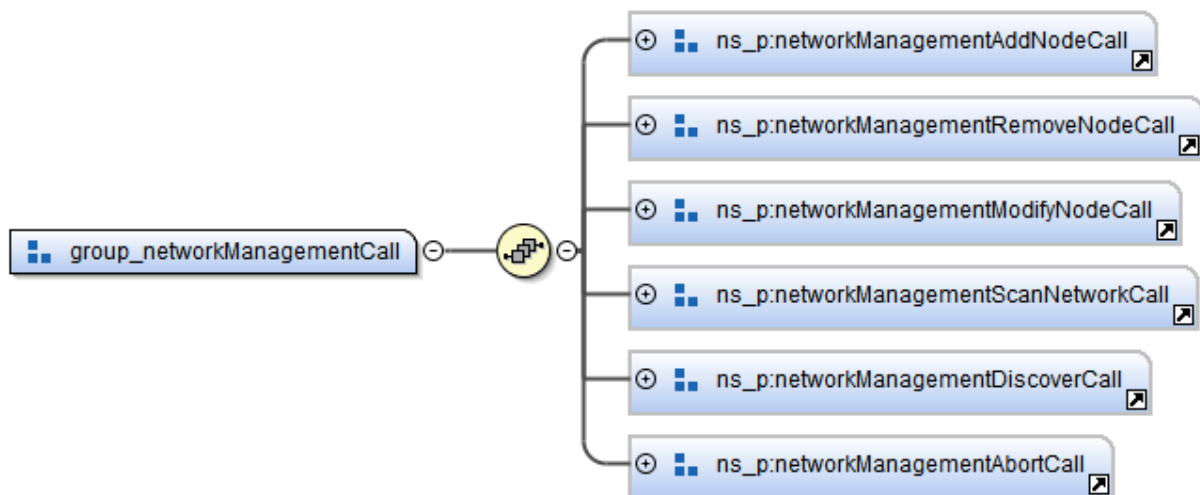
8301 The alarm device replies with a proper `messagingListData`:

```
8302 <function>messagingListData</function>
8303 <filter>
8304   <cmdControl>
8305     <partial/>
8306   </cmdControl>
8307 </filter>
8308 <messagingListData>
8309   <messagingData>
8310     <timestamp>2006-05-04T16:10:12.0Z</timestamp>
8311     <messagingNumber>2011</messagingNumber>
8312     <type>information</type>
8313     <text>Alarm disengaged.</text>
8314   </messagingData>
8315 </messagingListData>
```

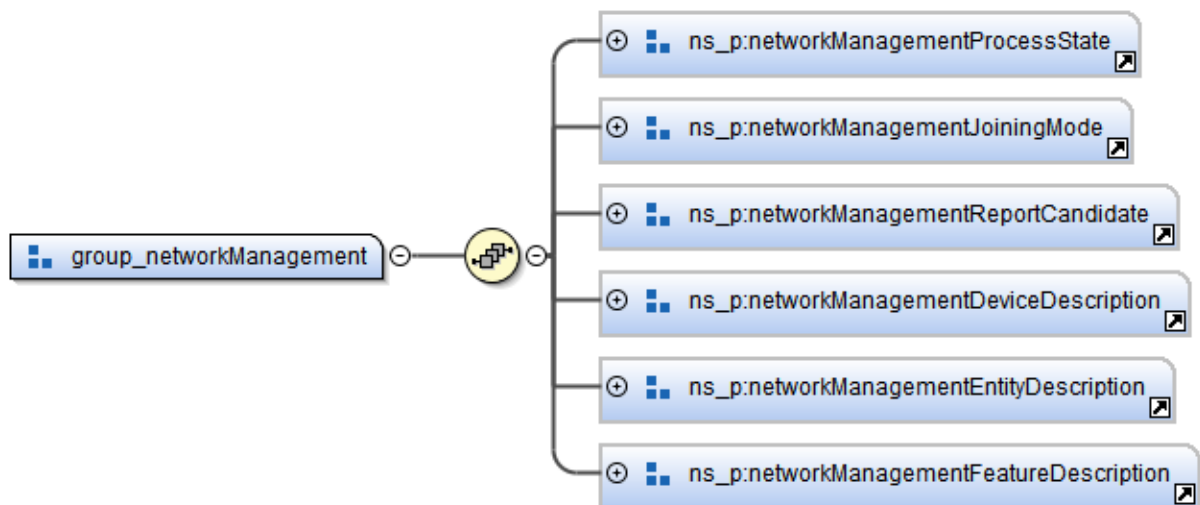
8316

8317 **5.3.17 NetworkManagement**8318 **5.3.17.1 Introduction**

8319 This class permits to model certain "knowledge" of other devices as well as the "modification" of this
 8320 knowledge. In terms of a communications technology these models can be used to add remote
 8321 nodes (devices, endpoints) to the own (SPINE) network, to remove them from the network, etc.



8322

8323 *Figure 126: NetworkManagement function-group overview (call)*

8324

8325 *Figure 127: NetworkManagement function-group overview (data)*

8326

8327 **5.3.17.1.1 SPINE address components, hierarchy, terminology**

8328 A SPINE address can contain these address elements:

- 8329 1. device
- 8330 2. entity
- 8331 3. feature

8332 The element "entity" can appear multiple times (i.e. as list). This is used to model an entity hierarchy,
8333 starting with a root entity as first item of the list. The consecutive list of "entity" elements finally
8334 denotes a specific entity.

8335 Altogether, the address elements are used to determine the proper functional component. Each
8336 device can contain several entities. Each entity can contain several features. Each addressable
8337 component is in general called a "node".

8338

8339 *5.3.17.1.2 Network Management Responsible*

8340 A so-called "Network Management Responsible" is defined in this class. This is an instance
8341 responsible to perform SPINE network management tasks for other nodes. The Network
8342 Management Responsible itself is considered a node with node address as well. A brief example
8343 explains this: A sophisticated "SPINE gateway" permits software applications to exchange SPINE
8344 XMLs with an (internal) interface that provides access to devices of different communications
8345 technologies. For each communications technology there is an (internal) implementation that
8346 realizes the translation between SPINE XML (from/to the aforementioned internal interface) and the
8347 dedicated communications technology interface/protocol (i.e. it makes use of the so-called
8348 "mapping"). We assume this "mapping implementation" is responsible to "find" devices accessible
8349 via its communications technology. In general, the implementation is responsible to control whether
8350 data can be exchanged between software applications and devices or not. This means these
8351 implementations are responsible for network management activities concerning their attached or
8352 accessible devices.

8353 Please note this example is just to explain the term "Network Management Responsible". The
8354 architecture sketched in the example shall not be considered to be the only permitted architecture.

8355

8356 *5.3.17.1.3 Use of "energy management gateway" in examples*

8357 Some communications technologies follow a "centralised" approach, whereas others are "de-
8358 centralised". The class "NetworkManagement" does not impose which approach has to be followed.
8359 In the interest of simplification, subsequent examples often assume an "energy management
8360 gateway" which is capable to connect with other SPINE devices. However, please be aware that these
8361 examples are neither a mandatory feature of an "energy management gateway", nor will a
8362 communications architecture be derived from it within this class description.

8363

8364 **5.3.17.2 networkManagement...Call**

8365 The *networkManagement...Calls* are used to add, modify or remove nodes (usually on Gateways) or
8366 to tell a node, that it shall perform a network scan, discover a specific node or abort the current task.

8367

8368 **5.3.17.3 networkManagementAddNodeCall**

8369 *5.3.17.3.1 General*

8370 This function can be used to add another node (device) to the own SPINE network.

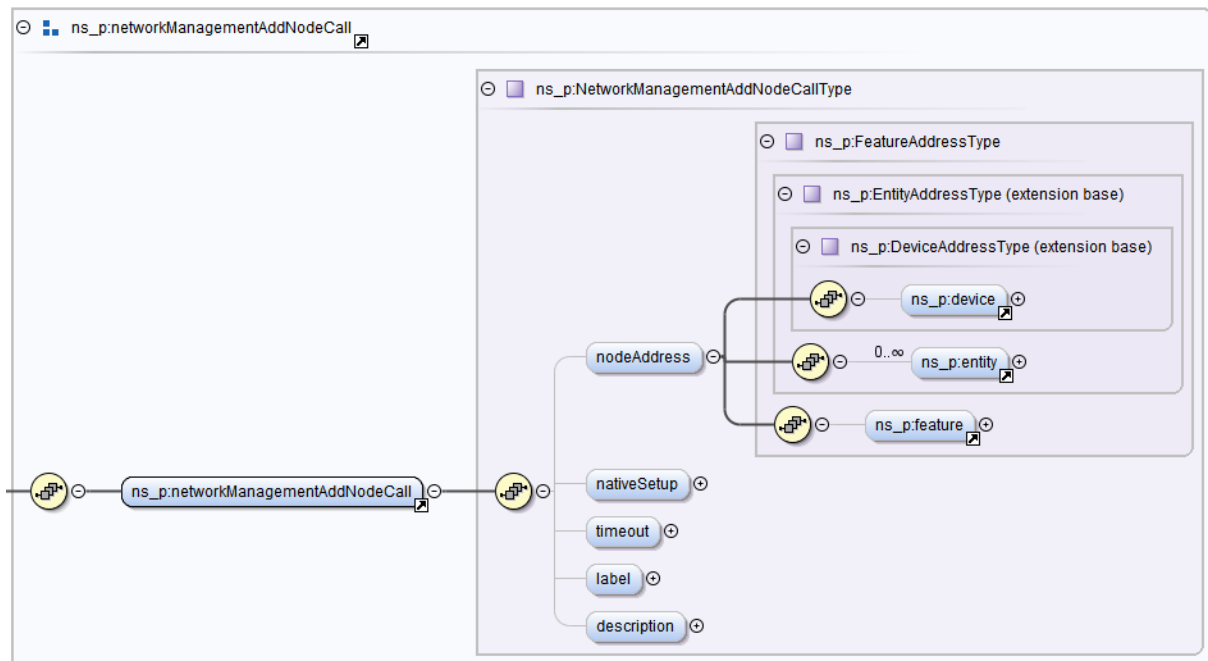


Figure 128: networkManagementAddNodeCall function overview

5.3.17.3.2 Detailed description of elements

Element	Type	Description
networkManagementAddNodeCall.		
nodeAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The node address of the node that shall be added. Contains elements for <i>device</i> , <i>entity</i> and <i>feature</i> .
nodeAddress.device	See section 3.10.3.6.	Device address part.
nodeAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.
nodeAddress.feature	See section 3.10.3.6.	Feature address part.
nativeSetup	Simple type "NetworkManagementNativeSetupType" (restriction of xs:string)	Technology dependent or implementation dependent configuration to identify (and maybe configure) the new node.
timeout	xs:duration (W3C standard type)	The procedure of adding the node shall be aborted latest after this duration.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the new node.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the new node.

Table 256: networkManagementAddNodeCall function element description

5.3.17.3.3 Available selectors

None.

8379

8380 **5.3.17.3.4 Examples**8381 **5.3.17.3.4.1 Call**

8382 The user interface of an energy management gateway permits adding a display that is accessible via a
8383 specific communications technology denoted here as "CP". In this example the native address of the
8384 display in "CP" is known in advance ("21.6.5.3.11", e.g.). Furthermore it is assumed that "CP" is a
8385 rather low-level protocol and the energy management gateway's interface to "CP" has to be
8386 configured in order to communicate with the display properly (configuration of profiles, e.g.). There
8387 is information available how to translate this configuration into "nativeSetup" properly. We assume
8388 the vendor of the energy management gateway provided a sophisticated user interface that
8389 simplifies this procedure for the user. Thus, the user just selects the technology interface to "CP",
8390 enters the native address and the label and then selects the right profile for a display for the
8391 interface to "CP". The user interface creates the subsequent XML and provides it to the Network
8392 Management Responsible of "CP":

```
8393 <networkManagementAddNodeCall>  
8394   <nativeSetup>21.6.5.3.11;R-2.7;at-42.344;pqtl-  
8395 27638162683172637812356813</nativeSetup>  
8396   <timeout>PT40S</timeout>  
8397   <label>Display</label>  
8398 </networkManagementAddNodeCall>
```

8399 In this example the *nodeAddress* is unset to let the Network Management Responsible of "CP" assign
8400 a proper address. Upon the successful completion of the process the communications technology
8401 interface provides a proper *networkManagementProcessStateData* notification to the user interface.
8402 It furthermore creates proper *networkManagementDeviceDescriptionListData*,
8403 *networkManagementEntityDescriptionListData* and *networkManagementFeatureDescriptionListData*
8404 notifications for internal purposes (information for further applications installed on the energy
8405 management gateway: notification of a new node).

8406

8407 **5.3.17.4 networkManagementRemoveNodeCall**8408 **5.3.17.4.1 General**

8409 This function can be used to remove a node from the own SPINE network.

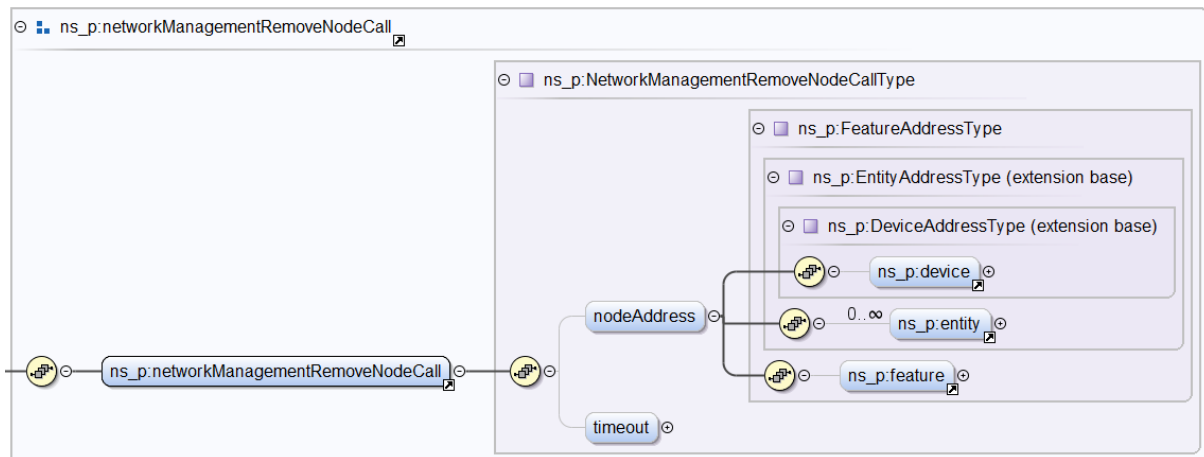


Figure 129: networkManagementRemoveNodeCall function overview

5.3.17.4.2 Detailed description of elements

Element	Type	Description
networkManagementRemoveNodeCall.		
nodeAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The node address of the node that will be removed. Contains elements for <i>device</i> , <i>entity</i> and <i>feature</i> .
nodeAddress.device	See section 3.10.3.6.	Device address part.
nodeAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.
nodeAddress.feature	See section 3.10.3.6.	Feature address part.
timeout	xs:duration (W3C standard type)	The procedure of removing the node will be aborted latest after this duration.

Table 257: networkManagementRemoveNodeCall function element description

5.3.17.4.3 Available selectors

None.

5.3.17.4.4 Examples

5.3.17.4.4.1 Call

We assume a display belongs to the SPINE network of an energy management gateway and is assigned the device address "d:_i:46925_CP_pqtl-27638162683172637812356813" (as a result of the example in 5.3.17.3.4, e.g.). The user interface of the energy management gateway permits removing the display from the gateway's network. The user selects the display on the user interface and requests its removal. The user interface creates the subsequent XML and provides it to the Network Management Responsible of this node:

```
<networkManagementRemoveNodeCall>
  <nodeAddress>
    <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
  </nodeAddress>
  <timeout>PT30S</timeout>
```

8432 </networkManagementRemoveNodeCall>

8433 Because the whole *device* will be removed (and not a specific *entity* or *feature*) only the *device*-part
8434 of the address is to be filled.

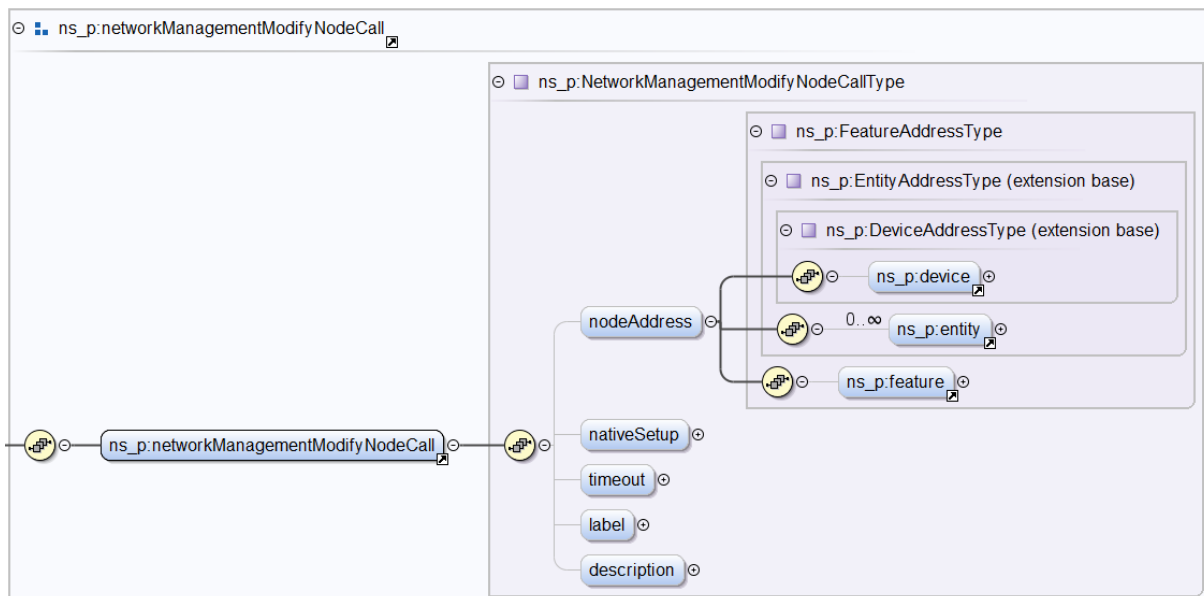
8435 Upon the successful completion of the process the Network Management Responsible provides a
8436 proper *networkManagementProcessStateData* notification to the user interface. It furthermore
8437 creates a proper *networkManagementDeviceDescriptionListData* notification for internal purposes
8438 (information for further applications installed on the energy management gateway: notification of a
8439 removed node).

8440

8441 5.3.17.5 *networkManagementModifyNodeCall*

8442 5.3.17.5.1 *General*

8443 This function can be used to modify an already "present" node of the own SPINE network. Here,
8444 "present" implies the node has a node address.



8445

8446 Figure 130: *networkManagementModifyNodeCall* function overview

8447

8448 5.3.17.5.2 *Detailed description of elements*

Element	Type	Description
networkManagementModifyNodeCall.		
nodeAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The node address of the node that will be modified. Contains elements for <i>device</i> , <i>entity</i> and <i>feature</i> .
nodeAddress.device	See section 3.10.3.6.	Device address part.
nodeAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.

nodeAddress.feature	See section 3.10.3.6.	Feature address part.
nativeSetup	Simple type "NetworkManagementNativeSetupType" (restriction of xs:string)	Similar to <i>nativeSetup</i> of <i>networkManagementAddNodeCall</i> . But for <i>networkManagementModifyNodeCall</i> all this element is not used to identify which node has to be modified (see element <i>nodeAddress</i> for this purpose).
timeout	xs:duration (W3C standard type)	The procedure of modifying the node will be aborted latest after this duration.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the node.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the node.

Table 258: networkManagementModifyNodeCall function element description

5.3.17.5.3 Available selectors

None.

5.3.17.5.4 Examples

5.3.17.5.4.1 Call

We assume a display belongs to the SPINE network of an energy management gateway and is assigned the device address "d:_i:46925_CP_pqtl-27638162683172637812356813" (as a result of the example in 5.3.17.3.4, e.g.). The user interface of the energy management gateway permits modifying the display's label. The user selects the display on the user interface and enters the new label "Kitchen display". The user interface creates the subsequent XML and provides it to its own interface to the Network Management Responsible of this node:

```
<networkManagementModifyNodeCall>
  <nodeAddress>
    <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
  </nodeAddress>
  <nativeSetup>21.6.5.3.11;R-2.7;at-42.344;pqtl-
27638162683172637812356813</nativeSetup>
  <timeout>PT15S</timeout>
  <label>Kitchen display</label>
  <description></description>
</networkManagementModifyNodeCall>
```

Because the label of the physical device will be changed (and not of a specific *entity* or *feature*) only the *device*-part of the address is to be filled.

Upon the successful completion of the process the Network Management Responsible provides a proper *networkManagementProcessStateData* notification to the user interface. It furthermore creates a proper *networkManagementDeviceDescriptionListData* notification for internal purposes

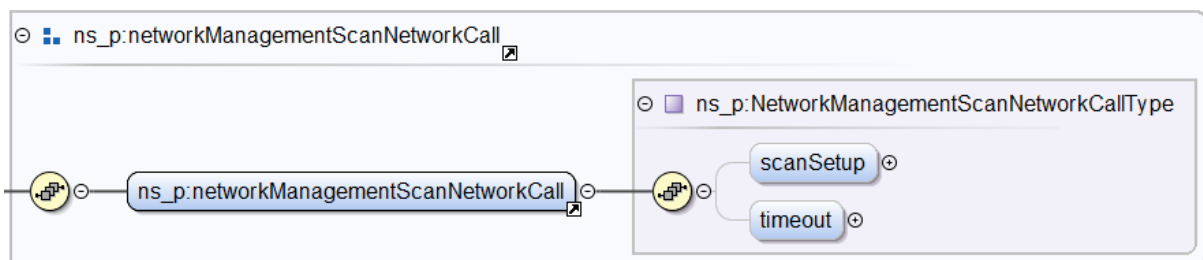
8477 (information for further applications installed on the energy management gateway: notification of a
8478 modified node).

8479

8480 **5.3.17.6 networkManagementScanNetworkCall**

8481 **5.3.17.6.1 General**

8482 Some communications technologies provide measures to scan which other devices are accessible.
8483 The function *networkManagementScanNetworkCall* can be used to trigger such an action, if
8484 supported by technology and implementation. This function is not intended to add discovered
8485 devices directly to the own SPINE network. Instead, it is intended just to get a report. This report can
8486 be modelled by *networkManagementReportCandidateData*. Of course, these reports may then be
8487 used to add discovered nodes.



8488

8489 *Figure 131: networkManagementScanNetworkCall function overview*

8490

8491 **5.3.17.6.2 Detailed description of elements**

Element	Type	Description
networkManagementScanNetworkCall.		
scanSetup	Simple type "NetworkManagementScanSetupType" (restriction of xs:string)	Technology dependent or implementation dependent configuration for the scanning procedure. This might contain parameters to search for specific types of devices, e.g.
timeout	xs:duration (W3C standard type)	The procedure of scanning will be aborted latest after this duration.

8492 *Table 259: networkManagementScanNetworkCall function element description*

8493

8494 **5.3.17.6.3 Available selectors**

8495 None.

8496

8497 **5.3.17.6.4 Examples**

8498 **5.3.17.6.4.1 Call**

8499 An energy management gateway includes (among others) a specific communications technology
8500 (denoted here as "CP"). The implementation provides scanning other devices. The user selects "CP"

8501 in the gateway's user interface and triggers "Scan". The user interface then creates the subsequent
 8502 XML and provides it to the Network Management Responsible of "CP":

```
8503 <networkManagementScanNetworkCall>
8504   <timeout>PT1M</timeout>
8505 </networkManagementScanNetworkCall>
```

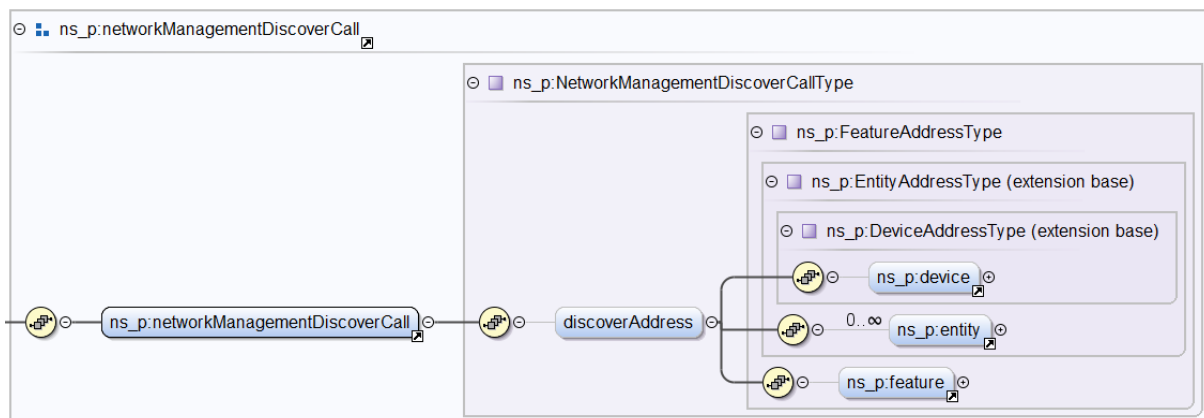
8506 In this example "CP" is assumed to not require "scanSetup". During the scan the Network
 8507 Management Responsible reports discovered devices as described above. Upon the successful
 8508 completion of the process the Network Management Responsible also provides a proper
 8509 *networkManagementProcessStateData* notification to the user interface.

8510

8511 5.3.17.7 *networkManagementDiscoverCall*

8512 5.3.17.7.1 *General*

8513 If a node changed its functionality, it can announce the changes with all details to all nodes
 8514 interested in that information. This approach can generate a lot of unneeded traffic as not every
 8515 recipient requires knowledge of all changes. Hence, the *networkManagementDiscoverCall* can be
 8516 sent instead to those nodes to tell them, which address to discover (again). Then the nodes can still
 8517 decide to query for details on the changes.



8518

8519 Figure 132: *networkManagementDiscoverCall* function overview

8520

8521 5.3.17.7.2 *Detailed description of elements*

Element	Type	Description
networkManagementDiscoverCall.		
discoverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The SPINE address of the node that will be discovered in detail. Contains elements for device, entity and feature.
discoverAddress.device	See section 3.10.3.6.	Device address part.
discoverAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.
discoverAddress.feature	See section 3.10.3.6.	Feature address part.

8522 Table 260: *networkManagementDiscoverCall* function element description

8523

8524 5.3.17.7.3 Available selectors

8525 None.

8526

8527 5.3.17.7.4 Examples

8528 5.3.17.7.4.1 Call

8529 If the functionality of a node has changed, it can send a *networkManagementDiscoverCall* to ask
 8530 other nodes to discover the changed node:

```

8531 <networkManagementDiscoverCall>
8532   <discoverAddress>
8533     <device>d: i:46925_CP\_pqt1-27638162683172637812356813</device>
8534     <entity>3</entity>
8535   </discoverAddress>
8536 </networkManagementDiscoverCall>

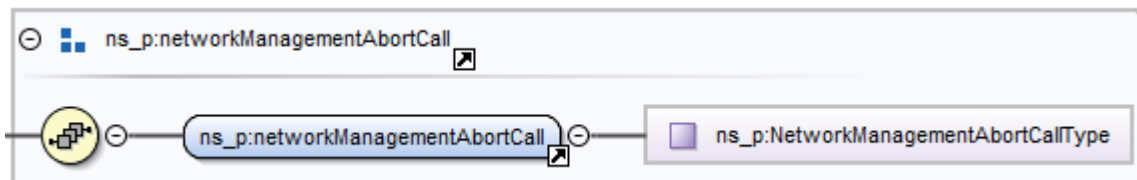
```

8537

8538 5.3.17.8 networkManagementAbortCall

8539 5.3.17.8.1 General

8540 This function can be used to model the request to abort all ongoing processes triggered by
 8541 *networkManagementAddNodeCall*, *networkManagementRemoveNodeCall*,
 8542 *networkManagementModifyNodeCall*, *networkManagementScanNetworkCall* or
 8543 *networkManagementDiscoverCall*.



8544

8545 Figure 133: networkManagementAbortCall function overview

8546

8547 5.3.17.8.2 Detailed description of elements

8548 In this version of the specification the function *networkManagementAbortCall* does not contain child
 8549 elements.

8550

8551 5.3.17.8.3 Available selectors

8552 None.

8553

8554 5.3.17.8.4 Examples

8555 5.3.17.8.4.1 Call

8556 We assume a user triggered a scan of the communications technology "CP" as described in
 8557 5.3.17.6.4. Furthermore, we assume the implementation provides the possibility to abort this

process. 25 seconds after the user initiated the scan he decides to abort the process and selects the "Abort" button on the gateway's user interface. The user interface then creates the subsequent XML and provides it to the Network Management Responsible of "CP":

```
<networkManagementAbortCall/>
```

Upon the abortion of the process the Network Management Responsible provides a proper *networkManagementProcessStateData* notification to the user interface.

5.3.17.9 *networkManagement...Data*

The *networkManagement...Data* functions inform about the process of actions triggered with the aforementioned calls, deliver or change the current joining mode or report new candidates (found via the *networkManagementScanNetworkCall*). The device, entity and feature descriptions deliver detailed information about nodes (in the meaning of a node-graph including all devices, entities and features) and specific information about entities and features.

5.3.17.10 *networkManagementProcessStateData*

5.3.17.10.1 *General*

This function can be used to model the state of a process triggered by *networkManagementAddNodeCall*, *networkManagementRemoveNodeCall*, *networkManagementModifyNodeCall*, *networkManagementScanNetworkCall* or *networkManagementDiscoverCall*.

The underlying revision of the SPINE data model supports only notification of this function (i.e. read, write, or reply mechanisms are not supported).

Please note this function does not support the identification of a specific process. This should be considered carefully in case an implementation supports parallel processes.

It is advised to notify an instance of *networkManagementProcessStateData* only to the originator of the related network management process. Considering the architecture example of section 5.3.17.1.2 this would mean only the software application that triggered a certain process would be notified by the "Network Management Responsible".

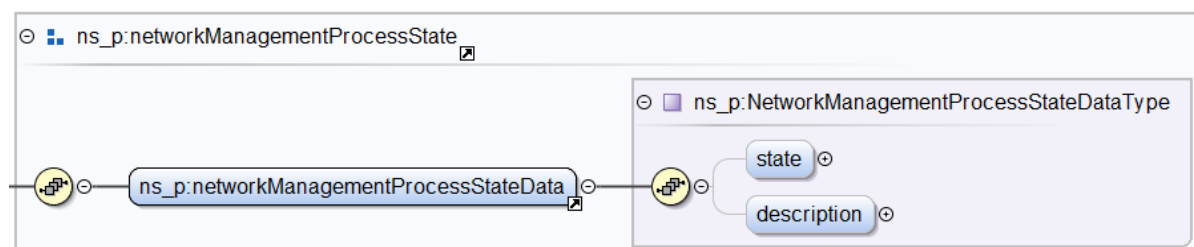


Figure 134: *networkManagementProcessStateData* function overview

8589 5.3.17.10.2 Detailed description of elements

Element	Type	Description
networkManagementProcessStateData.		
state	Enum (see Table 262): NetworkManagementProcessStateType	The state of the process.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information.

8590 Table 261: networkManagementProcessStateData function element description

8591 Enumeration **NetworkManagementProcessStateType**:

Value	Description
succeeded	The process completed successfully.
failed	The process failed.
aborted	The process was aborted.

8592 Table 262: Enumeration NetworkManagementProcessStateType

8593

8594 5.3.17.10.3 Available selectors

8595 None.

8596

8597 5.3.17.10.4 Examples

8598 5.3.17.10.4.1 Notify

8599 Section 5.3.17.3.4 described the creation of a *networkManagementProcessStateData* notification as
 8600 a result of the process of adding a display. This XML could look like this:

```

8601 <networkManagementProcessStateData>
8602     <state>succeeded</state>
8603     <description>Added node.</description>
8604 </networkManagementProcessStateData>

```

8605

8606 **5.3.17.11 networkManagementJoiningModeData**

8607 5.3.17.11.1 General

8608 Some communications technologies or implementations permit a temporary or permanent
 8609 acceptance of other devices or at least a report of discovery. I.e. upon an incoming joining request of
 8610 another device the local device may add it automatically to its list of accepted devices. In terms of
 8611 the SPINE network management this means a local device would be added automatically to the own
 8612 SPINE network. As mentioned before, the new device also might just be reported instead of
 8613 automatically added. The function *networkManagementJoiningModeData* permits to model the
 8614 configuration of the local node's joining behaviour.

8615 Please note joining does not apply to a "scan network" process (see 5.3.17.6). But an "add node"
 8616 process initiated on another device might lead to an incoming joining request on the local device,
 8617 e.g. Depending on the joining mode configuration this request can be skipped (rejected), reported
 8618 (see 5.3.17.12) or automatically added (see 5.3.17.13.4.2).

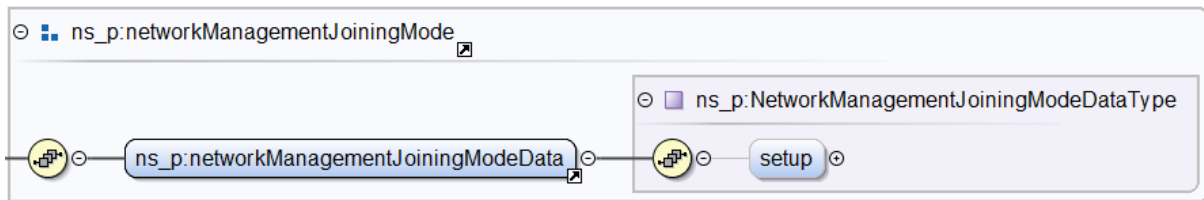


Figure 135: networkManagementJoiningModeData function overview

5.3.17.11.2 Detailed description of elements

Element	Type	Description
networkManagementJoiningModeData.		
setup	Simple type "NetworkManagementSetupType" (restriction of xs:string)	Technology dependent or implementation dependent configuration of the joining behaviour. In addition to the kind of joining procedure this could also contain parameters for a temporary enabling only, acceptance of specific devices only, etc., e.g.

Table 263: networkManagementJoiningModeData function element description

5.3.17.11.3 Available selectors

None.

5.3.17.11.4 Examples

5.3.17.11.4.1 Write

An energy management gateway includes (among others) a specific communications technology (denoted here as "CP"). The implementation provides so-called pairing with another device. In this example it will not be discussed which kind of joining procedure applies (adding the new device or just reporting it). But we assume "CP" requires some proprietary parameters.

The user chooses "CP" in the gateway's user interface and selects "Enable pairing". The user interface then creates the subsequent XML and provides it together with a *write* classifier to the Network Management Responsible of "CP" which configures its joining behaviour:

```
<networkManagementJoiningModeData>
  <setup>pairingOn;T=40;v=130.255</setup>
</networkManagementJoiningModeData>
```

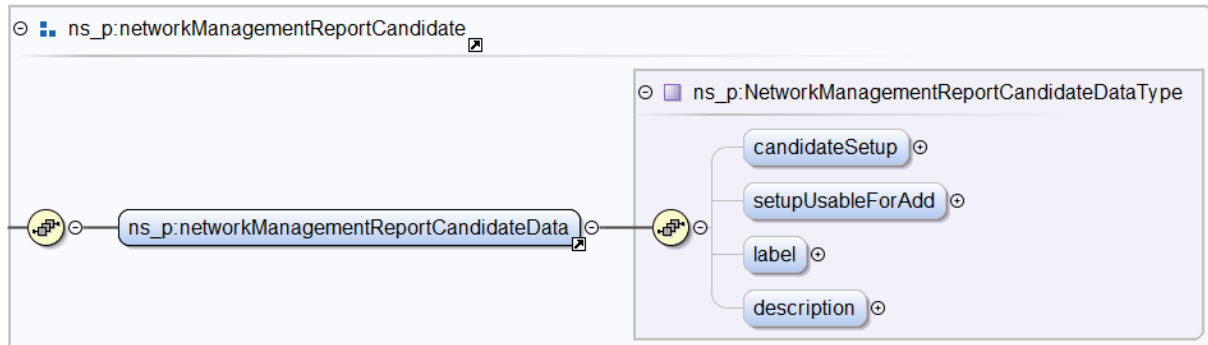
5.3.17.12 networkManagementReportCandidateData

5.3.17.12.1 General

As already described in 5.3.17.6.1 and 5.3.17.11.1 some processes might require the report of a potential (but not yet accepted/added) device. The function *networkManagementReportCandidateData* can be used to model such a report.

8646 Please note the reported data strongly depends on the implementation and possibilities of the
8647 related communications technology.

8648 The underlying revision of the SPINE data model supports only notifications of this function (i.e. read,
8649 write and reply mechanisms are not supported).



8650

8651 Figure 136: networkManagementReportCandidateData function overview

8652

8653 5.3.17.12.2 Detailed description of elements

Element	Type	Description
networkManagementReportCandidateData.		
candidateSetup	Simple type "NetworkManagementCandidateSetupType" (restriction of xs:string)	Technology dependent or implementation dependent information on the reported node. This element is intended to provide information usable for an "add node" process. Whether its content could be copied unchanged into an element <i>nativeSetup</i> of <i>networkManagementAddNodeCall</i> depends on the element <i>setupUsableForAdd</i> .
setupUsableForAdd	xs:boolean (W3C standard type)	Only if this element is set to <i>true</i> the content of <i>candidateSetup</i> could be used for <i>nativeSetup</i> of <i>networkManagementAddNodeCall</i> .
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the reported node.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the reported node.

8654 Table 264: networkManagementReportCandidateData function element description

8655

8656 5.3.17.12.3 Available selectors

8657 None.

8658

5.3.17.12.4 Examples

5.3.17.12.4.1 Notify

We assume the "scan network" example of 5.3.17.6.4 occurs on a device with implementation and communications technology capable of providing sufficient details of reported devices. Among others, the Network Management Responsible provides this XML as notification to the user interface due to a discovered device:

```
<networkManagementReportCandidateData>
  <candidateSetup>21.6.5.3.11;R-2.7;at-42.344;pqtl-
27638162683172637812356813</candidateSetup>
  <setupUsableForAdd>true</setupUsableForAdd>
  <label>Display</label>
  <description>Device type: Display/Generic</description>
</networkManagementReportCandidateData>
```

5.3.17.13 networkManagementDeviceDescriptionListData

5.3.17.13.1 General

This function can be used to model SPINE network management relevant detailed information about the "device level" of a node.

It is recommended to notify an instance of *networkManagementDeviceDescriptionListData* to all "parts" requiring knowledge of "any" added/removed/modified device. Considering the architecture example of section 5.3.17.1.2 this could mean all those software applications that want to react dynamically on such changes could be notified by the "Network Management Responsible". Thus, each of these applications could verify whether the reported change affects its own tasks.

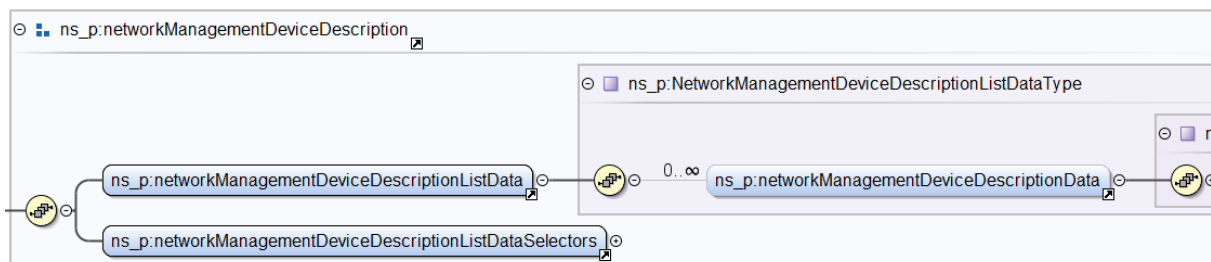


Figure 137: networkManagementDeviceDescriptionListData function overview, part 1

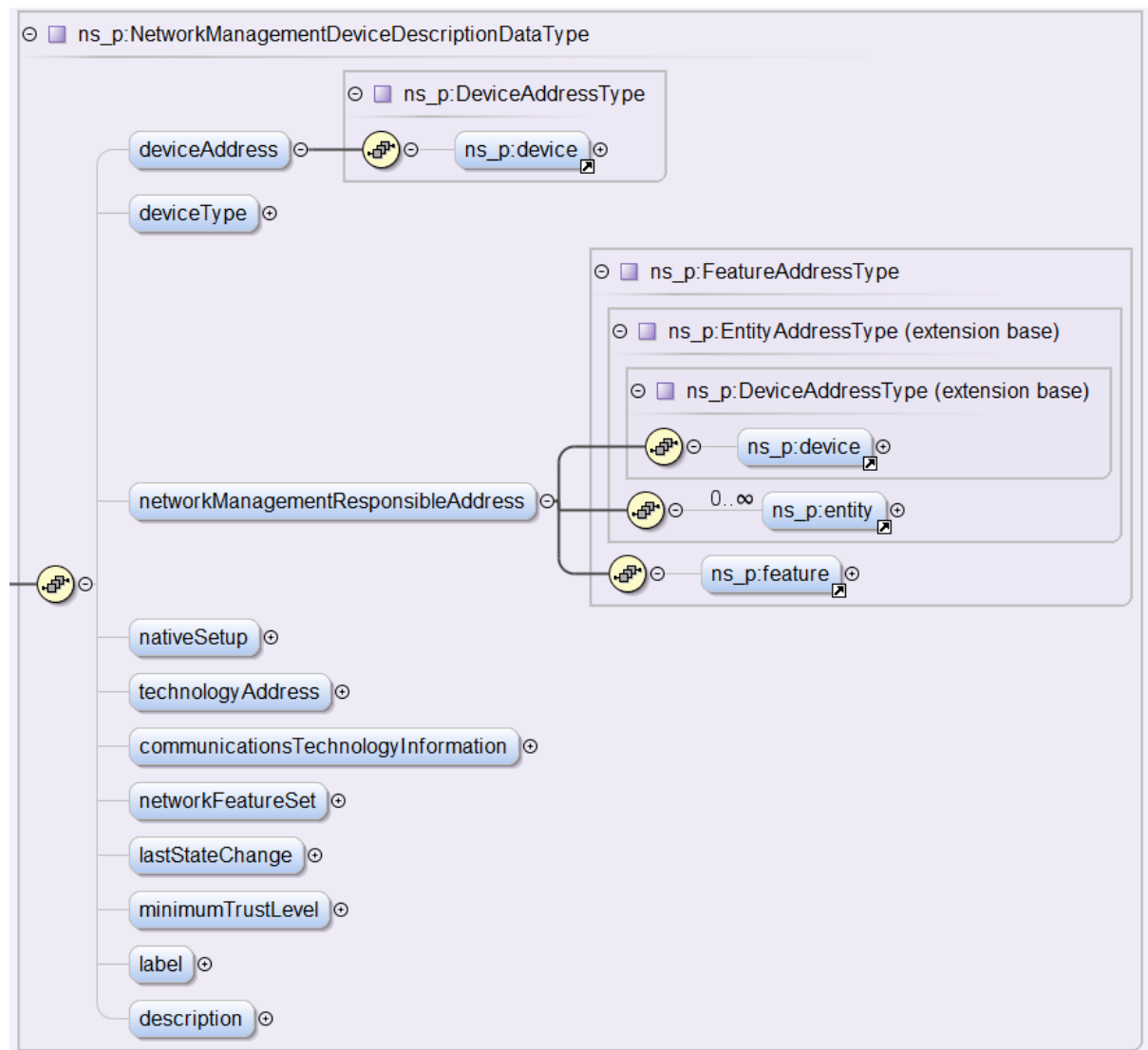


Figure 138: networkManagementDeviceDescriptionListData function overview, part 2

5.3.17.13.2 Detailed description of elements

Element	Type	Description
networkManagementDeviceDescriptionListData. networkManagementDeviceDescriptionData. (list)		
deviceAddress	Common data type "DeviceAddressType". See section 3.10.3.4.	The node address of the node. Contains elements for <i>device</i> .
deviceAddress.device	See section 3.10.3.4.	Device address part.
deviceType	Common data type "DeviceTypeType". See section 3.10.1.25.	The type of the device. E.g. Washer, Dryer, ...
networkManagementResponsibleAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The address of the so-called "Network Management Responsible" (see section 5.3.17.1.2) of <i>deviceAddress</i> . Contains elements for <i>device</i> , <i>entity</i> and <i>feature</i> .
networkManagementResponsibleAddress.device	See section 3.10.3.6.	Device address part.

networkManagementResponseAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.
networkManagementResponseAddress.feature	See section 3.10.3.6.	Feature address part.
nativeSetup	Simple type "NetworkManagementNativeSetupType" (restriction of xs:string)	Technology dependent or implementation dependent configuration to identify (and maybe configure) the device.
technologyAddress	Simple type "NetworkManagementTechnologyAddressType" (restriction of xs:string)	The address the device has in its own communications technology.
communicationsTechnologyInformation	Simple type "NetworkManagementCommunicationsTechnologyInformationType" (restriction of xs:string)	Technology dependent or implementation dependent information on the device with focus on communications aspects. <i>communicationsTechnologyInformation</i> is intended to be rather "read only" (but may change dependent on configuration, e.g.).
networkFeatureSet	Enum (see Table 266): NetworkManagementFeatureSetType	Specifies which network capabilities the device has (simple, smart, router, ...).
lastStateChange	Enum (see Table 267): NetworkManagementStateChangeType	This element can contain the last change of its network management state (<i>added, removed, etc.</i>).
minimumTrustLevel	Simple type "NetworkManagementMinimumTrustLevelType" (restriction of xs:string)	Contains the lowest trust level needed, to be allowed to communicate with this node.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the device.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the device.

8688 Table 265: networkManagementDeviceDescriptionListData function element description

8689 Enumeration **NetworkManagementFeatureSetType**:

Value	Description
gateway	Connects to devices with another communications technology; keeps a list of its "attached devices" and provides routing to them.
router	Reserved for future use.
smart	Keeps a list of devices of interest for its communication and may permit routing to them.
simple	Can only communicate with directly reachable devices (i.e. no communication via an intermediate SPINE device). Does not provide routing to other devices.

8690 Table 266: Enumeration NetworkManagementFeatureSetType

8691 Enumeration **NetworkManagementStateChangeType**:

Value	Description
added	The device was added. See also 5.3.17.3.
removed	The device was removed. See also 5.3.17.4.
modified	The device was modified. See also 5.3.17.5.

8692 *Table 267: Enumeration NetworkManagementStateChangeType*

8693

8694 *5.3.17.13.3 Available selectors*

8695 - deviceAddress.device

8696

8697 *5.3.17.13.4 Examples*

8698 *5.3.17.13.4.1 Notify (change of a "FridgeFreezer")*

8699 After the information of a device has changed, the updated device information is notified to all
 8700 subscribed instances (the "filter/partial" part to announce the "restricted function exchange" is
 8701 simplified by "..." just to improve the readability):

```

8702 ...
8703 <networkManagementDeviceDescriptionListData>
8704   <networkManagementDeviceDescriptionData>
8705     <deviceAddress>
8706       <device>d:_i:46925_CP\_irto-74629867356210956382438562</device>
8707     </deviceAddress>
8708     <deviceType>FridgeFreezer</deviceType>
8709     <technologyAddress>21.6.5.3.11</technologyAddress>
8710 <communicationsTechnologyInformation>CP;addr=21.6.5.3.11;shortAddr=114</com
8711 municationsTechnologyInformation>
8712   <networkFeatureSet>smart</networkFeatureSet>
8713 </networkManagementDeviceDescriptionData>
8714 </networkManagementDeviceDescriptionListData>

```

8715

8716 *5.3.17.13.4.2 Notify (added "Display")*

8717 We assume a user triggered adding a display as described in section 5.3.17.3.4. That section
 8718 mentions the notification of a proper *networkManagementDeviceDescriptionListData* instance. In
 8719 this example it is assumed the Network Management Responsible of "CP" has the device address
 8720 "d:_i:46925_CP_A873C6BC09EF4E50" and the display has been assigned the node address
 8721 "d:_i:46925_CP_pqtl-27638162683172637812356813". Thus, the subsequent XML could be created
 8722 (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..." just to
 8723 improve the readability):

```

8724 ...
8725 <networkManagementDeviceDescriptionListData>
8726   <networkManagementDeviceDescriptionData>
8727     <deviceAddress>
8728       <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
8729     </deviceAddress>
8730     <deviceType>Generic</deviceType>
8731     <networkManagementResponsibleAddress>
8732       <device>d:_i:46925_CP\_A873C6BC09EF4E50</device>

```

```

8733         <entity>0</entity>
8734     </networkManagementResponsibleAddress>
8735     <nativeSetup>21.6.5.3.11;R-2.7;at-42.344;pqtl-
8736 27638162683172637812356813</nativeSetup>
8737     <lastStateChange>added</lastStateChange>
8738     <minimumTrustLevel>8</minimumTrustLevel>
8739     <label>Display</label>
8740 </networkManagementDeviceDescriptionData>
8741 </networkManagementDeviceDescriptionListData>

```

8742

8743 5.3.17.13.4.3 Read-reply (from gateway with full device list information)

8744 We assume a user added a display as described in sections 5.3.17.3.4 and 5.3.17.13.4.2. In this
8745 example, "CP" provides a resource with aggregated information on all "device description details" of
8746 its attached devices and itself. We also assume only the display is currently attached via "CP". The
8747 user wants to see the list of devices from the described resource. He selects this option on the user
8748 interface. The user interface then creates a proper request to the Network Management
8749 Responsible's aggregating resource of "CP" and finally gets a proper response where the Network
8750 Management Responsible itself and the display are described:

```

8751 <networkManagementDeviceDescriptionListData>
8752   <networkManagementDeviceDescriptionData>
8753     <deviceAddress>
8754       <device>d:_i:46925_CP\_A873C6BC09EF4E50</device>
8755     </deviceAddress>
8756     <networkFeatureSet>gateway</networkFeatureSet>
8757     <lastStateChange>modified</lastStateChange>
8758     <minimumTrustLevel>8</minimumTrustLevel>
8759     <label>CP mapping</label>
8760   </networkManagementDeviceDescriptionData>
8761   <networkManagementDeviceDescriptionData>
8762     <deviceAddress>
8763       <device>d:_i:46925_CP\_pqtl-27638162683172637812356813</device>
8764     </deviceAddress>
8765     <deviceType>Generic</deviceType>
8766     <networkManagementResponsibleAddress>
8767       <device>d:_i:46925_CP\_A873C6BC09EF4E50</device>
8768       <entity>0</entity>
8769     </networkManagementResponsibleAddress>
8770     <nativeSetup>21.6.5.3.11;R-2.7;at-42.344;pqtl-
8771 27638162683172637812356813</nativeSetup>
8772     <lastStateChange>added</lastStateChange>
8773     <minimumTrustLevel>8</minimumTrustLevel>
8774     <label>Display</label>
8775   </networkManagementDeviceDescriptionData>
8776 </networkManagementDeviceDescriptionListData>

```

8777

8778 5.3.17.14 networkManagementEntityDescriptionListData

8779 5.3.17.14.1 General

8780 In addition to the *networkManagementDeviceDescriptionListData* function, this functions can be
8781 used to model SPINE network management relevant detailed information about an "entity level" of a
8782 node.

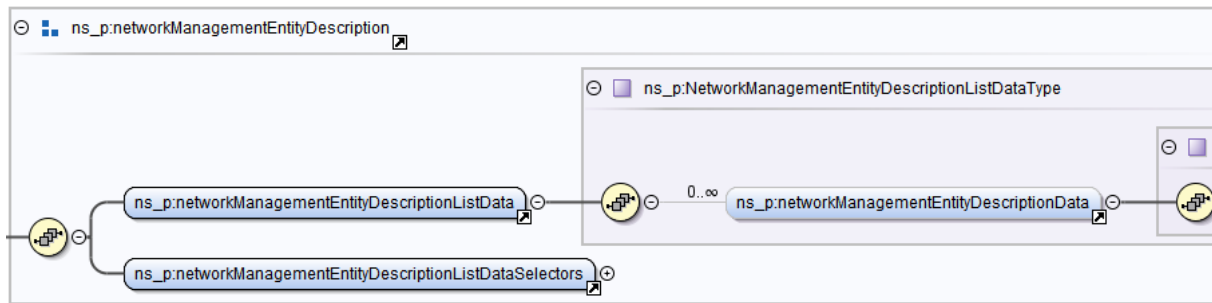


Figure 139: networkManagementEntityDescriptionListData function overview, part 1

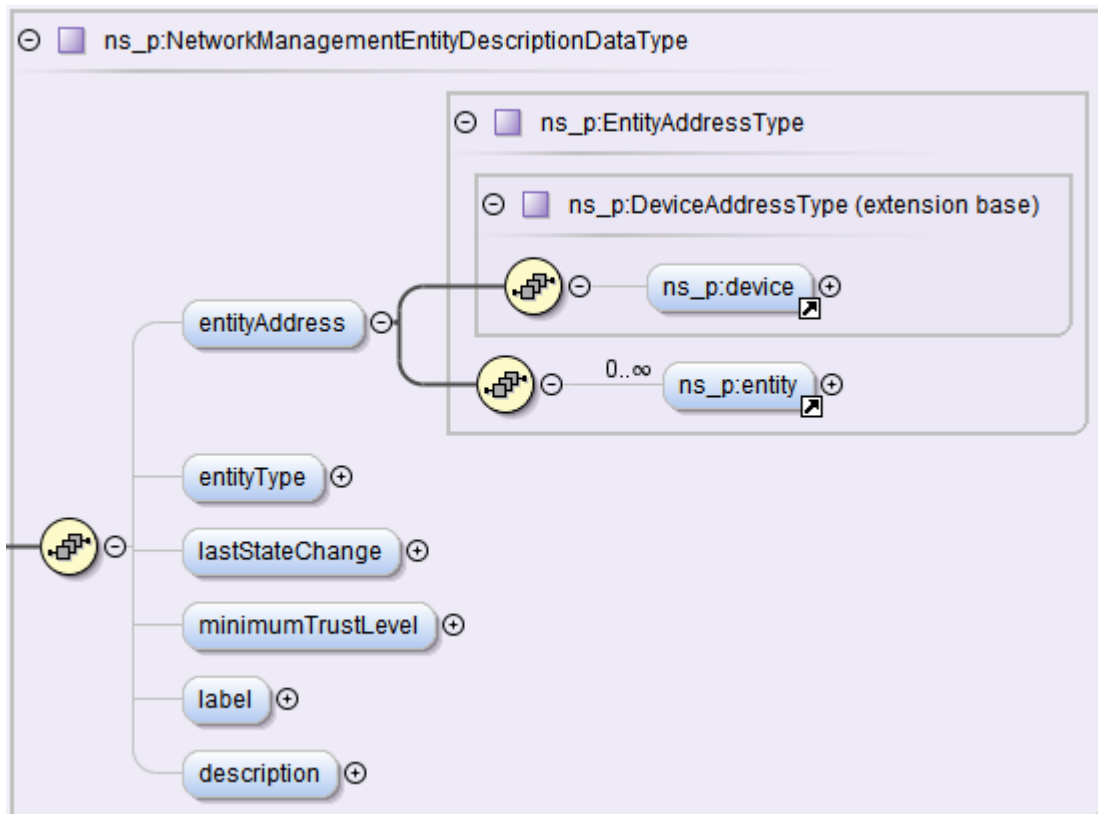


Figure 140: networkManagementEntityDescriptionListData function overview, part 2

5.3.17.14.2 Detailed description of elements

Element	Type	Description
networkManagementEntityDescriptionListData. networkManagementEntityDescriptionData. (list)		
entityAddress	Common data type "EntityType". See section 3.10.3.5.	The node address of the node. Contains elements for <i>device</i> and <i>entity</i> .
entityAddress.device	See section 3.10.3.5.	Device address part.
entityAddress.entity (list)	See section 3.10.3.5.	Entity (list) address part.
entityType	Common data type "EntityType". See section 3.10.1.27.	The type of the entity. E.g. Battery, Fan, ...
lastStateChange	Enum (see Table 267): NetworkManagement StateChangeType	This element can contain the last change of its network management state (<i>added</i> , <i>removed</i> , etc.).

minimumTrustLevel	<i>Simple type "NetworkManagementMinimumTrustLevelType" (restriction of xs:string)</i>	Contains the lowest trust level needed, to be allowed to communicate with this node.
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A short label of the entity.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Descriptive information on the entity.

Table 268: networkManagementEntityDescriptionListData function element description

8789

8790

8791 5.3.17.14.3 Available selectors

8792 - entityAddress

8793

8794 5.3.17.14.4 Examples

8795 5.3.17.14.4.1 Notify

8796 After updating information on its entities, a device notifies the corresponding information to all
8797 subscribed instances (the "filter/partial" part to announce the "restricted function exchange" is
8798 simplified by "... " just to improve the readability):

```

8799 ...
8800 <networkManagementEntityDescriptionListData>
8801   <networkManagementEntityDescriptionData>
8802     <nodeAddress>
8803       <device>d: i:46925_CP\_irto-74629867356210956382438562</device>
8804       <entity>1</entity>
8805     </nodeAddress>
8806     <entityType>Fridge</entityType>
8807     <label>Big fridge</label>
8808   </networkManagementEntityDescriptionData>
8809   <networkManagementEntityDescriptionData>
8810     <nodeAddress>
8811       <device>d: i:46925_CP\_irto-74629867356210956382438562</device>
8812       <entity>2</entity>
8813     </nodeAddress>
8814     <entityType>Freezer</entityType>
8815     <label>Small freezer compartment</label>
8816   </networkManagementEntityDescriptionData>
8817 </networkManagementEntityDescriptionListData>

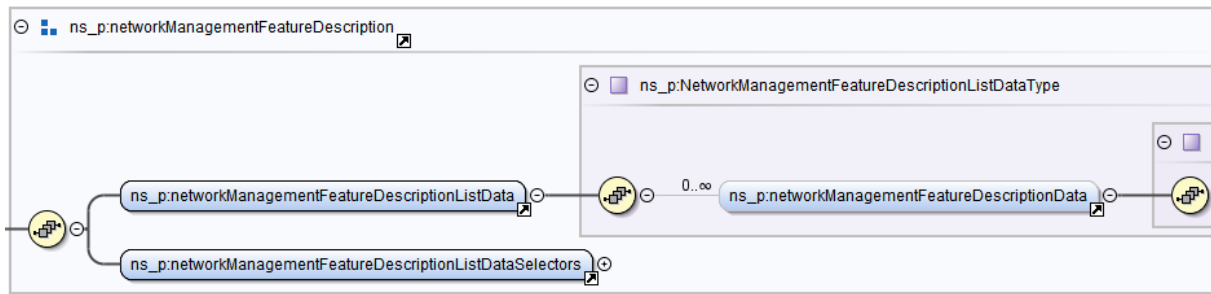
```

8818

8819 5.3.17.15 networkManagementFeatureDescriptionListData

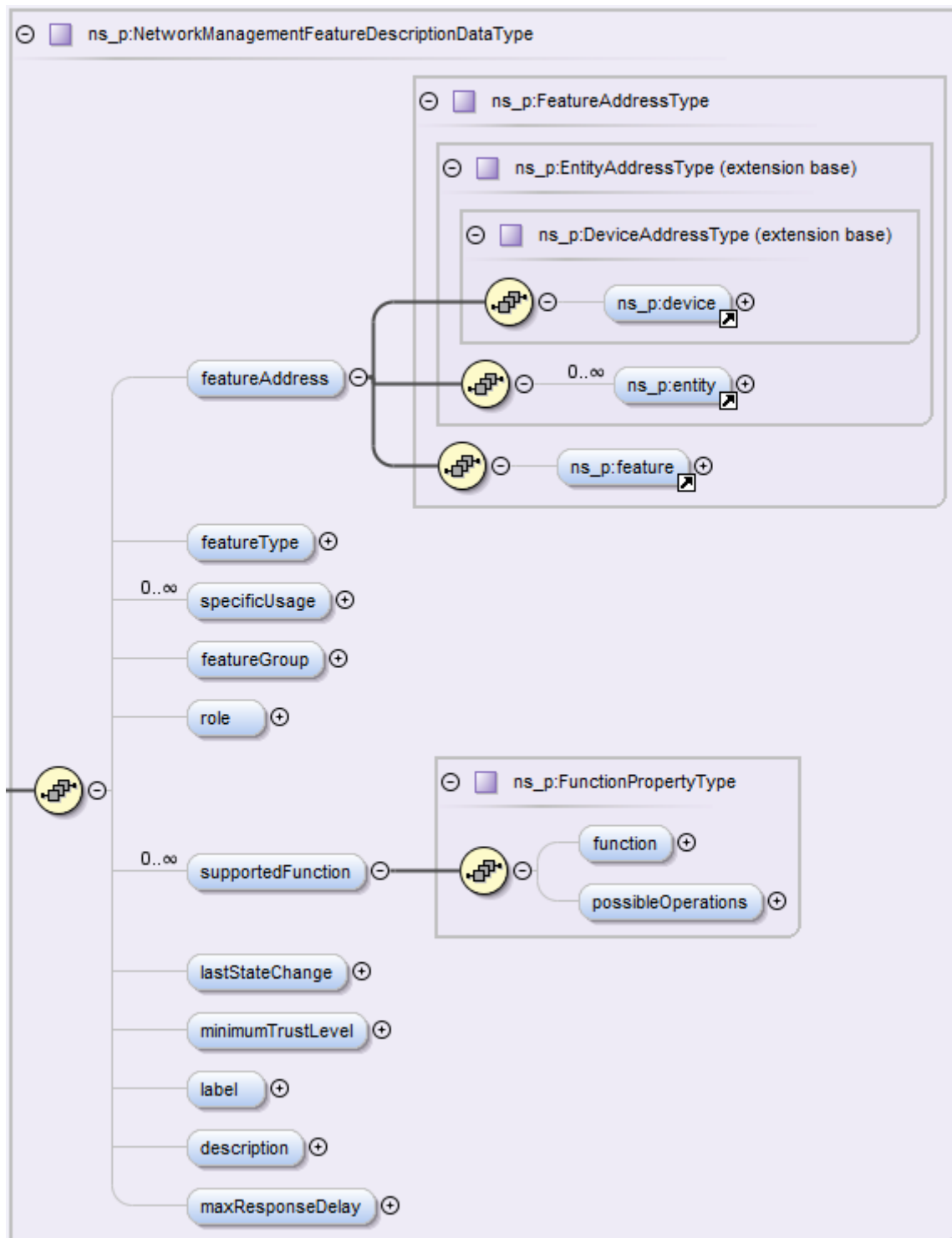
8820 5.3.17.15.1 General

8821 In addition to the *networkManagementDeviceDescriptionListData* function, this functions adds
8822 feature-specific information which is only needed for nodes on address hierarchy level *feature*.



8823

8824 *Figure 141: networkManagementFeatureDescriptionListData function overview, part 1*



8825

8826 *Figure 142: networkManagementFeatureDescriptionListData function overview, part 2*

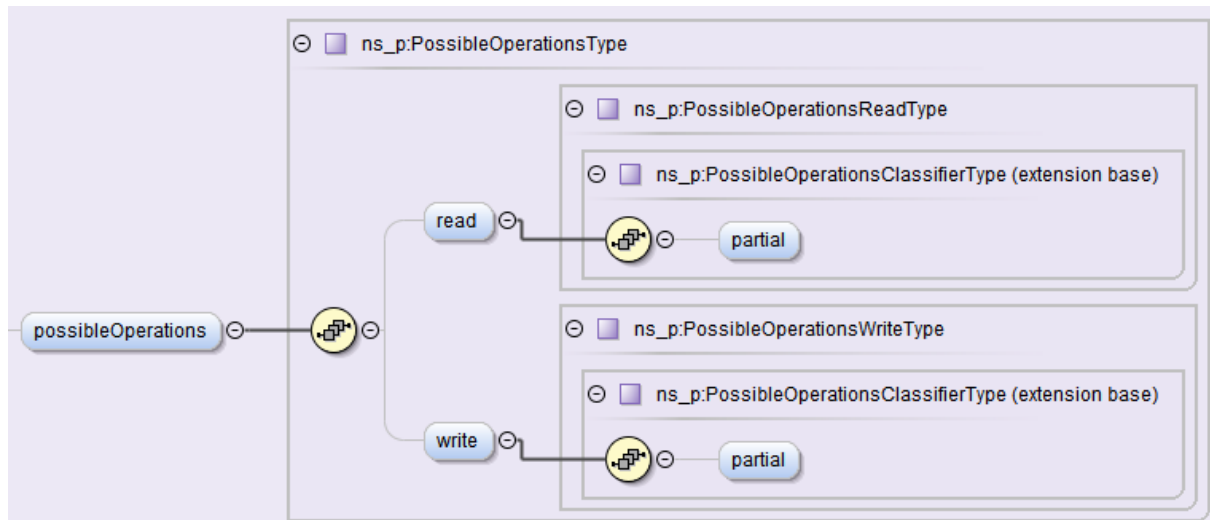


Figure 143: networkManagementFeatureDescriptionListData function overview, part 3

5.3.17.15.2 Detailed description of elements

Element	Type	Description
networkManagementFeatureDescriptionListData. networkManagementFeatureDescriptionData. (list)		
featureAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	The node address of the node. Contains elements for <i>device</i> , <i>entity</i> and <i>feature</i> .
featureAddress.device	See section 3.10.3.6.	Device address part.
featureAddress.entity (list)	See section 3.10.3.6.	Entity (list) address part.
featureAddress.feature	See section 3.10.3.6.	Feature address part.
featureType	Common data type "FeatureTypeType". See section 3.10.1.29.	The type of the feature. E.g. ActuatorSwitch, Sensing, ...
specificUsage (list)	Common data type "FeatureSpecificUsageType". See section 3.10.1.31.	Each occurrence of "specificUsage" denotes a specific usage (see section 2.1.5) that is implemented for this Feature Type. Please consider the proper definition of "featureType" for possible values.
featureGroup	Common data type "FeatureGroupType". See section 3.10.1.24.	Information to "link" to related features of this entity. See section 2.1.4 for details.
role	Common data type "RoleType". See section 3.10.1.23.	The functional role of this feature (i.e. "client" or "server" or "special").
supportedFunction (list)	Common data type "FunctionPropertyType". See section 3.10.1.41.	Each occurrence contains information on a function that is supported on this feature with the given role. Note: This element is most useful for features with the role "server" or "special" to express which functionality is offered for clients. But it can also be

		used if the role is "client" in order to express client functionality.
supportedFunction.function	See section 3.10.1.41.	Contains the name of a supported function.
supportedFunction.possibleOperations	See section 3.10.1.41.	Can be used to express some details on operations supported by the given function (which also means the element "function" has to be set properly) if the role is "server" or "special".
supportedFunction.possibleOperations.read	See section 3.10.1.41.	If present, it denotes the feature can receive "read" commands for the given function and creates proper replies.
supportedFunction.possibleOperations.read.partial	See section 3.10.1.41.	If present, it denotes the feature permits the creation of somehow "curtailed" responses (in contrast to full copies of its function instance).
supportedFunction.possibleOperations.write	See section 3.10.1.41.	If present, it denotes the feature can receive "write" commands for the given function and adjusts its own function instance accordingly.
supportedFunction.possibleOperations.write.partial	See section 3.10.1.41.	If present, it denotes the feature accepts also somehow "curtailed" "write" commands (in contrast to a full replacement of its own data).
lastStateChange	Enum (see Table 267): NetworkManagementStateChangeType	This element can contain the last change of its network management state (<i>added</i> , <i>removed</i> , etc.).
minimumTrustLevel	Simple type "NetworkManagementMinimumTrustLevelType" (restriction of xs:string)	Contains the lowest trust level needed, to be allowed to communicate with this node.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the feature.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the feature.
maxResponseDelay	Common data type "MaxResponseDelayType". See section 3.10.1.42.	The duration the Server usually needs to generate a proper reply.

Table 269: networkManagementFeatureDescriptionListData function element description

5.3.17.15.3 Available selectors
- featureAddress

8836 **5.3.17.15.4 Examples**8837 **5.3.17.15.4.1 Notify**

8838 After the functionality of an entity has changed, the new feature information is notified to all bound
 8839 instances (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..."
 8840 just to improve the readability):

```

8841 ...
8842 <networkManagementFeatureDescriptionListData>
8843   <networkManagementFeatureDescriptionData>
8844     <featureAddress>
8845       <device>d:_i:46925_CP\_pctl-27638162683172637812356813</device>
8846       <entity>1</entity>
8847       <feature>1</feature>
8848     </featureAddress>
8849     <featureType>ActuatorLevel</featureType>
8850     <role>server</role>
8851     <supportedFunction>
8852       <function>actuatorLevelData</function>
8853       <possibleOperations>
8854         <read/>
8855         <write/>
8856       </possibleOperations>
8857     </supportedFunction>
8858     <supportedFunction>
8859       <function>actuatorLevelDescriptionData</function>
8860       <possibleOperations>
8861         <read/>
8862       </possibleOperations>
8863     </supportedFunction>
8864     <lastStateChange>modified</lastStateChange>
8865     <label>Illumination</label>
8866     <maxResponseDelay>PT20S</maxResponseDelay>
8867   </networkManagementFeatureDescriptionData>
8868 </networkManagementFeatureDescriptionListData>

```

8869

8870 **5.3.18 OperatingConstraints**8871 **5.3.18.1 Introduction**

8872 This class is designed to model some constraints of an appliance (or functionality) with regards to its
 8873 energy consumption or production. I.e. this class is NOT intended to control a device. Rather, it
 8874 enables an appliance to express what needs to be considered especially for the case it permits being
 8875 controlled by another device. An example is a central energy management system that uses the class
 8876 DirectControl to control the appliance; in this case the energy management system needs to be
 8877 aware of the appliance's operating constraints.

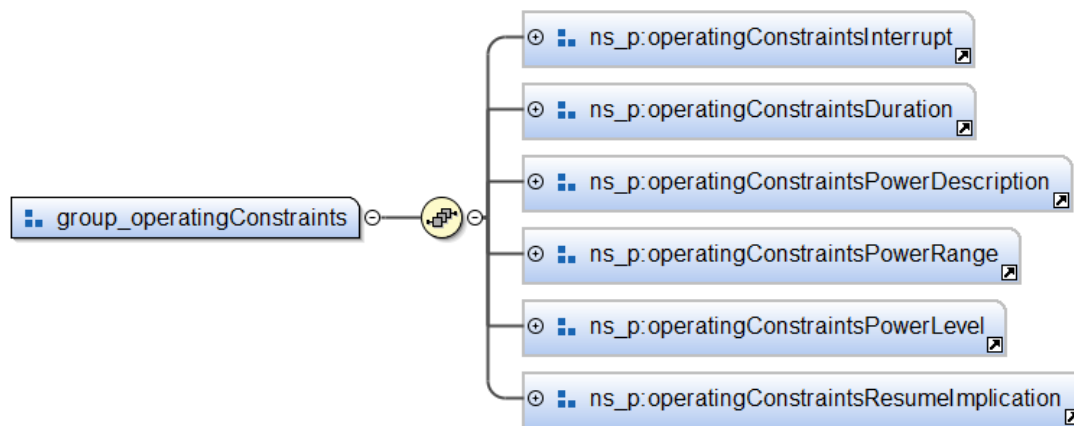


Figure 144: OperatingConstraints function-group overview

In general the class OperatingConstraints can be used with an without relation to the class PowerSequences. As can be seen in the following functions this is usually achieved with the (optional) element "sequenceld". In fact, most functions of this class are list-based, hence provide a possibility to model also constraints per power sequence.

Without a relation to a power sequence the OperatingConstraints functions usually relate to "now", i.e. the current state of the functionality. With the relation to a power sequence the corresponding constraints apply for the time of the related power sequence. Care should be taken if some constraints depend on the power sequence and other constraints are rather "global", i.e. independent from a power sequence. In every case the use of a feature with such a function should be clearly defined in order to avoid ambiguities. The following brief recommendations may be helpful to achieve this:

1. Avoid a mixture of operatingConstraintsInterruptData items with and without sequenceld within operatingConstraintsInterruptListData.
2. Within operatingConstraintsInterruptListData NO child element of operatingConstraintsInterruptData should depend on sequenceld in some operatingConstraintsInterruptData items and be independent of sequenceld in other operatingConstraintsInterruptData items.

5.3.18.2 operatingConstraintsInterruptListData

5.3.18.2.1 General

This function models basic constraints for an interrupt of a device's current task.

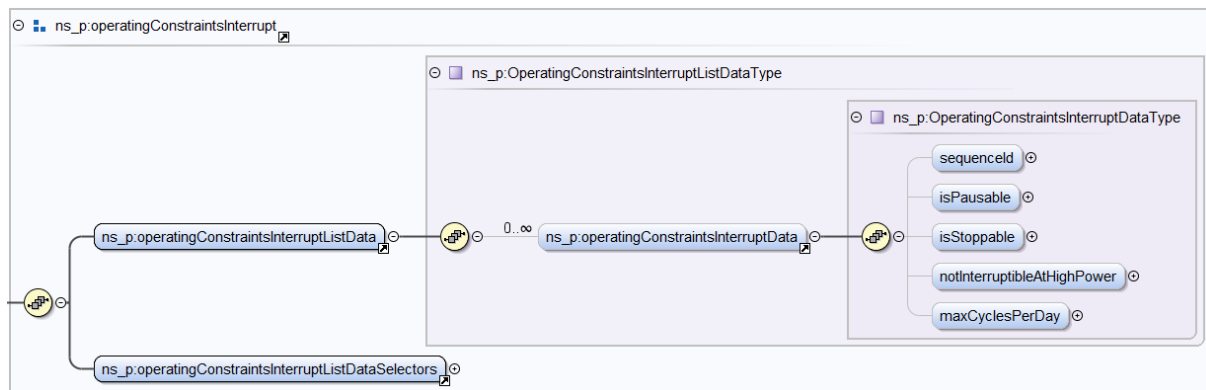


Figure 145: operatingConstraintsInterruptListData function overview

5.3.18.2.2 Detailed description of elements

Element	Type	Description
operatingConstraintsInterruptListData. operatingConstraintsInterruptData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	If interrupt constraints are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
isPausable	xs:boolean (W3C standard type)	States whether a functionality or task is pausable (true) or not (false).
isStoppable	xs:boolean (W3C standard type)	States whether a functionality or task is stoppable (true) or not (false).
notInterruptibleAtHighPower	xs:boolean (W3C standard type)	States whether a functionality or task is NOT interruptible while it is consuming (or producing) energy (true) or whether it can be interrupted nevertheless (false).
maxCyclesPerDay	xs:unsignedInt (W3C standard type)	If a functionality or task may be started only a maximum number of times per day, this value is denoted here.

Table 270: operatingConstraintsInterruptListData function element description

5.3.18.2.3 Available selectors

- sequenceId

5.3.18.2.4 Examples

5.3.18.2.4.1 Read-reply

We assume an energy management system wants to control an appliance. We also assume the appliance expresses details on its process with a power sequence (with sequenceId "1" in this example). The energy management system queries the appliance's operatingConstraintsInterrupt information and gets the following response:

```
<operatingConstraintsInterruptListData>
  <operatingConstraintsInterruptData>
    <sequenceId>1</sequenceId>
```

```

8919         <isPausable>true</isPausable>
8920         <notPausableAtHighPower>true</notPausableAtHighPower>
8921     </operatingConstraintsInterruptData>
8922 </operatingConstraintsInterruptListData>

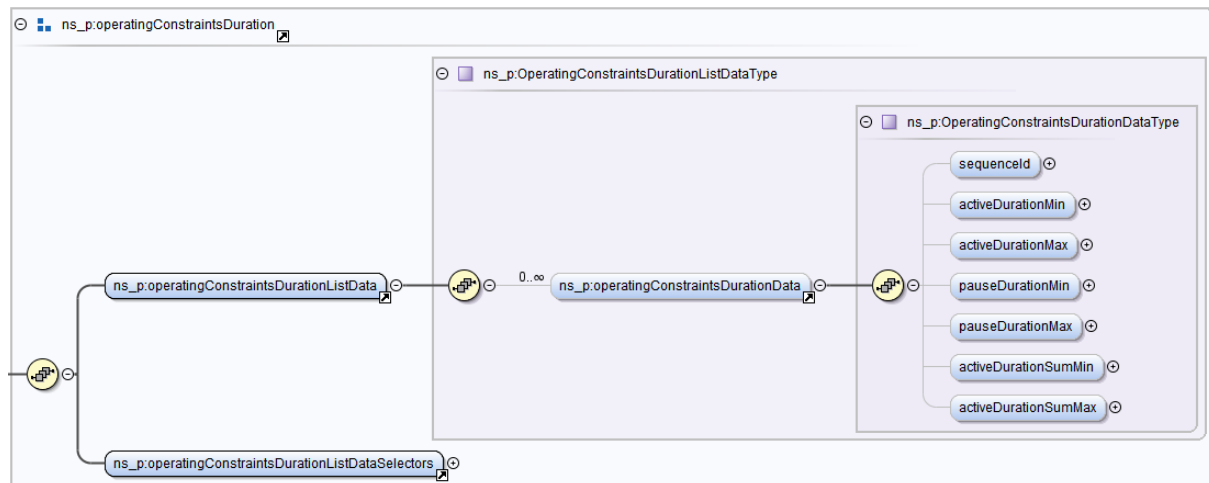
```

8923

8924 5.3.18.3 *operatingConstraintsDurationListData*

8925 5.3.18.3.1 *General*

8926 This function models constraints on the minimum/maximum duration of active ("on") and pause
 8927 ("off") phases. Example: A compressor can be damaged if it is started and then switched off too early.



8928

8929 Figure 146: *operatingConstraintsDurationListData* function overview

8930

8931 5.3.18.3.2 *Detailed description of elements*

Element	Type	Description
operatingConstraintsDurationListData. operatingConstraintsDurationData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	If duration constraints are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
activeDurationMin	xs:duration (W3C standard type)	This is the minimum duration a device has to run without interruption.
activeDurationMax	xs:duration (W3C standard type)	This is the maximum duration a device can run without interruption.
pauseDurationMin	xs:duration (W3C standard type)	This is the minimum duration a device has to pause after the end of an activity.
pauseDurationMax	xs:duration (W3C standard type)	This is the maximum duration a device can pause after the end of an activity.
activeDurationSumMin	xs:duration (W3C standard type)	This is the minimum duration a device has to run in total for a given task (summation of all active times).
activeDurationSumMax	xs:duration (W3C standard type)	This is the maximum duration a device can run in total for a given task (summation of all active times).

8932 Table 271: *operatingConstraintsDurationListData* function element description

8933

5.3.18.3.3 Available selectors

- `sequenceId`

5.3.18.3.4 Examples

5.3.18.3.4.1 Read-reply

This example continues the example of section 5.3.18.2.4.1. The energy management system queries for the appliance's duration constraints and gets the following response:

```
<operatingConstraintsDurationListData>
  <operatingConstraintsDurationData>
    <sequenceId>1</sequenceId>
    <activeDurationMin>PT10M</activeDurationMin>
    <activeDurationMax>PT1H</activeDurationMax>
    <pauseDurationMin>PT5M30S</pauseDurationMin>
    <pauseDurationMax>P1D</pauseDurationMax>
    <activeDurationSumMin>PT3H15M</activeDurationSumMin>
    <activeDurationSumMax>PT5H30M</activeDurationSumMax>
  </operatingConstraintsDurationData>
</operatingConstraintsDurationListData>
```

5.3.18.4 operatingConstraintsPowerDescriptionListData

5.3.18.4.1 General

This function models basic information relevant for *operatingConstraintsPowerRangeListData* and *operatingConstraintsPowerLevelListData*.

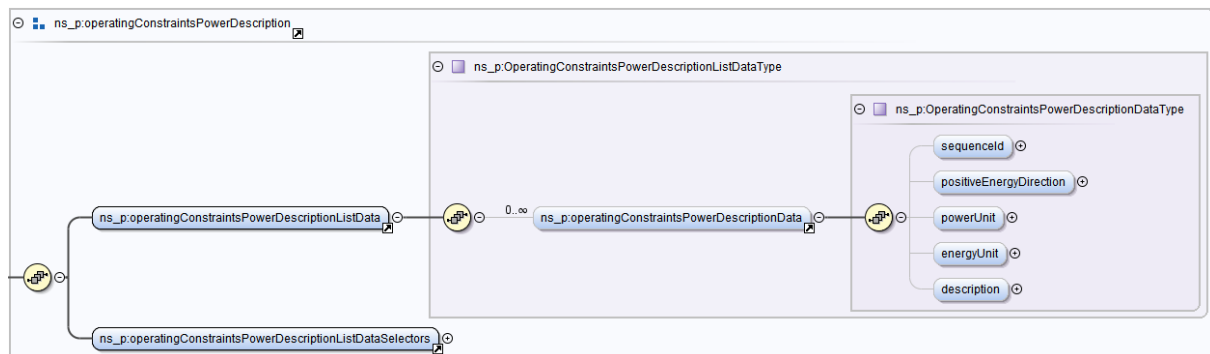


Figure 147: *operatingConstraintsPowerDescriptionListData* function overview

5.3.18.4.2 Detailed description of elements

Element	Type	Description
operatingConstraintsPowerDescriptionListData. operatingConstraintsPowerDescriptionData. (list)		
<code>sequenceId</code>	<i>Identifier</i> "PowerSequenceIdType" e" (see Table 358).	If function elements are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
<code>positiveEnergyDirection</code>	<i>Common data type</i> "EnergyDirectionType"	The <i>positiveEnergyDirection</i> states whether energy consumption or production is counted as positive value.

	". See section 3.10.1.13.	
powerUnit	Common data type "UnitOfMeasurement Type". See section 3.10.1.17.	The unit for all power values in operatingConstraintsPowerRangeListData and operatingConstraintsPowerLevelListData.
energyUnit	Common data type "UnitOfMeasurement Type". See section 3.10.1.17.	The unit for all energy values in operatingConstraintsPowerRangeListData and operatingConstraintsPowerLevelListData.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information.

Table 272: operatingConstraintsPowerDescriptionListData function element description

5.3.18.4.3 Available selectors

- sequenceId

5.3.18.4.4 Examples

5.3.18.4.4.1 Read-reply

This example continues the example of section 5.3.18.2.4.1. The energy management system queries for the appliance's description on constraints and gets the following response:

```
<operatingConstraintsPowerDescriptionListData>
  <operatingConstraintsPowerDescriptionData>
    <sequenceId>1</sequenceId>
    <positiveEnergyDirection>consume</positiveEnergyDirection>
    <powerUnit>W</powerUnit>
    <energyUnit>Wh</energyUnit>
    <description>Constraints for sequence 1</description>
  </operatingConstraintsPowerDescriptionData>
</operatingConstraintsPowerDescriptionListData>
```

5.3.18.5 operatingConstraintsPowerRangeListData

5.3.18.5.1 General

This function can be used to model a power/energy range(!) a device can operate. An alternative model for power/energy constraints is operatingConstraintsPowerLevelListData.

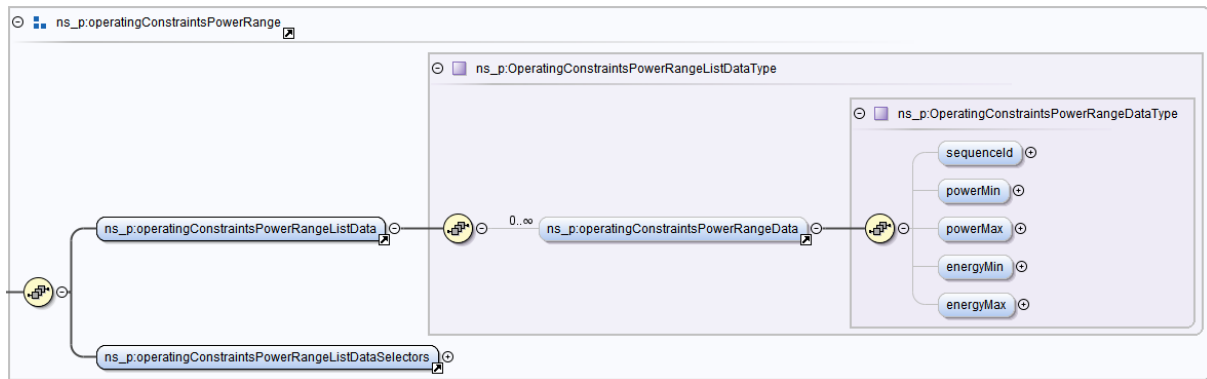


Figure 148: operatingConstraintsPowerRangeListData function overview

5.3.18.5.2 Detailed description of elements

Element	Type	Description
operatingConstraintsPowerRangeListData. operatingConstraintsPowerRangeData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	If the power range is related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.
powerMin	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum power a device can consume/produce.
powerMax	Common data type "ScaledNumberType". See section 3.10.1.8.	The maximum power a device can consume/produce.
energyMin	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum energy a device can consume/produce.
energyMax	Common data type "ScaledNumberType". See section 3.10.1.8.	The maximum energy a device can consume/produce.

Table 273: operatingConstraintsPowerRangeListData function element description

5.3.18.5.3 Available selectors

- sequenceId

5.3.18.5.4 Examples

5.3.18.5.4.1 Read-reply

This example continues the example of section 5.3.18.4.4.1. The energy management system queries for the appliance's power range constraints and gets the following response:

```

<operatingConstraintsPowerRangeListData>
  <operatingConstraintsPowerRangeData>
    <sequenceId>1</sequenceId>
    <powerMin>
      <number>0</number>
    </powerMin>
    <powerMax>
      <number>2</number>
      <scale>3</scale>
    </powerMax>
    <energyMin>

```

```

9008         <number>5</number>
9009         <scale>2</scale>
9010     </energyMin>
9011     <energyMax>
9012         <number>21</number>
9013         <scale>2</scale>
9014     </energyMax>
9015 </operatingConstraintsPowerRangeData>
9016 </operatingConstraintsPowerRangeListData>

```

5.3.18.6 operatingConstraintsPowerLevelListData

5.3.18.6.1 General

This function can be used to model a list of possible power values a device can operate. An alternative model for power constraints is operatingConstraintsPowerRangeListData, where a full range (not only discrete levels) can be modelled.

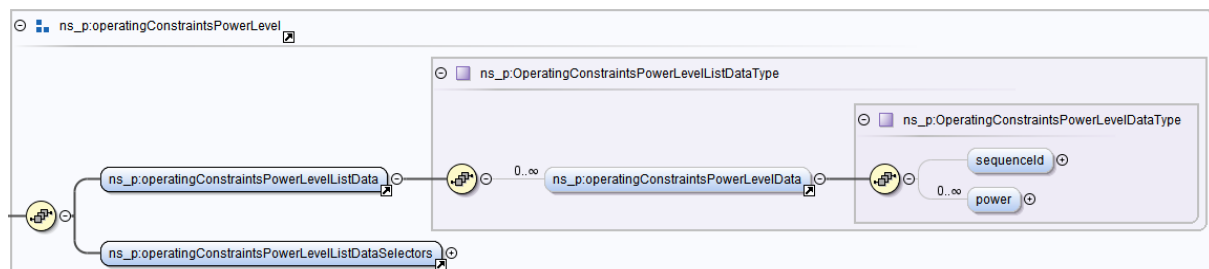


Figure 149: operatingConstraintsPowerLevelListData function overview

5.3.18.6.2 Detailed description of elements

Element	Type	Description
operatingConstraintsPowerLevelListData. operatingConstraintsPowerLevelData. (list)		
sequenced	Identifier "PowerSequencedType" (see Table 358).	If the power levels are related to a power sequence, the appropriate <i>sequenced</i> is stated in this element.
power (list)	Common data type "ScaledNumberType". See section 3.10.1.8.	Each "power" item contains a possible power level.

Table 274: operatingConstraintsPowerLevelListData element description

5.3.18.6.3 Available selectors

- sequenced

5.3.18.6.4 Examples

5.3.18.6.4.1 Read-reply

This example continues the example of section 5.3.18.4.4.1. The energy management system queries for the appliance's power level constraints and gets the following response:


```

9036 <operatingConstraintsPowerLevelListData>
9037   <operatingConstraintsPowerLevelData>
9038     <sequenceId>1</sequenceId>
9039     <power>
9040       <number>5</number>
9041       <scale>2</scale>
9042     </power>
9043     <power>
9044       <number>15</number>
9045       <scale>2</scale>
9046     </power>
9047     <power>
9048       <number>25</number>
9049       <scale>2</scale>
9050     </power>
9051   </operatingConstraintsPowerLevelData>
9052 </operatingConstraintsPowerLevelListData>

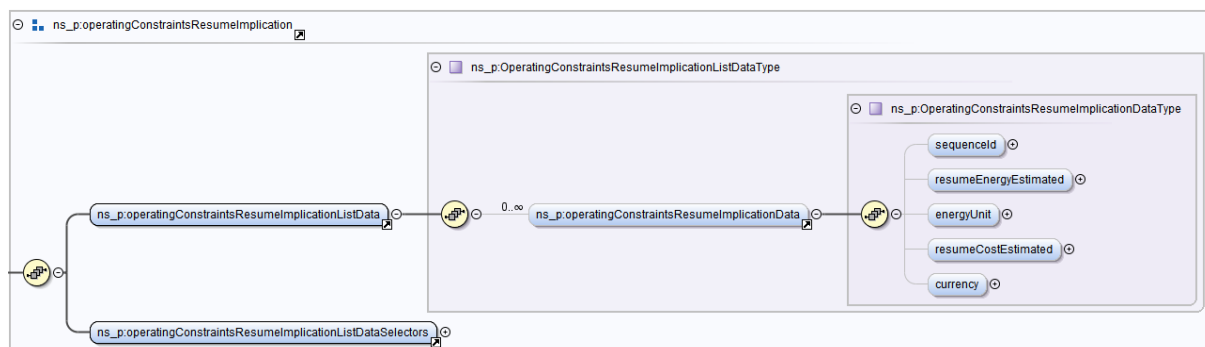
```

9053

9054 5.3.18.7 *operatingConstraintsResumeImplicationListData*

9055 5.3.18.7.1 *General*

9056 Pausing a device might be permitted according to above mentioned (duration) constraints.
 9057 Nevertheless, it can be expected that it has implications if a device's task is paused temporarily and
 9058 then resumed to continue the task. One possible implication is an additional mechanical stress which
 9059 reduces the device's lifetime and could be interpreted as additional cost. Such additional costs or
 9060 additional energy consumption can be modelled with this function in order to be considered for
 9061 energy management. However, these are typically empirical (thus rather constant) data, therefore
 9062 should not be considered to be precise.



9063

9064 Figure 150: *operatingConstraintsResumeImplicationListData* function overview

9065

9066 5.3.18.7.2 *Detailed description of elements*

Element	Type	Description
operatingConstraintsResumeImplicationListData. operatingConstraintsResumeImplicationData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	If the resume implications are related to a power sequence, the appropriate <i>sequenceId</i> is stated in this element.

resumeEnergyEstimated	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Additional energy a device consumes before resuming to its normal operation (after a pause).
energyUnit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	Unit for the energy stated in the element resumeEnergyEstimated.
resumeCostEstimated	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Additional costs for the resumption of a device to its normal operation.
currency	<i>Common data type "CurrencyType". See section 3.10.1.19.</i>	Currency for the price stated in the element resumeCostEstimated.

Table 275: operatingConstraintsResumeImplicationListData function element description

5.3.18.7.3 Available selectors

- sequenceId

5.3.18.7.4 Examples

5.3.18.7.4.1 Read-reply

This example continues the example of section 5.3.18.2.4.1. The energy management system queries for the appliance's resume implications and gets the following response:

```
<operatingConstraintsResumeImplicationListData>
  <operatingConstraintsResumeImplicationData>
    <sequenceId>1</sequenceId>
    <resumeEnergyEstimated>
      <number>5</number>
      <scale>1</scale>
    </resumeEnergyEstimated>
    <energyUnit>Wh</energyUnit>
    <resumeCostEstimated>
      <number>2</number>
      <scale>-2</scale>
    </resumeCostEstimated>
    <currency>EUR</currency>
  </operatingConstraintsResumeImplicationData>
</operatingConstraintsResumeImplicationListData>
```

5.3.19 PowerSequences

5.3.19.1 Introduction

This class primarily permits to model curves of power and energy over time. This is considered in the sub-class PowerTimeSlot (see 5.3.19.2). Secondly, the class provides definitions for the modelling of power scheduling including alternative plans.

5.3.19.1.1 Basic concepts

Throughout this class two terms are of major interest: *Sequence* and *Slot*. A sequence can be used to describe a "task". It is the most "coarse" view in this class, i.e. a sequence represents all single steps of a whole task. The single steps are represented by slots. Slots are the most "fine" view in this class with regards to the time resolution of a task.

Each sequence and slot can be associated an identifier or number, respectively. The slot numbers of two sequences should be considered independent from each other. I.e. slot number 7 of sequence 1 describes a different slot than slot number 7 of sequence 2. This basically means a slot is only "uniquely" identified in combination with a sequence ID. As formal notation this leads to the following relation:

- 1 sequence has 0..m slots
- 1 slot belongs to 0..1 sequence
- 1 pair of { sequenceId, slotNumber } uniquely identifies 1 slot

These relations are quite general as they permit modelling sequences with no slot and also the use of slots without sequences. However, for the most applications it can be expected that constraints are required where a sequence has at least 1 slot and a slot belongs to exactly one sequence.

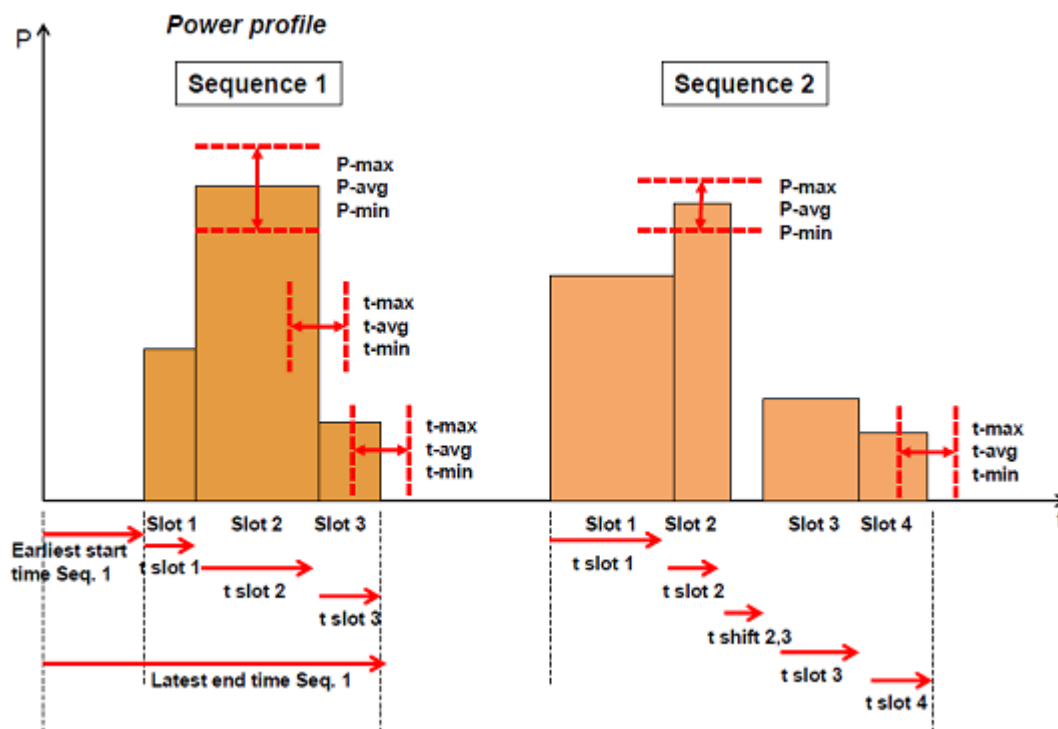


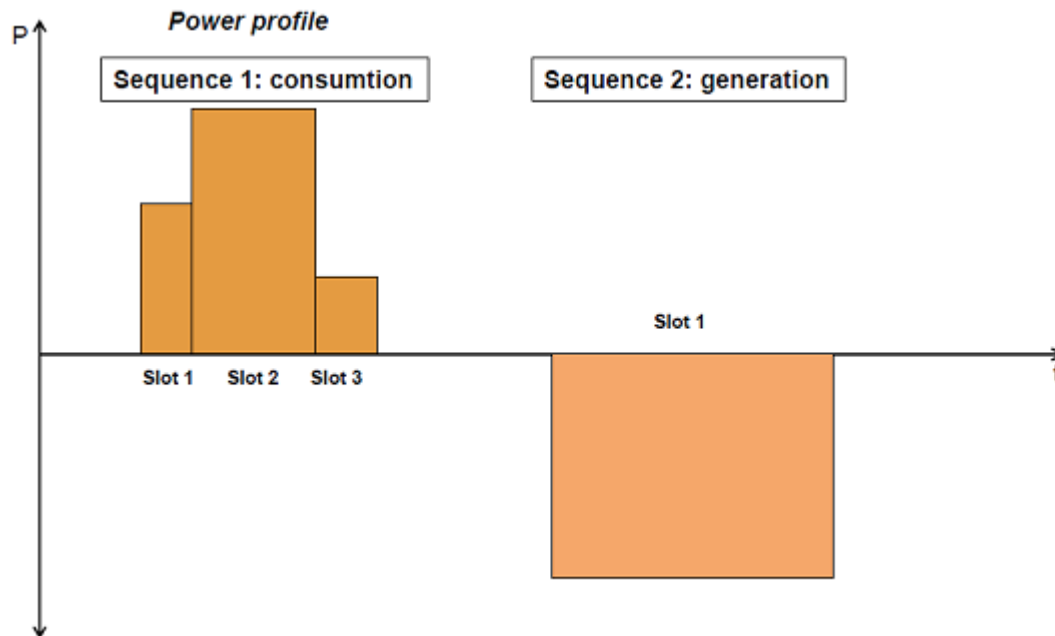
Figure 151: Concept overview of power sequences: Power P over time t ; sequences are constituted of slots

Irrespective of whether the Smart Device is a consumer or producer it needs to announce a kind of expected energy consumption or generation (power profile) to allow energy allocation within smart premises.

For example, a heat pump can ask for allocating 2 sequences per day, once in the morning, once in the afternoon, each sequence with different phases of power consumption. Alternatively, a smart device can offer multiple sequences as alternatives to each other (for the same time period, e.g.). An

9122 example for such alternative plans can be a battery pack offering charging or discharging. A customer
 9123 energy management (CEM) system may then choose between these possibilities. Some more details
 9124 on alternative plans will be described later on.

9125 Power sequences may exist for energy consumption or production.



9126

9127 *Figure 152: Example of a device that first consumes and later on generates power. In this case "consumption" is assigned a*
 9128 *positive value.*

9129 The easiest way to express the expected energy consumption / generation is one sequence with one
 9130 timeframe with one power value over timeslot.

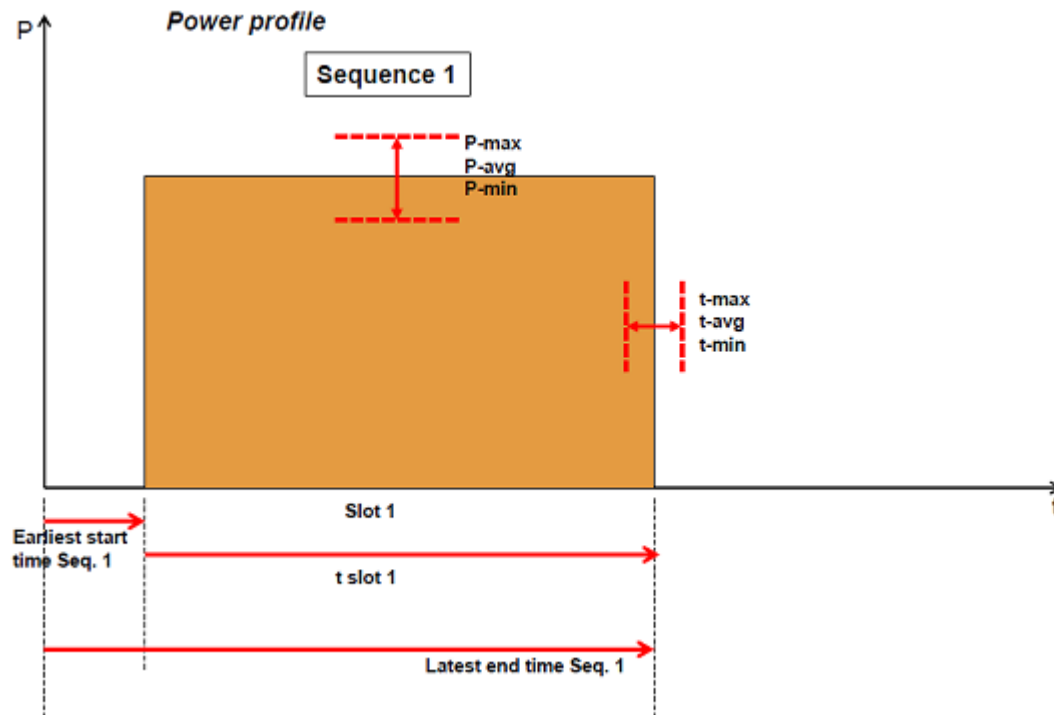


Figure 153: Example of a sequence with only one slot with some possible parameter boundaries.

5.3.19.1.2 Repetition concept

In its most basic form, a sequence can execute its explicitly announced slots one after the other and is completed when the last of these slots was completed. However, the PowerSequences also permits a "repetition concept" where a sequence is executed a given number of times.

As example we consider a sequence with three explicitly announced slots. As each slot represents a "phase" of the sequence, we denote the three phases "A", "B", and "C". In this example the sequence shall be executed two times. This finally means that this sequence of phases is executed: "A", "B", "C", "A", "B", "C".

5.3.19.1.3 Alternative sequences

As mentioned above, the class permits the modelling of "alternative plans". This means two or more sequences may be offered by an appliance to an energy management system (e.g.) to select none or exactly one of the alternatives for the execution. Among others, the so-called "state" of a power sequence can be useful to distinguish the selected sequence from the remaining alternatives (see section 5.3.19.9).

Some examples describe subsequent tasks (an appliance executes first sequence 1 and some time later it executes sequence 2, e.g.). The PowerSequences class permits as well the combination of alternatives and sequential plans. I.e. alternatives are grouped together and the groups are considered sequentially. Section 5.3.19.7 is useful to model groups of alternatives.

5.3.19.1.4 Overview on the class modules

The SPINE data model of this class follows the design principle of modular definitions. As a result the underlying revision of the SPINE data model contains almost no relation to slots within the sub-class PowerSequence. Thus, in theory it is possible to use PowerSequence without PowerTimeSlot. However, this is not the primary use case of PowerSequence and is not considered further in this document.

PowerTimeSlot, on the other hand, uses *sequenceId* in many functions, e.g. Therefore, using PowerTimeSlot without PowerSequences is not considered in detail.

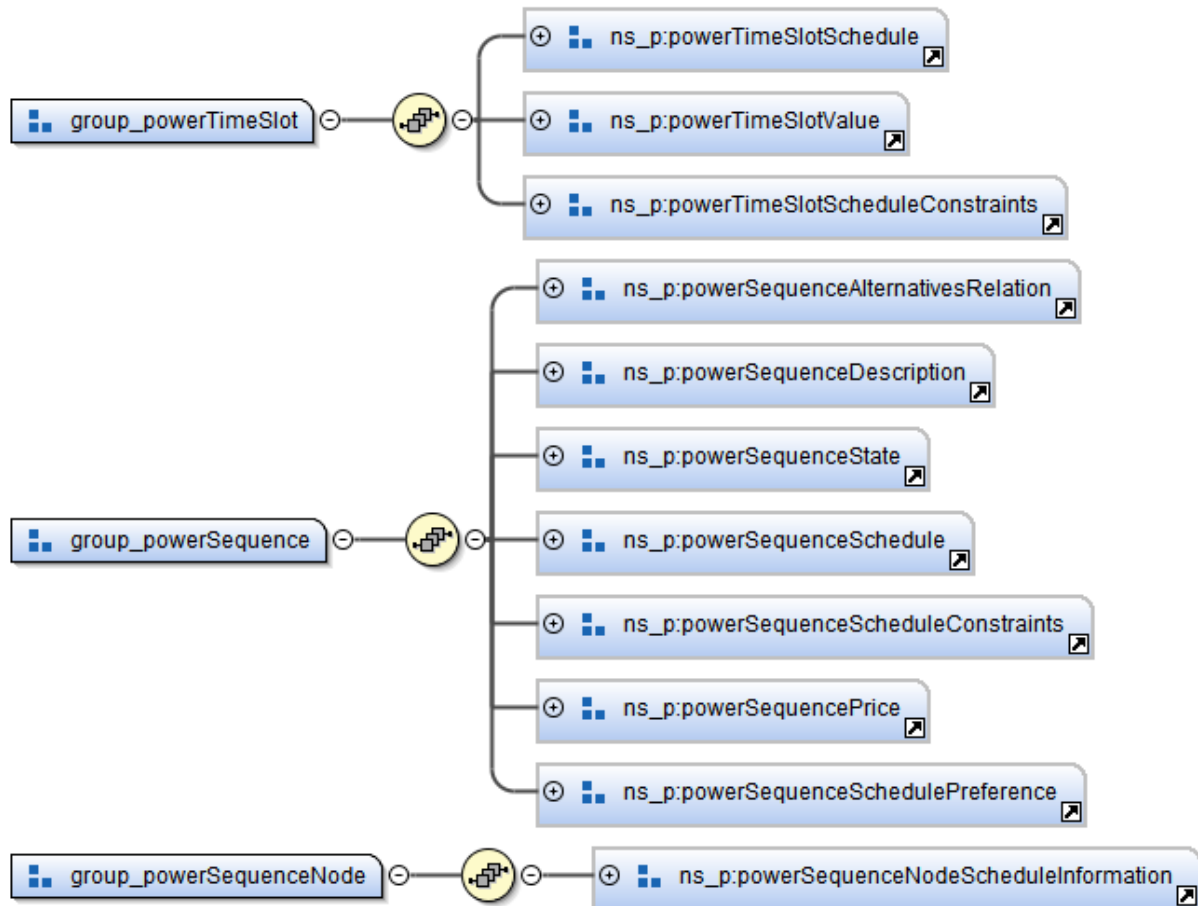


Figure 154: PowerSequences function-group overview (data)



Figure 155: PowerSequences function-group overview (call)

5.3.19.2 Sub-class PowerTimeSlot – Power values, facets, curves

The root of this sub-class is the consideration of single power or energy values. Each value can be "enriched" with miscellaneous facets like a timestamp, durations, dependency to other values,

9170 information on its "precision", etc. The combination of the single values or facets into lists permits
9171 modelling power curves over time, or modelling of "variants" at a given time, to give two examples.

9172 With regards to the time and dependency facets it is possible to model scheduling of periods or
9173 whole curves. Constraints on scheduling can be modelled, too.

9174 Please note this class basically supports modelling curves for "any purpose". One purpose might be
9175 the announcement of a device's planned energy consumption. Another purpose might be the
9176 presentation of a calculated or potential power availability. Esp. information from the
9177 *PowerSequence* class may provide further information on the use of slots and curves.

9178

9179 **5.3.19.3 *powerTimeSlotScheduleListData***

9180 **5.3.19.3.1 *General***

9181 This function can be used to model the schedule of a single slot. This information is intended to be
9182 used as actual information rather than schedule constraints (see 5.3.19.5). Together with a proper
9183 "write" classifier it can also be used to model the configuration of a slot schedule.

9184 As already mentioned in 5.3.19.2 the "purpose" or applicability of a slot might depend on further
9185 information.

9186 The function distinguishes between slots with "absolute" and "differential" schedule. The "absolute"
9187 schedule applies the element "timePeriod" (see below) and should be considered the most relevant
9188 information and the most common use. It can be used for curves with known or "already decided"
9189 time information, e.g. As timePeriod's sub-elements make use of the type
9190 *AbsoluteOrRelativeTimeType* it is possible to express all slot's time information with UTC or relative
9191 to "now" (or from a different point of view: "absolute" with "now" as reference time). A differential
9192 schedule can be used to model theoretical curves, e.g. The latter is useful for curves having no
9193 absolute schedule yet.

9194 Of course, the slots of a curve shall not overlap in time. On the other hand, subsequent slots may
9195 well be separated by a gap in time. Mathematically spoken the slots of a sequence shall be disjoint
9196 and may be non-contiguous with regards to their schedule.

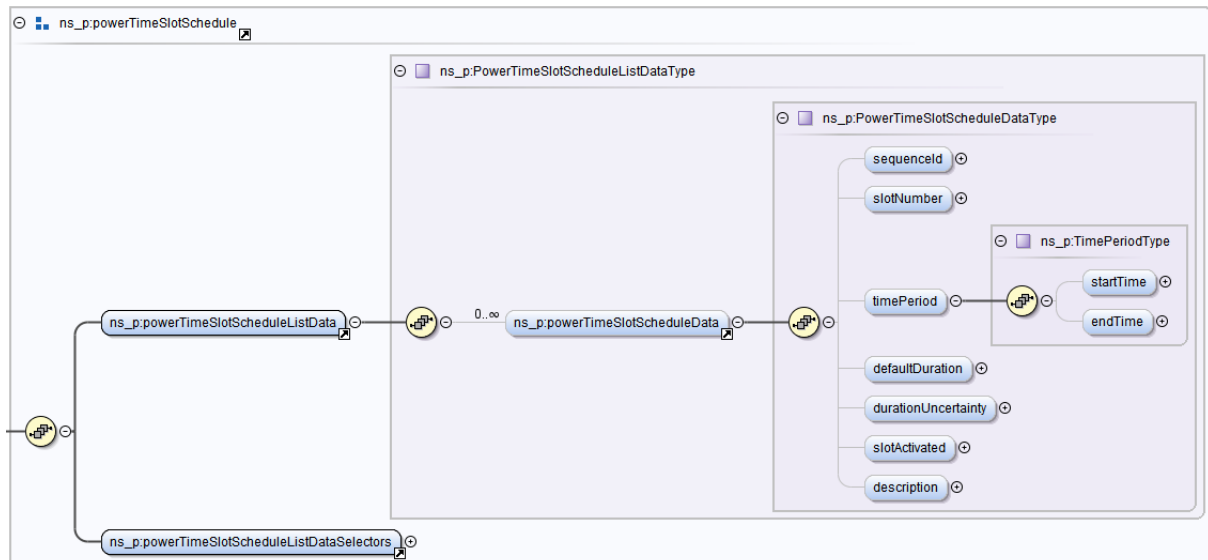


Figure 156: powerTimeSlotScheduleListData function overview

5.3.19.3.2 Detailed description of elements

Element	Type	Description
powerTimeSlotScheduleListData. powerTimeSlotScheduleData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
slotNumber	Identifier "PowerTimeSlotNumberType" (see Table 358).	The number of the slot.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	Time information on the slot.
timePeriod.startTime	See section 3.10.2.1.	Start time of the slot.
timePeriod.endTime	See section 3.10.2.1.	End time of the slot.
defaultDuration	xs:duration (W3C standard type)	The duration of this slot.
durationUncertainty	xs:duration (W3C standard type)	The defaultDuration may have some uncertainty. The actual duration then may be in the range from <i>defaultDuration - durationUncertainty</i> to <i>defaultDuration + durationUncertainty</i> .
slotActivated	xs:boolean (W3C standard type)	This element permits modelling the state of an "optional slot". Whether a slot is activated or not is specified here. The value "true" denotes the slot is used, whereas "false" means it is not used. See also element "optionalSlot" of function

		"powerTimeSlotScheduleConstraintsList Data" (see Table 279).
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Descriptive information on this slot.

Table 276: powerTimeSlotScheduleListData function element description

5.3.19.3.3 Available selectors

- sequenceId
- slotNumber

5.3.19.3.4 Examples

5.3.19.3.4.1 Read-reply

An energy management gateway was notified by a fridge about an upcoming task, referenced by sequence 2. The energy management gateway queries for powerTimeSlotScheduleListData schedule details on sequence 2 and receives a proper reply from the fridge:

```

<powerTimeSlotScheduleListData>
  <powerTimeSlotScheduleData>
    <sequenceId>2</sequenceId>
    <slotNumber>1</slotNumber>
    <timePeriod>
      <startTime>2006-05-04T16:15:00.0Z</startTime>
      <endTime>2006-05-04T16:22:00.0Z</endTime>
    </timePeriod>
    <defaultDuration>PT7M</defaultDuration>
    <durationUncertainty>PT0S</durationUncertainty>
    <slotActivated>true</slotActivated>
    <description>Super freeze</description>
  </powerTimeSlotScheduleData>
  <powerTimeSlotScheduleData>
    <sequenceId>2</sequenceId>
    <slotNumber>2</slotNumber>
    <timePeriod>
      <startTime>2006-05-04T16:25:00.0Z</startTime>
      <endTime>2006-05-04T16:29:30.0Z</endTime>
    </timePeriod>
    <defaultDuration>PT4M30S</defaultDuration>
    <durationUncertainty>PT5S</durationUncertainty>
    <slotActivated>true</slotActivated>
    <description>Freeze</description>
  </powerTimeSlotScheduleData>
</powerTimeSlotScheduleListData>

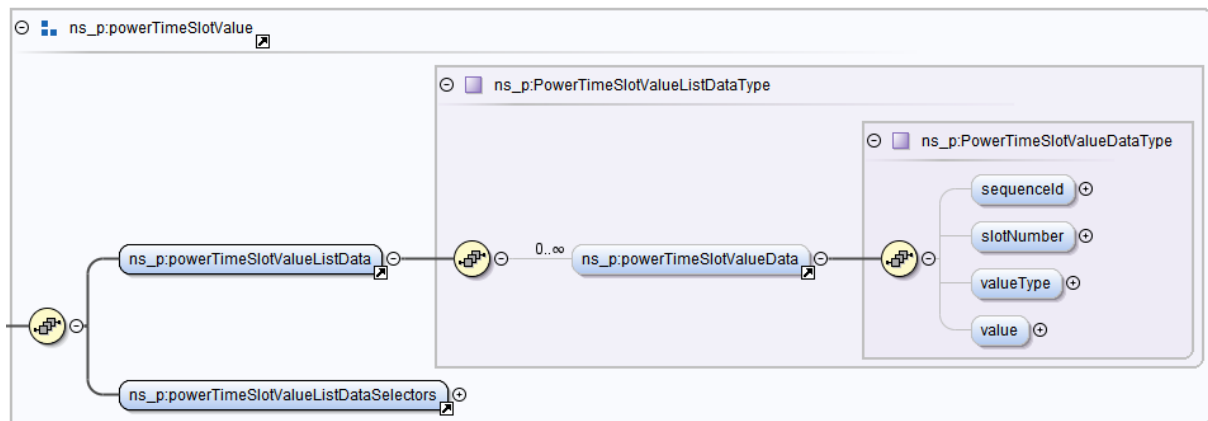
```

5.3.19.4 powerTimeSlotValueListData

5.3.19.4.1 General

This function can be used to model miscellaneous values associated to a single slot. The most essential values are certainly the "real" power and the "real" energy. In addition it is possible to model value facets like average, skewness, minimum, etc.

9244 Please note the units of energy and power values are modelled in the PowerSequence sub-class.



9245
9246 Figure 157: powerTimeSlotValueListData function overview

9247
9248 5.3.19.4.2 Detailed description of elements

Element	Type	Description
powerTimeSlotValueListData. powerTimeSlotValueData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
slotNumber	Identifier "PowerTimeSlotNumberType" (see Table 358).	The number of the slot.
valueType	Union "PowerTimeSlotValueTypeType": - Enum (see Table 278): PowerTimeSlotValueTypeEnumType - EnumExtendType (see section 3.10.1.5)	The kind of the value ("power", "powerMin", "energy", ...).
value	Common data type "ScaledNumberType". See section 3.10.1.8.	The magnitude of the value denoted by valueType.

9249 Table 277: powerTimeSlotValueListData function element description

9250 Enumeration **PowerTimeSlotValueTypeEnumType**:

Value	Description
power	Power
powerMin	Lower bound of power; minimum
powerMax	Upper bound of power; maximum
powerExpectedValue	Expected value (or "expectation" in terms of probability theory) of power; may be used to express the estimation of an average value.
powerStandardDeviation	Standard deviation (in terms of probability theory) of power
powerSkewness	Skewness (in terms of probability theory) of power
energy	Energy
energyMin	Lower bound of energy; minimum
energyMax	Upper bound of energy; maximum
energyExpectedValue	Expected value (or "expectation" in terms of probability theory) of energy; may be used to express the estimation of an average value.
energyStandardDeviation	Standard deviation (in terms of probability theory) of energy
energySkewness	Skewness (in terms of probability theory) of energy

9251 Table 278: Enumeration PowerTimeSlotValueTypeEnumType

9252

9253 **5.3.19.4.3 Available selectors**

- 9254 - sequenceId
- 9255 - slotNumber
- 9256 - valueType

9257

9258 **5.3.19.4.4 Examples**9259 **5.3.19.4.4.1 Read-reply**

9260 We continue with the example of section 5.3.19.3.4.1. The energy management gateway now
 9261 queries for all values of sequence 2 and receives a proper reply from the fridge:

```

9262 <powerTimeSlotValueListData>
9263   <powerTimeSlotValueData>
9264     <sequenceId>2</sequenceId>
9265     <slotNumber>1</slotNumber>
9266     <valueType>power</valueType>
9267     <value>
9268       <number>3</number>
9269       <scale>2</scale>
9270     </value>
9271   </powerTimeSlotValueData>
9272   <powerTimeSlotValueData>
9273     <sequenceId>2</sequenceId>
9274     <slotNumber>1</slotNumber>
9275     <valueType>powerMax</valueType>
9276     <value>
9277       <number>315</number>
9278     </value>
9279   </powerTimeSlotValueData>
9280   <powerTimeSlotValueData>
9281     <sequenceId>2</sequenceId>
9282     <slotNumber>1</slotNumber>
9283     <valueType>energyExpectedValue</valueType>
9284     <value>
9285       <number>5</number>
9286     </value>
9287   </powerTimeSlotValueData>
9288   <powerTimeSlotValueData>
9289     <sequenceId>2</sequenceId>
9290     <slotNumber>2</slotNumber>
9291     <valueType>power</valueType>
9292     <value>
9293       <number>12</number>
9294       <scale>1</scale>
9295     </value>
9296   </powerTimeSlotValueData>
9297   <powerTimeSlotValueData>
9298     <sequenceId>2</sequenceId>
9299     <slotNumber>2</slotNumber>
9300     <valueType>powerMax</valueType>
9301     <value>
9302       <number>14</number>
9303       <scale>1</scale>
9304     </value>
9305   </powerTimeSlotValueData>
9306   <powerTimeSlotValueData>
9307     <sequenceId>2</sequenceId>
9308     <slotNumber>2</slotNumber>

```

```

9309         <valueType>energyExpectedValue</valueType>
9310         <value>
9311             <number>2</number>
9312         </value>
9313     </powerTimeSlotValueData>
9314 </powerTimeSlotValueListData>

```

5.3.19.5 powerTimeSlotScheduleConstraintsListData

5.3.19.5.1 General

This function can be used to model constraints of a slot's schedule. It is useful in case a device supports the configuration or modification of a schedule. In this case the constraints can be used to determine the conditions a configuration needs to fulfil.

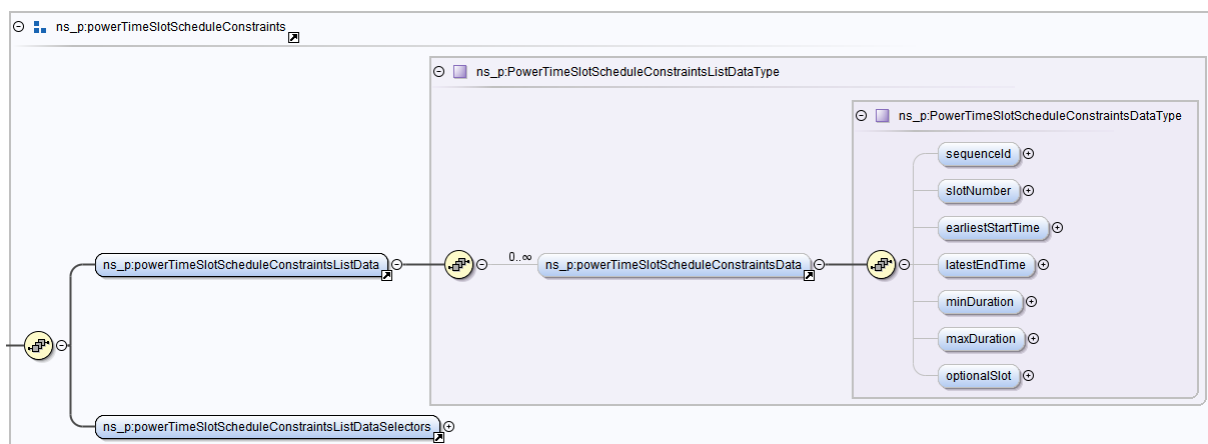


Figure 158: powerTimeSlotScheduleConstraintsListData function overview

5.3.19.5.2 Detailed description of elements

Element	Type	Description
powerTimeSlotScheduleConstraintsListData. powerTimeSlotScheduleConstraintsData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
slotNumber	Identifier "PowerTimeSlotNumberType" (see Table 358).	The number of the slot.
earliestStartTime	Common data type "AbsoluteOrRelativeTimeType" . See section 3.10.2.3.	The earliest start time of the slot (UTC or relative to "now").
latestEndTime	Common data type "AbsoluteOrRelativeTimeType" . See section 3.10.2.3.	The end time of the slot has to be not later than this time (UTC or relative to "now").
minDuration	xs:duration (W3C standard type)	This is the minimum duration the slot can run.
maxDuration	xs:duration (W3C standard type)	This is the maximum duration the slot has to run.

optionalSlot	xs:boolean (W3C standard type)	A slot is optional if this element is set to "true". See also element "slotActivated" of function "powerTimeSlotScheduleListData" (see Table 276) for the state of the slot (activated or deactivated).
--------------	--------------------------------	--

Table 279: powerTimeSlotScheduleConstraintsListData function element description

5.3.19.5.3 Available selectors

- sequenceId
- slotNumber

5.3.19.5.4 Examples

5.3.19.5.4.1 Read-reply

We continue with the examples of section 5.3.19.3.4.1 and section 5.3.19.4.4. The energy management gateway wants to figure out whether some slots of sequence 2 can be shifted somehow. It sends a request for powerTimeSlotScheduleConstraintsListData of sequence 2 and receives a proper reply from the fridge:

```

<powerTimeSlotScheduleConstraintsListData>
  <powerTimeSlotScheduleConstraintsData>
    <sequenceId>2</sequenceId>
    <slotNumber>1</slotNumber>
    <earliestStartTime>2006-05-04T16:15:00.0Z</earliestStartTime>
    <latestEndTime>2006-05-04T18:00:00.0Z</latestEndTime>
    <minDuration>PT1H10M</minDuration>
    <maxDuration>PT1H30M</maxDuration>
    <optionalSlot>false</optionalSlot>
  </powerTimeSlotScheduleConstraintsData>
  <powerTimeSlotScheduleConstraintsData>
    <sequenceId>2</sequenceId>
    <slotNumber>2</slotNumber>
    <earliestStartTime>2006-05-04T19:00:00.0Z</earliestStartTime>
    <latestEndTime>2006-05-04T19:45:00.0Z</latestEndTime>
    <minDuration>PT15M</minDuration>
    <maxDuration>PT20M</maxDuration>
    <optionalSlot>false</optionalSlot>
  </powerTimeSlotScheduleConstraintsData>
</powerTimeSlotScheduleConstraintsListData>

```

5.3.19.6 Sub-class PowerSequence

The PowerSequence sub-class delivers all information about the power sequences themselves. This covers descriptions, states, constraints for scheduling, scheduling and price.

5.3.19.7 powerSequenceAlternativesRelationListData

5.3.19.7.1 General

As already described in section 5.3.19.1.3 for some use cases it is necessary to consider power sequences representing alternative tasks, i.e. tasks where at maximum one can finally be chosen. In order to model such a group of alternative sequences the powerSequenceAlternativesRelation can be used. It permits to model the relation

- 1 alternatives group has 0..m sequences

though practical use cases should have at least one sequence in an alternatives group.

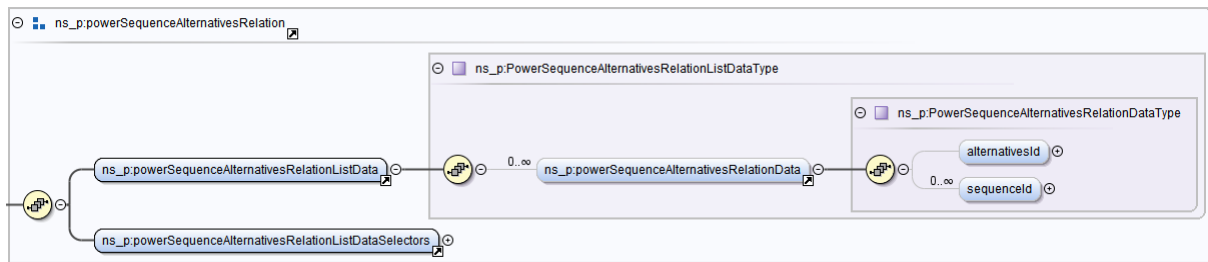


Figure 159: powerSequenceAlternativesRelationListData function overview

5.3.19.7.2 Detailed description of elements

Element	Type	Description
powerSequenceAlternativesRelationListData. powerSequenceAlternativesRelationData. (list)		
alternativesId	Identifier "AlternativesIdType" (see Table 358).	The identifier for the group of alternative sequences.
sequenceId (list)	Identifier "PowerSequenceIdType" (see Table 358).	A sequence identifier that belongs to the given alternatives group.

Table 280: powerSequenceAlternativesRelationListData function element description

5.3.19.7.3 Available selectors

- alternativesId
- sequenceId

5.3.19.7.4 Examples

5.3.19.7.4.1 Notify

A heating system announces some proposals for the afternoon and the evening: For the afternoon group 1 contains sequences 1 and 2 to choose from and for the evening group 2 permits to select sequence 3, 4, or 5. It sends this (complete) notification:

```
<powerSequenceAlternativesRelationListData>
  <powerSequenceAlternativesRelationData>
```

```

9387         <alternativesId>1</alternativesId>
9388         <sequenceId>1</sequenceId>
9389         <sequenceId>2</sequenceId>
9390     </powerSequenceAlternativesRelationData>
9391     <powerSequenceAlternativesRelationData>
9392         <alternativesId>2</alternativesId>
9393         <sequenceId>3</sequenceId>
9394         <sequenceId>4</sequenceId>
9395         <sequenceId>5</sequenceId>
9396     </powerSequenceAlternativesRelationData>
9397 </powerSequenceAlternativesRelationListData>

```

5.3.19.8 powerSequenceDescriptionListData

5.3.19.8.1 General

The *powerSequenceDescriptionData* function models descriptive (typically non-changing) information on sequences (one for each available sequence of a node).

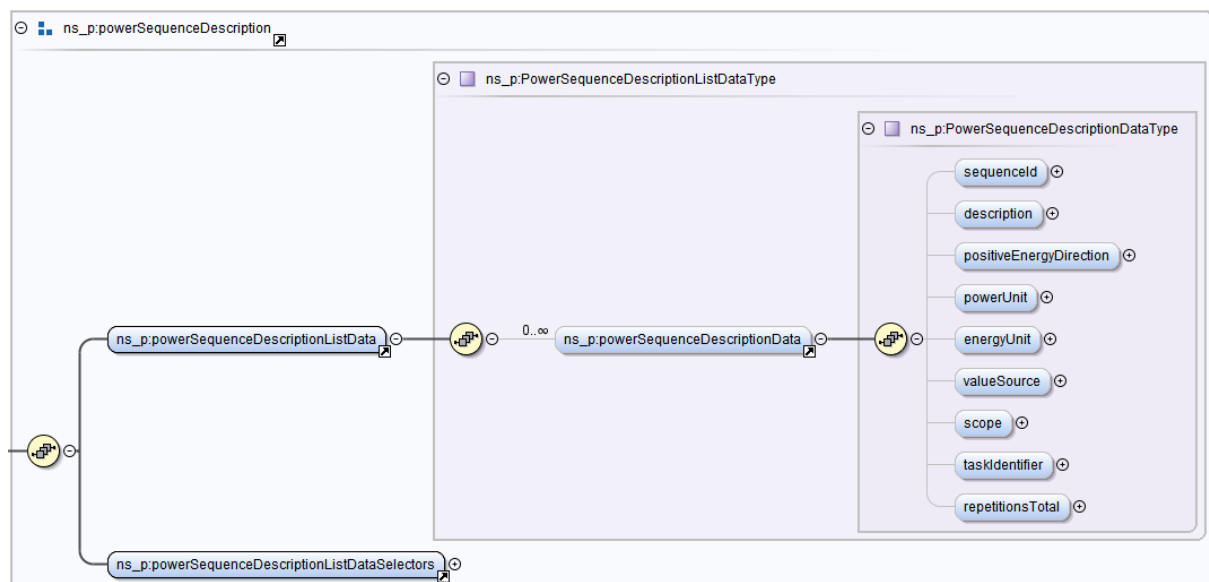


Figure 160: powerSequenceDescriptionListData function overview

5.3.19.8.2 Detailed description of elements

Element	Type	Description
powerSequenceDescriptionListData. powerSequenceDescriptionData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on this sequence.

positiveEnergyDirection	<i>Common data type "EnergyDirectionType". See section 3.10.1.13.</i>	The <i>positiveEnergyDirection</i> states whether energy consumption or production will be counted as positive value: If the element is set to "consume": energy values are positive during consumption and negative during production of energy. If the element is set to "produce": energy values are negative during consumption and positive during production of energy.
powerUnit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	This is the unit for all values related to power.
energyUnit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	This is the unit for all values concerning energy consumption and production.
valueSource	<i>Union "MeasurementValueSourceType":</i> - Enum (see Table 246): MeasurementValueSourceEnumType - EnumExtendType (see section 3.10.1.5)	This element enables to express whether the values of the given sequence are based upon measurement or calculation, e.g.
scope	<i>Union "PowerSequenceScopeType":</i> - Enum (see Table 282): PowerSequenceScopeEnumType - EnumExtendType (see section 3.10.1.5)	With this element it is possible to distinguish between sequences intended for different scopes (forecast, measurement, etc.). Note: Sequences with scope "forecast" or "measurement" typically originate from a device executing a task, whereas sequences with scope "recommendation" rather denote a configuration into a device that shall execute a task.
taskIdIdentifier	xs:unsignedInt (W3C standard type)	This "technical identifier" denotes which kind of task or condition applies to the power sequence. A device can provide identifiers in order to enable energy management applications recognize common (i.e. "typical" or frequent) tasks of a device. Note: This element should not be confused with "taskId" of the class TaskManagement.
repetitionsTotal	xs:unsignedInt (W3C standard type)	If the sequence applies the "repetition concept", this element denotes the total number of executions of the sequence. See also element "activeRepetitionNumber" of function "powerSequenceStateData".

Table 281: powerSequenceDescriptionListData function element description

Enumeration **PowerSequenceScopeEnumType**:

Value	Description
forecast	The sequence's values reflect the best guess of a device's task which typically begins in the future.
measurement	The sequence's values reflect the measurement of a device's task.
recommendation	The value "recommendation" can be used to model a "target power curve", i.e. a power profile a device SHOULD follow for a given task (if possible).

Table 282: Enumeration *PowerSequenceScopeEnumType*

5.3.19.8.3 Available selectors

- sequenceId

5.3.19.8.4 Examples

5.3.19.8.4.1 Read-reply

One is interested in the description of all available power sequences. A query for a device's *powerSequenceDescriptionListData* may lead to the following reply:

```

<powerSequenceDescriptionListData>
  <powerSequenceDescriptionData>
    <sequenceId>1</sequenceId>
    <description>Washing programme 1</description>
    <positiveEnergyDirection>consume</positiveEnergyDirection>
    <powerUnit>W</powerUnit>
    <energyUnit>Wh</energyUnit>
    <scope>forecast</scope>
    <taskIdentifier>7</taskIdentifier>
  </powerSequenceDescriptionData>
  <powerSequenceDescriptionData>
    <sequenceId>2</sequenceId>
    <description>Washing programme 2</description>
    <positiveEnergyDirection>consume</positiveEnergyDirection>
    <powerUnit>W</powerUnit>
    <energyUnit>Wh</energyUnit>
    <scope>forecast</scope>
    <taskIdentifier>8</taskIdentifier>
  </powerSequenceDescriptionData>
</powerSequenceDescriptionListData>

```

5.3.19.9 powerSequenceStateListData

5.3.19.9.1 General

Information on the current state of a power sequence can be modelled with the *powerSequenceStateData* function.

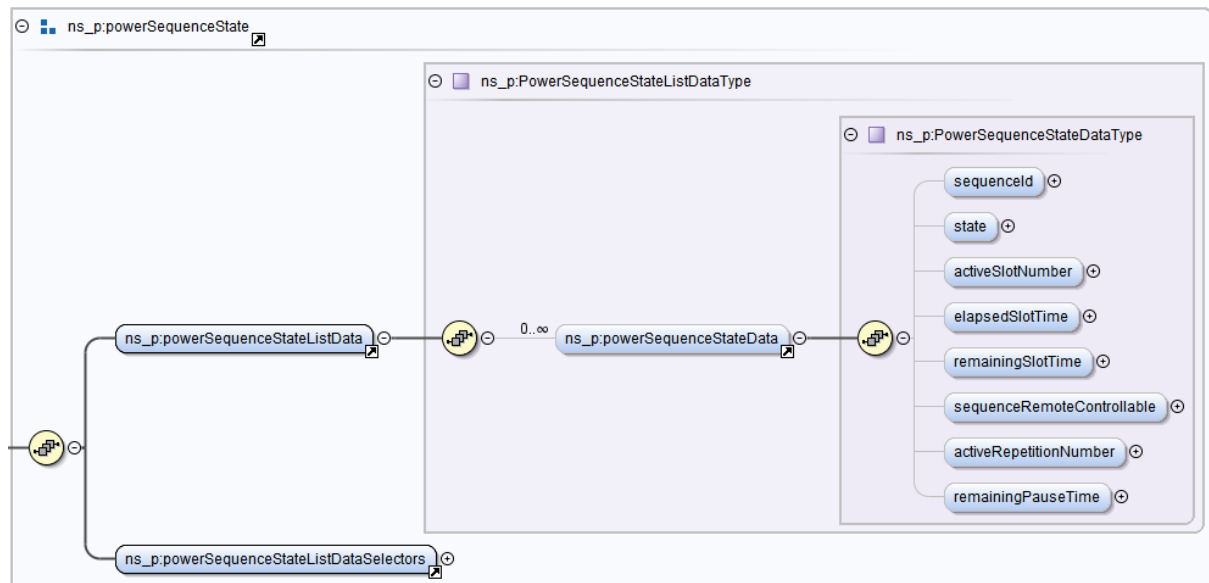


Figure 161: powerSequenceStateListData function overview

5.3.19.9.2 Detailed description of elements

Element	Type	Description
powerSequenceStateListData. powerSequenceStateData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
state	Union "PowerSequenceStateTy pe": - Enum (see Table 284): PowerSequenceStateEnu mType - EnumExtendType (see section 3.10.1.5)	The current state of the power sequence.
activeSlotNumber	Identifier "PowerTimeSlotNumberT ype" (see Table 358).	Contains the currently active slot.
elapsedSlotTime	xs:duration (W3C standard type)	Contains the time the slot has already been in "running" state (this also means the value remains constant during a "paused" state).
remainingSlotTime	xs:duration (W3C standard type)	Contains the time the slot still needs to be in "running" state (this also means the value remains constant during a "paused" state).
sequenceRemoteCo ntrollable	xs:boolean (W3C standard type)	If this element is set to "false" the sequence is executed autonomously and cannot be modified by SPINE commands received external nodes. Note: Consider also the element "nodeRemoteControllable" of function "powerSequenceNodeScheduleInformationDat a" in Table 289!

activeRepetitionNumber	xs:unsignedInt (W3C standard type)	If the sequence applies the "repetition concept", this element denotes the current number of the sequence execution.
remainingPauseTime	xs:duration (W3C standard type)	Denotes the duration the current slot (element "activeSlotNumber") permits being paused.

9447 Table 283: powerSequenceStateListData function element description

9448 Enumeration **PowerSequenceStateEnumType**:

Value	Description
running	The sequence is currently executed.
paused	The sequence is currently paused.
scheduled	The sequence is not executed at the moment but will be executed (i.e. gain state "running") by the power sequences server at the scheduled time.
scheduledPaused	Similar to the value "scheduled", but at the scheduled time the sequence will be set into the state "paused".
pending	The sequence is "recommended" (in terms of "preferred") by the power sequences server but will not be executed autonomously by the server. Remark: This state is most appropriate for devices announcing that a specific task (power sequence) is going to become "valid" but still lacks something / requires configuration or advice. It is most suitable in conjunction with the function powerSequenceScheduleConfigurationRequestCall.
inactive	The sequence denotes an "optional task". I.e. it is not invalid but it is also not running/scheduled/... so far. However, this sequence may become running or scheduled(Paused) by appropriate means.
completed	The sequence has been completed.
invalid	The sequence is invalid and shall not be considered anymore.

9449 Table 284: Enumeration PowerSequenceStateEnumType

9450

9451 5.3.19.9.3 Available selectors

9452 - sequenceId

9453

9454 5.3.19.9.4 Examples

9455 5.3.19.9.4.1 Notify

9456 A power sequences server executes sequence "2" and just completed slot number 1. Now, it begins
 9457 with slot number 2 and notifies this change (the "filter/partial" part to announce the "restricted
 9458 function exchange" is simplified by "..." just to improve the readability):

9459 ...

9460 <powerSequenceStateListData>

9461 <powerSequenceStateData>

9462 <sequenceId>2</sequenceId>

9463 <activeSlotNumber>2</activeSlotNumber>

9464 <elapsedSlotTime>PT0M</elapsedSlotTime>

9465 <remainingSlotTime>PT20M</remainingSlotTime>

9466 </powerSequenceStateData>

9467 </powerSequenceStateListData>

9468

5.3.19.10 powerSequenceScheduleListData

5.3.19.10.1 General

The scheduling of a sequence can be modelled with this function.

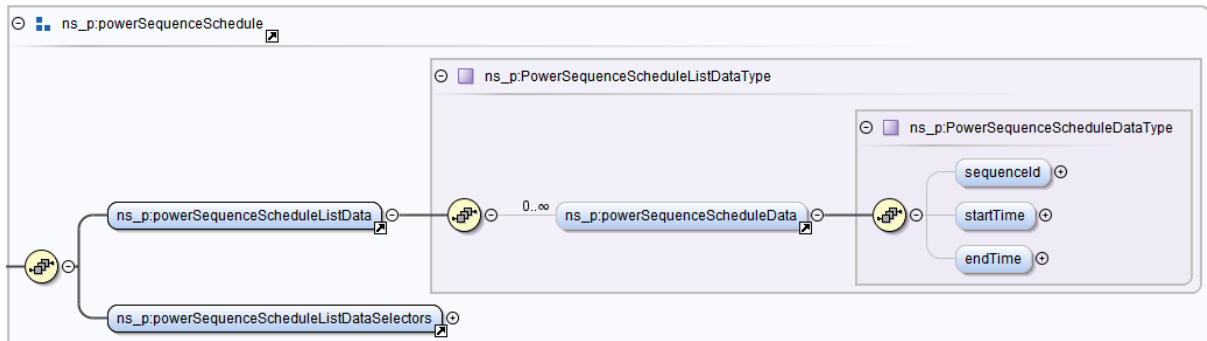


Figure 162: powerSequenceScheduleListData function overview

5.3.19.10.2 Detailed description of elements

Element	Type	Description
powerSequenceScheduleListData. powerSequenceScheduleData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
startTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The start time of the sequence.
endTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The end time of the sequence.

Table 285: powerSequenceScheduleListData function element description

5.3.19.10.3 Available selectors

- sequenceId

5.3.19.10.4 Examples

5.3.19.10.4.1 Read-reply

An Energy Management Software wants to get information on the current schedule of a specific sequence (2). It sends a proper request and receives this response:

```

<powerSequenceScheduleListData>
  <powerSequenceScheduleData>
    <sequenceId>2</sequenceId>
    <startTime>2013-07-13T18:11:00.0Z</startTime>
    <endTime>2013-07-14T16:55:00.0Z</endTime>
  </powerSequenceScheduleData>
</powerSequenceScheduleListData>

```

5.3.19.11 powerSequenceScheduleConstraintsListData

5.3.19.11.1 General

A power sequence of an appliance may have time-related constraints that are to be considered by an energy management software during demand optimization calculations. Otherwise the appliance could reject a schedule change requested by the energy management software.

This section deals with overall schedule constraints of a sequence. Schedule constraints of slots are discussed in section 5.3.19.5. An application may make use of both kinds of constraints together. Of course, contradictory information from constraints and schedules should be avoided.

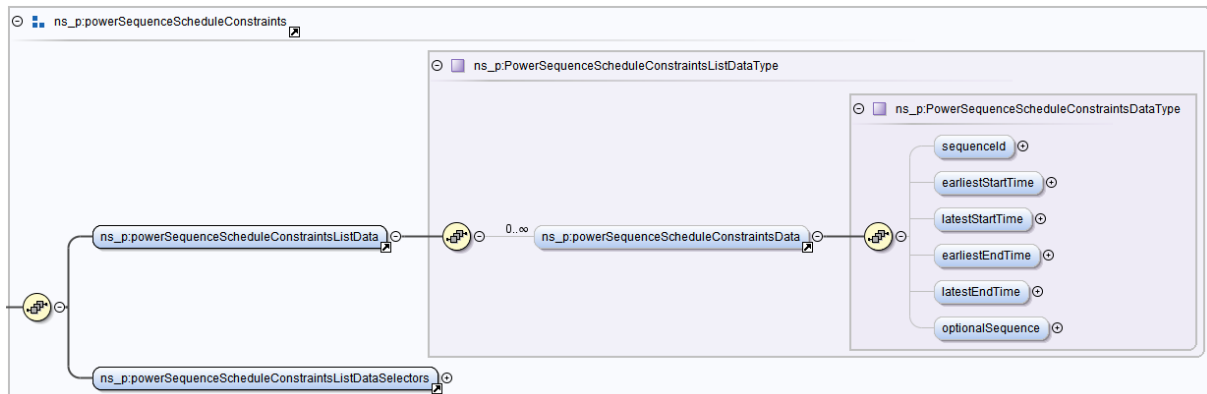


Figure 163: powerSequenceScheduleConstraintsListData function overview

5.3.19.11.2 Detailed description of elements

Element	Type	Description
powerSequenceScheduleConstraintsListData. powerSequenceScheduleConstraintsData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
earliestStartTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The earliest time the sequence can start.
latestStartTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The latest time the sequence can start.
earliestEndTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The earliest time the sequence can end.
latestEndTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The latest time the sequence can end.
optionalSequence	xs:boolean (W3C standard type)	For specific use cases a sequence can be tagged explicitly as "optional task". In this case this element can be set to <i>true</i> .

Table 286: powerSequenceScheduleConstraintsListData function element description

Please note that start and end time constraints just model start and end boundaries of a power sequence. I.e. in general the duration of a sequence can be shorter than the period given by the start/end time constraints.

5.3.19.11.3 Available selectors

- sequenceId

5.3.19.11.4 Examples

5.3.19.11.4.1 Read-reply

Before an Energy Management Software re-schedules a specific sequence (2) of an appliance it queries the appliance for the proper constraints. The appliance sends this response:

```
<powerSequenceScheduleConstraintsListData>
  <powerSequenceScheduleConstraintsData>
    <sequenceId>2</sequenceId>
    <earliestStartTime>2013-07-14T12:00:00.0Z</earliestStartTime>
    <latestStartTime>2013-07-14T12:45:00.0Z</latestStartTime>
    <earliestEndTime>2013-07-14T16:00:00.0Z</earliestEndTime>
    <latestEndTime>2013-07-14T18:00:00.0Z</latestEndTime>
  </powerSequenceScheduleConstraintsData>
</powerSequenceScheduleConstraintsListData>
```

5.3.19.12 powerSequencePriceListData

5.3.19.12.1 General

The execution of a power sequence may cause some cost or earnings due to the consumption or production of energy. This means a power sequence can be assigned a price. In fact, as the availability of energy may vary with time, the price of a power sequence could as well depend upon its time of execution (which may esp. be of interest if the scheduling of power sequences shall consider cost saving requirements). Therefore, the *powerSequencePriceData* function permits modelling a price per assumed start time of the sequence. Please note assumed start times may differ from the "final" start time.

How an appliance gains knowledge of the price associated to one of its sequences shall not be discussed in detail. Just a few possibilities shall be sketched briefly: Configuration (the sequence has a fixed price which is stored in the appliance), local calculation (the appliance obtains price information and applies it to the power consumption), or query to an external node (see *powerSequencePriceCalculationRequestCall* function in section 5.3.19.18).

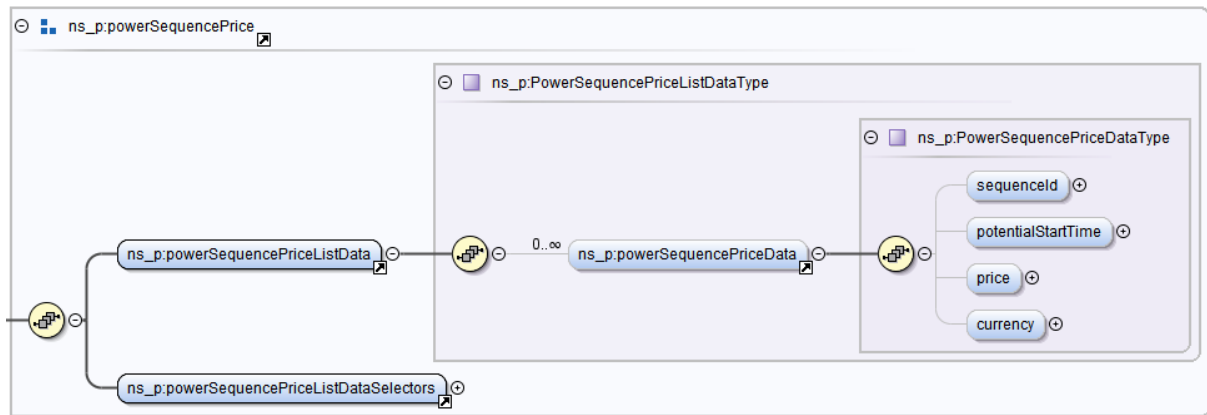


Figure 164: powerSequencePriceListData function overview

5.3.19.12.2 Detailed description of elements

Element	Type	Description
powerSequencePriceListData. powerSequencePriceData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
potentialStartTime	Common data type "AbsoluteOrRelativeTime Type". See section 3.10.2.3.	The price information belongs to the start time given with this element. Please note this can differ from the final start time of a sequence (in case the sequence has been shifted in time but the price has not been updated accordingly, e.g.). Of course, in this case the price denoted in this instance can differ from the "real" price.
price	Common data type "ScaledNumberType". See section 3.10.1.8.	The price of the sequence for the given potential start time.
currency	Common data type "CurrencyType". See section 3.10.1.19.	The currency of the price.

Table 287: powerSequencePriceListData function element description

5.3.19.12.3 Available selectors

- sequenceId
- potentialStartTimeInterval

5.3.19.12.4 Examples

5.3.19.12.4.1 Write

An energy management system got information about power sequence 2 of an appliance. The energy management system has also knowledge of the energy price and can calculate a proper cost. It submits the calculated price for a potential start time to the appliance and the appliance can then display the cost to the user:

```

9557 <powerSequencePriceListData>
9558   <powerSequencePriceData>
9559     <sequenceId>2</sequenceId>
9560     <potentialStartTime>2013-07-14T12:35:00.0Z</potentialStartTime>
9561     <price>
9562       <number>95</number>
9563       <scale>-2</scale>
9564     </price>
9565     <currency>EUR</currency>
9566   </powerSequencePriceData>
9567 </powerSequencePriceListData>

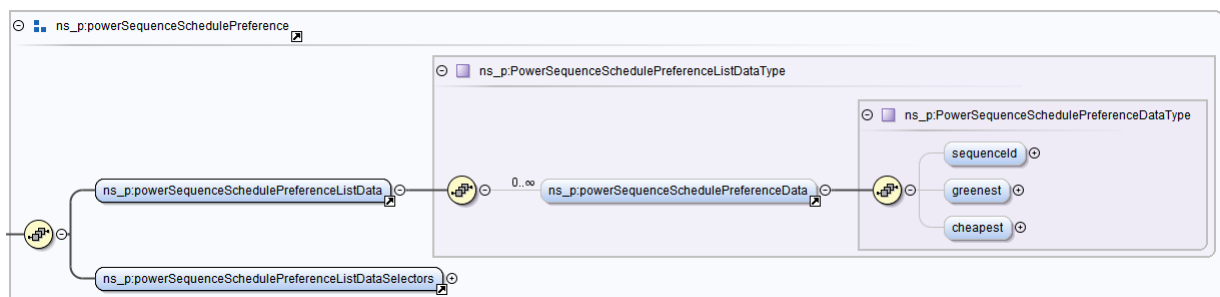
```

9568

9569 5.3.19.13 powerSequenceSchedulePreferenceListData

9570 5.3.19.13.1 General

9571 A "smart energy management" typically includes the consideration of a user's preference for the kind
 9572 of energy consumption optimisation. Some may prefer the optimisation to reduce the cost whereas
 9573 others may prefer first of all the use of "green energy". This kind of preference can be modelled using
 9574 *powerSequenceSchedulePreference*, where the preference can be assigned per sequence.



9575

9576 Figure 165: powerSequenceSchedulePreferenceListData function overview

9577

9578 5.3.19.13.2 Detailed description of elements

Element	Type	Description
powerSequenceSchedulePreferenceListData. powerSequenceSchedulePreferenceData. (list)		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier.
greenest	xs:boolean (W3C standard type)	For this sequence an optimisation towards the minimum usage of NON-renewable energy is preferred.
cheapest	xs:boolean (W3C standard type)	For this sequence an optimisation that minimises the user's energy bill is preferred.

9579 Table 288: powerSequenceSchedulePreferenceListData function element description

9580

9581 5.3.19.13.3 Available selectors

9582 - sequenceId

9583

5.3.19.13.4 Examples

5.3.19.13.4.1 Read-reply

A "smart washing machine" is connected with an energy management system. At the washing machine a user can press a button to enable the preference for renewable energy. Once the washing machine announced an upcoming task, the energy management system queries for the user's preference and receives this response:

```
<powerSequenceSchedulePreferenceListData>
  <powerSequenceSchedulePreferenceData>
    <sequenceId>3</sequenceId>
    <greenest>true</greenest>
  </powerSequenceSchedulePreferenceData>
</powerSequenceSchedulePreferenceListData>
```

5.3.19.14 Sub-class PowerSequenceNode

This sub-class includes PowerSequences information affecting the whole node rather than a specific sequence.

5.3.19.15 powerSequenceNodeScheduleInformationData

5.3.19.15.1 General

The *powerSequenceNodeScheduleInformationData* function permits modelling of some schedule information that applies to a whole node rather than to a specific sequence. This information can change during runtime (especially the *nodeRemoteControllable* element).

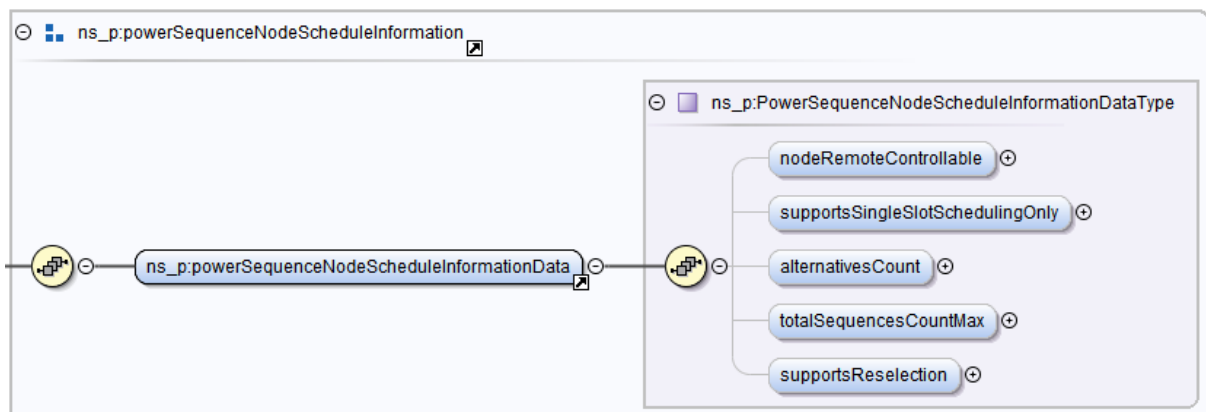


Figure 166: powerSequenceNodeScheduleInformationData function overview

5.3.19.15.2 Detailed description of elements

Element	Type	Description
powerSequenceNodeScheduleInformationData.		
nodeRemoteControl lable	xs:boolean (W3C standard type)	This element denotes whether a node permits being scheduled (value is "true") by a remote device or not (value is "false").

supportsSingleSlotSchedulingOnly	xs:boolean (W3C standard type)	If set to "true" the appliance does NOT permit the modification of more than one slot per configuration command: The sub-class PowerTimeSlot provides functions that can be used to configure slots of an appliance. Multiple slots may be configured with a single command (single function instance). However, some kind of appliances may support the scheduling of only a single slot per sequence as configuration request. This constraint can be modelled setting this element to <i>true</i> . In this case for each slot a separate configuration request (write command with the function properly set) for scheduling should be used.
alternativesCount	xs:unsignedInt (W3C standard type)	The total number of "alternatives groups" (see sections 5.3.19.1.3, 5.3.19.7). If powerSequenceAlternativesRelationListData is used, the element alternativesCount denotes the list size.
totalSequencesCountMax	xs:unsignedInt (W3C standard type)	The maximum total number of sequences the device may offer at any given moment. This information is primarily intended for energy management systems to get a notion of an appliance's schedule complexity. If an appliance counts all of its currently known sequences (i.e. across all alternatives groups and also regardless of the sequences' states) the result will not exceed totalSequencesCountMax.
supportsReselection	xs:boolean (W3C standard type)	As mentioned in section 5.3.19.1.3 an appliance may permit an energy management system to select one of its sequences from an alternatives group for execution. Some appliances may permit such a selection only one time per alternatives group. This can be expressed with "supportsReselection" setting to "false". If an appliance grants the energy management system to alter the selection (i.e. re-select) of the alternatives group, this can be expressed with the value "true".

Table 289: powerSequenceNodeScheduleInformationData function element description

9610

9611

9612 5.3.19.15.3 Available selectors

9613 None.

9614

9615 5.3.19.15.4 Examples

9616 5.3.19.15.4.1 Notify

9617 An Energy Management Software wants to get an overview on the current state of a washing

9618 machine, esp. in order to see whether it can be scheduled. It sends a requests and gets this response:

9619 <powerSequenceNodeScheduleInformationData>

9620 <nodeRemoteControllable>true</nodeRemoteControllable>

```

<supportsSingleSlotSchedulingOnly>true</supportsSingleSlotSchedulingOnly>
  <alternativesCount>1</alternativesCount>
  <totalSequencesCountMax>3</totalSequencesCountMax>
  <supportsReselection>false</supportsReselection>
</powerSequenceNodeScheduleInformationData>

```

5.3.19.16 PowerSequence calls

Subsequently, the calls available in the PowerSequence Class are described.

5.3.19.17 powerSequenceScheduleConfigurationRequestCall

5.3.19.17.1 General

Some functions of the sub-class PowerSequence can be used to model notification of new or changed sequences. Energy management gateways may listen to such notifications and take action on demand. However, the nature of notifications does not imply any mandatory action upon receipt. With the function *powerSequenceScheduleConfigurationRequestCall* the owner of a sequence (i.e. a server with regards to the PowerSequences class) can model an explicit request stating a sequence shall be configured. I.e. an Energy Management Software (i.e. a client with regards to the PowerSequences class) receiving this function would know it needs to take proper action (provided it is capable of such action, i.e. performing write operations with sequence configuration).

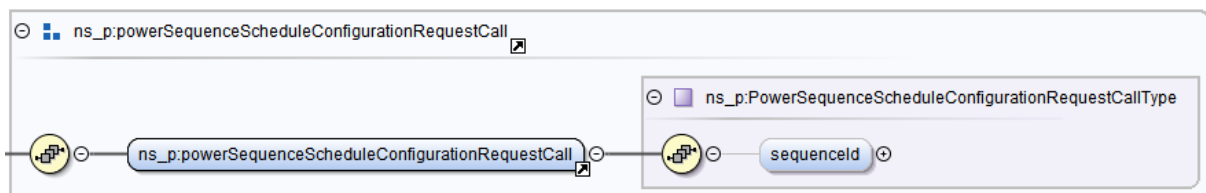


Figure 167: *powerSequenceScheduleConfigurationRequestCall* function overview

5.3.19.17.2 Detailed description of elements

Element	Type	Description
powerSequenceScheduleConfigurationRequestCall.		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier the owner of the power sequence wants to be configured.

Table 290: *powerSequenceScheduleConfigurationRequestCall* function element description

5.3.19.17.3 Available selectors

None.

5.3.19.17.4 Examples

5.3.19.17.4.1 Call

If a node wants one of its power sequences to be configured, it could send the following XML with a *call* classifier to an appropriate Energy Management Software:

```
<powerSequenceScheduleConfigurationRequestCall>
  <sequenceId>2</sequenceId>
</powerSequenceScheduleConfigurationRequestCall>
```

5.3.19.18 powerSequencePriceCalculationRequestCall

5.3.19.18.1 General

The owner of a power sequence can use this function in order to trigger the determination of a price for one of its sequences at an external node. This shall be explained in more detail now.

Section 5.3.19.12.1 describes how the owner of a sequence can express a price for a given sequence and start time. Before the owner is able to tell a price upon request it needs to get knowledge of the price by itself. The section also briefly mentions different possibilities to achieve this. One of these possibilities is the use of the *powerSequencePriceCalculationRequestCall* function: It requires a sequence is available that expresses its energy or power demand. It also requires the owner of the sequence knows of an external node that has the capability to determine a price for the owner's sequence. Then, the owner of the sequence can submit a *powerSequencePriceCalculationResponseCall* function to this external node in order to trigger the determination of the price. This function can take the *sequenceId* and also a potential start time. Finally, the external node may use the function *powerSequencePriceData* (see section 5.3.19.12) in order to return the calculated price to the owner of the sequence.

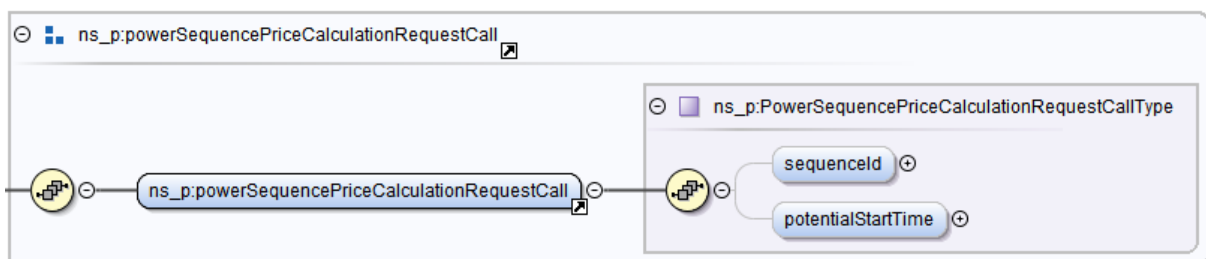


Figure 168: powerSequencePriceCalculationRequestCall function overview

5.3.19.18.2 Detailed description of elements

Element	Type	Description
powerSequencePriceCalculationRequestCall.		
sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	The sequence identifier the owner of the power sequence wants to get a price for.
potentialStartTime	Common data type "AbsoluteOrRelative"	The price shall be determined for a potential start time given in this element.

	<i>TimeType". See section 3.10.2.3.</i>	
--	---	--

Table 291: powerSequencePriceCalculationRequestCall function element description

5.3.19.18.3 Available selectors

None.

5.3.19.18.4 Examples

5.3.19.18.4.1 Call

The scenario assumes an external node (an energy management system, e.g.) that is capable of calculating prices based upon a given energy demand and schedule and also has knowledge of the tariff that is relevant for the required energy. A washing machine prepared a power sequence and wants to get a price for this sequence with a potential start time. It sends this XML with a *call* classifier to the energy management system:

```
<powerSequencePriceCalculationRequestCall>
  <sequenceId>2</sequenceId>
  <potentialStartTime>2013-07-14T12:35:00.0Z</potentialStartTime>
</powerSequencePriceCalculationRequestCall>
```

5.3.20 Sensing

5.3.20.1 Introduction

The *Sensing* class is used for modelling data obtained by sensors in the meaning of more state-based sensors like contact sensors, switches and buttons, etc., in contrast to value-based sensors like temperature or humidity sensors that are treated with the *Measurement* class.

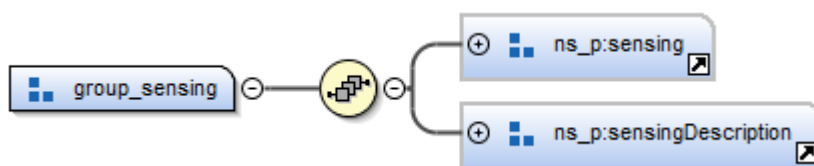


Figure 169: Sensing function-group overview

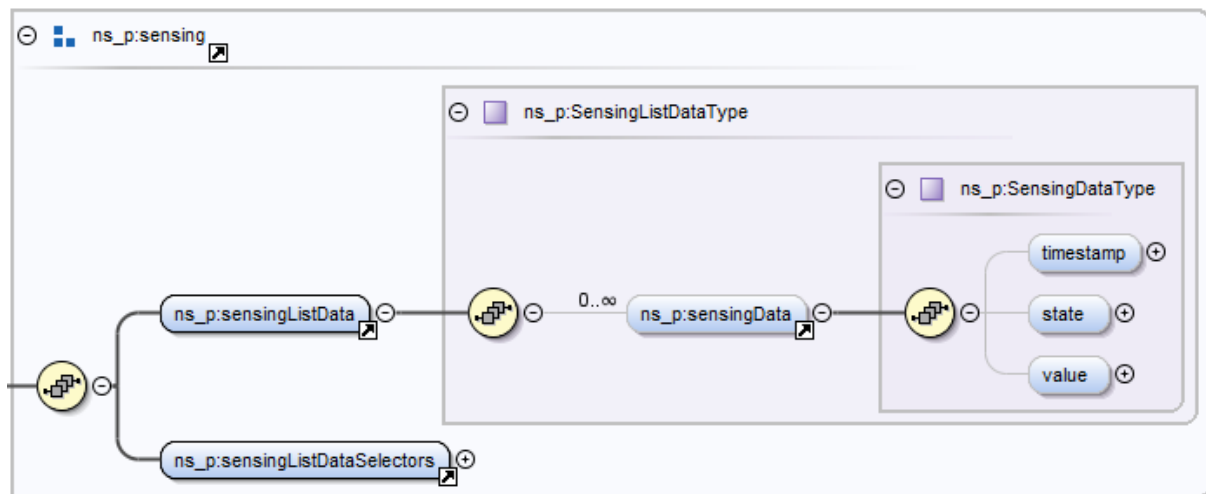
5.3.20.2 sensingListData

5.3.20.2.1 General

The list function of sensing can be used to model sensor information over time. This may be of interest for a motion detector (when someone was at the backdoor), a contact sensor (when was the window open in the last 24 hours), etc.

Some sensors may only store the latest value and therefore only support one entry in the list. To enable users and applications to query past changes, a communications technology interface may

9709 cache states/values of simple sensors that only support the single data function. That is no
 9710 mandatory feature, however.



9711
 9712 Figure 170: sensingListData function overview

9713

9714 5.3.20.2.2 Detailed description of elements

Element	Type	Description
sensingListData. sensingData. (list)		
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The creation time of the sensingData instance.
state	Union "SensingStateType": - Enum (see Table 293): SensingStateEnumType - EnumExtendType (see section 3.10.1.5)	The state of the sensor is denoted in this element.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	Some sensors deliver a value (e.g. a levelSwitch).

9715 Table 292: sensingListData function detailed description of elements

9716 Enumeration **SensingStateEnumType**:

Value	Description
on	"on" was detected, meaning that the according functionality was switched on.
off	"off" was detected, meaning that the according functionality was switched off.
toggle	"toggle" was detected, meaning that the according functionality was toggled (e.g. from "on" to "off" or vice versa).
level	"level" was detected, meaning that the according functionality changed its level (e.g. from 2.4 to 8.2) that can be found in the Element "value".
levelUp	"levelUp" was detected, meaning that the according functionality changed its level upwards.
levelDown	"levelDown" was detected, meaning that the according functionality changed its level downwards.

levelStart	"levelStart" was detected, meaning that the according functionality started to change its level.
levelStop	"levelStop" was detected, meaning that the according functionality stopped to change its level.
levelAbsolute	"levelAbsolute" was detected, meaning that the according functionality changed its level to an absolute value stated in the Element "value" (e.g. 1190).
levelRelative	"levelRelative" was detected, meaning that the according functionality moved its level in comparison to the old one. The relative change can be found in the Element "value" (e.g. a change from 22 to 33 would result in a content of the Element "value" of 11).
levelPercentageAbsolute	"levelPercentageAbsolute" was detected, meaning that the according functionality changed its level to an absolute percentage value stated in the Element "value" (e.g. 90%).
levelPercentageRelative	"levelPercentageRelative" was detected, meaning that the according functionality changed its percentage level relative to the old one. The relative change can be found in the Element "value" (e.g. a change from 40% to 50% would result in a content of the Element "value" of 10).
pressed	"pressed" was detected, meaning that the according functionality was pressed (e.g. a button, that was pressed).
longPressed	"longPressed" was detected, meaning that the according functionality was pressed longer (e.g. a button, that was pressed longer). The timing difference between "pressed" and "longPressed" is up to the vendor.
released	"released" was detected, meaning that the according functionality was released (e.g. a button, that was released).
changed	"changed" was detected, meaning that the according functionality changed.
started	"started" was detected, meaning that the according functionality started.
stopped	"stopped" was detected, meaning that the according functionality stopped.
paused	"paused" was detected, meaning that the according functionality paused.
middle	"middle" was detected, meaning that the according functionality (e.g. a scroll bar or a throttle) is in position "middle".
up	"up" was detected, meaning that the according functionality moves up.
down	"down" was detected, meaning that the according functionality moves down.
forward	"forward" was detected, meaning that the according functionality moves forward.
backwards	"backwards" was detected, meaning that the according functionality moves backwards.
open	"open" was detected, meaning that the according functionality is open (e.g. a door).
closed	"closed" was detected, meaning that the according functionality is closed (e.g. a door).
opening	"opening" was detected, meaning that the according functionality is opening (e.g. a door, that is changing from "closed" to "open").

closing	"closing" was detected, meaning that the according functionality is closing (e.g. a door, that is changing from "open" to "closed").
high	"high" was detected, meaning that the according functionality is at a high position or value.
low	"low" was detected, meaning that the according functionality is at a low position or value.
day	"day" was detected, meaning that the according functionality is in day-mode.
night	"night" was detected, meaning that the according functionality is in night-mode.
detected	"detected" was detected, meaning that the according functionality was detected (e.g. a motion sensor detected a motion).
notDetected	"notDetected" was detected, meaning that the according functionality was detected (e.g. a motion sensor reports that no motion is detected).
alarmed	"alarmed" was detected, meaning that the according functionality is in alarm mode (e.g. a smoke detector).
notAlarmed	"notAlarmed" was detected, meaning that the according functionality is not in alarm mode (e.g. a smoke detector).

Table 293: Enumeration SensingStateEnumType

9717

9718

9719 5.3.20.2.3 Available selectors

9720 - timestampInterval

9721

9722 5.3.20.2.4 Examples

9723 5.3.20.2.4.1 Notify

9724 If a contact sensor detects an opening (e.g. of a window), it will send a (complete) notification about
 9725 that occurrence:

```

9726 <sensingListData>
9727   <sensingData>
9728     <timestamp>2013-07-15T12:01:02.0Z</timestamp>
9729     <state>opened</state>
9730   </sensingData>
9731 </sensingListData>

```

9732

9733 5.3.20.2.4.2 Read-reply

9734 If one is interested in the activities of a switch for the time between 12:00 and 14:00 o'clock, he
 9735 could send the following read command:

```

9736 <function>sensingListData</function>
9737 <filter>
9738   <cmdControl>
9739     <partial/>
9740   </cmdControl>
9741   <sensingListDataSelectors>
9742     <timestampInterval>
9743       <startTime>2013-07-15T12:00:00.0Z</startTime>

```



```

9744         <endTime>2013-07-15T14:00:00.0Z</endTime>
9745     </timestampInterval>
9746 </sensingListDataSelectors>
9747 </filter>
9748 <sensingListData/>

```

9749 As a reply, something like the following could be returned:

```

9750 <function>sensingListData</function>
9751 <filter>
9752     <cmdControl>
9753         <partial/>
9754     </cmdControl>
9755 </filter>
9756 <sensingListData>
9757     <sensingData>
9758         <timestamp>2013-07-15T12:01:00.0Z</timestamp>
9759         <state>on</state>
9760     </sensingData>
9761     <sensingData>
9762         <timestamp>2013-07-15T12:50:32.0Z</timestamp>
9763         <state>off</state>
9764     </sensingData>
9765     <sensingData>
9766         <timestamp>2013-07-15T13:15:05.0Z</timestamp>
9767         <state>on</state>
9768     </sensingData>
9769 </sensingListData>

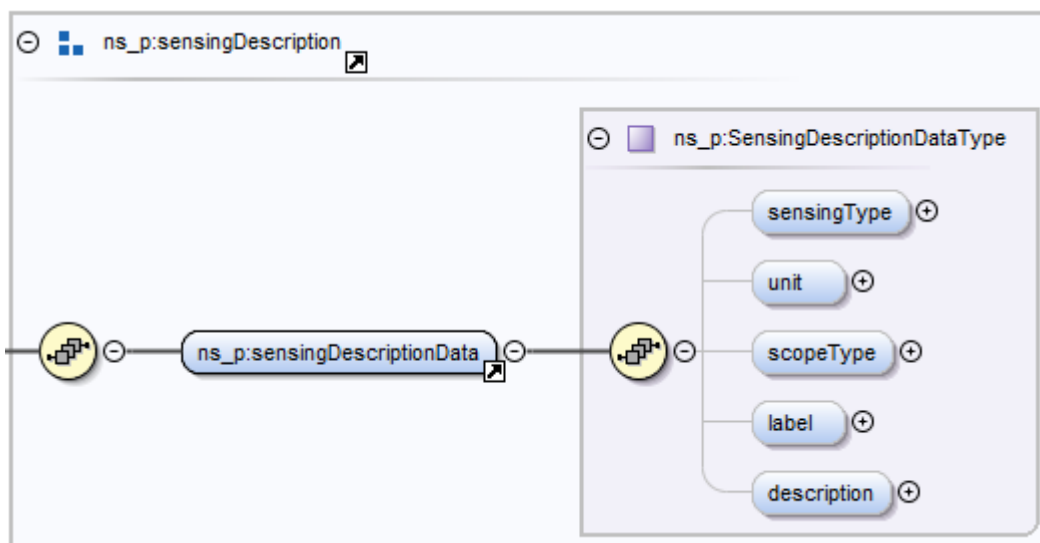
```

9770

9771 5.3.20.3 *sensingDescriptionData*

9772 5.3.20.3.1 *General*

9773 The rather static (non-changing) elements of the *Sensing* class are combined in the
 9774 *sensingDescriptionData* function. Typically, it is only requested once by a user or an application.



9775

9776 Figure 171: *sensingDescriptionData* function overview

9777

9778 5.3.20.3.2 Detailed description of elements

Element	Type	Description
sensingDescriptionData.		
sensingType	<i>Union "SensingTypeType":</i> - Enum (see Table 295): SensingTypeEnumType - EnumExtendType (see section 3.10.1.5)	Which type of sensor is modelled is denoted with this element.
unit	<i>Common data type</i> <i>"UnitOfMeasurementType". See section 3.10.1.17.</i>	The unit in which the values of the sensor are given. Not all sensor types need the value element and therefore the unit may be omitted as well.
scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	Specifies a more detailed meaning of the sensor (e.g. shutter).
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A short label of the sensor.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Descriptive information on the sensor.

9779 Table 294: sensingDescriptionData function detailed description of elements

9780 Enumeration **SensingTypeEnumType**:

Value	Description
switch	A switch that has usually two states "on" and "off".
button	A button that can be "pressed" (and eventually supports "longPressed" and "released")
level	Some level functionality supporting one or more of the level-related sensing states.
levelSwitch	Some level functionality supporting one or more of the level-related sensing states and a switch functionality ("on" / "off").
windowHandle	A window handle supporting different positions.
contactSensor	Sensor that can detect whether something is "open" or "closed".
occupancySensor	Sensor that can detect whether someone is present in a room ("detected") or not ("notDetected").
motionDetector	Sensor that can detect motion ("detected"/"notDetected").
fireDetector	A sensor that detects fire ("detected"/"notDetected").
smokeDetector	A sensor that detects smoke ("detected"/"notDetected").
heatDetector	A sensor that detects heat ("detected"/"notDetected").
waterDetector	A sensor that detects water ("detected"/"notDetected").
gasDetector	A sensor that detects gas ("detected"/"notDetected").
alarmSensor	A sensor providing alarm information ("alarmed" / "notAlarmed").
powerAlarmSensor	A sensor reporting power alarms ("alarmed" / "notAlarmed").
dayNightIndicator	A sensor detecting whether it is "day" or "night".

9781 Table 295: Enumeration SensingTypeEnumType

9782

9783 5.3.20.3.3 Available selectors

9784 None.

9785

9786 5.3.20.3.4 Examples

9787 5.3.20.3.4.1 Notify

9788 An example instance of *sensingDescriptionData* of a smoke detector, sent as notification:

```

9789 <sensingDescriptionData>
9790   <sensingType>smokeDetector</sensingType>
9791   <label>SmokeDetector Bedroom</label>
9792 </sensingDescriptionData>

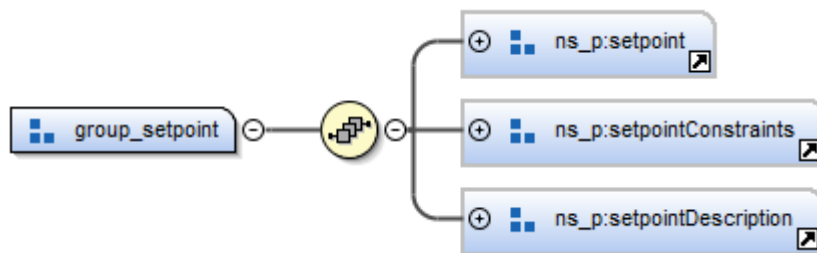
```

9793

9794 5.3.21 Setpoint

9795 5.3.21.1 Introduction

9796 Setting setpoint values is an often chosen way to control some functionality of a device. Especially in
 9797 the home automation and HVAC domain, several use cases depend on setpoints. This class handles
 9798 the setpoint itself. Relations to Measurement and TimeTable class enables the control of a
 9799 measurand for specific time frames.



9800

9801 Figure 172: Setpoint function-group overview

9802

9803 5.3.21.2 setpointListData

9804 5.3.21.2.1 General

9805 The setpoint values and (if needed) the identifier of related measurands or time tables are modelled
 9806 within this function. It is possible to either define an exact value, that should be reached or a range
 9807 which shall not be fallen below or exceeded. Absolute or relative tolerances can be defined, too.

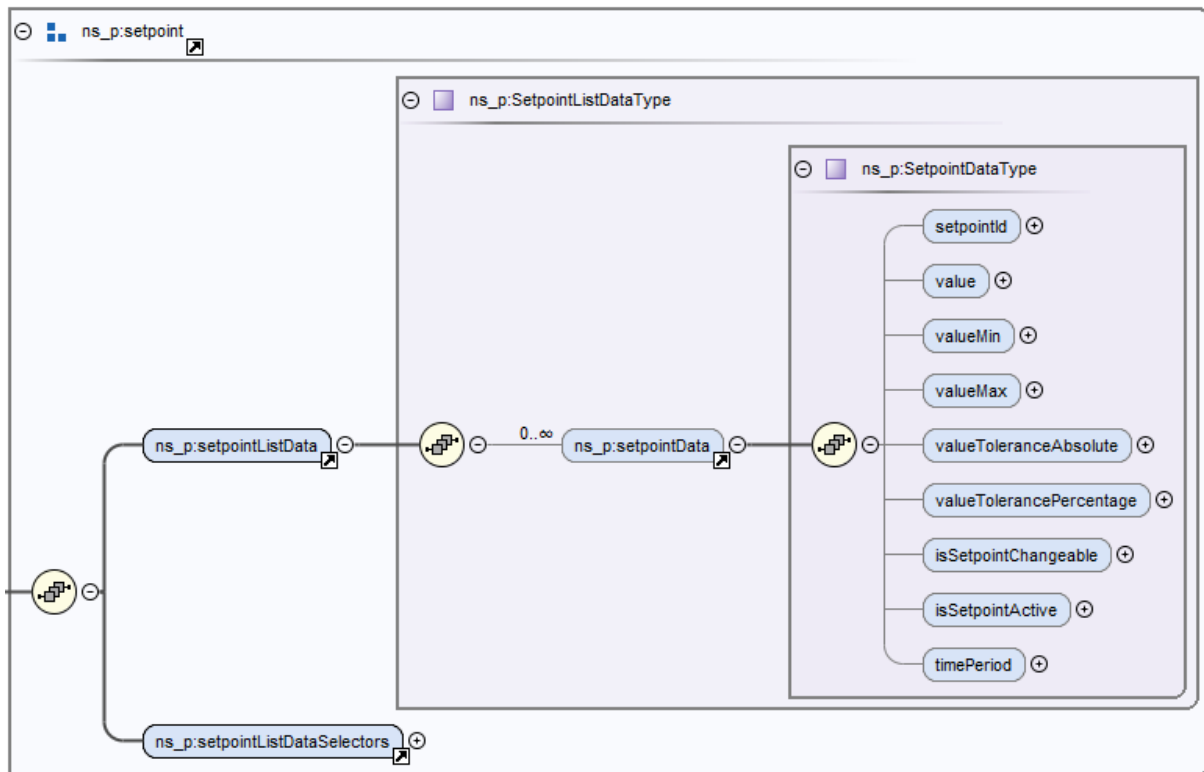


Figure 173: setpointListData function overview

5.3.21.2.2 Detailed description of elements

Element	Type	Description
setpointListData. setpointData. (list)		
setpointId	Identifier "SetpointIdType" (see Table 358).	Identifier of the setpoint.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	The setpoint value itself.
valueMin	Common data type "ScaledNumberType". See section 3.10.1.8.	Instead of a fixed value (see above) a value range can be set. This would be the lower limit.
valueMax	Common data type "ScaledNumberType". See section 3.10.1.8.	Instead of a fixed value (see above) a value range can be set. This would be the upper limit.
valueToleranceAbsolute	Common data type "ScaledNumberType". See section 3.10.1.8.	If the value may vary some absolute value around the desired setpoint, the maximum variation allowed (in both directions) can be stated here.
valueTolerancePercentage	Common data type "ScaledNumberType". See section 3.10.1.8.	If the value may vary some percentage around the desired setpoint, the maximum variation allowed (in both directions) can be stated here.
isSetpointChangeable	xs:boolean (W3C standard type)	States whether the setpoint is changeable by a client or not.

isSetpointActive	xs:boolean (W3C standard type)	Indicates whether the setpoint is currently active or not.
timePeriod	<i>Common data type "TimePeriodType". See section 3.10.2.1.</i>	The period where the setpoint shall be active.

Table 296: setpointListData function detailed description of elements

5.3.21.2.3 Available selectors

- setpointId

5.3.21.2.4 Examples

5.3.21.2.4.1 Read-reply

If one is interested in the setpoint with the *setpointId* "3", he could send the following read command:

```
<function>setpointListData</function>
<filter>
  <cmdControl>
    <partial/>
  </cmdControl>
  <setpointListDataSelectors>
    <setpointId>3</setpointId>
  </setpointListDataSelectors>
</filter>
<setpointListData/>
```

As a reply, something like the following could be returned:

```
<function>setpointListData</function>
<filter>
  <cmdControl>
    <partial/>
  </cmdControl>
</filter>
<setpointListData>
  <setpointData>
    <setpointId>3</setpointId>
    <value>
      <number>25</number>
    </value>
    <valueToleranceAbsolute>
      <number>1</number>
    </valueToleranceAbsolute>
    <isSetpointChangeable>true</isSetpointChangeable>
    <isSetpointActive>true</isSetpointActive>
    <timePeriod>
      <startTime>2020-08-17T12:00:00.0Z</startTime>
      <endTime>2020-08-17T15:15:00.0Z</endTime>
    </timePeriod>
  </setpointData>
</setpointListData>
```

5.3.21.3 setpointConstraintsListData

5.3.21.3.1 General

The constraints regarding the possible setpoints allowed are modelled within this function.

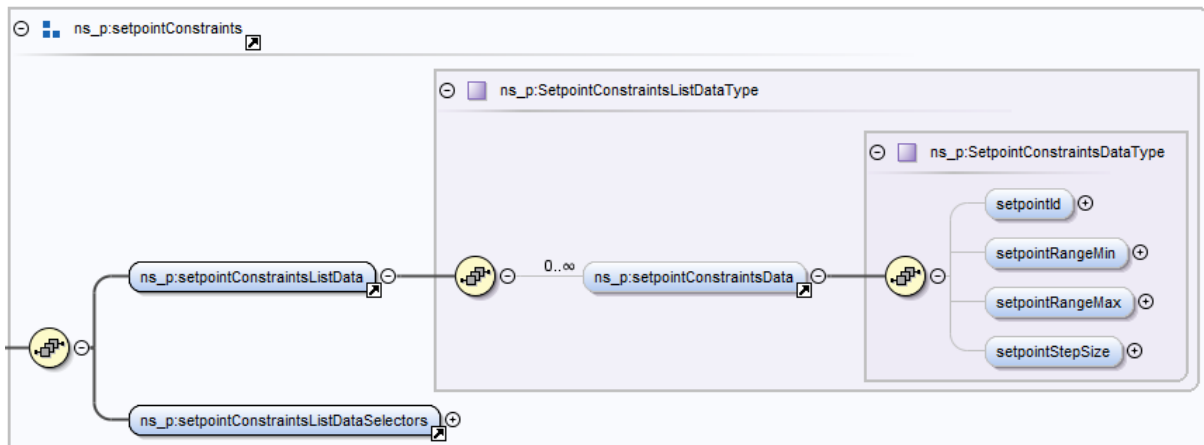


Figure 174: setpointConstraintsListData function overview

5.3.21.3.2 Detailed description of elements

Element	Type	Description
setpointConstraintsListData. setpointConstraintsData. (list)		
setpointId	Identifier "SetpointIdType" (see Table 358).	Identifier of the setpoint, these constraints belong to.
setpointRangeMin	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum value, the setpoint can be set to.
setpointRangeMax	Common data type "ScaledNumberType". See section 3.10.1.8.	The maximum value, the setpoint can be set to.
setpointStepSize	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum step size between two different values.

Table 297: setpointConstraintsListData function detailed description of elements

5.3.21.3.3 Available selectors

- setpointId

5.3.21.3.4 Examples

5.3.21.3.4.1 Read-reply

If one is interested in the setpoint constraints of all *setpointIds*, he could send the standard read command. A reply could be:

```
<setpointConstraintsListData>
  <setpointConstraintsData>
```

```

9874         <setpointId>1</setpointId>
9875         <setpointRangeMin>
9876             <number>0</number>
9877         </setpointRangeMin>
9878         <setpointRangeMax>
9879             <number>3</number>
9880             <scale>1</scale>
9881         </setpointRangeMax>
9882         <setpointStepSize>
9883             <number>5</number>
9884             <scale>-1</scale>
9885         </setpointStepSize>
9886     </setpointConstraintsData>
9887     <setpointConstraintsData>
9888         <setpointId>2</setpointId>
9889         <setpointRangeMin>
9890             <number>3</number>
9891             <scale>1</scale>
9892         </setpointRangeMin>
9893         <setpointRangeMax>
9894             <number>8</number>
9895             <scale>1</scale>
9896         </setpointRangeMax>
9897         <setpointStepSize>
9898             <number>1</number>
9899         </setpointStepSize>
9900     </setpointConstraintsData>
9901     <setpointConstraintsData>
9902         <setpointId>3</setpointId>
9903         <setpointRangeMin>
9904             <number>5</number>
9905         </setpointRangeMin>
9906         <setpointRangeMax>
9907             <number>2</number>
9908             <scale>2</scale>
9909         </setpointRangeMax>
9910         <setpointStepSize>
9911             <number>5</number>
9912         </setpointStepSize>
9913     </setpointConstraintsData>
9914 </setpointConstraintsListData>

```

5.3.21.4 *setpointDescriptionListData*

5.3.21.4.1 *General*

The rather static (non-changing) elements of the *Setpoint* class are combined in the *setpointDescriptionListData* function. Typically, it is only requested once by a user or an application.

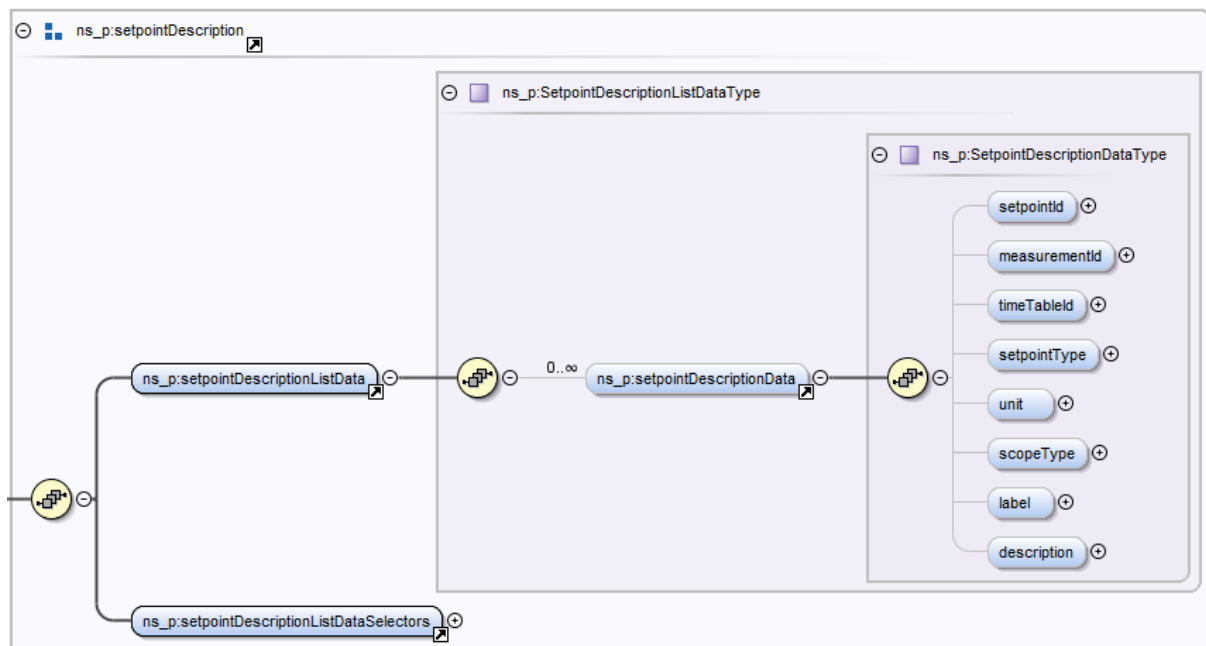


Figure 175: setpointDescriptionListData function overview

5.3.21.4.2 Detailed description of elements

Element	Type	Description
setpointDescriptionListData.setpointDescriptionData. (list)		
setpointId	Identifier "SetpointIdType" (see Table 358).	Identifier of the setpoint.
measurementId	Identifier "MeasurementIdType" (see Table 358).	If the setpoint relates to a measurand, the corresponding ID can be stated here.
timeTableId	Identifier "timeTableIdType" (see Table 358).	If the activation of the setpoint is time dependent, the ID of a corresponding time table can denoted in this element.
setpointType	Union "SetpointTypeType": - Enum (see Table 299): SetpointTypeEnumType - EnumExtendType (see section 3.10.1.5)	This is the type of setpoint used for this specific setpoint.
unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	The unit used for this setpoint (see SPINE data model for all possible values).
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	Specifies a more detailed meaning of the setpoint.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the setpoint.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the setpoint.

Table 298: setpointDescriptionListData function detailed description of elements

Enumeration **SetpointTypeEnumType**:

Value	Description
valueAbsolute	An absolute value, e.g. 28°C. A unit is needed in most cases.
valueRelative	A relative value in percentage. The unit is "pct" in most cases.

9926 Table 299: Enumeration SetpointTypeEnumType

9927

9928 5.3.21.4.3 Available selectors

- 9929 - setpointId
- 9930 - measurementId
- 9931 - timeTableId
- 9932 - setpointType
- 9933 - scopeType

9934

9935 5.3.21.4.4 Examples

9936 5.3.21.4.4.1 Read-reply

9937 If one is interested in the setpoints related to *measurementId* "3", he could send the following read
9938 command:

```
9939 <function>setpointDescriptionListData</function>
9940 <filter>
9941   <cmdControl>
9942     <partial/>
9943   </cmdControl>
9944   <setpointDescriptionListDataSelectors>
9945     <measurementId>3</measurementId>
9946   </setpointDescriptionListDataSelectors>
9947 </filter>
9948 <setpointDescriptionListData/>
```

9949 The reply could look like this:

```
9950 <function>setpointDescriptionListData</function>
9951 <filter>
9952   <cmdControl>
9953     <partial/>
9954   </cmdControl>
9955 </filter>
9956 <setpointDescriptionListData>
9957   <setpointDescriptionData>
9958     <setpointId>1</setpointId>
9959     <measurementId>3</measurementId>
9960     <timeTableId>2</timeTableId>
9961     <setpointType>valueAbsolute</setpointType>
9962     <unit>degC</unit>
9963     <label>Day heating</label>
9964   </setpointDescriptionData>
9965   <setpointDescriptionData>
9966     <setpointId>2</setpointId>
9967     <measurementId>3</measurementId>
9968     <timeTableId>4</timeTableId>
9969     <setpointType>valueAbsolute</setpointType>
9970     <unit>degC</unit>
9971     <label>Night heating</label>
9972   </setpointDescriptionData>
```

9973 </setpointDescriptionListData>

9974

9975 5.3.22 SubscriptionManagement

9976 5.3.22.1 Introduction

9977 Subscriptions are used to simplify communication between devices. Based on matching features,
9978 finding suitable communication partners is a lot easier than via manual configuration only.

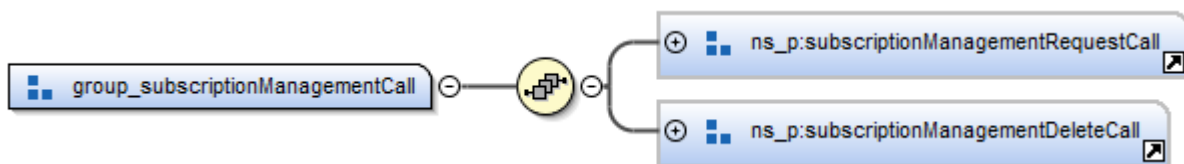
9979 Each subscription exists between a subscription client and a subscription server. Only clients may
9980 initiate new subscriptions. After a successful establishment of a subscription, the subscription server
9981 updates the client about changes of values or parameters of the corresponding feature, depending
9982 on the subscribed functionality. E.g. a subscription server sends new temperature values of a
9983 subscribed measurement feature to the subscription client. The exact definition of rules and
9984 proceedings is out of scope of this section.

9985 The *SubscriptionManagement* class provides functions for establishing, reading and deleting
9986 subscriptions.



9987

9988 Figure 176: SubscriptionManagement function-group overview (data)



9989

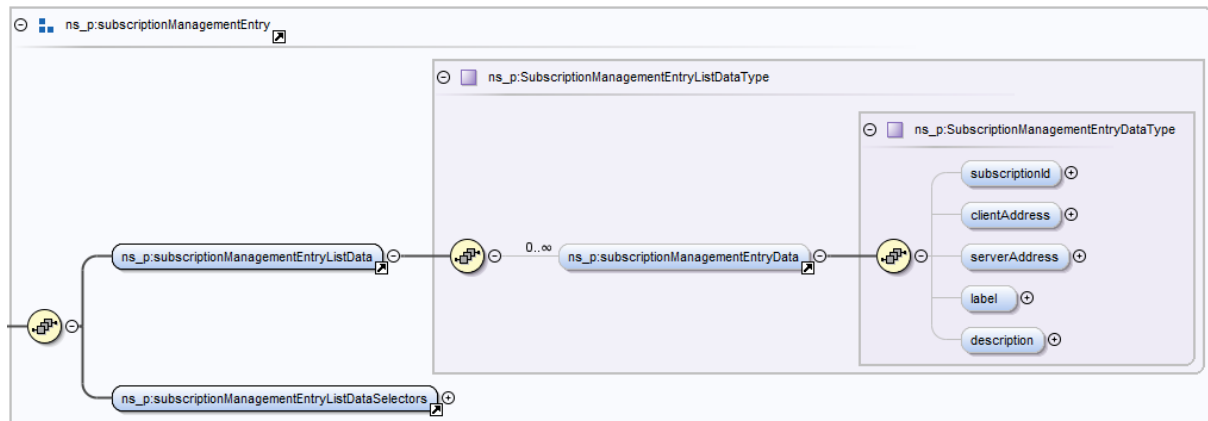
9990 Figure 177: SubscriptionManagement function-group overview (call)

9991

9992 5.3.22.2 subscriptionManagementEntryListData

9993 5.3.22.2.1 General

9994 All existing subscriptions can be listed in the *subscriptionManagementEntryListData* function. It may
9995 be used to store the entries in the device itself. If another device asks for the list of subscriptions, a
9996 device may decide whether it only replies the subscriptions with the requesting device or all available
9997 subscriptions.



9998

9999 Figure 178: subscriptionManagementEntryListData function overview

10000

10001 5.3.22.2.2 Detailed description of elements

Element	Type	Description
subscriptionManagementEntryListData. subscriptionManagementEntryData. (list)		
subscriptionId	Identifier "SubscriptionIdType" (see Table 358).	Internal identifier for the specific subscription entry.
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in this subscription.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in this subscription.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the subscription.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the subscription.

10002 Table 300: subscriptionManagementEntryListData function detailed description of elements

10003

10004 5.3.22.2.3 Available selectors

- 10005 - subscriptionId
- 10006 - clientAddress
- 10007 - serverAddress

10008

10009 5.3.22.2.4 Examples

10010 5.3.22.2.4.1 Read-reply

10011 If one is interested in all subscriptions of a device, he could send a standard read command to the
 10012 device, which could reply something like this:

```

10013 <subscriptionManagementEntryListData>
10014   <subscriptionManagementEntryData>
10015     <subscriptionId>1</subscriptionId>

```

```

10016      <clientAddress>
10017          <device>d_i:46925_CP\_pqt1-27638162683172637812356813</device>
10018          <entity>2</entity>
10019          <feature>1</feature>
10020      </clientAddress>
10021      <serverAddress>
10022          <device>d_i:46925_CP\_arqf-64986751356897215468975468</device>
10023          <entity>1</entity>
10024          <feature>1</feature>
10025      </serverAddress>
10026      <label>LightInfo_01</label>
10027  </subscriptionManagementEntryData>
10028  <subscriptionManagementEntryData>
10029      <subscriptionId>2</subscriptionId>
10030      <clientAddress>
10031          <device>d_i:46925_CP\_pqt1-27638162683172637812356813</device>
10032          <entity>2</entity>
10033          <feature>4</feature>
10034      </clientAddress>
10035      <serverAddress>
10036          <device>d_i:46925_CP\_arqf-64986751356897215468975468</device>
10037          <entity>1</entity>
10038          <feature>2</feature>
10039      </serverAddress>
10040      <label>LightInfo_02</label>
10041  </subscriptionManagementEntryData>
10042 </subscriptionManagementEntryListData>

```

5.3.22.3 *subscriptionManagementRequestCall*

5.3.22.3.1 *General*

To initiate a new subscription, an entity capable of node management on the client side, sends a *subscriptionManagementRequestCall* to an entity capable of node management on the server side, with the *clientAddress* element in the Function set to one of its features with the role *client* and the *serverAddress* element in the Function set to one of the other device's feature address with the role *server*. The receiving server device checks whether it will accept or decline the request and responds with an acknowledgement message (see [ProtocolSpecification], section "Acknowledgement message") with the *resultData* function and an according *errorNumber* set to 0 (accept subscription request) or some other value (definition of the error code is out of scope of this section; decline subscription request).

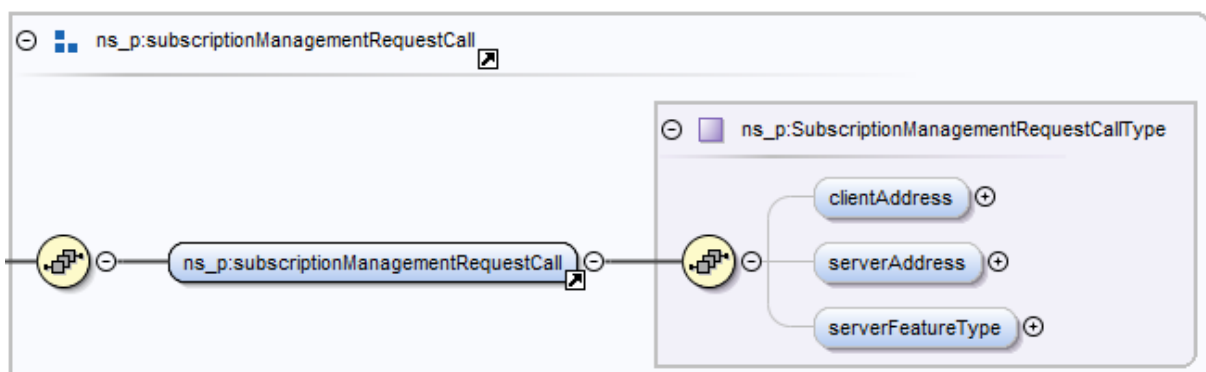


Figure 179: *subscriptionManagementRequestCall* function overview

10057

10058 **5.3.22.3.2 Detailed description of elements**

Element	Type	Description
subscriptionManagementRequestCall.		
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in the subscription request.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in the subscription request.
serverFeatureType	Common data type "FeatureTypeType". See section 3.10.1.29.	The <i>Feature Type</i> of the server feature. Stated for verification.

10059 Table 301: subscriptionManagementRequestCall function detailed description of elements

10060

10061 **5.3.22.3.3 Available selectors**

10062 None.

10063

10064 **5.3.22.3.4 Examples**10065 **5.3.22.3.4.1 Call**

10066 If a new subscription shall be initiated, the client device could send a command like this to the server
 10067 device:

```

10068 <subscriptionManagementRequestCall>
10069   <clientAddress>
10070     <device>d:_i:46925_CP\_pqt1-27638162683172637812356813</device>
10071     <entity>2</entity>
10072     <feature>1</feature>
10073   </clientAddress>
10074   <serverAddress>
10075     <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
10076     <entity>1</entity>
10077     <feature>1</feature>
10078   </serverAddress>
10079   <serverFeatureType>ActuatorSwitch</serverFeatureType>
10080 </subscriptionManagementRequestCall>

```

10081

10082 **5.3.22.4 subscriptionManagementDeleteCall**10083 **5.3.22.4.1 General**

10084 A device may delete an existing subscription with itself and another device. Therefore, this call is
 10085 used.

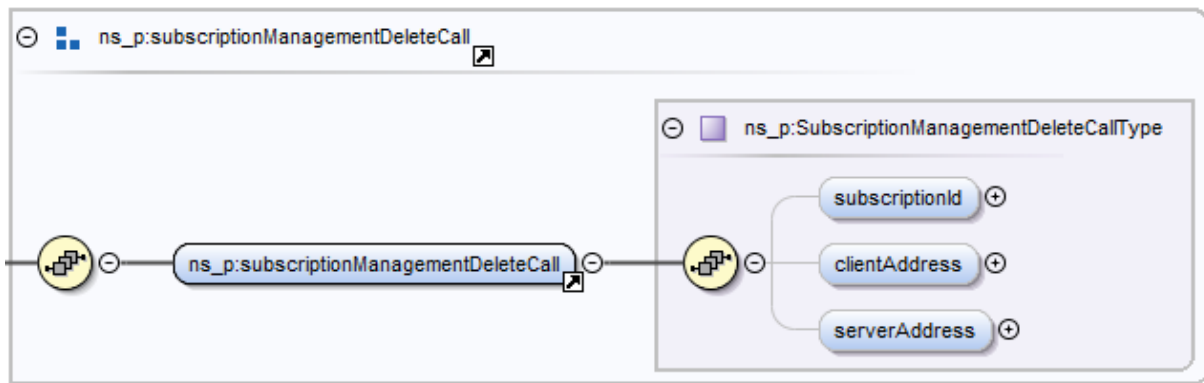


Figure 180: subscriptionManagementDeleteCall function overview

5.3.22.4.2 Detailed description of elements

Element	Type	Description
subscriptionManagementDeleteCall.		
subscriptionId	Identifier "SubscriptionIdType" (see Table 358).	Internal identifier for the specific subscription entry.
clientAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>client</i> in the subscription request.
serverAddress	Common data type "FeatureAddressType". See section 3.10.3.6.	Address of the feature with the role <i>server</i> in the subscription request.

Table 302: subscriptionManagementDeleteCall function detailed description of elements

5.3.22.4.3 Available selectors

None.

5.3.22.4.4 Examples

5.3.22.4.4.1 Call

If an existing subscription shall be deleted, one device could send a command like this to the other device:

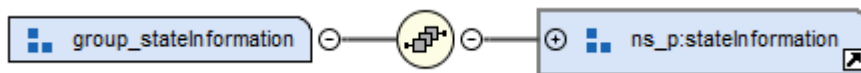
```
<subscriptionManagementDeleteCall>
  <subscriptionId>1</subscriptionId>
  <clientAddress>
    <device>d:_i:46925_CP\_pqt1-27638162683172637812356813</device>
    <entity>2</entity>
    <feature>1</feature>
  </clientAddress>
  <serverAddress>
    <device>d:_i:46925_CP\_arqf-64986751356897215468975468</device>
    <entity>1</entity>
    <feature>1</feature>
  </serverAddress>
</subscriptionManagementDeleteCall>
```

10112

10113 **5.3.23 StateInformation**10114 **5.3.23.1 Introduction**

10115 Information about the state of an Entity can help understand which operational mode or
 10116 functionality can be expected from the Entity or whether a failure occurred (which may lead to a
 10117 non-normal operation).

10118 Each entry within the state information list represents one possible functionality or failure. The
 10119 functionalities or failures then are marked as active or inactive.



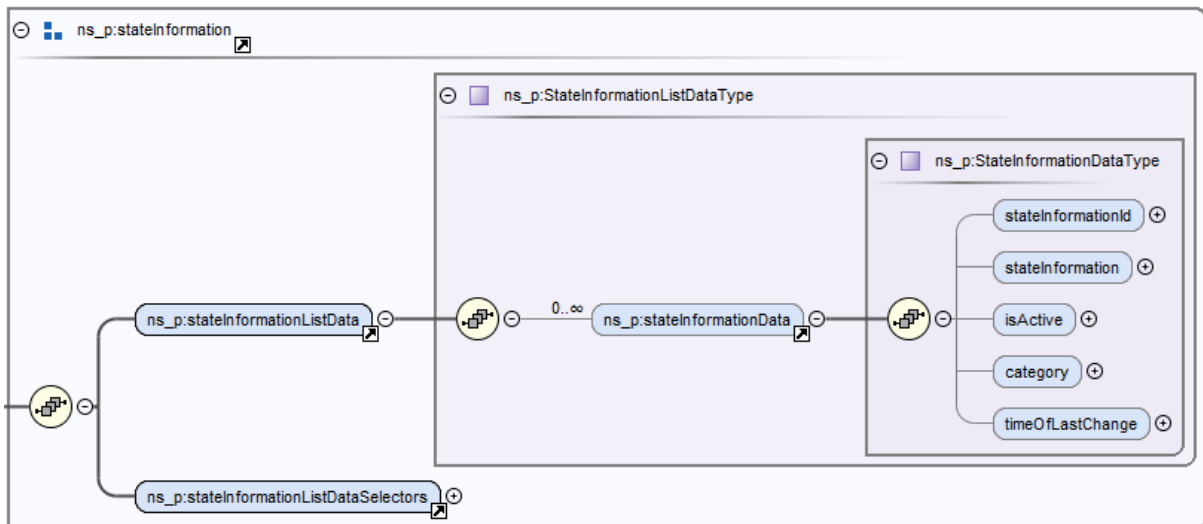
10120

10121 *Figure 181: StateInformation function-group overview*

10122

10123 **5.3.23.2 stateInformationListData**10124 **5.3.23.2.1 General**

10125 This function delivers all state information available at the Entity. A state information is not only
 10126 listed here when it is active, but always, as long as the state information is supported. Whether the
 10127 state information is active or not is marked with the "isActive" flag (see section 5.3.23.2.2).



10128

10129 *Figure 182: stateInformationListData function overview*

10130

10131 **5.3.23.2.2 Detailed description of elements**

Element	Type	Description
stateInformationListData. stateInformationData. (list)		
stateInformationId	Identifier "StateInformationIdType" (see Table 358).	Identifier of the state information.
stateInformation	Union "StateInformationType": - Enum (see Table 304):	The actual state information.

	StateInformationFunctionalityEnumType - Enum (see Table 305): StateInformationFailureEnumType - EnumExtendType (see section 3.10.1.5)	
isActive	xs:boolean (W3C standard type)	If the isActive flag is set to true, the according stateInformation is active. Otherwise it is inactive.
category	Union "StateInformationCategoryType": - Enum (see Table 306): StateInformationCategoryEnumType - EnumExtendType (see section 3.10.1.5)	Each state information has a category that classifies the kind of state.
timeOfLastChange	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The timestamp of the last change of the stateInformation.

10132 Table 303: stateInformationListData function detailed description of elements

10133 Enumeration **StateInformationFunctionalityEnumType**:

Value	Description
externalOverrideFromGrid	The appliance's functionality is overwritten by a command from grid-side.
autonomousGridSupport	The appliance supports grid stability.
islandingMode	The appliance works independent from the (electricity) grid.
balancing	The battery is in balancing mode.
trickleCharging	The battery is in trickle charging mode.
calibration	The appliance is calibrating itself.
commissioningMissing	The appliance waits for commissioning.
sleeping	The appliance is in sleeping mode.
starting	The appliance is starting.
mppt	The inverter is in MPPT (Maximum Power Point Tracker) mode.
throttled	The appliance throttled its consumption or production.
shuttingDown	The appliance is shutting down.
manualShutdown	The appliance is shutting down, initiated manually.

10134 Table 304: Enumeration StateInformationFunctionalityEnumType

10135 Enumeration **StateInformationFailureEnumType**:

Value	Description
inverterDefective	A defect of the inverter was detected.
batteryOvercurrentProtection	The battery is in overcurrent protection mode.
pvStringOvercurrentProtection	The PV string is in overcurrent protection mode.
gridFault	A fault of the grid connection was detected.
groundFault	A fault of the ground connection was detected.
acDisconnected	The AC connection was disconnected.
dcDisconnected	The DC connection was disconnected.
cabinetOpen	The cabinet is open.
overTemperature	The temperature is too high.
underTemperature	The temperature is too low.
frequencyAboveLimit	The frequency is above the limit.
frequencyBelowLimit	The frequency is below the limit.
acVoltageAboveLimit	The AC voltage is above the limit.
acVoltageBelowLimit	The AC voltage is below the limit.

dcVoltageAboveLimit	The DC voltage is above the limit.
dcVoltageBelowLimit	The DC voltage is below the limit.
hardwareTestFailure	A failure occurred during hardware test.
genericInternalError	Some internal error was detected.

Table 305: Enumeration StateInformationFailureEnumType

Enumeration **StateInformationCategoryEnumType**:

Value	Description
functionality	The state information category for functionalities (normal operation).
failure	The state information category for failures (most likely not normal operation).

Table 306: Enumeration StateInformationCategoryEnumType

5.3.23.2.3 Available selectors

- stateInformationId
- stateInformation
- isActive
- category

5.3.23.2.4 Examples

5.3.23.2.4.1 Notify

One of the supported state information has changed and the new data is sent to a bound client that may deliver this information to the user (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..." just to improve the readability):

```
...
<stateInformationListData>
  <stateInformationData>
    <stateInformationId>3</stateInformationId>
    <stateInformation>sleeping</stateInformation>
    <isActive>true</isActive>
    <category>functionality</category>
    <timeOfLastChange>2020-08-05T18:30:00.0Z</timeOfLastChange>
  </stateInformationData>
</stateInformationListData>
```

5.3.24 SupplyCondition

5.3.24.1 Introduction

Practice shows that the supply of electricity, water, etc. can be temporarily impaired. Some possible causes are announced maintenance work at the supply line, supply network overload, or depletion of a local energy resource (battery, e.g.). The Supply Condition concept permits indication of the supply condition by a household's SPINE data models.

The concept is less intended for any automatisms. It is rather intended for indications to users in order to prepare or apply measures in time.

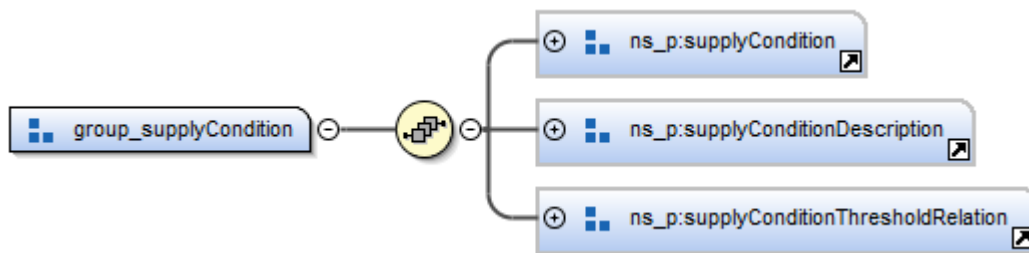


Figure 183: SupplyCondition function-group overview

5.3.24.2 supplyConditionListData

5.3.24.2.1 General

This function can be used to model the "events", i.e. the information on a specific supply condition. These events can, e.g., be sent from a DSO to a household. In this case, the household would be the data owner and the DSO would send (as a client) write commands to this resource.

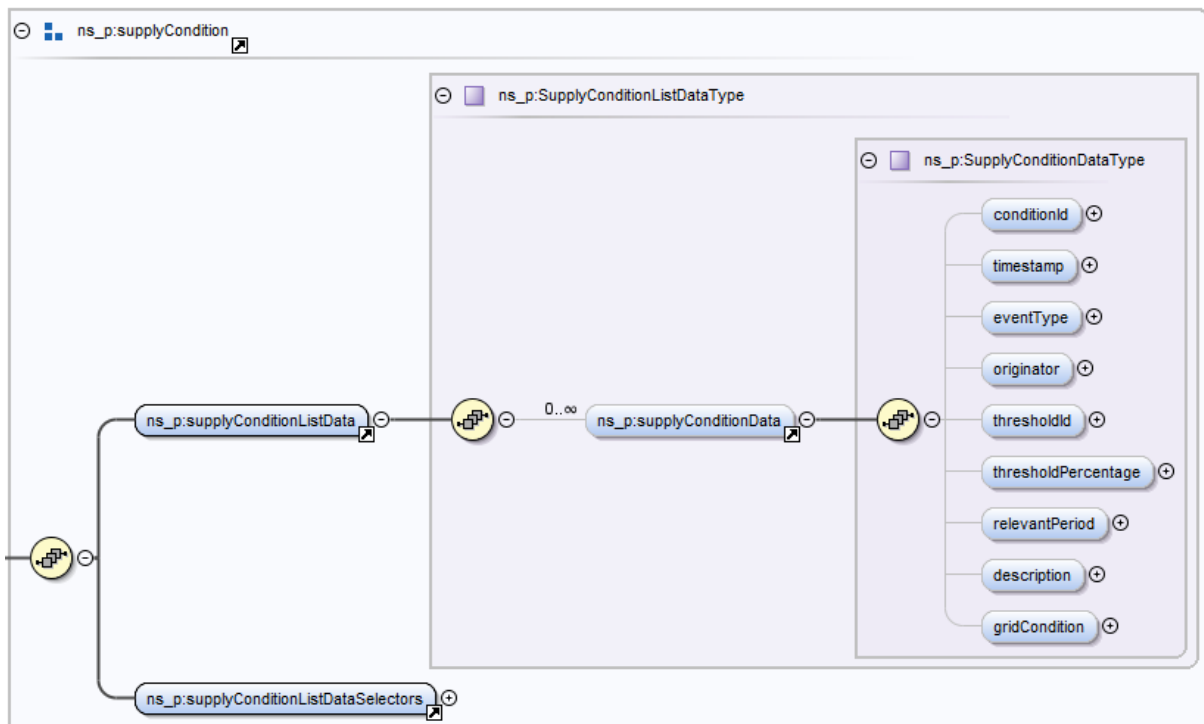


Figure 184: supplyConditionListData function overview

5.3.24.2.2 Detailed description of elements

Element	Type	Description
supplyConditionListData. supplyConditionData. (list)		
conditionId	Identifier "ConditionIdType" (see Table 358).	Identifier of the condition.
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	The timestamp of the event.

eventType	<i>Union</i> "SupplyConditionEventTypeType": - Enum (see Table 308): SupplyConditionEventTypeEnumType - EnumExtendType (see section 3.10.1.5)	The kind of the event.
originator	<i>Union</i> "SupplyConditionOriginatorType": - Enum (see Table 309): SupplyConditionOriginatorEnumType - EnumExtendType (see section 3.10.1.5)	The originator (in terms of instance, not in terms of address) of the information.
thresholdId	<i>Identifier</i> "ThresholdIdType" (see Table 358).	The identifier of this event's threshold (if available).
thresholdPercentage	<i>Common data type</i> "ScaledNumberType". See section 3.10.1.8.	Relation of current situation to (configured) threshold (if available).
relevantPeriod	<i>Common data type</i> "TimePeriodType". See section 3.10.2.1.	Esp. the announcement of maintenance can make use of this element in order to tell when a maintenance is planned.
description	<i>Common data type</i> "DescriptionType". See section 3.10.1.3.	Descriptive information on the event.
gridCondition	<i>Union</i> "GridConditionType": - Enum (see Table 310): GridConditionEnumType - EnumExtendType (see section 3.10.1.5)	Indicates the condition of the electricity grid.

10182 Table 307: supplyConditionListData function detailed description of elements

10183 Enumeration **SupplyConditionEventTypeEnumType**:

Value	Description
thresholdExceeded	The related threshold was exceeded (may be good or bad, see definition of the corresponding <i>thresholdType</i> element in section 5.3.27.4.2 for details).
fallenBelowThreshold	The related threshold was fallen below (may be good or bad, see definition of the corresponding <i>thresholdType</i> element in section 5.3.27.4.2 for details).
supplyInterrupt	The supply of the corresponding commodity was interrupted.
releaseOfLimitations	"Back to normal", no limitations or threshold violations.
otherProblem	Some problem with the supply without further details about the problem.

10184 Table 308: Enumeration SupplyConditionEventTypeEnumType

10185 Enumeration **SupplyConditionOriginatorEnumType**:

Value	Description
externDSO	The originator of the current condition "event" is the DSO (distribution system operator).
externSupplier	The originator of the current condition "event" is the energy supplier.
internalLimit	Some internal limit at the premises generated the current condition "event".

internalService	Due to an internal service at the premises, the current condition "event" was generated.
internalUser	An user interaction is the source of the current condition "event".

Table 309: Enumeration SupplyConditionOriginatorEnumType

Enumeration **GridConditionEnumType**:

Value	Description
consumptionRed	The electricity grid condition for consumption from the grid is "red".
consumptionYellow	The electricity grid condition for consumption from the grid is "yellow".
good	The electricity grid condition is "good".
productionYellow	The electricity grid condition for feeding into the grid is "yellow".
productionRed	The electricity grid condition for feeding into the grid is "red".

Table 310: Enumeration GridConditionEnumType

5.3.24.2.3 Available selectors

- conditionId
- timestampInterval

5.3.24.2.4 Examples

5.3.24.2.4.1 Notify

A user ordered a plumber and broadcasts this information to all displays (the "filter/partial" part to announce the "restricted function exchange" is simplified by "..." just to improve the readability):

```

...
<supplyConditionListData>
  <supplyConditionData>
    <conditionId>3</conditionId>
    <timestamp>2006-05-04T10:13:51.0Z</timestamp>
    <eventType>supplyInterrupt</eventType>
    <originator>internalService</originator>
    <relevantPeriod>
      <startTime>2006-05-04T14:00:00.0Z</startTime>
      <endTime>2006-05-04T15:00:00.0Z</endTime>
    </relevantPeriod>
    <description>Attention: No water supply from 2PM to
3PM</description>
    <gridCondition>good</gridCondition>
  </supplyConditionData>
</supplyConditionListData>

```

5.3.24.3 supplyConditionDescriptionListData

5.3.24.3.1 General

The more static information about a supply condition is modelled within this function.

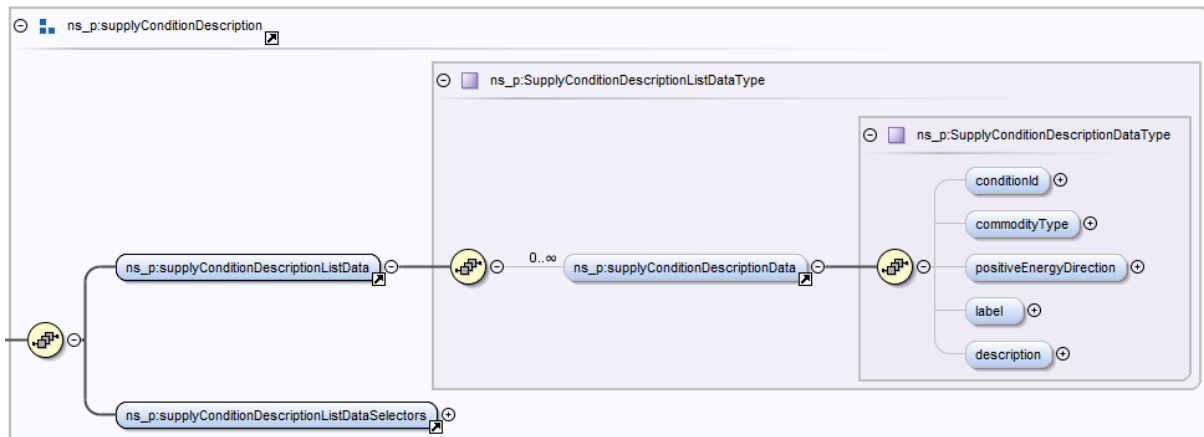


Figure 185: supplyConditionDescriptionListData function overview

5.3.24.3.2 Detailed description of elements

Element	Type	Description
supplyConditionDescriptionListData. supplyConditionDescriptionData. (list)		
conditionId	Identifier "ConditionIdType" (see Table 358).	Identifier of the condition.
commodityType	Common data type "CommodityTypeType". See section 3.10.1.9.	The type of the commodity handled by this condition is stated here.
positiveEnergyDirection	Common data type "EnergyDirectionType". See section 3.10.1.13.	The <i>positiveEnergyDirection</i> states whether energy consumption or production will be counted as positive value.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the condition.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the condition.

Table 311: supplyConditionDescriptionListData function detailed description of elements

5.3.24.3.3 Available selectors

- conditionId

5.3.24.3.4 Examples

5.3.24.3.4.1 Read-Reply

If one is interested in all supply conditions, he could send a standard read command to the server.
The reply could look like this:

```
<supplyConditionDescriptionListData>
  <supplyConditionDescriptionData>
    <conditionId>1</conditionId>
    <commodityType>electricity</commodityType>
```

```

10235         <positiveEnergyDirection>consume</positiveEnergyDirection>
10236         <label>Electricity supply</label>
10237         <description>Supply for electricity (3-phase
10238 connection).</description>
10239     </supplyConditionDescriptionData>
10240     <supplyConditionDescriptionData>
10241         <conditionId>2</conditionId>
10242         <commodityType>water</commodityType>
10243         <positiveEnergyDirection>consume</positiveEnergyDirection>
10244         <label>Water supply</label>
10245         <description>Supply for domestic water.</description>
10246     </supplyConditionDescriptionData>
10247 </supplyConditionDescriptionListData>

```

5.3.24.4 supplyConditionThresholdRelationListData

5.3.24.4.1 General

Multiple thresholds can be associated to a condition. This leads to the relation

- 1 conditionId -> 0..m thresholdId

The supplyConditionThresholdRelationListData function can be used to model concrete relations.

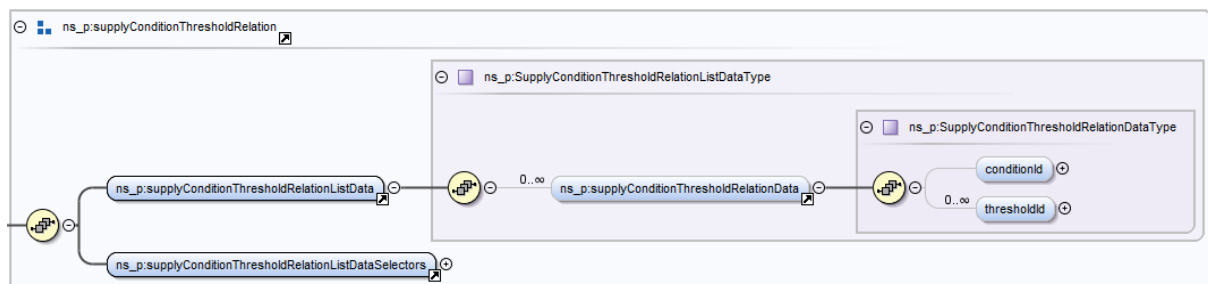


Figure 186: supplyConditionThresholdRelationListData function overview

5.3.24.4.2 Detailed description of elements

Element	Type	Description
supplyConditionThresholdRelationListData. supplyConditionThresholdRelationData. (list)		
conditionId	Identifier "ConditionIdType" (see Table 358).	Identifier of the condition.
thresholdId (list)	Identifier "ThresholdIdType" (see Table 358).	Related threshold identifier(s).

Table 312: supplyConditionThresholdRelationListData function detailed description of elements

5.3.24.4.3 Available selectors

- conditionId
- thresholdId

10264 5.3.24.4.4 Examples

10265 5.3.24.4.4.1 Read-reply

10266 An energy management gateway is able to keep relation information between condition and (supply)
10267 threshold and provide the list upon request.

10268 A user selects the energy management gateway on an external display and queries for information
10269 on conditionId 5. The display sends a read classifier together with the read command:

```
10270 <function>supplyConditionThresholdRelationListData</function>
10271 <filter>
10272     <cmdControl>
10273         <partial/>
10274     </cmdControl>
10275     <supplyConditionThresholdRelationListDataSelectors>
10276         <conditionId>5</conditionId>
10277     </supplyConditionThresholdRelationListDataSelectors>
10278 </filter>
10279 <messagingListData/>
```

10280 The energy management gateway replies with a proper supplyConditionThresholdRelationListData:

```
10281 <function>supplyConditionThresholdRelationListData</function>
10282 <filter>
10283     <cmdControl>
10284         <partial/>
10285     </cmdControl>
10286 </filter>
10287 <supplyConditionThresholdRelationListData>
10288     <supplyConditionThresholdRelationData>
10289         <conditionId>5</conditionId>
10290         <thresholdId>3</thresholdId>
10291         <thresholdId>4</thresholdId>
10292     </supplyConditionThresholdRelationData>
10293 </supplyConditionThresholdRelationListData>
```

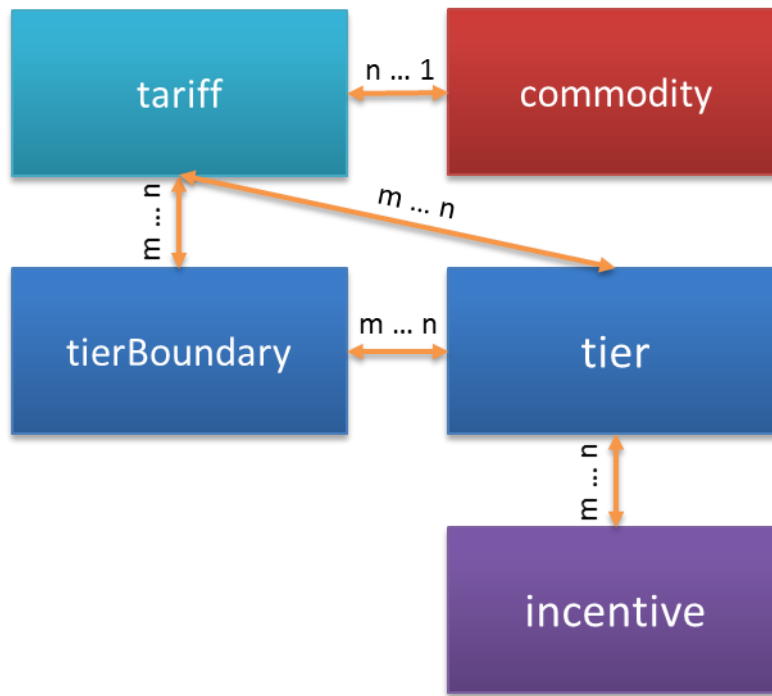
10294

10295 5.3.25 TariffInformation

10296 5.3.25.1 Introduction

10297 The TariffInformation Class provides functions for modelling simple or complex data related to tariffs.

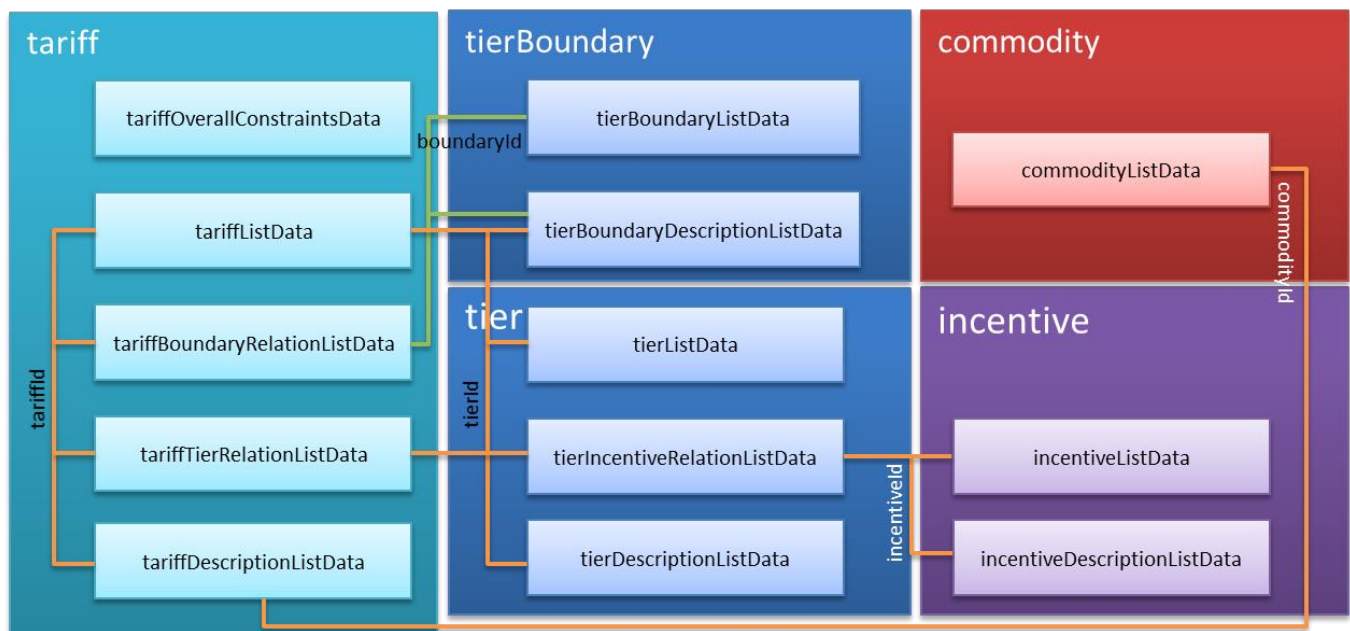
10298 The following figures visualize the dependencies of the different categories available in this class:



10299

10300 *Figure 187: TariffInformation relations, sub-class level*

10301 Basis is the *tariff* that has one *commodity* and some *tiers* with *tierBoundaries*. Each *tier* itself has
 10302 *incentives*. Identifiers are used for referencing the different aspects of the tariff model. They are
 10303 shown in the figure below:

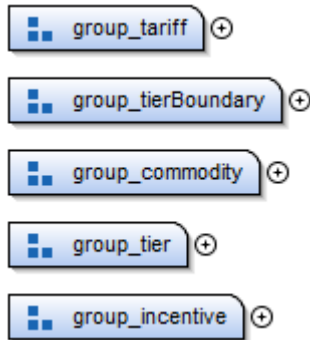


10304

10305 *Figure 188: TariffInformation relations, function level*

10306 As it can be seen, the **tariff** consists of some overall constraints (*tariffOverallConstraintsData*), a
 10307 dynamic data part (*tariffListData*) and the rather static part (*tariffDescriptionListData*). Additionally,
 10308 relations are given to the related boundaries (*tariffBoundaryRelationListData*) and tiers
 10309 (*tariffTierRelationListData*).

- 10310 The **tierBoundary** has a dynamic (*tierBoundaryListData*) and a rather static part
10311 (*tierBoundaryDescriptionListData*).
- 10312 The **tier** consists of the dynamic (*tierListData*), the rather static part (*tierDescriptionListData*) and a
10313 relation to related incentives (*tierIncentiveRelationListData*).
- 10314 The **incentive** has two functions, the dynamic part (*incentiveListData*) and the rather static part
10315 (*incentiveDescriptionListDat*).
- 10316 The **commodity** only has one information (*commodityListData*).



10317
10318 *Figure 189: TariffInformation Sub-class overview*

10319

10320 **5.3.25.2 Sub-class Tariff**

- 10321 The information related to the tariff itself are modelled within the functions of this sub-class.

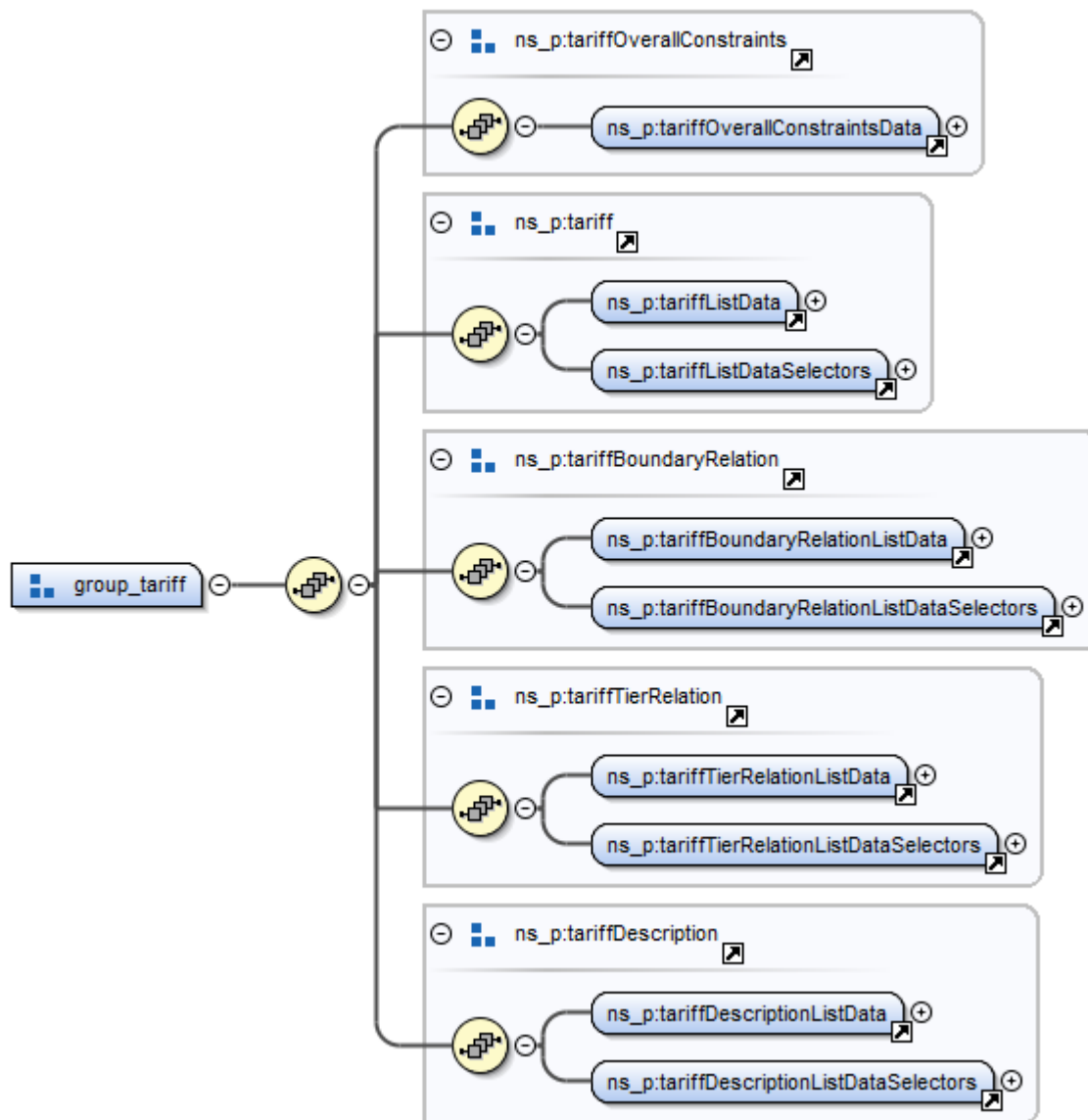


Figure 190: TariffInformation, sub-class Tariff, function-group overview

5.3.25.3 tariffOverallConstraintsData

5.3.25.3.1 General

Constraints that are valid for all tariffs on this particular feature are included in this function.

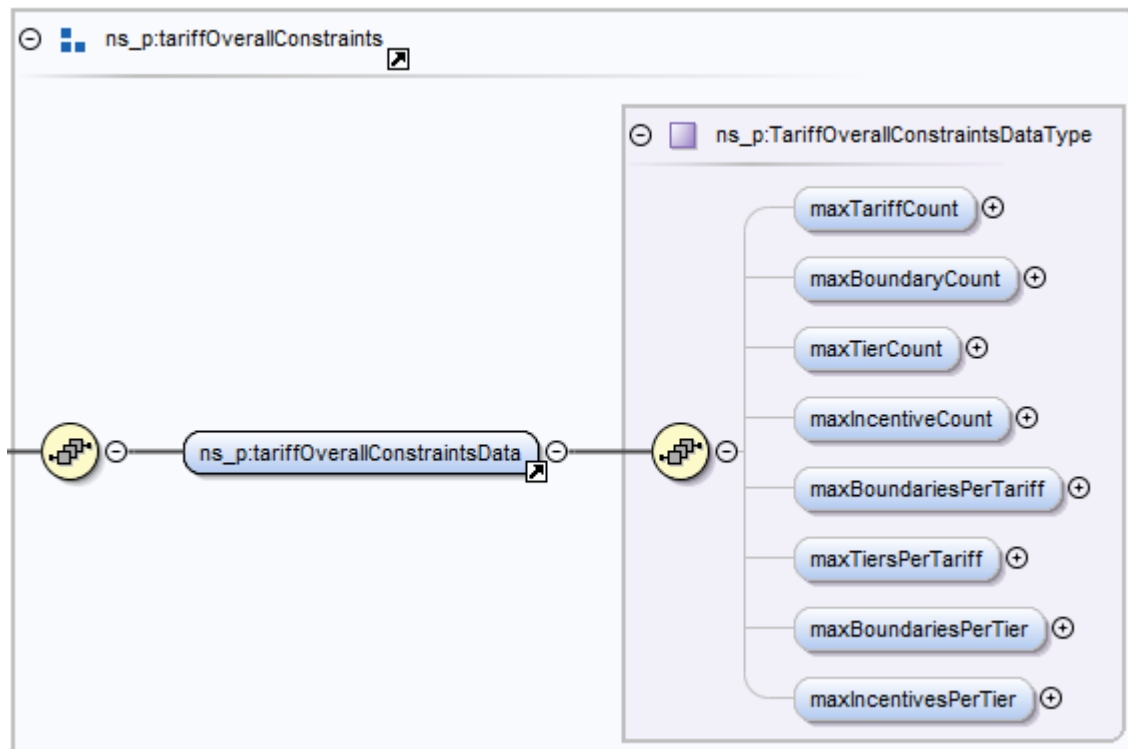


Figure 191: tariffOverallConstraintsData function overview

5.3.25.3.2 Detailed description of elements

Element	Type	Description
tariffOverallConstraintsData.		
maxTariffCount	Simple type "TariffCountType" (restriction of "TariffIdType" (see Table 358)).	Maximum amount of tariffs supported by the server.
maxBoundaryCount	Simple type "TierBoundaryCountType" (restriction of "TierBoundaryIdType" (see Table 358)).	Maximum amount of tier boundaries supported by the server.
maxTierCount	Simple type "TierCountType" (restriction of "TierIdType" (see Table 358)).	Maximum amount of tiers supported by the server.
maxIncentiveCount	Simple type "IncentiveCountType" (restriction of "IncentiveIdType" (see Table 358)).	Maximum amount of incentives supported by the server.
maxBoundariesPerTariff	Simple type "TierBoundaryCountType" (restriction of "TierBoundaryIdType" (see Table 358)).	Maximum amount of (tier) boundaries per tariff supported by the server.
maxTiersPerTariff	Simple type "TierCountType" (restriction of "TierIdType" (see Table 358)).	Maximum amount of tiers per tariff supported by the server.
maxBoundariesPerTier	Simple type "TierBoundaryCountType" (restriction of "TierBoundaryIdType" (see Table 358)).	Maximum amount of boundaries (different boundaryId values) per tier supported by the server.

maxIncentivesPerTier	Simple type "IncentiveCountType" (restriction of "IncentiveIdType" (see Table 358)).	Maximum amount of incentives per tier supported by the server.
----------------------	--	--

Table 313: tariffOverallConstraintsData function detailed description of elements

5.3.25.3.3 Available selectors

None.

5.3.25.3.4 Examples

5.3.25.3.4.1 Read-reply

If one is interested in the tariff overall constraints, he could send a standard read command to the server. The reply could look like this:

```
<tariffOverallConstraintsData>
  <maxTariffCount>5</maxTariffCount>
  <maxBoundaryCount>40</maxBoundaryCount>
  <maxTierCount>20</maxTierCount>
  <maxIncentiveCount>60</maxIncentiveCount>
  <maxBoundariesPerTariff>8</maxBoundariesPerTariff>
  <maxTiersPerTariff>4</maxTiersPerTariff>
  <maxBoundariesPerTier>2</maxBoundariesPerTier>
  <maxIncentivesPerTier>3</maxIncentivesPerTier>
</tariffOverallConstraintsData>
```

5.3.25.4 tariffListData

5.3.25.4.1 General

The dynamic part of the tariff sub-class delivers the information which tier(s) is/are currently active for this tariff.

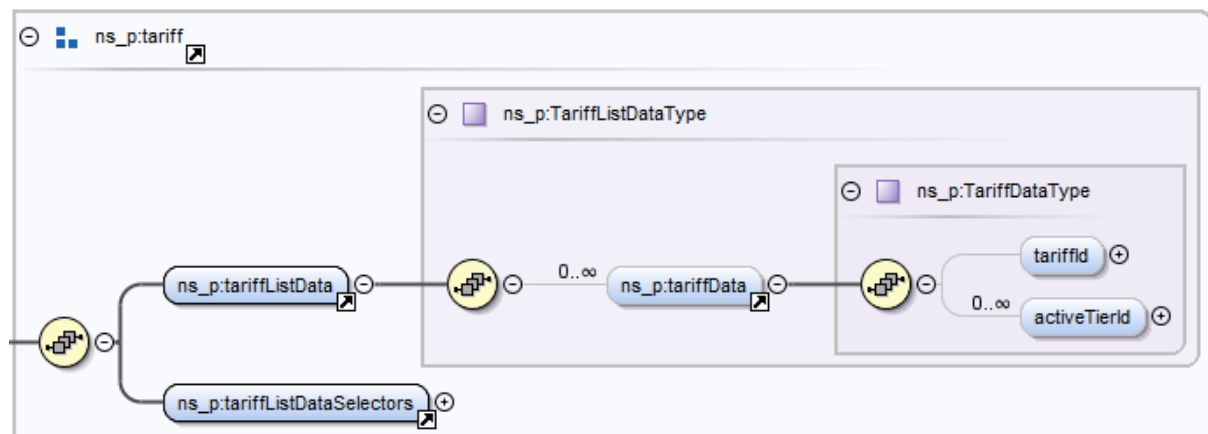


Figure 192: tariffListData function overview

5.3.25.4.2 Detailed description of elements

Element	Type	Description
tariffListData. tariffData. (list)		
tariffId	Identifier "TariffIdType" (see Table 358).	Primary identifier for the tariff.
activeTierId (list)	Identifier "TierIdType" (see Table 358).	Foreign key(s) of the active tier(s) for this tariff.

Table 314: tariffListData function detailed description of elements

5.3.25.4.3 Available selectors

- tariffId
- activeTierId

5.3.25.4.4 Examples

5.3.25.4.4.1 Read-reply

If one is interested in all dynamic tariff information, he could send a standard read command to the server. The reply could look like this:

```

<tariffListData>
  <tariffData>
    <tariffId>1</tariffId>
    <activeTierId>1</activeTierId>
    <activeTierId>2</activeTierId>
  </tariffData>
  <tariffData>
    <tariffId>2</tariffId>
    <activeTierId>3</activeTierId>
  </tariffData>
</tariffListData>

```

5.3.25.5 tariffBoundaryRelationListData

5.3.25.5.1 General

The boundaries related to a tariff are modelled within this function.

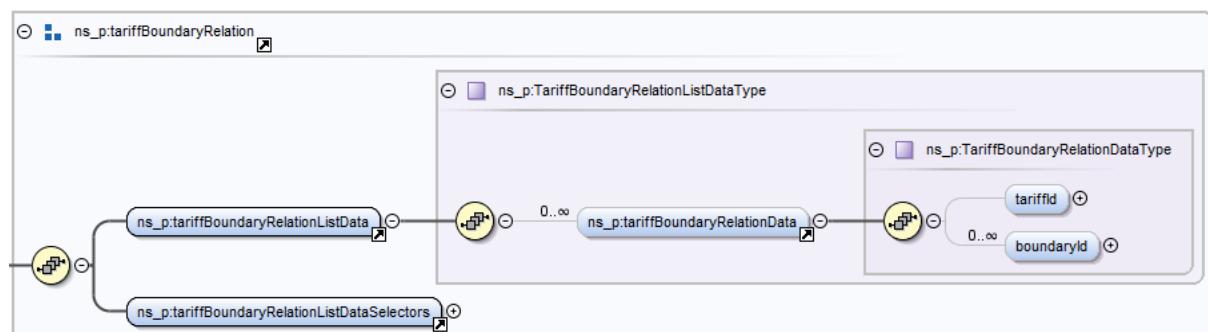


Figure 193: tariffBoundaryRelationListData function overview

5.3.25.5.2 Detailed description of elements

Element	Type	Description
tariffBoundaryRelationListData. tariffBoundaryRelationData. (list)		
tariffId	Identifier "TariffIdType" (see Table 358).	Primary identifier for the tariff.
boundaryId (list)	Identifier "TierBoundaryIdType" (see Table 358).	Foreign key(s) of the related boundary(/ies) for this tariff.

Table 315: tariffBoundaryRelationListData function detailed description of elements

5.3.25.5.3 Available selectors

- tariffId
- boundaryId

5.3.25.5.4 Examples

5.3.25.5.4.1 Read-reply

If one is interested in the possible boundaries of all tariffs, he could send a standard read command to the server. The reply could look like this:

```

<tariffBoundaryRelationListData>
  <tariffBoundaryRelationData>
    <tariffId>1</tariffId>
    <boundaryId>1</boundaryId>
    <boundaryId>2</boundaryId>
    <boundaryId>3</boundaryId>
  </tariffBoundaryRelationData>
  <tariffBoundaryRelationData>
    <tariffId>2</tariffId>
    <boundaryId>1</boundaryId>
    <boundaryId>3</boundaryId>
  </tariffBoundaryRelationData>
</tariffBoundaryRelationListData>

```

5.3.25.6 tariffTierRelationListData

5.3.25.6.1 General

The tiers related to a tariff are modelled within this function.

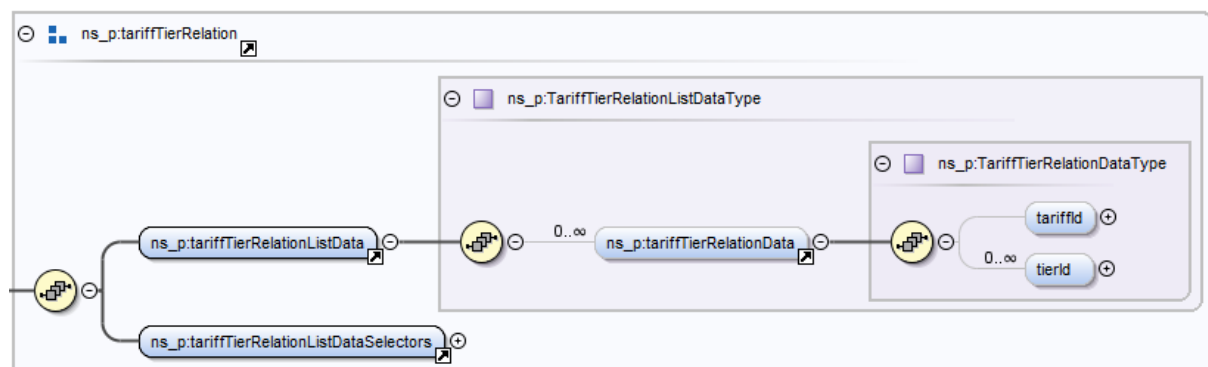


Figure 194: tariffTierRelationListData function overview

10418

10419 **5.3.25.6.2 Detailed description of elements**

Element	Type	Description
tariffTierRelationListData. tariffTierRelationData. (list)		
tariffId	Identifier "TariffIdType" (see Table 358).	Primary identifier for the tariff.
tierId (list)	Identifier "TierIdType" (see Table 358).	Foreign key(s) of the related tier(s) for this tariff.

10420 *Table 316: tariffTierRelationListData function detailed description of elements*

10421

10422 **5.3.25.6.3 Available selectors**

10423 - tariffId

10424 - tierId

10425

10426 **5.3.25.6.4 Examples**10427 **5.3.25.6.4.1 Read-reply**

10428 If one is interested in the possible tiers of all tariffs, he could send a standard read command to the
 10429 server. The reply could look like this:

```

10430 <tariffTierRelationListData>
10431   <tariffTierRelationData>
10432     <tariffId>1</tariffId>
10433     <tierId>1</tierId>
10434     <tierId>2</tierId>
10435     <tierId>3</tierId>
10436     <tierId>4</tierId>
10437   </tariffTierRelationData>
10438   <tariffTierRelationData>
10439     <tariffId>2</tariffId>
10440     <tierId>1</tierId>
10441     <tierId>2</tierId>
10442   </tariffTierRelationData>
10443 </tariffTierRelationListData>

```

10444

10445 **5.3.25.7 tariffDescriptionListData**10446 **5.3.25.7.1 General**

10447 The rather static information about a tariff (e.g. the references to a commodity or a measurand, the
 10448 scopeType, etc.) are modelled within this function.

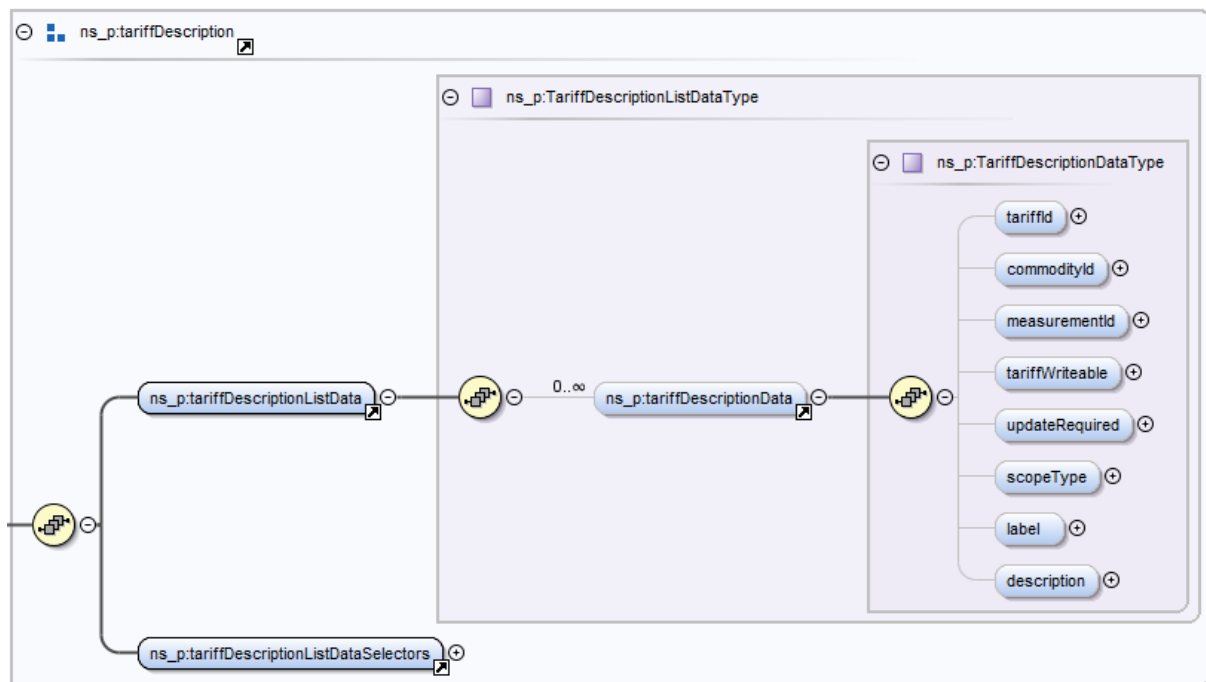


Figure 195: tariffDescriptionListData function overview

5.3.25.7.2 Detailed description of elements

Element	Type	Description
tariffDescriptionListData. tariffDescriptionData. (list)		
tariffId	Identifier "TariffIdType" (see Table 358).	Primary identifier for the tariff.
commodityId	Identifier "CommodityIdType" (see Table 358).	If the tariff is linked to a commodity, the corresponding identifier can be stated here.
measurementId	Identifier "MeasurementIdType" (see Table 358).	If the tariff is linked to a measurand, the corresponding identifier can be stated here.
tariffWriteable	xs:boolean (W3C standard type)	Whether the tariff is writeable or not can be denoted in this element.
updateRequired	xs:boolean (W3C standard type)	If an update for this tariff (by some bound client) is required, this element will be set to "true".
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	The scope type for this tariff.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the tariff.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the tariff.

Table 317: tariffDescriptionListData function detailed description of elements

5.3.25.7.3 Available selectors

- tariffId
- commodityId
- measurementId

10459 - scopeType

10460

10461 5.3.25.7.4 Examples

10462 5.3.25.7.4.1 Read-reply

10463 If one is interested in all tariff descriptions, he could send a standard read command to the server.

10464 The reply could look like this:

```

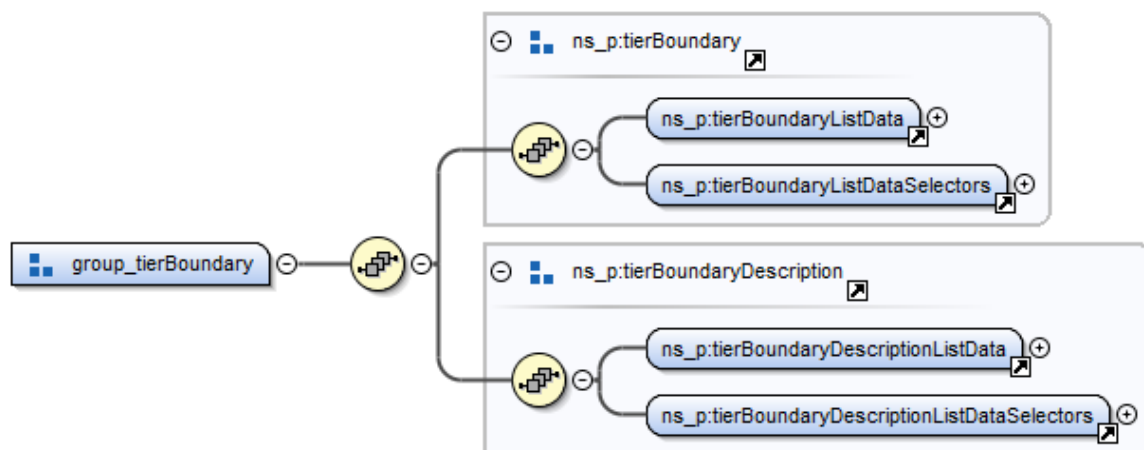
10465 <tariffDescriptionListData>
10466   <tariffDescriptionData>
10467     <tariffId>1</tariffId>
10468     <commodityId>1</commodityId>
10469     <measurementId>1</measurementId>
10470     <tariffWriteable>false</tariffWriteable>
10471     <updateRequired>true</updateRequired>
10472     <label>Tariff 1 - Basic</label>
10473   </tariffDescriptionData>
10474   <tariffDescriptionData>
10475     <tariffId>2</tariffId>
10476     <commodityId>1</commodityId>
10477     <measurementId>4</measurementId>
10478     <tariffWriteable>true</tariffWriteable>
10479     <updateRequired>false</updateRequired>
10480     <label>Tariff 2 - Eco Green</label>
10481   </tariffDescriptionData>
10482 </tariffDescriptionListData>

```

10483

10484 5.3.25.8 Sub-class TierBoundary

10485 All functions related to the boundaries of tiers are clustered in this *TierBoundary* sub-class.



10486

10487 Figure 196: TariffInformation, sub-class TierBoundary, function-group overview

10488

10489 5.3.25.9 tierBoundaryListData

10490 5.3.25.9.1 General

10491 Boundaries (for tiers) are needed to identify in which value range the tier is valid. The validity can
 10492 also depend on a time interval / time table.

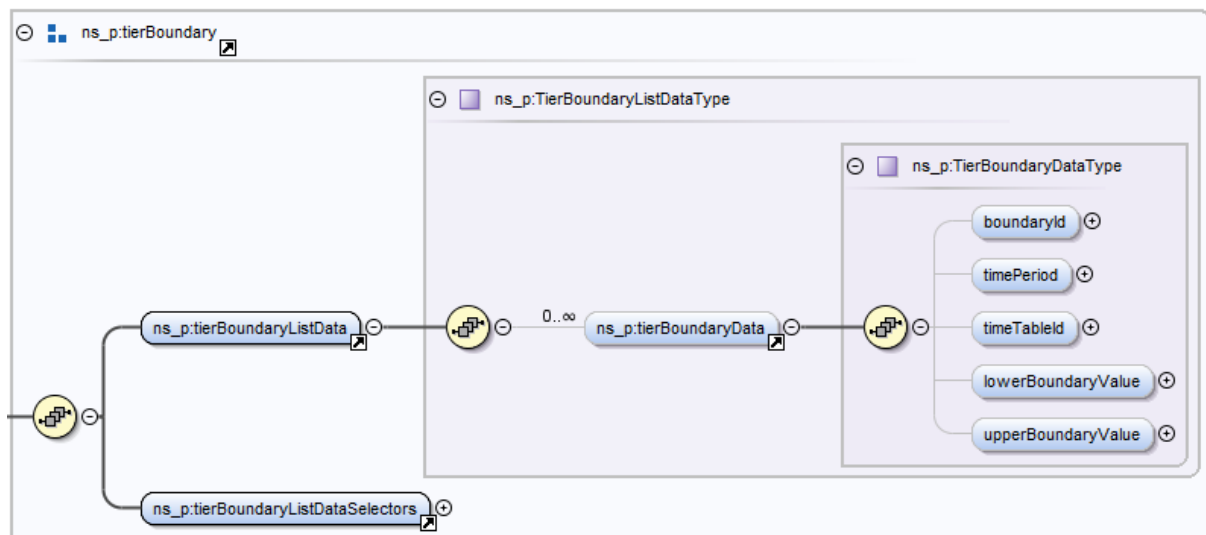


Figure 197: tierBoundaryListData function overview

5.3.25.9.2 Detailed description of elements

Element	Type	Description
tierBoundaryListData. tierBoundaryData. (list)		
boundaryId	Identifier "TierBoundaryIdType" (see Table 358).	Primary identifier for the boundary.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	If the boundary depends on a time period, it can be denoted here.
timeTableId	Identifier "TimeTableIdType" (see Table 358).	The identifier of a time table can be set here if the boundary depends on the given start and end times there.
lowerBoundaryValue	Common data type "ScaledNumberType". See section 3.10.1.8.	This represents to lower end value range boundary.
upperBoundaryValue	Common data type "ScaledNumberType". See section 3.10.1.8.	This represents to upper end value range boundary.

Table 318: tierBoundaryListData function detailed description of elements

5.3.25.9.3 Available selectors

- boundaryId

5.3.25.9.4 Examples

5.3.25.9.4.1 Read-reply

If one is interested in the dynamic parts of all tier boundaries, he could send a standard read command to the server. The reply could look like this:

```

10506 <tierBoundaryListData>
10507   <tierBoundaryData>
10508     <boundaryId>1</boundaryId>
10509     <timePeriod>
10510       <startTime>2018-05-06T10:00:00.0Z</startTime>
10511       <endTime>2018-05-06T20:00:00.0Z</endTime>
10512     </timePeriod>
10513     <lowerBoundaryValue>
10514       <number>0</number>
10515     </lowerBoundaryValue>
10516     <upperBoundaryValue>
10517       <number>20</number>
10518       <scale>3</scale>
10519     </upperBoundaryValue>
10520   </tierBoundaryData>
10521   <tierBoundaryData>
10522     <boundaryId>2</boundaryId>
10523     <timePeriod>
10524       <startTime>2018-05-06T20:00:00.0Z</startTime>
10525       <endTime>2018-05-07T10:00:00.0Z</endTime>
10526     </timePeriod>
10527     <lowerBoundaryValue>
10528       <number>0</number>
10529     </lowerBoundaryValue>
10530     <upperBoundaryValue>
10531       <number>20</number>
10532       <scale>3</scale>
10533     </upperBoundaryValue>
10534   </tierBoundaryData>
10535   <tierBoundaryData>
10536     <boundaryId>3</boundaryId>
10537     <timePeriod>
10538       <startTime>2018-05-06T10:00:00.0Z</startTime>
10539       <endTime>2018-05-06T20:00:00.0Z</endTime>
10540     </timePeriod>
10541     <lowerBoundaryValue>
10542       <number>20</number>
10543       <scale>3</scale>
10544     </lowerBoundaryValue>
10545   </tierBoundaryData>
10546   <tierBoundaryData>
10547     <boundaryId>4</boundaryId>
10548     <timePeriod>
10549       <startTime>2018-05-06T20:00:00.0Z</startTime>
10550       <endTime>2018-05-07T10:00:00.0Z</endTime>
10551     </timePeriod>
10552     <lowerBoundaryValue>
10553       <number>20</number>
10554       <scale>3</scale>
10555     </lowerBoundaryValue>
10556   </tierBoundaryData>
10557 </tierBoundaryListData>

```

10558

10559 **5.3.25.10 tierBoundaryDescriptionListData**

10560 **5.3.25.10.1 General**

10561 The rather static part of the (tier) boundary is modelled in this function.

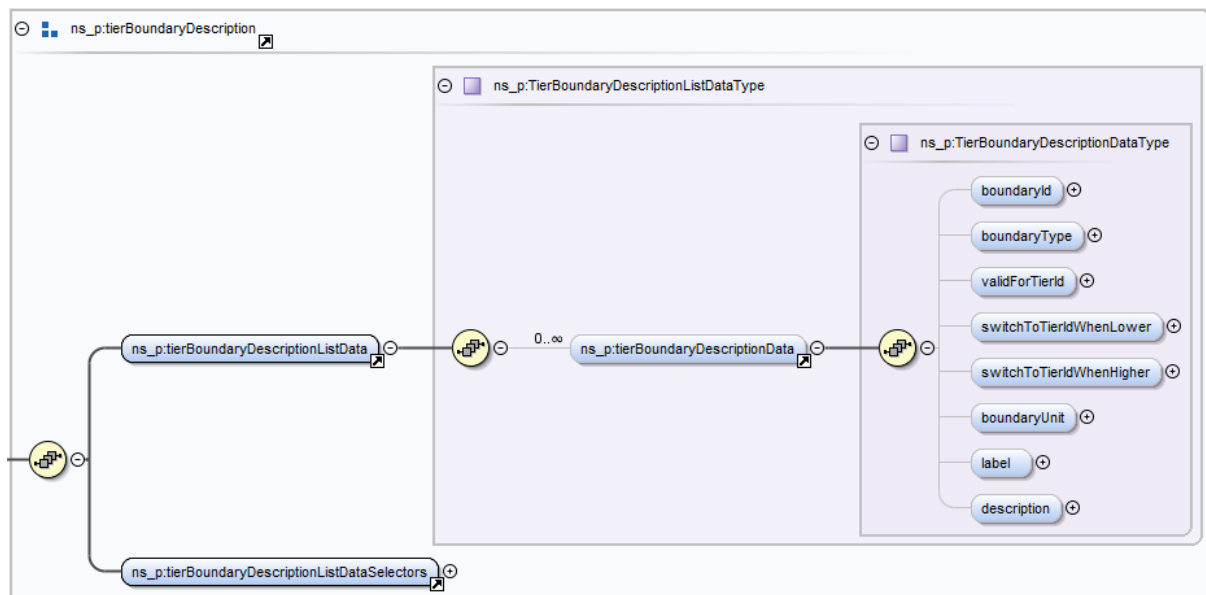


Figure 198: tierBoundaryDescriptionListData function overview

5.3.25.10.2 Detailed description of elements

Element	Type	Description
tierBoundaryDescriptionListData. tierBoundaryDescriptionData. (list)		
boundaryId	Identifier "TierBoundaryIdType" (see Table 358).	Primary identifier for the boundary.
boundaryType	Union "TierBoundaryTypeType": - Enum (see Table 320): TierBoundaryTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the boundary.
validForTierId	Identifier "TierIdType" (see Table 358).	Identifier of the tier this boundary is valid for.
switchToTierIdWhenLower	Identifier "TierIdType" (see Table 358).	Identifier of the tier that will become valid if the according value (defined in the Element "boundaryType") falls below the element lowerBoundaryValue in function tierBoundaryListData.
switchToTierIdWhenHigher	Identifier "TierIdType" (see Table 358).	Identifier of the tier that will become valid if the according value rises above the element upperBoundaryValue in function tierBoundaryListData.
boundaryUnit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	A unit for the boundary.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the tier boundary.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the tier boundary.

10566 *Table 319: tierBoundaryDescriptionListData function detailed description of elements*

10567 Enumeration **TierBoundaryTypeEnumType**:

Value	Description
powerBoundary	Provides boundaries for power.
energyBoundary	Provides boundaries for energy.
countBoundary	Provides boundaries for counts.

10568 *Table 320: Enumeration TierBoundaryTypeEnumType*

10569

10570 *5.3.25.10.3 Available selectors*

10571 - boundaryId

10572 - boundaryType

10573

10574 *5.3.25.10.4 Examples*

10575 *5.3.25.10.4.1 Read-reply*

10576 If one is interested in the rather static parts of all tier boundaries, he could send a standard read
10577 command to the server. The reply could look like this:

```

10578 <tierBoundaryDescriptionListData>
10579   <tierBoundaryDescriptionData>
10580     <boundaryId>1</boundaryId>
10581     <boundaryType>powerBoundary</boundaryType>
10582     <validForTierId>1</validForTierId>
10583     <switchToTierIdWhenHigher>3</switchToTierIdWhenHigher>
10584     <boundaryUnit>W</boundaryUnit>
10585   </tierBoundaryDescriptionData>
10586   <tierBoundaryDescriptionData>
10587     <boundaryId>2</boundaryId>
10588     <boundaryType>powerBoundary</boundaryType>
10589     <validForTierId>2</validForTierId>
10590     <switchToTierIdWhenHigher>4</switchToTierIdWhenHigher>
10591     <boundaryUnit>W</boundaryUnit>
10592   </tierBoundaryDescriptionData>
10593   <tierBoundaryDescriptionData>
10594     <boundaryId>3</boundaryId>
10595     <boundaryType>powerBoundary</boundaryType>
10596     <validForTierId>3</validForTierId>
10597     <switchToTierIdWhenLower>1</switchToTierIdWhenLower>
10598     <boundaryUnit>W</boundaryUnit>
10599   </tierBoundaryDescriptionData>
10600   <tierBoundaryDescriptionData>
10601     <boundaryId>4</boundaryId>
10602     <boundaryType>powerBoundary</boundaryType>
10603     <validForTierId>4</validForTierId>
10604     <switchToTierIdWhenLower>2</switchToTierIdWhenLower>
10605     <boundaryUnit>W</boundaryUnit>
10606   </tierBoundaryDescriptionData>
10607 </tierBoundaryDescriptionListData>

```

10608

5.3.25.11 Sub-class Commodity

Commodities (like "electricity", "water", etc.) are described within this sub-class.

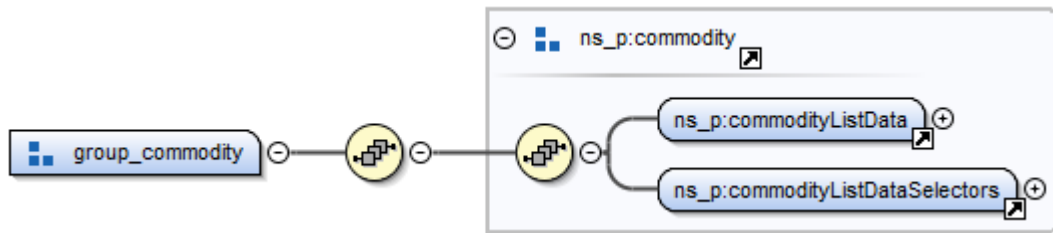


Figure 199: TariffInformation, sub-class Commodity, function-group overview

5.3.25.12 commodityListData

5.3.25.12.1 General

A tariff may be linked to a commodity. The commodity information is modelled with this function.

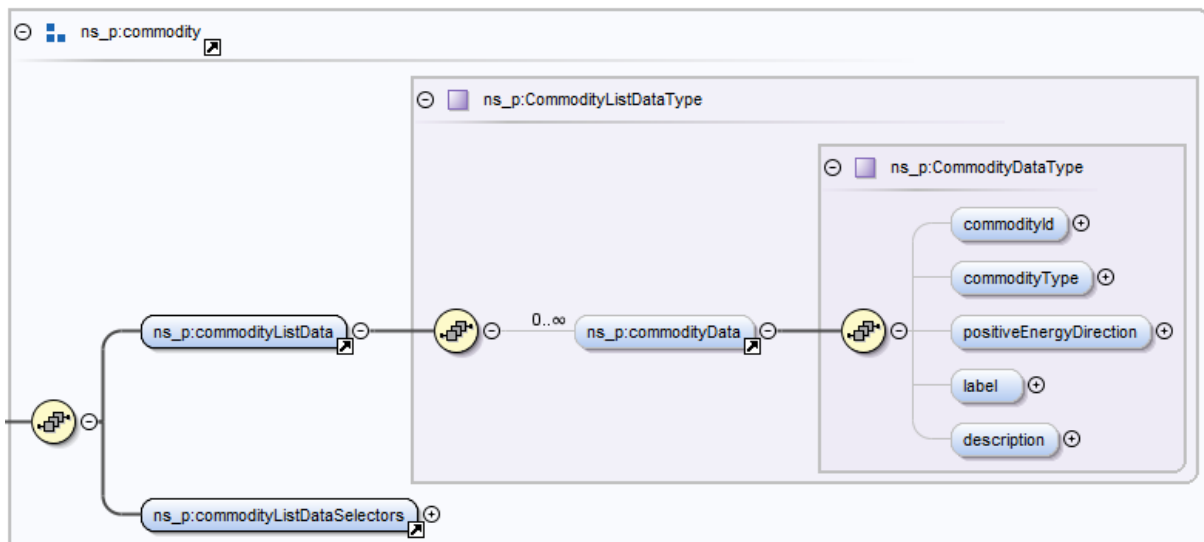


Figure 200: commodityListData function overview

5.3.25.12.2 Detailed description of elements

Element	Type	Description
commodityListData. commodityData. (list)		
commodityId	Identifier "CommodityIdType" (see Table 358).	Primary identifier for the commodity.
commodityType	Common data type "CommodityTypeType". See section 3.10.1.9.	The type of the commodity
positiveEnergyDirection	Common data type "EnergyDirectionType". See section 3.10.1.13.	States which direction counts positives values for this commodity.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the commodity.

description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Descriptive information on the commodity.
-------------	--	---

10621 *Table 321: commodityListData function detailed description of elements*

10622

10623 **5.3.25.12.3 Available selectors**

10624 - commodityId

10625 - commodityType

10626

10627 **5.3.25.12.4 Examples**

10628 **5.3.25.12.4.1 Read-reply**

10629 If one is interested in the information about all commodities, he could send a standard read
10630 command to the server. The reply could look like this:

```

10631 <commodityListData>
10632   <commodityData>
10633     <commodityId>1</commodityId>
10634     <commodityType>electricity</commodityType>
10635     <positiveEnergyDirection>consume</positiveEnergyDirection>
10636   </commodityData>
10637 </commodityListData>

```

10638

10639 **5.3.25.13 Sub-class Tier**

10640 Tiers are relevant for tariffs as they allow to different levels within the tariff (e.g. time based ("day"
10641 and "night" prices), boundary based ("the more you consume, the more you pay")).

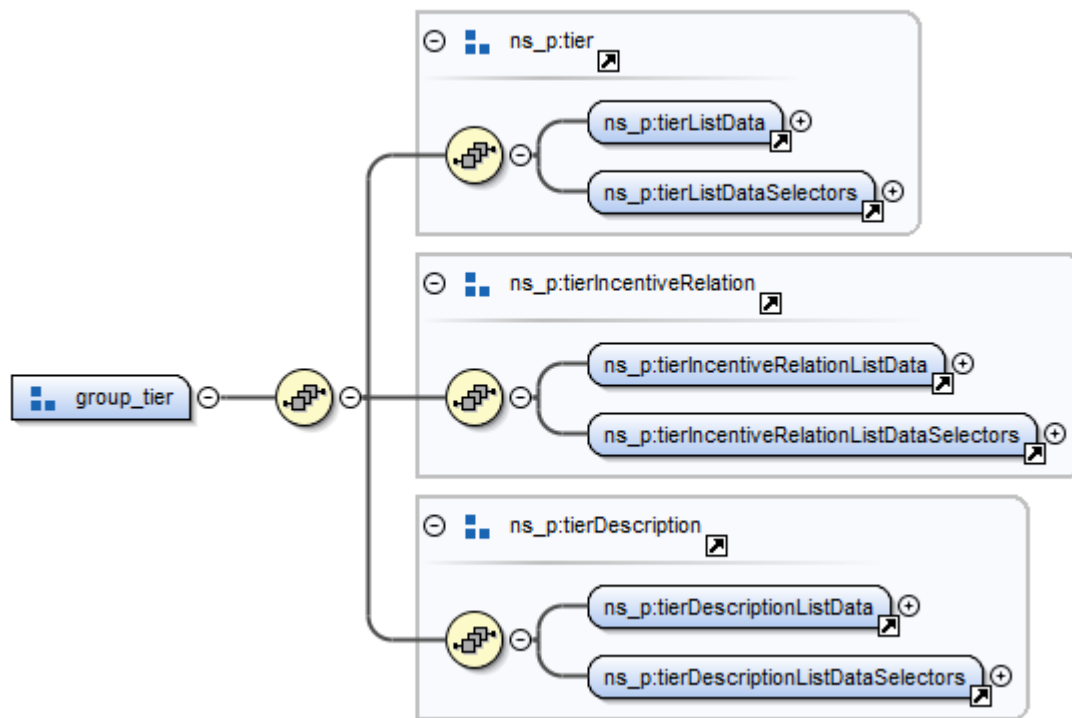


Figure 201: TariffInformation, sub-class Tier, function-group overview

5.3.25.14 tierListData

5.3.25.14.1 General

The dynamic parts (time dependency and active incentives) of the tiers is described within this function.

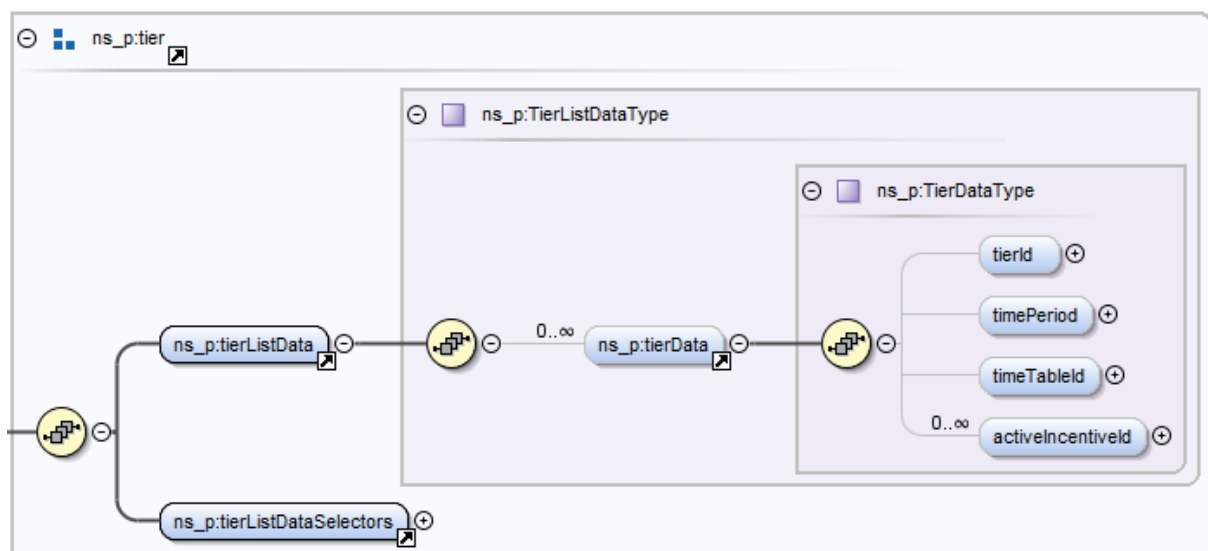


Figure 202: tierListData function overview

5.3.25.14.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

tierListData. tierData. (list)		
tierId	Identifier "TierIdType" (see Table 358).	Primary identifier for the tier.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	If the tier is only valid for some time period, this can be denoted here.
timeTableId	Identifier "TimeTableIdType" (see Table 358).	The foreign key of a time table can be set here if the tier is only valid during the given start and end times there.
activeIncentiveId (list)	Identifier "IncentiveIdType" (see Table 358).	Foreign key(s) of the currently active incentive(s) for this tier.

Table 322: tierListData function detailed description of elements

5.3.25.14.3 Available selectors

- tierId
- activeIncentiveId

5.3.25.14.4 Examples

5.3.25.14.4.1 Read-reply

If one is interested in the dynamic parts of all tiers, he could send a standard read command to the server. The reply could look like this:

```

<tierListData>
  <tierData>
    <tierId>1</tierId>
    <timeTableId>1</timeTableId>
    <activeIncentiveId>1</activeIncentiveId>
  </tierData>
  <tierData>
    <tierId>2</tierId>
    <timeTableId>2</timeTableId>
    <activeIncentiveId>1</activeIncentiveId>
  </tierData>
  <tierData>
    <tierId>3</tierId>
    <timeTableId>1</timeTableId>
    <activeIncentiveId>1</activeIncentiveId>
  </tierData>
  <tierData>
    <tierId>4</tierId>
    <timeTableId>2</timeTableId>
    <activeIncentiveId>1</activeIncentiveId>
  </tierData>
</tierListData>

```

5.3.25.15 tierIncentiveRelationListData

5.3.25.15.1 General

Which incentives can be used for specific tiers is modelled in the tierIncentiveRelationListData function.

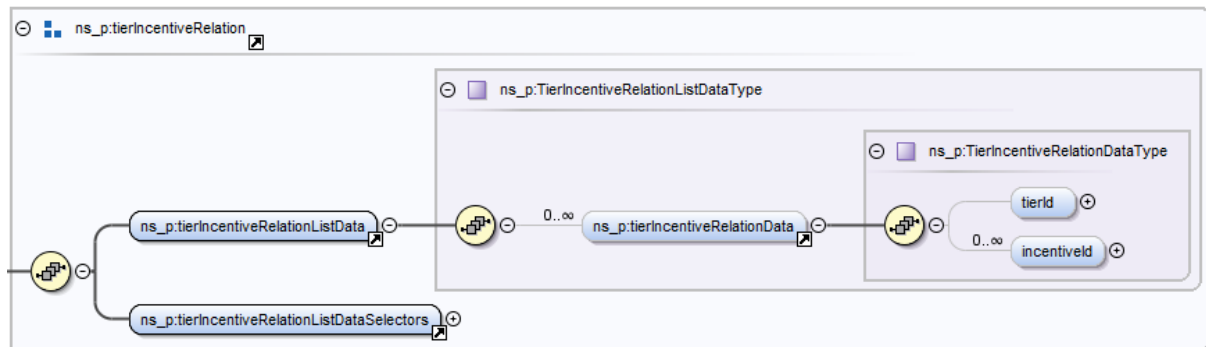


Figure 203: tierIncentiveRelationListData function overview

5.3.25.15.2 Detailed description of elements

Element	Type	Description
tierIncentiveRelationListData. tierIncentiveRelationData. (list)		
tierId	Identifier "TierIdType" (see Table 358).	Primary identifier for the tier.
incentiveId (list)	Identifier "IncentiveIdType" (see Table 358).	Foreign key(s) of the related incentive(s) for this tier.

Table 323: tierIncentiveRelationListData function detailed description of elements

5.3.25.15.3 Available selectors

- tierId
- incentiveId

5.3.25.15.4 Examples

5.3.25.15.4.1 Read-reply

If one is interested in the possible incentives of all tiers, he could send a standard read command to the server. The reply could look like this:

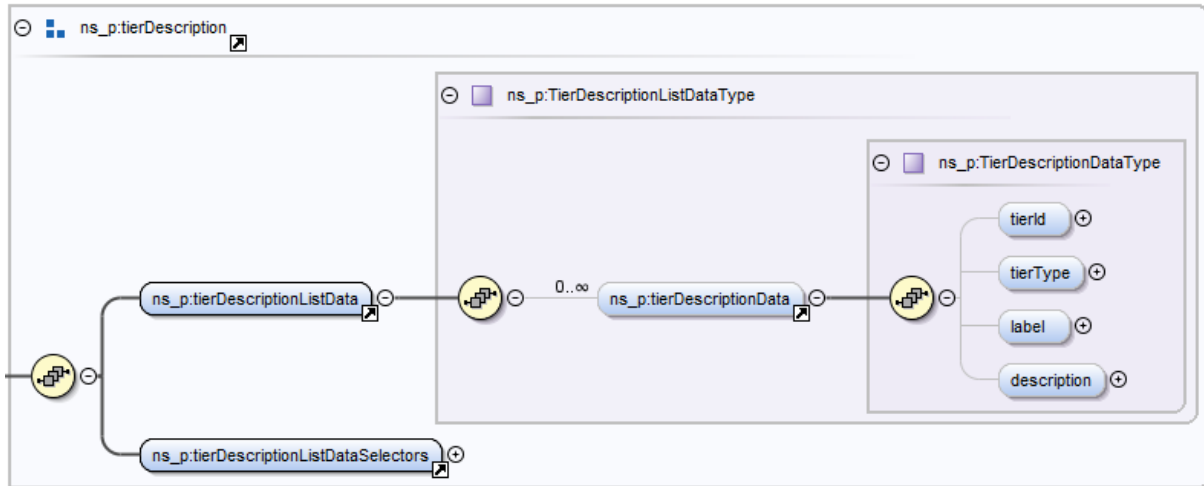
```

<tierIncentiveRelationListData>
  <tierIncentiveRelationData>
    <tierId>1</tierId>
    <incentiveId>1</incentiveId>
  </tierIncentiveRelationData>
  <tierIncentiveRelationData>
    <tierId>2</tierId>
    <incentiveId>1</incentiveId>
  </tierIncentiveRelationData>
  <tierIncentiveRelationData>
    <tierId>3</tierId>
    <incentiveId>1</incentiveId>
  </tierIncentiveRelationData>
  <tierIncentiveRelationData>
    <tierId>4</tierId>
    <incentiveId>1</incentiveId>
  </tierIncentiveRelationData>
</tierIncentiveRelationListData>

```

10723 **5.3.25.16 tierDescriptionListData**10724 **5.3.25.16.1 General**

10725 The rather static part of the tiers is modelled with this function.



10726

10727 *Figure 204: tierDescriptionListData function overview*

10728

10729 **5.3.25.16.2 Detailed description of elements**

Element	Type	Description
tierDescriptionListData. tierDescriptionData. (list)		
tierId	Identifier "TierIdType " (see Table 358).	Primary identifier for the tier.
tierType	Union "TierTypeType": - Enum (see Table 325): TierTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the tier.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the tier.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the tier.

10730 *Table 324: tierDescriptionListData function detailed description of elements*10731 Enumeration **TierTypeEnumType**:

Value	Description
fixedCost	The fixed cost have to be payed, independent from the consumed energy.
dynamicCost	The dynamic cost has a (linear) dependency to the consumed energy.

10732 *Table 325: Enumeration TierTypeEnumType*

10733

10734 **5.3.25.16.3 Available selectors**

- 10735 - tierId
- 10736 - tierType

10737

10738 **5.3.25.16.4Examples**10739 **5.3.25.16.4.1 Read-reply**

10740 If one is interested in the rather static parts of all tiers, he could send a standard read command to
 10741 the server. The reply could look like this:

```

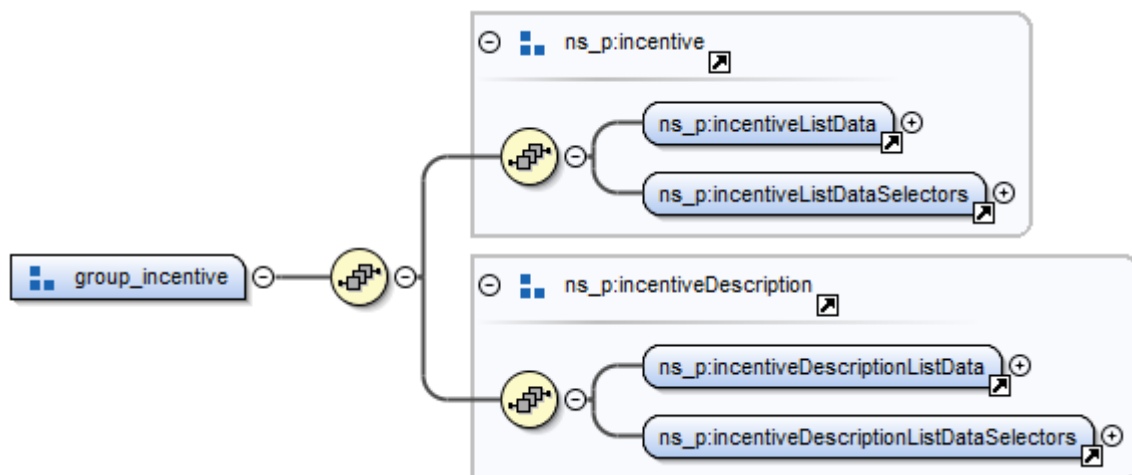
10742 <tierDescriptionListData>
10743   <tierDescriptionData>
10744     <tierId>1</tierId>
10745     <tierType>dynamicCost</tierType>
10746   </tierDescriptionData>
10747   <tierDescriptionData>
10748     <tierId>2</tierId>
10749     <tierType>dynamicCost</tierType>
10750   </tierDescriptionData>
10751   <tierDescriptionData>
10752     <tierId>3</tierId>
10753     <tierType>dynamicCost</tierType>
10754   </tierDescriptionData>
10755   <tierDescriptionData>
10756     <tierId>4</tierId>
10757     <tierType>dynamicCost</tierType>
10758   </tierDescriptionData>
10759 </tierDescriptionListData>

```

10760

10761 **5.3.25.17Sub-class Incentive**

10762 The actual costs are modelled as so-called incentives (e.g. real monetary costs, "co2Emission", etc.).



10763

10764 *Figure 205: TariffInformation, sub-class Incentive, function-group overview*

10765

10766 **5.3.25.18incentiveListData**10767 **5.3.25.18.1General**

10768 The value of the costs, together with a time dependency (if needed), is part of this dynamic incentive
 10769 function.

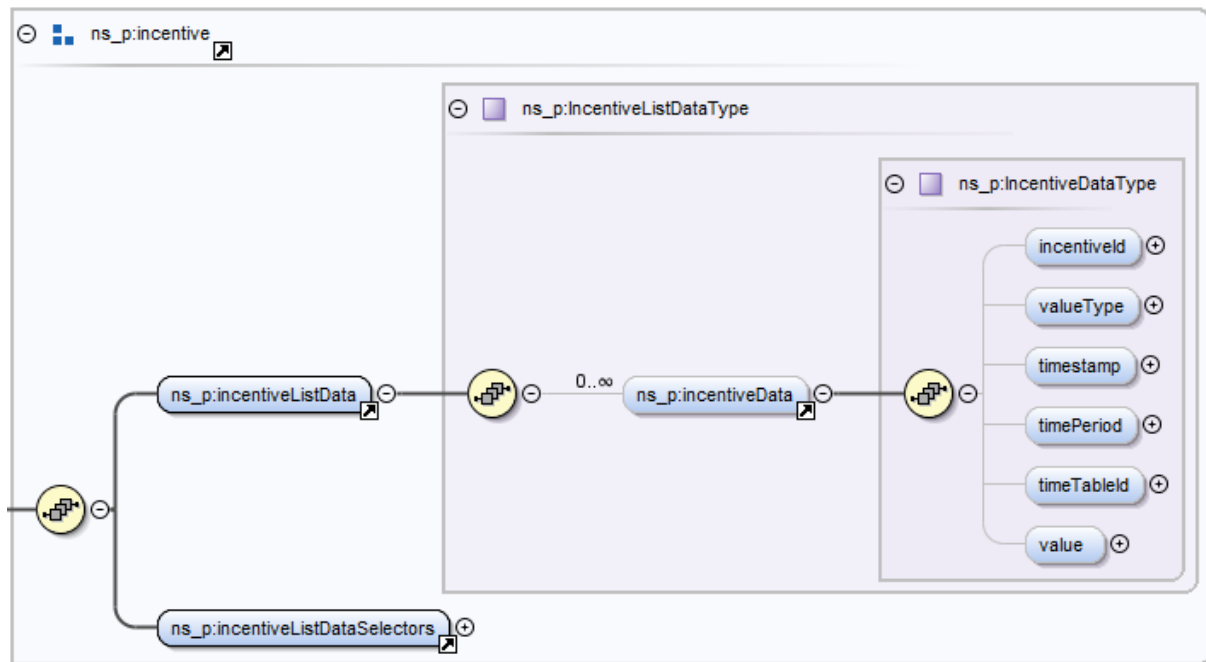


Figure 206: incentiveListData function overview

5.3.25.18.2 Detailed description of elements

Element	Type	Description
incentiveListData. incentiveData. (list)		
incentiveld	Identifier "IncentiveldType" (see Table 358).	Primary identifier for the incentive.
valueType	Union "IncentiveValueTypeType": - Enum (see Table 327): IncentiveValueTypeEnumType - EnumExtendType (see section 3.10.1.5)	A type for the value (e.g. "value", "minValue", etc.).
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	A timestamp for the value information.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	If there exists a time dependency for which the incentive value is valid, this can be denoted here.
timeTableId	Identifier "TimeTableIdType" (see Table 358).	The foreign key of a time table can be set here if the incentive value depends on the given start and end times there.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	The actual value for the incentive costs.

Table 326: incentiveListData function detailed description of elements

Enumeration IncentiveValueTypeEnumType:

Value	Description
value	Used for normal cost values for an incentive.
averageValue	Used if only an average cost value can be stated for an incentive.
minValue	Used for the minimum cost value of an incentive.

maxValue	Used for the maximum cost value of an incentive.
----------	--

Table 327: Enumeration IncentiveValueTypeEnumType

5.3.25.18.3 Available selectors

- incentiveId
- timestamp
- timestampInterval

5.3.25.18.4 Examples**5.3.25.18.4.1 Read-reply**

If one is interested in the dynamic parts of all incentives, he could send a standard read command to the server. The reply could look like this:

```
<incentiveListData>
  <incentiveData>
    <incentiveId>1</incentiveId>
    <valueType>value</valueType>
    <timestamp>2018-05-06T09:00:00.0Z</timestamp>
    <timePeriod>
      <startTime>2018-05-06T10:00:00.0Z</startTime>
      <endTime>2018-05-07T10:00:00.0Z</endTime>
    </timePeriod>
    <value>
      <number>28</number>
      <scale>-2</scale>
    </value>
  </incentiveData>
</incentiveListData>
```

5.3.25.19 incentiveDescriptionListData**5.3.25.19.1 General**

The rather static information about the incentives is modelled with this function.

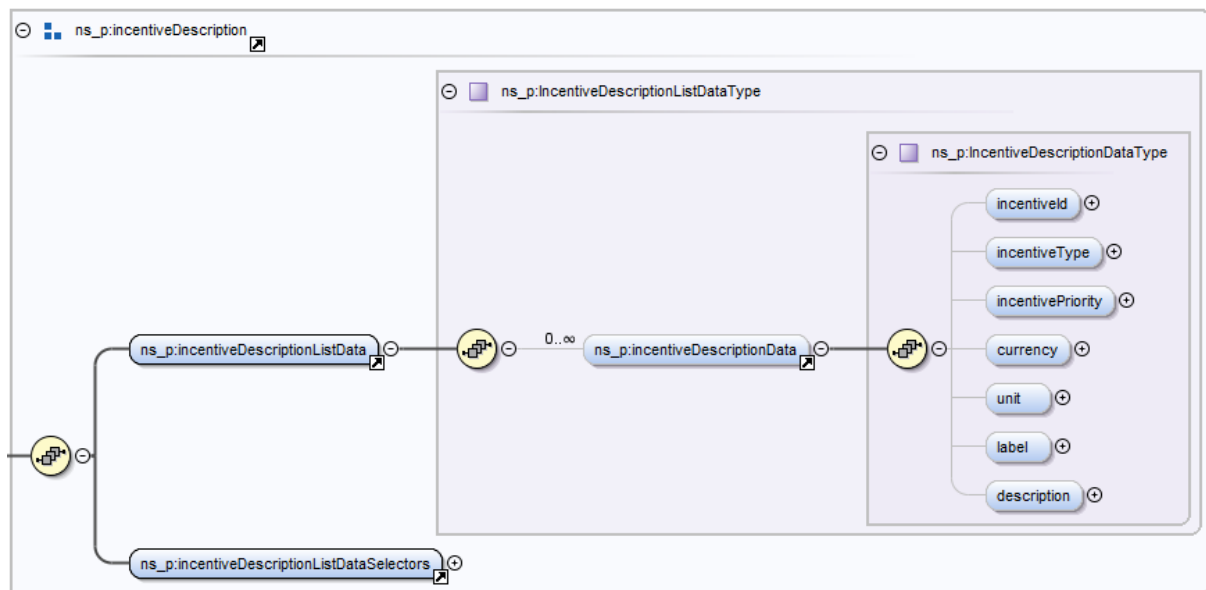


Figure 207: incentiveDescriptionListData function overview

5.3.25.19.2 Detailed description of elements

Element	Type	Description
incentiveDescriptionListData. incentiveDescriptionData. (list)		
incentiveId	Identifier "IncentiveIdType" (see Table 358).	Primary identifier for the incentive.
incentiveType	Union "IncentiveTypeType": - Enum (see Table 329): IncentiveTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the incentive.
incentivePriority	Simple type "IncentivePriorityType" (restriction of xs:unsignedInt)	A priority can be stated for an incentive.
currency	Common data type "CurrencyType". See section 3.10.1.19.	The currency for the incentive costs.
unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	A unit for the measured commodity this incentives models costs for.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the incentive.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the incentive.

Table 328: incentiveDescriptionListData function detailed description of elements

Enumeration IncentiveTypeEnumType:

Value	Description
absoluteCost	Absolute costs with a currency and a unit.
relativeCost	Relative costs that cannot be interpreted as some monetary costs.
renewableEnergyPercentage	Percentage of renewable energy of the total obtained energy. This is an ecological incentive that allows optimizing consumption of renewable energy instead of costs.
co2Emission	CO2 emission related to the obtained energy. This is an ecological incentive that allows optimizing the CO2 emission instead of costs.

10812 *Table 329: Enumeration IncentiveTypeEnumType*

10813

10814 *5.3.25.19.3 Available selectors*

10815 - incentiveId

10816 - incentiveType

10817

10818 *5.3.25.19.4 Examples*

10819 *5.3.25.19.4.1 Read-reply*

10820 If one is interested in the rather static parts of all incentives, he could send a standard read
10821 command to the server. The reply could look like this:

```
10822 <incentiveDescriptionListData>
10823   <incentiveDescriptionData>
10824     <incentiveId>1</incentiveId>
10825     <incentiveType>absoluteCost</incentiveType>
10826     <incentivePriority>1</incentivePriority>
10827     <currency>EUR</currency>
10828     <unit>Wh</unit>
10829   </incentiveDescriptionData>
10830 </incentiveDescriptionListData>
```

10831

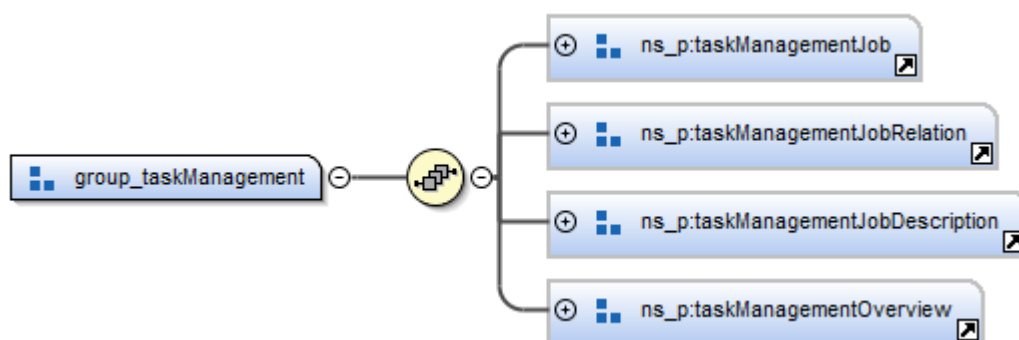
10832 **5.3.26 TaskManagement**

10833 **5.3.26.1 Introduction**

10834 The TaskManagement Class lists all "tasks" that are currently relevant on an entity. This enables one
10835 to have a quick look on the tasks of related classes, but not for their modification. Further
10836 information has to be gathered with the corresponding classes supporting the specific functionality.

10837 In this version of the specification, tasks can be "jobs" like in DirectControl, PowerSequences /
10838 SmartEnergyManagementPs, etc. (see the related function groups below).

10839 Remark: Future versions of this specification may add other kinds of tasks (i.e. tasks that are not
10840 covered by "jobs").



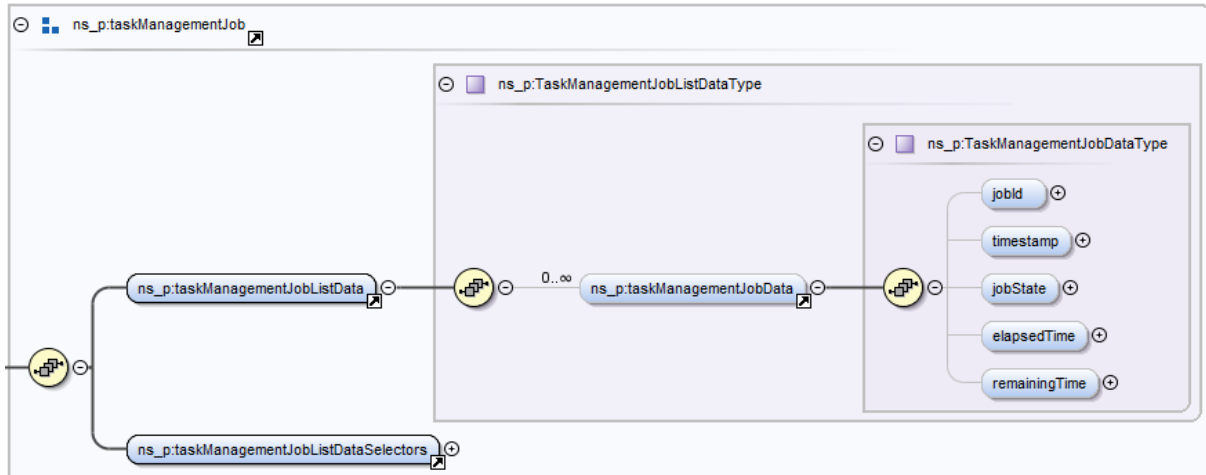
10841

10842 *Figure 208: TaskManagement function-group overview*

10843

10844 **5.3.26.2 taskManagementJobListData**10845 **5.3.26.2.1 General**

10846 All jobs that are available on the entity are listed here, together with their state and (if available) the
 10847 elapsed and remaining time.



10848
 10849 *Figure 209: taskManagementJobListData function overview*

10850

10851 **5.3.26.2.2 Detailed description of elements**

Element	Type	Description
taskManagementJobListData. taskManagementJobData. (list)		
jobId	Identifier "TaskManagementJobIdType" (see Table 358).	The <i>jobId</i> is used for indentifying different job information collected by this class.
timestamp	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	Point in time, this information was generated.
jobState	Union "TaskManagementJobStateType": - Enum (see Table 205): DirectControlActivityStateEnumType - Enum (see Table 227): HvacOverrunStatusEnumType - Enum (see Table 238): LoadControlEventStateEnumType - Enum (see Table 284): PowerSequenceStateEnumType - EnumExtendType (see section 3.10.1.5)	Current state of the job.
elapsedTime	xs:duration (W3C standard type)	The duration, this job was already active (normally, pause times are not counted).
remainingTime	xs:duration (W3C standard type)	The duration, this job will most likely be active before it ends.

10852 *Table 330: taskManagementJobListData function detailed description of elements*

10853

10854 **5.3.26.2.3 Available selectors**

- 10855 - jobId
- 10856 - jobState

10857

10858 **5.3.26.2.4 Examples**10859 **5.3.26.2.4.1 Read-reply**

10860 If one wants to request information about a specific job (jobId = "2"), he could sent the following
 10861 read command:

```

10862 <function>taskManagementJobListData</function>
10863 <filter>
10864   <cmdControl>
10865     <partial/>
10866   </cmdControl>
10867   <taskManagementJobListDataSelectors>
10868     <jobId>2</jobId>
10869   </taskManagementJobListDataSelectors>
10870 </filter>
10871 <taskManagementJobListData/>

```

10872 A response could be:

```

10873 <function>taskManagementJobListData</function>
10874 <filter>
10875   <cmdControl>
10876     <partial/>
10877   </cmdControl>
10878 </filter>
10879 <taskManagementJobListData>
10880   <taskManagementJobData>
10881     <jobId>2</jobId>
10882     <timestamp>2015-11-18T14:59:23.3Z</timestamp>
10883     <jobState>running</jobState>
10884     <elapsedTime>PT2H8M32S</elapsedTime>
10885     <remainingTime>PT21M28S</remainingTime>
10886   </taskManagementJobData>
10887 </taskManagementJobListData>

```

10888

10889 **5.3.26.3 taskManagementJobRelationListData**10890 **5.3.26.3.1 General**

10891 To identify where a job originates from, the *taskManagementJobRelationListData* function is used. If
 10892 the related class allows more than one job the corresponding identifier(s) are denoted.

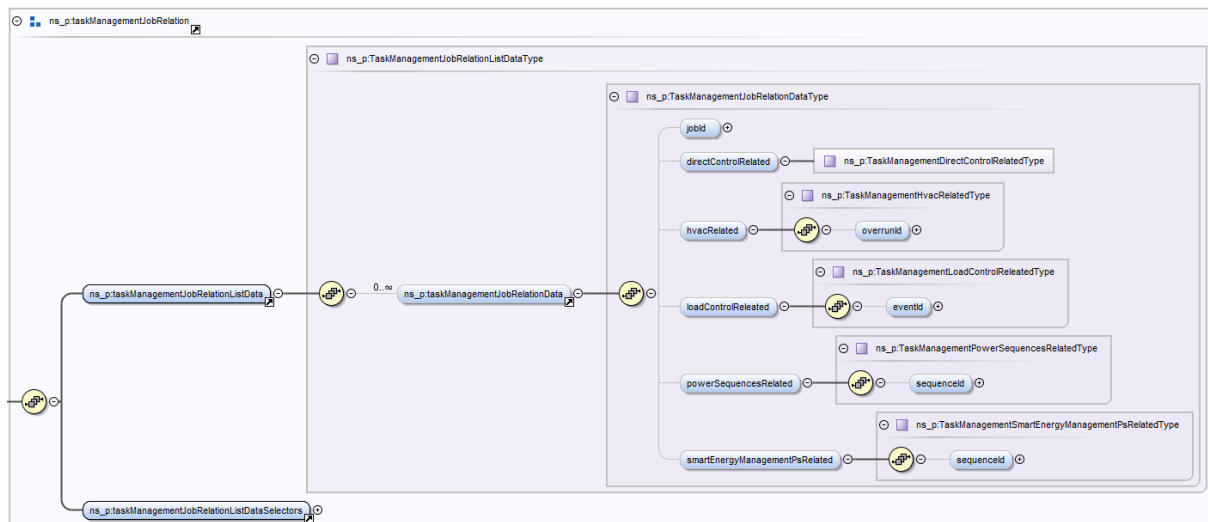


Figure 210: taskManagementJobRelationListData function overview

5.3.26.3.2 Detailed description of elements

Element	Type	Description
taskManagementJobRelationListData. taskManagementJobRelationData. (list)		
jobId	Identifier "TaskManagementJobIdType" (see Table 358).	The <i>jobId</i> is used for indentifying different job information collected by this class.
directControlRelated	ComplexType "TaskManagementDirectControlRelatedType"	Flag that indicates that the job is related to DirectControl.
hvacRelated	ComplexType "TaskManagementHvacRelatedType"	Flag that indicates that the job is related to HVAC.
hvacRelated. overrunId	Identifier "HvacOverrunIdType" (see Table 358).	Identifier for the related overrun.
loadControlRelated	ComplexType "TaskManagementLoadControlRelatedType"	Flag that indicates that the job is related to LoadControl.
loadControlRelated. eventId	Identifier "LoadControlEventIdType" (see Table 358).	Identifier for the related load control event.
powerSequencesRelated	ComplexType "TaskManagementPowerSequencesRelatedType"	Flag that indicates that the job is related to PowerSequences.
powerSequencesRelated. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	Identifier for the related power sequence.
smartEnergyManagementPsRelated	ComplexType "TaskManagementSmartEnergyManagementPsRelatedType"	Flag that indicates that the job is related to SmartEnergyManagementPs.
smartEnergyManagementPsRelated. sequenceId	Identifier "PowerSequenceIdType" (see Table 358).	Identifier for the related power sequence.

Table 331: taskManagementJobRelationListData function detailed description of elements

10898

10899 **5.3.26.3.3 Available selectors**

10900 - jobId

10901

10902 **5.3.26.3.4 Examples**10903 **5.3.26.3.4.1 Read-reply**

10904 If one requests all job relations, the reply could look like the following:

```

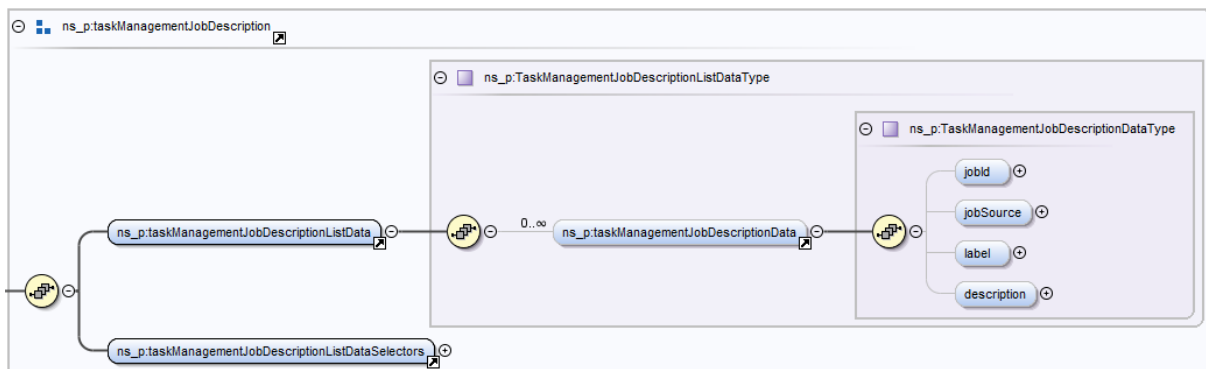
10905 <taskManagementJobRelationListData
10906   <taskManagementJobRelationData>
10907     <jobId>1</jobId>
10908     <hvacRelated>
10909       <overrunId>3</overrunId>
10910     </hvacRelated>
10911   </taskManagementJobRelationData>
10912   <taskManagementJobRelationData>
10913     <jobId>2</jobId>
10914     <directControlRelated/>
10915   </taskManagementJobRelationData>
10916 </taskManagementJobRelationListData>

```

10917

10918 **5.3.26.4 taskManagementJobDescriptionListData**10919 **5.3.26.4.1 General**

10920 The more static information about the jobs are communicated with this function.



10921

10922 *Figure 211: taskManagementJobDescriptionListData function overview*

10923

10924 **5.3.26.4.2 Detailed description of elements**

Element	Type	Description
taskManagementJobDescriptionListData. taskManagementJobDescriptionData. (list)		
jobId	Identifier "TaskManagementJobIdType" (see Table 358).	The <i>jobId</i> is used for indentifying different job information collected by this class.
jobSource	Union "TaskManagementJobSourceType": - Enum (see Table 333):	Specifies what kind of reason caused this job.

	TaskManagementJobSourceEnumType - EnumExtendType (see section 3.10.1.5)	
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	A short label of the job.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Descriptive information on the job.

Table 332: taskManagementJobDescriptionListData function detailed description of elements

Enumeration **TaskManagementJobSourceEnumType**:

Value	Description
internalMechanism	The job was started due to some device internal mechanism (e.g. a regular cleaning cycle).
userInteraction	The user activated some functionality directly via the UI of the device.
externalConfiguration	The job started, because it was triggered from some external device (e.g. an energy manager).

Table 333: Enumeration TaskManagementJobSourceEnumType

5.3.26.4.3 Available selectors

- jobId
- jobSource

5.3.26.4.4 Examples

5.3.26.4.4.1 Read-reply

If one requests all job descriptions, the reply could look like the following:

```

<taskManagementJobDescriptionListData>
  <taskManagementJobDescriptionData>
    <jobId>1</jobId>
    <jobSource>userInteraction</jobSource>
  </taskManagementJobDescriptionData>
  <taskManagementJobDescriptionData>
    <jobId>2</jobId>
    <jobSource>externalConfiguration</jobSource>
  </taskManagementJobDescriptionData>
</taskManagementJobDescriptionListData>

```

5.3.26.5 taskManagementOverviewData

5.3.26.5.1 General

A general overview is given with this function. It specifies whether a remote controlling is permitted at that moment and if currently jobs are active.

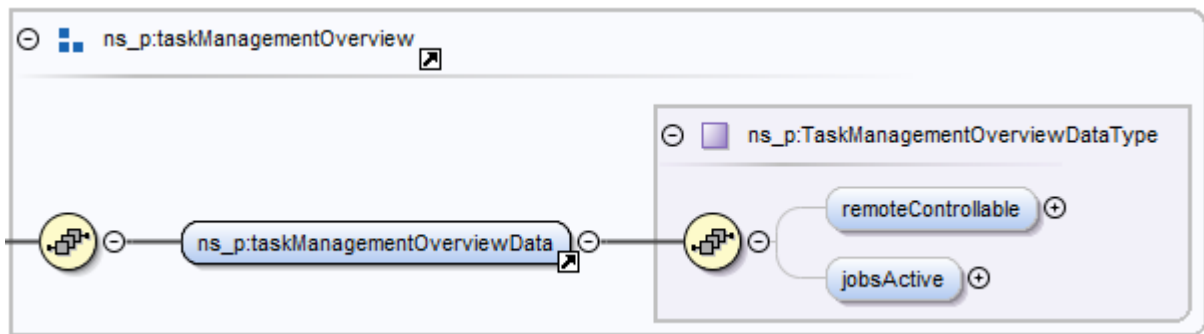


Figure 212: taskManagementOverviewData function overview

5.3.26.5.2 Detailed description of elements

Element	Type	Description
taskManagementOverviewData.		
remoteControllable	xs:boolean (W3C standard type)	States if remote controlling of the device or entity is permitted.
jobsActive	xs:boolean (W3C standard type)	Denotes whether jobs are active on the device or entity or not.

Table 334: taskManagementOverviewData function detailed description of elements

5.3.26.5.3 Available selectors

None.

5.3.26.5.4 Examples

5.3.26.5.4.1 Read-reply

If one requests the task management overview, a response could be:

```
<taskManagementOverviewData>
  <remoteControllable>true</remoteControllable>
  <jobsActive>true</jobsActive>
</taskManagementOverviewData>
```

5.3.27 Threshold

5.3.27.1 Introduction

Thresholds are helpful in several kind of use cases. E.g. a device or application can be configured to do some specific behaviour if a measured value is above or below a configured threshold. Some measurands need to stay in a certain value range, to ensure proper and safe working of devices. Automatic alarms after reaching a threshold help reacting on the occurrence immediately.

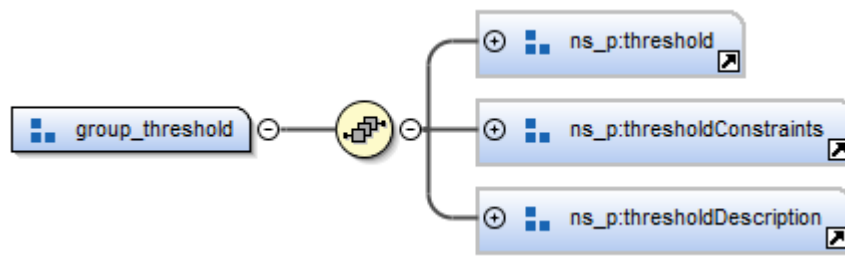


Figure 213: Threshold function-group overview

5.3.27.2 thresholdListData

5.3.27.2.1 General

Each threshold (identified by its *thresholdId*) has a configured value. This function is used to read or write the actual value.

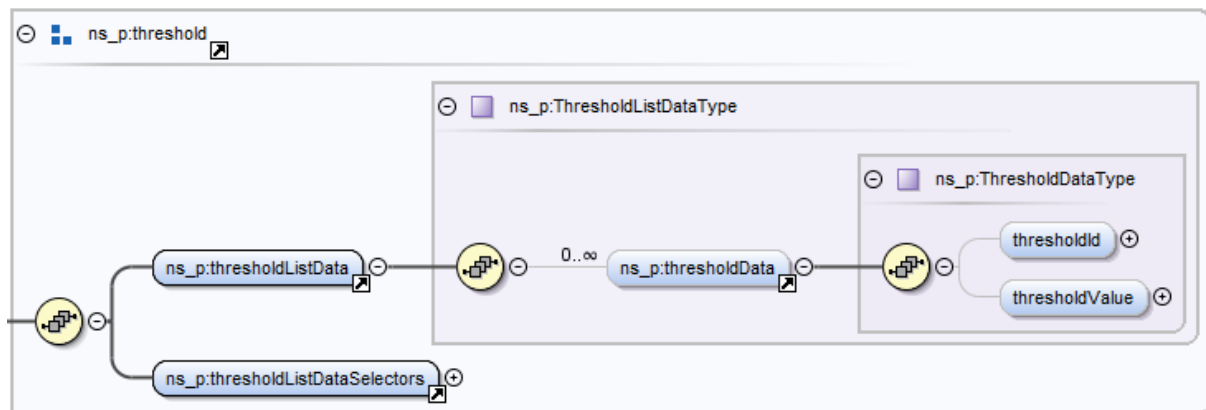


Figure 214: thresholdListData function overview

5.3.27.2.2 Detailed description of elements

Element	Type	Description
thresholdListData. thresholdData. (list)		
thresholdId	Identifier "ThresholdIdType" (see Table 358).	Identifier for the threshold.
thresholdValue	Common data type "ScaledNumberType". See section 3.10.1.8.	Value that should not be fallen below or exceeded (depends on the <i>thresholdType</i> , see section 5.3.27.4.2).

Table 335: thresholdListData function detailed description of elements

5.3.27.2.3 Available selectors

- thresholdId

10990 5.3.27.2.4 Examples

10991 5.3.27.2.4.1 Read-reply

10992 If one is interested in all thresholds, he could send the a standard read command to the device. The
 10993 device could respond with something like this:

```

10994 <thresholdListData>
10995   <thresholdData>
10996     <thresholdId>1</thresholdId>
10997     <thresholdValue>
10998       <number>155</number>
10999       <scale>-1</scale>
11000     </thresholdValue>
11001   </thresholdData>
11002   <thresholdData>
11003     <thresholdId>2</thresholdId>
11004     <thresholdValue>
11005       <number>23</number>
11006     </thresholdValue>
11007   </thresholdData>
11008 </thresholdListData>

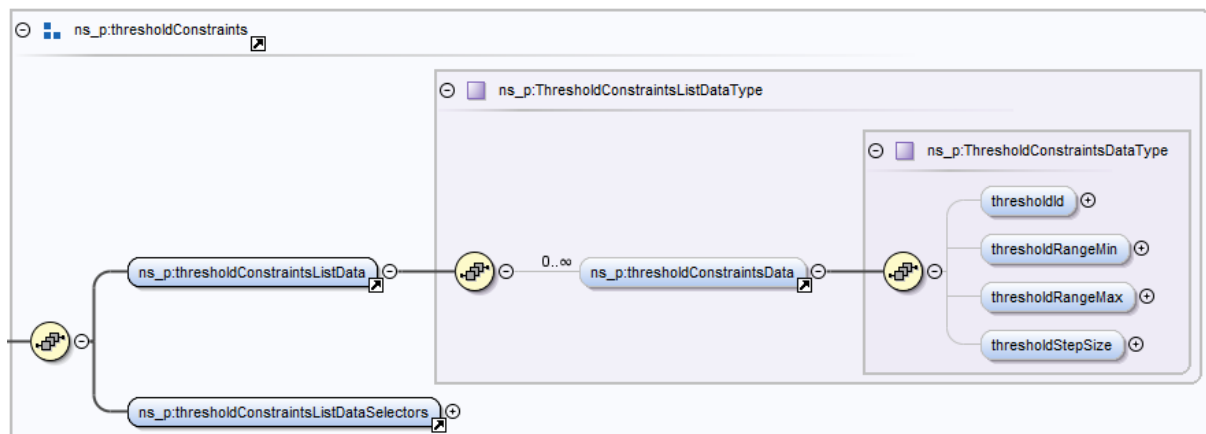
```

11009

11010 5.3.27.3 thresholdConstraintsListData

11011 5.3.27.3.1 General

11012 The value range from which a threshold value configuration may be chosen from may be restricted
 11013 by a minimum and a maximum value. Between these, not every value, but only ones with a specific
 11014 gap, may be used, specified with the step size.



11015

11016 Figure 215: thresholdConstraintsListData function overview

11017

11018 5.3.27.3.2 Detailed description of elements

Element	Type	Description
thresholdConstraintsListData. thresholdConstraintsData. (list)		
thresholdId	Identifier "ThresholdIdType" (see Table 358).	Identifier for the threshold.

thresholdRangeMin	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Minimum value that may be used to configure the element <i>thresholdValue</i> (see section 5.3.27.2.2).
thresholdRangeMax	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Maximum value that may be used to configure the element <i>thresholdValue</i> (see section 5.3.27.2.2).
thresholdStepSize	<i>Common data type "ScaledNumberType". See section 3.10.1.8.</i>	Step size for the configuration of the element <i>thresholdValue</i> (see section 5.3.27.2.2).

Table 336: *thresholdConstraintsListData* function detailed description of elements

5.3.27.3.3 Available selectors

- thresholdId

5.3.27.3.4 Examples

5.3.27.3.4.1 Read-reply

If one is interested in the constraints of all thresholds, he could send the a standard read command to the device. The device could respond with something like this:

```

<thresholdConstraintsListData>
  <thresholdConstraintsData>
    <thresholdId>1</thresholdId>
    <thresholdRangeMin>
      <number>5</number>
    </thresholdRangeMin>
    <thresholdRangeMax>
      <number>35</number>
    </thresholdRangeMax>
    <thresholdStepSize>
      <number>5</number>
      <scale>-1</scale>
    </thresholdStepSize>
  </thresholdConstraintsData>
  <thresholdConstraintsData>
    <thresholdId>2</thresholdId>
    <thresholdRangeMin>
      <number>5</number>
    </thresholdRangeMin>
    <thresholdRangeMax>
      <number>35</number>
    </thresholdRangeMax>
    <thresholdStepSize>
      <number>1</number>
    </thresholdStepSize>
  </thresholdConstraintsData>
</thresholdConstraintsListData>

```

5.3.27.4 thresholdDescriptionListData

5.3.27.4.1 General

The rather static information about thresholds is defined within this function.

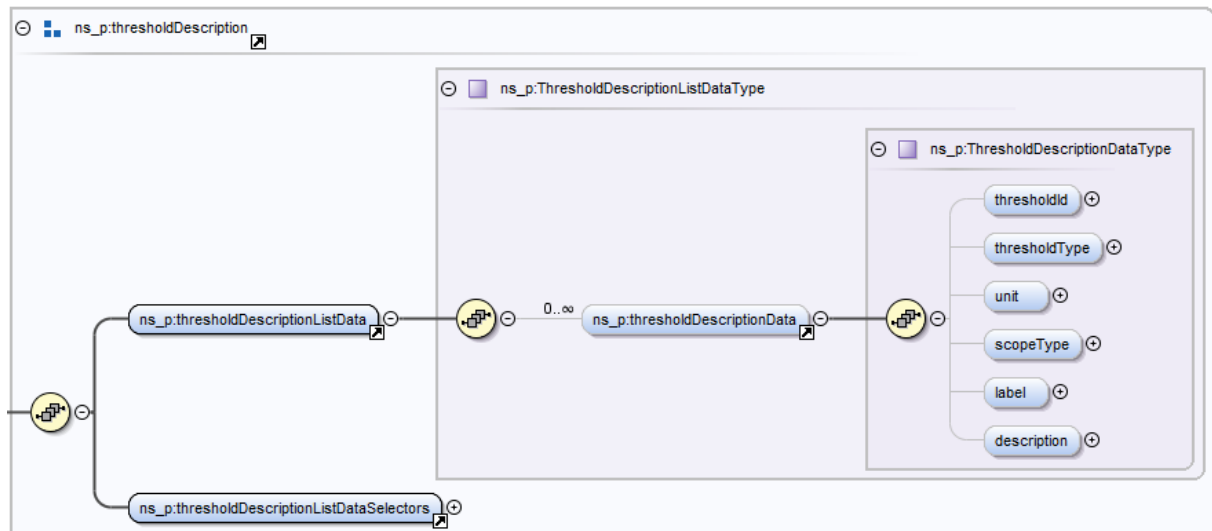


Figure 216: thresholdDescriptionListData function overview

5.3.27.4.2 Detailed description of elements

Element	Type	Description
thresholdDescriptionListData. thresholdDescriptionData. (list)		
thresholdId	Identifier "ThresholdIdType" (see Table 358).	Identifier for the threshold.
thresholdType	Union "ThresholdTypeType": - Enum (see Table 338): ThresholdTypeEnumType - EnumExtendType (see section 3.10.1.5)	The type of the treshold.
unit	Common data type "UnitOfMeasurementType". See section 3.10.1.17.	The unit in which the value of the threshold is denoted.
scopeType	Common data type "ScopeTypeType". See section 3.10.1.21.	Specifies a more detailed meaning of the threshold.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the threshold.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the threshold.

Table 337: thresholdDescriptionListData function detailed description of elements

Enumeration ThresholdTypeEnumType:

Value	Description
goodAbove	A positive reaction will follow after the value exceeds the threshold.
badAbove	A negative reaction will follow after the value exceeds the threshold.
goodBelow	A positive reaction will follow after the value falls below the threshold.
badBelow	A negative reaction will follow after the value falls below the threshold.

minValueThreshold	If this minimum threshold is underrun, some warning will be activated.
maxValueThreshold	If this maximum threshold is exceeded, some warning will be activated.
minValueThresholdExtreme	If this extreme minimum threshold is underrun, some important warning will be activated.
maxValueThresholdExtreme	If this extreme maximum threshold is exceeded, some important warning will be activated.
sagThreshold	If this sag threshold is underrun, some warning will be activated. Commonly used for electricity voltage sags.
swellThreshold	If this swell threshold is exceeded, some warning will be activated. Commonly used for electricity voltage swells.

Table 338: Enumeration ThresholdTypeEnumType

5.3.27.4.3 Available selectors

- thresholdId
- scopeType

5.3.27.4.4 Examples

5.3.27.4.4.1 Read-reply

If one is interested in the description of all thresholds, he could send the a standard read command to the device. The device could respond with something like this:

```
<thresholdDescriptionListData>
  <thresholdDescriptionData>
    <thresholdId>1</thresholdId>
    <thresholdType>badBelow</thresholdType>
    <unit>degC</unit>
    <label>Minimum temperature</label>
  </thresholdDescriptionData>
  <thresholdDescriptionData>
    <thresholdId>2</thresholdId>
    <thresholdType>badAbove</thresholdType>
    <unit>degC</unit>
    <label>Maximum temperature</label>
  </thresholdDescriptionData>
</thresholdDescriptionListData>
```

5.3.28 TimeInformation

5.3.28.1 Introduction

The exchange of time information is important for most smart grid and home automation systems. Please note the underlying revision of the SPINE data model does not specify a sophisticated synchronisation mechanism that includes a given precision. This is due to the complex of problems appearing with communication delays in the different technologies and hardware platforms. However, an application could measure all relevant delays and take care of a real synchronisation in the network using the *TimeInformation* class.

Independent of any synchronisation aspect the "owner concept" applies here as well: The "owner" of any given data (a measured value of a sensor, e.g.) applies its own SPINE time to its own data. This is relevant for the case of a substantial difference between the SPINE time of two nodes.

In order to avoid problems from such differences a typical sequence could be that a new device sends a *timeDistributorEnquiryCall* (with the *isTimeDistributor* element set to true) to all devices in the SPINE network. All time distributors would send a *timeDistributorData* instance as notification to the device. The new device would pick the time distributor with the highest priority and send a read command to the *timeInformationData* function of this distributor. The time distributor would reply with a *timeInformationData* containing its current time.

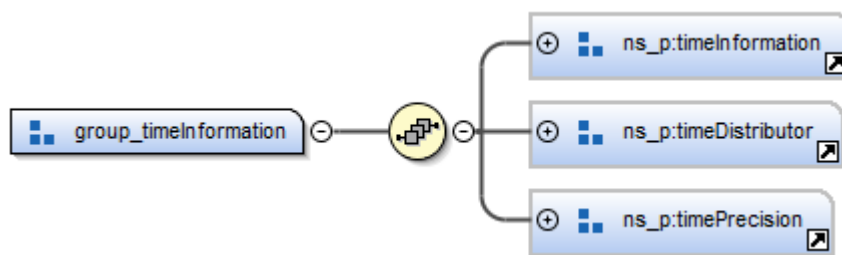


Figure 217: TimeInformation function-group overview (data)

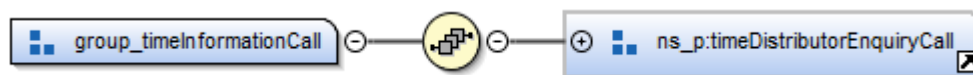


Figure 218: TimeInformation function-group overview (call)

5.3.28.2 timeInformationData

5.3.28.2.1 General

The *timeInformationData* command is used for exchanging the SPINE time.

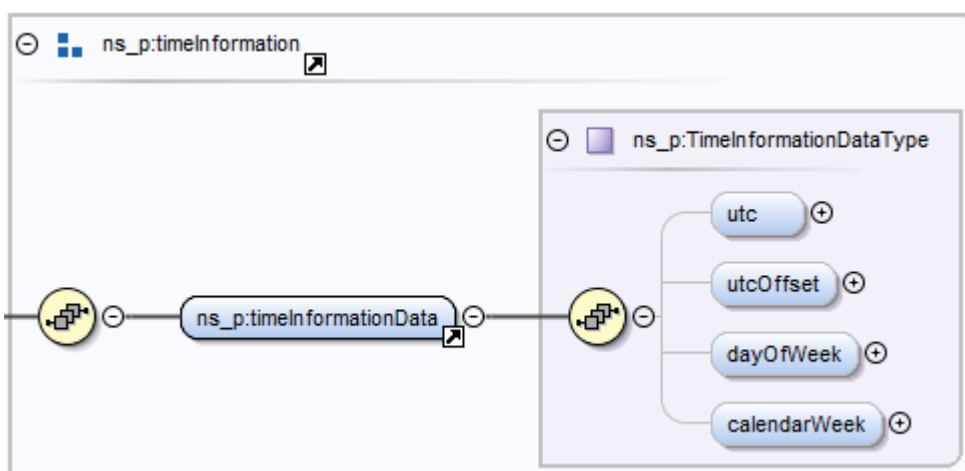


Figure 219: timeInformationData function overview

5.3.28.2.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

timeInformationData.		
utc	xs:dateTime (W3C standard type)	The current time in UTC.
utcOffset	xs:duration (W3C standard type)	The configured offset to UTC (e.g. PT2H [= +2 hours] for Central European Summer Time (CEST)).
dayOfWeek	<i>Common data type "DayOfWeekType". See section 3.10.2.9.</i>	To model the actual day of week (Monday to Sunday) this element is used. It refers to the local time. If there is no utcOffset given, it is assumed, that the local time is equivalent to UTC-time.
calendarWeek	<i>Common data type "CalendarWeekType". See section 3.10.2.8.</i>	A value between 1 and 53 stating the current calendar week number. This element refers to the local time. If there is no utcOffset given, it is assumed, that the local time is equivalent to UTC-time.

Table 339: timeInformationData function detailed description of elements

5.3.28.2.3 Available selectors

None.

5.3.28.2.4 Examples

5.3.28.2.4.1 Read-reply

A timeInformationData command sent as reply to a read command on timeInformationData:

```
<timeInformationData>
  <utc>2013-07-15T12:00:00.0Z</utc>
  <utcOffset>PT2H</utcOffset>
  <dayOfWeek>tuesday</dayOfWeek>
  <calendarWeek>17</calendarWeek>
</timeInformationData>
```

5.3.28.3 timeDistributorData

5.3.28.3.1 General

One or some devices in a SPINE network should act as time distributor. The information thereof and the dedicated priority can be modelled with the timeDistributorData command.

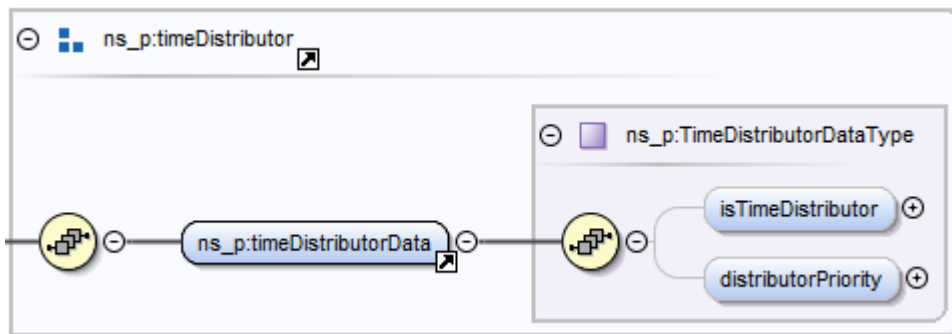


Figure 220: timeDistributorData function overview

5.3.28.3.2 Detailed description of elements

Element	Type	Description
timeDistributorData.		
isTimeDistributor	xs:boolean (W3C standard type)	Denotes whether a device can act as a time distributor in the SPINE network or not.
distributorPriority	xs:unsignedInt (W3C standard type)	A (configured) priority of the time distributor.

Table 340: timeDistributorData function detailed description of elements

5.3.28.3.3 Available selectors

None.

5.3.28.3.4 Examples

5.3.28.3.4.1 Notify

Example of a notification from a device, stating that it can act as a time distributor with a priority of "2":

```
<timeDistributorData>
  <isTimeDistributor>true</isTimeDistributor>
  <distributorPriority>2</distributorPriority>
</timeDistributorData>
```

Such notification could typically be send from a device that previously received a *timeDistributorEnquiryCall*.

5.3.28.4 timePrecisionData

5.3.28.4.1 General

Information regarding the precision of the time of a device can be modelled using *timePrecisionData*.

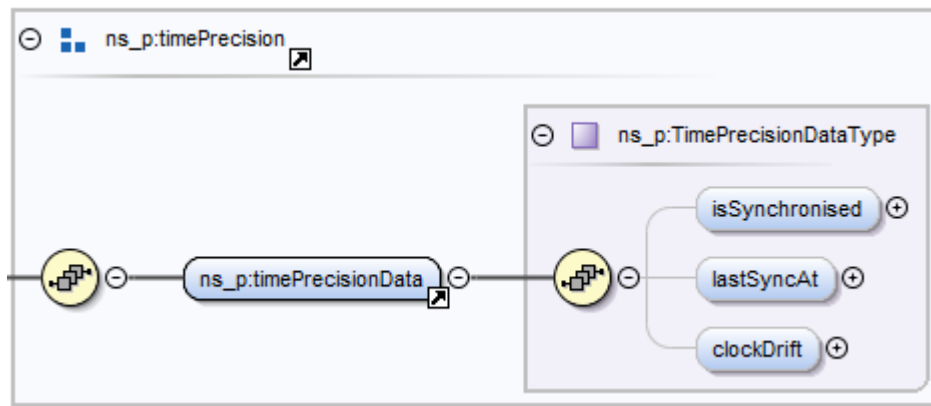


Figure 221: timePrecisionData function overview

5.3.28.4.2 Detailed description of elements

Element	Type	Description
timePrecisionData.		
isSynchronised	xs:boolean (W3C standard type)	This is a Boolean value, stating whether the device's SPINE time information is synchronized with the SPINE network or not, meaning that it could receive the SPINE time from a time distributor and has still a "sufficiently precise time".
lastSyncAt	xs:dateTime (W3C standard type)	This is the timestamp of the last synchronisation with a time distributor, or, in case of a time distributor itself, the timestamp of the last official date and time information (e.g. from an NTP server).
clockDrift	xs:integer (W3C standard type)	The drift of the internal clock (in seconds per day) is indicated by this element.

Table 341: timePrecisionData function detailed description of elements

5.3.28.4.3 Available selectors

None.

5.3.28.4.4 Examples

5.3.28.4.4.1 Read-reply

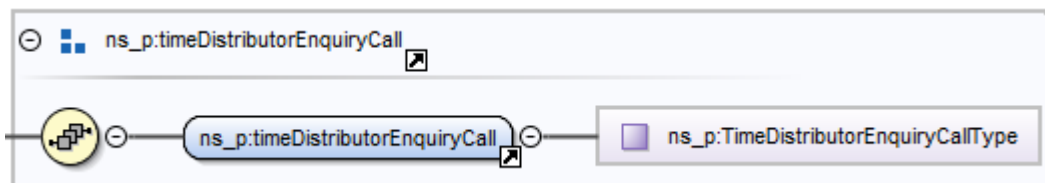
We assume a device had a synchronization with a time distributor more than two days ago. We also assume the device has a clock drift of 5 seconds per day and only 10 seconds deviation are acceptable to fulfil the network synchronization criteria. Then, if the device is requested for its current *timePrecisionData* information it could send the following reply:

```
<timePrecisionData>
  <isSynchronised>false</isSynchronised>
  <lastSyncAt>2013-07-14T15:00:00.0Z</lastSyncAt>
  <clockDrift>5</clockDrift>
</timePrecisionData>
```

11181

11182 **5.3.28.5 *timeDistributorEnquiryCall***11183 **5.3.28.5.1 *General***

11184 This function can be used for finding all time distributors in the network. Therefore, it can be sent to
 11185 all devices in the network that have their *isTimeDistributor* element in the function *timeDistributor*
 11186 (see section 5.3.28.3) set to true.



11187

11188 *Figure 222: timeDistributorEnquiryCall function overview*

11189

11190 **5.3.28.5.2 *Detailed description of elements***

11191 In this version of the specification the function *timeDistributorEnquiryCall* does not contain child
 11192 elements.

11193

11194 **5.3.28.5.3 *Available selectors***

11195 None.

11196

11197 **5.3.28.5.4 *Examples***11198 **5.3.28.5.4.1 *Call***

11199 A *timeDistributorEnquiryCall* instance, sent as call (the only classifier allowed for this function):

11200 `<timeDistributorEnquiryCall/>`

11201

11202 **5.3.29 *TimeSeries***11203 **5.3.29.1 *Introduction***

11204 TimeSeries is a versatile class that allows to express any kind of time dependent values as a time
 11205 series of values. It can be used to express electricity related values like power constraints or power
 11206 consumption over time, but also other values that are not electricity related like temperature or
 11207 humidity over time.

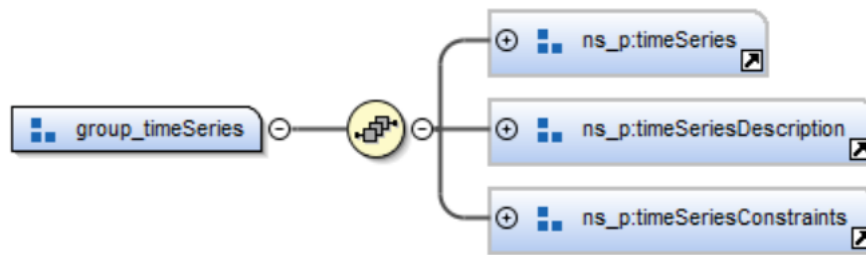


Figure 223: TimeSeries function-group overview

5.3.29.2 timeSeriesListData

5.3.29.2.1 General

The timeSeriesListData function is used to describe the value of a timeSeries over time.

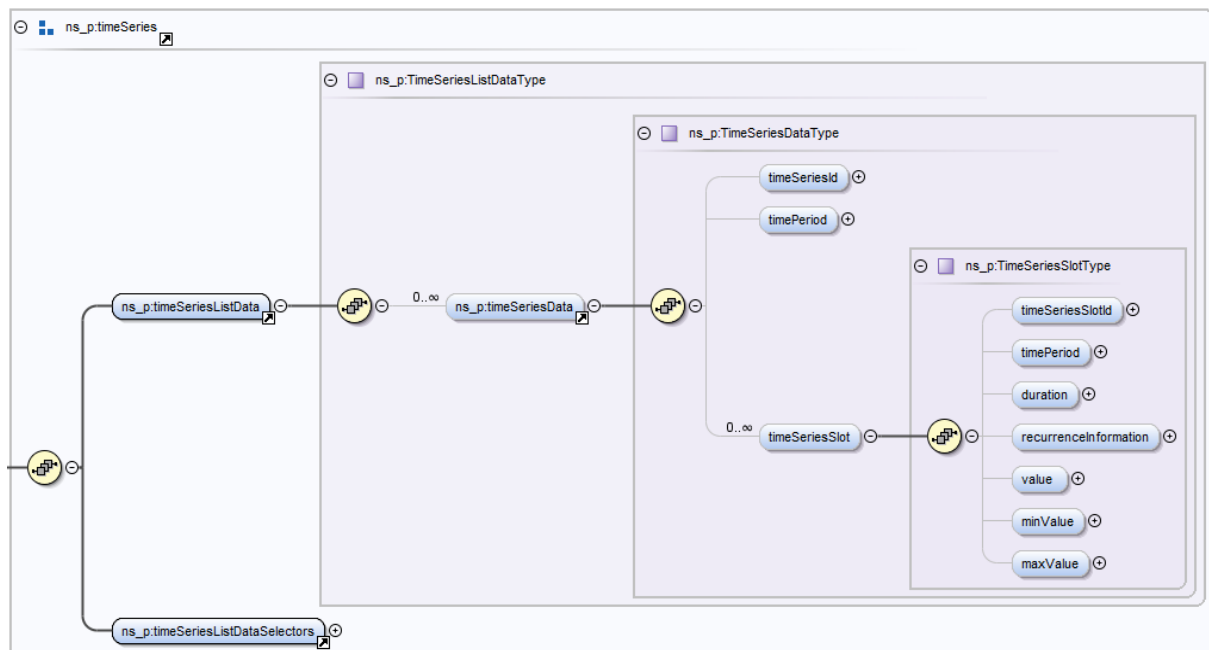


Figure 224: timeSeriesListData function overview

5.3.29.2.2 Detailed description of elements

Element	Type	Description
timeSeriesListData. timeSeriesData. (list)		
timeSeriesId	Identifier "TimeSeriesIdType" (see Table 358).	Allows the identification of a timeSeries.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	Time period of the complete timeSeries.
timeSeriesSlot (list)	Complex type "TimeSeriesSlotType". See Table 343.	Allows to define slots of a timeSeries.

Table 342: timeSeriesListData function detailed description of elements

11219 Complex type **TimeSeriesSlotType**:

Element	Type	Description
timeSeriesSlotId	Identifier "TimeSeriesSlotIdType" (see Table 358).	Allows identification of a timeSeriesSlot within a timeSeries.
timePeriod	Common data type "TimePeriodType". See section 3.10.2.1.	Allows to define a timePeriod of a timeSeriesSlot and to model time gaps between timeSeriesSlots.
duration	xs:duration (W3C standard type)	Allows to define a duration of a timeSeriesSlot.
recurrenceInformation	Common data type "RecurrenceInformationType". See section 3.10.2.14.	If the timeSeriesSlot is a recurring one, the corresponding information can be found within this element and its sub-elements.
value	Common data type "ScaledNumberType". See section 3.10.1.8.	Defines the expected value during a timeSeriesSlot.
minValue	Common data type "ScaledNumberType". See section 3.10.1.8.	Defines a lower value limit
maxValue	Common data type "ScaledNumberType". See section 3.10.1.8.	Defines an upper value limit

11220 Table 343: Complex type TimeSeriesSlotType

11221

11222 5.3.29.2.3 Available selectors

- 11223 - timeSeriesId
- 11224 - timeSeriesSlotId

11225

11226 5.3.29.2.4 Examples

11227 5.3.29.2.4.1 Read-reply

11228 To read the time series entry with the *timeSeriesId* "3", the following read command can be send:

```

11229 <function>timeSeriesListData</function>
11230 <filter>
11231   <cmdControl>
11232     <partial/>
11233   </cmdControl>
11234   <timeSeriesListDataSelectors>
11235     <timeSeriesId>3</timeSeriesId>
11236   </timeSeriesListDataSelectors>
11237 </filter>
11238 <timeSeriesListData/>

```

11239 A reply could look like this:

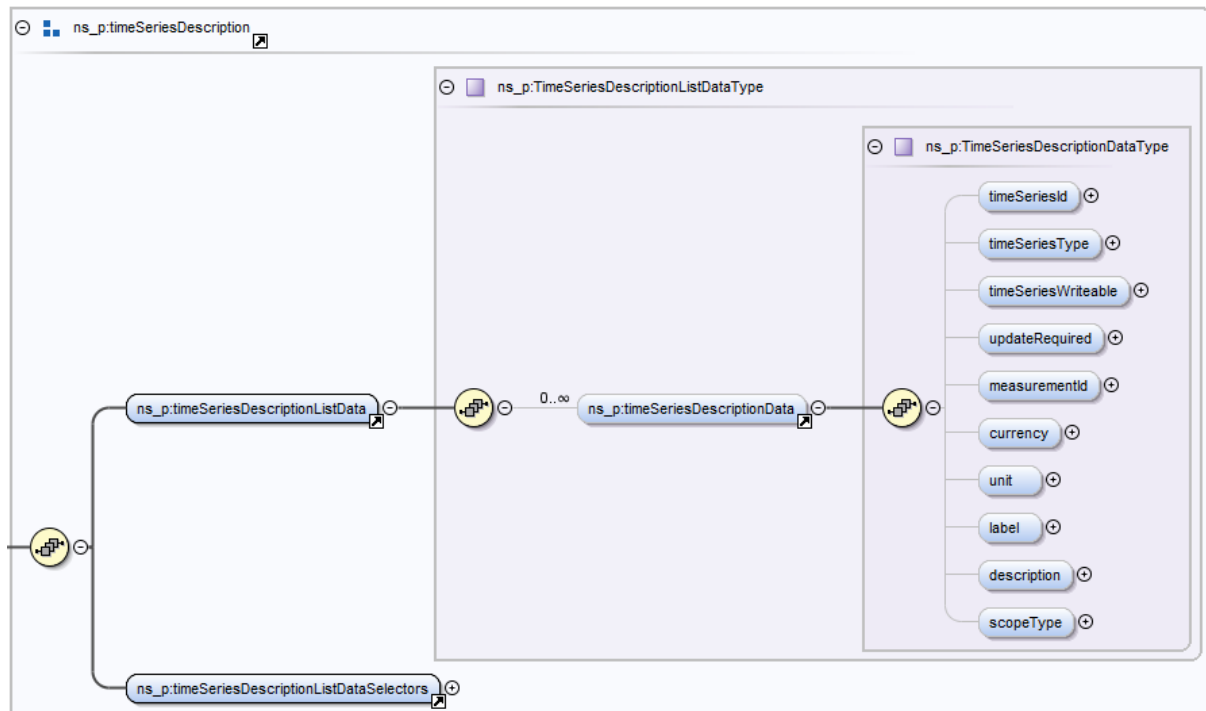
11240 <timeSeriesListData>

```
11241     <timeSeriesData>
11242         <timeSeriesId>3</timeSeriesId>
11243         <timePeriod>
11244             <startTime>2018-05-30T10:00:00.0Z</startTime>
11245             <endTime>2018-05-30T12:00:00.0Z</endTime>
11246         </timePeriod>
11247         <timeSeriesSlot>
11248             <timeSeriesSlotId>1</timeSeriesSlotId>
11249             <timePeriod>
11250                 <startTime>2018-05-30T10:00:00.0Z</startTime>
11251             </timePeriod>
11252             <duration>PT30M</duration>
11253             <recurrenceInformation>
11254                 <daysOfWeek>
11255                     <saturday/>
11256                     <sunday/>
11257                 </daysOfWeek>
11258             </recurrenceInformation>
11259             <value>
11260                 <number>21</number>
11261                 <scale>2</scale>
11262             </value>
11263         </timeSeriesSlot>
11264         <timeSeriesSlot>
11265             <timeSeriesSlotId>2</timeSeriesSlotId>
11266             <timePeriod>
11267                 <startTime>2018-05-30T11:00:00.0Z</startTime>
11268             </timePeriod>
11269             <duration>PT1H</duration>
11270             <recurrenceInformation>
11271                 <daysOfWeek>
11272                     <saturday/>
11273                     <sunday/>
11274                 </daysOfWeek>
11275             </recurrenceInformation>
11276             <value>
11277                 <number>35</number>
11278                 <scale>2</scale>
11279             </value>
11280         </timeSeriesSlot>
11281     </timeSeriesData>
11282 </timeSeriesListData>
```

11283

11284 **5.3.29.3 timeSeriesDescriptionListData**11285 **5.3.29.3.1 General**

11286 The timeSeriesDescriptionListData function is used to describe a timeSeries.



11287

11288 *Figure 225: timeSeriesDescriptionListData function overview*

11289

11290 **5.3.29.3.2 Detailed description of elements**

Element	Type	Description
timeSeriesDescriptionListData. timeSeriesDescriptionData. (list)		
timeSeriesId	Identifier "TimeSeriesIdType" (see Table 358).	Allows the identification of a timeSeries.
timeSeriesType	Union "TimeSeriesTypeType": - Enum (see Table 345): TimeSeriesTypeEnumType - EnumExtendType (see section 3.10.1.5)	Allows to define which timeSeries type is supported.
timeSeriesWriteable	xs:boolean (W3C standard type)	Allows to define if the timeSeries is writeable.
updateRequired	xs:boolean (W3C standard type)	Request a write to timeSeriesData from the bound client.
measurementId	Identifier "MeasurementIdType" (see Table 358).	Allows to link data of other Features to a timeSeries, e.g. Measurement or ElectricalConnection data.
currency	Common data type "CurrencyType". See section 3.10.1.19.	Allows to describe the currency of the value, if the value is a price.

unit	<i>Common data type "UnitOfMeasurementType". See section 3.10.1.17.</i>	Allows to describe the unit of the value, if the value has a unit.
label	<i>Common data type "LabelType". See section 3.10.1.2.</i>	Allows to define user friendly name for the timeSeries.
description	<i>Common data type "DescriptionType". See section 3.10.1.3.</i>	Allows to define a user friendly description for the timeSeries.
scopeType	<i>Common data type "ScopeTypeType". See section 3.10.1.21.</i>	Specifies a more detailed meaning of the time series.

11291 Table 344: timeSeriesDescriptionListData function detailed description of elements

11292 Enumeration **TimeSeriesTypeEnumType**:

Value	Description
plan	"plan" is used to model how a certain value changes over time..
singleDemand	"singleDemand" is used to model a single demand with a simple timeSeries (e.g. this can be used to model a simple energy demand).
constraints	"constraints" is used to model certain value constraints over time with a min- and/or max-Value curve.
energyRequest	"energyRequest" is used for energy consumption requests of an appliance.
dischargingEnergyRequest	"dischargingEnergyRequest" is used for energy production requests of an appliance (e.g., discharging a battery).
consumptionLimitCurve	"consumptionLimitCurve" is used for curves limiting the consumption.
productionLimitCurve	"productionLimitCurve" is used for curves limiting the production.

11293 Table 345: Enumeration TimeSeriesTypeEnumType

11294

11295 5.3.29.3.3 Available selectors

- 11296 - timeSeriesId
- 11297 - timeSeriesType
- 11298 - measurementId
- 11299 - scopeType

11300

11301 5.3.29.3.4 Examples

11302 5.3.29.3.4.1 Read-reply

11303 To read the time series description entry with the *timeSeriesId* "3", the following read command can
 11304 be send:

```

11305 <function>timeSeriesDescriptionListData</function>
11306 <filter>
11307   <cmdControl>
11308     <partial/>
11309   </cmdControl>
11310   <timeSeriesDescriptionListDataSelectors>
11311     <timeSeriesId>3</timeSeriesId>
11312   </timeSeriesDescriptionListDataSelectors>

```

11313 </filter>

11314 A reply could look like this:

```

11315 <timeSeriesDescriptionListData>
11316   <timeSeriesDescriptionData>
11317     <timeSeriesId>3</timeSeriesId>
11318     <timeSeriesType>plan</timeSeriesType>
11319     <timeSeriesWriteable>true</timeSeriesWriteable>
11320     <updateRequired>false</updateRequired>
11321     <measurementId>1</measurementId>
11322     <currency>EUR</currency>
11323     <unit>Wh</unit>
11324   </timeSeriesDescriptionData>
11325 </timeSeriesDescriptionListData>

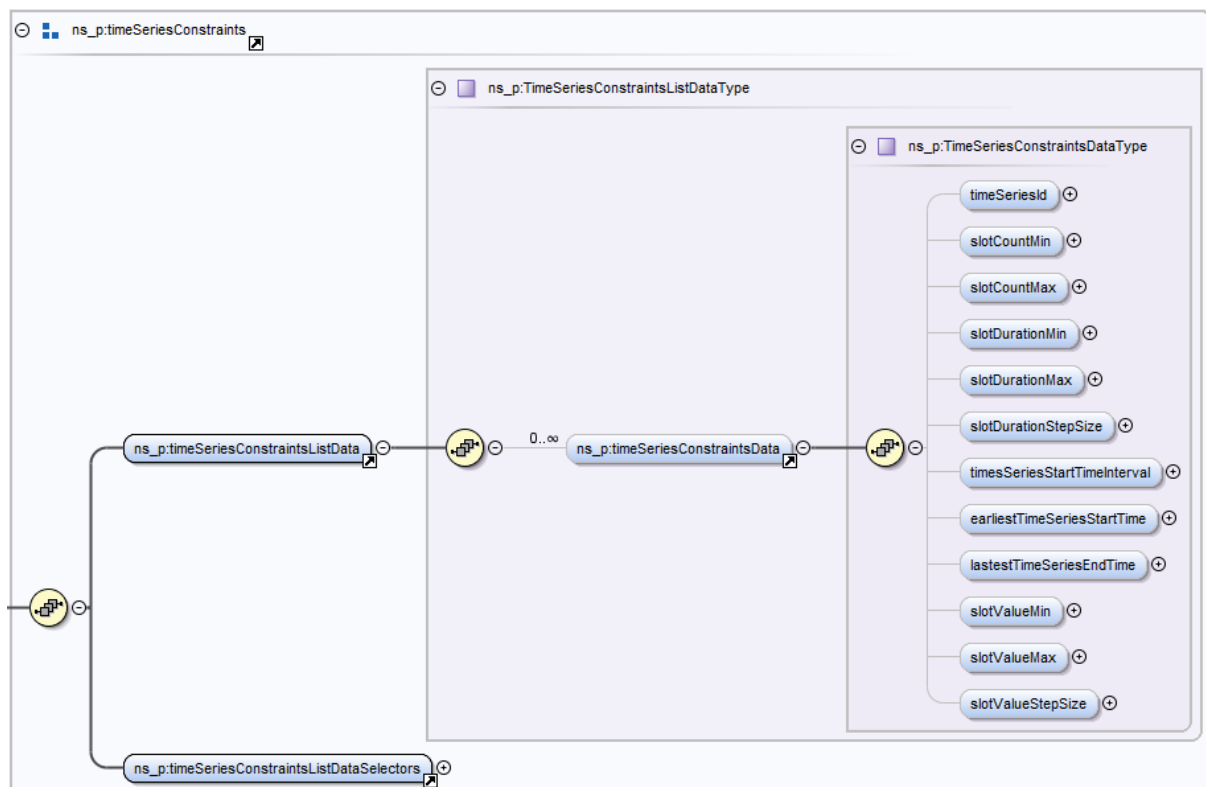
```

11326

11327 5.3.29.4 timeSeriesConstraintsListData

11328 5.3.29.4.1 General

11329 Allows to define constraints for the timeSeries, e.g. what is the maximum amount of slots allowed or
 11330 similar. This function can be especially valuable if timeSeriesData is writeable but the server has
 11331 certain constraints. The client can then match its own constraints with the server constraints, to form
 11332 a conforming write.



11333

11334 Figure 226: timeSeriesConstraintsListData function overview

11335

11336 5.3.29.4.2 Detailed description of elements

Element	Type	Description
---------	------	-------------

timeSeriesConstraintsListData. timeSeriesConstraintsData. (list)		
timeSeriesId	Identifier "TimeSeriesIdType" (see Table 358).	Allows the identification of a timeSeries.
slotCountMin	Simple type "TimeSeriesSlotCountType" (restriction of "TimeSeriesSlotIdType", see Table 343).	The minimum amount of timeSeriesSlots for a specific time series is stated in this element. Recurring slots only count as one.
slotCountMax	Simple type "TimeSeriesSlotCountType" (restriction of "TimeSeriesSlotIdType", see Table 343).	The maximum amount of timeSeriesSlots for a specific time series is stated in this element. Recurring timeSeriesSlots only count as one.
slotDurationMin	xs:duration (W3C standard type)	If the timeSeriesSlots of this time series have a minimum duration, it is denoted here.
slotDurationMax	xs:duration (W3C standard type)	If the timeSeriesSlots of this time series have a maximum duration, it is denoted here.
slotDurationStepSize	xs:duration (W3C standard type)	timeSeriesSlots can have a specific duration step size. This element holds the corresponding value.
earliestTimeSeriesStartTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	This is the earliest start time the corresponding time series is allowed to start.
latestTimeSeriesEndTime	Common data type "AbsoluteOrRelativeTimeType". See section 3.10.2.3.	This is the latest end time the corresponding times series is allowed to end.
slotValueMin	Common data type "ScaledNumberType". See section 3.10.1.8.	The values within the timeSeriesSlots of the corresponding time series are not allowed to be lower than this value.
slotValueMax	Common data type "ScaledNumberType". See section 3.10.1.8.	The values within the timeSeriesSlots of the corresponding time series are not allowed to be higher than this value.
slotValueStepSize	Common data type "ScaledNumberType". See section 3.10.1.8.	The minimum step size between two different values within the timeSeriesSlots of the corresponding time series.

Table 346: timeSeriesConstraintsListData function detailed description of elements

5.3.29.4.3 Available selectors

- timeSeriesId

11342 **5.3.29.4.4 Examples**11343 **5.3.29.4.4.1 Read-reply**11344 To read `timeSeriesId "3"`, the following read command can be send:

```

11345 <function>timeSeriesConstraintsListData</function>
11346 <filter>
11347     <cmdControl>
11348         <partial/>
11349     </cmdControl>
11350     <timeSeriesConstraintsListDataSelectors>
11351         <timeSeriesId>3</timeSeriesId>
11352     </timeSeriesConstraintsListDataSelectors>
11353 </filter>

```

11354 A reply could look like this:

```

11355 <timeSeriesConstraintsListData>
11356     <timeSeriesConstraintsData>
11357         <timeSeriesId>3</timeSeriesId>
11358         <slotCountMin>1</slotCountMin>
11359         <slotCountMax>50</slotCountMax>
11360         <slotDurationMin>PT1M</slotDurationMin>
11361         <slotDurationMax>PT24H</slotDurationMax>
11362         <slotDurationStepSize>PT1S</slotDurationStepSize>
11363         <earliestTimeSeriesStartTime>2018-10-
11364 15T10:00:00.0Z</earliestTimeSeriesStartTime>
11365         <latestTimeSeriesEndTime>2018-10-
11366 15T18:00:00.0Z</latestTimeSeriesEndTime>
11367         <slotValueMin>
11368             <number>0</number>
11369         </slotValueMin>
11370         <slotValueMax>
11371             <number>99999</number>
11372         </slotValueMax>
11373         <slotValueStepSize>
11374             <number>1</number>
11375             <scale>-1</scale>
11376         </slotValueStepSize>
11377     </timeSeriesConstraintsData>
11378 </timeSeriesConstraintsListData>

```

11379

11380 **5.3.30 TimeTable**11381 **5.3.30.1 Introduction**

11382 All time dependent functionality like setpoint-configuration, etc. makes use of this elementary class.
 11383 It provides a list of time tables, each containing some number of time slots. Each slot has a start and
 11384 end time. The times can either be absolute (*dateTime*) or recurring (*daysOfWeek* plus *time* and, if
 11385 required, the recurring interval step elements). The count of slots and the start and end times can be
 11386 configurable or static (device specific). Other classes that make use of time tables only reference to
 11387 the *timeTableId*, not to specific time slots.

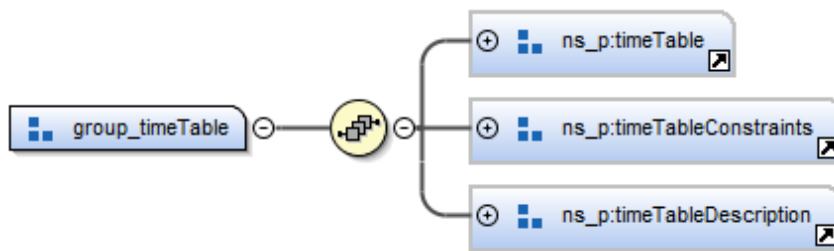


Figure 227: TimeTable function-group overview

5.3.30.2 timeTableListData

5.3.30.2.1 General

The list of time tables and time slots together with the start and end times are modelled in this function.

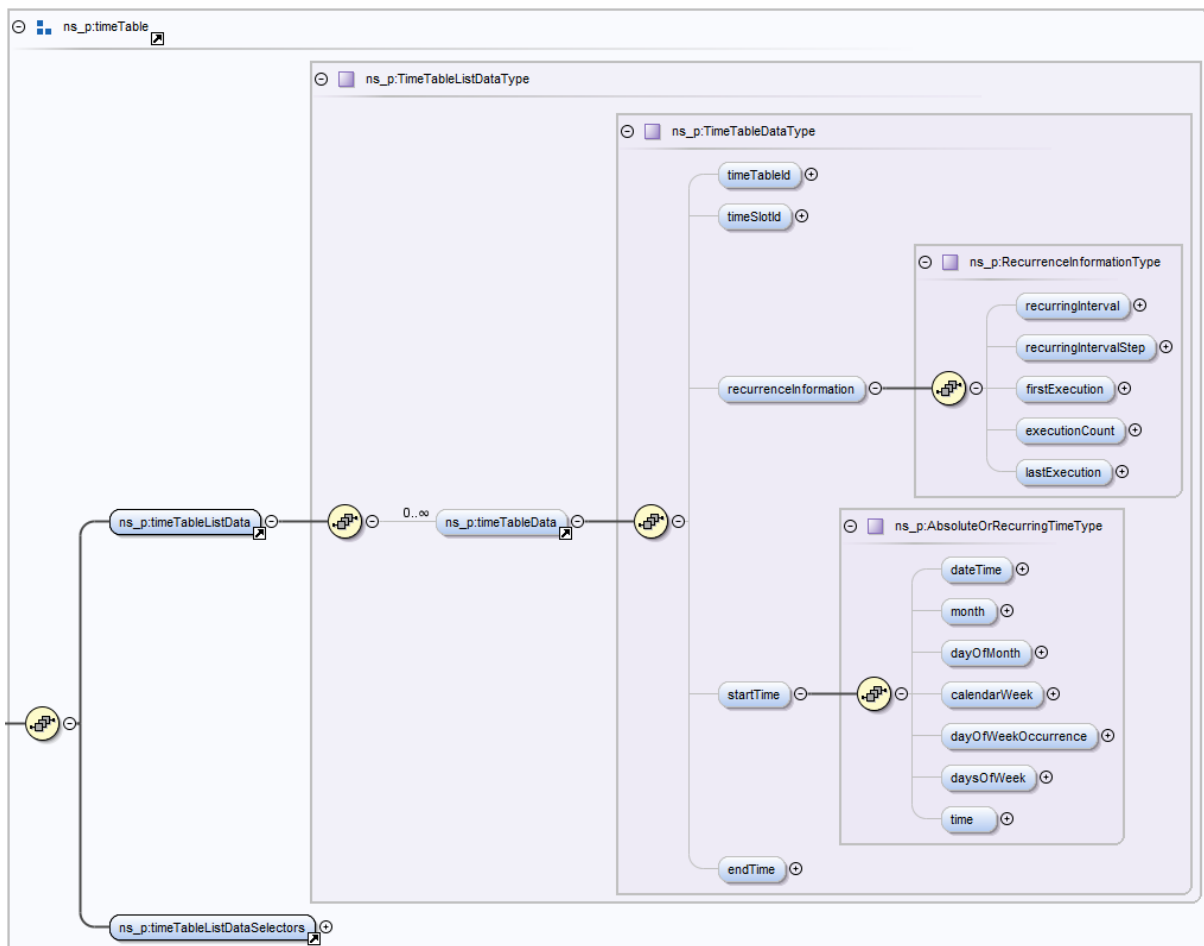


Figure 228: timeTableListData function overview

5.3.30.2.2 Detailed description of elements

Element	Type	Description
timeTableListData. timeTableData. (list)		

timeTableId	Identifier "TimeTableIdType" (see Table 358).	The <i>timeTableId</i> is used for identifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
timeSlotId	Identifier "TimeSlotIdType" (see Table 358).	Each slot of a time table has its own slot ID.
recurrenceInformation	Common data type "RecurrenceInformationType". See section 3.10.2.14.	Information about the recurrence of a slot.
startTime	Common data type "AbsoluteOrRecurringTimeType". See section 3.10.2.13.	Start time of one slot.
endTime	Common data type "AbsoluteOrRecurringTimeType". See section 3.10.2.13.	End time of one slot.

11399 Table 347: timeTableListData function detailed description of elements

11400

11401 5.3.30.2.3 Available selectors

- 11402 - timeTableId
- 11403 - timeSlotId

11404

11405 5.3.30.2.4 Examples

11406 5.3.30.2.4.1 Read-reply

11407 If one wants to request a specific time table ("2"), he could sent the following read command:

```

11408 <function>timeTableListData</function>
11409 <filter>
11410     <cmdControl>
11411         <partial/>
11412     </cmdControl>
11413     <timeTableListDataSelectors>
11414         <timeTableId>2</timeTableId>
11415     </timeTableListDataSelectors>
11416 </filter>
11417 <timeTableListData/>

```

11418 A response could be:

```

11419 <function>timeTableListData</function>
11420 <filter>
11421     <cmdControl>
11422         <partial/>
11423     </cmdControl>
11424 </filter>
11425 <timeTableListData>
11426     <timeTableData>
11427         <timeTableId>2</timeTableId>
11428         <timeSlotId>0</timeSlotId>
11429         <recurrenceInformation>
11430             <recurringInterval>weekly</recurringInterval>
11431         </recurrenceInformation>
11432         <startTime>

```

```

11433         <daysOfWeek>
11434             <monday/>
11435         </daysOfWeek>
11436         <time>12:00:00Z</time>
11437     </startTime>
11438     <endTime>
11439         <daysOfWeek>
11440             <monday/>
11441         </daysOfWeek>
11442         <time>14:00:00Z</time>
11443     </endTime>
11444 </timeTableData>
11445 <timeTableData>
11446     <timeTableId>2</timeTableId>
11447     <timeSlotId>1</timeSlotId>
11448     <recurrenceInformation>
11449         <recurringInterval>weekly</recurringInterval>
11450     </recurrenceInformation>
11451     <startTime>
11452         <daysOfWeek>
11453             <tuesday/>
11454         </daysOfWeek>
11455         <time>12:00:00Z</time>
11456     </startTime>
11457     <endTime>
11458         <daysOfWeek>
11459             <tuesday/>
11460         </daysOfWeek>
11461         <time>14:00:00Z</time>
11462     </endTime>
11463 </timeTableData>
11464 <timeTableData>
11465     <timeTableId>2</timeTableId>
11466     <timeSlotId>2</timeSlotId>
11467     <recurrenceInformation>
11468         <recurringInterval>weekly</recurringInterval>
11469     </recurrenceInformation>
11470     <startTime>
11471         <daysOfWeek>
11472             <friday/>
11473         </daysOfWeek>
11474         <time>11:00:00Z</time>
11475     </startTime>
11476     <endTime>
11477         <daysOfWeek>
11478             <friday/>
11479         </daysOfWeek>
11480         <time>17:00:00Z</time>
11481     </endTime>
11482 </timeTableData>
11483 </timeTableListData>

```

11484

11485 5.3.30.2.4.2 Write (partial)

11486 To write new times (if allowed) into slots, one could send a message with classifier *write* like this:

```

11487 <function>timeTableListData</function>
11488 <filter>
11489     <cmdControl>
11490         <partial/>
11491     </cmdControl>

```

```

11492 </filter>
11493 <timeTableListData>
11494   <timeTableData>
11495     <timeTableId>1</timeTableId>
11496     <timeSlotId>0</timeSlotId>
11497     <startTime>
11498       <daysOfWeek>
11499         <monday/>
11500         <tuesday/>
11501         <wednesday/>
11502         <thursday/>
11503         <friday/>
11504         <saturday/>
11505         <sunday/>
11506       </daysOfWeek>
11507       <time>10:00:00Z</time>
11508     </startTime>
11509     <endTime>
11510       <daysOfWeek>
11511         <monday/>
11512         <tuesday/>
11513         <wednesday/>
11514         <thursday/>
11515         <friday/>
11516         <saturday/>
11517         <sunday/>
11518       </daysOfWeek>
11519       <time>16:00:00Z</time>
11520     </endTime>
11521   </timeTableData>
11522 </timeTableListData>

```

11523

11524 **5.3.30.3 timeTableConstraintsListData**

11525 **5.3.30.3.1 General**

11526 If a time table has some constraints, they are denoted with the help of this function (e.g. the
 11527 minimum and maximum count of slots (may differ between the time tables)).

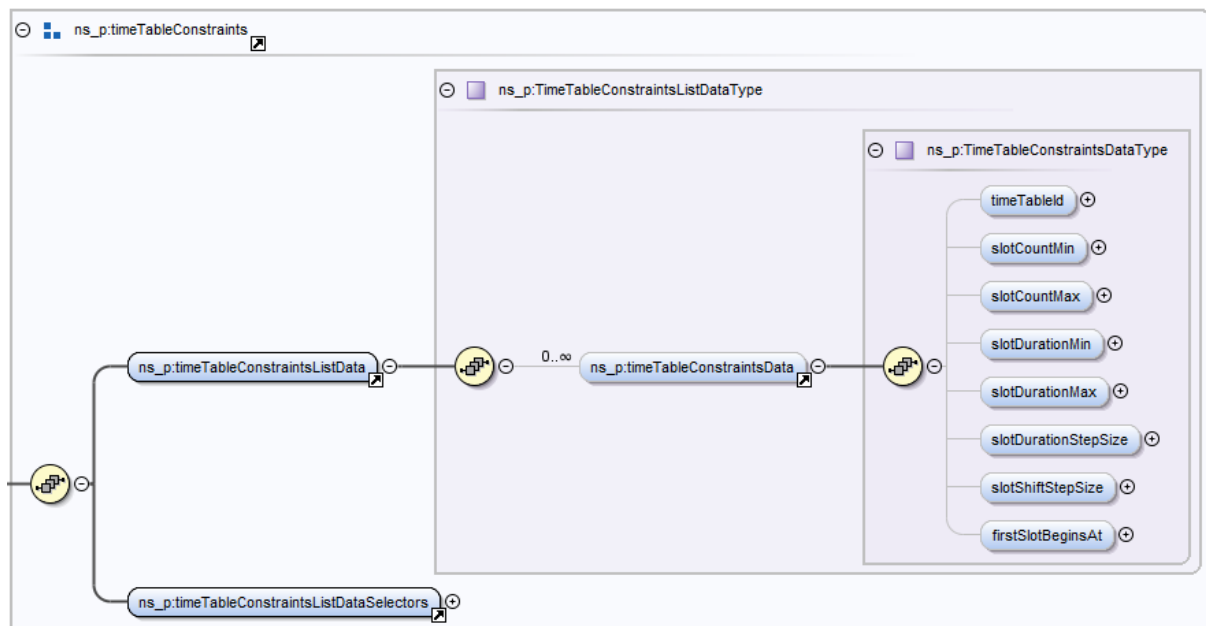


Figure 229: timeTableConstraintsListData function overview

5.3.30.3.2 Detailed description of elements

Element	Type	Description
timeTableConstraintsListData. timeTableConstraintsData. (list)		
timeTableId	Identifier "TimeTableIdType" (see Table 358).	The <i>timeTableId</i> is used for indentifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
slotCountMin	Simple type "TimeSlotCountType" (restriction of "TimeSlotIdType", see Table 347).	The minimum amount of slots for a specific time table is stated in this element. Recurring slots only count as one.
slotCountMax	Simple type "TimeSlotCountType" (restriction of "TimeSlotIdType", see Table 347).	The maximum amount of slots for a specific time table is stated in this element. Recurring slots only count as one.
slotDurationMin	xs:duration (W3C standard type)	If the slots of this time table have a minimum duration, it is denoted here.
slotDurationMax	xs:duration (W3C standard type)	If the slots of this time table have a maximum duration, it is denoted here.
slotDurationStepSize	xs:duration (W3C standard type)	Slots can have a specific duration step size. This element holds the corresponding value.
slotShiftStepSize	xs:duration (W3C standard type)	Slots can have a specific shift step size. This element holds the corresponding value.
firstSlotBeginsAt	xs:time (W3C standard type)	Especially time tables with fixed slot durations probably need this information to specify, where in time the slots are located (i.e. fixed 15 minute slots can start either at 00:00:00 (default) or e.g. at 00:10:00). This value only defines, when the time

		pattern (defined by the element <i>slotDurationStepSize</i>) begins on a day.
--	--	--

11532 *Table 348: timeTableConstraintsListData function detailed description of elements*

11533

11534 5.3.30.3.3 Available selectors

11535 - timeTableId

11536

11537 5.3.30.3.4 Examples

11538 5.3.30.3.4.1 Notify

11539 If some constraints data has changed, a device could send an update of its information with a *notify*
 11540 classifier to all bound devices on the corresponding feature (the "filter/partial" part to announce the
 11541 "restricted function exchange" is simplified by "..." just to improve the readability):

```

11542 ...
11543 <timeTableConstraintsListData>
11544   <timeTableConstraintsData>
11545     <timeTableId>3</timeTableId>
11546     <slotCountMin>1</slotCountMin>
11547     <slotCountMax>10</slotCountMax>
11548     <slotDurationMin>PT10M</slotDurationMin>
11549     <slotDurationMax>PT20M</slotDurationMax>
11550     <slotDurationStepSize>PT5M</slotDurationStepSize>
11551     <slotShiftStepSize>PT5M</slotShiftStepSize>
11552     <firstSlotBeginsAt>00:00:00</firstSlotBeginsAt>
11553   </timeTableConstraintsData>
11554 </timeTableConstraintsListData>

```

11555

11556 5.3.30.4 timeTableDescriptionListData

11557 5.3.30.4.1 General

11558 The more static information about the time tables are described in the *timeTableDescriptionData*
 11559 function.

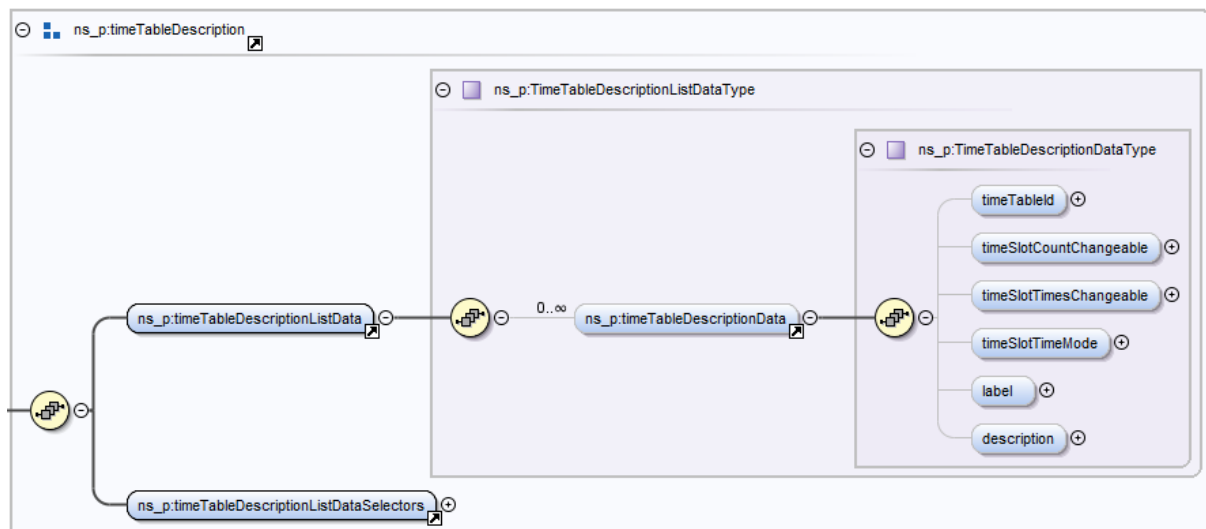


Figure 230: timeTableDescriptionListData function overview

5.3.30.4.2 Detailed description of elements

Element	Type	Description
timeTableDescriptionListData. timeTableDescriptionData. (list)		
timeTableId	Identifier "TimeTableIdType" (see Table 358).	The <i>timeTableId</i> is used for indentifying specific table information in the different functions and for relations in other classes like <i>Setpoint</i> .
timeSlotCountChangeable	xs:boolean (W3C standard type)	States whether the amount of slots is changeable by a client for a specific time table or not.
timeSlotTimesChangeable	xs:boolean (W3C standard type)	States whether the times of the slots of a specific time table are changeable by a client or not.
timeSlotTimeMode	Union "TimeSlotTimeModeType": - Enum (see Table 350): TimeSlotTimeModeEnumType - EnumExtendType (see section 3.10.1.5)	If either <i>absolute</i> or <i>recurring</i> time slots or <i>both</i> modes are supported is stated in this element.
label	Common data type "LabelType". See section 3.10.1.2.	A short label of the time table.
description	Common data type "DescriptionType". See section 3.10.1.3.	Descriptive information on the time table.

Table 349: timeTableDescriptionListData function detailed description of elements

Enumeration TimeSlotTimeModeEnumType:

Value	Description
absolute	Only absolute times are allowed for the time table.
recurring	Only recurring times are allowed for the time table.

both	Absolute and recurring times are allowed for the time table.
------	--

Table 350: Enumeration TimeSlotTimeModeEnumType

5.3.30.4.3 Available selectors

- timeTableId

5.3.30.4.4 Examples

5.3.30.4.4.1 Read-reply

If one is interested in the time table description of all available time tables, he would send the standard read command on the *timeTableDescriptionListData* resource. A response could look like this:

```
<timeTableDescriptionListData>
  <timeTableDescriptionData>
    <timeTableId>1</timeTableId>
    <timeSlotCountChangeable>false</timeSlotCountChangeable>
    <timeSlotTimesChangeable>true</timeSlotTimesChangeable>
    <timeSlotTimeMode>absolute</timeSlotTimeMode>
  </timeTableDescriptionData>
  <timeTableDescriptionData>
    <timeTableId>2</timeTableId>
    <timeSlotCountChangeable>false</timeSlotCountChangeable>
    <timeSlotTimesChangeable>true</timeSlotTimesChangeable>
    <timeSlotTimeMode>both</timeSlotTimeMode>
  </timeTableDescriptionData>
</timeTableDescriptionListData>
```

5.3.31 UseCaseInformation

5.3.31.1 Introduction

To model which Use Cases are supported by a node the UseCaseInformation Class can be used.



Figure 231: UseCaseInformation function-group overview

5.3.31.2 useCaseInformationListData

5.3.31.2.1 General

This function provides information about the supported Use Cases together with a version information and further relevant details.

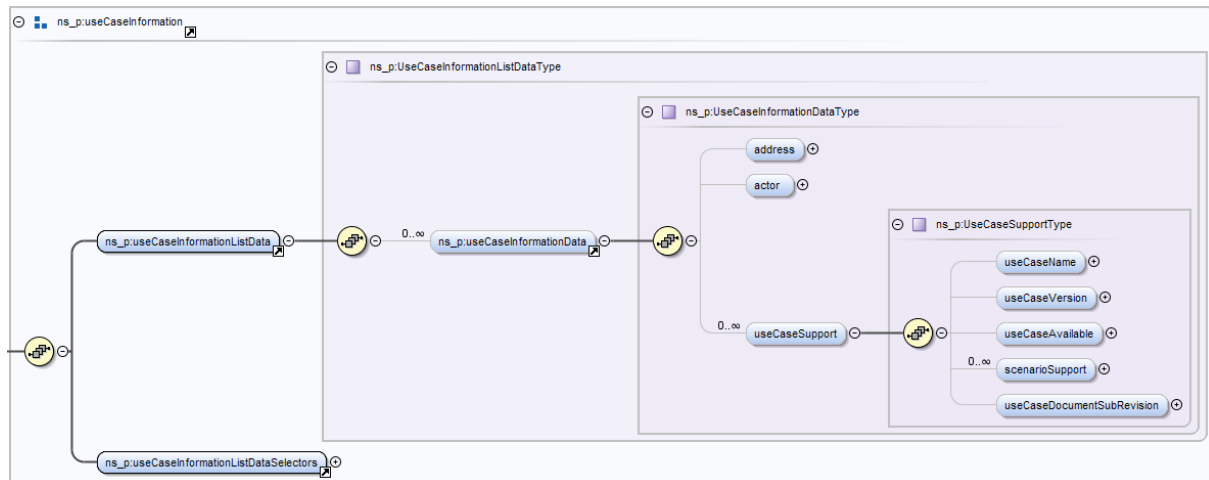


Figure 232: useCaseInformationListData function overview

5.3.31.2.2 Detailed description of elements

Element	Type	Description
useCaseInformationListData. useCaseInformationData. (list)		
address	Common data type "FeatureAddressType". See section 3.10.3.6.	The address where the supported Use Case is located. Note: not all sub-elements always need to be set here!
actor	Union "UseCaseActorType": - Simple type "UseCaseActorEnumType" (restriction of xs:string) - EnumExtendType (see section 3.10.1.5)	The actor of the supported Use Case located on the above stated address.
useCaseSupport (list)		List of supported Use Cases.
useCaseSupport. useCaseName	Union "UseCaseNameType": - Simple type "UseCaseNameEnumType" (restriction of xs:string) - EnumExtendType (see section 3.10.1.5)	The name of the supported Use Case.
useCaseSupport. useCaseVersion	Common data type "SpecificationVersionType". See section 3.10.1.4.	Version of the supported Use Case.
useCaseSupport. useCaseAvailable	xs:boolean (W3C standard type)	States whether the Use Case is available at the moment.
useCaseSupport. scenarioSupport (list)	Simple type "UseCaseScenarioSupportType" (restriction of "xs:unsignedInt")	List of supported scenarios of the Use Case.
useCaseSupport. useCaseDocumentSub Revision	xs:string (W3C standard type)	The element "useCaseDocumentSubRevision" serves as extension to "useCaseVersion". This extension contains the release state of an underlying Use Case specification, e.g. "draft1...". This permits

		identifying implementations that are based upon a pre-release.
--	--	--

Table 351: useCaseInformationListData function detailed description of elements

5.3.31.2.3 Available selectors

- address (with sub-elements, see above)
- actor
- useCaseSupport (with sub-elements, see above)

5.3.31.2.4 Examples

5.3.31.2.4.1 Read-reply

A useCaseInformationListData instance sent as reply to a standard read command:

```

<useCaseInformationListData>
  <useCaseInformationData>
    <address>
      <device>d: i:46925_TestDevice_1</device>
      <entity>1</entity>
      <entity>1</entity>
    </address>
    <actor>Compressor</actor>
    <useCaseSupport>
      <useCaseName>visualizationOfPowerConsumptionOfHeatPumpCompressor</useCaseName>
      <useCaseVersion>1.0.0</useCaseVersion>
      <useCaseAvailable>true</useCaseAvailable>
      <scenarioSupport>1</scenarioSupport>
    </useCaseSupport>
    <useCaseDocumentSubRevision>draft1_976</useCaseDocumentSubRevision>
  </useCaseInformationData>
  <useCaseInformationData>
    <address>
      <device>d: i:46925_TestDevice_1</device>
      <entity>2</entity>
      <feature>1</feature>
    </address>
    <actor>GridConnectionPointOfPremises</actor>
    <useCaseSupport>
      <useCaseName>optimizationForOptionalHeatPumpPowerConsumption</useCaseName>
      <useCaseVersion>1.0.0</useCaseVersion>
      <useCaseAvailable>false</useCaseAvailable>
      <scenarioSupport>1</scenarioSupport>
    </useCaseSupport>
    <useCaseDocumentSubRevision>RC2_7207_20210518</useCaseDocumentSubRevision>
  </useCaseInformationData>
  <useCaseInformationData>
    <address>
      <device>d: i:46925_TestDevice_1</device>
      <entity>2</entity>
      <feature>1</feature>
    </address>
    <actor>GridConnectionPointOfPremises</actor>
    <useCaseSupport>
      <useCaseName>remoteMonitoringOfElectricityGridConnectionPoint</useCaseName>
      <useCaseVersion>1.0.0</useCaseVersion>
      <useCaseAvailable>true</useCaseAvailable>

```

```

11655         <scenarioSupport>1</scenarioSupport>
11656         <scenarioSupport>2</scenarioSupport>
11657         <scenarioSupport>3</scenarioSupport>
11658         <scenarioSupport>4</scenarioSupport>
11659
11660 <useCaseDocumentSubRevision>RC7_7219_20210519</useCaseDocumentSubRevision>
11661     </useCaseSupport>
11662 </useCaseInformationData>
11663 </useCaseInformationListData>

```

11664

11665 5.3.32 Version

11666 5.3.32.1 Introduction

11667 This class can be used to model the version information of the SPINE specification. The version string
 11668 is divided into *major*, *minor* and *revision*, separated by a dot (e.g. "1.0.0"), see section 5.3.32.2.1.
 11669 The version information is also used within the "header" of "datagram" (see document
 11670 [ProtocolSpecification]). This section, however, provides the definition and proper usage of "Version"
 11671 as SPINE class.



11672

11673 Figure 233: Version function-group overview

11674

11675 5.3.32.2 specificationVersionListData

11676 5.3.32.2.1 General

11677 The knowledge of the supported SPINE data model versions of a communication partner is essential
 11678 for the proper interworking of devices that implemented different versions. Considering the
 11679 compatibility rules defined in [ProtocolSpecification] is only possible with this kind of knowledge.
 11680 Therefore, the *specificationVersionListData* function is used.

11681 The three parts of the version have the following meaning:

11682 **major**

11683 Typically, a specification's major part of the version number is increased by the proper authority in
 11684 case of "fundamental changes". Among others, changes that break mandatory compatibility
 11685 requirements belong to such kind of changes.

11686 **minor**

11687 Typically, a specification's minor part of the version number is increased by the proper authority in
 11688 case of optional additions to the predecessor of the specification version.

11689 **revision**

11690 Typically, a specification's revision part of the version number is increased by the proper authority in
 11691 case of "minor changes" where major and minor part can remain unchanged.

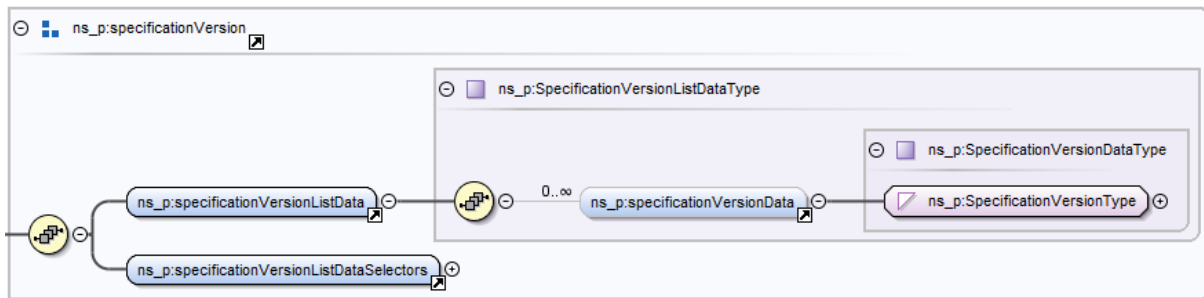


Figure 234: *specificationVersionListData* function overview

5.3.32.2.2 Detailed description of elements

Element	Type	Description
specificationVersionListData.		
specificationVersionData (list)	Common data type "SpecificationVersionType". See section 3.10.1.4.	One of the supported SPINE specification versions.

Table 352: *specificationVersionListData* function detailed description of elements

5.3.32.2.3 Available selectors

None.

5.3.32.2.4 Examples

5.3.32.2.4.1 Notify

A device sends the *specificationVersionListData* function (as (complete) notification) to another device, including all supported versions of the SPINE specification:

```
<specificationVersionListData>
  <specificationVersionData>1.0.0</specificationVersionData>
  <specificationVersionData>1.3.0</specificationVersionData>
</specificationVersionListData>
```

Annex A - SPINE XSDs

All necessary XSDs defined for SPINE are included in this annex. The so-called "overview" XSDs are left out as they only reference to the functions defined within the "normal" XSDs. An example for an overview XSD is given in section A.3.6 Overview example.

The EEBus Initiative e.V. also provides the XSD files for download (please consider the corresponding website [EEBus]). These XSD files should be preferred as the the printed XSDs in this annex contain additional line breaks due to page layout reasons.

A.1 Complex Classes

A.1.1 IncentiveTable

File name: EEBus_SPINE_TS_IncentiveTable.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
    Smart Premises Interoperable Neutral-Message Exchange (SPINE)
    Version 1.3.0 (final)
    2023-08-31
    Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
    Source: https://www.eebus.org/en/specifications/
-->
<xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
  blockDefault="#all" elementFormDefault="qualified">
  <xs:annotation>
    <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
    e.V. All rights reserved.</xs:documentation>
  </xs:annotation>
  <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
  <xs:include schemaLocation="EEBus_SPINE_TS_TariffInformation.xsd"/>
  <xs:complexType name="IncentiveTableType">
    <xs:sequence>
      <xs:element minOccurs="0" name="tariff">
        <xs:complexType>
          <xs:complexContent>
            <xs:restriction base="ns_p:TariffDataType">
              <xs:sequence>
                <xs:element minOccurs="0" name="tariffId"
type="ns_p:TariffIdType"/>
              </xs:sequence>
            </xs:restriction>
          </xs:complexContent>
        </xs:complexType>
      </xs:element>
      <xs:element maxOccurs="unbounded" minOccurs="0" name="incentiveSlot"
type="ns_p:IncentiveTableIncentiveSlotType"> </xs:element>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="IncentiveTableIncentiveSlotType">
    <xs:sequence>
      <xs:element minOccurs="0" name="timeInterval">
        <xs:complexType>
          <xs:complexContent>
            <xs:restriction base="ns_p:TimeTableDataType">
              <xs:sequence>
                <xs:element minOccurs="0" name="timeSlotId"
type="ns_p:TimeSlotIdType"/>
                <xs:element minOccurs="0" name="startTime">
                  <xs:complexType>
                    <xs:complexContent>
                      <xs:restriction
base="ns_p:AbsoluteOrRecurringTimeType">
                        <xs:sequence>
```

```
11773                                     <xs:element minOccurs="0" name="dateTime"
11774 type="xs:dateTime"/>
11775                                     <xs:element minOccurs="0" name="relative"
11776 type="xs:duration"/>
11777                                     </xs:sequence>
11778                                     </xs:restriction>
11779                                     </xs:complexContent>
11780                                     </xs:complexType>
11781                                     </xs:element>
11782                                     <xs:element minOccurs="0" name="endTime">
11783                                     <xs:complexType>
11784                                     <xs:complexContent>
11785                                     <xs:restriction
11786 base="ns_p:AbsoluteOrRecurringTimeType">
11787                                     <xs:sequence>
11788                                     <xs:element minOccurs="0" name="dateTime"
11789 type="xs:dateTime"/>
11790                                     <xs:element minOccurs="0" name="relative"
11791 type="xs:duration"/>
11792                                     </xs:sequence>
11793                                     </xs:restriction>
11794                                     </xs:complexContent>
11795                                     </xs:complexType>
11796                                     </xs:element>
11797                                     </xs:sequence>
11798                                     </xs:restriction>
11799                                     </xs:complexContent>
11800                                     </xs:complexType>
11801                                     </xs:element>
11802                                     <xs:element maxOccurs="unbounded" minOccurs="0" name="tier"
11803 type="ns_p:IncentiveTableTierType"> </xs:element>
11804                                     </xs:sequence>
11805                                     </xs:complexType>
11806                                     <xs:complexType name="IncentiveTableTierType">
11807                                     <xs:sequence>
11808                                     <xs:element minOccurs="0" name="tier">
11809                                     <xs:complexType>
11810                                     <xs:complexContent>
11811                                     <xs:restriction base="ns_p:TierDataType">
11812                                     <xs:sequence>
11813                                     <xs:element minOccurs="0" name="tierId"
11814 type="ns_p:TierIdType"/>
11815                                     </xs:sequence>
11816                                     </xs:restriction>
11817                                     </xs:complexContent>
11818                                     </xs:complexType>
11819                                     </xs:element>
11820                                     <xs:element maxOccurs="unbounded" minOccurs="0" name="boundary">
11821                                     <xs:complexType>
11822                                     <xs:complexContent>
11823                                     <xs:restriction base="ns_p:TierBoundaryDataType">
11824                                     <xs:sequence>
11825                                     <xs:element minOccurs="0" name="boundaryId"
11826 type="ns_p:TierBoundaryIdType"/>
11827                                     <xs:element minOccurs="0" name="lowerBoundaryValue"
11828 type="ns_p:ScaledNumberType"/>
11829                                     <xs:element minOccurs="0" name="upperBoundaryValue"
11830 type="ns_p:ScaledNumberType"/>
11831                                     </xs:sequence>
11832                                     </xs:restriction>
11833                                     </xs:complexContent>
11834                                     </xs:complexType>
11835                                     </xs:element>
11836                                     <xs:element minOccurs="0" name="incentive" maxOccurs="unbounded">
11837                                     <xs:complexType>
11838                                     <xs:complexContent>
11839                                     <xs:restriction base="ns_p:IncentiveDataType">
11840                                     <xs:sequence>
11841                                     <xs:element minOccurs="0" name="incentiveId"
11842 type="ns_p:IncentiveIdType"/>
11843                                     <xs:element minOccurs="0" name="value"
11844 type="ns_p:ScaledNumberType"/>
11845                                     </xs:sequence>
11846                                     </xs:restriction>
11847                                     </xs:complexContent>
11848                                     </xs:complexType>
11849                                     </xs:element>
11850                                     </xs:sequence>
```

```
11851     </xs:complexType>
11852     <xs:complexType name="IncentiveTableDataType">
11853       <xs:sequence>
11854         <xs:element maxOccurs="unbounded" minOccurs="0" name="incentiveTable"
11855 type="ns_p:IncentiveTableType"> </xs:element>
11856       </xs:sequence>
11857     </xs:complexType>
11858     <xs:element name="incentiveTableData" type="ns_p:IncentiveTableDataType"/>
11859     <xs:complexType name="IncentiveTableElementsType">
11860       <xs:sequence>
11861         <xs:element minOccurs="0" name="tariff">
11862           <xs:complexType>
11863             <xs:complexContent>
11864               <xs:restriction base="ns_p:TariffDataElementsType">
11865                 <xs:sequence>
11866                   <xs:element minOccurs="0" name="tariffId"
11867 type="ns_p:ElementTagType"/>
11868                 </xs:sequence>
11869               </xs:restriction>
11870             </xs:complexContent>
11871           </xs:complexType>
11872         </xs:element>
11873         <xs:element minOccurs="0" name="incentiveSlot"
11874 type="ns_p:IncentiveTableIncentiveSlotElementsType"> </xs:element>
11875       </xs:sequence>
11876     </xs:complexType>
11877     <xs:complexType name="IncentiveTableIncentiveSlotElementsType">
11878       <xs:sequence>
11879         <xs:element minOccurs="0" name="timeInterval">
11880           <xs:complexType>
11881             <xs:complexContent>
11882               <xs:restriction base="ns_p:TimeTableDataElementsType">
11883                 <xs:sequence>
11884                   <xs:element minOccurs="0" name="timeSlotId"
11885 type="ns_p:ElementTagType"/>
11886                   <xs:element minOccurs="0" name="startTime">
11887                     <xs:complexType>
11888                       <xs:complexContent>
11889                         <xs:restriction
11890 base="ns_p:AbsoluteOrRecurringTimeElementsType">
11891                           <xs:sequence>
11892                             <xs:element minOccurs="0" name="dateTime"
11893 type="ns_p:ElementTagType"/>
11894                             <xs:element minOccurs="0" name="relative"
11895 type="ns_p:ElementTagType"/>
11896                           </xs:sequence>
11897                         </xs:restriction>
11898                       </xs:complexContent>
11899                     </xs:complexType>
11900                   </xs:element>
11901                   <xs:element minOccurs="0" name="endTime">
11902                     <xs:complexType>
11903                       <xs:complexContent>
11904                         <xs:restriction
11905 base="ns_p:AbsoluteOrRecurringTimeElementsType">
11906                           <xs:sequence>
11907                             <xs:element minOccurs="0" name="dateTime"
11908 type="ns_p:ElementTagType"/>
11909                             <xs:element minOccurs="0" name="relative"
11910 type="ns_p:ElementTagType"/>
11911                           </xs:sequence>
11912                         </xs:restriction>
11913                       </xs:complexContent>
11914                     </xs:complexType>
11915                   </xs:element>
11916                 </xs:sequence>
11917               </xs:restriction>
11918             </xs:complexContent>
11919           </xs:complexType>
11920         </xs:element>
11921         <xs:element minOccurs="0" name="tier" type="ns_p:IncentiveTableTierElementsType">
11922 </xs:element>
11923       </xs:sequence>
11924     </xs:complexType>
11925     <xs:complexType name="IncentiveTableTierElementsType">
11926       <xs:sequence>
11927         <xs:element minOccurs="0" name="tier">
11928           <xs:complexType>
```

```
11929         <xs:complexContent>
11930             <xs:restriction base="ns_p:TierDataElementsType">
11931                 <xs:sequence>
11932                     <xs:element minOccurs="0" name="tierId"
type="ns_p:ElementTagType"/>
11933                 </xs:sequence>
11934             </xs:restriction>
11935         </xs:complexContent>
11936     </xs:complexType>
11937 </xs:element>
11938 <xs:element minOccurs="0" name="boundary">
11939     <xs:complexType>
11940         <xs:complexContent>
11941             <xs:restriction base="ns_p:TierBoundaryDataElementsType">
11942                 <xs:sequence>
11943                     <xs:element minOccurs="0" name="boundaryId"
type="ns_p:ElementTagType"/>
11944                     <xs:element minOccurs="0" name="lowerBoundaryValue"
type="ns_p:ScaledNumberElementsType"/>
11945                     <xs:element minOccurs="0" name="upperBoundaryValue"
type="ns_p:ScaledNumberElementsType"/>
11946                 </xs:sequence>
11947             </xs:restriction>
11948         </xs:complexContent>
11949     </xs:complexType>
11950 </xs:element>
11951 <xs:element minOccurs="0" name="incentive">
11952     <xs:complexType>
11953         <xs:complexContent>
11954             <xs:restriction base="ns_p:IncentiveDataElementsType">
11955                 <xs:sequence>
11956                     <xs:element minOccurs="0" name="incentiveId"
type="ns_p:ElementTagType"/>
11957                     <xs:element minOccurs="0" name="value"
type="ns_p:ElementTagType"/>
11958                 </xs:sequence>
11959             </xs:restriction>
11960         </xs:complexContent>
11961     </xs:complexType>
11962 </xs:element>
11963 <xs:complexType name="IncentiveTableDataElementsType">
11964     <xs:sequence>
11965         <xs:element minOccurs="0" name="incentiveTable"
type="ns_p:IncentiveTableElementsType"/> </xs:element>
11966     </xs:sequence>
11967 </xs:complexType>
11968 <xs:element name="incentiveTableDataElements" type="ns_p:IncentiveTableDataElementsType"/>
11969 <xs:complexType name="IncentiveTableDataSelectorsType">
11970     <xs:sequence>
11971         <xs:element minOccurs="0" name="tariff">
11972             <xs:complexType>
11973                 <xs:complexContent>
11974                     <xs:restriction base="ns_p:TariffListDataSelectorsType">
11975                         <xs:sequence>
11976                             <xs:element minOccurs="0" name="tariffId"
type="ns_p:TariffIdType"/>
11977                         </xs:sequence>
11978                     </xs:restriction>
11979                 </xs:complexContent>
11980             </xs:complexType>
11981         </xs:element>
11982     </xs:sequence>
11983 </xs:complexType>
11984 <xs:element name="incentiveTableDataSelectors"
type="ns_p:IncentiveTableDataSelectorsType"/>
11985 <xs:complexType name="IncentiveTableDescriptionType">
11986     <xs:sequence>
11987         <xs:element minOccurs="0" name="tariffDescription">
11988             <xs:complexType>
11989                 <xs:complexContent>
11990                     <xs:restriction base="ns_p:TariffDescriptionDataType">
11991                         <xs:sequence>
11992                             <xs:element minOccurs="0" name="tariffId"
type="ns_p:TariffIdType"/>
11993                             <xs:element minOccurs="0" name="measurementId"
type="ns_p:MeasurementIdType"/>
11994                         </xs:sequence>
11995                     </xs:restriction>
11996                 </xs:complexContent>
11997             </xs:complexType>
11998         </xs:element>
11999     </xs:sequence>
12000 </xs:complexType>
```



```

12007                                     <xs:element minOccurs="0" name="tariffWriteable"
12008 type="xs:boolean"/>
12009                                     <xs:element minOccurs="0" name="updateRequired"
12010 type="xs:boolean"/>
12011                                     <xs:element minOccurs="0" name="scopeType"
12012 type="ns_p:ScopeTypeType"/>
12013                                     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12014                                     <xs:element minOccurs="0" name="description"
12015 type="ns_p:DescriptionType"/>
12016                                     <xs:element minOccurs="0" name="slotIdSupport"
12017 type="xs:boolean"/>
12018                                     </xs:sequence>
12019                                 </xs:restriction>
12020                             </xs:complexContent>
12021                         </xs:complexType>
12022                     </xs:element>
12023                     <xs:element maxOccurs="unbounded" minOccurs="0" name="tier"
12024 type="ns_p:IncentiveTableDescriptionTierType"> </xs:element>
12025                 </xs:sequence>
12026             </xs:complexType>
12027             <xs:complexType name="IncentiveTableDescriptionTierType">
12028                 <xs:sequence>
12029                     <xs:element minOccurs="0" name="tierDescription">
12030                         <xs:complexType>
12031                             <xs:complexContent>
12032                                 <xs:restriction base="ns_p:TierDescriptionDataType">
12033                                     <xs:sequence>
12034                                         <xs:element minOccurs="0" name="tierId"
12035 type="ns_p:TierIdType"/>
12036                                         <xs:element minOccurs="0" name="tierType"
12037 type="ns_p:TierTypeType"/>
12038                                         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12039                                         <xs:element minOccurs="0" name="description"
12040 type="ns_p:DescriptionType"/>
12041                                     </xs:sequence>
12042                                 </xs:restriction>
12043                             </xs:complexContent>
12044                         </xs:complexType>
12045                     </xs:element>
12046                     <xs:element maxOccurs="unbounded" minOccurs="0" name="boundaryDescription">
12047                         <xs:complexType>
12048                             <xs:complexContent>
12049                                 <xs:restriction base="ns_p:TierBoundaryDescriptionDataType">
12050                                     <xs:sequence>
12051                                         <xs:element minOccurs="0" name="boundaryId"
12052 type="ns_p:TierBoundaryIdType"/>
12053                                         <xs:element minOccurs="0" name="boundaryType"
12054 type="ns_p:TierBoundaryTypeType"/>
12055                                         <xs:element minOccurs="0" name="switchToTierIdWhenLower"
12056 type="ns_p:TierIdType"/>
12057                                         <xs:element minOccurs="0" name="switchToTierIdWhenHigher"
12058 type="ns_p:TierIdType"/>
12059                                         <xs:element minOccurs="0" name="boundaryUnit"
12060 type="ns_p:UnitOfMeasurementType"/>
12061                                     </xs:sequence>
12062                                 </xs:restriction>
12063                             </xs:complexContent>
12064                         </xs:complexType>
12065                     </xs:element>
12066                     <xs:element minOccurs="0" name="incentiveDescription" maxOccurs="unbounded">
12067                         <xs:complexType>
12068                             <xs:complexContent>
12069                                 <xs:restriction base="ns_p:IncentiveDescriptionDataType">
12070                                     <xs:sequence>
12071                                         <xs:element minOccurs="0" name="incentiveId"
12072 type="ns_p:IncentiveIdType"/>
12073                                         <xs:element minOccurs="0" name="incentiveType"
12074 type="ns_p:IncentiveTypeType"/>
12075                                         <xs:element minOccurs="0" name="incentivePriority"
12076 type="ns_p:IncentivePriorityType"/>
12077                                         <xs:element minOccurs="0" name="currency"
12078 type="ns_p:CurrencyType"/>
12079                                         <xs:element minOccurs="0" name="unit"
12080 type="ns_p:UnitOfMeasurementType"/>
12081                                         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12082                                         <xs:element minOccurs="0" name="description"
12083 type="ns_p:DescriptionType"/>
12084                                     </xs:sequence>

```

```
12085         </xs:restriction>
12086     </xs:complexContent>
12087 </xs:complexType>
12088 </xs:element>
12089 </xs:sequence>
12090 </xs:complexType>
12091 <xs:complexType name="IncentiveTableDescriptionDataType">
12092     <xs:sequence>
12093         <xs:element maxOccurs="unbounded" minOccurs="0" name="incentiveTableDescription"
12094 type="ns_p:IncentiveTableDescriptionType"> </xs:element>
12095     </xs:sequence>
12096 </xs:complexType>
12097 <xs:element name="incentiveTableDescriptionData"
12098 type="ns_p:IncentiveTableDescriptionDataType"/>
12099 <xs:complexType name="IncentiveTableDescriptionElementsType">
12100     <xs:sequence>
12101         <xs:element minOccurs="0" name="tariffDescription">
12102             <xs:complexType>
12103                 <xs:complexContent>
12104                     <xs:restriction base="ns_p:TariffDescriptionDataElementsType">
12105                         <xs:sequence>
12106                             <xs:element minOccurs="0" name="tariffId"
12107 type="ns_p:ElementTagType"/>
12108                             <xs:element minOccurs="0" name="measurementId"
12109 type="ns_p:ElementTagType"/>
12110                             <xs:element minOccurs="0" name="tariffWriteable"
12111 type="ns_p:ElementTagType"/>
12112                             <xs:element minOccurs="0" name="updateRequired"
12113 type="ns_p:ElementTagType"/>
12114                             <xs:element minOccurs="0" name="scopeType"
12115 type="ns_p:ElementTagType"/>
12116                             <xs:element minOccurs="0" name="label"
12117 type="ns_p:ElementTagType"/>
12118                             <xs:element minOccurs="0" name="description"
12119 type="ns_p:ElementTagType"/>
12120                             <xs:element minOccurs="0" name="slotIdSupport"
12121 type="ns_p:ElementTagType"/>
12122                         </xs:sequence>
12123                     </xs:restriction>
12124                 </xs:complexContent>
12125             </xs:complexType>
12126         </xs:element>
12127         <xs:element minOccurs="0" name="tier"
12128 type="ns_p:IncentiveTableDescriptionTierElementsType"> </xs:element>
12129     </xs:sequence>
12130 </xs:complexType>
12131 <xs:complexType name="IncentiveTableDescriptionTierElementsType">
12132     <xs:sequence>
12133         <xs:element minOccurs="0" name="tierDescription">
12134             <xs:complexType>
12135                 <xs:complexContent>
12136                     <xs:restriction base="ns_p:TierDescriptionDataElementsType">
12137                         <xs:sequence>
12138                             <xs:element minOccurs="0" name="tierId"
12139 type="ns_p:ElementTagType"/>
12140                             <xs:element minOccurs="0" name="tierType"
12141 type="ns_p:ElementTagType"/>
12142                             <xs:element minOccurs="0" name="label"
12143 type="ns_p:ElementTagType"/>
12144                             <xs:element minOccurs="0" name="description"
12145 type="ns_p:ElementTagType"/>
12146                         </xs:sequence>
12147                     </xs:restriction>
12148                 </xs:complexContent>
12149             </xs:complexType>
12150         </xs:element>
12151         <xs:element minOccurs="0" name="boundaryDescription">
12152             <xs:complexType>
12153                 <xs:complexContent>
12154                     <xs:restriction base="ns_p:TierBoundaryDescriptionDataElementsType">
12155                         <xs:sequence>
12156                             <xs:element minOccurs="0" name="boundaryId"
12157 type="ns_p:ElementTagType"/>
12158                             <xs:element minOccurs="0" name="boundaryType"
12159 type="ns_p:ElementTagType"/>
12160                             <xs:element minOccurs="0" name="switchToTierIdWhenLower"
12161 type="ns_p:ElementTagType"/>

```

```
12162         <xs:element minOccurs="0" name="switchToTierIdWhenHigher"
12163 type="ns_p:ElementTagType"/>
12164         <xs:element minOccurs="0" name="boundaryUnit"
12165 type="ns_p:ElementTagType"/>
12166     </xs:sequence>
12167 </xs:restriction>
12168 </xs:complexContent>
12169 </xs:complexType>
12170 </xs:element>
12171 <xs:element minOccurs="0" name="incentiveDescription">
12172     <xs:complexType>
12173         <xs:complexContent>
12174             <xs:restriction base="ns_p:IncentiveDescriptionDataElementsType">
12175                 <xs:sequence>
12176                     <xs:element minOccurs="0" name="incentiveId"
12177 type="ns_p:ElementTagType"/>
12178                     <xs:element minOccurs="0" name="incentiveType"
12179 type="ns_p:ElementTagType"/>
12180                     <xs:element minOccurs="0" name="incentivePriority"
12181 type="ns_p:ElementTagType"/>
12182                     <xs:element minOccurs="0" name="currency"
12183 type="ns_p:ElementTagType"/>
12184                     <xs:element minOccurs="0" name="unit"
12185 type="ns_p:ElementTagType"/>
12186                     <xs:element minOccurs="0" name="label"
12187 type="ns_p:ElementTagType"/>
12188                     <xs:element minOccurs="0" name="description"
12189 type="ns_p:ElementTagType"/>
12190                 </xs:sequence>
12191             </xs:restriction>
12192         </xs:complexContent>
12193     </xs:complexType>
12194 </xs:element>
12195 </xs:sequence>
12196 </xs:complexType>
12197 <xs:complexType name="IncentiveTableDescriptionDataElementsType">
12198     <xs:sequence>
12199         <xs:element minOccurs="0" name="incentiveTableDescription"
12200 type="ns_p:IncentiveTableDescriptionElementsType"> </xs:element>
12201     </xs:sequence>
12202 </xs:complexType>
12203 <xs:element name="incentiveTableDescriptionDataElements"
12204 type="ns_p:IncentiveTableDescriptionDataElementsType"/>
12205 <xs:complexType name="IncentiveTableDescriptionDataSelectorsType">
12206     <xs:sequence>
12207         <xs:element minOccurs="0" name="tariffDescription">
12208             <xs:complexType>
12209                 <xs:complexContent>
12210                     <xs:restriction base="ns_p:TariffDescriptionListDataSelectorsType">
12211                         <xs:sequence>
12212                             <xs:element minOccurs="0" name="tariffId"
12213 type="ns_p:TariffIdType"/>
12214                             <xs:element minOccurs="0" name="scopeType"
12215 type="ns_p:ScopeTypeType"/>
12216                         </xs:sequence>
12217                     </xs:restriction>
12218                 </xs:complexContent>
12219             </xs:complexType>
12220 </xs:element>
12221 </xs:sequence>
12222 </xs:complexType>
12223 <xs:element name="incentiveTableDescriptionDataSelectors"
12224 type="ns_p:IncentiveTableDescriptionDataSelectorsType"/>
12225 <xs:complexType name="IncentiveTableConstraintsType">
12226     <xs:sequence>
12227         <xs:element minOccurs="0" name="tariff">
12228             <xs:complexType>
12229                 <xs:complexContent>
12230                     <xs:restriction base="ns_p:TariffDataType">
12231                         <xs:sequence>
12232                             <xs:element minOccurs="0" name="tariffId"
12233 type="ns_p:TariffIdType"/>
12234                         </xs:sequence>
12235                     </xs:restriction>
12236                 </xs:complexContent>
12237             </xs:complexType>
12238 </xs:element>
12239         <xs:element name="tariffConstraints" minOccurs="0">
```

```
12240         <xs:complexType>
12241             <xs:complexContent>
12242                 <xs:restriction base="ns_p:TariffOverallConstraintsDataType">
12243                     <xs:sequence>
12244                         <xs:element minOccurs="0" name="maxTiersPerTariff"
type="ns_p:TierCountType"/>
12245                         <xs:element minOccurs="0" name="maxBoundariesPerTier"
type="ns_p:TierBoundaryCountType"/>
12246                         <xs:element minOccurs="0" name="maxIncentivesPerTier"
type="ns_p:IncentiveCountType"/>
12247                     </xs:sequence>
12248                 </xs:restriction>
12249             </xs:complexContent>
12250         </xs:complexType>
12251     </xs:element>
12252     <xs:element name="incentiveSlotConstraints" minOccurs="0">
12253         <xs:complexType>
12254             <xs:complexContent>
12255                 <xs:restriction base="ns_p:TimeTableConstraintsDataType">
12256                     <xs:sequence>
12257                         <xs:element minOccurs="0" name="slotCountMax"
type="ns_p:TimeSlotCountType"/>
12258                     </xs:sequence>
12259                 </xs:restriction>
12260             </xs:complexContent>
12261         </xs:complexType>
12262     </xs:element>
12263     </xs:sequence>
12264 </xs:complexType>
12265 </xs:element>
12266 </xs:sequence>
12267 </xs:complexType>
12268 <xs:complexType name="IncentiveTableConstraintsDataType">
12269     <xs:sequence>
12270         <xs:element maxOccurs="unbounded" minOccurs="0" name="incentiveTableConstraints"
type="ns_p:IncentiveTableConstraintsType"/>
12271     </xs:sequence>
12272 </xs:complexType>
12273 <xs:element name="incentiveTableConstraintsData"
type="ns_p:IncentiveTableConstraintsDataType"/>
12274 <xs:complexType name="IncentiveTableConstraintsElementsType">
12275     <xs:sequence>
12276         <xs:element minOccurs="0" name="tariff">
12277             <xs:complexType>
12278                 <xs:complexContent>
12279                     <xs:restriction base="ns_p:TariffDataElementsType">
12280                         <xs:sequence>
12281                             <xs:element minOccurs="0" name="tariffId"
type="ns_p:ElementTagType"/>
12282                         </xs:sequence>
12283                     </xs:restriction>
12284                 </xs:complexContent>
12285             </xs:complexType>
12286         </xs:element>
12287         <xs:element name="tariffConstraints" minOccurs="0">
12288             <xs:complexType>
12289                 <xs:complexContent>
12290                     <xs:restriction base="ns_p:TariffOverallConstraintsDataElementsType">
12291                         <xs:sequence>
12292                             <xs:element minOccurs="0" name="maxTiersPerTariff"
type="ns_p:ElementTagType"/>
12293                             <xs:element minOccurs="0" name="maxBoundariesPerTier"
type="ns_p:ElementTagType"/>
12294                             <xs:element minOccurs="0" name="maxIncentivesPerTier"
type="ns_p:ElementTagType"/>
12295                         </xs:sequence>
12296                     </xs:restriction>
12297                 </xs:complexContent>
12298             </xs:complexType>
12299         </xs:element>
12300         <xs:element name="incentiveSlotConstraints" minOccurs="0">
12301             <xs:complexType>
12302                 <xs:complexContent>
12303                     <xs:restriction base="ns_p:TimeTableConstraintsDataElementsType">
12304                         <xs:sequence>
12305                             <xs:element minOccurs="0" name="slotCountMax"
type="ns_p:ElementTagType"/>
12306                         </xs:sequence>
12307                     </xs:restriction>
12308                 </xs:complexContent>
12309             </xs:complexType>
12310         </xs:element>
12311     </xs:sequence>
12312 </xs:complexType>
12313 </xs:element>
12314 </xs:sequence>
12315 </xs:complexType>
12316 </xs:complexType>
```

```
12318         </xs:element>
12319     </xs:sequence>
12320 </xs:complexType>
12321 <xs:complexType name="IncentiveTableConstraintsDataElementsType">
12322     <xs:sequence>
12323         <xs:element minOccurs="0" name="incentiveTableConstraints"
12324 type="ns_p:IncentiveTableConstraintsElementsType"/>
12325     </xs:sequence>
12326 </xs:complexType>
12327 <xs:element name="incentiveTableConstraintsDataElements"
12328 type="ns_p:IncentiveTableConstraintsDataElementsType"/>
12329 <xs:complexType name="IncentiveTableConstraintsDataSelectorsType">
12330     <xs:sequence>
12331         <xs:element minOccurs="0" name="tariff">
12332             <xs:complexType>
12333                 <xs:complexContent>
12334                     <xs:restriction base="ns_p:TariffListDataSelectorsType">
12335                         <xs:sequence>
12336                             <xs:element minOccurs="0" name="tariffId"
12337 type="ns_p:TariffIdType"/>
12338                         </xs:sequence>
12339                     </xs:restriction>
12340                 </xs:complexContent>
12341             </xs:complexType>
12342         </xs:element>
12343     </xs:sequence>
12344 </xs:complexType>
12345 <xs:element name="incentiveTableConstraintsDataSelectors"
12346 type="ns_p:IncentiveTableConstraintsDataSelectorsType"/>
12347 </xs:schema>
12348
12349
```

12350

12351 A.1.2 NodeManagement

12352 File name: EEBus_SPINE_TS_NodeManagement.xsd

```
12353
12354 <?xml version="1.0" encoding="UTF-8"?>
12355 <!--
12356     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
12357     Version 1.3.0 (final)
12358     2023-08-31
12359     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
12360     Source: https://www.eebus.org/en/specifications/
12361 -->
12362 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
12363 xmlns:xs="http://www.w3.org/2001/XMLSchema"
12364 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
12365 blockDefault="#all" elementFormDefault="qualified">
12366     <xs:annotation>
12367         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
12368 e.V. All rights reserved.</xs:documentation>
12369     </xs:annotation>
12370     <xs:include schemaLocation="EEBus_SPINE_TS_BindingManagement.xsd"/>
12371     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
12372     <xs:include schemaLocation="EEBus_SPINE_TS_NetworkManagement.xsd"/>
12373     <xs:include schemaLocation="EEBus_SPINE_TS_SubscriptionManagement.xsd"/>
12374     <xs:include schemaLocation="EEBus_SPINE_TS_UseCaseInformation.xsd"/>
12375     <xs:include schemaLocation="EEBus_SPINE_TS_Version.xsd"/>
12376     <xs:complexType name="NodeManagementSpecificationVersionListType">
12377         <xs:sequence>
12378             <xs:element maxOccurs="unbounded" minOccurs="0" name="specificationVersion"
12379 type="ns_p:SpecificationVersionDataType"/>
12380         </xs:sequence>
12381     </xs:complexType>
12382     <xs:complexType name="NodeManagementDetailedDiscoveryDeviceInformationType">
12383         <xs:sequence>
12384             <xs:element name="description" minOccurs="0">
12385                 <xs:complexType>
12386                     <xs:complexContent>
12387                         <xs:restriction
12388 base="ns_p:NetworkManagementDeviceDescriptionDataType">
12389                             <xs:sequence>
12390                                 <xs:element name="deviceAddress" minOccurs="0">
```

```
12391         <xs:complexType>
12392             <xs:complexContent>
12393                 <xs:restriction base="ns_p:DeviceAddressType">
12394                     <xs:sequence>
12395                         <xs:element ref="ns_p:device"
12396 minOccurs="0"/>
12397                     </xs:sequence>
12398                 </xs:restriction>
12399             </xs:complexContent>
12400         </xs:complexType>
12401     </xs:element>
12402     <xs:element name="deviceType" type="ns_p:DeviceTypeType"
12403 minOccurs="0"/>
12404     <xs:element minOccurs="0"
12405 name="networkManagementResponsibleAddress" type="ns_p:FeatureAddressType"/>
12406     <xs:element minOccurs="0" name="nativeSetup"
12407 type="ns_p:NetworkManagementNativeSetupType"/>
12408     <xs:element minOccurs="0" name="technologyAddress"
12409 type="ns_p:NetworkManagementTechnologyAddressType"/>
12410     <xs:element minOccurs="0"
12411 name="communicationsTechnologyInformation"
12412 type="ns_p:NetworkManagementCommunicationsTechnologyInformationType"/>
12413     <xs:element minOccurs="0" name="networkFeatureSet"
12414 type="ns_p:NetworkManagementFeatureSetType"/>
12415     <xs:element minOccurs="0" name="lastStateChange"
12416 type="ns_p:NetworkManagementStateChangeType"/>
12417     <xs:element minOccurs="0" name="minimumTrustLevel"
12418 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
12419     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12420     <xs:element minOccurs="0" name="description"
12421 type="ns_p:DescriptionType"/>
12422     </xs:sequence>
12423 </xs:restriction>
12424 </xs:complexContent>
12425 </xs:complexType>
12426 </xs:element>
12427 </xs:sequence>
12428 </xs:complexType>
12429 <xs:complexType name="NodeManagementDetailedDiscoveryEntityType">
12430     <xs:sequence>
12431         <xs:element name="description" minOccurs="0">
12432             <xs:complexType>
12433                 <xs:complexContent>
12434                     <xs:restriction
12435 base="ns_p:NetworkManagementEntityDescriptionDataType">
12436                         <xs:sequence>
12437                             <xs:element name="entityAddress" minOccurs="0">
12438                                 <xs:complexType>
12439                                     <xs:complexContent>
12440                                         <xs:restriction base="ns_p:EntityAddressType">
12441                                             <xs:sequence>
12442                                                 <xs:sequence>
12443                                                     <xs:element maxOccurs="unbounded"
12444 ref="ns_p:entity" minOccurs="0"/>
12445                                                 </xs:sequence>
12446                                             </xs:sequence>
12447                                         </xs:restriction>
12448                                     </xs:complexContent>
12449                                 </xs:complexType>
12450                             <xs:element name="entityType" type="ns_p:EntityTypeType"
12451 minOccurs="0"/>
12452                             <xs:element minOccurs="0" name="lastStateChange"
12453 type="ns_p:NetworkManagementStateChangeType"/>
12454                             <xs:element minOccurs="0" name="minimumTrustLevel"
12455 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
12456                             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12457                             <xs:element minOccurs="0" name="description"
12458 type="ns_p:DescriptionType"/>
12459                         </xs:sequence>
12460                     </xs:restriction>
12461                 </xs:complexContent>
12462             </xs:complexType>
12463         </xs:element>
12464     </xs:sequence>
12465 </xs:complexType>
12466 <xs:complexType name="NodeManagementDetailedDiscoveryFeatureInformationType">
12467     <xs:sequence>
```

```
12469         <xs:element name="description" minOccurs="0">
12470             <xs:complexType>
12471                 <xs:complexContent>
12472                     <xs:restriction
12473 base="ns_p:NetworkManagementFeatureDescriptionDataType">
12474                         <xs:sequence>
12475                             <xs:element name="featureAddress" minOccurs="0">
12476                                 <xs:complexType>
12477                                     <xs:complexContent>
12478                                         <xs:restriction base="ns_p:FeatureAddressType">
12479                                             <xs:sequence>
12480                                                 <xs:sequence>
12481                                                     <xs:element maxOccurs="unbounded"
12482 ref="ns_p:entity" minOccurs="0"/>
12483                                                 </xs:sequence>
12484                                             </xs:sequence>
12485                                             <xs:element ref="ns_p:feature"
12486 minOccurs="0"/>
12487                                         </xs:sequence>
12488                                     </xs:sequence>
12489                                 </xs:restriction>
12490                             </xs:complexContent>
12491                         </xs:complexType>
12492                     </xs:element>
12493                     <xs:element minOccurs="0" name="featureType"
12494 type="ns_p:FeatureTypeType"/>
12495                     <xs:element maxOccurs="unbounded" minOccurs="0"
12496 name="specificUsage" type="ns_p:FeatureSpecificUsageType"/>
12497                     <xs:element minOccurs="0" name="featureGroup"
12498 type="ns_p:FeatureGroupType"/>
12499                     <xs:element name="role" type="ns_p:RoleType" minOccurs="0"/>
12500                     <xs:element minOccurs="0" name="supportedFunction"
12501 type="ns_p:FunctionPropertyType" maxOccurs="unbounded"/>
12502                     <xs:element minOccurs="0" name="lastStateChange"
12503 type="ns_p:NetworkManagementStateChangeType"/>
12504                     <xs:element minOccurs="0" name="minimumTrustLevel"
12505 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
12506                     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12507                     <xs:element minOccurs="0" name="description"
12508 type="ns_p:DescriptionType"/>
12509                     <xs:element minOccurs="0" name="maxResponseDelay"
12510 type="ns_p:MaxResponseDelayType"/>
12511                 </xs:sequence>
12512             </xs:restriction>
12513         </xs:complexContent>
12514     </xs:complexType>
12515 </xs:element>
12516 </xs:sequence>
12517 </xs:complexType>
12518 <xs:complexType name="NodeManagementDetailedDiscoveryDataType">
12519     <xs:sequence>
12520         <xs:element maxOccurs="1" name="specificationVersionList" minOccurs="0"
12521 type="ns_p:NodeManagementSpecificationVersionListType"/>
12522         <xs:element name="deviceInformation"
12523 type="ns_p:NodeManagementDetailedDiscoveryDeviceInformationType" minOccurs="0"/>
12524         <xs:element name="entityInformation" maxOccurs="unbounded" minOccurs="0"
12525 type="ns_p:NodeManagementDetailedDiscoveryEntityInformationType"/>
12526         <xs:element name="featureInformation" maxOccurs="unbounded" minOccurs="0"
12527 type="ns_p:NodeManagementDetailedDiscoveryFeatureInformationType"/>
12528     </xs:sequence>
12529 </xs:complexType>
12530 <xs:element name="nodeManagementDetailedDiscoveryData"
12531 type="ns_p:NodeManagementDetailedDiscoveryDataType"/>
12532 <xs:complexType name="NodeManagementSpecificationVersionListElementsType">
12533     <xs:sequence>
12534         <xs:element maxOccurs="1" name="specificationVersion" minOccurs="0">
12535             <xs:complexType>
12536                 <xs:complexContent>
12537                     <xs:restriction base="ns_p:SpecificationVersionDataElementsType"/>
12538                 </xs:complexContent>
12539             </xs:complexType>
12540         </xs:element>
12541     </xs:sequence>
12542 </xs:complexType>
12543 <xs:complexType name="NodeManagementDetailedDiscoveryDeviceInformationElementsType">
12544     <xs:sequence>
12545         <xs:element name="description" minOccurs="0">
12546             <xs:complexType>
```

```
12547         <xs:complexContent>
12548             <xs:restriction
12549 base="ns_p:NetworkManagementDeviceDescriptionDataElementsType">
12550                 <xs:sequence>
12551                     <xs:element name="deviceAddress" minOccurs="0">
12552                         <xs:complexType>
12553                             <xs:complexContent>
12554                                 <xs:restriction
12555 base="ns_p:DeviceAddressElementsType">
12556                                     <xs:sequence>
12557                                         <xs:element minOccurs="0" name="device"
12558 type="ns_p:ElementTagType"/>
12559                                     </xs:sequence>
12560                                 </xs:restriction>
12561                             </xs:complexContent>
12562                         </xs:complexType>
12563                     </xs:element>
12564                     <xs:element name="deviceType" minOccurs="0"
12565 type="ns_p:ElementTagType"/>
12566                     <xs:element minOccurs="0"
12567 name="networkManagementResponsibleAddress" type="ns_p:ElementTagType"/>
12568                     <xs:element minOccurs="0" name="nativeSetup"
12569 type="ns_p:ElementTagType"/>
12570                     <xs:element minOccurs="0" name="technologyAddress"
12571 type="ns_p:ElementTagType"/>
12572                     <xs:element minOccurs="0"
12573 name="communicationsTechnologyInformation" type="ns_p:ElementTagType"/>
12574                     <xs:element minOccurs="0" name="networkFeatureSet"
12575 type="ns_p:ElementTagType"/>
12576                     <xs:element minOccurs="0" name="lastStateChange"
12577 type="ns_p:ElementTagType"/>
12578                     <xs:element minOccurs="0" name="minimumTrustLevel"
12579 type="ns_p:ElementTagType"/>
12580                     <xs:element minOccurs="0" name="label"
12581 type="ns_p:ElementTagType"/>
12582                     <xs:element minOccurs="0" name="description"
12583 type="ns_p:ElementTagType"/>
12584                 </xs:sequence>
12585             </xs:restriction>
12586         </xs:complexContent>
12587     </xs:complexType>
12588 </xs:element>
12589 </xs:sequence>
12590 </xs:complexType>
12591 <xs:complexType name="NodeManagementDetailedDiscoveryEntityInformationElementsType">
12592     <xs:sequence>
12593         <xs:element name="description" minOccurs="0">
12594             <xs:complexType>
12595                 <xs:complexContent>
12596                     <xs:restriction
12597 base="ns_p:NetworkManagementEntityDescriptionDataElementsType">
12598                         <xs:sequence>
12599                             <xs:element name="entityAddress" minOccurs="0">
12600                                 <xs:complexType>
12601                                     <xs:complexContent>
12602                                         <xs:restriction
12603 base="ns_p:EntityAddressElementsType">
12604                                             <xs:sequence>
12605                                                 <xs:element minOccurs="0" name="entity"
12606 type="ns_p:ElementTagType"/>
12607                                             </xs:sequence>
12608                                         </xs:restriction>
12609                                     </xs:complexContent>
12610                                 </xs:complexType>
12611                             </xs:element>
12612                             <xs:element name="entityType" minOccurs="0"
12613 type="ns_p:ElementTagType"/>
12614                             <xs:element minOccurs="0" name="lastStateChange"
12615 type="ns_p:ElementTagType"/>
12616                             <xs:element minOccurs="0" name="minimumTrustLevel"
12617 type="ns_p:ElementTagType"/>
12618                             <xs:element minOccurs="0" name="label"
12619 type="ns_p:ElementTagType"/>
12620                             <xs:element minOccurs="0" name="description"
12621 type="ns_p:ElementTagType"/>
12622                         </xs:sequence>
12623                     </xs:restriction>
12624                 </xs:complexContent>
```



```
12625         </xs:complexType>
12626     </xs:element>
12627 </xs:sequence>
12628 </xs:complexType>
12629 <xs:complexType name="NodeManagementDetailedDiscoveryFeatureInformationElementsType">
12630     <xs:sequence>
12631         <xs:element name="description" minOccurs="0">
12632             <xs:complexType>
12633                 <xs:complexContent>
12634                     <xs:restriction
12635 base="ns_p:NetworkManagementFeatureDescriptionDataElementsType">
12636                         <xs:sequence>
12637                             <xs:element name="featureAddress" minOccurs="0">
12638                                 <xs:complexType>
12639                                     <xs:complexContent>
12640                                         <xs:restriction
12641 base="ns_p:FeatureAddressElementsType">
12642                                             <xs:sequence>
12643                                                 <xs:element minOccurs="0" name="entity"
12644 type="ns_p:ElementTagType"/>
12645                                                 <xs:element minOccurs="0" name="feature"
12646 type="ns_p:ElementTagType"/>
12647                                             </xs:sequence>
12648                                         </xs:restriction>
12649                                     </xs:complexContent>
12650                                 </xs:complexType>
12651                             </xs:element>
12652                             <xs:element minOccurs="0" name="featureType"
12653 type="ns_p:ElementTagType"/>
12654                             <xs:element minOccurs="0" name="specificUsage"
12655 type="ns_p:ElementTagType"/>
12656                             <xs:element minOccurs="0" name="featureGroup"
12657 type="ns_p:ElementTagType"/>
12658                             <xs:element minOccurs="0" name="role"
12659 type="ns_p:ElementTagType"/>
12660                             <xs:element minOccurs="0" name="supportedFunction"
12661 type="ns_p:FunctionPropertyElementsType"/>
12662                             <xs:element minOccurs="0" name="lastStateChange"
12663 type="ns_p:ElementTagType"/>
12664                             <xs:element minOccurs="0" name="minimumTrustLevel"
12665 type="ns_p:ElementTagType"/>
12666                             <xs:element minOccurs="0" name="label"
12667 type="ns_p:ElementTagType"/>
12668                             <xs:element minOccurs="0" name="description"
12669 type="ns_p:ElementTagType"/>
12670                             <xs:element minOccurs="0" name="maxResponseDelay"
12671 type="ns_p:ElementTagType"/>
12672                         </xs:sequence>
12673                     </xs:restriction>
12674                 </xs:complexContent>
12675             </xs:complexType>
12676         </xs:element>
12677     </xs:sequence>
12678 </xs:complexType>
12679 <xs:complexType name="NodeManagementDetailedDiscoveryDataElementsType">
12680     <xs:sequence>
12681         <xs:element maxOccurs="1" name="specificationVersionList" minOccurs="0"
12682 type="ns_p:NodeManagementSpecificationVersionListElementsType"/>
12683         <xs:element name="deviceInformation"
12684 type="ns_p:NodeManagementDetailedDiscoveryDeviceInformationElementsType" minOccurs="0"/>
12685         <xs:element name="entityInformation" minOccurs="0"
12686 type="ns_p:NodeManagementDetailedDiscoveryEntityInformationElementsType"/>
12687         <xs:element name="featureInformation" minOccurs="0"
12688 type="ns_p:NodeManagementDetailedDiscoveryFeatureInformationElementsType"/>
12689     </xs:sequence>
12690 </xs:complexType>
12691 <xs:element name="nodeManagementDetailedDiscoveryDataElements"
12692 type="ns_p:NodeManagementDetailedDiscoveryDataElementsType"/>
12693 <xs:complexType name="NodeManagementDetailedDiscoveryDataSelectorsType">
12694     <xs:sequence>
12695         <xs:element minOccurs="0" name="deviceInformation">
12696             <xs:complexType>
12697                 <xs:complexContent>
12698                     <xs:restriction
12699 base="ns_p:NetworkManagementDeviceDescriptionListDataSelectorsType">
12700                         <xs:sequence>
12701                             <xs:element minOccurs="0" name="deviceAddress"
12702 type="ns_p:DeviceAddressType"/>
```

```

12703         <xs:element minOccurs="0" name="deviceType"
12704 type="ns_p:DeviceTypeType"/>
12705     </xs:sequence>
12706 </xs:restriction>
12707 </xs:complexContent>
12708 </xs:complexType>
12709 </xs:element>
12710 <xs:element minOccurs="0" name="entityInformation">
12711 <xs:complexType>
12712 <xs:complexContent>
12713 <xs:restriction
12714 base="ns_p:NetworkManagementEntityDescriptionListDataSelectorsType">
12715 <xs:sequence>
12716 <xs:element minOccurs="0" name="entityAddress"
12717 type="ns_p:EntityAddressType"/>
12718 <xs:element minOccurs="0" name="entityType"
12719 type="ns_p:EntityTypeType"/>
12720 </xs:sequence>
12721 </xs:restriction>
12722 </xs:complexContent>
12723 </xs:complexType>
12724 </xs:element>
12725 <xs:element minOccurs="0" name="featureInformation">
12726 <xs:complexType>
12727 <xs:complexContent>
12728 <xs:restriction
12729 base="ns_p:NetworkManagementFeatureDescriptionListDataSelectorsType">
12730 <xs:sequence>
12731 <xs:element minOccurs="0" name="featureAddress"
12732 type="ns_p:FeatureAddressType"/>
12733 <xs:element minOccurs="0" name="featureType"
12734 type="ns_p:FeatureTypeType"/>
12735 </xs:sequence>
12736 </xs:restriction>
12737 </xs:complexContent>
12738 </xs:complexType>
12739 </xs:element>
12740 </xs:sequence>
12741 </xs:complexType>
12742 <xs:element name="nodeManagementDetailedDiscoveryDataSelectors"
12743 type="ns_p:NodeManagementDetailedDiscoveryDataSelectorsType"/>
12744 <xs:complexType name="NodeManagementBindingDataType">
12745 <xs:sequence>
12746 <xs:element maxOccurs="unbounded" minOccurs="0" name="bindingEntry">
12747 <xs:complexType>
12748 <xs:complexContent>
12749 <xs:restriction base="ns_p:BindingManagementEntryDataType">
12750 <xs:sequence>
12751 <xs:element name="bindingId" type="ns_p:BindingIdType"
12752 minOccurs="0"/>
12753 <xs:element name="clientAddress" minOccurs="0"
12754 type="ns_p:FeatureAddressType"/>
12755 <xs:element name="serverAddress" minOccurs="0"
12756 type="ns_p:FeatureAddressType"/>
12757 <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12758 <xs:element minOccurs="0" name="description"
12759 type="ns_p:DescriptionType"/>
12760 </xs:sequence>
12761 </xs:restriction>
12762 </xs:complexContent>
12763 </xs:complexType>
12764 </xs:element>
12765 </xs:sequence>
12766 </xs:complexType>
12767 <xs:element name="nodeManagementBindingData" type="ns_p:NodeManagementBindingDataType"/>
12768 <xs:complexType name="NodeManagementBindingDataElementsType">
12769 <xs:sequence>
12770 <xs:element minOccurs="0" name="bindingEntry">
12771 <xs:complexType>
12772 <xs:complexContent>
12773 <xs:restriction base="ns_p:BindingManagementEntryDataElementsType">
12774 <xs:sequence>
12775 <xs:element minOccurs="0" name="bindingId"
12776 type="ns_p:ElementTagType"/>
12777 <xs:element minOccurs="0" name="clientAddress"
12778 type="ns_p:FeatureAddressElementsType"/>
12779 <xs:element minOccurs="0" name="serverAddress"
12780 type="ns_p:FeatureAddressElementsType"/>

```

```
12781         <xs:element minOccurs="0" name="label"
12782 type="ns_p:ElementTagType"/>
12783         <xs:element minOccurs="0" name="description"
12784 type="ns_p:ElementTagType"/>
12785     </xs:sequence>
12786 </xs:restriction>
12787 </xs:complexContent>
12788 </xs:complexType>
12789 </xs:element>
12790 </xs:sequence>
12791 </xs:complexType>
12792 <xs:element name="nodeManagementBindingDataElements"
12793 type="ns_p:NodeManagementBindingDataElementsType"/>
12794 <xs:complexType name="NodeManagementBindingDataSelectorsType">
12795     <xs:sequence>
12796         <xs:element minOccurs="0" name="bindingEntry">
12797             <xs:complexType>
12798                 <xs:complexContent>
12799                     <xs:restriction
12800 base="ns_p:BindingManagementEntryListDataSelectorsType"/>
12801                 </xs:complexContent>
12802             </xs:complexType>
12803         </xs:element>
12804     </xs:sequence>
12805 </xs:complexType>
12806 <xs:element name="nodeManagementBindingDataSelectors"
12807 type="ns_p:NodeManagementBindingDataSelectorsType"/>
12808 <xs:complexType name="NodeManagementBindingRequestCallType">
12809     <xs:sequence>
12810         <xs:element minOccurs="0" name="bindingRequest">
12811             <xs:complexType>
12812                 <xs:complexContent>
12813                     <xs:restriction base="ns_p:BindingManagementRequestCallType">
12814                         <xs:sequence>
12815                             <xs:element minOccurs="0" name="clientAddress"
12816 type="ns_p:FeatureAddressType"/>
12817                             <xs:element minOccurs="0" name="serverAddress"
12818 type="ns_p:FeatureAddressType"/>
12819                             <xs:element minOccurs="0" name="serverFeatureType"
12820 type="ns_p:FeatureTypeType"/>
12821                         </xs:sequence>
12822                     </xs:restriction>
12823                 </xs:complexContent>
12824             </xs:complexType>
12825         </xs:element>
12826     </xs:sequence>
12827 </xs:complexType>
12828 <xs:element name="nodeManagementBindingRequestCall"
12829 type="ns_p:NodeManagementBindingRequestCallType"/>
12830 <xs:complexType name="NodeManagementBindingRequestCallElementsType">
12831     <xs:sequence>
12832         <xs:element minOccurs="0" name="bindingRequest">
12833             <xs:complexType>
12834                 <xs:complexContent>
12835                     <xs:restriction base="ns_p:BindingManagementRequestCallElementsType">
12836                         <xs:sequence>
12837                             <xs:element minOccurs="0" name="clientAddress"
12838 type="ns_p:FeatureAddressElementsType"/>
12839                             <xs:element minOccurs="0" name="serverAddress"
12840 type="ns_p:FeatureAddressElementsType"/>
12841                             <xs:element minOccurs="0" name="serverFeatureType"
12842 type="ns_p:ElementTagType"/>
12843                         </xs:sequence>
12844                     </xs:restriction>
12845                 </xs:complexContent>
12846             </xs:complexType>
12847         </xs:element>
12848     </xs:sequence>
12849 </xs:complexType>
12850 <xs:element name="nodeManagementBindingRequestCallElements"
12851 type="ns_p:NodeManagementBindingRequestCallElementsType"/>
12852 <xs:complexType name="NodeManagementBindingDeleteCallType">
12853     <xs:sequence>
12854         <xs:element minOccurs="0" name="bindingDelete">
12855             <xs:complexType>
12856                 <xs:complexContent>
12857                     <xs:restriction base="ns_p:BindingManagementDeleteCallType">
12858                         <xs:sequence>
```

```
12859         <xs:element minOccurs="0" name="bindingId"
12860 type="ns_p:BindingIdType"/>
12861         <xs:element minOccurs="0" name="clientAddress"
12862 type="ns_p:FeatureAddressType"/>
12863         <xs:element minOccurs="0" name="serverAddress"
12864 type="ns_p:FeatureAddressType"/>
12865     </xs:sequence>
12866 </xs:restriction>
12867 </xs:complexContent>
12868 </xs:complexType>
12869 </xs:element>
12870 </xs:sequence>
12871 </xs:complexType>
12872 <xs:element name="nodeManagementBindingDeleteCall"
12873 type="ns_p:NodeManagementBindingDeleteCallType"/>
12874 <xs:complexType name="NodeManagementBindingDeleteCallElementsType">
12875     <xs:sequence>
12876         <xs:element minOccurs="0" name="bindingDelete">
12877             <xs:complexType>
12878                 <xs:complexContent>
12879                     <xs:restriction base="ns_p:BindingManagementDeleteCallElementsType">
12880                         <xs:sequence>
12881                             <xs:element minOccurs="0" name="bindingId"
12882 type="ns_p:ElementTagType"/>
12883                             <xs:element minOccurs="0" name="clientAddress"
12884 type="ns_p:FeatureAddressElementsType"/>
12885                             <xs:element minOccurs="0" name="serverAddress"
12886 type="ns_p:FeatureAddressElementsType"/>
12887                         </xs:sequence>
12888                     </xs:restriction>
12889                 </xs:complexContent>
12890             </xs:complexType>
12891         </xs:element>
12892     </xs:sequence>
12893 </xs:complexType>
12894 <xs:element name="nodeManagementBindingDeleteCallElements"
12895 type="ns_p:NodeManagementBindingDeleteCallElementsType"/>
12896 <xs:complexType name="NodeManagementSubscriptionDataType">
12897     <xs:sequence>
12898         <xs:element maxOccurs="unbounded" minOccurs="0" name="subscriptionEntry">
12899             <xs:complexType>
12900                 <xs:complexContent>
12901                     <xs:restriction base="ns_p:SubscriptionManagementEntryDataType">
12902                         <xs:sequence>
12903                             <xs:element name="subscriptionId"
12904 type="ns_p:SubscriptionIdType" minOccurs="0"/>
12905                             <xs:element name="clientAddress" minOccurs="0"
12906 type="ns_p:FeatureAddressType"/>
12907                             <xs:element name="serverAddress" minOccurs="0"
12908 type="ns_p:FeatureAddressType"/>
12909                             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
12910                             <xs:element minOccurs="0" name="description"
12911 type="ns_p:DescriptionType"/>
12912                         </xs:sequence>
12913                     </xs:restriction>
12914                 </xs:complexContent>
12915             </xs:complexType>
12916         </xs:element>
12917     </xs:sequence>
12918 </xs:complexType>
12919 <xs:element name="nodeManagementSubscriptionData"
12920 type="ns_p:NodeManagementSubscriptionDataType"/>
12921 <xs:complexType name="NodeManagementSubscriptionDataElementsType">
12922     <xs:sequence>
12923         <xs:element minOccurs="0" name="subscriptionEntry">
12924             <xs:complexType>
12925                 <xs:complexContent>
12926                     <xs:restriction
12927 base="ns_p:SubscriptionManagementEntryDataElementsType">
12928                         <xs:sequence>
12929                             <xs:element minOccurs="0" name="subscriptionId"
12930 type="ns_p:ElementTagType"/>
12931                             <xs:element minOccurs="0" name="clientAddress"
12932 type="ns_p:FeatureAddressElementsType"/>
12933                             <xs:element minOccurs="0" name="serverAddress"
12934 type="ns_p:FeatureAddressElementsType"/>
12935                             <xs:element minOccurs="0" name="label"
12936 type="ns_p:ElementTagType"/>
```

```
12937         <xs:element minOccurs="0" name="description"
12938 type="ns_p:ElementTagType"/>
12939     </xs:sequence>
12940 </xs:restriction>
12941 </xs:complexContent>
12942 </xs:complexType>
12943 </xs:element>
12944 </xs:sequence>
12945 </xs:complexType>
12946 <xs:element name="nodeManagementSubscriptionDataElements"
12947 type="ns_p:NodeManagementSubscriptionDataElementsType"/>
12948 <xs:complexType name="NodeManagementSubscriptionDataSelectorsType">
12949     <xs:sequence>
12950         <xs:element minOccurs="0" name="subscriptionEntry">
12951             <xs:complexType>
12952                 <xs:complexContent>
12953                     <xs:restriction
12954 base="ns_p:SubscriptionManagementEntryListDataSelectorsType">
12955                         <xs:sequence>
12956                             <xs:element minOccurs="0" name="subscriptionId"
12957 type="ns_p:SubscriptionIdType"/>
12958                             <xs:element minOccurs="0" name="clientAddress"
12959 type="ns_p:FeatureAddressType"/>
12960                             <xs:element minOccurs="0" name="serverAddress"
12961 type="ns_p:FeatureAddressType"/>
12962                         </xs:sequence>
12963                     </xs:restriction>
12964                 </xs:complexContent>
12965             </xs:complexType>
12966         </xs:element>
12967     </xs:sequence>
12968 </xs:complexType>
12969 <xs:element name="nodeManagementSubscriptionDataSelectors"
12970 type="ns_p:NodeManagementSubscriptionDataSelectorsType"/>
12971 <xs:complexType name="NodeManagementSubscriptionRequestCallType">
12972     <xs:sequence>
12973         <xs:element minOccurs="0" name="subscriptionRequest">
12974             <xs:complexType>
12975                 <xs:complexContent>
12976                     <xs:restriction base="ns_p:SubscriptionManagementRequestCallType">
12977                         <xs:sequence>
12978                             <xs:element minOccurs="0" name="clientAddress"
12979 type="ns_p:FeatureAddressType"/>
12980                             <xs:element minOccurs="0" name="serverAddress"
12981 type="ns_p:FeatureAddressType"/>
12982                             <xs:element minOccurs="0" name="serverFeatureType"
12983 type="ns_p:FeatureTypeType"/>
12984                         </xs:sequence>
12985                     </xs:restriction>
12986                 </xs:complexContent>
12987             </xs:complexType>
12988         </xs:element>
12989     </xs:sequence>
12990 </xs:complexType>
12991 <xs:element name="nodeManagementSubscriptionRequestCall"
12992 type="ns_p:NodeManagementSubscriptionRequestCallType"/>
12993 <xs:complexType name="NodeManagementSubscriptionRequestCallElementsType">
12994     <xs:sequence>
12995         <xs:element minOccurs="0" name="subscriptionRequest">
12996             <xs:complexType>
12997                 <xs:complexContent>
12998                     <xs:restriction
12999 base="ns_p:SubscriptionManagementRequestCallElementsType">
13000                         <xs:sequence>
13001                             <xs:element minOccurs="0" name="clientAddress"
13002 type="ns_p:FeatureAddressElementsType"/>
13003                             <xs:element minOccurs="0" name="serverAddress"
13004 type="ns_p:FeatureAddressElementsType"/>
13005                             <xs:element minOccurs="0" name="serverFeatureType"
13006 type="ns_p:ElementTagType"/>
13007                         </xs:sequence>
13008                     </xs:restriction>
13009                 </xs:complexContent>
13010             </xs:complexType>
13011         </xs:element>
13012     </xs:sequence>
13013 </xs:complexType>
```

```
13014     <xs:element name="nodeManagementSubscriptionRequestCallElements"
13015 type="ns_p:NodeManagementSubscriptionRequestCallElementsType"/>
13016     <xs:complexType name="NodeManagementSubscriptionDeleteCallType">
13017       <xs:sequence>
13018         <xs:element minOccurs="0" name="subscriptionDelete">
13019           <xs:complexType>
13020             <xs:complexContent>
13021               <xs:restriction base="ns_p:SubscriptionManagementDeleteCallType">
13022                 <xs:sequence>
13023                   <xs:element minOccurs="0" name="subscriptionId"
13024 type="ns_p:SubscriptionIdType"/>
13025                   <xs:element minOccurs="0" name="clientAddress"
13026 type="ns_p:FeatureAddressType"/>
13027                   <xs:element minOccurs="0" name="serverAddress"
13028 type="ns_p:FeatureAddressType"/>
13029                 </xs:sequence>
13030               </xs:restriction>
13031             </xs:complexContent>
13032           </xs:complexType>
13033         </xs:element>
13034       </xs:sequence>
13035     </xs:complexType>
13036     <xs:element name="nodeManagementSubscriptionDeleteCall"
13037 type="ns_p:NodeManagementSubscriptionDeleteCallType"/>
13038     <xs:complexType name="NodeManagementSubscriptionDeleteCallElementsType">
13039       <xs:sequence>
13040         <xs:element minOccurs="0" name="subscriptionDelete">
13041           <xs:complexType>
13042             <xs:complexContent>
13043               <xs:restriction
13044 base="ns_p:SubscriptionManagementDeleteCallElementsType">
13045                 <xs:sequence>
13046                   <xs:element minOccurs="0" name="subscriptionId"
13047 type="ns_p:ElementTagType"/>
13048                   <xs:element minOccurs="0" name="clientAddress"
13049 type="ns_p:FeatureAddressElementsType"/>
13050                   <xs:element minOccurs="0" name="serverAddress"
13051 type="ns_p:FeatureAddressElementsType"/>
13052                 </xs:sequence>
13053               </xs:restriction>
13054             </xs:complexContent>
13055           </xs:complexType>
13056         </xs:element>
13057       </xs:sequence>
13058     </xs:complexType>
13059     <xs:element name="nodeManagementSubscriptionDeleteCallElements"
13060 type="ns_p:NodeManagementSubscriptionDeleteCallElementsType"/>
13061     <xs:complexType name="NodeManagementDestinationDataType">
13062       <xs:sequence>
13063         <xs:element minOccurs="0" name="deviceDescription">
13064           <xs:complexType>
13065             <xs:complexContent>
13066               <xs:restriction
13067 base="ns_p:NetworkManagementDeviceDescriptionDataType">
13068                 <xs:sequence>
13069                   <xs:element minOccurs="0" name="deviceAddress"
13070 type="ns_p:DeviceAddressType"/>
13071                   <xs:element minOccurs="0"
13072 name="communicationsTechnologyInformation"
13073 type="ns_p:NetworkManagementCommunicationsTechnologyInformationType"/>
13074                   <xs:element minOccurs="0" name="networkFeatureSet"
13075 type="ns_p:NetworkManagementFeatureSetType"/>
13076                   <xs:element minOccurs="0" name="lastStateChange"
13077 type="ns_p:NetworkManagementStateChangeType"/>
13078                   <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
13079                 </xs:sequence>
13080               </xs:restriction>
13081             </xs:complexContent>
13082           </xs:complexType>
13083         </xs:element>
13084       </xs:sequence>
13085     </xs:complexType>
13086     <xs:element name="nodeManagementDestinationData"
13087 type="ns_p:NodeManagementDestinationDataType"/>
13088     <xs:complexType name="NodeManagementDestinationDataElementsType">
13089       <xs:sequence>
13090         <xs:element minOccurs="0" name="deviceDescription">
13091           <xs:complexType>
```

```
13092         <xs:complexContent>
13093             <xs:restriction
13094 base="ns_p:NetworkManagementDeviceDescriptionDataElementsType">
13095                 <xs:sequence>
13096                     <xs:element minOccurs="0" name="deviceAddress"
13097 type="ns_p:DeviceAddressElementsType"/>
13098                     <xs:element minOccurs="0"
13099 name="communicationsTechnologyInformation" type="ns_p:ElementTagType"/>
13100                     <xs:element minOccurs="0" name="networkFeatureSet"
13101 type="ns_p:ElementTagType"/>
13102                     <xs:element minOccurs="0" name="lastStateChange"
13103 type="ns_p:ElementTagType"/>
13104                     <xs:element minOccurs="0" name="label"
13105 type="ns_p:ElementTagType"/>
13106                 </xs:sequence>
13107             </xs:restriction>
13108         </xs:complexContent>
13109     </xs:complexType>
13110 </xs:element>
13111 </xs:sequence>
13112 </xs:complexType>
13113 <xs:element name="nodeManagementDestinationDataElements"
13114 type="ns_p:NodeManagementDestinationDataElementsType"/>
13115 <xs:complexType name="NodeManagementDestinationListDataType">
13116     <xs:sequence>
13117         <xs:element maxOccurs="unbounded" minOccurs="0"
13118 ref="ns_p:nodeManagementDestinationData"/>
13119     </xs:sequence>
13120 </xs:complexType>
13121 <xs:element name="nodeManagementDestinationListData"
13122 type="ns_p:NodeManagementDestinationListDataType"/>
13123 <xs:complexType name="NodeManagementDestinationListDataSelectorsType">
13124     <xs:sequence>
13125         <xs:element minOccurs="0" name="deviceDescription">
13126             <xs:complexType>
13127                 <xs:complexContent>
13128                     <xs:restriction
13129 base="ns_p:NetworkManagementDeviceDescriptionListDataSelectorsType">
13130                         <xs:sequence>
13131                             <xs:element minOccurs="0" name="deviceAddress"
13132 type="ns_p:DeviceAddressType"/>
13133                         </xs:sequence>
13134                     </xs:restriction>
13135                 </xs:complexContent>
13136             </xs:complexType>
13137         </xs:element>
13138     </xs:sequence>
13139 </xs:complexType>
13140 <xs:element name="nodeManagementDestinationListDataSelectors"
13141 type="ns_p:NodeManagementDestinationListDataSelectorsType"/>
13142 <xs:complexType name="NodeManagementUseCaseDataType">
13143     <xs:sequence>
13144         <xs:element maxOccurs="unbounded" minOccurs="0" name="useCaseInformation">
13145             <xs:complexType>
13146                 <xs:complexContent>
13147                     <xs:restriction base="ns_p:UseCaseInformationDataType">
13148                         <xs:sequence>
13149                             <xs:element minOccurs="0" name="address"
13150 type="ns_p:FeatureAddressType"/>
13151                             <xs:element minOccurs="0" name="actor"
13152 type="ns_p:UseCaseActorType"/>
13153                             <xs:element maxOccurs="unbounded" minOccurs="0"
13154 name="useCaseSupport">
13155                                 <xs:complexType>
13156                                     <xs:complexContent>
13157                                         <xs:restriction base="ns_p:UseCaseSupportType">
13158                                             <xs:sequence>
13159                                                 <xs:element minOccurs="0"
13160 name="useCaseName" type="ns_p:UseCaseNameType"/>
13161                                                 <xs:element minOccurs="0"
13162 name="useCaseVersion" type="ns_p:SpecificationVersionType"/>
13163                                                 <xs:element minOccurs="0"
13164 name="useCaseAvailable" type="xs:boolean"/>
13165                                                 <xs:element maxOccurs="unbounded"
13166 minOccurs="0" name="scenarioSupport" type="ns_p:UseCaseScenarioSupportType"/>
13167                                                 <xs:element minOccurs="0"
13168 name="useCaseDocumentSubRevision" type="xs:string"/>
13169                                             </xs:sequence>
```

```

13170         </xs:restriction>
13171     </xs:complexContent>
13172 </xs:complexType>
13173 </xs:element>
13174 </xs:sequence>
13175 </xs:restriction>
13176 </xs:complexContent>
13177 </xs:complexType>
13178 </xs:element>
13179 </xs:sequence>
13180 </xs:complexType>
13181 <xs:element name="nodeManagementUseCaseData" type="ns_p:NodeManagementUseCaseDataType"/>
13182 <xs:complexType name="NodeManagementUseCaseDataElementsType">
13183     <xs:sequence>
13184         <xs:element maxOccurs="1" minOccurs="0" name="useCaseInformation">
13185             <xs:complexType>
13186                 <xs:complexContent>
13187                     <xs:restriction base="ns_p:UseCaseInformationDataElementsType">
13188                         <xs:sequence>
13189                             <xs:element minOccurs="0" name="address"
13190 type="ns_p:FeatureAddressElementsType"/>
13191                             <xs:element minOccurs="0" name="actor"
13192 type="ns_p:ElementTagType"/>
13193                             <xs:element minOccurs="0" name="useCaseSupport">
13194                                 <xs:complexType>
13195                                     <xs:complexContent>
13196                                         <xs:restriction
13197 base="ns_p:UseCaseSupportElementsType">
13198                                             <xs:sequence>
13199                                                 <xs:element minOccurs="0"
13200 name="useCaseName" type="ns_p:ElementTagType"/>
13201                                                 <xs:element minOccurs="0"
13202 name="useCaseVersion" type="ns_p:ElementTagType"/>
13203                                                 <xs:element minOccurs="0"
13204 name="useCaseAvailable" type="ns_p:ElementTagType"/>
13205                                                 <xs:element minOccurs="0"
13206 name="scenarioSupport" type="ns_p:ElementTagType"/>
13207                                                 <xs:element minOccurs="0"
13208 name="useCaseDocumentSubRevision" type="ns_p:ElementTagType"/>
13209                                             </xs:sequence>
13210                                         </xs:restriction>
13211                                     </xs:complexContent>
13212                                 </xs:complexType>
13213                             </xs:element>
13214                         </xs:sequence>
13215                     </xs:restriction>
13216                 </xs:complexContent>
13217             </xs:complexType>
13218         </xs:element>
13219     </xs:sequence>
13220 </xs:complexType>
13221 <xs:element name="nodeManagementUseCaseDataElements"
13222 type="ns_p:NodeManagementUseCaseDataElementsType"/>
13223 <xs:complexType name="NodeManagementUseCaseDataSelectorsType">
13224     <xs:sequence>
13225         <xs:element minOccurs="0" name="useCaseInformation">
13226             <xs:complexType>
13227                 <xs:complexContent>
13228                     <xs:restriction base="ns_p:UseCaseInformationListDataSelectorsType">
13229                         <xs:sequence>
13230                             <xs:element minOccurs="0" name="address"
13231 type="ns_p:FeatureAddressType"/>
13232                             <xs:element minOccurs="0" name="actor"
13233 type="ns_p:UseCaseActorType"/>
13234                             <xs:element minOccurs="0" name="useCaseSupport">
13235                                 <xs:complexType>
13236                                     <xs:complexContent>
13237                                         <xs:restriction
13238 base="ns_p:UseCaseSupportSelectorsType">
13239                                             <xs:sequence>
13240                                                 <xs:element minOccurs="0"
13241 name="useCaseName" type="ns_p:UseCaseNameType"/>
13242                                                 <xs:element minOccurs="0"
13243 name="useCaseVersion" type="ns_p:SpecificationVersionType"/>
13244                                                 <xs:element minOccurs="0"
13245 name="scenarioSupport" type="ns_p:UseCaseScenarioSupportType"/>
13246                                             </xs:sequence>
13247                                         </xs:restriction>

```



```
13248         </xs:complexContent>
13249     </xs:complexType>
13250 </xs:element>
13251 </xs:sequence>
13252 </xs:restriction>
13253 </xs:complexContent>
13254 </xs:complexType>
13255 </xs:element>
13256 </xs:sequence>
13257 </xs:complexType>
13258 <xs:element name="nodeManagementUseCaseDataSelectors"
13259 type="ns_p:NodeManagementUseCaseDataSelectorsType"/>
13260 </xs:schema>
13261
13262
```

13263

13264 A.1.3 SmartEnergyManagementPs

13265 File name: EEBus_SPINE_TS_SmartEnergyManagementPs.xsd

```
13266
13267 <?xml version="1.0" encoding="UTF-8"?>
13268 <!--
13269     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
13270     Version 1.3.0 (final)
13271     2023-08-31
13272     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
13273     Source: https://www.eebus.org/en/specifications/
13274 -->
13275 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
13276 xmlns:xs="http://www.w3.org/2001/XMLSchema"
13277 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0" blockDefault="#all"
13278 elementFormDefault="qualified">
13279     <xs:annotation>
13280         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
13281 e.V. All rights reserved.</xs:documentation>
13282     </xs:annotation>
13283     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
13284     <xs:include schemaLocation="EEBus_SPINE_TS_OperatingConstraints.xsd"/>
13285     <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
13286     <xs:complexType name="SmartEnergyManagementPsAlternativesRelationType">
13287         <xs:complexContent>
13288             <xs:restriction base="ns_p:PowerSequenceAlternativesRelationDataType">
13289                 <xs:sequence>
13290                     <xs:element minOccurs="0" name="alternativesId"
13291 type="ns_p:AlternativesIdType"/>
13292                 </xs:sequence>
13293             </xs:restriction>
13294         </xs:complexContent>
13295     </xs:complexType>
13296     <xs:complexType name="SmartEnergyManagementPsAlternativesType">
13297         <xs:sequence>
13298             <xs:element minOccurs="0" name="relation"
13299 type="ns_p:SmartEnergyManagementPsAlternativesRelationType"/>
13300             <xs:element name="powerSequence" maxOccurs="unbounded" minOccurs="0"
13301 type="ns_p:SmartEnergyManagementPsPowerSequenceType"/>
13302         </xs:sequence>
13303     </xs:complexType>
13304     <xs:complexType name="SmartEnergyManagementPsPowerSequenceType">
13305         <xs:sequence>
13306             <xs:element name="description" minOccurs="0">
13307                 <xs:complexType>
13308                     <xs:complexContent>
13309                         <xs:restriction base="ns_p:PowerSequenceDescriptionDataType">
13310                             <xs:sequence>
13311                                 <xs:element name="sequenceId" type="ns_p:PowerSequenceIdType"
13312 minOccurs="0"/>
13313                                 <xs:element minOccurs="0" name="description">
13314                                     <xs:simpleType>
13315                                         <xs:restriction base="ns_p:DescriptionType">
13316                                             <xs:minLength value="1"/>
13317                                             <xs:maxLength value="60"/>
13318                                         </xs:restriction>
13319                                     </xs:simpleType>
13320                                 </xs:element>

```

```
13321         <xs:element minOccurs="0" name="powerUnit"
13322 type="ns_p:UnitOfMeasurementType"/>
13323         <xs:element minOccurs="0" name="energyUnit"
13324 type="ns_p:UnitOfMeasurementType"/>
13325         <xs:element minOccurs="0" name="valueSource"
13326 type="ns_p:MeasurementValueSourceType"/>
13327         <xs:element minOccurs="0" name="taskIdentifier"
13328 type="xs:unsignedInt"/>
13329         <xs:element minOccurs="0" name="repetitionsTotal"
13330 type="xs:unsignedInt"/>
13331     </xs:sequence>
13332 </xs:restriction>
13333 </xs:complexContent>
13334 </xs:complexType>
13335 </xs:element>
13336 <xs:element name="state" minOccurs="0">
13337     <xs:complexType>
13338         <xs:complexContent>
13339             <xs:restriction base="ns_p:PowerSequenceStateDataType">
13340                 <xs:sequence>
13341                     <xs:element name="state" type="ns_p:PowerSequenceStateType"
13342 minOccurs="0"/>
13343                     <xs:element minOccurs="0" name="activeSlotNumber"
13344 type="ns_p:PowerTimeSlotNumberType"/>
13345                     <xs:element minOccurs="0" name="elapsedSlotTime"
13346 type="xs:duration"/>
13347                     <xs:element minOccurs="0" name="remainingSlotTime"
13348 type="xs:duration"/>
13349                     <xs:element name="sequenceRemoteControllable"
13350 type="xs:boolean" minOccurs="0"/>
13351                     <xs:element minOccurs="0" name="activeRepetitionNumber"
13352 type="xs:unsignedInt"/>
13353                     <xs:element minOccurs="0" name="remainingPauseTime"
13354 type="xs:duration"/>
13355                 </xs:sequence>
13356             </xs:restriction>
13357         </xs:complexContent>
13358     </xs:complexType>
13359 </xs:element>
13360 <xs:element minOccurs="0" name="schedule">
13361     <xs:complexType>
13362         <xs:complexContent>
13363             <xs:restriction base="ns_p:PowerSequenceScheduleDataType">
13364                 <xs:sequence>
13365                     <xs:element name="startTime" type="xs:duration"
13366 minOccurs="0"/>
13367                     <xs:element name="endTime" type="xs:duration" minOccurs="0"/>
13368                 </xs:sequence>
13369             </xs:restriction>
13370         </xs:complexContent>
13371     </xs:complexType>
13372 </xs:element>
13373 <xs:element name="scheduleConstraints" minOccurs="0">
13374     <xs:complexType>
13375         <xs:complexContent>
13376             <xs:restriction base="ns_p:PowerSequenceScheduleConstraintsDataType">
13377                 <xs:sequence>
13378                     <xs:element name="earliestStartTime" type="xs:duration"
13379 minOccurs="0"/>
13380                     <xs:element name="latestEndTime" type="xs:duration"
13381 minOccurs="0"/>
13382                 </xs:sequence>
13383             </xs:restriction>
13384         </xs:complexContent>
13385     </xs:complexType>
13386 </xs:element>
13387 <xs:element minOccurs="0" name="schedulePreference">
13388     <xs:complexType>
13389         <xs:complexContent>
13390             <xs:restriction base="ns_p:PowerSequenceSchedulePreferenceDataType">
13391                 <xs:sequence>
13392                     <xs:element minOccurs="0" name="greenest" type="xs:boolean"/>
13393                     <xs:element minOccurs="0" name="cheapest" type="xs:boolean"/>
13394                 </xs:sequence>
13395             </xs:restriction>
13396         </xs:complexContent>
13397     </xs:complexType>
13398 </xs:element>
```

```
13399         <xs:element name="operatingConstraintsInterrupt" minOccurs="0">
13400             <xs:complexType>
13401                 <xs:complexContent>
13402                     <xs:restriction base="ns_p:OperatingConstraintsInterruptDataType">
13403                         <xs:sequence>
13404                             <xs:element minOccurs="0" name="isPausable"
13405 type="xs:boolean"/>
13406                             <xs:element minOccurs="0" name="isStoppable"
13407 type="xs:boolean"/>
13408                             <xs:element minOccurs="0" name="maxCyclesPerDay"
13409 type="xs:unsignedInt"/>
13410                         </xs:sequence>
13411                     </xs:restriction>
13412                 </xs:complexContent>
13413             </xs:complexType>
13414         </xs:element>
13415         <xs:element name="operatingConstraintsDuration" minOccurs="0">
13416             <xs:complexType>
13417                 <xs:complexContent>
13418                     <xs:restriction base="ns_p:OperatingConstraintsDurationDataType">
13419                         <xs:sequence>
13420                             <xs:element minOccurs="0" name="activeDurationMin"
13421 type="xs:duration"/>
13422                             <xs:element minOccurs="0" name="activeDurationMax"
13423 type="xs:duration"/>
13424                             <xs:element minOccurs="0" name="pauseDurationMin"
13425 type="xs:duration"/>
13426                             <xs:element minOccurs="0" name="pauseDurationMax"
13427 type="xs:duration"/>
13428                             <xs:element minOccurs="0" name="activeDurationSumMin"
13429 type="xs:duration"/>
13430                             <xs:element minOccurs="0" name="activeDurationSumMax"
13431 type="xs:duration"/>
13432                         </xs:sequence>
13433                     </xs:restriction>
13434                 </xs:complexContent>
13435             </xs:complexType>
13436         </xs:element>
13437         <xs:element name="operatingConstraintsResumeImplication" minOccurs="0">
13438             <xs:complexType>
13439                 <xs:complexContent>
13440                     <xs:restriction
13441 base="ns_p:OperatingConstraintsResumeImplicationDataType">
13442                         <xs:sequence>
13443                             <xs:element minOccurs="0" name="resumeEnergyEstimated"
13444 type="ns_p:ScaledNumberType"/>
13445                             <xs:element minOccurs="0" name="energyUnit"
13446 type="ns_p:UnitOfMeasurementType"/>
13447                             <xs:element minOccurs="0" name="resumeCostEstimated"
13448 type="ns_p:ScaledNumberType"/>
13449                             <xs:element minOccurs="0" name="currency"
13450 type="ns_p:CurrencyType"/>
13451                         </xs:sequence>
13452                     </xs:restriction>
13453                 </xs:complexContent>
13454             </xs:complexType>
13455         </xs:element>
13456         <xs:element maxOccurs="unbounded" minOccurs="0" name="powerTimeSlot"
13457 type="ns_p:SmartEnergyManagementPsPowerTimeSlotType"/>
13458     </xs:sequence>
13459 </xs:complexType>
13460 <xs:complexType name="SmartEnergyManagementPsPowerTimeSlotType">
13461     <xs:sequence>
13462         <xs:element name="schedule" minOccurs="0">
13463             <xs:complexType>
13464                 <xs:complexContent>
13465                     <xs:restriction base="ns_p:PowerTimeSlotScheduleDataType">
13466                         <xs:sequence>
13467                             <xs:element name="slotNumber"
13468 type="ns_p:PowerTimeSlotNumberType" minOccurs="0"/>
13469                             <xs:element name="timePeriod" minOccurs="0">
13470                                 <xs:complexType>
13471                                     <xs:complexContent>
13472                                         <xs:restriction base="ns_p:TimePeriodType">
13473                                             <xs:sequence>
13474                                                 <xs:element name="startTime"
13475 type="xs:duration" minOccurs="0"/>
```

```

13476                                     <xs:element name="endTime"
13477 type="xs:duration" minOccurs="0"/>
13478                                     </xs:sequence>
13479                                     </xs:restriction>
13480                                     </xs:complexContent>
13481                                     </xs:complexType>
13482                                   </xs:element>
13483                                   <xs:element name="defaultDuration" type="xs:duration"
13484 minOccurs="0"/>
13485                                   <xs:element minOccurs="0" name="durationUncertainty"
13486 type="xs:duration"/>
13487                                   <xs:element name="slotActivated" type="xs:boolean"
13488 minOccurs="0"/>
13489                                   <xs:element minOccurs="0" name="description">
13490                                     <xs:simpleType>
13491                                       <xs:restriction base="ns_p:DescriptionType">
13492                                         <xs:minLength value="1"/>
13493                                         <xs:maxLength value="60"/>
13494                                       </xs:restriction>
13495                                     </xs:simpleType>
13496                                   </xs:element>
13497                                 </xs:sequence>
13498                               </xs:restriction>
13499                             </xs:complexContent>
13500                           </xs:complexType>
13501                         </xs:element>
13502                         <xs:element minOccurs="0" name="valueList"
13503 type="ns_p:SmartEnergyManagementPsPowerTimeSlotValueListType"/>
13504                         <xs:element name="scheduleConstraints" minOccurs="0">
13505                           <xs:complexType>
13506                             <xs:complexContent>
13507                               <xs:restriction base="ns_p:PowerTimeSlotScheduleConstraintsDataType">
13508                                 <xs:sequence>
13509                                   <xs:element name="earliestStartTime" type="xs:duration"
13510 minOccurs="0"/>
13511                                   <xs:element name="latestEndTime" type="xs:duration"
13512 minOccurs="0"/>
13513                                   <xs:element minOccurs="0" name="minDuration"
13514 type="xs:duration"/>
13515                                   <xs:element minOccurs="0" name="maxDuration"
13516 type="xs:duration"/>
13517                                   <xs:element minOccurs="0" name="optionalSlot"
13518 type="xs:boolean"/>
13519                                 </xs:sequence>
13520                               </xs:restriction>
13521                             </xs:complexContent>
13522                           </xs:complexType>
13523                         </xs:element>
13524                       </xs:sequence>
13525                     </xs:complexType>
13526                     <xs:complexType name="SmartEnergyManagementPsPowerTimeSlotValueListType">
13527                       <xs:sequence>
13528                         <xs:element maxOccurs="unbounded" minOccurs="0" name="value">
13529                           <xs:complexType>
13530                             <xs:complexContent>
13531                               <xs:restriction base="ns_p:PowerTimeSlotValueDataType">
13532                                 <xs:sequence>
13533                                   <xs:element minOccurs="0" name="valueType"
13534 type="ns_p:PowerTimeSlotValueTypeType"/>
13535                                   <xs:element minOccurs="0" name="value"
13536 type="ns_p:ScaledNumberType"/>
13537                                 </xs:sequence>
13538                               </xs:restriction>
13539                             </xs:complexContent>
13540                           </xs:complexType>
13541                         </xs:element>
13542                       </xs:sequence>
13543                     </xs:complexType>
13544                     <xs:complexType name="SmartEnergyManagementPsDataType">
13545                       <xs:sequence>
13546                         <xs:element name="nodeScheduleInformation" minOccurs="0">
13547                           <xs:complexType>
13548                             <xs:complexContent>
13549                               <xs:restriction
13550 base="ns_p:PowerSequenceNodeScheduleInformationDataType">
13551                                 <xs:sequence>
13552                                   <xs:element name="nodeRemoteControllable" type="xs:boolean"
13553 minOccurs="0"/>

```

```
13554         <xs:element name="supportsSingleSlotSchedulingOnly"
13555 type="xs:boolean" minOccurs="0"/>
13556         <xs:element name="alternativesCount" type="xs:unsignedInt"
13557 minOccurs="0"/>
13558         <xs:element name="totalSequencesCountMax"
13559 type="xs:unsignedInt" minOccurs="0"/>
13560         <xs:element name="supportsReselection" type="xs:boolean"
13561 minOccurs="0"/>
13562     </xs:sequence>
13563 </xs:restriction>
13564 </xs:complexContent>
13565 </xs:complexType>
13566 </xs:element>
13567     <xs:element name="alternatives" maxOccurs="unbounded" minOccurs="0"
13568 type="ns_p:SmartEnergyManagementPsAlternativesType"/>
13569 </xs:sequence>
13570 </xs:complexType>
13571 <xs:element name="smartEnergyManagementPsData"
13572 type="ns_p:SmartEnergyManagementPsDataType"/>
13573 <xs:complexType name="SmartEnergyManagementPsAlternativesRelationElementsType">
13574     <xs:complexContent>
13575         <xs:restriction base="ns_p:PowerSequenceAlternativesRelationDataElementsType">
13576             <xs:sequence>
13577                 <xs:element minOccurs="0" name="alternativesId"
13578 type="ns_p:ElementTagType"/>
13579             </xs:sequence>
13580         </xs:restriction>
13581     </xs:complexContent>
13582 </xs:complexType>
13583 <xs:complexType name="SmartEnergyManagementPsAlternativesElementsType">
13584     <xs:sequence>
13585         <xs:element minOccurs="0" name="relation"
13586 type="ns_p:SmartEnergyManagementPsAlternativesRelationElementsType"/>
13587         <xs:element name="powerSequence" minOccurs="0"
13588 type="ns_p:SmartEnergyManagementPsPowerSequenceElementsType"/>
13589     </xs:sequence>
13590 </xs:complexType>
13591 <xs:complexType name="SmartEnergyManagementPsPowerSequenceElementsType">
13592     <xs:sequence>
13593         <xs:element name="description" minOccurs="0">
13594             <xs:complexType>
13595                 <xs:complexContent>
13596                     <xs:restriction base="ns_p:PowerSequenceDescriptionDataElementsType">
13597                         <xs:sequence>
13598                             <xs:element name="sequenceId" minOccurs="0"
13599 type="ns_p:ElementTagType"/>
13600                             <xs:element minOccurs="0" name="description"
13601 type="ns_p:ElementTagType"/>
13602                             <xs:element minOccurs="0" name="powerUnit"
13603 type="ns_p:ElementTagType"/>
13604                             <xs:element minOccurs="0" name="energyUnit"
13605 type="ns_p:ElementTagType"/>
13606                             <xs:element minOccurs="0" name="valueSource"
13607 type="ns_p:ElementTagType"/>
13608                             <xs:element minOccurs="0" name="taskIdIdentifier"
13609 type="ns_p:ElementTagType"/>
13610                             <xs:element minOccurs="0" name="repetitionsTotal"
13611 type="ns_p:ElementTagType"/>
13612                         </xs:sequence>
13613                     </xs:restriction>
13614                 </xs:complexContent>
13615             </xs:complexType>
13616         </xs:element>
13617         <xs:element name="state" minOccurs="0">
13618             <xs:complexType>
13619                 <xs:complexContent>
13620                     <xs:restriction base="ns_p:PowerSequenceStateDataElementsType">
13621                         <xs:sequence>
13622                             <xs:element name="state" minOccurs="0"
13623 type="ns_p:ElementTagType"/>
13624                             <xs:element minOccurs="0" name="activeSlotNumber"
13625 type="ns_p:ElementTagType"/>
13626                             <xs:element minOccurs="0" name="elapsedSlotTime"
13627 type="ns_p:ElementTagType"/>
13628                             <xs:element minOccurs="0" name="remainingSlotTime"
13629 type="ns_p:ElementTagType"/>
13630                             <xs:element name="sequenceRemoteControllable" minOccurs="0"
13631 type="ns_p:ElementTagType"/>

```

```

13632         <xs:element minOccurs="0" name="activeRepetitionNumber"
13633 type="ns_p:ElementTagType"/>
13634         <xs:element minOccurs="0" name="remainingPauseTime"
13635 type="ns_p:ElementTagType"/>
13636     </xs:sequence>
13637 </xs:restriction>
13638 </xs:complexContent>
13639 </xs:complexType>
13640 </xs:element>
13641 <xs:element minOccurs="0" name="schedule">
13642     <xs:complexType>
13643         <xs:complexContent>
13644             <xs:restriction base="ns_p:PowerSequenceScheduleDataElementsType">
13645                 <xs:sequence>
13646                     <xs:element name="startTime" minOccurs="0"
13647 type="ns_p:ElementTagType"/>
13648                     <xs:element name="endTime" minOccurs="0"
13649 type="ns_p:ElementTagType"/>
13650                 </xs:sequence>
13651             </xs:restriction>
13652         </xs:complexContent>
13653     </xs:complexType>
13654 </xs:element>
13655 <xs:element name="scheduleConstraints" minOccurs="0">
13656     <xs:complexType>
13657         <xs:complexContent>
13658             <xs:restriction
13659 base="ns_p:PowerSequenceScheduleConstraintsDataElementsType">
13660                 <xs:sequence>
13661                     <xs:element name="earliestStartTime" minOccurs="0"
13662 type="ns_p:ElementTagType"/>
13663                     <xs:element name="latestEndTime" minOccurs="0"
13664 type="ns_p:ElementTagType"/>
13665                 </xs:sequence>
13666             </xs:restriction>
13667         </xs:complexContent>
13668     </xs:complexType>
13669 </xs:element>
13670 <xs:element minOccurs="0" name="schedulePreference">
13671     <xs:complexType>
13672         <xs:complexContent>
13673             <xs:restriction
13674 base="ns_p:PowerSequenceSchedulePreferenceDataElementsType">
13675                 <xs:sequence>
13676                     <xs:element minOccurs="0" name="greenest"
13677 type="ns_p:ElementTagType"/>
13678                     <xs:element minOccurs="0" name="cheapest"
13679 type="ns_p:ElementTagType"/>
13680                 </xs:sequence>
13681             </xs:restriction>
13682         </xs:complexContent>
13683     </xs:complexType>
13684 </xs:element>
13685 <xs:element name="operatingConstraintsInterrupt" minOccurs="0">
13686     <xs:complexType>
13687         <xs:complexContent>
13688             <xs:restriction
13689 base="ns_p:OperatingConstraintsInterruptDataElementsType">
13690                 <xs:sequence>
13691                     <xs:element minOccurs="0" name="isPausable"
13692 type="ns_p:ElementTagType"/>
13693                     <xs:element minOccurs="0" name="isStoppable"
13694 type="ns_p:ElementTagType"/>
13695                     <xs:element minOccurs="0" name="maxCyclesPerDay"
13696 type="ns_p:ElementTagType"/>
13697                 </xs:sequence>
13698             </xs:restriction>
13699         </xs:complexContent>
13700     </xs:complexType>
13701 </xs:element>
13702 <xs:element name="operatingConstraintsDuration" minOccurs="0">
13703     <xs:complexType>
13704         <xs:complexContent>
13705             <xs:restriction
13706 base="ns_p:OperatingConstraintsDurationDataElementsType">
13707                 <xs:sequence>
13708                     <xs:element minOccurs="0" name="activeDurationMin"
13709 type="ns_p:ElementTagType"/>

```

```
13710 type="ns_p:ElementTagType"/> <xs:element minOccurs="0" name="activeDurationMax"
13711
13712 type="ns_p:ElementTagType"/> <xs:element minOccurs="0" name="pauseDurationMin"
13713
13714 type="ns_p:ElementTagType"/> <xs:element minOccurs="0" name="pauseDurationMax"
13715
13716 type="ns_p:ElementTagType"/> <xs:element minOccurs="0" name="activeDurationSumMin"
13717
13718 type="ns_p:ElementTagType"/> <xs:element minOccurs="0" name="activeDurationSumMax"
13719
13720 </xs:sequence>
13721 </xs:restriction>
13722 </xs:complexContent>
13723 </xs:complexType>
13724 </xs:element>
13725 <xs:element name="operatingConstraintsResumeImplication" minOccurs="0">
13726 <xs:complexType>
13727 <xs:complexContent>
13728 <xs:restriction
13729 base="ns_p:OperatingConstraintsResumeImplicationDataElementsType">
13730 <xs:sequence>
13731 <xs:element minOccurs="0" name="resumeEnergyEstimated"
13732 type="ns_p:ScaledNumberElementsType"/>
13733 <xs:element minOccurs="0" name="energyUnit"
13734 type="ns_p:ElementTagType"/>
13735 <xs:element minOccurs="0" name="resumeCostEstimated"
13736 type="ns_p:ScaledNumberElementsType"/>
13737 <xs:element minOccurs="0" name="currency"
13738 type="ns_p:ElementTagType"/>
13739 </xs:sequence>
13740 </xs:restriction>
13741 </xs:complexContent>
13742 </xs:complexType>
13743 </xs:element>
13744 <xs:element minOccurs="0" name="powerTimeSlot"
13745 type="ns_p:SmartEnergyManagementPsPowerTimeSlotElementsType"/>
13746 </xs:sequence>
13747 </xs:complexType>
13748 <xs:complexType name="SmartEnergyManagementPsPowerTimeSlotElementsType">
13749 <xs:sequence>
13750 <xs:element name="schedule" minOccurs="0">
13751 <xs:complexType>
13752 <xs:complexContent>
13753 <xs:restriction base="ns_p:PowerTimeSlotScheduleDataElementsType">
13754 <xs:sequence>
13755 <xs:element name="slotNumber" minOccurs="0"
13756 type="ns_p:ElementTagType"/>
13757 <xs:element name="timePeriod" minOccurs="0">
13758 <xs:complexType>
13759 <xs:complexContent>
13760 <xs:restriction
13761 base="ns_p:TimePeriodElementsType">
13762 <xs:sequence>
13763 <xs:element name="startTime" minOccurs="0"
13764 type="ns_p:ElementTagType"/>
13765 <xs:element name="endTime" minOccurs="0"
13766 type="ns_p:ElementTagType"/>
13767 </xs:sequence>
13768 </xs:restriction>
13769 </xs:complexContent>
13770 </xs:complexType>
13771 </xs:element>
13772 <xs:element name="defaultDuration" minOccurs="0"
13773 type="ns_p:ElementTagType"/>
13774 <xs:element minOccurs="0" name="durationUncertainty"
13775 type="ns_p:ElementTagType"/>
13776 <xs:element name="slotActivated" minOccurs="0"
13777 type="ns_p:ElementTagType"/>
13778 <xs:element minOccurs="0" name="description"
13779 type="ns_p:ElementTagType"/>
13780 </xs:sequence>
13781 </xs:restriction>
13782 </xs:complexContent>
13783 </xs:complexType>
13784 </xs:element>
13785 <xs:element name="valueList" minOccurs="0"
13786 type="ns_p:SmartEnergyManagementPsPowerTimeSlotValueListElementsType"/>
13787 <xs:element name="scheduleConstraints" minOccurs="0">
```

```
13788         <xs:complexType>
13789             <xs:complexContent>
13790                 <xs:restriction
13791 base="ns_p:PowerTimeSlotScheduleConstraintsDataElementsType">
13792                     <xs:sequence>
13793                         <xs:element name="earliestStartTime" minOccurs="0"
13794 type="ns_p:ElementTagType"/>
13795                         <xs:element name="latestEndTime" minOccurs="0"
13796 type="ns_p:ElementTagType"/>
13797                         <xs:element minOccurs="0" name="minDuration"
13798 type="ns_p:ElementTagType"/>
13799                         <xs:element minOccurs="0" name="maxDuration"
13800 type="ns_p:ElementTagType"/>
13801                         <xs:element minOccurs="0" name="optionalSlot"
13802 type="ns_p:ElementTagType"/>
13803                     </xs:sequence>
13804                 </xs:restriction>
13805             </xs:complexContent>
13806         </xs:complexType>
13807     </xs:element>
13808 </xs:sequence>
13809 </xs:complexType>
13810 <xs:complexType name="SmartEnergyManagementPsPowerTimeSlotValueListElementsType">
13811     <xs:sequence>
13812         <xs:element name="value" minOccurs="0">
13813             <xs:complexType>
13814                 <xs:complexContent>
13815                     <xs:restriction base="ns_p:PowerTimeSlotValueDataElementsType">
13816                         <xs:sequence>
13817                             <xs:element name="valueType" minOccurs="0"
13818 type="ns_p:ElementTagType"/>
13819                             <xs:element name="value" minOccurs="0"
13820 type="ns_p:ScaledNumberElementsType"/>
13821                         </xs:sequence>
13822                     </xs:restriction>
13823                 </xs:complexContent>
13824             </xs:complexType>
13825         </xs:element>
13826     </xs:sequence>
13827 </xs:complexType>
13828 <xs:complexType name="SmartEnergyManagementPsDataElementsType">
13829     <xs:sequence>
13830         <xs:element name="nodeScheduleInformation" minOccurs="0">
13831             <xs:complexType>
13832                 <xs:complexContent>
13833                     <xs:restriction
13834 base="ns_p:PowerSequenceNodeScheduleInformationDataElementsType">
13835                         <xs:sequence>
13836                             <xs:element name="nodeRemoteControllable" minOccurs="0"
13837 type="ns_p:ElementTagType"/>
13838                             <xs:element name="supportsSingleSlotSchedulingOnly"
13839 minOccurs="0" type="ns_p:ElementTagType"/>
13840                             <xs:element name="alternativesCount" minOccurs="0"
13841 type="ns_p:ElementTagType"/>
13842                             <xs:element name="totalSequencesCountMax" minOccurs="0"
13843 type="ns_p:ElementTagType"/>
13844                             <xs:element name="supportsReselection" minOccurs="0"
13845 type="ns_p:ElementTagType"/>
13846                         </xs:sequence>
13847                     </xs:restriction>
13848                 </xs:complexContent>
13849             </xs:complexType>
13850         </xs:element>
13851         <xs:element name="alternatives" minOccurs="0"
13852 type="ns_p:SmartEnergyManagementPsAlternativesElementsType"/>
13853     </xs:sequence>
13854 </xs:complexType>
13855 <xs:element name="smartEnergyManagementPsDataElements"
13856 type="ns_p:SmartEnergyManagementPsDataElementsType"/>
13857 <xs:complexType name="SmartEnergyManagementPsDataSelectorsType">
13858     <xs:sequence>
13859         <xs:element minOccurs="0" name="alternativesRelation">
13860             <xs:complexType>
13861                 <xs:complexContent>
13862                     <xs:restriction
13863 base="ns_p:PowerSequenceAlternativesRelationListDataSelectorsType">
13864                         <xs:sequence>
```



```
13865         <xs:element minOccurs="0" name="alternativesId"
13866 type="ns_p:AlternativesIdType"/>
13867     </xs:sequence>
13868 </xs:restriction>
13869 </xs:complexContent>
13870 </xs:complexType>
13871 </xs:element>
13872 <xs:element minOccurs="0" name="powerSequenceDescription">
13873     <xs:complexType>
13874         <xs:complexContent>
13875             <xs:restriction
13876 base="ns_p:PowerSequenceDescriptionListDataSelectorsType">
13877                 <xs:sequence>
13878                     <xs:element minOccurs="0" name="sequenceId"
13879 type="ns_p:PowerSequenceIdType"/>
13880                 </xs:sequence>
13881             </xs:restriction>
13882         </xs:complexContent>
13883     </xs:complexType>
13884 </xs:element>
13885 <xs:element minOccurs="0" name="powerTimeSlotSchedule">
13886     <xs:complexType>
13887         <xs:complexContent>
13888             <xs:restriction
13889 base="ns_p:PowerTimeSlotScheduleListDataSelectorsType">
13890                 <xs:sequence>
13891                     <xs:element minOccurs="0" name="slotNumber"
13892 type="ns_p:PowerTimeSlotNumberType"/>
13893                 </xs:sequence>
13894             </xs:restriction>
13895         </xs:complexContent>
13896     </xs:complexType>
13897 </xs:element>
13898 <xs:element minOccurs="0" name="powerTimeSlotValue">
13899     <xs:complexType>
13900         <xs:complexContent>
13901             <xs:restriction base="ns_p:PowerTimeSlotValueListDataSelectorsType">
13902                 <xs:sequence>
13903                     <xs:element minOccurs="0" name="valueType"
13904 type="ns_p:PowerTimeSlotValueTypeType"/>
13905                 </xs:sequence>
13906             </xs:restriction>
13907         </xs:complexContent>
13908     </xs:complexType>
13909 </xs:element>
13910 </xs:sequence>
13911 </xs:complexType>
13912 <xs:element name="smartEnergyManagementPsDataSelectors"
13913 type="ns_p:SmartEnergyManagementPsDataSelectorsType"/>
13914 <xs:complexType name="SmartEnergyManagementPsPriceDataType">
13915     <xs:sequence>
13916         <xs:element maxOccurs="unbounded" minOccurs="0" name="price">
13917             <xs:complexType>
13918                 <xs:complexContent>
13919                     <xs:restriction base="ns_p:PowerSequencePriceDataType">
13920                         <xs:sequence>
13921                             <xs:element minOccurs="0" name="sequenceId"
13922 type="ns_p:PowerSequenceIdType"/>
13923                             <xs:element minOccurs="0" name="price"
13924 type="ns_p:ScaledNumberType"/>
13925                             <xs:element minOccurs="0" name="currency"
13926 type="ns_p:CurrencyType"/>
13927                         </xs:sequence>
13928                     </xs:restriction>
13929                 </xs:complexContent>
13930             </xs:complexType>
13931 </xs:element>
13932 </xs:sequence>
13933 </xs:complexType>
13934 <xs:element name="smartEnergyManagementPsPriceData"
13935 type="ns_p:SmartEnergyManagementPsPriceDataType"/>
13936 <xs:complexType name="SmartEnergyManagementPsPriceDataElementsType">
13937     <xs:sequence>
13938         <xs:element minOccurs="0" name="price">
13939             <xs:complexType>
13940                 <xs:complexContent>
13941                     <xs:restriction base="ns_p:PowerSequencePriceDataElementsType">
13942                         <xs:sequence>
```

```
13943         <xs:element minOccurs="0" name="sequenceId"
13944 type="ns_p:ElementTagType"/>
13945         <xs:element minOccurs="0" name="price"
13946 type="ns_p:ScaledNumberElementsType"/>
13947         <xs:element minOccurs="0" name="currency"
13948 type="ns_p:ElementTagType"/>
13949     </xs:sequence>
13950 </xs:restriction>
13951 </xs:complexContent>
13952 </xs:complexType>
13953 </xs:element>
13954 </xs:sequence>
13955 </xs:complexType>
13956 <xs:element name="smartEnergyManagementPsPriceDataElements"
13957 type="ns_p:SmartEnergyManagementPsPriceDataElementsType"/>
13958 <xs:complexType name="SmartEnergyManagementPsPriceDataSelectorsType">
13959     <xs:sequence>
13960         <xs:element minOccurs="0" name="price">
13961             <xs:complexType>
13962                 <xs:complexContent>
13963                     <xs:restriction base="ns_p:PowerSequencePriceListDataSelectorsType">
13964                         <xs:sequence>
13965                             <xs:element minOccurs="0" name="sequenceId"
13966 type="ns_p:PowerSequenceIdType"/>
13967                         </xs:sequence>
13968                     </xs:restriction>
13969                 </xs:complexContent>
13970             </xs:complexType>
13971         </xs:element>
13972     </xs:sequence>
13973 </xs:complexType>
13974 <xs:element name="smartEnergyManagementPsPriceDataSelectors"
13975 type="ns_p:SmartEnergyManagementPsPriceDataSelectorsType"/>
13976 <xs:complexType name="SmartEnergyManagementPsConfigurationRequestCallType">
13977     <xs:sequence>
13978         <xs:element minOccurs="0" name="scheduleConfigurationRequest">
13979             <xs:complexType>
13980                 <xs:complexContent>
13981                     <xs:restriction
13982 base="ns_p:PowerSequenceScheduleConfigurationRequestCallType">
13983                         <xs:sequence>
13984                             <xs:element minOccurs="0" name="sequenceId"
13985 type="ns_p:PowerSequenceIdType"/>
13986                         </xs:sequence>
13987                     </xs:restriction>
13988                 </xs:complexContent>
13989             </xs:complexType>
13990         </xs:element>
13991     </xs:sequence>
13992 </xs:complexType>
13993 <xs:element name="smartEnergyManagementPsConfigurationRequestCall"
13994 type="ns_p:SmartEnergyManagementPsConfigurationRequestCallType"/>
13995 <xs:complexType name="SmartEnergyManagementPsConfigurationRequestCallElementsType">
13996     <xs:sequence>
13997         <xs:element minOccurs="0" name="scheduleConfigurationRequest">
13998             <xs:complexType>
13999                 <xs:complexContent>
14000                     <xs:restriction
14001 base="ns_p:PowerSequenceScheduleConfigurationRequestCallElementsType">
14002                         <xs:sequence>
14003                             <xs:element minOccurs="0" name="sequenceId"
14004 type="ns_p:ElementTagType"/>
14005                         </xs:sequence>
14006                     </xs:restriction>
14007                 </xs:complexContent>
14008             </xs:complexType>
14009         </xs:element>
14010     </xs:sequence>
14011 </xs:complexType>
14012 <xs:element name="smartEnergyManagementPsConfigurationRequestCallElements"
14013 type="ns_p:SmartEnergyManagementPsConfigurationRequestCallElementsType"/>
14014 <xs:complexType name="SmartEnergyManagementPsPriceCalculationRequestCallType">
14015     <xs:sequence>
14016         <xs:element minOccurs="0" name="priceCalculationRequest">
14017             <xs:complexType>
14018                 <xs:complexContent>
14019                     <xs:restriction
14020 base="ns_p:PowerSequencePriceCalculationRequestCallType">
```

```
14021         <xs:sequence>
14022             <xs:element minOccurs="0" name="sequenceId"
14023 type="ns_p:PowerSequenceIdType"/>
14024         </xs:sequence>
14025     </xs:restriction>
14026 </xs:complexType>
14027 </xs:complexType>
14028 </xs:element>
14029 </xs:sequence>
14030 </xs:complexType>
14031 <xs:element name="smartEnergyManagementPsPriceCalculationRequestCall"
14032 type="ns_p:SmartEnergyManagementPsPriceCalculationRequestCallType"/>
14033 <xs:complexType name="SmartEnergyManagementPsPriceCalculationRequestCallElementsType">
14034     <xs:sequence>
14035         <xs:element minOccurs="0" name="priceCalculationRequest">
14036             <xs:complexType>
14037                 <xs:complexContent>
14038                     <xs:restriction
14039 base="ns_p:PowerSequencePriceCalculationRequestCallElementsType">
14040                         <xs:sequence>
14041                             <xs:element minOccurs="0" name="sequenceId"
14042 type="ns_p:ElementTagType"/>
14043                         </xs:sequence>
14044                     </xs:restriction>
14045                 </xs:complexContent>
14046             </xs:complexType>
14047         </xs:element>
14048     </xs:sequence>
14049 </xs:complexType>
14050 <xs:element name="smartEnergyManagementPsPriceCalculationRequestCallElements"
14051 type="ns_p:SmartEnergyManagementPsPriceCalculationRequestCallElementsType"/>
14052 </xs:schema>
```

A.2 Standard Classes

A.2.1 ActuatorLevel

File name: EEBus_SPINE_TS_ActuatorLevel.xsd

```
14059 <?xml version="1.0" encoding="UTF-8"?>
14060 <!--
14061     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14062     Version 1.3.0 (final)
14063     2023-08-31
14064     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14065     Source: https://www.eebus.org/en/specifications/
14066 -->
14067 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14068 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14069 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14070 blockDefault="#all" elementFormDefault="qualified">
14071     <xs:annotation>
14072         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14073 e.V. All rights reserved.</xs:documentation>
14074     </xs:annotation>
14075     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14076     <xs:simpleType name="ActuatorLevelFctType">
14077         <xs:union memberTypes="ns_p:ActuatorLevelFctEnumType ns_p:EnumExtendType"/>
14078     </xs:simpleType>
14079     <xs:simpleType name="ActuatorLevelFctEnumType">
14080         <xs:restriction base="xs:string">
14081             <xs:enumeration value="start"/>
14082             <xs:enumeration value="up"/>
14083             <xs:enumeration value="down"/>
14084             <xs:enumeration value="stop"/>
14085             <xs:enumeration value="percentageAbsolute"/>
14086             <xs:enumeration value="percentageRelative"/>
14087             <xs:enumeration value="absolute"/>
14088             <xs:enumeration value="relative"/>
14089         </xs:restriction>
14090     </xs:simpleType>
```

```

14092     <xs:complexType name="ActuatorLevelDataType">
14093       <xs:sequence>
14094         <xs:element name="function" type="ns_p:ActuatorLevelFctType" minOccurs="0"/>
14095         <xs:element name="value" type="ns_p:ScaledNumberType" minOccurs="0"/>
14096       </xs:sequence>
14097     </xs:complexType>
14098     <xs:element name="actuatorLevelData" type="ns_p:ActuatorLevelDataType"/>
14099     <xs:complexType name="ActuatorLevelDataElementsType">
14100       <xs:sequence>
14101         <xs:element name="function" minOccurs="0" type="ns_p:ElementTagType"/>
14102         <xs:element name="value" type="ns_p:ScaledNumberElementsType" minOccurs="0"/>
14103       </xs:sequence>
14104     </xs:complexType>
14105     <xs:element name="actuatorLevelDataElements" type="ns_p:ActuatorLevelDataElementsType"/>
14106     <xs:complexType name="ActuatorLevelDescriptionDataType">
14107       <xs:sequence>
14108         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14109         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
14110         <xs:element minOccurs="0" name="levelDefaultUnit"
14111 type="ns_p:UnitOfMeasurementType"/>
14112       </xs:sequence>
14113     </xs:complexType>
14114     <xs:element name="actuatorLevelDescriptionData"
14115 type="ns_p:ActuatorLevelDescriptionDataType"/>
14116     <xs:complexType name="ActuatorLevelDescriptionDataElementsType">
14117       <xs:sequence>
14118         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
14119         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
14120         <xs:element minOccurs="0" name="levelDefaultUnit" type="ns_p:ElementTagType"/>
14121       </xs:sequence>
14122     </xs:complexType>
14123     <xs:element name="actuatorLevelDescriptionDataElements"
14124 type="ns_p:ActuatorLevelDescriptionDataElementsType"/>
14125   </xs:schema>
14126
14127

```

14128 A.2.2 ActuatorSwitch

14129 File name: EEBus_SPINE_TS_ActuatorSwitch.xsd

```

14131 <?xml version="1.0" encoding="UTF-8"?>
14132 <!--
14133   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14134   Version 1.3.0 (final)
14135   2023-08-31
14136   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14137   Source: https://www.eebus.org/en/specifications/
14138 -->
14139 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14140 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14141 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14142 blockDefault="#all" elementFormDefault="qualified">
14143   <xs:annotation>
14144     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14145 e.V. All rights reserved.</xs:documentation>
14146   </xs:annotation>
14147   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14148   <xs:simpleType name="ActuatorSwitchFctType">
14149     <xs:union memberTypes="ns_p:ActuatorSwitchFctEnumType ns_p:EnumExtendType"/>
14150   </xs:simpleType>
14151   <xs:simpleType name="ActuatorSwitchFctEnumType">
14152     <xs:restriction base="xs:string">
14153       <xs:enumeration value="on"/>
14154       <xs:enumeration value="off"/>
14155       <xs:enumeration value="toggle"/>
14156     </xs:restriction>
14157   </xs:simpleType>
14158   <xs:complexType name="ActuatorSwitchDataType">
14159     <xs:sequence>
14160       <xs:element name="function" type="ns_p:ActuatorSwitchFctType" minOccurs="0"/>
14161     </xs:sequence>
14162   </xs:complexType>
14163   <xs:element name="actuatorSwitchData" type="ns_p:ActuatorSwitchDataType"/>
14164

```

```
14165     <xs:complexType name="ActuatorSwitchDataElementsType">
14166       <xs:sequence>
14167         <xs:element name="function" minOccurs="0" type="ns_p:ElementTagType"/>
14168       </xs:sequence>
14169     </xs:complexType>
14170     <xs:element name="actuatorSwitchDataElements" type="ns_p:ActuatorSwitchDataElementsType"/>
14171     <xs:complexType name="ActuatorSwitchDescriptionDataType">
14172       <xs:sequence>
14173         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14174         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
14175       </xs:sequence>
14176     </xs:complexType>
14177     <xs:element name="actuatorSwitchDescriptionData"
14178 type="ns_p:ActuatorSwitchDescriptionDataType"/>
14179     <xs:complexType name="ActuatorSwitchDescriptionDataElementsType">
14180       <xs:sequence>
14181         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
14182         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
14183       </xs:sequence>
14184     </xs:complexType>
14185     <xs:element name="actuatorSwitchDescriptionDataElements"
14186 type="ns_p:ActuatorSwitchDescriptionDataElementsType"/>
14187   </xs:schema>
14188
14189
```

14190

14191 A.2.3 Alarm

14192 File name: EEBus_SPINE_TS_Alarm.xsd

```
14193
14194 <?xml version="1.0" encoding="UTF-8"?>
14195 <!--
14196   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14197   Version 1.3.0 (final)
14198   2023-08-31
14199   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14200   Source: https://www.eebus.org/en/specifications/
14201 -->
14202 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14203 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14204 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14205 blockDefault="#all" elementFormDefault="qualified">
14206   <xs:annotation>
14207     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14208 e.V. All rights reserved.</xs:documentation>
14209   </xs:annotation>
14210   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14211   <xs:include schemaLocation="EEBus_SPINE_TS_Threshold.xsd"/>
14212   <xs:simpleType name="AlarmIdType">
14213     <xs:restriction base="xs:unsignedInt"/>
14214   </xs:simpleType>
14215   <xs:simpleType name="AlarmTypeType">
14216     <xs:union memberTypes="ns_p:AlarmTypeEnumType ns_p:EnumExtendType"/>
14217   </xs:simpleType>
14218   <xs:simpleType name="AlarmTypeEnumType">
14219     <xs:restriction base="xs:string">
14220       <xs:enumeration value="alarmCancelled"/>
14221       <xs:enumeration value="underThreshold"/>
14222       <xs:enumeration value="overThreshold"/>
14223     </xs:restriction>
14224   </xs:simpleType>
14225   <xs:complexType name="AlarmDataType">
14226     <xs:sequence>
14227       <xs:element name="alarmId" minOccurs="0" type="ns_p:AlarmIdType"/>
14228       <xs:element minOccurs="0" name="thresholdId" type="ns_p:ThresholdIdType"/>
14229       <xs:element minOccurs="0" name="timestamp"
14230 type="ns_p:AbsoluteOrRelativeTimeType"/>
14231       <xs:element name="alarmType" minOccurs="0" type="ns_p:AlarmTypeType"/>
14232       <xs:element minOccurs="0" name="measuredValue" type="ns_p:ScaledNumberType"/>
14233       <xs:element minOccurs="0" name="evaluationPeriod" type="ns_p:TimePeriodType"/>
14234       <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
14235       <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14236       <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
14237     </xs:sequence>

```

```

14238     </xs:complexType>
14239     <xs:element name="alarmData" type="ns_p:AlarmDataType"/>
14240     <xs:complexType name="AlarmDataElementsType">
14241         <xs:sequence>
14242             <xs:element name="alarmId" minOccurs="0" type="ns_p:ElementTagType"/>
14243             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ElementTagType"/>
14244             <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
14245             <xs:element name="alarmType" minOccurs="0" type="ns_p:ElementTagType"/>
14246             <xs:element minOccurs="0" name="measuredValue"
type="ns_p:ScaledNumberElementsType"/>
14247             <xs:element minOccurs="0" name="evaluationPeriod"
type="ns_p:TimePeriodElementsType"/>
14248             <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
14249             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
14250             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
14251         </xs:sequence>
14252     </xs:complexType>
14253     <xs:element name="alarmDataElements" type="ns_p:AlarmDataElementsType"/>
14254     <xs:complexType name="AlarmListDataType">
14255         <xs:sequence>
14256             <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:alarmData"/>
14257         </xs:sequence>
14258     </xs:complexType>
14259     <xs:element name="alarmListData" type="ns_p:AlarmListDataType"/>
14260     <xs:complexType name="AlarmListDataSelectorsType">
14261         <xs:sequence>
14262             <xs:element name="alarmId" minOccurs="0" type="ns_p:AlarmIdType"/>
14263             <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
14264         </xs:sequence>
14265     </xs:complexType>
14266     <xs:element name="alarmListDataSelectors" type="ns_p:AlarmListDataSelectorsType"/>
14267 </xs:schema>
14268
14269
14270
14271

```

A.2.4 Bill

File name: EEBus_SPINE_TS_Bill.xsd

```

14275 <?xml version="1.0" encoding="UTF-8"?>
14276 <!--
14277     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14278     Version 1.3.0 (final)
14279     2023-08-31
14280     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14281     Source: https://www.eebus.org/en/specifications/
14282 -->
14283 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14284     xmlns:xs="http://www.w3.org/2001/XMLSchema"
14285     targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14286     blockDefault="#all" elementFormDefault="qualified">
14287     <xs:annotation>
14288         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14289 e.V. All rights reserved.</xs:documentation>
14290     </xs:annotation>
14291     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14292     <xs:include schemaLocation="EEBus_SPINE_TS_Identification.xsd"/>
14293     <xs:simpleType name="BillIdType">
14294         <xs:restriction base="xs:unsignedInt"/>
14295     </xs:simpleType>
14296     <xs:simpleType name="BillTypeType">
14297         <xs:union memberTypes="ns_p:BillTypeEnumType ns_p:EnumExtendType"/>
14298     </xs:simpleType>
14299     <xs:simpleType name="BillTypeEnumType">
14300         <xs:restriction base="xs:string">
14301             <xs:enumeration value="chargingSummary"/>
14302         </xs:restriction>
14303     </xs:simpleType>
14304     <xs:simpleType name="BillPositionIdType">
14305         <xs:restriction base="xs:unsignedInt"/>
14306     </xs:simpleType>
14307     <xs:simpleType name="BillPositionCountType">
14308         <xs:restriction base="ns_p:BillPositionIdType"/>
14309     </xs:simpleType>
14310

```

```
14311 <xs:simpleType name="BillPositionTypeType">
14312   <xs:union memberTypes="ns_p:BillPositionTypeEnumType ns_p:EnumExtendType"/>
14313 </xs:simpleType>
14314 <xs:simpleType name="BillPositionTypeEnumType">
14315   <xs:restriction base="xs:string">
14316     <xs:enumeration value="gridElectricEnergy"/>
14317     <xs:enumeration value="selfProducedElectricEnergy"/>
14318   </xs:restriction>
14319 </xs:simpleType>
14320 <xs:simpleType name="BillValueIdType">
14321   <xs:restriction base="xs:unsignedInt"/>
14322 </xs:simpleType>
14323 <xs:simpleType name="BillCostIdType">
14324   <xs:restriction base="xs:unsignedInt"/>
14325 </xs:simpleType>
14326 <xs:simpleType name="BillCostTypeType">
14327   <xs:union memberTypes="ns_p:BillCostTypeEnumType ns_p:EnumExtendType"/>
14328 </xs:simpleType>
14329 <xs:simpleType name="BillCostTypeEnumType">
14330   <xs:restriction base="xs:string">
14331     <xs:enumeration value="absolutePrice"/>
14332     <xs:enumeration value="relativePrice"/>
14333     <xs:enumeration value="co2Emission"/>
14334     <xs:enumeration value="renewableEnergy"/>
14335     <xs:enumeration value="radioactiveWaste"/>
14336   </xs:restriction>
14337 </xs:simpleType>
14338 <xs:complexType name="BillValueType">
14339   <xs:sequence>
14340     <xs:element minOccurs="0" name="valueId" type="ns_p:BillValueIdType"/>
14341     <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
14342     <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
14343     <xs:element minOccurs="0" name="valuePercentage" type="ns_p:ScaledNumberType"/>
14344   </xs:sequence>
14345 </xs:complexType>
14346 <xs:complexType name="BillValueElementsType">
14347   <xs:sequence>
14348     <xs:element minOccurs="0" name="valueId" type="ns_p:ElementTagType"/>
14349     <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
14350     <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
14351     <xs:element minOccurs="0" name="valuePercentage"/>
14352 type="ns_p:ScaledNumberElementsType"/>
14353   </xs:sequence>
14354 </xs:complexType>
14355 <xs:complexType name="BillCostType">
14356   <xs:sequence>
14357     <xs:element minOccurs="0" name="costId" type="ns_p:BillCostIdType"/>
14358     <xs:element minOccurs="0" name="costType" type="ns_p:BillCostTypeType"/>
14359     <xs:element minOccurs="0" name="valueId" type="ns_p:BillValueIdType"/>
14360     <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
14361     <xs:element minOccurs="0" name="currency" type="ns_p:CurrencyType"/>
14362     <xs:element minOccurs="0" name="cost" type="ns_p:ScaledNumberType"/>
14363     <xs:element minOccurs="0" name="costPercentage" type="ns_p:ScaledNumberType"/>
14364   </xs:sequence>
14365 </xs:complexType>
14366 <xs:complexType name="BillCostElementsType">
14367   <xs:sequence>
14368     <xs:element minOccurs="0" name="costId" type="ns_p:ElementTagType"/>
14369     <xs:element minOccurs="0" name="costType" type="ns_p:ElementTagType"/>
14370     <xs:element minOccurs="0" name="valueId" type="ns_p:ElementTagType"/>
14371     <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
14372     <xs:element minOccurs="0" name="currency" type="ns_p:ElementTagType"/>
14373     <xs:element minOccurs="0" name="cost" type="ns_p:ScaledNumberElementsType"/>
14374     <xs:element minOccurs="0" name="costPercentage"/>
14375 type="ns_p:ScaledNumberElementsType"/>
14376   </xs:sequence>
14377 </xs:complexType>
14378 <xs:complexType name="BillPositionType">
14379   <xs:sequence>
14380     <xs:element minOccurs="0" name="positionId" type="ns_p:BillPositionIdType"/>
14381     <xs:element minOccurs="0" name="positionType" type="ns_p:BillPositionTypeType"/>
14382     <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
14383     <xs:element maxOccurs="unbounded" minOccurs="0" name="value"
14384 type="ns_p:BillValueType"/>
14385     <xs:element minOccurs="0" name="cost" maxOccurs="unbounded"
14386 type="ns_p:BillCostType"/>
14387     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14388     <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
```

```

14389         </xs:sequence>
14390     </xs:complexType>
14391     <xs:complexType name="BillPositionElementsType">
14392         <xs:sequence>
14393             <xs:element minOccurs="0" name="positionId" type="ns_p:ElementTagType"/>
14394             <xs:element minOccurs="0" name="positionType" type="ns_p:ElementTagType"/>
14395             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
14396             <xs:element minOccurs="0" name="value" type="ns_p:BillValueElementsType"/>
14397             <xs:element minOccurs="0" name="cost" type="ns_p:BillCostElementsType"/>
14398             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
14399             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
14400         </xs:sequence>
14401     </xs:complexType>
14402     <xs:complexType name="BillDataType">
14403         <xs:sequence>
14404             <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14405             <xs:element minOccurs="0" name="billType" type="ns_p:BillTypeType"/>
14406             <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
14407             <xs:element minOccurs="0" name="total">
14408                 <xs:complexType>
14409                     <xs:complexContent>
14410                         <xs:restriction base="ns_p:BillPositionType">
14411                             <xs:sequence>
14412                                 <xs:element minOccurs="0" name="timePeriod"
14413 type="ns_p:TimePeriodType"/>
14414                                 <xs:element maxOccurs="unbounded" minOccurs="0" name="value">
14415                                     <xs:complexType>
14416                                         <xs:complexContent>
14417                                             <xs:restriction base="ns_p:BillValueType">
14418                                                 <xs:sequence>
14419                                                     <xs:element minOccurs="0" name="valueId"
14420 type="ns_p:BillValueIdType"/>
14421                                                     <xs:element minOccurs="0" name="unit"
14422 type="ns_p:UnitOfMeasurementType"/>
14423                                                     <xs:element minOccurs="0" name="value"
14424 type="ns_p:ScaledNumberType"/>
14425                                                 </xs:sequence>
14426                                             </xs:restriction>
14427                                         </xs:complexContent>
14428                                     </xs:complexType>
14429                                 </xs:element>
14430                                 <xs:element minOccurs="0" name="cost" maxOccurs="unbounded">
14431                                     <xs:complexType>
14432                                         <xs:complexContent>
14433                                             <xs:restriction base="ns_p:BillCostType">
14434                                                 <xs:sequence>
14435                                                     <xs:element minOccurs="0" name="costId"
14436 type="ns_p:BillCostIdType"/>
14437                                                     <xs:element minOccurs="0" name="costType"
14438 type="ns_p:BillCostTypeType"/>
14439                                                     <xs:element minOccurs="0" name="valueId"
14440 type="ns_p:BillValueIdType"/>
14441                                                     <xs:element minOccurs="0" name="unit"
14442 type="ns_p:UnitOfMeasurementType"/>
14443                                                     <xs:element minOccurs="0" name="currency"
14444 type="ns_p:CurrencyType"/>
14445                                                     <xs:element minOccurs="0" name="cost"
14446 type="ns_p:ScaledNumberType"/>
14447                                                 </xs:sequence>
14448                                             </xs:restriction>
14449                                         </xs:complexContent>
14450                                     </xs:complexType>
14451                                 </xs:element>
14452                                 <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14453                                 <xs:element minOccurs="0" name="description"
14454 type="ns_p:DescriptionType"/>
14455                             </xs:sequence>
14456                         </xs:restriction>
14457                     </xs:complexContent>
14458                 </xs:complexType>
14459             </xs:element>
14460             <xs:element maxOccurs="unbounded" minOccurs="0" name="position">
14461                 <xs:complexType>
14462                     <xs:complexContent>
14463                         <xs:restriction base="ns_p:BillPositionType">
14464                             <xs:sequence>
14465                                 <xs:element minOccurs="0" name="positionId"
14466 type="ns_p:BillPositionIdType"/>

```



```
14467         <xs:element minOccurs="0" name="positionType"
14468 type="ns_p:BillPositionTypeType"/>
14469         <xs:element minOccurs="0" name="timePeriod"
14470 type="ns_p:TimePeriodType"/>
14471         <xs:element maxOccurs="unbounded" minOccurs="0" name="value">
14472           <xs:complexType>
14473             <xs:complexContent>
14474               <xs:restriction base="ns_p:BillValueType">
14475                 <xs:sequence>
14476                   <xs:element minOccurs="0" name="valueId"
14477 type="ns_p:BillValueIdType"/>
14478                   <xs:element minOccurs="0" name="value"
14479 type="ns_p:ScaledNumberType"/>
14480                   <xs:element minOccurs="0"
14481 name="valuePercentage" type="ns_p:ScaledNumberType"/>
14482                 </xs:sequence>
14483               </xs:restriction>
14484             </xs:complexContent>
14485           </xs:complexType>
14486         </xs:element>
14487         <xs:element minOccurs="0" name="cost" maxOccurs="unbounded">
14488           <xs:complexType>
14489             <xs:complexContent>
14490               <xs:restriction base="ns_p:BillCostType">
14491                 <xs:sequence>
14492                   <xs:element minOccurs="0" name="costId"
14493 type="ns_p:BillCostIdType"/>
14494                   <xs:element minOccurs="0" name="cost"
14495 type="ns_p:ScaledNumberType"/>
14496                   <xs:element minOccurs="0"
14497 name="costPercentage" type="ns_p:ScaledNumberType"/>
14498                 </xs:sequence>
14499               </xs:restriction>
14500             </xs:complexContent>
14501           </xs:complexType>
14502         </xs:element>
14503         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14504         <xs:element minOccurs="0" name="description"
14505 type="ns_p:DescriptionType"/>
14506       </xs:sequence>
14507     </xs:restriction>
14508   </xs:complexContent>
14509 </xs:complexType>
14510 </xs:element>
14511 </xs:sequence>
14512 </xs:complexType>
14513 <xs:element name="billData" type="ns_p:BillDataType"/>
14514 <xs:complexType name="BillDataElementsType">
14515   <xs:sequence>
14516     <xs:element minOccurs="0" name="billId" type="ns_p:ElementTagType"/>
14517     <xs:element minOccurs="0" name="billType" type="ns_p:ElementTagType"/>
14518     <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
14519     <xs:element minOccurs="0" name="total">
14520       <xs:complexType>
14521         <xs:complexContent>
14522           <xs:restriction base="ns_p:BillPositionElementsType">
14523             <xs:sequence>
14524               <xs:element minOccurs="0" name="timePeriod"
14525 type="ns_p:TimePeriodElementsType"/>
14526               <xs:element minOccurs="0" name="value">
14527                 <xs:complexType>
14528                   <xs:complexContent>
14529                     <xs:restriction base="ns_p:BillValueElementsType">
14530                       <xs:sequence>
14531                         <xs:element minOccurs="0" name="valueId"
14532 type="ns_p:ElementTagType"/>
14533                         <xs:element minOccurs="0" name="unit"
14534 type="ns_p:ElementTagType"/>
14535                         <xs:element minOccurs="0" name="value"
14536 type="ns_p:ScaledNumberElementsType"/>
14537                       </xs:sequence>
14538                     </xs:restriction>
14539                   </xs:complexContent>
14540                 </xs:complexType>
14541               </xs:element>
14542               <xs:element minOccurs="0" name="cost">
14543                 <xs:complexType>
14544                   <xs:complexContent>
```

```
14545         <xs:restriction base="ns_p:BillCostElementsType">
14546             <xs:sequence>
14547                 <xs:element minOccurs="0" name="costId"
14548 type="ns_p:ElementTagType"/>
14549                 <xs:element minOccurs="0" name="costType"
14550 type="ns_p:ElementTagType"/>
14551                 <xs:element minOccurs="0" name="valueId"
14552 type="ns_p:ElementTagType"/>
14553                 <xs:element minOccurs="0" name="unit"
14554 type="ns_p:ElementTagType"/>
14555                 <xs:element minOccurs="0" name="currency"
14556 type="ns_p:ElementTagType"/>
14557                 <xs:element minOccurs="0" name="cost"
14558 type="ns_p:ScaledNumberElementsType"/>
14559             </xs:sequence>
14560         </xs:restriction>
14561     </xs:complexContent>
14562 </xs:complexType>
14563 </xs:element>
14564 <xs:element minOccurs="0" name="label"
14565 type="ns_p:ElementTagType"/>
14566 <xs:element minOccurs="0" name="description"
14567 type="ns_p:ElementTagType"/>
14568 </xs:sequence>
14569 </xs:restriction>
14570 </xs:complexContent>
14571 </xs:complexType>
14572 </xs:element>
14573 <xs:element minOccurs="0" name="position">
14574     <xs:complexType>
14575         <xs:complexContent>
14576             <xs:restriction base="ns_p:BillPositionElementsType">
14577                 <xs:sequence>
14578                     <xs:element minOccurs="0" name="positionId"
14579 type="ns_p:ElementTagType"/>
14580                     <xs:element minOccurs="0" name="positionType"
14581 type="ns_p:ElementTagType"/>
14582                     <xs:element minOccurs="0" name="timePeriod"
14583 type="ns_p:TimePeriodElementsType"/>
14584                     <xs:element minOccurs="0" name="value">
14585                         <xs:complexType>
14586                             <xs:complexContent>
14587                                 <xs:restriction base="ns_p:BillValueElementsType">
14588                                     <xs:sequence>
14589                                         <xs:element minOccurs="0" name="valueId"
14590 type="ns_p:ElementTagType"/>
14591                                         <xs:element minOccurs="0" name="value"
14592 type="ns_p:ScaledNumberElementsType"/>
14593                                         <xs:element minOccurs="0"
14594 name="valuePercentage" type="ns_p:ScaledNumberElementsType"/>
14595                                     </xs:sequence>
14596                                 </xs:restriction>
14597                             </xs:complexContent>
14598                         </xs:complexType>
14599                     </xs:element>
14600                     <xs:element minOccurs="0" name="cost">
14601                         <xs:complexType>
14602                             <xs:complexContent>
14603                                 <xs:restriction base="ns_p:BillCostElementsType">
14604                                     <xs:sequence>
14605                                         <xs:element minOccurs="0" name="costId"
14606 type="ns_p:ElementTagType"/>
14607                                         <xs:element minOccurs="0" name="cost"
14608 type="ns_p:ScaledNumberElementsType"/>
14609                                         <xs:element minOccurs="0"
14610 name="costPercentage" type="ns_p:ScaledNumberElementsType"/>
14611                                     </xs:sequence>
14612                                 </xs:restriction>
14613                             </xs:complexContent>
14614                         </xs:complexType>
14615                     </xs:element>
14616                     <xs:element minOccurs="0" name="label"
14617 type="ns_p:ElementTagType"/>
14618                     <xs:element minOccurs="0" name="description"
14619 type="ns_p:ElementTagType"/>
14620                 </xs:sequence>
14621             </xs:restriction>
14622         </xs:complexContent>
```

```
14623         </xs:complexType>
14624     </xs:element>
14625 </xs:sequence>
14626 </xs:complexType>
14627 <xs:element name="billDataElements" type="ns_p:BillDataElementsType"/>
14628 <xs:complexType name="BillListDataType">
14629     <xs:sequence>
14630         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:billData"/>
14631     </xs:sequence>
14632 </xs:complexType>
14633 <xs:element name="billListData" type="ns_p:BillListDataType"/>
14634 <xs:complexType name="BillListDataSelectorsType">
14635     <xs:sequence>
14636         <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14637         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
14638     </xs:sequence>
14639 </xs:complexType>
14640 <xs:element name="billListDataSelectors" type="ns_p:BillListDataSelectorsType"/>
14641 <xs:complexType name="BillConstraintsDataType">
14642     <xs:sequence>
14643         <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14644         <xs:element minOccurs="0" name="positionCountMin"
14645 type="ns_p:BillPositionCountType"/>
14646         <xs:element minOccurs="0" name="positionCountMax"
14647 type="ns_p:BillPositionCountType"/>
14648     </xs:sequence>
14649 </xs:complexType>
14650 <xs:element name="billConstraintsData" type="ns_p:BillConstraintsDataType"/>
14651 <xs:complexType name="BillConstraintsDataElementsType">
14652     <xs:sequence>
14653         <xs:element minOccurs="0" name="billId" type="ns_p:ElementTagType"/>
14654         <xs:element minOccurs="0" name="positionCountMin" type="ns_p:ElementTagType"/>
14655         <xs:element minOccurs="0" name="positionCountMax" type="ns_p:ElementTagType"/>
14656     </xs:sequence>
14657 </xs:complexType>
14658 <xs:element name="billConstraintsDataElements"
14659 type="ns_p:BillConstraintsDataElementsType"/>
14660 <xs:complexType name="BillConstraintsListDataType">
14661     <xs:sequence>
14662         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:billConstraintsData"/>
14663     </xs:sequence>
14664 </xs:complexType>
14665 <xs:element name="billConstraintsListData" type="ns_p:BillConstraintsListDataType"/>
14666 <xs:complexType name="BillConstraintsListDataSelectorsType">
14667     <xs:sequence>
14668         <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14669     </xs:sequence>
14670 </xs:complexType>
14671 <xs:element name="billConstraintsListDataSelectors"
14672 type="ns_p:BillConstraintsListDataSelectorsType"/>
14673 <xs:complexType name="BillDescriptionDataType">
14674     <xs:sequence>
14675         <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14676         <xs:element minOccurs="0" name="billWriteable" type="xs:boolean"/>
14677         <xs:element minOccurs="0" name="updateRequired" type="xs:boolean"/>
14678         <xs:element maxOccurs="unbounded" minOccurs="0" name="supportedBillType"
14679 type="ns_p:BillTypeType"/>
14680         <xs:element minOccurs="0" name="sessionId" type="ns_p:SessionIdType"/>
14681     </xs:sequence>
14682 </xs:complexType>
14683 <xs:element name="billDescriptionData" type="ns_p:BillDescriptionDataType"/>
14684 <xs:complexType name="BillDescriptionDataElementsType">
14685     <xs:sequence>
14686         <xs:element minOccurs="0" name="billId" type="ns_p:ElementTagType"/>
14687         <xs:element minOccurs="0" name="billWriteable" type="ns_p:ElementTagType"/>
14688         <xs:element minOccurs="0" name="updateRequired" type="ns_p:ElementTagType"/>
14689         <xs:element minOccurs="0" name="supportedBillType" type="ns_p:ElementTagType"/>
14690         <xs:element minOccurs="0" name="sessionId" type="ns_p:ElementTagType"/>
14691     </xs:sequence>
14692 </xs:complexType>
14693 <xs:element name="billDescriptionDataElements"
14694 type="ns_p:BillDescriptionDataElementsType"/>
14695 <xs:complexType name="BillDescriptionListDataType">
14696     <xs:sequence>
14697         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:billDescriptionData"/>
14698     </xs:sequence>
14699 </xs:complexType>
14700 <xs:element name="billDescriptionListData" type="ns_p:BillDescriptionListDataType"/>
```

```
14701     <xs:complexType name="BillDescriptionListDataSelectorsType">
14702       <xs:sequence>
14703         <xs:element minOccurs="0" name="billId" type="ns_p:BillIdType"/>
14704       </xs:sequence>
14705     </xs:complexType>
14706     <xs:element name="billDescriptionListDataSelectors"
14707 type="ns_p:BillDescriptionListDataSelectorsType"/>
14708 </xs:schema>
```

14712 A.2.5 BindingManagement

14713 File name: EEBus_SPINE_TS_BindingManagement.xsd

```
14714 <?xml version="1.0" encoding="UTF-8"?>
14715 <!--
14716   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14717   Version 1.3.0 (final)
14718   2023-08-31
14719   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14720   Source: https://www.eebus.org/en/specifications/
14721 -->
14722 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14723 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14724 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14725 blockDefault="#all" elementFormDefault="qualified">
14726   <xs:annotation>
14727     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14728 e.V. All rights reserved.</xs:documentation>
14729   </xs:annotation>
14730   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14731   <xs:simpleType name="BindingIdType">
14732     <xs:restriction base="xs:unsignedInt"/>
14733   </xs:simpleType>
14734   <xs:complexType name="BindingManagementEntryDataType">
14735     <xs:sequence>
14736       <xs:element minOccurs="0" name="bindingId" type="ns_p:BindingIdType"/>
14737       <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
14738       <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
14739       <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
14740       <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
14741     </xs:sequence>
14742   </xs:complexType>
14743   <xs:element name="bindingManagementEntryData" type="ns_p:BindingManagementEntryDataType"/>
14744   <xs:complexType name="BindingManagementEntryDataElementsType">
14745     <xs:sequence>
14746       <xs:element minOccurs="0" name="bindingId" type="ns_p:ElementTagType"/>
14747       <xs:element minOccurs="0" name="clientAddress"
14748 type="ns_p:FeatureAddressElementsType"/>
14749       <xs:element minOccurs="0" name="serverAddress"
14750 type="ns_p:FeatureAddressElementsType"/>
14751       <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
14752       <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
14753     </xs:sequence>
14754   </xs:complexType>
14755   <xs:element name="bindingManagementEntryDataElements"
14756 type="ns_p:BindingManagementEntryDataElementsType"/>
14757   <xs:complexType name="BindingManagementEntryListDataType">
14758     <xs:sequence>
14759       <xs:element maxOccurs="unbounded" minOccurs="0"
14760 ref="ns_p:bindingManagementEntryData"/>
14761     </xs:sequence>
14762   </xs:complexType>
14763   <xs:element name="bindingManagementEntryListData"
14764 type="ns_p:BindingManagementEntryListDataType"/>
14765   <xs:complexType name="BindingManagementEntryListDataSelectorsType">
14766     <xs:sequence>
14767       <xs:element minOccurs="0" name="bindingId" type="ns_p:BindingIdType"/>
14768       <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
14769       <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
14770     </xs:sequence>
14771   </xs:complexType>
```

```

14773     <xs:element name="bindingManagementEntryListDataSelectors"
14774 type="ns_p:BindingManagementEntryListDataSelectorsType"/>
14775     <xs:complexType name="BindingManagementRequestCallType">
14776       <xs:sequence>
14777         <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
14778         <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
14779         <xs:element minOccurs="0" name="serverFeatureType" type="ns_p:FeatureTypeType"/>
14780       </xs:sequence>
14781     </xs:complexType>
14782     <xs:element name="bindingManagementRequestCall"
14783 type="ns_p:BindingManagementRequestCallType"/>
14784     <xs:complexType name="BindingManagementRequestCallElementsType">
14785       <xs:sequence>
14786         <xs:element minOccurs="0" name="clientAddress"
14787 type="ns_p:FeatureAddressElementsType"/>
14788         <xs:element minOccurs="0" name="serverAddress"
14789 type="ns_p:FeatureAddressElementsType"/>
14790         <xs:element minOccurs="0" name="serverFeatureType" type="ns_p:ElementTagType"/>
14791       </xs:sequence>
14792     </xs:complexType>
14793     <xs:element name="bindingManagementRequestCallElements"
14794 type="ns_p:BindingManagementRequestCallElementsType"/>
14795     <xs:complexType name="BindingManagementDeleteCallType">
14796       <xs:sequence>
14797         <xs:element minOccurs="0" name="bindingId" type="ns_p:BindingIdType"/>
14798         <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
14799         <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
14800       </xs:sequence>
14801     </xs:complexType>
14802     <xs:element name="bindingManagementDeleteCall"
14803 type="ns_p:BindingManagementDeleteCallType"/>
14804     <xs:complexType name="BindingManagementDeleteCallElementsType">
14805       <xs:sequence>
14806         <xs:element minOccurs="0" name="bindingId" type="ns_p:ElementTagType"/>
14807         <xs:element minOccurs="0" name="clientAddress"
14808 type="ns_p:FeatureAddressElementsType"/>
14809         <xs:element minOccurs="0" name="serverAddress"
14810 type="ns_p:FeatureAddressElementsType"/>
14811       </xs:sequence>
14812     </xs:complexType>
14813     <xs:element name="bindingManagementDeleteCallElements"
14814 type="ns_p:BindingManagementDeleteCallElementsType"/>
14815   </xs:schema>

```

14819 A.2.6 DataTunneling

14820 File name: EEBus_SPINE_TS_DataTunneling.xsd

```

14821 <?xml version="1.0" encoding="UTF-8"?>
14822 <!--
14823     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14824     Version 1.3.0 (final)
14825     2023-08-31
14826     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14827     Source: https://www.eebus.org/en/specifications/
14828 -->
14829 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14830 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14831 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14832 blockDefault="#all" elementFormDefault="qualified">
14833   <xs:annotation>
14834     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14835 e.V. All rights reserved.</xs:documentation>
14836   </xs:annotation>
14837   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14838   <xs:simpleType name="PurposeIdType">
14839     <xs:restriction base="xs:string"/>
14840   </xs:simpleType>
14841   <xs:simpleType name="ChannelIdType">
14842     <xs:restriction base="xs:unsignedInt"/>
14843   </xs:simpleType>
14844   <xs:complexType name="DataTunnelingHeaderType">

```

```
14846     <xs:sequence>
14847       <xs:element name="purposeId" type="ns_p:PurposeIdType" minOccurs="0"/>
14848       <xs:element name="channelId" type="ns_p:ChannelIdType" minOccurs="0"/>
14849       <xs:element name="sequenceId" type="xs:unsignedInt" minOccurs="0"/>
14850     </xs:sequence>
14851   </xs:complexType>
14852   <xs:complexType name="DataTunnelingHeaderElementsType">
14853     <xs:sequence>
14854       <xs:element name="purposeId" minOccurs="0" type="ns_p:ElementTagType"/>
14855       <xs:element name="channelId" minOccurs="0" type="ns_p:ElementTagType"/>
14856       <xs:element name="sequenceId" minOccurs="0" type="ns_p:ElementTagType"/>
14857     </xs:sequence>
14858   </xs:complexType>
14859   <xs:complexType name="DataTunnelingCallType">
14860     <xs:sequence>
14861       <xs:element name="header" minOccurs="0" type="ns_p:DataTunnelingHeaderType"/>
14862       <xs:element name="payload" type="xs:hexBinary" minOccurs="0"/>
14863     </xs:sequence>
14864   </xs:complexType>
14865   <xs:element name="dataTunnelingCall" type="ns_p:DataTunnelingCallType"/>
14866   <xs:complexType name="DataTunnelingCallElementsType">
14867     <xs:sequence>
14868       <xs:element name="header" minOccurs="0"
14869 type="ns_p:DataTunnelingHeaderElementsType"/>
14870       <xs:element name="payload" minOccurs="0" type="ns_p:ElementTagType"/>
14871     </xs:sequence>
14872   </xs:complexType>
14873   <xs:element name="dataTunnelingCallElements" type="ns_p:DataTunnelingCallElementsType"/>
14874 </xs:schema>
14875
14876
```

14877

14878 A.2.7 DeviceClassification

14879 File name: EEBus_SPINE_TS_DeviceClassification.xsd

```
14880 <?xml version="1.0" encoding="UTF-8"?>
14881 <!--
14882   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
14883   Version 1.3.0 (final)
14884   2023-08-31
14885   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14886   Source: https://www.eebus.org/en/specifications/
14887 -->
14888 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14889 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14890 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14891 blockDefault="#all" elementFormDefault="qualified">
14892   <xs:annotation>
14893     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
14894 e.V. All rights reserved.</xs:documentation>
14895   </xs:annotation>
14896   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
14897   <xs:simpleType name="DeviceClassificationStringType">
14898     <xs:restriction base="xs:string"/>
14899   </xs:simpleType>
14900   <xs:simpleType name="PowerSourceType">
14901     <xs:union memberTypes="ns_p:PowerSourceEnumType ns_p:EnumExtendType"/>
14902   </xs:simpleType>
14903   <xs:simpleType name="PowerSourceEnumType">
14904     <xs:restriction base="xs:string">
14905       <xs:enumeration value="unknown"/>
14906       <xs:enumeration value="mainsSinglePhase"/>
14907       <xs:enumeration value="mains3Phase"/>
14908       <xs:enumeration value="battery"/>
14909       <xs:enumeration value="dc"/>
14910     </xs:restriction>
14911   </xs:simpleType>
14912   <xs:complexType name="DeviceClassificationManufacturerDataType">
14913     <xs:sequence>
14914       <xs:element minOccurs="0" name="deviceName"
14915 type="ns_p:DeviceClassificationStringType"/>
14916       <xs:element minOccurs="0" name="deviceCode"
14917 type="ns_p:DeviceClassificationStringType"/>
14918
```

```
14919         <xs:element minOccurs="0" name="serialNumber"
14920 type="ns_p:DeviceClassificationStringType"/>
14921         <xs:element minOccurs="0" name="softwareRevision"
14922 type="ns_p:DeviceClassificationStringType"/>
14923         <xs:element minOccurs="0" name="hardwareRevision"
14924 type="ns_p:DeviceClassificationStringType"/>
14925         <xs:element minOccurs="0" name="vendorName"
14926 type="ns_p:DeviceClassificationStringType"/>
14927         <xs:element minOccurs="0" name="vendorCode"
14928 type="ns_p:DeviceClassificationStringType"/>
14929         <xs:element minOccurs="0" name="brandName"
14930 type="ns_p:DeviceClassificationStringType"/>
14931         <xs:element minOccurs="0" name="powerSource" type="ns_p:PowerSourceType"/>
14932         <xs:element minOccurs="0" name="manufacturerNodeIdentification"
14933 type="ns_p:DeviceClassificationStringType"/>
14934         <xs:element minOccurs="0" name="manufacturerLabel" type="ns_p:LabelType"/>
14935         <xs:element minOccurs="0" name="manufacturerDescription"
14936 type="ns_p:DescriptionType"/>
14937     </xs:sequence>
14938 </xs:complexType>
14939 <xs:element name="deviceClassificationManufacturerData"
14940 type="ns_p:DeviceClassificationManufacturerDataType"/>
14941 <xs:complexType name="DeviceClassificationManufacturerDataElementsType">
14942     <xs:sequence>
14943         <xs:element minOccurs="0" name="deviceName" type="ns_p:ElementTagType"/>
14944         <xs:element minOccurs="0" name="deviceCode" type="ns_p:ElementTagType"/>
14945         <xs:element minOccurs="0" name="serialNumber" type="ns_p:ElementTagType"/>
14946         <xs:element minOccurs="0" name="softwareRevision" type="ns_p:ElementTagType"/>
14947         <xs:element minOccurs="0" name="hardwareRevision" type="ns_p:ElementTagType"/>
14948         <xs:element minOccurs="0" name="vendorName" type="ns_p:ElementTagType"/>
14949         <xs:element minOccurs="0" name="vendorCode" type="ns_p:ElementTagType"/>
14950         <xs:element minOccurs="0" name="brandName" type="ns_p:ElementTagType"/>
14951         <xs:element minOccurs="0" name="powerSource" type="ns_p:ElementTagType"/>
14952         <xs:element minOccurs="0" name="manufacturerNodeIdentification"
14953 type="ns_p:ElementTagType"/>
14954         <xs:element minOccurs="0" name="manufacturerLabel" type="ns_p:ElementTagType"/>
14955         <xs:element minOccurs="0" name="manufacturerDescription"
14956 type="ns_p:ElementTagType"/>
14957     </xs:sequence>
14958 </xs:complexType>
14959 <xs:element name="deviceClassificationManufacturerDataElements"
14960 type="ns_p:DeviceClassificationManufacturerDataElementsType"/>
14961 <xs:complexType name="DeviceClassificationUserDataTypes">
14962     <xs:sequence>
14963         <xs:element minOccurs="0" name="userNodeIdentification"
14964 type="ns_p:DeviceClassificationStringType"/>
14965         <xs:element minOccurs="0" name="userLabel" type="ns_p:LabelType"/>
14966         <xs:element minOccurs="0" name="userDescription" type="ns_p:DescriptionType"/>
14967     </xs:sequence>
14968 </xs:complexType>
14969 <xs:element name="deviceClassificationUserData"
14970 type="ns_p:DeviceClassificationUserDataTypes"/>
14971 <xs:complexType name="DeviceClassificationUserDataElementsType">
14972     <xs:sequence>
14973         <xs:element minOccurs="0" name="userNodeIdentification"
14974 type="ns_p:ElementTagType"/>
14975         <xs:element minOccurs="0" name="userLabel" type="ns_p:ElementTagType"/>
14976         <xs:element minOccurs="0" name="userDescription" type="ns_p:ElementTagType"/>
14977     </xs:sequence>
14978 </xs:complexType>
14979 <xs:element name="deviceClassificationUserDataElements"
14980 type="ns_p:DeviceClassificationUserDataElementsType"/>
14981 </xs:schema>
```

A.2.8 DeviceConfiguration

File name: EEBus_SPINE_TS_DeviceConfiguration.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
    Smart Premises Interoperable Neutral-Message Exchange (SPINE)
    Version 1.3.0 (final)
```

```
14992 2023-08-31
14993 Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
14994 Source: https://www.eebus.org/en/specifications/
14995 -->
14996 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
14997 xmlns:xs="http://www.w3.org/2001/XMLSchema"
14998 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
14999 blockDefault="#all" elementFormDefault="qualified">
15000 <xs:annotation>
15001 <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
15002 e.V. All rights reserved.</xs:documentation>
15003 </xs:annotation>
15004 <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
15005 <xs:simpleType name="DeviceConfigurationKeyIdType">
15006 <xs:restriction base="xs:unsignedInt"/>
15007 </xs:simpleType>
15008 <xs:simpleType name="DeviceConfigurationKeyValueStringType">
15009 <xs:restriction base="xs:string"/>
15010 </xs:simpleType>
15011 <xs:simpleType name="DeviceConfigurationKeyNameType">
15012 <xs:union memberTypes="ns_p:DeviceConfigurationKeyNameEnumType ns_p:EnumExtendType"/>
15013 </xs:simpleType>
15014 <xs:simpleType name="DeviceConfigurationKeyNameEnumType">
15015 <xs:restriction base="xs:string">
15016 <xs:enumeration value="peakPowerOfPvSystem"/>
15017 <xs:enumeration value="pvCurtaimentLimitFactor"/>
15018 <xs:enumeration value="asymmetricChargingSupported"/>
15019 <xs:enumeration value="communicationsStandard"/>
15020 <xs:enumeration value="inverterGridCode"/>
15021 <xs:enumeration value="pvStringAvailabilityStatus"/>
15022 <xs:enumeration value="batteryAvailabilityStatus"/>
15023 <xs:enumeration value="gridConnectionStatus"/>
15024 <xs:enumeration value="timeToAcChargePowerMax"/>
15025 <xs:enumeration value="timeToAcDischargePowerMax"/>
15026 <xs:enumeration value="tilt"/>
15027 <xs:enumeration value="azimuth"/>
15028 <xs:enumeration value="batteryType"/>
15029 <xs:enumeration value="maxCycleCountPerDay"/>
15030 <xs:enumeration value="failsafeConsumptionActivePowerLimit"/>
15031 <xs:enumeration value="failsafeProductionActivePowerLimit"/>
15032 <xs:enumeration value="failsafePositiveReactivePowerLimit"/>
15033 <xs:enumeration value="failsafeNegativeReactivePowerLimit"/>
15034 <xs:enumeration value="failsafePositiveCosPhiLimit"/>
15035 <xs:enumeration value="failsafeNegativeCosPhiLimit"/>
15036 <xs:enumeration value="maxAcChargePower"/>
15037 <xs:enumeration value="maxAcDischargePower"/>
15038 <xs:enumeration value="maxDcChargePower"/>
15039 <xs:enumeration value="maxDcDischargePower"/>
15040 <xs:enumeration value="batteryActiveControlMode"/>
15041 <xs:enumeration value="defaultAcPower"/>
15042 <xs:enumeration value="defaultDcPower"/>
15043 <xs:enumeration value="defaultPccPower"/>
15044 <xs:enumeration value="failsafeAcPowerSetpoint"/>
15045 <xs:enumeration value="failsafeDcPowerSetpoint"/>
15046 <xs:enumeration value="failsafePccPowerSetpoint"/>
15047 <xs:enumeration value="failsafeDurationMinimum"/>
15048 <xs:enumeration value="dischargingBelowTargetEnergyRequestPermitted"/>
15049 <xs:enumeration value="incentivesSimulationCyclesMax"/>
15050 <xs:enumeration value="incentivesSimulationConcurrent"/>
15051 <xs:enumeration value="incentivesTimeoutIncentiveRequest"/>
15052 <xs:enumeration value="incentivesWaitIncentiveWriteable"/>
15053 </xs:restriction>
15054 </xs:simpleType>
15055 <xs:simpleType name="DeviceConfigurationKeyValueValueType">
15056 <xs:restriction base="xs:string">
15057 <xs:enumeration value="boolean"/>
15058 <xs:enumeration value="date"/>
15059 <xs:enumeration value="dateTime"/>
15060 <xs:enumeration value="duration"/>
15061 <xs:enumeration value="string"/>
15062 <xs:enumeration value="time"/>
15063 <xs:enumeration value="scaledNumber"/>
15064 <xs:enumeration value="integer"/>
15065 </xs:restriction>
15066 </xs:simpleType>
15067 <xs:complexType name="DeviceConfigurationKeyValueValueType">
15068 <xs:sequence>
15069 <xs:element minOccurs="0" name="boolean" type="xs:boolean"/>
```



```
15070         <xs:element minOccurs="0" name="date" type="xs:date"/>
15071         <xs:element minOccurs="0" name="dateTime" type="xs:dateTime"/>
15072         <xs:element minOccurs="0" name="duration" type="xs:duration"/>
15073         <xs:element minOccurs="0" name="string"
15074 type="ns_p:DeviceConfigurationKeyValueStringType"/>
15075         <xs:element minOccurs="0" name="time" type="xs:time"/>
15076         <xs:element minOccurs="0" name="scaledNumber" type="ns_p:ScaledNumberType"/>
15077         <xs:element minOccurs="0" name="integer" type="xs:long"/>
15078     </xs:sequence>
15079 </xs:complexType>
15080 <xs:complexType name="DeviceConfigurationKeyValueValueElementsType">
15081     <xs:sequence>
15082         <xs:element minOccurs="0" name="boolean" type="ns_p:ElementTagType"/>
15083         <xs:element minOccurs="0" name="date" type="ns_p:ElementTagType"/>
15084         <xs:element minOccurs="0" name="dateTime" type="ns_p:ElementTagType"/>
15085         <xs:element minOccurs="0" name="duration" type="ns_p:ElementTagType"/>
15086         <xs:element minOccurs="0" name="string" type="ns_p:ElementTagType"/>
15087         <xs:element minOccurs="0" name="time" type="ns_p:ElementTagType"/>
15088         <xs:element minOccurs="0" name="scaledNumber"
15089 type="ns_p:ScaledNumberElementsType"/>
15090     </xs:sequence>
15091 </xs:complexType>
15092 <xs:complexType name="DeviceConfigurationKeyValueDataType">
15093     <xs:sequence>
15094         <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15095         <xs:element minOccurs="0" name="value"
15096 type="ns_p:DeviceConfigurationKeyValueValueType"/>
15097         <xs:element minOccurs="0" name="isValueChangeable" type="xs:boolean"/>
15098     </xs:sequence>
15099 </xs:complexType>
15100 <xs:element name="deviceConfigurationKeyValueData"
15101 type="ns_p:DeviceConfigurationKeyValueDataType"/>
15102 <xs:complexType name="DeviceConfigurationKeyValueDataElementsType">
15103     <xs:sequence>
15104         <xs:element minOccurs="0" name="keyId" type="ns_p:ElementTagType"/>
15105         <xs:element minOccurs="0" name="value"
15106 type="ns_p:DeviceConfigurationKeyValueValueElementsType"/>
15107         <xs:element minOccurs="0" name="isValueChangeable" type="ns_p:ElementTagType"/>
15108     </xs:sequence>
15109 </xs:complexType>
15110 <xs:element name="deviceConfigurationKeyValueDataElements"
15111 type="ns_p:DeviceConfigurationKeyValueDataElementsType"/>
15112 <xs:complexType name="DeviceConfigurationKeyValueListDataType">
15113     <xs:sequence>
15114         <xs:element maxOccurs="unbounded" minOccurs="0"
15115 ref="ns_p:deviceConfigurationKeyValueData"/>
15116     </xs:sequence>
15117 </xs:complexType>
15118 <xs:element name="deviceConfigurationKeyValueListData"
15119 type="ns_p:DeviceConfigurationKeyValueListDataType"/>
15120 <xs:complexType name="DeviceConfigurationKeyValueListDataSelectorsType">
15121     <xs:sequence>
15122         <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15123     </xs:sequence>
15124 </xs:complexType>
15125 <xs:element name="deviceConfigurationKeyValueListDataSelectors"
15126 type="ns_p:DeviceConfigurationKeyValueListDataSelectorsType"/>
15127 <xs:complexType name="DeviceConfigurationKeyValueDescriptionDataType">
15128     <xs:sequence>
15129         <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15130         <xs:element minOccurs="0" name="keyName"
15131 type="ns_p:DeviceConfigurationKeyNameType"/>
15132         <xs:element minOccurs="0" name="valueType"
15133 type="ns_p:DeviceConfigurationKeyValueType"/>
15134         <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
15135         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
15136         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
15137     </xs:sequence>
15138 </xs:complexType>
15139 <xs:element name="deviceConfigurationKeyValueDescriptionData"
15140 type="ns_p:DeviceConfigurationKeyValueDescriptionDataType"/>
15141 <xs:complexType name="DeviceConfigurationKeyValueDescriptionDataElementsType">
15142     <xs:sequence>
15143         <xs:element minOccurs="0" name="keyId" type="ns_p:ElementTagType"/>
15144         <xs:element minOccurs="0" name="keyName" type="ns_p:ElementTagType"/>
15145         <xs:element minOccurs="0" name="valueType" type="ns_p:ElementTagType"/>
15146         <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
15147         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
```

```

15148         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
15149     </xs:sequence>
15150 </xs:complexType>
15151     <xs:element name="deviceConfigurationKeyValueDescriptionDataElements"
15152 type="ns_p:DeviceConfigurationKeyValueDescriptionDataElementsType"/>
15153     <xs:complexType name="DeviceConfigurationKeyValueDescriptionListDataType">
15154         <xs:sequence>
15155             <xs:element maxOccurs="unbounded" minOccurs="0"
15156 ref="ns_p:deviceConfigurationKeyValueDescriptionData"/>
15157         </xs:sequence>
15158     </xs:complexType>
15159     <xs:element name="deviceConfigurationKeyValueDescriptionListData"
15160 type="ns_p:DeviceConfigurationKeyValueDescriptionListDataType"/>
15161     <xs:complexType name="DeviceConfigurationKeyValueDescriptionListDataSelectorsType">
15162         <xs:sequence>
15163             <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15164             <xs:element minOccurs="0" name="keyName"
15165 type="ns_p:DeviceConfigurationKeyNameType"/>
15166         </xs:sequence>
15167     </xs:complexType>
15168     <xs:element name="deviceConfigurationKeyValueDescriptionListDataSelectors"
15169 type="ns_p:DeviceConfigurationKeyValueDescriptionListDataSelectorsType"/>
15170     <xs:complexType name="DeviceConfigurationKeyValueConstraintsDataType">
15171         <xs:sequence>
15172             <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15173             <xs:element minOccurs="0" name="valueRangeMin"
15174 type="ns_p:DeviceConfigurationKeyValueValueType"/>
15175             <xs:element minOccurs="0" name="valueRangeMax"
15176 type="ns_p:DeviceConfigurationKeyValueValueType"/>
15177             <xs:element minOccurs="0" name="valueStepSize"
15178 type="ns_p:DeviceConfigurationKeyValueValueType"/>
15179         </xs:sequence>
15180     </xs:complexType>
15181     <xs:element name="deviceConfigurationKeyValueConstraintsData"
15182 type="ns_p:DeviceConfigurationKeyValueConstraintsDataType"/>
15183     <xs:complexType name="DeviceConfigurationKeyValueConstraintsDataElementsType">
15184         <xs:sequence>
15185             <xs:element minOccurs="0" name="keyId" type="ns_p:ElementTagType"/>
15186             <xs:element minOccurs="0" name="valueRangeMin"
15187 type="ns_p:DeviceConfigurationKeyValueValueElementsType"/>
15188             <xs:element minOccurs="0" name="valueRangeMax"
15189 type="ns_p:DeviceConfigurationKeyValueValueElementsType"/>
15190             <xs:element minOccurs="0" name="valueStepSize"
15191 type="ns_p:DeviceConfigurationKeyValueValueElementsType"/>
15192         </xs:sequence>
15193     </xs:complexType>
15194     <xs:element name="deviceConfigurationKeyValueConstraintsDataElements"
15195 type="ns_p:DeviceConfigurationKeyValueConstraintsDataElementsType"/>
15196     <xs:complexType name="DeviceConfigurationKeyValueConstraintsListDataType">
15197         <xs:sequence>
15198             <xs:element maxOccurs="unbounded" minOccurs="0"
15199 ref="ns_p:deviceConfigurationKeyValueConstraintsData"/>
15200         </xs:sequence>
15201     </xs:complexType>
15202     <xs:element name="deviceConfigurationKeyValueConstraintsListData"
15203 type="ns_p:DeviceConfigurationKeyValueConstraintsListDataType"/>
15204     <xs:complexType name="DeviceConfigurationKeyValueConstraintsListDataSelectorsType">
15205         <xs:sequence>
15206             <xs:element minOccurs="0" name="keyId" type="ns_p:DeviceConfigurationKeyIdType"/>
15207         </xs:sequence>
15208     </xs:complexType>
15209     <xs:element name="deviceConfigurationKeyValueConstraintsListDataSelectors"
15210 type="ns_p:DeviceConfigurationKeyValueConstraintsListDataSelectorsType"/>
15211 </xs:schema>
15212
15213

```

15215 A.2.9 DeviceDiagnosis

15216 File name: EEBus_SPINE_TS_DeviceDiagnosis.xsd

```

15217 <?xml version="1.0" encoding="UTF-8"?>
15218 <!--
15219     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
15220

```

```
15221     Version 1.3.0 (final)
15222     2023-08-31
15223     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
15224     Source: https://www.eebus.org/en/specifications/
15225 -->
15226 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
15227   xmlns:xs="http://www.w3.org/2001/XMLSchema"
15228   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
15229   blockDefault="#all" elementFormDefault="qualified">
15230   <xs:annotation>
15231     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
15232     e.V. All rights reserved.</xs:documentation>
15233   </xs:annotation>
15234   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
15235   <xs:simpleType name="VendorStateCodeType">
15236     <xs:restriction base="xs:string"/>
15237   </xs:simpleType>
15238   <xs:simpleType name="LastErrorCodeType">
15239     <xs:restriction base="xs:string"/>
15240   </xs:simpleType>
15241   <xs:simpleType name="DeviceDiagnosisOperatingStateType">
15242     <xs:union memberTypes="ns_p:DeviceDiagnosisOperatingStateEnumType
15243 ns_p:EnumExtendType"/>
15244   </xs:simpleType>
15245   <xs:simpleType name="DeviceDiagnosisOperatingStateEnumType">
15246     <xs:restriction base="xs:string">
15247       <xs:enumeration value="normalOperation"/>
15248       <xs:enumeration value="standby"/>
15249       <xs:enumeration value="failure"/>
15250       <xs:enumeration value="serviceNeeded"/>
15251       <xs:enumeration value="overrideDetected"/>
15252       <xs:enumeration value="inAlarm"/>
15253       <xs:enumeration value="notReachable"/>
15254       <xs:enumeration value="finished"/>
15255       <xs:enumeration value="temporarilyNotReady"/>
15256       <xs:enumeration value="off"/>
15257     </xs:restriction>
15258   </xs:simpleType>
15259   <xs:simpleType name="PowerSupplyConditionType">
15260     <xs:union memberTypes="ns_p:PowerSupplyConditionEnumType ns_p:EnumExtendType"/>
15261   </xs:simpleType>
15262   <xs:simpleType name="PowerSupplyConditionEnumType">
15263     <xs:restriction base="xs:string">
15264       <xs:enumeration value="good"/>
15265       <xs:enumeration value="low"/>
15266       <xs:enumeration value="critical"/>
15267       <xs:enumeration value="unknown"/>
15268       <xs:enumeration value="error"/>
15269     </xs:restriction>
15270   </xs:simpleType>
15271   <xs:complexType name="DeviceDiagnosisStateDataType">
15272     <xs:sequence>
15273       <xs:element name="timestamp" type="ns_p:AbsoluteOrRelativeTimeType"
15274 minOccurs="0"/>
15275       <xs:element name="operatingState" type="ns_p:DeviceDiagnosisOperatingStateType"
15276 minOccurs="0"/>
15277       <xs:element name="vendorStateCode" type="ns_p:VendorStateCodeType" minOccurs="0"/>
15278       <xs:element name="lastErrorCode" type="ns_p:LastErrorCodeType" minOccurs="0"/>
15279       <xs:element name="upTime" type="xs:duration" minOccurs="0"/>
15280       <xs:element name="totalUpTime" type="xs:duration" minOccurs="0"/>
15281       <xs:element name="powerSupplyCondition" minOccurs="0"
15282 type="ns_p:PowerSupplyConditionType"/>
15283     </xs:sequence>
15284   </xs:complexType>
15285   <xs:element name="deviceDiagnosisStateData" type="ns_p:DeviceDiagnosisStateDataType"/>
15286   <xs:complexType name="DeviceDiagnosisStateDataElementsType">
15287     <xs:sequence>
15288       <xs:element name="timestamp" minOccurs="0" type="ns_p:ElementTagType"/>
15289       <xs:element name="operatingState" minOccurs="0" type="ns_p:ElementTagType"/>
15290       <xs:element name="vendorStateCode" minOccurs="0" type="ns_p:ElementTagType"/>
15291       <xs:element name="lastErrorCode" minOccurs="0" type="ns_p:ElementTagType"/>
15292       <xs:element name="upTime" minOccurs="0" type="ns_p:ElementTagType"/>
15293       <xs:element name="totalUpTime" minOccurs="0" type="ns_p:ElementTagType"/>
15294       <xs:element name="powerSupplyCondition" minOccurs="0" type="ns_p:ElementTagType"/>
15295     </xs:sequence>
15296   </xs:complexType>
15297   <xs:element name="deviceDiagnosisStateDataElements"
15298 type="ns_p:DeviceDiagnosisStateDataElementsType"/>
```

```
15299     <xs:complexType name="DeviceDiagnosisHeartbeatDataType">
15300       <xs:sequence>
15301         <xs:element name="timestamp" type="ns_p:AbsoluteOrRelativeTimeType"
15302 minOccurs="0"/>
15303         <xs:element name="heartbeatCounter" type="xs:unsignedLong" minOccurs="0"/>
15304         <xs:element name="heartbeatTimeout" type="xs:duration" minOccurs="0"/>
15305       </xs:sequence>
15306     </xs:complexType>
15307     <xs:element name="deviceDiagnosisHeartbeatData"
15308 type="ns_p:DeviceDiagnosisHeartbeatDataType"/>
15309     <xs:complexType name="DeviceDiagnosisHeartbeatDataElementsType">
15310       <xs:sequence>
15311         <xs:element name="timestamp" minOccurs="0" type="ns_p:ElementTagType"/>
15312         <xs:element name="heartbeatCounter" minOccurs="0" type="ns_p:ElementTagType"/>
15313         <xs:element name="heartbeatTimeout" minOccurs="0" type="ns_p:ElementTagType"/>
15314       </xs:sequence>
15315     </xs:complexType>
15316     <xs:element name="deviceDiagnosisHeartbeatDataElements"
15317 type="ns_p:DeviceDiagnosisHeartbeatDataElementsType"/>
15318     <xs:complexType name="DeviceDiagnosisServiceDataType">
15319       <xs:sequence>
15320         <xs:element name="timestamp" type="ns_p:AbsoluteOrRelativeTimeType"
15321 minOccurs="0"/>
15322         <xs:element name="installationTime" type="ns_p:AbsoluteOrRelativeTimeType"
15323 minOccurs="0"/>
15324         <xs:element name="bootCounter" type="xs:unsignedLong" minOccurs="0"/>
15325         <xs:element name="nextService" type="ns_p:AbsoluteOrRelativeTimeType"
15326 minOccurs="0"/>
15327       </xs:sequence>
15328     </xs:complexType>
15329     <xs:element name="deviceDiagnosisServiceData" type="ns_p:DeviceDiagnosisServiceDataType"/>
15330     <xs:complexType name="DeviceDiagnosisServiceDataElementsType">
15331       <xs:sequence>
15332         <xs:element name="timestamp" minOccurs="0" type="ns_p:ElementTagType"/>
15333         <xs:element name="installationTime" minOccurs="0" type="ns_p:ElementTagType"/>
15334         <xs:element name="bootCounter" minOccurs="0" type="ns_p:ElementTagType"/>
15335         <xs:element name="nextService" minOccurs="0" type="ns_p:ElementTagType"/>
15336       </xs:sequence>
15337     </xs:complexType>
15338     <xs:element name="deviceDiagnosisServiceDataElements"
15339 type="ns_p:DeviceDiagnosisServiceDataElementsType"/>
15340 </xs:schema>
```

A.2.10 DirectControl

File name: EEBus_SPINE_TS_DirectControl.xsd

```
15346 <?xml version="1.0" encoding="UTF-8"?>
15347 <!--
15348   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
15349   Version 1.3.0 (final)
15350   2023-08-31
15351   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
15352   Source: https://www.eebus.org/en/specifications/
15353 -->
15354 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
15355 xmlns:xs="http://www.w3.org/2001/XMLSchema"
15356 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
15357 blockDefault="#all" elementFormDefault="qualified">
15358   <xs:annotation>
15359     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
15360 e.V. All rights reserved.</xs:documentation>
15361   </xs:annotation>
15362   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
15363   <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
15364   <xs:simpleType name="DirectControlActivityStateType">
15365     <xs:union memberTypes="ns_p:DirectControlActivityStateEnumType ns_p:EnumExtendType"/>
15366   </xs:simpleType>
15367   <xs:simpleType name="DirectControlActivityStateEnumType">
15368     <xs:restriction base="xs:string">
15369       <xs:enumeration value="running"/>
15370       <xs:enumeration value="paused"/>
```

```
15372         <xs:enumeration value="inactive"/>
15373     </xs:restriction>
15374 </xs:simpleType>
15375 <xs:complexType name="DirectControlActivityDataType">
15376     <xs:sequence>
15377         <xs:element minOccurs="0" name="timestamp"
15378 type="ns_p:AbsoluteOrRelativeTimeType"/>
15379         <xs:element minOccurs="0" name="activityState"
15380 type="ns_p:DirectControlActivityStateType"/>
15381         <xs:element minOccurs="0" name="isActivityStateChangeable" type="xs:boolean"/>
15382         <xs:element minOccurs="0" name="energyMode" type="ns_p:EnergyModeType"/>
15383         <xs:element minOccurs="0" name="isEnergyModeChangeable" type="xs:boolean"/>
15384         <xs:element name="power" type="ns_p:ScaledNumberType" minOccurs="0"/>
15385         <xs:element minOccurs="0" name="isPowerChangeable" type="xs:boolean"/>
15386         <xs:element name="energy" type="ns_p:ScaledNumberType" minOccurs="0"/>
15387         <xs:element minOccurs="0" name="isEnergyChangeable" type="xs:boolean"/>
15388         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
15389     </xs:sequence>
15390 </xs:complexType>
15391 <xs:element name="directControlActivityData" type="ns_p:DirectControlActivityDataType"/>
15392 <xs:complexType name="DirectControlActivityDataElementsType">
15393     <xs:sequence>
15394         <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
15395         <xs:element minOccurs="0" name="activityState" type="ns_p:ElementTagType"/>
15396         <xs:element minOccurs="0" name="isActivityStateChangeable"
15397 type="ns_p:ElementTagType"/>
15398         <xs:element minOccurs="0" name="energyMode" type="ns_p:ElementTagType"/>
15399         <xs:element minOccurs="0" name="isEnergyModeChangeable"
15400 type="ns_p:ElementTagType"/>
15401         <xs:element name="power" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
15402         <xs:element minOccurs="0" name="isPowerChangeable" type="ns_p:ElementTagType"/>
15403         <xs:element name="energy" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
15404         <xs:element minOccurs="0" name="isEnergyChangeable" type="ns_p:ElementTagType"/>
15405         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
15406     </xs:sequence>
15407 </xs:complexType>
15408 <xs:element name="directControlActivityDataElements"
15409 type="ns_p:DirectControlActivityDataElementsType"/>
15410 <xs:complexType name="DirectControlActivityListDataType">
15411     <xs:sequence>
15412         <xs:element maxOccurs="unbounded" minOccurs="0"
15413 ref="ns_p:directControlActivityData"/>
15414     </xs:sequence>
15415 </xs:complexType>
15416 <xs:element name="directControlActivityListData"
15417 type="ns_p:DirectControlActivityListDataType"/>
15418 <xs:complexType name="DirectControlActivityListDataSelectorsType">
15419     <xs:sequence>
15420         <xs:element minOccurs="0" name="timestampInterval"
15421 type="ns_p:TimestampIntervalType"/>
15422     </xs:sequence>
15423 </xs:complexType>
15424 <xs:element name="directControlActivityListDataSelectors"
15425 type="ns_p:DirectControlActivityListDataSelectorsType"/>
15426 <xs:complexType name="DirectControlDescriptionDataType">
15427     <xs:sequence>
15428         <xs:element minOccurs="0" name="positiveEnergyDirection"
15429 type="ns_p:EnergyDirectionType"/>
15430         <xs:element minOccurs="0" name="powerUnit" type="ns_p:UnitOfMeasurementType"/>
15431         <xs:element minOccurs="0" name="energyUnit" type="ns_p:UnitOfMeasurementType"/>
15432     </xs:sequence>
15433 </xs:complexType>
15434 <xs:element name="directControlDescriptionData"
15435 type="ns_p:DirectControlDescriptionDataType"/>
15436 <xs:complexType name="DirectControlDescriptionDataElementsType">
15437     <xs:sequence>
15438         <xs:element minOccurs="0" name="positiveEnergyDirection"
15439 type="ns_p:ElementTagType"/>
15440         <xs:element minOccurs="0" name="powerUnit" type="ns_p:ElementTagType"/>
15441         <xs:element minOccurs="0" name="energyUnit" type="ns_p:ElementTagType"/>
15442     </xs:sequence>
15443 </xs:complexType>
15444 <xs:element name="directControlDescriptionDataElements"
15445 type="ns_p:DirectControlDescriptionDataElementsType"/>
15446 </xs:schema>
15447
15448
```

15449

15450 **A.2.11 ElectricalConnection**

15451 File name: EEBus_SPINE_TS_ElectricalConnection.xsd

15452

15453

<?xml version="1.0" encoding="UTF-8"?>

15454

<!--

15455

Smart Premises Interoperable Neutral-Message Exchange (SPINE)

15456

Version 1.3.0 (final)

15457

2023-08-31

15458

Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.

15459

Source: <https://www.eebus.org/en/specifications/>

15460

-->

15461

<xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"

15462

xmlns:xs="http://www.w3.org/2001/XMLSchema"

15463

targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"

15464

blockDefault="#all" elementFormDefault="qualified">

15465

<xs:annotation>

15466

<xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative

15467

e.V. All rights reserved.</xs:documentation>

15468

</xs:annotation>

15469

<xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>

15470

<xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>

15471

<xs:simpleType name="ElectricalConnectionIdType">

15472

<xs:restriction base="xs:unsignedInt"/>

15473

</xs:simpleType>

15474

<xs:simpleType name="ElectricalConnectionParameterIdType">

15475

<xs:restriction base="xs:unsignedInt"/>

15476

</xs:simpleType>

15477

<xs:simpleType name="ElectricalConnectionMeasurandVariantType">

15478

<xs:union memberTypes="ns_p:ElectricalConnectionMeasurandVariantEnumType

15479

ns_p:EnumExtendType"/>

15480

</xs:simpleType>

15481

<xs:simpleType name="ElectricalConnectionMeasurandVariantEnumType">

15482

<xs:restriction base="xs:string">

15483

<xs:enumeration value="amplitude"/>

15484

<xs:enumeration value="rms"/>

15485

<xs:enumeration value="instantaneous"/>

15486

<xs:enumeration value="angle"/>

15487

<xs:enumeration value="cosPhi"/>

15488

</xs:restriction>

15489

</xs:simpleType>

15490

<xs:simpleType name="ElectricalConnectionVoltageTypeType">

15491

<xs:union memberTypes="ns_p:ElectricalConnectionVoltageTypeEnumType

15492

ns_p:EnumExtendType"/>

15493

</xs:simpleType>

15494

<xs:simpleType name="ElectricalConnectionVoltageTypeEnumType">

15495

<xs:restriction base="xs:string">

15496

<xs:enumeration value="ac"/>

15497

<xs:enumeration value="dc"/>

15498

</xs:restriction>

15499

</xs:simpleType>

15500

<xs:simpleType name="ElectricalConnectionAcMeasurementTypeType">

15501

<xs:union memberTypes="ns_p:ElectricalConnectionAcMeasurementTypeEnumType

15502

ns_p:EnumExtendType"/>

15503

</xs:simpleType>

15504

<xs:simpleType name="ElectricalConnectionAcMeasurementTypeEnumType">

15505

<xs:restriction base="xs:string">

15506

<xs:enumeration value="real"/>

15507

<xs:enumeration value="reactive"/>

15508

<xs:enumeration value="apparent"/>

15509

<xs:enumeration value="phase"/>

15510

</xs:restriction>

15511

</xs:simpleType>

15512

<xs:simpleType name="ElectricalConnectionPhaseNameType">

15513

<xs:union memberTypes="ns_p:ElectricalConnectionPhaseNameEnumType

15514

ns_p:EnumExtendType"/>

15515

</xs:simpleType>

15516

<xs:simpleType name="ElectricalConnectionPhaseNameEnumType">

15517

<xs:restriction base="xs:string">

15518

<xs:enumeration value="a"/>

15519

<xs:enumeration value="b"/>

15520

<xs:enumeration value="c"/>

15521

<xs:enumeration value="ab"/>

15522

<xs:enumeration value="bc"/>

```

15523         <xs:enumeration value="ac"/>
15524         <xs:enumeration value="abc"/>
15525         <xs:enumeration value="neutral"/>
15526         <xs:enumeration value="ground"/>
15527         <xs:enumeration value="none"/>
15528     </xs:restriction>
15529 </xs:simpleType>
15530 <xs:simpleType name="ElectricalConnectionConnectionPointType">
15531     <xs:restriction base="xs:string">
15532         <xs:enumeration value="grid"/>
15533         <xs:enumeration value="home"/>
15534         <xs:enumeration value="pv"/>
15535         <xs:enumeration value="sd"/>
15536         <xs:enumeration value="other"/>
15537     </xs:restriction>
15538 </xs:simpleType>
15539 <xs:simpleType name="ElectricalConnectionCharacteristicIdType">
15540     <xs:restriction base="xs:unsignedInt"/>
15541 </xs:simpleType>
15542 <xs:simpleType name="ElectricalConnectionCharacteristicContextType">
15543     <xs:union memberTypes="ns_p:ElectricalConnectionCharacteristicContextEnumType
15544 ns_p:EnumExtendType"/>
15545 </xs:simpleType>
15546 <xs:simpleType name="ElectricalConnectionCharacteristicContextEnumType">
15547     <xs:restriction base="xs:string">
15548         <xs:enumeration value="device"/>
15549         <xs:enumeration value="entity"/>
15550         <xs:enumeration value="inverter"/>
15551         <xs:enumeration value="pvString"/>
15552         <xs:enumeration value="battery"/>
15553     </xs:restriction>
15554 </xs:simpleType>
15555 <xs:simpleType name="ElectricalConnectionCharacteristicTypeType">
15556     <xs:union memberTypes="ns_p:ElectricalConnectionCharacteristicTypeEnumType
15557 ns_p:EnumExtendType"/>
15558 </xs:simpleType>
15559 <xs:simpleType name="ElectricalConnectionCharacteristicTypeEnumType">
15560     <xs:restriction base="xs:string">
15561         <xs:enumeration value="powerConsumptionMin"/>
15562         <xs:enumeration value="powerConsumptionMax"/>
15563         <xs:enumeration value="powerConsumptionNominalMin"/>
15564         <xs:enumeration value="powerConsumptionNominalMax"/>
15565         <xs:enumeration value="powerProductionMin"/>
15566         <xs:enumeration value="powerProductionMax"/>
15567         <xs:enumeration value="powerProductionNominalMin"/>
15568         <xs:enumeration value="powerProductionNominalMax"/>
15569         <xs:enumeration value="energyCapacityNominalMax"/>
15570         <xs:enumeration value="contractualConsumptionNominalMax"/>
15571         <xs:enumeration value="contractualProductionNominalMax"/>
15572         <xs:enumeration value="apparentPowerProductionNominalMax"/>
15573         <xs:enumeration value="apparentPowerConsumptionNominalMax"/>
15574     </xs:restriction>
15575 </xs:simpleType>
15576 <xs:complexType name="ElectricalConnectionParameterDescriptionDataType">
15577     <xs:sequence>
15578         <xs:element name="electricalConnectionId" minOccurs="0"
15579 type="ns_p:ElectricalConnectionIdType"/>
15580         <xs:element name="parameterId" minOccurs="0"
15581 type="ns_p:ElectricalConnectionParameterIdType"/>
15582         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
15583         <xs:element minOccurs="0" name="voltageType"
15584 type="ns_p:ElectricalConnectionVoltageTypeType"/>
15585         <xs:element name="acMeasuredPhases" minOccurs="0"
15586 type="ns_p:ElectricalConnectionPhaseNameType"/>
15587         <xs:element name="acMeasuredInReferenceTo" minOccurs="0"
15588 type="ns_p:ElectricalConnectionPhaseNameType"/>
15589         <xs:element name="acMeasurementType"
15590 type="ns_p:ElectricalConnectionAcMeasurementTypeType" minOccurs="0"/>
15591         <xs:element name="acMeasurementVariant" minOccurs="0"
15592 type="ns_p:ElectricalConnectionMeasurendVariantType"/>
15593         <xs:element minOccurs="0" name="acMeasuredHarmonic" type="xs:unsignedByte"/>
15594         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
15595         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
15596         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
15597     </xs:sequence>
15598 </xs:complexType>
15599 <xs:element name="electricalConnectionParameterDescriptionData"
15600 type="ns_p:ElectricalConnectionParameterDescriptionDataType"/>

```

```
15601     <xs:complexType name="ElectricalConnectionParameterDescriptionDataElementsType">
15602       <xs:sequence>
15603         <xs:element name="electricalConnectionId" minOccurs="0"
15604 type="ns_p:ElementTagType"/>
15605         <xs:element name="parameterId" minOccurs="0" type="ns_p:ElementTagType"/>
15606         <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
15607         <xs:element minOccurs="0" name="voltageType" type="ns_p:ElementTagType"/>
15608         <xs:element name="acMeasuredPhases" minOccurs="0" type="ns_p:ElementTagType"/>
15609         <xs:element name="acMeasuredInReferenceTo" minOccurs="0"
15610 type="ns_p:ElementTagType"/>
15611         <xs:element name="acMeasurementType" minOccurs="0" type="ns_p:ElementTagType"/>
15612         <xs:element name="acMeasurementVariant" minOccurs="0" type="ns_p:ElementTagType"/>
15613         <xs:element minOccurs="0" name="acMeasuredHarmonic" type="ns_p:ElementTagType"/>
15614         <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
15615         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
15616         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
15617       </xs:sequence>
15618     </xs:complexType>
15619     <xs:element name="electricalConnectionParameterDescriptionDataElements"
15620 type="ns_p:ElectricalConnectionParameterDescriptionDataElementsType"/>
15621     <xs:complexType name="ElectricalConnectionParameterDescriptionListDataType">
15622       <xs:sequence>
15623         <xs:element maxOccurs="unbounded" minOccurs="0"
15624 ref="ns_p:electricalConnectionParameterDescriptionData"/>
15625       </xs:sequence>
15626     </xs:complexType>
15627     <xs:element name="electricalConnectionParameterDescriptionListData"
15628 type="ns_p:ElectricalConnectionParameterDescriptionListDataType"/>
15629     <xs:complexType name="ElectricalConnectionParameterDescriptionListDataSelectorsType">
15630       <xs:sequence>
15631         <xs:element name="electricalConnectionId" minOccurs="0"
15632 type="ns_p:ElectricalConnectionIdType"/>
15633         <xs:element name="parameterId" minOccurs="0"
15634 type="ns_p:ElectricalConnectionParameterIdType"/>
15635         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
15636         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
15637       </xs:sequence>
15638     </xs:complexType>
15639     <xs:element name="electricalConnectionParameterDescriptionListDataSelectors"
15640 type="ns_p:ElectricalConnectionParameterDescriptionListDataSelectorsType"/>
15641     <xs:complexType name="ElectricalConnectionPermittedValueSetDataType">
15642       <xs:sequence>
15643         <xs:element name="electricalConnectionId" minOccurs="0"
15644 type="ns_p:ElectricalConnectionIdType"/>
15645         <xs:element name="parameterId" minOccurs="0"
15646 type="ns_p:ElectricalConnectionParameterIdType"/>
15647         <xs:element minOccurs="0" name="permittedValueSet" type="ns_p:ScaledNumberSetType"
15648 maxOccurs="unbounded"/>
15649       </xs:sequence>
15650     </xs:complexType>
15651     <xs:element name="electricalConnectionPermittedValueSetData"
15652 type="ns_p:ElectricalConnectionPermittedValueSetDataType"/>
15653     <xs:complexType name="ElectricalConnectionPermittedValueSetDataElementsType">
15654       <xs:sequence>
15655         <xs:element name="electricalConnectionId" minOccurs="0"
15656 type="ns_p:ElementTagType"/>
15657         <xs:element name="parameterId" minOccurs="0" type="ns_p:ElementTagType"/>
15658         <xs:element minOccurs="0" name="permittedValueSet"
15659 type="ns_p:ScaledNumberSetElementsType"/>
15660       </xs:sequence>
15661     </xs:complexType>
15662     <xs:element name="electricalConnectionPermittedValueSetDataElements"
15663 type="ns_p:ElectricalConnectionPermittedValueSetDataElementsType"/>
15664     <xs:complexType name="ElectricalConnectionPermittedValueSetListDataType">
15665       <xs:sequence>
15666         <xs:element maxOccurs="unbounded" minOccurs="0"
15667 ref="ns_p:electricalConnectionPermittedValueSetData"/>
15668       </xs:sequence>
15669     </xs:complexType>
15670     <xs:element name="electricalConnectionPermittedValueSetListData"
15671 type="ns_p:ElectricalConnectionPermittedValueSetListDataType"/>
15672     <xs:complexType name="ElectricalConnectionPermittedValueSetListDataSelectorsType">
15673       <xs:sequence>
15674         <xs:element name="electricalConnectionId" minOccurs="0"
15675 type="ns_p:ElectricalConnectionIdType"/>
15676         <xs:element name="parameterId" minOccurs="0"
15677 type="ns_p:ElectricalConnectionParameterIdType"/>
15678       </xs:sequence>
```



```
15679     </xs:complexType>
15680     <xs:element name="electricalConnectionPermittedValueSetListDataSelectors"
15681 type="ns_p:ElectricalConnectionPermittedValueSetListDataSelectorsType"/>
15682     <xs:complexType name="ElectricalConnectionCharacteristicDataType">
15683       <xs:sequence>
15684         <xs:element name="electricalConnectionId" minOccurs="0"
15685 type="ns_p:ElectricalConnectionIdType"/>
15686         <xs:element name="parameterId" minOccurs="0"
15687 type="ns_p:ElectricalConnectionParameterIdType"/>
15688         <xs:element name="characteristicId" minOccurs="0"
15689 type="ns_p:ElectricalConnectionCharacteristicIdType"/>
15690         <xs:element name="characteristicContext" minOccurs="0"
15691 type="ns_p:ElectricalConnectionCharacteristicContextType"/>
15692         <xs:element name="characteristicType" minOccurs="0"
15693 type="ns_p:ElectricalConnectionCharacteristicTypeType"/>
15694         <xs:element name="value" minOccurs="0" type="ns_p:ScaledNumberType"/>
15695         <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
15696       </xs:sequence>
15697     </xs:complexType>
15698     <xs:element name="electricalConnectionCharacteristicData"
15699 type="ns_p:ElectricalConnectionCharacteristicDataType"/>
15700     <xs:complexType name="ElectricalConnectionCharacteristicDataElementsType">
15701       <xs:sequence>
15702         <xs:element name="electricalConnectionId" minOccurs="0"
15703 type="ns_p:ElementTagType"/>
15704         <xs:element name="parameterId" minOccurs="0" type="ns_p:ElementTagType"/>
15705         <xs:element name="characteristicId" minOccurs="0" type="ns_p:ElementTagType"/>
15706         <xs:element name="characteristicContext" minOccurs="0"
15707 type="ns_p:ElementTagType"/>
15708         <xs:element name="characteristicType" minOccurs="0" type="ns_p:ElementTagType"/>
15709         <xs:element name="value" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
15710         <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
15711       </xs:sequence>
15712     </xs:complexType>
15713     <xs:element name="electricalConnectionCharacteristicDataElements"
15714 type="ns_p:ElectricalConnectionCharacteristicDataElementsType"/>
15715     <xs:complexType name="ElectricalConnectionCharacteristicListDataType">
15716       <xs:sequence>
15717         <xs:element maxOccurs="unbounded" minOccurs="0"
15718 ref="ns_p:electricalConnectionCharacteristicData"/>
15719       </xs:sequence>
15720     </xs:complexType>
15721     <xs:element name="electricalConnectionCharacteristicListData"
15722 type="ns_p:ElectricalConnectionCharacteristicListDataType"/>
15723     <xs:complexType name="ElectricalConnectionCharacteristicListDataSelectorsType">
15724       <xs:sequence>
15725         <xs:element name="electricalConnectionId" minOccurs="0"
15726 type="ns_p:ElectricalConnectionIdType"/>
15727         <xs:element name="parameterId" minOccurs="0"
15728 type="ns_p:ElectricalConnectionParameterIdType"/>
15729         <xs:element name="characteristicId" minOccurs="0"
15730 type="ns_p:ElectricalConnectionCharacteristicIdType"/>
15731         <xs:element name="characteristicContext" minOccurs="0"
15732 type="ns_p:ElectricalConnectionCharacteristicContextType"/>
15733         <xs:element name="characteristicType" minOccurs="0"
15734 type="ns_p:ElectricalConnectionCharacteristicTypeType"/>
15735       </xs:sequence>
15736     </xs:complexType>
15737     <xs:element name="electricalConnectionCharacteristicListDataSelectors"
15738 type="ns_p:ElectricalConnectionCharacteristicListDataSelectorsType"/>
15739     <xs:complexType name="ElectricalConnectionStateDataType">
15740       <xs:sequence>
15741         <xs:element name="electricalConnectionId" minOccurs="0"
15742 type="ns_p:ElectricalConnectionIdType"/>
15743         <xs:element minOccurs="0" name="timestamp"
15744 type="ns_p:AbsoluteOrRelativeTimeType"/>
15745         <xs:element name="currentEnergyMode" minOccurs="0" type="ns_p:EnergyModeType"/>
15746         <xs:element minOccurs="0" name="consumptionTime" type="xs:duration"/>
15747         <xs:element minOccurs="0" name="productionTime" type="xs:duration"/>
15748         <xs:element minOccurs="0" name="totalConsumptionTime" type="xs:duration"/>
15749         <xs:element minOccurs="0" name="totalProductionTime" type="xs:duration"/>
15750       </xs:sequence>
15751     </xs:complexType>
15752     <xs:element name="electricalConnectionStateData"
15753 type="ns_p:ElectricalConnectionStateDataType"/>
15754     <xs:complexType name="ElectricalConnectionStateDataElementsType">
15755       <xs:sequence>
```

```
15756         <xs:element name="electricalConnectionId" minOccurs="0"
15757 type="ns_p:ElementTagType"/>
15758         <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
15759         <xs:element name="currentEnergyMode" minOccurs="0" type="ns_p:ElementTagType"/>
15760         <xs:element minOccurs="0" name="consumptionTime" type="ns_p:ElementTagType"/>
15761         <xs:element minOccurs="0" name="productionTime" type="ns_p:ElementTagType"/>
15762         <xs:element minOccurs="0" name="totalConsumptionTime" type="ns_p:ElementTagType"/>
15763         <xs:element minOccurs="0" name="totalProductionTime" type="ns_p:ElementTagType"/>
15764     </xs:sequence>
15765 </xs:complexType>
15766 <xs:element name="electricalConnectionStateDataElements"
15767 type="ns_p:ElectricalConnectionStateDataElementsType"/>
15768 <xs:complexType name="ElectricalConnectionStateListDataType">
15769     <xs:sequence>
15770         <xs:element maxOccurs="unbounded" minOccurs="0"
15771 ref="ns_p:electricalConnectionStateData"/>
15772     </xs:sequence>
15773 </xs:complexType>
15774 <xs:element name="electricalConnectionStateListData"
15775 type="ns_p:ElectricalConnectionStateListDataType"/>
15776 <xs:complexType name="ElectricalConnectionStateListDataSelectorsType">
15777     <xs:sequence>
15778         <xs:element name="electricalConnectionId" minOccurs="0"
15779 type="ns_p:ElectricalConnectionIdType"/>
15780     </xs:sequence>
15781 </xs:complexType>
15782 <xs:element name="electricalConnectionStateListDataSelectors"
15783 type="ns_p:ElectricalConnectionStateListDataSelectorsType"/>
15784 <xs:complexType name="ElectricalConnectionDescriptionDataType">
15785     <xs:sequence>
15786         <xs:element name="electricalConnectionId" minOccurs="0"
15787 type="ns_p:ElectricalConnectionIdType"/>
15788         <xs:element name="powerSupplyType" type="ns_p:ElectricalConnectionVoltageTypeType"
15789 minOccurs="0"/>
15790         <xs:element name="acConnectedPhases" minOccurs="0" type="xs:unsignedInt"/>
15791         <xs:element minOccurs="0" name="acRmsPeriodDuration" type="xs:duration"/>
15792         <xs:element minOccurs="0" name="positiveEnergyDirection"
15793 type="ns_p:EnergyDirectionType"/>
15794         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
15795         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
15796         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
15797     </xs:sequence>
15798 </xs:complexType>
15799 <xs:element name="electricalConnectionDescriptionData"
15800 type="ns_p:ElectricalConnectionDescriptionDataType"/>
15801 <xs:complexType name="ElectricalConnectionDescriptionDataElementsType">
15802     <xs:sequence>
15803         <xs:element name="electricalConnectionId" minOccurs="0"
15804 type="ns_p:ElementTagType"/>
15805         <xs:element name="powerSupplyType" minOccurs="0" type="ns_p:ElementTagType"/>
15806         <xs:element name="acConnectedPhases" minOccurs="0" type="ns_p:ElementTagType"/>
15807         <xs:element minOccurs="0" name="acRmsPeriodDuration" type="ns_p:ElementTagType"/>
15808         <xs:element minOccurs="0" name="positiveEnergyDirection"
15809 type="ns_p:ElementTagType"/>
15810         <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
15811         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
15812         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
15813     </xs:sequence>
15814 </xs:complexType>
15815 <xs:element name="electricalConnectionDescriptionDataElements"
15816 type="ns_p:ElectricalConnectionDescriptionDataElementsType"/>
15817 <xs:complexType name="ElectricalConnectionDescriptionListDataType">
15818     <xs:sequence>
15819         <xs:element maxOccurs="unbounded" minOccurs="0"
15820 ref="ns_p:electricalConnectionDescriptionData"/>
15821     </xs:sequence>
15822 </xs:complexType>
15823 <xs:element name="electricalConnectionDescriptionListData"
15824 type="ns_p:ElectricalConnectionDescriptionListDataType"/>
15825 <xs:complexType name="ElectricalConnectionDescriptionListDataSelectorsType">
15826     <xs:sequence>
15827         <xs:element name="electricalConnectionId" minOccurs="0"
15828 type="ns_p:ElectricalConnectionIdType"/>
15829         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
15830     </xs:sequence>
15831 </xs:complexType>
15832 <xs:element name="electricalConnectionDescriptionListDataSelectors"
15833 type="ns_p:ElectricalConnectionDescriptionListDataSelectorsType"/>
```

```
15834 </xs:schema>
15835
15836
```

15837

15838 A.2.12 HVAC

15839 File name: EEBus_SPINE_TS_HVAC.xsd

```
15840
15841 <?xml version="1.0" encoding="UTF-8"?>
15842 <!--
15843     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
15844     Version 1.3.0 (final)
15845     2023-08-31
15846     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
15847     Source: https://www.eebus.org/en/specifications/
15848 -->
15849 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
15850 xmlns:xs="http://www.w3.org/2001/XMLSchema"
15851 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
15852 blockDefault="#all" elementFormDefault="qualified">
15853     <xs:annotation>
15854         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
15855 e.V. All rights reserved.</xs:documentation>
15856     </xs:annotation>
15857     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
15858     <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
15859     <xs:include schemaLocation="EEBus_SPINE_TS_Setpoint.xsd"/>
15860     <xs:simpleType name="HvacSystemFunctionIdType">
15861         <xs:restriction base="xs:unsignedInt"/>
15862     </xs:simpleType>
15863     <xs:simpleType name="HvacSystemFunctionTypeType">
15864         <xs:union memberTypes="ns_p:HvacSystemFunctionTypeEnumType ns_p:EnumExtendType"/>
15865     </xs:simpleType>
15866     <xs:simpleType name="HvacSystemFunctionTypeEnumType">
15867         <xs:restriction base="xs:string">
15868             <xs:enumeration value="heating"/>
15869             <xs:enumeration value="cooling"/>
15870             <xs:enumeration value="ventilation"/>
15871             <xs:enumeration value="dhw"/>
15872         </xs:restriction>
15873     </xs:simpleType>
15874     <xs:simpleType name="HvacOperationModeIdType">
15875         <xs:restriction base="xs:unsignedInt"/>
15876     </xs:simpleType>
15877     <xs:simpleType name="HvacOperationModeTypeType">
15878         <xs:union memberTypes="ns_p:HvacOperationModeTypeEnumType ns_p:EnumExtendType"/>
15879     </xs:simpleType>
15880     <xs:simpleType name="HvacOperationModeTypeEnumType">
15881         <xs:restriction base="xs:string">
15882             <xs:enumeration value="auto"/>
15883             <xs:enumeration value="on"/>
15884             <xs:enumeration value="off"/>
15885             <xs:enumeration value="eco"/>
15886         </xs:restriction>
15887     </xs:simpleType>
15888     <xs:simpleType name="HvacOverrunIdType">
15889         <xs:restriction base="xs:unsignedInt"/>
15890     </xs:simpleType>
15891     <xs:simpleType name="HvacOverrunTypeType">
15892         <xs:union memberTypes="ns_p:HvacOverrunTypeEnumType ns_p:EnumExtendType"/>
15893     </xs:simpleType>
15894     <xs:simpleType name="HvacOverrunTypeEnumType">
15895         <xs:restriction base="xs:string">
15896             <xs:enumeration value="oneTimeDhw"/>
15897             <xs:enumeration value="party"/>
15898             <xs:enumeration value="sgReadyCondition1"/>
15899             <xs:enumeration value="sgReadyCondition3"/>
15900             <xs:enumeration value="sgReadyCondition4"/>
15901             <xs:enumeration value="oneDayAway"/>
15902             <xs:enumeration value="oneDayAtHome"/>
15903             <xs:enumeration value="oneTimeVentilation"/>
15904             <xs:enumeration value="hvacSystemOff"/>
15905             <xs:enumeration value="valveKick"/>
15906         </xs:restriction>
```

```
15907 </xs:simpleType>
15908 <xs:simpleType name="HvacOverrunStatusType">
15909   <xs:union memberTypes="ns_p:HvacOverrunStatusEnumType ns_p:EnumExtendType"/>
15910 </xs:simpleType>
15911 <xs:simpleType name="HvacOverrunStatusEnumType">
15912   <xs:restriction base="xs:string">
15913     <xs:enumeration value="active"/>
15914     <xs:enumeration value="running"/>
15915     <xs:enumeration value="finished"/>
15916     <xs:enumeration value="inactive"/>
15917   </xs:restriction>
15918 </xs:simpleType>
15919 <xs:complexType name="HvacSystemFunctionDataType">
15920   <xs:sequence>
15921     <xs:element minOccurs="0" name="systemFunctionId"
15922 type="ns_p:HvacSystemFunctionIdType"/>
15923     <xs:element minOccurs="0" name="currentOperationModeId"
15924 type="ns_p:HvacOperationModeIdType"/>
15925     <xs:element minOccurs="0" name="isOperationModeIdChangeable" type="xs:boolean"/>
15926     <xs:element minOccurs="0" name="currentSetpointId" type="ns_p:SetpointIdType"/>
15927     <xs:element minOccurs="0" name="isSetpointIdChangeable" type="xs:boolean"/>
15928     <xs:element minOccurs="0" name="isOverrunActive" type="xs:boolean"/>
15929   </xs:sequence>
15930 </xs:complexType>
15931 <xs:element name="hvacSystemFunctionData" type="ns_p:HvacSystemFunctionDataType"/>
15932 <xs:complexType name="HvacSystemFunctionDataElementsType">
15933   <xs:sequence>
15934     <xs:element minOccurs="0" name="systemFunctionId" type="ns_p:ElementTagType"/>
15935     <xs:element minOccurs="0" name="currentOperationModeId"
15936 type="ns_p:ElementTagType"/>
15937     <xs:element minOccurs="0" name="isOperationModeIdChangeable"
15938 type="ns_p:ElementTagType"/>
15939     <xs:element minOccurs="0" name="currentSetpointId" type="ns_p:ElementTagType"/>
15940     <xs:element minOccurs="0" name="isSetpointIdChangeable"
15941 type="ns_p:ElementTagType"/>
15942     <xs:element minOccurs="0" name="isOverrunActive" type="ns_p:ElementTagType"/>
15943   </xs:sequence>
15944 </xs:complexType>
15945 <xs:element name="hvacSystemFunctionDataElements"
15946 type="ns_p:HvacSystemFunctionDataElementsType"/>
15947 <xs:complexType name="HvacSystemFunctionListDataType">
15948   <xs:sequence>
15949     <xs:element maxOccurs="unbounded" minOccurs="0"
15950 ref="ns_p:hvacSystemFunctionData"/>
15951   </xs:sequence>
15952 </xs:complexType>
15953 <xs:element name="hvacSystemFunctionListData" type="ns_p:HvacSystemFunctionListDataType"/>
15954 <xs:complexType name="HvacSystemFunctionListDataSelectorsType">
15955   <xs:sequence>
15956     <xs:element minOccurs="0" name="systemFunctionId"
15957 type="ns_p:HvacSystemFunctionIdType"/>
15958   </xs:sequence>
15959 </xs:complexType>
15960 <xs:element name="hvacSystemFunctionListDataSelectors"
15961 type="ns_p:HvacSystemFunctionListDataSelectorsType"/>
15962 <xs:complexType name="HvacSystemFunctionOperationModeRelationDataType">
15963   <xs:sequence>
15964     <xs:element minOccurs="0" name="systemFunctionId"
15965 type="ns_p:HvacSystemFunctionIdType"/>
15966     <xs:element minOccurs="0" name="operationModeId"
15967 type="ns_p:HvacOperationModeIdType" maxOccurs="unbounded"/>
15968   </xs:sequence>
15969 </xs:complexType>
15970 <xs:element name="hvacSystemFunctionOperationModeRelationData"
15971 type="ns_p:HvacSystemFunctionOperationModeRelationDataType"/>
15972 <xs:complexType name="HvacSystemFunctionOperationModeRelationDataElementsType">
15973   <xs:sequence>
15974     <xs:element minOccurs="0" name="systemFunctionId" type="ns_p:ElementTagType"/>
15975     <xs:element minOccurs="0" name="operationModeId" type="ns_p:ElementTagType"/>
15976   </xs:sequence>
15977 </xs:complexType>
15978 <xs:element name="hvacSystemFunctionOperationModeRelationDataElements"
15979 type="ns_p:HvacSystemFunctionOperationModeRelationDataElementsType"/>
15980 <xs:complexType name="HvacSystemFunctionOperationModeRelationListDataType">
15981   <xs:sequence>
15982     <xs:element maxOccurs="unbounded" minOccurs="0"
15983 ref="ns_p:hvacSystemFunctionOperationModeRelationData"/>
15984   </xs:sequence>
```

```
15985     </xs:complexType>
15986     <xs:element name="hvacSystemFunctionOperationModeRelationListData"
15987 type="ns_p:HvacSystemFunctionOperationModeRelationListDataType"/>
15988     <xs:complexType name="HvacSystemFunctionOperationModeRelationListDataSelectorsType">
15989       <xs:sequence>
15990         <xs:element minOccurs="0" name="systemFunctionId"
15991 type="ns_p:HvacSystemFunctionIdType"/>
15992       </xs:sequence>
15993     </xs:complexType>
15994     <xs:element name="hvacSystemFunctionOperationModeRelationListDataSelectors"
15995 type="ns_p:HvacSystemFunctionOperationModeRelationListDataSelectorsType"/>
15996     <xs:complexType name="HvacSystemFunctionSetpointRelationDataType">
15997       <xs:sequence>
15998         <xs:element minOccurs="0" name="systemFunctionId"
15999 type="ns_p:HvacSystemFunctionIdType"/>
16000         <xs:element minOccurs="0" name="operationModeId"
16001 type="ns_p:HvacOperationModeIdType"/>
16002         <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"
16003 maxOccurs="unbounded"/>
16004       </xs:sequence>
16005     </xs:complexType>
16006     <xs:element name="hvacSystemFunctionSetpointRelationData"
16007 type="ns_p:HvacSystemFunctionSetpointRelationDataType"/>
16008     <xs:complexType name="HvacSystemFunctionSetpointRelationDataElementsType">
16009       <xs:sequence>
16010         <xs:element minOccurs="0" name="systemFunctionId" type="ns_p:ElementTagType"/>
16011         <xs:element minOccurs="0" name="operationModeId" type="ns_p:ElementTagType"/>
16012         <xs:element minOccurs="0" name="setpointId" type="ns_p:ElementTagType"/>
16013       </xs:sequence>
16014     </xs:complexType>
16015     <xs:element name="hvacSystemFunctionSetpointRelationDataElements"
16016 type="ns_p:HvacSystemFunctionSetpointRelationDataElementsType"/>
16017     <xs:complexType name="HvacSystemFunctionSetpointRelationListDataType">
16018       <xs:sequence>
16019         <xs:element maxOccurs="unbounded" minOccurs="0"
16020 ref="ns_p:hvacSystemFunctionSetpointRelationData"/>
16021       </xs:sequence>
16022     </xs:complexType>
16023     <xs:element name="hvacSystemFunctionSetpointRelationListData"
16024 type="ns_p:HvacSystemFunctionSetpointRelationListDataType"/>
16025     <xs:complexType name="HvacSystemFunctionSetpointRelationListDataSelectorsType">
16026       <xs:sequence>
16027         <xs:element minOccurs="0" name="systemFunctionId"
16028 type="ns_p:HvacSystemFunctionIdType"/>
16029         <xs:element minOccurs="0" name="operationModeId"
16030 type="ns_p:HvacOperationModeIdType"/>
16031       </xs:sequence>
16032     </xs:complexType>
16033     <xs:element name="hvacSystemFunctionSetpointRelationListDataSelectors"
16034 type="ns_p:HvacSystemFunctionSetpointRelationListDataSelectorsType"/>
16035     <xs:complexType name="HvacSystemFunctionPowerSequenceRelationDataType">
16036       <xs:sequence>
16037         <xs:element minOccurs="0" name="systemFunctionId"
16038 type="ns_p:HvacSystemFunctionIdType"/>
16039         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"
16040 maxOccurs="unbounded"/>
16041       </xs:sequence>
16042     </xs:complexType>
16043     <xs:element name="hvacSystemFunctionPowerSequenceRelationData"
16044 type="ns_p:HvacSystemFunctionPowerSequenceRelationDataType"/>
16045     <xs:complexType name="HvacSystemFunctionPowerSequenceRelationDataElementsType">
16046       <xs:sequence>
16047         <xs:element minOccurs="0" name="systemFunctionId" type="ns_p:ElementTagType"/>
16048         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
16049       </xs:sequence>
16050     </xs:complexType>
16051     <xs:element name="hvacSystemFunctionPowerSequenceRelationDataElements"
16052 type="ns_p:HvacSystemFunctionPowerSequenceRelationDataElementsType"/>
16053     <xs:complexType name="HvacSystemFunctionPowerSequenceRelationListDataType">
16054       <xs:sequence>
16055         <xs:element maxOccurs="unbounded" minOccurs="0"
16056 ref="ns_p:hvacSystemFunctionPowerSequenceRelationData"/>
16057       </xs:sequence>
16058     </xs:complexType>
16059     <xs:element name="hvacSystemFunctionPowerSequenceRelationListData"
16060 type="ns_p:HvacSystemFunctionPowerSequenceRelationListDataType"/>
16061     <xs:complexType name="HvacSystemFunctionPowerSequenceRelationListDataSelectorsType">
16062       <xs:sequence>
```

```
16063         <xs:element minOccurs="0" name="systemFunctionId"
16064 type="ns_p:HvacSystemFunctionIdType"/>
16065     </xs:sequence>
16066 </xs:complexType>
16067     <xs:element name="hvacSystemFunctionPowerSequenceRelationListDataSelectors"
16068 type="ns_p:HvacSystemFunctionPowerSequenceRelationListDataSelectorsType"/>
16069     <xs:complexType name="HvacSystemFunctionDescriptionDataType">
16070         <xs:sequence>
16071             <xs:element minOccurs="0" name="systemFunctionId"
16072 type="ns_p:HvacSystemFunctionIdType"/>
16073             <xs:element minOccurs="0" name="systemFunctionType"
16074 type="ns_p:HvacSystemFunctionTypeType"/>
16075             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
16076             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
16077         </xs:sequence>
16078     </xs:complexType>
16079     <xs:element name="hvacSystemFunctionDescriptionData"
16080 type="ns_p:HvacSystemFunctionDescriptionDataType"/>
16081     <xs:complexType name="HvacSystemFunctionDescriptionDataElementsType">
16082         <xs:sequence>
16083             <xs:element minOccurs="0" name="systemFunctionId" type="ns_p:ElementTagType"/>
16084             <xs:element minOccurs="0" name="systemFunctionType" type="ns_p:ElementTagType"/>
16085             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
16086             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
16087         </xs:sequence>
16088     </xs:complexType>
16089     <xs:element name="hvacSystemFunctionDescriptionDataElements"
16090 type="ns_p:HvacSystemFunctionDescriptionDataElementsType"/>
16091     <xs:complexType name="HvacSystemFunctionDescriptionListDataType">
16092         <xs:sequence>
16093             <xs:element maxOccurs="unbounded" minOccurs="0"
16094 ref="ns_p:hvacSystemFunctionDescriptionData"/>
16095         </xs:sequence>
16096     </xs:complexType>
16097     <xs:element name="hvacSystemFunctionDescriptionListData"
16098 type="ns_p:HvacSystemFunctionDescriptionListDataType"/>
16099     <xs:complexType name="HvacSystemFunctionDescriptionListDataSelectorsType">
16100         <xs:sequence>
16101             <xs:element minOccurs="0" name="systemFunctionId"
16102 type="ns_p:HvacSystemFunctionIdType"/>
16103         </xs:sequence>
16104     </xs:complexType>
16105     <xs:element name="hvacSystemFunctionDescriptionListDataSelectors"
16106 type="ns_p:HvacSystemFunctionDescriptionListDataSelectorsType"/>
16107     <xs:complexType name="HvacOperationModeDescriptionDataType">
16108         <xs:sequence>
16109             <xs:element minOccurs="0" name="operationModeId"
16110 type="ns_p:HvacOperationModeIdType"/>
16111             <xs:element minOccurs="0" name="operationModeType"
16112 type="ns_p:HvacOperationModeTypeType"/>
16113             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
16114             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
16115         </xs:sequence>
16116     </xs:complexType>
16117     <xs:element name="hvacOperationModeDescriptionData"
16118 type="ns_p:HvacOperationModeDescriptionDataType"/>
16119     <xs:complexType name="HvacOperationModeDescriptionDataElementsType">
16120         <xs:sequence>
16121             <xs:element minOccurs="0" name="operationModeId" type="ns_p:ElementTagType"/>
16122             <xs:element minOccurs="0" name="operationModeType" type="ns_p:ElementTagType"/>
16123             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
16124             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
16125         </xs:sequence>
16126     </xs:complexType>
16127     <xs:element name="hvacOperationModeDescriptionDataElements"
16128 type="ns_p:HvacOperationModeDescriptionDataElementsType"/>
16129     <xs:complexType name="HvacOperationModeDescriptionListDataType">
16130         <xs:sequence>
16131             <xs:element maxOccurs="unbounded" minOccurs="0"
16132 ref="ns_p:hvacOperationModeDescriptionData"/>
16133         </xs:sequence>
16134     </xs:complexType>
16135     <xs:element name="hvacOperationModeDescriptionListData"
16136 type="ns_p:HvacOperationModeDescriptionListDataType"/>
16137     <xs:complexType name="HvacOperationModeDescriptionListDataSelectorsType">
16138         <xs:sequence>
16139             <xs:element minOccurs="0" name="operationModeId"
16140 type="ns_p:HvacOperationModeIdType"/>
```

```
16141         </xs:sequence>
16142     </xs:complexType>
16143     <xs:element name="hvacOperationModeDescriptionListDataSelectors"
16144 type="ns_p:HvacOperationModeDescriptionListDataSelectorsType"/>
16145     <xs:complexType name="HvacOverrunDataType">
16146         <xs:sequence>
16147             <xs:element name="overrunId" minOccurs="0" type="ns_p:HvacOverrunIdType"/>
16148             <xs:element name="overrunStatus" minOccurs="0" type="ns_p:HvacOverrunStatusType"/>
16149             <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
16150             <xs:element minOccurs="0" name="isOverrunStatusChangeable" type="xs:boolean"/>
16151         </xs:sequence>
16152     </xs:complexType>
16153     <xs:element name="hvacOverrunData" type="ns_p:HvacOverrunDataType"/>
16154     <xs:complexType name="HvacOverrunDataElementsType">
16155         <xs:sequence>
16156             <xs:element name="overrunId" minOccurs="0" type="ns_p:ElementTagType"/>
16157             <xs:element name="overrunStatus" minOccurs="0" type="ns_p:ElementTagType"/>
16158             <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
16159             <xs:element minOccurs="0" name="isOverrunStatusChangeable"
16160 type="ns_p:ElementTagType"/>
16161         </xs:sequence>
16162     </xs:complexType>
16163     <xs:element name="hvacOverrunDataElements" type="ns_p:HvacOverrunDataElementsType"/>
16164     <xs:complexType name="HvacOverrunListDataType">
16165         <xs:sequence>
16166             <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:hvacOverrunData"/>
16167         </xs:sequence>
16168     </xs:complexType>
16169     <xs:element name="hvacOverrunListData" type="ns_p:HvacOverrunListDataType"/>
16170     <xs:complexType name="HvacOverrunListDataSelectorsType">
16171         <xs:sequence>
16172             <xs:element name="overrunId" minOccurs="0" type="ns_p:HvacOverrunIdType"/>
16173         </xs:sequence>
16174     </xs:complexType>
16175     <xs:element name="hvacOverrunListDataSelectors"
16176 type="ns_p:HvacOverrunListDataSelectorsType"/>
16177     <xs:complexType name="HvacOverrunDescriptionDataType">
16178         <xs:sequence>
16179             <xs:element name="overrunId" minOccurs="0" type="ns_p:HvacOverrunIdType"/>
16180             <xs:element minOccurs="0" name="overrunType" type="ns_p:HvacOverrunTypeType"/>
16181             <xs:element minOccurs="0" name="affectedSystemFunctionId"
16182 type="ns_p:HvacSystemFunctionIdType" maxOccurs="unbounded"/>
16183             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
16184             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
16185         </xs:sequence>
16186     </xs:complexType>
16187     <xs:element name="hvacOverrunDescriptionData" type="ns_p:HvacOverrunDescriptionDataType"/>
16188     <xs:complexType name="HvacOverrunDescriptionDataElementsType">
16189         <xs:sequence>
16190             <xs:element name="overrunId" minOccurs="0" type="ns_p:ElementTagType"/>
16191             <xs:element minOccurs="0" name="overrunType" type="ns_p:ElementTagType"/>
16192             <xs:element minOccurs="0" name="affectedSystemFunctionId"
16193 type="ns_p:ElementTagType"/>
16194             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
16195             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
16196         </xs:sequence>
16197     </xs:complexType>
16198     <xs:element name="hvacOverrunDescriptionDataElements"
16199 type="ns_p:HvacOverrunDescriptionDataElementsType"/>
16200     <xs:complexType name="HvacOverrunDescriptionListDataType">
16201         <xs:sequence>
16202             <xs:element maxOccurs="unbounded" minOccurs="0"
16203 ref="ns_p:hvacOverrunDescriptionData"/>
16204         </xs:sequence>
16205     </xs:complexType>
16206     <xs:element name="hvacOverrunDescriptionListData"
16207 type="ns_p:HvacOverrunDescriptionListDataType"/>
16208     <xs:complexType name="HvacOverrunDescriptionListDataSelectorsType">
16209         <xs:sequence>
16210             <xs:element name="overrunId" minOccurs="0" type="ns_p:HvacOverrunIdType"/>
16211         </xs:sequence>
16212     </xs:complexType>
16213     <xs:element name="hvacOverrunDescriptionListDataSelectors"
16214 type="ns_p:HvacOverrunDescriptionListDataSelectorsType"/>
16215 </xs:schema>
16216
16217
```

16218

16219 **A.2.13 Identification**

16220 File name: EEBus_SPINE_TS_Identification.xsd

```
16221 <?xml version="1.0" encoding="UTF-8"?>
16222 <!--
16223     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
16224     Version 1.3.0 (final)
16225     2023-08-31
16226     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
16227     Source: https://www.eebus.org/en/specifications/
16228 -->
16229 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
16230   xmlns:xs="http://www.w3.org/2001/XMLSchema"
16231   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
16232   blockDefault="#all" elementFormDefault="qualified">
16233   <xs:annotation>
16234     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
16235     e.V. All rights reserved.</xs:documentation>
16236   </xs:annotation>
16237   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
16238   <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
16239   <xs:simpleType name="IdentificationIdType">
16240     <xs:restriction base="xs:unsignedInt"/>
16241   </xs:simpleType>
16242   <xs:simpleType name="IdentificationTypeType">
16243     <xs:union memberTypes="ns_p:IdentificationTypeEnumType ns_p:EnumExtendType"/>
16244   </xs:simpleType>
16245   <xs:simpleType name="IdentificationTypeEnumType">
16246     <xs:restriction base="xs:string">
16247       <xs:enumeration value="eui48"/>
16248       <xs:enumeration value="eui64"/>
16249       <xs:enumeration value="userRfidTag"/>
16250     </xs:restriction>
16251   </xs:simpleType>
16252   <xs:simpleType name="IdentificationValueType">
16253     <xs:restriction base="xs:string"/>
16254   </xs:simpleType>
16255   <xs:simpleType name="SessionIdType">
16256     <xs:restriction base="xs:unsignedInt"/>
16257   </xs:simpleType>
16258   <xs:complexType name="IdentificationDataType">
16259     <xs:sequence>
16260       <xs:element minOccurs="0" name="identificationId"
16261         type="ns_p:IdentificationIdType"/>
16262       <xs:element minOccurs="0" name="identificationType"
16263         type="ns_p:IdentificationTypeType"/>
16264       <xs:element minOccurs="0" name="identificationValue"
16265         type="ns_p:IdentificationValueType"/>
16266       <xs:element minOccurs="0" name="authorized" type="xs:boolean"/>
16267     </xs:sequence>
16268   </xs:complexType>
16269   <xs:element name="identificationData" type="ns_p:IdentificationDataType"/>
16270   <xs:complexType name="IdentificationDataElementsType">
16271     <xs:sequence>
16272       <xs:element minOccurs="0" name="identificationId" type="ns_p:ElementTagType"/>
16273       <xs:element minOccurs="0" name="identificationType" type="ns_p:ElementTagType"/>
16274       <xs:element minOccurs="0" name="identificationValue" type="ns_p:ElementTagType"/>
16275       <xs:element minOccurs="0" name="authorized" type="ns_p:ElementTagType"/>
16276     </xs:sequence>
16277   </xs:complexType>
16278   <xs:element name="identificationDataElements" type="ns_p:IdentificationDataElementsType"/>
16279   <xs:complexType name="IdentificationListDataType">
16280     <xs:sequence>
16281       <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:identificationData"/>
16282     </xs:sequence>
16283   </xs:complexType>
16284   <xs:element name="identificationListData" type="ns_p:IdentificationListDataType"/>
16285   <xs:complexType name="IdentificationListDataSelectorsType">
16286     <xs:sequence>
16287       <xs:element minOccurs="0" name="identificationId"
16288         type="ns_p:IdentificationIdType"/>
16289       <xs:element minOccurs="0" name="identificationType"
16290         type="ns_p:IdentificationTypeType"/>
16291
```



```
16292         </xs:sequence>
16293     </xs:complexType>
16294     <xs:element name="identificationListDataSelectors"
16295 type="ns_p:IdentificationListDataSelectorsType"/>
16296     <xs:complexType name="SessionIdentificationDataType">
16297         <xs:sequence>
16298             <xs:element minOccurs="0" name="sessionId" type="ns_p:SessionIdType"/>
16299             <xs:element minOccurs="0" name="identificationId"
16300 type="ns_p:IdentificationIdType"/>
16301             <xs:element minOccurs="0" name="isLatestSession" type="xs:boolean"/>
16302             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
16303         </xs:sequence>
16304     </xs:complexType>
16305     <xs:element name="sessionIdentificationData" type="ns_p:SessionIdentificationDataType"/>
16306     <xs:complexType name="SessionIdentificationDataElementsType">
16307         <xs:sequence>
16308             <xs:element minOccurs="0" name="sessionId" type="ns_p:ElementTagType"/>
16309             <xs:element minOccurs="0" name="identificationId" type="ns_p:ElementTagType"/>
16310             <xs:element minOccurs="0" name="isLatestSession" type="ns_p:ElementTagType"/>
16311             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
16312         </xs:sequence>
16313     </xs:complexType>
16314     <xs:element name="sessionIdentificationDataElements"
16315 type="ns_p:SessionIdentificationDataElementsType"/>
16316     <xs:complexType name="SessionIdentificationListDataType">
16317         <xs:sequence>
16318             <xs:element maxOccurs="unbounded" minOccurs="0"
16319 ref="ns_p:sessionIdentificationData"/>
16320         </xs:sequence>
16321     </xs:complexType>
16322     <xs:element name="sessionIdentificationListData"
16323 type="ns_p:SessionIdentificationListDataType"/>
16324     <xs:complexType name="SessionIdentificationListDataSelectorsType">
16325         <xs:sequence>
16326             <xs:element minOccurs="0" name="sessionId" type="ns_p:SessionIdType"/>
16327             <xs:element minOccurs="0" name="identificationId"
16328 type="ns_p:IdentificationIdType"/>
16329             <xs:element minOccurs="0" name="isLatestSession" type="xs:boolean"/>
16330             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
16331         </xs:sequence>
16332     </xs:complexType>
16333     <xs:element name="sessionIdentificationListDataSelectors"
16334 type="ns_p:SessionIdentificationListDataSelectorsType"/>
16335     <xs:complexType name="SessionMeasurementRelationDataType">
16336         <xs:sequence>
16337             <xs:element minOccurs="0" name="sessionId" type="ns_p:SessionIdType"/>
16338             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"
16339 maxOccurs="unbounded"/>
16340         </xs:sequence>
16341     </xs:complexType>
16342     <xs:element name="sessionMeasurementRelationData"
16343 type="ns_p:SessionMeasurementRelationDataType"/>
16344     <xs:complexType name="SessionMeasurementRelationDataElementsType">
16345         <xs:sequence>
16346             <xs:element minOccurs="0" name="sessionId" type="ns_p:ElementTagType"/>
16347             <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16348         </xs:sequence>
16349     </xs:complexType>
16350     <xs:element name="sessionMeasurementRelationDataElements"
16351 type="ns_p:SessionMeasurementRelationDataElementsType"/>
16352     <xs:complexType name="SessionMeasurementRelationListDataType">
16353         <xs:sequence>
16354             <xs:element maxOccurs="unbounded" minOccurs="0"
16355 ref="ns_p:sessionMeasurementRelationData"/>
16356         </xs:sequence>
16357     </xs:complexType>
16358     <xs:element name="sessionMeasurementRelationListData"
16359 type="ns_p:SessionMeasurementRelationListDataType"/>
16360     <xs:complexType name="SessionMeasurementRelationListDataSelectorsType">
16361         <xs:sequence>
16362             <xs:element minOccurs="0" name="sessionId" type="ns_p:SessionIdType"/>
16363             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16364         </xs:sequence>
16365     </xs:complexType>
16366     <xs:element name="sessionMeasurementRelationListDataSelectors"
16367 type="ns_p:SessionMeasurementRelationListDataSelectorsType"/>
16368 </xs:schema>
16369
```

16370

16371

16372 **A.2.14 LoadControl**

16373 File name: EEBus_SPINE_TS_LoadControl.xsd

16374

16375

<?xml version="1.0" encoding="UTF-8"?>

16376

<!--

16377

Smart Premises Interoperable Neutral-Message Exchange (SPINE)

16378

Version 1.3.0 (final)

16379

2023-08-31

16380

Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.

16381

Source: <https://www.eebus.org/en/specifications/>

16382

-->

16383

<xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"

16384

xmlns:xs="http://www.w3.org/2001/XMLSchema"

16385

targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"

16386

blockDefault="#all" elementFormDefault="qualified">

16387

<xs:annotation>

16388

<xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative

16389

e.V. All rights reserved.</xs:documentation>

16390

</xs:annotation>

16391

<xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>

16392

<xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>

16393

<xs:simpleType name="LoadControlEventIdType">

16394

<xs:restriction base="xs:unsignedInt"/>

16395

</xs:simpleType>

16396

<xs:simpleType name="LoadControlEventActionType">

16397

<xs:union memberTypes="ns_p:LoadControlEventActionEnumType ns_p:EnumExtendType"/>

16398

</xs:simpleType>

16399

<xs:simpleType name="LoadControlEventActionEnumType">

16400

<xs:restriction base="xs:string">

16401

<xs:enumeration value="pause"/>

16402

<xs:enumeration value="resume"/>

16403

<xs:enumeration value="reduce"/>

16404

<xs:enumeration value="increase"/>

16405

<xs:enumeration value="emergency"/>

16406

<xs:enumeration value="normal"/>

16407

</xs:restriction>

16408

</xs:simpleType>

16409

<xs:simpleType name="LoadControlEventStateType">

16410

<xs:union memberTypes="ns_p:LoadControlEventStateEnumType ns_p:EnumExtendType"/>

16411

</xs:simpleType>

16412

<xs:simpleType name="LoadControlEventStateEnumType">

16413

<xs:restriction base="xs:string">

16414

<xs:enumeration value="eventAccepted"/>

16415

<xs:enumeration value="eventStarted"/>

16416

<xs:enumeration value="eventStopped"/>

16417

<xs:enumeration value="eventRejected"/>

16418

<xs:enumeration value="eventCancelled"/>

16419

<xs:enumeration value="eventError"/>

16420

</xs:restriction>

16421

</xs:simpleType>

16422

<xs:simpleType name="LoadControlLimitIdType">

16423

<xs:restriction base="xs:unsignedInt"/>

16424

</xs:simpleType>

16425

<xs:simpleType name="LoadControlLimitTypeType">

16426

<xs:union memberTypes="ns_p:LoadControlLimitTypeEnumType ns_p:EnumExtendType"/>

16427

</xs:simpleType>

16428

<xs:simpleType name="LoadControlLimitTypeEnumType">

16429

<xs:restriction base="xs:string">

16430

<xs:enumeration value="minValueLimit"/>

16431

<xs:enumeration value="maxValueLimit"/>

16432

<xs:enumeration value="signDependentAbsValueLimit"/>

16433

</xs:restriction>

16434

</xs:simpleType>

16435

<xs:simpleType name="LoadControlCategoryType">

16436

<xs:union memberTypes="ns_p:LoadControlCategoryEnumType ns_p:EnumExtendType"/>

16437

</xs:simpleType>

16438

<xs:simpleType name="LoadControlCategoryEnumType">

16439

<xs:restriction base="xs:string">

16440

<xs:enumeration value="obligation"/>

16441

<xs:enumeration value="recommendation"/>

16442

<xs:enumeration value="optimization"/>

```
16443         </xs:restriction>
16444     </xs:simpleType>
16445     <xs:complexType name="LoadControlNodeDataType">
16446         <xs:sequence>
16447             <xs:element minOccurs="0" name="isNodeRemoteControllable" type="xs:boolean"/>
16448         </xs:sequence>
16449     </xs:complexType>
16450     <xs:element name="loadControlNodeData" type="ns_p:LoadControlNodeDataType"/>
16451     <xs:complexType name="LoadControlNodeDataElementsType">
16452         <xs:sequence>
16453             <xs:element minOccurs="0" name="isNodeRemoteControllable"
16454 type="ns_p:ElementTagType"/>
16455         </xs:sequence>
16456     </xs:complexType>
16457     <xs:element name="loadControlNodeDataElements"
16458 type="ns_p:LoadControlNodeDataElementsType"/>
16459     <xs:complexType name="LoadControlEventDataDataType">
16460         <xs:sequence>
16461             <xs:element minOccurs="0" name="timestamp"
16462 type="ns_p:AbsoluteOrRelativeTimeType"/>
16463             <xs:element minOccurs="0" name="eventId" type="ns_p:LoadControlEventIdType"/>
16464             <xs:element minOccurs="0" name="eventActionConsume"
16465 type="ns_p:LoadControlEventActionType"/>
16466             <xs:element minOccurs="0" name="eventActionProduce"
16467 type="ns_p:LoadControlEventActionType"/>
16468             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
16469         </xs:sequence>
16470     </xs:complexType>
16471     <xs:element name="loadControlEventData" type="ns_p:LoadControlEventDataDataType"/>
16472     <xs:complexType name="LoadControlEventDataElementsType">
16473         <xs:sequence>
16474             <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
16475             <xs:element minOccurs="0" name="eventId" type="ns_p:ElementTagType"/>
16476             <xs:element minOccurs="0" name="eventActionConsume" type="ns_p:ElementTagType"/>
16477             <xs:element minOccurs="0" name="eventActionProduce" type="ns_p:ElementTagType"/>
16478             <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
16479         </xs:sequence>
16480     </xs:complexType>
16481     <xs:element name="loadControlEventDataElements"
16482 type="ns_p:LoadControlEventDataElementsType"/>
16483     <xs:complexType name="LoadControlEventListDataType">
16484         <xs:sequence>
16485             <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:loadControlEventData"/>
16486         </xs:sequence>
16487     </xs:complexType>
16488     <xs:element name="loadControlEventListData" type="ns_p:LoadControlEventListDataType"/>
16489     <xs:complexType name="LoadControlEventListDataSelectorsType">
16490         <xs:sequence>
16491             <xs:element minOccurs="0" name="timestampInterval"
16492 type="ns_p:TimestampIntervalType"/>
16493             <xs:element minOccurs="0" name="eventId" type="ns_p:LoadControlEventIdType"/>
16494         </xs:sequence>
16495     </xs:complexType>
16496     <xs:element name="loadControlEventListDataSelectors"
16497 type="ns_p:LoadControlEventListDataSelectorsType"/>
16498     <xs:complexType name="LoadControlStateDataType">
16499         <xs:sequence>
16500             <xs:element minOccurs="0" name="timestamp"
16501 type="ns_p:AbsoluteOrRelativeTimeType"/>
16502             <xs:element minOccurs="0" name="eventId" type="ns_p:LoadControlEventIdType"/>
16503             <xs:element minOccurs="0" name="eventStateConsume"
16504 type="ns_p:LoadControlEventStateType"/>
16505             <xs:element minOccurs="0" name="appliedEventActionConsume"
16506 type="ns_p:LoadControlEventActionType"/>
16507             <xs:element minOccurs="0" name="eventStateProduce"
16508 type="ns_p:LoadControlEventStateType"/>
16509             <xs:element minOccurs="0" name="appliedEventActionProduce"
16510 type="ns_p:LoadControlEventActionType"/>
16511         </xs:sequence>
16512     </xs:complexType>
16513     <xs:element name="loadControlStateData" type="ns_p:LoadControlStateDataType"/>
16514     <xs:complexType name="LoadControlStateDataElementsType">
16515         <xs:sequence>
16516             <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
16517             <xs:element minOccurs="0" name="eventId" type="ns_p:ElementTagType"/>
16518             <xs:element minOccurs="0" name="eventStateConsume" type="ns_p:ElementTagType"/>
16519             <xs:element minOccurs="0" name="appliedEventActionConsume"
16520 type="ns_p:ElementTagType"/>
```

```
16521         <xs:element minOccurs="0" name="eventStateProduce" type="ns_p:ElementTagType"/>
16522         <xs:element minOccurs="0" name="appliedEventActionProduce"
16523 type="ns_p:ElementTagType"/>
16524     </xs:sequence>
16525 </xs:complexType>
16526 <xs:element name="loadControlStateDataElements"
16527 type="ns_p:LoadControlStateDataElementsType"/>
16528 <xs:complexType name="LoadControlStateListDataType">
16529     <xs:sequence>
16530         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:loadControlStateData"/>
16531     </xs:sequence>
16532 </xs:complexType>
16533 <xs:element name="loadControlStateListData" type="ns_p:LoadControlStateListDataType"/>
16534 <xs:complexType name="LoadControlStateListDataSelectorsType">
16535     <xs:sequence>
16536         <xs:element minOccurs="0" name="timestampInterval"
16537 type="ns_p:TimestampIntervalType"/>
16538         <xs:element minOccurs="0" name="eventId" type="ns_p:LoadControlEventIdType"/>
16539     </xs:sequence>
16540 </xs:complexType>
16541 <xs:element name="loadControlStateListDataSelectors"
16542 type="ns_p:LoadControlStateListDataSelectorsType"/>
16543 <xs:complexType name="LoadControlLimitDataType">
16544     <xs:sequence>
16545         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16546         <xs:element name="isLimitChangeable" minOccurs="0" type="xs:boolean"/>
16547         <xs:element name="isLimitActive" minOccurs="0" type="xs:boolean"/>
16548         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
16549         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
16550     </xs:sequence>
16551 </xs:complexType>
16552 <xs:element name="loadControlLimitData" type="ns_p:LoadControlLimitDataType"/>
16553 <xs:complexType name="LoadControlLimitDataElementsType">
16554     <xs:sequence>
16555         <xs:element minOccurs="0" name="limitId" type="ns_p:ElementTagType"/>
16556         <xs:element name="isLimitChangeable" minOccurs="0" type="ns_p:ElementTagType"/>
16557         <xs:element name="isLimitActive" minOccurs="0" type="ns_p:ElementTagType"/>
16558         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
16559         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
16560     </xs:sequence>
16561 </xs:complexType>
16562 <xs:element name="loadControlLimitDataElements"
16563 type="ns_p:LoadControlLimitDataElementsType"/>
16564 <xs:complexType name="LoadControlLimitListDataType">
16565     <xs:sequence>
16566         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:loadControlLimitData"/>
16567     </xs:sequence>
16568 </xs:complexType>
16569 <xs:element name="loadControlLimitListData" type="ns_p:LoadControlLimitListDataType"/>
16570 <xs:complexType name="LoadControlLimitListDataSelectorsType">
16571     <xs:sequence>
16572         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16573     </xs:sequence>
16574 </xs:complexType>
16575 <xs:element name="loadControlLimitListDataSelectors"
16576 type="ns_p:LoadControlLimitListDataSelectorsType"/>
16577 <xs:complexType name="LoadControlLimitConstraintsDataType">
16578     <xs:sequence>
16579         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16580         <xs:element minOccurs="0" name="valueRangeMin" type="ns_p:ScaledNumberType"/>
16581         <xs:element minOccurs="0" name="valueRangeMax" type="ns_p:ScaledNumberType"/>
16582         <xs:element minOccurs="0" name="valueStepSize" type="ns_p:ScaledNumberType"/>
16583     </xs:sequence>
16584 </xs:complexType>
16585 <xs:element name="loadControlLimitConstraintsData"
16586 type="ns_p:LoadControlLimitConstraintsDataType"/>
16587 <xs:complexType name="LoadControlLimitConstraintsDataElementsType">
16588     <xs:sequence>
16589         <xs:element minOccurs="0" name="limitId" type="ns_p:ElementTagType"/>
16590         <xs:element minOccurs="0" name="valueRangeMin"
16591 type="ns_p:ScaledNumberElementsType"/>
16592         <xs:element minOccurs="0" name="valueRangeMax"
16593 type="ns_p:ScaledNumberElementsType"/>
16594         <xs:element minOccurs="0" name="valueStepSize"
16595 type="ns_p:ScaledNumberElementsType"/>
16596     </xs:sequence>
16597 </xs:complexType>
```

```
16598     <xs:element name="loadControlLimitConstraintsDataElements"
16599 type="ns_p:LoadControlLimitConstraintsDataElementsType"/>
16600     <xs:complexType name="LoadControlLimitConstraintsListDataType">
16601       <xs:sequence>
16602         <xs:element maxOccurs="unbounded" minOccurs="0"
16603 ref="ns_p:loadControlLimitConstraintsData"/>
16604       </xs:sequence>
16605     </xs:complexType>
16606     <xs:element name="loadControlLimitConstraintsListData"
16607 type="ns_p:LoadControlLimitConstraintsListDataType"/>
16608     <xs:complexType name="LoadControlLimitConstraintsListDataSelectorsType">
16609       <xs:sequence>
16610         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16611       </xs:sequence>
16612     </xs:complexType>
16613     <xs:element name="loadControlLimitConstraintsListDataSelectors"
16614 type="ns_p:LoadControlLimitConstraintsListDataSelectorsType"/>
16615     <xs:complexType name="LoadControlLimitDescriptionDataType">
16616       <xs:sequence>
16617         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16618         <xs:element minOccurs="0" name="limitType" type="ns_p:LoadControlLimitTypeType"/>
16619         <xs:element minOccurs="0" name="limitCategory"
16620 type="ns_p:LoadControlCategoryType"/>
16621         <xs:element minOccurs="0" name="limitDirection" type="ns_p:EnergyDirectionType"/>
16622         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16623         <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
16624         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
16625         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
16626         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
16627       </xs:sequence>
16628     </xs:complexType>
16629     <xs:element name="loadControlLimitDescriptionData"
16630 type="ns_p:LoadControlLimitDescriptionDataType"/>
16631     <xs:complexType name="LoadControlLimitDescriptionDataElementsType">
16632       <xs:sequence>
16633         <xs:element minOccurs="0" name="limitId" type="ns_p:ElementTagType"/>
16634         <xs:element minOccurs="0" name="limitType" type="ns_p:ElementTagType"/>
16635         <xs:element minOccurs="0" name="limitCategory" type="ns_p:ElementTagType"/>
16636         <xs:element minOccurs="0" name="limitDirection" type="ns_p:ElementTagType"/>
16637         <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16638         <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
16639         <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
16640         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
16641         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
16642       </xs:sequence>
16643     </xs:complexType>
16644     <xs:element name="loadControlLimitDescriptionDataElements"
16645 type="ns_p:LoadControlLimitDescriptionDataElementsType"/>
16646     <xs:complexType name="LoadControlLimitDescriptionListDataType">
16647       <xs:sequence>
16648         <xs:element maxOccurs="unbounded" minOccurs="0"
16649 ref="ns_p:loadControlLimitDescriptionData"/>
16650       </xs:sequence>
16651     </xs:complexType>
16652     <xs:element name="loadControlLimitDescriptionListData"
16653 type="ns_p:LoadControlLimitDescriptionListDataType"/>
16654     <xs:complexType name="LoadControlLimitDescriptionListDataSelectorsType">
16655       <xs:sequence>
16656         <xs:element minOccurs="0" name="limitId" type="ns_p:LoadControlLimitIdType"/>
16657         <xs:element minOccurs="0" name="limitType" type="ns_p:LoadControlLimitTypeType"/>
16658         <xs:element minOccurs="0" name="limitDirection" type="ns_p:EnergyDirectionType"/>
16659         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16660         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
16661       </xs:sequence>
16662     </xs:complexType>
16663     <xs:element name="loadControlLimitDescriptionListDataSelectors"
16664 type="ns_p:LoadControlLimitDescriptionListDataSelectorsType"/>
16665 </xs:schema>
```

A.2.15 Measurement

File name: EEBus_SPINE_TS_Measurement.xsd

```
<?xml version="1.0" encoding="UTF-8"?>
<!--
    Smart Premises Interoperable Neutral-Message Exchange (SPINE)
    Version 1.3.0 (final)
    2023-08-31
    Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
    Source: https://www.eebus.org/en/specifications/
-->
<xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
  blockDefault="#all" elementFormDefault="qualified">
  <xs:annotation>
    <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
    e.V. All rights reserved.</xs:documentation>
  </xs:annotation>
  <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
  <xs:include schemaLocation="EEBus_SPINE_TS_Threshold.xsd"/>
  <xs:simpleType name="MeasurementIdType">
    <xs:restriction base="xs:unsignedInt"/>
  </xs:simpleType>
  <xs:simpleType name="MeasurementTypeType">
    <xs:union memberTypes="ns_p:MeasurementTypeEnumType ns_p:EnumExtendType"/>
  </xs:simpleType>
  <xs:simpleType name="MeasurementTypeEnumType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="acceleration"/>
      <xs:enumeration value="angle"/>
      <xs:enumeration value="angularVelocity"/>
      <xs:enumeration value="area"/>
      <xs:enumeration value="atmosphericPressure"/>
      <xs:enumeration value="capacity"/>
      <xs:enumeration value="concentration"/>
      <xs:enumeration value="count"/>
      <xs:enumeration value="current"/>
      <xs:enumeration value="density"/>
      <xs:enumeration value="distance"/>
      <xs:enumeration value="electricField"/>
      <xs:enumeration value="energy"/>
      <xs:enumeration value="force"/>
      <xs:enumeration value="frequency"/>
      <xs:enumeration value="harmonicDistortion"/>
      <xs:enumeration value="heat"/>
      <xs:enumeration value="heatFlux"/>
      <xs:enumeration value="illuminance"/>
      <xs:enumeration value="impulse"/>
      <xs:enumeration value="level"/>
      <xs:enumeration value="magneticField"/>
      <xs:enumeration value="mass"/>
      <xs:enumeration value="massFlow"/>
      <xs:enumeration value="particles"/>
      <xs:enumeration value="percentage"/>
      <xs:enumeration value="power"/>
      <xs:enumeration value="powerFactor"/>
      <xs:enumeration value="pressure"/>
      <xs:enumeration value="radonActivity"/>
      <xs:enumeration value="relativeHumidity"/>
      <xs:enumeration value="resistance"/>
      <xs:enumeration value="solarRadiation"/>
      <xs:enumeration value="speed"/>
      <xs:enumeration value="temperature"/>
      <xs:enumeration value="time"/>
      <xs:enumeration value="torque"/>
      <xs:enumeration value="unknown"/>
      <xs:enumeration value="velocity"/>
      <xs:enumeration value="voltage"/>
      <xs:enumeration value="volume"/>
      <xs:enumeration value="volumetricFlow"/>
    </xs:restriction>
  </xs:simpleType>
  <xs:simpleType name="MeasurementValueTypeType">
    <xs:union memberTypes="ns_p:MeasurementValueTypeEnumType ns_p:EnumExtendType"/>
  </xs:simpleType>
  <xs:simpleType name="MeasurementValueTypeEnumType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="value"/>
      <xs:enumeration value="averageValue"/>
    </xs:restriction>
  </xs:simpleType>
```

```
16749         <xs:enumeration value="minValue"/>
16750         <xs:enumeration value="maxValue"/>
16751         <xs:enumeration value="standardDeviation"/>
16752     </xs:restriction>
16753 </xs:simpleType>
16754 <xs:simpleType name="MeasurementValueSourceType">
16755     <xs:union memberTypes="ns_p:MeasurementValueSourceEnumType ns_p:EnumExtendType"/>
16756 </xs:simpleType>
16757 <xs:simpleType name="MeasurementValueSourceEnumType">
16758     <xs:restriction base="xs:string">
16759         <xs:enumeration value="measuredValue"/>
16760         <xs:enumeration value="calculatedValue"/>
16761         <xs:enumeration value="empiricalValue"/>
16762     </xs:restriction>
16763 </xs:simpleType>
16764 <xs:simpleType name="MeasurementValueTendencyType">
16765     <xs:union memberTypes="ns_p:MeasurementValueTendencyEnumType ns_p:EnumExtendType"/>
16766 </xs:simpleType>
16767 <xs:simpleType name="MeasurementValueTendencyEnumType">
16768     <xs:restriction base="xs:string">
16769         <xs:enumeration value="rising"/>
16770         <xs:enumeration value="stable"/>
16771         <xs:enumeration value="falling"/>
16772     </xs:restriction>
16773 </xs:simpleType>
16774 <xs:simpleType name="MeasurementValueStateType">
16775     <xs:union memberTypes="ns_p:MeasurementValueStateEnumType ns_p:EnumExtendType"/>
16776 </xs:simpleType>
16777 <xs:simpleType name="MeasurementValueStateEnumType">
16778     <xs:restriction base="xs:string">
16779         <xs:enumeration value="normal"/>
16780         <xs:enumeration value="outOfRange"/>
16781         <xs:enumeration value="error"/>
16782     </xs:restriction>
16783 </xs:simpleType>
16784 <xs:complexType name="MeasurementDataType">
16785     <xs:sequence>
16786         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16787         <xs:element minOccurs="0" name="valueType" type="ns_p:MeasurementValueTypeType"/>
16788         <xs:element minOccurs="0" name="timestamp"
16789 type="ns_p:AbsoluteOrRelativeTimeType"/>
16790         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
16791         <xs:element minOccurs="0" name="evaluationPeriod" type="ns_p:TimePeriodType"/>
16792         <xs:element minOccurs="0" name="valueSource"
16793 type="ns_p:MeasurementValueSourceType"/>
16794         <xs:element minOccurs="0" name="valueTendency"
16795 type="ns_p:MeasurementValueTendencyType"/>
16796         <xs:element minOccurs="0" name="valueState"
16797 type="ns_p:MeasurementValueStateType"/>
16798     </xs:sequence>
16799 </xs:complexType>
16800 <xs:element name="measurementData" type="ns_p:MeasurementDataType"/>
16801 <xs:complexType name="MeasurementDataElementsType">
16802     <xs:sequence>
16803         <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16804         <xs:element minOccurs="0" name="valueType" type="ns_p:ElementTagType"/>
16805         <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
16806         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
16807         <xs:element minOccurs="0" name="evaluationPeriod"
16808 type="ns_p:TimePeriodElementsType"/>
16809         <xs:element minOccurs="0" name="valueSource" type="ns_p:ElementTagType"/>
16810         <xs:element minOccurs="0" name="valueTendency" type="ns_p:ElementTagType"/>
16811         <xs:element minOccurs="0" name="valueState" type="ns_p:ElementTagType"/>
16812     </xs:sequence>
16813 </xs:complexType>
16814 <xs:element name="measurementDataElements" type="ns_p:MeasurementDataElementsType"/>
16815 <xs:complexType name="MeasurementListDataType">
16816     <xs:sequence>
16817         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:measurementData"/>
16818     </xs:sequence>
16819 </xs:complexType>
16820 <xs:element name="measurementListData" type="ns_p:MeasurementListDataType"/>
16821 <xs:complexType name="MeasurementListDataSelectorsType">
16822     <xs:sequence>
16823         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16824         <xs:element minOccurs="0" name="valueType" type="ns_p:MeasurementValueTypeType"/>
16825         <xs:element minOccurs="0" name="timestampInterval"
16826 type="ns_p:TimestampIntervalType"/>
```

```
16827         </xs:sequence>
16828     </xs:complexType>
16829     <xs:element name="measurementListDataSelectors"
16830 type="ns_p:MeasurementListDataSelectorsType"/>
16831     <xs:complexType name="MeasurementSeriesDataType">
16832         <xs:sequence>
16833             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16834             <xs:element minOccurs="0" name="valueType" type="ns_p:MeasurementValueTypeType"/>
16835             <xs:element minOccurs="0" name="timestamp"
16836 type="ns_p:AbsoluteOrRelativeTimeType"/>
16837             <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
16838             <xs:element minOccurs="0" name="evaluationPeriod" type="ns_p:TimePeriodType"/>
16839             <xs:element minOccurs="0" name="valueSource"
16840 type="ns_p:MeasurementValueSourceType"/>
16841             <xs:element minOccurs="0" name="valueTendency"
16842 type="ns_p:MeasurementValueTendencyType"/>
16843             <xs:element minOccurs="0" name="valueState"
16844 type="ns_p:MeasurementValueStateType"/>
16845         </xs:sequence>
16846     </xs:complexType>
16847     <xs:element name="measurementSeriesData" type="ns_p:MeasurementSeriesDataType"/>
16848     <xs:complexType name="MeasurementSeriesDataElementsType">
16849         <xs:sequence>
16850             <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16851             <xs:element minOccurs="0" name="valueType" type="ns_p:ElementTagType"/>
16852             <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
16853             <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
16854             <xs:element minOccurs="0" name="evaluationPeriod"
16855 type="ns_p:TimePeriodElementsType"/>
16856             <xs:element minOccurs="0" name="valueSource" type="ns_p:ElementTagType"/>
16857             <xs:element minOccurs="0" name="valueTendency" type="ns_p:ElementTagType"/>
16858             <xs:element minOccurs="0" name="valueState" type="ns_p:ElementTagType"/>
16859         </xs:sequence>
16860     </xs:complexType>
16861     <xs:element name="measurementSeriesDataElements"
16862 type="ns_p:MeasurementSeriesDataElementsType"/>
16863     <xs:complexType name="MeasurementSeriesListDataType">
16864         <xs:sequence>
16865             <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:measurementSeriesData"/>
16866         </xs:sequence>
16867     </xs:complexType>
16868     <xs:element name="measurementSeriesListData" type="ns_p:MeasurementSeriesListDataType"/>
16869     <xs:complexType name="MeasurementSeriesListDataSelectorsType">
16870         <xs:sequence>
16871             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16872             <xs:element minOccurs="0" name="valueType" type="ns_p:MeasurementValueTypeType"/>
16873             <xs:element minOccurs="0" name="timestampInterval"
16874 type="ns_p:TimestampIntervalType"/>
16875         </xs:sequence>
16876     </xs:complexType>
16877     <xs:element name="measurementSeriesListDataSelectors"
16878 type="ns_p:MeasurementSeriesListDataSelectorsType"/>
16879     <xs:complexType name="MeasurementConstraintsDataType">
16880         <xs:sequence>
16881             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16882             <xs:element minOccurs="0" name="valueRangeMin" type="ns_p:ScaledNumberType"/>
16883             <xs:element minOccurs="0" name="valueRangeMax" type="ns_p:ScaledNumberType"/>
16884             <xs:element minOccurs="0" name="valueStepSize" type="ns_p:ScaledNumberType"/>
16885         </xs:sequence>
16886     </xs:complexType>
16887     <xs:element name="measurementConstraintsData" type="ns_p:MeasurementConstraintsDataType"/>
16888     <xs:complexType name="MeasurementConstraintsDataElementsType">
16889         <xs:sequence>
16890             <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16891             <xs:element minOccurs="0" name="valueRangeMin"
16892 type="ns_p:ScaledNumberElementsType"/>
16893             <xs:element minOccurs="0" name="valueRangeMax"
16894 type="ns_p:ScaledNumberElementsType"/>
16895             <xs:element minOccurs="0" name="valueStepSize"
16896 type="ns_p:ScaledNumberElementsType"/>
16897         </xs:sequence>
16898     </xs:complexType>
16899     <xs:element name="measurementConstraintsDataElements"
16900 type="ns_p:MeasurementConstraintsDataElementsType"/>
16901     <xs:complexType name="MeasurementConstraintsListDataType">
16902         <xs:sequence>
16903             <xs:element maxOccurs="unbounded" minOccurs="0"
16904 ref="ns_p:measurementConstraintsData"/>
```



```
16905         </xs:sequence>
16906     </xs:complexType>
16907     <xs:element name="measurementConstraintsListData"
16908 type="ns_p:MeasurementConstraintsListDataType"/>
16909     <xs:complexType name="MeasurementConstraintsListDataSelectorsType">
16910         <xs:sequence>
16911             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16912         </xs:sequence>
16913     </xs:complexType>
16914     <xs:element name="measurementConstraintsListDataSelectors"
16915 type="ns_p:MeasurementConstraintsListDataSelectorsType"/>
16916     <xs:complexType name="MeasurementDescriptionDataType">
16917         <xs:sequence>
16918             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16919             <xs:element minOccurs="0" name="measurementType" type="ns_p:MeasurementTypeType"/>
16920             <xs:element minOccurs="0" name="commodityType" type="ns_p:CommodityTypeType"/>
16921             <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
16922             <xs:element minOccurs="0" name="calibrationValue" type="ns_p:ScaledNumberType"/>
16923             <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
16924             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
16925             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
16926         </xs:sequence>
16927     </xs:complexType>
16928     <xs:element name="measurementDescriptionData" type="ns_p:MeasurementDescriptionDataType"/>
16929     <xs:complexType name="MeasurementDescriptionDataElementsType">
16930         <xs:sequence>
16931             <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16932             <xs:element minOccurs="0" name="measurementType" type="ns_p:ElementTagType"/>
16933             <xs:element minOccurs="0" name="commodityType" type="ns_p:ElementTagType"/>
16934             <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
16935             <xs:element minOccurs="0" name="calibrationValue"
16936 type="ns_p:ScaledNumberElementsType"/>
16937             <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
16938             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
16939             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
16940         </xs:sequence>
16941     </xs:complexType>
16942     <xs:element name="measurementDescriptionDataElements"
16943 type="ns_p:MeasurementDescriptionDataElementsType"/>
16944     <xs:complexType name="MeasurementDescriptionListDataType">
16945         <xs:sequence>
16946             <xs:element maxOccurs="unbounded" minOccurs="0"
16947 ref="ns_p:measurementDescriptionData"/>
16948         </xs:sequence>
16949     </xs:complexType>
16950     <xs:element name="measurementDescriptionListData"
16951 type="ns_p:MeasurementDescriptionListDataType"/>
16952     <xs:complexType name="MeasurementDescriptionListDataSelectorsType">
16953         <xs:sequence>
16954             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16955             <xs:element minOccurs="0" name="measurementType" type="ns_p:MeasurementTypeType"/>
16956             <xs:element minOccurs="0" name="commodityType" type="ns_p:CommodityTypeType"/>
16957             <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
16958         </xs:sequence>
16959     </xs:complexType>
16960     <xs:element name="measurementDescriptionListDataSelectors"
16961 type="ns_p:MeasurementDescriptionListDataSelectorsType"/>
16962     <xs:complexType name="MeasurementThresholdRelationDataType">
16963         <xs:sequence>
16964             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16965             <xs:element maxOccurs="unbounded" minOccurs="0" name="thresholdId"
16966 type="ns_p:ThresholdIdType"/>
16967         </xs:sequence>
16968     </xs:complexType>
16969     <xs:element name="measurementThresholdRelationData"
16970 type="ns_p:MeasurementThresholdRelationDataType"/>
16971     <xs:complexType name="MeasurementThresholdRelationDataElementsType">
16972         <xs:sequence>
16973             <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
16974             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ElementTagType"/>
16975         </xs:sequence>
16976     </xs:complexType>
16977     <xs:element name="measurementThresholdRelationDataElements"
16978 type="ns_p:MeasurementThresholdRelationDataElementsType"/>
16979     <xs:complexType name="MeasurementThresholdRelationListDataType">
16980         <xs:sequence>
16981             <xs:element maxOccurs="unbounded" minOccurs="0"
16982 ref="ns_p:measurementThresholdRelationData"/>
16983     </xs:sequence>
```

```
16983         </xs:sequence>
16984     </xs:complexType>
16985     <xs:element name="measurementThresholdRelationListData"
16986 type="ns_p:MeasurementThresholdRelationListDataType"/>
16987     <xs:complexType name="MeasurementThresholdRelationListDataSelectorsType">
16988         <xs:sequence>
16989             <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
16990             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ThresholdIdType"/>
16991         </xs:sequence>
16992     </xs:complexType>
16993     <xs:element name="measurementThresholdRelationListDataSelectors"
16994 type="ns_p:MeasurementThresholdRelationListDataSelectorsType"/>
16995 </xs:schema>
16996
16997
```

16998

16999 **A.2.16 Messaging**

17000 File name: EEBus_SPINE_TS_Messaging.xsd

```
17001 <?xml version="1.0" encoding="UTF-8"?>
17002 <!--
17003     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
17004     Version 1.3.0 (final)
17005     2023-08-31
17006     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
17007     Source: https://www.eebus.org/en/specifications/
17008 -->
17009 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
17010 xmlns:xs="http://www.w3.org/2001/XMLSchema"
17011 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
17012 blockDefault="#all" elementFormDefault="qualified">
17013     <xs:annotation>
17014         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
17015 e.V. All rights reserved.</xs:documentation>
17016     </xs:annotation>
17017     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
17018     <xs:simpleType name="MessagingNumberType">
17019         <xs:restriction base="xs:unsignedInt"/>
17020     </xs:simpleType>
17021     <xs:simpleType name="MessagingTypeType">
17022         <xs:union memberTypes="ns_p:MessagingTypeEnumType ns_p:EnumExtendType"/>
17023     </xs:simpleType>
17024     <xs:simpleType name="MessagingDataTextType">
17025         <xs:restriction base="xs:string"/>
17026     </xs:simpleType>
17027     <xs:simpleType name="MessagingTypeEnumType">
17028         <xs:restriction base="xs:string">
17029             <xs:enumeration value="logging"/>
17030             <xs:enumeration value="information"/>
17031             <xs:enumeration value="warning"/>
17032             <xs:enumeration value="alarm"/>
17033             <xs:enumeration value="emergency"/>
17034             <xs:enumeration value="obsolete"/>
17035         </xs:restriction>
17036     </xs:simpleType>
17037     <xs:complexType name="MessagingDataType">
17038         <xs:sequence>
17039             <xs:element minOccurs="0" name="timestamp"
17040 type="ns_p:AbsoluteOrRelativeTimeType"/>
17041             <xs:element minOccurs="0" name="messagingNumber" type="ns_p:MessagingNumberType"/>
17042             <xs:element minOccurs="0" name="type" type="ns_p:MessagingTypeType"/>
17043             <xs:element minOccurs="0" name="text" type="ns_p:MessagingDataTextType"/>
17044         </xs:sequence>
17045     </xs:complexType>
17046     <xs:element name="messagingData" type="ns_p:MessagingDataType"/>
17047     <xs:complexType name="MessagingDataElementsType">
17048         <xs:sequence>
17049             <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
17050             <xs:element minOccurs="0" name="messagingNumber" type="ns_p:ElementTagType"/>
17051             <xs:element minOccurs="0" name="type" type="ns_p:ElementTagType"/>
17052             <xs:element minOccurs="0" name="text" type="ns_p:ElementTagType"/>
17053         </xs:sequence>
17054     </xs:complexType>
17055
```

```
17056 <xs:element name="messagingDataElements" type="ns_p:MessagingDataElementsType"/>
17057 <xs:complexType name="MessagingListDataType">
17058   <xs:sequence>
17059     <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:messagingData"/>
17060   </xs:sequence>
17061 </xs:complexType>
17062 <xs:element name="messagingListData" type="ns_p:MessagingListDataType"/>
17063 <xs:complexType name="MessagingListDataSelectorsType">
17064   <xs:sequence>
17065     <xs:element minOccurs="0" name="timestampInterval"
17066 type="ns_p:TimestampIntervalType"/>
17067     <xs:element minOccurs="0" name="messagingNumber" type="ns_p:MessagingNumberType"/>
17068   </xs:sequence>
17069 </xs:complexType>
17070 <xs:element name="messagingListDataSelectors" type="ns_p:MessagingListDataSelectorsType"/>
17071 </xs:schema>
17072
17073
```

17074

17075 A.2.17 NetworkManagement

17076 File name: EEBus_SPINE_TS_NetworkManagement.xsd

```
17077 <?xml version="1.0" encoding="UTF-8"?>
17078 <!--
17079   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
17080   Version 1.3.0 (final)
17081   2023-08-31
17082   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
17083   Source: https://www.eebus.org/en/specifications/
17084 -->
17085 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
17086   xmlns:xs="http://www.w3.org/2001/XMLSchema"
17087   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
17088   blockDefault="#all" elementFormDefault="qualified">
17089   <xs:annotation>
17090     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
17091 e.V. All rights reserved.</xs:documentation>
17092   </xs:annotation>
17093   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
17094   <xs:simpleType name="NetworkManagementNativeSetupType">
17095     <xs:restriction base="xs:string"/>
17096   </xs:simpleType>
17097   <xs:simpleType name="NetworkManagementScanSetupType">
17098     <xs:restriction base="xs:string"/>
17099   </xs:simpleType>
17100   <xs:simpleType name="NetworkManagementSetupType">
17101     <xs:restriction base="xs:string"/>
17102   </xs:simpleType>
17103   <xs:simpleType name="NetworkManagementCandidateSetupType">
17104     <xs:restriction base="xs:string"/>
17105   </xs:simpleType>
17106   <xs:simpleType name="NetworkManagementTechnologyAddressType">
17107     <xs:restriction base="xs:string"/>
17108   </xs:simpleType>
17109   <xs:simpleType name="NetworkManagementCommunicationsTechnologyInformationType">
17110     <xs:restriction base="xs:string"/>
17111   </xs:simpleType>
17112   <xs:simpleType name="NetworkManagementMinimumTrustLevelType">
17113     <xs:restriction base="xs:string"/>
17114   </xs:simpleType>
17115   <xs:simpleType name="NetworkManagementProcessTimeoutType">
17116     <xs:restriction base="xs:duration"/>
17117   </xs:simpleType>
17118   <xs:simpleType name="NetworkManagementFeatureSetType">
17119     <xs:restriction base="xs:string">
17120       <xs:enumeration value="gateway"/>
17121       <xs:enumeration value="router"/>
17122       <xs:enumeration value="smart"/>
17123       <xs:enumeration value="simple"/>
17124     </xs:restriction>
17125   </xs:simpleType>
17126   <xs:simpleType name="NetworkManagementProcessStateType">
17127     <xs:restriction base="xs:string">
17128
```

```
17129         <xs:enumeration value="succeeded"/>
17130         <xs:enumeration value="failed"/>
17131         <xs:enumeration value="aborted"/>
17132     </xs:restriction>
17133 </xs:simpleType>
17134 <xs:simpleType name="NetworkManagementStateChangeType">
17135     <xs:restriction base="xs:string">
17136         <xs:enumeration value="added"/>
17137         <xs:enumeration value="removed"/>
17138         <xs:enumeration value="modified"/>
17139     </xs:restriction>
17140 </xs:simpleType>
17141 <xs:complexType name="NetworkManagementAddNodeCallType">
17142     <xs:sequence>
17143         <xs:element minOccurs="0" name="nodeAddress" type="ns_p:FeatureAddressType"/>
17144         <xs:element minOccurs="0" name="nativeSetup"
17145 type="ns_p:NetworkManagementNativeSetupType"/>
17146         <xs:element minOccurs="0" name="timeout"
17147 type="ns_p:NetworkManagementProcessTimeoutType"/>
17148         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17149         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17150     </xs:sequence>
17151 </xs:complexType>
17152 <xs:element name="networkManagementAddNodeCall"
17153 type="ns_p:NetworkManagementAddNodeCallType"/>
17154 <xs:complexType name="NetworkManagementAddNodeCallElementsType">
17155     <xs:sequence>
17156         <xs:element minOccurs="0" name="nodeAddress"
17157 type="ns_p:FeatureAddressElementsType"/>
17158         <xs:element minOccurs="0" name="nativeSetup" type="ns_p:ElementTagType"/>
17159         <xs:element minOccurs="0" name="timeout" type="ns_p:ElementTagType"/>
17160         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17161         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17162     </xs:sequence>
17163 </xs:complexType>
17164 <xs:element name="networkManagementAddNodeCallElements"
17165 type="ns_p:NetworkManagementAddNodeCallElementsType"/>
17166 <xs:complexType name="NetworkManagementRemoveNodeCallType">
17167     <xs:sequence>
17168         <xs:element minOccurs="0" name="nodeAddress" type="ns_p:FeatureAddressType"/>
17169         <xs:element minOccurs="0" name="timeout"
17170 type="ns_p:NetworkManagementProcessTimeoutType"/>
17171     </xs:sequence>
17172 </xs:complexType>
17173 <xs:element name="networkManagementRemoveNodeCall"
17174 type="ns_p:NetworkManagementRemoveNodeCallType"/>
17175 <xs:complexType name="NetworkManagementRemoveNodeCallElementsType">
17176     <xs:sequence>
17177         <xs:element minOccurs="0" name="nodeAddress"
17178 type="ns_p:FeatureAddressElementsType"/>
17179         <xs:element minOccurs="0" name="timeout" type="ns_p:ElementTagType"/>
17180     </xs:sequence>
17181 </xs:complexType>
17182 <xs:element name="networkManagementRemoveNodeCallElements"
17183 type="ns_p:NetworkManagementRemoveNodeCallElementsType"/>
17184 <xs:complexType name="NetworkManagementModifyNodeCallType">
17185     <xs:sequence>
17186         <xs:element minOccurs="0" name="nodeAddress" type="ns_p:FeatureAddressType"/>
17187         <xs:element minOccurs="0" name="nativeSetup"
17188 type="ns_p:NetworkManagementNativeSetupType"/>
17189         <xs:element minOccurs="0" name="timeout"
17190 type="ns_p:NetworkManagementProcessTimeoutType"/>
17191         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17192         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17193     </xs:sequence>
17194 </xs:complexType>
17195 <xs:element name="networkManagementModifyNodeCall"
17196 type="ns_p:NetworkManagementModifyNodeCallType"/>
17197 <xs:complexType name="NetworkManagementModifyNodeCallElementsType">
17198     <xs:sequence>
17199         <xs:element minOccurs="0" name="nodeAddress"
17200 type="ns_p:FeatureAddressElementsType"/>
17201         <xs:element minOccurs="0" name="nativeSetup" type="ns_p:ElementTagType"/>
17202         <xs:element minOccurs="0" name="timeout" type="ns_p:ElementTagType"/>
17203         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17204         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17205     </xs:sequence>
17206 </xs:complexType>
```

```
17207     <xs:element name="networkManagementModifyNodeCallElements"
17208 type="ns_p:NetworkManagementModifyNodeCallElementsType"/>
17209     <xs:complexType name="NetworkManagementScanNetworkCallType">
17210       <xs:sequence>
17211         <xs:element minOccurs="0" name="scanSetup"
17212 type="ns_p:NetworkManagementScanSetupType"/>
17213         <xs:element minOccurs="0" name="timeout"
17214 type="ns_p:NetworkManagementProcessTimeoutType"/>
17215       </xs:sequence>
17216     </xs:complexType>
17217     <xs:element name="networkManagementScanNetworkCall"
17218 type="ns_p:NetworkManagementScanNetworkCallType"/>
17219     <xs:complexType name="NetworkManagementScanNetworkCallElementsType">
17220       <xs:sequence>
17221         <xs:element minOccurs="0" name="scanSetup" type="ns_p:ElementTagType"/>
17222         <xs:element minOccurs="0" name="timeout" type="ns_p:ElementTagType"/>
17223       </xs:sequence>
17224     </xs:complexType>
17225     <xs:element name="networkManagementScanNetworkCallElements"
17226 type="ns_p:NetworkManagementScanNetworkCallElementsType"/>
17227     <xs:complexType name="NetworkManagementDiscoverCallType">
17228       <xs:sequence>
17229         <xs:element minOccurs="0" name="discoverAddress" type="ns_p:FeatureAddressType"/>
17230       </xs:sequence>
17231     </xs:complexType>
17232     <xs:element name="networkManagementDiscoverCall"
17233 type="ns_p:NetworkManagementDiscoverCallType"/>
17234     <xs:complexType name="NetworkManagementDiscoverCallElementsType">
17235       <xs:sequence>
17236         <xs:element minOccurs="0" name="discoverAddress"
17237 type="ns_p:FeatureAddressElementsType"/>
17238       </xs:sequence>
17239     </xs:complexType>
17240     <xs:element name="networkManagementDiscoverCallElements"
17241 type="ns_p:NetworkManagementDiscoverCallElementsType"/>
17242     <xs:complexType name="NetworkManagementAbortCallType">
17243       <xs:element name="networkManagementAbortCall" type="ns_p:NetworkManagementAbortCallType"/>
17244     <xs:complexType name="NetworkManagementAbortCallElementsType">
17245       <xs:element name="networkManagementAbortCallElements"
17246 type="ns_p:NetworkManagementAbortCallElementsType"/>
17247     <xs:complexType name="NetworkManagementProcessStateDataType">
17248       <xs:sequence>
17249         <xs:element minOccurs="0" name="state"
17250 type="ns_p:NetworkManagementProcessStateStateType"/>
17251         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17252       </xs:sequence>
17253     </xs:complexType>
17254     <xs:element name="networkManagementProcessStateData"
17255 type="ns_p:NetworkManagementProcessStateDataType"/>
17256     <xs:complexType name="NetworkManagementProcessStateDataElementsType">
17257       <xs:sequence>
17258         <xs:element minOccurs="0" name="state" type="ns_p:ElementTagType"/>
17259         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17260       </xs:sequence>
17261     </xs:complexType>
17262     <xs:element name="networkManagementProcessStateDataElements"
17263 type="ns_p:NetworkManagementProcessStateDataElementsType"/>
17264     <xs:complexType name="NetworkManagementJoiningModeDataType">
17265       <xs:sequence>
17266         <xs:element minOccurs="0" name="setup" type="ns_p:NetworkManagementSetupType"/>
17267       </xs:sequence>
17268     </xs:complexType>
17269     <xs:element name="networkManagementJoiningModeData"
17270 type="ns_p:NetworkManagementJoiningModeDataType"/>
17271     <xs:complexType name="NetworkManagementJoiningModeDataElementsType">
17272       <xs:sequence>
17273         <xs:element minOccurs="0" name="setup" type="ns_p:ElementTagType"/>
17274       </xs:sequence>
17275     </xs:complexType>
17276     <xs:element name="networkManagementJoiningModeDataElements"
17277 type="ns_p:NetworkManagementJoiningModeDataElementsType"/>
17278     <xs:complexType name="NetworkManagementReportCandidateDataType">
17279       <xs:sequence>
17280         <xs:element minOccurs="0" name="candidateSetup"
17281 type="ns_p:NetworkManagementCandidateSetupType"/>
17282         <xs:element minOccurs="0" name="setupUsableForAdd" type="xs:boolean"/>
17283         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17284         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
```

```
17285         </xs:sequence>
17286     </xs:complexType>
17287     <xs:element name="networkManagementReportCandidateData"
17288 type="ns_p:NetworkManagementReportCandidateDataType"/>
17289     <xs:complexType name="NetworkManagementReportCandidateDataElementsType">
17290         <xs:sequence>
17291             <xs:element minOccurs="0" name="candidateSetup" type="ns_p:ElementTagType"/>
17292             <xs:element minOccurs="0" name="setupUsableForAdd" type="ns_p:ElementTagType"/>
17293             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17294             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17295         </xs:sequence>
17296     </xs:complexType>
17297     <xs:element name="networkManagementReportCandidateDataElements"
17298 type="ns_p:NetworkManagementReportCandidateDataElementsType"/>
17299     <xs:complexType name="NetworkManagementDeviceDescriptionDataType">
17300         <xs:sequence>
17301             <xs:element minOccurs="0" name="deviceAddress" type="ns_p:DeviceAddressType"/>
17302             <xs:element minOccurs="0" name="deviceType" type="ns_p:DeviceTypeType"/>
17303             <xs:element minOccurs="0" name="networkManagementResponsibleAddress"
17304 type="ns_p:FeatureAddressType"/>
17305             <xs:element minOccurs="0" name="nativeSetup"
17306 type="ns_p:NetworkManagementNativeSetupType"/>
17307             <xs:element minOccurs="0" name="technologyAddress"
17308 type="ns_p:NetworkManagementTechnologyAddressType"/>
17309             <xs:element minOccurs="0" name="communicationsTechnologyInformation"
17310 type="ns_p:NetworkManagementCommunicationsTechnologyInformationType"/>
17311             <xs:element minOccurs="0" name="networkFeatureSet"
17312 type="ns_p:NetworkManagementFeatureSetType"/>
17313             <xs:element minOccurs="0" name="lastStateChange"
17314 type="ns_p:NetworkManagementStateChangeType"/>
17315             <xs:element minOccurs="0" name="minimumTrustLevel"
17316 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
17317             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17318             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17319         </xs:sequence>
17320     </xs:complexType>
17321     <xs:element name="networkManagementDeviceDescriptionData"
17322 type="ns_p:NetworkManagementDeviceDescriptionDataType"/>
17323     <xs:complexType name="NetworkManagementDeviceDescriptionDataElementsType">
17324         <xs:sequence>
17325             <xs:element minOccurs="0" name="deviceAddress"
17326 type="ns_p:DeviceAddressElementsType"/>
17327             <xs:element minOccurs="0" name="deviceType" type="ns_p:ElementTagType"/>
17328             <xs:element minOccurs="0" name="networkManagementResponsibleAddress"
17329 type="ns_p:ElementTagType"/>
17330             <xs:element minOccurs="0" name="nativeSetup" type="ns_p:ElementTagType"/>
17331             <xs:element minOccurs="0" name="technologyAddress" type="ns_p:ElementTagType"/>
17332             <xs:element minOccurs="0" name="communicationsTechnologyInformation"
17333 type="ns_p:ElementTagType"/>
17334             <xs:element minOccurs="0" name="networkFeatureSet" type="ns_p:ElementTagType"/>
17335             <xs:element minOccurs="0" name="lastStateChange" type="ns_p:ElementTagType"/>
17336             <xs:element minOccurs="0" name="minimumTrustLevel" type="ns_p:ElementTagType"/>
17337             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17338             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17339         </xs:sequence>
17340     </xs:complexType>
17341     <xs:element name="networkManagementDeviceDescriptionDataElements"
17342 type="ns_p:NetworkManagementDeviceDescriptionDataElementsType"/>
17343     <xs:complexType name="NetworkManagementDeviceDescriptionListDataType">
17344         <xs:sequence>
17345             <xs:element maxOccurs="unbounded" minOccurs="0"
17346 ref="ns_p:networkManagementDeviceDescriptionData"/>
17347         </xs:sequence>
17348     </xs:complexType>
17349     <xs:element name="networkManagementDeviceDescriptionListData"
17350 type="ns_p:NetworkManagementDeviceDescriptionListDataType"/>
17351     <xs:complexType name="NetworkManagementDeviceDescriptionListDataSelectorsType">
17352         <xs:sequence>
17353             <xs:element minOccurs="0" name="deviceAddress" type="ns_p:DeviceAddressType"/>
17354             <xs:element minOccurs="0" name="deviceType" type="ns_p:DeviceTypeType"/>
17355         </xs:sequence>
17356     </xs:complexType>
17357     <xs:element name="networkManagementDeviceDescriptionListDataSelectors"
17358 type="ns_p:NetworkManagementDeviceDescriptionListDataSelectorsType"/>
17359     <xs:complexType name="NetworkManagementEntityDescriptionDataType">
17360         <xs:sequence>
17361             <xs:element minOccurs="0" name="entityAddress" type="ns_p:EntityAddressType"/>
17362             <xs:element minOccurs="0" name="entityType" type="ns_p:EntityTypeType"/>
```

```
17363         <xs:element minOccurs="0" name="lastStateChange"
17364 type="ns_p:NetworkManagementStateChangeType"/>
17365         <xs:element minOccurs="0" name="minimumTrustLevel"
17366 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
17367         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17368         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17369     </xs:sequence>
17370 </xs:complexType>
17371 <xs:element name="networkManagementEntityDescriptionData"
17372 type="ns_p:NetworkManagementEntityDescriptionData"/>
17373 <xs:complexType name="NetworkManagementEntityDescriptionDataElementsType">
17374     <xs:sequence>
17375         <xs:element minOccurs="0" name="entityAddress"
17376 type="ns_p:EntityAddressElementsType"/>
17377         <xs:element minOccurs="0" name="entityType" type="ns_p:ElementTagType"/>
17378         <xs:element minOccurs="0" name="lastStateChange" type="ns_p:ElementTagType"/>
17379         <xs:element minOccurs="0" name="minimumTrustLevel" type="ns_p:ElementTagType"/>
17380         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17381         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17382     </xs:sequence>
17383 </xs:complexType>
17384 <xs:element name="networkManagementEntityDescriptionDataElements"
17385 type="ns_p:NetworkManagementEntityDescriptionDataElementsType"/>
17386 <xs:complexType name="NetworkManagementEntityDescriptionListDataType">
17387     <xs:sequence>
17388         <xs:element maxOccurs="unbounded" minOccurs="0"
17389 ref="ns_p:networkManagementEntityDescriptionData"/>
17390     </xs:sequence>
17391 </xs:complexType>
17392 <xs:element name="networkManagementEntityDescriptionListData"
17393 type="ns_p:NetworkManagementEntityDescriptionListDataType"/>
17394 <xs:complexType name="NetworkManagementEntityDescriptionListDataSelectorsType">
17395     <xs:sequence>
17396         <xs:element minOccurs="0" name="entityAddress" type="ns_p:EntityAddressType"/>
17397         <xs:element minOccurs="0" name="entityType" type="ns_p:EntityTypeType"/>
17398     </xs:sequence>
17399 </xs:complexType>
17400 <xs:element name="networkManagementEntityDescriptionListDataSelectors"
17401 type="ns_p:NetworkManagementEntityDescriptionListDataSelectorsType"/>
17402 <xs:complexType name="NetworkManagementFeatureDescriptionDataType">
17403     <xs:sequence>
17404         <xs:element minOccurs="0" name="featureAddress" type="ns_p:FeatureAddressType"/>
17405         <xs:element minOccurs="0" name="featureType" type="ns_p:FeatureTypeType"/>
17406         <xs:element maxOccurs="unbounded" minOccurs="0" name="specificUsage"
17407 type="ns_p:FeatureSpecificUsageType"/>
17408         <xs:element minOccurs="0" name="featureGroup" type="ns_p:FeatureGroupType"/>
17409         <xs:element minOccurs="0" name="role" type="ns_p:RoleType"/>
17410         <xs:element maxOccurs="unbounded" minOccurs="0" name="supportedFunction"
17411 type="ns_p:FunctionPropertyType"/>
17412         <xs:element minOccurs="0" name="lastStateChange"
17413 type="ns_p:NetworkManagementStateChangeType"/>
17414         <xs:element minOccurs="0" name="minimumTrustLevel"
17415 type="ns_p:NetworkManagementMinimumTrustLevelType"/>
17416         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
17417         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17418         <xs:element minOccurs="0" name="maxResponseDelay"
17419 type="ns_p:MaxResponseDelayType"/>
17420     </xs:sequence>
17421 </xs:complexType>
17422 <xs:element name="networkManagementFeatureDescriptionData"
17423 type="ns_p:NetworkManagementFeatureDescriptionDataType"/>
17424 <xs:complexType name="NetworkManagementFeatureDescriptionDataElementsType">
17425     <xs:sequence>
17426         <xs:element minOccurs="0" name="featureAddress"
17427 type="ns_p:FeatureAddressElementsType"/>
17428         <xs:element minOccurs="0" name="featureType" type="ns_p:ElementTagType"/>
17429         <xs:element minOccurs="0" name="specificUsage" type="ns_p:ElementTagType"/>
17430         <xs:element minOccurs="0" name="featureGroup" type="ns_p:ElementTagType"/>
17431         <xs:element minOccurs="0" name="role" type="ns_p:ElementTagType"/>
17432         <xs:element minOccurs="0" name="supportedFunction"
17433 type="ns_p:FunctionPropertyElementsType"/>
17434         <xs:element minOccurs="0" name="lastStateChange" type="ns_p:ElementTagType"/>
17435         <xs:element minOccurs="0" name="minimumTrustLevel" type="ns_p:ElementTagType"/>
17436         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
17437         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17438         <xs:element minOccurs="0" name="maxResponseDelay" type="ns_p:ElementTagType"/>
17439     </xs:sequence>
17440 </xs:complexType>
```

```
17441 <xs:element name="networkManagementFeatureDescriptionDataElements"
17442 type="ns_p:NetworkManagementFeatureDescriptionDataElementsType"/>
17443 <xs:complexType name="NetworkManagementFeatureDescriptionListDataType">
17444 <xs:sequence>
17445 <xs:element maxOccurs="unbounded" minOccurs="0"
17446 ref="ns_p:networkManagementFeatureDescriptionData"/>
17447 </xs:sequence>
17448 </xs:complexType>
17449 <xs:element name="networkManagementFeatureDescriptionListData"
17450 type="ns_p:NetworkManagementFeatureDescriptionListDataType"/>
17451 <xs:complexType name="NetworkManagementFeatureDescriptionListDataSelectorsType">
17452 <xs:sequence>
17453 <xs:element minOccurs="0" name="featureAddress" type="ns_p:FeatureAddressType"/>
17454 <xs:element minOccurs="0" name="featureType" type="ns_p:FeatureTypeType"/>
17455 </xs:sequence>
17456 </xs:complexType>
17457 <xs:element name="networkManagementFeatureDescriptionListDataSelectors"
17458 type="ns_p:NetworkManagementFeatureDescriptionListDataSelectorsType"/>
17459 </xs:schema>
17460
17461
```

17462

17463 A.2.18 OperatingConstraints

17464 File name: EEBus_SPINE_TS_OperatingConstraints.xsd

```
17465
17466 <?xml version="1.0" encoding="UTF-8"?>
17467 <!--
17468 Smart Premises Interoperable Neutral-Message Exchange (SPINE)
17469 Version 1.3.0 (final)
17470 2023-08-31
17471 Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
17472 Source: https://www.eebus.org/en/specifications/
17473 -->
17474 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
17475 xmlns:xs="http://www.w3.org/2001/XMLSchema"
17476 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
17477 blockDefault="#all" elementFormDefault="qualified">
17478 <xs:annotation>
17479 <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
17480 e.V. All rights reserved.</xs:documentation>
17481 </xs:annotation>
17482 <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
17483 <xs:complexType name="OperatingConstraintsInterruptDataType">
17484 <xs:sequence>
17485 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17486 <xs:element name="isPausable" type="xs:boolean" minOccurs="0"/>
17487 <xs:element name="isStoppable" type="xs:boolean" minOccurs="0"/>
17488 <xs:element name="notInterruptibleAtHighPower" type="xs:boolean" minOccurs="0"/>
17489 <xs:element name="maxCyclesPerDay" type="xs:unsignedInt" minOccurs="0"/>
17490 </xs:sequence>
17491 </xs:complexType>
17492 <xs:element name="operatingConstraintsInterruptData"
17493 type="ns_p:OperatingConstraintsInterruptDataType"/>
17494 <xs:complexType name="OperatingConstraintsInterruptDataElementsType">
17495 <xs:sequence>
17496 <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17497 <xs:element name="isPausable" minOccurs="0" type="ns_p:ElementTagType"/>
17498 <xs:element name="isStoppable" minOccurs="0" type="ns_p:ElementTagType"/>
17499 <xs:element name="notInterruptibleAtHighPower" minOccurs="0"
17500 type="ns_p:ElementTagType"/>
17501 <xs:element name="maxCyclesPerDay" minOccurs="0" type="ns_p:ElementTagType"/>
17502 </xs:sequence>
17503 </xs:complexType>
17504 <xs:element name="operatingConstraintsInterruptDataElements"
17505 type="ns_p:OperatingConstraintsInterruptDataElementsType"/>
17506 <xs:complexType name="OperatingConstraintsInterruptListDataType">
17507 <xs:sequence>
17508 <xs:element maxOccurs="unbounded" minOccurs="0"
17509 ref="ns_p:operatingConstraintsInterruptData"/>
17510 </xs:sequence>
17511 </xs:complexType>
17512 <xs:element name="operatingConstraintsInterruptListData"
17513 type="ns_p:OperatingConstraintsInterruptListDataType"/>
```



```
17514     <xs:complexType name="OperatingConstraintsInterruptListDataSelectorsType">
17515       <xs:sequence>
17516         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17517       </xs:sequence>
17518     </xs:complexType>
17519     <xs:element name="operatingConstraintsInterruptListDataSelectors"
17520 type="ns_p:OperatingConstraintsInterruptListDataSelectorsType"/>
17521     <xs:complexType name="OperatingConstraintsDurationDataType">
17522       <xs:sequence>
17523         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17524         <xs:element name="activeDurationMin" type="xs:duration" minOccurs="0"/>
17525         <xs:element name="activeDurationMax" type="xs:duration" minOccurs="0"/>
17526         <xs:element name="pauseDurationMin" type="xs:duration" minOccurs="0"/>
17527         <xs:element name="pauseDurationMax" type="xs:duration" minOccurs="0"/>
17528         <xs:element name="activeDurationSumMin" type="xs:duration" minOccurs="0"/>
17529         <xs:element name="activeDurationSumMax" type="xs:duration" minOccurs="0"/>
17530       </xs:sequence>
17531     </xs:complexType>
17532     <xs:element name="operatingConstraintsDurationData"
17533 type="ns_p:OperatingConstraintsDurationDataType"/>
17534     <xs:complexType name="OperatingConstraintsDurationDataElementsType">
17535       <xs:sequence>
17536         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17537         <xs:element name="activeDurationMin" minOccurs="0" type="ns_p:ElementTagType"/>
17538         <xs:element name="activeDurationMax" minOccurs="0" type="ns_p:ElementTagType"/>
17539         <xs:element name="pauseDurationMin" minOccurs="0" type="ns_p:ElementTagType"/>
17540         <xs:element name="pauseDurationMax" minOccurs="0" type="ns_p:ElementTagType"/>
17541         <xs:element name="activeDurationSumMin" minOccurs="0" type="ns_p:ElementTagType"/>
17542         <xs:element name="activeDurationSumMax" minOccurs="0" type="ns_p:ElementTagType"/>
17543       </xs:sequence>
17544     </xs:complexType>
17545     <xs:element name="operatingConstraintsDurationDataElements"
17546 type="ns_p:OperatingConstraintsDurationDataElementsType"/>
17547     <xs:complexType name="OperatingConstraintsDurationListDataType">
17548       <xs:sequence>
17549         <xs:element maxOccurs="unbounded" minOccurs="0"
17550 ref="ns_p:operatingConstraintsDurationData"/>
17551       </xs:sequence>
17552     </xs:complexType>
17553     <xs:element name="operatingConstraintsDurationListData"
17554 type="ns_p:OperatingConstraintsDurationListDataType"/>
17555     <xs:complexType name="OperatingConstraintsDurationListDataSelectorsType">
17556       <xs:sequence>
17557         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17558       </xs:sequence>
17559     </xs:complexType>
17560     <xs:element name="operatingConstraintsDurationListDataSelectors"
17561 type="ns_p:OperatingConstraintsDurationListDataSelectorsType"/>
17562     <xs:complexType name="OperatingConstraintsPowerDescriptionDataType">
17563       <xs:sequence>
17564         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17565         <xs:element minOccurs="0" name="positiveEnergyDirection"
17566 type="ns_p:EnergyDirectionType"/>
17567         <xs:element minOccurs="0" name="powerUnit" type="ns_p:UnitOfMeasurementType"/>
17568         <xs:element minOccurs="0" name="energyUnit" type="ns_p:UnitOfMeasurementType"/>
17569         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17570       </xs:sequence>
17571     </xs:complexType>
17572     <xs:element name="operatingConstraintsPowerDescriptionData"
17573 type="ns_p:OperatingConstraintsPowerDescriptionDataType"/>
17574     <xs:complexType name="OperatingConstraintsPowerDescriptionDataElementsType">
17575       <xs:sequence>
17576         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17577         <xs:element minOccurs="0" name="positiveEnergyDirection"
17578 type="ns_p:ElementTagType"/>
17579         <xs:element minOccurs="0" name="powerUnit" type="ns_p:ElementTagType"/>
17580         <xs:element minOccurs="0" name="energyUnit" type="ns_p:ElementTagType"/>
17581         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17582       </xs:sequence>
17583     </xs:complexType>
17584     <xs:element name="operatingConstraintsPowerDescriptionDataElements"
17585 type="ns_p:OperatingConstraintsPowerDescriptionDataElementsType"/>
17586     <xs:complexType name="OperatingConstraintsPowerDescriptionListDataType">
17587       <xs:sequence>
17588         <xs:element maxOccurs="unbounded" minOccurs="0"
17589 ref="ns_p:operatingConstraintsPowerDescriptionData"/>
17590       </xs:sequence>
17591     </xs:complexType>
```

```
17592 <xs:element name="operatingConstraintsPowerDescriptionListData"
17593 type="ns_p:OperatingConstraintsPowerDescriptionListDataType"/>
17594 <xs:complexType name="OperatingConstraintsPowerDescriptionListDataSelectorsType">
17595 <xs:sequence>
17596 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17597 </xs:sequence>
17598 </xs:complexType>
17599 <xs:element name="operatingConstraintsPowerDescriptionListDataSelectors"
17600 type="ns_p:OperatingConstraintsPowerDescriptionListDataSelectorsType"/>
17601 <xs:complexType name="OperatingConstraintsPowerRangeDataType">
17602 <xs:sequence>
17603 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17604 <xs:element name="powerMin" type="ns_p:ScaledNumberType" minOccurs="0"/>
17605 <xs:element name="powerMax" type="ns_p:ScaledNumberType" minOccurs="0"/>
17606 <xs:element name="energyMin" type="ns_p:ScaledNumberType" minOccurs="0"/>
17607 <xs:element name="energyMax" type="ns_p:ScaledNumberType" minOccurs="0"/>
17608 </xs:sequence>
17609 </xs:complexType>
17610 <xs:element name="operatingConstraintsPowerRangeData"
17611 type="ns_p:OperatingConstraintsPowerRangeDataType"/>
17612 <xs:complexType name="OperatingConstraintsPowerRangeDataElementsType">
17613 <xs:sequence>
17614 <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17615 <xs:element name="powerMin" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
17616 <xs:element name="powerMax" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
17617 <xs:element name="energyMin" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
17618 <xs:element name="energyMax" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
17619 </xs:sequence>
17620 </xs:complexType>
17621 <xs:element name="operatingConstraintsPowerRangeDataElements"
17622 type="ns_p:OperatingConstraintsPowerRangeDataElementsType"/>
17623 <xs:complexType name="OperatingConstraintsPowerRangeListDataType">
17624 <xs:sequence>
17625 <xs:element maxOccurs="unbounded" minOccurs="0"
17626 ref="ns_p:operatingConstraintsPowerRangeData"/>
17627 </xs:sequence>
17628 </xs:complexType>
17629 <xs:element name="operatingConstraintsPowerRangeListData"
17630 type="ns_p:OperatingConstraintsPowerRangeListDataType"/>
17631 <xs:complexType name="OperatingConstraintsPowerRangeListDataSelectorsType">
17632 <xs:sequence>
17633 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17634 </xs:sequence>
17635 </xs:complexType>
17636 <xs:element name="operatingConstraintsPowerRangeListDataSelectors"
17637 type="ns_p:OperatingConstraintsPowerRangeListDataSelectorsType"/>
17638 <xs:complexType name="OperatingConstraintsPowerLevelDataType">
17639 <xs:sequence>
17640 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17641 <xs:element name="power" type="ns_p:ScaledNumberType" minOccurs="0"
17642 maxOccurs="unbounded"/>
17643 </xs:sequence>
17644 </xs:complexType>
17645 <xs:element name="operatingConstraintsPowerLevelData"
17646 type="ns_p:OperatingConstraintsPowerLevelDataType"/>
17647 <xs:complexType name="OperatingConstraintsPowerLevelDataElementsType">
17648 <xs:sequence>
17649 <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17650 <xs:element name="power" minOccurs="0" type="ns_p:ScaledNumberElementsType"/>
17651 </xs:sequence>
17652 </xs:complexType>
17653 <xs:element name="operatingConstraintsPowerLevelDataElements"
17654 type="ns_p:OperatingConstraintsPowerLevelDataElementsType"/>
17655 <xs:complexType name="OperatingConstraintsPowerLevelListDataType">
17656 <xs:sequence>
17657 <xs:element maxOccurs="unbounded" minOccurs="0"
17658 ref="ns_p:operatingConstraintsPowerLevelData"/>
17659 </xs:sequence>
17660 </xs:complexType>
17661 <xs:element name="operatingConstraintsPowerLevelListData"
17662 type="ns_p:OperatingConstraintsPowerLevelListDataType"/>
17663 <xs:complexType name="OperatingConstraintsPowerLevelListDataSelectorsType">
17664 <xs:sequence>
17665 <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17666 </xs:sequence>
17667 </xs:complexType>
17668 <xs:element name="operatingConstraintsPowerLevelListDataSelectors"
17669 type="ns_p:OperatingConstraintsPowerLevelListDataSelectorsType"/>
```

```
17670     <xs:complexType name="OperatingConstraintsResumeImplicationDataType">
17671       <xs:sequence>
17672         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17673         <xs:element name="resumeEnergyEstimated" type="ns_p:ScaledNumberType"
17674 minOccurs="0"/>
17675         <xs:element name="energyUnit" minOccurs="0" type="ns_p:UnitOfMeasurementType"/>
17676         <xs:element name="resumeCostEstimated" type="ns_p:ScaledNumberType"
17677 minOccurs="0"/>
17678         <xs:element minOccurs="0" name="currency" type="ns_p:CurrencyType"/>
17679       </xs:sequence>
17680     </xs:complexType>
17681     <xs:element name="operatingConstraintsResumeImplicationData"
17682 type="ns_p:OperatingConstraintsResumeImplicationDataType"/>
17683     <xs:complexType name="OperatingConstraintsResumeImplicationDataElementsType">
17684       <xs:sequence>
17685         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17686         <xs:element name="resumeEnergyEstimated" minOccurs="0"
17687 type="ns_p:ScaledNumberElementsType"/>
17688         <xs:element name="energyUnit" minOccurs="0" type="ns_p:ElementTagType"/>
17689         <xs:element name="resumeCostEstimated" minOccurs="0"
17690 type="ns_p:ScaledNumberElementsType"/>
17691         <xs:element minOccurs="0" name="currency" type="ns_p:ElementTagType"/>
17692       </xs:sequence>
17693     </xs:complexType>
17694     <xs:element name="operatingConstraintsResumeImplicationDataElements"
17695 type="ns_p:OperatingConstraintsResumeImplicationDataElementsType"/>
17696     <xs:complexType name="OperatingConstraintsResumeImplicationListDataType">
17697       <xs:sequence>
17698         <xs:element maxOccurs="unbounded" minOccurs="0"
17699 ref="ns_p:operatingConstraintsResumeImplicationData"/>
17700       </xs:sequence>
17701     </xs:complexType>
17702     <xs:element name="operatingConstraintsResumeImplicationListData"
17703 type="ns_p:OperatingConstraintsResumeImplicationListDataType"/>
17704     <xs:complexType name="OperatingConstraintsResumeImplicationListDataSelectorsType">
17705       <xs:sequence>
17706         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17707       </xs:sequence>
17708     </xs:complexType>
17709     <xs:element name="operatingConstraintsResumeImplicationListDataSelectors"
17710 type="ns_p:OperatingConstraintsResumeImplicationListDataSelectorsType"/>
17711 </xs:schema>
17712
17713
```

17714

17715 A.2.19 PowerSequences

17716 File name: EEBus_SPINE_TS_PowerSequences.xsd

```
17717
17718 <?xml version="1.0" encoding="UTF-8"?>
17719 <!--
17720   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
17721   Version 1.3.0 (final)
17722   2023-08-31
17723   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
17724   Source: https://www.eebus.org/en/specifications/
17725 -->
17726 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
17727 xmlns:xs="http://www.w3.org/2001/XMLSchema"
17728 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
17729 blockDefault="#all" elementFormDefault="qualified">
17730   <xs:annotation>
17731     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
17732 e.V. All rights reserved.</xs:documentation>
17733   </xs:annotation>
17734   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
17735   <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
17736   <xs:simpleType name="AlternativesIdType">
17737     <xs:restriction base="xs:unsignedInt"/>
17738   </xs:simpleType>
17739   <xs:simpleType name="PowerSequenceIdType">
17740     <xs:restriction base="xs:unsignedInt"/>
17741   </xs:simpleType>
17742   <xs:simpleType name="PowerTimeSlotNumberType">
```

```
17743     <xs:restriction base="xs:unsignedInt"/>
17744 </xs:simpleType>
17745 <xs:simpleType name="PowerTimeSlotValueTypeType">
17746   <xs:union memberTypes="ns_p:PowerTimeSlotValueTypeEnumType ns_p:EnumExtendType"/>
17747 </xs:simpleType>
17748 <xs:simpleType name="PowerTimeSlotValueTypeEnumType">
17749   <xs:restriction base="xs:string">
17750     <xs:enumeration value="power"/>
17751     <xs:enumeration value="powerMin"/>
17752     <xs:enumeration value="powerMax"/>
17753     <xs:enumeration value="powerExpectedValue"/>
17754     <xs:enumeration value="powerStandardDeviation"/>
17755     <xs:enumeration value="powerSkewness"/>
17756     <xs:enumeration value="energy"/>
17757     <xs:enumeration value="energyMin"/>
17758     <xs:enumeration value="energyMax"/>
17759     <xs:enumeration value="energyExpectedValue"/>
17760     <xs:enumeration value="energyStandardDeviation"/>
17761     <xs:enumeration value="energySkewness"/>
17762   </xs:restriction>
17763 </xs:simpleType>
17764 <xs:simpleType name="PowerSequenceScopeType">
17765   <xs:union memberTypes="ns_p:PowerSequenceScopeEnumType ns_p:EnumExtendType"/>
17766 </xs:simpleType>
17767 <xs:simpleType name="PowerSequenceScopeEnumType">
17768   <xs:restriction base="xs:string">
17769     <xs:enumeration value="forecast"/>
17770     <xs:enumeration value="measurement"/>
17771     <xs:enumeration value="recommendation"/>
17772   </xs:restriction>
17773 </xs:simpleType>
17774 <xs:simpleType name="PowerSequenceStateType">
17775   <xs:union memberTypes="ns_p:PowerSequenceStateEnumType ns_p:EnumExtendType"/>
17776 </xs:simpleType>
17777 <xs:simpleType name="PowerSequenceStateEnumType">
17778   <xs:restriction base="xs:string">
17779     <xs:enumeration value="running"/>
17780     <xs:enumeration value="paused"/>
17781     <xs:enumeration value="scheduled"/>
17782     <xs:enumeration value="scheduledPaused"/>
17783     <xs:enumeration value="pending"/>
17784     <xs:enumeration value="inactive"/>
17785     <xs:enumeration value="completed"/>
17786     <xs:enumeration value="invalid"/>
17787   </xs:restriction>
17788 </xs:simpleType>
17789 <xs:complexType name="PowerTimeSlotScheduleDataType">
17790   <xs:sequence>
17791     <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17792     <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17793     <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
17794     <xs:element minOccurs="0" name="defaultDuration" type="xs:duration"/>
17795     <xs:element minOccurs="0" name="durationUncertainty" type="xs:duration"/>
17796     <xs:element minOccurs="0" name="slotActivated" type="xs:boolean"/>
17797     <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17798   </xs:sequence>
17799 </xs:complexType>
17800 <xs:element name="powerTimeSlotScheduleData" type="ns_p:PowerTimeSlotScheduleDataType"/>
17801 <xs:complexType name="PowerTimeSlotScheduleDataElementsType">
17802   <xs:sequence>
17803     <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17804     <xs:element minOccurs="0" name="slotNumber" type="ns_p:ElementTagType"/>
17805     <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
17806     <xs:element minOccurs="0" name="defaultDuration" type="ns_p:ElementTagType"/>
17807     <xs:element minOccurs="0" name="durationUncertainty" type="ns_p:ElementTagType"/>
17808     <xs:element minOccurs="0" name="slotActivated" type="ns_p:ElementTagType"/>
17809     <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17810   </xs:sequence>
17811 </xs:complexType>
17812 <xs:element name="powerTimeSlotScheduleDataElements"
17813 type="ns_p:PowerTimeSlotScheduleDataElementsType"/>
17814 <xs:complexType name="PowerTimeSlotScheduleListDataType">
17815   <xs:sequence>
17816     <xs:element maxOccurs="unbounded" minOccurs="0"
17817 ref="ns_p:powerTimeSlotScheduleData"/>
17818   </xs:sequence>
17819 </xs:complexType>
```

```
17820     <xs:element name="powerTimeSlotScheduleListData"
17821 type="ns_p:PowerTimeSlotScheduleListDataType"/>
17822     <xs:complexType name="PowerTimeSlotScheduleListDataSelectorsType">
17823       <xs:sequence>
17824         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17825         <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17826       </xs:sequence>
17827     </xs:complexType>
17828     <xs:element name="powerTimeSlotScheduleListDataSelectors"
17829 type="ns_p:PowerTimeSlotScheduleListDataSelectorsType"/>
17830     <xs:complexType name="PowerTimeSlotValueDataType">
17831       <xs:sequence>
17832         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17833         <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17834         <xs:element minOccurs="0" name="valueType" type="ns_p:PowerTimeSlotValueTypeType"/>
17835       </xs:sequence>
17836     </xs:complexType>
17837     <xs:element name="powerTimeSlotValueData" type="ns_p:PowerTimeSlotValueDataType"/>
17838     <xs:complexType name="PowerTimeSlotValueDataElementsType">
17839       <xs:sequence>
17840         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17841         <xs:element minOccurs="0" name="slotNumber" type="ns_p:ElementTagType"/>
17842         <xs:element minOccurs="0" name="valueType" type="ns_p:ElementTagType"/>
17843         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
17844       </xs:sequence>
17845     </xs:complexType>
17846     <xs:element name="powerTimeSlotValueDataElements"
17847 type="ns_p:PowerTimeSlotValueDataElementsType"/>
17848     <xs:complexType name="PowerTimeSlotValueListDataType">
17849       <xs:sequence>
17850         <xs:element maxOccurs="unbounded" minOccurs="0"
17851 ref="ns_p:powerTimeSlotValueData"/>
17852       </xs:sequence>
17853     </xs:complexType>
17854     <xs:element name="powerTimeSlotValueListData" type="ns_p:PowerTimeSlotValueListDataType"/>
17855     <xs:complexType name="PowerTimeSlotValueListDataSelectorsType">
17856       <xs:sequence>
17857         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17858         <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17859         <xs:element minOccurs="0" name="valueType" type="ns_p:PowerTimeSlotValueTypeType"/>
17860       </xs:sequence>
17861     </xs:complexType>
17862     <xs:element name="powerTimeSlotValueListDataSelectors"
17863 type="ns_p:PowerTimeSlotValueListDataSelectorsType"/>
17864     <xs:complexType name="PowerTimeSlotScheduleConstraintsDataType">
17865       <xs:sequence>
17866         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17867         <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17868         <xs:element minOccurs="0" name="earliestStartTime" type="ns_p:AbsoluteOrRelativeTimeType"/>
17869         <xs:element minOccurs="0" name="latestEndTime" type="ns_p:AbsoluteOrRelativeTimeType"/>
17870         <xs:element minOccurs="0" name="minDuration" type="xs:duration"/>
17871         <xs:element minOccurs="0" name="maxDuration" type="xs:duration"/>
17872         <xs:element minOccurs="0" name="optionalSlot" type="xs:boolean"/>
17873       </xs:sequence>
17874     </xs:complexType>
17875     <xs:element name="powerTimeSlotScheduleConstraintsData"
17876 type="ns_p:PowerTimeSlotScheduleConstraintsDataType"/>
17877     <xs:complexType name="PowerTimeSlotScheduleConstraintsDataElementsType">
17878       <xs:sequence>
17879         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
17880         <xs:element minOccurs="0" name="slotNumber" type="ns_p:ElementTagType"/>
17881         <xs:element minOccurs="0" name="earliestStartTime" type="ns_p:ElementTagType"/>
17882         <xs:element minOccurs="0" name="latestEndTime" type="ns_p:ElementTagType"/>
17883         <xs:element minOccurs="0" name="minDuration" type="ns_p:ElementTagType"/>
17884         <xs:element minOccurs="0" name="maxDuration" type="ns_p:ElementTagType"/>
17885         <xs:element minOccurs="0" name="optionalSlot" type="ns_p:ElementTagType"/>
17886       </xs:sequence>
17887     </xs:complexType>
17888     <xs:element name="powerTimeSlotScheduleConstraintsDataElements"
17889 type="ns_p:PowerTimeSlotScheduleConstraintsDataElementsType"/>
17890     <xs:complexType name="PowerTimeSlotScheduleConstraintsListDataType">
17891       <xs:sequence>
```

```
17897         <xs:element maxOccurs="unbounded" minOccurs="0"
17898 ref="ns_p:powerTimeSlotScheduleConstraintsData"/>
17899       </xs:sequence>
17900     </xs:complexType>
17901     <xs:element name="powerTimeSlotScheduleConstraintsListData"
17902 type="ns_p:PowerTimeSlotScheduleConstraintsListDataType"/>
17903     <xs:complexType name="PowerTimeSlotScheduleConstraintsListDataSelectorsType">
17904       <xs:sequence>
17905         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17906         <xs:element minOccurs="0" name="slotNumber" type="ns_p:PowerTimeSlotNumberType"/>
17907       </xs:sequence>
17908     </xs:complexType>
17909     <xs:element name="powerTimeSlotScheduleConstraintsListDataSelectors"
17910 type="ns_p:PowerTimeSlotScheduleConstraintsListDataSelectorsType"/>
17911     <xs:complexType name="PowerSequenceAlternativesRelationDataType">
17912       <xs:sequence>
17913         <xs:element minOccurs="0" name="alternativesId" type="ns_p:AlternativesIdType"/>
17914         <xs:element name="sequenceId" type="ns_p:PowerSequenceIdType" minOccurs="0"
17915 maxOccurs="unbounded"/>
17916       </xs:sequence>
17917     </xs:complexType>
17918     <xs:element name="powerSequenceAlternativesRelationData"
17919 type="ns_p:PowerSequenceAlternativesRelationDataType"/>
17920     <xs:complexType name="PowerSequenceAlternativesRelationDataElementsType">
17921       <xs:sequence>
17922         <xs:element minOccurs="0" name="alternativesId" type="ns_p:ElementTagType"/>
17923         <xs:element name="sequenceId" minOccurs="0" type="ns_p:ElementTagType"/>
17924       </xs:sequence>
17925     </xs:complexType>
17926     <xs:element name="powerSequenceAlternativesRelationDataElements"
17927 type="ns_p:PowerSequenceAlternativesRelationDataElementsType"/>
17928     <xs:complexType name="PowerSequenceAlternativesRelationListDataType">
17929       <xs:sequence>
17930         <xs:element maxOccurs="unbounded" minOccurs="0"
17931 ref="ns_p:powerSequenceAlternativesRelationData"/>
17932       </xs:sequence>
17933     </xs:complexType>
17934     <xs:element name="powerSequenceAlternativesRelationListData"
17935 type="ns_p:PowerSequenceAlternativesRelationListDataType"/>
17936     <xs:complexType name="PowerSequenceAlternativesRelationListDataSelectorsType">
17937       <xs:sequence>
17938         <xs:element minOccurs="0" name="alternativesId" type="ns_p:AlternativesIdType"/>
17939         <xs:element name="sequenceId" type="ns_p:PowerSequenceIdType" minOccurs="0"/>
17940       </xs:sequence>
17941     </xs:complexType>
17942     <xs:element name="powerSequenceAlternativesRelationListDataSelectors"
17943 type="ns_p:PowerSequenceAlternativesRelationListDataSelectorsType"/>
17944     <xs:complexType name="PowerSequenceDescriptionDataType">
17945       <xs:sequence>
17946         <xs:element name="sequenceId" type="ns_p:PowerSequenceIdType" minOccurs="0"/>
17947         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
17948         <xs:element minOccurs="0" name="positiveEnergyDirection"
17949 type="ns_p:EnergyDirectionType"/>
17950         <xs:element minOccurs="0" name="powerUnit" type="ns_p:UnitOfMeasurementType"/>
17951         <xs:element minOccurs="0" name="energyUnit" type="ns_p:UnitOfMeasurementType"/>
17952         <xs:element minOccurs="0" name="valueSource"
17953 type="ns_p:MeasurementValueSourceType"/>
17954         <xs:element minOccurs="0" name="scope" type="ns_p:PowerSequenceScopeType"/>
17955         <xs:element minOccurs="0" name="taskIdIdentifier" type="xs:unsignedInt"/>
17956         <xs:element minOccurs="0" name="repetitionsTotal" type="xs:unsignedInt"/>
17957       </xs:sequence>
17958     </xs:complexType>
17959     <xs:element name="powerSequenceDescriptionData"
17960 type="ns_p:PowerSequenceDescriptionDataType"/>
17961     <xs:complexType name="PowerSequenceDescriptionDataElementsType">
17962       <xs:sequence>
17963         <xs:element name="sequenceId" minOccurs="0" type="ns_p:ElementTagType"/>
17964         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
17965         <xs:element minOccurs="0" name="positiveEnergyDirection"
17966 type="ns_p:ElementTagType"/>
17967         <xs:element minOccurs="0" name="powerUnit" type="ns_p:ElementTagType"/>
17968         <xs:element minOccurs="0" name="energyUnit" type="ns_p:ElementTagType"/>
17969         <xs:element minOccurs="0" name="valueSource" type="ns_p:ElementTagType"/>
17970         <xs:element minOccurs="0" name="scope" type="ns_p:ElementTagType"/>
17971         <xs:element minOccurs="0" name="taskIdIdentifier" type="ns_p:ElementTagType"/>
17972         <xs:element minOccurs="0" name="repetitionsTotal" type="ns_p:ElementTagType"/>
17973       </xs:sequence>
17974     </xs:complexType>
```

```
17975     <xs:element name="powerSequenceDescriptionDataElements"
17976 type="ns_p:PowerSequenceDescriptionDataElementsType"/>
17977     <xs:complexType name="PowerSequenceDescriptionListDataType">
17978       <xs:sequence>
17979         <xs:element maxOccurs="unbounded" minOccurs="0"
17980 ref="ns_p:powerSequenceDescriptionData"/>
17981       </xs:sequence>
17982     </xs:complexType>
17983     <xs:element name="powerSequenceDescriptionListData"
17984 type="ns_p:PowerSequenceDescriptionListDataType"/>
17985     <xs:complexType name="PowerSequenceDescriptionListDataSelectorsType">
17986       <xs:sequence>
17987         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17988       </xs:sequence>
17989     </xs:complexType>
17990     <xs:element name="powerSequenceDescriptionListDataSelectors"
17991 type="ns_p:PowerSequenceDescriptionListDataSelectorsType"/>
17992     <xs:complexType name="PowerSequenceStateDataType">
17993       <xs:sequence>
17994         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
17995         <xs:element minOccurs="0" name="state" type="ns_p:PowerSequenceStateType"/>
17996         <xs:element minOccurs="0" name="activeSlotNumber"
17997 type="ns_p:PowerTimeSlotNumberType"/>
17998         <xs:element minOccurs="0" name="elapsedSlotTime" type="xs:duration"/>
17999         <xs:element minOccurs="0" name="remainingSlotTime" type="xs:duration"/>
18000         <xs:element minOccurs="0" name="sequenceRemoteControllable" type="xs:boolean"/>
18001         <xs:element minOccurs="0" name="activeRepetitionNumber" type="xs:unsignedInt"/>
18002         <xs:element minOccurs="0" name="remainingPauseTime" type="xs:duration"/>
18003       </xs:sequence>
18004     </xs:complexType>
18005     <xs:element name="powerSequenceStateData" type="ns_p:PowerSequenceStateDataType"/>
18006     <xs:complexType name="PowerSequenceStateDataElementsType">
18007       <xs:sequence>
18008         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18009         <xs:element minOccurs="0" name="state" type="ns_p:ElementTagType"/>
18010         <xs:element minOccurs="0" name="activeSlotNumber" type="ns_p:ElementTagType"/>
18011         <xs:element minOccurs="0" name="elapsedSlotTime" type="ns_p:ElementTagType"/>
18012         <xs:element minOccurs="0" name="remainingSlotTime" type="ns_p:ElementTagType"/>
18013         <xs:element minOccurs="0" name="sequenceRemoteControllable"
18014 type="ns_p:ElementTagType"/>
18015         <xs:element minOccurs="0" name="activeRepetitionNumber"
18016 type="ns_p:ElementTagType"/>
18017         <xs:element minOccurs="0" name="remainingPauseTime" type="ns_p:ElementTagType"/>
18018       </xs:sequence>
18019     </xs:complexType>
18020     <xs:element name="powerSequenceStateDataElements"
18021 type="ns_p:PowerSequenceStateDataElementsType"/>
18022     <xs:complexType name="PowerSequenceStateListDataType">
18023       <xs:sequence>
18024         <xs:element maxOccurs="unbounded" minOccurs="0"
18025 ref="ns_p:powerSequenceStateData"/>
18026       </xs:sequence>
18027     </xs:complexType>
18028     <xs:element name="powerSequenceStateListData" type="ns_p:PowerSequenceStateListDataType"/>
18029     <xs:complexType name="PowerSequenceStateListDataSelectorsType">
18030       <xs:sequence>
18031         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18032       </xs:sequence>
18033     </xs:complexType>
18034     <xs:element name="powerSequenceStateListDataSelectors"
18035 type="ns_p:PowerSequenceStateListDataSelectorsType"/>
18036     <xs:complexType name="PowerSequenceScheduleDataType">
18037       <xs:sequence>
18038         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18039         <xs:element minOccurs="0" name="startTime"
18040 type="ns_p:AbsoluteOrRelativeTimeType"/>
18041         <xs:element minOccurs="0" name="endTime" type="ns_p:AbsoluteOrRelativeTimeType"/>
18042       </xs:sequence>
18043     </xs:complexType>
18044     <xs:element name="powerSequenceScheduleData" type="ns_p:PowerSequenceScheduleDataType"/>
18045     <xs:complexType name="PowerSequenceScheduleDataElementsType">
18046       <xs:sequence>
18047         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18048         <xs:element minOccurs="0" name="startTime" type="ns_p:ElementTagType"/>
18049         <xs:element minOccurs="0" name="endTime" type="ns_p:ElementTagType"/>
18050       </xs:sequence>
18051     </xs:complexType>
```

```
18052     <xs:element name="powerSequenceScheduleDataElements"
18053 type="ns_p:PowerSequenceScheduleDataElementsType"/>
18054     <xs:complexType name="PowerSequenceScheduleListDataType">
18055       <xs:sequence>
18056         <xs:element maxOccurs="unbounded" minOccurs="0"
18057 ref="ns_p:powerSequenceScheduleData"/>
18058       </xs:sequence>
18059     </xs:complexType>
18060     <xs:element name="powerSequenceScheduleListData"
18061 type="ns_p:PowerSequenceScheduleListDataType"/>
18062     <xs:complexType name="PowerSequenceScheduleListDataSelectorsType">
18063       <xs:sequence>
18064         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18065       </xs:sequence>
18066     </xs:complexType>
18067     <xs:element name="powerSequenceScheduleListDataSelectors"
18068 type="ns_p:PowerSequenceScheduleListDataSelectorsType"/>
18069     <xs:complexType name="PowerSequenceScheduleConstraintsDataType">
18070       <xs:sequence>
18071         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18072         <xs:element minOccurs="0" name="earliestStartTime"
18073 type="ns_p:AbsoluteOrRelativeTimeType"/>
18074         <xs:element minOccurs="0" name="latestStartTime"
18075 type="ns_p:AbsoluteOrRelativeTimeType"/>
18076         <xs:element minOccurs="0" name="earliestEndTime"
18077 type="ns_p:AbsoluteOrRelativeTimeType"/>
18078         <xs:element minOccurs="0" name="latestEndTime"
18079 type="ns_p:AbsoluteOrRelativeTimeType"/>
18080         <xs:element minOccurs="0" name="optionalSequence" type="xs:boolean"/>
18081       </xs:sequence>
18082     </xs:complexType>
18083     <xs:element name="powerSequenceScheduleConstraintsData"
18084 type="ns_p:PowerSequenceScheduleConstraintsDataType"/>
18085     <xs:complexType name="PowerSequenceScheduleConstraintsDataElementsType">
18086       <xs:sequence>
18087         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18088         <xs:element minOccurs="0" name="earliestStartTime" type="ns_p:ElementTagType"/>
18089         <xs:element minOccurs="0" name="latestStartTime" type="ns_p:ElementTagType"/>
18090         <xs:element minOccurs="0" name="earliestEndTime" type="ns_p:ElementTagType"/>
18091         <xs:element minOccurs="0" name="latestEndTime" type="ns_p:ElementTagType"/>
18092         <xs:element minOccurs="0" name="optionalSequence" type="ns_p:ElementTagType"/>
18093       </xs:sequence>
18094     </xs:complexType>
18095     <xs:element name="powerSequenceScheduleConstraintsDataElements"
18096 type="ns_p:PowerSequenceScheduleConstraintsDataElementsType"/>
18097     <xs:complexType name="PowerSequenceScheduleConstraintsListDataType">
18098       <xs:sequence>
18099         <xs:element maxOccurs="unbounded" minOccurs="0"
18100 ref="ns_p:powerSequenceScheduleConstraintsData"/>
18101       </xs:sequence>
18102     </xs:complexType>
18103     <xs:element name="powerSequenceScheduleConstraintsListData"
18104 type="ns_p:PowerSequenceScheduleConstraintsListDataType"/>
18105     <xs:complexType name="PowerSequenceScheduleConstraintsListDataSelectorsType">
18106       <xs:sequence>
18107         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18108       </xs:sequence>
18109     </xs:complexType>
18110     <xs:element name="powerSequenceScheduleConstraintsListDataSelectors"
18111 type="ns_p:PowerSequenceScheduleConstraintsListDataSelectorsType"/>
18112     <xs:complexType name="PowerSequencePriceDataType">
18113       <xs:sequence>
18114         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18115         <xs:element minOccurs="0" name="potentialStartTime"
18116 type="ns_p:AbsoluteOrRelativeTimeType"/>
18117         <xs:element minOccurs="0" name="price" type="ns_p:ScaledNumberType"/>
18118         <xs:element minOccurs="0" name="currency" type="ns_p:CurrencyType"/>
18119       </xs:sequence>
18120     </xs:complexType>
18121     <xs:element name="powerSequencePriceData" type="ns_p:PowerSequencePriceDataType"/>
18122     <xs:complexType name="PowerSequencePriceDataElementsType">
18123       <xs:sequence>
18124         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18125         <xs:element minOccurs="0" name="potentialStartTime" type="ns_p:ElementTagType"/>
18126         <xs:element minOccurs="0" name="price" type="ns_p:ScaledNumberElementsType"/>
18127         <xs:element minOccurs="0" name="currency" type="ns_p:ElementTagType"/>
18128       </xs:sequence>
18129     </xs:complexType>
```



```
18130     <xs:element name="powerSequencePriceDataElements"
18131 type="ns_p:PowerSequencePriceDataElementsType"/>
18132     <xs:complexType name="PowerSequencePriceListDataType">
18133       <xs:sequence>
18134         <xs:element maxOccurs="unbounded" minOccurs="0"
18135 ref="ns_p:powerSequencePriceData"/>
18136       </xs:sequence>
18137     </xs:complexType>
18138     <xs:element name="powerSequencePriceListData" type="ns_p:PowerSequencePriceListDataType"/>
18139     <xs:complexType name="PowerSequencePriceListDataSelectorsType">
18140       <xs:sequence>
18141         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18142         <xs:element minOccurs="0" name="potentialStartTimeInterval"
18143 type="ns_p:TimestampIntervalType"/>
18144       </xs:sequence>
18145     </xs:complexType>
18146     <xs:element name="powerSequencePriceListDataSelectors"
18147 type="ns_p:PowerSequencePriceListDataSelectorsType"/>
18148     <xs:complexType name="PowerSequenceSchedulePreferenceDataType">
18149       <xs:sequence>
18150         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18151         <xs:element minOccurs="0" name="greenest" type="xs:boolean"/>
18152         <xs:element minOccurs="0" name="cheapest" type="xs:boolean"/>
18153       </xs:sequence>
18154     </xs:complexType>
18155     <xs:element name="powerSequenceSchedulePreferenceData"
18156 type="ns_p:PowerSequenceSchedulePreferenceDataType"/>
18157     <xs:complexType name="PowerSequenceSchedulePreferenceDataElementsType">
18158       <xs:sequence>
18159         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18160         <xs:element minOccurs="0" name="greenest" type="ns_p:ElementTagType"/>
18161         <xs:element minOccurs="0" name="cheapest" type="ns_p:ElementTagType"/>
18162       </xs:sequence>
18163     </xs:complexType>
18164     <xs:element name="powerSequenceSchedulePreferenceDataElements"
18165 type="ns_p:PowerSequenceSchedulePreferenceDataElementsType"/>
18166     <xs:complexType name="PowerSequenceSchedulePreferenceListDataType">
18167       <xs:sequence>
18168         <xs:element maxOccurs="unbounded" minOccurs="0"
18169 ref="ns_p:powerSequenceSchedulePreferenceData"/>
18170       </xs:sequence>
18171     </xs:complexType>
18172     <xs:element name="powerSequenceSchedulePreferenceListData"
18173 type="ns_p:PowerSequenceSchedulePreferenceListDataType"/>
18174     <xs:complexType name="PowerSequenceSchedulePreferenceListDataSelectorsType">
18175       <xs:sequence>
18176         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18177       </xs:sequence>
18178     </xs:complexType>
18179     <xs:element name="powerSequenceSchedulePreferenceListDataSelectors"
18180 type="ns_p:PowerSequenceSchedulePreferenceListDataSelectorsType"/>
18181     <xs:complexType name="PowerSequenceNodeScheduleInformationDataType">
18182       <xs:sequence>
18183         <xs:element minOccurs="0" name="nodeRemoteControllable" type="xs:boolean"/>
18184         <xs:element minOccurs="0" name="supportsSingleSlotSchedulingOnly"
18185 type="xs:boolean"/>
18186         <xs:element minOccurs="0" name="alternativesCount" type="xs:unsignedInt"/>
18187         <xs:element minOccurs="0" name="totalSequencesCountMax" type="xs:unsignedInt"/>
18188         <xs:element minOccurs="0" name="supportsReselection" type="xs:boolean"/>
18189       </xs:sequence>
18190     </xs:complexType>
18191     <xs:element name="powerSequenceNodeScheduleInformationData"
18192 type="ns_p:PowerSequenceNodeScheduleInformationDataType"/>
18193     <xs:complexType name="PowerSequenceNodeScheduleInformationDataElementsType">
18194       <xs:sequence>
18195         <xs:element minOccurs="0" name="nodeRemoteControllable"
18196 type="ns_p:ElementTagType"/>
18197         <xs:element minOccurs="0" name="supportsSingleSlotSchedulingOnly"
18198 type="ns_p:ElementTagType"/>
18199         <xs:element minOccurs="0" name="alternativesCount" type="ns_p:ElementTagType"/>
18200         <xs:element minOccurs="0" name="totalSequencesCountMax"
18201 type="ns_p:ElementTagType"/>
18202         <xs:element minOccurs="0" name="supportsReselection" type="ns_p:ElementTagType"/>
18203       </xs:sequence>
18204     </xs:complexType>
18205     <xs:element name="powerSequenceNodeScheduleInformationDataElements"
18206 type="ns_p:PowerSequenceNodeScheduleInformationDataElementsType"/>
18207     <xs:complexType name="PowerSequenceScheduleConfigurationRequestCallType">
```

```
18208         <xs:sequence>
18209             <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18210         </xs:sequence>
18211     </xs:complexType>
18212     <xs:element name="powerSequenceScheduleConfigurationRequestCall"
18213 type="ns_p:PowerSequenceScheduleConfigurationRequestCallType"/>
18214     <xs:complexType name="PowerSequenceScheduleConfigurationRequestCallElementsType">
18215         <xs:sequence>
18216             <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18217         </xs:sequence>
18218     </xs:complexType>
18219     <xs:element name="powerSequenceScheduleConfigurationRequestCallElements"
18220 type="ns_p:PowerSequenceScheduleConfigurationRequestCallElementsType"/>
18221     <xs:complexType name="PowerSequencePriceCalculationRequestCallType">
18222         <xs:sequence>
18223             <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
18224             <xs:element minOccurs="0" name="potentialStartTime"
18225 type="ns_p:AbsoluteOrRelativeTimeType"/>
18226         </xs:sequence>
18227     </xs:complexType>
18228     <xs:element name="powerSequencePriceCalculationRequestCall"
18229 type="ns_p:PowerSequencePriceCalculationRequestCallType"/>
18230     <xs:complexType name="PowerSequencePriceCalculationRequestCallElementsType">
18231         <xs:sequence>
18232             <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
18233             <xs:element minOccurs="0" name="potentialStartTime" type="ns_p:ElementTagType"/>
18234         </xs:sequence>
18235     </xs:complexType>
18236     <xs:element name="powerSequencePriceCalculationRequestCallElements"
18237 type="ns_p:PowerSequencePriceCalculationRequestCallElementsType"/>
18238 </xs:schema>
18239
18240
```

18241

18242 A.2.20 Sensing

18243 File name: EEBus_SPINE_TS_Sensing.xsd

```
18244
18245 <?xml version="1.0" encoding="UTF-8"?>
18246 <!--
18247     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18248     Version 1.3.0 (final)
18249     2023-08-31
18250     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18251     Source: https://www.eebus.org/en/specifications/
18252 -->
18253 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18254 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18255 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18256 blockDefault="#all" elementFormDefault="qualified">
18257     <xs:annotation>
18258         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18259 e.V. All rights reserved.</xs:documentation>
18260     </xs:annotation>
18261     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18262     <xs:simpleType name="SensingStateType">
18263         <xs:union memberTypes="ns_p:SensingStateEnumType ns_p:EnumExtendType"/>
18264     </xs:simpleType>
18265     <xs:simpleType name="SensingStateEnumType">
18266         <xs:restriction base="xs:string">
18267             <xs:enumeration value="on"/>
18268             <xs:enumeration value="off"/>
18269             <xs:enumeration value="toggle"/>
18270             <xs:enumeration value="level"/>
18271             <xs:enumeration value="levelUp"/>
18272             <xs:enumeration value="levelDown"/>
18273             <xs:enumeration value="levelStart"/>
18274             <xs:enumeration value="levelStop"/>
18275             <xs:enumeration value="levelAbsolute"/>
18276             <xs:enumeration value="levelRelative"/>
18277             <xs:enumeration value="levelPercentageAbsolute"/>
18278             <xs:enumeration value="levelPercentageRelative"/>
18279             <xs:enumeration value="pressed"/>
18280             <xs:enumeration value="longPressed"/>

```

```

18281         <xs:enumeration value="released"/>
18282         <xs:enumeration value="changed"/>
18283         <xs:enumeration value="started"/>
18284         <xs:enumeration value="stopped"/>
18285         <xs:enumeration value="paused"/>
18286         <xs:enumeration value="middle"/>
18287         <xs:enumeration value="up"/>
18288         <xs:enumeration value="down"/>
18289         <xs:enumeration value="forward"/>
18290         <xs:enumeration value="backwards"/>
18291         <xs:enumeration value="open"/>
18292         <xs:enumeration value="closed"/>
18293         <xs:enumeration value="opening"/>
18294         <xs:enumeration value="closing"/>
18295         <xs:enumeration value="high"/>
18296         <xs:enumeration value="low"/>
18297         <xs:enumeration value="day"/>
18298         <xs:enumeration value="night"/>
18299         <xs:enumeration value="detected"/>
18300         <xs:enumeration value="notDetected"/>
18301         <xs:enumeration value="alarmed"/>
18302         <xs:enumeration value="notAlarmed"/>
18303     </xs:restriction>
18304 </xs:simpleType>
18305 <xs:simpleType name="SensingTypeType">
18306     <xs:union memberTypes="ns_p:SensingTypeEnumType ns_p:EnumExtendType"/>
18307 </xs:simpleType>
18308 <xs:simpleType name="SensingTypeEnumType">
18309     <xs:restriction base="xs:string">
18310         <xs:enumeration value="switch"/>
18311         <xs:enumeration value="button"/>
18312         <xs:enumeration value="level"/>
18313         <xs:enumeration value="levelSwitch"/>
18314         <xs:enumeration value="windowHandle"/>
18315         <xs:enumeration value="contactSensor"/>
18316         <xs:enumeration value="occupancySensor"/>
18317         <xs:enumeration value="motionDetector"/>
18318         <xs:enumeration value="fireDetector"/>
18319         <xs:enumeration value="smokeDetector"/>
18320         <xs:enumeration value="heatDetector"/>
18321         <xs:enumeration value="waterDetector"/>
18322         <xs:enumeration value="gasDetector"/>
18323         <xs:enumeration value="alarmSensor"/>
18324         <xs:enumeration value="powerAlarmSensor"/>
18325         <xs:enumeration value="dayNightIndicator"/>
18326     </xs:restriction>
18327 </xs:simpleType>
18328 <xs:complexType name="SensingDataType">
18329     <xs:sequence>
18330         <xs:element minOccurs="0" name="timestamp"
18331 type="ns_p:AbsoluteOrRelativeTimeType"/>
18332         <xs:element minOccurs="0" name="state" type="ns_p:SensingStateType"/>
18333         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
18334     </xs:sequence>
18335 </xs:complexType>
18336 <xs:element name="sensingData" type="ns_p:SensingDataType"/>
18337 <xs:complexType name="SensingDataElementsType">
18338     <xs:sequence>
18339         <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
18340         <xs:element minOccurs="0" name="state" type="ns_p:ElementTagType"/>
18341         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
18342     </xs:sequence>
18343 </xs:complexType>
18344 <xs:element name="sensingDataElements" type="ns_p:SensingDataElementsType"/>
18345 <xs:complexType name="SensingListDataType">
18346     <xs:sequence>
18347         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:sensingData"/>
18348     </xs:sequence>
18349 </xs:complexType>
18350 <xs:element name="sensingListData" type="ns_p:SensingListDataType"/>
18351 <xs:complexType name="SensingListDataSelectorsType">
18352     <xs:sequence>
18353         <xs:element minOccurs="0" name="timestampInterval"
18354 type="ns_p:TimestampIntervalType"/>
18355     </xs:sequence>
18356 </xs:complexType>
18357 <xs:element name="sensingListDataSelectors" type="ns_p:SensingListDataSelectorsType"/>
18358 <xs:complexType name="SensingDescriptionDataType">

```

```
18359     <xs:sequence>
18360       <xs:element minOccurs="0" name="sensingType" type="ns_p:SensingTypeType"/>
18361       <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
18362       <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
18363       <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
18364       <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
18365     </xs:sequence>
18366   </xs:complexType>
18367   <xs:element name="sensingDescriptionData" type="ns_p:SensingDescriptionDataType"/>
18368   <xs:complexType name="SensingDescriptionDataElementsType">
18369     <xs:sequence>
18370       <xs:element minOccurs="0" name="sensingType" type="ns_p:ElementTagType"/>
18371       <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
18372       <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
18373       <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
18374       <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
18375     </xs:sequence>
18376   </xs:complexType>
18377   <xs:element name="sensingDescriptionDataElements"
18378 type="ns_p:SensingDescriptionDataElementsType"/>
18379 </xs:schema>
18380
18381
```

A.2.21 Setpoint

File name: EEBus_SPINE_TS_Setpoint.xsd

```
18385 <?xml version="1.0" encoding="UTF-8"?>
18386 <!--
18387   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18388   Version 1.3.0 (final)
18389   2023-08-31
18390   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18391   Source: https://www.eebus.org/en/specifications/
18392 -->
18393 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18394 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18395 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18396 blockDefault="#all" elementFormDefault="qualified">
18397   <xs:annotation>
18398     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18399 e.V. All rights reserved.</xs:documentation>
18400   </xs:annotation>
18401   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18402   <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
18403   <xs:include schemaLocation="EEBus_SPINE_TS_TimeTable.xsd"/>
18404   <xs:simpleType name="SetpointIdType">
18405     <xs:restriction base="xs:unsignedInt"/>
18406   </xs:simpleType>
18407   <xs:simpleType name="SetpointTypeType">
18408     <xs:union memberTypes="ns_p:SetpointTypeEnumType ns_p:EnumExtendType"/>
18409   </xs:simpleType>
18410   <xs:simpleType name="SetpointTypeEnumType">
18411     <xs:restriction base="xs:string">
18412       <xs:enumeration value="valueAbsolute"/>
18413       <xs:enumeration value="valueRelative"/>
18414     </xs:restriction>
18415   </xs:simpleType>
18416   <xs:complexType name="SetpointDataType">
18417     <xs:sequence>
18418       <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18419       <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
18420       <xs:element minOccurs="0" name="valueMin" type="ns_p:ScaledNumberType"/>
18421       <xs:element minOccurs="0" name="valueMax" type="ns_p:ScaledNumberType"/>
18422       <xs:element minOccurs="0" name="valueToleranceAbsolute"
18423 type="ns_p:ScaledNumberType"/>
18424       <xs:element minOccurs="0" name="valueTolerancePercentage"
18425 type="ns_p:ScaledNumberType"/>
18426       <xs:element minOccurs="0" name="isSetpointChangeable" type="xs:boolean"/>
18427       <xs:element minOccurs="0" name="isSetpointActive" type="xs:boolean"/>
18428       <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
18429     </xs:sequence>
18430   </xs:complexType>
18431
```

```
18432 <xs:element name="setpointData" type="ns_p:SetpointDataType"/>
18433 <xs:complexType name="SetpointDataElementsType">
18434   <xs:sequence>
18435     <xs:element minOccurs="0" name="setpointId" type="ns_p:ElementTagType"/>
18436     <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
18437     <xs:element minOccurs="0" name="valueMin" type="ns_p:ScaledNumberElementsType"/>
18438     <xs:element minOccurs="0" name="valueMax" type="ns_p:ScaledNumberElementsType"/>
18439     <xs:element minOccurs="0" name="valueToleranceAbsolute"
18440 type="ns_p:ScaledNumberElementsType"/>
18441     <xs:element minOccurs="0" name="valueTolerancePercentage"
18442 type="ns_p:ScaledNumberElementsType"/>
18443     <xs:element minOccurs="0" name="isSetpointChangeable" type="ns_p:ElementTagType"/>
18444     <xs:element minOccurs="0" name="isSetpointActive" type="ns_p:ElementTagType"/>
18445     <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
18446   </xs:sequence>
18447 </xs:complexType>
18448 <xs:element name="setpointDataElements" type="ns_p:SetpointDataElementsType"/>
18449 <xs:complexType name="SetpointListDataType">
18450   <xs:sequence>
18451     <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:setpointData"/>
18452   </xs:sequence>
18453 </xs:complexType>
18454 <xs:element name="setpointListData" type="ns_p:SetpointListDataType"/>
18455 <xs:complexType name="SetpointListDataSelectorsType">
18456   <xs:sequence>
18457     <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18458   </xs:sequence>
18459 </xs:complexType>
18460 <xs:element name="setpointListDataSelectors" type="ns_p:SetpointListDataSelectorsType"/>
18461 <xs:complexType name="SetpointConstraintsDataType">
18462   <xs:sequence>
18463     <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18464     <xs:element minOccurs="0" name="setpointRangeMin" type="ns_p:ScaledNumberType"/>
18465     <xs:element minOccurs="0" name="setpointRangeMax" type="ns_p:ScaledNumberType"/>
18466     <xs:element minOccurs="0" name="setpointStepSize" type="ns_p:ScaledNumberType"/>
18467   </xs:sequence>
18468 </xs:complexType>
18469 <xs:element name="setpointConstraintsData" type="ns_p:SetpointConstraintsDataType"/>
18470 <xs:complexType name="SetpointConstraintsDataElementsType">
18471   <xs:sequence>
18472     <xs:element minOccurs="0" name="setpointId" type="ns_p:ElementTagType"/>
18473     <xs:element minOccurs="0" name="setpointRangeMin"
18474 type="ns_p:ScaledNumberElementsType"/>
18475     <xs:element minOccurs="0" name="setpointRangeMax"
18476 type="ns_p:ScaledNumberElementsType"/>
18477     <xs:element minOccurs="0" name="setpointStepSize"
18478 type="ns_p:ScaledNumberElementsType"/>
18479   </xs:sequence>
18480 </xs:complexType>
18481 <xs:element name="setpointConstraintsDataElements"
18482 type="ns_p:SetpointConstraintsDataElementsType"/>
18483 <xs:complexType name="SetpointConstraintsListDataType">
18484   <xs:sequence>
18485     <xs:element maxOccurs="unbounded" minOccurs="0"
18486 ref="ns_p:setpointConstraintsData"/>
18487   </xs:sequence>
18488 </xs:complexType>
18489 <xs:element name="setpointConstraintsListData"
18490 type="ns_p:SetpointConstraintsListDataType"/>
18491 <xs:complexType name="SetpointConstraintsListDataSelectorsType">
18492   <xs:sequence>
18493     <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18494   </xs:sequence>
18495 </xs:complexType>
18496 <xs:element name="setpointConstraintsListDataSelectors"
18497 type="ns_p:SetpointConstraintsListDataSelectorsType"/>
18498 <xs:complexType name="SetpointDescriptionDataType">
18499   <xs:sequence>
18500     <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18501     <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
18502     <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
18503     <xs:element minOccurs="0" name="setpointType" type="ns_p:SetpointTypeType"/>
18504     <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
18505     <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
18506     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
18507     <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
18508   </xs:sequence>
18509 </xs:complexType>
```

```
18510 <xs:element name="setpointDescriptionData" type="ns_p:SetpointDescriptionDataType"/>
18511 <xs:complexType name="SetpointDescriptionDataElementsType">
18512   <xs:sequence>
18513     <xs:element minOccurs="0" name="setpointId" type="ns_p:ElementTagType"/>
18514     <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
18515     <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
18516     <xs:element minOccurs="0" name="setpointType" type="ns_p:ElementTagType"/>
18517     <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
18518     <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
18519     <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
18520     <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
18521   </xs:sequence>
18522 </xs:complexType>
18523 <xs:element name="setpointDescriptionDataElements"
18524 type="ns_p:SetpointDescriptionDataElementsType"/>
18525 <xs:complexType name="SetpointDescriptionListDataType">
18526   <xs:sequence>
18527     <xs:element maxOccurs="unbounded" minOccurs="0"
18528 ref="ns_p:setpointDescriptionData"/>
18529   </xs:sequence>
18530 </xs:complexType>
18531 <xs:element name="setpointDescriptionListData"
18532 type="ns_p:SetpointDescriptionListDataType"/>
18533 <xs:complexType name="SetpointDescriptionListDataSelectorsType">
18534   <xs:sequence>
18535     <xs:element minOccurs="0" name="setpointId" type="ns_p:SetpointIdType"/>
18536     <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
18537     <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
18538     <xs:element minOccurs="0" name="setpointType" type="ns_p:SetpointTypeType"/>
18539     <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
18540   </xs:sequence>
18541 </xs:complexType>
18542 <xs:element name="setpointDescriptionListDataSelectors"
18543 type="ns_p:SetpointDescriptionListDataSelectorsType"/>
18544 </xs:schema>
```

A.2.22 StateInformation

File name: EEBus_SPINE_TS_StateInformation.xsd

```
18550 <?xml version="1.0" encoding="UTF-8"?>
18551 <!--
18552   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18553   Version 1.3.0 (final)
18554   2023-08-31
18555   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18556   Source: https://www.eebus.org/en/specifications/
18557 -->
18558 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18559 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18560 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18561 blockDefault="#all" elementFormDefault="qualified">
18562   <xs:annotation>
18563     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18564 e.V. All rights reserved.</xs:documentation>
18565   </xs:annotation>
18566   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18567   <xs:simpleType name="stateInformationIdType">
18568     <xs:restriction base="xs:unsignedInt"/>
18569   </xs:simpleType>
18570   <xs:simpleType name="StateInformationType">
18571     <xs:union memberTypes="ns_p:StateInformationFunctionalityEnumType
18572 ns_p:StateInformationFailureEnumType ns_p:EnumExtendType"/>
18573   </xs:simpleType>
18574   <xs:simpleType name="StateInformationFunctionalityEnumType">
18575     <xs:restriction base="xs:string">
18576       <xs:enumeration value="externalOverrideFromGrid"/>
18577       <xs:enumeration value="autonomousGridSupport"/>
18578       <xs:enumeration value="islandingMode"/>
18579       <xs:enumeration value="balancing"/>
18580       <xs:enumeration value="trickleCharging"/>
18581       <xs:enumeration value="calibration"/>
```

```
18583         <xs:enumeration value="commissioningMissing"/>
18584         <xs:enumeration value="sleeping"/>
18585         <xs:enumeration value="starting"/>
18586         <xs:enumeration value="mppt"/>
18587         <xs:enumeration value="throttled"/>
18588         <xs:enumeration value="shuttingDown"/>
18589         <xs:enumeration value="manualShutdown"/>
18590     </xs:restriction>
18591 </xs:simpleType>
18592 <xs:simpleType name="StateInformationFailureEnumType">
18593     <xs:restriction base="xs:string">
18594         <xs:enumeration value="inverterDefective"/>
18595         <xs:enumeration value="batteryOvercurrentProtection"/>
18596         <xs:enumeration value="pvStringOvercurrentProtection"/>
18597         <xs:enumeration value="gridFault"/>
18598         <xs:enumeration value="groundFault"/>
18599         <xs:enumeration value="acDisconnected"/>
18600         <xs:enumeration value="dcDisconnected"/>
18601         <xs:enumeration value="cabinetOpen"/>
18602         <xs:enumeration value="overTemperature"/>
18603         <xs:enumeration value="underTemperature"/>
18604         <xs:enumeration value="frequencyAboveLimit"/>
18605         <xs:enumeration value="frequencyBelowLimit"/>
18606         <xs:enumeration value="acVoltageAboveLimit"/>
18607         <xs:enumeration value="acVoltageBelowLimit"/>
18608         <xs:enumeration value="dcVoltageAboveLimit"/>
18609         <xs:enumeration value="dcVoltageBelowLimit"/>
18610         <xs:enumeration value="hardwareTestFailure"/>
18611         <xs:enumeration value="genericInternalError"/>
18612     </xs:restriction>
18613 </xs:simpleType>
18614 <xs:simpleType name="StateInformationCategoryType">
18615     <xs:union memberTypes="ns_p:StateInformationCategoryEnumType ns_p:EnumExtendType"/>
18616 </xs:simpleType>
18617 <xs:simpleType name="StateInformationCategoryEnumType">
18618     <xs:restriction base="xs:string">
18619         <xs:enumeration value="functionality"/>
18620         <xs:enumeration value="failure"/>
18621     </xs:restriction>
18622 </xs:simpleType>
18623 <xs:complexType name="StateInformationDataType">
18624     <xs:sequence>
18625         <xs:element minOccurs="0" name="stateInformationId"
18626 type="ns_p:StateInformationIdType"/>
18627         <xs:element minOccurs="0" name="stateInformation"
18628 type="ns_p:StateInformationType"/>
18629         <xs:element minOccurs="0" name="isActive" type="xs:boolean"/>
18630         <xs:element minOccurs="0" name="category"
18631 type="ns_p:StateInformationCategoryType"/>
18632         <xs:element minOccurs="0" name="timeOfLastChange"
18633 type="ns_p:AbsoluteOrRelativeTimeType"/>
18634     </xs:sequence>
18635 </xs:complexType>
18636 <xs:element name="stateInformationData" type="ns_p:StateInformationDataType"/>
18637 <xs:complexType name="StateInformationDataElementsType">
18638     <xs:sequence>
18639         <xs:element minOccurs="0" name="stateInformationId" type="ns_p:ElementTagType"/>
18640         <xs:element minOccurs="0" name="stateInformation" type="ns_p:ElementTagType"/>
18641         <xs:element minOccurs="0" name="isActive" type="ns_p:ElementTagType"/>
18642         <xs:element minOccurs="0" name="category" type="ns_p:ElementTagType"/>
18643         <xs:element minOccurs="0" name="timeOfLastChange" type="ns_p:ElementTagType"/>
18644     </xs:sequence>
18645 </xs:complexType>
18646 <xs:element name="stateInformationDataElements"
18647 type="ns_p:StateInformationDataElementsType"/>
18648 <xs:complexType name="StateInformationListDataType">
18649     <xs:sequence>
18650         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:stateInformationData"/>
18651     </xs:sequence>
18652 </xs:complexType>
18653 <xs:element name="stateInformationListData" type="ns_p:StateInformationListDataType"/>
18654 <xs:complexType name="StateInformationListDataSelectorsType">
18655     <xs:sequence>
18656         <xs:element minOccurs="0" name="stateInformationId"
18657 type="ns_p:StateInformationIdType"/>
18658         <xs:element minOccurs="0" name="stateInformation"
18659 type="ns_p:StateInformationType"/>
18660         <xs:element minOccurs="0" name="isActive" type="xs:boolean"/>
```

```
18661         <xs:element minOccurs="0" name="category"
18662 type="ns_p:StateInformationCategoryType"/>
18663     </xs:sequence>
18664 </xs:complexType>
18665     <xs:element name="stateInformationListDataSelectors"
18666 type="ns_p:StateInformationListDataSelectorsType"/>
18667 </xs:schema>
```

A.2.23 SubscriptionManagement

File name: EEBus_SPINE_TS_SubscriptionManagement.xsd

```
18673 <?xml version="1.0" encoding="UTF-8"?>
18674 <!--
18675     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18676     Version 1.3.0 (final)
18677     2023-08-31
18678     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18679     Source: https://www.eebus.org/en/specifications/
18680 -->
18681 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18682 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18683 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18684 blockDefault="#all" elementFormDefault="qualified">
18685     <xs:annotation>
18686         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18687 e.V. All rights reserved.</xs:documentation>
18688     </xs:annotation>
18689     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18690     <xs:simpleType name="SubscriptionIdType">
18691         <xs:restriction base="xs:unsignedInt"/>
18692     </xs:simpleType>
18693     <xs:complexType name="SubscriptionManagementEntryDataType">
18694         <xs:sequence>
18695             <xs:element minOccurs="0" name="subscriptionId" type="ns_p:SubscriptionIdType"/>
18696             <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
18697             <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
18698             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
18699             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
18700         </xs:sequence>
18701     </xs:complexType>
18702     <xs:element name="subscriptionManagementEntryData"
18703 type="ns_p:SubscriptionManagementEntryDataType"/>
18704     <xs:complexType name="SubscriptionManagementEntryDataElementsType">
18705         <xs:sequence>
18706             <xs:element minOccurs="0" name="subscriptionId" type="ns_p:ElementTagType"/>
18707             <xs:element minOccurs="0" name="clientAddress"
18708 type="ns_p:FeatureAddressElementsType"/>
18709             <xs:element minOccurs="0" name="serverAddress"
18710 type="ns_p:FeatureAddressElementsType"/>
18711             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
18712             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
18713         </xs:sequence>
18714     </xs:complexType>
18715     <xs:element name="subscriptionManagementEntryDataElements"
18716 type="ns_p:SubscriptionManagementEntryDataElementsType"/>
18717     <xs:complexType name="SubscriptionManagementEntryListDataType">
18718         <xs:sequence>
18719             <xs:element maxOccurs="unbounded" minOccurs="0"
18720 ref="ns_p:subscriptionManagementEntryData"/>
18721         </xs:sequence>
18722     </xs:complexType>
18723     <xs:element name="subscriptionManagementEntryListData"
18724 type="ns_p:SubscriptionManagementEntryListDataType"/>
18725     <xs:complexType name="SubscriptionManagementEntryListDataSelectorsType">
18726         <xs:sequence>
18727             <xs:element minOccurs="0" name="subscriptionId" type="ns_p:SubscriptionIdType"/>
18728             <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
18729             <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
18730         </xs:sequence>
18731     </xs:complexType>
```



```

18733     <xs:element name="subscriptionManagementEntryListDataSelectors"
18734 type="ns_p:SubscriptionManagementEntryListDataSelectorsType"/>
18735     <xs:complexType name="SubscriptionManagementRequestCallType">
18736       <xs:sequence>
18737         <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
18738         <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
18739         <xs:element minOccurs="0" name="serverFeatureType" type="ns_p:FeatureTypeType"/>
18740       </xs:sequence>
18741     </xs:complexType>
18742     <xs:element name="subscriptionManagementRequestCall"
18743 type="ns_p:SubscriptionManagementRequestCallType"/>
18744     <xs:complexType name="SubscriptionManagementRequestCallElementsType">
18745       <xs:sequence>
18746         <xs:element minOccurs="0" name="clientAddress"
18747 type="ns_p:FeatureAddressElementsType"/>
18748         <xs:element minOccurs="0" name="serverAddress"
18749 type="ns_p:FeatureAddressElementsType"/>
18750         <xs:element minOccurs="0" name="serverFeatureType" type="ns_p:ElementTagType"/>
18751       </xs:sequence>
18752     </xs:complexType>
18753     <xs:element name="subscriptionManagementRequestCallElements"
18754 type="ns_p:SubscriptionManagementRequestCallElementsType"/>
18755     <xs:complexType name="SubscriptionManagementDeleteCallType">
18756       <xs:sequence>
18757         <xs:element minOccurs="0" name="subscriptionId" type="ns_p:SubscriptionIdType"/>
18758         <xs:element minOccurs="0" name="clientAddress" type="ns_p:FeatureAddressType"/>
18759         <xs:element minOccurs="0" name="serverAddress" type="ns_p:FeatureAddressType"/>
18760       </xs:sequence>
18761     </xs:complexType>
18762     <xs:element name="subscriptionManagementDeleteCall"
18763 type="ns_p:SubscriptionManagementDeleteCallType"/>
18764     <xs:complexType name="SubscriptionManagementDeleteCallElementsType">
18765       <xs:sequence>
18766         <xs:element minOccurs="0" name="subscriptionId" type="ns_p:ElementTagType"/>
18767         <xs:element minOccurs="0" name="clientAddress"
18768 type="ns_p:FeatureAddressElementsType"/>
18769         <xs:element minOccurs="0" name="serverAddress"
18770 type="ns_p:FeatureAddressElementsType"/>
18771       </xs:sequence>
18772     </xs:complexType>
18773     <xs:element name="subscriptionManagementDeleteCallElements"
18774 type="ns_p:SubscriptionManagementDeleteCallElementsType"/>
18775   </xs:schema>
18776
18777

```

18779 A.2.24 SupplyCondition

18780 File name: EEBus_SPINE_TS_SupplyCondition.xsd

```

18781 <?xml version="1.0" encoding="UTF-8"?>
18782 <!--
18783     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18784     Version 1.3.0 (final)
18785     2023-08-31
18786     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18787     Source: https://www.eebus.org/en/specifications/
18788 -->
18789 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18790 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18791 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18792 blockDefault="#all" elementFormDefault="qualified">
18793   <xs:annotation>
18794     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18795 e.V. All rights reserved.</xs:documentation>
18796   </xs:annotation>
18797   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18798   <xs:include schemaLocation="EEBus_SPINE_TS_Threshold.xsd"/>
18799   <xs:simpleType name="ConditionIdType">
18800     <xs:restriction base="xs:unsignedInt"/>
18801   </xs:simpleType>
18802   <xs:simpleType name="SupplyConditionEventTypeType">
18803     <xs:union memberTypes="ns_p:SupplyConditionEventTypeEnumType ns_p:EnumExtendType"/>
18804   </xs:simpleType>
18805

```

```
18806 <xs:simpleType name="SupplyConditionEventTypeEnumType">
18807 <xs:restriction base="xs:string">
18808 <xs:enumeration value="thresholdExceeded"/>
18809 <xs:enumeration value="fallenBelowThreshold"/>
18810 <xs:enumeration value="supplyInterrupt"/>
18811 <xs:enumeration value="releaseOfLimitations"/>
18812 <xs:enumeration value="otherProblem"/>
18813 <xs:enumeration value="gridConditionUpdate"/>
18814 </xs:restriction>
18815 </xs:simpleType>
18816 <xs:simpleType name="SupplyConditionOriginatorType">
18817 <xs:union memberTypes="ns_p:SupplyConditionOriginatorEnumType ns_p:EnumExtendType"/>
18818 </xs:simpleType>
18819 <xs:simpleType name="SupplyConditionOriginatorEnumType">
18820 <xs:restriction base="xs:string">
18821 <xs:enumeration value="externDSO"/>
18822 <xs:enumeration value="externSupplier"/>
18823 <xs:enumeration value="internalLimit"/>
18824 <xs:enumeration value="internalService"/>
18825 <xs:enumeration value="internalUser"/>
18826 </xs:restriction>
18827 </xs:simpleType>
18828 <xs:simpleType name="GridConditionType">
18829 <xs:union memberTypes="ns_p:GridConditionEnumType ns_p:EnumExtendType"/>
18830 </xs:simpleType>
18831 <xs:simpleType name="GridConditionEnumType">
18832 <xs:restriction base="xs:string">
18833 <xs:enumeration value="consumptionRed"/>
18834 <xs:enumeration value="consumptionYellow"/>
18835 <xs:enumeration value="good"/>
18836 <xs:enumeration value="productionYellow"/>
18837 <xs:enumeration value="productionRed"/>
18838 </xs:restriction>
18839 </xs:simpleType>
18840 <xs:complexType name="SupplyConditionDataType">
18841 <xs:sequence>
18842 <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
18843 <xs:element minOccurs="0" name="timestamp"
18844 type="ns_p:AbsoluteOrRelativeTimeType"/>
18845 <xs:element minOccurs="0" name="eventType"
18846 type="ns_p:SupplyConditionEventTypeType"/>
18847 <xs:element minOccurs="0" name="originator"
18848 type="ns_p:SupplyConditionOriginatorType"/>
18849 <xs:element minOccurs="0" name="thresholdId" type="ns_p:ThresholdIdType"/>
18850 <xs:element minOccurs="0" name="thresholdPercentage"
18851 type="ns_p:ScaledNumberType"/>
18852 <xs:element minOccurs="0" name="relevantPeriod" type="ns_p:TimePeriodType"/>
18853 <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
18854 <xs:element minOccurs="0" name="gridCondition" type="ns_p:GridConditionType"/>
18855 </xs:sequence>
18856 </xs:complexType>
18857 <xs:element name="supplyConditionData" type="ns_p:SupplyConditionDataType"/>
18858 <xs:complexType name="SupplyConditionDataElementsType">
18859 <xs:sequence>
18860 <xs:element minOccurs="0" name="conditionId" type="ns_p:ElementTagType"/>
18861 <xs:element minOccurs="0" name="timestamp" type="ns_p:ElementTagType"/>
18862 <xs:element minOccurs="0" name="eventType" type="ns_p:ElementTagType"/>
18863 <xs:element minOccurs="0" name="originator" type="ns_p:ElementTagType"/>
18864 <xs:element minOccurs="0" name="thresholdId" type="ns_p:ElementTagType"/>
18865 <xs:element minOccurs="0" name="thresholdPercentage"
18866 type="ns_p:ScaledNumberElementsType"/>
18867 <xs:element minOccurs="0" name="relevantPeriod"
18868 type="ns_p:TimePeriodElementsType"/>
18869 <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
18870 <xs:element minOccurs="0" name="gridCondition" type="ns_p:ElementTagType"/>
18871 </xs:sequence>
18872 </xs:complexType>
18873 <xs:element name="supplyConditionDataElements"
18874 type="ns_p:SupplyConditionDataElementsType"/>
18875 <xs:complexType name="SupplyConditionListDataType">
18876 <xs:sequence>
18877 <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:supplyConditionData"/>
18878 </xs:sequence>
18879 </xs:complexType>
18880 <xs:element name="supplyConditionListData" type="ns_p:SupplyConditionListDataType"/>
18881 <xs:complexType name="SupplyConditionListDataSelectorsType">
18882 <xs:sequence>
18883 <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
```

```
18884         <xs:element minOccurs="0" name="timestampInterval"
18885 type="ns_p:TimestampIntervalType"/>
18886         <xs:element minOccurs="0" name="eventType"
18887 type="ns_p:SupplyConditionEventTypeType"/>
18888         <xs:element minOccurs="0" name="originator"
18889 type="ns_p:SupplyConditionOriginatorType"/>
18890     </xs:sequence>
18891 </xs:complexType>
18892     <xs:element name="supplyConditionListDataSelectors"
18893 type="ns_p:SupplyConditionListDataSelectorsType"/>
18894     <xs:complexType name="SupplyConditionDescriptionDataType">
18895         <xs:sequence>
18896             <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
18897             <xs:element minOccurs="0" name="commodityType" type="ns_p:CommodityTypeType"/>
18898             <xs:element minOccurs="0" name="positiveEnergyDirection"
18899 type="ns_p:EnergyDirectionType"/>
18900             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
18901             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
18902         </xs:sequence>
18903     </xs:complexType>
18904     <xs:element name="supplyConditionDescriptionData"
18905 type="ns_p:SupplyConditionDescriptionDataType"/>
18906     <xs:complexType name="SupplyConditionDescriptionDataElementsType">
18907         <xs:sequence>
18908             <xs:element minOccurs="0" name="conditionId" type="ns_p:ElementTagType"/>
18909             <xs:element minOccurs="0" name="commodityType" type="ns_p:ElementTagType"/>
18910             <xs:element minOccurs="0" name="positiveEnergyDirection"
18911 type="ns_p:ElementTagType"/>
18912             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
18913             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
18914         </xs:sequence>
18915     </xs:complexType>
18916     <xs:element name="supplyConditionDescriptionDataElements"
18917 type="ns_p:SupplyConditionDescriptionDataElementsType"/>
18918     <xs:complexType name="SupplyConditionDescriptionListDataType">
18919         <xs:sequence>
18920             <xs:element maxOccurs="unbounded" minOccurs="0"
18921 ref="ns_p:supplyConditionDescriptionData"/>
18922         </xs:sequence>
18923     </xs:complexType>
18924     <xs:element name="supplyConditionDescriptionListData"
18925 type="ns_p:SupplyConditionDescriptionListDataType"/>
18926     <xs:complexType name="SupplyConditionDescriptionListDataSelectorsType">
18927         <xs:sequence>
18928             <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
18929         </xs:sequence>
18930     </xs:complexType>
18931     <xs:element name="supplyConditionDescriptionListDataSelectors"
18932 type="ns_p:SupplyConditionDescriptionListDataSelectorsType"/>
18933     <xs:complexType name="SupplyConditionThresholdRelationDataType">
18934         <xs:sequence>
18935             <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
18936             <xs:element maxOccurs="unbounded" minOccurs="0" name="thresholdId"
18937 type="ns_p:ThresholdIdType"/>
18938         </xs:sequence>
18939     </xs:complexType>
18940     <xs:element name="supplyConditionThresholdRelationData"
18941 type="ns_p:SupplyConditionThresholdRelationDataType"/>
18942     <xs:complexType name="SupplyConditionThresholdRelationDataElementsType">
18943         <xs:sequence>
18944             <xs:element minOccurs="0" name="conditionId" type="ns_p:ElementTagType"/>
18945             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ElementTagType"/>
18946         </xs:sequence>
18947     </xs:complexType>
18948     <xs:element name="supplyConditionThresholdRelationDataElements"
18949 type="ns_p:SupplyConditionThresholdRelationDataElementsType"/>
18950     <xs:complexType name="SupplyConditionThresholdRelationListDataType">
18951         <xs:sequence>
18952             <xs:element maxOccurs="unbounded" minOccurs="0"
18953 ref="ns_p:supplyConditionThresholdRelationData"/>
18954         </xs:sequence>
18955     </xs:complexType>
18956     <xs:element name="supplyConditionThresholdRelationListData"
18957 type="ns_p:SupplyConditionThresholdRelationListDataType"/>
18958     <xs:complexType name="SupplyConditionThresholdRelationListDataSelectorsType">
18959         <xs:sequence>
18960             <xs:element minOccurs="0" name="conditionId" type="ns_p:ConditionIdType"/>
18961             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ThresholdIdType"/>
```

```
18962         </xs:sequence>
18963     </xs:complexType>
18964     <xs:element name="supplyConditionThresholdRelationListDataSelectors"
18965 type="ns_p:SupplyConditionThresholdRelationListDataSelectorsType"/>
18966 </xs:schema>
18967
18968
```

18970 A.2.25 TariffInformation

18971 File name: EEBus_SPINE_TS_TariffInformation.xsd

```
18972 <?xml version="1.0" encoding="UTF-8"?>
18973 <!--
18974 Smart Premises Interoperable Neutral-Message Exchange (SPINE)
18975 Version 1.3.0 (final)
18976 2023-08-31
18977 Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
18978 Source: https://www.eebus.org/en/specifications/
18979 -->
18980 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
18981 xmlns:xs="http://www.w3.org/2001/XMLSchema"
18982 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
18983 blockDefault="#all" elementFormDefault="qualified">
18984     <xs:annotation>
18985         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
18986 e.V. All rights reserved.</xs:documentation>
18987     </xs:annotation>
18988     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
18989     <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
18990     <xs:include schemaLocation="EEBus_SPINE_TS_TimeTable.xsd"/>
18991     <xs:simpleType name="TariffIdType">
18992         <xs:restriction base="xs:unsignedInt"/>
18993     </xs:simpleType>
18994     <xs:simpleType name="TariffCountType">
18995         <xs:restriction base="ns_p:TariffIdType"/>
18996     </xs:simpleType>
18997     <xs:simpleType name="TierBoundaryIdType">
18998         <xs:restriction base="xs:unsignedInt"/>
18999     </xs:simpleType>
19000     <xs:simpleType name="TierBoundaryCountType">
19001         <xs:restriction base="ns_p:TierBoundaryIdType"/>
19002     </xs:simpleType>
19003     <xs:simpleType name="TierBoundaryTypeEnumType">
19004         <xs:union memberTypes="ns_p:TierBoundaryTypeEnumType ns_p:EnumExtendType"/>
19005     </xs:simpleType>
19006     <xs:simpleType name="TierBoundaryTypeEnumType">
19007         <xs:restriction base="xs:string">
19008             <xs:enumeration value="powerBoundary"/>
19009             <xs:enumeration value="energyBoundary"/>
19010             <xs:enumeration value="countBoundary"/>
19011         </xs:restriction>
19012     </xs:simpleType>
19013     <xs:simpleType name="CommodityIdType">
19014         <xs:restriction base="xs:unsignedInt"/>
19015     </xs:simpleType>
19016     <xs:simpleType name="TierIdType">
19017         <xs:restriction base="xs:unsignedInt"/>
19018     </xs:simpleType>
19019     <xs:simpleType name="TierCountType">
19020         <xs:restriction base="ns_p:TierIdType"/>
19021     </xs:simpleType>
19022     <xs:simpleType name="TierTypeEnumType">
19023         <xs:union memberTypes="ns_p:TierTypeEnumType ns_p:EnumExtendType"/>
19024     </xs:simpleType>
19025     <xs:simpleType name="TierTypeEnumType">
19026         <xs:restriction base="xs:string">
19027             <xs:enumeration value="fixedCost"/>
19028             <xs:enumeration value="dynamicCost"/>
19029         </xs:restriction>
19030     </xs:simpleType>
19031     <xs:simpleType name="IncentiveIdType">
19032         <xs:restriction base="xs:unsignedInt"/>
19033     </xs:simpleType>
19034
```

```
19035 <xs:simpleType name="IncentiveCountType">
19036 <xs:restriction base="ns_p:IncentiveIdType"/>
19037 </xs:simpleType>
19038 <xs:simpleType name="IncentiveTypeType">
19039 <xs:union memberTypes="ns_p:IncentiveTypeEnumType ns_p:EnumExtendType"/>
19040 </xs:simpleType>
19041 <xs:simpleType name="IncentiveTypeEnumType">
19042 <xs:restriction base="xs:string">
19043 <xs:enumeration value="absoluteCost"/>
19044 <xs:enumeration value="relativeCost"/>
19045 <xs:enumeration value="renewableEnergyPercentage"/>
19046 <xs:enumeration value="co2Emission"/>
19047 </xs:restriction>
19048 </xs:simpleType>
19049 <xs:simpleType name="IncentivePriorityType">
19050 <xs:restriction base="xs:unsignedInt"/>
19051 </xs:simpleType>
19052 <xs:simpleType name="IncentiveValueTypeType">
19053 <xs:union memberTypes="ns_p:IncentiveValueTypeEnumType ns_p:EnumExtendType"/>
19054 </xs:simpleType>
19055 <xs:simpleType name="IncentiveValueTypeEnumType">
19056 <xs:restriction base="xs:string">
19057 <xs:enumeration value="value"/>
19058 <xs:enumeration value="averageValue"/>
19059 <xs:enumeration value="minValue"/>
19060 <xs:enumeration value="maxValue"/>
19061 </xs:restriction>
19062 </xs:simpleType>
19063 <xs:complexType name="TariffOverallConstraintsDataType">
19064 <xs:sequence>
19065 <xs:element minOccurs="0" name="maxTariffCount" type="ns_p:TariffCountType"/>
19066 <xs:element minOccurs="0" name="maxBoundaryCount"
19067 type="ns_p:TierBoundaryCountType"/>
19068 <xs:element minOccurs="0" name="maxTierCount" type="ns_p:TierCountType"/>
19069 <xs:element minOccurs="0" name="maxIncentiveCount"
19070 type="ns_p:IncentiveCountType"/>
19071 <xs:element minOccurs="0" name="maxBoundariesPerTariff"
19072 type="ns_p:TierBoundaryCountType"/>
19073 <xs:element minOccurs="0" name="maxTiersPerTariff" type="ns_p:TierCountType"/>
19074 <xs:element minOccurs="0" name="maxBoundariesPerTier"
19075 type="ns_p:TierBoundaryCountType"/>
19076 <xs:element minOccurs="0" name="maxIncentivesPerTier"
19077 type="ns_p:IncentiveCountType"/>
19078 </xs:sequence>
19079 </xs:complexType>
19080 <xs:element name="tariffOverallConstraintsData"
19081 type="ns_p:TariffOverallConstraintsDataType"/>
19082 <xs:complexType name="TariffOverallConstraintsDataElementsType">
19083 <xs:sequence>
19084 <xs:element minOccurs="0" name="maxTariffCount" type="ns_p:ElementTagType"/>
19085 <xs:element minOccurs="0" name="maxBoundaryCount" type="ns_p:ElementTagType"/>
19086 <xs:element minOccurs="0" name="maxTierCount" type="ns_p:ElementTagType"/>
19087 <xs:element minOccurs="0" name="maxIncentiveCount" type="ns_p:ElementTagType"/>
19088 <xs:element minOccurs="0" name="maxBoundariesPerTariff"
19089 type="ns_p:ElementTagType"/>
19090 <xs:element minOccurs="0" name="maxTiersPerTariff" type="ns_p:ElementTagType"/>
19091 <xs:element minOccurs="0" name="maxBoundariesPerTier" type="ns_p:ElementTagType"/>
19092 <xs:element minOccurs="0" name="maxIncentivesPerTier" type="ns_p:ElementTagType"/>
19093 </xs:sequence>
19094 </xs:complexType>
19095 <xs:element name="tariffOverallConstraintsDataElements"
19096 type="ns_p:TariffOverallConstraintsDataElementsType"/>
19097 <xs:complexType name="TariffDataType">
19098 <xs:sequence>
19099 <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19100 <xs:element name="activeTierId" type="ns_p:TierIdType" minOccurs="0"
19101 maxOccurs="unbounded"/>
19102 </xs:sequence>
19103 </xs:complexType>
19104 <xs:element name="tariffData" type="ns_p:TariffDataType"/>
19105 <xs:complexType name="TariffDataElementsType">
19106 <xs:sequence>
19107 <xs:element name="tariffId" type="ns_p:ElementTagType" minOccurs="0"/>
19108 <xs:element name="activeTierId" type="ns_p:ElementTagType" minOccurs="0"/>
19109 </xs:sequence>
19110 </xs:complexType>
19111 <xs:element name="tariffDataElements" type="ns_p:TariffDataElementsType"/>
19112 <xs:complexType name="TariffListDataType">
```

```
19113     <xs:sequence>
19114         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:tariffData"/>
19115     </xs:sequence>
19116 </xs:complexType>
19117 <xs:element name="tariffListData" type="ns_p:TariffListDataType"/>
19118 <xs:complexType name="TariffListDataSelectorsType">
19119     <xs:sequence>
19120         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19121         <xs:element name="activeTierId" type="ns_p:TierIdType" minOccurs="0"/>
19122     </xs:sequence>
19123 </xs:complexType>
19124 <xs:element name="tariffListDataSelectors" type="ns_p:TariffListDataSelectorsType"/>
19125 <xs:complexType name="TariffTierRelationDataType">
19126     <xs:sequence>
19127         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19128         <xs:element name="tierId" maxOccurs="unbounded" type="ns_p:TierIdType"
19129 minOccurs="0"/>
19130     </xs:sequence>
19131 </xs:complexType>
19132 <xs:element name="tariffTierRelationData" type="ns_p:TariffTierRelationDataType"/>
19133 <xs:complexType name="TariffTierRelationDataElementsType">
19134     <xs:sequence>
19135         <xs:element name="tariffId" type="ns_p:ElementTagType" minOccurs="0"/>
19136         <xs:element name="tierId" type="ns_p:ElementTagType" minOccurs="0"/>
19137     </xs:sequence>
19138 </xs:complexType>
19139 <xs:element name="tariffTierRelationDataElements"
19140 type="ns_p:TariffTierRelationDataElementsType"/>
19141 <xs:complexType name="TariffTierRelationListDataType">
19142     <xs:sequence>
19143         <xs:element maxOccurs="unbounded" minOccurs="0"
19144 ref="ns_p:tariffTierRelationData"/>
19145     </xs:sequence>
19146 </xs:complexType>
19147 <xs:element name="tariffTierRelationListData" type="ns_p:TariffTierRelationListDataType"/>
19148 <xs:complexType name="TariffTierRelationListDataSelectorsType">
19149     <xs:sequence>
19150         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19151         <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19152     </xs:sequence>
19153 </xs:complexType>
19154 <xs:element name="tariffTierRelationListDataSelectors"
19155 type="ns_p:TariffTierRelationListDataSelectorsType"/>
19156 <xs:complexType name="TariffBoundaryRelationDataType">
19157     <xs:sequence>
19158         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19159         <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"
19160 maxOccurs="unbounded"/>
19161     </xs:sequence>
19162 </xs:complexType>
19163 <xs:element name="tariffBoundaryRelationData" type="ns_p:TariffBoundaryRelationDataType"/>
19164 <xs:complexType name="TariffBoundaryRelationDataElementsType">
19165     <xs:sequence>
19166         <xs:element name="tariffId" type="ns_p:ElementTagType" minOccurs="0"/>
19167         <xs:element name="boundaryId" type="ns_p:ElementTagType" minOccurs="0"/>
19168     </xs:sequence>
19169 </xs:complexType>
19170 <xs:element name="tariffBoundaryRelationDataElements"
19171 type="ns_p:TariffBoundaryRelationDataElementsType"/>
19172 <xs:complexType name="TariffBoundaryRelationListDataType">
19173     <xs:sequence>
19174         <xs:element maxOccurs="unbounded" minOccurs="0"
19175 ref="ns_p:tariffBoundaryRelationData"/>
19176     </xs:sequence>
19177 </xs:complexType>
19178 <xs:element name="tariffBoundaryRelationListData"
19179 type="ns_p:TariffBoundaryRelationListDataType"/>
19180 <xs:complexType name="TariffBoundaryRelationListDataSelectorsType">
19181     <xs:sequence>
19182         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19183         <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"/>
19184     </xs:sequence>
19185 </xs:complexType>
19186 <xs:element name="tariffBoundaryRelationListDataSelectors"
19187 type="ns_p:TariffBoundaryRelationListDataSelectorsType"/>
19188 <xs:complexType name="TariffDescriptionDataType">
19189     <xs:sequence>
19190         <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
```

```
19191      <xs:element name="commodityId" type="ns_p:CommodityIdType" minOccurs="0"/>
19192      <xs:element name="measurementId" type="ns_p:MeasurementIdType" minOccurs="0"/>
19193      <xs:element name="tariffWriteable" type="xs:boolean" minOccurs="0"/>
19194      <xs:element name="updateRequired" type="xs:boolean" minOccurs="0"/>
19195      <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
19196      <xs:element name="label" type="ns_p:LabelType" minOccurs="0"/>
19197      <xs:element name="description" type="ns_p:DescriptionType" minOccurs="0"/>
19198      <xs:element name="slotIdSupport" type="xs:boolean" minOccurs="0"/>
19199    </xs:sequence>
19200  </xs:complexType>
19201  <xs:element name="tariffDescriptionData" type="ns_p:TariffDescriptionDataType"/>
19202  <xs:complexType name="TariffDescriptionDataElementsType">
19203    <xs:sequence>
19204      <xs:element name="tariffId" type="ns_p:ElementTagType" minOccurs="0"/>
19205      <xs:element name="commodityId" type="ns_p:ElementTagType" minOccurs="0"/>
19206      <xs:element name="measurementId" type="ns_p:ElementTagType" minOccurs="0"/>
19207      <xs:element name="tariffWriteable" type="ns_p:ElementTagType" minOccurs="0"/>
19208      <xs:element name="updateRequired" type="ns_p:ElementTagType" minOccurs="0"/>
19209      <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
19210      <xs:element name="label" type="ns_p:ElementTagType" minOccurs="0"/>
19211      <xs:element name="description" type="ns_p:ElementTagType" minOccurs="0"/>
19212      <xs:element name="slotIdSupport" type="ns_p:ElementTagType" minOccurs="0"/>
19213    </xs:sequence>
19214  </xs:complexType>
19215  <xs:element name="tariffDescriptionDataElements"
19216    type="ns_p:TariffDescriptionDataElementsType"/>
19217  <xs:complexType name="TariffDescriptionListDataType">
19218    <xs:sequence>
19219      <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:tariffDescriptionData"/>
19220    </xs:sequence>
19221  </xs:complexType>
19222  <xs:element name="tariffDescriptionListData" type="ns_p:TariffDescriptionListDataType"/>
19223  <xs:complexType name="TariffDescriptionListDataSelectorsType">
19224    <xs:sequence>
19225      <xs:element name="tariffId" type="ns_p:TariffIdType" minOccurs="0"/>
19226      <xs:element name="commodityId" type="ns_p:CommodityIdType" minOccurs="0"/>
19227      <xs:element name="measurementId" type="ns_p:MeasurementIdType" minOccurs="0"/>
19228      <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
19229    </xs:sequence>
19230  </xs:complexType>
19231  <xs:element name="tariffDescriptionListDataSelectors"
19232    type="ns_p:TariffDescriptionListDataSelectorsType"/>
19233  <xs:complexType name="TierBoundaryDataType">
19234    <xs:sequence>
19235      <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"/>
19236      <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
19237      <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
19238      <xs:element minOccurs="0" name="lowerBoundaryValue" type="ns_p:ScaledNumberType"/>
19239      <xs:element minOccurs="0" name="upperBoundaryValue" type="ns_p:ScaledNumberType"/>
19240    </xs:sequence>
19241  </xs:complexType>
19242  <xs:element name="tierBoundaryData" type="ns_p:TierBoundaryDataType"/>
19243  <xs:complexType name="TierBoundaryDataElementsType">
19244    <xs:sequence>
19245      <xs:element minOccurs="0" name="boundaryId" type="ns_p:ElementTagType"/>
19246      <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
19247      <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
19248      <xs:element minOccurs="0" name="lowerBoundaryValue"
19249        type="ns_p:ScaledNumberElementsType"/>
19250      <xs:element minOccurs="0" name="upperBoundaryValue"
19251        type="ns_p:ScaledNumberElementsType"/>
19252    </xs:sequence>
19253  </xs:complexType>
19254  <xs:element name="tierBoundaryDataElements" type="ns_p:TierBoundaryDataElementsType"/>
19255  <xs:complexType name="TierBoundaryListDataType">
19256    <xs:sequence>
19257      <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:tierBoundaryData"/>
19258    </xs:sequence>
19259  </xs:complexType>
19260  <xs:element name="tierBoundaryListData" type="ns_p:TierBoundaryListDataType"/>
19261  <xs:complexType name="TierBoundaryListDataSelectorsType">
19262    <xs:sequence>
19263      <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"/>
19264    </xs:sequence>
19265  </xs:complexType>
19266  <xs:element name="tierBoundaryListDataSelectors"
19267    type="ns_p:TierBoundaryListDataSelectorsType"/>
19268  <xs:complexType name="TierBoundaryDescriptionDataType">
```

```
19269     <xs:sequence>
19270       <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"/>
19271       <xs:element minOccurs="0" name="boundaryType" type="ns_p:TierBoundaryTypeType"/>
19272       <xs:element minOccurs="0" name="validForTierId" type="ns_p:TierIdType"/>
19273       <xs:element minOccurs="0" name="switchToTierIdWhenLower" type="ns_p:TierIdType"/>
19274       <xs:element minOccurs="0" name="switchToTierIdWhenHigher" type="ns_p:TierIdType"/>
19275       <xs:element minOccurs="0" name="boundaryUnit" type="ns_p:UnitOfMeasurementType"/>
19276       <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
19277       <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
19278     </xs:sequence>
19279   </xs:complexType>
19280   <xs:element name="tierBoundaryDescriptionData"
19281     type="ns_p:TierBoundaryDescriptionDataType"/>
19282   <xs:complexType name="TierBoundaryDescriptionDataElementsType">
19283     <xs:sequence>
19284       <xs:element minOccurs="0" name="boundaryId" type="ns_p:ElementTagType"/>
19285       <xs:element minOccurs="0" name="boundaryType" type="ns_p:ElementTagType"/>
19286       <xs:element minOccurs="0" name="validForTierId" type="ns_p:ElementTagType"/>
19287       <xs:element minOccurs="0" name="switchToTierIdWhenLower"
19288     type="ns_p:ElementTagType"/>
19289       <xs:element minOccurs="0" name="switchToTierIdWhenHigher"
19290     type="ns_p:ElementTagType"/>
19291       <xs:element minOccurs="0" name="boundaryUnit" type="ns_p:ElementTagType"/>
19292       <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
19293       <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
19294     </xs:sequence>
19295   </xs:complexType>
19296   <xs:element name="tierBoundaryDescriptionDataElements"
19297     type="ns_p:TierBoundaryDescriptionDataElementsType"/>
19298   <xs:complexType name="TierBoundaryDescriptionListDataType">
19299     <xs:sequence>
19300       <xs:element maxOccurs="unbounded" minOccurs="0"
19301     ref="ns_p:tierBoundaryDescriptionData"/>
19302     </xs:sequence>
19303   </xs:complexType>
19304   <xs:element name="tierBoundaryDescriptionListData"
19305     type="ns_p:TierBoundaryDescriptionListDataType"/>
19306   <xs:complexType name="TierBoundaryDescriptionListDataSelectorsType">
19307     <xs:sequence>
19308       <xs:element minOccurs="0" name="boundaryId" type="ns_p:TierBoundaryIdType"/>
19309       <xs:element minOccurs="0" name="boundaryType" type="ns_p:TierBoundaryTypeType"/>
19310     </xs:sequence>
19311   </xs:complexType>
19312   <xs:element name="tierBoundaryDescriptionListDataSelectors"
19313     type="ns_p:TierBoundaryDescriptionListDataSelectorsType"/>
19314   <xs:complexType name="CommodityDataType">
19315     <xs:sequence>
19316       <xs:element name="commodityId" type="ns_p:CommodityIdType" minOccurs="0"/>
19317       <xs:element name="commodityType" type="ns_p:CommodityTypeType" minOccurs="0"/>
19318       <xs:element name="positiveEnergyDirection" type="ns_p:EnergyDirectionType"
19319     minOccurs="0"/>
19320       <xs:element name="label" type="ns_p:LabelType" minOccurs="0"/>
19321       <xs:element name="description" type="ns_p:DescriptionType" minOccurs="0"/>
19322     </xs:sequence>
19323   </xs:complexType>
19324   <xs:element name="commodityData" type="ns_p:CommodityDataType"/>
19325   <xs:complexType name="CommodityDataElementsType">
19326     <xs:sequence>
19327       <xs:element name="commodityId" type="ns_p:ElementTagType" minOccurs="0"/>
19328       <xs:element name="commodityType" type="ns_p:ElementTagType" minOccurs="0"/>
19329       <xs:element name="positiveEnergyDirection" type="ns_p:ElementTagType"
19330     minOccurs="0"/>
19331       <xs:element name="label" type="ns_p:ElementTagType" minOccurs="0"/>
19332       <xs:element name="description" type="ns_p:ElementTagType" minOccurs="0"/>
19333     </xs:sequence>
19334   </xs:complexType>
19335   <xs:element name="commodityDataElements" type="ns_p:CommodityDataElementsType"/>
19336   <xs:complexType name="CommodityListDataType">
19337     <xs:sequence>
19338       <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:commodityData"/>
19339     </xs:sequence>
19340   </xs:complexType>
19341   <xs:element name="commodityListData" type="ns_p:CommodityListDataType"/>
19342   <xs:complexType name="CommodityListDataSelectorsType">
19343     <xs:sequence>
19344       <xs:element name="commodityId" type="ns_p:CommodityIdType" minOccurs="0"/>
19345       <xs:element name="commodityType" type="ns_p:CommodityTypeType" minOccurs="0"/>
19346     </xs:sequence>
```



```
19347 </xs:complexType>
19348 <xs:element name="commodityListDataSelectors" type="ns_p:CommodityListDataSelectorsType"/>
19349 <xs:complexType name="TierDataType">
19350 <xs:sequence>
19351 <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19352 <xs:element name="timePeriod" type="ns_p:TimePeriodType" minOccurs="0"/>
19353 <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
19354 <xs:element name="activeIncentiveId" type="ns_p:IncentiveIdType" minOccurs="0"
19355 maxOccurs="unbounded"/>
19356 </xs:sequence>
19357 </xs:complexType>
19358 <xs:element name="tierData" type="ns_p:TierDataType"/>
19359 <xs:complexType name="TierDataElementsType">
19360 <xs:sequence>
19361 <xs:element name="tierId" type="ns_p:ElementTagType" minOccurs="0"/>
19362 <xs:element name="timePeriod" type="ns_p:TimePeriodElementsType" minOccurs="0"/>
19363 <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
19364 <xs:element name="activeIncentiveId" type="ns_p:ElementTagType" minOccurs="0"/>
19365 </xs:sequence>
19366 </xs:complexType>
19367 <xs:element name="tierDataElements" type="ns_p:TierDataElementsType"/>
19368 <xs:complexType name="TierListDataType">
19369 <xs:sequence>
19370 <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:tierData"/>
19371 </xs:sequence>
19372 </xs:complexType>
19373 <xs:element name="tierListData" type="ns_p:TierListDataType"/>
19374 <xs:complexType name="TierListDataSelectorsType">
19375 <xs:sequence>
19376 <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19377 <xs:element name="activeIncentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19378 </xs:sequence>
19379 </xs:complexType>
19380 <xs:element name="tierListDataSelectors" type="ns_p:TierListDataSelectorsType"/>
19381 <xs:complexType name="TierIncentiveRelationDataType">
19382 <xs:sequence>
19383 <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19384 <xs:element name="incentiveId" maxOccurs="unbounded" type="ns_p:IncentiveIdType"
19385 minOccurs="0"/>
19386 </xs:sequence>
19387 </xs:complexType>
19388 <xs:element name="tierIncentiveRelationData" type="ns_p:TierIncentiveRelationDataType"/>
19389 <xs:complexType name="TierIncentiveRelationDataElementsType">
19390 <xs:sequence>
19391 <xs:element name="tierId" type="ns_p:ElementTagType" minOccurs="0"/>
19392 <xs:element name="incentiveId" type="ns_p:ElementTagType" minOccurs="0"/>
19393 </xs:sequence>
19394 </xs:complexType>
19395 <xs:element name="tierIncentiveRelationDataElements"
19396 type="ns_p:TierIncentiveRelationDataElementsType"/>
19397 <xs:complexType name="TierIncentiveRelationListDataType">
19398 <xs:sequence>
19399 <xs:element maxOccurs="unbounded" minOccurs="0"
19400 ref="ns_p:tierIncentiveRelationData"/>
19401 </xs:sequence>
19402 </xs:complexType>
19403 <xs:element name="tierIncentiveRelationListData"
19404 type="ns_p:TierIncentiveRelationListDataType"/>
19405 <xs:complexType name="TierIncentiveRelationListDataSelectorsType">
19406 <xs:sequence>
19407 <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19408 <xs:element name="incentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19409 </xs:sequence>
19410 </xs:complexType>
19411 <xs:element name="tierIncentiveRelationListDataSelectors"
19412 type="ns_p:TierIncentiveRelationListDataSelectorsType"/>
19413 <xs:complexType name="TierDescriptionDataType">
19414 <xs:sequence>
19415 <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19416 <xs:element name="tierType" minOccurs="0" type="ns_p:TierTypeType"/>
19417 <xs:element name="label" type="ns_p:LabelType" minOccurs="0"/>
19418 <xs:element name="description" type="ns_p:DescriptionType" minOccurs="0"/>
19419 </xs:sequence>
19420 </xs:complexType>
19421 <xs:element name="tierDescriptionData" type="ns_p:TierDescriptionDataType"/>
19422 <xs:complexType name="TierDescriptionDataElementsType">
19423 <xs:sequence>
19424 <xs:element name="tierId" type="ns_p:ElementTagType" minOccurs="0"/>
```

```
19425         <xs:element name="tierType" minOccurs="0" type="ns_p:ElementTagType"/>
19426         <xs:element name="label" type="ns_p:ElementTagType" minOccurs="0"/>
19427         <xs:element name="description" type="ns_p:ElementTagType" minOccurs="0"/>
19428     </xs:sequence>
19429 </xs:complexType>
19430 <xs:element name="tierDescriptionDataElements"
19431 type="ns_p:TierDescriptionDataElementsType"/>
19432 <xs:complexType name="TierDescriptionListDataType">
19433     <xs:sequence>
19434         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:tierDescriptionData"/>
19435     </xs:sequence>
19436 </xs:complexType>
19437 <xs:element name="tierDescriptionListData" type="ns_p:TierDescriptionListDataType"/>
19438 <xs:complexType name="TierDescriptionListDataSelectorsType">
19439     <xs:sequence>
19440         <xs:element name="tierId" type="ns_p:TierIdType" minOccurs="0"/>
19441         <xs:element name="tierType" minOccurs="0" type="ns_p:TierTypeType"/>
19442     </xs:sequence>
19443 </xs:complexType>
19444 <xs:element name="tierDescriptionListDataSelectors"
19445 type="ns_p:TierDescriptionListDataSelectorsType"/>
19446 <xs:complexType name="IncentiveDataType">
19447     <xs:sequence>
19448         <xs:element name="incentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19449         <xs:element name="valueType" type="ns_p:IncentiveValueTypeType" minOccurs="0"/>
19450         <xs:element name="timestamp" type="ns_p:AbsoluteOrRelativeTimeType"
19451 minOccurs="0"/>
19452         <xs:element name="timePeriod" type="ns_p:TimePeriodType" minOccurs="0"/>
19453         <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
19454         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
19455     </xs:sequence>
19456 </xs:complexType>
19457 <xs:element name="incentiveData" type="ns_p:IncentiveDataType"/>
19458 <xs:complexType name="IncentiveDataElementsType">
19459     <xs:sequence>
19460         <xs:element name="incentiveId" type="ns_p:ElementTagType" minOccurs="0"/>
19461         <xs:element name="valueType" type="ns_p:ElementTagType" minOccurs="0"/>
19462         <xs:element name="timestamp" type="ns_p:ElementTagType" minOccurs="0"/>
19463         <xs:element name="timePeriod" type="ns_p:TimePeriodElementsType" minOccurs="0"/>
19464         <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
19465         <xs:element minOccurs="0" name="value" type="ns_p:ElementTagType"/>
19466     </xs:sequence>
19467 </xs:complexType>
19468 <xs:element name="incentiveDataElements" type="ns_p:IncentiveDataElementsType"/>
19469 <xs:complexType name="IncentiveListDataType">
19470     <xs:sequence>
19471         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:incentiveData"/>
19472     </xs:sequence>
19473 </xs:complexType>
19474 <xs:element name="incentiveListData" type="ns_p:IncentiveListDataType"/>
19475 <xs:complexType name="IncentiveListDataSelectorsType">
19476     <xs:sequence>
19477         <xs:element name="incentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19478         <xs:element name="valueType" type="ns_p:IncentiveValueTypeType" minOccurs="0"/>
19479         <xs:element minOccurs="0" name="timestampInterval"
19480 type="ns_p:TimestampIntervalType"/>
19481     </xs:sequence>
19482 </xs:complexType>
19483 <xs:element name="incentiveListDataSelectors" type="ns_p:IncentiveListDataSelectorsType"/>
19484 <xs:complexType name="IncentiveDescriptionDataType">
19485     <xs:sequence>
19486         <xs:element name="incentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19487         <xs:element name="incentiveType" type="ns_p:IncentiveTypeType" minOccurs="0"/>
19488         <xs:element name="incentivePriority" type="ns_p:IncentivePriorityType"
19489 minOccurs="0"/>
19490         <xs:element name="currency" type="ns_p:CurrencyType" minOccurs="0"/>
19491         <xs:element name="unit" minOccurs="0" type="ns_p:UnitOfMeasurementType"/>
19492         <xs:element name="label" type="ns_p:LabelType" minOccurs="0"/>
19493         <xs:element name="description" type="ns_p:DescriptionType" minOccurs="0"/>
19494     </xs:sequence>
19495 </xs:complexType>
19496 <xs:element name="incentiveDescriptionData" type="ns_p:IncentiveDescriptionDataType"/>
19497 <xs:complexType name="IncentiveDescriptionDataElementsType">
19498     <xs:sequence>
19499         <xs:element name="incentiveId" type="ns_p:ElementTagType" minOccurs="0"/>
19500         <xs:element name="incentiveType" type="ns_p:ElementTagType" minOccurs="0"/>
19501         <xs:element name="incentivePriority" type="ns_p:ElementTagType" minOccurs="0"/>
19502         <xs:element name="currency" type="ns_p:ElementTagType" minOccurs="0"/>
```

```

19503         <xs:element name="unit" minOccurs="0" type="ns_p:ElementTagType"/>
19504         <xs:element name="label" type="ns_p:ElementTagType" minOccurs="0"/>
19505         <xs:element name="description" type="ns_p:ElementTagType" minOccurs="0"/>
19506     </xs:sequence>
19507 </xs:complexType>
19508     <xs:element name="incentiveDescriptionDataElements"
19509 type="ns_p:IncentiveDescriptionDataElementsType"/>
19510     <xs:complexType name="IncentiveDescriptionListDataType">
19511         <xs:sequence>
19512             <xs:element maxOccurs="unbounded" minOccurs="0"
19513 ref="ns_p:incentiveDescriptionData"/>
19514         </xs:sequence>
19515     </xs:complexType>
19516     <xs:element name="incentiveDescriptionListData"
19517 type="ns_p:IncentiveDescriptionListDataType"/>
19518     <xs:complexType name="IncentiveDescriptionListDataSelectorsType">
19519         <xs:sequence>
19520             <xs:element name="incentiveId" type="ns_p:IncentiveIdType" minOccurs="0"/>
19521             <xs:element name="incentiveType" type="ns_p:IncentiveTypeType" minOccurs="0"/>
19522         </xs:sequence>
19523     </xs:complexType>
19524     <xs:element name="incentiveDescriptionListDataSelectors"
19525 type="ns_p:IncentiveDescriptionListDataSelectorsType"/>
19526 </xs:schema>
19527
19528

```

19529

19530 A.2.26 TaskManagement

19531 File name: EEBus_SPINE_TS_TaskManagement.xsd

```

19532
19533 <?xml version="1.0" encoding="UTF-8"?>
19534 <!--
19535     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
19536     Version 1.3.0 (final)
19537     2023-08-31
19538     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
19539     Source: https://www.eebus.org/en/specifications/
19540 -->
19541 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
19542 xmlns:xs="http://www.w3.org/2001/XMLSchema"
19543 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
19544 blockDefault="#all" elementFormDefault="qualified">
19545     <xs:annotation>
19546         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
19547 e.V. All rights reserved.</xs:documentation>
19548     </xs:annotation>
19549     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
19550     <xs:include schemaLocation="EEBus_SPINE_TS_DirectControl.xsd"/>
19551     <xs:include schemaLocation="EEBus_SPINE_TS_HVAC.xsd"/>
19552     <xs:include schemaLocation="EEBus_SPINE_TS_LoadControl.xsd"/>
19553     <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
19554     <xs:simpleType name="TaskManagementJobIdType">
19555         <xs:restriction base="xs:unsignedInt"/>
19556     </xs:simpleType>
19557     <xs:simpleType name="TaskManagementJobStateType">
19558         <xs:union memberTypes="ns_p:DirectControlActivityStateEnumType
19559 ns_p:HvacOverrunStatusEnumType ns_p:LoadControlEventStateEnumType
19560 ns_p:PowerSequenceStateEnumType ns_p:EnumExtendType"/>
19561     </xs:simpleType>
19562     <xs:simpleType name="TaskManagementJobSourceType">
19563         <xs:union memberTypes="ns_p:TaskManagementJobSourceEnumType ns_p:EnumExtendType"/>
19564     </xs:simpleType>
19565     <xs:simpleType name="TaskManagementJobSourceEnumType">
19566         <xs:restriction base="xs:string">
19567             <xs:enumeration value="internalMechanism"/>
19568             <xs:enumeration value="userInteraction"/>
19569             <xs:enumeration value="externalConfiguration"/>
19570         </xs:restriction>
19571     </xs:simpleType>
19572     <xs:complexType name="TaskManagementDirectControlRelatedType"/>
19573     <xs:complexType name="TaskManagementDirectControlRelatedElementsType"/>
19574     <xs:complexType name="TaskManagementHvacRelatedType">
19575         <xs:sequence>

```

```
19576         <xs:element minOccurs="0" name="overrunId" type="ns_p:HvacOverrunIdType"/>
19577     </xs:sequence>
19578 </xs:complexType>
19579 <xs:complexType name="TaskManagementHvacRelatedElementsType">
19580     <xs:sequence>
19581         <xs:element minOccurs="0" name="overrunId" type="ns_p:ElementTagType"/>
19582     </xs:sequence>
19583 </xs:complexType>
19584 <xs:complexType name="TaskManagementLoadControlRelatedType">
19585     <xs:sequence>
19586         <xs:element minOccurs="0" name="eventId" type="ns_p:LoadControlEventIdType"/>
19587     </xs:sequence>
19588 </xs:complexType>
19589 <xs:complexType name="TaskManagementLoadControlRelatedElementsType">
19590     <xs:sequence>
19591         <xs:element minOccurs="0" name="eventId" type="ns_p:ElementTagType"/>
19592     </xs:sequence>
19593 </xs:complexType>
19594 <xs:complexType name="TaskManagementPowerSequencesRelatedType">
19595     <xs:sequence>
19596         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
19597     </xs:sequence>
19598 </xs:complexType>
19599 <xs:complexType name="TaskManagementPowerSequencesRelatedElementsType">
19600     <xs:sequence>
19601         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
19602     </xs:sequence>
19603 </xs:complexType>
19604 <xs:complexType name="TaskManagementSmartEnergyManagementPsRelatedType">
19605     <xs:sequence>
19606         <xs:element minOccurs="0" name="sequenceId" type="ns_p:PowerSequenceIdType"/>
19607     </xs:sequence>
19608 </xs:complexType>
19609 <xs:complexType name="TaskManagementSmartEnergyManagementPsRelatedElementsType">
19610     <xs:sequence>
19611         <xs:element minOccurs="0" name="sequenceId" type="ns_p:ElementTagType"/>
19612     </xs:sequence>
19613 </xs:complexType>
19614 <xs:complexType name="TaskManagementJobDataType">
19615     <xs:sequence>
19616         <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
19617         <xs:element name="timestamp" type="ns_p:AbsoluteOrRelativeTimeType"
19618 minOccurs="0"/>
19619         <xs:element minOccurs="0" name="jobState" type="ns_p:TaskManagementJobStateType"/>
19620         <xs:element name="elapsedTime" type="xs:duration" minOccurs="0"/>
19621         <xs:element name="remainingTime" type="xs:duration" minOccurs="0"/>
19622     </xs:sequence>
19623 </xs:complexType>
19624 <xs:element name="taskManagementJobData" type="ns_p:TaskManagementJobDataType"/>
19625 <xs:complexType name="TaskManagementJobDataElementsType">
19626     <xs:sequence>
19627         <xs:element minOccurs="0" name="jobId" type="ns_p:ElementTagType"/>
19628         <xs:element name="timestamp" minOccurs="0" type="ns_p:ElementTagType"/>
19629         <xs:element minOccurs="0" name="jobState" type="ns_p:ElementTagType"/>
19630         <xs:element name="elapsedTime" minOccurs="0" type="ns_p:ElementTagType"/>
19631         <xs:element name="remainingTime" minOccurs="0" type="ns_p:ElementTagType"/>
19632     </xs:sequence>
19633 </xs:complexType>
19634 <xs:element name="taskManagementJobDataElements"
19635 type="ns_p:TaskManagementJobDataElementsType"/>
19636 <xs:complexType name="TaskManagementJobListDataType">
19637     <xs:sequence>
19638         <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:taskManagementJobData"/>
19639     </xs:sequence>
19640 </xs:complexType>
19641 <xs:element name="taskManagementJobListData" type="ns_p:TaskManagementJobListDataType"/>
19642 <xs:complexType name="TaskManagementJobListDataSelectorsType">
19643     <xs:sequence>
19644         <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
19645         <xs:element minOccurs="0" name="jobState" type="ns_p:TaskManagementJobStateType"/>
19646     </xs:sequence>
19647 </xs:complexType>
19648 <xs:element name="taskManagementJobListDataSelectors"
19649 type="ns_p:TaskManagementJobListDataSelectorsType"/>
19650 <xs:complexType name="TaskManagementJobRelationDataType">
19651     <xs:sequence>
19652         <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
```

```
19653         <xs:element minOccurs="0" name="directControlRelated"
19654 type="ns_p:TaskManagementDirectControlRelatedType"/>
19655         <xs:element minOccurs="0" name="hvacRelated"
19656 type="ns_p:TaskManagementHvacRelatedType"/>
19657         <xs:element minOccurs="0" name="loadControlRelated"
19658 type="ns_p:TaskManagementLoadControlRelatedType"/>
19659         <xs:element minOccurs="0" name="powerSequencesRelated"
19660 type="ns_p:TaskManagementPowerSequencesRelatedType"/>
19661         <xs:element minOccurs="0" name="smartEnergyManagementPsRelated"
19662 type="ns_p:TaskManagementSmartEnergyManagementPsRelatedType"/>
19663     </xs:sequence>
19664 </xs:complexType>
19665     <xs:element name="taskManagementJobRelationData"
19666 type="ns_p:TaskManagementJobRelationDataType"/>
19667     <xs:complexType name="TaskManagementJobRelationDataElementsType">
19668         <xs:sequence>
19669             <xs:element minOccurs="0" name="jobId" type="ns_p:ElementTagType"/>
19670             <xs:element minOccurs="0" name="directControlRelated"
19671 type="ns_p:TaskManagementDirectControlRelatedElementsType"/>
19672             <xs:element minOccurs="0" name="hvacRelated"
19673 type="ns_p:TaskManagementHvacRelatedElementsType"/>
19674             <xs:element minOccurs="0" name="loadControlRelated"
19675 type="ns_p:TaskManagementLoadControlRelatedElementsType"/>
19676             <xs:element minOccurs="0" name="powerSequencesRelated"
19677 type="ns_p:TaskManagementPowerSequencesRelatedElementsType"/>
19678             <xs:element minOccurs="0" name="smartEnergyManagementPsRelated"
19679 type="ns_p:TaskManagementSmartEnergyManagementPsRelatedElementsType"/>
19680         </xs:sequence>
19681     </xs:complexType>
19682     <xs:element name="taskManagementJobRelationDataElements"
19683 type="ns_p:TaskManagementJobRelationDataElementsType"/>
19684     <xs:complexType name="TaskManagementJobRelationListDataType">
19685         <xs:sequence>
19686             <xs:element maxOccurs="unbounded" minOccurs="0"
19687 ref="ns_p:taskManagementJobRelationData"/>
19688         </xs:sequence>
19689     </xs:complexType>
19690     <xs:element name="taskManagementJobRelationListData"
19691 type="ns_p:TaskManagementJobRelationListDataType"/>
19692     <xs:complexType name="TaskManagementJobRelationListDataSelectorsType">
19693         <xs:sequence>
19694             <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
19695         </xs:sequence>
19696     </xs:complexType>
19697     <xs:element name="taskManagementJobRelationListDataSelectors"
19698 type="ns_p:TaskManagementJobRelationListDataSelectorsType"/>
19699     <xs:complexType name="TaskManagementJobDescriptionDataType">
19700         <xs:sequence>
19701             <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
19702             <xs:element minOccurs="0" name="jobSource"
19703 type="ns_p:TaskManagementJobSourceType"/>
19704             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
19705             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
19706         </xs:sequence>
19707     </xs:complexType>
19708     <xs:element name="taskManagementJobDescriptionData"
19709 type="ns_p:TaskManagementJobDescriptionDataType"/>
19710     <xs:complexType name="TaskManagementJobDescriptionDataElementsType">
19711         <xs:sequence>
19712             <xs:element minOccurs="0" name="jobId" type="ns_p:ElementTagType"/>
19713             <xs:element minOccurs="0" name="jobSource" type="ns_p:ElementTagType"/>
19714             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
19715             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
19716         </xs:sequence>
19717     </xs:complexType>
19718     <xs:element name="taskManagementJobDescriptionDataElements"
19719 type="ns_p:TaskManagementJobDescriptionDataElementsType"/>
19720     <xs:complexType name="TaskManagementJobDescriptionListDataType">
19721         <xs:sequence>
19722             <xs:element maxOccurs="unbounded" minOccurs="0"
19723 ref="ns_p:taskManagementJobDescriptionData"/>
19724         </xs:sequence>
19725     </xs:complexType>
19726     <xs:element name="taskManagementJobDescriptionListData"
19727 type="ns_p:TaskManagementJobDescriptionListDataType"/>
19728     <xs:complexType name="TaskManagementJobDescriptionListDataSelectorsType">
19729         <xs:sequence>
19730             <xs:element minOccurs="0" name="jobId" type="ns_p:TaskManagementJobIdType"/>
```

```

19731         <xs:element minOccurs="0" name="jobSource"
19732 type="ns_p:TaskManagementJobSourceType"/>
19733     </xs:sequence>
19734 </xs:complexType>
19735     <xs:element name="taskManagementJobDescriptionListDataSelectors"
19736 type="ns_p:TaskManagementJobDescriptionListDataSelectorsType"/>
19737     <xs:complexType name="TaskManagementOverviewDataType">
19738         <xs:sequence>
19739             <xs:element minOccurs="0" name="remoteControllable" type="xs:boolean"/>
19740             <xs:element minOccurs="0" name="jobsActive" type="xs:boolean"/>
19741         </xs:sequence>
19742     </xs:complexType>
19743     <xs:element name="taskManagementOverviewData" type="ns_p:TaskManagementOverviewDataType"/>
19744     <xs:complexType name="TaskManagementOverviewDataElementsType">
19745         <xs:sequence>
19746             <xs:element minOccurs="0" name="remoteControllable" type="ns_p:ElementTagType"/>
19747             <xs:element minOccurs="0" name="jobsActive" type="ns_p:ElementTagType"/>
19748         </xs:sequence>
19749     </xs:complexType>
19750     <xs:element name="taskManagementOverviewDataElements"
19751 type="ns_p:TaskManagementOverviewDataElementsType"/>
19752 </xs:schema>
19753
19754

```

19755

19756 A.2.27 Threshold

19757 File name: EEBus_SPINE_TS_Threshold.xsd

```

19758 <?xml version="1.0" encoding="UTF-8"?>
19759 <!--
19760     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
19761     Version 1.3.0 (final)
19762     2023-08-31
19763     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
19764     Source: https://www.eebus.org/en/specifications/
19765 -->
19766 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
19767 xmlns:xs="http://www.w3.org/2001/XMLSchema"
19768 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
19769 blockDefault="#all" elementFormDefault="qualified">
19770     <xs:annotation>
19771         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
19772 e.V. All rights reserved.</xs:documentation>
19773     </xs:annotation>
19774     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
19775     <xs:simpleType name="ThresholdIdType">
19776         <xs:restriction base="xs:unsignedInt"/>
19777     </xs:simpleType>
19778     <xs:simpleType name="ThresholdTypeEnumType">
19779         <xs:union memberTypes="ns_p:ThresholdTypeEnumType ns_p:EnumExtendType"/>
19780     </xs:simpleType>
19781     <xs:simpleType name="ThresholdTypeEnumType">
19782         <xs:restriction base="xs:string">
19783             <xs:enumeration value="goodAbove"/>
19784             <xs:enumeration value="badAbove"/>
19785             <xs:enumeration value="goodBelow"/>
19786             <xs:enumeration value="badBelow"/>
19787             <xs:enumeration value="minValueThreshold"/>
19788             <xs:enumeration value="maxValueThreshold"/>
19789             <xs:enumeration value="minValueThresholdExtreme"/>
19790             <xs:enumeration value="maxValueThresholdExtreme"/>
19791             <xs:enumeration value="sagThreshold"/>
19792             <xs:enumeration value="swellThreshold"/>
19793         </xs:restriction>
19794     </xs:simpleType>
19795     <xs:complexType name="ThresholdDataType">
19796         <xs:sequence>
19797             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ThresholdIdType"/>
19798             <xs:element minOccurs="0" name="thresholdValue" type="ns_p:ScaledNumberType"/>
19799         </xs:sequence>
19800     </xs:complexType>
19801     <xs:element name="thresholdData" type="ns_p:ThresholdDataType"/>
19802     <xs:complexType name="ThresholdDataElementsType">

```

```
19804         <xs:sequence>
19805             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ElementTagType"/>
19806             <xs:element minOccurs="0" name="thresholdValue"
19807 type="ns_p:ScaledNumberElementsType"/>
19808         </xs:sequence>
19809     </xs:complexType>
19810     <xs:element name="thresholdDataElements" type="ns_p:ThresholdDataElementsType"/>
19811     <xs:complexType name="ThresholdListDataType">
19812         <xs:sequence>
19813             <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:thresholdData"/>
19814         </xs:sequence>
19815     </xs:complexType>
19816     <xs:element name="thresholdListData" type="ns_p:ThresholdListDataType"/>
19817     <xs:complexType name="ThresholdListDataSelectorsType">
19818         <xs:sequence>
19819             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ThresholdIdType"/>
19820         </xs:sequence>
19821     </xs:complexType>
19822     <xs:element name="thresholdListDataSelectors" type="ns_p:ThresholdListDataSelectorsType"/>
19823     <xs:complexType name="ThresholdConstraintsDataType">
19824         <xs:sequence>
19825             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ThresholdIdType"/>
19826             <xs:element minOccurs="0" name="thresholdRangeMin" type="ns_p:ScaledNumberType"/>
19827             <xs:element minOccurs="0" name="thresholdRangeMax" type="ns_p:ScaledNumberType"/>
19828             <xs:element minOccurs="0" name="thresholdStepSize" type="ns_p:ScaledNumberType"/>
19829         </xs:sequence>
19830     </xs:complexType>
19831     <xs:element name="thresholdConstraintsData" type="ns_p:ThresholdConstraintsDataType"/>
19832     <xs:complexType name="ThresholdConstraintsDataElementsType">
19833         <xs:sequence>
19834             <xs:element minOccurs="0" name="thresholdId" type="ns_p:ElementTagType"/>
19835             <xs:element minOccurs="0" name="thresholdRangeMin"
19836 type="ns_p:ScaledNumberElementsType"/>
19837             <xs:element minOccurs="0" name="thresholdRangeMax"
19838 type="ns_p:ScaledNumberElementsType"/>
19839             <xs:element minOccurs="0" name="thresholdStepSize"
19840 type="ns_p:ScaledNumberElementsType"/>
19841         </xs:sequence>
19842     </xs:complexType>
19843     <xs:element name="thresholdConstraintsDataElements"
19844 type="ns_p:ThresholdConstraintsDataElementsType"/>
19845     <xs:complexType name="ThresholdConstraintsListDataType">
19846         <xs:sequence>
19847             <xs:element maxOccurs="unbounded" minOccurs="0"
19848 ref="ns_p:thresholdConstraintsData"/>
19849         </xs:sequence>
19850     </xs:complexType>
19851     <xs:element name="thresholdConstraintsListData"
19852 type="ns_p:ThresholdConstraintsListDataType"/>
19853     <xs:complexType name="ThresholdConstraintsListDataSelectorsType">
19854         <xs:sequence>
19855             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ThresholdIdType"/>
19856         </xs:sequence>
19857     </xs:complexType>
19858     <xs:element name="thresholdConstraintsListDataSelectors"
19859 type="ns_p:ThresholdConstraintsListDataSelectorsType"/>
19860     <xs:complexType name="ThresholdDescriptionDataType">
19861         <xs:sequence>
19862             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ThresholdIdType"/>
19863             <xs:element name="thresholdType" minOccurs="0" type="ns_p:ThresholdType"/>
19864             <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
19865             <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeType"/>
19866             <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
19867             <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
19868         </xs:sequence>
19869     </xs:complexType>
19870     <xs:element name="thresholdDescriptionData" type="ns_p:ThresholdDescriptionDataType"/>
19871     <xs:complexType name="ThresholdDescriptionDataElementsType">
19872         <xs:sequence>
19873             <xs:element name="thresholdId" minOccurs="0" type="ns_p:ElementTagType"/>
19874             <xs:element name="thresholdType" minOccurs="0" type="ns_p:ElementTagType"/>
19875             <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
19876             <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
19877             <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
19878             <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
19879         </xs:sequence>
19880     </xs:complexType>
```

```
19881     <xs:element name="thresholdDescriptionDataElements"
19882 type="ns_p:ThresholdDescriptionDataElementsType"/>
19883     <xs:complexType name="ThresholdDescriptionListDataType">
19884       <xs:sequence>
19885         <xs:element maxOccurs="unbounded" minOccurs="0"
19886 ref="ns_p:thresholdDescriptionData"/>
19887       </xs:sequence>
19888     </xs:complexType>
19889     <xs:element name="thresholdDescriptionListData"
19890 type="ns_p:ThresholdDescriptionListDataType"/>
19891     <xs:complexType name="ThresholdDescriptionListDataSelectorsType">
19892       <xs:sequence>
19893         <xs:element name="thresholdId" minOccurs="0" type="ns_p:ThresholdIdType"/>
19894         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
19895       </xs:sequence>
19896     </xs:complexType>
19897     <xs:element name="thresholdDescriptionListDataSelectors"
19898 type="ns_p:ThresholdDescriptionListDataSelectorsType"/>
19899 </xs:schema>
19900
19901
```

19902

19903 **A.2.28 TimeInformation**

19904 File name: EEBus_SPINE_TS_TimeInformation.xsd

```
19905
19906 <?xml version="1.0" encoding="UTF-8"?>
19907 <!--
19908   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
19909   Version 1.3.0 (final)
19910   2023-08-31
19911   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
19912   Source: https://www.eebus.org/en/specifications/
19913 -->
19914 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
19915 xmlns:xs="http://www.w3.org/2001/XMLSchema"
19916 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0" blockDefault="#all"
19917 elementFormDefault="qualified">
19918   <xs:annotation>
19919     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
19920 e.V. All rights reserved.</xs:documentation>
19921   </xs:annotation>
19922   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
19923   <xs:complexType name="TimeInformationDataType">
19924     <xs:sequence>
19925       <xs:element minOccurs="0" name="utc" type="xs:dateTime"/>
19926       <xs:element minOccurs="0" name="utcOffset" type="xs:duration"/>
19927       <xs:element minOccurs="0" name="dayOfWeek" type="ns_p:DayOfWeekType"/>
19928       <xs:element minOccurs="0" name="calendarWeek" type="ns_p:CalendarWeekType"/>
19929     </xs:sequence>
19930   </xs:complexType>
19931   <xs:element name="timeInformationData" type="ns_p:TimeInformationDataType"/>
19932   <xs:complexType name="TimeInformationDataElementsType">
19933     <xs:sequence>
19934       <xs:element minOccurs="0" name="utc" type="ns_p:ElementTagType"/>
19935       <xs:element minOccurs="0" name="utcOffset" type="ns_p:ElementTagType"/>
19936       <xs:element minOccurs="0" name="dayOfWeek" type="ns_p:ElementTagType"/>
19937       <xs:element minOccurs="0" name="calendarWeek" type="ns_p:ElementTagType"/>
19938     </xs:sequence>
19939   </xs:complexType>
19940   <xs:element name="timeInformationDataElements"
19941 type="ns_p:TimeInformationDataElementsType"/>
19942   <xs:complexType name="TimeDistributorDataType">
19943     <xs:sequence>
19944       <xs:element minOccurs="0" name="isTimeDistributor" type="xs:boolean"/>
19945       <xs:element minOccurs="0" name="distributorPriority" type="xs:unsignedInt"/>
19946     </xs:sequence>
19947   </xs:complexType>
19948   <xs:element name="timeDistributorData" type="ns_p:TimeDistributorDataType"/>
19949   <xs:complexType name="TimeDistributorDataElementsType">
19950     <xs:sequence>
19951       <xs:element minOccurs="0" name="isTimeDistributor" type="ns_p:ElementTagType"/>
19952       <xs:element minOccurs="0" name="distributorPriority" type="ns_p:ElementTagType"/>
19953     </xs:sequence>

```



```

19954     </xs:complexType>
19955     <xs:element name="timeDistributorDataElements"
19956 type="ns_p:TimeDistributorDataElementsType"/>
19957     <xs:complexType name="TimePrecisionDataType">
19958       <xs:sequence>
19959         <xs:element minOccurs="0" name="isSynchronised" type="xs:boolean"/>
19960         <xs:element minOccurs="0" name="lastSyncAt" type="xs:dateTime"/>
19961         <xs:element minOccurs="0" name="clockDrift" type="xs:integer"/>
19962       </xs:sequence>
19963     </xs:complexType>
19964     <xs:element name="timePrecisionData" type="ns_p:TimePrecisionDataType"/>
19965     <xs:complexType name="TimePrecisionDataElementsType">
19966       <xs:sequence>
19967         <xs:element minOccurs="0" name="isSynchronised" type="ns_p:ElementTagType"/>
19968         <xs:element minOccurs="0" name="lastSyncAt" type="ns_p:ElementTagType"/>
19969         <xs:element minOccurs="0" name="clockDrift" type="ns_p:ElementTagType"/>
19970       </xs:sequence>
19971     </xs:complexType>
19972     <xs:element name="timePrecisionDataElements" type="ns_p:TimePrecisionDataElementsType"/>
19973     <xs:complexType name="TimeDistributorEnquiryCallType"/>
19974     <xs:element name="timeDistributorEnquiryCall" type="ns_p:TimeDistributorEnquiryCallType"/>
19975     <xs:complexType name="TimeDistributorEnquiryCallElementsType"/>
19976     <xs:element name="timeDistributorEnquiryCallElements"
19977 type="ns_p:TimeDistributorEnquiryCallElementsType"/>
19978   </xs:schema>
19979
19980

```

19981

19982 A.2.29 TimeSeries

19983 File name: EEBus_SPINE_TS_TimeSeries.xsd

```

19984
19985 <?xml version="1.0" encoding="UTF-8"?>
19986 <!--
19987   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
19988   Version 1.3.0 (final)
19989   2023-08-31
19990   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
19991   Source: https://www.eebus.org/en/specifications/
19992 -->
19993 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
19994 xmlns:xs="http://www.w3.org/2001/XMLSchema"
19995 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
19996 blockDefault="#all" elementFormDefault="qualified">
19997   <xs:annotation>
19998     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
19999 e.V. All rights reserved.</xs:documentation>
20000   </xs:annotation>
20001   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
20002   <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
20003   <xs:simpleType name="TimeSeriesIdType">
20004     <xs:restriction base="xs:unsignedInt"/>
20005   </xs:simpleType>
20006   <xs:simpleType name="TimeSeriesSlotIdType">
20007     <xs:restriction base="xs:unsignedInt"/>
20008   </xs:simpleType>
20009   <xs:simpleType name="TimeSeriesSlotCountType">
20010     <xs:restriction base="ns_p:TimeSeriesSlotIdType"/>
20011   </xs:simpleType>
20012   <xs:simpleType name="TimeSeriesTypeType">
20013     <xs:union memberTypes="ns_p:TimeSeriesTypeEnumType ns_p:EnumExtendType"/>
20014   </xs:simpleType>
20015   <xs:simpleType name="TimeSeriesTypeEnumType">
20016     <xs:restriction base="xs:string">
20017       <xs:enumeration value="plan"/>
20018       <xs:enumeration value="singleDemand"/>
20019       <xs:enumeration value="constraints"/>
20020       <xs:enumeration value="energyRequest"/>
20021       <xs:enumeration value="dischargingEnergyRequest"/>
20022       <xs:enumeration value="consumptionLimitCurve"/>
20023       <xs:enumeration value="productionLimitCurve"/>
20024     </xs:restriction>
20025   </xs:simpleType>
20026   <xs:complexType name="TimeSeriesSlotType">

```

```
20027     <xs:sequence>
20028         <xs:element minOccurs="0" name="timeSeriesSlotId"
20029 type="ns_p:TimeSeriesSlotIdType"/>
20030         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
20031         <xs:element minOccurs="0" name="duration" type="xs:duration"/>
20032         <xs:element minOccurs="0" name="recurrenceInformation"
20033 type="ns_p:AbsoluteOrRecurringTimeType"/>
20034         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberType"/>
20035         <xs:element minOccurs="0" name="minValue" type="ns_p:ScaledNumberType"/>
20036         <xs:element minOccurs="0" name="maxValue" type="ns_p:ScaledNumberType"/>
20037     </xs:sequence>
20038 </xs:complexType>
20039 <xs:complexType name="TimeSeriesSlotElementsType">
20040     <xs:sequence>
20041         <xs:element minOccurs="0" name="timeSeriesSlotId" type="ns_p:ElementTagType"/>
20042         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
20043         <xs:element minOccurs="0" name="duration" type="ns_p:ElementTagType"/>
20044         <xs:element minOccurs="0" name="recurrenceInformation"
20045 type="ns_p:AbsoluteOrRecurringTimeElementsType"/>
20046         <xs:element minOccurs="0" name="value" type="ns_p:ElementTagType"/>
20047         <xs:element minOccurs="0" name="minValue" type="ns_p:ElementTagType"/>
20048         <xs:element minOccurs="0" name="maxValue" type="ns_p:ElementTagType"/>
20049     </xs:sequence>
20050 </xs:complexType>
20051 <xs:complexType name="TimeSeriesDataType">
20052     <xs:sequence>
20053         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20054         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodType"/>
20055         <xs:element minOccurs="0" name="timeSeriesSlot"
20056 type="ns_p:TimeSeriesSlotType"/>
20057     </xs:sequence>
20058 </xs:complexType>
20059 <xs:element name="timeSeriesData" type="ns_p:TimeSeriesDataType"/>
20060 <xs:complexType name="TimeSeriesDataElementsType">
20061     <xs:sequence>
20062         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:ElementTagType"/>
20063         <xs:element minOccurs="0" name="timePeriod" type="ns_p:TimePeriodElementsType"/>
20064         <xs:element minOccurs="0" name="timeSeriesSlot"
20065 type="ns_p:TimeSeriesSlotElementsType"/>
20066     </xs:sequence>
20067 </xs:complexType>
20068 <xs:element name="timeSeriesDataElements" type="ns_p:TimeSeriesDataElementsType"/>
20069 <xs:complexType name="TimeSeriesListDataType">
20070     <xs:sequence>
20071         <xs:element minOccurs="0" name="timeSeriesData" ref="ns_p:timeSeriesData"/>
20072     </xs:sequence>
20073 </xs:complexType>
20074 <xs:element name="timeSeriesListData" type="ns_p:TimeSeriesListDataType"/>
20075 <xs:complexType name="TimeSeriesListDataSelectorsType">
20076     <xs:sequence>
20077         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20078         <xs:element minOccurs="0" name="timeSeriesSlotId"
20079 type="ns_p:TimeSeriesSlotIdType"/>
20080     </xs:sequence>
20081 </xs:complexType>
20082 <xs:element name="timeSeriesListDataSelectors"
20083 type="ns_p:TimeSeriesListDataSelectorsType"/>
20084 <xs:complexType name="TimeSeriesDescriptionDataType">
20085     <xs:sequence>
20086         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20087         <xs:element minOccurs="0" name="timeSeriesType" type="ns_p:TimeSeriesTypeType"/>
20088         <xs:element minOccurs="0" name="timeSeriesWriteable" type="xs:boolean"/>
20089         <xs:element minOccurs="0" name="updateRequired" type="xs:boolean"/>
20090         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
20091         <xs:element minOccurs="0" name="currency" type="ns_p:CurrencyType"/>
20092         <xs:element minOccurs="0" name="unit" type="ns_p:UnitOfMeasurementType"/>
20093         <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
20094         <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
20095         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
20096     </xs:sequence>
20097 </xs:complexType>
20098 <xs:element name="timeSeriesDescriptionData" type="ns_p:TimeSeriesDescriptionDataType"/>
20099 <xs:complexType name="TimeSeriesDescriptionDataElementsType">
20100     <xs:sequence>
20101         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:ElementTagType"/>
20102         <xs:element minOccurs="0" name="timeSeriesType" type="ns_p:ElementTagType"/>
20103         <xs:element minOccurs="0" name="timeSeriesWriteable" type="ns_p:ElementTagType"/>
20104         <xs:element minOccurs="0" name="updateRequired" type="ns_p:ElementTagType"/>
```

```
20105         <xs:element minOccurs="0" name="measurementId" type="ns_p:ElementTagType"/>
20106         <xs:element minOccurs="0" name="currency" type="ns_p:ElementTagType"/>
20107         <xs:element minOccurs="0" name="unit" type="ns_p:ElementTagType"/>
20108         <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
20109         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
20110         <xs:element minOccurs="0" name="scopeType" type="ns_p:ElementTagType"/>
20111     </xs:sequence>
20112 </xs:complexType>
20113 <xs:element name="timeSeriesDescriptionDataElements"
20114 type="ns_p:TimeSeriesDescriptionDataElementsType"/>
20115 <xs:complexType name="TimeSeriesDescriptionListDataType">
20116     <xs:sequence>
20117         <xs:element maxOccurs="unbounded" minOccurs="0"
20118 ref="ns_p:timeSeriesDescriptionData"/>
20119     </xs:sequence>
20120 </xs:complexType>
20121 <xs:element name="timeSeriesDescriptionListData"
20122 type="ns_p:TimeSeriesDescriptionListDataType"/>
20123 <xs:complexType name="TimeSeriesDescriptionListDataSelectorsType">
20124     <xs:sequence>
20125         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20126         <xs:element minOccurs="0" name="timeSeriesType" type="ns_p:TimeSeriesTypeType"/>
20127         <xs:element minOccurs="0" name="measurementId" type="ns_p:MeasurementIdType"/>
20128         <xs:element minOccurs="0" name="scopeType" type="ns_p:ScopeTypeType"/>
20129     </xs:sequence>
20130 </xs:complexType>
20131 <xs:element name="timeSeriesDescriptionListDataSelectors"
20132 type="ns_p:TimeSeriesDescriptionListDataSelectorsType"/>
20133 <xs:complexType name="TimeSeriesConstraintsDataType">
20134     <xs:sequence>
20135         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20136         <xs:element minOccurs="0" name="slotCountMin"
20137 type="ns_p:TimeSeriesSlotCountType"/>
20138         <xs:element minOccurs="0" name="slotCountMax"
20139 type="ns_p:TimeSeriesSlotCountType"/>
20140         <xs:element minOccurs="0" name="slotDurationMin" type="xs:duration"/>
20141         <xs:element minOccurs="0" name="slotDurationMax" type="xs:duration"/>
20142         <xs:element minOccurs="0" name="slotDurationStepSize" type="xs:duration"/>
20143         <xs:element minOccurs="0" name="earliestTimeSeriesStartTime"
20144 type="ns_p:AbsoluteOrRelativeTimeType"/>
20145         <xs:element minOccurs="0" name="latestTimeSeriesEndTime"
20146 type="ns_p:AbsoluteOrRelativeTimeType"/>
20147         <xs:element minOccurs="0" name="slotValueMin" type="ns_p:ScaledNumberType"/>
20148         <xs:element minOccurs="0" name="slotValueMax" type="ns_p:ScaledNumberType"/>
20149         <xs:element minOccurs="0" name="slotValueStepSize" type="ns_p:ScaledNumberType"/>
20150     </xs:sequence>
20151 </xs:complexType>
20152 <xs:element name="timeSeriesConstraintsData" type="ns_p:TimeSeriesConstraintsDataType"/>
20153 <xs:complexType name="TimeSeriesConstraintsDataElementsType">
20154     <xs:sequence>
20155         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:ElementTagType"/>
20156         <xs:element minOccurs="0" name="slotCountMin" type="ns_p:ElementTagType"/>
20157         <xs:element minOccurs="0" name="slotCountMax" type="ns_p:ElementTagType"/>
20158         <xs:element minOccurs="0" name="slotDurationMin" type="ns_p:ElementTagType"/>
20159         <xs:element minOccurs="0" name="slotDurationMax" type="ns_p:ElementTagType"/>
20160         <xs:element minOccurs="0" name="slotDurationStepSize" type="ns_p:ElementTagType"/>
20161         <xs:element minOccurs="0" name="earliestTimeSeriesStartTime"
20162 type="ns_p:ElementTagType"/>
20163         <xs:element minOccurs="0" name="latestTimeSeriesEndTime"
20164 type="ns_p:ElementTagType"/>
20165         <xs:element minOccurs="0" name="slotValueMin"
20166 type="ns_p:ScaledNumberElementsType"/>
20167         <xs:element minOccurs="0" name="slotValueMax"
20168 type="ns_p:ScaledNumberElementsType"/>
20169         <xs:element minOccurs="0" name="slotValueStepSize"
20170 type="ns_p:ScaledNumberElementsType"/>
20171     </xs:sequence>
20172 </xs:complexType>
20173 <xs:element name="timeSeriesConstraintsDataElements"
20174 type="ns_p:TimeSeriesConstraintsDataElementsType"/>
20175 <xs:complexType name="TimeSeriesConstraintsListDataType">
20176     <xs:sequence>
20177         <xs:element maxOccurs="unbounded" minOccurs="0"
20178 ref="ns_p:timeSeriesConstraintsData"/>
20179     </xs:sequence>
20180 </xs:complexType>
20181 <xs:element name="timeSeriesConstraintsListData"
20182 type="ns_p:TimeSeriesConstraintsListDataType"/>
```

```
20183     <xs:complexType name="TimeSeriesConstraintsListDataSelectorsType">
20184       <xs:sequence>
20185         <xs:element minOccurs="0" name="timeSeriesId" type="ns_p:TimeSeriesIdType"/>
20186       </xs:sequence>
20187     </xs:complexType>
20188     <xs:element name="timeSeriesConstraintsListDataSelectors"
20189 type="ns_p:TimeSeriesConstraintsListDataSelectorsType"/>
20190 </xs:schema>
20191
20192
```

20193

20194 **A.2.30 TimeTable**

20195 File name: EEBus_SPINE_TS_TimeTable.xsd

```
20196
20197 <?xml version="1.0" encoding="UTF-8"?>
20198 <!--
20199   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
20200   Version 1.3.0 (final)
20201   2023-08-31
20202   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
20203   Source: https://www.eebus.org/en/specifications/
20204 -->
20205 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
20206 xmlns:xs="http://www.w3.org/2001/XMLSchema"
20207 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
20208   blockDefault="#all" elementFormDefault="qualified">
20209   <xs:annotation>
20210     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
20211 e.V. All rights reserved.</xs:documentation>
20212   </xs:annotation>
20213   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
20214   <xs:simpleType name="TimeTableIdType">
20215     <xs:restriction base="xs:unsignedInt"/>
20216   </xs:simpleType>
20217   <xs:simpleType name="TimeSlotIdType">
20218     <xs:restriction base="xs:unsignedInt"/>
20219   </xs:simpleType>
20220   <xs:simpleType name="TimeSlotCountType">
20221     <xs:restriction base="ns_p:TimeSlotIdType"/>
20222   </xs:simpleType>
20223   <xs:simpleType name="TimeSlotTimeModeType">
20224     <xs:union memberTypes="ns_p:TimeSlotTimeModeEnumType ns_p:EnumExtendType"/>
20225   </xs:simpleType>
20226   <xs:simpleType name="TimeSlotTimeModeEnumType">
20227     <xs:restriction base="xs:string">
20228       <xs:enumeration value="absolute"/>
20229       <xs:enumeration value="recurring"/>
20230       <xs:enumeration value="both"/>
20231     </xs:restriction>
20232   </xs:simpleType>
20233   <xs:complexType name="TimeTableDataType">
20234     <xs:sequence>
20235       <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
20236       <xs:element minOccurs="0" name="timeSlotId" type="ns_p:TimeSlotIdType"/>
20237       <xs:element name="recurrenceInformation" minOccurs="0"
20238 type="ns_p:RecurrenceInformationType"/>
20239       <xs:element minOccurs="0" name="startTime"
20240 type="ns_p:AbsoluteOrRecurringTimeType"/>
20241       <xs:element minOccurs="0" name="endTime" type="ns_p:AbsoluteOrRecurringTimeType"/>
20242     </xs:sequence>
20243   </xs:complexType>
20244   <xs:element name="timeTableData" type="ns_p:TimeTableDataType"/>
20245   <xs:complexType name="TimeTableDataElementsType">
20246     <xs:sequence>
20247       <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
20248       <xs:element minOccurs="0" name="timeSlotId" type="ns_p:ElementTagType"/>
20249       <xs:element minOccurs="0" name="recurrenceInformation"
20250 type="ns_p:RecurrenceInformationElementsType"/>
20251       <xs:element minOccurs="0" name="startTime"
20252 type="ns_p:AbsoluteOrRecurringTimeElementsType"/>
20253       <xs:element minOccurs="0" name="endTime"
20254 type="ns_p:AbsoluteOrRecurringTimeElementsType"/>
20255     </xs:sequence>
```

```
20256 </xs:complexType>
20257 <xs:element name="timeTableDataElements" type="ns_p:TimeTableDataElementsType"/>
20258 <xs:complexType name="TimeTableListDataType">
20259   <xs:sequence>
20260     <xs:element maxOccurs="unbounded" minOccurs="0" ref="ns_p:timeTableData"/>
20261   </xs:sequence>
20262 </xs:complexType>
20263 <xs:element name="timeTableListData" type="ns_p:TimeTableListDataType"/>
20264 <xs:complexType name="TimeTableListDataSelectorsType">
20265   <xs:sequence>
20266     <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
20267     <xs:element minOccurs="0" name="timeSlotId" type="ns_p:TimeSlotIdType"/>
20268   </xs:sequence>
20269 </xs:complexType>
20270 <xs:element name="timeTableListDataSelectors" type="ns_p:TimeTableListDataSelectorsType"/>
20271 <xs:complexType name="TimeTableConstraintsDataType">
20272   <xs:sequence>
20273     <xs:element minOccurs="0" name="timeTableId" type="xs:unsignedInt"/>
20274     <xs:element minOccurs="0" name="slotCountMin" type="ns_p:TimeSlotCountType"/>
20275     <xs:element minOccurs="0" name="slotCountMax" type="ns_p:TimeSlotCountType"/>
20276     <xs:element minOccurs="0" name="slotDurationMin" type="xs:duration"/>
20277     <xs:element minOccurs="0" name="slotDurationMax" type="xs:duration"/>
20278     <xs:element minOccurs="0" name="slotDurationStepSize" type="xs:duration"/>
20279     <xs:element minOccurs="0" name="slotShiftStepSize" type="xs:duration"/>
20280     <xs:element minOccurs="0" name="firstSlotBeginsAt" type="xs:time"/>
20281   </xs:sequence>
20282 </xs:complexType>
20283 <xs:element name="timeTableConstraintsData" type="ns_p:TimeTableConstraintsDataType"/>
20284 <xs:complexType name="TimeTableConstraintsDataElementsType">
20285   <xs:sequence>
20286     <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
20287     <xs:element minOccurs="0" name="slotCountMin" type="ns_p:ElementTagType"/>
20288     <xs:element minOccurs="0" name="slotCountMax" type="ns_p:ElementTagType"/>
20289     <xs:element minOccurs="0" name="slotDurationMin" type="ns_p:ElementTagType"/>
20290     <xs:element minOccurs="0" name="slotDurationMax" type="ns_p:ElementTagType"/>
20291     <xs:element minOccurs="0" name="slotDurationStepSize" type="ns_p:ElementTagType"/>
20292     <xs:element minOccurs="0" name="slotShiftStepSize" type="ns_p:ElementTagType"/>
20293     <xs:element minOccurs="0" name="firstSlotBeginsAt" type="ns_p:ElementTagType"/>
20294   </xs:sequence>
20295 </xs:complexType>
20296 <xs:element name="timeTableConstraintsDataElements"
20297 type="ns_p:TimeTableConstraintsDataElementsType"/>
20298 <xs:complexType name="TimeTableConstraintsListDataType">
20299   <xs:sequence>
20300     <xs:element maxOccurs="unbounded" minOccurs="0"
20301 ref="ns_p:timeTableConstraintsData"/>
20302   </xs:sequence>
20303 </xs:complexType>
20304 <xs:element name="timeTableConstraintsListData"
20305 type="ns_p:TimeTableConstraintsListDataType"/>
20306 <xs:complexType name="TimeTableConstraintsListDataSelectorsType">
20307   <xs:sequence>
20308     <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
20309   </xs:sequence>
20310 </xs:complexType>
20311 <xs:element name="timeTableConstraintsListDataSelectors"
20312 type="ns_p:TimeTableConstraintsListDataSelectorsType"/>
20313 <xs:complexType name="TimeTableDescriptionDataType">
20314   <xs:sequence>
20315     <xs:element minOccurs="0" name="timeTableId" type="xs:unsignedInt"/>
20316     <xs:element minOccurs="0" name="timeSlotCountChangeable" type="xs:boolean"/>
20317     <xs:element minOccurs="0" name="timeSlotTimesChangeable" type="xs:boolean"/>
20318     <xs:element minOccurs="0" name="timeSlotTimeMode"
20319 type="ns_p:TimeSlotTimeModeType"/>
20320     <xs:element minOccurs="0" name="label" type="ns_p:LabelType"/>
20321     <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
20322   </xs:sequence>
20323 </xs:complexType>
20324 <xs:element name="timeTableDescriptionData" type="ns_p:TimeTableDescriptionDataType"/>
20325 <xs:complexType name="TimeTableDescriptionDataElementsType">
20326   <xs:sequence>
20327     <xs:element minOccurs="0" name="timeTableId" type="ns_p:ElementTagType"/>
20328     <xs:element minOccurs="0" name="timeSlotCountChangeable"
20329 type="ns_p:ElementTagType"/>
20330     <xs:element minOccurs="0" name="timeSlotTimesChangeable"
20331 type="ns_p:ElementTagType"/>
20332     <xs:element minOccurs="0" name="timeSlotTimeMode" type="ns_p:ElementTagType"/>
20333     <xs:element minOccurs="0" name="label" type="ns_p:ElementTagType"/>
```

```
20334         <xs:element minOccurs="0" name="description" type="ns_p:ElementTagType"/>
20335     </xs:sequence>
20336 </xs:complexType>
20337     <xs:element name="timeTableDescriptionDataElements"
20338 type="ns_p:TimeTableDescriptionDataElementsType"/>
20339     <xs:complexType name="TimeTableDescriptionListDataType">
20340         <xs:sequence>
20341             <xs:element maxOccurs="unbounded" minOccurs="0"
20342 ref="ns_p:timeTableDescriptionData"/>
20343         </xs:sequence>
20344     </xs:complexType>
20345     <xs:element name="timeTableDescriptionListData"
20346 type="ns_p:TimeTableDescriptionListDataType"/>
20347     <xs:complexType name="TimeTableDescriptionListDataSelectorsType">
20348         <xs:sequence>
20349             <xs:element minOccurs="0" name="timeTableId" type="ns_p:TimeTableIdType"/>
20350         </xs:sequence>
20351     </xs:complexType>
20352     <xs:element name="timeTableDescriptionListDataSelectors"
20353 type="ns_p:TimeTableDescriptionListDataSelectorsType"/>
20354 </xs:schema>
20355
20356
```

20357

20358 A.2.31 UseCaseInformation

20359 File name: EEBus_SPINE_TS_UseCaseInformation.xsd

```
20360
20361 <?xml version="1.0" encoding="UTF-8"?>
20362 <!--
20363     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
20364     Version 1.3.0 (final)
20365     2023-08-31
20366     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
20367     Source: https://www.eebus.org/en/specifications/
20368 -->
20369 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
20370 xmlns:xs="http://www.w3.org/2001/XMLSchema"
20371 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
20372 blockDefault="#all" elementFormDefault="qualified">
20373     <xs:annotation>
20374         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
20375 e.V. All rights reserved.</xs:documentation>
20376     </xs:annotation>
20377     <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
20378     <xs:include schemaLocation="EEBus_SPINE_TS_Version.xsd"/>
20379     <xs:simpleType name="UseCaseActorType">
20380         <xs:union memberTypes="ns_p:UseCaseActorEnumType ns_p:EnumExtendType"/>
20381     </xs:simpleType>
20382     <xs:simpleType name="UseCaseActorEnumType">
20383         <xs:restriction base="xs:string"/>
20384     </xs:simpleType>
20385     <xs:simpleType name="UseCaseNameType">
20386         <xs:union memberTypes="ns_p:UseCaseNameEnumType ns_p:EnumExtendType"/>
20387     </xs:simpleType>
20388     <xs:simpleType name="UseCaseNameEnumType">
20389         <xs:restriction base="xs:string"/>
20390     </xs:simpleType>
20391     <xs:simpleType name="UseCaseScenarioSupportType">
20392         <xs:restriction base="xs:unsignedInt"/>
20393     </xs:simpleType>
20394     <xs:complexType name="UseCaseSupportType">
20395         <xs:sequence>
20396             <xs:element minOccurs="0" name="useCaseName" type="ns_p:UseCaseNameType"/>
20397             <xs:element minOccurs="0" name="useCaseVersion"
20398 type="ns_p:SpecificationVersionType"/>
20399             <xs:element minOccurs="0" name="useCaseAvailable" type="xs:boolean"/>
20400             <xs:element maxOccurs="unbounded" minOccurs="0" name="scenarioSupport"
20401 type="ns_p:UseCaseScenarioSupportType"/>
20402             <xs:element minOccurs="0" name="useCaseDocumentSubRevision" type="xs:string"/>
20403         </xs:sequence>
20404     </xs:complexType>
20405     <xs:complexType name="UseCaseSupportElementsType">
20406         <xs:sequence>
```

```

20407         <xs:element minOccurs="0" name="useCaseName" type="ns_p:ElementTagType"/>
20408         <xs:element minOccurs="0" name="useCaseVersion" type="ns_p:ElementTagType"/>
20409         <xs:element minOccurs="0" name="useCaseAvailable" type="ns_p:ElementTagType"/>
20410         <xs:element minOccurs="0" name="scenarioSupport" type="ns_p:ElementTagType"/>
20411         <xs:element minOccurs="0" name="useCaseDocumentSubRevision"
20412 type="ns_p:ElementTagType"/>
20413     </xs:sequence>
20414 </xs:complexType>
20415 <xs:complexType name="UseCaseSupportSelectorsType">
20416     <xs:sequence>
20417         <xs:element minOccurs="0" name="useCaseName" type="ns_p:UseCaseNameType"/>
20418         <xs:element minOccurs="0" name="useCaseVersion"
20419 type="ns_p:SpecificationVersionType"/>
20420         <xs:element minOccurs="0" name="scenarioSupport"
20421 type="ns_p:UseCaseScenarioSupportType"/>
20422     </xs:sequence>
20423 </xs:complexType>
20424 <xs:complexType name="UseCaseInformationDataType">
20425     <xs:sequence>
20426         <xs:element minOccurs="0" name="address" type="ns_p:FeatureAddressType"/>
20427         <xs:element minOccurs="0" name="actor" type="ns_p:UseCaseActorType"/>
20428         <xs:element maxOccurs="unbounded" minOccurs="0" name="useCaseSupport"
20429 type="ns_p:UseCaseSupportType"/>
20430     </xs:sequence>
20431 </xs:complexType>
20432 <xs:element name="useCaseInformationData" type="ns_p:UseCaseInformationDataType"/>
20433 <xs:complexType name="UseCaseInformationDataElementsType">
20434     <xs:sequence>
20435         <xs:element minOccurs="0" name="address" type="ns_p:FeatureAddressElementsType"/>
20436         <xs:element minOccurs="0" name="actor" type="ns_p:ElementTagType"/>
20437         <xs:element minOccurs="0" name="useCaseSupport"
20438 type="ns_p:UseCaseSupportElementsType"/>
20439     </xs:sequence>
20440 </xs:complexType>
20441 <xs:element name="useCaseInformationDataElements"
20442 type="ns_p:UseCaseInformationDataElementsType"/>
20443 <xs:complexType name="UseCaseInformationListDataType">
20444     <xs:sequence>
20445         <xs:element maxOccurs="unbounded" minOccurs="0"
20446 ref="ns_p:useCaseInformationData"/>
20447     </xs:sequence>
20448 </xs:complexType>
20449 <xs:element name="useCaseInformationListData" type="ns_p:UseCaseInformationListDataType"/>
20450 <xs:complexType name="UseCaseInformationListDataSelectorsType">
20451     <xs:sequence>
20452         <xs:element minOccurs="0" name="address" type="ns_p:FeatureAddressType"/>
20453         <xs:element minOccurs="0" name="actor" type="ns_p:UseCaseActorType"/>
20454         <xs:element minOccurs="0" name="useCaseSupport"
20455 type="ns_p:UseCaseSupportSelectorsType"/>
20456     </xs:sequence>
20457 </xs:complexType>
20458 <xs:element name="useCaseInformationListDataSelectors"
20459 type="ns_p:UseCaseInformationListDataSelectorsType"/>
20460 </xs:schema>
20461
20462
20463

```

A.2.32 Version

File name: EEBus_SPINE_TS_Version.xsd

```

20466 <?xml version="1.0" encoding="UTF-8"?>
20467 <!--
20468     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
20469     Version 1.3.0 (final)
20470     2023-08-31
20471     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
20472     Source: https://www.eebus.org/en/specifications/
20473 -->
20474 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
20475 xmlns:xs="http://www.w3.org/2001/XMLSchema"
20476 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0" blockDefault="#all"
20477 elementFormDefault="qualified">
20478     <xs:annotation>
20479

```

```
20480     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
20481 e.V. All rights reserved.</xs:documentation>
20482   </xs:annotation>
20483   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
20484   <xs:complexType name="SpecificationVersionDataType">
20485     <xs:simpleContent>
20486       <xs:extension base="ns_p:SpecificationVersionType"/>
20487     </xs:simpleContent>
20488   </xs:complexType>
20489   <xs:element name="specificationVersionData" type="ns_p:SpecificationVersionDataType"/>
20490   <xs:complexType name="SpecificationVersionDataElementsType"/>
20491   <xs:element name="specificationVersionDataElements"
20492 type="ns_p:SpecificationVersionDataElementsType"/>
20493   <xs:complexType name="SpecificationVersionListDataType">
20494     <xs:sequence>
20495       <xs:element maxOccurs="unbounded" minOccurs="0"
20496 ref="ns_p:specificationVersionData"/>
20497     </xs:sequence>
20498   </xs:complexType>
20499   <xs:element name="specificationVersionListData"
20500 type="ns_p:SpecificationVersionListDataType"/>
20501   <xs:complexType name="SpecificationVersionListDataSelectorsType"/>
20502   <xs:element name="specificationVersionListDataSelectors"
20503 type="ns_p:SpecificationVersionListDataSelectorsType"/>
20504 </xs:schema>
20505
20506
```

20507

20508 A.3 Further XSDs

20509 A.3.1 CommandCommonDefinitions

20510 File name: EEBus_SPINE_TS_CommandCommonDefinitions.xsd

```
20511 <?xml version="1.0" encoding="UTF-8"?>
20512 <!--
20513   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
20514   Version 1.3.0 (final)
20515   2023-08-31
20516   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
20517   Source: https://www.eebus.org/en/specifications/
20518 -->
20519 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
20520 xmlns:xs="http://www.w3.org/2001/XMLSchema"
20521 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
20522 blockDefault="#all" elementFormDefault="qualified">
20523   <xs:annotation>
20524     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
20525 e.V. All rights reserved.</xs:documentation>
20526   </xs:annotation>
20527   <xs:include schemaLocation="EEBus_SPINE_TS_ActuatorLevel.xsd"/>
20528   <xs:include schemaLocation="EEBus_SPINE_TS_ActuatorSwitch.xsd"/>
20529   <xs:include schemaLocation="EEBus_SPINE_TS_Alarm.xsd"/>
20530   <xs:include schemaLocation="EEBus_SPINE_TS_Bill.xsd"/>
20531   <xs:include schemaLocation="EEBus_SPINE_TS_BindingManagement.xsd"/>
20532   <xs:include schemaLocation="EEBus_SPINE_TS_DataTunneling.xsd"/>
20533   <xs:include schemaLocation="EEBus_SPINE_TS_DeviceClassification.xsd"/>
20534   <xs:include schemaLocation="EEBus_SPINE_TS_DeviceConfiguration.xsd"/>
20535   <xs:include schemaLocation="EEBus_SPINE_TS_DeviceDiagnosis.xsd"/>
20536   <xs:include schemaLocation="EEBus_SPINE_TS_DirectControl.xsd"/>
20537   <xs:include schemaLocation="EEBus_SPINE_TS_ElectricalConnection.xsd"/>
20538   <xs:include schemaLocation="EEBus_SPINE_TS_HVAC.xsd"/>
20539   <xs:include schemaLocation="EEBus_SPINE_TS_Identification.xsd"/>
20540   <xs:include schemaLocation="EEBus_SPINE_TS_IncentiveTable.xsd"/>
20541   <xs:include schemaLocation="EEBus_SPINE_TS_LoadControl.xsd"/>
20542   <xs:include schemaLocation="EEBus_SPINE_TS_Measurement.xsd"/>
20543   <xs:include schemaLocation="EEBus_SPINE_TS_Messaging.xsd"/>
20544   <xs:include schemaLocation="EEBus_SPINE_TS_NetworkManagement.xsd"/>
20545   <xs:include schemaLocation="EEBus_SPINE_TS_NodeManagement.xsd"/>
20546   <xs:include schemaLocation="EEBus_SPINE_TS_OperatingConstraints.xsd"/>
20547   <xs:include schemaLocation="EEBus_SPINE_TS_PowerSequences.xsd"/>
20548   <xs:include schemaLocation="EEBus_SPINE_TS_Result.xsd"/>
20549   <xs:include schemaLocation="EEBus_SPINE_TS_Sensing.xsd"/>
20550
```



```
20551 <xs:include schemaLocation="EEBus_SPINE_TS_Setpoint.xsd"/>
20552 <xs:include schemaLocation="EEBus_SPINE_TS_SmartEnergyManagementPs.xsd"/>
20553 <xs:include schemaLocation="EEBus_SPINE_TS_StateInformation.xsd"/>
20554 <xs:include schemaLocation="EEBus_SPINE_TS_SubscriptionManagement.xsd"/>
20555 <xs:include schemaLocation="EEBus_SPINE_TS_SupplyCondition.xsd"/>
20556 <xs:include schemaLocation="EEBus_SPINE_TS_TariffInformation.xsd"/>
20557 <xs:include schemaLocation="EEBus_SPINE_TS_TaskManagement.xsd"/>
20558 <xs:include schemaLocation="EEBus_SPINE_TS_Threshold.xsd"/>
20559 <xs:include schemaLocation="EEBus_SPINE_TS_TimeInformation.xsd"/>
20560 <xs:include schemaLocation="EEBus_SPINE_TS_TimeSeries.xsd"/>
20561 <xs:include schemaLocation="EEBus_SPINE_TS_TimeTable.xsd"/>
20562 <xs:include schemaLocation="EEBus_SPINE_TS_UseCaseInformation.xsd"/>
20563 <xs:include schemaLocation="EEBus_SPINE_TS_Version.xsd"/>
20564 <xs:group name="DataChoiceGroup">
20565   <xs:choice>
20566     <xs:element ref="ns_p:actuatorLevelData"/>
20567     <xs:element ref="ns_p:actuatorLevelDescriptionData"/>
20568     <xs:element ref="ns_p:actuatorSwitchData"/>
20569     <xs:element ref="ns_p:actuatorSwitchDescriptionData"/>
20570     <xs:element ref="ns_p:alarmListData"/>
20571     <xs:element ref="ns_p:billConstraintsListData"/>
20572     <xs:element ref="ns_p:billDescriptionListData"/>
20573     <xs:element ref="ns_p:billListData"/>
20574     <xs:element ref="ns_p:bindingManagementDeleteCall"/>
20575     <xs:element ref="ns_p:bindingManagementEntryListData"/>
20576     <xs:element ref="ns_p:bindingManagementRequestCall"/>
20577     <xs:element ref="ns_p:commodityListData"/>
20578     <xs:element ref="ns_p:dataTunnelingCall"/>
20579     <xs:element ref="ns_p:deviceClassificationManufacturerData"/>
20580     <xs:element ref="ns_p:deviceClassificationUserData"/>
20581     <xs:element ref="ns_p:deviceConfigurationKeyValueConstraintsListData"/>
20582     <xs:element ref="ns_p:deviceConfigurationKeyValueDescriptionListData"/>
20583     <xs:element ref="ns_p:deviceConfigurationKeyValueListData"/>
20584     <xs:element ref="ns_p:deviceDiagnosisHeartbeatData"/>
20585     <xs:element ref="ns_p:deviceDiagnosisServiceData"/>
20586     <xs:element ref="ns_p:deviceDiagnosisStateData"/>
20587     <xs:element ref="ns_p:directControlActivityListData"/>
20588     <xs:element ref="ns_p:directControlDescriptionData"/>
20589     <xs:element ref="ns_p:electricalConnectionCharacteristicListData"/>
20590     <xs:element ref="ns_p:electricalConnectionDescriptionListData"/>
20591     <xs:element ref="ns_p:electricalConnectionParameterDescriptionListData"/>
20592     <xs:element ref="ns_p:electricalConnectionPermittedValueSetListData"/>
20593     <xs:element ref="ns_p:electricalConnectionStateListData"/>
20594     <xs:element ref="ns_p:hvacOperationModeDescriptionListData"/>
20595     <xs:element ref="ns_p:hvacOverrunDescriptionListData"/>
20596     <xs:element ref="ns_p:hvacOverrunListData"/>
20597     <xs:element ref="ns_p:hvacSystemFunctionDescriptionListData"/>
20598     <xs:element ref="ns_p:hvacSystemFunctionListData"/>
20599     <xs:element ref="ns_p:hvacSystemFunctionOperationModeRelationListData"/>
20600     <xs:element ref="ns_p:hvacSystemFunctionPowerSequenceRelationListData"/>
20601     <xs:element ref="ns_p:hvacSystemFunctionSetpointRelationListData"/>
20602     <xs:element ref="ns_p:identificationListData"/>
20603     <xs:element ref="ns_p:incentiveDescriptionListData"/>
20604     <xs:element ref="ns_p:incentiveListData"/>
20605     <xs:element ref="ns_p:incentiveTableConstraintsData"/>
20606     <xs:element ref="ns_p:incentiveTableData"/>
20607     <xs:element ref="ns_p:incentiveTableDescriptionData"/>
20608     <xs:element ref="ns_p:loadControlEventListData"/>
20609     <xs:element ref="ns_p:loadControlLimitConstraintsListData"/>
20610     <xs:element ref="ns_p:loadControlLimitDescriptionListData"/>
20611     <xs:element ref="ns_p:loadControlLimitListData"/>
20612     <xs:element ref="ns_p:loadControlNodeData"/>
20613     <xs:element ref="ns_p:loadControlStateListData"/>
20614     <xs:element ref="ns_p:measurementConstraintsListData"/>
20615     <xs:element ref="ns_p:measurementDescriptionListData"/>
20616     <xs:element ref="ns_p:measurementListData"/>
20617     <xs:element ref="ns_p:measurementSeriesListData"/>
20618     <xs:element ref="ns_p:measurementThresholdRelationListData"/>
20619     <xs:element ref="ns_p:messagingListData"/>
20620     <xs:element ref="ns_p:networkManagementAbortCall"/>
20621     <xs:element ref="ns_p:networkManagementAddNodeCall"/>
20622     <xs:element ref="ns_p:networkManagementDeviceDescriptionListData"/>
20623     <xs:element ref="ns_p:networkManagementDiscoverCall"/>
20624     <xs:element ref="ns_p:networkManagementEntityDescriptionListData"/>
20625     <xs:element ref="ns_p:networkManagementFeatureDescriptionListData"/>
20626     <xs:element ref="ns_p:networkManagementJoiningModeData"/>
20627     <xs:element ref="ns_p:networkManagementModifyNodeCall"/>
20628     <xs:element ref="ns_p:networkManagementProcessStateData"/>
```

20629 <xs:element ref="ns_p:networkManagementRemoveNodeCall"/>
20630 <xs:element ref="ns_p:networkManagementReportCandidateData"/>
20631 <xs:element ref="ns_p:networkManagementScanNetworkCall"/>
20632 <xs:element ref="ns_p:nodeManagementBindingData"/>
20633 <xs:element ref="ns_p:nodeManagementBindingDeleteCall"/>
20634 <xs:element ref="ns_p:nodeManagementBindingRequestCall"/>
20635 <xs:element ref="ns_p:nodeManagementDestinationListData"/>
20636 <xs:element ref="ns_p:nodeManagementDetailedDiscoveryData"/>
20637 <xs:element ref="ns_p:nodeManagementSubscriptionData"/>
20638 <xs:element ref="ns_p:nodeManagementSubscriptionDeleteCall"/>
20639 <xs:element ref="ns_p:nodeManagementSubscriptionRequestCall"/>
20640 <xs:element ref="ns_p:nodeManagementUseCaseData"/>
20641 <xs:element ref="ns_p:operatingConstraintsDurationListData"/>
20642 <xs:element ref="ns_p:operatingConstraintsInterruptListData"/>
20643 <xs:element ref="ns_p:operatingConstraintsPowerDescriptionListData"/>
20644 <xs:element ref="ns_p:operatingConstraintsPowerLevelListData"/>
20645 <xs:element ref="ns_p:operatingConstraintsPowerRangeListData"/>
20646 <xs:element ref="ns_p:operatingConstraintsResumeImplicationListData"/>
20647 <xs:element ref="ns_p:powerSequenceAlternativesRelationListData"/>
20648 <xs:element ref="ns_p:powerSequenceDescriptionListData"/>
20649 <xs:element ref="ns_p:powerSequenceNodeScheduleInformationData"/>
20650 <xs:element ref="ns_p:powerSequencePriceCalculationRequestCall"/>
20651 <xs:element ref="ns_p:powerSequencePriceListData"/>
20652 <xs:element ref="ns_p:powerSequenceScheduleConfigurationRequestCall"/>
20653 <xs:element ref="ns_p:powerSequenceScheduleConstraintsListData"/>
20654 <xs:element ref="ns_p:powerSequenceScheduleListData"/>
20655 <xs:element ref="ns_p:powerSequenceSchedulePreferenceListData"/>
20656 <xs:element ref="ns_p:powerSequenceStateListData"/>
20657 <xs:element ref="ns_p:powerTimeSlotScheduleConstraintsListData"/>
20658 <xs:element ref="ns_p:powerTimeSlotScheduleListData"/>
20659 <xs:element ref="ns_p:powerTimeSlotValueListData"/>
20660 <xs:element ref="ns_p:resultData"/>
20661 <xs:element ref="ns_p:sensingDescriptionData"/>
20662 <xs:element ref="ns_p:sensingListData"/>
20663 <xs:element ref="ns_p:sessionIdentificationListData"/>
20664 <xs:element ref="ns_p:sessionMeasurementRelationListData"/>
20665 <xs:element ref="ns_p:setpointConstraintsListData"/>
20666 <xs:element ref="ns_p:setpointDescriptionListData"/>
20667 <xs:element ref="ns_p:setpointListData"/>
20668 <xs:element ref="ns_p:smartEnergyManagementPsConfigurationRequestCall"/>
20669 <xs:element ref="ns_p:smartEnergyManagementPsData"/>
20670 <xs:element ref="ns_p:smartEnergyManagementPsPriceCalculationRequestCall"/>
20671 <xs:element ref="ns_p:smartEnergyManagementPsPriceData"/>
20672 <xs:element ref="ns_p:specificationVersionListData"/>
20673 <xs:element ref="ns_p:stateInformationListData"/>
20674 <xs:element ref="ns_p:subscriptionManagementDeleteCall"/>
20675 <xs:element ref="ns_p:subscriptionManagementEntryListData"/>
20676 <xs:element ref="ns_p:subscriptionManagementRequestCall"/>
20677 <xs:element ref="ns_p:supplyConditionDescriptionListData"/>
20678 <xs:element ref="ns_p:supplyConditionListData"/>
20679 <xs:element ref="ns_p:supplyConditionThresholdRelationListData"/>
20680 <xs:element ref="ns_p:tariffBoundaryRelationListData"/>
20681 <xs:element ref="ns_p:tariffDescriptionListData"/>
20682 <xs:element ref="ns_p:tariffListData"/>
20683 <xs:element ref="ns_p:tariffOverallConstraintsData"/>
20684 <xs:element ref="ns_p:tariffTierRelationListData"/>
20685 <xs:element ref="ns_p:taskManagementJobDescriptionListData"/>
20686 <xs:element ref="ns_p:taskManagementJobListData"/>
20687 <xs:element ref="ns_p:taskManagementJobRelationListData"/>
20688 <xs:element ref="ns_p:taskManagementOverviewData"/>
20689 <xs:element ref="ns_p:thresholdConstraintsListData"/>
20690 <xs:element ref="ns_p:thresholdDescriptionListData"/>
20691 <xs:element ref="ns_p:thresholdListData"/>
20692 <xs:element ref="ns_p:tierBoundaryDescriptionListData"/>
20693 <xs:element ref="ns_p:tierBoundaryListData"/>
20694 <xs:element ref="ns_p:tierDescriptionListData"/>
20695 <xs:element ref="ns_p:tierIncentiveRelationListData"/>
20696 <xs:element ref="ns_p:tierListData"/>
20697 <xs:element ref="ns_p:timeDistributorData"/>
20698 <xs:element ref="ns_p:timeDistributorEnquiryCall"/>
20699 <xs:element ref="ns_p:timeInformationData"/>
20700 <xs:element ref="ns_p:timePrecisionData"/>
20701 <xs:element ref="ns_p:timeSeriesConstraintsListData"/>
20702 <xs:element ref="ns_p:timeSeriesDescriptionListData"/>
20703 <xs:element ref="ns_p:timeSeriesListData"/>
20704 <xs:element ref="ns_p:timeTableConstraintsListData"/>
20705 <xs:element ref="ns_p:timeTableDescriptionListData"/>
20706 <xs:element ref="ns_p:timeTableListData"/>

```
20707     <xs:element ref="ns_p:useCaseInformationListData"/>
20708   </xs:choice>
20709 </xs:group>
20710 <xs:group name="DataElementsChoiceGroup">
20711   <xs:choice>
20712     <xs:element ref="ns_p:actuatorLevelDataElements"/>
20713     <xs:element ref="ns_p:actuatorLevelDescriptionDataElements"/>
20714     <xs:element ref="ns_p:actuatorSwitchDataElements"/>
20715     <xs:element ref="ns_p:actuatorSwitchDescriptionDataElements"/>
20716     <xs:element ref="ns_p:alarmDataElements"/>
20717     <xs:element ref="ns_p:billConstraintsDataElements"/>
20718     <xs:element ref="ns_p:billDataElements"/>
20719     <xs:element ref="ns_p:billDescriptionDataElements"/>
20720     <xs:element ref="ns_p:bindingManagementDeleteCallElements"/>
20721     <xs:element ref="ns_p:bindingManagementEntryDataElements"/>
20722     <xs:element ref="ns_p:bindingManagementRequestCallElements"/>
20723     <xs:element ref="ns_p:commodityDataElements"/>
20724     <xs:element ref="ns_p:dataTunnelingCallElements"/>
20725     <xs:element ref="ns_p:deviceClassificationManufacturerDataElements"/>
20726     <xs:element ref="ns_p:deviceClassificationUserDataElements"/>
20727     <xs:element ref="ns_p:deviceConfigurationKeyValueConstraintsDataElements"/>
20728     <xs:element ref="ns_p:deviceConfigurationKeyValueDataElements"/>
20729     <xs:element ref="ns_p:deviceConfigurationKeyValueDescriptionDataElements"/>
20730     <xs:element ref="ns_p:deviceDiagnosisHeartbeatDataElements"/>
20731     <xs:element ref="ns_p:deviceDiagnosisServiceDataElements"/>
20732     <xs:element ref="ns_p:deviceDiagnosisStateDataElements"/>
20733     <xs:element ref="ns_p:directControlActivityDataElements"/>
20734     <xs:element ref="ns_p:directControlDescriptionDataElements"/>
20735     <xs:element ref="ns_p:electricalConnectionCharacteristicDataElements"/>
20736     <xs:element ref="ns_p:electricalConnectionDescriptionDataElements"/>
20737     <xs:element ref="ns_p:electricalConnectionParameterDescriptionDataElements"/>
20738     <xs:element ref="ns_p:electricalConnectionPermittedValueSetDataElements"/>
20739     <xs:element ref="ns_p:electricalConnectionStateDataElements"/>
20740     <xs:element ref="ns_p:hvacOperationModeDescriptionDataElements"/>
20741     <xs:element ref="ns_p:hvacOverrunDataElements"/>
20742     <xs:element ref="ns_p:hvacOverrunDescriptionDataElements"/>
20743     <xs:element ref="ns_p:hvacSystemFunctionDataElements"/>
20744     <xs:element ref="ns_p:hvacSystemFunctionDescriptionDataElements"/>
20745     <xs:element ref="ns_p:hvacSystemFunctionOperationModeRelationDataElements"/>
20746     <xs:element ref="ns_p:hvacSystemFunctionPowerSequenceRelationDataElements"/>
20747     <xs:element ref="ns_p:hvacSystemFunctionSetpointRelationDataElements"/>
20748     <xs:element ref="ns_p:identificationDataElements"/>
20749     <xs:element ref="ns_p:incentiveDataElements"/>
20750     <xs:element ref="ns_p:incentiveDescriptionDataElements"/>
20751     <xs:element ref="ns_p:incentiveTableConstraintsDataElements"/>
20752     <xs:element ref="ns_p:incentiveTableDataElements"/>
20753     <xs:element ref="ns_p:incentiveTableDescriptionDataElements"/>
20754     <xs:element ref="ns_p:loadControlEventDataElements"/>
20755     <xs:element ref="ns_p:loadControlLimitConstraintsDataElements"/>
20756     <xs:element ref="ns_p:loadControlLimitDataElements"/>
20757     <xs:element ref="ns_p:loadControlLimitDescriptionDataElements"/>
20758     <xs:element ref="ns_p:loadControlNodeDataElements"/>
20759     <xs:element ref="ns_p:loadControlStateDataElements"/>
20760     <xs:element ref="ns_p:measurementConstraintsDataElements"/>
20761     <xs:element ref="ns_p:measurementDataElements"/>
20762     <xs:element ref="ns_p:measurementDescriptionDataElements"/>
20763     <xs:element ref="ns_p:measurementSeriesDataElements"/>
20764     <xs:element ref="ns_p:measurementThresholdRelationDataElements"/>
20765     <xs:element ref="ns_p:messagingDataElements"/>
20766     <xs:element ref="ns_p:networkManagementAbortCallElements"/>
20767     <xs:element ref="ns_p:networkManagementAddNodeCallElements"/>
20768     <xs:element ref="ns_p:networkManagementDeviceDescriptionDataElements"/>
20769     <xs:element ref="ns_p:networkManagementDiscoverCallElements"/>
20770     <xs:element ref="ns_p:networkManagementEntityDescriptionDataElements"/>
20771     <xs:element ref="ns_p:networkManagementFeatureDescriptionDataElements"/>
20772     <xs:element ref="ns_p:networkManagementJoiningModeDataElements"/>
20773     <xs:element ref="ns_p:networkManagementModifyNodeCallElements"/>
20774     <xs:element ref="ns_p:networkManagementProcessStateDataElements"/>
20775     <xs:element ref="ns_p:networkManagementRemoveNodeCallElements"/>
20776     <xs:element ref="ns_p:networkManagementReportCandidateDataElements"/>
20777     <xs:element ref="ns_p:networkManagementScanNetworkCallElements"/>
20778     <xs:element ref="ns_p:nodeManagementBindingDataElements"/>
20779     <xs:element ref="ns_p:nodeManagementBindingDeleteCallElements"/>
20780     <xs:element ref="ns_p:nodeManagementBindingRequestCallElements"/>
20781     <xs:element ref="ns_p:nodeManagementDestinationDataElements"/>
20782     <xs:element ref="ns_p:nodeManagementDetailedDiscoveryDataElements"/>
20783     <xs:element ref="ns_p:nodeManagementSubscriptionDataElements"/>
20784     <xs:element ref="ns_p:nodeManagementSubscriptionDeleteCallElements"/>
```

```
20785 <xs:element ref="ns_p:nodeManagementSubscriptionRequestCallElements"/>
20786 <xs:element ref="ns_p:nodeManagementUseCaseDataElements"/>
20787 <xs:element ref="ns_p:operatingConstraintsDurationDataElements"/>
20788 <xs:element ref="ns_p:operatingConstraintsInterruptDataElements"/>
20789 <xs:element ref="ns_p:operatingConstraintsPowerDescriptionDataElements"/>
20790 <xs:element ref="ns_p:operatingConstraintsPowerLevelDataElements"/>
20791 <xs:element ref="ns_p:operatingConstraintsPowerRangeDataElements"/>
20792 <xs:element ref="ns_p:operatingConstraintsResumeImplicationDataElements"/>
20793 <xs:element ref="ns_p:powerSequenceAlternativesRelationDataElements"/>
20794 <xs:element ref="ns_p:powerSequenceDescriptionDataElements"/>
20795 <xs:element ref="ns_p:powerSequenceNodeScheduleInformationDataElements"/>
20796 <xs:element ref="ns_p:powerSequencePriceCalculationRequestCallElements"/>
20797 <xs:element ref="ns_p:powerSequencePriceDataElements"/>
20798 <xs:element ref="ns_p:powerSequenceScheduleConfigurationRequestCallElements"/>
20799 <xs:element ref="ns_p:powerSequenceScheduleConstraintsDataElements"/>
20800 <xs:element ref="ns_p:powerSequenceScheduleDataElements"/>
20801 <xs:element ref="ns_p:powerSequenceSchedulePreferenceDataElements"/>
20802 <xs:element ref="ns_p:powerSequenceStateDataElements"/>
20803 <xs:element ref="ns_p:powerTimeSlotScheduleConstraintsDataElements"/>
20804 <xs:element ref="ns_p:powerTimeSlotScheduleDataElements"/>
20805 <xs:element ref="ns_p:powerTimeSlotValueDataElements"/>
20806 <xs:element ref="ns_p:sensingDataElements"/>
20807 <xs:element ref="ns_p:sensingDescriptionDataElements"/>
20808 <xs:element ref="ns_p:sessionIdentificationDataElements"/>
20809 <xs:element ref="ns_p:sessionMeasurementRelationDataElements"/>
20810 <xs:element ref="ns_p:setpointConstraintsDataElements"/>
20811 <xs:element ref="ns_p:setpointDataElements"/>
20812 <xs:element ref="ns_p:setpointDescriptionDataElements"/>
20813 <xs:element ref="ns_p:smartEnergyManagementPsConfigurationRequestCallElements"/>
20814 <xs:element ref="ns_p:smartEnergyManagementPsDataElements"/>
20815 <xs:element
20816 ref="ns_p:smartEnergyManagementPsPriceCalculationRequestCallElements"/>
20817 <xs:element ref="ns_p:smartEnergyManagementPsPriceDataElements"/>
20818 <xs:element ref="ns_p:specificationVersionDataElements"/>
20819 <xs:element ref="ns_p:stateInformationDataElements"/>
20820 <xs:element ref="ns_p:subscriptionManagementDeleteCallElements"/>
20821 <xs:element ref="ns_p:subscriptionManagementEntryDataElements"/>
20822 <xs:element ref="ns_p:subscriptionManagementRequestCallElements"/>
20823 <xs:element ref="ns_p:supplyConditionDataElements"/>
20824 <xs:element ref="ns_p:supplyConditionDescriptionDataElements"/>
20825 <xs:element ref="ns_p:supplyConditionThresholdRelationDataElements"/>
20826 <xs:element ref="ns_p:tariffBoundaryRelationDataElements"/>
20827 <xs:element ref="ns_p:tariffDataElements"/>
20828 <xs:element ref="ns_p:tariffDescriptionDataElements"/>
20829 <xs:element ref="ns_p:tariffOverallConstraintsDataElements"/>
20830 <xs:element ref="ns_p:tariffTierRelationDataElements"/>
20831 <xs:element ref="ns_p:taskManagementJobDataElements"/>
20832 <xs:element ref="ns_p:taskManagementJobDescriptionDataElements"/>
20833 <xs:element ref="ns_p:taskManagementJobRelationDataElements"/>
20834 <xs:element ref="ns_p:taskManagementOverviewDataElements"/>
20835 <xs:element ref="ns_p:thresholdConstraintsDataElements"/>
20836 <xs:element ref="ns_p:thresholdDataElements"/>
20837 <xs:element ref="ns_p:thresholdDescriptionDataElements"/>
20838 <xs:element ref="ns_p:tierBoundaryDataElements"/>
20839 <xs:element ref="ns_p:tierBoundaryDescriptionDataElements"/>
20840 <xs:element ref="ns_p:tierDataElements"/>
20841 <xs:element ref="ns_p:tierDescriptionDataElements"/>
20842 <xs:element ref="ns_p:tierIncentiveRelationDataElements"/>
20843 <xs:element ref="ns_p:timeDistributorDataElements"/>
20844 <xs:element ref="ns_p:timeDistributorEnquiryCallElements"/>
20845 <xs:element ref="ns_p:timeInformationDataElements"/>
20846 <xs:element ref="ns_p:timePrecisionDataElements"/>
20847 <xs:element ref="ns_p:timeSeriesConstraintsDataElements"/>
20848 <xs:element ref="ns_p:timeSeriesDataElements"/>
20849 <xs:element ref="ns_p:timeSeriesDescriptionDataElements"/>
20850 <xs:element ref="ns_p:timeTableConstraintsDataElements"/>
20851 <xs:element ref="ns_p:timeTableDataElements"/>
20852 <xs:element ref="ns_p:timeTableDescriptionDataElements"/>
20853 <xs:element ref="ns_p:useCaseInformationDataElements"/>
20854 </xs:choice>
20855 </xs:group>
20856 <xs:group name="DataSelectorsChoiceGroup">
20857 <xs:choice>
20858 <xs:element ref="ns_p:alarmListDataSelectors"/>
20859 <xs:element ref="ns_p:billConstraintsListDataSelectors"/>
20860 <xs:element ref="ns_p:billDescriptionListDataSelectors"/>
20861 <xs:element ref="ns_p:billListDataSelectors"/>
20862 <xs:element ref="ns_p:bindingManagementEntryListDataSelectors"/>
```

```
20863 <xs:element ref="ns_p:commodityListDataSelectors"/>
20864 <xs:element ref="ns_p:deviceConfigurationKeyValueConstraintsListDataSelectors"/>
20865 <xs:element ref="ns_p:deviceConfigurationKeyValueDescriptionListDataSelectors"/>
20866 <xs:element ref="ns_p:deviceConfigurationKeyValueListDataSelectors"/>
20867 <xs:element ref="ns_p:directControlActivityListDataSelectors"/>
20868 <xs:element ref="ns_p:electricalConnectionCharacteristicListDataSelectors"/>
20869 <xs:element ref="ns_p:electricalConnectionDescriptionListDataSelectors"/>
20870 <xs:element ref="ns_p:electricalConnectionParameterDescriptionListDataSelectors"/>
20871 <xs:element ref="ns_p:electricalConnectionPermittedValueSetListDataSelectors"/>
20872 <xs:element ref="ns_p:electricalConnectionStateListDataSelectors"/>
20873 <xs:element ref="ns_p:hvacOperationModeDescriptionListDataSelectors"/>
20874 <xs:element ref="ns_p:hvacOverrunDescriptionListDataSelectors"/>
20875 <xs:element ref="ns_p:hvacOverrunListDataSelectors"/>
20876 <xs:element ref="ns_p:hvacSystemFunctionDescriptionListDataSelectors"/>
20877 <xs:element ref="ns_p:hvacSystemFunctionListDataSelectors"/>
20878 <xs:element ref="ns_p:hvacSystemFunctionOperationModeRelationListDataSelectors"/>
20879 <xs:element ref="ns_p:hvacSystemFunctionPowerSequenceRelationListDataSelectors"/>
20880 <xs:element ref="ns_p:hvacSystemFunctionSetpointRelationListDataSelectors"/>
20881 <xs:element ref="ns_p:identificationListDataSelectors"/>
20882 <xs:element ref="ns_p:incentiveDescriptionListDataSelectors"/>
20883 <xs:element ref="ns_p:incentiveListDataSelectors"/>
20884 <xs:element ref="ns_p:incentiveTableConstraintsDataSelectors"/>
20885 <xs:element ref="ns_p:incentiveTableDataSelectors"/>
20886 <xs:element ref="ns_p:incentiveTableDescriptionDataSelectors"/>
20887 <xs:element ref="ns_p:loadControlEventListDataSelectors"/>
20888 <xs:element ref="ns_p:loadControlLimitConstraintsListDataSelectors"/>
20889 <xs:element ref="ns_p:loadControlLimitDescriptionListDataSelectors"/>
20890 <xs:element ref="ns_p:loadControlLimitListDataSelectors"/>
20891 <xs:element ref="ns_p:loadControlStateListDataSelectors"/>
20892 <xs:element ref="ns_p:measurementConstraintsListDataSelectors"/>
20893 <xs:element ref="ns_p:measurementDescriptionListDataSelectors"/>
20894 <xs:element ref="ns_p:measurementListDataSelectors"/>
20895 <xs:element ref="ns_p:measurementSeriesListDataSelectors"/>
20896 <xs:element ref="ns_p:measurementThresholdRelationListDataSelectors"/>
20897 <xs:element ref="ns_p:messagingListDataSelectors"/>
20898 <xs:element ref="ns_p:networkManagementDeviceDescriptionListDataSelectors"/>
20899 <xs:element ref="ns_p:networkManagementEntityDescriptionListDataSelectors"/>
20900 <xs:element ref="ns_p:networkManagementFeatureDescriptionListDataSelectors"/>
20901 <xs:element ref="ns_p:nodeManagementBindingDataSelectors"/>
20902 <xs:element ref="ns_p:nodeManagementDestinationListDataSelectors"/>
20903 <xs:element ref="ns_p:nodeManagementDetailedDiscoveryDataSelectors"/>
20904 <xs:element ref="ns_p:nodeManagementSubscriptionDataSelectors"/>
20905 <xs:element ref="ns_p:nodeManagementUseCaseDataSelectors"/>
20906 <xs:element ref="ns_p:operatingConstraintsDurationListDataSelectors"/>
20907 <xs:element ref="ns_p:operatingConstraintsInterruptListDataSelectors"/>
20908 <xs:element ref="ns_p:operatingConstraintsPowerDescriptionListDataSelectors"/>
20909 <xs:element ref="ns_p:operatingConstraintsPowerLevelListDataSelectors"/>
20910 <xs:element ref="ns_p:operatingConstraintsPowerRangeListDataSelectors"/>
20911 <xs:element ref="ns_p:operatingConstraintsResumeImplicationListDataSelectors"/>
20912 <xs:element ref="ns_p:powerSequenceAlternativesRelationListDataSelectors"/>
20913 <xs:element ref="ns_p:powerSequenceDescriptionListDataSelectors"/>
20914 <xs:element ref="ns_p:powerSequencePriceListDataSelectors"/>
20915 <xs:element ref="ns_p:powerSequenceScheduleConstraintsListDataSelectors"/>
20916 <xs:element ref="ns_p:powerSequenceScheduleListDataSelectors"/>
20917 <xs:element ref="ns_p:powerSequenceSchedulePreferenceListDataSelectors"/>
20918 <xs:element ref="ns_p:powerSequenceStateListDataSelectors"/>
20919 <xs:element ref="ns_p:powerTimeSlotScheduleConstraintsListDataSelectors"/>
20920 <xs:element ref="ns_p:powerTimeSlotScheduleListDataSelectors"/>
20921 <xs:element ref="ns_p:powerTimeSlotValueListDataSelectors"/>
20922 <xs:element ref="ns_p:sensingListDataSelectors"/>
20923 <xs:element ref="ns_p:sessionIdentificationListDataSelectors"/>
20924 <xs:element ref="ns_p:sessionMeasurementRelationListDataSelectors"/>
20925 <xs:element ref="ns_p:setpointConstraintsListDataSelectors"/>
20926 <xs:element ref="ns_p:setpointDescriptionListDataSelectors"/>
20927 <xs:element ref="ns_p:setpointListDataSelectors"/>
20928 <xs:element ref="ns_p:smartEnergyManagementPsDataSelectors"/>
20929 <xs:element ref="ns_p:smartEnergyManagementPsPriceDataSelectors"/>
20930 <xs:element ref="ns_p:specificationVersionListDataSelectors"/>
20931 <xs:element ref="ns_p:stateInformationListDataSelectors"/>
20932 <xs:element ref="ns_p:subscriptionManagementEntryListDataSelectors"/>
20933 <xs:element ref="ns_p:supplyConditionDescriptionListDataSelectors"/>
20934 <xs:element ref="ns_p:supplyConditionListDataSelectors"/>
20935 <xs:element ref="ns_p:supplyConditionThresholdRelationListDataSelectors"/>
20936 <xs:element ref="ns_p:tariffBoundaryRelationListDataSelectors"/>
20937 <xs:element ref="ns_p:tariffDescriptionListDataSelectors"/>
20938 <xs:element ref="ns_p:tariffListDataSelectors"/>
20939 <xs:element ref="ns_p:tariffTierRelationListDataSelectors"/>
20940 <xs:element ref="ns_p:taskManagementJobDescriptionListDataSelectors"/>
```

```
20941      <xs:element ref="ns_p:taskManagementJobListDataSelectors"/>
20942      <xs:element ref="ns_p:taskManagementJobRelationListDataSelectors"/>
20943      <xs:element ref="ns_p:thresholdConstraintsListDataSelectors"/>
20944      <xs:element ref="ns_p:thresholdDescriptionListDataSelectors"/>
20945      <xs:element ref="ns_p:thresholdListDataSelectors"/>
20946      <xs:element ref="ns_p:tierBoundaryDescriptionListDataSelectors"/>
20947      <xs:element ref="ns_p:tierBoundaryListDataSelectors"/>
20948      <xs:element ref="ns_p:tierDescriptionListDataSelectors"/>
20949      <xs:element ref="ns_p:tierIncentiveRelationListDataSelectors"/>
20950      <xs:element ref="ns_p:tierListDataSelectors"/>
20951      <xs:element ref="ns_p:timeSeriesConstraintsListDataSelectors"/>
20952      <xs:element ref="ns_p:timeSeriesDescriptionListDataSelectors"/>
20953      <xs:element ref="ns_p:timeSeriesListDataSelectors"/>
20954      <xs:element ref="ns_p:timeTableConstraintsListDataSelectors"/>
20955      <xs:element ref="ns_p:timeTableDescriptionListDataSelectors"/>
20956      <xs:element ref="ns_p:timeTableListDataSelectors"/>
20957      <xs:element ref="ns_p:useCaseInformationListDataSelectors"/>
20958    </xs:choice>
20959  </xs:group>
20960 </xs:schema>
20961
20962
```

20963

20964 **A.3.2 CommandFrame**

20965 File name: EEBus_SPINE_TS_CommandFrame.xsd

```
20966
20967 <?xml version="1.0" encoding="UTF-8"?>
20968 <!--
20969   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
20970   Version 1.3.0 (final)
20971   2023-08-31
20972   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
20973   Source: https://www.eebus.org/en/specifications/
20974 -->
20975 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
20976   xmlns:xs="http://www.w3.org/2001/XMLSchema"
20977   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
20978   blockDefault="#all" elementFormDefault="qualified">
20979   <xs:annotation>
20980     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
20981     e.V. All rights reserved.</xs:documentation>
20982   </xs:annotation>
20983   <xs:include schemaLocation="EEBus_SPINE_TS_CommandCommonDefinitions.xsd"/>
20984   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
20985   <xs:simpleType name="MsgCounterType">
20986     <xs:restriction base="xs:unsignedLong"> </xs:restriction>
20987   </xs:simpleType>
20988   <xs:simpleType name="CmdClassifierType">
20989     <xs:restriction base="xs:string">
20990       <xs:enumeration value="read"/>
20991       <xs:enumeration value="reply"/>
20992       <xs:enumeration value="notify"/>
20993       <xs:enumeration value="write"/>
20994       <xs:enumeration value="call"/>
20995       <xs:enumeration value="result"/>
20996     </xs:restriction>
20997   </xs:simpleType>
20998   <xs:element name="cmdClassifier" type="ns_p:CmdClassifierType"/>
20999   <xs:element name="manufacturerSpecificExtension" type="xs:hexBinary"/>
21000   <xs:element name="lastUpdateAt" type="ns_p:AbsoluteOrRelativeTimeType"/>
21001   <xs:element name="function" type="ns_p:FunctionType"/>
21002   <xs:simpleType name="FilterIdType">
21003     <xs:restriction base="xs:unsignedInt"/>
21004   </xs:simpleType>
21005   <xs:complexType name="FilterType">
21006     <xs:sequence>
21007       <xs:element minOccurs="0" name="filterId" type="ns_p:FilterIdType"/>
21008       <xs:element minOccurs="0" ref="ns_p:cmdControl"/>
21009       <xs:group ref="ns_p:DataSelectorsChoiceGroup" minOccurs="0"
21010 maxOccurs="unbounded"/>
21011       <xs:group ref="ns_p:DataElementsChoiceGroup" minOccurs="0"/>
21012     </xs:sequence>
21013   </xs:complexType>
```

```
21014 <xs:element name="filter" type="ns_p:FilterType"/>
21015 <xs:complexType name="CmdControlType">
21016   <xs:sequence>
21017     <xs:element name="delete" minOccurs="0" type="ns_p:ElementTagType"/>
21018     <xs:element name="partial" minOccurs="0" type="ns_p:ElementTagType"/>
21019   </xs:sequence>
21020 </xs:complexType>
21021 <xs:element name="cmdControl" type="ns_p:CmdControlType"/>
21022 <xs:group name="CmdOptionGroup">
21023   <xs:sequence>
21024     <xs:element minOccurs="0" ref="ns_p:function"/>
21025     <xs:element minOccurs="0" ref="ns_p:filter" maxOccurs="unbounded"/>
21026   </xs:sequence>
21027 </xs:group>
21028 <xs:group name="DataExtendGroup">
21029   <xs:sequence>
21030     <xs:element minOccurs="0" ref="ns_p:manufacturerSpecificExtension"/>
21031     <xs:element minOccurs="0" ref="ns_p:lastUpdateAt"/>
21032   </xs:sequence>
21033 </xs:group>
21034 <xs:group name="PayloadContributionGroup">
21035   <xs:sequence>
21036     <xs:group ref="ns_p:CmdOptionGroup" minOccurs="0"/>
21037     <xs:group ref="ns_p:DataChoiceGroup" minOccurs="0"/>
21038     <xs:group minOccurs="0" ref="ns_p:DataExtendGroup"/>
21039   </xs:sequence>
21040 </xs:group>
21041 <xs:complexType name="CmdType">
21042   <xs:sequence>
21043     <xs:group ref="ns_p:PayloadContributionGroup"/>
21044   </xs:sequence>
21045 </xs:complexType>
21046 </xs:schema>
```

A.3.3 CommonDataTypes

File name: EEBus_SPINE_TS_CommonDataTypes.xsd

```
21052 <?xml version="1.0" encoding="UTF-8"?>
21053 <!--
21054   Smart Premises Interoperable Neutral-Message Exchange (SPINE)
21055   Version 1.3.0 (final)
21056   2023-08-31
21057   Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
21058   Source: https://www.eebus.org/en/specifications/
21059 -->
21060 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
21061   xmlns:xs="http://www.w3.org/2001/XMLSchema"
21062   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
21063   blockDefault="#all" elementFormDefault="qualified">
21064   <xs:annotation>
21065     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
21066     e.V. All rights reserved.</xs:documentation>
21067   </xs:annotation>
21068   <xs:complexType name="ElementTagType"/>
21069   <xs:simpleType name="LabelType">
21070     <xs:restriction base="xs:string"> </xs:restriction>
21071   </xs:simpleType>
21072   <xs:simpleType name="DescriptionType">
21073     <xs:restriction base="xs:string"> </xs:restriction>
21074   </xs:simpleType>
21075   <xs:simpleType name="SpecificationVersionType">
21076     <xs:restriction base="xs:string">
21077       <xs:pattern value="[1-9][0-9]*\.[0-9]+\.[0-9]+" />
21078       <xs:maxLength value="128" />
21079     </xs:restriction>
21080   </xs:simpleType>
21081   <xs:complexType name="TimePeriodType">
21082     <xs:sequence>
21083       <xs:element minOccurs="0" name="startTime"
21084         type="ns_p:AbsoluteOrRelativeTimeType"/>
21085       <xs:element minOccurs="0" name="endTime" type="ns_p:AbsoluteOrRelativeTimeType"/>
21086     </xs:sequence>
```

```
21087         </xs:sequence>
21088     </xs:complexType>
21089     <xs:complexType name="TimePeriodElementsType">
21090         <xs:sequence>
21091             <xs:element minOccurs="0" name="startTime" type="ns_p:ElementTagType"/>
21092             <xs:element minOccurs="0" name="endTime" type="ns_p:ElementTagType"/>
21093         </xs:sequence>
21094     </xs:complexType>
21095     <xs:complexType name="TimestampIntervalType">
21096         <xs:sequence>
21097             <xs:element name="startTime" type="ns_p:AbsoluteOrRelativeTimeType"
21098 minOccurs="0"/>
21099             <xs:element name="endTime" type="ns_p:AbsoluteOrRelativeTimeType" minOccurs="0"/>
21100         </xs:sequence>
21101     </xs:complexType>
21102     <xs:simpleType name="EnumExtendType">
21103         <xs:restriction base="xs:string">
21104             <xs:pattern value="_([1-9][0-9]*)|n:[a-zA-Z0-9-+)]_([p{Cc}\p{Cf}\p{Z}])+"/>
21105             <xs:maxLength value="256"/>
21106         </xs:restriction>
21107     </xs:simpleType>
21108     <xs:simpleType name="AbsoluteOrRelativeTimeType">
21109         <xs:union memberTypes="xs:duration xs:dateTime"/>
21110     </xs:simpleType>
21111     <xs:simpleType name="RecurringIntervalType">
21112         <xs:union memberTypes="ns_p:RecurringIntervalEnumType ns_p:EnumExtendType"/>
21113     </xs:simpleType>
21114     <xs:simpleType name="RecurringIntervalEnumType">
21115         <xs:restriction base="xs:string">
21116             <xs:enumeration value="yearly"/>
21117             <xs:enumeration value="monthly"/>
21118             <xs:enumeration value="weekly"/>
21119             <xs:enumeration value="daily"/>
21120             <xs:enumeration value="hourly"/>
21121             <xs:enumeration value="everyMinute"/>
21122             <xs:enumeration value="everySecond"/>
21123         </xs:restriction>
21124     </xs:simpleType>
21125     <xs:simpleType name="MonthType">
21126         <xs:restriction base="xs:string">
21127             <xs:enumeration value="january"/>
21128             <xs:enumeration value="february"/>
21129             <xs:enumeration value="march"/>
21130             <xs:enumeration value="april"/>
21131             <xs:enumeration value="may"/>
21132             <xs:enumeration value="june"/>
21133             <xs:enumeration value="july"/>
21134             <xs:enumeration value="august"/>
21135             <xs:enumeration value="september"/>
21136             <xs:enumeration value="october"/>
21137             <xs:enumeration value="november"/>
21138             <xs:enumeration value="december"/>
21139         </xs:restriction>
21140     </xs:simpleType>
21141     <xs:simpleType name="DayOfMonthType">
21142         <xs:restriction base="xs:unsignedByte">
21143             <xs:minInclusive value="1"/>
21144             <xs:maxInclusive value="31"/>
21145         </xs:restriction>
21146     </xs:simpleType>
21147     <xs:simpleType name="CalendarWeekType">
21148         <xs:restriction base="xs:unsignedByte">
21149             <xs:minInclusive value="1"/>
21150             <xs:maxInclusive value="53"/>
21151         </xs:restriction>
21152     </xs:simpleType>
21153     <xs:simpleType name="DayOfWeekType">
21154         <xs:restriction base="xs:string">
21155             <xs:enumeration value="monday"/>
21156             <xs:enumeration value="tuesday"/>
21157             <xs:enumeration value="wednesday"/>
21158             <xs:enumeration value="thursday"/>
21159             <xs:enumeration value="friday"/>
21160             <xs:enumeration value="saturday"/>
21161             <xs:enumeration value="sunday"/>
21162         </xs:restriction>
21163     </xs:simpleType>
21164     <xs:complexType name="DaysOfWeekType">
```



```
21165     <xs:sequence>
21166       <xs:element name="monday" minOccurs="0" type="ns_p:ElementTagType"/>
21167       <xs:element name="tuesday" minOccurs="0" type="ns_p:ElementTagType"/>
21168       <xs:element name="wednesday" minOccurs="0" type="ns_p:ElementTagType"/>
21169       <xs:element name="thursday" minOccurs="0" type="ns_p:ElementTagType"/>
21170       <xs:element name="friday" minOccurs="0" type="ns_p:ElementTagType"/>
21171       <xs:element name="saturday" minOccurs="0" type="ns_p:ElementTagType"/>
21172       <xs:element name="sunday" minOccurs="0" type="ns_p:ElementTagType"/>
21173     </xs:sequence>
21174   </xs:complexType>
21175   <xs:simpleType name="OccurrenceType">
21176     <xs:union memberTypes="ns_p:OccurrenceEnumType ns_p:EnumExtendType"/>
21177   </xs:simpleType>
21178   <xs:simpleType name="OccurrenceEnumType">
21179     <xs:restriction base="xs:string">
21180       <xs:enumeration value="first"/>
21181       <xs:enumeration value="second"/>
21182       <xs:enumeration value="third"/>
21183       <xs:enumeration value="fourth"/>
21184       <xs:enumeration value="last"/>
21185     </xs:restriction>
21186   </xs:simpleType>
21187   <xs:complexType name="AbsoluteOrRecurringTimeType">
21188     <xs:sequence>
21189       <xs:element minOccurs="0" name="dateTime" type="xs:dateTime"/>
21190       <xs:element name="month" minOccurs="0" type="ns_p:MonthType"/>
21191       <xs:element name="dayOfMonth" minOccurs="0" type="ns_p:DayOfMonthType"/>
21192       <xs:element minOccurs="0" name="calendarWeek" type="ns_p:CalendarWeekType"/>
21193       <xs:element minOccurs="0" name="dayOfWeekOccurrence" type="ns_p:OccurrenceType"/>
21194       <xs:element minOccurs="0" name="daysOfWeek" type="ns_p:DaysOfWeekType"/>
21195       <xs:element minOccurs="0" name="time" type="xs:time"/>
21196       <xs:element minOccurs="0" name="relative" type="xs:duration"/>
21197     </xs:sequence>
21198   </xs:complexType>
21199   <xs:complexType name="AbsoluteOrRecurringTimeElementsType">
21200     <xs:sequence>
21201       <xs:element minOccurs="0" name="dateTime" type="ns_p:ElementTagType"/>
21202       <xs:element name="month" minOccurs="0" type="ns_p:ElementTagType"/>
21203       <xs:element name="dayOfMonth" minOccurs="0" type="ns_p:ElementTagType"/>
21204       <xs:element minOccurs="0" name="calendarWeek" type="ns_p:ElementTagType"/>
21205       <xs:element minOccurs="0" name="dayOfWeekOccurrence" type="ns_p:ElementTagType"/>
21206       <xs:element minOccurs="0" name="daysOfWeek" type="ns_p:ElementTagType"/>
21207       <xs:element minOccurs="0" name="time" type="ns_p:ElementTagType"/>
21208       <xs:element minOccurs="0" name="relative" type="ns_p:ElementTagType"/>
21209     </xs:sequence>
21210   </xs:complexType>
21211   <xs:complexType name="RecurrenceInformationType">
21212     <xs:sequence>
21213       <xs:element minOccurs="0" name="recurringInterval"
21214 type="ns_p:RecurringIntervalType"/>
21215       <xs:element minOccurs="0" name="recurringIntervalStep" type="xs:unsignedInt"/>
21216       <xs:element minOccurs="0" name="firstExecution" type="xs:dateTime"/>
21217       <xs:element minOccurs="0" name="executionCount" type="xs:unsignedInt"/>
21218       <xs:element minOccurs="0" name="lastExecution" type="xs:dateTime"/>
21219     </xs:sequence>
21220   </xs:complexType>
21221   <xs:complexType name="RecurrenceInformationElementsType">
21222     <xs:sequence>
21223       <xs:element minOccurs="0" name="recurringInterval" type="ns_p:ElementTagType"/>
21224       <xs:element minOccurs="0" name="recurringIntervalStep"
21225 type="ns_p:ElementTagType"/>
21226       <xs:element minOccurs="0" name="firstExecution" type="ns_p:ElementTagType"/>
21227       <xs:element minOccurs="0" name="executionCount" type="ns_p:ElementTagType"/>
21228       <xs:element minOccurs="0" name="lastExecution" type="ns_p:ElementTagType"/>
21229     </xs:sequence>
21230   </xs:complexType>
21231   <xs:complexType name="ScaledNumberRangeType">
21232     <xs:sequence>
21233       <xs:element minOccurs="0" name="min" type="ns_p:ScaledNumberType"/>
21234       <xs:element minOccurs="0" name="max" type="ns_p:ScaledNumberType"/>
21235     </xs:sequence>
21236   </xs:complexType>
21237   <xs:complexType name="ScaledNumberRangeElementsType">
21238     <xs:sequence>
21239       <xs:element minOccurs="0" name="min" type="ns_p:ScaledNumberElementsType"/>
21240       <xs:element minOccurs="0" name="max" type="ns_p:ScaledNumberElementsType"/>
21241     </xs:sequence>
21242   </xs:complexType>
```

```
21243     <xs:complexType name="ScaledNumberSetType">
21244       <xs:sequence>
21245         <xs:element maxOccurs="unbounded" minOccurs="0" name="value"
21246 type="ns_p:ScaledNumberType"/>
21247         <xs:element maxOccurs="unbounded" minOccurs="0" name="range"
21248 type="ns_p:ScaledNumberRangeType"/>
21249       </xs:sequence>
21250     </xs:complexType>
21251     <xs:complexType name="ScaledNumberSetElementsType">
21252       <xs:sequence>
21253         <xs:element minOccurs="0" name="value" type="ns_p:ScaledNumberElementsType"/>
21254         <xs:element minOccurs="0" name="range" type="ns_p:ScaledNumberRangeElementsType"/>
21255       </xs:sequence>
21256     </xs:complexType>
21257     <xs:simpleType name="NumberType">
21258       <xs:restriction base="xs:long"/>
21259     </xs:simpleType>
21260     <xs:simpleType name="ScaleType">
21261       <xs:restriction base="xs:short"/>
21262     </xs:simpleType>
21263     <xs:complexType name="ScaledNumberType">
21264       <xs:sequence>
21265         <xs:element minOccurs="0" name="number" type="ns_p:NumberType"/>
21266         <xs:element minOccurs="0" name="scale" type="ns_p:ScaleType"/>
21267       </xs:sequence>
21268     </xs:complexType>
21269     <xs:complexType name="ScaledNumberElementsType">
21270       <xs:sequence>
21271         <xs:element minOccurs="0" name="number" type="ns_p:ElementTagType"/>
21272         <xs:element minOccurs="0" name="scale" type="ns_p:ElementTagType"/>
21273       </xs:sequence>
21274     </xs:complexType>
21275     <xs:simpleType name="MaxResponseDelayType">
21276       <xs:restriction base="xs:duration"/>
21277     </xs:simpleType>
21278     <xs:simpleType name="CommodityTypeType">
21279       <xs:union memberTypes="ns_p:CommodityTypeEnumType ns_p:EnumExtendType"/>
21280     </xs:simpleType>
21281     <xs:simpleType name="CommodityTypeEnumType">
21282       <xs:restriction base="xs:string">
21283         <xs:enumeration value="electricity"/>
21284         <xs:enumeration value="gas"/>
21285         <xs:enumeration value="oil"/>
21286         <xs:enumeration value="water"/>
21287         <xs:enumeration value="wasteWater"/>
21288         <xs:enumeration value="domesticHotWater"/>
21289         <xs:enumeration value="heatingWater"/>
21290         <xs:enumeration value="steam"/>
21291         <xs:enumeration value="heat"/>
21292         <xs:enumeration value="coolingLoad"/>
21293         <xs:enumeration value="air"/>
21294       </xs:restriction>
21295     </xs:simpleType>
21296     <xs:simpleType name="EnergyDirectionType">
21297       <xs:union memberTypes="ns_p:EnergyDirectionEnumType ns_p:EnumExtendType"/>
21298     </xs:simpleType>
21299     <xs:simpleType name="EnergyDirectionEnumType">
21300       <xs:restriction base="xs:string">
21301         <xs:enumeration value="consume"/>
21302         <xs:enumeration value="produce"/>
21303       </xs:restriction>
21304     </xs:simpleType>
21305     <xs:simpleType name="EnergyModeType">
21306       <xs:union memberTypes="ns_p:EnergyModeEnumType ns_p:EnumExtendType"/>
21307     </xs:simpleType>
21308     <xs:simpleType name="EnergyModeEnumType">
21309       <xs:restriction base="xs:string">
21310         <xs:enumeration value="consume"/>
21311         <xs:enumeration value="produce"/>
21312         <xs:enumeration value="idle"/>
21313         <xs:enumeration value="auto"/>
21314       </xs:restriction>
21315     </xs:simpleType>
21316     <xs:simpleType name="UnitOfMeasurementType">
21317       <xs:union memberTypes="ns_p:UnitOfMeasurementEnumType ns_p:EnumExtendType"/>
21318     </xs:simpleType>
21319     <xs:simpleType name="UnitOfMeasurementEnumType">
21320       <xs:restriction base="xs:string">
```

21321	<xs:enumeration value="unknown"/>
21322	<xs:enumeration value="1"/>
21323	<xs:enumeration value="m"/>
21324	<xs:enumeration value="kg"/>
21325	<xs:enumeration value="s"/>
21326	<xs:enumeration value="A"/>
21327	<xs:enumeration value="K"/>
21328	<xs:enumeration value="mol"/>
21329	<xs:enumeration value="cd"/>
21330	<xs:enumeration value="V"/>
21331	<xs:enumeration value="W"/>
21332	<xs:enumeration value="Wh"/>
21333	<xs:enumeration value="VA"/>
21334	<xs:enumeration value="VAh"/>
21335	<xs:enumeration value="var"/>
21336	<xs:enumeration value="varh"/>
21337	<xs:enumeration value="degC"/>
21338	<xs:enumeration value="degF"/>
21339	<xs:enumeration value="lm"/>
21340	<xs:enumeration value="lx"/>
21341	<xs:enumeration value="Ohm"/>
21342	<xs:enumeration value="Hz"/>
21343	<xs:enumeration value="dB"/>
21344	<xs:enumeration value="dBm"/>
21345	<xs:enumeration value="pct"/>
21346	<xs:enumeration value="ppm"/>
21347	<xs:enumeration value="l"/>
21348	<xs:enumeration value="l/s"/>
21349	<xs:enumeration value="l/h"/>
21350	<xs:enumeration value="deg"/>
21351	<xs:enumeration value="rad"/>
21352	<xs:enumeration value="rad/s"/>
21353	<xs:enumeration value="sr"/>
21354	<xs:enumeration value="Gy"/>
21355	<xs:enumeration value="Bq"/>
21356	<xs:enumeration value="Bq/m^3"/>
21357	<xs:enumeration value="Sv"/>
21358	<xs:enumeration value="Rd"/>
21359	<xs:enumeration value="C"/>
21360	<xs:enumeration value="F"/>
21361	<xs:enumeration value="H"/>
21362	<xs:enumeration value="J"/>
21363	<xs:enumeration value="N"/>
21364	<xs:enumeration value="N_m"/>
21365	<xs:enumeration value="N_s"/>
21366	<xs:enumeration value="Wb"/>
21367	<xs:enumeration value="T"/>
21368	<xs:enumeration value="Pa"/>
21369	<xs:enumeration value="bar"/>
21370	<xs:enumeration value="atm"/>
21371	<xs:enumeration value="psi"/>
21372	<xs:enumeration value="mmHg"/>
21373	<xs:enumeration value="m^2"/>
21374	<xs:enumeration value="m^3"/>
21375	<xs:enumeration value="m^3/h"/>
21376	<xs:enumeration value="m/s"/>
21377	<xs:enumeration value="m/s^2"/>
21378	<xs:enumeration value="m^3/s"/>
21379	<xs:enumeration value="m/m^3"/>
21380	<xs:enumeration value="kg/m^3"/>
21381	<xs:enumeration value="kg_m"/>
21382	<xs:enumeration value="m^2/s"/>
21383	<xs:enumeration value="W/m_K"/>
21384	<xs:enumeration value="J/K"/>
21385	<xs:enumeration value="l/s"/>
21386	<xs:enumeration value="W/m^2"/>
21387	<xs:enumeration value="J/m^2"/>
21388	<xs:enumeration value="S"/>
21389	<xs:enumeration value="S/m"/>
21390	<xs:enumeration value="K/s"/>
21391	<xs:enumeration value="Pa/s"/>
21392	<xs:enumeration value="J/kg_K"/>
21393	<xs:enumeration value="Vs"/>
21394	<xs:enumeration value="V/m"/>
21395	<xs:enumeration value="V/Hz"/>
21396	<xs:enumeration value="As"/>
21397	<xs:enumeration value="A/m"/>
21398	<xs:enumeration value="Hz/s"/>

```
21399         <xs:enumeration value="kg/s"/>
21400         <xs:enumeration value="kg_m^2"/>
21401         <xs:enumeration value="J/Wh"/>
21402         <xs:enumeration value="W/s"/>
21403         <xs:enumeration value="ft^3"/>
21404         <xs:enumeration value="ft^3/h"/>
21405         <xs:enumeration value="ccf"/>
21406         <xs:enumeration value="ccf/h"/>
21407         <xs:enumeration value="US.liq.gal"/>
21408         <xs:enumeration value="US.liq.gal/h"/>
21409         <xs:enumeration value="Imp.gal"/>
21410         <xs:enumeration value="Imp.gal/h"/>
21411         <xs:enumeration value="Btu"/>
21412         <xs:enumeration value="Btu/h"/>
21413         <xs:enumeration value="Ah"/>
21414         <xs:enumeration value="kg/Wh"/>
21415     </xs:restriction>
21416 </xs:simpleType>
21417 <xs:simpleType name="CurrencyType">
21418     <xs:union memberTypes="ns_p:CurrencyEnumType ns_p:EnumExtendType"/>
21419 </xs:simpleType>
21420 <xs:simpleType name="CurrencyEnumType">
21421     <xs:restriction base="xs:string">
21422         <xs:enumeration value="AED"/>
21423         <xs:enumeration value="AFN"/>
21424         <xs:enumeration value="ALL"/>
21425         <xs:enumeration value="AMD"/>
21426         <xs:enumeration value="ANG"/>
21427         <xs:enumeration value="AOA"/>
21428         <xs:enumeration value="ARS"/>
21429         <xs:enumeration value="AUD"/>
21430         <xs:enumeration value="AWG"/>
21431         <xs:enumeration value="AZN"/>
21432         <xs:enumeration value="BAM"/>
21433         <xs:enumeration value="BBD"/>
21434         <xs:enumeration value="BDT"/>
21435         <xs:enumeration value="BGN"/>
21436         <xs:enumeration value="BHD"/>
21437         <xs:enumeration value="BIF"/>
21438         <xs:enumeration value="BMD"/>
21439         <xs:enumeration value="BND"/>
21440         <xs:enumeration value="BOB"/>
21441         <xs:enumeration value="BOV"/>
21442         <xs:enumeration value="BRL"/>
21443         <xs:enumeration value="BSD"/>
21444         <xs:enumeration value="BTN"/>
21445         <xs:enumeration value="BWP"/>
21446         <xs:enumeration value="BYR"/>
21447         <xs:enumeration value="BZD"/>
21448         <xs:enumeration value="CAD"/>
21449         <xs:enumeration value="CDF"/>
21450         <xs:enumeration value="CHE"/>
21451         <xs:enumeration value="CHF"/>
21452         <xs:enumeration value="CHW"/>
21453         <xs:enumeration value="CLF"/>
21454         <xs:enumeration value="CLP"/>
21455         <xs:enumeration value="CNY"/>
21456         <xs:enumeration value="COP"/>
21457         <xs:enumeration value="COU"/>
21458         <xs:enumeration value="CRC"/>
21459         <xs:enumeration value="CUC"/>
21460         <xs:enumeration value="CUP"/>
21461         <xs:enumeration value="CVE"/>
21462         <xs:enumeration value="CZK"/>
21463         <xs:enumeration value="DJF"/>
21464         <xs:enumeration value="DKK"/>
21465         <xs:enumeration value="DOP"/>
21466         <xs:enumeration value="DZD"/>
21467         <xs:enumeration value="EGP"/>
21468         <xs:enumeration value="ERN"/>
21469         <xs:enumeration value="ETB"/>
21470         <xs:enumeration value="EUR"/>
21471         <xs:enumeration value="FJD"/>
21472         <xs:enumeration value="FKP"/>
21473         <xs:enumeration value="GBP"/>
21474         <xs:enumeration value="GEL"/>
21475         <xs:enumeration value="GHS"/>
21476         <xs:enumeration value="GIP"/>
```

21477	<xs:enumeration value="GMD"/>
21478	<xs:enumeration value="GNF"/>
21479	<xs:enumeration value="GTQ"/>
21480	<xs:enumeration value="GYD"/>
21481	<xs:enumeration value="HKD"/>
21482	<xs:enumeration value="HNL"/>
21483	<xs:enumeration value="HRK"/>
21484	<xs:enumeration value="HTG"/>
21485	<xs:enumeration value="HUF"/>
21486	<xs:enumeration value="IDR"/>
21487	<xs:enumeration value="ILS"/>
21488	<xs:enumeration value="INR"/>
21489	<xs:enumeration value="IQD"/>
21490	<xs:enumeration value="IRR"/>
21491	<xs:enumeration value="ISK"/>
21492	<xs:enumeration value="JMD"/>
21493	<xs:enumeration value="JOD"/>
21494	<xs:enumeration value="JPY"/>
21495	<xs:enumeration value="KES"/>
21496	<xs:enumeration value="KGS"/>
21497	<xs:enumeration value="KHR"/>
21498	<xs:enumeration value="KMF"/>
21499	<xs:enumeration value="KPW"/>
21500	<xs:enumeration value="KRW"/>
21501	<xs:enumeration value="KWD"/>
21502	<xs:enumeration value="KYD"/>
21503	<xs:enumeration value="KZT"/>
21504	<xs:enumeration value="LAK"/>
21505	<xs:enumeration value="LBP"/>
21506	<xs:enumeration value="LKR"/>
21507	<xs:enumeration value="LRD"/>
21508	<xs:enumeration value="LSL"/>
21509	<xs:enumeration value="LYD"/>
21510	<xs:enumeration value="MAD"/>
21511	<xs:enumeration value="MDL"/>
21512	<xs:enumeration value="MGA"/>
21513	<xs:enumeration value="MKD"/>
21514	<xs:enumeration value="MMK"/>
21515	<xs:enumeration value="MNT"/>
21516	<xs:enumeration value="MOP"/>
21517	<xs:enumeration value="MRO"/>
21518	<xs:enumeration value="MUR"/>
21519	<xs:enumeration value="MVR"/>
21520	<xs:enumeration value="MWK"/>
21521	<xs:enumeration value="MXN"/>
21522	<xs:enumeration value="MXV"/>
21523	<xs:enumeration value="MYR"/>
21524	<xs:enumeration value="MZN"/>
21525	<xs:enumeration value="NAD"/>
21526	<xs:enumeration value="NGN"/>
21527	<xs:enumeration value="NIO"/>
21528	<xs:enumeration value="NOK"/>
21529	<xs:enumeration value="NPR"/>
21530	<xs:enumeration value="NZD"/>
21531	<xs:enumeration value="OMR"/>
21532	<xs:enumeration value="PAB"/>
21533	<xs:enumeration value="PEN"/>
21534	<xs:enumeration value="PGK"/>
21535	<xs:enumeration value="PHP"/>
21536	<xs:enumeration value="PKR"/>
21537	<xs:enumeration value="PLN"/>
21538	<xs:enumeration value="PYG"/>
21539	<xs:enumeration value="QAR"/>
21540	<xs:enumeration value="RON"/>
21541	<xs:enumeration value="RSD"/>
21542	<xs:enumeration value="RUB"/>
21543	<xs:enumeration value="RWF"/>
21544	<xs:enumeration value="SAR"/>
21545	<xs:enumeration value="SBD"/>
21546	<xs:enumeration value="SCR"/>
21547	<xs:enumeration value="SDG"/>
21548	<xs:enumeration value="SEK"/>
21549	<xs:enumeration value="SGD"/>
21550	<xs:enumeration value="SHP"/>
21551	<xs:enumeration value="SLL"/>
21552	<xs:enumeration value="SOS"/>
21553	<xs:enumeration value="SRD"/>
21554	<xs:enumeration value="SSP"/>

```
21555 <xs:enumeration value="STD"/>
21556 <xs:enumeration value="SVC"/>
21557 <xs:enumeration value="SYP"/>
21558 <xs:enumeration value="SZL"/>
21559 <xs:enumeration value="THB"/>
21560 <xs:enumeration value="TJS"/>
21561 <xs:enumeration value="TMT"/>
21562 <xs:enumeration value="TND"/>
21563 <xs:enumeration value="TOP"/>
21564 <xs:enumeration value="TRY"/>
21565 <xs:enumeration value="TTD"/>
21566 <xs:enumeration value="TWD"/>
21567 <xs:enumeration value="TZS"/>
21568 <xs:enumeration value="UAH"/>
21569 <xs:enumeration value="UGX"/>
21570 <xs:enumeration value="USD"/>
21571 <xs:enumeration value="USN"/>
21572 <xs:enumeration value="UYI"/>
21573 <xs:enumeration value="UYU"/>
21574 <xs:enumeration value="UZS"/>
21575 <xs:enumeration value="VEF"/>
21576 <xs:enumeration value="VND"/>
21577 <xs:enumeration value="VUV"/>
21578 <xs:enumeration value="WST"/>
21579 <xs:enumeration value="XAF"/>
21580 <xs:enumeration value="XAG"/>
21581 <xs:enumeration value="XAU"/>
21582 <xs:enumeration value="XBA"/>
21583 <xs:enumeration value="XBB"/>
21584 <xs:enumeration value="XBC"/>
21585 <xs:enumeration value="XBD"/>
21586 <xs:enumeration value="XCD"/>
21587 <xs:enumeration value="XDR"/>
21588 <xs:enumeration value="XOF"/>
21589 <xs:enumeration value="XPD"/>
21590 <xs:enumeration value="XPF"/>
21591 <xs:enumeration value="XPT"/>
21592 <xs:enumeration value="XSU"/>
21593 <xs:enumeration value="XTS"/>
21594 <xs:enumeration value="XUA"/>
21595 <xs:enumeration value="XXX"/>
21596 <xs:enumeration value="YER"/>
21597 <xs:enumeration value="ZAR"/>
21598 <xs:enumeration value="ZMW"/>
21599 <xs:enumeration value="ZWL"/>
21600 </xs:restriction>
21601 </xs:simpleType>
21602 <xs:simpleType name="AddressDeviceType">
21603 <xs:restriction base="xs:string">
21604 <xs:pattern value="d:_(i:[1-9][0-9]*|n:[a-zA-Z0-9-]+)_[^\\p{Cc}\\p{Cf}\\p{Z}]+"/>
21605 <xs:maxLength value="256"/>
21606 </xs:restriction>
21607 </xs:simpleType>
21608 <xs:simpleType name="AddressEntityType">
21609 <xs:restriction base="xs:unsignedInt"/>
21610 </xs:simpleType>
21611 <xs:simpleType name="AddressFeatureType">
21612 <xs:restriction base="xs:unsignedInt"/>
21613 </xs:simpleType>
21614 <xs:element name="device" type="ns_p:AddressDeviceType"/>
21615 <xs:element name="entity" type="ns_p:AddressEntityType"/>
21616 <xs:element name="feature" type="ns_p:AddressFeatureType"/>
21617 <xs:complexType name="DeviceAddressType">
21618 <xs:sequence>
21619 <xs:element minOccurs="0" ref="ns_p:device"/>
21620 </xs:sequence>
21621 </xs:complexType>
21622 <xs:complexType name="DeviceAddressElementsType">
21623 <xs:sequence>
21624 <xs:element minOccurs="0" name="device" type="ns_p:ElementTagType"/>
21625 </xs:sequence>
21626 </xs:complexType>
21627 <xs:complexType name="EntityAddressType">
21628 <xs:complexContent>
21629 <xs:extension base="ns_p:DeviceAddressType">
21630 <xs:sequence>
21631 <xs:element minOccurs="0" ref="ns_p:entity" maxOccurs="unbounded"/>
21632 </xs:sequence>
```

```
21633         </xs:extension>
21634     </xs:complexContent>
21635 </xs:complexType>
21636 <xs:complexType name="EntityAddressElementsType">
21637     <xs:complexContent>
21638         <xs:extension base="ns_p:DeviceAddressElementsType">
21639             <xs:sequence>
21640                 <xs:element minOccurs="0" name="entity" type="ns_p:ElementTagType"/>
21641             </xs:sequence>
21642         </xs:extension>
21643     </xs:complexContent>
21644 </xs:complexType>
21645 <xs:complexType name="FeatureAddressType">
21646     <xs:complexContent>
21647         <xs:extension base="ns_p:EntityAddressType">
21648             <xs:sequence>
21649                 <xs:element minOccurs="0" ref="ns_p:feature"/>
21650             </xs:sequence>
21651         </xs:extension>
21652     </xs:complexContent>
21653 </xs:complexType>
21654 <xs:complexType name="FeatureAddressElementsType">
21655     <xs:complexContent>
21656         <xs:extension base="ns_p:EntityAddressElementsType">
21657             <xs:sequence>
21658                 <xs:element minOccurs="0" name="feature" type="ns_p:ElementTagType"/>
21659             </xs:sequence>
21660         </xs:extension>
21661     </xs:complexContent>
21662 </xs:complexType>
21663 <xs:simpleType name="ScopeTypeType">
21664     <xs:union memberTypes="ns_p:ScopeTypeEnumType ns_p:EnumExtendType"/>
21665 </xs:simpleType>
21666 <xs:simpleType name="ScopeTypeEnumType">
21667     <xs:restriction base="xs:string">
21668         <xs:enumeration value="ac"/>
21669         <xs:enumeration value="acCosPhiGrid"/>
21670         <xs:enumeration value="acCurrentA"/>
21671         <xs:enumeration value="acCurrentB"/>
21672         <xs:enumeration value="acCurrentC"/>
21673         <xs:enumeration value="acFrequencyGrid"/>
21674         <xs:enumeration value="acPowerA"/>
21675         <xs:enumeration value="acPowerB"/>
21676         <xs:enumeration value="acPowerC"/>
21677         <xs:enumeration value="acPowerLimitPct"/>
21678         <xs:enumeration value="acPowerTotal"/>
21679         <xs:enumeration value="acVoltageA"/>
21680         <xs:enumeration value="acVoltageB"/>
21681         <xs:enumeration value="acVoltageC"/>
21682         <xs:enumeration value="acYieldDay"/>
21683         <xs:enumeration value="acYieldTotal"/>
21684         <xs:enumeration value="dcCurrent"/>
21685         <xs:enumeration value="dcPower"/>
21686         <xs:enumeration value="dcString1"/>
21687         <xs:enumeration value="dcString2"/>
21688         <xs:enumeration value="dcString3"/>
21689         <xs:enumeration value="dcString4"/>
21690         <xs:enumeration value="dcString5"/>
21691         <xs:enumeration value="dcString6"/>
21692         <xs:enumeration value="dcTotal"/>
21693         <xs:enumeration value="dcVoltage"/>
21694         <xs:enumeration value="dhwTemperature"/>
21695         <xs:enumeration value="flowTemperature"/>
21696         <xs:enumeration value="outsideAirTemperature"/>
21697         <xs:enumeration value="returnTemperature"/>
21698         <xs:enumeration value="roomAirTemperature"/>
21699         <xs:enumeration value="charge"/>
21700         <xs:enumeration value="stateOfCharge"/>
21701         <xs:enumeration value="discharge"/>
21702         <xs:enumeration value="gridConsumption"/>
21703         <xs:enumeration value="gridFeedIn"/>
21704         <xs:enumeration value="selfConsumption"/>
21705         <xs:enumeration value="overloadProtection"/>
21706         <xs:enumeration value="acPower"/>
21707         <xs:enumeration value="acEnergy"/>
21708         <xs:enumeration value="acCurrent"/>
21709         <xs:enumeration value="acVoltage"/>
21710         <xs:enumeration value="batteryControl"/>
```

```
21711 <xs:enumeration value="simpleIncentiveTable"/>
21712 <xs:enumeration value="stateOfHealth"/>
21713 <xs:enumeration value="travelRange"/>
21714 <xs:enumeration value="nominalEnergyCapacity"/>
21715 <xs:enumeration value="acPowerReal"/>
21716 <xs:enumeration value="acPowerApparent"/>
21717 <xs:enumeration value="acPowerReactive"/>
21718 <xs:enumeration value="acYieldMonth"/>
21719 <xs:enumeration value="acYieldYear"/>
21720 <xs:enumeration value="acFrequency"/>
21721 <xs:enumeration value="acCosPhi"/>
21722 <xs:enumeration value="dcEnergy"/>
21723 <xs:enumeration value="insulationResistance"/>
21724 <xs:enumeration value="stateOfEnergy"/>
21725 <xs:enumeration value="useableCapacity"/>
21726 <xs:enumeration value="dcChargeEnergy"/>
21727 <xs:enumeration value="dcDischargeEnergy"/>
21728 <xs:enumeration value="loadCycleCount"/>
21729 <xs:enumeration value="componentTemperature"/>
21730 <xs:enumeration value="gridLimit"/>
21731 <xs:enumeration value="gridLimitFallback"/>
21732 <xs:enumeration value="acPowerApparentTotal"/>
21733 <xs:enumeration value="acPowerReactiveTotal"/>
21734 <xs:enumeration value="acCurrentTotal"/>
21735 <xs:enumeration value="acEnergyConsumed"/>
21736 <xs:enumeration value="acEnergyProduced"/>
21737 <xs:enumeration value="batteryAcPower"/>
21738 <xs:enumeration value="batteryAcPowerPhaseSpecific"/>
21739 <xs:enumeration value="batteryDcPower"/>
21740 <xs:enumeration value="pccPower"/>
21741 <xs:enumeration value="activePowerLimit"/>
21742 <xs:enumeration value="activePowerLimitPercentage"/>
21743 <xs:enumeration value="simpleCommittedIncentiveTable"/>
21744 <xs:enumeration value="simplePreliminaryIncentiveTable"/>
21745 <xs:enumeration value="committedPowerPlan"/>
21746 <xs:enumeration value="preliminaryPowerPlan"/>
21747 <xs:enumeration value="incentiveTableEnConsWithPoETF"/>
21748 <xs:enumeration value="incentiveTableEnProdWithPoETF"/>
21749 <xs:enumeration value="incentiveTableEnConsWithPoE"/>
21750 <xs:enumeration value="incentiveTableEnProdWithPoE"/>
21751 <xs:enumeration value="incentiveTableEnConsWithTF"/>
21752 <xs:enumeration value="incentiveTableEnProdWithTF"/>
21753 <xs:enumeration value="activePowerForecast"/>
21754 </xs:restriction>
21755 </xs:simpleType>
21756 <xs:simpleType name="RoleType">
21757 <xs:restriction base="xs:string">
21758 <xs:enumeration value="client"/>
21759 <xs:enumeration value="server"/>
21760 <xs:enumeration value="special"/>
21761 </xs:restriction>
21762 </xs:simpleType>
21763 <xs:simpleType name="FeatureGroupType">
21764 <xs:restriction base="xs:string">
21765 <xs:pattern value="([1-9][0-9]*)"/>
21766 <xs:maxLength value="128"/>
21767 </xs:restriction>
21768 </xs:simpleType>
21769 <xs:simpleType name="DeviceTypeType">
21770 <xs:union memberTypes="ns_p:DeviceTypeEnumType ns_p:EnumExtendType"/>
21771 </xs:simpleType>
21772 <xs:simpleType name="DeviceTypeEnumType">
21773 <xs:restriction base="xs:string">
21774 <xs:enumeration value="Dishwasher"/>
21775 <xs:enumeration value="Dryer"/>
21776 <xs:enumeration value="EnvironmentSensor"/>
21777 <xs:enumeration value="Generic"/>
21778 <xs:enumeration value="HeatGenerationSystem"/>
21779 <xs:enumeration value="HeatSinkSystem"/>
21780 <xs:enumeration value="HeatStorageSystem"/>
21781 <xs:enumeration value="HVACController"/>
21782 <xs:enumeration value="SubMeter"/>
21783 <xs:enumeration value="Washer"/>
21784 <xs:enumeration value="ElectricitySupplySystem"/>
21785 <xs:enumeration value="EnergyManagementSystem"/>
21786 <xs:enumeration value="Inverter"/>
21787 <xs:enumeration value="ChargingStation"/>
21788 </xs:restriction>
```



```
21789 </xs:simpleType>
21790 <xs:simpleType name="EntityTypeType">
21791   <xs:union memberTypes="ns_p:EntityTypeEnumType ns_p:EnumExtendType"/>
21792 </xs:simpleType>
21793 <xs:simpleType name="EntityTypeEnumType">
21794   <xs:restriction base="xs:string">
21795     <xs:enumeration value="Battery"/>
21796     <xs:enumeration value="Compressor"/>
21797     <xs:enumeration value="DeviceInformation"/>
21798     <xs:enumeration value="DHWcircuit"/>
21799     <xs:enumeration value="DHWStorage"/>
21800     <xs:enumeration value="Dishwasher"/>
21801     <xs:enumeration value="Dryer"/>
21802     <xs:enumeration value="ElectricalImmersionHeater"/>
21803     <xs:enumeration value="Fan"/>
21804     <xs:enumeration value="GasHeatingAppliance"/>
21805     <xs:enumeration value="Generic"/>
21806     <xs:enumeration value="HeatingBufferStorage"/>
21807     <xs:enumeration value="HeatingCircuit"/>
21808     <xs:enumeration value="HeatingObject"/>
21809     <xs:enumeration value="HeatingZone"/>
21810     <xs:enumeration value="HeatPumpAppliance"/>
21811     <xs:enumeration value="HeatSinkCircuit"/>
21812     <xs:enumeration value="HeatSourceCircuit"/>
21813     <xs:enumeration value="HeatSourceUnit"/>
21814     <xs:enumeration value="HVACController"/>
21815     <xs:enumeration value="HVACRoom"/>
21816     <xs:enumeration value="InstantDHWHeater"/>
21817     <xs:enumeration value="Inverter"/>
21818     <xs:enumeration value="OilHeatingAppliance"/>
21819     <xs:enumeration value="Pump"/>
21820     <xs:enumeration value="RefrigerantCircuit"/>
21821     <xs:enumeration value="SmartEnergyAppliance"/>
21822     <xs:enumeration value="SolarDHWStorage"/>
21823     <xs:enumeration value="SolarThermalCircuit"/>
21824     <xs:enumeration value="SubMeterElectricity"/>
21825     <xs:enumeration value="TemperatureSensor"/>
21826     <xs:enumeration value="Washer"/>
21827     <xs:enumeration value="BatterySystem"/>
21828     <xs:enumeration value="ElectricityGenerationSystem"/>
21829     <xs:enumeration value="ElectricityStorageSystem"/>
21830     <xs:enumeration value="GridConnectionPointOfPremises"/>
21831     <xs:enumeration value="Household"/>
21832     <xs:enumeration value="PVSystem"/>
21833     <xs:enumeration value="EV"/>
21834     <xs:enumeration value="EVSE"/>
21835     <xs:enumeration value="ChargingOutlet"/>
21836     <xs:enumeration value="CEM"/>
21837     <xs:enumeration value="PV"/>
21838     <xs:enumeration value="PVESHybrid"/>
21839     <xs:enumeration value="ElectricalStorage"/>
21840     <xs:enumeration value="PVString"/>
21841     <xs:enumeration value="GridGuard"/>
21842     <xs:enumeration value="ControllableSystem"/>
21843   </xs:restriction>
21844 </xs:simpleType>
21845 <xs:simpleType name="FeatureTypeType">
21846   <xs:union memberTypes="ns_p:FeatureTypeEnumType ns_p:EnumExtendType"/>
21847 </xs:simpleType>
21848 <xs:simpleType name="FeatureTypeEnumType">
21849   <xs:restriction base="xs:string">
21850     <xs:enumeration value="ActuatorLevel"/>
21851     <xs:enumeration value="ActuatorSwitch"/>
21852     <xs:enumeration value="Alarm"/>
21853     <xs:enumeration value="DataTunneling"/>
21854     <xs:enumeration value="DeviceClassification"/>
21855     <xs:enumeration value="DeviceDiagnosis"/>
21856     <xs:enumeration value="DirectControl"/>
21857     <xs:enumeration value="ElectricalConnection"/>
21858     <xs:enumeration value="Generic"/>
21859     <xs:enumeration value="HVAC"/>
21860     <xs:enumeration value="LoadControl"/>
21861     <xs:enumeration value="Measurement"/>
21862     <xs:enumeration value="Messaging"/>
21863     <xs:enumeration value="NetworkManagement"/>
21864     <xs:enumeration value="NodeManagement"/>
21865     <xs:enumeration value="OperatingConstraints"/>
21866     <xs:enumeration value="PowerSequences"/>
```

```
21867         <xs:enumeration value="Sensing"/>
21868         <xs:enumeration value="Setpoint"/>
21869         <xs:enumeration value="SmartEnergyManagementPs"/>
21870         <xs:enumeration value="TaskManagement"/>
21871         <xs:enumeration value="Threshold"/>
21872         <xs:enumeration value="TimeInformation"/>
21873         <xs:enumeration value="TimeTable"/>
21874         <xs:enumeration value="DeviceConfiguration"/>
21875         <xs:enumeration value="SupplyCondition"/>
21876         <xs:enumeration value="TimeSeries"/>
21877         <xs:enumeration value="TariffInformation"/>
21878         <xs:enumeration value="IncentiveTable"/>
21879         <xs:enumeration value="Bill"/>
21880         <xs:enumeration value="Identification"/>
21881         <xs:enumeration value="StateInformation"/>
21882     </xs:restriction>
21883 </xs:simpleType>
21884 <xs:simpleType name="FeatureSpecificUsageType">
21885     <xs:union memberTypes="ns_p:FeatureSpecificUsageEnumType ns_p:EnumExtendType"/>
21886 </xs:simpleType>
21887 <xs:simpleType name="FeatureSpecificUsageEnumType">
21888     <xs:union
21889         memberTypes="ns_p:FeatureDirectControlSpecificUsageEnumType
21890 ns_p:FeatureHvacSpecificUsageEnumType ns_p:FeatureMeasurementSpecificUsageEnumType
21891 ns_p:FeatureSetpointSpecificUsageEnumType
21892 ns_p:FeatureSmartEnergyManagementPsSpecificUsageEnumType"
21893     />
21894 </xs:simpleType>
21895 <xs:simpleType name="FeatureDirectControlSpecificUsageEnumType">
21896     <xs:restriction base="xs:string">
21897         <xs:enumeration value="History"/>
21898         <xs:enumeration value="RealTime"/>
21899     </xs:restriction>
21900 </xs:simpleType>
21901 <xs:simpleType name="FeatureHvacSpecificUsageEnumType">
21902     <xs:restriction base="xs:string">
21903         <xs:enumeration value="OperationMode"/>
21904         <xs:enumeration value="Overrun"/>
21905     </xs:restriction>
21906 </xs:simpleType>
21907 <xs:simpleType name="FeatureMeasurementSpecificUsageEnumType">
21908     <xs:restriction base="xs:string">
21909         <xs:enumeration value="Contact"/>
21910         <xs:enumeration value="Electrical"/>
21911         <xs:enumeration value="Heat"/>
21912         <xs:enumeration value="Level"/>
21913         <xs:enumeration value="Pressure"/>
21914         <xs:enumeration value="Temperature"/>
21915     </xs:restriction>
21916 </xs:simpleType>
21917 <xs:simpleType name="FeatureSetpointSpecificUsageEnumType">
21918     <xs:restriction base="ns_p:FeatureMeasurementSpecificUsageEnumType"/>
21919 </xs:simpleType>
21920 <xs:simpleType name="FeatureSmartEnergyManagementPsSpecificUsageEnumType">
21921     <xs:restriction base="xs:string">
21922         <xs:enumeration value="FixedForecast"/>
21923         <xs:enumeration value="FlexibleChosenForecast"/>
21924         <xs:enumeration value="FlexibleOptionalForecast"/>
21925         <xs:enumeration value="OptionalSequenceBasedImmediateControl"/>
21926     </xs:restriction>
21927 </xs:simpleType>
21928 <xs:simpleType name="FunctionType">
21929     <xs:union memberTypes="ns_p:FunctionEnumType ns_p:EnumExtendType"/>
21930 </xs:simpleType>
21931 <xs:simpleType name="FunctionEnumType">
21932     <xs:restriction base="xs:string">
21933         <xs:enumeration value="actuatorLevelData"/>
21934         <xs:enumeration value="actuatorLevelDescriptionData"/>
21935         <xs:enumeration value="actuatorSwitchData"/>
21936         <xs:enumeration value="actuatorSwitchDescriptionData"/>
21937         <xs:enumeration value="alarmListData"/>
21938         <xs:enumeration value="bindingManagementDeleteCall"/>
21939         <xs:enumeration value="bindingManagementEntryListData"/>
21940         <xs:enumeration value="bindingManagementRequestCall"/>
21941         <xs:enumeration value="dataTunnelingCall"/>
21942         <xs:enumeration value="deviceClassificationManufacturerData"/>
21943         <xs:enumeration value="deviceClassificationUserData"/>
21944         <xs:enumeration value="deviceDiagnosisHeartbeatData"/>
```

```
21945 <xs:enumeration value="deviceDiagnosisServiceData"/>
21946 <xs:enumeration value="deviceDiagnosisStateData"/>
21947 <xs:enumeration value="directControlActivityListData"/>
21948 <xs:enumeration value="directControlDescriptionData"/>
21949 <xs:enumeration value="electricalConnectionDescriptionListData"/>
21950 <xs:enumeration value="electricalConnectionParameterDescriptionListData"/>
21951 <xs:enumeration value="electricalConnectionStateListData"/>
21952 <xs:enumeration value="hvacOperationModeDescriptionListData"/>
21953 <xs:enumeration value="hvacOverrunDescriptionListData"/>
21954 <xs:enumeration value="hvacOverrunListData"/>
21955 <xs:enumeration value="hvacSystemFunctionDescriptionListData"/>
21956 <xs:enumeration value="hvacSystemFunctionListData"/>
21957 <xs:enumeration value="hvacSystemFunctionOperationModeRelationListData"/>
21958 <xs:enumeration value="hvacSystemFunctionPowerSequenceRelationListData"/>
21959 <xs:enumeration value="hvacSystemFunctionSetpointRelationListData"/>
21960 <xs:enumeration value="loadControlEventListData"/>
21961 <xs:enumeration value="loadControlStateListData"/>
21962 <xs:enumeration value="measurementConstraintsListData"/>
21963 <xs:enumeration value="measurementDescriptionListData"/>
21964 <xs:enumeration value="measurementListData"/>
21965 <xs:enumeration value="measurementThresholdRelationListData"/>
21966 <xs:enumeration value="messagingListData"/>
21967 <xs:enumeration value="networkManagementAbortCall"/>
21968 <xs:enumeration value="networkManagementAddNodeCall"/>
21969 <xs:enumeration value="networkManagementDeviceDescriptionListData"/>
21970 <xs:enumeration value="networkManagementDiscoverCall"/>
21971 <xs:enumeration value="networkManagementEntityDescriptionListData"/>
21972 <xs:enumeration value="networkManagementFeatureDescriptionListData"/>
21973 <xs:enumeration value="networkManagementJoiningModeData"/>
21974 <xs:enumeration value="networkManagementModifyNodeCall"/>
21975 <xs:enumeration value="networkManagementProcessStateData"/>
21976 <xs:enumeration value="networkManagementRemoveNodeCall"/>
21977 <xs:enumeration value="networkManagementReportCandidateData"/>
21978 <xs:enumeration value="networkManagementScanNetworkCall"/>
21979 <xs:enumeration value="nodeManagementBindingData"/>
21980 <xs:enumeration value="nodeManagementBindingDeleteCall"/>
21981 <xs:enumeration value="nodeManagementBindingRequestCall"/>
21982 <xs:enumeration value="nodeManagementDestinationListData"/>
21983 <xs:enumeration value="nodeManagementDetailedDiscoveryData"/>
21984 <xs:enumeration value="nodeManagementSubscriptionData"/>
21985 <xs:enumeration value="nodeManagementSubscriptionDeleteCall"/>
21986 <xs:enumeration value="nodeManagementSubscriptionRequestCall"/>
21987 <xs:enumeration value="operatingConstraintsDurationListData"/>
21988 <xs:enumeration value="operatingConstraintsInterruptListData"/>
21989 <xs:enumeration value="operatingConstraintsPowerDescriptionListData"/>
21990 <xs:enumeration value="operatingConstraintsPowerLevelListData"/>
21991 <xs:enumeration value="operatingConstraintsPowerRangeListData"/>
21992 <xs:enumeration value="operatingConstraintsResumeImplicationListData"/>
21993 <xs:enumeration value="powerSequenceAlternativesRelationListData"/>
21994 <xs:enumeration value="powerSequenceDescriptionListData"/>
21995 <xs:enumeration value="powerSequenceNodeScheduleInformationData"/>
21996 <xs:enumeration value="powerSequencePriceCalculationRequestCall"/>
21997 <xs:enumeration value="powerSequencePriceListData"/>
21998 <xs:enumeration value="powerSequenceScheduleConfigurationRequestCall"/>
21999 <xs:enumeration value="powerSequenceScheduleConstraintsListData"/>
22000 <xs:enumeration value="powerSequenceScheduleListData"/>
22001 <xs:enumeration value="powerSequenceSchedulePreferenceListData"/>
22002 <xs:enumeration value="powerSequenceStateListData"/>
22003 <xs:enumeration value="powerTimeSlotScheduleConstraintsListData"/>
22004 <xs:enumeration value="powerTimeSlotScheduleListData"/>
22005 <xs:enumeration value="powerTimeSlotValueListData"/>
22006 <xs:enumeration value="resultData"/>
22007 <xs:enumeration value="sensingDescriptionData"/>
22008 <xs:enumeration value="sensingListData"/>
22009 <xs:enumeration value="sessionIdentificationListData"/>
22010 <xs:enumeration value="sessionMeasurementRelationListData"/>
22011 <xs:enumeration value="setpointConstraintsListData"/>
22012 <xs:enumeration value="setpointDescriptionListData"/>
22013 <xs:enumeration value="setpointListData"/>
22014 <xs:enumeration value="smartEnergyManagementPsConfigurationRequestCall"/>
22015 <xs:enumeration value="smartEnergyManagementPsData"/>
22016 <xs:enumeration value="smartEnergyManagementPsPriceCalculationRequestCall"/>
22017 <xs:enumeration value="smartEnergyManagementPsPriceData"/>
22018 <xs:enumeration value="specificationVersionListData"/>
22019 <xs:enumeration value="subscriptionManagementDeleteCall"/>
22020 <xs:enumeration value="subscriptionManagementEntryListData"/>
22021 <xs:enumeration value="subscriptionManagementRequestCall"/>
22022 <xs:enumeration value="supplyConditionDescriptionListData"/>
```

```
22023 <xs:enumeration value="supplyConditionListData"/>
22024 <xs:enumeration value="supplyConditionThresholdRelationListData"/>
22025 <xs:enumeration value="taskManagementJobDescriptionListData"/>
22026 <xs:enumeration value="taskManagementJobListData"/>
22027 <xs:enumeration value="taskManagementJobRelationListData"/>
22028 <xs:enumeration value="taskManagementOverviewData"/>
22029 <xs:enumeration value="thresholdConstraintsListData"/>
22030 <xs:enumeration value="thresholdDescriptionListData"/>
22031 <xs:enumeration value="thresholdListData"/>
22032 <xs:enumeration value="timeDistributorData"/>
22033 <xs:enumeration value="timeDistributorEnquiryCall"/>
22034 <xs:enumeration value="timeInformationData"/>
22035 <xs:enumeration value="timePrecisionData"/>
22036 <xs:enumeration value="timeTableConstraintsListData"/>
22037 <xs:enumeration value="timeTableDescriptionListData"/>
22038 <xs:enumeration value="timeTableListData"/>
22039 <xs:enumeration value="deviceConfigurationKeyValueConstraintsListData"/>
22040 <xs:enumeration value="deviceConfigurationKeyValueListData"/>
22041 <xs:enumeration value="deviceConfigurationKeyValueDescriptionListData"/>
22042 <xs:enumeration value="loadControlLimitConstraintsListData"/>
22043 <xs:enumeration value="loadControlLimitDescriptionListData"/>
22044 <xs:enumeration value="loadControlLimitListData"/>
22045 <xs:enumeration value="loadControlNodeData"/>
22046 <xs:enumeration value="timeSeriesConstraintsListData"/>
22047 <xs:enumeration value="timeSeriesDescriptionListData"/>
22048 <xs:enumeration value="timeSeriesListData"/>
22049 <xs:enumeration value="tariffOverallConstraintsData"/>
22050 <xs:enumeration value="tariffListData"/>
22051 <xs:enumeration value="tariffBoundaryRelationListData"/>
22052 <xs:enumeration value="tariffTierRelationListData"/>
22053 <xs:enumeration value="tariffDescriptionListData"/>
22054 <xs:enumeration value="tierBoundaryListData"/>
22055 <xs:enumeration value="tierBoundaryDescriptionListData"/>
22056 <xs:enumeration value="commodityListData"/>
22057 <xs:enumeration value="tierListData"/>
22058 <xs:enumeration value="tierIncentiveRelationListData"/>
22059 <xs:enumeration value="tierDescriptionListData"/>
22060 <xs:enumeration value="incentiveListData"/>
22061 <xs:enumeration value="incentiveDescriptionListData"/>
22062 <xs:enumeration value="incentiveTableData"/>
22063 <xs:enumeration value="incentiveTableDescriptionData"/>
22064 <xs:enumeration value="incentiveTableConstraintsData"/>
22065 <xs:enumeration value="electricalConnectionPermittedValueSetListData"/>
22066 <xs:enumeration value="useCaseInformationListData"/>
22067 <xs:enumeration value="nodeManagementUseCaseData"/>
22068 <xs:enumeration value="billConstraintsListData"/>
22069 <xs:enumeration value="billDescriptionListData"/>
22070 <xs:enumeration value="billListData"/>
22071 <xs:enumeration value="identificationListData"/>
22072 <xs:enumeration value="measurementSeriesListData"/>
22073 <xs:enumeration value="electricalConnectionCharacteristicListData"/>
22074 <xs:enumeration value="stateInformationListData"/>
22075 </xs:restriction>
22076 </xs:simpleType>
22077 <xs:complexType name="PossibleOperationsClassifierType">
22078 <xs:sequence>
22079 <xs:element minOccurs="0" name="partial" type="ns_p:ElementTagType"/>
22080 </xs:sequence>
22081 </xs:complexType>
22082 <xs:complexType name="PossibleOperationsReadType">
22083 <xs:complexContent>
22084 <xs:extension base="ns_p:PossibleOperationsClassifierType"/>
22085 </xs:complexContent>
22086 </xs:complexType>
22087 <xs:complexType name="PossibleOperationsWriteType">
22088 <xs:complexContent>
22089 <xs:extension base="ns_p:PossibleOperationsClassifierType"/>
22090 </xs:complexContent>
22091 </xs:complexType>
22092 <xs:complexType name="PossibleOperationsType">
22093 <xs:sequence>
22094 <xs:element name="read" minOccurs="0" type="ns_p:PossibleOperationsReadType"/>
22095 <xs:element name="write" minOccurs="0" type="ns_p:PossibleOperationsWriteType"/>
22096 </xs:sequence>
22097 </xs:complexType>
22098 <xs:complexType name="PossibleOperationsElementsType">
22099 <xs:sequence>
22100 <xs:element name="read" minOccurs="0" type="ns_p:ElementTagType"/>
```

```
22101         <xs:element name="write" minOccurs="0" type="ns_p:ElementTagType"/>
22102     </xs:sequence>
22103 </xs:complexType>
22104 <xs:complexType name="FunctionPropertyType">
22105     <xs:sequence>
22106         <xs:element minOccurs="0" name="function" type="ns_p:FunctionType"/>
22107         <xs:element minOccurs="0" name="possibleOperations"
22108 type="ns_p:PossibleOperationsType"/>
22109     </xs:sequence>
22110 </xs:complexType>
22111 <xs:complexType name="FunctionPropertyElementsType">
22112     <xs:sequence>
22113         <xs:element minOccurs="0" name="function" type="ns_p:ElementTagType"/>
22114         <xs:element minOccurs="0" name="possibleOperations"
22115 type="ns_p:PossibleOperationsElementsType"/>
22116     </xs:sequence>
22117 </xs:complexType>
22118 </xs:schema>
22119
22120
```

22121

22122 A.3.4 Datagram

22123 File name: EEBus_SPINE_TS_Datagram.xsd

```
22124
22125 <?xml version="1.0" encoding="UTF-8"?>
22126 <!--
22127     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
22128     Version 1.3.0 (final)
22129     2023-08-31
22130     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
22131     Source: https://www.eebus.org/en/specifications/
22132 -->
22133 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
22134 xmlns:xs="http://www.w3.org/2001/XMLSchema"
22135 targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0" blockDefault="#all"
22136 elementFormDefault="qualified">
22137     <xs:annotation>
22138         <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
22139 e.V. All rights reserved.</xs:documentation>
22140     </xs:annotation>
22141     <xs:include schemaLocation="EEBus_SPINE_TS_CommandFrame.xsd"/>
22142     <xs:complexType name="HeaderType">
22143         <xs:sequence>
22144             <xs:element minOccurs="0" name="specificationVersion"
22145 type="ns_p:SpecificationVersionType"/>
22146             <xs:element minOccurs="0" name="addressSource" type="ns_p:FeatureAddressType"/>
22147             <xs:element minOccurs="0" name="addressDestination"
22148 type="ns_p:FeatureAddressType"/>
22149             <xs:element minOccurs="0" name="addressOriginator"
22150 type="ns_p:FeatureAddressType"/>
22151             <xs:element minOccurs="0" name="msgCounter" type="ns_p:MsgCounterType"/>
22152             <xs:element minOccurs="0" name="msgCounterReference" type="ns_p:MsgCounterType"/>
22153             <xs:element minOccurs="0" name="cmdClassifier" type="ns_p:CmdClassifierType"/>
22154             <xs:element minOccurs="0" name="ackRequest" type="xs:boolean"/>
22155             <xs:element minOccurs="0" name="timestamp"
22156 type="ns_p:AbsoluteOrRelativeTimeType"/>
22157         </xs:sequence>
22158     </xs:complexType>
22159     <xs:element name="header" type="ns_p:HeaderType"/>
22160     <xs:complexType name="PayloadType">
22161         <xs:sequence>
22162             <xs:element maxOccurs="unbounded" minOccurs="0" name="cmd" type="ns_p:CmdType"/>
22163         </xs:sequence>
22164     </xs:complexType>
22165     <xs:element name="payload" type="ns_p:PayloadType"/>
22166     <xs:complexType name="DatagramType">
22167         <xs:sequence>
22168             <xs:element ref="ns_p:header"/>
22169             <xs:element ref="ns_p:payload"/>
22170         </xs:sequence>
22171     </xs:complexType>
22172     <xs:element name="datagram" type="ns_p:DatagramType"/>
22173 </xs:schema>
```

22174
22175

22176

22177 **A.3.5 Result**

22178 File name: EEBus_SPINE_TS_Result.xsd

```
22179
22180 <?xml version="1.0" encoding="UTF-8"?>
22181 <!--
22182     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
22183     Version 1.3.0 (final)
22184     2023-08-31
22185     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
22186     Source: https://www.eebus.org/en/specifications/
22187 -->
22188 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
22189   xmlns:xs="http://www.w3.org/2001/XMLSchema"
22190   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0"
22191   blockDefault="#all" elementFormDefault="qualified">
22192   <xs:annotation>
22193     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
22194     e.V. All rights reserved.</xs:documentation>
22195   </xs:annotation>
22196   <xs:include schemaLocation="EEBus_SPINE_TS_CommonDataTypes.xsd"/>
22197   <xs:simpleType name="ErrorNumberType">
22198     <xs:restriction base="xs:unsignedInt"/>
22199   </xs:simpleType>
22200   <xs:complexType name="ResultDataType">
22201     <xs:sequence>
22202       <xs:element minOccurs="0" name="errorNumber" type="ns_p:ErrorNumberType"/>
22203       <xs:element minOccurs="0" name="description" type="ns_p:DescriptionType"/>
22204     </xs:sequence>
22205   </xs:complexType>
22206   <xs:element name="resultData" type="ns_p:ResultDataType"/>
22207 </xs:schema>
22208
22209
22210
```

22211

22212 **A.3.6 Overview example**

22213 For each SPINE class, an additional *_overview.xsd is defined that references the functions defined
22214 by this class. The following example shows this for the class ActuatorLevel.

22215 File name: EEBus_SPINE_TS_ActuatorLevel_overview.xsd

```
22216
22217 <?xml version="1.0" encoding="UTF-8"?>
22218 <!--
22219     Smart Premises Interoperable Neutral-Message Exchange (SPINE)
22220     Version 1.3.0 (final)
22221     2023-08-31
22222     Copyright (c) 2023 EEBus Initiative e.V. All Rights Reserved.
22223     Source: https://www.eebus.org/en/specifications/
22224 -->
22225 <xs:schema xmlns:ns_p="http://docs.eebus.org/spine/xsd/v1"
22226   xmlns:xs="http://www.w3.org/2001/XMLSchema"
22227   targetNamespace="http://docs.eebus.org/spine/xsd/v1" version="1.3.0" blockDefault="#all"
22228   elementFormDefault="qualified">
22229   <xs:annotation>
22230     <xs:documentation>EEBus SPINE Specification schema. Copyright 2023 EEBus Initiative
22231     e.V. All rights reserved.</xs:documentation>
22232   </xs:annotation>
22233   <xs:include schemaLocation="EEBus_SPINE_TS_ActuatorLevel.xsd"/>
22234   <xs:group name="group_actuatorLevel">
22235     <xs:sequence>
22236       <xs:group ref="ns_p:actuatorLevel"/>
22237       <xs:group ref="ns_p:actuatorLevelDescription"/>
22238     </xs:sequence>
22239   </xs:group>
22240 </xs:schema>
```

```
22238         </xs:sequence>
22239     </xs:group>
22240     <xs:group name="actuatorLevel">
22241         <xs:sequence>
22242             <xs:element ref="ns_p:actuatorLevelData"/>
22243         </xs:sequence>
22244     </xs:group>
22245     <xs:group name="actuatorLevelDescription">
22246         <xs:sequence>
22247             <xs:element ref="ns_p:actuatorLevelDescriptionData"/>
22248         </xs:sequence>
22249     </xs:group>
22250 </xs:schema>
22251
22252
```

22253

Annex B - Resource Definitions Overview

B.1 Introduction

All Well Known Identifiers, defined by the EEBus Initiative e.V., are listed in this chapter. Detailed specifications can be found in the referenced sections.

B.2 Device Types

For detailed descriptions, please consider the corresponding device type definitions.

Device Type	Section	Description
ChargingStation	4.1.1	Represents a Charging Station, typically used in the context of electrical vehicle charging.
Dishwasher	4.1.2	A machine that cleans the dishes.
Dryer	4.1.3	A dryer for clothes (laundry) after they were washed.
ElectricitySupplySystem	4.1.4	Represents a set of sources and storages of electric power of a premises.
EnergyManagementSystem	4.1.5	Represents an Energy Management System (EMS).
EnvironmentSensor	4.1.6	Represents an environment sensor appliance.
Generic	4.1.7	Device with no specific content. May be used for developing, testing or evaluating purposes.
HeatGenerationSystem	4.1.8	Represents a heat generation system consisting of one or more heat generation appliances (for example gas boilers, heat pumps, solar thermal systems).
HeatSinkSystem	4.1.9	Represents the end user relevant HVAC system parts (heating circuits, DHW circuits, ventilation system) and system functions (heating, cooling, DHW, ventilation).
HeatStorageSystem	4.1.10	Represents a system of heat storages consisting of one or more heat storage appliances.
HVACController	4.1.11	Represents (as SPINE Device) the central controller appliance of the HVAC System.
Inverter	4.1.12	Represents a power Inverter.
SubMeter	4.1.13	Metering device for households or devices that is not billing relevant.
Washer	4.1.14	A washer for clothes (laundry).

Table 353: Well known device types

B.3 Entity Types

For detailed descriptions and rules, please consider the corresponding entity type definitions.

Entity Type	Section	Description
Battery	4.2.1	Appliance for storing (electrical) energy.
BatterySystem	4.2.2	A system consisting of one or more battery storages and one or more battery inverters.

CEM	4.2.3	Abbreviation for Customer Energy Manager and typically used to represent an Energy Manager within SPINE.
ChargingOutlet	4.2.4	Represents a Charging Outlet of a Charging Station.
Compressor	4.2.5	Represents the compressor of (e.g.) a heat pump appliance.
ControllableSystem	4.2.6	An appliance or a system of appliances controllable by an energy manager.
DeviceInformation	4.2.7	A special entity representing the complete device! All features that are included here hold information about the device and all entities.
DHWCircuit	4.2.8	Represents all elements for domestic hot water supply like cold water line, domestic hot water line, circulation line, tapping points. Should not include DHW storages, these should be represented as Entity Type "DHWStorage".
DHWStorage	4.2.9	Represents a heat storage appliance for storing domestic hot water.
Dishwasher	4.2.10	A functionality that cleans the dishes.
Dryer	4.2.11	A dryer for clothes (laundry) after they were washed.
ElectricalImmersionHeater	4.2.12	Represents an electrical immersion heater (electrical resistance heater) for example as part of a heat pump appliance, a heat source unit or a DHW storage.
ElectricalStorage	4.2.13	Entity used to categorize an inverter as (pure) battery inverter (no (e.g.) PV string(s) connected to the inverter).
ElectricityGenerationSystem	4.2.14	A system that is capable of producing electric power through local generation.
ElectricityStorageSystem	4.2.15	A system that is capable of storing electric power and releasing the majority of it as electric power.
EV	4.2.16	Represents an Electrical Vehicle.
EVSE	4.2.17	Abbreviation for Electrical Vehicle Supply Equipment, typically used to charge an Electrical Vehicle.
Fan	4.2.18	Represents an air fan, for example of an air water heat pump unit or a ventilation unit.
GasHeatingAppliance	4.2.19	Represents a heating appliance that generates heat (for example for heating or domestic hot water) by burning natural gas.
Generic	4.2.20	Entity with no specific content. May be used for developing, testing or evaluating purposes. Client functionality may be implemented within this entity, too.
GridConnectionPointOfPremises	4.2.21	The electrical connection point between the utility electricity grid and the electric circuit of a premises.
GridGuard	4.2.22	An energy manager located inside or outside the house or premises that limits or controls appliances.
HeatingBufferStorage	4.2.23	Represents a heat storage appliance for storing heating water.

HeatingCircuit	4.2.24	Represents a single heating circuit for space heating, including flow and return lines, optional mixing units (for mixed heating circuits) and the heat distribution: a) directly to one or more heated rooms (see HVACRoom) b) to one or more heating zones (see HeatingZone).
HeatingObject	4.2.25	A more general entity representing functionalities that convert (electrical) energy into heat.
HeatingZone	4.2.26	Represents a heating zone that provides heating energy to one or more heated rooms (see HVACRoom) One or more heating zones can be part of a heating circuit.
HeatPumpAppliance	4.2.27	Represents either a complete heat pump appliance (for example a brine water heat pump, indoor; an air water heat pump, completely indoor or completely outdoor) or the indoor part of a refrigerant split heat pump combination.
HeatSinkCircuit	4.2.28	Represents the circuit of a heat generator that delivers the heat energy to the heat sinks (for example to a heating circuit).
HeatSourceCircuit	4.2.29	Represents the circuit that delivers heat energy from an environmental heat source to a heat pump appliance (for example a brine circuit).
HeatSourceUnit	4.2.30	Represents an additional unit that provides the heat from the environmental heat source to the heat pump appliance (for example an outdoor ventilation unit that transfers the heat energy from the outside air to the heat source circuit of the heat pump appliance).
Household	4.2.31	A more virtual entity representing a complete household.
HVACController	4.2.32	Represents the central controller appliance of the HVAC System (as a SPINE Entity).
HVACRoom	4.2.33	Represents one physical room in a building/house that can be heated (H) / cooled, ventilated (V) or air conditioned (AC).
InstantDHWHeater	4.2.34	Represents an appliance for instant generation of domestic hot water (for example by using heat from a heating buffer storage).
Inverter	4.2.35	Represents an appliance that converts the DC-voltage (e.g. of a PV string) into AC-voltage.
OilHeatingAppliance	4.2.36	Represents a heating appliance that generates heat (for example for heating or domestic hot water) by burning oil.
Pump	4.2.37	Represents a pump to generate a flow rate (for example in a heating circuit).
PV	4.2.38	Entity used to categorize an inverter as (pure) PV inverter (no (e.g.) battery connected to the inverter).

PVESHybrid	4.2.39	Entity used to categorize an inverter as PV/battery hybrid inverter (battery or batteries and PV panel(s) connected to the inverter).
PVString	4.2.40	This Entity represents a (DC connected) PV panel connected to an inverter.
PVSystem	4.2.41	A system consisting of one or more PV inverters with one or more PV modules. The PVSystem provides aggregated PV related system parts of the house or premises.
RefrigerantCircuit	4.2.42	Represents the Refrigerant circuit of e.g. a heat pump appliance or a fridge or freezer.
SmartEnergyAppliance	4.2.43	Represents an appliance that allows shifting of its load via the SmartEnergyManagementPs concepts.
SolarDHWStorage	4.2.44	Represents a heat storage appliance for storing domestic hot water. The heat energy can be provided especially by solar thermal energy.
SolarThermalCircuit	4.2.45	Represents a single circuit of a solar thermal system including a solar thermal collector field and flow and return pipes to/from the collector field and pumps.
SubMeterElectricity	4.2.46	Electricity metering for households or functionalities that is not billing relevant.
TemperatureSensor	4.2.47	Represents a sensor that can provide a temperature measurement value.
Washer	4.2.48	A washer for clothes (laundry).

Table 354: Well known entity types

B.4 Feature Types

For detailed descriptions and rules, please consider the corresponding feature type definitions.

Feature Type	Section	Description
ActuatorLevel	4.3.1	Standard Feature Type for the Standard Class ActuatorLevel.
ActuatorSwitch	4.3.2	Standard Feature Type for the Standard Class ActuatorSwitch.
Alarm	4.3.3	Standard Feature Type for the Standard Class Alarm.
Bill	4.3.4	Standard Feature Type for the Standard Class Bill.
DataTunneling	4.3.5	Standard Feature Type for the Standard Class DataTunneling.
DeviceClassification	4.3.6	Standard Feature Type for the Standard Class DeviceClassification.
DeviceConfiguration	4.3.7	Standard Feature Type for the Standard Class DeviceConfiguration.
DeviceDiagnosis	4.3.8	Standard Feature Type for the Standard Class DeviceDiagnosis.
DirectControl	4.3.9	Standard Feature Type for the Standard Class DirectControl.
ElectricalConnection	4.3.10	Standard Feature Type for the Standard Class ElectricalConnection.

Generic	4.3.11	Feature with no specific content. May be used for developing, testing or evaluating purposes. Client functionality may be implemented within this feature, too.
HVAC	4.3.12	Standard Feature Type for the Standard Class HVAC.
Identification	4.3.13	Standard Feature Type for the Standard Class Identification.
IncentiveTable	4.3.14	Standard Feature Type for the Complex Class IncentiveTable.
LoadControl	4.3.15	Standard Feature Type for the Standard Class LoadControl.
Measurement	4.3.16	Standard Feature Type for the Standard Class Measurement.
Messaging	4.3.17	Standard Feature Type for the Standard Class Messaging.
NetworkManagement	4.3.18	Standard Feature Type for the Standard Class NetworkManagement.
NodeManagement	4.3.19	Standard Feature Type for the Complex Class NodeManagement.
OperatingConstraints	4.3.20	Standard Feature Type for the Standard Class OperatingConstraints.
PowerSequences	4.3.21	Standard Feature Type for the Standard Class PowerSequences.
Sensing	4.3.22	Standard Feature Type for the Standard Class Sensing.
Setpoint	4.3.23	Standard Feature Type for the Standard Class Setpoint.
SmartEnergyManagementPs	4.3.24	Standard Feature Type for the Complex Class SmartEnergyManagementPs.
StateInformation	4.3.25	Standard Feature Type for the Standard Class StateInformation.
SupplyCondition	4.3.26	Standard Feature Type for the Standard Class SupplyCondition.
TariffInformation	4.3.27	Standard Feature Type for the Standard Class TariffInformation.
TaskManagement	4.3.28	Standard Feature Type for the Standard Class TaskManagement.
Threshold	4.3.29	Standard Feature Type for the Standard Class Threshold.
TimeInformation	4.3.30	Standard Feature Type for the Standard Class TimeInformation.
TimeSeries	4.3.31	Standard Feature Type for the Standard Class TimeSeries.
TimeTable	4.3.32	Standard Feature Type for the Standard Class TimeTable.

Table 355: Well known feature types

B.5 Complex Classes

For detailed descriptions and rules, please consider the corresponding complex class definitions.

Complex Class	Section	Description
---------------	---------	-------------

IncentiveTable	5.2.1	The IncentiveTable data model allows to define tariffs with tiers and different incentives and boundaries.
NodeManagement	5.2.2	All data needed to manage nodes is combined in this complex class. Functions from the standard classes <i>NetworkManagement</i> , <i>BindingManagement</i> , <i>SubscriptionManagement</i> and <i>Version</i> are used.
SmartEnergyManagementPs	5.2.3	Shifting the load of devices in some given constraints is done with this complex class. Functions from the standard classes <i>PowerSequences</i> and <i>OperatingConstraints</i> are used.

22273 Table 356: Well known complex classes

22274

22275 **B.6 Standard Classes**

22276 For detailed descriptions and rules, please consider the corresponding standard class definitions.

Standard Class	Section	Description
ActuatorLevel	5.3.1	Dimming actuators, actuators with adjustability in stages.
ActuatorSwitch	5.3.2	Switch actuators.
Alarm	5.3.3	Alarms, generated due to exceeding thresholds (see corresponding class) or other events.
Bill	5.3.4	Bills for costs and profits of different purposes (e.g. a charging summary).
BindingManagement	5.3.5	Request / delete bindings or list existing bindings.
DataTunneling	5.3.6	Tunnels any (unspecified) data.
DeviceClassification	5.3.7	Free text information about a device (name, serial, etc.) and some enumerations like device type.
DeviceConfiguration	5.3.8	Used to model configuration data needed for interoperable functionality not fitting any other SPINE class.
DeviceDiagnosis	5.3.9	Information about the state of a device, heartbeat counting and service data.
DirectControl	5.3.10	Notification of instantaneous consumption/generation as well as instantaneous control.
ElectricalConnection	5.3.11	Information about specific parameters used for electrical connections, with relation to Measurement.
HVAC	5.3.12	Monitoring and controlling HVAC system functions (heating, cooling, ventilation, air conditioning, ...) and corresponding operation modes and overrun functionalities.
Identification	5.3.13	The Identification Class can be used to express certain identification values of the parent Entity.
LoadControl	5.3.14	Enabling (partial) (external) management of the load of a household or device by (e.g.) a DSO or some energy manager.
Measurement	5.3.15	General measurements (e.g. temperature, voltage, ...).
Messaging	5.3.16	Text messages (such as error messages, information, warnings, ...).
NetworkManagement	5.3.17	Add devices, remove etc., List of Inventory.
OperatingConstraints	5.3.18	Information on the power/energy constraints and implications.

PowerSequences	5.3.19	PowerSequences can be used to model energy management on a schedule basis.
Sensing	5.3.20	Various sensors, switches, contact sensors (e.g. window contact), smoke detectors, etc.
Setpoint	5.3.21	Setting desired values to control some functionality (with reference to Measurement and TimeTable).
SubscriptionManagement	5.3.22	Request / delete subscriptions or list existing subscriptions.
StateInformation	5.3.23	Information about different states the appliance supports (functionalities and failures).
SupplyCondition	5.3.24	Condition of power supply, limitations, etc.
TariffInformation	5.3.25	The TariffInformation Class provides functions for modelling tariffs.
TaskManagement	5.3.26	General status of device- or entity-wide tasks (triggered by user, internal mechanisms or external configuration).
Threshold	5.3.27	Thresholds for automatic behaviour of a device (may generate alarms for exceeding values).
TimeInformation	5.3.28	Time information (including the approximate synchronisation).
TimeSeries	5.3.29	TimeSeries is a versatile class that allows to express time dependent values as a time series of values.
TimeTable	5.3.30	Providing some amount of tables and slots to be used by other classes (e.g. Setpoint).
UseCaseInformation	5.3.31	To model which Use Cases are supported by a node the UseCaseInformation Class can be used.
Version	5.3.32	SPINE version information.

Table 357: Well known standard classes

B.7 Identifier list

The following table list all identifiers that belong to the identifiers concept as described in section 3.4.

Type Name	Used in elements	Base Type
AbsoluteOrRelativeTimeType	timestamp	<i>Union of xs:duration and xs:dateTime</i>
AddressDeviceType	device	xs:string
AddressEntityType	entity	xs:unsignedInt
AddressFeatureType	feature	xs:unsignedInt
AlarmIdType	alarmId	xs:unsignedInt
AlternativesIdType	alternativesId	xs:unsignedInt
BillCostIdType	costId	xs:unsignedInt
BillIdType	billId	xs:unsignedInt
BillPositionIdType	positionId	xs:unsignedInt
BillValueIdType	valueId	xs:unsignedInt
BindingIdType	bindingId	xs:unsignedInt
CommodityIdType	commodityId	xs:unsignedInt
ConditionIdType	conditionId	xs:unsignedInt
DeviceConfigurationKeyIdType	keyId	xs:unsignedInt

ElectricalConnectionIdType	electricalConnectionId	xs:unsignedInt
ElectricalConnectionCharacteristicIdType	characteristicId	xs:unsignedInt
ElectricalConnectionParameterIdType	parameterId	xs:unsignedInt
HvacOperationModelIdType	operationModelId	xs:unsignedInt
HvacOverrunIdType	overrunId	xs:unsignedInt
HvacSystemFunctionIdType	systemFunctionId	xs:unsignedInt
IdentificationIdType	identificationId	xs:unsignedInt
IncentiveIdType	incentiveId	xs:unsignedInt
LoadControlEventIdType	eventId	xs:unsignedInt
LoadControlLimitIdType	limitId	xs:unsignedInt
MeasurementIdType	measurementId	xs:unsignedInt
MessagingNumberType	messagingNumber	xs:unsignedInt
PowerSequenceIdType	sequenceId	xs:unsignedInt
PowerTimeSlotNumberType	slotNumber	xs:unsignedInt
SessionIdType	sessionId	xs:unsignedInt
SetpointIdType	setpointId	xs:unsignedInt
StateInformationIdType	stateInformationId	xs:unsignedInt
SubscriptionIdType	subscriptionId	xs:unsignedInt
TariffIdType	tariffId	xs:unsignedInt
TaskManagementJobIdType	jobId	xs:unsignedInt
ThresholdIdType	thresholdId	xs:unsignedInt
TierBoundaryIdType	boundaryId	xs:unsignedInt
TierIdType	tierId	xs:unsignedInt
TimeSeriesIdType	timeSeriesId	xs:unsignedInt
TimeSeriesSlotIdType	timeSeriesSlotId	xs:unsignedInt
TimeSlotIdType	timeSlotId	xs:unsignedInt
TimeTableIdType	timeTableId	xs:unsignedInt

Table 358: Identifier overview